# Namespace Lib9c

## Classes

Currencies

SerializeKeys

# Class Currencies

Namespace: Lib9c

Assembly: Lib9c.dll

```
public static class Currencies
```

**Inheritance**

object ← Currencies

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Crystal

```
public static readonly Currency Crystal
```

### Field Value

Currency

## DailyRewardRune

```
public static readonly Currency DailyRewardRune
```

### Field Value

Currency

## FreyaBlessingRune

```
public static readonly Currency FreyaBlessingRune
```

### Field Value

Currency

## FreyaLiberationRune

```
public static readonly Currency FreyaLiberationRune
```

### Field Value

Currency

## Garage

```
public static readonly Currency Garage
```

### Field Value

Currency

## GuildGold

```
public static readonly Currency GuildGold
```

### Field Value

Currency

## ItemCurrencyTickerPrefix

Ticker prefix shared by every item currency.

```
public const string ItemCurrencyTickerPrefix = "Item_"
```

### Field Value

string

## Mead

```
public static readonly Currency Mead
```

### Field Value

Currency

## OdinWeaknessRune

```
public static readonly Currency OdinWeaknessRune
```

### Field Value

Currency

## OdinWisdomRune

```
public static readonly Currency OdinWisdomRune
```

### Field Value

Currency

## StakeRune

```
public static readonly Currency StakeRune
```

### Field Value

Currency

## WrappedCurrencyTickerPrefix

Ticker prefix GetWrappedCurrency(Currency) puts in front of the wrapped ticker.

```
public const string WrappedCurrencyTickerPrefix = "FAV__"
```

### Field Value

string

# Methods

## GetCurrencyByTicker(string)

Gets the appropriate currency object for a given ticker. Uses GetMinterlessCurrency for supported tickers (CRYSTAL, GARAGE, Rune, Soulstone). NCG is not supported as it may have minters and requires special handling.

```
public static Currency GetCurrencyByTicker(string ticker)
```

### Parameters

`ticker` string

The currency ticker string

### Returns

Currency

The currency object for the given ticker

### Exceptions

ArgumentException

Thrown when ticker is NCG or not supported by GetMinterlessCurrency

## GetItemCurrency(int, bool)

Returns the currency that stands in for an item, which `ClaimItems` burns back into inventory items.

```
public static Currency GetItemCurrency(int itemId, bool tradable)
```

### Parameters

`itemId` int

The item the currency stands in for.

`tradable` bool

Whether the claimed item should be tradable. Note that this only takes effect for material rows — `InventoryExtensions.MintItem` ignores it for every other item type, so a costume claimed through a non tradable ticker is tradable all the same.

### Returns

Currency

The currency for the item.

## GetMinterlessCurrency(string?)

Covers the reward.CurrencyTicker is following cases: - Currencies.Crystal.Ticker - Currencies.Garage.Ticker - lower case is starting with "rune_" or "runestone_" - lower case is starting with "soulstone_"

```
public static Currency GetMinterlessCurrency(string? ticker)
```

### Parameters

`ticker` string

### Returns

Currency

### Exceptions

ArgumentNullException

ArgumentException

## GetRune(string?)

```
public static Currency GetRune(string? ticker)
```

### Parameters

`ticker` string

### Returns

Currency

## GetRunes(RuneSheet?)

```
public static IOrderedEnumerable<Currency> GetRunes(RuneSheet? sheet)
```

### Parameters

`sheet` RuneSheet

### Returns

IOrderedEnumerable<Currency>

## GetRunes(params string?[])

```
public static IOrderedEnumerable<Currency> GetRunes(params string?[] tickers)
```

### Parameters

`tickers` string[]

### Returns

IOrderedEnumerable<Currency>

## GetSoulStone(string?)

```
public static Currency GetSoulStone(string? ticker)
```

### Parameters

`ticker` string

### Returns

Currency

## GetSoulStones(PetSheet?)

```
public static IOrderedEnumerable<Currency> GetSoulStones(PetSheet? sheet)
```

### Parameters

`sheet` PetSheet

### Returns

IOrderedEnumerable<Currency>

## GetSoulStones(params string?[])

```
public static IOrderedEnumerable<Currency> GetSoulStones(params string?[] tickers)
```

### Parameters

`tickers` string[]

### Returns

IOrderedEnumerable<Currency>

## GetUnwrappedCurrency(Currency)

```
public static Currency GetUnwrappedCurrency(Currency currency)
```

### Parameters

`currency` Currency

### Returns

Currency

## GetWrappedCurrency(Currency)

Returns the wrapped counterpart of the currency: a minterless currency whose ticker carries WrappedCurrencyTickerPrefix. GetUnwrappedCurrency(Currency) reverses it.

```
public static Currency GetWrappedCurrency(Currency currency)
```

### Parameters

`currency` Currency

The currency to wrap.

### Returns

Currency

The wrapped currency.

## IsItemCurrencyTicker(string)

Whether the ticker denotes an item wrapped into a currency by GetItemCurrency(int, bool), however many times it was wrapped afterwards. Such a ticker exists per item id, so a policy sheet cannot enumerate them as rows.

```
public static bool IsItemCurrencyTicker(string ticker)
```

### Parameters

`ticker` string

The ticker to test.

### Returns

bool

## IsRuneTicker(string)

```
public static bool IsRuneTicker(string ticker)
```

### Parameters

`ticker` string

### Returns

bool

## IsSoulstoneTicker(string)

```
public static bool IsSoulstoneTicker(string ticker)
```

### Parameters

`ticker` string

### Returns

bool

## IsWrappedCurrency(Currency)

```
public static bool IsWrappedCurrency(Currency currency)
```

### Parameters

`currency` Currency

### Returns

bool

## ParseItemCurrency(Currency)

```
public static (bool tradable, int itemId) ParseItemCurrency(Currency currency)
```

### Parameters

`currency` Currency

### Returns

(bool tradable, int itemId)

## PickAddress(Currency, Address, Address)

pick address by currency ticker

```
public static Address PickAddress(Currency currency, Address agentAddress,
Address avatarAddress)
```

### Parameters

`currency` Currency

Libplanet.Types.Assets.Currency

`agentAddress` Address

Libplanet.Crypto.Address

`avatarAddress` Address

Libplanet.Crypto.Address

### Returns

Address

agentAddress when ticker is NCG or Crystal or Garage or Mead else avatarAddress

## UnwrapTicker(string)

Strips every WrappedCurrencyTickerPrefix the ticker was wrapped with, so that a policy keyed by the plain ticker also covers the wrapped form. A wrapper is not a different asset — `claim_items` unwraps it straight back.

```
public static string UnwrapTicker(string ticker)
```

### Parameters

`ticker` string

The ticker to unwrap.

### Returns

string

# Class SerializeKeys

Namespace: Lib9c

Assembly: Lib9c.dll

```
public static class SerializeKeys
```

**Inheritance**

object ← SerializeKeys

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Fields

### AchievementsKey

```
public const string AchievementsKey = "ach"
```

#### Field Value

string

### ActionPointKey

```
public const string ActionPointKey = "ap"
```

#### Field Value

string

### AdditionalStatValueKey

```
public const string AdditionalStatValueKey = "av"
```

### Field Value

string

## AddressKey

```
public const string AddressKey = "a"
```

### Field Value

string

## AgentAddressKey

```
public const string AgentAddressKey = "aga"
```

### Field Value

string

## AmountKey

```
public const string AmountKey = "am"
```

### Field Value

string

## ArenaProviderKey

```
public const string ArenaProviderKey = "arp"
```

### Field Value

string

## AvatarAddressKey

```
public const string AvatarAddressKey = "aa"
```

### Field Value

string

## BlockIndexKey

```
public const string BlockIndexKey = "bi"
```

### Field Value

string

## BuffSkillsKey

```
public const string BuffSkillsKey = "bs"
```

### Field Value

string

## BuyerAgentAddressKey

```
public const string BuyerAgentAddressKey = "bga"
```

### Field Value

string

## BuyerAvatarAddressKey

```
public const string BuyerAvatarAddressKey = "ba"
```

### Field Value

string

## ByCustomCraftKey

```
public const string ByCustomCraftKey = "bcc"
```

### Field Value

string

## CancellableBlockIndexKey

```
public const string CancellableBlockIndexKey = "cbi"
```

### Field Value

string

## ChampionshipIdKey

```
public const string ChampionshipIdKey = "chi"
```

### Field Value

string

## ChanceKey

```
public const string ChanceKey = "ch"
```

### Field Value

string

## CharacterIdKey

```
public const string CharacterIdKey = "ci"
```

### Field Value

string

## ChargeApKey

```
public const string ChargeApKey = "cha"
```

### Field Value

string

## ClaimDataKey

```
public const string ClaimDataKey = "cd"
```

### Field Value

string

## CombatPointKey

```
public const string CombatPointKey = "cp"
```

### Field Value

string

## CombinationSlotAddressesKey

```
public const string CombinationSlotAddressesKey = "csa"
```

### Field Value

string

## CompletedQuestIdsKeyDeprecated

```
public const string CompletedQuestIdsKeyDeprecated = "completedQuestIds"
```

### Field Value

string

## CostumeKey

```
public const string CostumeKey = "c"
```

### Field Value

string

## CostumesKey

```
public const string CostumesKey = "cs"
```

### Field Value

string

## CouponWalletKey

```
public const string CouponWalletKey = "coupon_wallet"
```

### Field Value

string

## CraftWithRandomKey

```
public const string CraftWithRandomKey = "cwr"
```

### Field Value

string

## DailyRewardReceivedIndexKey

```
public const string DailyRewardReceivedIndexKey = "dri"
```

### Field Value

string

## EarKey

```
public const string EarKey = "e"
```

### Field Value

string

## ElementalTypeKey

```
public const string ElementalTypeKey = "et"
```

### Field Value

string

## EndKey

```
public const string EndKey = "ed"
```

### Field Value

string

## EnemyAvatarAddressKey

```
public const string EnemyAvatarAddressKey = "eaa"
```

### Field Value

string

## EquipmentExpKey

```
public const string EquipmentExpKey = "eq_exp"
```

### Field Value

string

## EquipmentIconIdKey

```
public const string EquipmentIconIdKey = "icon_id"
```

### Field Value

string

## EquipmentsKey

```
public const string EquipmentsKey = "es"
```

### Field Value

string

## ErrorCodeKey

```
public const string ErrorCodeKey = "ec"
```

### Field Value

string

## EventMapKey

```
public const string EventMapKey = "em"
```

### Field Value

string

## ExpKey

```
public const string ExpKey = "exp"
```

### Field Value

string

## ExpiredBlockIndexKey

```
public const string ExpiredBlockIndexKey = "ebi"
```

### Field Value

string

## GradeKey

```
public const string GradeKey = "g"
```

### Field Value

string

## GrandFinaleIdKey

```
public const string GrandFinaleIdKey = "gfi"
```

### Field Value

string

## HairKey

```
public const string HairKey = "h"
```

### Field Value

string

## HasRandomOnlyIconKey

```
public const string HasRandomOnlyIconKey = "hroi"
```

### Field Value

string

## IdKey

```
public const string IdKey = "id"
```

### Field Value

string

## ItemCountKey

```
public const string ItemCountKey = "item_count"
```

### Field Value

string

## ItemIdKey

```
public const string ItemIdKey = "ii"
```

### Field Value

string

## ItemMapKey

```
public const string ItemMapKey = "im"
```

### Field Value

string

## ItemSubTypeKey

```
public const string ItemSubTypeKey = "ist"
```

### Field Value

string

## ItemTypeKey

```
public const string ItemTypeKey = "it"
```

### Field Value

string

## LegacyActionPointKey

```
public const string LegacyActionPointKey = "actionPoint"
```

### Field Value

string

## LegacyAdditionalStatValueKey

```
public const string LegacyAdditionalStatValueKey = "additionalValue"
```

### Field Value

string

## LegacyAddressKey

```
public const string LegacyAddressKey = "address"
```

### Field Value

string

## LegacyAgentAddressKey

```
public const string LegacyAgentAddressKey = "agentAddress"
```

### Field Value

string

## LegacyBlockIndexKey

```
public const string LegacyBlockIndexKey = "blockIndex"
```

### Field Value

string

## LegacyBuffSkillsKey

```
public const string LegacyBuffSkillsKey = "buffSkills"
```

### Field Value

string

## LegacyChanceKey

```
public const string LegacyChanceKey = "chance"
```

### Field Value

string

## LegacyCharacterIdKey

```
public const string LegacyCharacterIdKey = "characterId"
```

### Field Value

string

## LegacyCombinationSlotAddressesKey

```
public const string LegacyCombinationSlotAddressesKey = "combinationSlotAddresses"
```

### Field Value

string

## LegacyCostumeItemIdKey

```
public const string LegacyCostumeItemIdKey = "item_id"
```

### Field Value

string

## LegacyCostumeKey

```
public const string LegacyCostumeKey = "costume"
```

### Field Value

string

## LegacyDailyRewardReceivedIndexKey

```
public const string LegacyDailyRewardReceivedIndexKey = "dailyRewardReceivedIndex"
```

### Field Value

string

## LegacyEarKey

```
public const string LegacyEarKey = "ear"
```

### Field Value

string

## LegacyElementalTypeKey

```
public const string LegacyElementalTypeKey = "elemental_type"
```

### Field Value

string

## LegacyEquippedKey

```
public const string LegacyEquippedKey = "equipped"
```

### Field Value

string

## LegacyEventMapKey

```
public const string LegacyEventMapKey = "eventMap"
```

### Field Value

string

## LegacyGradeKey

```
public const string LegacyGradeKey = "grade"
```

### Field Value

string

## LegacyHairKey

```
public const string LegacyHairKey = "hair"
```

### Field Value

string

## LegacyInventoryKey

```
public const string LegacyInventoryKey = "inventory"
```

### Field Value

string

## LegacyItemIdKey

```
public const string LegacyItemIdKey = "itemId"
```

### Field Value

string

## LegacyItemMapKey

```
public const string LegacyItemMapKey = "itemMap"
```

### Field Value

string

## LegacyItemSubTypeKey

```
public const string LegacyItemSubTypeKey = "item_sub_type"
```

### Field Value

string

## LegacyItemTypeKey

```
public const string LegacyItemTypeKey = "item_type"
```

### Field Value

string

## LegacyItemUsableKey

```
public const string LegacyItemUsableKey = "itemUsable"
```

### Field Value

string

## LegacyLevelKey

```
public const string LegacyLevelKey = "level"
```

### Field Value

string

## LegacyMailBoxKey

```
public const string LegacyMailBoxKey = "mailBox"
```

### Field Value

string

## LegacyMonsterMapKey

```
public const string LegacyMonsterMapKey = "monsterMap"
```

### Field Value

string

## LegacyNameKey

```
public const string LegacyNameKey = "name"
```

### Field Value

string

## LegacyNonceKey

```
public const string LegacyNonceKey = "nonce"
```

### Field Value

string

## LegacyPowerKey

```
public const string LegacyPowerKey = "power"
```

### Field Value

string

## LegacyPriceKey

```
public const string LegacyPriceKey = "price"
```

### Field Value

string

## LegacyProductIdKey

```
public const string LegacyProductIdKey = "productId"
```

### Field Value

string

## LegacyProductsKey

```
public const string LegacyProductsKey = "products"
```

### Field Value

string

## LegacyQuestListKey

```
public const string LegacyQuestListKey = "questList"
```

### Field Value

string

## LegacyRankingMapAddressKey

```
public const string LegacyRankingMapAddressKey = "ranking_map_address"
```

### Field Value

string

## LegacyRequiredBlockIndexKey

```
public const string LegacyRequiredBlockIndexKey = "requiredBlockIndex"
```

### Field Value

string

## LegacySellerAgentAddressKey

```
public const string LegacySellerAgentAddressKey = "sellerAgentAddress"
```

### Field Value

string

## LegacySellerAvatarAddressKey

```
public const string LegacySellerAvatarAddressKey = "sellerAvatarAddress"
```

### Field Value

string

## LegacySetIdKey

```
public const string LegacySetIdKey = "set_id"
```

### Field Value

string

## LegacySkillRowKey

```
public const string LegacySkillRowKey = "skillRow"
```

### Field Value

string

## LegacySkillsKey

```
public const string LegacySkillsKey = "skills"
```

### Field Value

string

## LegacySpineResourcePathKey

```
public const string LegacySpineResourcePathKey = "spine_resource_path"
```

### Field Value

string

## LegacyStageMapKey

```
public const string LegacyStageMapKey = "stageMap"
```

### Field Value

string

## LegacyStatKey

```
public const string LegacyStatKey = "stat"
```

### Field Value

string

## LegacyStatTypeKey

```
public const string LegacyStatTypeKey = "statType"
```

### Field Value

string

## LegacyStatValueKey

```
public const string LegacyStatValueKey = "value"
```

### Field Value

string

## LegacyStatsMapKey

```
public const string LegacyStatsMapKey = "statsMap"
```

### Field Value

string

## LegacyTailKey

```
public const string LegacyTailKey = "tail"
```

### Field Value

string

## LegacyUpdatedAtKey

```
public const string LegacyUpdatedAtKey = "updatedAt"
```

### Field Value

string

## LegacyWorldInformationKey

```
public const string LegacyWorldInformationKey = "worldInformation"
```

### Field Value

string

## LensKey

```
public const string LensKey = "lens"
```

### Field Value

string

## LevelKey

```
public const string LevelKey = "l"
```

### Field Value

string

## MadeWithMimisbrunnrRecipeKey

```
public const string MadeWithMimisbrunnrRecipeKey = "mwmr"
```

### Field Value

string

## MailBoxKey

```
public const string MailBoxKey = "mb"
```

### Field Value

string

## MemoKey

```
public const string MemoKey = "m"
```

### Field Value

string

## MonsterCollectionResultKey

```
public const string MonsterCollectionResultKey = "mcr"
```

### Field Value

string

## MonsterCollectionRoundKey

```
public const string MonsterCollectionRoundKey = "s"
```

### Field Value

string

## MonsterMapKey

```
public const string MonsterMapKey = "mm"
```

### Field Value

string

## MyAvatarAddressKey

```
public const string MyAvatarAddressKey = "maa"
```

### Field Value

string

## NameKey

```
public const string NameKey = "n"
```

### Field Value

string

## OptionCountFromCombinationKey

```
public const string OptionCountFromCombinationKey = "oc"
```

### Field Value

string

## OrderDigestListKey

```
public const string OrderDigestListKey = "odl"
```

### Field Value

string

## OrderIdKey

```
public const string OrderIdKey = "oi"
```

### Field Value

string

## OrderReceiptListKey

```
public const string OrderReceiptListKey = "orl"
```

### Field Value

string

## OrderTypeKey

```
public const string OrderTypeKey = "ot"
```

### Field Value

string

## PowerKey

```
public const string PowerKey = "pw"
```

### Field Value

string

## PriceKey

```
public const string PriceKey = "p"
```

### Field Value

string

## ProductIdKey

```
public const string ProductIdKey = "pi"
```

### Field Value

string

## ProductsKey

```
public const string ProductsKey = "pr"
```

### Field Value

string

## PurchaseInfosKey

```
public const string PurchaseInfosKey = "pis"
```

### Field Value

string

## PurchaseResultsKey

```
public const string PurchaseResultsKey = "prs"
```

### Field Value

string

## QuantityKey

```
public const string QuantityKey = "q"
```

### Field Value

string

## QuestsKeyDeprecated

```
public const string QuestsKeyDeprecated = "quests"
```

### Field Value

string

## RankingMapAddressKey

```
public const string RankingMapAddressKey = "rma"
```

### Field Value

string

## RateKey

```
public const string RateKey = "r"
```

### Field Value

string

## ReceivedBlockIndexKey

```
public const string ReceivedBlockIndexKey = "rbi2"
```

### Field Value

string

## RequiredBlockIndexKey

```
public const string RequiredBlockIndexKey = "rbi"
```

### Field Value

string

## RequiredCharacterLevelKey

```
public const string RequiredCharacterLevelKey = "rc"
```

### Field Value

string

## RewardLevelKey

```
public const string RewardLevelKey = "rl"
```

### Field Value

string

## RewardLevelMapKey

```
public const string RewardLevelMapKey = "rlm"
```

### Field Value

string

## RewardMapKey

```
public const string RewardMapKey = "rm"
```

### Field Value

string

## RoundKey

```
public const string RoundKey = "rd"
```

### Field Value

string

## RuneInfos

```
public const string RuneInfos = "ri"
```

### Field Value

string

## SellerAgentAddressKey

```
public const string SellerAgentAddressKey = "sga"
```

### Field Value

string

## SellerAvatarAddressKey

```
public const string SellerAvatarAddressKey = "sva"
```

### Field Value

string

## SellerResultsKey

```
public const string SellerResultsKey = "srs"
```

### Field Value

string

## SetIdKey

```
public const string SetIdKey = "si"
```

### Field Value

string

## SkillsKey

```
public const string SkillsKey = "sk"
```

### Field Value

string

## SpineResourcePathKey

```
public const string SpineResourcePathKey = "srp"
```

### Field Value

string

## StageMapKey

```
public const string StageMapKey = "sm"
```

### Field Value

string

## StakeAvatarAddressKey

```
public const string StakeAvatarAddressKey = "saa"
```

### Field Value

string

## StartedBlockIndexKey

```
public const string StartedBlockIndexKey = "sbi"
```

### Field Value

string

## StatKey

```
public const string StatKey = "st"
```

### Field Value

string

## StatTypeKey

```
public const string StatTypeKey = "stt"
```

### Field Value

string

## StatValueKey

```
public const string StatValueKey = "v"
```

### Field Value

string

## TailKey

```
public const string TailKey = "t"
```

### Field Value

string

## TicketKey

```
public const string TicketKey = "tk"
```

### Field Value

string

## TradableFungibleItemCountKey

```
public const string TradableFungibleItemCountKey = "tradable_fungible_item_count"
```

### Field Value

string

## TradableFungibleItemKey

```
public const string TradableFungibleItemKey = "tradable_fungible_item"
```

### Field Value

string

## TradableIdKey

```
public const string TradableIdKey = "ti"
```

### Field Value

string

## UpdateSellInfoKey

```
public const string UpdateSellInfoKey = "usi"
```

### Field Value

string

## UpdatedAtKey

```
public const string UpdatedAtKey = "ua"
```

### Field Value

string

## updateSellOrderIdKey

```
public const string updateSellOrderIdKey = "roi"
```

## Field Value

string

# Namespace Lib9c.Exceptions

## Classes

InvalidStateTypeException

# Class InvalidStateTypeException

Namespace: Lib9c.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidStateTypeException : Exception, ISerializable
```

### Inheritance

object ← Exception ← InvalidStateTypeException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidStateTypeException(string)

```
public InvalidStateTypeException(string message)
```

#### Parameters

`message` string

# Namespace Lib9c.Model.Order

## Classes

FungibleOrder

NonFungibleOrder

Order

OrderBase

OrderDigest

OrderDigestListState

OrderFactory

OrderReceipt

## Enums

Order.OrderType

# Class FungibleOrder

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class FungibleOrder : Order
```

### Inheritance

object ← OrderBase ← Order ← FungibleOrder

### Inherited Members

Order.ExpirationInterval , Order.DeriveAddress(Guid) , Order.SellerAgentAddress , Order.SellerAvatarAddress , Order.Price , Order.ItemSubType , Order.GetTax() , Order.Cancel(AvatarState, long) , Order.Equals(Order) , OrderBase.OrderId , OrderBase.TradableId , OrderBase.StartedBlockIndex , OrderBase.ExpiredBlockIndex , OrderBase.Equals(OrderBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### FungibleOrder(Dictionary)

```
public FungibleOrder(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### FungibleOrder(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType, int)

```
public FungibleOrder(Address sellerAgentAddress, Address sellerAvatarAddress, Guid
orderId, FungibleAssetValue price, Guid tradableId, long startedBlockIndex,
ItemSubType itemSubType, int itemCount)
```

### Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`tradableId` Guid

`startedBlockIndex` long

`itemSubType` ItemSubType

`itemCount` int

## Fields

### ItemCount

```
public readonly int ItemCount
```

#### Field Value

int

## Properties

### Type

```
public override Order.OrderType Type { get; }
```

#### Property Value

Order.OrderType

# Methods

## Cancel2(AvatarState, long)

```
[Obsolete("Use Cancel")]
public override ITradableItem Cancel2(AvatarState avatarState, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`blockIndex` long

### Returns

ITradableItem

## Digest(AvatarState, CostumeStatSheet)

```
public override OrderDigest Digest(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

### Returns

OrderDigest

## Digest2(AvatarState, CostumeStatSheet)

```
[Obsolete("Use Digest")]
public override OrderDigest Digest2(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

### Returns

OrderDigest

## Equals(FungibleOrder)

```
protected bool Equals(FungibleOrder other)
```

### Parameters

`other` FungibleOrder

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Sell(AvatarState)

```
[Obsolete("Not Used")]
public override ITradableItem Sell(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell2(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell2(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell3(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell3(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell4(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell4(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Transfer(AvatarState, AvatarState, long)

```
public override OrderReceipt Transfer(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Transfer2(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public override OrderReceipt Transfer2(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Transfer3(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public override OrderReceipt Transfer3(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Validate(AvatarState, int)

```
public override void Validate(AvatarState avatarState, int count)
```

### Parameters

`avatarState` AvatarState

`count` int

## ValidateCancelOrder(AvatarState, Guid)

```
public override void ValidateCancelOrder(AvatarState avatarState, Guid tradableId)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

## ValidateCancelOrder2(AvatarState, Guid)

```
[Obsolete("Use ValidateCancelOrder")]
public override void ValidateCancelOrder2(AvatarState avatarState, Guid tradableId)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

## ValidateTransfer(AvatarState, Guid, FungibleAssetValue, long)

```
public override int ValidateTransfer(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

### Returns

int

## ValidateTransfer2(AvatarState, Guid, FungibleAssetValue, long)

```
[Obsolete("Use ValidateTransfer")]
public override int ValidateTransfer2(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

## Returns

int

# Class NonFungibleOrder

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class NonFungibleOrder : Order
```

**Inheritance**

object ← OrderBase ← Order ← NonFungibleOrder

**Inherited Members**

Order.ExpirationInterval , Order.DeriveAddress(Guid) , Order.SellerAgentAddress , Order.SellerAvatarAddress , Order.Price , Order.ItemSubType , Order.GetTax() , Order.Cancel(AvatarState, long) , Order.Serialize() , Order.Equals(Order) , Order.Equals(object) , Order.GetHashCode() , OrderBase.OrderId , OrderBase.TradableId , OrderBase.StartedBlockIndex , OrderBase.ExpiredBlockIndex , OrderBase.Equals(OrderBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### NonFungibleOrder(Dictionary)

```
public NonFungibleOrder(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### NonFungibleOrder(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType)

```
public NonFungibleOrder(Address sellerAgentAddress, Address sellerAvatarAddress,
Guid orderId, FungibleAssetValue price, Guid tradableId, long startedBlockIndex,
```

```
ItemSubType itemSubType)
```

### Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`tradableId` Guid

`startedBlockIndex` long

`itemSubType` ItemSubType

## Properties

### Type

```
public override Order.OrderType Type { get; }
```

### Property Value

Order.OrderType

## Methods

### Cancel2(AvatarState, long)

```
[Obsolete("Use Cancel")]
public override ITradableItem Cancel2(AvatarState avatarState, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`blockIndex` long

### Returns

ITradableItem

## Digest(AvatarState, CostumeStatSheet)

```
public override OrderDigest Digest(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

### Returns

OrderDigest

## Digest2(AvatarState, CostumeStatSheet)

```
[Obsolete("Use Digest")]
public override OrderDigest Digest2(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

### Returns

OrderDigest

## Sell(AvatarState)

```
public override ITradableItem Sell(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell2(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell2(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell3(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell3(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell4(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell4(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Transfer(AvatarState, AvatarState, long)

```
public override OrderReceipt Transfer(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Transfer2(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public override OrderReceipt Transfer2(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Transfer3(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public override OrderReceipt Transfer3(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Validate(AvatarState, int)

```
public override void Validate(AvatarState avatarState, int count)
```

### Parameters

`avatarState` AvatarState

`count` int

## ValidateCancelOrder(AvatarState, Guid)

```
public override void ValidateCancelOrder(AvatarState avatarState, Guid tradableId)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

## ValidateCancelOrder2(AvatarState, Guid)

```
[Obsolete("Use ValidateCancelOrder")]
public override void ValidateCancelOrder2(AvatarState avatarState, Guid tradableId)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

## ValidateTransfer(AvatarState, Guid, FungibleAssetValue, long)

```
public override int ValidateTransfer(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

## Returns

int

# ValidateTransfer2(AvatarState, Guid, FungibleAssetValue, long)

```
[Obsolete("Use ValidateTransfer")]
public override int ValidateTransfer2(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

## Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

## Returns

int

# Class Order

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Order : OrderBase
```

**Inheritance**

object ← OrderBase ← Order

**Derived**

FungibleOrder, NonFungibleOrder

**Inherited Members**

OrderBase.OrderId , OrderBase.TradableId , OrderBase.StartedBlockIndex , OrderBase.ExpiredBlockIndex , OrderBase.Equals(OrderBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Order(Dictionary)

```
protected Order(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### Order(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType)

```
protected Order(Address sellerAgentAddress, Address sellerAvatarAddress, Guid
orderId, FungibleAssetValue price, Guid tradableId, long startedBlockIndex,
ItemSubType itemSubType)
```

### Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`tradableId` Guid

`startedBlockIndex` long

`itemSubType` ItemSubType

## Fields

### ExpirationInterval

```
public const long ExpirationInterval = 36000
```

#### Field Value

long

## Properties

### ItemSubType

```
public ItemSubType ItemSubType { get; }
```

#### Property Value

ItemSubType

### Price

```
public FungibleAssetValue Price { get; }
```

### Property Value

FungibleAssetValue

## SellerAgentAddress

```
public Address SellerAgentAddress { get; }
```

### Property Value

Address

## SellerAvatarAddress

```
public Address SellerAvatarAddress { get; }
```

### Property Value

Address

## Type

```
public abstract Order.OrderType Type { get; }
```

### Property Value

Order.OrderType

# Methods

## Cancel(AvatarState, long)

```
public ITradableItem Cancel(AvatarState avatarState, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`blockIndex` long

### Returns

ITradableItem

## Cancel2(AvatarState, long)

```
[Obsolete("Use Cancel")]
public abstract ITradableItem Cancel2(AvatarState avatarState, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`blockIndex` long

### Returns

ITradableItem

## DeriveAddress(Guid)

```
public static Address DeriveAddress(Guid orderId)
```

### Parameters

`orderId` Guid

### Returns

## Digest(AvatarState, CostumeStatSheet)

```
public abstract OrderDigest Digest(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

### Returns

OrderDigest

## Digest2(AvatarState, CostumeStatSheet)

```
[Obsolete("Use Digest")]
public abstract OrderDigest Digest2(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

### Returns

OrderDigest

## Equals(Order)

```
protected bool Equals(Order other)
```

### Parameters

`other` Order

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## GetTax()

```
public FungibleAssetValue GetTax()
```

### Returns

FungibleAssetValue

## Sell(AvatarState)

```
public abstract ITradableItem Sell(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell2(AvatarState)

```
[Obsolete("Use Sell")]
public abstract ITradableItem Sell2(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell3(AvatarState)

```
[Obsolete("Use Sell")]
public abstract ITradableItem Sell3(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Sell4(AvatarState)

```
[Obsolete("Use Sell")]
public abstract ITradableItem Sell4(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

### Returns

ITradableItem

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Transfer(AvatarState, AvatarState, long)

```
public abstract OrderReceipt Transfer(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Transfer2(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public abstract OrderReceipt Transfer2(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Transfer3(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public abstract OrderReceipt Transfer3(AvatarState seller, AvatarState buyer,
long blockIndex)
```

### Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

### Returns

OrderReceipt

## Validate(AvatarState, int)

```
public virtual void Validate(AvatarState avatarState, int count)
```

### Parameters

`avatarState` AvatarState

`count` int

## ValidateCancelOrder(AvatarState, Guid)

```
public virtual void ValidateCancelOrder(AvatarState avatarState, Guid tradableId)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

## ValidateCancelOrder2(AvatarState, Guid)

```
[Obsolete("Use ValidateCancelOrder")]
public virtual void ValidateCancelOrder2(AvatarState avatarState, Guid tradableId)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

## ValidateTransfer(AvatarState, Guid, FungibleAssetValue, long)

```
public virtual int ValidateTransfer(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

### Returns

int

## ValidateTransfer2(AvatarState, Guid, FungibleAssetValue, long)

```
[Obsolete("Use ValidateTransfer")]
public virtual int ValidateTransfer2(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

## Returns

int

# Enum Order.OrderType

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
public enum Order.OrderType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Fungible = 0`

`NonFungible = 1`

# Class OrderBase

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class OrderBase
```

**Inheritance**

object ← OrderBase

**Derived**

Order, OrderDigest

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### OrderBase(Dictionary)

```
public OrderBase(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### OrderBase(Guid, Guid, long, long)

```
public OrderBase(Guid orderId, Guid tradableId, long startedBlockIndex,
long expiredBlockIndex)
```

#### Parameters

`orderId` Guid

`tradableId` Guid

`startedBlockIndex` long

`expiredBlockIndex` long

# Fields

## OrderId

```
public readonly Guid OrderId
```

### Field Value

Guid

## TradableId

```
public readonly Guid TradableId
```

### Field Value

Guid

# Properties

## ExpiredBlockIndex

```
public long ExpiredBlockIndex { get; }
```

### Property Value

long

## StartedBlockIndex

```
public long StartedBlockIndex { get; }
```

### Property Value

long

# Methods

## Equals(OrderBase)

```
protected bool Equals(OrderBase other)
```

### Parameters

`other` OrderBase

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

```
public virtual IValue Serialize()
```

### Returns

IValue

# Class OrderDigest

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class OrderDigest : OrderBase
```

**Inheritance**

object ← OrderBase ← OrderDigest

**Inherited Members**

OrderBase.OrderId , OrderBase.TradableId , OrderBase.StartedBlockIndex , OrderBase.ExpiredBlockIndex , OrderBase.Equals(OrderBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### OrderDigest(Dictionary)

```
public OrderDigest(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### OrderDigest(Address, long, long, Guid, Guid, FungibleAssetValue, long, int, int, int)

```
public OrderDigest(Address sellerAgentAddress, long startedBlockIndex, long
expiredBlockIndex, Guid orderId, Guid tradableId, FungibleAssetValue price, long
combatPoint, int level, int itemId, int itemCount)
```

#### Parameters

`sellerAgentAddress` Address

`startedBlockIndex` long

`expiredBlockIndex` long

`orderId` Guid

`tradableId` Guid

`price` FungibleAssetValue

`combatPoint` long

`level` int

`itemId` int

`itemCount` int

# Fields

## CombatPoint

```
public readonly long CombatPoint
```

### Field Value

long

## ItemCount

```
public readonly int ItemCount
```

### Field Value

int

## ItemId

```
public readonly int ItemId
```

### Field Value

int

## Level

```
public readonly int Level
```

### Field Value

int

## Price

```
public readonly FungibleAssetValue Price
```

### Field Value

FungibleAssetValue

## SellerAgentAddress

```
public readonly Address SellerAgentAddress
```

### Field Value

Address

## Methods

## Equals(OrderDigest)

```
protected bool Equals(OrderDigest other)
```

### Parameters

`other` OrderDigest

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class OrderDigestListState

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class OrderDigestListState
```

**Inheritance**

object ← OrderDigestListState

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### OrderDigestListState(Dictionary)

```
public OrderDigestListState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### OrderDigestListState(Address)

```
public OrderDigestListState(Address address)
```

#### Parameters

`address` Address

## Fields

## Address

```
public readonly Address Address
```

### Field Value

Address

# Properties

## OrderDigestList

```
public IReadOnlyList<OrderDigest> OrderDigestList { get; }
```

### Property Value

IReadOnlyList<OrderDigest>

# Methods

## Add(OrderDigest)

```
public void Add(OrderDigest orderDigest)
```

### Parameters

`orderDigest` OrderDigest

## DeriveAddress(Address)

```
public static Address DeriveAddress(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

### Returns

Address

## Equals(OrderDigestListState)

```
protected bool Equals(OrderDigestListState other)
```

### Parameters

`other` OrderDigestListState

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Remove(Guid)

```
public void Remove(Guid orderId)
```

### Parameters

`orderId` Guid

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class OrderFactory

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
public static class OrderFactory
```

**Inheritance**

object ← OrderFactory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### Create(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType, int)

```
public static Order Create(Address agentAddress, Address avatarAddress, Guid
orderId, FungibleAssetValue price, Guid tradableId, long startedIndex, ItemSubType
itemSubType, int count)
```

#### Parameters

`agentAddress` Address

`avatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`tradableId` Guid

`startedIndex` long

`itemSubType` ItemSubType

`count` int

### Returns

Order

## CreateFungibleOrder(Address, Address, Guid, FungibleAssetValue, Guid, long, int, ItemSubType)

```
public static FungibleOrder CreateFungibleOrder(Address sellerAgentAddress, Address
sellerAvatarAddress, Guid orderId, FungibleAssetValue price, Guid itemId, long
startedBlockIndex, int count, ItemSubType itemSubType)
```

### Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`itemId` Guid

`startedBlockIndex` long

`count` int

`itemSubType` ItemSubType

### Returns

FungibleOrder

## CreateNonFungibleOrder(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType)

```
public static NonFungibleOrder CreateNonFungibleOrder(Address sellerAgentAddress,
Address sellerAvatarAddress, Guid orderId, FungibleAssetValue price, Guid itemId,
long startedBlockIndex, ItemSubType itemSubType)
```

### Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`itemId` Guid

`startedBlockIndex` long

`itemSubType` ItemSubType

### Returns

NonFungibleOrder

## Deserialize(Dictionary)

```
public static Order Deserialize(Dictionary dictionary)
```

### Parameters

`dictionary` Dictionary

### Returns

Order

# Class OrderReceipt

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class OrderReceipt
```

**Inheritance**

object ← OrderReceipt

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### OrderReceipt(Dictionary)

```
public OrderReceipt(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### OrderReceipt(Guid, Address, Address, long)

```
public OrderReceipt(Guid orderId, Address buyerAgentAddress, Address
buyerAvatarAddress, long transferredBlockIndex)
```

#### Parameters

`orderId` Guid

`buyerAgentAddress` Address

`buyerAvatarAddress` Address

`transferredBlockIndex` long

# Fields

## BuyerAgentAddress

```
public readonly Address BuyerAgentAddress
```

### Field Value

Address

## BuyerAvatarAddress

```
public readonly Address BuyerAvatarAddress
```

### Field Value

Address

## OrderId

```
public readonly Guid OrderId
```

### Field Value

Guid

## TransferredBlockIndex

```
public readonly long TransferredBlockIndex
```

### Field Value

long

## Methods

## DeriveAddress(Guid)

```
public static Address DeriveAddress(Guid orderId)
```

### Parameters

`orderId` Guid

### Returns

Address

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Namespace Nekoyume

## Classes

ActionObsoleteAttributeExtension

ActionObsoleteConfig

Addresses

CurrencyExtensions

GameConfig

GameConfig.MaxEquipmentSlotCount

GameConfig.RequireCharacterLevel

GameConfig.RequireClearedStageLevel

MeadConfig

PlanetExtension

StoreUtils

## Structs

AppProtocolVersionExtra

## Enums

Planet

# Class ActionObsoleteAttributeExtension

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class ActionObsoleteAttributeExtension
```

**Inheritance**

object ← ActionObsoleteAttributeExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### IsObsolete(IEnumerable<ActionObsoleteAttribute>, Planet, long)

```
public static bool IsObsolete(this IEnumerable<ActionObsoleteAttribute> attrs,
Planet planet, long blockIndex)
```

#### Parameters

`attrs` IEnumerable<ActionObsoleteAttribute>

`planet` Planet

`blockIndex` long

#### Returns

bool

# Class ActionObsoleteConfig

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class ActionObsoleteConfig
```

**Inheritance**

object ← ActionObsoleteConfig

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## V100080ObsoleteIndex

```
public const long V100080ObsoleteIndex = 2448000
```

### Field Value

long

## V100081ObsoleteIndex

```
public const long V100081ObsoleteIndex = 2550000
```

### Field Value

long

## V100083ObsoleteIndex

```
public const long V100083ObsoleteIndex = 2680000
```

### Field Value

long

## V100086ObsoleteIndex

```
public const long V100086ObsoleteIndex = 2800001
```

### Field Value

long

## V100089ObsoleteIndex

```
public const long V100089ObsoleteIndex = 2908000
```

### Field Value

long

## V100093ObsoleteIndex

```
public const long V100093ObsoleteIndex = 3050000
```

### Field Value

long

## V100095ObsoleteIndex

```
public const long V100095ObsoleteIndex = 3317632
```

### Field Value

long

## V100096ObsoleteIndex

```
public const long V100096ObsoleteIndex = 3317632
```

### Field Value

long

## V100170ObsoleteIndex

```
public const long V100170ObsoleteIndex = 3810000
```

### Field Value

long

## V100190ObsoleteIndex

```
public const long V100190ObsoleteIndex = 4204863
```

### Field Value

long

## V100193ObsoleteIndex

```
public const long V100193ObsoleteIndex = 3975929
```

### Field Value

long

## V100200ObsoleteIndex

```
public const long V100200ObsoleteIndex = 4246225
```

### Field Value

long

## V100210ObsoleteIndex

```
public const long V100210ObsoleteIndex = 4366622
```

### Field Value

long

## V100220ObsoleteIndex

```
public const long V100220ObsoleteIndex = 4390272
```

### Field Value

long

## V100230ObsoleteIndex

```
public const long V100230ObsoleteIndex = 4558559
```

### Field Value

long

## V100240ObsoleteIndex

```
public const long V100240ObsoleteIndex = 4578480
```

### Field Value

long

## V100260ObsoleteIndex

```
public const long V100260ObsoleteIndex = 4596180
```

### Field Value

long

## V100270ObsoleteIndex

```
public const long V100270ObsoleteIndex = 4841774
```

### Field Value

long

## V100282ObsoleteIndex

```
public const long V100282ObsoleteIndex = 4835445
```

### Field Value

long

## V100290ObsoleteIndex

```
public const long V100290ObsoleteIndex = 4913153
```

### Field Value

long

## V100300ObsoleteIndex

```
public const long V100300ObsoleteIndex = 5150000
```

### Field Value

long

## V100301ExecutedBlockIndex

```
public const long V100301ExecutedBlockIndex = 5048399
```

### Field Value

long

## V100310ExecutedBlockIndex

```
public const long V100310ExecutedBlockIndex = 5217577
```

### Field Value

long

## V100310ObsoleteIndex

```
public const long V100310ObsoleteIndex = 5300000
```

### Field Value

long

## V100320ObsoleteIndex

```
public const long V100320ObsoleteIndex = 5398001
```

### Field Value

long

## V100340ObsoleteIndex

```
public const long V100340ObsoleteIndex = 5800000
```

### Field Value

long

## V100351ObsoleteIndex

```
public const long V100351ObsoleteIndex = 5828750
```

### Field Value

long

## V100360ObsoleteIndex

```
public const long V100360ObsoleteIndex = 6020000
```

### Field Value

long

## V200010ObsoleteIndex

```
public const long V200010ObsoleteIndex = 6642114
```

### Field Value

long

## V200020AccidentObsoleteIndex

```
public const long V200020AccidentObsoleteIndex = 7000000
```

### Field Value

long

## V200020ObsoleteIndex

```
public const long V200020ObsoleteIndex = 6856587
```

### Field Value

long

## V200030ObsoleteIndex

```
public const long V200030ObsoleteIndex = 7200000
```

### Field Value

long

## V200031ObsoleteIndex

```
public const long V200031ObsoleteIndex = 7206000
```

### Field Value

long

## V200040ObsoleteIndex

```
public const long V200040ObsoleteIndex = 7330000
```

### Field Value

long

## V200060ObsoleteIndex

```
public const long V200060ObsoleteIndex = 7649999
```

### Field Value

long

## V200061ObsoleteIndex

```
public const long V200061ObsoleteIndex = 7660000
```

### Field Value

long

## V200063ObsoleteIndex

```
public const long V200063ObsoleteIndex = 7700000
```

### Field Value

long

## V200070ObsoleteIndex

```
public const long V200070ObsoleteIndex = 7716400
```

### Field Value

long

## V200071ObsoleteIndex

```
public const long V200071ObsoleteIndex = 7718878
```

### Field Value

long

## V200080ObsoleteIndex

```
public const long V200080ObsoleteIndex = 7983895
```

### Field Value

long

## V200090ObsoleteIndex

```
public const long V200090ObsoleteIndex = 8070865
```

### Field Value

long

## V200092ObsoleteIndex

```
public const long V200092ObsoleteIndex = 8324909
```

### Field Value

long

# Class Addresses

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class Addresses
```

**Inheritance**

object ← Addresses

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AbstainHistory

An address of an account having AbstainHistory.

```
public static readonly Address AbstainHistory
```

### Field Value

Address

## ActionPoint

```
public static readonly Address ActionPoint
```

### Field Value

Address

## ActivatedAccount

```
public static readonly Address ActivatedAccount
```

### Field Value

Address

## Admin

```
public static readonly Address Admin
```

### Field Value

Address

## AdventureBoss

```
public static readonly Address AdventureBoss
```

### Field Value

Address

## AdventureCp

```
public static readonly Address AdventureCp
```

### Field Value

Address

## Agent

```
public static readonly Address Agent
```

### Field Value

Address

## Arena

```
public static readonly Address Arena
```

### Field Value

Address

## ArenaCp

```
public static readonly Address ArenaCp
```

### Field Value

Address

## AssetMinters

```
public static readonly Address AssetMinters
```

### Field Value

Address

## AuthorizedMiners

```
public static readonly Address AuthorizedMiners
```

### Field Value

Address

## Avatar

```
public static readonly Address Avatar
```

### Field Value

Address

## Battle

```
public static readonly Address Battle
```

### Field Value

Address

## Blacksmith

```
public static readonly Address Blacksmith
```

### Field Value

Address

## BountyBoard

```
public static readonly Address BountyBoard
```

### Field Value

Address

## ClaimedGiftIds

```
public static readonly Address ClaimedGiftIds
```

### Field Value

Address

## Collection

```
public static readonly Address Collection
```

### Field Value

Address

## CombinationSlot

```
public static readonly Address CombinationSlot
```

### Field Value

Address

## CommunityPool

An address for community fund.

```
public static readonly Address CommunityPool
```

### Field Value

Address

## Credits

```
public static readonly Address Credits
```

### Field Value

Address

## DailyReward

```
public static readonly Address DailyReward
```

### Field Value

Address

## EmptyAccountAddress

```
public static readonly Address EmptyAccountAddress
```

### Field Value

Address

## EventDungeon

```
public static readonly Address EventDungeon
```

### Field Value

Address

## ExploreBoard

```
public static readonly Address ExploreBoard
```

### Field Value

Address

## ExplorerList

```
public static readonly Address ExplorerList
```

### Field Value

Address

## GameConfig

```
public static readonly Address GameConfig
```

### Field Value

Address

## GarageWallet

```
public static readonly Address GarageWallet
```

### Field Value

Address

## GasPool

```
public static readonly Address GasPool
```

### Field Value

Address

## GoldCurrency

```
public static readonly Address GoldCurrency
```

### Field Value

Address

## GoldDistribution

```
public static readonly Address GoldDistribution
```

### Field Value

Address

## Guild

An address of an account having Guild.

```
public static readonly Address Guild
```

### Field Value

Address

## GuildApplication

An address of an account having Nekoyume.Model.Guild.GuildApplication.

```
public static readonly Address GuildApplication
```

### Field Value

Address

## GuildBond

An address of an account having Bond.

```
public static readonly Address GuildBond
```

### Field Value

Address

## GuildDelegateeMetadata

An address of an account having DelegateeMetadata.

```
public static readonly Address GuildDelegateeMetadata
```

### Field Value

Address

## GuildDelegatorMetadata

An address of an account having DelegatorMetadata.

```
public static readonly Address GuildDelegatorMetadata
```

### Field Value

Address

## GuildLumpSumRewardsRecord

An address of an account having LumpSumRewardsRecord.

```
public static readonly Address GuildLumpSumRewardsRecord
```

### Field Value

Address

## GuildMemberCounter

An address of an account having Bencodex.Types.Integer which means the number of the guild.

```
public static readonly Address GuildMemberCounter
```

### Field Value

Address

## GuildParticipant

An address of an account having GuildParticipant

```
public static readonly Address GuildParticipant
```

### Field Value

Address

## GuildRebondGrace

An address of an account having RebondGrace.

```
public static readonly Address GuildRebondGrace
```

### Field Value

Address

## GuildRejoinCooldown

An address of an account having Nekoyume.Model.Guild.GuildRejoinCooldown

```
public static readonly Address GuildRejoinCooldown
```

### Field Value

Address

## GuildRewardBase

An address of an account having RewardBase.

```
public static readonly Address GuildRewardBase
```

### Field Value

Address

## GuildUnbondLockIn

An address of an account having UnbondLockIn.

```
public static readonly Address GuildUnbondLockIn
```

### Field Value

Address

## GuildUnbondingSet

An address of an account having Delegation.UnbondingSet.

```
public static readonly Address GuildUnbondingSet
```

### Field Value

Address

## InfiniteTowerBoard

Address for the infinite tower board state.

```
public static readonly Address InfiniteTowerBoard
```

### Field Value

Address

## InfiniteTowerCp

```
public static readonly Address InfiniteTowerCp
```

### Field Value

Address

## InfiniteTowerInfo

Address for the infinite tower info state.

```
public static readonly Address InfiniteTowerInfo
```

### Field Value

Address

## Inventory

```
public static readonly Address Inventory
```

### Field Value

Address

## Market

```
public static readonly Address Market
```

### Field Value

Address

## MaterialCost

```
public static readonly Address MaterialCost
```

### Field Value

Address

## Migration

An account address for migration.

```
public static readonly Address Migration
```

### Field Value

Address

## MortgagePool

```
public static readonly Address MortgagePool
```

### Field Value

Address

## NonValidatorDelegatee

An address for non validator.

```
[Obsolete("It is not used anymore.")]
public static readonly Address NonValidatorDelegatee
```

### Field Value

Address

## PatrolReward

```
public static readonly Address PatrolReward
```

### Field Value

Address

## PayMaster

```
public static readonly Address PayMaster
```

### Field Value

Address

## PendingActivation

```
public static readonly Address PendingActivation
```

### Field Value

Address

## QuestList

```
public static readonly Address QuestList
```

### Field Value

Address

## Raid

```
public static readonly Address Raid
```

### Field Value

Address

## RaidCp

```
public static readonly Address RaidCp
```

### Field Value

Address

## Ranking

```
public static readonly Address Ranking
```

### Field Value

Address

## RedeemCode

```
public static readonly Address RedeemCode
```

### Field Value

Address

## Relationship

```
public static readonly Address Relationship
```

### Field Value

Address

## RewardPool

An address for reward pool of validators.

```
public static readonly Address RewardPool
```

### Field Value

Address

## Rune

```
public static readonly Address Rune
```

### Field Value

Address

## RuneState

```
public static readonly Address RuneState
```

### Field Value

Address

## Shop

```
public static readonly Address Shop
```

### Field Value

Address

## StageRandomBuff

```
public static readonly Address StageRandomBuff
```

### Field Value

Address

## SuperCraft

```
public static readonly Address SuperCraft
```

### Field Value

Address

## SwapPool

An address of an account having SwapPool.

```
public static readonly Address SwapPool
```

### Field Value

Address

## TableSheet

```
public static readonly Address TableSheet
```

### Field Value

Address

## UnlockEquipmentRecipe

```
public static readonly Address UnlockEquipmentRecipe
```

### Field Value

Address

## UnlockWorld

```
public static readonly Address UnlockWorld
```

### Field Value

Address

## ValidatorBond

An address of an account having Bond.

```
public static readonly Address ValidatorBond
```

### Field Value

Address

## ValidatorDelegatee

An address of an account having ValidatorDelegatee.

```
public static readonly Address ValidatorDelegatee
```

### Field Value

Address

## ValidatorDelegateeMetadata

An address of an account having DelegateeMetadata.

```
public static readonly Address ValidatorDelegateeMetadata
```

### Field Value

Address

## ValidatorDelegator

An address of an account having ValidatorDelegator.

```
public static readonly Address ValidatorDelegator
```

### Field Value

Address

## ValidatorDelegatorMetadata

An address of an account having DelegatorMetadata.

```
public static readonly Address ValidatorDelegatorMetadata
```

### Field Value

Address

## ValidatorList

An address of an account having ValidatorList.

```
public static readonly Address ValidatorList
```

### Field Value

Address

## ValidatorLumpSumRewardsRecord

An address of an account having LumpSumRewardsRecord.

```
public static readonly Address ValidatorLumpSumRewardsRecord
```

### Field Value

Address

## ValidatorRebondGrace

An address of an account having RebondGrace.

```
public static readonly Address ValidatorRebondGrace
```

### Field Value

Address

## ValidatorRewardBase

An address of an account having RewardBase.

```
public static readonly Address ValidatorRewardBase
```

### Field Value

Address

## ValidatorUnbondLockIn

An address of an account having UnbondLockIn.

```
public static readonly Address ValidatorUnbondLockIn
```

### Field Value

Address

## ValidatorUnbondingSet

An address of an account having Delegation.UnbondingSet.

```
public static readonly Address ValidatorUnbondingSet
```

### Field Value

Address

## WeeklyArena

```
public static readonly Address WeeklyArena
```

### Field Value

Address

## WorldInformation

```
public static readonly Address WorldInformation
```

### Field Value

Address

# Methods

## AdventureSeasonAddress(long)

```
public static Address AdventureSeasonAddress(long season)
```

### Parameters

`season` long

### Returns

Address

## CheckAgentHasPermissionOnBalanceAddr(Address, Address)

```
public static bool CheckAgentHasPermissionOnBalanceAddr(Address agentAddr,
Address balanceAddr)
```

### Parameters

`agentAddr` Address

`balanceAddr` Address

### Returns

bool

## CheckAvatarAddrIsContainedInAgent(Address, Address)

```
public static bool CheckAvatarAddrIsContainedInAgent(Address agentAddr,
Address avatarAddr)
```

### Parameters

`agentAddr` Address

`avatarAddr` Address

### Returns

bool

## CheckInventoryAddrIsContainedInAgent(Address, Address)

```
public static bool CheckInventoryAddrIsContainedInAgent(Address agentAddr,
Address inventoryAddr)
```

### Parameters

`agentAddr` Address

`inventoryAddr` Address

### Returns

bool

## GetArenaParticipantAccountAddress(int, int)

Get the account address of an arena participant.

```
public static Address GetArenaParticipantAccountAddress(int championshipId,
int round)
```

### Parameters

`championshipId` int

`round` int

### Returns

Address

The account address of an arena participant.
"0100000000000000000000000000000000000000" ~
"0199999999999999999999999999999999999999".

## GetAvatarAddress(Address, int)

```
public static Address GetAvatarAddress(Address agentAddr, int index)
```

### Parameters

`agentAddr` Address

`index` int

### Returns

Address

## GetCombinationSlotAddress(Address, int)

```
public static Address GetCombinationSlotAddress(Address avatarAddr, int index)
```

### Parameters

`avatarAddr` Address

`index` int

### Returns

Address

## GetCpAccountAddress(BattleType)

Gets the account address for storing CP based on the battle type.

```
public static Address GetCpAccountAddress(BattleType battleType)
```

### Parameters

`battleType` BattleType

The type of battle (Adventure, Raid, or Arena)

### Returns

Address

The address where CP is stored for the given battle type

### Exceptions

ArgumentException

Thrown when an invalid battle type is provided

## GetDailyCrystalCostAddress(int)

```
public static Address GetDailyCrystalCostAddress(int index)
```

### Parameters

`index` int

### Returns

Address

## GetGarageAddress(Address, HashDigest<SHA256>)

```
public static Address GetGarageAddress(Address agentAddr,
HashDigest<SHA256> fungibleId)
```

### Parameters

`agentAddr` Address

`fungibleId` HashDigest<SHA256>

### Returns

Address

## GetGarageBalanceAddress(Address)

```
public static Address GetGarageBalanceAddress(Address agentAddr)
```

### Parameters

`agentAddr` Address

### Returns

Address

## GetGuildBanAccountAddress(Address)

Build an Libplanet.Crypto.Address of an Libplanet.Action.State.Account, represented as `agentAddress` ↔ Bencodex.Types.Boolean, indicates whether the `agentAddress` is banned.

```
public static Address GetGuildBanAccountAddress(Address guildAddress)
```

### Parameters

`guildAddress` Address

The guild address.

### Returns

Address

An account address.

## GetHammerPointStateAddress(Address, int)

```
public static Address GetHammerPointStateAddress(Address avatarAddress,
int recipeId)
```

### Parameters

`avatarAddress` Address

`recipeId` int

### Returns

Address

## GetInventoryAddress(Address, int)

```
public static Address GetInventoryAddress(Address agentAddr, int avatarIndex)
```

### Parameters

`agentAddr` Address

`avatarIndex` int

### Returns

Address

## GetItemAddress(Guid)

```
public static Address GetItemAddress(Guid itemId)
```

### Parameters

`itemId` Guid

### Returns

Address

## GetRaiderAddress(Address, int)

```
public static Address GetRaiderAddress(Address avatarAddress, int raidId)
```

### Parameters

`avatarAddress` Address

`raidId` int

### Returns

Address

## GetRaiderListAddress(int)

```
public static Address GetRaiderListAddress(int raidId)
```

### Parameters

`raidId` int

### Returns

Address

## GetSheetAddress(string)

```
public static Address GetSheetAddress(string sheetName)
```

### Parameters

`sheetName` string

### Returns

Address

## GetSheetAddress<T>()

```
public static Address GetSheetAddress<T>() where T : ISheet
```

### Returns

Address

### Type Parameters

`T`

## GetSkillStateAddressFromAvatarAddress(Address)

```
public static Address GetSkillStateAddressFromAvatarAddress(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

### Returns

Address

## GetWeeklyCrystalCostAddress(int)

```
public static Address GetWeeklyCrystalCostAddress(int index)
```

### Parameters

`index` int

### Returns

Address

## GetWorldBossAddress(int)

```
public static Address GetWorldBossAddress(int raidId)
```

### Parameters

`raidId` int

### Returns

Address

## GetWorldBossKillRewardRecordAddress(Address, int)

```
public static Address GetWorldBossKillRewardRecordAddress(Address avatarAddress,
int raidId)
```

### Parameters

`avatarAddress` Address

`raidId` int

### Returns

Address

## IsArenaParticipantAccountAddress(Address)

Check if the given address is within the range of arena participant account addresses. "0100000000000000000000000000000000000000" ~ "0199999999999999999999999999999999999999".

```
public static bool IsArenaParticipantAccountAddress(Address address)
```

## Parameters

`address` Address

## Returns

bool

# Struct AppProtocolVersionExtra

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public readonly struct AppProtocolVersionExtra
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### AppProtocolVersionExtra(Dictionary)

```
public AppProtocolVersionExtra(Dictionary dictionary)
```

#### Parameters

`dictionary` Dictionary

### AppProtocolVersionExtra(string, string, DateTimeOffset)

```
public AppProtocolVersionExtra(string macOSBinaryUrl, string windowsBinaryUrl,
DateTimeOffset timestamp)
```

#### Parameters

`macOSBinaryUrl` string

`windowsBinaryUrl` string

`timestamp` DateTimeOffset

# Fields

## MacOSBinaryUrl

```
public readonly string MacOSBinaryUrl
```

### Field Value

string

## Timestamp

```
public readonly DateTimeOffset Timestamp
```

### Field Value

DateTimeOffset

## WindowsBinaryUrl

```
public readonly string WindowsBinaryUrl
```

### Field Value

string

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class CurrencyExtensions

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class CurrencyExtensions
```

**Inheritance**

object ← CurrencyExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### Deserialize(Dictionary)

```
public static Currency Deserialize(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

#### Returns

Currency

### Serialize(Currency)

```
public static Dictionary Serialize(this Currency currency)
```

#### Parameters

`currency` Currency

## Returns

Dictionary

# Class GameConfig

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class GameConfig
```

**Inheritance**

object ← GameConfig

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ActionCostAP

```
public const int ActionCostAP = 5
```

### Field Value

int

## ArenaChallengeCountMax

```
public const int ArenaChallengeCountMax = 5
```

### Field Value

int

## ArenaScoreDefault

```
public const int ArenaScoreDefault = 1000
```

### Field Value

int

## AvatarNickNamePattern

```
public const string AvatarNickNamePattern = "^[0-9a-zA-Z]{2,20}$"
```

### Field Value

string

## DefaultAttackId

```
public const int DefaultAttackId = 100000
```

### Field Value

int

## DefaultAvatarArmorId

```
public const int DefaultAvatarArmorId = 10200000
```

### Field Value

int

## DefaultAvatarCharacterId

```
public const int DefaultAvatarCharacterId = 100010
```

### Field Value

int

## DefaultAvatarWeaponId

```
public const int DefaultAvatarWeaponId = 10100000
```

### Field Value

int

## EnhanceEquipmentCostAP

```
public const int EnhanceEquipmentCostAP = 0
```

### Field Value

int

## IsEditor

```
public static readonly bool IsEditor
```

### Field Value

bool

## MaximumProbability

```
public const int MaximumProbability = 10000
```

### Field Value

int

## MimisbrunnrStartStageId

```
public const int MimisbrunnrStartStageId = 10000001
```

### Field Value

int

## MimisbrunnrWorldId

```
public const int MimisbrunnrWorldId = 10001
```

### Field Value

int

## RequiredAppraiseBlock

```
[Obsolete("Use GameConfigState.RequiredAppraiseBlock")]
public const int RequiredAppraiseBlock = 50
```

### Field Value

int

## SlotCount

```
public const int SlotCount = 3
```

## Field Value

int

# Class GameConfig.MaxEquipmentSlotCount

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class GameConfig.MaxEquipmentSlotCount
```

**Inheritance**

object ← GameConfig.MaxEquipmentSlotCount

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Armor

```
public const int Armor = 1
```

### Field Value

int

## Aura

```
public const int Aura = 1
```

### Field Value

int

## Belt

```
public const int Belt = 1
```

### Field Value

int

## Grimoire

```
public const int Grimoire = 1
```

### Field Value

int

## Necklace

```
public const int Necklace = 1
```

### Field Value

int

## Ring

```
public const int Ring = 2
```

### Field Value

int

## Weapon

```
public const int Weapon = 1
```

## Field Value

int

# Class GameConfig.RequireCharacterLevel

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class GameConfig.RequireCharacterLevel
```

**Inheritance**

object ← GameConfig.RequireCharacterLevel

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Fields

### CharacterConsumableSlot1

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot1")]
public const int CharacterConsumableSlot1 = 1
```

#### Field Value

int

### CharacterConsumableSlot2

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot2")]
public const int CharacterConsumableSlot2 = 35
```

#### Field Value

int

## CharacterConsumableSlot3

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot3")]
public const int CharacterConsumableSlot3 = 100
```

### Field Value

int

## CharacterConsumableSlot4

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot4")]
public const int CharacterConsumableSlot4 = 200
```

### Field Value

int

## CharacterConsumableSlot5

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot5")]
public const int CharacterConsumableSlot5 = 350
```

### Field Value

int

## CharacterEarCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EarCostumeSlot")]
public const int CharacterEarCostumeSlot = 2
```

### Field Value

int

## CharacterEquipmentSlotArmor

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotArmor")]
public const int CharacterEquipmentSlotArmor = 3
```

### Field Value

int

## CharacterEquipmentSlotAura

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotAura")]
public const int CharacterEquipmentSlotAura = 1
```

### Field Value

int

## CharacterEquipmentSlotBelt

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotBelt")]
public const int CharacterEquipmentSlotBelt = 5
```

### Field Value

int

## CharacterEquipmentSlotNecklace

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotNecklace")]
public const int CharacterEquipmentSlotNecklace = 8
```

### Field Value

int

## CharacterEquipmentSlotRing1

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotRing1")]
public const int CharacterEquipmentSlotRing1 = 13
```

### Field Value

int

## CharacterEquipmentSlotRing2

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotRing2")]
public const int CharacterEquipmentSlotRing2 = 46
```

### Field Value

int

## CharacterEquipmentSlotWeapon

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotWeapon")]
public const int CharacterEquipmentSlotWeapon = 1
```

### Field Value

int

## CharacterEyeCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EyeCostumeSlot")]
public const int CharacterEyeCostumeSlot = 2
```

### Field Value

int

## CharacterFullCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_FullCostumeSlot")]
public const int CharacterFullCostumeSlot = 2
```

### Field Value

int

## CharacterHairCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_HairCostumeSlot")]
public const int CharacterHairCostumeSlot = 2
```

### Field Value

int

## CharacterTailCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_TailCostumeSlot")]
public const int CharacterTailCostumeSlot = 2
```

### Field Value

int

## CharacterTitleSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_TitleSlot")]
```

```
public const int CharacterTitleSlot = 1
```

## Field Value

int

# Class GameConfig.RequireClearedStageLevel

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class GameConfig.RequireClearedStageLevel
```

**Inheritance**

object ← GameConfig.RequireClearedStageLevel

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Fields

### ActionsInMimisbrunnr

```
[Obsolete("Not used anymore since v200092")]
public const int ActionsInMimisbrunnr = 100
```

#### Field Value

int

### ActionsInRaid

```
[Obsolete("Not used anymore since v200092")]
public const int ActionsInRaid = 50
```

#### Field Value

int

## ActionsInRankingBoard

```
[Obsolete("Not used anymore since v200092")]
public const int ActionsInRankingBoard = 25
```

### Field Value

int

## ActionsInShop

```
[Obsolete("Not used anymore since v200092")]
public const int ActionsInShop = 17
```

### Field Value

int

## CombinationConsumableAction

```
[Obsolete("Not used anymore since v200092")]
public const int CombinationConsumableAction = 6
```

### Field Value

int

## CombinationEquipmentAction

```
[Obsolete("Not used anymore since v200092")]
public const int CombinationEquipmentAction = 3
```

### Field Value

int

## ItemEnhancementAction

```
[Obsolete("Not used anymore since v200092")]
public const int ItemEnhancementAction = 9
```

### Field Value

int

# Class MeadConfig

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class MeadConfig
```

**Inheritance**

object ← MeadConfig

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## PatronAddress

```
public static readonly Address PatronAddress
```

### Field Value

Address

# Enum Planet

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public enum Planet : byte
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

```
Heimdall = 1
```

```
HeimdallInternal = 5
```

```
Idun = 2
```

```
Odin = 0
```

```
OdinInternal = 4
```

```
Thor = 3
```

```
ThorInternal = 6
```

# Class PlanetExtension

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class PlanetExtension
```

**Inheritance**

object ← PlanetExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### DeterminePlanet(ITransaction)

```
public static Planet? DeterminePlanet(this ITransaction tx)
```

#### Parameters

`tx` ITransaction

#### Returns

Planet?

# Class StoreUtils

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class StoreUtils
```

**Inheritance**

object ← StoreUtils

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BackupNameFormat

```
public const string BackupNameFormat = "{0}_{1:yyyyMMdd_HHmmss}"
```

### Field Value

string

## BackupNamePattern

```
public const string BackupNamePattern = "{0}_*"
```

### Field Value

string

# Methods

## ClearBackupStores(string)

```
public static void ClearBackupStores(string storePath)
```

### Parameters

`storePath` string

## ResetStore(string)

```
public static void ResetStore(string storePath)
```

### Parameters

`storePath` string

# Namespace Nekoyume.Action

## Classes

ActionBase

ActionContextExtensions

ActionObsoleteAttribute

An attribute on an Libplanet.Action.IAction to indicate that such Libplanet.Action.IAction is obsoleted after a certain index.

Due to a bug introduced earlier and to keep backward compatibility, an Libplanet.Action.IAction with this attribute is obsoleted, i.e. cannot be included, starting from Obsolete Index + 2.

ActionObsoletedException

ActionPointExceededException

ActionUnavailableException

ActivateCollection

ActivatedAccountsDoesNotExistsException

ActivationException

AddRedeemCode

AddressExtension

AdminPermissionException

AgentStateNotContainsAvatarAddressException

AlreadyActivatedException

AlreadyClaimedGiftsException

AlreadyContractedException

AlreadyReceivedException

AlreadyRecipeUnlockedException

AlreadyWorldUnlockedException

AppraiseBlockNotReachedException

ApprovePledge

ArenaNotEndedException

AssetInfo

AttachmentActionResult

AuraSummon

AvatarIndexAlreadyUsedException

AvatarIndexOutOfRangeException

BalanceDoesNotExistsException

BattleArena
Introduce at https://github.com/planetarium/lib9c/pull/2229 Changed at https://github.com/planetarium/lib9c/pull/2242

BurnAsset

Buy
Updated at https://github.com/planetarium/lib9c/pull/1164

Buy7

Buy7.BuyerMultipleResult

Buy7.BuyerResult

Buy7.PurchaseResult

Buy7.SellerMultipleResult

Buy7.SellerResult

BuyMultiple
Introduced at https://github.com/planetarium/lib9c/pull/331 Updated at many pull requests Obsoleted at https://github.com/planetarium/lib9c/pull/487 Updated at many pull requests Updated at https://github.com/planetarium/lib9c/pull/957

BuyMultiple.BuyerResult

BuyMultiple.PurchaseInfo

BuyMultiple.PurchaseResult

BuyMultiple.SellerResult

BuyProduct

CancelProductRegistration

ChargeActionPoint
Hard forked at https://github.com/planetarium/lib9c/pull/430 Updated at https://github.com/planetarium/lib9c/pull/474 Updated at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

ClaimGifts
An action to claim gifts using the Gift Id.
Refer ClaimableGiftsSheet to find gifts data.
It can only claim once at the specified block index.

ClaimItems

ClaimPatrolReward
Claim patrol reward

ClaimRaidReward

ClaimStakeReward
Hard forked at https://github.com/planetarium/lib9c/pull/2097

ClaimStakeReward2

ClaimStakeReward8

ClaimWordBossKillReward

ClaimWorldBossReward
The ClaimWorldBossReward class is an action that allows a player to claim rewards for their contribution in defeating a world boss. This action ensures that only the owner of the avatar can claim the reward and calculates the reward based on the player's contribution.

ClaimableGiftsNotAvailableException

CombinationConsumable

Hard forked at https://github.com/planetarium/lib9c/pull/2195

CombinationConsumable5

CombinationConsumable5.ResultModel

CombinationEquipment
Hard forked at https://github.com/planetarium/lib9c/pull/2195

CombinationEquipment5

CombinationSlotResultNullException

CombinationSlotUnlockException

ConsumableSlotOutOfRangeException

ConsumableSlotUnlockException

CostumeSlotUnlockException

CostumeSummon

CreateAvatar
Hard forked at https://github.com/planetarium/lib9c/pull/2195

CreatePendingActivation
Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at https://github.com/planetarium/lib9c/pull/957

CreatePendingActivations
Updated at https://github.com/planetarium/lib9c/pull/746 Updated at https://github.com/planetarium/lib9c/pull/957

CreatePledge
Admin action for pledge contract

DailyReward
Hard forked at https://github.com/planetarium/lib9c/pull/1828 Updated at https://github.com/planetarium/lib9c/pull/1828

DailyReward2

DailyReward2.DailyRewardResult

DuplicateCostumeException

DuplicateEquipmentException

DuplicateMaterialException

DuplicateOrderIdException

EndPledge

EquipmentLevelExceededException

EquipmentSlotUnlockException

EventConsumableItemCrafts

Hard forked at https://github.com/planetarium/lib9c/pull/2195

EventDungeonBattle

Action for participating in an event dungeon battle. This action allows an avatar to battle in a specific stage of an event dungeon, consuming tickets and receiving rewards upon victory. Hard forked at https://github.com/planetarium/lib9c/pull/2195

EventDungeonBattleSweep

Represents a sweep action that automatically completes multiple event dungeon battles without player interaction.

This action allows players to automatically repeat battles on already cleared event dungeon stages, consuming tickets and receiving rewards without manual combat interaction.

Key Features: - Only works on previously cleared stages (no stage clearing) - No ticket purchase functionality (uses existing tickets only) - Automatic reward and experience calculation - Multiple battle execution in a single action

This action modifies the following states: - AvatarState: Updates player's stats, inventory, and experience - EventDungeonInfo: Updates event dungeon progress and ticket usage - RuneState: Updates rune effects and bonuses - ItemSlotState: Updates equipped items - RuneSlotState: Updates equipped runes - CP: Updates character's combat power

Execution Flow: 1. Validate input parameters and avatar state 2. Load and validate game sheets and configurations 3. Validate event dungeon info and ticket availability 4. Check stage clearance requirements 5. Update rune and item slot states 6. Calculate combat power (CP) 7. Generate rewards and experience 8. Update avatar state and return modified world state

Hard forked at https://github.com/planetarium/lib9c/pull/2195

EventMaterialItemCrafts

Hard forked at https://github.com/planetarium/lib9c/pull/2195

FailedLoadSheetException

FailedLoadStateException

FavProductInfo

GameAction

GoldDistribution

GrantItems

Grants items/assets to target avatars from token-like inputs. This action is restricted to a fixed allow-list of signers and the current admin signer. Even if the signer does not have sufficient token balance, the grant will proceed and will burn only the available amount of the token.

Grinding

HackAndSlash

Represents a battle action in the game where a player engages in combat with monsters. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- WorldBossKillRewardRecord: Records boss kill rewards
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power Hard forked at https://github.com/planetarium/lib9c/pull/2195

HackAndSlashRandomBuff

Created at https://github.com/planetarium/lib9c/pull/1031

HackAndSlashSweep

Represents a sweep action that automatically completes multiple battles without player interaction. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses

- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power Hard forked at https://github.com/planetarium/lib9c/pull/2195

InfiniteTowerBattle

Represents a battle action in the infinite tower where players engage in combat with various conditions and restrictions. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- InfiniteTowerInfo: Updates tower progress and tickets
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

InitializeStates

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at NCG distribution(7e4515b6e14cc5d6eb716d5ebb587ab04b4246f9) Updated at https://github.com/planetarium/lib9c/pull/19 Updated at https://github.com/planetarium/lib9c/pull/36 Updated at https://github.com/planetarium/lib9c/pull/42 Updated at https://github.com/planetarium/lib9c/pull/55 Updated at https://github.com/planetarium/lib9c/pull/57 Updated at https://github.com/planetarium/lib9c/pull/60 Updated at https://github.com/planetarium/lib9c/pull/102 Updated at https://github.com/planetarium/lib9c/pull/128 Updated at https://github.com/planetarium/lib9c/pull/167 Updated at https://github.com/planetarium/lib9c/pull/422 Updated at https://github.com/planetarium/lib9c/pull/747 Updated at https://github.com/planetarium/lib9c/pull/798 Updated at https://github.com/planetarium/lib9c/pull/957

InvalidAddressException

InvalidClaimException

InvalidCurrencyException

InvalidElementalException

InvalidEquipmentException

InvalidItemCountException

InvalidItemIdException

Represents an exception that is thrown when an invalid item ID is used.

InvalidItemTypeException

InvalidLevelException

InvalidMaterialException

InvalidMinterException

InvalidMonsterCollectionRoundException

InvalidNamePatternException

InvalidPriceException

InvalidProductTypeException

InvalidRecipeIdException

InvalidRepeatPlayException

InvalidShopItemException

InvalidSignatureException

InvalidSlotIndexException

InvalidStageException

InvalidTradableIdException

InvalidTradableItemException

InvalidTransferCurrencyException

InvalidTransferMinterException

InvalidTransferRecipientException

InvalidTransferSignerException

InvalidTransferUnactivatedRecipientException

InvalidWorldException

IssueToken

IssueTokensFromGarage

ItemDoesNotExistException

ItemEnhancement
  Updated at https://github.com/planetarium/lib9c/pull/2195

ItemEnhancement10
  Updated at https://github.com/planetarium/lib9c/pull/1176

ItemEnhancement10.ResultModel

ItemEnhancement11
  Updated at https://github.com/planetarium/lib9c/pull/1164

ItemEnhancement11.ResultModel

ItemEnhancement12
  Updated at https://github.com/planetarium/lib9c/pull/2068

ItemEnhancement12.ResultModel

ItemEnhancement13
  Updated at https://github.com/planetarium/lib9c/pull/2068

ItemEnhancement13.ResultModel

ItemEnhancement7

ItemEnhancement7.ResultModel

ItemEnhancement9

ItemEnhancement9.ResultModel

ItemProductInfo

JoinArena
  Introduced at https://github.com/planetarium/lib9c/pull/2195

JoinArena1
  Introduced at https://github.com/planetarium/lib9c/pull/1495

JoinArena3
  Introduced at https://github.com/planetarium/lib9c/pull/1663

MemoLengthOverflowException

MigrateAgentAvatar

MigrateFee

MigrateMonsterCollection
An action to claim remained monster collection rewards and to migrate MonsterCollectionState into LegacyStakeState without cancellation, to keep its staked period.

MigrationActivatedAccountsState

MintAssets

MonsterCollectionExistingClaimableException

MonsterCollectionExistingException
An exception thrown when there is unexpected monster collection.

MonsterCollectionExpiredException

MonsterCollectionLevelException

NotEnoughActionPointException

NotEnoughAvatarLevelException

NotEnoughClearedStageLevelException

NotEnoughCombatPointException

NotEnoughFungibleAssetValueException

NotEnoughHammerPointException

NotEnoughMaterialException

NotEnoughMedalException

NotEnoughRankException

NotEnoughStarException

NotEnoughWeeklyArenaChallengeCountException

OrderIdDoesNotExistException

PatchTableSheet

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at https://github.com/planetarium/lib9c/pull/42 Updated at https://github.com/planetarium/lib9c/pull/101 Updated at https://github.com/planetarium/lib9c/pull/287 Updated at https://github.com/planetarium/lib9c/pull/315 Updated at https://github.com/planetarium/lib9c/pull/957 Updated at https://github.com/planetarium/lib9c/pull/1560

PatchTableSheetCompressed

Compressed version of PatchTableSheet to reduce transaction size for large CSV files. Uses GZip compression to minimize the size of table data in transactions.

This action is particularly useful for large CSV files like RuneOptionSheet.csv which can be compressed from ~842KB to ~154KB (81% reduction).

Example usage:

```
var compressedAction = new PatchTableSheetCompressed
{
    TableName = "RuneOptionSheet",
    CompressedTableCsv = PatchTableSheetCompressed.CompressCsv(csvData),
};
```

PendingActivationDoesNotExistsException

PermissionDeniedException

PetEnhancement

PetIsLockedException

PlayCountIsZeroException

PolicyExpiredException

PrepareRewardAssets

ProductNotFoundException

PurchaseInfo

PurchaseInfo0

Raid

Represents a raid action where players participate in a world boss battle. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- WorldBossKillRewardRecord: Records boss kill rewards
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

RankingBattle

Hard forked at https://github.com/planetarium/lib9c/pull/941 Updated at https://github.com/planetarium/lib9c/pull/1135

RankingBattle11

Updated at https://github.com/planetarium/lib9c/pull/1176

RankingExceededException

RapidCombination

Hard forked at https://github.com/planetarium/lib9c/pull/2195

RapidCombination0

RapidCombination0.ResultModel

RapidCombination5

RapidCombination5.ResultModel

ReRegisterProduct

RedeemCode

Hard forked at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

RedeemCode2

RegisterInfo

RegisterProduct

RegisterProduct0

RemoveAddressState

Action to remove state for multiple addresses.

RenewAdminState

Hard forked at https://github.com/planetarium/lib9c/pull/805 Updated at https://github.com/planetarium/lib9c/pull/815 Updated at https://github.com/planetarium/lib9c/pull/957

RequestPledge

RequiredBlockIndexException

RequiredBlockIntervalException

RetrieveAvatarAssets

RewardGold

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at many pull requests Updated at https://github.com/planetarium/lib9c/pull/1135

RuneEnhancement

RuneSlotInfo

Represents rune slot information for a character.

RuneSummon

SecureMiningReward

Sell

Hard forked at https://github.com/planetarium/lib9c/pull/1640

Sell6

SellCancellation

Hard forked at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/609 Updated at https://github.com/planetarium/lib9c/pull/620 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

SellCancellation.Result

SetAddressState

Action to set state for multiple addresses. Throws an exception if the state already exists for the target address.

SetAddressStateCompressed

Compressed version of SetAddressState to reduce transaction size for large state data. Uses GZip compression to minimize the size of state data in transactions.

This action is particularly useful when setting large state data that would exceed transaction policy size limits. The compression can reduce state data size by 60-80% depending on the content structure and redundancy.

Example usage:

```
var states = new List<(Address, Address, IValue)>
{
    (accountAddress, targetAddress, largeStateValue),
};
var compressedAction = new SetAddressStateCompressed(states);
```

ShopItemExpiredException

SlotAlreadyUnlockedException

StageNotClearedException

Stake

Stake2

StakeExistingClaimableException

Swap

Swap action swaps one currency to another currency.

Synthesize

Synthesize action is a type of action that synthesizes items. MaterialIds: Id list of items as material
ChargeAp: Whether to charge action points with action execution

TotalSupplyDoesNotExistException

TransferAsset

Hard forked at https://github.com/planetarium/lib9c/pull/2143 Updated at https://github.com/planetarium/lib9c/pull/2143

TransferAsset3

TransferAssets
Hard forked at https://github.com/planetarium/lib9c/pull/636 Updated at https://github.com/planetarium/lib9c/pull/957

UnlockCombinationSlot

UnlockEquipmentRecipe

UnlockRuneSlot

UnlockWorld
Unlocks one or more worlds for an avatar and records the result to the legacy `world_ids` state. The required CRYSTAL is calculated from WorldUnlockSheet and transferred to UnlockWorld. If the calculated cost is zero, no asset transfer occurs.

UpdateSell
Hard forked at https://github.com/planetarium/lib9c/pull/1640

UpdateSellInfo

UsageLimitExceedException

ValidatorSetOperate

ValidatorSetOperatorTypeExtensions

WeeklyArenaStateAlreadyEndedException

WeeklyArenaStateNotContainsAvatarAddressException

# Structs

IssueTokensFromGarage.Spec

MintAssets.MintSpec

RuneEnhancement.LevelUpResult

# Interfaces

IBuy0
Common interface used before IBuy5.

IBuy5
Common interface used after IBuy5.

IClaimItems

IClaimStakeReward

IHackAndSlashV10
Interface for hack and slash action version 10.

IProductInfo

IPurchaseInfo

IRegisterInfo

ITransferAsset

ITransferAssets

## Enums

ItemEnhancement10.EnhancementResult

ItemEnhancement11.EnhancementResult

ItemEnhancement12.EnhancementResult

ItemEnhancement13.EnhancementResult

ItemEnhancement9.EnhancementResult

ShopErrorType

ValidatorSetOperatorType

# Class ActionBase

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ActionBase : IAction
```

### Inheritance

object ← ActionBase

### Implements

IAction

### Derived

ApprovePledge, BurnAsset, ClaimPatrolReward, CreatePendingActivation, CreatePendingActivations, CreatePledge, CustomEquipmentCraft, EndPledge, GameAction, BanGuildMember, ClaimGuildReward, ClaimReward, ClaimUnbonded, JoinGuild, MakeGuild, MigrateDelegationHeight, MigratePlanetariumValidator, MigrateStakeAndJoinGuild, MoveGuild, QuitGuild, RemoveGuild, UnbanGuildMember, IssueToken, IssueTokensFromGarage, MigrateAgentAvatar, MigrateFee, MigrateMonsterCollection, MintAssets, PrepareRewardAssets, RemoveAddressState, RequestPledge, RetrieveAvatarAssets, RewardGold, SecureMiningReward, SetAddressState, SetAddressStateCompressed, Swap, TransferAsset, TransferAssets, AllocateGuildReward, AllocateReward, ClaimValidatorRewardSelf, DelegateValidator, Mortgage, PromoteValidator, RecordProposer, Refund, Reward, SetValidatorCommission, SlashValidator, UndelegateValidator, UnjailValidator, UpdateValidators, ValidatorSetOperate

### Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### PlainValue

```
public abstract IValue PlainValue { get; }
```

### Property Value

IValue

## Methods

## CheckObsolete(long, IActionContext)

```
protected void CheckObsolete(long obsoleteIndex, IActionContext ctx)
```

### Parameters

`obsoleteIndex` long

`ctx` IActionContext

## CheckPermission(IActionContext)

```
protected void CheckPermission(IActionContext ctx)
```

### Parameters

`ctx` IActionContext

## Execute(IActionContext)

```
public abstract IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# GetSignerAndOtherAddressesHex(IActionContext, params Address[])

returns "[Signer Address, AvatarState Address, ...]"

```
protected string GetSignerAndOtherAddressesHex(IActionContext ctx, params
Address[] addresses)
```

## Parameters

`ctx` IActionContext

`addresses` Address[]

## Returns

string

# LoadPlainValue(IValue)

```
public abstract void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# TryGetAdminState(IActionContext, out AdminState)

```
protected bool TryGetAdminState(IActionContext ctx, out AdminState state)
```

## Parameters

`ctx` IActionContext

`state` AdminState

## Returns

bool

# Class ActionContextExtensions

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class ActionContextExtensions
```

**Inheritance**

object ← ActionContextExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### IsMainNet(IActionContext)

```
public static bool IsMainNet(this IActionContext context)
```

#### Parameters

`context` IActionContext

#### Returns

bool

### Since(IActionContext, long)

```
public static bool Since(this IActionContext context, long blockIndex)
```

#### Parameters

`context` IActionContext

`blockIndex` long

## Returns

bool

# Class ActionObsoleteAttribute

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

An attribute on an Libplanet.Action.IAction to indicate that such Libplanet.Action.IAction is obsoleted after a certain index.

Due to a bug introduced earlier and to keep backward compatibility, an Libplanet.Action.IAction with this attribute is obsoleted, i.e. cannot be included, starting from ObsoleteIndex + 2.

```
[AttributeUsage(AttributeTargets.Class, AllowMultiple = true)]
public class ActionObsoleteAttribute : Attribute
```

### Inheritance

object ← Attribute ← ActionObsoleteAttribute

### Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) ,
Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) ,
Attribute.GetCustomAttributes(Assembly) ,
Attribute.GetCustomAttributes(Assembly, bool) ,
Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) ,
Attribute.GetCustomAttributes(MemberInfo) ,
Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) ,
Attribute.GetCustomAttributes(Module) , Attribute.GetCustomAttributes(Module, bool) ,
Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) ,
Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,

Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) ,
Attribute.TypeId , object.Equals(object, object) , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ActionObsoleteAttribute(Planet, long)

```
public ActionObsoleteAttribute(Planet planet, long obsoleteIndex)
```

### Parameters

`planet` Planet

`obsoleteIndex` long

## ActionObsoleteAttribute(long)

```
public ActionObsoleteAttribute(long obsoleteIndex)
```

### Parameters

`obsoleteIndex` long

# Fields

## AffectedPlanet

```
public readonly Planet? AffectedPlanet
```

### Field Value

Planet?

## ObsoleteIndex

```
public readonly long ObsoleteIndex
```

### Field Value

long

# Class ActionObsoletedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ActionObsoletedException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ActionObsoletedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ActionObsoletedException()

```
public ActionObsoletedException()
```

### ActionObsoletedException(SerializationInfo, StreamingContext)

```
protected ActionObsoletedException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## ActionObsoletedException(string)

```
public ActionObsoletedException(string s)
```

### Parameters

`s` string

# Class ActionPointExceededException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ActionPointExceededException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ActionPointExceededException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ActionPointExceededException()

```
public ActionPointExceededException()
```

### ActionPointExceededException(SerializationInfo, StreamingContext)

```
protected ActionPointExceededException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## ActionPointExceededException(string)

```
public ActionPointExceededException(string s)
```

### Parameters

`s` string

# Class ActionUnavailableException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ActionUnavailableException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ActionUnavailableException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ActionUnavailableException()

```
public ActionUnavailableException()
```

### ActionUnavailableException(SerializationInfo, StreamingContext)

```
protected ActionUnavailableException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## ActionUnavailableException(string)

```
public ActionUnavailableException(string s)
```

### Parameters

`s` string

# Class ActivateCollection

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("activate_collection")]
public class ActivateCollection : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← ActivateCollection

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## CollectionData

```
public List<(int collectionId, List<ICollectionMaterial> materials)> CollectionData
```

### Field Value

List<(int collectionId, List<ICollectionMaterial> materials)>

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ActivatedAccountsDoesNotExistsException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ActivatedAccountsDoesNotExistsException : ActivationException,
ISerializable
```

### Inheritance

object ← Exception ← ActivationException ← ActivatedAccountsDoesNotExistsException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ActivatedAccountsDoesNotExistsException()

```
public ActivatedAccountsDoesNotExistsException()
```

### ActivatedAccountsDoesNotExistsException(SerializationInfo, StreamingContext)

```
public ActivatedAccountsDoesNotExistsException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## ActivatedAccountsDoesNotExistsException(string)

```
public ActivatedAccountsDoesNotExistsException(string message)
```

### Parameters

`message` string

# Class ActivationException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public abstract class ActivationException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← ActivationException

**Implements**

ISerializable

**Derived**

ActivatedAccountsDoesNotExistsException, InvalidSignatureException, PendingActivationDoesNotExistsException

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ActivationException()

```
protected ActivationException()
```

### ActivationException(SerializationInfo, StreamingContext)

```
protected ActivationException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## ActivationException(string)

```
protected ActivationException(string message)
```

### Parameters

`message` string

# Class AddRedeemCode

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("add_redeem_code")]
public class AddRedeemCode : GameAction, IAction, IAddRedeemCodeV1
```

### Inheritance

object ← ActionBase ← GameAction ← AddRedeemCode

### Implements

IAction, IAddRedeemCodeV1

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## redeemCsv

```
public string redeemCsv
```

### Field Value

string

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class AddressExtension

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class AddressExtension
```

**Inheritance**

object ← AddressExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### Derive(Address, byte[])

```
public static Address Derive(this Address address, byte[] key)
```

#### Parameters

`address` Address

`key` byte[]

#### Returns

Address

### Derive(Address, string)

```
public static Address Derive(this Address address, string key)
```

### Parameters

`address` Address

`key` string

### Returns

Address

## GetPledgeAddress(Address)

```
public static Address GetPledgeAddress(this Address address)
```

### Parameters

`address` Address

### Returns

Address

# Class AdminPermissionException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public abstract class AdminPermissionException : Exception, ISerializable
```

### Inheritance

object ← Exception ← AdminPermissionException

### Implements

ISerializable

### Derived

PermissionDeniedException, PolicyExpiredException

### Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AdminPermissionException(AdminState)

```
protected AdminPermissionException(AdminState policy)
```

#### Parameters

`policy` AdminState

## AdminPermissionException(SerializationInfo, StreamingContext)

```
protected AdminPermissionException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## AdminPermissionException(string)

```
protected AdminPermissionException(string message)
```

### Parameters

`message` string

# Properties

## Policy

```
public AdminState Policy { get; }
```

### Property Value

AdminState

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

## Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

## Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class AgentStateNotContainsAvatarAddressException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AgentStateNotContainsAvatarAddressException : Exception, ISerializable
```

### Inheritance

object ← Exception ← AgentStateNotContainsAvatarAddressException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AgentStateNotContainsAvatarAddressException(SerializationInfo, StreamingContext)

```
public AgentStateNotContainsAvatarAddressException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## AgentStateNotContainsAvatarAddressException(string)

```
public AgentStateNotContainsAvatarAddressException(string message)
```

### Parameters

`message` string

# Class AlreadyActivatedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyActivatedException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← AlreadyActivatedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## AlreadyActivatedException(SerializationInfo, StreamingContext)

```
protected AlreadyActivatedException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# AlreadyActivatedException(string)

```
public AlreadyActivatedException(string s)
```

## Parameters

s [string]

# Class AlreadyClaimedGiftsException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyClaimedGiftsException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← AlreadyClaimedGiftsException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AlreadyClaimedGiftsException()

```
public AlreadyClaimedGiftsException()
```

### AlreadyClaimedGiftsException(SerializationInfo, StreamingContext)

```
public AlreadyClaimedGiftsException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## AlreadyClaimedGiftsException(string)

```
public AlreadyClaimedGiftsException(string msg)
```

### Parameters

`msg` string

# Class AlreadyContractedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyContractedException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← AlreadyContractedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## AlreadyContractedException(SerializationInfo, StreamingContext)

```
protected AlreadyContractedException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# AlreadyContractedException(string)

```
public AlreadyContractedException(string s)
```

## Parameters

s string

# Class AlreadyReceivedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyReceivedException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← AlreadyReceivedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## AlreadyReceivedException(SerializationInfo, StreamingContext)

```
protected AlreadyReceivedException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# AlreadyReceivedException(string)

```
public AlreadyReceivedException(string s)
```

## Parameters

s string

# Class AlreadyRecipeUnlockedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyRecipeUnlockedException : Exception, ISerializable
```

### Inheritance

object ← Exception ← AlreadyRecipeUnlockedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AlreadyRecipeUnlockedException()

```
public AlreadyRecipeUnlockedException()
```

### AlreadyRecipeUnlockedException(SerializationInfo, StreamingContext)

```
protected AlreadyRecipeUnlockedException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## AlreadyRecipeUnlockedException(string)

```
public AlreadyRecipeUnlockedException(string msg)
```

### Parameters

`msg` string

# Class AlreadyWorldUnlockedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyWorldUnlockedException : Exception, ISerializable
```

### Inheritance

object ← Exception ← AlreadyWorldUnlockedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AlreadyWorldUnlockedException()

```
public AlreadyWorldUnlockedException()
```

### AlreadyWorldUnlockedException(SerializationInfo, StreamingContext)

```
protected AlreadyWorldUnlockedException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## AlreadyWorldUnlockedException(string)

```
public AlreadyWorldUnlockedException(string msg)
```

### Parameters

`msg` string

# Class AppraiseBlockNotReachedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AppraiseBlockNotReachedException : Exception, ISerializable
```

### Inheritance

object ← Exception ← AppraiseBlockNotReachedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AppraiseBlockNotReachedException(SerializationInfo, StreamingContext)

```
protected AppraiseBlockNotReachedException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# AppraiseBlockNotReachedException(string)

```
public AppraiseBlockNotReachedException(string message)
```

## Parameters

`message` string

# Class ApprovePledge

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("approve_pledge")]
public class ApprovePledge : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← ApprovePledge

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ApprovePledge()

```
public ApprovePledge()
```

## Fields

### PatronAddress

```
public Address PatronAddress
```

#### Field Value

Address

## TypeIdentifier

```
public const string TypeIdentifier = "approve_pledge"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class ArenaNotEndedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaNotEndedException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ArenaNotEndedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## ArenaNotEndedException(SerializationInfo, StreamingContext)

```
protected ArenaNotEndedException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# ArenaNotEndedException(string)

```
public ArenaNotEndedException(string s)
```

## Parameters

s string

# Class AssetInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class AssetInfo : IRegisterInfo
```

**Inheritance**

object ← AssetInfo

**Implements**

IRegisterInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AssetInfo()

```
public AssetInfo()
```

### AssetInfo(List)

```
public AssetInfo(List serialized)
```

#### Parameters

`serialized` List

## Properties

## Asset

```
public FungibleAssetValue Asset { get; set; }
```

### Property Value

FungibleAssetValue

## AvatarAddress

```
public Address AvatarAddress { get; set; }
```

### Property Value

Address

## Price

```
public FungibleAssetValue Price { get; set; }
```

### Property Value

FungibleAssetValue

## Type

```
public ProductType Type { get; set; }
```

### Property Value

ProductType

## Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Validate()

```
public void Validate()
```

## ValidateAddress(Address)

```
public void ValidateAddress(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

## ValidatePrice(Currency)

```
public void ValidatePrice(Currency ncg)
```

### Parameters

`ncg` Currency

# Class AttachmentActionResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public abstract class AttachmentActionResult : IState
```

**Inheritance**

object ← AttachmentActionResult

**Implements**

IState

**Derived**

Buy7.BuyerResult, Buy7.SellerResult, CombinationConsumable5.ResultModel, DailyReward2.DailyRewardResult, ItemEnhancement10.ResultModel, ItemEnhancement11.ResultModel, ItemEnhancement12.ResultModel, ItemEnhancement13.ResultModel, ItemEnhancement7.ResultModel, ItemEnhancement9.ResultModel, RapidCombination5.ResultModel, SellCancellation.Result, MonsterCollectionResult

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AttachmentActionResult()

```
protected AttachmentActionResult()
```

## AttachmentActionResult(Dictionary)

```
protected AttachmentActionResult(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## Fields

### costume

```
public Costume costume
```

#### Field Value

Costume

### itemUsable

```
public ItemUsable itemUsable
```

#### Field Value

ItemUsable

### tradableFungibleItem

```
public ITradableFungibleItem tradableFungibleItem
```

#### Field Value

ITradableFungibleItem

### tradableFungibleItemCount

```
public int tradableFungibleItemCount
```

### Field Value

int

## Properties

### TypeId

```
protected abstract string TypeId { get; }
```

### Property Value

string

## Methods

### Deserialize(Dictionary)

```
public static AttachmentActionResult Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

AttachmentActionResult

### Serialize()

```
public virtual IValue Serialize()
```

### Returns

IValue

# SerializeBackup1()

```
public virtual IValue SerializeBackup1()
```

## Returns

IValue

# Class AuraSummon

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("aura_summon")]
public class AuraSummon : GameAction, IAction, IAuraSummonV1
```

**Inheritance**

object ← ActionBase ← GameAction ← AuraSummon

**Implements**

IAction, IAuraSummonV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AuraSummon()

```
public AuraSummon()
```

### AuraSummon(Address, int, int)

```
public AuraSummon(Address avatarAddress, int groupId, int summonCount)
```

#### Parameters

`avatarAddress` Address

`groupId` int

`summonCount` int

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## AvatarAddressKey

```
public const string AvatarAddressKey = "aa"
```

### Field Value

string

## GroupId

```
public int GroupId
```

### Field Value

int

## GroupIdKey

```
public const string GroupIdKey = "gid"
```

### Field Value

string

## SummonCount

```
public int SummonCount
```

### Field Value

int

## SummonCountKey

```
public const string SummonCountKey = "sc"
```

### Field Value

string

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## AddAndUnlockOption(Equipment, IRandom, Row, EquipmentItemOptionSheet, SkillSheet)

```
public static void AddAndUnlockOption(Equipment equipment, IRandom random,
EquipmentItemSubRecipeSheetV2.Row subRecipe, EquipmentItemOptionSheet optionSheet,
SkillSheet skillSheet)
```

### Parameters

`equipment` Equipment

`random` IRandom

`subRecipe` EquipmentItemSubRecipeSheetV2.Row

`optionSheet` EquipmentItemOptionSheet

`skillSheet` SkillSheet

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# SimulateSummon(string, EquipmentItemRecipeSheet, EquipmentItemSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, SkillSheet, Row, int, IRandom, long, Row)

Simulates aura equipment summoning with optional grade guarantee system. Applies 10+1 bonus rule and uses grade guarantee settings from SummonSheet.Row when enabled. When grade guarantee is enabled, ensures minimum grade items are obtained based on summon count.

```
public static IEnumerable<(int, Equipment)> SimulateSummon(string addressesHex,
EquipmentItemRecipeSheet recipeSheet, EquipmentItemSheet equipmentItemSheet,
EquipmentItemSubRecipeSheetV2 equipmentItemSubRecipeSheetV2,
EquipmentItemOptionSheet optionSheet, SkillSheet skillSheet, SummonSheet.Row
summonRow, int summonCount, IRandom random, long blockIndex, BoostScheduleSheet.Row
guaranteeBoost = null)
```

## Parameters

`addressesHex` string

Addresses hex for error reporting

`recipeSheet` EquipmentItemRecipeSheet

Equipment item recipe sheet

`equipmentItemSheet` EquipmentItemSheet

Equipment item sheet

`equipmentItemSubRecipeSheetV2` EquipmentItemSubRecipeSheetV2

Equipment item sub-recipe sheet

`optionSheet` EquipmentItemOptionSheet

Equipment item option sheet

`skillSheet` SkillSheet

Skill sheet

`summonRow` SummonSheet.Row

Summon configuration row with recipes and guarantee settings

`summonCount` int

Number of items to summon (before 10+1 bonus)

`random` IRandom

Random number generator

`blockIndex` long

Current block index for item creation

`guaranteeBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this group's guarantee count, or `null` for none. A caller replaying a summon must pass what the action resolved.

## Returns

IEnumerable<(int, Equipment)>

Collection of (recipe ID, equipment) pairs

# Class AvatarIndexAlreadyUsedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AvatarIndexAlreadyUsedException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← AvatarIndexAlreadyUsedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AvatarIndexAlreadyUsedException(SerializationInfo, StreamingContext)

```
protected AvatarIndexAlreadyUsedException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# AvatarIndexAlreadyUsedException(string)

```
public AvatarIndexAlreadyUsedException(string s)
```

## Parameters

s [string]

# Class AvatarIndexOutOfRangeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class AvatarIndexOutOfRangeException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← AvatarIndexOutOfRangeException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AvatarIndexOutOfRangeException(SerializationInfo, StreamingContext)

```
protected AvatarIndexOutOfRangeException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# AvatarIndexOutOfRangeException(string)

```
public AvatarIndexOutOfRangeException(string message)
```

## Parameters

`message` string

# Class BalanceDoesNotExistsException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BalanceDoesNotExistsException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← BalanceDoesNotExistsException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### BalanceDoesNotExistsException(Address, Currency)

```
public BalanceDoesNotExistsException(Address address, Currency currency)
```

#### Parameters

`address` Address

`currency` Currency

### BalanceDoesNotExistsException(SerializationInfo, StreamingContext)

```
protected BalanceDoesNotExistsException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## BalanceDoesNotExistsException(string)

```
public BalanceDoesNotExistsException(string message)
```

### Parameters

`message` string

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## Currency

```
public Currency Currency { get; }
```

### Property Value

Currency

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

### Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class BattleArena

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduce at https://github.com/planetarium/lib9c/pull/2229 Changed at https://github.com/planetarium/lib9c/pull/2242

```
[Serializable]
[ActionType("battle_arena15")]
public class BattleArena : GameAction, IAction, IBattleArenaV1
```

**Inheritance**

object ← ActionBase ← GameAction ← BattleArena

**Implements**

IAction, IBattleArenaV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## HpIncreasingModifier

```
public const int HpIncreasingModifier = 5
```

### Field Value

int

## PurchasedCountKey

```
public const string PurchasedCountKey = "purchased_count_during_interval"
```

### Field Value

string

## championshipId

```
public int championshipId
```

### Field Value

int

## costumes

```
public List<Guid> costumes
```

### Field Value

List<Guid>

## enemyAvatarAddress

```
public Address enemyAvatarAddress
```

### Field Value

Address

## equipments

```
public List<Guid> equipments
```

### Field Value

List<Guid>

## myAvatarAddress

```
public Address myAvatarAddress
```

### Field Value

Address

## round

```
public int round
```

### Field Value

int

## runeInfos

```
public List<RuneSlotInfo> runeInfos
```

### Field Value

List<RuneSlotInfo>

## ticket

```
public int ticket
```

### Field Value

int

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class BurnAsset

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("burn_asset")]
public class BurnAsset : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← BurnAsset

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BurnAsset()

```
public BurnAsset()
```

### BurnAsset(Address, FungibleAssetValue, string)

```
public BurnAsset(Address owner, FungibleAssetValue amount, string memo)
```

#### Parameters

`owner` Address

`amount` FungibleAssetValue

`memo` string

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "burn_asset"
```

### Field Value

string

# Properties

## Amount

```
public FungibleAssetValue Amount { get; }
```

### Property Value

FungibleAssetValue

## Memo

```
public string Memo { get; }
```

### Property Value

string

## Owner

```
public Address Owner { get; }
```

### Property Value

Address

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class Buy

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/1164

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("buy12")]
public class Buy : GameAction, IBuy5, IAction, IBuyV2
```

**Inheritance**

object ← ActionBase ← GameAction ← Buy

**Implements**

IBuy5, IAction, IBuyV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ErrorCodeDuplicateSell

```
public const int ErrorCodeDuplicateSell = 10
```

### Field Value

int

## ErrorCodeFailedLoadingState

```
public const int ErrorCodeFailedLoadingState = 1
```

### Field Value

int

## ErrorCodeInsufficientBalance

```
public const int ErrorCodeInsufficientBalance = 4
```

### Field Value

int

## ErrorCodeInvalidAddress

```
public const int ErrorCodeInvalidAddress = 5
```

### Field Value

int

## ErrorCodeInvalidItemType

```
public const int ErrorCodeInvalidItemType = 9
```

### Field Value

int

## ErrorCodeInvalidOrderId

```
public const int ErrorCodeInvalidOrderId = 7
```

### Field Value

int

## ErrorCodeInvalidPrice

```
public const int ErrorCodeInvalidPrice = 6
```

### Field Value

int

## ErrorCodeInvalidTradableId

```
public const int ErrorCodeInvalidTradableId = 8
```

### Field Value

int

## ErrorCodeItemDoesNotExist

```
public const int ErrorCodeItemDoesNotExist = 2
```

### Field Value

int

## ErrorCodeShopItemExpired

```
public const int ErrorCodeShopItemExpired = 3
```

### Field Value

int

## TaxRate

```
public const int TaxRate = 8
```

### Field Value

int

## errors

```
public List<(Guid orderId, int errorCode)> errors
```

### Field Value

List<(Guid orderId, int errorCode)>

## purchaseInfos

```
public IEnumerable<PurchaseInfo> purchaseInfos
```

### Field Value

IEnumerable<PurchaseInfo>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## buyerAvatarAddress

```
public Address buyerAvatarAddress { get; set; }
```

### Property Value

Address

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class Buy7

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("buy7")]
public class Buy7 : GameAction, IBuy5, IAction, IBuyV2
```

### Inheritance

object ← ActionBase ← GameAction ← Buy7

### Implements

IBuy5, IAction, IBuyV2

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ErrorCodeFailedLoadingState

```
public const int ErrorCodeFailedLoadingState = 1
```

### Field Value

int

## ErrorCodeInsufficientBalance

```
public const int ErrorCodeInsufficientBalance = 4
```

### Field Value

int

## ErrorCodeInvalidAddress

```
public const int ErrorCodeInvalidAddress = 5
```

### Field Value

int

## ErrorCodeInvalidPrice

```
public const int ErrorCodeInvalidPrice = 6
```

### Field Value

int

## ErrorCodeItemDoesNotExist

```
public const int ErrorCodeItemDoesNotExist = 2
```

### Field Value

int

## ErrorCodeShopItemExpired

```
public const int ErrorCodeShopItemExpired = 3
```

### Field Value

int

## TaxRate

```
public const int TaxRate = 8
```

### Field Value

int

## buyerMultipleResult

```
public Buy7.BuyerMultipleResult buyerMultipleResult
```

### Field Value

Buy7.BuyerMultipleResult

## purchaseInfos

```
public IEnumerable<PurchaseInfo0> purchaseInfos
```

### Field Value

IEnumerable<PurchaseInfo0>

## sellerMultipleResult

```
public Buy7.SellerMultipleResult sellerMultipleResult
```

### Field Value

Buy7.SellerMultipleResult

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

### buyerAvatarAddress

```
public Address buyerAvatarAddress { get; set; }
```

### Property Value

Address

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class Buy7.BuyerMultipleResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.BuyerMultipleResult
```

**Inheritance**

object ← Buy7.BuyerMultipleResult

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BuyerMultipleResult()

```
public BuyerMultipleResult()
```

### BuyerMultipleResult(Dictionary)

```
public BuyerMultipleResult(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## Fields

### purchaseResults

```
public IEnumerable<Buy7.PurchaseResult> purchaseResults
```

### Field Value

IEnumerable<Buy7.PurchaseResult>

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class Buy7.BuyerResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.BuyerResult : AttachmentActionResult, IState
```

### Inheritance

object ← AttachmentActionResult ← Buy7.BuyerResult

### Implements

IState

### Derived

Buy7.PurchaseResult, BuyMultiple.PurchaseResult

### Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BuyerResult()

```
public BuyerResult()
```

### BuyerResult(Dictionary)

```
public BuyerResult(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## Fields

### id

```
public Guid id
```

#### Field Value

Guid

### shopItem

```
public ShopItem shopItem
```

#### Field Value

ShopItem

## Properties

### TypeId

```
protected override string TypeId { get; }
```

#### Property Value

string

## Methods

# Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Class Buy7.PurchaseResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.PurchaseResult : Buy7.BuyerResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← Buy7.BuyerResult ← Buy7.PurchaseResult

**Implements**

IState

**Inherited Members**

Buy7.BuyerResult.shopItem , Buy7.BuyerResult.id , Buy7.BuyerResult.TypeId , AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## PurchaseResult(Dictionary)

```
public PurchaseResult(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## PurchaseResult(Guid)

```
public PurchaseResult(Guid shopProductId)
```

### Parameters

`shopProductId` Guid

## Fields

### errorCode

```
public int errorCode
```

### Field Value

int

### productId

```
public readonly Guid productId
```

### Field Value

Guid

## Methods

### Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class Buy7.SellerMultipleResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.SellerMultipleResult
```

**Inheritance**

object ← Buy7.SellerMultipleResult

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SellerMultipleResult()

```
public SellerMultipleResult()
```

### SellerMultipleResult(Dictionary)

```
public SellerMultipleResult(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## Fields

### sellerResults

```
public IEnumerable<Buy7.SellerResult> sellerResults
```

### Field Value

IEnumerable<Buy7.SellerResult>

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class Buy7.SellerResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.SellerResult : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← Buy7.SellerResult

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SellerResult()

```
public SellerResult()
```

## SellerResult(Dictionary)

```
public SellerResult(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

# Fields

## gold

```
public FungibleAssetValue gold
```

### Field Value

FungibleAssetValue

## id

```
public Guid id
```

### Field Value

Guid

## shopItem

```
public ShopItem shopItem
```

### Field Value

ShopItem

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class BuyMultiple

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/331 Updated at many pull requests Obsoleted at https://github.com/planetarium/lib9c/pull/487 Updated at many pull requests Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("buy_multiple")]
public class BuyMultiple : GameAction, IAction, IBuyMultipleV1
```

**Inheritance**

object ← ActionBase ← GameAction ← BuyMultiple

**Implements**

IAction, IBuyMultipleV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ERROR_CODE_FAILED_LOADING_STATE

```
public const int ERROR_CODE_FAILED_LOADING_STATE = 1
```

### Field Value

int

## ERROR_CODE_INSUFFICIENT_BALANCE

```
public const int ERROR_CODE_INSUFFICIENT_BALANCE = 4
```

### Field Value

int

## ERROR_CODE_ITEM_DOES_NOT_EXIST

```
public const int ERROR_CODE_ITEM_DOES_NOT_EXIST = 2
```

### Field Value

int

## ERROR_CODE_SHOPITEM_EXPIRED

```
public const int ERROR_CODE_SHOPITEM_EXPIRED = 3
```

### Field Value

int

## buyerAvatarAddress

```
public Address buyerAvatarAddress
```

### Field Value

Address

## buyerResult

```
public BuyMultiple.BuyerResult buyerResult
```

### Field Value

BuyMultiple.BuyerResult

## purchaseInfos

```
public IEnumerable<BuyMultiple.PurchaseInfo> purchaseInfos
```

### Field Value

IEnumerable<BuyMultiple.PurchaseInfo>

## sellerResult

```
public BuyMultiple.SellerResult sellerResult
```

### Field Value

BuyMultiple.SellerResult

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class BuyMultiple.BuyerResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BuyMultiple.BuyerResult
```

**Inheritance**

object ← BuyMultiple.BuyerResult

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BuyerResult()

```
public BuyerResult()
```

### BuyerResult(Dictionary)

```
public BuyerResult(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## Fields

### purchaseResults

```
public IEnumerable<BuyMultiple.PurchaseResult> purchaseResults
```

### Field Value

IEnumerable<BuyMultiple.PurchaseResult>

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class BuyMultiple.PurchaseInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BuyMultiple.PurchaseInfo : IComparable<BuyMultiple.PurchaseInfo>,
IComparable
```

**Inheritance**

object ← BuyMultiple.PurchaseInfo

**Implements**

IComparable<BuyMultiple.PurchaseInfo>, IComparable

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PurchaseInfo(Dictionary)

```
public PurchaseInfo(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### PurchaseInfo(Guid, Address, Address)

```
public PurchaseInfo(Guid productId, Address sellerAgentAddress,
Address sellerAvatarAddress)
```

#### Parameters

`productId` Guid

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

# Fields

## productId

```
public Guid productId
```

### Field Value

Guid

## sellerAgentAddress

```
public Address sellerAgentAddress
```

### Field Value

Address

## sellerAvatarAddress

```
public Address sellerAvatarAddress
```

### Field Value

Address

# Methods

# CompareTo(PurchaseInfo)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(BuyMultiple.PurchaseInfo other)
```

## Parameters

`other` BuyMultiple.PurchaseInfo

An object to compare with this instance.

## Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `other` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `other`. |
| **Greater than zero** | This instance follows `other` in the sort order. |

# CompareTo(object)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(object obj)
```

### Parameters

`obj` object

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `obj` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `obj`. |
| **Greater than zero** | This instance follows `obj` in the sort order. |

### Exceptions

ArgumentException

`obj` is not the same type as this instance.

## Equals(PurchaseInfo)

```
protected bool Equals(BuyMultiple.PurchaseInfo other)
```

### Parameters

`other` BuyMultiple.PurchaseInfo

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Operators

## operator ==(PurchaseInfo, PurchaseInfo)

```
public static bool operator ==(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

### Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

### Returns

bool

## operator >(PurchaseInfo, PurchaseInfo)

```
public static bool operator >(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

### Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

### Returns

bool

## operator >=(PurchaseInfo, PurchaseInfo)

```
public static bool operator >=(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

### Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

### Returns

bool

## operator !=(PurchaseInfo, PurchaseInfo)

```
public static bool operator !=(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

### Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

### Returns

bool

## operator <(PurchaseInfo, PurchaseInfo)

```
public static bool operator <(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

### Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

### Returns

bool

# operator <=(PurchaseInfo, PurchaseInfo)

```
public static bool operator <=(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

## Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

## Returns

bool

# Class BuyMultiple.PurchaseResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BuyMultiple.PurchaseResult : Buy7.BuyerResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← Buy7.BuyerResult ← BuyMultiple.PurchaseResult

**Implements**

IState

**Inherited Members**

Buy7.BuyerResult.shopItem , Buy7.BuyerResult.id , Buy7.BuyerResult.TypeId , AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## PurchaseResult(Dictionary)

```
public PurchaseResult(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## PurchaseResult(Guid)

```
public PurchaseResult(Guid shopProductId)
```

### Parameters

`shopProductId` Guid

## Fields

### errorCode

```
public int errorCode
```

### Field Value

int

### productId

```
public Guid productId
```

### Field Value

Guid

## Methods

### Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class BuyMultiple.SellerResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BuyMultiple.SellerResult
```

**Inheritance**

object ← BuyMultiple.SellerResult

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SellerResult()

```
public SellerResult()
```

### SellerResult(Dictionary)

```
public SellerResult(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## Fields

### sellerResults

```
public IEnumerable<Buy7.SellerResult> sellerResults
```

### Field Value

IEnumerable<Buy7.SellerResult>

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class BuyProduct

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("buy_product3")]
public class BuyProduct : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← BuyProduct

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Capacity

```
public const int Capacity = 20
```

### Field Value

int

## ProductInfos

```
public IEnumerable<IProductInfo> ProductInfos
```

### Field Value

IEnumerable<IProductInfo>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class CancelProductRegistration

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("cancel_product_registration2")]
public class CancelProductRegistration : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← CancelProductRegistration

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Capacity

```
public const int Capacity = 100
```

### Field Value

int

## ChargeAp

```
public bool ChargeAp
```

### Field Value

bool

## CostAp

```
public const int CostAp = 5
```

### Field Value

int

## ProductInfos

```
public List<IProductInfo> ProductInfos
```

### Field Value

List<IProductInfo>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

## Cancel(ProductsState, IProductInfo, IWorld, AvatarState, IActionContext)

```
public static IWorld Cancel(ProductsState productsState, IProductInfo productInfo,
IWorld states, AvatarState avatarState, IActionContext context)
```

### Parameters

`productsState` ProductsState

`productInfo` IProductInfo

`states` IWorld

`avatarState` AvatarState

`context` IActionContext

### Returns

IWorld

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ChargeActionPoint

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/430 Updated at https://github.com/planetarium/lib9c/pull/474 Updated at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("charge_action_point3")]
public class ChargeActionPoint : GameAction, IAction, IChargeActionPointV1
```

**Inheritance**

object ← ActionBase ← GameAction ← ChargeActionPoint

**Implements**

IAction, IChargeActionPointV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ClaimGifts

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

An action to claim gifts using the Gift Id.
Refer ClaimableGiftsSheet to find gifts data.
It can only claim once at the specified block index.

```
[ActionType("claim_gifts")]
public class ClaimGifts : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← ClaimGifts

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ClaimGifts()

```
public ClaimGifts()
```

## ClaimGifts(Address, int)

```
public ClaimGifts(Address avatarAddress, int giftId)
```

### Parameters

`avatarAddress` Address

`giftId` int

## Properties

## AvatarAddress

The address of the avatar claiming the gift.

```
public Address AvatarAddress { get; }
```

### Property Value

Address

## GiftId

The ID of the gift to be claimed. This ID is used in the ClaimableGiftsSheet.

```
public int GiftId { get; }
```

### Property Value

int

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### Exceptions

FailedLoadStateException

Thrown when the inventory could not be loaded.

SheetRowNotFoundException

Thrown when the gift ID is not found in the ClaimableGiftsSheet.

ClaimableGiftsNotAvailableException

Thrown when the gift is not available at the current block index.

AlreadyClaimedGiftsException

Thrown when the gift has already been claimed.

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ClaimItems

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("claim_items")]
public class ClaimItems : GameAction, IAction, IClaimItems
```

**Inheritance**

object ← ActionBase ← GameAction ← ClaimItems

**Implements**

IAction, IClaimItems

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ClaimItems()

```
public ClaimItems()
```

### ClaimItems(IReadOnlyList<(Address, IReadOnlyList<FungibleAssetValue>)>, string)

```
public ClaimItems(IReadOnlyList<(Address, IReadOnlyList<FungibleAssetValue>)>
claimData, string memo = null)
```

### Parameters

`claimData` IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

`memo` string

## Fields

### Memo

```
public string Memo
```

### Field Value

string

## Properties

### ClaimData

```
public IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue>
fungibleAssetValues)> ClaimData { get; }
```

### Property Value

IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ClaimPatrolReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Claim patrol reward

```
[Serializable]
[ActionType("claim_patrol_reward")]
public class ClaimPatrolReward : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← ClaimPatrolReward

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ClaimPatrolReward()

```
public ClaimPatrolReward()
```

## ClaimPatrolReward(Address)

```
public ClaimPatrolReward(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

# Fields

## AvatarAddress

The address of the avatar to receive the patrol reward.

```
public Address AvatarAddress
```

### Field Value

Address

## TypeIdentifier

```
public const string TypeIdentifier = "claim_patrol_reward"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class ClaimRaidReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("claim_raid_reward")]
public class ClaimRaidReward : GameAction, IAction, IClaimRaidRewardV1
```

**Inheritance**

object ← ActionBase ← GameAction ← ClaimRaidReward

**Implements**

IAction, IClaimRaidRewardV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ClaimRaidReward()

```
public ClaimRaidReward()
```

### ClaimRaidReward(Address)

```
public ClaimRaidReward(Address avatarAddress)
```

#### Parameters

`avatarAddress` Address

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ClaimStakeReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2097

```
[ActionType("claim_stake_reward9")]
public class ClaimStakeReward : GameAction, IClaimStakeReward,
IAction, IClaimStakeRewardV1
```

**Inheritance**

object ← ActionBase ← GameAction ← ClaimStakeReward

**Implements**

IClaimStakeReward, IAction, IClaimStakeRewardV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ClaimStakeReward()

```
public ClaimStakeReward()
```

## ClaimStakeReward(Address)

```
public ClaimStakeReward(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

## Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

## Claim(IWorld, IActionContext, Address, Address, AvatarState, StakeState)

```
public static IWorld Claim(IWorld states, IActionContext context, Address
avatarAddress, Address stakeStateAddress, AvatarState avatarState,
StakeState stakeStateV2)
```

### Parameters

`states` IWorld

`context` IActionContext

`avatarAddress` Address

`stakeStateAddress` Address

`avatarState` AvatarState

`stakeStateV2` StakeState

### Returns

IWorld

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ClaimStakeReward2

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class ClaimStakeReward2
```

**Inheritance**

object ← ClaimStakeReward2

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ObsoletedIndex

```
public const long ObsoletedIndex = 5549200
```

### Field Value

long

# Class ClaimStakeReward8

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class ClaimStakeReward8
```

**Inheritance**

object ← ClaimStakeReward8

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ObsoleteBlockIndex

```
public const long ObsoleteBlockIndex = 7983895
```

### Field Value

long

# Class ClaimWordBossKillReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("claim_world_boss_kill_reward")]
public class ClaimWordBossKillReward : GameAction,
IAction, IClaimWordBossKillRewardV1
```

**Inheritance**

object ← ActionBase ← GameAction ← ClaimWordBossKillReward

**Implements**

IAction, IClaimWordBossKillRewardV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ClaimWorldBossReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

The ClaimWorldBossReward class is an action that allows a player to claim rewards for their contribution in defeating a world boss. This action ensures that only the owner of the avatar can claim the reward and calculates the reward based on the player's contribution.

```
[Serializable]
[ActionType("claim_world_boss_reward")]
public class ClaimWorldBossReward : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← ClaimWorldBossReward

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ClaimWorldBossReward()

Initializes a new instance of the ClaimWorldBossReward class.

```
public ClaimWorldBossReward()
```

## ClaimWorldBossReward(Address)

Initializes a new instance with the specified avatar address.

```
public ClaimWorldBossReward(Address address)
```

### Parameters

`address` Address

The address of the avatar.

## Fields

## AvatarAddress

The address of the avatar claiming the reward.

```
public Address AvatarAddress
```

### Field Value

Address

## MinimumContribution

The minimum contribution percentage required to be eligible for a reward.

```
public const decimal MinimumContribution = 0.0001
```

### Field Value

decimal

## TypeIdentifier

A string identifier for this action type.

```
public const string TypeIdentifier = "claim_world_boss_reward"
```

### Field Value

string

## Properties

### PlainValueInternal

Serializes the action's data into a plain value format.

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

Executes the action to claim the world boss reward.

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

The action context.

### Returns

IWorld

The updated world state.

### Exceptions

InvalidAddressException

Thrown if the avatar address does not belong to the signer.

AlreadyClaimedException

Thrown if the reward has already been claimed.

InvalidClaimException

Thrown if no valid rewards are calculated.

InvalidOperationException

Thrown if the contribution does not meet the minimum requirement.

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

Deserializes the plain value back into the action's data.

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

The serialized plain value.

# Class ClaimableGiftsNotAvailableException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ClaimableGiftsNotAvailableException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ClaimableGiftsNotAvailableException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ClaimableGiftsNotAvailableException()

```
public ClaimableGiftsNotAvailableException()
```

### ClaimableGiftsNotAvailableException(SerializationInfo, StreamingContext)

```
public ClaimableGiftsNotAvailableException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## ClaimableGiftsNotAvailableException(string)

```
public ClaimableGiftsNotAvailableException(string msg)
```

### Parameters

`msg` string

# Class CombinationConsumable

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("combination_consumable9")]
public class CombinationConsumable : GameAction, IAction, ICombinationConsumableV1
```

**Inheritance**

object ← ActionBase ← GameAction ← CombinationConsumable

**Implements**

IAction, ICombinationConsumableV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddressKey

```
public const string AvatarAddressKey = "a"
```

### Field Value

string

## RecipeIdKey

```
public const string RecipeIdKey = "r"
```

### Field Value

string

## SlotIndexKey

```
public const string SlotIndexKey = "s"
```

### Field Value

string

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## recipeId

```
public int recipeId
```

### Field Value

int

## slotIndex

```
public int slotIndex
```

### Field Value

int

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class CombinationConsumable5

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("combination_consumable5")]
public class CombinationConsumable5 : GameAction, IAction, ICombinationConsumableV1
```

**Inheritance**

object ← ActionBase ← GameAction ← CombinationConsumable5

**Implements**

IAction, ICombinationConsumableV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## recipeId

```
public int recipeId
```

### Field Value

int

## slotIndex

```
public int slotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class CombinationConsumable5.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationConsumable5.ResultModel : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← CombinationConsumable5.ResultModel

**Implements**

IState

**Derived**

RapidCombination0.ResultModel

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ResultModel()

```
public ResultModel()
```

### ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## Fields

### actionPoint

```
public int actionPoint
```

### Field Value

int

### gold

```
public BigInteger gold
```

### Field Value

BigInteger

### id

```
public Guid id
```

### Field Value

Guid

### itemType

```
public ItemType itemType
```

### Field Value

ItemType

## materials

```
public Dictionary<Material, int> materials
```

### Field Value

Dictionary<Material, int>

## recipeId

```
public int recipeId
```

### Field Value

int

## subRecipeId

```
public int? subRecipeId
```

### Field Value

int?

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class CombinationEquipment

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("combination_equipment17")]
public class CombinationEquipment : GameAction, IAction, ICombinationEquipmentV4
```

**Inheritance**

object ← ActionBase ← GameAction ← CombinationEquipment

**Implements**

IAction, ICombinationEquipmentV4

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddressKey

```
public const string AvatarAddressKey = "a"
```

### Field Value

string

## BasicSubRecipeHammerPoint

```
public const int BasicSubRecipeHammerPoint = 1
```

### Field Value

int

## PayByCrystalKey

```
public const string PayByCrystalKey = "p"
```

### Field Value

string

## PetIdKey

```
public const string PetIdKey = "pid"
```

### Field Value

string

## RecipeIdKey

```
public const string RecipeIdKey = "r"
```

### Field Value

string

## SlotIndexKey

```
public const string SlotIndexKey = "s"
```

### Field Value

string

## SpecialSubRecipeHammerPoint

```
public const int SpecialSubRecipeHammerPoint = 2
```

### Field Value

int

## SubRecipeIdKey

```
public const string SubRecipeIdKey = "i"
```

### Field Value

string

## UseHammerPointKey

```
public const string UseHammerPointKey = "h"
```

### Field Value

string

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## payByCrystal

```
public bool payByCrystal
```

### Field Value

bool

## petId

```
public int? petId
```

### Field Value

int?

## recipeId

```
public int recipeId
```

### Field Value

int

## slotIndex

```
public int slotIndex
```

### Field Value

int

## subRecipeId

```
public int? subRecipeId
```

### Field Value

int?

## useHammerPoint

```
public bool useHammerPoint
```

### Field Value

bool

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

# AddAndUnlockOption(AgentState, PetState, Equipment, IRandom, Row, EquipmentItemOptionSheet, PetOptionSheet, SkillSheet)

```
public static void AddAndUnlockOption(AgentState agentState, PetState petState,
Equipment equipment, IRandom random, EquipmentItemSubRecipeSheetV2.Row
subRecipe, EquipmentItemOptionSheet optionSheet, PetOptionSheet petOptionSheet,
SkillSheet skillSheet)
```

## Parameters

`agentState` AgentState

`petState` PetState

`equipment` Equipment

`random` IRandom

`subRecipe` EquipmentItemSubRecipeSheetV2.Row

`optionSheet` EquipmentItemOptionSheet

`petOptionSheet` PetOptionSheet

`skillSheet` SkillSheet

# AddSkillOption(AgentState, Equipment, IRandom, Row, EquipmentItemOptionSheet, SkillSheet)

```
public static void AddSkillOption(AgentState agentState, Equipment equipment,
IRandom random, EquipmentItemSubRecipeSheetV2.Row subRecipe,
EquipmentItemOptionSheet optionSheet, SkillSheet skillSheet)
```

## Parameters

`agentState` AgentState

`equipment` Equipment

`random` IRandom

`subRecipe` EquipmentItemSubRecipeSheetV2.Row

`optionSheet` EquipmentItemOptionSheet

`skillSheet` SkillSheet

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetSkill(Row, SkillSheet, IRandom)

```
public static Skill GetSkill(EquipmentItemOptionSheet.Row row, SkillSheet
skillSheet, IRandom random)
```

### Parameters

`row` EquipmentItemOptionSheet.Row

`skillSheet` SkillSheet

`random` IRandom

### Returns

Skill

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class CombinationEquipment5

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class CombinationEquipment5
```

**Inheritance**

object ← CombinationEquipment5

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BlacksmithAddress

```
public static readonly Address BlacksmithAddress
```

### Field Value

Address

# Methods

## GetSkill(Row, SkillSheet, IRandom)

```
public static Skill GetSkill(EquipmentItemOptionSheet.Row row, SkillSheet
skillSheet, IRandom random)
```

### Parameters

`row` EquipmentItemOptionSheet.Row

`skillSheet` SkillSheet

`random` IRandom

### Returns

Skill

## GetStat(Row, IRandom)

```
public static DecimalStat GetStat(EquipmentItemOptionSheet.Row row, IRandom random)
```

### Parameters

`row` EquipmentItemOptionSheet.Row

`random` IRandom

### Returns

DecimalStat

## SelectOption(EquipmentItemOptionSheet, SkillSheet, Row, IRandom, Equipment)

```
public static HashSet<int> SelectOption(EquipmentItemOptionSheet optionSheet,
SkillSheet skillSheet, EquipmentItemSubRecipeSheet.Row subRecipe, IRandom random,
Equipment equipment)
```

### Parameters

`optionSheet` EquipmentItemOptionSheet

`skillSheet` SkillSheet

`subRecipe` EquipmentItemSubRecipeSheet.Row

`random` IRandom

`equipment` Equipment

## Returns

HashSet<int>

# Class CombinationSlotResultNullException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationSlotResultNullException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← CombinationSlotResultNullException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### CombinationSlotResultNullException(SerializationInfo, StreamingContext)

```
public CombinationSlotResultNullException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# CombinationSlotResultNullException(string)

```
public CombinationSlotResultNullException(string message)
```

## Parameters

`message` string

# Class CombinationSlotUnlockException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationSlotUnlockException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← CombinationSlotUnlockException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### CombinationSlotUnlockException(SerializationInfo, StreamingContext)

```
protected CombinationSlotUnlockException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# CombinationSlotUnlockException(string)

```
public CombinationSlotUnlockException(string s)
```

## Parameters

s [string](#)

# Class ConsumableSlotOutOfRangeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableSlotOutOfRangeException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ConsumableSlotOutOfRangeException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ConsumableSlotOutOfRangeException()

```
public ConsumableSlotOutOfRangeException()
```

### ConsumableSlotOutOfRangeException(SerializationInfo, StreamingContext)

```
protected ConsumableSlotOutOfRangeException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## ConsumableSlotOutOfRangeException(string)

```
public ConsumableSlotOutOfRangeException(string message)
```

### Parameters

`message` string

# Class ConsumableSlotUnlockException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableSlotUnlockException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ConsumableSlotUnlockException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ConsumableSlotUnlockException(SerializationInfo, StreamingContext)

```
protected ConsumableSlotUnlockException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ConsumableSlotUnlockException(string)

```
public ConsumableSlotUnlockException(string s)
```

## Parameters

s [string]

# Class CostumeSlotUnlockException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeSlotUnlockException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← CostumeSlotUnlockException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## CostumeSlotUnlockException(SerializationInfo, StreamingContext)

```
protected CostumeSlotUnlockException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# CostumeSlotUnlockException(string)

```
public CostumeSlotUnlockException(string s)
```

## Parameters

s string

# Class CostumeSummon

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("costume_summon")]
public class CostumeSummon : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← CostumeSummon

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CostumeSummon()

```
public CostumeSummon()
```

## CostumeSummon(Address, int, int)

```
public CostumeSummon(Address avatarAddress, int groupId, int summonCount)
```

### Parameters

`avatarAddress` Address

`groupId` int

`summonCount` int

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## AvatarAddressKey

```
public const string AvatarAddressKey = "aa"
```

### Field Value

string

## GroupId

```
public int GroupId
```

### Field Value

int

## GroupIdKey

```
public const string GroupIdKey = "gid"
```

### Field Value

string

## SummonCount

```
public int SummonCount
```

### Field Value

int

## SummonCountKey

```
public const string SummonCountKey = "sc"
```

### Field Value

string

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

# Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

## Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# SimulateSummon(string, CostumeItemSheet, Row, int, IRandom, Row)

Simulates costume summoning with optional grade guarantee system. Applies 10+1 bonus rule and uses grade guarantee settings from SummonSheet.Row when enabled. When grade guarantee is enabled, ensures minimum grade costumes are obtained based on summon count.

```
public static IEnumerable<Costume> SimulateSummon(string addressesHex,
CostumeItemSheet costumeItemSheet, SummonSheet.Row summonRow, int summonCount,
IRandom random, BoostScheduleSheet.Row guaranteeBoost = null)
```

## Parameters

`addressesHex` string

Addresses hex for error reporting

`costumeItemSheet` CostumeItemSheet

Costume item sheet containing costume information

`summonRow` SummonSheet.Row

Summon configuration row with recipes and guarantee settings

`summonCount` int

Number of costumes to summon (before 10+1 bonus)

`random` IRandom

Random number generator

`guaranteeBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this group's guarantee count, or `null` for none. A caller replaying a summon must pass what the action resolved.

## Returns

IEnumerable<Costume>

Collection of summoned costumes

# Class CreateAvatar

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("create_avatar11")]
public class CreateAvatar : GameAction, IAction, ICreateAvatarV2
```

**Inheritance**

object ← ActionBase ← GameAction ← CreateAvatar

**Implements**

IAction, ICreateAvatarV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## DeriveFormat

```
public const string DeriveFormat = "avatar-state-{0}"
```

### Field Value

string

## ear

```
public int ear
```

### Field Value

int

## hair

```
public int hair
```

### Field Value

int

## index

```
public int index
```

### Field Value

int

## lens

```
public int lens
```

### Field Value

int

## name

```
public string name
```

### Field Value

string

## tail

```
public int tail
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## AddItem(ItemSheet, CreateAvatarItemSheet, AvatarState, IRandom)

```
public static void AddItem(ItemSheet itemSheet, CreateAvatarItemSheet
createAvatarItemSheet, AvatarState avatarState, IRandom random)
```

### Parameters

`itemSheet` ItemSheet

`createAvatarItemSheet` CreateAvatarItemSheet

`avatarState` AvatarState

`random` IRandom

## CreateAvatarState(string, Address, IActionContext, Address)

```
public static AvatarState CreateAvatarState(string name, Address avatarAddress,
IActionContext ctx, Address rankingMapAddress)
```

### Parameters

`name` string

`avatarAddress` Address

`ctx` IActionContext

`rankingMapAddress` Address

### Returns

AvatarState

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

## MintAsset(CreateAvatarFavSheet, AvatarState, IWorld, IActionContext)

```
public static IWorld MintAsset(CreateAvatarFavSheet favSheet, AvatarState
avatarState, IWorld states, IActionContext context)
```

### Parameters

`favSheet` CreateAvatarFavSheet

`avatarState` AvatarState

`states` IWorld

`context` IActionContext

### Returns

IWorld

# Class CreatePendingActivation

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("create_pending_activation")]
[ActionObsolete(7200000)]
public class CreatePendingActivation : ActionBase,
IAction, ICreatePendingActivationV1
```

**Inheritance**

object ← ActionBase ← CreatePendingActivation

**Implements**

IAction, ICreatePendingActivationV1

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CreatePendingActivation()

```
public CreatePendingActivation()
```

## CreatePendingActivation(PendingActivationState)

```
public CreatePendingActivation(PendingActivationState activationKey)
```

### Parameters

`activationKey` PendingActivationState

## Properties

### PendingActivation

```
public PendingActivationState PendingActivation { get; }
```

### Property Value

PendingActivationState

### PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class CreatePendingActivations

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/746 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("create_pending_activations")]
[ActionObsolete(7200000)]
public class CreatePendingActivations : ActionBase,
IAction, ICreatePendingActivationsV1
```

**Inheritance**

object ← ActionBase ← CreatePendingActivations

**Implements**

IAction, ICreatePendingActivationsV1

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CreatePendingActivations()

```
public CreatePendingActivations()
```

## CreatePendingActivations(IEnumerable<PendingActivationState>)

```
public CreatePendingActivations(IEnumerable<PendingActivationState> states)
```

### Parameters

`states` IEnumerable<PendingActivationState>

## Properties

### PendingActivations

```
public IList<(byte[] Address, byte[] Nonce, byte[] PublicKey)> PendingActivations {
get; }
```

### Property Value

IList<(byte[] Address, byte[] Nonce, byte[] PublicKey)>

### PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class CreatePledge

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Admin action for pledge contract

```
[ActionType("create_pledge")]
public class CreatePledge : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← CreatePledge

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AgentAddresses

```
public IEnumerable<(Address, Address)> AgentAddresses
```

### Field Value

IEnumerable<(Address, Address)>

## Mead

```
public int Mead
```

### Field Value

int

## PatronAddress

```
public Address PatronAddress
```

### Field Value

Address

## TypeIdentifier

```
public const string TypeIdentifier = "create_pledge"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class DailyReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/1828 Updated at https://github.com/planetarium/lib9c/pull/1828

```
[Serializable]
[ActionType("daily_reward7")]
public class DailyReward : GameAction, IAction, IDailyRewardV1
```

**Inheritance**

object ← ActionBase ← GameAction ← DailyReward

**Implements**

IAction, IDailyRewardV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ActionPointMax

```
public const int ActionPointMax = 120
```

### Field Value

int

## AvatarAddressKey

```
public const string AvatarAddressKey = "a"
```

### Field Value

string

## DailyRewardInterval

```
public const long DailyRewardInterval = 2550
```

### Field Value

long

## DailyRuneRewardAmount

```
public const int DailyRuneRewardAmount = 1
```

### Field Value

int

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class DailyReward2

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("daily_reward2")]
public class DailyReward2 : GameAction, IAction, IDailyRewardV1
```

**Inheritance**

object ← ActionBase ← GameAction ← DailyReward2

**Implements**

IAction, IDailyRewardV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## dailyRewardResult

```
public DailyReward2.DailyRewardResult dailyRewardResult
```

### Field Value

DailyReward2.DailyRewardResult

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class DailyReward2.DailyRewardResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DailyReward2.DailyRewardResult : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← DailyReward2.DailyRewardResult

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## DailyRewardResult()

```
public DailyRewardResult()
```

## DailyRewardResult(Dictionary)

```
public DailyRewardResult(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

# Fields

## id

```
public Guid id
```

### Field Value

Guid

## materials

```
public Dictionary<Material, int> materials
```

### Field Value

Dictionary<Material, int>

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Class DuplicateCostumeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateCostumeException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateCostumeException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## DuplicateCostumeException(SerializationInfo, StreamingContext)

```
protected DuplicateCostumeException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicateCostumeException(string)

```
public DuplicateCostumeException(string s)
```

## Parameters

s string

# Class DuplicateEquipmentException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateEquipmentException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateEquipmentException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## DuplicateEquipmentException(SerializationInfo, StreamingContext)

```
protected DuplicateEquipmentException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicateEquipmentException(string)

```
public DuplicateEquipmentException(string s)
```

## Parameters

s string

# Class DuplicateMaterialException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateMaterialException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateMaterialException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## DuplicateMaterialException(SerializationInfo, StreamingContext)

```
protected DuplicateMaterialException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicateMaterialException(string)

```
public DuplicateMaterialException(string s)
```

## Parameters

s string

# Class DuplicateOrderIdException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateOrderIdException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateOrderIdException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## DuplicateOrderIdException(SerializationInfo, StreamingContext)

```
protected DuplicateOrderIdException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicateOrderIdException(string)

```
public DuplicateOrderIdException(string s)
```

## Parameters

s [string]

# Class EndPledge

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("end_pledge")]
public class EndPledge : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← EndPledge

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EndPledge()

```
public EndPledge()
```

# Fields

## AgentAddress

```
public Address AgentAddress
```

### Field Value

Address

## TypeIdentifier

```
public const string TypeIdentifier = "end_pledge"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class EquipmentLevelExceededException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentLevelExceededException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← EquipmentLevelExceededException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### EquipmentLevelExceededException(SerializationInfo, StreamingContext)

```
protected EquipmentLevelExceededException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# EquipmentLevelExceededException(string)

```
public EquipmentLevelExceededException(string s)
```

## Parameters

s string

# Class EquipmentSlotUnlockException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentSlotUnlockException : Exception, ISerializable
```

### Inheritance

object ← Exception ← EquipmentSlotUnlockException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### EquipmentSlotUnlockException()

```
public EquipmentSlotUnlockException()
```

### EquipmentSlotUnlockException(SerializationInfo, StreamingContext)

```
protected EquipmentSlotUnlockException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## EquipmentSlotUnlockException(string)

```
public EquipmentSlotUnlockException(string msg)
```

### Parameters

`msg` string

# Class EventConsumableItemCrafts

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("event_consumable_item_crafts2")]
public class EventConsumableItemCrafts : GameAction,
IAction, IEventConsumableItemCraftsV1
```

**Inheritance**

object ← ActionBase ← GameAction ← EventConsumableItemCrafts

**Implements**

IAction, IEventConsumableItemCraftsV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## EventConsumableItemRecipeId

```
public int EventConsumableItemRecipeId
```

### Field Value

int

## EventScheduleId

```
public int EventScheduleId
```

### Field Value

int

## SlotIndex

```
public int SlotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class EventDungeonBattle

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Action for participating in an event dungeon battle. This action allows an avatar to battle in a specific stage of an event dungeon, consuming tickets and receiving rewards upon victory. Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("event_dungeon_battle6")]
public class EventDungeonBattle : GameAction, IAction, IEventDungeonBattleV2
```

**Inheritance**

object ← ActionBase ← GameAction ← EventDungeonBattle

**Implements**

IAction, IEventDungeonBattleV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

The address of the avatar that will participate in the event dungeon battle.

```
public Address AvatarAddress
```

### Field Value

Address

## BuyTicketIfNeeded

Whether to automatically purchase a ticket if the avatar doesn't have enough tickets. If true and tickets are insufficient, the action will attempt to buy a ticket using currency.

```
public bool BuyTicketIfNeeded
```

### Field Value

bool

## Costumes

A list of costume item GUIDs to be equipped for the battle.

```
public List<Guid> Costumes
```

### Field Value

List<Guid>

## Equipments

A list of equipment item GUIDs to be equipped for the battle.

```
public List<Guid> Equipments
```

### Field Value

List<Guid>

## EventDungeonId

The ID of the event dungeon to battle in.

```
public int EventDungeonId
```

### Field Value

int

## EventDungeonStageId

The ID of the specific stage within the event dungeon to battle.

```
public int EventDungeonStageId
```

### Field Value

int

## EventScheduleId

The ID of the event schedule that defines when the event dungeon is available.

```
public int EventScheduleId
```

### Field Value

int

## Foods

A list of food item GUIDs to be consumed during the battle.

```
public List<Guid> Foods
```

### Field Value

List<Guid>

## PlayCount

The number of battles to play in a single action execution. Currently fixed to 1 for this action type.

```
public const int PlayCount = 1
```

### Field Value

int

## RuneInfos

A list of rune slot information specifying which runes to use in the battle.

```
public List<RuneSlotInfo> RuneInfos
```

### Field Value

List<RuneSlotInfo>

# Properties

## PlainValueInternal

Serializes the action's data into a Bencodex dictionary. The data is stored as a list under the key "l" containing: AvatarAddress, EventScheduleId, EventDungeonId, EventDungeonStageId, Equipments, Costumes, Foods, BuyTicketIfNeeded, and RuneInfos.

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

An immutable dictionary containing the serialized action data.

# Methods

## Execute(IActionContext)

Executes the event dungeon battle action. This method performs the following operations:

1. Validates the avatar state and event dungeon information
2. Checks and consumes event dungeon tickets (or purchases if needed)
3. Validates equipment, costumes, foods, and runes
4. Updates rune and item slot states
5. Simulates the battle using StageSimulator
6. Applies battle results to the avatar state (experience, rewards, etc.)
7. Updates the event dungeon info if the stage is cleared

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

The action context containing block index, signer, and random seed.

### Returns

IWorld

The updated world state after executing the battle.

### Exceptions

FailedLoadStateException

Thrown when the avatar state cannot be loaded.

NotEnoughEventDungeonTicketsException

Thrown when the avatar doesn't have enough tickets and BuyTicketIfNeeded is false.

StageNotClearedException

Thrown when attempting to battle a stage without clearing the previous stage.

SheetRowNotFoundException

Thrown when required sheet data cannot be found.

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

Deserializes the action's data from a Bencodex dictionary. Expects a list under the key "l" containing at least 9 items in the order: AvatarAddress, EventScheduleId, EventDungeonId, EventDungeonStageId, Equipments, Costumes, Foods, BuyTicketIfNeeded, and RuneInfos.

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

The Bencodex dictionary containing the serialized action data.

### Exceptions

ArgumentException

Thrown when the plainValue doesn't contain the expected key "l", when "l" is not a Bencodex list, or when the list has fewer than 9 items.

# Class EventDungeonBattleSweep

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a sweep action that automatically completes multiple event dungeon battles without player interaction.

This action allows players to automatically repeat battles on already cleared event dungeon stages, consuming tickets and receiving rewards without manual combat interaction.

Key Features: - Only works on previously cleared stages (no stage clearing) - No ticket purchase functionality (uses existing tickets only) - Automatic reward and experience calculation - Multiple battle execution in a single action

This action modifies the following states: - AvatarState: Updates player's stats, inventory, and experience - EventDungeonInfo: Updates event dungeon progress and ticket usage - RuneState: Updates rune effects and bonuses - ItemSlotState: Updates equipped items - RuneSlotState: Updates equipped runes - CP: Updates character's combat power

Execution Flow: 1. Validate input parameters and avatar state 2. Load and validate game sheets and configurations 3. Validate event dungeon info and ticket availability 4. Check stage clearance requirements 5. Update rune and item slot states 6. Calculate combat power (CP) 7. Generate rewards and experience 8. Update avatar state and return modified world state

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("event_dungeon_battle_sweep")]
public class EventDungeonBattleSweep : GameAction, IAction, IEventDungeonBattleSweep
```

### Inheritance

object ← ActionBase ← GameAction ← EventDungeonBattleSweep

### Implements

IAction, IEventDungeonBattleSweep

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,

ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

The address of the avatar performing the sweep action.

```
public Address AvatarAddress
```

### Field Value

Address

## Costumes

List of costume item IDs to use during the sweep.

```
public List<Guid> Costumes
```

### Field Value

List<Guid>

## Equipments

List of equipment item IDs to use during the sweep.

```
public List<Guid> Equipments
```

### Field Value

List<Guid>

## EventDungeonId

The ID of the event dungeon to sweep.

```
public int EventDungeonId
```

### Field Value

int

## EventDungeonStageId

The ID of the event dungeon stage to sweep.

```
public int EventDungeonStageId
```

### Field Value

int

## EventScheduleId

The ID of the event schedule for the dungeon.

```
public int EventScheduleId
```

### Field Value

int

## Foods

List of food item IDs to use during the sweep.

```
public List<Guid> Foods
```

### Field Value

List<Guid>

## PlayCount

The number of times to perform the sweep (must be between 1 and MaxPlayCount).

```
public int PlayCount
```

### Field Value

int

## RuneInfos

List of rune slot information for equipped runes.

```
public List<RuneSlotInfo> RuneInfos
```

### Field Value

List<RuneSlotInfo>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

Executes the event dungeon battle sweep action.

This method performs the following steps: 1. Validates input parameters and loads avatar state 2. Loads and validates all required game sheets 3. Validates event dungeon configuration and ticket availability 4. Checks stage clearance requirements 5. Updates rune and item slot states 6. Calculates combat power (CP) 7. Generates rewards and experience 8. Updates all relevant states and returns the modified world

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

Action context containing previous state, signer, and block information

### Returns

IWorld

Modified world state with updated avatar, dungeon info, and other states

## GetRewardItems(IRandom, int, Row, MaterialItemSheet)

Gets reward items for event dungeon sweep based on play count and stage configuration.

This method simulates multiple battle rewards without actually performing combat. It generates items based on the stage's drop configuration and the number of plays requested.

Process: 1. Determine the number of items to drop per play (random between min and max) 2. For each play, generate rewards using the stage's item selector 3. Collect all rewards and sort them by item ID for consistency

```
public static List<ItemBase> GetRewardItems(IRandom random, int playCount,
EventDungeonStageSheet.Row stageRow, MaterialItemSheet materialItemSheet)
```

### Parameters

`random` IRandom

Random number generator for deterministic reward generation

`playCount` int

Number of plays to simulate (determines total reward count)

`stageRow` EventDungeonStageSheet.Row

Event dungeon stage configuration containing drop rates and item pools

`materialItemSheet` MaterialItemSheet

Material item sheet for reward generation and validation

### Returns

List<ItemBase>

List of reward items sorted by item ID

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class EventMaterialItemCrafts

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("event_material_item_crafts2")]
public class EventMaterialItemCrafts : GameAction,
IAction, IEventMaterialItemCraftsV1
```

**Inheritance**

object ← ActionBase ← GameAction ← EventMaterialItemCrafts

**Implements**

IAction, IEventMaterialItemCraftsV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## EventMaterialItemRecipeId

```
public int EventMaterialItemRecipeId
```

### Field Value

int

## EventScheduleId

```
public int EventScheduleId
```

### Field Value

int

## MaterialsToUse

```
public Dictionary<int, int> MaterialsToUse
```

### Field Value

Dictionary<int, int>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class FailedLoadSheetException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class FailedLoadSheetException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← FailedLoadSheetException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### FailedLoadSheetException(SerializationInfo, StreamingContext)

```
protected FailedLoadSheetException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## FailedLoadSheetException(string)

```
public FailedLoadSheetException(string message)
```

### Parameters

`message` string

## FailedLoadSheetException(Type)

```
public FailedLoadSheetException(Type sheetType)
```

### Parameters

`sheetType` Type

# Class FailedLoadStateException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class FailedLoadStateException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← FailedLoadStateException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### FailedLoadStateException(Address, Type, Exception)

```
public FailedLoadStateException(Address address, Type stateType, Exception
innerException = null)
```

#### Parameters

`address` Address

`stateType` Type

`innerException` Exception

## FailedLoadStateException(SerializationInfo, StreamingContext)

```
protected FailedLoadStateException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## FailedLoadStateException(string)

```
public FailedLoadStateException(string message)
```

### Parameters

`message` string

## FailedLoadStateException(string, Exception)

```
public FailedLoadStateException(string message, Exception innerException = null)
```

### Parameters

`message` string

`innerException` Exception

## FailedLoadStateException(string, string, string, Exception)

```
public FailedLoadStateException(string actionType, string addressesHex, string
message, Exception innerException = null)
```

### Parameters

`actionType` string

`addressesHex` string

`message` string

`innerException` Exception

## FailedLoadStateException(string, string, Type, Address)

```
public FailedLoadStateException(string actionType, string addressesHex, Type
stateType, Address address)
```

### Parameters

`actionType` string

`addressesHex` string

`stateType` Type

`address` Address

# Class FavProductInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class FavProductInfo : IProductInfo
```

**Inheritance**

object ← FavProductInfo

**Implements**

IProductInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### FavProductInfo()

```
public FavProductInfo()
```

### FavProductInfo(List)

```
public FavProductInfo(List serialized)
```

#### Parameters

`serialized` List

## Properties

## AgentAddress

```
public Address AgentAddress { get; set; }
```

### Property Value

Address

## AvatarAddress

```
public Address AvatarAddress { get; set; }
```

### Property Value

Address

## Price

```
public FungibleAssetValue Price { get; set; }
```

### Property Value

FungibleAssetValue

## ProductId

```
public Guid ProductId { get; set; }
```

### Property Value

Guid

## Type

```
public ProductType Type { get; set; }
```

### Property Value

ProductType

## Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## ValidateType()

```
public void ValidateType()
```

# Class GameAction

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public abstract class GameAction : ActionBase, IAction
```

### Inheritance

object ← ActionBase ← GameAction

### Implements

IAction

### Derived

ActivateCollection, AddRedeemCode, ClaimAdventureBossReward, ExploreAdventureBoss, SweepAdventureBoss, UnlockFloor, Wanted, Battle, AuraSummon, BattleArena, Buy, Buy7, BuyMultiple, BuyProduct, CancelProductRegistration, ChargeActionPoint, ClaimGifts, ClaimItems, ClaimRaidReward, ClaimStakeReward, ClaimWordBossKillReward, ClaimWorldBossReward, CombinationConsumable, CombinationConsumable5, CombinationEquipment, CostumeSummon, IssueCoupons, RedeemCoupon, TransferCoupons, CreateAvatar, DailyReward, DailyReward2, EventConsumableItemCrafts, EventDungeonBattle, EventDungeonBattleSweep, EventMaterialItemCrafts, BulkUnloadFromGarages, DeliverToOthersGarages, LoadIntoMyGarages, UnloadFromMyGarages, GrantItems, Grinding, HackAndSlash, HackAndSlashRandomBuff, HackAndSlashSweep, InfiniteTowerBattle, InitializeStates, ItemEnhancement, ItemEnhancement10, ItemEnhancement11, ItemEnhancement12, ItemEnhancement13, ItemEnhancement7, ItemEnhancement9, JoinArena, JoinArena1, JoinArena3, MigrationActivatedAccountsState, PatchTableSheet, PatchTableSheetCompressed, PetEnhancement, Raid, RankingBattle, RankingBattle11, RapidCombination, RapidCombination0, RapidCombination5, ReRegisterProduct, RedeemCode, RedeemCode2, RegisterProduct, RegisterProduct0, RenewAdminState, RuneEnhancement, RuneSummon, Sell, Sell6, SellCancellation, Stake, Stake2, Synthesize, UnlockCombinationSlot, UnlockEquipmentRecipe, UnlockRuneSlot, UnlockWorld, UpdateSell

### Inherited Members

ActionBase.Execute(IActionContext) ,
ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,

ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GameAction()

```
protected GameAction()
```

# Properties

## Id

```
public Guid Id { get; }
```

### Property Value

Guid

## PlainValue

```
public override sealed IValue PlainValue { get; }
```

### Property Value

IValue

## PlainValueInternal

```
protected abstract IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## LoadPlainValue(IValue)

```
public override sealed void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected abstract void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class GoldDistribution

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class GoldDistribution : IEquatable<GoldDistribution>
```

**Inheritance**

object ← GoldDistribution

**Implements**

IEquatable<GoldDistribution>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GoldDistribution()

```
public GoldDistribution()
```

### GoldDistribution(Dictionary)

```
public GoldDistribution(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### GoldDistribution(IValue)

```
public GoldDistribution(IValue serialized)
```

### Parameters

`serialized` IValue

## Fields

### Address

```
[Ignore]
public Address Address
```

### Field Value

Address

## Properties

### AddressString

```
[Index(0, -1)]
public string AddressString { get; set; }
```

### Property Value

string

### AmountPerBlock

```
[Index(1, -1)]
public BigInteger AmountPerBlock { get; set; }
```

### Property Value

BigInteger

## EndBlock

```
[Index(3, -1)]
public long EndBlock { get; set; }
```

### Property Value

long

## StartBlock

```
[Index(2, -1)]
public long StartBlock { get; set; }
```

### Property Value

long

# Methods

## Equals(GoldDistribution)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GoldDistribution other)
```

### Parameters

`other` GoldDistribution

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetAmount(long)

```
public BigInteger GetAmount(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

BigInteger

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## LoadInDescendingEndBlockOrder(string)

```
public static GoldDistribution[] LoadInDescendingEndBlockOrder(string csvPath)
```

### Parameters

`csvPath` string

### Returns

GoldDistribution[]

## Serialize()

```
public Dictionary Serialize()
```

### Returns

Dictionary

# Class GrantItems

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Grants items/assets to target avatars from token-like inputs. This action is restricted to a fixed allow-list of signers and the current admin signer. Even if the signer does not have sufficient token balance, the grant will proceed and will burn only the available amount of the token.

```
[ActionType("grant_items")]
public class GrantItems : GameAction, IAction, IClaimItems
```

**Inheritance**

object ← ActionBase ← GameAction ← GrantItems

**Implements**

IAction, IClaimItems

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GrantItems()

Creates an empty action instance.

```
public GrantItems()
```

# GrantItems(IReadOnlyList<(Address, IReadOnlyList<FungibleAssetValue>)>, string)

Creates an action instance with the given claim data.

```
public GrantItems(IReadOnlyList<(Address, IReadOnlyList<FungibleAssetValue>)>
claimData, string memo = null)
```

## Parameters

`claimData` IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

Grant targets and token-like inputs.

`memo` string

Optional memo written to the resulting mail.

# Fields

## Memo

Optional memo written to the resulting mail.

```
public string Memo
```

## Field Value

string

# Properties

## ClaimData

```
public IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue>
fungibleAssetValues)> ClaimData { get; }
```

### Property Value

IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class Grinding

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("grinding2")]
public class Grinding : GameAction, IAction, IGrindingV1
```

**Inheritance**

object ← ActionBase ← GameAction ← Grinding

**Implements**

IAction, IGrindingV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## ChargeAp

```
public bool ChargeAp
```

### Field Value

bool

## CostAp

```
public const int CostAp = 5
```

### Field Value

int

## EquipmentIds

```
public List<Guid> EquipmentIds
```

### Field Value

List<Guid>

## Limit

```
public const int Limit = 50
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

## CalculateMaterialReward(IEnumerable<Equipment>, CrystalEquipmentGrindingSheet, MaterialItemSheet)

```
public static Dictionary<Material, int>
CalculateMaterialReward(IEnumerable<Equipment> equipmentList,
CrystalEquipmentGrindingSheet crystalEquipmentGrindingSheet,
MaterialItemSheet materialItemSheet)
```

### Parameters

`equipmentList` IEnumerable<Equipment>

`crystalEquipmentGrindingSheet` CrystalEquipmentGrindingSheet

`materialItemSheet` MaterialItemSheet

### Returns

Dictionary<Material, int>

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class HackAndSlash

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a battle action in the game where a player engages in combat with monsters. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- WorldBossKillRewardRecord: Records boss kill rewards
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("hack_and_slash22")]
public class HackAndSlash : GameAction, IAction, IHackAndSlashV10
```

### Inheritance

object ← ActionBase ← GameAction ← HackAndSlash

### Implements

IAction, IHackAndSlashV10

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Fields

## ApStoneCount

```
public int ApStoneCount
```

### Field Value

int

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Costumes

```
public List<Guid> Costumes
```

### Field Value

List<Guid>

## EntryCostItemCount

Gets or sets the declared total entry cost material item count for this action execution, matching EntryCostItemCount × TotalPlayCount. Required when the stage has an entry material cost; must match the computed value exactly.

```
public int EntryCostItemCount
```

### Field Value

int

## EntryCostItemId

Gets or sets the declared entry cost material item ID, matching EntryCostItemId. Required when the stage has an entry material cost; must match the sheet value exactly.

```
public int EntryCostItemId
```

### Field Value

int

## Equipments

```
public List<Guid> Equipments
```

### Field Value

List<Guid>

## Foods

```
public List<Guid> Foods
```

### Field Value

List<Guid>

## RuneInfos

```
public List<RuneSlotInfo> RuneInfos
```

### Field Value

List<RuneSlotInfo>

## StageBuffId

```
public int? StageBuffId
```

### Field Value

int?

## StageId

```
public int StageId
```

### Field Value

int

## TotalPlayCount

```
public int TotalPlayCount
```

### Field Value

int

## UsableApStoneCount

```
public const int UsableApStoneCount = 10
```

### Field Value

int

## WorldId

```
public int WorldId
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## Execute(IWorld, Address, long, IRandom, IActionContext)

```
public IWorld Execute(IWorld states, Address signer, long blockIndex, IRandom
random, IActionContext context = null)
```

### Parameters

`states` IWorld

`signer` Address

`blockIndex` long

`random` IRandom

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class HackAndSlashRandomBuff

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Created at https://github.com/planetarium/lib9c/pull/1031

```
[Serializable]
[ActionType("hack_and_slash_random_buff")]
public class HackAndSlashRandomBuff : GameAction, IAction, IHackAndSlashRandomBuffV1
```

**Inheritance**

object ← ActionBase ← GameAction ← HackAndSlashRandomBuff

**Implements**

IAction, IHackAndSlashRandomBuffV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AdvancedGacha

```
public bool AdvancedGacha
```

### Field Value

bool

## AdvancedGachaCount

```
public const int AdvancedGachaCount = 10
```

### Field Value

int

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## NormalGachaCount

```
public const int NormalGachaCount = 5
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class HackAndSlashSweep

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a sweep action that automatically completes multiple battles without player interaction. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("hack_and_slash_sweep10")]
public class HackAndSlashSweep : GameAction, IAction, IHackAndSlashSweepV3
```

**Inheritance**

object ← ActionBase ← GameAction ← HackAndSlashSweep

**Implements**

IAction, IHackAndSlashSweepV3

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## UsableApStoneCount

```
public const int UsableApStoneCount = 100
```

### Field Value

int

## actionPoint

```
public int actionPoint
```

### Field Value

int

## apStoneCount

```
public int apStoneCount
```

### Field Value

int

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## costumes

```
public List<Guid> costumes
```

### Field Value

List<Guid>

## entryCostItemCount

Declared total entry cost material item count for this sweep execution, matching EntryCostItemCount × playCount. Required when the stage has an entry material cost; must match the computed value exactly.

```
public int entryCostItemCount
```

### Field Value

int

## entryCostItemId

Declared entry cost material item ID, matching EntryCostItemId. Required when the stage has an entry material cost; must match the sheet value exactly.

```
public int entryCostItemId
```

### Field Value

int

## equipments

```
public List<Guid> equipments
```

### Field Value

List<Guid>

## runeInfos

```
public List<RuneSlotInfo> runeInfos
```

### Field Value

List<RuneSlotInfo>

## stageId

```
public int stageId
```

### Field Value

int

## worldId

```
public int worldId
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetRewardItems(IRandom, int, Row, MaterialItemSheet, Row)

Draws the item rewards of `playCount` sweeps of `stageRow`, one object per unit.

```
public static List<ItemBase> GetRewardItems(IRandom random, int playCount,
StageSheet.Row stageRow, MaterialItemSheet materialItemSheet, BoostScheduleSheet.Row
itemRewardBoost = null)
```

### Parameters

`random` IRandom

The random number generator.

`playCount` int

The number of clears swept.

`stageRow` StageSheet.Row

The stage being swept.

`materialItemSheet` MaterialItemSheet

The sheet the rewards are created from.

`itemRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting each clear's items, or `null` for none. Applied per clear exactly as `HackAndSlash` applies it; see ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet) and GetBoostedItemRewardCount(Row, int). It never changes the random draws. A caller replaying a sweep must pass what the action resolved.

### Returns

List<ItemBase>

The items of every clear, one object per unit, ordered by item id. Their count per item id is exactly what the action grants, although the action adds the units a boost adds with one count per item id instead of creating these objects.

## GetRewardItems(IRandom, int, Row, MaterialItemSheet, Row, IDictionary<int, int>)

Draws the item rewards of `playCount` sweeps of `stageRow`, counting the units a boost adds instead of creating them.

```
public static List<ItemBase> GetRewardItems(IRandom random, int playCount,
StageSheet.Row stageRow, MaterialItemSheet materialItemSheet, BoostScheduleSheet.Row
itemRewardBoost, IDictionary<int, int> extraCounts)
```

### Parameters

`random` IRandom

The random number generator.

`playCount` int

The number of clears swept.

`stageRow` StageSheet.Row

The stage being swept.

`materialItemSheet` MaterialItemSheet

The sheet the rewards are created from.

`itemRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting each clear's items, or `null` for none. See SplitItemRewardBoost(List<ItemBase>, Row, IDictionary<int, int>).

`extraCounts` IDictionary<int, int>

Accumulates, per item id, the units the boost adds on top of the returned items over every clear. Stays empty without a row.

### Returns

List<ItemBase>

The drawn items that remain after the boost, one object per unit, ordered by item id. Without a row, exactly the items the sweep drew.

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Interface IBuy0

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Common interface used before IBuy5.

```
public interface IBuy0 : IAction
```

**Inherited Members**
IAction.LoadPlainValue(IValue) , IAction.Execute(IActionContext) , IAction.PlainValue

## Properties

### buyerAvatarAddress

```
Address buyerAvatarAddress { get; }
```

#### Property Value

Address

### productId

```
Guid productId { get; }
```

#### Property Value

Guid

### sellerAgentAddress

```
Address sellerAgentAddress { get; }
```

### Property Value

Address

## sellerAvatarAddress

```
Address sellerAvatarAddress { get; }
```

### Property Value

Address

## See Also

IBuy5

# Interface IBuy5

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Common interface used after IBuy5.

```
public interface IBuy5 : IAction
```

**Inherited Members**
IAction.LoadPlainValue(IValue) , IAction.Execute(IActionContext) , IAction.PlainValue

# Properties

## buyerAvatarAddress

```
Address buyerAvatarAddress { get; }
```

### Property Value

Address

## purchaseInfos

```
IEnumerable<IPurchaseInfo> purchaseInfos { get; }
```

### Property Value

IEnumerable<IPurchaseInfo>

# See Also

IBuy5

# Interface IClaimItems

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IClaimItems
```

## Properties

### ClaimData

```
IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue>
fungibleAssetValues)> ClaimData { get; }
```

#### Property Value

IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

# Interface IClaimStakeReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IClaimStakeReward : IAction
```

### Inherited Members

IAction.LoadPlainValue(IValue) , IAction.Execute(IActionContext) , IAction.PlainValue

# Interface IHackAndSlashV10

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Interface for hack and slash action version 10.

```
public interface IHackAndSlashV10
```

# Properties

## ApStoneCount

Gets the number of AP stones to use.

```
int ApStoneCount { get; }
```

### Property Value

int

## AvatarAddress

Gets the avatar address for the battle.

```
Address AvatarAddress { get; }
```

### Property Value

Address

## Costumes

Gets the costume IDs to use in battle.

```
IEnumerable<Guid> Costumes { get; }
```

### Property Value

IEnumerable<Guid>

## Equipments

Gets the equipment IDs to use in battle.

```
IEnumerable<Guid> Equipments { get; }
```

### Property Value

IEnumerable<Guid>

## Foods

Gets the food IDs to use in battle.

```
IEnumerable<Guid> Foods { get; }
```

### Property Value

IEnumerable<Guid>

## RuneSlotInfos

Gets the rune slot information for the battle.

```
IEnumerable<IValue> RuneSlotInfos { get; }
```

### Property Value

IEnumerable<IValue>

## StageBuffId

Gets the stage buff ID to apply.

```
int? StageBuffId { get; }
```

### Property Value

int?

## StageId

Gets the stage ID to challenge.

```
int StageId { get; }
```

### Property Value

int

## TotalPlayCount

Gets the total number of plays.

```
int TotalPlayCount { get; }
```

### Property Value

int

## WorldId

Gets the world ID for the stage.

```
int WorldId { get; }
```

## Property Value

int

# Interface IProductInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IProductInfo
```

## Properties

### AgentAddress

```
Address AgentAddress { get; set; }
```

#### Property Value

Address

### AvatarAddress

```
Address AvatarAddress { get; set; }
```

#### Property Value

Address

### Price

```
FungibleAssetValue Price { get; set; }
```

#### Property Value

FungibleAssetValue

## ProductId

```
Guid ProductId { get; set; }
```

### Property Value

Guid

## Type

```
ProductType Type { get; set; }
```

### Property Value

ProductType

# Methods

## Serialize()

```
IValue Serialize()
```

### Returns

IValue

## ValidateType()

```
void ValidateType()
```

# Interface IPurchaseInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IPurchaseInfo
```

## Properties

### ItemSubType

```
ItemSubType ItemSubType { get; }
```

#### Property Value

ItemSubType

### OrderId

```
Guid? OrderId { get; }
```

#### Property Value

Guid?

### Price

```
FungibleAssetValue Price { get; }
```

#### Property Value

FungibleAssetValue

# SellerAgentAddress

```
Address SellerAgentAddress { get; }
```

## Property Value

Address

# SellerAvatarAddress

```
Address SellerAvatarAddress { get; }
```

## Property Value

Address

# Interface IRegisterInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IRegisterInfo
```

## Properties

### AvatarAddress

```
Address AvatarAddress { get; set; }
```

#### Property Value

Address

### Price

```
FungibleAssetValue Price { get; set; }
```

#### Property Value

FungibleAssetValue

### Type

```
ProductType Type { get; set; }
```

#### Property Value

ProductType

# Methods

## Serialize()

```
IValue Serialize()
```

### Returns

IValue

## Validate()

```
void Validate()
```

## ValidateAddress(Address)

```
void ValidateAddress(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

## ValidatePrice(Currency)

```
void ValidatePrice(Currency ncg)
```

### Parameters

`ncg` Currency

# Interface ITransferAsset

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface ITransferAsset
```

## Properties

### Amount

```
FungibleAssetValue Amount { get; }
```

#### Property Value

FungibleAssetValue

### Memo

```
string? Memo { get; }
```

#### Property Value

string

### Recipient

```
Address Recipient { get; }
```

#### Property Value

Address

# Sender

```
Address Sender { get; }
```

## Property Value

Address

# Interface ITransferAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface ITransferAssets
```

## Properties

### Memo

```
string? Memo { get; }
```

#### Property Value

string

### Recipients

```
List<(Address recipient, FungibleAssetValue amount)> Recipients { get; }
```

#### Property Value

List<(Address recipient, FungibleAssetValue amount)>

### Sender

```
Address Sender { get; }
```

#### Property Value

Address

# Class InfiniteTowerBattle

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a battle action in the infinite tower where players engage in combat with various conditions and restrictions. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- InfiniteTowerInfo: Updates tower progress and tickets
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

```
[Serializable]
[ActionType("infinite_tower_battle")]
public class InfiniteTowerBattle : GameAction, IAction, IInfiniteTowerBattleV1
```

**Inheritance**

object ← ActionBase ← GameAction ← InfiniteTowerBattle

**Implements**

IAction, IInfiniteTowerBattleV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

The avatar address for the battle.

```
public Address AvatarAddress
```

### Field Value

Address

## BuyTicketIfNeeded

Whether to buy a ticket if needed.

```
public bool BuyTicketIfNeeded
```

### Field Value

bool

## Costumes

The costume IDs to use in battle.

```
public List<Guid> Costumes
```

### Field Value

List<Guid>

## Equipments

The equipment IDs to use in battle.

```
public List<Guid> Equipments
```

### Field Value

List<Guid>

## FloorId

The floor ID to challenge.

```
public int FloorId
```

### Field Value

int

## Foods

The food IDs to use in battle.

```
public List<Guid> Foods
```

### Field Value

List<Guid>

## InfiniteTowerId

The infinite tower ID.

```
public int InfiniteTowerId
```

### Field Value

int

## PlayCount

The number of times this action can be played.

```
public const int PlayCount = 1
```

### Field Value

int

## RuneInfos

The rune slot information for the battle.

```
public List<RuneSlotInfo> RuneInfos
```

### Field Value

List<RuneSlotInfo>

## UseNcgForTicket

Whether to use NCG for buying tickets.

```
public bool UseNcgForTicket
```

### Field Value

bool

# Properties

## PlainValueInternal

Gets the plain value representation of this action for serialization.

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

Executes the infinite tower battle action.

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

The action context containing state and block information.

### Returns

IWorld

The updated world state after the battle.

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

Loads the action data from a plain value dictionary.

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

The plain value dictionary containing action data.

## PerformFullReset(IWorld, Address, int, long)

Helper method to manually trigger full reset for testing purposes.

```
public static void PerformFullReset(IWorld states, Address avatarAddress, int
infiniteTowerId, long currentBlockIndex)
```

### Parameters

`states` IWorld

`avatarAddress` Address

`infiniteTowerId` int

`currentBlockIndex` long

## RefillDailyTickets(IWorld, Address, int, long)

Helper method to manually trigger daily ticket refill for testing purposes.

```
public static void RefillDailyTickets(IWorld states, Address avatarAddress, int
infiniteTowerId, long currentBlockIndex)
```

### Parameters

`states` IWorld

`avatarAddress` Address

`infiniteTowerId` int

`currentBlockIndex` long

# Class InitializeStates

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at NCG distribution(7e4515b6e14cc5d6eb716d5ebb587ab04b4246f9) Updated at https://github.com/planetarium/lib9c/pull/19 Updated at https://github.com/planetarium/lib9c/pull/36 Updated at https://github.com/planetarium/lib9c/pull/42 Updated at https://github.com/planetarium/lib9c/pull/55 Updated at https://github.com/planetarium/lib9c/pull/57 Updated at https://github.com/planetarium/lib9c/pull/60 Updated at https://github.com/planetarium/lib9c/pull/102 Updated at https://github.com/planetarium/lib9c/pull/128 Updated at https://github.com/planetarium/lib9c/pull/167 Updated at https://github.com/planetarium/lib9c/pull/422 Updated at https://github.com/planetarium/lib9c/pull/747 Updated at https://github.com/planetarium/lib9c/pull/798 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("initialize_states")]
public class InitializeStates : GameAction, IAction, IInitializeStatesV1
```

### Inheritance

object ← ActionBase ← GameAction ← InitializeStates

### Implements

IAction, IInitializeStatesV1

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## InitializeStates()

```
public InitializeStates()
```

## InitializeStates(ValidatorSet, RankingState0, ShopState, Dictionary<string, string>, GameConfigState, RedeemCodeState, ActivatedAccountsState, GoldCurrencyState, GoldDistribution[], PendingActivationState[], AdminState, AuthorizedMinersState, CreditsState, ISet<Address>)

```
public InitializeStates(ValidatorSet validatorSet, RankingState0 rankingState,
ShopState shopState, Dictionary<string, string> tableSheets, GameConfigState
gameConfigState, RedeemCodeState redeemCodeState, ActivatedAccountsState
activatedAccountsState, GoldCurrencyState goldCurrencyState, GoldDistribution[]
goldDistributions, PendingActivationState[] pendingActivationStates, AdminState
adminAddressState = null, AuthorizedMinersState authorizedMinersState = null,
CreditsState creditsState = null, ISet<Address> assetMinters = null)
```

### Parameters

`validatorSet` ValidatorSet

`rankingState` RankingState0

`shopState` ShopState

`tableSheets` Dictionary<string, string>

`gameConfigState` GameConfigState

`redeemCodeState` RedeemCodeState

`activatedAccountsState` ActivatedAccountsState

`goldCurrencyState` GoldCurrencyState

`goldDistributions` GoldDistribution[]

`pendingActivationStates` PendingActivationState[]

`adminAddressState` AdminState

`authorizedMinersState` AuthorizedMinersState

`creditsState` CreditsState

`assetMinters` ISet<Address>

# Properties

## ActivatedAccounts

```
public Dictionary ActivatedAccounts { get; set; }
```

### Property Value

Dictionary

## AdminAddressState

```
public Dictionary AdminAddressState { get; set; }
```

### Property Value

Dictionary

## AssetMinters

```
public ISet<Address> AssetMinters { get; set; }
```

### Property Value

ISet<Address>

## AuthorizedMiners

```
public Dictionary AuthorizedMiners { get; set; }
```

### Property Value

Dictionary

## Credits

```
public Dictionary Credits { get; set; }
```

### Property Value

Dictionary

## GameConfig

```
public Dictionary GameConfig { get; set; }
```

### Property Value

Dictionary

## GoldCurrency

```
public Dictionary GoldCurrency { get; set; }
```

### Property Value

Dictionary

## GoldDistributions

```
public List GoldDistributions { get; set; }
```

### Property Value

List

## PendingActivations

```
public List PendingActivations { get; set; }
```

### Property Value

List

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Ranking

```
public Dictionary Ranking { get; set; }
```

### Property Value

Dictionary

## RedeemCode

```
public Dictionary RedeemCode { get; set; }
```

### Property Value

Dictionary

## Shop

```
public Dictionary Shop { get; set; }
```

### Property Value

Dictionary

## TableSheets

```
public Dictionary<string, string> TableSheets { get; set; }
```

### Property Value

Dictionary<string, string>

## ValidatorSet

```
public IValue ValidatorSet { get; set; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class InvalidAddressException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidAddressException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidAddressException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidAddressException()

```
public InvalidAddressException()
```

### InvalidAddressException(SerializationInfo, StreamingContext)

```
protected InvalidAddressException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidAddressException(string)

```
public InvalidAddressException(string msg)
```

### Parameters

`msg` string

# Class InvalidClaimException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidClaimException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidClaimException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidClaimException()

```
public InvalidClaimException()
```

### InvalidClaimException(SerializationInfo, StreamingContext)

```
protected InvalidClaimException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidClaimException(string)

```
public InvalidClaimException(string msg)
```

### Parameters

`msg` string

# Class InvalidCurrencyException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidCurrencyException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidCurrencyException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## InvalidCurrencyException(SerializationInfo, StreamingContext)

```
protected InvalidCurrencyException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidCurrencyException(string)

```
public InvalidCurrencyException(string msg)
```

## Parameters

`msg` string

# Class InvalidElementalException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidElementalException : Exception, ISerializable
```

### Inheritance

object ← Exception ← InvalidElementalException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidElementalException()

```
public InvalidElementalException()
```

### InvalidElementalException(SerializationInfo, StreamingContext)

```
protected InvalidElementalException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidElementalException(string)

```
public InvalidElementalException(string message)
```

### Parameters

`message` string

# Class InvalidEquipmentException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidEquipmentException : Exception, ISerializable
```

### Inheritance

object ← Exception ← InvalidEquipmentException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidEquipmentException()

```
public InvalidEquipmentException()
```

### InvalidEquipmentException(SerializationInfo, StreamingContext)

```
protected InvalidEquipmentException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidEquipmentException(string)

```
public InvalidEquipmentException(string msg)
```

### Parameters

`msg` string

# Class InvalidItemCountException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidItemCountException : ArgumentOutOfRangeException, ISerializable
```

**Inheritance**

object ← Exception ← SystemException ← ArgumentException ← ArgumentOutOfRangeException ← InvalidItemCountException

**Implements**

ISerializable

**Inherited Members**

ArgumentOutOfRangeException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentOutOfRangeException.ActualValue , ArgumentOutOfRangeException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidItemCountException()

```
public InvalidItemCountException()
```

### InvalidItemCountException(SerializationInfo, StreamingContext)

```
protected InvalidItemCountException(SerializationInfo info,
```

```
  StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidItemCountException(string)

```
  public InvalidItemCountException(string msg)
```

### Parameters

`msg` string

# Class InvalidItemIdException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents an exception that is thrown when an invalid item ID is used.

```
[Serializable]
public class InvalidItemIdException : ArgumentOutOfRangeException, ISerializable
```

**Inheritance**

object ← Exception ← SystemException ← ArgumentException ← ArgumentOutOfRangeException ← InvalidItemIdException

**Implements**

ISerializable

**Inherited Members**

ArgumentOutOfRangeException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentOutOfRangeException.ActualValue , ArgumentOutOfRangeException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Constructors

## InvalidItemIdException()

Initializes a new instance of the InvalidItemIdException class.

```
public InvalidItemIdException()
```

# InvalidItemIdException(SerializationInfo, StreamingContext)

Initializes a new instance of the InvalidItemIdException class with a specified error message

```
protected InvalidItemIdException(SerializationInfo info, StreamingContext context)
```

## Parameters

`info` SerializationInfo

SerializationInfo

`context` StreamingContext

StreamingContext

# InvalidItemIdException(string)

Initializes a new instance of the InvalidItemIdException class with a specified error message.

```
public InvalidItemIdException(string msg)
```

## Parameters

`msg` string

The message that describes the error.

# Class InvalidItemTypeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidItemTypeException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidItemTypeException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## InvalidItemTypeException(SerializationInfo, StreamingContext)

```
protected InvalidItemTypeException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidItemTypeException(string)

```
public InvalidItemTypeException(string msg)
```

## Parameters

`msg` string

# Class InvalidLevelException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidLevelException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidLevelException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidLevelException(SerializationInfo, StreamingContext)

```
protected InvalidLevelException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidLevelException(string)

```
public InvalidLevelException(string s)
```

## Parameters

s string

# Class InvalidMaterialException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidMaterialException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidMaterialException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## InvalidMaterialException(SerializationInfo, StreamingContext)

```
protected InvalidMaterialException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidMaterialException(string)

```
public InvalidMaterialException(string s)
```

## Parameters

s [string]

# Class InvalidMinterException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidMinterException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidMinterException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidMinterException()

```
public InvalidMinterException()
```

### InvalidMinterException(Address)

```
public InvalidMinterException(Address signer)
```

#### Parameters

`signer` Address

## InvalidMinterException(SerializationInfo, StreamingContext)

```
protected InvalidMinterException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidMinterException(string)

```
public InvalidMinterException(string message)
```

### Parameters

`message` string

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

## Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class InvalidMonsterCollectionRoundException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidMonsterCollectionRoundException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidMonsterCollectionRoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidMonsterCollectionRoundException(SerializationInfo, StreamingContext)

```
protected InvalidMonsterCollectionRoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidMonsterCollectionRoundException(string)

```
public InvalidMonsterCollectionRoundException(string msg)
```

### Parameters

`msg` string

# Class InvalidNamePatternException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidNamePatternException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidNamePatternException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## InvalidNamePatternException(SerializationInfo, StreamingContext)

```
protected InvalidNamePatternException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidNamePatternException(string)

```
public InvalidNamePatternException(string s)
```

## Parameters

s string

# Class InvalidPriceException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidPriceException : ArgumentOutOfRangeException, ISerializable
```

**Inheritance**

object ← Exception ← SystemException ← ArgumentException ← ArgumentOutOfRangeException ← InvalidPriceException

**Implements**

ISerializable

**Inherited Members**

ArgumentOutOfRangeException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentOutOfRangeException.ActualValue , ArgumentOutOfRangeException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidPriceException(SerializationInfo, StreamingContext)

```
protected InvalidPriceException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidPriceException(string)

```
public InvalidPriceException(string msg)
```

## Parameters

`msg` string

# Class InvalidProductTypeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidProductTypeException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidProductTypeException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## InvalidProductTypeException(SerializationInfo, StreamingContext)

```
protected InvalidProductTypeException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidProductTypeException(string)

```
public InvalidProductTypeException(string msg)
```

## Parameters

`msg` string

# Class InvalidRecipeIdException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidRecipeIdException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidRecipeIdException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidRecipeIdException()

```
public InvalidRecipeIdException()
```

### InvalidRecipeIdException(SerializationInfo, StreamingContext)

```
protected InvalidRecipeIdException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidRecipeIdException(string)

```
public InvalidRecipeIdException(string msg)
```

### Parameters

`msg` string

# Class InvalidRepeatPlayException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidRepeatPlayException : Exception, ISerializable
```

### Inheritance

object ← Exception ← InvalidRepeatPlayException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidRepeatPlayException()

```
public InvalidRepeatPlayException()
```

### InvalidRepeatPlayException(SerializationInfo, StreamingContext)

```
protected InvalidRepeatPlayException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidRepeatPlayException(string)

```
public InvalidRepeatPlayException(string msg)
```

### Parameters

`msg` string

# Class InvalidShopItemException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidShopItemException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidShopItemException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## InvalidShopItemException(SerializationInfo, StreamingContext)

```
protected InvalidShopItemException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidShopItemException(string)

```
public InvalidShopItemException(string s)
```

## Parameters

s [string]

# Class InvalidSignatureException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidSignatureException : ActivationException, ISerializable
```

**Inheritance**

object ← Exception ← ActivationException ← InvalidSignatureException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidSignatureException(PendingActivationState, byte[])

```
public InvalidSignatureException(PendingActivationState pending, byte[] signature)
```

#### Parameters

`pending` PendingActivationState

`signature` byte[]

## InvalidSignatureException(SerializationInfo, StreamingContext)

```
public InvalidSignatureException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidSignatureException(string)

```
public InvalidSignatureException(string message)
```

### Parameters

`message` string

# Properties

## Pending

```
public PendingActivationState Pending { get; }
```

### Property Value

PendingActivationState

## Signature

```
public byte[] Signature { get; }
```

### Property Value

byte[]

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

### Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class InvalidSlotIndexException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidSlotIndexException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidSlotIndexException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## InvalidSlotIndexException(SerializationInfo, StreamingContext)

```
protected InvalidSlotIndexException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidSlotIndexException(string)

```
public InvalidSlotIndexException(string s)
```

## Parameters

s [string]

# Class InvalidStageException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidStageException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidStageException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidStageException()

```
public InvalidStageException()
```

### InvalidStageException(SerializationInfo, StreamingContext)

```
protected InvalidStageException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidStageException(string)

```
public InvalidStageException(string msg)
```

### Parameters

`msg` string

# Class InvalidTradableIdException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTradableIdException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidTradableIdException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## InvalidTradableIdException(SerializationInfo, StreamingContext)

```
protected InvalidTradableIdException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidTradableIdException(string)

```
public InvalidTradableIdException(string message)
```

## Parameters

`message` string

# Class InvalidTradableItemException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class InvalidTradableItemException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidTradableItemException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidTradableItemException(SerializationInfo, StreamingContext)

```
protected InvalidTradableItemException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidTradableItemException(string)

```
public InvalidTradableItemException(string message)
```

## Parameters

`message` string

# Class InvalidTransferCurrencyException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferCurrencyException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidTransferCurrencyException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidTransferCurrencyException(SerializationInfo, StreamingContext)

```
protected InvalidTransferCurrencyException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidTransferCurrencyException(string)

```
public InvalidTransferCurrencyException(string message)
```

## Parameters

`message` string

# Class InvalidTransferMinterException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferMinterException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidTransferMinterException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidTransferMinterException(IEnumerable<Address>, Address, Address)

```
public InvalidTransferMinterException(IEnumerable<Address> minters, Address sender,
Address recipient)
```

#### Parameters

`minters` IEnumerable<Address>

`sender` Address

`recipient` Address

## InvalidTransferMinterException(SerializationInfo, StreamingContext)

```
protected InvalidTransferMinterException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidTransferMinterException(string)

```
public InvalidTransferMinterException(string message)
```

### Parameters

`message` string

# Properties

## Minters

```
public HashSet<Address> Minters { get; }
```

### Property Value

HashSet<Address>

## Recipient

```
public Address Recipient { get; }
```

### Property Value

Address

## Sender

```
public Address Sender { get; }
```

### Property Value

Address

# Class InvalidTransferRecipientException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferRecipientException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidTransferRecipientException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidTransferRecipientException(Address, Address)

```
public InvalidTransferRecipientException(Address sender, Address recipient)
```

#### Parameters

`sender` Address

`recipient` Address

### InvalidTransferRecipientException(SerializationInfo, StreamingContext)

```
public InvalidTransferRecipientException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidTransferRecipientException(string)

```
public InvalidTransferRecipientException(string message)
```

### Parameters

`message` string

# Properties

## Recipient

```
public Address Recipient { get; }
```

### Property Value

Address

## Sender

```
public Address Sender { get; }
```

### Property Value

Address

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

### Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class InvalidTransferSignerException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferSignerException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidTransferSignerException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidTransferSignerException(Address, Address, Address)

```
public InvalidTransferSignerException(Address txSigner, Address sender,
Address recipient)
```

#### Parameters

`txSigner` Address

`sender` Address

`recipient` Address

## InvalidTransferSignerException(SerializationInfo, StreamingContext)

```
public InvalidTransferSignerException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidTransferSignerException(string)

```
public InvalidTransferSignerException(string message)
```

### Parameters

`message` string

# Properties

## Recipient

```
public Address Recipient { get; }
```

### Property Value

Address

## Sender

```
public Address Sender { get; }
```

### Property Value

Address

## TxSigner

```
public Address TxSigner { get; }
```

### Property Value

Address

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

### Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class InvalidTransferUnactivatedRecipientException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferUnactivatedRecipientException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidTransferUnactivatedRecipientException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidTransferUnactivatedRecipientException(Address, Address)

```
public InvalidTransferUnactivatedRecipientException(Address sender,
Address recipient)
```

#### Parameters

`sender` Address

`recipient` Address

## InvalidTransferUnactivatedRecipientException(SerializationInfo, StreamingContext)

```
public InvalidTransferUnactivatedRecipientException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidTransferUnactivatedRecipientException(string)

```
public InvalidTransferUnactivatedRecipientException(string message)
```

### Parameters

`message` string

# Properties

## Recipient

```
public Address Recipient { get; }
```

### Property Value

Address

## Sender

```
public Address Sender { get; }
```

### Property Value

Address

## Methods

### GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

### Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class InvalidWorldException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidWorldException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidWorldException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidWorldException()

```
public InvalidWorldException()
```

### InvalidWorldException(SerializationInfo, StreamingContext)

```
protected InvalidWorldException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidWorldException(string)

```
public InvalidWorldException(string msg)
```

### Parameters

`msg` string

# Class IssueToken

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("issue_token")]
public class IssueToken : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← IssueToken

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## IssueToken()

```
public IssueToken()
```

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## FungibleAssetValues

```
public List<FungibleAssetValue> FungibleAssetValues
```

### Field Value

List<FungibleAssetValue>

## Items

```
public List<(int id, int count, bool tradable)> Items
```

### Field Value

List<(int id, int count, bool tradable)>

## TypeIdentifier

```
public const string TypeIdentifier = "issue_token"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class IssueTokensFromGarage

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("issue_tokens_from_garage")]
public class IssueTokensFromGarage : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← IssueTokensFromGarage

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### IssueTokensFromGarage()

```
public IssueTokensFromGarage()
```

### IssueTokensFromGarage(IEnumerable<Spec>)

```
public IssueTokensFromGarage(IEnumerable<IssueTokensFromGarage.Spec> specs)
```

#### Parameters

`specs` IEnumerable<IssueTokensFromGarage.Spec>

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "issue_tokens_from_garage"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Specs

```
public IEnumerable<IssueTokensFromGarage.Spec> Specs { get; }
```

### Property Value

IEnumerable<IssueTokensFromGarage.Spec>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Struct IssueTokensFromGarage.Spec

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public readonly struct IssueTokensFromGarage.Spec
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### Spec(List)

```
public Spec(List bencoded)
```

#### Parameters

`bencoded` List

## Properties

### Assets

```
public FungibleAssetValue? Assets { get; }
```

#### Property Value

FungibleAssetValue?

### Items

```
public FungibleItemValue? Items { get; }
```

### Property Value

FungibleItemValue?

## Methods

### FromFungibleAssetValue(FungibleAssetValue)

```
public static IssueTokensFromGarage.Spec FromFungibleAssetValue(FungibleAssetValue
assets)
```

### Parameters

`assets` FungibleAssetValue

### Returns

IssueTokensFromGarage.Spec

### FromFungibleItemValue(FungibleItemValue)

```
public static IssueTokensFromGarage.Spec FromFungibleItemValue(FungibleItemValue
items)
```

### Parameters

`items` FungibleItemValue

### Returns

IssueTokensFromGarage.Spec

### Serialize()

```
public IValue Serialize()
```

## Returns

IValue

# Class ItemDoesNotExistException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemDoesNotExistException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ItemDoesNotExistException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## ItemDoesNotExistException(SerializationInfo, StreamingContext)

```
public ItemDoesNotExistException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# ItemDoesNotExistException(string)

```
public ItemDoesNotExistException(string msg)
```

## Parameters

`msg` string

# Class ItemEnhancement

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("item_enhancement14")]
public class ItemEnhancement : GameAction, IAction, IItemEnhancementV5
```

**Inheritance**

object ← ActionBase ← GameAction ← ItemEnhancement

**Implements**

IAction, IItemEnhancementV5

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## HammerBannedTypes

```
public static readonly IReadOnlyCollection<ItemSubType> HammerBannedTypes
```

### Field Value

IReadOnlyCollection<ItemSubType>

## HammerIds

```
public static readonly IReadOnlyCollection<int> HammerIds
```

### Field Value

IReadOnlyCollection<int>

## MaterialCountLimit

```
public const int MaterialCountLimit = 50
```

### Field Value

int

## MaxHammerExpBoostFactor

Upper bound of a EnhancementHammerExp row: the adjusted hammer experience never exceeds this many times the configured one, so a mistyped multiplier cannot hand out the top hammer's 500,000,000 experience tenfold.

```
public const long MaxHammerExpBoostFactor = 3
```

### Field Value

long

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## hammers

```
public Dictionary<int, int> hammers
```

### Field Value

Dictionary<int, int>

## itemId

```
public Guid itemId
```

### Field Value

Guid

## materialIds

```
public List<Guid> materialIds
```

### Field Value

List<Guid>

## slotIndex

```
public int slotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV3)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV3 sheet)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

### Returns

int

## GetHammerExp(EnhancementCostSheetV3, int, Row)

Experience one hammer gives, adjusted by a EnhancementHammerExp row.

```
public static long GetHammerExp(EnhancementCostSheetV3 enhancementCostSheet, int
hammerId, BoostScheduleSheet.Row hammerExpBoost)
```

### Parameters

`enhancementCostSheet` EnhancementCostSheetV3

The sheet configuring each hammer's experience.

`hammerId` int

The hammer item id.

`hammerExpBoost` BoostScheduleSheet.Row

The row active for `hammerId` at the block, or `null` for none. A client previewing an enhancement must pass the row for the block it expects, or it shows less experience, a lower level and a lower NCG cost than the action produces.

### Returns

long

The configured experience adjusted by the row, rounded down and kept between the configured experience and MaxHammerExpBoostFactor times it.

## GetRequiredBlockCount(Equipment, Equipment, EnhancementCostSheetV3)

```
public static int GetRequiredBlockCount(Equipment preEquipment, Equipment
targetEquipment, EnhancementCostSheetV3 sheet)
```

### Parameters

`preEquipment` Equipment

`targetEquipment` Equipment

`sheet` EnhancementCostSheetV3

### Returns

int

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

## TryGetRow(Equipment, EnhancementCostSheetV3, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV3 sheet, out
EnhancementCostSheetV3.Row row)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

`row` EnhancementCostSheetV3.Row

### Returns

bool

# Class ItemEnhancement10

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/1176

```
[Serializable]
[ActionType("item_enhancement10")]
[ActionObsolete(7200000)]
public class ItemEnhancement10 : GameAction, IAction, IItemEnhancementV2
```

**Inheritance**

object ← ActionBase ← GameAction ← ItemEnhancement10

**Implements**

IAction, IItemEnhancementV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## itemId

```
public Guid itemId
```

### Field Value

Guid

## materialId

```
public Guid materialId
```

### Field Value

Guid

## slotIndex

```
public int slotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetEnhancementResult(Row, IRandom)

```
public static ItemEnhancement10.EnhancementResult
GetEnhancementResult(EnhancementCostSheetV2.Row row, IRandom random)
```

### Parameters

`row` EnhancementCostSheetV2.Row

`random` IRandom

### Returns

ItemEnhancement10.EnhancementResult

## GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV2)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV2 sheet)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

### Returns

int

## GetFeeStoreAddress()

```
public static Address GetFeeStoreAddress()
```

### Returns

Address

## GetRequiredAp()

```
public static int GetRequiredAp()
```

### Returns

int

## GetRequiredBlockCount(Row, EnhancementResult)

```
public static int GetRequiredBlockCount(EnhancementCostSheetV2.Row row,
ItemEnhancement10.EnhancementResult result)
```

### Parameters

`row` EnhancementCostSheetV2.Row

`result` ItemEnhancement10.EnhancementResult

### Returns

int

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

## TryGetRow(Equipment, EnhancementCostSheetV2, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV2 sheet, out
EnhancementCostSheetV2.Row row)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

`row` EnhancementCostSheetV2.Row

### Returns

bool

# Enum ItemEnhancement10.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement10.EnhancementResult
```

**Extension Methods**

StateExtensions.Serialize(Enum)

## Fields

`Fail = 2`

`GreatSuccess = 0`

`Success = 1`

# Class ItemEnhancement10.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement10.ResultModel : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← ItemEnhancement10.ResultModel

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ResultModel()

```
public ResultModel()
```

### ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

# Fields

## actionPoint

```
public int actionPoint
```

### Field Value

int

## enhancementResult

```
public ItemEnhancement10.EnhancementResult enhancementResult
```

### Field Value

ItemEnhancement10.EnhancementResult

## gold

```
public BigInteger gold
```

### Field Value

BigInteger

## id

```
public Guid id
```

### Field Value

Guid

## materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

### Field Value

IEnumerable<Guid>

## preItemUsable

```
public ItemUsable preItemUsable
```

### Field Value

ItemUsable

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

# Returns

IValue

# Class ItemEnhancement11

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/1164

```
[Serializable]
[ActionType("item_enhancement11")]
public class ItemEnhancement11 : GameAction, IAction, IItemEnhancementV2
```

**Inheritance**

object ← ActionBase ← GameAction ← ItemEnhancement11

**Implements**

IAction, IItemEnhancementV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## itemId

```
public Guid itemId
```

### Field Value

Guid

## materialId

```
public Guid materialId
```

### Field Value

Guid

## slotIndex

```
public int slotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetEnhancementResult(Row, IRandom)

```
public static ItemEnhancement11.EnhancementResult
GetEnhancementResult(EnhancementCostSheetV2.Row row, IRandom random)
```

### Parameters

`row` EnhancementCostSheetV2.Row

`random` IRandom

### Returns

ItemEnhancement11.EnhancementResult

## GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV2)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV2 sheet)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

### Returns

int

## GetRequiredAp()

```
public static int GetRequiredAp()
```

### Returns

int

## GetRequiredBlockCount(Row, EnhancementResult)

```
public static int GetRequiredBlockCount(EnhancementCostSheetV2.Row row,
ItemEnhancement11.EnhancementResult result)
```

### Parameters

`row` EnhancementCostSheetV2.Row

`result` ItemEnhancement11.EnhancementResult

### Returns

int

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

## TryGetRow(Equipment, EnhancementCostSheetV2, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV2 sheet, out
EnhancementCostSheetV2.Row row)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

`row` EnhancementCostSheetV2.Row

### Returns

bool

# Enum ItemEnhancement11.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement11.EnhancementResult
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Fail = 2`

`GreatSuccess = 0`

`Success = 1`

# Class ItemEnhancement11.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement11.ResultModel : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← ItemEnhancement11.ResultModel

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ResultModel()

```
public ResultModel()
```

## ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

# Fields

## CRYSTAL

```
public FungibleAssetValue CRYSTAL
```

### Field Value

FungibleAssetValue

## actionPoint

```
public int actionPoint
```

### Field Value

int

## enhancementResult

```
public ItemEnhancement11.EnhancementResult enhancementResult
```

### Field Value

ItemEnhancement11.EnhancementResult

## gold

```
public BigInteger gold
```

### Field Value

BigInteger

## id

```
public Guid id
```

### Field Value

Guid

## materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

### Field Value

IEnumerable<Guid>

## preItemUsable

```
public ItemUsable preItemUsable
```

### Field Value

ItemUsable

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ItemEnhancement12

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/2068

```
[Serializable]
[ActionObsolete(7718878)]
[ActionType("item_enhancement12")]
public class ItemEnhancement12 : GameAction, IAction, IItemEnhancementV4
```

**Inheritance**

object ← ActionBase ← GameAction ← ItemEnhancement12

**Implements**

IAction, IItemEnhancementV4

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## MaterialCountLimit

```
public const int MaterialCountLimit = 50
```

### Field Value

int

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## itemId

```
public Guid itemId
```

### Field Value

Guid

## materialIds

```
public List<Guid> materialIds
```

### Field Value

List<Guid>

## slotIndex

```
public int slotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV3)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV3 sheet)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

### Returns

int

## GetRequiredAp()

```
public static int GetRequiredAp()
```

### Returns

int

## GetRequiredBlockCount(Equipment, Equipment, EnhancementCostSheetV3)

```
public static int GetRequiredBlockCount(Equipment preEquipment, Equipment
targetEquipment, EnhancementCostSheetV3 sheet)
```

### Parameters

`preEquipment` Equipment

`targetEquipment` Equipment

`sheet` EnhancementCostSheetV3

### Returns

int

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# TryGetRow(Equipment, EnhancementCostSheetV3, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV3 sheet, out
EnhancementCostSheetV3.Row row)
```

## Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

`row` EnhancementCostSheetV3.Row

## Returns

bool

# Enum ItemEnhancement12.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement12.EnhancementResult
```

**Extension Methods**

StateExtensions.Serialize(Enum)

## Fields

`Success = 1`

# Class ItemEnhancement12.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement12.ResultModel : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← ItemEnhancement12.ResultModel

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ResultModel()

```
public ResultModel()
```

### ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

# Fields

## CRYSTAL

```
public FungibleAssetValue CRYSTAL
```

### Field Value

FungibleAssetValue

## actionPoint

```
public int actionPoint
```

### Field Value

int

## enhancementResult

```
public ItemEnhancement12.EnhancementResult enhancementResult
```

### Field Value

ItemEnhancement12.EnhancementResult

## gold

```
public BigInteger gold
```

### Field Value

BigInteger

## id

```
public Guid id
```

### Field Value

Guid

## materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

### Field Value

IEnumerable<Guid>

## preItemUsable

```
public ItemUsable preItemUsable
```

### Field Value

ItemUsable

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ItemEnhancement13

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/2068

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("item_enhancement13")]
public class ItemEnhancement13 : GameAction, IAction, IItemEnhancementV4
```

**Inheritance**

object ← ActionBase ← GameAction ← ItemEnhancement13

**Implements**

IAction, IItemEnhancementV4

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## MaterialCountLimit

```
public const int MaterialCountLimit = 50
```

### Field Value

int

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## itemId

```
public Guid itemId
```

### Field Value

Guid

## materialIds

```
public List<Guid> materialIds
```

### Field Value

List<Guid>

## slotIndex

```
public int slotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV3)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV3 sheet)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

### Returns

int

## GetRequiredAp()

```
public static int GetRequiredAp()
```

### Returns

int

## GetRequiredBlockCount(Equipment, Equipment, EnhancementCostSheetV3)

```
public static int GetRequiredBlockCount(Equipment preEquipment, Equipment
targetEquipment, EnhancementCostSheetV3 sheet)
```

### Parameters

`preEquipment` Equipment

`targetEquipment` Equipment

`sheet` EnhancementCostSheetV3

### Returns

int

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# TryGetRow(Equipment, EnhancementCostSheetV3, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV3 sheet, out
EnhancementCostSheetV3.Row row)
```

## Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

`row` EnhancementCostSheetV3.Row

## Returns

bool

# Enum ItemEnhancement13.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement13.EnhancementResult
```

**Extension Methods**

StateExtensions.Serialize(Enum)

## Fields

`Success = 1`

# Class ItemEnhancement13.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement13.ResultModel : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← ItemEnhancement13.ResultModel

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ResultModel()

```
public ResultModel()
```

## ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

# Fields

## CRYSTAL

```
public FungibleAssetValue CRYSTAL
```

### Field Value

FungibleAssetValue

## actionPoint

```
public int actionPoint
```

### Field Value

int

## enhancementResult

```
public ItemEnhancement13.EnhancementResult enhancementResult
```

### Field Value

ItemEnhancement13.EnhancementResult

## gold

```
public BigInteger gold
```

### Field Value

BigInteger

## id

```
public Guid id
```

### Field Value

Guid

## materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

### Field Value

IEnumerable<Guid>

## preItemUsable

```
public ItemUsable preItemUsable
```

### Field Value

ItemUsable

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ItemEnhancement7

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("item_enhancement7")]
public class ItemEnhancement7 : GameAction, IAction, IItemEnhancementV2
```

**Inheritance**

object ← ActionBase ← GameAction ← ItemEnhancement7

**Implements**

IAction, IItemEnhancementV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BlacksmithAddress

```
public static readonly Address BlacksmithAddress
```

### Field Value

Address

## RequiredBlockCount

```
public const int RequiredBlockCount = 1
```

### Field Value

int

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## itemId

```
public Guid itemId
```

### Field Value

Guid

## materialId

```
public Guid materialId
```

### Field Value

Guid

## slotIndex

```
public int slotIndex
```

### Field Value

int

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### GetRequiredAp()

```
public static int GetRequiredAp()
```

### Returns

int

## GetRequiredNCG(EnhancementCostSheet, int, int)

```
public static BigInteger GetRequiredNCG(EnhancementCostSheet costSheet, int grade,
int level)
```

### Parameters

`costSheet` EnhancementCostSheet

`grade` int

`level` int

### Returns

BigInteger

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

## UpgradeEquipment(Equipment)

```
public static Equipment UpgradeEquipment(Equipment equipment)
```

### Parameters

`equipment` Equipment

## Returns

Equipment

# Class ItemEnhancement7.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement7.ResultModel : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← ItemEnhancement7.ResultModel

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ResultModel()

```
public ResultModel()
```

### ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

# Fields

## actionPoint

```
public int actionPoint
```

### Field Value

int

## gold

```
public BigInteger gold
```

### Field Value

BigInteger

## id

```
public Guid id
```

### Field Value

Guid

## materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

### Field Value

IEnumerable<Guid>

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ItemEnhancement9

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("item_enhancement9")]
public class ItemEnhancement9 : GameAction, IAction, IItemEnhancementV2
```

### Inheritance

object ← ActionBase ← GameAction ← ItemEnhancement9

### Implements

IAction, IItemEnhancementV2

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BlacksmithAddress

```
public static readonly Address BlacksmithAddress
```

### Field Value

Address

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## itemId

```
public Guid itemId
```

### Field Value

Guid

## materialId

```
public Guid materialId
```

### Field Value

Guid

## slotIndex

```
public int slotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

#### Parameters

`context` IActionContext

#### Returns

IWorld

### GetEnhancementResult(Row, IRandom)

```
public static ItemEnhancement9.EnhancementResult
GetEnhancementResult(EnhancementCostSheetV2.Row row, IRandom random)
```

#### Parameters

`row` EnhancementCostSheetV2.Row

`random` IRandom

#### Returns

ItemEnhancement9.EnhancementResult

## GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV2)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV2 sheet)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

### Returns

int

## GetRequiredAp()

```
public static int GetRequiredAp()
```

### Returns

int

## GetRequiredBlockCount(Row, EnhancementResult)

```
public static int GetRequiredBlockCount(EnhancementCostSheetV2.Row row,
ItemEnhancement9.EnhancementResult result)
```

### Parameters

`row` EnhancementCostSheetV2.Row

`result` ItemEnhancement9.EnhancementResult

### Returns

int

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

## TryGetRow(Equipment, EnhancementCostSheetV2, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV2 sheet, out
EnhancementCostSheetV2.Row row)
```

### Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

`row` EnhancementCostSheetV2.Row

### Returns

bool

# Enum ItemEnhancement9.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement9.EnhancementResult
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Fail = 2`

`GreatSuccess = 0`

`Success = 1`

# Class ItemEnhancement9.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement9.ResultModel : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← ItemEnhancement9.ResultModel

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ResultModel()

```
public ResultModel()
```

### ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

# Fields

## actionPoint

```
public int actionPoint
```

### Field Value

int

## enhancementResult

```
public ItemEnhancement9.EnhancementResult enhancementResult
```

### Field Value

ItemEnhancement9.EnhancementResult

## gold

```
public BigInteger gold
```

### Field Value

BigInteger

## id

```
public Guid id
```

### Field Value

Guid

## materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

### Field Value

IEnumerable<Guid>

## preItemUsable

```
public ItemUsable preItemUsable
```

### Field Value

ItemUsable

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

# Returns

IValue

# Class ItemProductInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class ItemProductInfo : IProductInfo
```

**Inheritance**

object ← ItemProductInfo

**Implements**

IProductInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ItemProductInfo()

```
public ItemProductInfo()
```

### ItemProductInfo(List)

```
public ItemProductInfo(List serialized)
```

#### Parameters

`serialized` List

## Properties

## AgentAddress

```
public Address AgentAddress { get; set; }
```

### Property Value

Address

## AvatarAddress

```
public Address AvatarAddress { get; set; }
```

### Property Value

Address

## ItemSubType

```
public ItemSubType ItemSubType { get; set; }
```

### Property Value

ItemSubType

## Legacy

```
public bool Legacy { get; set; }
```

### Property Value

bool

## Price

```
public FungibleAssetValue Price { get; set; }
```

### Property Value

FungibleAssetValue

## ProductId

```
public Guid ProductId { get; set; }
```

### Property Value

Guid

## TradableId

```
public Guid TradableId { get; set; }
```

### Property Value

Guid

## Type

```
public ProductType Type { get; set; }
```

### Property Value

ProductType

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## ValidateType()

```
public void ValidateType()
```

# Class JoinArena

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("join_arena4")]
public class JoinArena : GameAction, IAction, IJoinArenaV1
```

**Inheritance**

object ← ActionBase ← GameAction ← JoinArena

**Implements**

IAction, IJoinArenaV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## championshipId

```
public int championshipId
```

### Field Value

int

## costumes

```
public List<Guid> costumes
```

### Field Value

List<Guid>

## equipments

```
public List<Guid> equipments
```

### Field Value

List<Guid>

## round

```
public int round
```

### Field Value

int

## runeInfos

```
public List<RuneSlotInfo> runeInfos
```

### Field Value

List<RuneSlotInfo>

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class JoinArena1

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1495

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("join_arena")]
public class JoinArena1 : GameAction, IAction, IJoinArenaV1
```

### Inheritance

object ← ActionBase ← GameAction ← JoinArena1

### Implements

IAction, IJoinArenaV1

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## championshipId

```
public int championshipId
```

### Field Value

int

## costumes

```
public List<Guid> costumes
```

### Field Value

List<Guid>

## equipments

```
public List<Guid> equipments
```

### Field Value

List<Guid>

## round

```
public int round
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class JoinArena3

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1663

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("join_arena3")]
public class JoinArena3 : GameAction, IAction, IJoinArenaV1
```

**Inheritance**

object ← ActionBase ← GameAction ← JoinArena3

**Implements**

IAction, IJoinArenaV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## championshipId

```
public int championshipId
```

### Field Value

int

## costumes

```
public List<Guid> costumes
```

### Field Value

List<Guid>

## equipments

```
public List<Guid> equipments
```

### Field Value

List<Guid>

## round

```
public int round
```

### Field Value

int

## runeInfos

```
public List<RuneSlotInfo> runeInfos
```

### Field Value

List<RuneSlotInfo>

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class MemoLengthOverflowException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class MemoLengthOverflowException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← MemoLengthOverflowException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### MemoLengthOverflowException(SerializationInfo, StreamingContext)

```
protected MemoLengthOverflowException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# MemoLengthOverflowException(string)

```
public MemoLengthOverflowException(string message)
```

## Parameters

`message` string

# Class MigrateAgentAvatar

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("migrate_agent_avatar")]
public class MigrateAgentAvatar : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← MigrateAgentAvatar

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## MigrateAgentAvatar()

```
public MigrateAgentAvatar()
```

# Fields

## AgentAddresses

```
public List<Address> AgentAddresses
```

### Field Value

List<Address>

## TypeIdentifier

```
public const string TypeIdentifier = "migrate_agent_avatar"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class MigrateFee

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("migrate_fee")]
public class MigrateFee : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← MigrateFee

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MigrateFee()

```
public MigrateFee()
```

## Fields

### Memo

```
public string Memo
```

#### Field Value

string

## TransferData

```
public List<(Address sender, Address recipient, BigInteger amount)> TransferData
```

### Field Value

List<(Address sender, Address recipient, BigInteger amount)>

## TypeIdentifier

```
public const string TypeIdentifier = "migrate_fee"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class MigrateMonsterCollection

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

An action to claim remained monster collection rewards and to migrate MonsterCollectionState into LegacyStakeState without cancellation, to keep its staked period.

```
[ActionType("migrate_monster_collection")]
public class MigrateMonsterCollection : ActionBase,
IAction, IMigrateMonsterCollectionV1
```

**Inheritance**

object ← ActionBase ← MigrateMonsterCollection

**Implements**

IAction, IMigrateMonsterCollectionV1

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MigrateMonsterCollection()

```
public MigrateMonsterCollection()
```

### MigrateMonsterCollection(Address)

```
public MigrateMonsterCollection(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

## Properties

### AvatarAddress

```
public Address AvatarAddress { get; }
```

### Property Value

Address

### PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class MigrationActivatedAccountsState

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("migration_activated_accounts_state")]
public class MigrationActivatedAccountsState : GameAction, IAction,
IMigrationActivatedAccountsStateV1
```

**Inheritance**

object ← ActionBase ← GameAction ← MigrationActivatedAccountsState

**Implements**

IAction, IMigrationActivatedAccountsStateV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

# Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

## Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class MintAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("mint_assets")]
public class MintAssets : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← MintAssets

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MintAssets()

```
public MintAssets()
```

### MintAssets(IEnumerable<MintSpec>, string?)

```
public MintAssets(IEnumerable<MintAssets.MintSpec> specs, string? memo)
```

#### Parameters

`specs` IEnumerable<MintAssets.MintSpec>

memo string

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "mint_assets"
```

### Field Value

string

# Properties

## Memo

```
public string? Memo { get; }
```

### Property Value

string

## MintSpecs

```
public List<MintAssets.MintSpec>? MintSpecs { get; }
```

### Property Value

List<MintAssets.MintSpec>

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Struct MintAssets.MintSpec

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public readonly struct MintAssets.MintSpec
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### MintSpec(List)

```
public MintSpec(List bencoded)
```

#### Parameters

`bencoded` List

### MintSpec(Address, FungibleAssetValue?, FungibleItemValue?)

```
public MintSpec(Address recipient, FungibleAssetValue? assets,
FungibleItemValue? items)
```

#### Parameters

`recipient` Address

`assets` FungibleAssetValue?

`items` FungibleItemValue?

# Properties

## Assets

```
public FungibleAssetValue? Assets { get; }
```

### Property Value

FungibleAssetValue?

## Items

```
public FungibleItemValue? Items { get; }
```

### Property Value

FungibleItemValue?

## Recipient

```
public Address Recipient { get; }
```

### Property Value

Address

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class MonsterCollectionExistingClaimableException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionExistingClaimableException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← MonsterCollectionExistingClaimableException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### MonsterCollectionExistingClaimableException()

```
public MonsterCollectionExistingClaimableException()
```

### MonsterCollectionExistingClaimableException(SerializationInfo, StreamingContext)

```
protected MonsterCollectionExistingClaimableException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## MonsterCollectionExistingClaimableException(string)

```
public MonsterCollectionExistingClaimableException(string message)
```

### Parameters

`message` string

## MonsterCollectionExistingClaimableException(string, Exception)

```
public MonsterCollectionExistingClaimableException(string message,
Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

# Class MonsterCollectionExistingException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

An exception thrown when there is unexpected monster collection.

```
[Serializable]
public class MonsterCollectionExistingException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← MonsterCollectionExistingException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### MonsterCollectionExistingException()

```
public MonsterCollectionExistingException()
```

### MonsterCollectionExistingException(SerializationInfo, StreamingContext)

```
protected MonsterCollectionExistingException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## MonsterCollectionExistingException(string)

```
public MonsterCollectionExistingException(string message)
```

### Parameters

`message` string

## MonsterCollectionExistingException(string, Exception)

```
public MonsterCollectionExistingException(string message, Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

# Class MonsterCollectionExpiredException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionExpiredException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← MonsterCollectionExpiredException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### MonsterCollectionExpiredException(SerializationInfo, StreamingContext)

```
protected MonsterCollectionExpiredException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# MonsterCollectionExpiredException(string)

```
public MonsterCollectionExpiredException(string msg)
```

## Parameters

`msg` string

# Class MonsterCollectionLevelException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionLevelException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← MonsterCollectionLevelException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### MonsterCollectionLevelException()

```
public MonsterCollectionLevelException()
```

### MonsterCollectionLevelException(SerializationInfo, StreamingContext)

```
protected MonsterCollectionLevelException(SerializationInfo info,
```

```
  StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## MonsterCollectionLevelException(string)

```
  public MonsterCollectionLevelException(string msg)
```

### Parameters

`msg` string

# Class NotEnoughActionPointException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughActionPointException : Exception, ISerializable
```

### Inheritance

object ← Exception ← NotEnoughActionPointException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughActionPointException(SerializationInfo, StreamingContext)

```
public NotEnoughActionPointException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# NotEnoughActionPointException(string)

```
public NotEnoughActionPointException(string msg)
```

## Parameters

`msg` string

# Class NotEnoughAvatarLevelException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughAvatarLevelException : Exception, ISerializable
```

### Inheritance

object ← Exception ← NotEnoughAvatarLevelException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughAvatarLevelException(int, bool, int, int)

```
public NotEnoughAvatarLevelException(int itemId, bool isMadeWithMimisbrunnrRecipe,
int require, int current)
```

#### Parameters

`itemId` int

`isMadeWithMimisbrunnrRecipe` bool

`require` int

`current` int

## NotEnoughAvatarLevelException(int, int)

```
public NotEnoughAvatarLevelException(int require, int current)
```

### Parameters

`require` int

`current` int

## NotEnoughAvatarLevelException(SerializationInfo, StreamingContext)

```
public NotEnoughAvatarLevelException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## NotEnoughAvatarLevelException(string)

```
public NotEnoughAvatarLevelException(string message)
```

### Parameters

`message` string

# Class NotEnoughClearedStageLevelException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughClearedStageLevelException : Exception, ISerializable
```

### Inheritance

object ← Exception ← NotEnoughClearedStageLevelException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughClearedStageLevelException(SerializationInfo, StreamingContext)

```
public NotEnoughClearedStageLevelException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## NotEnoughClearedStageLevelException(string)

```
public NotEnoughClearedStageLevelException(string message)
```

### Parameters

`message` string

## NotEnoughClearedStageLevelException(string, int, int)

```
public NotEnoughClearedStageLevelException(string addressesHex, int require,
int current)
```

### Parameters

`addressesHex` string

`require` int

`current` int

## NotEnoughClearedStageLevelException(string, string, int, int)

```
public NotEnoughClearedStageLevelException(string actionType, string addressesHex,
int require, int current)
```

### Parameters

`actionType` string

`addressesHex` string

`require` int

`current` int

# Class NotEnoughCombatPointException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class NotEnoughCombatPointException : InvalidOperationException,
ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← NotEnoughCombatPointException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## NotEnoughCombatPointException(SerializationInfo, StreamingContext)

```
protected NotEnoughCombatPointException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# NotEnoughCombatPointException(string)

```
public NotEnoughCombatPointException(string s)
```

## Parameters

s [string]

# Class NotEnoughFungibleAssetValueException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughFungibleAssetValueException : Exception, ISerializable
```

### Inheritance

object ← Exception ← NotEnoughFungibleAssetValueException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughFungibleAssetValueException(SerializationInfo, StreamingContext)

```
public NotEnoughFungibleAssetValueException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## NotEnoughFungibleAssetValueException(string)

```
public NotEnoughFungibleAssetValueException(string message)
```

### Parameters

`message` string

## NotEnoughFungibleAssetValueException(string, Exception)

```
public NotEnoughFungibleAssetValueException(string message,
Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

## NotEnoughFungibleAssetValueException(string, BigInteger, FungibleAssetValue)

```
public NotEnoughFungibleAssetValueException(string addressesHex, BigInteger require,
FungibleAssetValue current)
```

### Parameters

`addressesHex` string

`require` BigInteger

`current` FungibleAssetValue

## NotEnoughFungibleAssetValueException(string, string, FungibleAssetValue, FungibleAssetValue)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
FungibleAssetValue require, FungibleAssetValue current)
```

### Parameters

`actionType` string

`addressesHex` string

`require` FungibleAssetValue

`current` FungibleAssetValue

## NotEnoughFungibleAssetValueException(string, string, FungibleAssetValue, BigInteger)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
FungibleAssetValue require, BigInteger current)
```

### Parameters

`actionType` string

`addressesHex` string

`require` FungibleAssetValue

`current` BigInteger

## NotEnoughFungibleAssetValueException(string, string, BigInteger, FungibleAssetValue)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
BigInteger require, FungibleAssetValue current)
```

### Parameters

`actionType` string

`addressesHex` string

`require` BigInteger

`current` FungibleAssetValue

## NotEnoughFungibleAssetValueException(string, string, BigInteger, BigInteger)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
BigInteger require, BigInteger current)
```

### Parameters

`actionType` string

`addressesHex` string

`require` BigInteger

`current` BigInteger

## NotEnoughFungibleAssetValueException(string, string, string, Exception)

```
public NotEnoughFungibleAssetValueException(string addressesHex, string require,
string current, Exception innerException = null)
```

### Parameters

`addressesHex` string

`require` string

`current` string

`innerException` Exception

## NotEnoughFungibleAssetValueException(string, string, string, string, Exception)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
string require, string current, Exception innerException = null)
```

### Parameters

`actionType` string

`addressesHex` string

`require` string

`current` string

`innerException` Exception

# Class NotEnoughHammerPointException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class NotEnoughHammerPointException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← NotEnoughHammerPointException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughHammerPointException(SerializationInfo, StreamingContext)

```
protected NotEnoughHammerPointException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# NotEnoughHammerPointException(string)

```
public NotEnoughHammerPointException(string s)
```

## Parameters

s [string]

# Class NotEnoughMaterialException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughMaterialException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← NotEnoughMaterialException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughMaterialException(SerializationInfo, StreamingContext)

```
protected NotEnoughMaterialException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# NotEnoughMaterialException(string)

```
public NotEnoughMaterialException(string s)
```

## Parameters

s [string]

# Class NotEnoughMedalException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughMedalException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← NotEnoughMedalException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughMedalException(SerializationInfo, StreamingContext)

```
public NotEnoughMedalException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# NotEnoughMedalException(string)

```
public NotEnoughMedalException(string msg)
```

## Parameters

`msg` string

# Class NotEnoughRankException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughRankException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← NotEnoughRankException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughRankException()

```
public NotEnoughRankException()
```

### NotEnoughRankException(SerializationInfo, StreamingContext)

```
public NotEnoughRankException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## NotEnoughRankException(string)

```
public NotEnoughRankException(string message)
```

### Parameters

`message` string

# Class NotEnoughStarException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class NotEnoughStarException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← NotEnoughStarException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughStarException(SerializationInfo, StreamingContext)

```
protected NotEnoughStarException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

### NotEnoughStarException(string)

```
public NotEnoughStarException(string s)
```

## Parameters

s [string](#)

# Class NotEnoughWeeklyArenaChallengeCountException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughWeeklyArenaChallengeCountException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← NotEnoughWeeklyArenaChallengeCountException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughWeeklyArenaChallengeCountException()

```
public NotEnoughWeeklyArenaChallengeCountException()
```

### NotEnoughWeeklyArenaChallengeCountException(SerializationInfo, StreamingContext)

```
public NotEnoughWeeklyArenaChallengeCountException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## NotEnoughWeeklyArenaChallengeCountException(string)

```
public NotEnoughWeeklyArenaChallengeCountException(string message)
```

### Parameters

`message` string

# Fields

## BaseMessage

```
public const string BaseMessage = "Aborted as the arena state reached the
daily limit."
```

### Field Value

string

# Class OrderIdDoesNotExistException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class OrderIdDoesNotExistException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← OrderIdDoesNotExistException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## OrderIdDoesNotExistException(SerializationInfo, StreamingContext)

```
protected OrderIdDoesNotExistException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# OrderIdDoesNotExistException(string)

```
public OrderIdDoesNotExistException(string msg)
```

## Parameters

`msg` string

# Class PatchTableSheet

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at https://github.com/planetarium/lib9c/pull/42 Updated at https://github.com/planetarium/lib9c/pull/101 Updated at https://github.com/planetarium/lib9c/pull/287 Updated at https://github.com/planetarium/lib9c/pull/315 Updated at https://github.com/planetarium/lib9c/pull/957 Updated at https://github.com/planetarium/lib9c/pull/1560

```
[Serializable]
[ActionType("patch_table_sheet")]
public class PatchTableSheet : GameAction, IAction, IPatchTableSheetV1
```

**Inheritance**

object ← ActionBase ← GameAction ← PatchTableSheet

**Implements**

IAction, IPatchTableSheetV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Operator

The operator address that has special permission to operation actions. When the action is signed by this operator, permission check is skipped. This is a temporary solution until

v100340, after which this concept should be eliminated or justified differently. For any questions about this account, please consult Nine Chronicles Dev team.

```
public static readonly Address Operator
```

### Field Value

Address

## TableCsv

```
public string TableCsv
```

### Field Value

string

## TableName

```
public string TableName
```

### Field Value

string

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class PatchTableSheetCompressed

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Compressed version of PatchTableSheet to reduce transaction size for large CSV files. Uses GZip compression to minimize the size of table data in transactions.

This action is particularly useful for large CSV files like RuneOptionSheet.csv which can be compressed from ~842KB to ~154KB (81% reduction).

Example usage:

```
var compressedAction = new PatchTableSheetCompressed
{
    TableName = "RuneOptionSheet",
    CompressedTableCsv = PatchTableSheetCompressed.CompressCsv(csvData),
};
```

```
[Serializable]
[ActionType("patch_table_sheet_compressed")]
public class PatchTableSheetCompressed : GameAction,
IAction, IPatchTableSheetCompressedV1
```

### Inheritance

object ← ActionBase ← GameAction ← PatchTableSheetCompressed

### Implements

IAction, IPatchTableSheetCompressedV1

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Operator

The operator address that has special permission to operation actions. When the action is signed by this operator, permission check is skipped. This is a temporary solution until v100340, after which this concept should be eliminated or justified differently. For any questions about this account, please consult Nine Chronicles Dev team.

```
public static readonly Address Operator
```

### Field Value

Address

# Properties

## CompressedTableCsv

Gets or sets the compressed CSV data as a byte array. This should be compressed using the CompressCsv method.

```
public byte[] CompressedTableCsv { get; set; }
```

### Property Value

byte[]

### Remarks

The compression can reduce file size by 60-80% depending on the CSV content. For example, RuneOptionSheet.csv (842KB) can be compressed to ~154KB.

## PlainValueInternal

Serializes the action data for blockchain storage. Contains the table name and compressed CSV data.

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## TableName

Gets or sets the name of the table sheet to patch. This should match the name of the CSV file without the .csv extension.

```
public string TableName { get; set; }
```

### Property Value

string

### Examples

"RuneOptionSheet" for RuneOptionSheet.csv "ItemSheet" for ItemSheet.csv

# Methods

## CompressCsv(string)

Compresses CSV data using GZip compression.

This method is used to prepare CSV data for the CompressedTableCsv property. GZip compression typically reduces CSV file size by 60-80%.

Example:

```
var csvData = File.ReadAllText("RuneOptionSheet.csv");
var compressedData = PatchTableSheetCompressed.CompressCsv(csvData);
// compressedData can now be used in CompressedTableCsv property
```

```
public static byte[] CompressCsv(string csvData)
```

### Parameters

`csvData` string

The CSV data to compress. Must not be null.

### Returns

byte[]

Compressed byte array using GZip compression.

### Exceptions

ArgumentNullException

Thrown when csvData is null.

## DecompressCsv(byte[])

Decompresses CSV data using GZip compression.

This method is used internally to restore the original CSV data from the compressed byte array stored in CompressedTableCsv.

Example:

```
var originalCsv = PatchTableSheetCompressed.DecompressCsv(compressedData);
// originalCsv now contains the decompressed CSV string
```

```
public static string DecompressCsv(byte[] compressedData)
```

### Parameters

`compressedData` byte[]

The compressed data to decompress. Must not be null.

### Returns

string

Decompressed CSV string in UTF-8 encoding.

### Exceptions

ArgumentNullException

Thrown when compressedData is null.

InvalidDataException

Thrown when the compressed data is corrupted or not valid GZip format.

## Execute(IActionContext)

Executes the compressed table sheet patching action.

This method:

1. Validates permissions (unless signed by operator)
2. Decompresses the CSV data
3. Updates the table sheet state
4. Handles special cases like GameConfigSheet

The action reduces transaction size significantly compared to PatchTableSheet while maintaining the same functionality.

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

The action context containing state and transaction information.

### Returns

IWorld

The updated world state with the patched table sheet.

### Exceptions

InvalidDataException

Thrown when the compressed data is corrupted or invalid.

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

Deserializes the action data from blockchain storage. Restores the table name and compressed CSV data.

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

The serialized action data.

# Class PendingActivationDoesNotExistsException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PendingActivationDoesNotExistsException : ActivationException,
ISerializable
```

### Inheritance

object ← Exception ← ActivationException ← PendingActivationDoesNotExistsException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### PendingActivationDoesNotExistsException(Address)

```
public PendingActivationDoesNotExistsException(Address pendingAddress)
```

#### Parameters

`pendingAddress` Address

## PendingActivationDoesNotExistsException(SerializationInfo, StreamingContext)

```
public PendingActivationDoesNotExistsException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## PendingActivationDoesNotExistsException(string)

```
public PendingActivationDoesNotExistsException(string message)
```

### Parameters

`message` string

# Properties

## PendingAddress

```
public Address PendingAddress { get; }
```

### Property Value

Address

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

## Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

## Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class PermissionDeniedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PermissionDeniedException : AdminPermissionException, ISerializable
```

**Inheritance**

object ← Exception ← AdminPermissionException ← PermissionDeniedException

**Implements**

ISerializable

**Inherited Members**

AdminPermissionException.Policy , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### PermissionDeniedException(AdminState, Address)

```
public PermissionDeniedException(AdminState policy, Address signer)
```

#### Parameters

`policy` AdminState

`signer` Address

### PermissionDeniedException(SerializationInfo, StreamingContext)

```
public PermissionDeniedException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## PermissionDeniedException(string)

```
public PermissionDeniedException(string message)
```

### Parameters

`message` string

# Properties

## Signer

```
public Address Signer { get; }
```

### Property Value

Address

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

## Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

## Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class PetEnhancement

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("pet_enhancement")]
public class PetEnhancement : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← PetEnhancement

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ActionTypeIdentifier

```
public const string ActionTypeIdentifier = "pet_enhancement"
```

### Field Value

string

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## PetId

```
public int PetId
```

### Field Value

int

## TargetLevel

```
public int TargetLevel
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

# Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

## Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class PetIsLockedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PetIsLockedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← PetIsLockedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### PetIsLockedException(SerializationInfo, StreamingContext)

```
public PetIsLockedException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# PetIsLockedException(string)

```
public PetIsLockedException(string msg)
```

## Parameters

`msg` string

# Class PlayCountIsZeroException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PlayCountIsZeroException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← PlayCountIsZeroException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### PlayCountIsZeroException(SerializationInfo, StreamingContext)

```
public PlayCountIsZeroException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# PlayCountIsZeroException(string)

```
public PlayCountIsZeroException(string msg)
```

## Parameters

`msg` string

# Class PolicyExpiredException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PolicyExpiredException : AdminPermissionException, ISerializable
```

**Inheritance**

object ← Exception ← AdminPermissionException ← PolicyExpiredException

**Implements**

ISerializable

**Inherited Members**

AdminPermissionException.Policy , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### PolicyExpiredException(AdminState, long)

```
public PolicyExpiredException(AdminState policy, long blockIndex)
```

#### Parameters

`policy` AdminState

`blockIndex` long

### PolicyExpiredException(SerializationInfo, StreamingContext)

```
public PolicyExpiredException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## PolicyExpiredException(string)

```
public PolicyExpiredException(string message)
```

### Parameters

`message` string

# Properties

## BlockIndex

```
public long BlockIndex { get; }
```

### Property Value

long

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

## Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

## Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class PrepareRewardAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("prepare_reward_assets")]
public class PrepareRewardAssets : ActionBase, IAction, IPrepareRewardAssetsV1
```

**Inheritance**

object ← ActionBase ← PrepareRewardAssets

**Implements**

IAction, IPrepareRewardAssetsV1

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PrepareRewardAssets()

```
public PrepareRewardAssets()
```

### PrepareRewardAssets(Address, List<FungibleAssetValue>)

```
public PrepareRewardAssets(Address rewardPoolAddress, List<FungibleAssetValue>
assets)
```

#### Parameters

`rewardPoolAddress` Address

`assets` List<FungibleAssetValue>

# Fields

## Assets

```
public List<FungibleAssetValue> Assets
```

### Field Value

List<FungibleAssetValue>

## RewardPoolAddress

```
public Address RewardPoolAddress
```

### Field Value

Address

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class ProductNotFoundException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ProductNotFoundException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ProductNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ProductNotFoundException(SerializationInfo, StreamingContext)

```
public ProductNotFoundException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ProductNotFoundException(string)

```
public ProductNotFoundException(string msg)
```

## Parameters

`msg` string

# Class PurchaseInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PurchaseInfo : IComparable<PurchaseInfo>, IComparable, IPurchaseInfo
```

**Inheritance**

object ← PurchaseInfo

**Implements**

IComparable<PurchaseInfo>, IComparable, IPurchaseInfo

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PurchaseInfo(Dictionary)

```
public PurchaseInfo(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### PurchaseInfo(Guid, Guid, Address, Address, ItemSubType, FungibleAssetValue)

```
public PurchaseInfo(Guid orderId, Guid tradableId, Address agentAddress, Address
avatarAddress, ItemSubType type, FungibleAssetValue itemPrice)
```

#### Parameters

`orderId` Guid

`tradableId` Guid

`agentAddress` Address

`avatarAddress` Address

`type` ItemSubType

`itemPrice` FungibleAssetValue

# Fields

## OrderId

```
public readonly Guid OrderId
```

### Field Value

Guid

## TradableId

```
public readonly Guid TradableId
```

### Field Value

Guid

# Properties

## ItemSubType

```
public ItemSubType ItemSubType { get; }
```

### Property Value

ItemSubType

## Price

```
public FungibleAssetValue Price { get; }
```

### Property Value

FungibleAssetValue

## SellerAgentAddress

```
public Address SellerAgentAddress { get; }
```

### Property Value

Address

## SellerAvatarAddress

```
public Address SellerAvatarAddress { get; }
```

### Property Value

Address

# Methods

## CompareTo(PurchaseInfo)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(PurchaseInfo other)
```

### Parameters

`other` PurchaseInfo

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `other` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `other`. |
| **Greater than zero** | This instance follows `other` in the sort order. |

## CompareTo(object)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(object obj)
```

### Parameters

`obj` object

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `obj` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `obj`. |
| **Greater than zero** | This instance follows `obj` in the sort order. |

### Exceptions

ArgumentException

`obj` is not the same type as this instance.

## Equals(PurchaseInfo)

```
protected bool Equals(PurchaseInfo other)
```

### Parameters

`other` PurchaseInfo

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Operators

## operator ==(PurchaseInfo, PurchaseInfo)

```
public static bool operator ==(PurchaseInfo left, PurchaseInfo right)
```

### Parameters

`left` PurchaseInfo

`right` PurchaseInfo

### Returns

bool

## operator >(PurchaseInfo, PurchaseInfo)

```
public static bool operator >(PurchaseInfo left, PurchaseInfo right)
```

### Parameters

`left` PurchaseInfo

`right` PurchaseInfo

### Returns

bool

## operator >=(PurchaseInfo, PurchaseInfo)

```
public static bool operator >=(PurchaseInfo left, PurchaseInfo right)
```

### Parameters

`left` PurchaseInfo

`right` PurchaseInfo

### Returns

bool

## operator !=(PurchaseInfo, PurchaseInfo)

```
public static bool operator !=(PurchaseInfo left, PurchaseInfo right)
```

### Parameters

`left` PurchaseInfo

`right` PurchaseInfo

### Returns

bool

## operator <(PurchaseInfo, PurchaseInfo)

```
public static bool operator <(PurchaseInfo left, PurchaseInfo right)
```

### Parameters

`left` PurchaseInfo

`right` PurchaseInfo

### Returns

bool

## operator <=(PurchaseInfo, PurchaseInfo)

```
public static bool operator <=(PurchaseInfo left, PurchaseInfo right)
```

### Parameters

`left` PurchaseInfo

`right` PurchaseInfo

## Returns

bool

# Class PurchaseInfo0

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PurchaseInfo0 : IComparable<PurchaseInfo0>, IComparable, IPurchaseInfo
```

**Inheritance**

object ← PurchaseInfo0

**Implements**

IComparable<PurchaseInfo0>, IComparable, IPurchaseInfo

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PurchaseInfo0(Dictionary)

```
public PurchaseInfo0(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### PurchaseInfo0(Guid, Address, Address, ItemSubType, FungibleAssetValue)

```
public PurchaseInfo0(Guid id, Address agentAddress, Address avatarAddress,
ItemSubType type, FungibleAssetValue itemPrice = default)
```

#### Parameters

`id` Guid

`agentAddress` Address

`avatarAddress` Address

`type` ItemSubType

`itemPrice` FungibleAssetValue

# Fields

## itemSubType

```
public readonly ItemSubType itemSubType
```

### Field Value

ItemSubType

## price

```
public readonly FungibleAssetValue price
```

### Field Value

FungibleAssetValue

## productId

```
public readonly Guid productId
```

### Field Value

Guid

## sellerAgentAddress

```
public readonly Address sellerAgentAddress
```

### Field Value

Address

## sellerAvatarAddress

```
public readonly Address sellerAvatarAddress
```

### Field Value

Address

# Methods

## CompareTo(PurchaseInfo0)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(PurchaseInfo0 other)
```

### Parameters

`other` PurchaseInfo0

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `other` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `other`. |
| **Greater than zero** | This instance follows `other` in the sort order. |

## CompareTo(object)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(object obj)
```

### Parameters

`obj` object

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `obj` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `obj`. |

| Value | Meaning |
| --- | --- |
| **Greater than zero** | This instance follows `obj` in the sort order. |

### Exceptions

ArgumentException

`obj` is not the same type as this instance.

## Equals(PurchaseInfo0)

```
protected bool Equals(PurchaseInfo0 other)
```

### Parameters

`other` PurchaseInfo0

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Operators

## operator ==(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator ==(PurchaseInfo0 left, PurchaseInfo0 right)
```

### Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

### Returns

bool

## operator >(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator >(PurchaseInfo0 left, PurchaseInfo0 right)
```

### Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

### Returns

bool

## operator >=(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator >=(PurchaseInfo0 left, PurchaseInfo0 right)
```

### Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

### Returns

bool

## operator !=(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator !=(PurchaseInfo0 left, PurchaseInfo0 right)
```

### Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

### Returns

bool

## operator <(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator <(PurchaseInfo0 left, PurchaseInfo0 right)
```

### Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

### Returns

bool

## operator <=(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator <=(PurchaseInfo0 left, PurchaseInfo0 right)
```

### Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

### Returns

bool

# Class Raid

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a raid action where players participate in a world boss battle. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- WorldBossKillRewardRecord: Records boss kill rewards
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

```
[Serializable]
[ActionType("raid7")]
public class Raid : GameAction, IAction, IRaidV2
```

**Inheritance**

object ← ActionBase ← GameAction ← Raid

**Implements**

IAction, IRaidV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## CostumeIds

```
public List<Guid> CostumeIds
```

### Field Value

List<Guid>

## EquipmentIds

```
public List<Guid> EquipmentIds
```

### Field Value

List<Guid>

## FoodIds

```
public List<Guid> FoodIds
```

### Field Value

List<Guid>

## PayNcg

```
public bool PayNcg
```

### Field Value

bool

## RuneInfos

```
public List<RuneSlotInfo> RuneInfos
```

### Field Value

List<RuneSlotInfo>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RankingBattle

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/941 Updated at https://github.com/planetarium/lib9c/pull/1135

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("ranking_battle12")]
public class RankingBattle : GameAction, IAction, IRankingBattleV2
```

**Inheritance**

object ← ActionBase ← GameAction ← RankingBattle

**Implements**

IAction, IRankingBattleV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## PreviousArenaInfo

```
public ArenaInfo PreviousArenaInfo
```

### Field Value

ArenaInfo

## PreviousEnemyArenaInfo

```
public ArenaInfo PreviousEnemyArenaInfo
```

### Field Value

ArenaInfo

## PreviousEnemyPlayerDigest

```
public EnemyPlayerDigest PreviousEnemyPlayerDigest
```

### Field Value

EnemyPlayerDigest

## StageId

```
public const int StageId = 999999
```

### Field Value

int

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## costumeIds

```
public List<Guid> costumeIds
```

### Field Value

List<Guid>

## enemyAddress

```
public Address enemyAddress
```

### Field Value

Address

## equipmentIds

```
public List<Guid> equipmentIds
```

### Field Value

List<Guid>

## weeklyArenaAddress

```
public Address weeklyArenaAddress
```

### Field Value

Address

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

#### Parameters

`context` IActionContext

#### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

#### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RankingBattle11

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/1176

```
[Serializable]
[ActionType("ranking_battle11")]
[ActionObsolete(7200000)]
public class RankingBattle11 : GameAction, IAction, IRankingBattleV2
```

**Inheritance**

object ← ActionBase ← GameAction ← RankingBattle11

**Implements**

IAction, IRankingBattleV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ArenaInfo

```
public ArenaInfo ArenaInfo
```

### Field Value

ArenaInfo

## EnemyArenaInfo

```
public ArenaInfo EnemyArenaInfo
```

### Field Value

ArenaInfo

## EnemyPlayerDigest

```
public EnemyPlayerDigest EnemyPlayerDigest
```

### Field Value

EnemyPlayerDigest

## EntranceFee

```
public static readonly BigInteger EntranceFee
```

### Field Value

BigInteger

## StageId

```
public const int StageId = 999999
```

### Field Value

int

## UpdateTargetBlockIndex

```
public const long UpdateTargetBlockIndex = 3808000
```

### Field Value

long

## UpdateTargetWeeklyArenaIndex

```
public const int UpdateTargetWeeklyArenaIndex = 68
```

### Field Value

int

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## costumeIds

```
public List<Guid> costumeIds
```

### Field Value

List<Guid>

## enemyAddress

```
public Address enemyAddress
```

### Field Value

Address

## equipmentIds

```
public List<Guid> equipmentIds
```

### Field Value

List<Guid>

## weeklyArenaAddress

```
public Address weeklyArenaAddress
```

### Field Value

Address

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

# Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

## Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RankingExceededException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RankingExceededException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← RankingExceededException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RankingExceededException()

```
public RankingExceededException()
```

### RankingExceededException(SerializationInfo, StreamingContext)

```
protected RankingExceededException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## RankingExceededException(string)

```
public RankingExceededException(string message)
```

### Parameters

`message` string

# Class RapidCombination

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("rapid_combination10")]
public class RapidCombination : GameAction, IAction, IRapidCombinationV2
```

**Inheritance**

object ← ActionBase ← GameAction ← RapidCombination

**Implements**

IAction, IRapidCombinationV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## slotIndexList

```
public List<int> slotIndexList
```

### Field Value

List<int>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RapidCombination0

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("rapid_combination")]
public class RapidCombination0 : GameAction, IAction, IRapidCombinationV1
```

**Inheritance**

object ← ActionBase ← GameAction ← RapidCombination0

**Implements**

IAction, IRapidCombinationV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## slotIndex

```
public int slotIndex
```

### Field Value

int

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### CalculateHourglassCount(GameConfigState, long)

```
public static int CalculateHourglassCount(GameConfigState state, long diff)
```

### Parameters

`state` GameConfigState

`diff` long

### Returns

int

### CalculateHourglassCount(decimal, long)

```
public static int CalculateHourglassCount(decimal hourglassPerBlock, long diff)
```

### Parameters

`hourglassPerBlock` decimal

`diff` long

### Returns

int

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RapidCombination0.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RapidCombination0.ResultModel : CombinationConsumable5.ResultModel,
IState
```

### Inheritance

object ← AttachmentActionResult ← CombinationConsumable5.ResultModel ← RapidCombination0.ResultModel

### Implements

IState

### Inherited Members

CombinationConsumable5.ResultModel.materials , CombinationConsumable5.ResultModel.id , CombinationConsumable5.ResultModel.gold , CombinationConsumable5.ResultModel.actionPoint , CombinationConsumable5.ResultModel.recipeId , CombinationConsumable5.ResultModel.subRecipeId , CombinationConsumable5.ResultModel.itemType , AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

# Fields

## cost

```
public Dictionary<Material, int> cost
```

### Field Value

Dictionary<Material, int>

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class RapidCombination5

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("rapid_combination5")]
public class RapidCombination5 : GameAction, IAction, IRapidCombinationV1
```

**Inheritance**

object ← ActionBase ← GameAction ← RapidCombination5

**Implements**

IAction, IRapidCombinationV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## slotIndex

```
public int slotIndex
```

### Field Value

int

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RapidCombination5.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RapidCombination5.ResultModel : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← RapidCombination5.ResultModel

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## Fields

### cost

```
public Dictionary<Material, int> cost
```

### Field Value

Dictionary<Material, int>

## id

```
public Guid id
```

### Field Value

Guid

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ReRegisterProduct

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("re_register_product2")]
public class ReRegisterProduct : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← ReRegisterProduct

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Capacity

```
public const int Capacity = 100
```

### Field Value

int

## ChargeAp

```
public bool ChargeAp
```

### Field Value

bool

## CostAp

```
public const int CostAp = 5
```

### Field Value

int

## ReRegisterInfos

```
public List<(IProductInfo, IRegisterInfo)> ReRegisterInfos
```

### Field Value

List<(IProductInfo, IRegisterInfo)>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

#### Parameters

`context` IActionContext

#### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

#### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RedeemCode

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("redeem_code3")]
public class RedeemCode : GameAction, IAction, IRedeemCodeV1
```

**Inheritance**

object ← ActionBase ← GameAction ← RedeemCode

**Implements**

IAction, IRedeemCodeV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RedeemCode()

```
public RedeemCode()
```

## RedeemCode(string, Address)

```
public RedeemCode(string code, Address avatarAddress)
```

### Parameters

`code` string

`avatarAddress` Address

## Properties

## AvatarAddress

```
public Address AvatarAddress { get; }
```

### Property Value

Address

## Code

```
public string Code { get; }
```

### Property Value

string

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RedeemCode2

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("redeem_code2")]
public class RedeemCode2 : GameAction, IAction, IRedeemCodeV1
```

**Inheritance**

object ← ActionBase ← GameAction ← RedeemCode2

**Implements**

IAction, IRedeemCodeV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RedeemCode2()

```
public RedeemCode2()
```

### RedeemCode2(string, Address)

```
public RedeemCode2(string code, Address avatarAddress)
```

#### Parameters

`code` string

`avatarAddress` Address

# Properties

## AvatarAddress

```
public Address AvatarAddress { get; }
```

### Property Value

Address

## Code

```
public string Code { get; }
```

### Property Value

string

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RegisterInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class RegisterInfo : IRegisterInfo
```

**Inheritance**

object ← RegisterInfo

**Implements**

IRegisterInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RegisterInfo()

```
public RegisterInfo()
```

### RegisterInfo(List)

```
public RegisterInfo(List serialized)
```

#### Parameters

`serialized` List

## Properties

## AvatarAddress

```
public Address AvatarAddress { get; set; }
```

### Property Value

Address

## ItemCount

```
public int ItemCount { get; set; }
```

### Property Value

int

## Price

```
public FungibleAssetValue Price { get; set; }
```

### Property Value

FungibleAssetValue

## TradableId

```
public Guid TradableId { get; set; }
```

### Property Value

Guid

## Type

```
public ProductType Type { get; set; }
```

### Property Value

ProductType

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

### Validate()

```
public void Validate()
```

### ValidateAddress(Address)

```
public void ValidateAddress(Address avatarAddress)
```

### Parameters

avatarAddress Address

### ValidatePrice(Currency)

```
public void ValidatePrice(Currency ncg)
```

## Parameters

ncg Currency

# Class RegisterProduct

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("register_product3")]
public class RegisterProduct : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← RegisterProduct

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Capacity

```
public const int Capacity = 100
```

### Field Value

int

## ChargeAp

```
public bool ChargeAp
```

### Field Value

bool

## CostAp

```
public const int CostAp = 5
```

### Field Value

int

## NonTradableTickerCurrencies

```
public static readonly IReadOnlyCollection<Currency> NonTradableTickerCurrencies
```

### Field Value

IReadOnlyCollection<Currency>

## RegisterInfos

```
public IEnumerable<IRegisterInfo> RegisterInfos
```

### Field Value

IEnumerable<IRegisterInfo>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Register(IActionContext, IRegisterInfo, AvatarState, ProductsState, IWorld, IRandom, RestrictionSheet)

```
public static IWorld Register(IActionContext context, IRegisterInfo info,
AvatarState avatarState, ProductsState productsState, IWorld states, IRandom random,
RestrictionSheet restrictionSheet)
```

## Parameters

`context` IActionContext

The action context the registration runs in.

`info` IRegisterInfo

What to register: an item or a fungible asset.

`avatarState` AvatarState

The seller's avatar, whose inventory the item leaves.

`productsState` ProductsState

The seller's product list the new product id joins.

`states` IWorld

The world the product is written into.

`random` IRandom

Source of the product id.

`restrictionSheet` RestrictionSheet

Restrictions on top of the hardcoded ones, or null on a chain that has not been patched with RestrictionSheet yet. The sheet only adds: a patched sheet that forgets an entry must not lift a restriction that code already enforces. Required rather than optional so that a new call site has to decide instead of silently registering without the policy.

## Returns

IWorld

# Class RegisterProduct0

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("register_product")]
[ActionObsolete(8324909)]
public class RegisterProduct0 : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← RegisterProduct0

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Capacity

```
public const int Capacity = 100
```

### Field Value

int

## ChargeAp

```
public bool ChargeAp
```

### Field Value

bool

## CostAp

```
public const int CostAp = 5
```

### Field Value

int

## RegisterInfos

```
public IEnumerable<IRegisterInfo> RegisterInfos
```

### Field Value

IEnumerable<IRegisterInfo>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

## Register(IActionContext, IRegisterInfo, AvatarState, ProductsState, IWorld, IRandom)

```
public static IWorld Register(IActionContext context, IRegisterInfo info,
AvatarState avatarState, ProductsState productsState, IWorld states, IRandom random)
```

## Parameters

`context` IActionContext

`info` IRegisterInfo

`avatarState` AvatarState

`productsState` ProductsState

`states` IWorld

`random` IRandom

## Returns

IWorld

# Class RemoveAddressState

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Action to remove state for multiple addresses.

```
[ActionType("remove_address_state")]
public class RemoveAddressState : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← RemoveAddressState

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RemoveAddressState()

Default constructor.

```
public RemoveAddressState()
```

### RemoveAddressState(IReadOnlyList<(Address accountAddress, Address targetAddress)>)

Constructor that initializes with a list of removals.

```
public RemoveAddressState(IReadOnlyList<(Address accountAddress, Address
targetAddress)> removals)
```

### Parameters

`removals` IReadOnlyList<(Address accountAddress, Address targetAddress)>

List of tuples to remove state for

## Fields

### TypeId

The type identifier of the action.

```
public const string TypeId = "remove_address_state"
```

### Field Value

string

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Removals

List of tuples (account address, target address) to remove state for.

```
public IReadOnlyList<(Address accountAddress, Address targetAddress)> Removals {
get; }
```

### Property Value

IReadOnlyList<(Address accountAddress, Address targetAddress)>

## Methods

## Execute(IActionContext)

Executes the action to remove state for the specified addresses.

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

Action context

### Returns

IWorld

The updated world state

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class RenewAdminState

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/805 Updated at https://github.com/planetarium/lib9c/pull/815 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("renew_admin_state")]
public class RenewAdminState : GameAction, IAction, IRenewAdminStateV1
```

**Inheritance**

object ← ActionBase ← GameAction ← RenewAdminState

**Implements**

IAction, IRenewAdminStateV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RenewAdminState()

```
public RenewAdminState()
```

## RenewAdminState(long)

```
public RenewAdminState(long newValidUntil)
```

### Parameters

`newValidUntil` long

## Properties

### NewValidUntil

```
public long NewValidUntil { get; }
```

### Property Value

long

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RequestPledge

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("request_pledge")]
public class RequestPledge : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← RequestPledge

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RequestPledge()

```
public RequestPledge()
```

## Fields

### AgentAddress

```
public Address AgentAddress
```

#### Field Value

## DefaultRefillMead

```
public const int DefaultRefillMead = 4
```

### Field Value

int

## RefillMead

```
public int RefillMead
```

### Field Value

int

## TypeIdentifier

```
public const string TypeIdentifier = "request_pledge"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class RequiredBlockIndexException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RequiredBlockIndexException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← RequiredBlockIndexException

**Implements**

ISerializable

**Derived**

RequiredBlockIntervalException

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RequiredBlockIndexException()

```
public RequiredBlockIndexException()
```

### RequiredBlockIndexException(SerializationInfo, StreamingContext)

```
protected RequiredBlockIndexException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## RequiredBlockIndexException(string)

```
public RequiredBlockIndexException(string msg)
```

### Parameters

`msg` string

## RequiredBlockIndexException(string, string, long)

```
public RequiredBlockIndexException(string actionType, string addressesHex,
long currentBlockIndex)
```

### Parameters

`actionType` string

`addressesHex` string

`currentBlockIndex` long

## RequiredBlockIndexException(string, string, long, long)

```
public RequiredBlockIndexException(string actionType, string addressesHex, long
requiredBlockIndex, long currentBlockIndex)
```

## Parameters

`actionType` string

`addressesHex` string

`requiredBlockIndex` long

`currentBlockIndex` long

# Class RequiredBlockIntervalException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RequiredBlockIntervalException : RequiredBlockIndexException,
ISerializable
```

### Inheritance

object ← Exception ← RequiredBlockIndexException ← RequiredBlockIntervalException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RequiredBlockIntervalException()

```
public RequiredBlockIntervalException()
```

### RequiredBlockIntervalException(SerializationInfo, StreamingContext)

```
protected RequiredBlockIntervalException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## RequiredBlockIntervalException(string)

```
public RequiredBlockIntervalException(string msg)
```

### Parameters

`msg` string

## RequiredBlockIntervalException(string, string, long)

```
public RequiredBlockIntervalException(string actionType, string addressesHex,
long currentBlockIndex)
```

### Parameters

`actionType` string

`addressesHex` string

`currentBlockIndex` long

## RequiredBlockIntervalException(string, string, long, long)

```
public RequiredBlockIntervalException(string actionType, string addressesHex, long
requiredBlockIndex, long currentBlockIndex)
```

### Parameters

`actionType` string

`addressesHex` string

`requiredBlockIndex` long

`currentBlockIndex` long

# Class RetrieveAvatarAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("retrieve_avatar_assets")]
public class RetrieveAvatarAssets : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← RetrieveAvatarAssets

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RetrieveAvatarAssets()

```
public RetrieveAvatarAssets()
```

### RetrieveAvatarAssets(Address)

```
public RetrieveAvatarAssets(Address avatarAddress)
```

#### Parameters

`avatarAddress` Address

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## TypeIdentifier

```
public const string TypeIdentifier = "retrieve_avatar_assets"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class RewardGold

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at many pull requests Updated at https://github.com/planetarium/lib9c/pull/1135

```
[Serializable]
public class RewardGold : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← RewardGold

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## FilterInactiveArenaInfoBlockIndex

```
public const long FilterInactiveArenaInfoBlockIndex = 3976000
```

### Field Value

long

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GenesisGoldDistribution(IActionContext, IWorld)

```
public IWorld GenesisGoldDistribution(IActionContext ctx, IWorld states)
```

### Parameters

`ctx` IActionContext

`states` IWorld

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

## MinerReward(IActionContext, IWorld)

```
public IWorld MinerReward(IActionContext ctx, IWorld states)
```

### Parameters

`ctx` IActionContext

`states` IWorld

### Returns

IWorld

## PrepareNextArena(IActionContext, IWorld)

```
public IWorld PrepareNextArena(IActionContext ctx, IWorld states)
```

### Parameters

`ctx` IActionContext

`states` IWorld

### Returns

IWorld

## ResetChallengeCount(IActionContext, IWorld)

```
public IWorld ResetChallengeCount(IActionContext ctx, IWorld states)
```

### Parameters

`ctx` IActionContext

`states` IWorld

### Returns

IWorld

## WeeklyArenaRankingBoard(IActionContext, IWorld)

```
[Obsolete("Use WeeklyArenaRankingBoard2 for performance.")]
public IWorld WeeklyArenaRankingBoard(IActionContext ctx, IWorld states)
```

### Parameters

`ctx` IActionContext

`states` IWorld

### Returns

IWorld

## WeeklyArenaRankingBoard2(IActionContext, IWorld)

```
public IWorld WeeklyArenaRankingBoard2(IActionContext ctx, IWorld states)
```

### Parameters

`ctx` IActionContext

`states` IWorld

## Returns

IWorld

# Class RuneEnhancement

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("runeEnhancement2")]
public class RuneEnhancement : GameAction, IAction, IRuneEnhancementV1
```

### Inheritance

object ← ActionBase ← GameAction ← RuneEnhancement

### Implements

IAction, IRuneEnhancementV1

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## RuneId

```
public int RuneId
```

### Field Value

int

## TryCount

```
public int TryCount
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Struct RuneEnhancement.LevelUpResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public struct RuneEnhancement.LevelUpResult
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Properties

### CrystalCost

```
public int CrystalCost { readonly get; set; }
```

#### Property Value

int

### LevelUpCount

```
public int LevelUpCount { readonly get; set; }
```

#### Property Value

int

### NcgCost

```
public int NcgCost { readonly get; set; }
```

### Property Value

int

## RuneCost

```
public int RuneCost { readonly get; set; }
```

### Property Value

int

# Methods

## ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

### Returns

string

The fully qualified type name.

# Class RuneSlotInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents rune slot information for a character.

```
[Serializable]
public class RuneSlotInfo
```

**Inheritance**

object ← RuneSlotInfo

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RuneSlotInfo(List)

Initializes a new instance of the RuneSlotInfo class from serialized data.

```
public RuneSlotInfo(List serialized)
```

#### Parameters

`serialized` List

The serialized data.

### RuneSlotInfo(int, int)

Initializes a new instance of the RuneSlotInfo class.

```
public RuneSlotInfo(int slotIndex, int runeId)
```

### Parameters

`slotIndex` int

The slot index.

`runeId` int

The rune ID.

## Properties

### RuneId

Gets the rune ID equipped in this slot.

```
public int RuneId { get; }
```

#### Property Value

int

### SlotIndex

Gets the slot index for the rune.

```
public int SlotIndex { get; }
```

#### Property Value

int

## Methods

### Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

True if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

Serializes the rune slot info to Bencodex format.

```
public IValue Serialize()
```

### Returns

IValue

The serialized data.

# Class RuneSummon

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("rune_summon")]
public class RuneSummon : GameAction, IAction, IRuneSummonV1
```

**Inheritance**

object ← ActionBase ← GameAction ← RuneSummon

**Implements**

IAction, IRuneSummonV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## AvatarAddressKey

```
public const string AvatarAddressKey = "aa"
```

### Field Value

string

## GroupId

```
public int GroupId
```

### Field Value

int

## GroupIdKey

```
public const string GroupIdKey = "gid"
```

### Field Value

string

## RuneQuantity

Number of runes obtained per summon.

```
public const int RuneQuantity = 10
```

### Field Value

int

## SummonCount

```
public int SummonCount
```

### Field Value

int

## SummonCountKey

```
public const string SummonCountKey = "sc"
```

### Field Value

string

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# SimulateSummon(RuneSheet, Row, int, IRandom, RuneListSheet, Row)

Simulates rune summoning with optional grade guarantee system. Applies 10+1 bonus rule and uses grade guarantee settings from SummonSheet.Row when enabled. When grade guarantee is enabled, ensures minimum grade runes are obtained based on summon count. Each summon yields 10 runes of the selected type.

```
public static Dictionary<Currency, int> SimulateSummon(RuneSheet runeSheet,
SummonSheet.Row summonRow, int summonCount, IRandom random, RuneListSheet
runeListSheet, BoostScheduleSheet.Row guaranteeBoost = null)
```

## Parameters

`runeSheet` RuneSheet

Rune sheet containing rune information

`summonRow` SummonSheet.Row

Summon configuration row with recipes and guarantee settings

`summonCount` int

Number of summons to perform (before 10+1 bonus)

`random` IRandom

Random number generator

`runeListSheet` RuneListSheet

Rune list sheet for grade information

`guaranteeBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this group's guarantee count, or `null` for none. A caller replaying a summon must pass what the action resolved.

### Returns

Dictionary<Currency, int>

Dictionary mapping rune currencies to quantities

## Summon(IActionContext, Address, RuneSheet, Row, int, IRandom, IWorld, RuneListSheet, Row)

```
public static IWorld Summon(IActionContext context, Address avatarAddress, RuneSheet runeSheet, SummonSheet.Row summonRow, int summonCount, IRandom random, IWorld states, RuneListSheet runeListSheet, BoostScheduleSheet.Row guaranteeBoost = null)
```

### Parameters

`context` IActionContext

`avatarAddress` Address

`runeSheet` RuneSheet

`summonRow` SummonSheet.Row

`summonCount` int

`random` IRandom

`states` IWorld

`runeListSheet` RuneListSheet

`guaranteeBoost` BoostScheduleSheet.Row

### Returns

# Class SecureMiningReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("secure_mining_reward")]
public class SecureMiningReward : ActionBase, IAction, ISecureMiningReward
```

**Inheritance**

object ← ActionBase ← SecureMiningReward

**Implements**

IAction, ISecureMiningReward

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SecureMiningReward()

```
public SecureMiningReward()
```

## SecureMiningReward(Address)

```
public SecureMiningReward(Address recipient)
```

### Parameters

`recipient` Address

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Recipient

```
public Address Recipient { get; }
```

### Property Value

Address

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class Sell

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/1640

```
[Serializable]
[ActionType("sell12")]
[ActionObsolete(7200000)]
public class Sell : GameAction, IAction, ISellV2
```

**Inheritance**

object ← ActionBase ← GameAction ← Sell

**Implements**

IAction, ISellV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## count

```
public int count
```

### Field Value

int

## itemSubType

```
public ItemSubType itemSubType
```

### Field Value

ItemSubType

## orderId

```
public Guid orderId
```

### Field Value

Guid

## price

```
public FungibleAssetValue price
```

### Field Value

FungibleAssetValue

## sellerAvatarAddress

```
public Address sellerAvatarAddress
```

### Field Value

Address

## tradableId

```
public Guid tradableId
```

### Field Value

Guid

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class Sell6

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("sell6")]
public class Sell6 : GameAction, IAction, ISellV2
```

### Inheritance

object ← ActionBase ← GameAction ← Sell6

### Implements

IAction, ISellV2

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ExpiredBlockIndex

```
public const long ExpiredBlockIndex = 16000
```

### Field Value

long

## count

```
public int count
```

### Field Value

int

## itemSubType

```
public ItemSubType itemSubType
```

### Field Value

ItemSubType

## price

```
public FungibleAssetValue price
```

### Field Value

FungibleAssetValue

## sellerAvatarAddress

```
public Address sellerAvatarAddress
```

### Field Value

Address

## tradableId

```
public Guid tradableId
```

### Field Value

Guid

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class SellCancellation

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/609 Updated at https://github.com/planetarium/lib9c/pull/620 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("sell_cancellation9")]
public class SellCancellation : GameAction, IAction, ISellCancellationV3
```

**Inheritance**

object ← ActionBase ← GameAction ← SellCancellation

**Implements**

IAction, ISellCancellationV3

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## itemSubType

```
public ItemSubType itemSubType
```

### Field Value

ItemSubType

## orderId

```
public Guid orderId
```

### Field Value

Guid

## sellerAvatarAddress

```
public Address sellerAvatarAddress
```

### Field Value

Address

## tradableId

```
public Guid tradableId
```

### Field Value

Guid

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Cancel(IActionContext, IWorld, AvatarState, string, Order)

```
public static IWorld Cancel(IActionContext context, IWorld states, AvatarState
avatarState, string addressesHex, Order order)
```

### Parameters

`context` IActionContext

`states` IWorld

`avatarState` AvatarState

`addressesHex` string

`order` Order

### Returns

IWorld

## CancelV2(IActionContext, IWorld, AvatarState, string, Order, Guid, ItemSubType)

```
public static IWorld CancelV2(IActionContext context, IWorld states,
AvatarState avatarState, string addressesHex, Order order, Guid tradableId,
ItemSubType itemSubType)
```

### Parameters

`context` IActionContext

`states` IWorld

`avatarState` AvatarState

`addressesHex` string

`order` Order

`tradableId` Guid

`itemSubType` ItemSubType

### Returns

IWorld

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class SellCancellation.Result

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class SellCancellation.Result : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← SellCancellation.Result

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Result()

```
public Result()
```

### Result(Dictionary)

```
public Result(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

# Fields

## id

```
public Guid id
```

### Field Value

Guid

## shopItem

```
public ShopItem shopItem
```

### Field Value

ShopItem

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Class SetAddressState

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Action to set state for multiple addresses. Throws an exception if the state already exists for the target address.

```
[ActionType("set_address_state")]
public class SetAddressState : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← SetAddressState

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SetAddressState()

Default constructor.

```
public SetAddressState()
```

### SetAddressState(IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>)

Constructor that initializes with a list of states to set.

```
public SetAddressState(IReadOnlyList<(Address accountAddress, Address targetAddress,
IValue state)> states)
```

### Parameters

`states` IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>

List of tuples to set state for

### Exceptions

ArgumentNullException

Thrown if states is null or any state value is null

## Fields

### TypeId

The type identifier of the action.

```
public const string TypeId = "set_address_state"
```

### Field Value

string

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## States

List of tuples (account address, target address, state value) to set state for.

```
public IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>
States { get; }
```

### Property Value

IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>

# Methods

## Execute(IActionContext)

Executes the action to set state for the specified addresses. Throws an exception if the state already exists for the target address.

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

Action context

### Returns

IWorld

The updated world state

### Exceptions

InvalidOperationException

Thrown if the state already exists for the target address

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class SetAddressStateCompressed

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Compressed version of SetAddressState to reduce transaction size for large state data. Uses GZip compression to minimize the size of state data in transactions.

This action is particularly useful when setting large state data that would exceed transaction policy size limits. The compression can reduce state data size by 60-80% depending on the content structure and redundancy.

Example usage:

```
var states = new List<(Address, Address, IValue)>
{
    (accountAddress, targetAddress, largeStateValue),
};
var compressedAction = new SetAddressStateCompressed(states);
```

```
[ActionType("set_address_state_compressed")]
public class SetAddressStateCompressed : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← SetAddressStateCompressed

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

# SetAddressStateCompressed()

Default constructor.

```
public SetAddressStateCompressed()
```

# SetAddressStateCompressed(IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>)

Constructor that initializes with a list of states to set. The states will be automatically compressed using GZip compression.

```
public SetAddressStateCompressed(IReadOnlyList<(Address accountAddress, Address
targetAddress, IValue state)> states)
```

## Parameters

`states` IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>

List of tuples to set state for

## Exceptions

ArgumentNullException

Thrown if states is null or any state value is null

ArgumentException

Thrown if states list is empty

# SetAddressStateCompressed(IReadOnlyList<(Address accountAddress, Address targetAddress, byte[] compressedState)>)

Constructor that initializes with already compressed state data.

```
public SetAddressStateCompressed(IReadOnlyList<(Address accountAddress, Address
targetAddress, byte[] compressedState)> compressedStates)
```

### Parameters

`compressedStates` IReadOnlyList<(Address accountAddress, Address targetAddress, byte[] compressedState)>

List of tuples with compressed state data

### Exceptions

ArgumentNullException

Thrown if compressedStates is null or any compressed state is null

ArgumentException

Thrown if compressedStates list is empty

## Fields

## TypeId

The type identifier of the action.

```
public const string TypeId = "set_address_state_compressed"
```

### Field Value

string

## Properties

## CompressedStates

List of tuples (account address, target address, compressed state data) to set state for. The state data is compressed using GZip compression.

```
public IReadOnlyList<(Address accountAddress, Address targetAddress, byte[]
compressedState)> CompressedStates { get; }
```

### Property Value

IReadOnlyList<(Address accountAddress, Address targetAddress, byte[] compressedState)>

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## CompressState(IValue)

Compresses state data using GZip compression.

This method is used to prepare state data for the CompressedStates property. GZip compression typically reduces state data size by 60-80% depending on content.

Example:

```
var state = (Text)"large state data";
var compressedData = SetAddressStateCompressed.CompressState(state);
// compressedData can now be used in CompressedStates property
```

```
public static byte[] CompressState(IValue state)
```

### Parameters

`state` IValue

The state data to compress. Must not be null.

### Returns

byte[]

Compressed byte array using GZip compression.

### Exceptions

ArgumentNullException

Thrown when state is null.

InvalidOperationException

Thrown when compression fails.

## DecompressState(byte[])

Decompresses state data using GZip compression.

This method is used internally to restore the original state data from the compressed byte array stored in CompressedStates.

Example:

```
var originalState = SetAddressStateCompressed.DecompressState(compressedData);
// originalState now contains the decompressed IValue
```

```
public static IValue DecompressState(byte[] compressedData)
```

### Parameters

`compressedData` byte[]

The compressed data to decompress. Must not be null.

### Returns

IValue

Decompressed IValue state data.

### Exceptions

ArgumentNullException

Thrown when compressedData is null.

ArgumentException

Thrown when compressedData is empty.

InvalidDataException

Thrown when the compressed data is corrupted or not valid GZip format.

InvalidOperationException

Thrown when decompression fails.

## Execute(IActionContext)

Executes the action to set state for the specified addresses. Throws an exception if the state already exists for the target address.

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

Action context

### Returns

IWorld

The updated world state

### Exceptions

ArgumentNullException

Thrown if context is null

InvalidOperationException

Thrown if the state already exists for the target address

InvalidDataException

Thrown when the compressed data is corrupted or invalid

UnauthorizedAccessException

Thrown when the signer does not have permission to execute this action

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Enum ShopErrorType

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ShopErrorType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

ERROR_CODE_DUPLICATE_SELL = 10

ERROR_CODE_FAILED_LOADING_STATE = 1

ERROR_CODE_INSUFFICIENT_BALANCE = 4

ERROR_CODE_INVALID_ADDRESS = 5

ERROR_CODE_INVALID_ITEMTYPE = 9

ERROR_CODE_INVALID_ORDERID = 7

ERROR_CODE_INVALID_PRICE = 6

ERROR_CODE_INVALID_TRADABLEID = 8

ERROR_CODE_ITEM_DOES_NOT_EXIST = 2

ERROR_CODE_SHOPITEM_EXPIRED = 3

# Class ShopItemExpiredException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ShopItemExpiredException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← ShopItemExpiredException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ShopItemExpiredException()

```
public ShopItemExpiredException()
```

### ShopItemExpiredException(SerializationInfo, StreamingContext)

```
protected ShopItemExpiredException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## ShopItemExpiredException(string)

```
public ShopItemExpiredException(string msg)
```

### Parameters

`msg` string

# Class SlotAlreadyUnlockedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class SlotAlreadyUnlockedException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← SlotAlreadyUnlockedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SlotAlreadyUnlockedException(SerializationInfo, StreamingContext)

```
protected SlotAlreadyUnlockedException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# SlotAlreadyUnlockedException(string)

```
public SlotAlreadyUnlockedException(string s)
```

## Parameters

s [string]

# Class StageNotClearedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class StageNotClearedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← StageNotClearedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### StageNotClearedException()

```
public StageNotClearedException()
```

### StageNotClearedException(SerializationInfo, StreamingContext)

```
protected StageNotClearedException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## StageNotClearedException(string)

```
public StageNotClearedException(string message)
```

### Parameters

`message` string

## StageNotClearedException(string, Exception)

```
public StageNotClearedException(string message, Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

## StageNotClearedException(string, string, int, int)

```
public StageNotClearedException(string actionType, string addressesHex, int
requiredToClearedStage, int currentClearedStage)
```

### Parameters

`actionType` string

`addressesHex` string

`requiredToClearedStage` int

`currentClearedStage` int

# Class Stake

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("stake3")]
public class Stake : GameAction, IAction, IStakeV1
```

**Inheritance**

object ← ActionBase ← GameAction ← Stake

**Implements**

IAction, IStakeV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Stake()

```
public Stake()
```

## Stake(BigInteger)

```
public Stake(BigInteger amount)
```

### Parameters

`amount` BigInteger

## Stake(BigInteger, Address)

```
public Stake(BigInteger amount, Address avatarAddress)
```

### Parameters

`amount` BigInteger

`avatarAddress` Address

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class Stake2

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("stake2")]
[ActionObsolete(7983895)]
public class Stake2 : GameAction, IAction, IStakeV1
```

### Inheritance

object ← ActionBase ← GameAction ← Stake2

### Implements

IAction, IStakeV1

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Stake2()

```
public Stake2()
```

### Stake2(BigInteger)

```
public Stake2(BigInteger amount)
```

#### Parameters

`amount` BigInteger

# Fields

## ObsoleteBlockIndex

```
public const long ObsoleteBlockIndex = 7983895
```

### Field Value

long

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class StakeExistingClaimableException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class StakeExistingClaimableException : Exception, ISerializable
```

### Inheritance

object ← Exception ← StakeExistingClaimableException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### StakeExistingClaimableException()

```
public StakeExistingClaimableException()
```

### StakeExistingClaimableException(SerializationInfo, StreamingContext)

```
protected StakeExistingClaimableException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## StakeExistingClaimableException(string)

```
public StakeExistingClaimableException(string message)
```

### Parameters

`message` string

## StakeExistingClaimableException(string, Exception)

```
public StakeExistingClaimableException(string message, Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

# Class Swap

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Swap action swaps one currency to another currency.

```
[Serializable]
[ActionType("swap")]
public class Swap : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← Swap

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Swap()

Initializes a new instance of the Swap class.

```
public Swap()
```

## Swap(FungibleAssetValue, Currency)

Initializes a new instance of the Swap class.

```
public Swap(FungibleAssetValue from, Currency to)
```

### Parameters

`from` FungibleAssetValue

The Libplanet.Types.Assets.FungibleAssetValue to swap from.

`to` Currency

The Libplanet.Types.Assets.Currency to swap to.

## Fields

### TypeIdentifier

The type identifier for the Swap class.

```
public const string TypeIdentifier = "swap"
```

### Field Value

string

## Properties

### From

The Libplanet.Types.Assets.FungibleAssetValue to swap from.

```
public FungibleAssetValue From { get; }
```

### Property Value

FungibleAssetValue

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## To

The Libplanet.Types.Assets.Currency to swap to.

```
public Currency To { get; }
```

### Property Value

Currency

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class Synthesize

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Synthesize action is a type of action that synthesizes items. MaterialIds: Id list of items as material
ChargeAp: Whether to charge action points with action execution

```
[Serializable]
[ActionType("synthesize")]
public class Synthesize : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← Synthesize

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

AvatarAddress of the signer.

```
public Address AvatarAddress
```

### Field Value

Address

## ChargeAp

Whether to charge action points with action execution.

```
public bool ChargeAp
```

### Field Value

bool

## InvalidMaterialItemId

The list of invalid item ids for material.

```
public static readonly int[] InvalidMaterialItemId
```

### Field Value

int[]

## MaterialGradeId

MaterialGrade of the material item.

```
public int MaterialGradeId
```

### Field Value

int

## MaterialIds

Id list of items as material.

```
public List<Guid> MaterialIds
```

### Field Value

List<Guid>

## MaterialItemSubTypeId

ItemSubType of the material item.

```
public int MaterialItemSubTypeId
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

Execute Synthesize action, Input MaterialIds and AvatarAddress. Depending on the probability, you can get different items. Success: Return new AvatarState with synthesized item in inventory.

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class TotalSupplyDoesNotExistException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class TotalSupplyDoesNotExistException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← TotalSupplyDoesNotExistException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### TotalSupplyDoesNotExistException(Currency)

```
public TotalSupplyDoesNotExistException(Currency currency)
```

#### Parameters

`currency` Currency

### TotalSupplyDoesNotExistException(SerializationInfo, StreamingContext)

```
protected TotalSupplyDoesNotExistException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## TotalSupplyDoesNotExistException(string)

```
public TotalSupplyDoesNotExistException(string message)
```

### Parameters

`message` string

# Properties

## Currency

```
public Currency Currency { get; }
```

### Property Value

Currency

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

## Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

## Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

# Class TransferAsset

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2143 Updated at https://github.com/planetarium/lib9c/pull/2143

```
[Serializable]
[ActionType("transfer_asset5")]
public class TransferAsset : ActionBase, IAction, ISerializable,
ITransferAsset, ITransferAssetV1
```

**Inheritance**

object ← ActionBase ← TransferAsset

**Implements**

IAction, ISerializable, ITransferAsset, ITransferAssetV1

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## TransferAsset()

```
public TransferAsset()
```

## TransferAsset(Address, Address, FungibleAssetValue, string)

```
public TransferAsset(Address sender, Address recipient, FungibleAssetValue amount,
string memo = null)
```

### Parameters

`sender` Address

`recipient` Address

`amount` FungibleAssetValue

`memo` string

## TransferAsset(SerializationInfo, StreamingContext)

```
protected TransferAsset(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "transfer_asset5"
```

### Field Value

string

# Properties

## Amount

```
public FungibleAssetValue Amount { get; }
```

### Property Value

FungibleAssetValue

## Memo

```
public string Memo { get; }
```

### Property Value

string

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Recipient

```
public Address Recipient { get; }
```

### Property Value

Address

## Sender

```
public Address Sender { get; }
```

### Property Value

Address

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

#### Parameters

`context` IActionContext

#### Returns

IWorld

### GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

#### Exceptions

SecurityException

The caller does not have the required permission.

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

## ThrowIfStakeState(IWorld, Address)

```
public static void ThrowIfStakeState(IWorld state, Address recipient)
```

### Parameters

`state` IWorld

`recipient` Address

# Class TransferAsset3

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class TransferAsset3
```

**Inheritance**

object ← TransferAsset3

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AllowedCrystalTransfers

```
public static readonly IReadOnlyList<Address> AllowedCrystalTransfers
```

### Field Value

IReadOnlyList<Address>

## CrystalTransferringRestrictionStartIndex

```
public const long CrystalTransferringRestrictionStartIndex = 6220000
```

### Field Value

long

# Methods

# CheckCrystalSender(Currency, long, Address)

```
public static void CheckCrystalSender(Currency currency, long blockIndex,
Address sender)
```

## Parameters

`currency` Currency

`blockIndex` long

`sender` Address

# Class TransferAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/636 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("transfer_assets3")]
public class TransferAssets : ActionBase, IAction, ISerializable,
ITransferAssets, ITransferAssetsV1
```

**Inheritance**

object ← ActionBase ← TransferAssets

**Implements**

IAction, ISerializable, ITransferAssets, ITransferAssetsV1

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## TransferAssets()

```
public TransferAssets()
```

## TransferAssets(Address, List<(Address, FungibleAssetValue)>, string)

```
public TransferAssets(Address sender, List<(Address, FungibleAssetValue)>
recipients, string memo = null)
```

### Parameters

`sender` Address

`recipients` List<(Address recipient, FungibleAssetValue amount)>

`memo` string

## TransferAssets(SerializationInfo, StreamingContext)

```
protected TransferAssets(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Fields

## RecipientsCapacity

```
public const int RecipientsCapacity = 100
```

### Field Value

int

## TypeIdentifier

```
public const string TypeIdentifier = "transfer_assets3"
```

### Field Value

string

## Properties

## Memo

```
public string Memo { get; }
```

### Property Value

string

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Recipients

```
public List<(Address recipient, FungibleAssetValue amount)> Recipients { get; }
```

### Property Value

List<(Address recipient, FungibleAssetValue amount)>

## Sender

```
public Address Sender { get; }
```

### Property Value

Address

## Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class UnlockCombinationSlot

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("unlock_combination_slot")]
public class UnlockCombinationSlot : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← UnlockCombinationSlot

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## GoldenDustId

```
public const int GoldenDustId = 600201
```

### Field Value

int

## RubyDustId

```
public const int RubyDustId = 600202
```

### Field Value

int

## SlotIndex

```
public int SlotIndex
```

### Field Value

int

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

# Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

## Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class UnlockEquipmentRecipe

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("unlock_equipment_recipe2")]
public class UnlockEquipmentRecipe : GameAction, IAction, IUnlockEquipmentRecipeV1
```

**Inheritance**

object ← ActionBase ← GameAction ← UnlockEquipmentRecipe

**Implements**

IAction, IUnlockEquipmentRecipeV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## RecipeIds

```
public List<int> RecipeIds
```

### Field Value

List<int>

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# UnlockedIds(IWorld, Address, EquipmentItemRecipeSheet, WorldInformation, List<int>)

```
public static List<int> UnlockedIds(IWorld states, Address unlockedRecipeIdsAddress,
EquipmentItemRecipeSheet equipmentRecipeSheet, WorldInformation worldInformation,
List<int> recipeIds)
```

## Parameters

`states` IWorld

`unlockedRecipeIdsAddress` Address

`equipmentRecipeSheet` EquipmentItemRecipeSheet

`worldInformation` WorldInformation

`recipeIds` List<int>

## Returns

List<int>

# Class UnlockRuneSlot

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("unlock_rune_slot")]
public class UnlockRuneSlot : GameAction, IAction, IUnlockRuneSlotV1
```

### Inheritance

object ← ActionBase ← GameAction ← UnlockRuneSlot

### Implements

IAction, IUnlockRuneSlotV1

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## SlotIndex

```
public int SlotIndex
```

### Field Value

int

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class UnlockWorld

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Unlocks one or more worlds for an avatar and records the result to the legacy `world_ids` state. The required CRYSTAL is calculated from WorldUnlockSheet and transferred to UnlockWorld. If the calculated cost is zero, no asset transfer occurs.

```
[ActionType("unlock_world2")]
public class UnlockWorld : GameAction, IAction, IUnlockWorldV1
```

**Inheritance**

object ← ActionBase ← GameAction ← UnlockWorld

**Implements**

IAction, IUnlockWorldV1

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

Hard-forked at https://github.com/planetarium/lib9c/pull/1309.

# Fields

## AvatarAddress

The avatar address to apply the unlock to.

```
public Address AvatarAddress
```

### Field Value

Address

## WorldIds

Target world IDs to unlock.

```
public List<int> WorldIds
```

### Field Value

List<int>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class UpdateSell

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/1640

```
[Serializable]
[ActionType("update_sell5")]
[ActionObsolete(7200000)]
public class UpdateSell : GameAction, IAction, IUpdateSellV2
```

**Inheritance**

object ← ActionBase ← GameAction ← UpdateSell

**Implements**

IAction, IUpdateSellV2

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## sellerAvatarAddress

```
public Address sellerAvatarAddress
```

### Field Value

Address

## updateSellInfos

```
public IEnumerable<UpdateSellInfo> updateSellInfos
```

### Field Value

IEnumerable<UpdateSellInfo>

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Cancel(IWorld, UpdateSellInfo, string, AvatarState, OrderDigestListState, IActionContext, Address)

```
public static IWorld Cancel(IWorld states, UpdateSellInfo updateSellInfo, string
addressesHex, AvatarState avatarState, OrderDigestListState digestList,
IActionContext context, Address sellerAvatarAddress)
```

### Parameters

`states` IWorld

`updateSellInfo` UpdateSellInfo

`addressesHex` string

`avatarState` AvatarState

`digestList` OrderDigestListState

`context` IActionContext

`sellerAvatarAddress` Address

### Returns

IWorld

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class UpdateSellInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class UpdateSellInfo
```

**Inheritance**

object ← UpdateSellInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### UpdateSellInfo(List)

```
public UpdateSellInfo(List serialized)
```

#### Parameters

`serialized` List

### UpdateSellInfo(Guid, Guid, Guid, ItemSubType, FungibleAssetValue, int)

```
public UpdateSellInfo(Guid orderId, Guid updateSellOrderId, Guid tradableId,
ItemSubType itemSubType, FungibleAssetValue price, int count)
```

#### Parameters

`orderId` Guid

`updateSellOrderId` Guid

`tradableId` Guid

`itemSubType` ItemSubType

`price` FungibleAssetValue

`count` int

# Fields

## count

```
public int count
```

### Field Value

int

## itemSubType

```
public ItemSubType itemSubType
```

### Field Value

ItemSubType

## orderId

```
public Guid orderId
```

### Field Value

Guid

## price

```
public FungibleAssetValue price
```

### Field Value

FungibleAssetValue

## tradableId

```
public Guid tradableId
```

### Field Value

Guid

## updateSellOrderId

```
public Guid updateSellOrderId
```

### Field Value

Guid

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class UsageLimitExceedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class UsageLimitExceedException : InvalidOperationException, ISerializable
```

**Inheritance**

object ← Exception ← SystemException ← InvalidOperationException ← UsageLimitExceedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### UsageLimitExceedException()

```
public UsageLimitExceedException()
```

### UsageLimitExceedException(SerializationInfo, StreamingContext)

```
protected UsageLimitExceedException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## UsageLimitExceedException(string)

```
public UsageLimitExceedException(string message)
```

### Parameters

`message` string

# Class ValidatorSetOperate

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("op_validator_set")]
public sealed class ValidatorSetOperate : ActionBase, IAction,
IEquatable<ValidatorSetOperate>
```

**Inheritance**

object ← ActionBase ← ValidatorSetOperate

**Implements**

IAction, IEquatable<ValidatorSetOperate>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ValidatorSetOperate()

```
public ValidatorSetOperate()
```

### ValidatorSetOperate(ValidatorSetOperatorType, Validator)

```
public ValidatorSetOperate(ValidatorSetOperatorType @operator, Validator operand)
```

#### Parameters

`operator` ValidatorSetOperatorType

`operand` Validator

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "op_validator_set"
```

### Field Value

string

## ValidatorSetOperateKey

```
public static readonly string ValidatorSetOperateKey
```

### Field Value

string

# Properties

## Error

```
public string Error { get; }
```

### Property Value

string

## Operand

```
public Validator Operand { get; }
```

### Property Value

Validator

## Operator

```
public ValidatorSetOperatorType Operator { get; }
```

### Property Value

ValidatorSetOperatorType

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Append(Validator)

```
public static ValidatorSetOperate Append(Validator operand)
```

### Parameters

`operand` Validator

### Returns

ValidatorSetOperate

# Equals(ValidatorSetOperate)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(ValidatorSetOperate other)
```

## Parameters

`other` ValidatorSetOperate

An object to compare with this object.

## Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

# Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

## Parameters

`obj` object

The object to compare with the current object.

## Returns

bool

true if the specified object is equal to the current object; otherwise, false.

# Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

## Remove(Validator)

```
public static ValidatorSetOperate Remove(Validator operand)
```

### Parameters

`operand` Validator

### Returns

ValidatorSetOperate

## Update(Validator)

```
public static ValidatorSetOperate Update(Validator operand)
```

### Parameters

`operand` Validator

### Returns

ValidatorSetOperate

# Enum ValidatorSetOperatorType

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ValidatorSetOperatorType
```

**Extension Methods**
ValidatorSetOperatorTypeExtensions.ToFunc(ValidatorSetOperatorType) ,
StateExtensions.Serialize(Enum)

## Fields

`Append = 0`

Append the validator. no-op if validator public key already exists.

`Remove = 1`

Set validator's power to 0.

`Update = 2`

Update validator. if not exists, append it.

# Class ValidatorSetOperatorTypeExtensions

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class ValidatorSetOperatorTypeExtensions
```

**Inheritance**

object ← ValidatorSetOperatorTypeExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## ToFunc(ValidatorSetOperatorType)

```
public static Func<ValidatorSet, Validator, Validator> ToFunc(this
ValidatorSetOperatorType @operator)
```

### Parameters

`operator` ValidatorSetOperatorType

### Returns

Func<ValidatorSet, Validator, Validator>

# Class WeeklyArenaStateAlreadyEndedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaStateAlreadyEndedException : Exception, ISerializable
```

### Inheritance

object ← Exception ← WeeklyArenaStateAlreadyEndedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### WeeklyArenaStateAlreadyEndedException()

```
public WeeklyArenaStateAlreadyEndedException()
```

### WeeklyArenaStateAlreadyEndedException(SerializationInfo, StreamingContext)

```
public WeeklyArenaStateAlreadyEndedException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## WeeklyArenaStateAlreadyEndedException(string)

```
public WeeklyArenaStateAlreadyEndedException(string message)
```

### Parameters

`message` string

# Fields

## BaseMessage

```
public const string BaseMessage = "Aborted as the weekly arena state already ended."
```

### Field Value

string

# Class WeeklyArenaStateNotContainsAvatarAddressException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaStateNotContainsAvatarAddressException :
Exception, ISerializable
```

### Inheritance

object ← Exception ← WeeklyArenaStateNotContainsAvatarAddressException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### WeeklyArenaStateNotContainsAvatarAddressException(SerializationInfo, StreamingContext)

```
public WeeklyArenaStateNotContainsAvatarAddressException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## WeeklyArenaStateNotContainsAvatarAddressException(string)

```
public WeeklyArenaStateNotContainsAvatarAddressException(string message)
```

### Parameters

`message` string

## WeeklyArenaStateNotContainsAvatarAddressException(string, Address)

```
public WeeklyArenaStateNotContainsAvatarAddressException(string addressesHex,
Address avatarAddress)
```

### Parameters

`addressesHex` string

`avatarAddress` Address

# Namespace Nekoyume.Action.AdventureBoss

## Classes

ClaimAdventureBossReward

ExploreAdventureBoss

Represents an action where a player explores and battles through adventure boss stages. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- ExploreBoard: Updates exploration progress and points
- Explorer: Updates player's exploration status and achievements
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

SweepAdventureBoss

Represents a sweep action that automatically completes multiple adventure boss stages without player interaction. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- ExploreBoard: Updates exploration progress and points
- Explorer: Updates player's exploration status and achievements
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

UnlockFloor

Wanted

# Class ClaimAdventureBossReward

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("claim_adventure_boss_reward")]
public class ClaimAdventureBossReward : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← ClaimAdventureBossReward

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## TypeIdentifier

```
public const string TypeIdentifier = "claim_adventure_boss_reward"
```

### Field Value

string

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class ExploreAdventureBoss

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

Represents an action where a player explores and battles through adventure boss stages.
This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- ExploreBoard: Updates exploration progress and points
- Explorer: Updates player's exploration status and achievements
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

```
[Serializable]
[ActionType("explore_adventure_boss")]
public class ExploreAdventureBoss : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← ExploreAdventureBoss

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Costumes

```
public List<Guid> Costumes
```

### Field Value

List<Guid>

## Equipments

```
public List<Guid> Equipments
```

### Field Value

List<Guid>

## Foods

```
public List<Guid> Foods
```

### Field Value

List<Guid>

## RuneInfos

```
public List<RuneSlotInfo> RuneInfos
```

### Field Value

List<RuneSlotInfo>

## Season

```
public int Season
```

### Field Value

int

## StageBuffId

```
public int? StageBuffId
```

### Field Value

int?

## TypeIdentifier

```
public const string TypeIdentifier = "explore_adventure_boss"
```

### Field Value

string

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

#### Parameters

`context` IActionContext

#### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

#### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class SweepAdventureBoss

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

Represents a sweep action that automatically completes multiple adventure boss stages without player interaction. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- ExploreBoard: Updates exploration progress and points
- Explorer: Updates player's exploration status and achievements
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

```
[Serializable]
[ActionType("sweep_adventure_boss")]
public class SweepAdventureBoss : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← SweepAdventureBoss

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Costumes

```
public List<Guid> Costumes
```

### Field Value

List<Guid>

## Equipments

```
public List<Guid> Equipments
```

### Field Value

List<Guid>

## RuneInfos

```
public List<RuneSlotInfo> RuneInfos
```

### Field Value

List<RuneSlotInfo>

## Season

```
public int Season
```

### Field Value

int

## TypeIdentifier

```
public const string TypeIdentifier = "sweep_adventure_boss"
```

### Field Value

string

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class UnlockFloor

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("unlock_floor")]
public class UnlockFloor : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← UnlockFloor

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## GoldenDustId

```
public const int GoldenDustId = 600201
```

### Field Value

int

## OpeningFloor

```
public const int OpeningFloor = 5
```

### Field Value

int

## Season

```
public int Season
```

### Field Value

int

## TotalFloor

```
public const int TotalFloor = 30
```

### Field Value

int

## TypeIdentifier

```
public const string TypeIdentifier = "unlock_floor"
```

### Field Value

string

## UseNcg

```
public bool UseNcg
```

### Field Value

bool

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class Wanted

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("wanted")]
public class Wanted : GameAction, IAction
```

### Inheritance

object ← ActionBase ← GameAction ← Wanted

### Implements

IAction

### Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Bounty

```
public FungibleAssetValue Bounty
```

### Field Value

FungibleAssetValue

## Season

```
public int Season
```

### Field Value

int

## TypeIdentifier

```
public const string TypeIdentifier = "wanted"
```

### Field Value

string

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

# Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

## Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Namespace Nekoyume.Action.Arena

## Classes

Battle

Temp

## Enums

ArenaProvider

Temp

# Enum ArenaProvider

Namespace: Nekoyume.Action.Arena

Assembly: Lib9c.dll

Temp

```
public enum ArenaProvider
```

**Extension Methods**
StateExtensions.Serialize(Enum)

# Fields

```
PLANETARIUM = 0
```

# Class Battle

Namespace: Nekoyume.Action.Arena

Assembly: Lib9c.dll

Temp

```
[Serializable]
[ActionType("battle")]
public class Battle : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← Battle

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## CostAp

```
public const int CostAp = 5
```

### Field Value

int

## HpIncreasingModifier

```
public const int HpIncreasingModifier = 5
```

### Field Value

int

## arenaProvider

```
public ArenaProvider arenaProvider
```

### Field Value

ArenaProvider

## chargeAp

```
public bool chargeAp
```

### Field Value

bool

## costumes

```
public List<Guid> costumes
```

### Field Value

List<Guid>

## enemyAvatarAddress

```
public Address enemyAvatarAddress
```

### Field Value

Address

## equipments

```
public List<Guid> equipments
```

### Field Value

List<Guid>

## memo

```
public string memo
```

### Field Value

string

## myAvatarAddress

```
public Address myAvatarAddress
```

### Field Value

Address

## runeInfos

```
public List<RuneSlotInfo> runeInfos
```

### Field Value

List<RuneSlotInfo>

## Properties

### PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Namespace Nekoyume.Action.Coupons

## Classes

IssueCoupons

RedeemCoupon

TransferCoupons

# Class IssueCoupons

Namespace: Nekoyume.Action.Coupons

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("issue_coupons")]
public sealed class IssueCoupons : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← IssueCoupons

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### IssueCoupons()

```
public IssueCoupons()
```

### IssueCoupons(IImmutableDictionary<RewardSet, uint>, Address)

```
public IssueCoupons(IImmutableDictionary<RewardSet, uint> rewards,
Address recipient)
```

#### Parameters

`rewards` IImmutableDictionary<RewardSet, uint>

`recipient` Address

# Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## Recipient

```
public Address Recipient { get; }
```

### Property Value

Address

## Rewards

```
public IImmutableDictionary<RewardSet, uint> Rewards { get; }
```

### Property Value

IImmutableDictionary<RewardSet, uint>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class RedeemCoupon

Namespace: Nekoyume.Action.Coupons

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("redeem_coupon")]
public sealed class RedeemCoupon : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← RedeemCoupon

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RedeemCoupon()

```
public RedeemCoupon()
```

### RedeemCoupon(Guid, Address)

```
public RedeemCoupon(Guid couponId, Address avatarAddress)
```

#### Parameters

`couponId` Guid

`avatarAddress` Address

# Properties

## AvatarAddress

```
public Address AvatarAddress { get; }
```

### Property Value

Address

## CouponId

```
public Guid CouponId { get; }
```

### Property Value

Guid

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class TransferCoupons

Namespace: Nekoyume.Action.Coupons

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("transfer_coupons")]
public sealed class TransferCoupons : GameAction, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← TransferCoupons

**Implements**

IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### TransferCoupons()

```
public TransferCoupons()
```

### TransferCoupons(IImmutableDictionary<Address, IImmutableSet<Guid>>)

```
public TransferCoupons(IImmutableDictionary<Address, IImmutableSet<Guid>>
couponsPerRecipient)
```

#### Parameters

`couponsPerRecipient` IImmutableDictionary<Address, IImmutableSet<Guid>>

# Properties

## CouponsPerRecipient

```
public IImmutableDictionary<Address, IImmutableSet<Guid>> CouponsPerRecipient {
get; }
```

### Property Value

IImmutableDictionary<Address, IImmutableSet<Guid>>

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Namespace Nekoyume.Action.Custom EquipmentCraft

## Classes

CustomEquipmentCraft

## Structs

CustomCraftData

# Struct CustomCraftData

Namespace: Nekoyume.Action.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public struct CustomCraftData
```

**Inherited Members**

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Fields

### IconId

```
public int IconId
```

#### Field Value

int

### RecipeId

```
public int RecipeId
```

#### Field Value

int

### SlotIndex

```
public int SlotIndex
```

### Field Value

int

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class CustomEquipmentCraft

Namespace: Nekoyume.Action.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("custom_equipment_craft")]
public class CustomEquipmentCraft : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← CustomEquipmentCraft

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## CraftList

```
public List<CustomCraftData> CraftList
```

### Field Value

List<CustomCraftData>

## RandomIconId

```
public const int RandomIconId = 0
```

### Field Value

int

## TypeIdentifier

```
public const string TypeIdentifier = "custom_equipment_craft"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Namespace Nekoyume.Action.Exceptions

## Classes

CombatPowerException
: Exception thrown when player's combat power doesn't meet the floor requirements.

DuplicatedCraftSlotIndexException

EmptyRewardException

FloorAlreadyClearedException
: Exception thrown when trying to re-enter a floor that has already been cleared in the current season.

InfiniteTowerScheduleEndedException
: Exception thrown when infinite tower schedule has ended.

InfiniteTowerScheduleNotActiveException
: Exception thrown when infinite tower schedule is not active.

InfiniteTowerScheduleNotStartedException
: Exception thrown when infinite tower schedule has not started yet.

InvalidItemGradeException
: Exception thrown when an item grade doesn't meet the floor requirements.

InvalidItemLevelException
: Exception thrown when an item level doesn't meet the floor requirements.

InvalidSummonCountException

NotEnoughInfiniteTowerTicketsException
: Exception thrown when player doesn't have enough infinite tower tickets.

StageNotClearedException
: Exception thrown when trying to access a stage that hasn't been cleared yet.

# Class CombatPowerException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when player's combat power doesn't meet the floor requirements.

```
public class CombatPowerException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← CombatPowerException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### CombatPowerException(string, string, long, long, bool)

Initializes a new instance of the CombatPowerException class.

```
public CombatPowerException(string actionType, string addressesHex, long requiredCp,
long currentCp, bool isInsufficient = true)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`requiredCp` long

The required combat power.

`currentCp` long

The current combat power.

`isInsufficient` bool

Whether the combat power is insufficient (true) or excessive (false).

# Class DuplicatedCraftSlotIndexException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicatedCraftSlotIndexException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← DuplicatedCraftSlotIndexException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### DuplicatedCraftSlotIndexException(SerializationInfo, StreamingContext)

```
protected DuplicatedCraftSlotIndexException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicatedCraftSlotIndexException(string)

```
public DuplicatedCraftSlotIndexException(string message)
```

## Parameters

`message` string

# Class EmptyRewardException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class EmptyRewardException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← EmptyRewardException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### EmptyRewardException()

```
public EmptyRewardException()
```

### EmptyRewardException(SerializationInfo, StreamingContext)

```
protected EmptyRewardException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## EmptyRewardException(string)

```
public EmptyRewardException(string msg)
```

### Parameters

`msg` string

# Class FloorAlreadyClearedException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when trying to re-enter a floor that has already been cleared in the current season.

```
public class FloorAlreadyClearedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← FloorAlreadyClearedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### FloorAlreadyClearedException(string, string, int, int)

Initializes a new instance of the FloorAlreadyClearedException class.

```
public FloorAlreadyClearedException(string actionType, string addressesHex, int
floorId, int clearedFloor)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`floorId` int

The floor that was attempted.

`clearedFloor` int

The highest cleared floor.

# Class InfiniteTowerScheduleEndedException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when infinite tower schedule has ended.

```
public class InfiniteTowerScheduleEndedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InfiniteTowerScheduleEndedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InfiniteTowerScheduleEndedException(string, string, int, long, long)

Initializes a new instance of the InfiniteTowerScheduleEndedException class.

```
public InfiniteTowerScheduleEndedException(string actionType, string addressesHex,
int infiniteTowerId, long currentBlockIndex, long endBlockIndex)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`infiniteTowerId` int

The infinite tower ID.

`currentBlockIndex` long

The current block index.

`endBlockIndex` long

The block index when the schedule ended.

# Class InfiniteTowerScheduleNotActiveException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when infinite tower schedule is not active.

```
public class InfiniteTowerScheduleNotActiveException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InfiniteTowerScheduleNotActiveException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InfiniteTowerScheduleNotActiveException(string, string, int, long, long, long)

Initializes a new instance of the InfiniteTowerScheduleNotActiveException class.

```
public InfiniteTowerScheduleNotActiveException(string actionType, string
addressesHex, int infiniteTowerId, long currentBlockIndex, long startBlockIndex,
long endBlockIndex)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`infiniteTowerId` int

The infinite tower ID.

`currentBlockIndex` long

The current block index.

`startBlockIndex` long

The block index when the schedule starts.

`endBlockIndex` long

The block index when the schedule ends.

# Class InfiniteTowerScheduleNotStartedException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when infinite tower schedule has not started yet.

```
public class InfiniteTowerScheduleNotStartedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InfiniteTowerScheduleNotStartedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InfiniteTowerScheduleNotStartedException(string, string, int, long, long)

Initializes a new instance of the InfiniteTowerScheduleNotStartedException class.

```
public InfiniteTowerScheduleNotStartedException(string actionType, string
addressesHex, int infiniteTowerId, long currentBlockIndex, long startBlockIndex)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`infiniteTowerId` int

The infinite tower ID.

`currentBlockIndex` long

The current block index.

`startBlockIndex` long

The block index when the schedule starts.

# Class InvalidItemGradeException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when an item grade doesn't meet the floor requirements.

```
public class InvalidItemGradeException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidItemGradeException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidItemGradeException(string, string, int, int, bool)

Initializes a new instance of the InvalidItemGradeException class.

```
public InvalidItemGradeException(string actionType, string addressesHex, int
itemGrade, int requiredGrade, bool isMinimum = true)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`itemGrade` int

The item grade that failed validation.

`requiredGrade` int

The required grade.

`isMinimum` bool

Whether the requirement is a minimum (true) or maximum (false).

# Class InvalidItemLevelException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when an item level doesn't meet the floor requirements.

```
public class InvalidItemLevelException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidItemLevelException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidItemLevelException(string, string, int, int, bool)

Initializes a new instance of the InvalidItemLevelException class.

```
public InvalidItemLevelException(string actionType, string addressesHex, int
itemLevel, int requiredLevel, bool isMinimum = true)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`itemLevel` int

The item level that failed validation.

`requiredLevel` int

The required level.

`isMinimum` bool

Whether the requirement is a minimum (true) or maximum (false).

# Class InvalidSummonCountException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

```
public class InvalidSummonCountException : ArgumentException, ISerializable
```

**Inheritance**

object ← Exception ← SystemException ← ArgumentException ← InvalidSummonCountException

**Implements**

ISerializable

**Inherited Members**

ArgumentException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidSummonCountException()

```
public InvalidSummonCountException()
```

### InvalidSummonCountException(SerializationInfo, StreamingContext)

```
protected InvalidSummonCountException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidSummonCountException(string)

```
public InvalidSummonCountException(string msg)
```

### Parameters

`msg` string

# Class NotEnoughInfiniteTowerTicketsException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when player doesn't have enough infinite tower tickets.

```
public class NotEnoughInfiniteTowerTicketsException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← NotEnoughInfiniteTowerTicketsException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughInfiniteTowerTicketsException(string, string, int, int)

Initializes a new instance of the NotEnoughInfiniteTowerTicketsException class.

```
public NotEnoughInfiniteTowerTicketsException(string actionType, string
addressesHex, int requiredTickets, int availableTickets)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`requiredTickets` int

The number of tickets required.

`availableTickets` int

The number of tickets available.

# Class StageNotClearedException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when trying to access a stage that hasn't been cleared yet.

```
public class StageNotClearedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← StageNotClearedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### StageNotClearedException(string, string, int, int)

Initializes a new instance of the StageNotClearedException class.

```
public StageNotClearedException(string actionType, string addressesHex, int
requiredStage, int clearedStage)
```

#### Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`requiredStage` int

The stage that needs to be cleared.

`clearedStage` int

The highest stage that has been cleared.

# Namespace Nekoyume.Action.Exceptions.AdventureBoss

## Classes

AlreadyClaimedException

ClaimExpiredException

InsufficientStakingException

InvalidAdventureBossSeasonException

InvalidBountyException

MaxInvestmentCountExceededException

PreviousBountyException

SeasonInProgressException

# Class AlreadyClaimedException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyClaimedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← AlreadyClaimedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AlreadyClaimedException()

```
public AlreadyClaimedException()
```

### AlreadyClaimedException(SerializationInfo, StreamingContext)

```
protected AlreadyClaimedException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## AlreadyClaimedException(string)

```
public AlreadyClaimedException(string msg)
```

### Parameters

`msg` string

# Class ClaimExpiredException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class ClaimExpiredException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← ClaimExpiredException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ClaimExpiredException()

```
public ClaimExpiredException()
```

### ClaimExpiredException(SerializationInfo, StreamingContext)

```
protected ClaimExpiredException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## ClaimExpiredException(string)

```
public ClaimExpiredException(string msg)
```

### Parameters

`msg` string

# Class InsufficientStakingException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class InsufficientStakingException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InsufficientStakingException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InsufficientStakingException()

```
public InsufficientStakingException()
```

### InsufficientStakingException(SerializationInfo, StreamingContext)

```
protected InsufficientStakingException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InsufficientStakingException(string)

```
public InsufficientStakingException(string msg)
```

### Parameters

`msg` string

# Class InvalidAdventureBossSeasonException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidAdventureBossSeasonException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidAdventureBossSeasonException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidAdventureBossSeasonException()

```
public InvalidAdventureBossSeasonException()
```

### InvalidAdventureBossSeasonException(SerializationInfo, StreamingContext)

```
protected InvalidAdventureBossSeasonException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidAdventureBossSeasonException(string)

```
public InvalidAdventureBossSeasonException(string message)
```

### Parameters

`message` string

# Class InvalidBountyException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidBountyException : Exception, ISerializable
```

### Inheritance

object ← Exception ← InvalidBountyException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidBountyException()

```
public InvalidBountyException()
```

### InvalidBountyException(SerializationInfo, StreamingContext)

```
protected InvalidBountyException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidBountyException(string)

```
public InvalidBountyException(string msg)
```

### Parameters

`msg` string

# Class MaxInvestmentCountExceededException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class MaxInvestmentCountExceededException : Exception, ISerializable
```

### Inheritance

object ← Exception ← MaxInvestmentCountExceededException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### MaxInvestmentCountExceededException()

```
public MaxInvestmentCountExceededException()
```

### MaxInvestmentCountExceededException(SerializationInfo, StreamingContext)

```
protected MaxInvestmentCountExceededException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## MaxInvestmentCountExceededException(string)

```
public MaxInvestmentCountExceededException(string message)
```

### Parameters

`message` string

# Class PreviousBountyException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class PreviousBountyException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← PreviousBountyException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### PreviousBountyException()

```
public PreviousBountyException()
```

### PreviousBountyException(SerializationInfo, StreamingContext)

```
protected PreviousBountyException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## PreviousBountyException(string)

```
public PreviousBountyException(string msg)
```

### Parameters

`msg` string

# Class SeasonInProgressException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class SeasonInProgressException : Exception, ISerializable
```

### Inheritance

object ← Exception ← SeasonInProgressException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SeasonInProgressException()

```
public SeasonInProgressException()
```

### SeasonInProgressException(SerializationInfo, StreamingContext)

```
protected SeasonInProgressException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## SeasonInProgressException(string)

```
public SeasonInProgressException(string msg)
```

### Parameters

`msg` string

# Namespace Nekoyume.Action.Exceptions.Arena

## Classes

AlreadyJoinedArenaException

# Class AlreadyJoinedArenaException

Namespace: Nekoyume.Action.Exceptions.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyJoinedArenaException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← AlreadyJoinedArenaException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Constructors

## AlreadyJoinedArenaException(int, int, Address)

```
public AlreadyJoinedArenaException(int championshipId, int round,
Address avatarAddress)
```

### Parameters

`championshipId` int

`round` int

`avatarAddress` Address

# AlreadyJoinedArenaException(SerializationInfo, StreamingContext)

```
protected AlreadyJoinedArenaException(SerializationInfo info,
StreamingContext context)
```

## Parameters

`info` SerializationInfo

`context` StreamingContext

# AlreadyJoinedArenaException(string)

```
public AlreadyJoinedArenaException(string msg)
```

## Parameters

`msg` string

# Namespace Nekoyume.Action.Exceptions.CustomEquipmentCraft

## Classes

NotEnoughRelationshipException

RandomOnlyIconException

# Class NotEnoughRelationshipException

Namespace: Nekoyume.Action.Exceptions.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughRelationshipException : Exception, ISerializable
```

### Inheritance

object ← Exception ← NotEnoughRelationshipException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughRelationshipException(SerializationInfo, StreamingContext)

```
protected NotEnoughRelationshipException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# NotEnoughRelationshipException(string)

```
public NotEnoughRelationshipException(string s)
```

## Parameters

s [string]

# Class RandomOnlyIconException

Namespace: Nekoyume.Action.Exceptions.CustomEquipmentCraft

Assembly: Lib9c.dll

```
public class RandomOnlyIconException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← RandomOnlyIconException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RandomOnlyIconException(int)

```
public RandomOnlyIconException(int iconId)
```

#### Parameters

`iconId` int

### RandomOnlyIconException(SerializationInfo, StreamingContext)

```
protected RandomOnlyIconException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## RandomOnlyIconException(string)

```
public RandomOnlyIconException(string s)
```

### Parameters

`s` string

# Namespace Nekoyume.Action.Factory

## Classes

ClaimStakeRewardFactory

# Class ClaimStakeRewardFactory

Namespace: Nekoyume.Action.Factory

Assembly: Lib9c.dll

```
public static class ClaimStakeRewardFactory
```

**Inheritance**

object ← ClaimStakeRewardFactory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## CreateByBlockIndex(long, Address)

```
public static IClaimStakeReward CreateByBlockIndex(long blockIndex,
Address avatarAddress)
```

### Parameters

`blockIndex` long

`avatarAddress` Address

### Returns

IClaimStakeReward

## CreateByVersion(int, Address)

```
public static IClaimStakeReward CreateByVersion(int version, Address avatarAddress)
```

## Parameters

`version` int

`avatarAddress` Address

## Returns

IClaimStakeReward

# Namespace Nekoyume.Action.Garages

## Classes

BulkUnloadFromGarages

DeliverToOthersGarages

GarageUtils

LoadIntoMyGarages

UnloadFromMyGarages

# Class BulkUnloadFromGarages

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
[ActionType("bulk_unload_from_garages")]
public class BulkUnloadFromGarages : GameAction, IBulkUnloadFromGaragesV1, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← BulkUnloadFromGarages

**Implements**

IBulkUnloadFromGaragesV1, IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BulkUnloadFromGarages()

```
public BulkUnloadFromGarages()
```

### BulkUnloadFromGarages(IReadOnlyList<(Address recipientAvatarAddress, IEnumerable<(Address balanceAddress, FungibleAssetValue value)>? fungibleAssetValues, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>? fungibleIdAndCounts, string? memo)>)

```
public BulkUnloadFromGarages(IReadOnlyList<(Address recipientAvatarAddress,
IEnumerable<(Address balanceAddress, FungibleAssetValue value)>?
fungibleAssetValues, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
fungibleIdAndCounts, string? memo)> unloadData)
```

### Parameters

`unloadData` IReadOnlyList<(Address recipientAvatarAddress, IEnumerable<(Address balanceAddress, FungibleAssetValue value)> fungibleAssetValues, IEnumerable<(HashDigest<SHA256> fungibleId, int count)> fungibleIdAndCounts, string memo)>

## Properties

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## UnloadData

```
public IReadOnlyList<(Address recipientAvatarAddress, IOrderedEnumerable<(Address
balanceAddress, FungibleAssetValue value)>? fungibleAssetValues,
IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>? fungibleIdAndCounts,
string? memo)> UnloadData { get; }
```

### Property Value

IReadOnlyList<(Address recipientAvatarAddress, IOrderedEnumerable<(Address balanceAddress, FungibleAssetValue value)> fungibleAssetValues, IOrdered Enumerable<(HashDigest<SHA256> fungibleId, int count)> fungibleIdAndCounts , string memo)>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class DeliverToOthersGarages

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
[ActionType("deliver_to_others_garages")]
public class DeliverToOthersGarages : GameAction, IDeliverToOthersGaragesV1, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← DeliverToOthersGarages

**Implements**

IDeliverToOthersGaragesV1, IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### DeliverToOthersGarages()

```
public DeliverToOthersGarages()
```

### DeliverToOthersGarages(Address, IEnumerable<FungibleAssetValue>?, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?, string?)

```
public DeliverToOthersGarages(Address recipientAgentAddr,
IEnumerable<FungibleAssetValue>? fungibleAssetValues,
IEnumerable<(HashDigest<SHA256> fungibleId, int count)>? fungibleIdAndCounts,
string? memo)
```

### Parameters

`recipientAgentAddr` Address

`fungibleAssetValues` IEnumerable<FungibleAssetValue>

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

`memo` string

## Properties

### FungibleAssetValues

```
public IOrderedEnumerable<FungibleAssetValue>? FungibleAssetValues { get; }
```

#### Property Value

IOrderedEnumerable<FungibleAssetValue>

### FungibleIdAndCounts

```
public IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
FungibleIdAndCounts { get; }
```

#### Property Value

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

### Memo

```
public string? Memo { get; }
```

### Property Value

string

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## RecipientAgentAddr

```
public Address RecipientAgentAddr { get; }
```

### Property Value

Address

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class GarageUtils

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
public static class GarageUtils
```

**Inheritance**

object ← GarageUtils

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### ConvertToFungibleItemGarage(IValue?, Address, HashDigest<SHA256>)

```
public static FungibleItemGarage ConvertToFungibleItemGarage(IValue? garageState,
Address garageAddr, HashDigest<SHA256> fungibleId)
```

#### Parameters

`garageState` IValue

`garageAddr` Address

`fungibleId` HashDigest<SHA256>

#### Returns

FungibleItemGarage

# MergeAndSort(IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?)

```
public static IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int
count)>? MergeAndSort(IEnumerable<(HashDigest<SHA256> fungibleId, int
count)>? fungibleIdAndCounts)
```

## Parameters

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

## Returns

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

# MergeAndSort(IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?)

```
public static IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue
value)>? MergeAndSort(IEnumerable<(Address balanceAddr, FungibleAssetValue
value)>? fungibleAssetValues)
```

## Parameters

`fungibleAssetValues` IEnumerable<(Address balanceAddr, FungibleAssetValue value)>

## Returns

IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>

# WithGarageStateTuples(Address, IWorldState, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>)

```
public static IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count,
Address garageAddr, IValue? garageState)> WithGarageStateTuples(Address
agentAddr, IWorldState states, IEnumerable<(HashDigest<SHA256> fungibleId, int
count)> fungibleIdAndCounts)
```

### Parameters

`agentAddr` Address

`states` IWorldState

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

### Returns

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count, Address garageAddr, IValue garageState)>

## WithGarageStateTuples(Address, Address, IWorldState, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>)

```
public static IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count, Address
senderGarageAddr, FungibleItemGarage senderGarage, Address recipientGarageAddr,
IValue? recipientGarageState)> WithGarageStateTuples(Address senderAgentAddr,
Address recipientAgentAddr, IWorldState states, IEnumerable<(HashDigest<SHA256>
fungibleId, int count)> fungibleIdAndCounts)
```

### Parameters

`senderAgentAddr` Address

`recipientAgentAddr` Address

`states` IWorldState

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

## Returns

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count, Address senderGarageAddr, FungibleItemGarage senderGarage, Address recipientGarageAddr, IValue recipientGarageState)>

# WithGarageTuples(Address, IWorldState, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>)

```
public static IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count,
Address garageAddr, FungibleItemGarage garage)> WithGarageTuples(Address
agentAddr, IWorldState states, IEnumerable<(HashDigest<SHA256> fungibleId, int
count)> fungibleIdAndCounts)
```

## Parameters

`agentAddr` Address

`states` IWorldState

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

## Returns

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count, Address garageAddr, FungibleItemGarage garage)>

# Class LoadIntoMyGarages

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
[ActionType("load_into_my_garages")]
public class LoadIntoMyGarages : GameAction, ILoadIntoMyGaragesV1, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← LoadIntoMyGarages

**Implements**

ILoadIntoMyGaragesV1, IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### LoadIntoMyGarages()

```
public LoadIntoMyGarages()
```

### LoadIntoMyGarages(IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?, Address?, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?, string?)

```
public LoadIntoMyGarages(IEnumerable<(Address balanceAddr, FungibleAssetValue
value)>? fungibleAssetValues, Address? avatarAddr, IEnumerable<(HashDigest<SHA256>
fungibleId, int count)>? fungibleIdAndCounts, string? memo)
```

### Parameters

`fungibleAssetValues` IEnumerable<(Address balanceAddr, FungibleAssetValue value)>

`avatarAddr` Address?

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

`memo` string

## Properties

## AvatarAddr

This address should belong to one of the signer's avatars. If the avatar state is v1, there is no separate inventory, so it should be execute another action first to migrate the avatar state to v2. And then, the inventory address will be set.

```
public Address? AvatarAddr { get; }
```

### Property Value

Address?

## FungibleAssetValues

```
public IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>?
FungibleAssetValues { get; }
```

### Property Value

IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>

## FungibleIdAndCounts

```
public IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
FungibleIdAndCounts { get; }
```

### Property Value

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

## Memo

```
public string? Memo { get; }
```

### Property Value

string

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

### Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Class UnloadFromMyGarages

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
[ActionType("unload_from_my_garages")]
public class UnloadFromMyGarages : GameAction, IUnloadFromMyGaragesV1, IAction
```

**Inheritance**

object ← ActionBase ← GameAction ← UnloadFromMyGarages

**Implements**

IUnloadFromMyGaragesV1, IAction

**Inherited Members**

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### UnloadFromMyGarages()

```
public UnloadFromMyGarages()
```

### UnloadFromMyGarages(Address, IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?, string?)

```
public UnloadFromMyGarages(Address recipientAvatarAddr, IEnumerable<(Address
balanceAddr, FungibleAssetValue value)>? fungibleAssetValues,
IEnumerable<(HashDigest<SHA256> fungibleId, int count)>? fungibleIdAndCounts,
string? memo)
```

### Parameters

`recipientAvatarAddr` Address

`fungibleAssetValues` IEnumerable<(Address balanceAddr, FungibleAssetValue value)>

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

`memo` string

## Properties

## FungibleAssetValues

```
public IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>?
FungibleAssetValues { get; }
```

### Property Value

IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>

## FungibleIdAndCounts

```
public IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
FungibleIdAndCounts { get; }
```

### Property Value

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

## Memo

```
public string? Memo { get; }
```

### Property Value

string

## PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

### Property Value

IImmutableDictionary<string, IValue>

## RecipientAvatarAddr

If the avatar state is v1, there is no separate inventory, so it should be execute another action first to migrate the avatar state to v2. And then, the inventory address will be set.

```
public Address RecipientAvatarAddr { get; }
```

### Property Value

Address

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

## Returns

IWorld

# LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

## Parameters

`plainValue` IImmutableDictionary<string, IValue>

# Namespace Nekoyume.Action.Guild

## Classes

BanGuildMember

ClaimGuildReward

ClaimReward

ClaimUnbonded
: An action to claim unbonded assets. This action can be executed only when the unbonding period is over.

GuildConfig

JoinGuild

MakeGuild

MoveGuild

QuitGuild

RemoveGuild
: An action to remove the guild.

UnbanGuildMember

# Class BanGuildMember

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("ban_guild_member")]
public class BanGuildMember : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← BanGuildMember

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BanGuildMember()

```
public BanGuildMember()
```

### BanGuildMember(AgentAddress)

```
public BanGuildMember(AgentAddress target)
```

#### Parameters

`target` AgentAddress

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "ban_guild_member"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Target

```
public AgentAddress Target { get; }
```

### Property Value

AgentAddress

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class ClaimGuildReward

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("claim_guild_reward")]
public sealed class ClaimGuildReward : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← ClaimGuildReward

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ClaimGuildReward()

```
public ClaimGuildReward()
```

## Fields

### TypeIdentifier

```
public const string TypeIdentifier = "claim_guild_reward"
```

#### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class ClaimReward

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("claim_reward")]
public sealed class ClaimReward : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← ClaimReward

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ClaimReward()

```
public ClaimReward()
```

## Fields

### TypeIdentifier

```
public const string TypeIdentifier = "claim_reward"
```

#### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class ClaimUnbonded

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

An action to claim unbonded assets. This action can be executed only when the unbonding period is over.

```
[ActionType("claim_unbonded")]
public sealed class ClaimUnbonded : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← ClaimUnbonded

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ClaimUnbonded()

```
public ClaimUnbonded()
```

## Fields

### TypeIdentifier

```
public const string TypeIdentifier = "claim_unbonded"
```

#### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class GuildConfig

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
public static class GuildConfig
```

**Inheritance**

object ← GuildConfig

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## PlanetariumGuildOwner

```
public static readonly AgentAddress PlanetariumGuildOwner
```

### Field Value

AgentAddress

# Class JoinGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("join_guild")]
public class JoinGuild : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← JoinGuild

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### JoinGuild()

```
public JoinGuild()
```

### JoinGuild(GuildAddress)

```
public JoinGuild(GuildAddress guildAddress)
```

#### Parameters

`guildAddress` GuildAddress

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "join_guild"
```

### Field Value

string

# Properties

## GuildAddress

```
public GuildAddress GuildAddress { get; }
```

### Property Value

GuildAddress

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class MakeGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("make_guild")]
public class MakeGuild : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← MakeGuild

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MakeGuild()

```
public MakeGuild()
```

### MakeGuild(Address)

```
public MakeGuild(Address validatorAddress)
```

#### Parameters

`validatorAddress` Address

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "make_guild"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## ValidatorAddress

```
public Address ValidatorAddress { get; }
```

### Property Value

Address

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class MoveGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("move_guild")]
public class MoveGuild : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← MoveGuild

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MoveGuild()

```
public MoveGuild()
```

### MoveGuild(GuildAddress)

```
public MoveGuild(GuildAddress guildAddress)
```

#### Parameters

`guildAddress` GuildAddress

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "move_guild"
```

### Field Value

string

# Properties

## GuildAddress

```
public GuildAddress GuildAddress { get; }
```

### Property Value

GuildAddress

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class QuitGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("quit_guild")]
public class QuitGuild : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← QuitGuild

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "quit_guild"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class RemoveGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

An action to remove the guild.

```
[ActionType("remove_guild")]
public class RemoveGuild : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← RemoveGuild

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "remove_guild"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class UnbanGuildMember

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("unban_guild_member")]
public class UnbanGuildMember : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← UnbanGuildMember

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## UnbanGuildMember()

```
public UnbanGuildMember()
```

## UnbanGuildMember(Address)

```
public UnbanGuildMember(Address target)
```

### Parameters

`target` Address

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "unban_guild_member"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Target

```
public Address Target { get; }
```

### Property Value

Address

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Namespace Nekoyume.Action.Guild.Migration

## Classes

MigrateDelegationHeight

An action to migrate the delegation height. This action triggers the migration of the delegation height.

MigratePlanetariumValidator

An action to migrate the planetarium validator. With this migration, planetarium validator is now active, and bonded FAVs are moved to active address.

MigrateStakeAndJoinGuild

An action to migrate guild delegation and join guild. After migration is done, guild participant now have delegation with validator.

# Class MigrateDelegationHeight

Namespace: Nekoyume.Action.Guild.Migration

Assembly: Lib9c.dll

An action to migrate the delegation height. This action triggers the migration of the delegation height.

```
[ActionType("migrate_delegation_height")]
public class MigrateDelegationHeight : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← MigrateDelegationHeight

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MigrateDelegationHeight()

```
public MigrateDelegationHeight()
```

### MigrateDelegationHeight(long)

```
public MigrateDelegationHeight(long height)
```

#### Parameters

`height` long

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "migrate_delegation_height"
```

### Field Value

string

# Properties

## Height

```
public long Height { get; }
```

### Property Value

long

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class MigratePlanetariumValidator

Namespace: Nekoyume.Action.Guild.Migration

Assembly: Lib9c.dll

An action to migrate the planetarium validator. With this migration, planetarium validator is now active, and bonded FAVs are moved to active address.

```
[ActionType("migrate_planetarium_validator")]
public class MigratePlanetariumValidator : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← MigratePlanetariumValidator

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## MigratePlanetariumValidator()

```
public MigratePlanetariumValidator()
```

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "migrate_planetarium_validator"
```

### Field Value

string

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

### LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class MigrateStakeAndJoinGuild

Namespace: Nekoyume.Action.Guild.Migration

Assembly: Lib9c.dll

An action to migrate guild delegation and join guild. After migration is done, guild participant now have delegation with validator.

```
[ActionType("migrate_stake_and_join_guild")]
public class MigrateStakeAndJoinGuild : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← MigrateStakeAndJoinGuild

**Implements**

IAction

**Inherited Members**

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MigrateStakeAndJoinGuild()

Initializes a new instance of the MigrateStakeAndJoinGuild class.

```
public MigrateStakeAndJoinGuild()
```

### MigrateStakeAndJoinGuild(AgentAddress)

Initializes a new instance of the MigrateStakeAndJoinGuild class.

```
public MigrateStakeAndJoinGuild(AgentAddress target)
```

### Parameters

`target` AgentAddress

## Fields

### TypeIdentifier

The type identifier for this action.

```
public const string TypeIdentifier = "migrate_stake_and_join_guild"
```

### Field Value

string

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

### Target

The target address to migrate.

```
public AgentAddress Target { get; }
```

### Property Value

AgentAddress

## Methods

### Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

#### Parameters

`context` IActionContext

#### Returns

IWorld

### LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

#### Parameters

`plainValue` IValue

# Namespace Nekoyume.Action.Guild.Migration.LegacyModels

## Classes

LegacyGuildParticipant
: The legacy model for GuildParticipant.

MigrationModule
: The module for delegation height for migration.

# Class LegacyGuildParticipant

Namespace: Nekoyume.Action.Guild.Migration.LegacyModels

Assembly: Lib9c.dll

The legacy model for GuildParticipant.

```
public class LegacyGuildParticipant : IBencodable,
IEquatable<LegacyGuildParticipant>
```

**Inheritance**

object ← LegacyGuildParticipant

**Implements**

IBencodable, IEquatable<LegacyGuildParticipant>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## LegacyGuildParticipant(List)

Constructor of LegacyGuildParticipant.

```
public LegacyGuildParticipant(List list)
```

### Parameters

`list` List

The serialized data.

### Exceptions

InvalidCastException

Throws when the deserialization failed.

FailedLoadStateException

Throws when the state is un-deserializable.

## LegacyGuildParticipant(GuildAddress)

Constructor of LegacyGuildParticipant.

```
public LegacyGuildParticipant(GuildAddress guildAddress)
```

### Parameters

`guildAddress` GuildAddress

The guild address.

# Fields

## GuildAddress

The guild address.

```
public readonly GuildAddress GuildAddress
```

### Field Value

GuildAddress

# Properties

## Bencoded

Serialize the state.

```
public List Bencoded { get; }
```

### Property Value

List

## Methods

## Equals(LegacyGuildParticipant)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(LegacyGuildParticipant other)
```

### Parameters

`other` LegacyGuildParticipant

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

# Class MigrationModule

Namespace: Nekoyume.Action.Guild.Migration.LegacyModels

Assembly: Lib9c.dll

The module for delegation height for migration.

```
public static class MigrationModule
```

**Inheritance**

object ← MigrationModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## DelegationMigrationHeight

An address for delegation height migration.

```
public static readonly Address DelegationMigrationHeight
```

### Field Value

Address

# Methods

## GetDelegationMigrationHeight(IWorldState)

```
public static long? GetDelegationMigrationHeight(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

long?

## SetDelegationMigrationHeight(IWorld, long)

```
public static IWorld SetDelegationMigrationHeight(this IWorld world, long height)
```

### Parameters

`world` IWorld

`height` long

### Returns

IWorld

# Namespace Nekoyume.Action.Loader

## Classes

NCActionLoader

An Libplanet.Action.Loader.IActionLoader implementation for Nine Chronicles's BlockChain. This is a simple wrapper around a Libplanet.Action.Loader.TypedActionLoader for loading all Libplanet.Action.IAction classes within the same assembly as the Action Base class, inheriting from the ActionBase class, and has an Libplanet.Action.ActionType Attribute from Bencodex.Types.IValues

# Class NCActionLoader

Namespace: Nekoyume.Action.Loader

Assembly: Lib9c.dll

An Libplanet.Action.Loader.IActionLoader implementation for Nine Chronicles's BlockChain. This is a simple wrapper around a Libplanet.Action.Loader.TypedActionLoader for loading all Libplanet.Action.IAction classes within the same assembly as the ActionBase class, inheriting from the ActionBase class, and has an Libplanet.Action.ActionTypeAttribute from Bencodex.Types.IValues

```
public class NCActionLoader : IActionLoader
```

**Inheritance**

object ← NCActionLoader

**Implements**

IActionLoader

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### NCActionLoader()

```
public NCActionLoader()
```

## Methods

### LoadAction(long, IValue)

```
public IAction LoadAction(long index, IValue value)
```

## Parameters

`index` long

`value` IValue

## Returns

IAction

# See Also

TypedActionLoader

# Namespace Nekoyume.Action.Validator Delegation

## Classes

AllocateGuildReward

AllocateReward

ClaimValidatorRewardSelf

DelegateValidator

Mortgage
An action for mortgage gas fee for a transaction. Should be executed at the beginning of the tx.

PromoteValidator

RecordProposer
An action for recording proposer of the block to use in next block's reward distribution.

Refund
An action for refund gas fee for a transaction. Should be executed at the beginning of the tx.

Reward
An action for reward for a transaction. Should be executed at the beginning of the tx.

SetValidatorCommission
Set the commission percentage of the validator.

SlashValidator

UndelegateValidator

UnjailValidator

UpdateValidators

# Class AllocateGuildReward

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class AllocateGuildReward : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← AllocateGuildReward

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AllocateGuildReward()

```
public AllocateGuildReward()
```

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

#### Property Value

IValue

## Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class AllocateReward

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class AllocateReward : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← AllocateReward

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AllocateReward()

```
public AllocateReward()
```

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

#### Property Value

IValue

## Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class ClaimValidatorRewardSelf

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("claim_validator_reward_self")]
public sealed class ClaimValidatorRewardSelf : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← ClaimValidatorRewardSelf

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ClaimValidatorRewardSelf()

```
public ClaimValidatorRewardSelf()
```

## Fields

### TypeIdentifier

```
public const string TypeIdentifier = "claim_validator_reward_self"
```

#### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## ValidatorDelegatee

```
public Address ValidatorDelegatee { get; }
```

### Property Value

Address

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class DelegateValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("delegate_validator")]
public sealed class DelegateValidator : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← DelegateValidator

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## DelegateValidator()

```
public DelegateValidator()
```

## DelegateValidator(FungibleAssetValue)

```
public DelegateValidator(FungibleAssetValue fav)
```

### Parameters

`fav` FungibleAssetValue

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "delegate_validator"
```

### Field Value

string

# Properties

## FAV

```
public FungibleAssetValue FAV { get; }
```

### Property Value

FungibleAssetValue

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class Mortgage

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

An action for mortgage gas fee for a transaction. Should be executed at the beginning of the tx.

```
public sealed class Mortgage : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← Mortgage

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Mortgage()

Creates a new instance of Mortgage.

```
public Mortgage()
```

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

#### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class PromoteValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("promote_validator")]
public sealed class PromoteValidator : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← PromoteValidator

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PromoteValidator()

```
public PromoteValidator()
```

### PromoteValidator(PublicKey, FungibleAssetValue)

```
public PromoteValidator(PublicKey publicKey, FungibleAssetValue fav)
```

#### Parameters

`publicKey` PublicKey

`fav` FungibleAssetValue

# PromoteValidator(PublicKey, FungibleAssetValue, BigInteger)

```
public PromoteValidator(PublicKey publicKey, FungibleAssetValue fav,
BigInteger commissionPercentage)
```

## Parameters

`publicKey` PublicKey

`fav` FungibleAssetValue

`commissionPercentage` BigInteger

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "promote_validator"
```

### Field Value

string

# Properties

## CommissionPercentage

```
public BigInteger CommissionPercentage { get; }
```

### Property Value

BigInteger

## FAV

```
public FungibleAssetValue FAV { get; }
```

### Property Value

FungibleAssetValue

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## PublicKey

```
public PublicKey PublicKey { get; }
```

### Property Value

PublicKey

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## ExecutePublic(IActionContext)

```
public IWorld ExecutePublic(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class RecordProposer

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

An action for recording proposer of the block to use in next block's reward distribution.

```
public sealed class RecordProposer : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← RecordProposer

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RecordProposer()

Creates a new instance of RecordProposer.

```
public RecordProposer()
```

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

#### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class Refund

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

An action for refund gas fee for a transaction. Should be executed at the beginning of the tx.

```
public sealed class Refund : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← Refund

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Refund()

Creates a new instance of Refund.

```
public Refund()
```

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

#### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class Reward

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

An action for reward for a transaction. Should be executed at the beginning of the tx.

```
public sealed class Reward : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← Reward

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Reward()

Creates a new instance of Refund.

```
public Reward()
```

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

#### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class SetValidatorCommission

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

Set the commission percentage of the validator.

```
[ActionType("set_validator_commission")]
public sealed class SetValidatorCommission : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← SetValidatorCommission

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SetValidatorCommission()

```
public SetValidatorCommission()
```

## SetValidatorCommission(BigInteger)

```
public SetValidatorCommission(BigInteger commissionPercentage)
```

### Parameters

`commissionPercentage` BigInteger

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "set_validator_commission"
```

### Field Value

string

# Properties

## CommissionPercentage

```
public BigInteger CommissionPercentage { get; }
```

### Property Value

BigInteger

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class SlashValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class SlashValidator : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← SlashValidator

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SlashValidator()

```
public SlashValidator()
```

## Properties

### AbstainJailTime

```
public static long AbstainJailTime { get; }
```

#### Property Value

long

### DuplicateVoteSlashFactor

```
public static BigInteger DuplicateVoteSlashFactor { get; }
```

### Property Value

BigInteger

## LivenessSlashFactor

```
public static BigInteger LivenessSlashFactor { get; }
```

### Property Value

BigInteger

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

# LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

## Parameters

`plainValue` IValue

# Class UndelegateValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("undelegate_validator")]
public sealed class UndelegateValidator : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← UndelegateValidator

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## UndelegateValidator()

```
public UndelegateValidator()
```

## UndelegateValidator(BigInteger)

```
public UndelegateValidator(BigInteger share)
```

### Parameters

`share` BigInteger

# Fields

## TypeIdentifier

```
public const string TypeIdentifier = "undelegate_validator"
```

### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

## Share

```
public BigInteger Share { get; }
```

### Property Value

BigInteger

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class UnjailValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("unjail_validator")]
public sealed class UnjailValidator : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← UnjailValidator

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### UnjailValidator()

```
public UnjailValidator()
```

## Fields

### TypeIdentifier

```
public const string TypeIdentifier = "unjail_validator"
```

#### Field Value

string

# Properties

## PlainValue

```
public override IValue PlainValue { get; }
```

### Property Value

IValue

# Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Class UpdateValidators

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class UpdateValidators : ActionBase, IAction
```

**Inheritance**

object ← ActionBase ← UpdateValidators

**Implements**

IAction

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### UpdateValidators()

```
public UpdateValidators()
```

## Properties

### PlainValue

```
public override IValue PlainValue { get; }
```

#### Property Value

IValue

## Methods

## Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

### Parameters

`context` IActionContext

### Returns

IWorld

## LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

### Parameters

`plainValue` IValue

# Namespace Nekoyume.Arena

## Classes

ArenaHelper

There are only things that don't change

ArenaSimulator

Changed at https://github.com/planetarium/lib9c/pull/2229

ArenaSimulator2

This class is only for previewnet.

## Interfaces

IArenaSimulator

# Class ArenaHelper

Namespace: Nekoyume.Arena

Assembly: Lib9c.dll

There are only things that don't change

```
public static class ArenaHelper
```

**Inheritance**

object ← ArenaHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ScoreLimits

```
public static readonly IReadOnlyDictionary<ArenaType, (int upper, int
lower)> ScoreLimits
```

### Field Value

IReadOnlyDictionary<ArenaType, (int, int)>

## ScoreLimitsV1

```
[Obsolete("Use `ScoreLimits` instead.")]
public static readonly IReadOnlyDictionary<ArenaType, (int, int)> ScoreLimitsV1
```

### Field Value

IReadOnlyDictionary<ArenaType, (int, int)>

## ScoreLimitsV2

```
[Obsolete("Use `ScoreLimits` instead.")]
public static readonly IReadOnlyDictionary<ArenaType, (int upper, int
lower)> ScoreLimitsV2
```

### Field Value

IReadOnlyDictionary<ArenaType, (int, int)>

## ScoreLimitsV3

```
[Obsolete("Use `ScoreLimits` instead.")]
public static readonly IReadOnlyDictionary<ArenaType, (int upper, int
lower)> ScoreLimitsV3
```

### Field Value

IReadOnlyDictionary<ArenaType, (int, int)>

# Methods

## DeriveArenaAddress(int, int)

```
public static Address DeriveArenaAddress(int championshipId, int round)
```

### Parameters

`championshipId` int

`round` int

### Returns

Address

## GetCurrentTicketResetCount(long, long, int)

```
public static int GetCurrentTicketResetCount(long currentBlockIndex, long
roundStartBlockIndex, int interval)
```

### Parameters

`currentBlockIndex` long

`roundStartBlockIndex` long

`interval` int

### Returns

int

## GetEntranceFee(RoundData, long, int)

```
public static FungibleAssetValue GetEntranceFee(ArenaSheet.RoundData roundData, long
currentBlockIndex, int avatarLevel)
```

### Parameters

`roundData` ArenaSheet.RoundData

`currentBlockIndex` long

`avatarLevel` int

### Returns

FungibleAssetValue

## GetMaxPurchasedTicketCount(RoundData)

```
[Obsolete("not use since v100320, battle_arena6. Use
`ArenaSheet.RoundData.MaxPurchaseCount`")]
public static long GetMaxPurchasedTicketCount(ArenaSheet.RoundData roundData)
```

### Parameters

`roundData` ArenaSheet.RoundData

### Returns

long

## GetMedalTotalCount(Row, AvatarState)

```
public static int GetMedalTotalCount(ArenaSheet.Row row, AvatarState avatarState)
```

### Parameters

`row` ArenaSheet.Row

`avatarState` AvatarState

### Returns

int

## GetRewardCount(int)

```
public static int GetRewardCount(int score)
```

### Parameters

`score` int

### Returns

int

## GetScores(int, int)

```
public static (int myWinScore, int myDefeatScore, int enemyDefeatScore)
GetScores(int myScore, int enemyScore)
```

### Parameters

`myScore` int

`enemyScore` int

### Returns

(int myWinScore, int myDefeatScore, int enemyDefeatScore)

## GetScoresV1(int, int)

```
[Obsolete("Use `GetScores()` instead.")]
public static (int myWinScore, int myDefeatScore, int enemyDefeatScore)
GetScoresV1(int myScore, int enemyScore)
```

### Parameters

`myScore` int

`enemyScore` int

### Returns

(int myWinScore, int myDefeatScore, int enemyDefeatScore)

## GetTicketPrice(RoundData, ArenaInformation, Currency)

```
public static FungibleAssetValue GetTicketPrice(ArenaSheet.RoundData roundData,
ArenaInformation arenaInformation, Currency currency)
```

### Parameters

`roundData` ArenaSheet.RoundData

`arenaInformation` ArenaInformation

`currency` Currency

### Returns

FungibleAssetValue

## ValidateScoreDifference(IReadOnlyDictionary<ArenaType, (int, int)>, ArenaType, int, int)

```
public static bool ValidateScoreDifference(IReadOnlyDictionary<ArenaType, (int,
int)> scoreLimits, ArenaType arenaType, int myScore, int enemyScore)
```

### Parameters

`scoreLimits` IReadOnlyDictionary<ArenaType, (int, int)>

`arenaType` ArenaType

`myScore` int

`enemyScore` int

### Returns

bool

## ValidateScoreDifferenceV1(IReadOnlyDictionary<ArenaType, (int, int)>, ArenaType, int, int)

```
[Obsolete("Use `ValidateScoreDifference()` instead.")]
public static bool ValidateScoreDifferenceV1(IReadOnlyDictionary<ArenaType, (int,
int)> scoreLimits, ArenaType arenaType, int myScore, int enemyScore)
```

### Parameters

`scoreLimits` IReadOnlyDictionary<ArenaType, (int, int)>

`arenaType` ArenaType

`myScore` int

`enemyScore` int

### Returns

bool

## ValidateScoreDifferenceV2(IReadOnlyDictionary<ArenaType, (int, int)>, ArenaType, int, int)

```
[Obsolete("Use `ValidateScoreDifference()` instead.")]
public static bool ValidateScoreDifferenceV2(IReadOnlyDictionary<ArenaType, (int, int)> scoreLimits, ArenaType arenaType, int myScore, int enemyScore)
```

### Parameters

`scoreLimits` IReadOnlyDictionary<ArenaType, (int, int)>

`arenaType` ArenaType

`myScore` int

`enemyScore` int

### Returns

bool

# Class ArenaSimulator

Namespace: Nekoyume.Arena

Assembly: Lib9c.dll

Changed at https://github.com/planetarium/lib9c/pull/2229

```
public class ArenaSimulator : IArenaSimulator
```

**Inheritance**

object ← ArenaSimulator

**Implements**

IArenaSimulator

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaSimulator(IRandom, int, long)

```
public ArenaSimulator(IRandom random, int hpModifier = 2, long
shatterStrikeMaxDamage = 400000)
```

#### Parameters

`random` IRandom

`hpModifier` int

`shatterStrikeMaxDamage` long

## Properties

## BuffLimitSheet

```
public BuffLimitSheet BuffLimitSheet { get; }
```

### Property Value

BuffLimitSheet

## BuffLinkSheet

```
public BuffLinkSheet BuffLinkSheet { get; set; }
```

### Property Value

BuffLinkSheet

## HpModifier

```
public int HpModifier { get; }
```

### Property Value

int

## Log

```
public ArenaLog Log { get; }
```

### Property Value

ArenaLog

## Random

```
public IRandom Random { get; }
```

### Property Value

IRandom

## ShatterStrikeMaxDamage

```
public long ShatterStrikeMaxDamage { get; }
```

### Property Value

long

## Turn

```
public int Turn { get; }
```

### Property Value

int

# Methods

## Simulate(ArenaPlayerDigest, ArenaPlayerDigest, ArenaSimulatorSheets, List<StatModifier>, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, bool)

```
public ArenaLog Simulate(ArenaPlayerDigest challenger, ArenaPlayerDigest enemy,
ArenaSimulatorSheets sheets, List<StatModifier> challengerCollectionModifiers,
List<StatModifier> enemyCollectionModifiers, BuffLimitSheet buffLimitSheet,
BuffLinkSheet buffLinkSheet, bool setExtraValueBuffBeforeGetBuffs = false)
```

### Parameters

`challenger` ArenaPlayerDigest

`enemy` ArenaPlayerDigest

`sheets` ArenaSimulatorSheets

`challengerCollectionModifiers` List<StatModifier>

`enemyCollectionModifiers` List<StatModifier>

`buffLimitSheet` BuffLimitSheet

`buffLinkSheet` BuffLinkSheet

`setExtraValueBuffBeforeGetBuffs` bool

## Returns

ArenaLog

# Class ArenaSimulator2

Namespace: Nekoyume.Arena

Assembly: Lib9c.dll

This class is only for previewnet.

```
public class ArenaSimulator2 : Simulator, ISimulator
```

**Inheritance**

object ← Simulator ← ArenaSimulator2

**Implements**

ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaSimulator2(IRandom, ArenaPlayerDigest, ArenaPlayerDigest, ArenaSimulatorSheetsV1)

```
public ArenaSimulator2(IRandom random, ArenaPlayerDigest myDigest, ArenaPlayerDigest
enemyDigest, ArenaSimulatorSheetsV1 simulatorSheets)
```

### Parameters

`random` IRandom

`myDigest` ArenaPlayerDigest

`enemyDigest` ArenaPlayerDigest

`simulatorSheets` ArenaSimulatorSheetsV1

## Properties

### Reward

do not use it

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

## Methods

### Simulate()

```
public Player Simulate()
```

### Returns

Player

# Interface IArenaSimulator

Namespace: Nekoyume.Arena

Assembly: Lib9c.dll

```
public interface IArenaSimulator
```

## Properties

### BuffLimitSheet

```
BuffLimitSheet BuffLimitSheet { get; }
```

#### Property Value

BuffLimitSheet

### BuffLinkSheet

```
BuffLinkSheet BuffLinkSheet { get; set; }
```

#### Property Value

BuffLinkSheet

### Log

```
ArenaLog Log { get; }
```

#### Property Value

ArenaLog

## Random

```
IRandom Random { get; }
```

### Property Value

IRandom

## ShatterStrikeMaxDamage

```
long ShatterStrikeMaxDamage { get; }
```

### Property Value

long

## Turn

```
int Turn { get; }
```

### Property Value

int

# Namespace Nekoyume.Battle

## Classes

ArenaScoreHelper

AttackCountHelper

CPHelper

CriticalHelper

DamageHelper

HitHelper

InfiniteTowerLog
: Log for infinite tower battle results. Contains information about the battle outcome and player performance.

InfiniteTowerSimulator
: Simulator for infinite tower battles with conditions and restrictions. Handles turn-based combat with wave progression, special conditions, and reward processing.

InfiniteTowerWave
: Represents a wave in the infinite tower. Contains information about the enemies that will spawn in this wave.

InvalidCountException

ListEmptyException

RaidSimulator

RaidSimulatorV1

RaidSimulatorV2

RankingSimulator

RankingSimulatorV1

RewardSelector

Simulator

StageRewardExpHelper

StageSimulator
: Simulator for stage battles that handles wave-based combat.

StageSimulatorV1

StageSimulatorV2

StageSimulatorV3

TestSimulator

Wave
: Represents a wave of enemies in a battle.

WaveResult
: Result of a single wave battle. Contains information about the wave completion status and battle outcome.

WeightedSelector<T>

## Structs

EnemyPlayerDigest

## Interfaces

ISimulator

IStageSimulator

## Enums

InfiniteTowerLog.InfiniteTowerResult
: Represents the possible results of an infinite tower battle.

# Class ArenaScoreHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class ArenaScoreHelper
```

**Inheritance**

object ← ArenaScoreHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## DifferLowerLimit

```
[Obsolete("Only to used for V1")]
public const int DifferLowerLimit = -1000
```

### Field Value

int

## DifferUpperLimit

```
[Obsolete("Only to used for V1")]
public const int DifferUpperLimit = 1000
```

### Field Value

int

## LoseScoreMax

```
[Obsolete("Only to used for V1")]
public const int LoseScoreMax = -30
```

### Field Value

int

## LoseScoreMin

```
[Obsolete("Only to used for V1")]
public const int LoseScoreMin = -5
```

### Field Value

int

## WinScoreMax

```
[Obsolete("Only to used for V1")]
public const int WinScoreMax = 60
```

### Field Value

int

## WinScoreMin

```
[Obsolete("Only to used for V1")]
public const int WinScoreMin = 1
```

### Field Value

int

# Methods

## GetScore(int, int, Result)

```
public static (int challengerScoreDelta, int defenderScoreDelta) GetScore(int
challengerRating, int defenderRating, BattleLog.Result result)
```

### Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

### Returns

(int challengerScoreDelta, int defenderScoreDelta)

## GetScoreV1(int, int, Result)

```
[Obsolete("Use GetScore()")]
public static int GetScoreV1(int challengerRating, int defenderRating,
BattleLog.Result result)
```

### Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

### Returns

int

# GetScoreV2(int, int, Result)

```
[Obsolete("Use GetScore()")]
public static int GetScoreV2(int challengerRating, int defenderRating,
BattleLog.Result result)
```

## Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

## Returns

int

# GetScoreV3(int, int, Result)

```
[Obsolete("Use GetScore()")]
public static (int challengerScore, int defenderScore) GetScoreV3(int
challengerRating, int defenderRating, BattleLog.Result result)
```

## Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

## Returns

(int challengerScoreDelta, int defenderScoreDelta)

# GetScoreV4(int, int, Result)

```
[Obsolete("Use GetScore()")]
public static (int challengerScoreDelta, int defenderScoreDelta) GetScoreV4(int
challengerRating, int defenderRating, BattleLog.Result result)
```

## Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

## Returns

(int challengerScoreDelta, int defenderScoreDelta)

# Class AttackCountHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class AttackCountHelper
```

**Inheritance**

object ← AttackCountHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## CountMaxLowerLimit

```
public const int CountMaxLowerLimit = 2
```

### Field Value

int

## CountMaxUpperLimit

```
public const int CountMaxUpperLimit = 5
```

### Field Value

int

# Methods

## GetAdditionalCriticalChance(int, int)

```
public static int GetAdditionalCriticalChance(int attackCount, int attackCountMax)
```

### Parameters

`attackCount` int

`attackCountMax` int

### Returns

int

## GetCountMax(int)

```
public static int GetCountMax(int level)
```

### Parameters

`level` int

### Returns

int

## GetDamageMultiplier(int, int)

```
public static int GetDamageMultiplier(int attackCount, int attackCountMax)
```

### Parameters

`attackCount` int

`attackCountMax` int

### Returns

int

# Class CPHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class CPHelper
```

**Inheritance**

object ← CPHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## ConvertCpToStat(StatType, decimal, int)

```
public static decimal ConvertCpToStat(StatType statType, decimal cp,
int characterLevel)
```

### Parameters

`statType` StatType

`cp` decimal

`characterLevel` int

### Returns

decimal

## DecimalToLong(decimal)

```
public static long DecimalToLong(decimal value)
```

### Parameters

`value` decimal

### Returns

long

## GetCP(Costume, CostumeStatSheet)

```
public static long GetCP(Costume costume, CostumeStatSheet sheet)
```

### Parameters

`costume` Costume

`sheet` CostumeStatSheet

### Returns

long

## GetCP(INonFungibleItem, CostumeStatSheet)

```
[Obsolete("Use GetCp")]
public static long GetCP(INonFungibleItem tradableItem, CostumeStatSheet sheet)
```

### Parameters

`tradableItem` INonFungibleItem

`sheet` CostumeStatSheet

### Returns

long

## GetCP(ItemUsable)

```
public static long GetCP(ItemUsable itemUsable)
```

### Parameters

`itemUsable` ItemUsable

### Returns

long

## GetCP(AvatarState, CharacterSheet)

```
[Obsolete("Use TotalCP")]
public static long GetCP(AvatarState avatarState, CharacterSheet characterSheet)
```

### Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

### Returns

long

## GetCPOfATK(decimal)

```
public static decimal GetCPOfATK(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetCPOfArmorPenetration(decimal)

```
public static decimal GetCPOfArmorPenetration(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetCPOfCDMG(decimal, int)

```
public static decimal GetCPOfCDMG(decimal value, int characterLevel)
```

### Parameters

`value` decimal

`characterLevel` int

### Returns

decimal

## GetCPOfCRI(decimal, int)

```
public static decimal GetCPOfCRI(decimal value, int characterLevel)
```

### Parameters

`value` decimal

`characterLevel` int

### Returns

decimal

## GetCPOfDEF(decimal)

```
public static decimal GetCPOfDEF(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetCPOfDRR(decimal, int)

```
public static decimal GetCPOfDRR(decimal value, int characterLevel)
```

### Parameters

`value` decimal

`characterLevel` int

### Returns

decimal

## GetCPOfDRV(decimal)

```
public static decimal GetCPOfDRV(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetCPOfHIT(decimal)

```
public static decimal GetCPOfHIT(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetCPOfHP(decimal)

```
public static decimal GetCPOfHP(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetCPOfSPD(decimal)

```
public static decimal GetCPOfSPD(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetCPOfThorn(decimal)

```
public static decimal GetCPOfThorn(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetCPV2(AvatarState, CharacterSheet, CostumeStatSheet)

```
[Obsolete("Use TotalCP")]
public static long GetCPV2(AvatarState avatarState, CharacterSheet characterSheet,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

### Returns

long

## GetSkillsMultiplier(int)

```
public static decimal GetSkillsMultiplier(int skillsCount)
```

### Parameters

`skillsCount` int

### Returns

decimal

## GetStatCP(StatType, decimal, int)

```
public static decimal GetStatCP(StatType statType, decimal statValue, int
characterLevel = 1)
```

### Parameters

`statType` StatType

`statValue` decimal

`characterLevel` int

### Returns

decimal

## GetStatsCP(IStats, int)

```
public static decimal GetStatsCP(IStats stats, int characterLevel = 1)
```

### Parameters

`stats` IStats

`characterLevel` int

### Returns

decimal

## TotalCP(IReadOnlyCollection<Equipment>, IReadOnlyCollection<Costume>, IReadOnlyCollection<RuneOptionInfo>, int, Row, CostumeStatSheet, List<StatModifier>, int)

```
public static long TotalCP(IReadOnlyCollection<Equipment> equipments,
IReadOnlyCollection<Costume> costumes,
IReadOnlyCollection<RuneOptionSheet.Row.RuneOptionInfo> runeOptions, int level,
CharacterSheet.Row row, CostumeStatSheet costumeStatSheet, List<StatModifier>
collectionStatModifiers, int runeLevelBonus)
```

### Parameters

`equipments` IReadOnlyCollection<Equipment>

`costumes` IReadOnlyCollection<Costume>

`runeOptions` IReadOnlyCollection<RuneOptionSheet.Row.RuneOptionInfo>

`level` int

`row` CharacterSheet.Row

`costumeStatSheet` CostumeStatSheet

`collectionStatModifiers` List<StatModifier>

`runeLevelBonus` int

### Returns

long

# Class CriticalHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class CriticalHelper
```

**Inheritance**

object ← CriticalHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetCriticalDamage(CharacterBase, long)

```
public static long GetCriticalDamage(CharacterBase caster, long originalDamage)
```

#### Parameters

`caster` CharacterBase

`originalDamage` long

#### Returns

long

### GetCriticalDamageForArena(ArenaCharacter, long)

```
public static long GetCriticalDamageForArena(ArenaCharacter caster,
long originalDamage)
```

## Parameters

`caster` ArenaCharacter

`originalDamage` long

## Returns

long

# Class DamageHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class DamageHelper
```

**Inheritance**

object ← DamageHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetDamageReductionRate(long)

calculate damage reduction rate.

```
public static decimal GetDamageReductionRate(long drr)
```

### Parameters

`drr` long

enemy DRR stats

### Returns

decimal

damage reduction rate.

## GetFinalDefense(long, long)

Calculates final defense after applying armor penetration. Uses Math.Clamp to ensure the result is between 0 and long.MaxValue, preventing overflow issues when dealing with large defense values.

```
public static long GetFinalDefense(long targetDefense, long armorPenetration)
```

### Parameters

`targetDefense` long

enemy DEF stats

`armorPenetration` long

caster ArmorPenetration stats

### Returns

long

calculated final defense.

## GetReducedDamage(long, long, long)

Calculates reduced damage after applying damage reduction values (DRV) and damage reduction rate (DRR). Uses NumberConversionHelper.SafeDecimalToInt64 to prevent overflow when dealing with large damage values. Ensures minimum damage of 1 to prevent zero damage attacks.

```
public static long GetReducedDamage(long damage, long drv, long drr)
```

### Parameters

`damage` long

damage

`drv` long

enemy DRV stats

`drr` long

enemy DRR stats

## Returns

long

calculated damage.

# Struct EnemyPlayerDigest

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public readonly struct EnemyPlayerDigest : IState
```

**Implements**
IState

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### EnemyPlayerDigest(List)

```
public EnemyPlayerDigest(List serialized)
```

#### Parameters

`serialized` List

### EnemyPlayerDigest(AvatarState)

```
public EnemyPlayerDigest(AvatarState enemyAvatarState)
```

#### Parameters

`enemyAvatarState` AvatarState

## Fields

## CharacterId

```
public readonly int CharacterId
```

### Field Value

int

## Costumes

```
public readonly IReadOnlyList<Costume> Costumes
```

### Field Value

IReadOnlyList<Costume>

## EarIndex

```
public readonly int EarIndex
```

### Field Value

int

## Equipments

```
public readonly IReadOnlyList<Equipment> Equipments
```

### Field Value

IReadOnlyList<Equipment>

## HairIndex

```
public readonly int HairIndex
```

### Field Value

int

## LensIndex

```
public readonly int LensIndex
```

### Field Value

int

## Level

```
public readonly int Level
```

### Field Value

int

## NameWithHash

```
public readonly string NameWithHash
```

### Field Value

string

## TailIndex

```
public readonly int TailIndex
```

### Field Value

int

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class HitHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class HitHelper
```

**Inheritance**

object ← HitHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## GetHitStep1CorrectionMax

```
public const long GetHitStep1CorrectionMax = 50
```

### Field Value

long

## GetHitStep1CorrectionMin

```
public const long GetHitStep1CorrectionMin = -5
```

### Field Value

long

## GetHitStep1LevelDiffMax

```
public const long GetHitStep1LevelDiffMax = 10
```

### Field Value

long

## GetHitStep1LevelDiffMin

```
public const long GetHitStep1LevelDiffMin = -14
```

### Field Value

long

## GetHitStep2AdditionalCorrectionMax

```
public const long GetHitStep2AdditionalCorrectionMax = 50
```

### Field Value

long

## GetHitStep2AdditionalCorrectionMin

```
public const long GetHitStep2AdditionalCorrectionMin = 0
```

### Field Value

long

## GetHitStep3CorrectionMax

```
public const long GetHitStep3CorrectionMax = 90
```

### Field Value

long

## GetHitStep3CorrectionMin

```
public const long GetHitStep3CorrectionMin = 10
```

### Field Value

long

# Methods

## GetHitStep1(long, long)

```
public static long GetHitStep1(long attackerLevel, long defenderLevel)
```

### Parameters

`attackerLevel` long

`defenderLevel` long

### Returns

long

## GetHitStep2(long, long)

```
public static long GetHitStep2(long attackerHit, long defenderHit)
```

### Parameters

`attackerHit` long

`defenderHit` long

### Returns

long

## GetHitStep3(long)

```
public static long GetHitStep3(long correction)
```

### Parameters

`correction` long

### Returns

long

## GetHitStep4(long, long)

```
public static bool GetHitStep4(long lowLimitChance, long correction)
```

### Parameters

`lowLimitChance` long

`correction` long

### Returns

bool

## IsHit(long, long, long, long, long)

```
public static bool IsHit(long attackerLevel, long attackerHit, long defenderLevel,
long defenderHit, long lowLimitChance)
```

### Parameters

`attackerLevel` long

`attackerHit` long

`defenderLevel` long

`defenderHit` long

`lowLimitChance` long

### Returns

bool

## IsHitWithoutLevelCorrection(long, long, long, long, long)

```
public static bool IsHitWithoutLevelCorrection(long attackerLevel, long attackerHit,
long defenderLevel, long defenderHit, long lowLimitChance)
```

### Parameters

`attackerLevel` long

`attackerHit` long

`defenderLevel` long

`defenderHit` long

`lowLimitChance` long

### Returns

bool

# Interface ISimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public interface ISimulator
```

## Properties

### Characters

```
SimplePriorityQueue<CharacterBase, decimal> Characters { get; }
```

#### Property Value

SimplePriorityQueue<CharacterBase, decimal>

### Log

```
BattleLog Log { get; }
```

#### Property Value

BattleLog

### LogEvent

```
bool LogEvent { get; }
```

#### Property Value

bool

## Player

```
Player Player { get; }
```

### Property Value

Player

## Reward

```
IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

## WaveNumber

```
int WaveNumber { get; }
```

### Property Value

int

## WaveTurn

```
int WaveTurn { get; }
```

### Property Value

int

# Interface IStageSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public interface IStageSimulator : ISimulator
```

**Inherited Members**
ISimulator.Player , ISimulator.Log , ISimulator.Characters , ISimulator.Reward , ISimulator.WaveNumber , ISimulator.WaveTurn , ISimulator.LogEvent

## Properties

### EnemySkillSheet

```
EnemySkillSheet EnemySkillSheet { get; }
```

#### Property Value

EnemySkillSheet

### ItemMap

```
CollectionMap ItemMap { get; }
```

#### Property Value

CollectionMap

### StageId

```
int StageId { get; }
```

## Property Value

int

# Class InfiniteTowerLog

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Log for infinite tower battle results. Contains information about the battle outcome and player performance.

```
public class InfiniteTowerLog
```

**Inheritance**

object ← InfiniteTowerLog

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ClearedWaveNumber

Gets or sets the number of waves cleared.

```
public int ClearedWaveNumber { get; set; }
```

### Property Value

int

## EnemiesDefeated

Gets or sets the number of enemies defeated.

```
public int EnemiesDefeated { get; set; }
```

### Property Value

int

## IsClear

Gets or sets whether the floor was cleared.

```
public bool IsClear { get; set; }
```

### Property Value

bool

## PlayerHpRemaining

Gets or sets the player's remaining HP after battle.

```
public int PlayerHpRemaining { get; set; }
```

### Property Value

int

## Result

Gets or sets the battle result (Win/Lose).

```
public InfiniteTowerLog.InfiniteTowerResult Result { get; set; }
```

### Property Value

InfiniteTowerLog.InfiniteTowerResult

# Enum InfiniteTowerLog.InfiniteTowerResult

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Represents the possible results of an infinite tower battle.

```
public enum InfiniteTowerLog.InfiniteTowerResult
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Lose = 1`

The player lost the battle.

`Win = 0`

The player won the battle.

# Class InfiniteTowerSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Simulator for infinite tower battles with conditions and restrictions. Handles turn-based combat with wave progression, special conditions, and reward processing.

```
public class InfiniteTowerSimulator : Simulator, IStageSimulator, ISimulator
```

**Inheritance**

object ← Simulator ← InfiniteTowerSimulator

**Implements**

IStageSimulator, ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### InfiniteTowerSimulator(IRandom, AvatarState, List<Guid>, AllRuneState, RuneSlotState, int, int, Row, List<WaveData>, bool, int, SimulatorSheets,

# EnemySkillSheet, CostumeStatSheet, ItemSheet, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, List<InfiniteTowerCondition>, int, int, bool)

Initializes a new instance of the InfiniteTowerSimulator class.

```
public InfiniteTowerSimulator(IRandom random, AvatarState avatarState, List<Guid>
foods, AllRuneState runeStates, RuneSlotState runeSlotState, int infiniteTowerId,
int floorId, InfiniteTowerFloorSheet.Row floorRow,
List<InfiniteTowerFloorWaveSheet.WaveData> waveRows, bool isCleared, int exp,
SimulatorSheets simulatorSheets, EnemySkillSheet enemySkillSheet, CostumeStatSheet
costumeStatSheet, ItemSheet itemSheet, List<StatModifier> collectionModifiers,
BuffLimitSheet buffLimitSheet, BuffLinkSheet buffLinkSheet,
List<InfiniteTowerCondition> conditions, int shatterStrikeMaxDamage, int turnLimit =
150, bool logEvent = false)
```

## Parameters

`random` IRandom

Random number generator for battle calculations.

`avatarState` AvatarState

The player's avatar state.

`foods` List<Guid>

List of food items to be consumed.

`runeStates` AllRuneState

Player's rune states.

`runeSlotState` RuneSlotState

Player's rune slot configuration.

`infiniteTowerId` int

The infinite tower ID.

`floorId` int

The floor ID being challenged.

`floorRow` InfiniteTowerFloorSheet.Row

Floor data containing rewards and requirements.

`waveRows` List<InfiniteTowerFloorWaveSheet.WaveData>

List of wave data for this floor.

`isCleared` bool

Whether this floor has been cleared before.

`exp` int

Experience reward (always 0 for infinite tower).

`simulatorSheets` SimulatorSheets

Game data sheets for simulation.

`enemySkillSheet` EnemySkillSheet

Enemy skill data.

`costumeStatSheet` CostumeStatSheet

Costume stat bonuses.

`itemSheet` ItemSheet

Item data for reward creation.

`collectionModifiers` List<StatModifier>

Collection-based stat modifiers.

`buffLimitSheet` BuffLimitSheet

Buff limit data.

`buffLinkSheet` BuffLinkSheet

Buff link data.

`conditions` List<InfiniteTowerCondition>

Infinite tower conditions for this floor.

`shatterStrikeMaxDamage` int

Maximum damage for shatter strike.

`turnLimit` int

Turn limit for all waves combined (default: 200).

`logEvent` bool

Whether to log battle events (default: true).

# Properties

## CollectionModifiers

Gets the collection modifiers for stat bonuses.

```
public List<StatModifier> CollectionModifiers { get; }
```

### Property Value

List<StatModifier>

## Conditions

Gets the infinite tower conditions that apply to this floor.

```
public List<InfiniteTowerCondition> Conditions { get; }
```

### Property Value

List<InfiniteTowerCondition>

## CostumeStatSheet

Gets the costume stat sheet for costume bonuses.

```
public CostumeStatSheet CostumeStatSheet { get; }
```

### Property Value

CostumeStatSheet

## EnemySkillSheet

Gets the enemy skill sheet for enemy abilities.

```
public EnemySkillSheet EnemySkillSheet { get; }
```

### Property Value

EnemySkillSheet

## Exp

Gets the experience reward (always 0 for infinite tower).

```
public int Exp { get; }
```

### Property Value

int

## FloorId

Gets the floor ID being challenged.

```
public int FloorId { get; }
```

### Property Value

int

## FloorRow

Gets the floor data containing rewards and requirements.

```
public InfiniteTowerFloorSheet.Row FloorRow { get; }
```

### Property Value

InfiniteTowerFloorSheet.Row

## Foods

Gets the list of food items used by the player.

```
public List<Guid> Foods { get; }
```

### Property Value

List<Guid>

## FungibleAssetRewards

Gets the fungible asset rewards (NCG, Crystal, etc.) obtained from this floor.

```
public Dictionary<string, int> FungibleAssetRewards { get; }
```

### Property Value

Dictionary<string, int>

## InfiniteTowerId

Gets the infinite tower ID.

```
public int InfiniteTowerId { get; }
```

### Property Value

int

## IsCleared

Gets whether this floor has been cleared before.

```
public bool IsCleared { get; }
```

### Property Value

bool

## ItemMap

Gets the item map for collection tracking (not used in infinite tower).

```
public CollectionMap ItemMap { get; }
```

### Property Value

CollectionMap

## ItemSheet

Gets the item sheet for creating reward items.

```
public ItemSheet ItemSheet { get; }
```

### Property Value

ItemSheet

## Reward

Gets the reward items obtained from this floor. Returns the list of items that were dropped during the battle.

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

## RewardItems

Gets the list of reward items obtained from this floor.

```
public List<ItemBase> RewardItems { get; }
```

### Property Value

List<ItemBase>

## RuneSlotState

Gets the rune slot state for the player.

```
public RuneSlotState RuneSlotState { get; }
```

### Property Value

RuneSlotState

## RuneStates

Gets the rune states for the player.

```
public AllRuneState RuneStates { get; }
```

### Property Value

AllRuneState

## SimulatorSheets

Gets the simulator sheets containing game data.

```
public SimulatorSheets SimulatorSheets { get; }
```

### Property Value

SimulatorSheets

## StageId

Gets the stage ID, which is the same as the floor ID for infinite tower.

```
public int StageId { get; }
```

### Property Value

int

## TurnLimit

Gets the turn limit for all waves combined.

```
public int TurnLimit { get; }
```

### Property Value

int

## WaveRows

Gets the list of wave data for this floor.

```
public List<InfiniteTowerFloorWaveSheet.WaveData> WaveRows { get; }
```

### Property Value

List<InfiniteTowerFloorWaveSheet.WaveData>

## Methods

### ApplyConditionsToCharacter(CharacterBase, bool)

Applies infinite tower conditions to any character (Player, Enemy, Boss, etc.). This unified method works for all CharacterBase subclasses and applies stat modifiers based on the condition's targeting rules.

```
public void ApplyConditionsToCharacter(CharacterBase character, bool isPlayer)
```

#### Parameters

`character` CharacterBase

The character to apply conditions to.

`isPlayer` bool

Whether the character is a player (affects targeting rules).

### SetConditions(List<InfiniteTowerCondition>)

Sets the infinite tower conditions for this floor. This method allows setting conditions after simulator creation. The conditions will be applied when Simulate() is called.

```
public void SetConditions(List<InfiniteTowerCondition> conditions)
```

#### Parameters

`conditions` List<InfiniteTowerCondition>

The infinite tower conditions to apply.

## Simulate()

Simulates the infinite tower battle with wave progression and turn-based combat. Processes all waves sequentially with integrated turn limit and applies conditions.

```
public void Simulate()
```

# Class InfiniteTowerWave

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Represents a wave in the infinite tower. Contains information about the enemies that will spawn in this wave.

```
public class InfiniteTowerWave
```

**Inheritance**

object ← InfiniteTowerWave

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### InfiniteTowerWave(int, int, int)

Initializes a new instance of the InfiniteTowerWave class.

```
public InfiniteTowerWave(int waveNumber, int enemyId, int enemyCount)
```

#### Parameters

`waveNumber` int

The wave number.

`enemyId` int

The enemy character ID.

`enemyCount` int

The number of enemies to spawn.

# Properties

## EnemyCount

Gets the number of enemies to spawn in this wave.

```
public int EnemyCount { get; }
```

### Property Value

int

## EnemyId

Gets the enemy character ID for this wave.

```
public int EnemyId { get; }
```

### Property Value

int

## WaveNumber

Gets the wave number.

```
public int WaveNumber { get; }
```

### Property Value

int

# Class InvalidCountException

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class InvalidCountException : InvalidOperationException, ISerializable
```

**Inheritance**

object ← Exception ← SystemException ← InvalidOperationException ← InvalidCountException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidCountException()

```
public InvalidCountException()
```

### InvalidCountException(string)

```
public InvalidCountException(string message)
```

#### Parameters

`message` string

# Class ListEmptyException

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class ListEmptyException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ListEmptyException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ListEmptyException(string)

```
public ListEmptyException(string message = "list is empty. add value first")
```

#### Parameters

`message` string

# Class RaidSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RaidSimulator : Simulator, ISimulator
```

**Inheritance**

object ← Simulator ← RaidSimulator

**Implements**

ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RaidSimulator(int, IRandom, AvatarState, List<Guid>, AllRuneState, RuneSlotState, RaidSimulatorSheets, CostumeStatSheet, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, long, Row)

```
public RaidSimulator(int bossId, IRandom random, AvatarState avatarState, List<Guid>
foods, AllRuneState runeStates, RuneSlotState runeSlotState, RaidSimulatorSheets
simulatorSheets, CostumeStatSheet costumeStatSheet, List<StatModifier>
collectionModifiers, BuffLimitSheet buffLimitSheet, BuffLinkSheet buffLinkSheet,
long shatterStrikeMaxDamage = 400000, BoostScheduleSheet.Row battleRewardBoost
= null)
```

### Parameters

`bossId` int

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`runeStates` AllRuneState

`runeSlotState` RuneSlotState

`simulatorSheets` RaidSimulatorSheets

`costumeStatSheet` CostumeStatSheet

`collectionModifiers` List<StatModifier>

`buffLimitSheet` BuffLimitSheet

`buffLinkSheet` BuffLinkSheet

`shatterStrikeMaxDamage` long

`battleRewardBoost` BoostScheduleSheet.Row

## Properties

### AssetReward

```
public List<FungibleAssetValue> AssetReward { get; }
```

### Property Value

List<FungibleAssetValue>

## BossId

```
public int BossId { get; }
```

### Property Value

int

## DamageDealt

```
public long DamageDealt { get; }
```

### Property Value

long

## Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

# Methods

## Simulate()

```
public BattleLog Simulate()
```

### Returns

BattleLog

## SpawnBoss(RaidBoss)

```
public void SpawnBoss(RaidBoss raidBoss)
```

### Parameters

`raidBoss` RaidBoss

# Class RaidSimulatorV1

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RaidSimulatorV1 : Simulator, ISimulator
```

**Inheritance**

object ← Simulator ← RaidSimulatorV1

**Implements**

ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RaidSimulatorV1(int, IRandom, AvatarState, List<Guid>, RaidSimulatorSheetsV1, CostumeStatSheet)

```
public RaidSimulatorV1(int bossId, IRandom random, AvatarState
avatarState, List<Guid> foods, RaidSimulatorSheetsV1 simulatorSheets,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`bossId` int

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`simulatorSheets` RaidSimulatorSheetsV1

`costumeStatSheet` CostumeStatSheet

## Properties

## AssetReward

```
public List<FungibleAssetValue> AssetReward { get; }
```

### Property Value

List<FungibleAssetValue>

## BossId

```
public int BossId { get; }
```

### Property Value

int

## DamageDealt

```
public long DamageDealt { get; }
```

### Property Value

long

## Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

# Methods

## Simulate()

```
public BattleLog Simulate()
```

### Returns

BattleLog

## SpawnBoss(RaidBoss)

```
public void SpawnBoss(RaidBoss raidBoss)
```

### Parameters

`raidBoss` RaidBoss

# Class RaidSimulatorV2

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RaidSimulatorV2 : Simulator, ISimulator
```

**Inheritance**

object ← Simulator ← RaidSimulatorV2

**Implements**

ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RaidSimulatorV2(int, IRandom, AvatarState, List<Guid>, List<RuneState>, RaidSimulatorSheets, CostumeStatSheet)

```
public RaidSimulatorV2(int bossId, IRandom random, AvatarState avatarState,
List<Guid> foods, List<RuneState> runeStates, RaidSimulatorSheets simulatorSheets,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`bossId` int

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`runeStates` List<RuneState>

`simulatorSheets` RaidSimulatorSheets

`costumeStatSheet` CostumeStatSheet

## Properties

### AssetReward

```
public List<FungibleAssetValue> AssetReward { get; }
```

#### Property Value

List<FungibleAssetValue>

### BossId

```
public int BossId { get; }
```

#### Property Value

int

## DamageDealt

```
public long DamageDealt { get; }
```

### Property Value

long

## Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

# Methods

## Simulate()

```
public BattleLog Simulate()
```

### Returns

BattleLog

## SpawnBoss(RaidBoss)

```
public void SpawnBoss(RaidBoss raidBoss)
```

### Parameters

`raidBoss` RaidBoss

# Class RankingSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RankingSimulator : Simulator, ISimulator
```

**Inheritance**

object ← Simulator ← RankingSimulator

**Implements**

ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingSimulator(IRandom, Player, EnemyPlayerDigest, List<Guid>, RankingSimulatorSheetsV1, int, CostumeStatSheet)

```
public RankingSimulator(IRandom random, Player player, EnemyPlayerDigest
enemyPlayerDigest, List<Guid> foods, RankingSimulatorSheetsV1
rankingSimulatorSheets, int stageId, CostumeStatSheet costumeStatSheet)
```

### Parameters

`random` IRandom

`player` Player

`enemyPlayerDigest` EnemyPlayerDigest

`foods` List<Guid>

`rankingSimulatorSheets` RankingSimulatorSheetsV1

`stageId` int

`costumeStatSheet` CostumeStatSheet

## RankingSimulator(IRandom, AvatarState, AvatarState, List<Guid>, RankingSimulatorSheetsV1, int)

```
public RankingSimulator(IRandom random, AvatarState avatarState, AvatarState
enemyAvatarState, List<Guid> foods, RankingSimulatorSheetsV1 rankingSimulatorSheets,
int stageId)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`enemyAvatarState` AvatarState

`foods` List<Guid>

`rankingSimulatorSheets` RankingSimulatorSheetsV1

`stageId` int

# RankingSimulator(IRandom, AvatarState, AvatarState, List<Guid>, RankingSimulatorSheetsV1, int, CostumeStatSheet)

```
public RankingSimulator(IRandom random, AvatarState avatarState, AvatarState
enemyAvatarState, List<Guid> foods, RankingSimulatorSheetsV1 rankingSimulatorSheets,
int stageId, CostumeStatSheet costumeStatSheet)
```

## Parameters

`random` IRandom

`avatarState` AvatarState

`enemyAvatarState` AvatarState

`foods` List<Guid>

`rankingSimulatorSheets` RankingSimulatorSheetsV1

`stageId` int

`costumeStatSheet` CostumeStatSheet

# Properties

## Reward

Use this property after invoke the `PostSimulate(rewards)` function. This property will returns `null` basically.

```
public override IEnumerable<ItemBase> Reward { get; }
```

## Property Value

IEnumerable<ItemBase>

# Methods

# PostSimulate(List<ItemBase>, int, int)

This function should invoke after the `Simulate()` function invoked.

```
public void PostSimulate(List<ItemBase> rewards, int challengerScoreDelta,
int challengerScore)
```

## Parameters

`rewards` List<ItemBase>

`challengerScoreDelta` int

`challengerScore` int

# Simulate()

```
public Player Simulate()
```

## Returns

Player

# Class RankingSimulatorV1

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RankingSimulatorV1 : Simulator, ISimulator
```

**Inheritance**

object ← Simulator ← RankingSimulatorV1

**Implements**

ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingSimulatorV1(IRandom, Player, EnemyPlayerDigest, List<Guid>, RankingSimulatorSheetsV1, int, ArenaInfo, ArenaInfo, CostumeStatSheet)

```
public RankingSimulatorV1(IRandom random, Player player, EnemyPlayerDigest
enemyPlayerDigest, List<Guid> foods, RankingSimulatorSheetsV1
rankingSimulatorSheets, int stageId, ArenaInfo arenaInfo, ArenaInfo enemyInfo,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`random` IRandom

`player` Player

`enemyPlayerDigest` EnemyPlayerDigest

`foods` List<Guid>

`rankingSimulatorSheets` RankingSimulatorSheetsV1

`stageId` int

`arenaInfo` ArenaInfo

`enemyInfo` ArenaInfo

`costumeStatSheet` CostumeStatSheet

## RankingSimulatorV1(IRandom, AvatarState, AvatarState, List<Guid>, RankingSimulatorSheetsV1, int, ArenaInfo, ArenaInfo)

```
public RankingSimulatorV1(IRandom random, AvatarState avatarState, AvatarState
enemyAvatarState, List<Guid> foods, RankingSimulatorSheetsV1 rankingSimulatorSheets,
int stageId, ArenaInfo arenaInfo, ArenaInfo enemyInfo)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`enemyAvatarState` AvatarState

foods List<Guid>

rankingSimulatorSheets RankingSimulatorSheetsV1

stageId int

arenaInfo ArenaInfo

enemyInfo ArenaInfo

## RankingSimulatorV1(IRandom, AvatarState, AvatarState, List<Guid>, RankingSimulatorSheetsV1, int, ArenaInfo, ArenaInfo, CostumeStatSheet)

```
public RankingSimulatorV1(IRandom random, AvatarState avatarState,
AvatarState enemyAvatarState, List<Guid> foods, RankingSimulatorSheetsV1
rankingSimulatorSheets, int stageId, ArenaInfo arenaInfo, ArenaInfo enemyInfo,
CostumeStatSheet costumeStatSheet)
```

### Parameters

random IRandom

avatarState AvatarState

enemyAvatarState AvatarState

foods List<Guid>

rankingSimulatorSheets RankingSimulatorSheetsV1

stageId int

arenaInfo ArenaInfo

enemyInfo ArenaInfo

costumeStatSheet CostumeStatSheet

## Fields

# WeeklyArenaRewardSheet

```
public readonly WeeklyArenaRewardSheet WeeklyArenaRewardSheet
```

## Field Value

WeeklyArenaRewardSheet

# Properties

## Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

## Property Value

IEnumerable<ItemBase>

# Methods

## Simulate()

```
public Player Simulate()
```

## Returns

Player

## SimulateV1()

```
[Obsolete("Use Simulate")]
public Player SimulateV1()
```

## Returns

Player

## SimulateV2()

```
[Obsolete("Use Simulate")]
public Player SimulateV2()
```

### Returns

Player

## SimulateV3()

```
[Obsolete("Use Simulate")]
public Player SimulateV3()
```

### Returns

Player

## SimulateV4()

```
[Obsolete("use Simulate")]
public Player SimulateV4()
```

### Returns

Player

# Class RewardSelector

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class RewardSelector
```

**Inheritance**

object ← RewardSelector

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### Select(IRandom, WeeklyArenaRewardSheet, MaterialItemSheet, int, int)

```
public static List<ItemBase> Select(IRandom random, WeeklyArenaRewardSheet
weeklyArenaRewardSheet, MaterialItemSheet materialItemSheet, int playerLevel, int
maxCount = 1)
```

#### Parameters

`random` IRandom

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

`materialItemSheet` MaterialItemSheet

`playerLevel` int

`maxCount` int

#### Returns

List<ItemBase>

# Class Simulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public abstract class Simulator : ISimulator
```

**Inheritance**

object ← Simulator

**Implements**

ISimulator

**Derived**

ArenaSimulator2, AdventureBossSimulator, InfiniteTowerSimulator, RaidSimulator, RaidSimulatorV1, RaidSimulatorV2, RankingSimulator, RankingSimulatorV1, StageSimulator, StageSimulatorV1, StageSimulatorV2, StageSimulatorV3, TestSimulator

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Simulator(IRandom, Player, List<Guid>, SimulatorSheets)

```
protected Simulator(IRandom random, Player player, List<Guid> foods,
SimulatorSheets simulatorSheets)
```

#### Parameters

`random` IRandom

`player` Player

`foods` List<Guid>

`simulatorSheets` SimulatorSheets

## Simulator(IRandom, Player, List<Guid>, SimulatorSheetsV1)

```
protected Simulator(IRandom random, Player player, List<Guid> foods,
SimulatorSheetsV1 simulatorSheets)
```

### Parameters

`random` IRandom

`player` Player

`foods` List<Guid>

`simulatorSheets` SimulatorSheetsV1

## Simulator(IRandom, AvatarState, List<Guid>, SimulatorSheets, bool, long)

```
protected Simulator(IRandom random, AvatarState avatarState, List<Guid> foods,
SimulatorSheets simulatorSheets, bool logEvent = true, long shatterStrikeMaxDamage
= 400000)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`simulatorSheets` SimulatorSheets

`logEvent` bool

`shatterStrikeMaxDamage` long

# Simulator(IRandom, AvatarState, List<Guid>, SimulatorSheetsV1, bool)

```
protected Simulator(IRandom random, AvatarState avatarState, List<Guid> foods,
SimulatorSheetsV1 simulatorSheets, bool logEvent = true)
```

## Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`simulatorSheets` SimulatorSheetsV1

`logEvent` bool

# Fields

## ActionBuffSheet

```
public readonly ActionBuffSheet ActionBuffSheet
```

## Field Value

ActionBuffSheet

## CharacterLevelSheet

```
public readonly CharacterLevelSheet CharacterLevelSheet
```

## Field Value

CharacterLevelSheet

## CharacterSheet

```
public readonly CharacterSheet CharacterSheet
```

### Field Value

CharacterSheet

## EquipmentItemSetEffectSheet

```
public readonly EquipmentItemSetEffectSheet EquipmentItemSetEffectSheet
```

### Field Value

EquipmentItemSetEffectSheet

## MaterialItemSheet

```
public readonly MaterialItemSheet MaterialItemSheet
```

### Field Value

MaterialItemSheet

## MaxTurn

```
protected const int MaxTurn = 200
```

### Field Value

int

## Random

```
public readonly IRandom Random
```

### Field Value

IRandom

## SkillActionBuffSheet

```
public readonly SkillActionBuffSheet SkillActionBuffSheet
```

### Field Value

SkillActionBuffSheet

## SkillBuffSheet

```
public readonly SkillBuffSheet SkillBuffSheet
```

### Field Value

SkillBuffSheet

## SkillSheet

```
public readonly SkillSheet SkillSheet
```

### Field Value

SkillSheet

## StatBuffSheet

```
public readonly StatBuffSheet StatBuffSheet
```

### Field Value

StatBuffSheet

## TurnNumber

```
public int TurnNumber
```

### Field Value

int

## TurnPriority

```
public const decimal TurnPriority = 100
```

### Field Value

decimal

# Properties

## BuffLimitSheet

```
public BuffLimitSheet BuffLimitSheet { get; protected set; }
```

### Property Value

BuffLimitSheet

## BuffLinkSheet

```
public BuffLinkSheet BuffLinkSheet { get; set; }
```

### Property Value

BuffLinkSheet

## Characters

```
public SimplePriorityQueue<CharacterBase, decimal> Characters { get; set; }
```

### Property Value

SimplePriorityQueue<CharacterBase, decimal>

## Log

```
public BattleLog Log { get; }
```

### Property Value

BattleLog

## LogEvent

```
public bool LogEvent { get; protected set; }
```

### Property Value

bool

## Player

```
public Player Player { get; }
```

### Property Value

Player

## Result

```
public BattleLog.Result Result { get; protected set; }
```

### Property Value

BattleLog.Result

## Reward

```
public abstract IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

## ShatterStrikeMaxDamage

```
public long ShatterStrikeMaxDamage { get; }
```

### Property Value

long

## WaveNumber

```
public int WaveNumber { get; protected set; }
```

### Property Value

int

## WaveTurn

```
public int WaveTurn { get; set; }
```

### Property Value

int

## Methods

### SetReward(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet)

```
public static List<ItemBase> SetReward(WeightedSelector<StageSheet.RewardData>
itemSelector, int maxCount, IRandom random, MaterialItemSheet materialItemSheet)
```

### Parameters

`itemSelector` WeightedSelector<StageSheet.RewardData>

`maxCount` int

`random` IRandom

`materialItemSheet` MaterialItemSheet

### Returns

List<ItemBase>

# SetRewardV2(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet)

```
public static List<ItemBase> SetRewardV2(WeightedSelector<StageSheet.RewardData>
itemSelector, int maxCount, IRandom random, MaterialItemSheet materialItemSheet)
```

## Parameters

`itemSelector` WeightedSelector<StageSheet.RewardData>

`maxCount` int

`random` IRandom

`materialItemSheet` MaterialItemSheet

## Returns

List<ItemBase>

# Class StageRewardExpHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class StageRewardExpHelper
```

**Inheritance**

object ← StageRewardExpHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## CachedExp

key: differ (stage number - character level) value: exp

```
public static readonly IReadOnlyDictionary<int, int> CachedExp
```

### Field Value

IReadOnlyDictionary<int, int>

## DifferLowerLimit

```
public const int DifferLowerLimit = -15
```

### Field Value

int

## DifferUpperLimit

```
public const int DifferUpperLimit = 10
```

### Field Value

int

## RewardExpMax

```
public const int RewardExpMax = 300
```

### Field Value

int

## RewardExpMin

```
public const int RewardExpMin = 0
```

### Field Value

int

# Methods

## GetExp(int, int)

```
public static int GetExp(int characterLevel, int stageNumber)
```

### Parameters

`characterLevel` int

`stageNumber` int

## Returns

int

# Class StageSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Simulator for stage battles that handles wave-based combat.

```
public class StageSimulator : Simulator, IStageSimulator, ISimulator
```

**Inheritance**

object ← Simulator ← StageSimulator

**Implements**

IStageSimulator, ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StageSimulator(IRandom, AvatarState, List<Guid>, AllRuneState, RuneSlotState, List<Skill>, int, int, Row, Row, bool, int, SimulatorSheets, EnemySkillSheet,

# CostumeStatSheet, List<ItemBase>, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, bool, long, Row)

Initializes a new instance of the StageSimulator class.

```
public StageSimulator(IRandom random, AvatarState avatarState, List<Guid> foods,
AllRuneState runeStates, RuneSlotState runeSlotState, List<Skill> skillsOnWaveStart,
int worldId, int stageId, StageSheet.Row stageRow, StageWaveSheet.Row stageWaveRow,
bool isCleared, int exp, SimulatorSheets simulatorSheets, EnemySkillSheet
enemySkillSheet, CostumeStatSheet costumeStatSheet, List<ItemBase> waveRewards,
List<StatModifier> collectionModifiers, BuffLimitSheet buffLimitSheet, BuffLinkSheet
buffLinkSheet, bool logEvent = true, long shatterStrikeMaxDamage = 400000,
BoostScheduleSheet.Row favRewardBoost = null)
```

## Parameters

`random` IRandom

The random number generator.

`avatarState` AvatarState

The avatar state.

`foods` List<Guid>

The food items to use.

`runeStates` AllRuneState

The rune states.

`runeSlotState` RuneSlotState

The rune slot state.

`skillsOnWaveStart` List<Skill>

The skills to use at wave start.

`worldId` int

The world ID.

`stageId` int

The stage ID.

`stageRow` StageSheet.Row

The stage row data.

`stageWaveRow` StageWaveSheet.Row

The stage wave row data.

`isCleared` bool

Whether the stage is cleared.

`exp` int

The experience points.

`simulatorSheets` SimulatorSheets

The simulator sheets.

`enemySkillSheet` EnemySkillSheet

The enemy skill sheet.

`costumeStatSheet` CostumeStatSheet

The costume stat sheet.

`waveRewards` List<ItemBase>

The item rewards already drawn for this clear.

`collectionModifiers` List<StatModifier>

The stat modifiers from activated collections.

`buffLimitSheet` BuffLimitSheet

The buff limit sheet.

`buffLinkSheet` BuffLinkSheet

The buff link sheet.

`logEvent` bool

Whether to record the battle log.

`shatterStrikeMaxDamage` long

Maximum damage for shatter strike.

`favRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this stage's fungible asset rewards, or `null` for none. See ApplyFavRewardBoost(List<(string ticker, int amount)>, Row). A caller replaying a battle must pass what the action resolved.

# Fields

## MaxItemRewardBoostFactor

Upper bound of the boosted count of each distinct item in one clear, as a multiple of how many times it was drawn in that clear: a BoostScheduleSheet row grants at most this many times the drawn items per clear, whatever its value. It bounds a mistyped value (e.g. `MUL 200` for `MUL 2.00`) to a tenfold over-reward, and the objects ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet) creates per clear to this many times `DropItemMax`. It is not a per-transaction bound; `HackAndSlashSweep` adds its extra units with one count per item id instead of one object per unit, so its cost does not grow with the value.

```
public const int MaxItemRewardBoostFactor = 10
```

### Field Value

int

# Properties

## EnemySkillSheet

Gets the enemy skill sheet.

```
public EnemySkillSheet EnemySkillSheet { get; }
```

### Property Value

EnemySkillSheet

## FungibleAssetRewards

Gets the fungible asset rewards (e.g., Crystal) obtained from wave 2 clear.

```
public Dictionary<string, int> FungibleAssetRewards { get; }
```

### Property Value

Dictionary<string, int>

## ItemMap

Gets the collection map for items.

```
public CollectionMap ItemMap { get; }
```

### Property Value

CollectionMap

## Reward

Gets the rewards from the battle.

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

## StageId

Gets the stage ID.

```
public int StageId { get; }
```

### Property Value

int

# Methods

## ApplyFavRewardBoost(List<(string ticker, int amount)>, Row)

Applies a BoostScheduleSheet row to the fungible asset rewards of one clear.

```
public static List<(string ticker, int amount)> ApplyFavRewardBoost(List<(string
ticker, int amount)> rewards, BoostScheduleSheet.Row favRewardBoost)
```

### Parameters

`rewards` List<(string ticker, int amount)>

The amount drawn for each ticker.

`favRewardBoost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

### Returns

List<(string ticker, int amount)>

`rewards` itself when `favRewardBoost` is `null`; otherwise a new list with each positive amount replaced by Apply(int) of it, in the same order. A ticker whose adjusted amount is 0 is left out so that nothing mints a zero amount.

### Remarks

Draws nothing, so the random sequence is the same as without a row. A boost adjusts what was drawn and never creates a reward: a ticker that was not drawn, or drawn as 0, stays as it is.

## ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet)

Applies a BoostScheduleSheet row to the item rewards of one clear, one object per unit.

```
public static List<ItemBase> ApplyItemRewardBoost(List<ItemBase> rewards,
BoostScheduleSheet.Row itemRewardBoost, MaterialItemSheet materialItemSheet)
```

### Parameters

`rewards` List<ItemBase>

The items drawn in one clear, one object per unit.

`itemRewardBoost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`materialItemSheet` MaterialItemSheet

The sheet the rewards were created from.

### Returns

List<ItemBase>

`rewards` itself when `itemRewardBoost` is `null`; otherwise a new list, ordered by item id, in which each distinct item drawn `n` times appears GetBoostedItemRewardCount(Row, int) (row, `n`) times, at most MaxItemRewardBoostFactor × `n`.

### Remarks

Draws nothing, so the random sequence is the same as without a row. `MUL 2` grants exactly twice each drawn item; `ADD 1` one more of each distinct item; a multiplier below 1 takes items away. A `MUL` above 1 turns an item drawn once into at least 2, so `MUL 1.5` grants 2 of a single drop (and 3 of a double). The count can exceed the stage's `DropItemMax`, which only limits the draws. Added items are created the way

SetRewardV2(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet) creates them, so a circle is tradable as usual. Equivalent to SplitItemRewardBoost(List<ItemBase>, Row, IDictionary<int, int>) followed by CreateItemRewardBoostExtras(IReadOnlyDictionary<int, int>, MaterialItemSheet).

## CreateItemRewardBoostExtras(IReadOnlyDictionary<int, int>, MaterialItemSheet)

Creates the items SplitItemRewardBoost(List<ItemBase>, Row, IDictionary<int, int>) counted, one object per unit, for a caller that needs objects (a battle's drop log, a result view).

```
public static List<ItemBase> CreateItemRewardBoostExtras(IReadOnlyDictionary<int,
int> extraCounts, MaterialItemSheet materialItemSheet)
```

### Parameters

`extraCounts` IReadOnlyDictionary<int, int>

Units to create per item id.

`materialItemSheet` MaterialItemSheet

The sheet the rewards were created from.

### Returns

List<ItemBase>

The items, ordered by item id.

## CreateItemRewardBoostItem(Row)

Creates one unit of a boosted item reward the way SetRewardV2(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet) creates a drawn one, so a circle is tradable as usual. A fungible item, so one object may stand for any number of units when added with a count.

```
public static ItemBase CreateItemRewardBoostItem(MaterialItemSheet.Row materialRow)
```

### Parameters

`materialRow` MaterialItemSheet.Row

The item's row.

### Returns

ItemBase

The item.

## GetBoostedItemRewardCount(Row, int)

The boosted count of an item drawn `count` times in one clear.

```
public static int GetBoostedItemRewardCount(BoostScheduleSheet.Row itemRewardBoost,
int count)
```

### Parameters

`itemRewardBoost` BoostScheduleSheet.Row

The row to apply.

`count` int

How many times the item was drawn in the clear.

### Returns

int

Apply(int)(`count`), except that a Mul above 1 turns an item drawn once into at least 2, so that a fractional multiplier still rewards the single drops most stages make. A count of 2 or more keeps the plain floor (`MUL 1.2` on 2 stays 2). Capped at MaxItemRewardBoost Factor × `count`.

## GetFavWaveRewards(IRandom, Row, Row)

Draws the fungible asset rewards of one clear of `stageRow`.

```
public static List<(string ticker, int amount)> GetFavWaveRewards(IRandom random,
StageSheet.Row stageRow, BoostScheduleSheet.Row favRewardBoost = null)
```

### Parameters

`random` IRandom

The random number generator.

`stageRow` StageSheet.Row

The stage being cleared.

`favRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting the amounts, or `null` for none. See ApplyFavRewardBoost(List<(string ticker, int amount)>, Row). It never changes the random draws.

### Returns

List<(string ticker, int amount)>

The amount drawn for each ticker.

## GetWaveRewards(IRandom, Row, MaterialItemSheet, int, Row)

Draws the item rewards of `playCount` clears of `stageRow`.

```
public static List<ItemBase> GetWaveRewards(IRandom random, StageSheet.Row stageRow,
MaterialItemSheet materialItemSheet, int playCount = 1, BoostScheduleSheet.Row
itemRewardBoost = null)
```

### Parameters

`random` IRandom

The random number generator.

`stageRow` StageSheet.Row

The stage being cleared.

`materialItemSheet` MaterialItemSheet

The sheet the rewards are created from.

`playCount` int

The number of clears.

`itemRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting each clear's items, or `null` for none. See ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet). It never changes the random draws. A caller replaying a battle must pass what the action resolved.

### Returns

List<ItemBase>

The items, one object per unit, each clear's ordered by item id.

## Simulate()

```
public Player Simulate()
```

### Returns

Player

## SplitItemRewardBoost(List<ItemBase>, Row, IDictionary<int, int>)

Applies a BoostScheduleSheet row to the item rewards of one clear without creating an object per added unit: units taken away are dropped from the returned list, and units added are only counted into `extraCounts`.

```
public static List<ItemBase> SplitItemRewardBoost(List<ItemBase> rewards,
BoostScheduleSheet.Row itemRewardBoost, IDictionary<int, int> extraCounts)
```

## Parameters

`rewards` List<ItemBase>

The items drawn in one clear, one object per unit.

`itemRewardBoost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`extraCounts` IDictionary<int, int>

Accumulates, per item id, how many units the row adds on top of the returned list. Left untouched when nothing is added, so one dictionary can collect several clears.

## Returns

List<ItemBase>

`rewards` itself when `itemRewardBoost` is `null`; otherwise the drawn items that remain, ordered by item id. Together with `extraCounts` they add up to exactly what ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet) grants for the clear.

# Class StageSimulatorV1

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class StageSimulatorV1 : Simulator, IStageSimulator, ISimulator
```

**Inheritance**

object ← Simulator ← StageSimulatorV1

**Implements**

IStageSimulator, ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StageSimulatorV1(IRandom, AvatarState, List<Guid>, List<Skill>, int, int, StageSimulatorSheetsV1, CostumeStatSheet, int, int)

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
List<Skill> skillsOnWaveStart, int worldId, int stageId, StageSimulatorSheetsV1
stageSimulatorSheets, CostumeStatSheet costumeStatSheet, int constructorVersion, int
playCount = 1)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`skillsOnWaveStart` List<Skill>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`costumeStatSheet` CostumeStatSheet

`constructorVersion` int

`playCount` int

## StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1)

Do not use anymore since v100025.

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

## StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, Skill)

Do not use anymore since v100025.

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets, Skill skill)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`skill` Skill

## StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, Skill, int, int)

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets, Skill skill,
int constructorVersion, int playCount)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`skill` Skill

`constructorVersion` int

`playCount` int

## StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, CostumeStatSheet)

Do not use anymore since v100025.

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`costumeStatSheet` CostumeStatSheet

## StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, CostumeStatSheet, int, int)

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets,
CostumeStatSheet costumeStatSheet, int constructorVersion, int playCount = 1)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`costumeStatSheet` CostumeStatSheet

`constructorVersion` int

`playCount` int

## StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, int, int)

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets, int
constructorVersion, int playCount)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`constructorVersion` int

`playCount` int

# Fields

## ConstructorVersionDefault

```
public const int ConstructorVersionDefault = 1
```

### Field Value

int

## ConstructorVersionV100025

```
public const int ConstructorVersionV100025 = 2
```

### Field Value

int

## ConstructorVersionV100080

```
public const int ConstructorVersionV100080 = 3
```

### Field Value

int

## Properties

### EnemySkillSheet

```
public EnemySkillSheet EnemySkillSheet { get; }
```

### Property Value

EnemySkillSheet

### ItemMap

```
public CollectionMap ItemMap { get; }
```

### Property Value

CollectionMap

### Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

### StageId

```
public int StageId { get; }
```

### Property Value

int

## Methods

### SetItemSelector(Row, IRandom)

```
public static WeightedSelector<StageSheet.RewardData> SetItemSelector(StageSheet.Row
stageRow, IRandom random)
```

#### Parameters

`stageRow` StageSheet.Row

`random` IRandom

#### Returns

WeightedSelector<StageSheet.RewardData>

### Simulate(int)

```
public Player Simulate(int playCount)
```

#### Parameters

`playCount` int

#### Returns

Player

### SimulateV1()

```
[Obsolete("Use Simulate")]
```

```
public Player SimulateV1()
```

### Returns

Player

## SimulateV2()

```
[Obsolete("Use Simulate")]
public Player SimulateV2()
```

### Returns

Player

## SimulateV3()

```
[Obsolete("Use Simulate")]
public Player SimulateV3()
```

### Returns

Player

## SimulateV4()

```
[Obsolete("Use Simulate")]
public Player SimulateV4()
```

### Returns

Player

## SimulateV5(int)

```
[Obsolete("Use Simulate")]
public Player SimulateV5(int playCount)
```

## Parameters

`playCount` int

## Returns

Player

# Class StageSimulatorV2

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class StageSimulatorV2 : Simulator, IStageSimulator, ISimulator
```

**Inheritance**

object ← Simulator ← StageSimulatorV2

**Implements**

IStageSimulator, ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StageSimulatorV2(IRandom, AvatarState, List<Guid>, List<Skill>, int, int, Row, Row, bool, int, SimulatorSheetsV1, EnemySkillSheet, CostumeStatSheet, List<ItemBase>)

```
public StageSimulatorV2(IRandom random, AvatarState avatarState, List<Guid> foods,
List<Skill> skillsOnWaveStart, int worldId, int stageId, StageSheet.Row stageRow,
StageWaveSheet.Row stageWaveRow, bool isCleared, int exp, SimulatorSheetsV1
simulatorSheets, EnemySkillSheet enemySkillSheet, CostumeStatSheet costumeStatSheet,
List<ItemBase> waveRewards)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`skillsOnWaveStart` List<Skill>

`worldId` int

`stageId` int

`stageRow` StageSheet.Row

`stageWaveRow` StageWaveSheet.Row

`isCleared` bool

`exp` int

`simulatorSheets` SimulatorSheetsV1

`enemySkillSheet` EnemySkillSheet

`costumeStatSheet` CostumeStatSheet

`waveRewards` List<ItemBase>

## Properties

## EnemySkillSheet

```
public EnemySkillSheet EnemySkillSheet { get; }
```

### Property Value

EnemySkillSheet

## ItemMap

```
public CollectionMap ItemMap { get; }
```

### Property Value

CollectionMap

## Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

## StageId

```
public int StageId { get; }
```

### Property Value

int

# Methods

## GetWaveRewards(IRandom, Row, MaterialItemSheet, int)

```
public static List<ItemBase> GetWaveRewards(IRandom random, StageSheet.Row stageRow,
MaterialItemSheet materialItemSheet, int playCount = 1)
```

### Parameters

`random` IRandom

`stageRow` StageSheet.Row

`materialItemSheet` MaterialItemSheet

`playCount` int

### Returns

List<ItemBase>

## Simulate()

```
public Player Simulate()
```

### Returns

Player

# Class StageSimulatorV3

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class StageSimulatorV3 : Simulator, IStageSimulator, ISimulator
```

**Inheritance**

object ← Simulator ← StageSimulatorV3

**Implements**

IStageSimulator, ISimulator

**Inherited Members**

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StageSimulatorV3(IRandom, AvatarState, List<Guid>, List<RuneState>, List<Skill>, int, int, Row, Row, bool, int, SimulatorSheets, EnemySkillSheet, CostumeStatSheet, List<ItemBase>)

```
public StageSimulatorV3(IRandom random, AvatarState avatarState, List<Guid> foods,
List<RuneState> runeStates, List<Skill> skillsOnWaveStart, int worldId, int stageId,
StageSheet.Row stageRow, StageWaveSheet.Row stageWaveRow, bool isCleared, int exp,
SimulatorSheets simulatorSheets, EnemySkillSheet enemySkillSheet, CostumeStatSheet
costumeStatSheet, List<ItemBase> waveRewards)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`runeStates` List<RuneState>

`skillsOnWaveStart` List<Skill>

`worldId` int

`stageId` int

`stageRow` StageSheet.Row

`stageWaveRow` StageWaveSheet.Row

`isCleared` bool

`exp` int

`simulatorSheets` SimulatorSheets

`enemySkillSheet` EnemySkillSheet

`costumeStatSheet` CostumeStatSheet

`waveRewards` List<ItemBase>

## Properties

### EnemySkillSheet

```
public EnemySkillSheet EnemySkillSheet { get; }
```

### Property Value

EnemySkillSheet

## ItemMap

```
public CollectionMap ItemMap { get; }
```

### Property Value

CollectionMap

## Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

## StageId

```
public int StageId { get; }
```

### Property Value

int

## Methods

# GetWaveRewards(IRandom, Row, MaterialItemSheet, int)

```
public static List<ItemBase> GetWaveRewards(IRandom random, StageSheet.Row stageRow,
MaterialItemSheet materialItemSheet, int playCount = 1)
```

## Parameters

`random` IRandom

`stageRow` StageSheet.Row

`materialItemSheet` MaterialItemSheet

`playCount` int

## Returns

List<ItemBase>

# Simulate()

```
public Player Simulate()
```

## Returns

Player

# Class TestSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class TestSimulator : Simulator, ISimulator
```

### Inheritance

object ← Simulator ← TestSimulator

### Implements

ISimulator

### Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### TestSimulator(IRandom, AvatarState, List<Guid>, SimulatorSheets)

```
public TestSimulator(IRandom random, AvatarState avatarState, List<Guid> foods,
SimulatorSheets simulatorSheets)
```

### Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`simulatorSheets` SimulatorSheets

## Properties

## Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

# Class Wave

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Represents a wave of enemies in a battle.

```
public class Wave
```

**Inheritance**

object ← Wave

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## HasBoss

Gets or sets whether this wave contains a boss enemy.

```
public bool HasBoss
```

### Field Value

bool

# Methods

## Add(Enemy)

Adds an enemy to this wave.

```
public void Add(Enemy enemy)
```

### Parameters

`enemy` Enemy

The enemy to add.

## Spawn(ISimulator)

Spawns all enemies in this wave into the simulator.

```
public void Spawn(ISimulator simulator)
```

### Parameters

`simulator` ISimulator

The simulator to spawn enemies into.

## SpawnV1(ISimulator)

```
[Obsolete("Use Spawn")]
public void SpawnV1(ISimulator simulator)
```

### Parameters

`simulator` ISimulator

## SpawnV2(ISimulator)

```
[Obsolete("Use Spawn")]
public void SpawnV2(ISimulator simulator)
```

### Parameters

`simulator` ISimulator

# Class WaveResult

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Result of a single wave battle. Contains information about the wave completion status and battle outcome.

```
public class WaveResult
```

**Inheritance**

object ← WaveResult

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### EnemiesDefeated

Gets or sets the number of enemies defeated in this wave.

```
public int EnemiesDefeated { get; set; }
```

#### Property Value

int

### IsClear

Gets or sets whether the wave was cleared.

```
public bool IsClear { get; set; }
```

### Property Value

bool

## PlayerHpRemaining

Gets or sets the player's remaining HP after the wave.

```
public int PlayerHpRemaining { get; set; }
```

### Property Value

int

## Result

Gets or sets the battle result for this wave.

```
public BattleLog.Result Result { get; set; }
```

### Property Value

BattleLog.Result

# Class WeightedSelector<T>

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class WeightedSelector<T>
```

## Type Parameters

T

**Inheritance**

object ← WeightedSelector<T>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WeightedSelector(IRandom)

```
public WeightedSelector(IRandom random)
```

## Parameters

random IRandom

# Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## Methods

## Add(T, decimal)

```
public void Add(T item, decimal weight)
```

### Parameters

`item` T

`weight` decimal

## Clear()

```
public void Clear()
```

## Select(int)

```
public IEnumerable<T> Select(int count)
```

### Parameters

`count` int

### Returns

IEnumerable<T>

## SelectV1(int)

```
[Obsolete("Use Select")]
public IEnumerable<T> SelectV1(int count)
```

### Parameters

`count` int

### Returns

IEnumerable<T>

## SelectV2(int)

```
[Obsolete("Use Select")]
public IEnumerable<T> SelectV2(int count)
```

### Parameters

`count` int

### Returns

IEnumerable<T>

# Namespace Nekoyume.Battle.AdventureBoss

## Classes

AdventureBossSimulator

# Class AdventureBossSimulator

Namespace: Nekoyume.Battle.AdventureBoss

Assembly: Lib9c.dll

```
public class AdventureBossSimulator : Simulator, IStageSimulator, ISimulator
```

### Inheritance

object ← Simulator ← AdventureBossSimulator

### Implements

IStageSimulator, ISimulator

### Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AdventureBossSimulator(int, int, IRandom, AvatarState, SimulatorSheets, bool, long)

```
public AdventureBossSimulator(int bossId, int floorId, IRandom random, AvatarState
avatarState, SimulatorSheets simulatorSheets, bool logEvent = true, long
```

```
  shatterStrikeMaxDamage = 400000)
```

## Parameters

`bossId` int

`floorId` int

`random` IRandom

`avatarState` AvatarState

`simulatorSheets` SimulatorSheets

`logEvent` bool

`shatterStrikeMaxDamage` long

# AdventureBossSimulator(int, int, IRandom, AvatarState, List<Guid>, AllRuneState, RuneSlotState, Row, Row, SimulatorSheets, EnemySkillSheet, CostumeStatSheet, List<ItemBase>, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, bool, long)

```
  public AdventureBossSimulator(int bossId, int floorId, IRandom random, AvatarState
  avatarState, List<Guid> foods, AllRuneState runeStates, RuneSlotState runeSlotState,
  AdventureBossFloorSheet.Row floorRow, AdventureBossFloorWaveSheet.Row floorWaveRow,
  SimulatorSheets simulatorSheets, EnemySkillSheet enemySkillSheet, CostumeStatSheet
  costumeStatSheet, List<ItemBase> waveRewards, List<StatModifier>
  collectionModifiers, BuffLimitSheet buffLimitSheet, BuffLinkSheet buffLinkSheet,
  bool logEvent = true, long shatterStrikeMaxDamage = 400000)
```

## Parameters

`bossId` int

`floorId` int

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`runeStates` AllRuneState

`runeSlotState` RuneSlotState

`floorRow` AdventureBossFloorSheet.Row

`floorWaveRow` AdventureBossFloorWaveSheet.Row

`simulatorSheets` SimulatorSheets

`enemySkillSheet` EnemySkillSheet

`costumeStatSheet` CostumeStatSheet

`waveRewards` List<ItemBase>

`collectionModifiers` List<StatModifier>

`buffLimitSheet` BuffLimitSheet

`buffLinkSheet` BuffLinkSheet

`logEvent` bool

`shatterStrikeMaxDamage` long

# Properties

## BossId

```
public int BossId { get; }
```

### Property Value

int

## EnemySkillSheet

```
public EnemySkillSheet EnemySkillSheet { get; }
```

### Property Value

EnemySkillSheet

## FloorId

```
public int FloorId { get; }
```

### Property Value

int

## ItemMap

```
public CollectionMap ItemMap { get; }
```

### Property Value

CollectionMap

## Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

### Property Value

IEnumerable<ItemBase>

## StageId

```
public int StageId { get; }
```

### Property Value

int

## Methods

### AddBreakthrough(List<int>, AdventureBossFloorWaveSheet)

```
public void AddBreakthrough(List<int> floorIdList, AdventureBossFloorWaveSheet
adventureBossFloorWaveSheet)
```

### Parameters

`floorIdList` List<int>

`adventureBossFloorWaveSheet` AdventureBossFloorWaveSheet

### GetWaveRewards(IRandom, Row, MaterialItemSheet, int)

```
public static List<ItemBase> GetWaveRewards(IRandom random,
AdventureBossFloorSheet.Row floorRow, MaterialItemSheet materialItemSheet, int
playCount = 1)
```

### Parameters

`random` IRandom

`floorRow` AdventureBossFloorSheet.Row

`materialItemSheet` MaterialItemSheet

`playCount` int

### Returns

List<ItemBase>

## SetItemSelector(Row, IRandom)

```
public static WeightedSelector<AdventureBossFloorSheet.RewardData>
SetItemSelector(AdventureBossFloorSheet.Row floorRow, IRandom random)
```

### Parameters

`floorRow` AdventureBossFloorSheet.Row

`random` IRandom

### Returns

WeightedSelector<AdventureBossFloorSheet.RewardData>

## SetReward(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet)

```
public static List<ItemBase>
SetReward(WeightedSelector<AdventureBossFloorSheet.RewardData> itemSelector, int
maxCount, IRandom random, MaterialItemSheet materialItemSheet)
```

### Parameters

`itemSelector` WeightedSelector<AdventureBossFloorSheet.RewardData>

`maxCount` int

`random` IRandom

`materialItemSheet` MaterialItemSheet

### Returns

List<ItemBase>

# Simulate()

```
public Player Simulate()
```

## Returns

Player

# Namespace Nekoyume.Data

## Classes

AdventureBossGameData

## Structs

AdventureBossGameData.ClaimableReward

# Class AdventureBossGameData

Namespace: Nekoyume.Data

Assembly: Lib9c.dll

```
public static class AdventureBossGameData
```

**Inheritance**

object ← AdventureBossGameData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AdventureBossOperationalAddress

```
public static readonly Address AdventureBossOperationalAddress
```

### Field Value

Address

# Struct AdventureBossGameData.ClaimableReward

Namespace: Nekoyume.Data

Assembly: Lib9c.dll

```
public struct AdventureBossGameData.ClaimableReward
```

**Inherited Members**

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Fields

### FavReward

```
public Dictionary<int, int> FavReward
```

#### Field Value

Dictionary<int, int>

### ItemReward

```
public Dictionary<int, int> ItemReward
```

#### Field Value

Dictionary<int, int>

# NcgReward

```
public FungibleAssetValue? NcgReward
```

## Field Value

FungibleAssetValue?

# Methods

## IsEmpty()

```
public bool IsEmpty()
```

## Returns

bool

# Namespace Nekoyume.Delegation

## Classes

Bond

CurrencyComparer
Represents a comparer for Libplanet.Types.Assets.Currency.

DelegateeMetadata

Delegatee<TRepository, TDelegatee, TDelegator>

DelegationAddress

DelegationRepository<TRepository, TDelegatee, TDelegator>

DelegatorMetadata

Delegator<TRepository, TDelegatee, TDelegator>

LumpSumRewardsRecord

RebondGrace

RewardBase
RewardBase is a class that represents the base of the reward. If it's multiplied by the number of shares, it will be the reward for the period. Also, it holds the significant figure to calculate the reward.

UnbondLockIn

UnbondingEntry

UnbondingEntry.Comparer

UnbondingFactory

UnbondingRef

## Interfaces

IDelegatee

IDelegateeMetadata

IDelegationRepository

IDelegator

IDelegatorMetadata

IUnbonding

## Enums

UnbondingType

# Class Bond

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public sealed class Bond : IBencodable, IEquatable<Bond>
```

**Inheritance**

object ← Bond

**Implements**

IBencodable, IEquatable<Bond>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Bond(Address)

```
public Bond(Address address)
```

#### Parameters

`address` Address

### Bond(Address, IValue)

```
public Bond(Address address, IValue bencoded)
```

#### Parameters

`address` Address

`bencoded` IValue

## Bond(Address, List)

```
public Bond(Address address, List bencoded)
```

### Parameters

`address` Address

`bencoded` List

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## IsEmpty

```
public bool IsEmpty { get; }
```

### Property Value

bool

## LastDistributeHeight

```
public long? LastDistributeHeight { get; }
```

### Property Value

long?

## Share

```
public BigInteger Share { get; }
```

### Property Value

BigInteger

# Methods

## Equals(Bond?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(Bond? other)
```

### Parameters

`other` Bond

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

# Class CurrencyComparer

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

Represents a comparer for Libplanet.Types.Assets.Currency.

```
public abstract class CurrencyComparer : IComparer<Currency>
```

**Inheritance**

object ← CurrencyComparer

**Implements**

IComparer<Currency>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### HashBytes

Gets a CurrencyComparer that compares Libplanet.Types.Assets.Currency by their hash byte representation.

```
public static CurrencyComparer HashBytes { get; }
```

#### Property Value

CurrencyComparer

## Methods

### Compare(Currency, Currency)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public abstract int Compare(Currency x, Currency y)
```

## Parameters

x Currency

The first object to compare.

y Currency

The second object to compare.

## Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

| Value | Meaning |
| --- | --- |
| **Less than zero** | x is less than y. |
| **Zero** | x equals y. |
| **Greater than zero** | x is greater than y. |

# Class DelegateeMetadata

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class DelegateeMetadata : IDelegateeMetadata, IBencodable
```

**Inheritance**

object ← DelegateeMetadata

**Implements**

IDelegateeMetadata, IBencodable

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### DelegateeMetadata(Address, Address, IValue)

```
public DelegateeMetadata(Address delegateeAddress, Address delegateeAccountAddress,
IValue bencoded)
```

#### Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`bencoded` IValue

### DelegateeMetadata(Address, Address, List)

```
public DelegateeMetadata(Address delegateeAddress, Address delegateeAccountAddress,
List bencoded)
```

### Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`bencoded` List

## DelegateeMetadata(Address, Address, Currency, IEnumerable<Currency>, Address, Address, Address, Address, long, int, int)

Create a new instance of DelegateeMetadata.

```
public DelegateeMetadata(Address delegateeAddress, Address delegateeAccountAddress,
Currency delegationCurrency, IEnumerable<Currency> rewardCurrencies, Address
delegationPoolAddress, Address rewardPoolAddress, Address
rewardRemainderPoolAddress, Address slashedPoolAddress, long unbondingPeriod, int
maxUnbondLockInEntries, int maxRebondGraceEntries)
```

### Parameters

`delegateeAddress` Address

The Address of the Delegatee<T, TSelf>.

`delegateeAccountAddress` Address

The Address of the account of the Delegatee<T, TSelf>.

`delegationCurrency` Currency

The Libplanet.Types.Assets.Currency used for delegation.

`rewardCurrencies` IEnumerable<Currency>

The enumerable of Libplanet.Types.Assets.Currencys used for reward.

`delegationPoolAddress` Address

The Address of the delegation pool that stores delegated Libplanet.Types.Assets.FungibleAssetValues.

`rewardPoolAddress` Address

The Address of the reward pool that gathers rewards to be distributed.

`rewardRemainderPoolAddress` Address

The Address of the reward remainder pool to sends the remainder of the rewards to.

`slashedPoolAddress` Address

The Address of the pool that sends the slashed Libplanet.Types.Assets.FungibleAsset Values to.

`unbondingPeriod` long

The period in blocks that the unbonded Libplanet.Types.Assets.FungibleAssetValues can be withdrawn.

`maxUnbondLockInEntries` int

The maximum number of entries that can be locked in for unbonding.

`maxRebondGraceEntries` int

The maximum number of entries that can be locked in for rebonding.

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## DelegateeAccountAddress

```
public Address DelegateeAccountAddress { get; }
```

### Property Value

Address

## DelegateeAddress

```
public Address DelegateeAddress { get; }
```

### Property Value

Address

## DelegationCurrency

```
public Currency DelegationCurrency { get; }
```

### Property Value

Currency

## DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

### Property Value

Address

## Jailed

```
public bool Jailed { get; }
```

### Property Value

bool

## JailedUntil

```
public long JailedUntil { get; }
```

### Property Value

long

## MaxRebondGraceEntries

```
public int MaxRebondGraceEntries { get; }
```

### Property Value

int

## MaxUnbondLockInEntries

```
public int MaxUnbondLockInEntries { get; }
```

### Property Value

int

## RewardCurrencies

```
public ImmutableSortedSet<Currency> RewardCurrencies { get; }
```

### Property Value

ImmutableSortedSet<Currency>

## RewardPoolAddress

```
public Address RewardPoolAddress { get; }
```

### Property Value

Address

## RewardRemainderPoolAddress

```
public Address RewardRemainderPoolAddress { get; }
```

### Property Value

Address

## SlashedPoolAddress

```
public Address SlashedPoolAddress { get; }
```

### Property Value

Address

## Tombstoned

```
public bool Tombstoned { get; }
```

### Property Value

bool

## TotalDelegatedFAV

```
public FungibleAssetValue TotalDelegatedFAV { get; }
```

### Property Value

FungibleAssetValue

## TotalShares

```
public BigInteger TotalShares { get; }
```

### Property Value

BigInteger

## UnbondingPeriod

```
public long UnbondingPeriod { get; }
```

### Property Value

long

## UnbondingRefs

```
public ImmutableSortedSet<UnbondingRef> UnbondingRefs { get; }
```

### Property Value

ImmutableSortedSet<UnbondingRef>

## Methods

### AddDelegatedFAV(FungibleAssetValue)

```
public void AddDelegatedFAV(FungibleAssetValue fav)
```

#### Parameters

`fav` FungibleAssetValue

### AddShare(BigInteger)

```
public void AddShare(BigInteger share)
```

#### Parameters

`share` BigInteger

### AddUnbondingRef(UnbondingRef)

```
public void AddUnbondingRef(UnbondingRef unbondingRef)
```

#### Parameters

`unbondingRef` UnbondingRef

### BondAddress(Address)

```
public Address BondAddress(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

Address

## CurrentLumpSumRewardsRecordAddress()

Get the Address of the current lump sum rewards record. This will be removed after the migration is done.

```
public virtual Address CurrentLumpSumRewardsRecordAddress()
```

### Returns

Address

Address of the current lump sum rewards record.

## CurrentRewardBaseAddress()

Get the Address of the current RewardBase.

```
public virtual Address CurrentRewardBaseAddress()
```

### Returns

Address

Address of the current RewardBase.

## DistributionPoolAddress()

Get the Address of the distribution pool where the rewards are distributed from.

```
public virtual Address DistributionPoolAddress()
```

### Returns

Address

Address of the distribution pool.

## Equals(IDelegateeMetadata?)

Check if the given IDelegateeMetadata is equal to this instance.

```
public virtual bool Equals(IDelegateeMetadata? other)
```

### Parameters

`other` IDelegateeMetadata

The IDelegateeMetadata to compare with this instance.

### Returns

bool

true if the given IDelegateeMetadata is equal to this instance;

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## FAVFromShare(BigInteger)

```
public FungibleAssetValue FAVFromShare(BigInteger share)
```

### Parameters

`share` BigInteger

### Returns

FungibleAssetValue

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## LumpSumRewardsRecordAddress(long)

Get the Address of the lump sum rewards record at the given height. This will be removed after the migration is done.

```
public virtual Address LumpSumRewardsRecordAddress(long height)
```

### Parameters

`height` long

The height of the lump sum rewards record.

### Returns

Address

Address of the lump sum rewards record at the given height.

## RebondGraceAddress(Address)

```
public virtual Address RebondGraceAddress(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

Address

## RemoveDelegatedFAV(FungibleAssetValue)

```
public void RemoveDelegatedFAV(FungibleAssetValue fav)
```

### Parameters

`fav` FungibleAssetValue

## RemoveShare(BigInteger)

```
public void RemoveShare(BigInteger share)
```

### Parameters

`share` BigInteger

## RemoveUnbondingRef(UnbondingRef)

```
public void RemoveUnbondingRef(UnbondingRef unbondingRef)
```

### Parameters

`unbondingRef` UnbondingRef

## RewardBaseAddress(long)

Get the Address of the RewardBase at the given height.

```
public virtual Address RewardBaseAddress(long height)
```

### Parameters

`height` long

### Returns

Address

Address of the RewardBase at the given height.

## ShareFromFAV(FungibleAssetValue)

```
public BigInteger ShareFromFAV(FungibleAssetValue fav)
```

### Parameters

`fav` FungibleAssetValue

### Returns

BigInteger

## UnbondLockInAddress(Address)

```
public Address UnbondLockInAddress(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

Address

## UpdateUnbondingPeriod(long)

Update the unbonding period.

```
public void UpdateUnbondingPeriod(long unbondingPeriod)
```

### Parameters

`unbondingPeriod` long

The new unbonding period.

# Class Delegatee<TRepository, TDelegatee, TDelegator>

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public abstract class Delegatee<TRepository, TDelegatee, TDelegator> : IDelegatee
where TRepository : DelegationRepository<TRepository, TDelegatee, TDelegator> where
TDelegatee : Delegatee<TRepository, TDelegatee, TDelegator> where TDelegator :
Delegator<TRepository, TDelegatee, TDelegator>
```

## Type Parameters

`TRepository`

`TDelegatee`

`TDelegator`

**Inheritance**

object ← Delegatee<TRepository, TDelegatee, TDelegator>

**Implements**

IDelegatee

**Derived**

GuildDelegatee, ValidatorDelegatee

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Delegatee(Address, Address, Currency, IEnumerable<Currency>, Address, Address, Address, Address, long, int, int, TRepository)

```
public Delegatee(Address address, Address accountAddress, Currency
delegationCurrency, IEnumerable<Currency> rewardCurrencies, Address
delegationPoolAddress, Address rewardPoolAddress, Address
rewardRemainderPoolAddress, Address slashedPoolAddress, long unbondingPeriod, int
maxUnbondLockInEntries, int maxRebondGraceEntries, TRepository repository)
```

### Parameters

`address` Address

`accountAddress` Address

`delegationCurrency` Currency

`rewardCurrencies` IEnumerable<Currency>

`delegationPoolAddress` Address

`rewardPoolAddress` Address

`rewardRemainderPoolAddress` Address

`slashedPoolAddress` Address

`unbondingPeriod` long

`maxUnbondLockInEntries` int

`maxRebondGraceEntries` int

`repository` TRepository

## Delegatee(Address, TRepository)

```
public Delegatee(Address address, TRepository repository)
```

### Parameters

`address` Address

`repository` TRepository

# Properties

## AccountAddress

```
public Address AccountAddress { get; }
```

### Property Value

Address

## Address

```
public Address Address { get; }
```

### Property Value

Address

## DelegationCurrency

```
public Currency DelegationCurrency { get; }
```

### Property Value

Currency

## DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

### Property Value

Address

## Jailed

```
public bool Jailed { get; }
```

### Property Value

bool

## JailedUntil

```
public long JailedUntil { get; }
```

### Property Value

long

## MaxRebondGraceEntries

```
public int MaxRebondGraceEntries { get; }
```

### Property Value

int

## MaxUnbondLockInEntries

```
public int MaxUnbondLockInEntries { get; }
```

### Property Value

int

## Metadata

```
public DelegateeMetadata Metadata { get; }
```

### Property Value

DelegateeMetadata

## MetadataAddress

```
public Address MetadataAddress { get; }
```

### Property Value

Address

## MetadataBencoded

```
public List MetadataBencoded { get; }
```

### Property Value

List

## Repository

```
public TRepository Repository { get; }
```

### Property Value

TRepository

## RewardCurrencies

```
public ImmutableSortedSet<Currency> RewardCurrencies { get; }
```

### Property Value

ImmutableSortedSet<Currency>

## RewardPoolAddress

```
public Address RewardPoolAddress { get; }
```

### Property Value

Address

## RewardRemainderPoolAddress

```
public Address RewardRemainderPoolAddress { get; }
```

### Property Value

Address

## SlashedPoolAddress

```
public Address SlashedPoolAddress { get; }
```

### Property Value

Address

## Tombstoned

```
public bool Tombstoned { get; }
```

### Property Value

bool

## TotalDelegated

```
public FungibleAssetValue TotalDelegated { get; }
```

### Property Value

FungibleAssetValue

## TotalShares

```
public BigInteger TotalShares { get; }
```

### Property Value

BigInteger

## UnbondingPeriod

```
public long UnbondingPeriod { get; }
```

### Property Value

long

# Methods

## Bond(TDelegator, FungibleAssetValue, long)

```
public virtual BigInteger Bond(TDelegator delegator, FungibleAssetValue fav,
long height)
```

### Parameters

`delegator` TDelegator

`fav` FungibleAssetValue

`height` long

### Returns

BigInteger

## BondAddress(Address)

```
public Address BondAddress(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

Address

## CalculateRewards(BigInteger, RewardBase, RewardBase?)

```
public IEnumerable<FungibleAssetValue> CalculateRewards(BigInteger share, RewardBase
currentRewardBase, RewardBase? lastRewardBase)
```

### Parameters

`share` BigInteger

`currentRewardBase` RewardBase

`lastRewardBase` RewardBase

### Returns

IEnumerable<FungibleAssetValue>

## CollectRewards(long)

```
public void CollectRewards(long height)
```

### Parameters

`height` long

## CurrentLumpSumRewardsRecordAddress()

```
public Address CurrentLumpSumRewardsRecordAddress()
```

### Returns

Address

## CurrentRewardBaseAddress()

Get the Address of the current RewardBase.

```
public Address CurrentRewardBaseAddress()
```

### Returns

Address

Address of the current RewardBase.

## DistributeReward(TDelegator, long)

```
public void DistributeReward(TDelegator delegator, long height)
```

### Parameters

`delegator` TDelegator

`height` long

## DistributionPoolAddress()

Get the Address of the distribution pool where the rewards are distributed from.

```
public Address DistributionPoolAddress()
```

### Returns

Address

Address of the distribution pool.

## FAVFromShare(BigInteger)

```
public FungibleAssetValue FAVFromShare(BigInteger share)
```

### Parameters

`share` BigInteger

### Returns

FungibleAssetValue

## Jail(long)

```
public void Jail(long releaseHeight)
```

### Parameters

`releaseHeight` long

## LumpSumRewardsRecordAddress(long)

```
public Address LumpSumRewardsRecordAddress(long height)
```

### Parameters

`height` long

### Returns

Address

## OnDelegationChanged(long)

```
protected virtual void OnDelegationChanged(long height)
```

### Parameters

`height` long

## OnEnjailed()

```
protected virtual void OnEnjailed()
```

## OnUnjailed()

```
protected virtual void OnUnjailed()
```

## RebondGraceAddress(Address)

```
public Address RebondGraceAddress(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

Address

## RewardBaseAddress(long)

Get the Address of the RewardBase at the given height.

```
public Address RewardBaseAddress(long height)
```

### Parameters

`height` long

### Returns

Address

Address of the RewardBase at the given height.

## ShareFromFAV(FungibleAssetValue)

```
public BigInteger ShareFromFAV(FungibleAssetValue fav)
```

### Parameters

`fav` FungibleAssetValue

### Returns

BigInteger

## Slash(BigInteger, long, long)

```
public virtual void Slash(BigInteger slashFactor, long infractionHeight,
long height)
```

### Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

## StartNewRewardPeriod(long)

Start a new reward period. It generates a new RewardBase and archives the current one.

```
public void StartNewRewardPeriod(long height)
```

### Parameters

`height` long

The height of the block where the new reward period starts.

## Tombstone()

```
public void Tombstone()
```

# Unbond(TDelegator, BigInteger, long)

```
public FungibleAssetValue Unbond(TDelegator delegator, BigInteger share,
long height)
```

## Parameters

`delegator` TDelegator

`share` BigInteger

`height` long

## Returns

FungibleAssetValue

# UnbondLockInAddress(Address)

```
public Address UnbondLockInAddress(Address delegatorAddress)
```

## Parameters

`delegatorAddress` Address

## Returns

Address

# Unjail(long)

```
public void Unjail(long height)
```

## Parameters

`height` long

# Class DelegationAddress

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public static class DelegationAddress
```

**Inheritance**

object ← DelegationAddress

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## BondAddress(Address, Address)

```
public static Address BondAddress(Address delegateeMetadataAddress,
Address delegatorAddress)
```

### Parameters

`delegateeMetadataAddress` Address

`delegatorAddress` Address

### Returns

Address

## BondAddress(Address, Address, Address)

```
public static Address BondAddress(Address delegateeAddress, Address
delegateeAccountAddress, Address delegatorAddress)
```

### Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`delegatorAddress` Address

### Returns

Address

## CurrentRewardBaseAddress(Address)

Get the Libplanet.Crypto.Address of the current RewardBase.

```
public static Address CurrentRewardBaseAddress(Address delegateeMetadataAddress)
```

### Parameters

`delegateeMetadataAddress` Address

Libplanet.Crypto.Address of the DelegateeMetadata.

### Returns

Address

Libplanet.Crypto.Address of the current RewardBase.

## CurrentRewardBaseAddress(Address, Address)

Get the Libplanet.Crypto.Address of the current RewardBase.

```
public static Address CurrentRewardBaseAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

### Parameters

`delegateeAddress` Address

Libplanet.Crypto.Address of the Delegatee<T, TSelf>.

`delegateeAccountAddress` Address

Libplanet.Crypto.Address of the account of the Delegatee<T, TSelf>.

### Returns

Address

Libplanet.Crypto.Address of the current RewardBase.

## DelegateeMetadataAddress(Address, Address)

```
public static Address DelegateeMetadataAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

### Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

### Returns

Address

## DelegationPoolAddress(Address)

```
public static Address DelegationPoolAddress(Address delegateeMetadataAddress)
```

### Parameters

`delegateeMetadataAddress` Address

### Returns

Address

## DelegationPoolAddress(Address, Address)

```
public static Address DelegationPoolAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

### Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

### Returns

Address

## DelegatorMetadataAddress(Address, Address)

```
public static Address DelegatorMetadataAddress(Address delegatorAddress,
Address delegatorAccountAddress)
```

### Parameters

`delegatorAddress` Address

`delegatorAccountAddress` Address

### Returns

Address

## DistributionPoolAddress(Address)

Get the Libplanet.Crypto.Address of the distribution pool where the rewards are distributed from.

```
public static Address DistributionPoolAddress(Address delegateeMetadataAddress)
```

### Parameters

`delegateeMetadataAddress` Address

Libplanet.Crypto.Address of the DelegateeMetadata.

### Returns

Address

Libplanet.Crypto.Address of the distribution pool.

## DistributionPoolAddress(Address, Address)

Get the Libplanet.Crypto.Address of the distribution pool where the rewards are distributed from.

```
public static Address DistributionPoolAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

### Parameters

`delegateeAddress` Address

Libplanet.Crypto.Address of the Delegatee<T, TSelf>.

`delegateeAccountAddress` Address

Libplanet.Crypto.Address of the account of the Delegatee<T, TSelf>.

### Returns

Address

Libplanet.Crypto.Address of the distribution pool.

## RebondGraceAddress(Address, Address)

```
public static Address RebondGraceAddress(Address delegateeMetadataAddress,
Address delegatorAddress)
```

### Parameters

`delegateeMetadataAddress` Address

`delegatorAddress` Address

### Returns

Address

## RebondGraceAddress(Address, Address, Address)

```
public static Address RebondGraceAddress(Address delegateeAddress, Address
delegateeAccountAddress, Address delegatorAddress)
```

### Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`delegatorAddress` Address

### Returns

Address

## RewardBaseAddress(Address, Address, long)

Get the Libplanet.Crypto.Address of the RewardBase at the given height.

```
public static Address RewardBaseAddress(Address delegateeAddress, Address
delegateeAccountAddress, long height)
```

### Parameters

`delegateeAddress` Address

Libplanet.Crypto.Address of the Delegatee<T, TSelf>.

`delegateeAccountAddress` Address

Libplanet.Crypto.Address of the account of the Delegatee<T, TSelf>.

`height` long

The height of the RewardBase.

### Returns

Address

Libplanet.Crypto.Address of the RewardBase at the given height.

## RewardBaseAddress(Address, long)

Get the Libplanet.Crypto.Address of the RewardBase at the given height.

```
public static Address RewardBaseAddress(Address delegateeMetadataAddress,
long height)
```

### Parameters

`delegateeMetadataAddress` Address

Libplanet.Crypto.Address of the DelegateeMetadata.

`height` long

The height of the RewardBase.

### Returns

Address

Libplanet.Crypto.Address of the RewardBase at the given height.

## RewardPoolAddress(Address)

```
public static Address RewardPoolAddress(Address delegateeMetadataAddress)
```

### Parameters

`delegateeMetadataAddress` Address

### Returns

Address

## RewardPoolAddress(Address, Address)

```
public static Address RewardPoolAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

### Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

### Returns

Address

## UnbondLockInAddress(Address, Address)

```
public static Address UnbondLockInAddress(Address delegateeMetadataAddress,
Address delegatorAddress)
```

### Parameters

`delegateeMetadataAddress` Address

`delegatorAddress` Address

## Returns

Address

# UnbondLockInAddress(Address, Address, Address)

```
public static Address UnbondLockInAddress(Address delegateeAddress, Address
delegateeAccountAddress, Address delegatorAddress)
```

## Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`delegatorAddress` Address

## Returns

Address

# Class DelegationRepository<TRepository, TDelegatee, TDelegator>

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public abstract class DelegationRepository<TRepository, TDelegatee, TDelegator> :
IDelegationRepository where TRepository : DelegationRepository<TRepository,
TDelegatee, TDelegator> where TDelegatee : Delegatee<TRepository, TDelegatee,
TDelegator> where TDelegator : Delegator<TRepository, TDelegatee, TDelegator>
```

## Type Parameters

`TRepository`

`TDelegatee`

`TDelegator`

**Inheritance**

object ← DelegationRepository<TRepository, TDelegatee, TDelegator>

**Implements**

IDelegationRepository

**Derived**

GuildRepository, ValidatorRepository

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## DelegationRepository(IWorld, IActionContext, Address, Address, Address, Address, Address, Address, Address, Address, Address)

```
public DelegationRepository(IWorld world, IActionContext actionContext, Address
delegateeAccountAddress, Address delegatorAccountAddress, Address
delegateeMetadataAccountAddress, Address delegatorMetadataAccountAddress, Address
bondAccountAddress, Address unbondLockInAccountAddress, Address
rebondGraceAccountAddress, Address unbondingSetAccountAddress, Address
rewardBaseAccountAddress, Address lumpSumRewardRecordAccountAddress)
```

### Parameters

`world` IWorld

`actionContext` IActionContext

`delegateeAccountAddress` Address

`delegatorAccountAddress` Address

`delegateeMetadataAccountAddress` Address

`delegatorMetadataAccountAddress` Address

`bondAccountAddress` Address

`unbondLockInAccountAddress` Address

`rebondGraceAccountAddress` Address

`unbondingSetAccountAddress` Address

`rewardBaseAccountAddress` Address

`lumpSumRewardRecordAccountAddress` Address

## Fields

## bondAccount

```
protected IAccount bondAccount
```

### Field Value

IAccount

## delegateeAccount

```
protected IAccount delegateeAccount
```

### Field Value

IAccount

## delegateeMetadataAccount

```
protected IAccount delegateeMetadataAccount
```

### Field Value

IAccount

## delegatorAccount

```
protected IAccount delegatorAccount
```

### Field Value

IAccount

## delegatorMetadataAccount

```
protected IAccount delegatorMetadataAccount
```

### Field Value

IAccount

# lumpSumRewardsRecordAccount

```
protected IAccount lumpSumRewardsRecordAccount
```

## Field Value

IAccount

# previousWorld

```
protected IWorld previousWorld
```

## Field Value

IWorld

# rebondGraceAccount

```
protected IAccount rebondGraceAccount
```

## Field Value

IAccount

# rewardBaseAccount

```
protected IAccount rewardBaseAccount
```

## Field Value

IAccount

# unbondLockInAccount

```
protected IAccount unbondLockInAccount
```

### Field Value

IAccount

## unbondingSetAccount

```
protected IAccount unbondingSetAccount
```

### Field Value

IAccount

# Properties

## ActionContext

Libplanet.Action.IActionContext of the current action.

```
public IActionContext ActionContext { get; }
```

### Property Value

IActionContext

## BondAccountAddress

Libplanet.Crypto.Address of the Bond account.

```
public Address BondAccountAddress { get; }
```

### Property Value

Address

## DelegateeAccountAddress

```
public Address DelegateeAccountAddress { get; }
```

### Property Value

Address

## DelegateeMetadataAccountAddress

```
public Address DelegateeMetadataAccountAddress { get; }
```

### Property Value

Address

## DelegatorAccountAddress

```
public Address DelegatorAccountAddress { get; }
```

### Property Value

Address

## DelegatorMetadataAccountAddress

```
public Address DelegatorMetadataAccountAddress { get; }
```

### Property Value

Address

## LumpSumRewardsRecordAccountAddress

Libplanet.Crypto.Address of the LumpSumRewardsRecord account.

```
public Address LumpSumRewardsRecordAccountAddress { get; }
```

### Property Value

Address

## RebondGraceAccountAddress

Libplanet.Crypto.Address of the RebondGrace account

```
public Address RebondGraceAccountAddress { get; }
```

### Property Value

Address

## RewardBaseAccountAddress

Libplanet.Crypto.Address of the RewardBase account.

```
public Address RewardBaseAccountAddress { get; }
```

### Property Value

Address

## UnbondLockInAccountAddress

Libplanet.Crypto.Address of the UnbondLockIn account.

```
public Address UnbondLockInAccountAddress { get; }
```

### Property Value

Address

## UnbondingSetAccountAddress

Libplanet.Crypto.Address of the UnbondingSet account.

```
public Address UnbondingSetAccountAddress { get; }
```

### Property Value

Address

## World

```
public virtual IWorld World { get; }
```

### Property Value

IWorld

# Methods

## GetBalance(Address, Currency)

```
public FungibleAssetValue GetBalance(Address address, Currency currency)
```

### Parameters

`address` Address

`currency` Currency

### Returns

## GetBond(TDelegatee, Address)

```
public Bond GetBond(TDelegatee delegatee, Address delegatorAddress)
```

### Parameters

`delegatee` TDelegatee

`delegatorAddress` Address

### Returns

Bond

## GetCurrentLumpSumRewardsRecord(TDelegatee)

```
public LumpSumRewardsRecord? GetCurrentLumpSumRewardsRecord(TDelegatee delegatee)
```

### Parameters

`delegatee` TDelegatee

### Returns

LumpSumRewardsRecord

## GetCurrentRewardBase(TDelegatee)

```
public RewardBase? GetCurrentRewardBase(TDelegatee delegatee)
```

### Parameters

`delegatee` TDelegatee

### Returns

RewardBase

## GetDelegatee(Address)

```
public abstract TDelegatee GetDelegatee(Address address)
```

### Parameters

`address` Address

### Returns

TDelegatee

## GetDelegateeMetadata(Address)

```
public DelegateeMetadata GetDelegateeMetadata(Address delegateeAddress)
```

### Parameters

`delegateeAddress` Address

### Returns

DelegateeMetadata

## GetDelegator(Address)

```
public abstract TDelegator GetDelegator(Address address)
```

### Parameters

`address` Address

### Returns

TDelegator

## GetDelegatorMetadata(Address)

```
public DelegatorMetadata GetDelegatorMetadata(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

DelegatorMetadata

## GetLumpSumRewardsRecord(TDelegatee, long)

```
public LumpSumRewardsRecord? GetLumpSumRewardsRecord(TDelegatee delegatee,
long height)
```

### Parameters

`delegatee` TDelegatee

`height` long

### Returns

LumpSumRewardsRecord

## GetRebondGrace(TDelegatee, Address)

```
public RebondGrace GetRebondGrace(TDelegatee delegatee, Address delegatorAddress)
```

### Parameters

`delegatee` TDelegatee

`delegatorAddress` Address

### Returns

RebondGrace

## GetRewardBase(TDelegatee, long)

```
public RewardBase? GetRewardBase(TDelegatee delegatee, long height)
```

### Parameters

`delegatee` TDelegatee

`height` long

### Returns

RewardBase

## GetUnbondLockIn(TDelegatee, Address)

```
public UnbondLockIn GetUnbondLockIn(TDelegatee delegatee, Address delegatorAddress)
```

### Parameters

`delegatee` TDelegatee

`delegatorAddress` Address

### Returns

UnbondLockIn

## GetUnlimitedRebondGrace(Address)

```
public RebondGrace GetUnlimitedRebondGrace(Address address)
```

### Parameters

`address` Address

### Returns

RebondGrace

## GetUnlimitedUnbondLockIn(Address)

```
public UnbondLockIn GetUnlimitedUnbondLockIn(Address address)
```

### Parameters

`address` Address

### Returns

UnbondLockIn

## RemoveLumpSumRewardsRecord(LumpSumRewardsRecord)

Remove the LumpSumRewardsRecord from the IDelegatee. This is used when the LumpSumRewardsRecord is no longer needed. This can be removed when the migration for RewardBase is done.

```
public void RemoveLumpSumRewardsRecord(LumpSumRewardsRecord lumpSumRewardsRecord)
```

### Parameters

`lumpSumRewardsRecord` LumpSumRewardsRecord

The LumpSumRewardsRecord to remove.

## SetBond(Bond)

```
public void SetBond(Bond bond)
```

### Parameters

`bond` Bond

## SetDelegatee(TDelegatee)

```
public abstract void SetDelegatee(TDelegatee delegatee)
```

### Parameters

`delegatee` TDelegatee

## SetDelegateeMetadata(DelegateeMetadata)

```
public void SetDelegateeMetadata(DelegateeMetadata delegateeMetadata)
```

### Parameters

`delegateeMetadata` DelegateeMetadata

## SetDelegator(TDelegator)

```
public abstract void SetDelegator(TDelegator delegator)
```

### Parameters

`delegator` TDelegator

## SetDelegatorMetadata(DelegatorMetadata)

```
public void SetDelegatorMetadata(DelegatorMetadata delegatorMetadata)
```

### Parameters

`delegatorMetadata` DelegatorMetadata

## SetLumpSumRewardsRecord(LumpSumRewardsRecord)

```
public void SetLumpSumRewardsRecord(LumpSumRewardsRecord lumpSumRewardsRecord)
```

### Parameters

`lumpSumRewardsRecord` LumpSumRewardsRecord

## SetRebondGrace(RebondGrace)

```
public void SetRebondGrace(RebondGrace rebondGrace)
```

### Parameters

`rebondGrace` RebondGrace

## SetRewardBase(RewardBase)

Set the RewardBase of the IDelegatee.

```
public void SetRewardBase(RewardBase rewardBase)
```

### Parameters

`rewardBase` RewardBase

The RewardBase to set.

## SetUnbondLockIn(UnbondLockIn)

```
public void SetUnbondLockIn(UnbondLockIn unbondLockIn)
```

### Parameters

`unbondLockIn` UnbondLockIn

## TransferAsset(Address, Address, FungibleAssetValue)

```
public void TransferAsset(Address sender, Address recipient,
FungibleAssetValue value)
```

### Parameters

`sender` Address

`recipient` Address

`value` FungibleAssetValue

## UpdateWorld(IWorld)

```
public virtual void UpdateWorld(IWorld world)
```

### Parameters

`world` IWorld

# Class DelegatorMetadata

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class DelegatorMetadata : IDelegatorMetadata, IBencodable
```

**Inheritance**

object ← DelegatorMetadata

**Implements**

IDelegatorMetadata, IBencodable

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### DelegatorMetadata(Address, Address, IValue)

```
public DelegatorMetadata(Address address, Address accountAddress, IValue bencoded)
```

#### Parameters

`address` Address

`accountAddress` Address

`bencoded` IValue

### DelegatorMetadata(Address, Address, List)

```
public DelegatorMetadata(Address delegatorAddress, Address delegatorAccountAddress,
List bencoded)
```

### Parameters

`delegatorAddress` Address

`delegatorAccountAddress` Address

`bencoded` List

## DelegatorMetadata(Address, Address, Address, Address)

```
public DelegatorMetadata(Address address, Address accountAddress, Address
delegationPoolAddress, Address rewardAddress)
```

### Parameters

`address` Address

`accountAddress` Address

`delegationPoolAddress` Address

`rewardAddress` Address

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## Delegatees

```
public ImmutableSortedSet<Address> Delegatees { get; }
```

### Property Value

ImmutableSortedSet<Address>

## DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

### Property Value

Address

## DelegatorAccountAddress

```
public Address DelegatorAccountAddress { get; }
```

### Property Value

Address

## DelegatorAddress

```
public Address DelegatorAddress { get; }
```

### Property Value

Address

## RewardAddress

```
public Address RewardAddress { get; }
```

### Property Value

Address

## UnbondingRefs

```
public ImmutableSortedSet<UnbondingRef> UnbondingRefs { get; }
```

### Property Value

ImmutableSortedSet<UnbondingRef>

## Methods

## AddDelegatee(Address)

```
public void AddDelegatee(Address delegatee)
```

### Parameters

`delegatee` Address

## AddUnbondingRef(UnbondingRef)

```
public void AddUnbondingRef(UnbondingRef unbondingRef)
```

### Parameters

`unbondingRef` UnbondingRef

## Equals(IDelegatorMetadata?)

Check if the given IDelegatorMetadata is equal to this instance.

```
public virtual bool Equals(IDelegatorMetadata? other)
```

### Parameters

`other` IDelegatorMetadata

The IDelegatorMetadata to compare with this instance.

### Returns

bool

true if the given IDelegatorMetadata is equal to this instance;

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## RemoveDelegatee(Address)

```
public void RemoveDelegatee(Address delegatee)
```

### Parameters

`delegatee` Address

## RemoveUnbondingRef(UnbondingRef)

```
public void RemoveUnbondingRef(UnbondingRef unbondingRef)
```

### Parameters

`unbondingRef` UnbondingRef

# Class Delegator<TRepository, TDelegatee, TDelegator>

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public abstract class Delegator<TRepository, TDelegatee, TDelegator> : IDelegator
where TRepository : DelegationRepository<TRepository, TDelegatee, TDelegator> where
TDelegatee : Delegatee<TRepository, TDelegatee, TDelegator> where TDelegator :
Delegator<TRepository, TDelegatee, TDelegator>
```

## Type Parameters

`TRepository`

`TDelegatee`

`TDelegator`

**Inheritance**

object ← Delegator<TRepository, TDelegatee, TDelegator>

**Implements**

IDelegator

**Derived**

GuildDelegator, ValidatorDelegator

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Delegator(Address, Address, Address, Address, TRepository)

```
public Delegator(Address address, Address accountAddress, Address
delegationPoolAddress, Address rewardAddress, TRepository repository)
```

### Parameters

`address` Address

`accountAddress` Address

`delegationPoolAddress` Address

`rewardAddress` Address

`repository` TRepository

## Delegator(Address, TRepository)

```
public Delegator(Address address, TRepository repository)
```

### Parameters

`address` Address

`repository` TRepository

# Properties

## AccountAddress

```
public Address AccountAddress { get; }
```

### Property Value

Address

## Address

```
public Address Address { get; }
```

### Property Value

Address

## Delegatees

```
public ImmutableSortedSet<Address> Delegatees { get; }
```

### Property Value

ImmutableSortedSet<Address>

## DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

### Property Value

Address

## Metadata

```
public DelegatorMetadata Metadata { get; }
```

### Property Value

DelegatorMetadata

## MetadataAddress

```
public Address MetadataAddress { get; }
```

### Property Value

Address

## MetadataBencoded

```
public List MetadataBencoded { get; }
```

### Property Value

List

## Repository

```
public TRepository Repository { get; }
```

### Property Value

TRepository

## RewardAddress

```
public Address RewardAddress { get; }
```

### Property Value

Address

## UnbondingRefs

```
public ImmutableSortedSet<UnbondingRef> UnbondingRefs { get; }
```

### Property Value

ImmutableSortedSet<UnbondingRef>

## Methods

## CancelUndelegate(TDelegatee, FungibleAssetValue, long)

```
public void CancelUndelegate(TDelegatee delegatee, FungibleAssetValue fav,
long height)
```

### Parameters

`delegatee` TDelegatee

`fav` FungibleAssetValue

`height` long

## ClaimReward(TDelegatee, long)

```
public void ClaimReward(TDelegatee delegatee, long height)
```

### Parameters

`delegatee` TDelegatee

`height` long

## Delegate(TDelegatee, FungibleAssetValue, long)

```
public virtual void Delegate(TDelegatee delegatee, FungibleAssetValue fav,
long height)
```

### Parameters

`delegatee` TDelegatee

`fav` FungibleAssetValue

`height` long

## OnUnbondingReleased(long, IUnbonding, FungibleAssetValue?)

```
protected virtual void OnUnbondingReleased(long height, IUnbonding
releasedUnbonding, FungibleAssetValue? releasedFAV)
```

### Parameters

`height` long

`releasedUnbonding` IUnbonding

`releasedFAV` FungibleAssetValue?

## Redelegate(TDelegatee, TDelegatee, BigInteger, long)

```
public virtual void Redelegate(TDelegatee srcDelegatee, TDelegatee dstDelegatee,
BigInteger share, long height)
```

### Parameters

`srcDelegatee` TDelegatee

`dstDelegatee` TDelegatee

`share` BigInteger

`height` long

## ReleaseUnbondings(long)

```
public void ReleaseUnbondings(long height)
```

### Parameters

`height` long

## Undelegate(TDelegatee, BigInteger, long)

```
public virtual void Undelegate(TDelegatee delegatee, BigInteger share, long height)
```

### Parameters

`delegatee` TDelegatee

`share` BigInteger

`height` long

# Interface IDelegatee

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegatee
```

## Properties

### AccountAddress

```
Address AccountAddress { get; }
```

#### Property Value

Address

### Address

```
Address Address { get; }
```

#### Property Value

Address

### DelegationCurrency

```
Currency DelegationCurrency { get; }
```

#### Property Value

Currency

## DelegationPoolAddress

```
Address DelegationPoolAddress { get; }
```

### Property Value

Address

## Jailed

```
bool Jailed { get; }
```

### Property Value

bool

## JailedUntil

```
long JailedUntil { get; }
```

### Property Value

long

## MaxRebondGraceEntries

```
int MaxRebondGraceEntries { get; }
```

### Property Value

int

## MaxUnbondLockInEntries

```
int MaxUnbondLockInEntries { get; }
```

### Property Value

int

## RewardCurrencies

```
ImmutableSortedSet<Currency> RewardCurrencies { get; }
```

### Property Value

ImmutableSortedSet<Currency>

## RewardPoolAddress

```
Address RewardPoolAddress { get; }
```

### Property Value

Address

## RewardRemainderPoolAddress

```
Address RewardRemainderPoolAddress { get; }
```

### Property Value

Address

## Tombstoned

```
bool Tombstoned { get; }
```

### Property Value

bool

## TotalDelegated

```
FungibleAssetValue TotalDelegated { get; }
```

### Property Value

FungibleAssetValue

## TotalShares

```
BigInteger TotalShares { get; }
```

### Property Value

BigInteger

## UnbondingPeriod

```
long UnbondingPeriod { get; }
```

### Property Value

long

# Methods

## Bond(IDelegator, FungibleAssetValue, long)

```
BigInteger Bond(IDelegator delegator, FungibleAssetValue fav, long height)
```

### Parameters

`delegator` IDelegator

`fav` FungibleAssetValue

`height` long

### Returns

BigInteger

## BondAddress(Address)

```
Address BondAddress(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

Address

## CollectRewards(long)

```
void CollectRewards(long height)
```

### Parameters

`height` long

## CurrentLumpSumRewardsRecordAddress()

```
Address CurrentLumpSumRewardsRecordAddress()
```

### Returns

Address

## CurrentRewardBaseAddress()

Get the Address of the current RewardBase.

```
Address CurrentRewardBaseAddress()
```

### Returns

Address

The Address of the current RewardBase.

## DistributeReward(IDelegator, long)

```
void DistributeReward(IDelegator delegator, long height)
```

### Parameters

`delegator` IDelegator

`height` long

## FAVFromShare(BigInteger)

```
FungibleAssetValue FAVFromShare(BigInteger share)
```

### Parameters

`share` BigInteger

### Returns

FungibleAssetValue

## Jail(long)

```
void Jail(long releaseHeight)
```

### Parameters

`releaseHeight` long

## LumpSumRewardsRecordAddress(long)

```
Address LumpSumRewardsRecordAddress(long height)
```

### Parameters

`height` long

### Returns

Address

## RebondGraceAddress(Address)

```
Address RebondGraceAddress(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

Address

# RewardBaseAddress(long)

Get the Address of the RewardBase at the given height.

```
Address RewardBaseAddress(long height)
```

## Parameters

`height` long

The height of the RewardBase.

## Returns

Address

The Address of the RewardBase at the given height.

# ShareFromFAV(FungibleAssetValue)

```
BigInteger ShareFromFAV(FungibleAssetValue fav)
```

## Parameters

`fav` FungibleAssetValue

## Returns

BigInteger

# Slash(BigInteger, long, long)

```
void Slash(BigInteger slashFactor, long infractionHeight, long height)
```

## Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

## Tombstone()

```
void Tombstone()
```

## Unbond(IDelegator, BigInteger, long)

```
FungibleAssetValue Unbond(IDelegator delegator, BigInteger share, long height)
```

### Parameters

`delegator` IDelegator

`share` BigInteger

`height` long

### Returns

FungibleAssetValue

## UnbondLockInAddress(Address)

```
Address UnbondLockInAddress(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

Address

# Unjail(long)

```
void Unjail(long height)
```

## Parameters

`height` long

# Interface IDelegateeMetadata

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegateeMetadata : IBencodable
```

**Inherited Members**
IBencodable.Bencoded

## Properties

### Address

```
Address Address { get; }
```

#### Property Value

Address

### DelegateeAccountAddress

```
Address DelegateeAccountAddress { get; }
```

#### Property Value

Address

### DelegateeAddress

```
Address DelegateeAddress { get; }
```

#### Property Value

Address

## DelegationCurrency

```
Currency DelegationCurrency { get; }
```

### Property Value

Currency

## DelegationPoolAddress

```
Address DelegationPoolAddress { get; }
```

### Property Value

Address

## Jailed

```
bool Jailed { get; }
```

### Property Value

bool

## JailedUntil

```
long JailedUntil { get; }
```

### Property Value

long

## MaxRebondGraceEntries

```
int MaxRebondGraceEntries { get; }
```

### Property Value

int

## MaxUnbondLockInEntries

```
int MaxUnbondLockInEntries { get; }
```

### Property Value

int

## RewardCurrencies

```
ImmutableSortedSet<Currency> RewardCurrencies { get; }
```

### Property Value

ImmutableSortedSet<Currency>

## RewardPoolAddress

```
Address RewardPoolAddress { get; }
```

### Property Value

Address

## RewardRemainderPoolAddress

```
Address RewardRemainderPoolAddress { get; }
```

### Property Value

Address

## Tombstoned

```
bool Tombstoned { get; }
```

### Property Value

bool

## TotalDelegatedFAV

```
FungibleAssetValue TotalDelegatedFAV { get; }
```

### Property Value

FungibleAssetValue

## TotalShares

```
BigInteger TotalShares { get; }
```

### Property Value

BigInteger

## UnbondingPeriod

```
long UnbondingPeriod { get; }
```

### Property Value

long

## Methods

### FAVFromShare(BigInteger)

```
FungibleAssetValue FAVFromShare(BigInteger share)
```

#### Parameters

`share` BigInteger

#### Returns

FungibleAssetValue

### ShareFromFAV(FungibleAssetValue)

```
BigInteger ShareFromFAV(FungibleAssetValue fav)
```

#### Parameters

`fav` FungibleAssetValue

#### Returns

BigInteger

# Interface IDelegationRepository

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegationRepository
```

## Properties

### ActionContext

```
IActionContext ActionContext { get; }
```

#### Property Value

IActionContext

### DelegateeAccountAddress

```
Address DelegateeAccountAddress { get; }
```

#### Property Value

Address

### DelegatorAccountAddress

```
Address DelegatorAccountAddress { get; }
```

#### Property Value

Address

## World

```
IWorld World { get; }
```

### Property Value

IWorld

# Methods

## GetBalance(Address, Currency)

```
FungibleAssetValue GetBalance(Address address, Currency currency)
```

### Parameters

`address` Address

`currency` Currency

### Returns

FungibleAssetValue

## GetBond(IDelegatee, Address)

```
Bond GetBond(IDelegatee delegatee, Address delegatorAddress)
```

### Parameters

`delegatee` IDelegatee

`delegatorAddress` Address

### Returns

Bond

## GetCurrentLumpSumRewardsRecord(IDelegatee)

```
LumpSumRewardsRecord? GetCurrentLumpSumRewardsRecord(IDelegatee delegatee)
```

### Parameters

`delegatee` IDelegatee

### Returns

LumpSumRewardsRecord

## GetCurrentRewardBase(IDelegatee)

Get the current RewardBase of the `delegatee`.

```
RewardBase? GetCurrentRewardBase(IDelegatee delegatee)
```

### Parameters

`delegatee` IDelegatee

The IDelegatee to get the current RewardBase.

### Returns

RewardBase

The current RewardBase of the `delegatee`.

## GetDelegatee(Address)

```
IDelegatee GetDelegatee(Address address)
```

### Parameters

`address` Address

### Returns

IDelegatee

## GetDelegateeMetadata(Address)

```
DelegateeMetadata GetDelegateeMetadata(Address delegateeAddress)
```

### Parameters

`delegateeAddress` Address

### Returns

DelegateeMetadata

## GetDelegator(Address)

```
IDelegator GetDelegator(Address address)
```

### Parameters

`address` Address

### Returns

IDelegator

## GetDelegatorMetadata(Address)

```
DelegatorMetadata GetDelegatorMetadata(Address delegatorAddress)
```

### Parameters

`delegatorAddress` Address

### Returns

DelegatorMetadata

## GetLumpSumRewardsRecord(IDelegatee, long)

```
LumpSumRewardsRecord? GetLumpSumRewardsRecord(IDelegatee delegatee, long height)
```

### Parameters

`delegatee` IDelegatee

`height` long

### Returns

LumpSumRewardsRecord

## GetRebondGrace(IDelegatee, Address)

```
RebondGrace GetRebondGrace(IDelegatee delegatee, Address delegatorAddress)
```

### Parameters

`delegatee` IDelegatee

`delegatorAddress` Address

### Returns

RebondGrace

## GetRewardBase(IDelegatee, long)

Get the RewardBase of the `delegatee` at the given `height`.

```
RewardBase? GetRewardBase(IDelegatee delegatee, long height)
```

### Parameters

`delegatee` IDelegatee

The IDelegatee to get the RewardBase of.

`height` long

The height to get the RewardBase at.

### Returns

RewardBase

The RewardBase of the `delegatee` at the given `height`.

## GetUnbondLockIn(IDelegatee, Address)

```
UnbondLockIn GetUnbondLockIn(IDelegatee delegatee, Address delegatorAddress)
```

### Parameters

`delegatee` IDelegatee

`delegatorAddress` Address

### Returns

UnbondLockIn

## GetUnlimitedRebondGrace(Address)

```
RebondGrace GetUnlimitedRebondGrace(Address address)
```

### Parameters

`address` Address

### Returns

RebondGrace

## GetUnlimitedUnbondLockIn(Address)

```
UnbondLockIn GetUnlimitedUnbondLockIn(Address address)
```

### Parameters

`address` Address

### Returns

UnbondLockIn

## RemoveLumpSumRewardsRecord(LumpSumRewardsRecord)

Remove the LumpSumRewardsRecord from the IDelegatee. This is used when the LumpSumRewardsRecord is no longer needed. This can be removed when the migration for RewardBase is done.

```
void RemoveLumpSumRewardsRecord(LumpSumRewardsRecord lumpSumRewardsRecord)
```

### Parameters

`lumpSumRewardsRecord` LumpSumRewardsRecord

The LumpSumRewardsRecord to remove.

## SetBond(Bond)

```
void SetBond(Bond bond)
```

### Parameters

`bond` Bond

## SetDelegatee(IDelegatee)

```
void SetDelegatee(IDelegatee delegatee)
```

### Parameters

`delegatee` IDelegatee

## SetDelegateeMetadata(DelegateeMetadata)

```
void SetDelegateeMetadata(DelegateeMetadata delegateeMetadata)
```

### Parameters

`delegateeMetadata` DelegateeMetadata

## SetDelegator(IDelegator)

```
void SetDelegator(IDelegator delegator)
```

### Parameters

`delegator` IDelegator

## SetDelegatorMetadata(DelegatorMetadata)

```
void SetDelegatorMetadata(DelegatorMetadata delegatorMetadata)
```

### Parameters

`delegatorMetadata` DelegatorMetadata

## SetLumpSumRewardsRecord(LumpSumRewardsRecord)

```
void SetLumpSumRewardsRecord(LumpSumRewardsRecord lumpSumRewardsRecord)
```

### Parameters

`lumpSumRewardsRecord` LumpSumRewardsRecord

## SetRebondGrace(RebondGrace)

```
void SetRebondGrace(RebondGrace rebondGrace)
```

### Parameters

`rebondGrace` RebondGrace

## SetRewardBase(RewardBase)

Set the RewardBase of the IDelegatee.

```
void SetRewardBase(RewardBase rewardBase)
```

### Parameters

`rewardBase` RewardBase

The RewardBase to set.

## SetUnbondLockIn(UnbondLockIn)

```
void SetUnbondLockIn(UnbondLockIn unbondLockIn)
```

### Parameters

`unbondLockIn` UnbondLockIn

## TransferAsset(Address, Address, FungibleAssetValue)

```
void TransferAsset(Address sender, Address recipient, FungibleAssetValue value)
```

### Parameters

`sender` Address

`recipient` Address

`value` FungibleAssetValue

## UpdateWorld(IWorld)

```
void UpdateWorld(IWorld world)
```

### Parameters

`world` IWorld

# Interface IDelegator

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegator
```

## Properties

### AccountAddress

```
Address AccountAddress { get; }
```

#### Property Value

Address

### Address

```
Address Address { get; }
```

#### Property Value

Address

### Delegatees

```
ImmutableSortedSet<Address> Delegatees { get; }
```

#### Property Value

ImmutableSortedSet<Address>

## DelegationPoolAddress

```
Address DelegationPoolAddress { get; }
```

### Property Value

Address

## RewardAddress

```
Address RewardAddress { get; }
```

### Property Value

Address

# Methods

## CancelUndelegate(IDelegatee, FungibleAssetValue, long)

```
void CancelUndelegate(IDelegatee delegatee, FungibleAssetValue fav, long height)
```

### Parameters

`delegatee` IDelegatee

`fav` FungibleAssetValue

`height` long

## ClaimReward(IDelegatee, long)

```
void ClaimReward(IDelegatee delegatee, long height)
```

### Parameters

`delegatee` IDelegatee

`height` long

## Delegate(IDelegatee, FungibleAssetValue, long)

```
void Delegate(IDelegatee delegatee, FungibleAssetValue fav, long height)
```

### Parameters

`delegatee` IDelegatee

`fav` FungibleAssetValue

`height` long

## Redelegate(IDelegatee, IDelegatee, BigInteger, long)

```
void Redelegate(IDelegatee srcDelegatee, IDelegatee dstDelegatee, BigInteger share,
long height)
```

### Parameters

`srcDelegatee` IDelegatee

`dstDelegatee` IDelegatee

`share` BigInteger

`height` long

## Undelegate(IDelegatee, BigInteger, long)

```
void Undelegate(IDelegatee delegatee, BigInteger share, long height)
```

## Parameters

`delegatee` IDelegatee

`share` BigInteger

`height` long

# Interface IDelegatorMetadata

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegatorMetadata : IBencodable
```

**Inherited Members**
IBencodable.Bencoded

## Properties

### Address

```
Address Address { get; }
```

#### Property Value

Address

### Delegatees

```
ImmutableSortedSet<Address> Delegatees { get; }
```

#### Property Value

ImmutableSortedSet<Address>

### DelegationPoolAddress

```
Address DelegationPoolAddress { get; }
```

#### Property Value

Address

## DelegatorAccountAddress

```
Address DelegatorAccountAddress { get; }
```

### Property Value

Address

## DelegatorAddress

```
Address DelegatorAddress { get; }
```

### Property Value

Address

# Methods

## AddDelegatee(Address)

```
void AddDelegatee(Address delegatee)
```

### Parameters

`delegatee` Address

## RemoveDelegatee(Address)

```
void RemoveDelegatee(Address delegatee)
```

### Parameters

`delegatee` Address

# Interface IUnbonding

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IUnbonding
```

## Properties

### Address

```
Address Address { get; }
```

#### Property Value

Address

### DelegateeAddress

```
Address DelegateeAddress { get; }
```

#### Property Value

Address

### DelegatorAddress

```
Address DelegatorAddress { get; }
```

#### Property Value

Address

## IsEmpty

```
bool IsEmpty { get; }
```

### Property Value

bool

## IsFull

```
bool IsFull { get; }
```

### Property Value

bool

## LowestExpireHeight

```
long LowestExpireHeight { get; }
```

### Property Value

long

# Methods

## Release(long, out FungibleAssetValue?)

```
IUnbonding Release(long height, out FungibleAssetValue? releasedFAV)
```

### Parameters

`height` long

`releasedFAV` FungibleAssetValue?

## Returns

IUnbonding

# Slash(BigInteger, long, long, Address)

```
IUnbonding Slash(BigInteger slashFactor, long infractionHeight, long height,
Address slashedPoolAddress)
```

## Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

`slashedPoolAddress` Address

## Returns

IUnbonding

# Class LumpSumRewardsRecord

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class LumpSumRewardsRecord : IBencodable, IEquatable<LumpSumRewardsRecord>
```

**Inheritance**

object ← LumpSumRewardsRecord

**Implements**

IBencodable, IEquatable<LumpSumRewardsRecord>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### LumpSumRewardsRecord(Address, IValue)

```
public LumpSumRewardsRecord(Address address, IValue bencoded)
```

#### Parameters

`address` Address

`bencoded` IValue

### LumpSumRewardsRecord(Address, List)

```
public LumpSumRewardsRecord(Address address, List bencoded)
```

#### Parameters

`address` Address

`bencoded` List

## LumpSumRewardsRecord(Address, long, BigInteger, IEnumerable<Currency>)

```
public LumpSumRewardsRecord(Address address, long startHeight, BigInteger
totalShares, IEnumerable<Currency> currencies)
```

### Parameters

`address` Address

`startHeight` long

`totalShares` BigInteger

`currencies` IEnumerable<Currency>

## LumpSumRewardsRecord(Address, long, BigInteger, IEnumerable<Currency>, long?)

```
public LumpSumRewardsRecord(Address address, long startHeight, BigInteger
totalShares, IEnumerable<Currency> currencies, long? lastStartHeight)
```

### Parameters

`address` Address

`startHeight` long

`totalShares` BigInteger

`currencies` IEnumerable<Currency>

`lastStartHeight` long?

# LumpSumRewardsRecord(Address, long, BigInteger, IEnumerable<FungibleAssetValue>, long?)

```
public LumpSumRewardsRecord(Address address, long startHeight, BigInteger
totalShares, IEnumerable<FungibleAssetValue> lumpSumRewards, long? lastStartHeight)
```

## Parameters

`address` Address

`startHeight` long

`totalShares` BigInteger

`lumpSumRewards` IEnumerable<FungibleAssetValue>

`lastStartHeight` long?

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## LastStartHeight

```
public long? LastStartHeight { get; }
```

### Property Value

long?

## LumpSumRewards

```
public ImmutableDictionary<Currency, FungibleAssetValue> LumpSumRewards { get; }
```

### Property Value

ImmutableDictionary<Currency, FungibleAssetValue>

## StartHeight

```
public long StartHeight { get; }
```

### Property Value

long

## TotalShares

```
public BigInteger TotalShares { get; }
```

### Property Value

BigInteger

# Methods

## AddLumpSumRewards(FungibleAssetValue)

```
public LumpSumRewardsRecord AddLumpSumRewards(FungibleAssetValue rewards)
```

### Parameters

`rewards` FungibleAssetValue

### Returns

LumpSumRewardsRecord

## AddLumpSumRewards(LumpSumRewardsRecord, FungibleAssetValue)

```
public static LumpSumRewardsRecord AddLumpSumRewards(LumpSumRewardsRecord record,
FungibleAssetValue rewards)
```

### Parameters

`record` LumpSumRewardsRecord

`rewards` FungibleAssetValue

### Returns

LumpSumRewardsRecord

## AddLumpSumRewards(IEnumerable<FungibleAssetValue>)

```
public LumpSumRewardsRecord AddLumpSumRewards(IEnumerable<FungibleAssetValue>
rewards)
```

### Parameters

`rewards` IEnumerable<FungibleAssetValue>

### Returns

LumpSumRewardsRecord

## Equals(LumpSumRewardsRecord?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(LumpSumRewardsRecord? other)
```

### Parameters

`other` LumpSumRewardsRecord

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## MoveAddress(Address)

```
public LumpSumRewardsRecord MoveAddress(Address address)
```

### Parameters

`address` Address

### Returns

LumpSumRewardsRecord

## RewardsDuringPeriod(BigInteger)

```
public ImmutableSortedDictionary<Currency, FungibleAssetValue>
RewardsDuringPeriod(BigInteger share)
```

### Parameters

`share` BigInteger

### Returns

ImmutableSortedDictionary<Currency, FungibleAssetValue>

# RewardsDuringPeriod(BigInteger, Currency)

```
public FungibleAssetValue RewardsDuringPeriod(BigInteger share, Currency currency)
```

## Parameters

`share` BigInteger

`currency` Currency

## Returns

FungibleAssetValue

# Class RebondGrace

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public sealed class RebondGrace : IUnbonding, IBencodable, IEquatable<RebondGrace>
```

**Inheritance**

object ← RebondGrace

**Implements**

IUnbonding, IBencodable, IEquatable<RebondGrace>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RebondGrace(Address, int, IValue, IDelegationRepository?)

```
public RebondGrace(Address address, int maxEntries, IValue bencoded,
IDelegationRepository? repository = null)
```

#### Parameters

`address` Address

`maxEntries` int

`bencoded` IValue

`repository` IDelegationRepository

### RebondGrace(Address, int, List, IDelegationRepository?)

```
public RebondGrace(Address address, int maxEntries, List bencoded,
IDelegationRepository? repository = null)
```

### Parameters

`address` Address

`maxEntries` int

`bencoded` List

`repository` IDelegationRepository

## RebondGrace(Address, int, Address, Address, IDelegationRepository?)

```
public RebondGrace(Address address, int maxEntries, Address delegateeAddress,
Address delegatorAddress, IDelegationRepository? repository = null)
```

### Parameters

`address` Address

`maxEntries` int

`delegateeAddress` Address

`delegatorAddress` Address

`repository` IDelegationRepository

## RebondGrace(Address, int, Address, Address, IEnumerable<UnbondingEntry>, IDelegationRepository?)

```
public RebondGrace(Address address, int maxEntries, Address delegateeAddress,
Address delegatorAddress, IEnumerable<UnbondingEntry> entries,
IDelegationRepository? repository = null)
```

### Parameters

`address` Address

`maxEntries` int

`delegateeAddress` Address

`delegatorAddress` Address

`entries` IEnumerable<UnbondingEntry>

`repository` IDelegationRepository

## Properties

### Address

```
public Address Address { get; }
```

#### Property Value

Address

### Bencoded

```
public List Bencoded { get; }
```

#### Property Value

List

### DelegateeAddress

```
public Address DelegateeAddress { get; }
```

### Property Value

Address

## DelegatorAddress

```
public Address DelegatorAddress { get; }
```

### Property Value

Address

## Entries

```
public ImmutableSortedDictionary<long, ImmutableList<UnbondingEntry>> Entries {
get; }
```

### Property Value

ImmutableSortedDictionary<long, ImmutableList<UnbondingEntry>>

## FlattenedEntries

```
public ImmutableArray<UnbondingEntry> FlattenedEntries { get; }
```

### Property Value

ImmutableArray<UnbondingEntry>

## IsEmpty

```
public bool IsEmpty { get; }
```

### Property Value

bool

## IsFull

```
public bool IsFull { get; }
```

### Property Value

bool

## LowestExpireHeight

```
public long LowestExpireHeight { get; }
```

### Property Value

long

## MaxEntries

```
public int MaxEntries { get; }
```

### Property Value

int

## Repository

```
public IDelegationRepository? Repository { get; }
```

### Property Value

IDelegationRepository

# Methods

## Equals(RebondGrace?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(RebondGrace? other)
```

### Parameters

`other` RebondGrace

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Release(long, out FungibleAssetValue?)

```
public RebondGrace Release(long height, out FungibleAssetValue? releasedFAV)
```

### Parameters

`height` long

`releasedFAV` FungibleAssetValue?

### Returns

RebondGrace

## Slash(BigInteger, long, long, Address)

```
public RebondGrace Slash(BigInteger slashFactor, long infractionHeight, long height,
Address slashedPoolAddress)
```

### Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

`slashedPoolAddress` Address

## Returns

RebondGrace

# Class RewardBase

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

RewardBase is a class that represents the base of the reward. If it's multiplied by the number of shares, it will be the reward for the period. Also, it holds the significant figure to calculate the reward.

```
public class RewardBase : IBencodable, IEquatable<RewardBase>
```

**Inheritance**

object ← RewardBase

**Implements**

IBencodable, IEquatable<RewardBase>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardBase(Address, IValue)

```
public RewardBase(Address address, IValue bencoded)
```

#### Parameters

`address` Address

`bencoded` IValue

### RewardBase(Address, List)

```
public RewardBase(Address address, List bencoded)
```

### Parameters

`address` Address

`bencoded` List

## RewardBase(Address, BigInteger, IEnumerable<Currency>)

```
public RewardBase(Address address, BigInteger totalShares,
IEnumerable<Currency> currencies)
```

### Parameters

`address` Address

`totalShares` BigInteger

`currencies` IEnumerable<Currency>

# Fields

## Margin

Margin for significant figure. It's used to calculate the significant figure of the reward base.

```
public const int Margin = 2
```

### Field Value

int

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## RewardPortion

Cumulative reward portion of RewardBase. When it's multiplied by the number of shares, it will be the reward for the period.

```
public ImmutableSortedDictionary<Currency, BigInteger> RewardPortion { get; }
```

### Property Value

ImmutableSortedDictionary<Currency, BigInteger>

## SigFig

Significant figure of RewardBase.

```
public int SigFig { get; }
```

### Property Value

int

## StartHeight

Start height of RewardBase that attached when archived.

```
public long? StartHeight { get; }
```

### Property Value

long?

# Methods

## AddReward(FungibleAssetValue, BigInteger)

```
public RewardBase AddReward(FungibleAssetValue reward, BigInteger totalShares)
```

### Parameters

`reward` FungibleAssetValue

`totalShares` BigInteger

### Returns

RewardBase

## AddRewards(IEnumerable<FungibleAssetValue>, BigInteger)

Add rewards to the RewardBase.

```
public RewardBase AddRewards(IEnumerable<FungibleAssetValue> rewards,
BigInteger totalShares)
```

### Parameters

`rewards` IEnumerable<FungibleAssetValue>

Rewards to add.

`totalShares` BigInteger

TotalShares used as denominator of the portion.

### Returns

RewardBase

New RewardBase with added rewards.

## AttachHeight(Address, long)

```
public RewardBase AttachHeight(Address address, long startHeight)
```

### Parameters

`address` Address

`startHeight` long

### Returns

RewardBase

## CumulativeRewardDuringPeriod(BigInteger)

Calculate the cumulative reward during the period.

```
public ImmutableSortedDictionary<Currency, FungibleAssetValue>
CumulativeRewardDuringPeriod(BigInteger share)
```

### Parameters

`share` BigInteger

The number of shares to calculate the reward.

### Returns

ImmutableSortedDictionary<Currency, FungibleAssetValue>

Cumulative reward during the period.

## CumulativeRewardDuringPeriod(BigInteger, Currency)

Calculate the cumulative reward during the period, for the specific currency.

```
public FungibleAssetValue CumulativeRewardDuringPeriod(BigInteger share,
Currency currency)
```

### Parameters

`share` BigInteger

The number of shares to calculate the reward.

`currency` Currency

The currency to calculate the reward.

### Returns

FungibleAssetValue

Cumulative reward during the period, for the specific currency.

### Exceptions

ArgumentException

Thrown when the `currency` is not in the RewardPortion.

## Equals(RewardBase?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(RewardBase? other)
```

### Parameters

`other` RewardBase

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## RecommendedSigFig(BigInteger)

```
public static int RecommendedSigFig(BigInteger totalShares)
```

### Parameters

`totalShares` BigInteger

### Returns

int

## UpdateSigFig(BigInteger)

```
public RewardBase UpdateSigFig(BigInteger totalShares)
```

### Parameters

`totalShares` BigInteger

### Returns

RewardBase

# Class UnbondLockIn

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public sealed class UnbondLockIn : IUnbonding, IBencodable, IEquatable<UnbondLockIn>
```

**Inheritance**

object ← UnbondLockIn

**Implements**

IUnbonding, IBencodable, IEquatable<UnbondLockIn>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### UnbondLockIn(Address, int, IValue, IDelegationRepository?)

```
public UnbondLockIn(Address address, int maxEntries, IValue bencoded,
IDelegationRepository? repository = null)
```

#### Parameters

`address` Address

`maxEntries` int

`bencoded` IValue

`repository` IDelegationRepository

### UnbondLockIn(Address, int, List,

## IDelegationRepository?)

```
public UnbondLockIn(Address address, int maxEntries, List bencoded,
IDelegationRepository? repository = null)
```

### Parameters

`address` Address

`maxEntries` int

`bencoded` List

`repository` IDelegationRepository

## UnbondLockIn(Address, int, Address, Address, IDelegationRepository?)

```
public UnbondLockIn(Address address, int maxEntries, Address delegateeAddress,
Address delegatorAddress, IDelegationRepository? repository)
```

### Parameters

`address` Address

`maxEntries` int

`delegateeAddress` Address

`delegatorAddress` Address

`repository` IDelegationRepository

## UnbondLockIn(Address, int, Address, Address, IEnumerable<UnbondingEntry>, IDelegationRepository?)

```
public UnbondLockIn(Address address, int maxEntries, Address delegateeAddress,
Address delegatorAddress, IEnumerable<UnbondingEntry> entries,
IDelegationRepository? repository = null)
```

### Parameters

`address` Address

`maxEntries` int

`delegateeAddress` Address

`delegatorAddress` Address

`entries` IEnumerable<UnbondingEntry>

`repository` IDelegationRepository

## Properties

### Address

```
public Address Address { get; }
```

#### Property Value

Address

### Bencoded

```
public List Bencoded { get; }
```

#### Property Value

List

## DelegateeAddress

```
public Address DelegateeAddress { get; }
```

### Property Value

Address

## DelegatorAddress

```
public Address DelegatorAddress { get; }
```

### Property Value

Address

## Entries

```
public ImmutableSortedDictionary<long, ImmutableList<UnbondingEntry>> Entries {
get; }
```

### Property Value

ImmutableSortedDictionary<long, ImmutableList<UnbondingEntry>>

## FlattenedEntries

```
public ImmutableArray<UnbondingEntry> FlattenedEntries { get; }
```

### Property Value

ImmutableArray<UnbondingEntry>

## IsEmpty

```
public bool IsEmpty { get; }
```

### Property Value

bool

## IsFull

```
public bool IsFull { get; }
```

### Property Value

bool

## LowestExpireHeight

```
public long LowestExpireHeight { get; }
```

### Property Value

long

## MaxEntries

```
public int MaxEntries { get; }
```

### Property Value

int

## Repository

```
public IDelegationRepository? Repository { get; }
```

### Property Value

IDelegationRepository

## Methods

### Equals(UnbondLockIn?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(UnbondLockIn? other)
```

### Parameters

`other` UnbondLockIn

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

### Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Release(long, out FungibleAssetValue?)

```
public UnbondLockIn Release(long height, out FungibleAssetValue? releasedFAV)
```

### Parameters

`height` long

`releasedFAV` FungibleAssetValue?

### Returns

UnbondLockIn

## Slash(BigInteger, long, long, Address)

```
public UnbondLockIn Slash(BigInteger slashFactor, long infractionHeight, long
height, Address slashedPoolAddress)
```

## Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

`slashedPoolAddress` Address

## Returns

UnbondLockIn

# Class UnbondingEntry

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class UnbondingEntry : IBencodable, IEquatable<UnbondingEntry>
```

**Inheritance**

object ← UnbondingEntry

**Implements**

IBencodable, IEquatable<UnbondingEntry>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### UnbondingEntry(IValue)

```
public UnbondingEntry(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### UnbondingEntry(Address, FungibleAssetValue, FungibleAssetValue, long, long)

```
public UnbondingEntry(Address unbondeeAddress, FungibleAssetValue
initialUnbondingFAV, FungibleAssetValue unbondingFAV, long creationHeight,
long expireHeight)
```

#### Parameters

`unbondeeAddress` Address

`initialUnbondingFAV` FungibleAssetValue

`unbondingFAV` FungibleAssetValue

`creationHeight` long

`expireHeight` long

## UnbondingEntry(Address, FungibleAssetValue, long, long)

```
public UnbondingEntry(Address unbondeeAddress, FungibleAssetValue unbondingFAV, long
creationHeight, long expireHeight)
```

### Parameters

`unbondeeAddress` Address

`unbondingFAV` FungibleAssetValue

`creationHeight` long

`expireHeight` long

# Properties

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## CreationHeight

```
public long CreationHeight { get; }
```

### Property Value

long

## ExpireHeight

```
public long ExpireHeight { get; }
```

### Property Value

long

## InitialUnbondingFAV

```
public FungibleAssetValue InitialUnbondingFAV { get; }
```

### Property Value

FungibleAssetValue

## UnbondeeAddress

```
public Address UnbondeeAddress { get; }
```

### Property Value

Address

## UnbondingFAV

```
public FungibleAssetValue UnbondingFAV { get; }
```

### Property Value

FungibleAssetValue

## Methods

### Equals(UnbondingEntry?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(UnbondingEntry? other)
```

### Parameters

`other` UnbondingEntry

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

### Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Slash(BigInteger, long, out FungibleAssetValue)

```
public UnbondingEntry Slash(BigInteger slashFactor, long infractionHeight, out
FungibleAssetValue slashedFAV)
```

### Parameters

`slashFactor` BigInteger

`infractionHeight` long

`slashedFAV` FungibleAssetValue

### Returns

UnbondingEntry

# Class UnbondingEntry.Comparer

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class UnbondingEntry.Comparer : IComparer<UnbondingEntry>
```

**Inheritance**

object ← UnbondingEntry.Comparer

**Implements**

IComparer<UnbondingEntry>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## Compare(UnbondingEntry?, UnbondingEntry?)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public int Compare(UnbondingEntry? x, UnbondingEntry? y)
```

### Parameters

x UnbondingEntry

The first object to compare.

y UnbondingEntry

The second object to compare.

### Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

| Value | Meaning |
| --- | --- |
| **Less than zero** | x is less than y. |
| **Zero** | x equals y. |
| **Greater than zero** | x is greater than y. |

# Class UnbondingFactory

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public static class UnbondingFactory
```

**Inheritance**

object ← UnbondingFactory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetUnbondingFromRef(IValue, IDelegationRepository)

```
public static IUnbonding GetUnbondingFromRef(IValue bencoded,
IDelegationRepository repository)
```

#### Parameters

`bencoded` IValue

`repository` IDelegationRepository

#### Returns

IUnbonding

### GetUnbondingFromRef(UnbondingRef, IDelegationRepository)

```
public static IUnbonding GetUnbondingFromRef(UnbondingRef reference,
IDelegationRepository repository)
```

### Parameters

`reference` UnbondingRef

`repository` IDelegationRepository

### Returns

IUnbonding

## ToReference(IUnbonding)

```
public static UnbondingRef ToReference(IUnbonding unbonding)
```

### Parameters

`unbonding` IUnbonding

### Returns

UnbondingRef

# Class UnbondingRef

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class UnbondingRef : IEquatable<UnbondingRef>, IComparable<UnbondingRef>,
IComparable, IBencodable
```

**Inheritance**

object ← UnbondingRef

**Implements**

IEquatable<UnbondingRef>, IComparable<UnbondingRef>, IComparable, IBencodable

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### UnbondingRef(IValue)

```
public UnbondingRef(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### UnbondingRef(List)

```
public UnbondingRef(List list)
```

#### Parameters

`list` List

# UnbondingRef(Address, UnbondingType)

```
public UnbondingRef(Address address, UnbondingType unbondingType)
```

## Parameters

`address` Address

`unbondingType` UnbondingType

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## UnbondingType

```
public UnbondingType UnbondingType { get; }
```

### Property Value

UnbondingType

# Methods

## CompareTo(UnbondingRef?)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(UnbondingRef? other)
```

### Parameters

`other` UnbondingRef

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `other` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `other`. |
| **Greater than zero** | This instance follows `other` in the sort order. |

## CompareTo(object?)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(object? obj)
```

### Parameters

`obj` object

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
|---|---|
| **Less than zero** | This instance precedes `obj` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `obj`. |
| **Greater than zero** | This instance follows `obj` in the sort order. |

### Exceptions

ArgumentException

`obj` is not the same type as this instance.

## Equals(UnbondingRef?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(UnbondingRef? other)
```

### Parameters

`other` UnbondingRef

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

# Operators

## operator ==(UnbondingRef?, UnbondingRef?)

```
public static bool operator ==(UnbondingRef? left, UnbondingRef? right)
```

### Parameters

`left` UnbondingRef

`right` UnbondingRef

### Returns

bool

## operator >(UnbondingRef, UnbondingRef)

```
public static bool operator >(UnbondingRef left, UnbondingRef right)
```

### Parameters

`left` UnbondingRef

`right` UnbondingRef

### Returns

bool

## operator >=(UnbondingRef, UnbondingRef)

```
public static bool operator >=(UnbondingRef left, UnbondingRef right)
```

### Parameters

`left` UnbondingRef

`right` UnbondingRef

### Returns

bool

## operator !=(UnbondingRef?, UnbondingRef?)

```
public static bool operator !=(UnbondingRef? left, UnbondingRef? right)
```

### Parameters

`left` UnbondingRef

`right` UnbondingRef

### Returns

bool

## operator <(UnbondingRef, UnbondingRef)

```
public static bool operator <(UnbondingRef left, UnbondingRef right)
```

### Parameters

`left` UnbondingRef

`right` UnbondingRef

### Returns

bool

## operator <=(UnbondingRef, UnbondingRef)

```
public static bool operator <=(UnbondingRef left, UnbondingRef right)
```

## Parameters

`left` UnbondingRef

`right` UnbondingRef

## Returns

bool

# Enum UnbondingType

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public enum UnbondingType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`RebondGrace = 1`

`UnbondLockIn = 0`

# Namespace Nekoyume.Exceptions

## Classes

AlreadyFoughtAvatarException

BencodexTypesExtensions

InvalidActionFieldException

InvalidItemIdException

InvalidMaterialCountException

ItemNotFoundException

MedalIdNotFoundException

NotEnoughEventDungeonTicketsException

NotEnoughItemException

PetCostNotFoundException

StateNullException

UnsupportedStateException

# Class AlreadyFoughtAvatarException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyFoughtAvatarException : Exception, ISerializable
```

### Inheritance

object ← Exception ← AlreadyFoughtAvatarException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AlreadyFoughtAvatarException()

```
public AlreadyFoughtAvatarException()
```

### AlreadyFoughtAvatarException(string)

```
public AlreadyFoughtAvatarException(string message)
```

#### Parameters

`message` string

# Class BencodexTypesExtensions

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
public static class BencodexTypesExtensions
```

**Inheritance**

object ← BencodexTypesExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## Replace(List, int, IValue)

Returns a new list with the value replaced at the specified index.

```
public static List Replace(this List list, int index, IValue value)
```

### Parameters

`list` List

`index` int

`value` IValue

### Returns

List

### Exceptions

ArgumentOutOfRangeException

# Class InvalidActionFieldException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidActionFieldException : Exception, ISerializable
```

### Inheritance

object ← Exception ← InvalidActionFieldException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidActionFieldException(SerializationInfo, StreamingContext)

```
protected InvalidActionFieldException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidActionFieldException(string?)

```
public InvalidActionFieldException(string? message = null)
```

### Parameters

`message` string

## InvalidActionFieldException(string?, Exception?)

```
public InvalidActionFieldException(string? message = null, Exception? innerException
= null)
```

### Parameters

`message` string

`innerException` Exception

## InvalidActionFieldException(string, string, string, string, Exception?)

```
public InvalidActionFieldException(string actionType, string addressesHex, string
fieldName, string message, Exception? innerException = null)
```

### Parameters

`actionType` string

`addressesHex` string

`fieldName` string

`message` string

`innerException` Exception

# Class InvalidItemIdException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidItemIdException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidItemIdException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidItemIdException()

```
public InvalidItemIdException()
```

### InvalidItemIdException(SerializationInfo, StreamingContext)

```
protected InvalidItemIdException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidItemIdException(string)

```
public InvalidItemIdException(string message)
```

### Parameters

`message` string

## InvalidItemIdException(string, Exception)

```
public InvalidItemIdException(string message, Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

## InvalidItemIdException(string, string, int)

```
public InvalidItemIdException(string actionType, string addressesHex, int itemId)
```

### Parameters

`actionType` string

`addressesHex` string

`itemId` int

# Class InvalidMaterialCountException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidMaterialCountException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← InvalidMaterialCountException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Constructors

## InvalidMaterialCountException(string)

```
public InvalidMaterialCountException(string message)
```

### Parameters

`message` string

## InvalidMaterialCountException(string, string, int, int)

```
public InvalidMaterialCountException(string actionType, string addressesHex, int
required, int playerHas)
```

## Parameters

`actionType` string

`addressesHex` string

`required` int

`playerHas` int

# Class ItemNotFoundException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class ItemNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ItemNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ItemNotFoundException()

```
public ItemNotFoundException()
```

### ItemNotFoundException(Address, HashDigest<SHA256>, Exception?)

```
public ItemNotFoundException(Address inventoryAddr, HashDigest<SHA256> fungibleId,
Exception? innerException = null)
```

### Parameters

`inventoryAddr` Address

`fungibleId` HashDigest<SHA256>

`innerException` Exception

## ItemNotFoundException(Address, Guid, Exception?)

```
public ItemNotFoundException(Address inventoryAddr, Guid nonFungibleId, Exception?
innerException = null)
```

### Parameters

`inventoryAddr` Address

`nonFungibleId` Guid

`innerException` Exception

## ItemNotFoundException(SerializationInfo, StreamingContext)

```
protected ItemNotFoundException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## ItemNotFoundException(string?)

```
public ItemNotFoundException(string? message)
```

### Parameters

`message` string

## ItemNotFoundException(string?, Exception?)

```
public ItemNotFoundException(string? message, Exception? innerException)
```

### Parameters

`message` string

`innerException` Exception

# Class MedalIdNotFoundException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class MedalIdNotFoundException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← MedalIdNotFoundException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### MedalIdNotFoundException()

```
public MedalIdNotFoundException()
```

### MedalIdNotFoundException(SerializationInfo, StreamingContext)

```
protected MedalIdNotFoundException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## MedalIdNotFoundException(string)

```
public MedalIdNotFoundException(string message)
```

### Parameters

`message` string

## MedalIdNotFoundException(string, Exception)

```
public MedalIdNotFoundException(string message, Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

## MedalIdNotFoundException(string, string, int, int)

```
public MedalIdNotFoundException(string actionType, string addressesHex, int
requiredAmount, int currentAmount)
```

### Parameters

`actionType` string

`addressesHex` string

`requiredAmount` int

`currentAmount` int

# Class NotEnoughEventDungeonTicketsException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughEventDungeonTicketsException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← NotEnoughEventDungeonTicketsException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughEventDungeonTicketsException()

```
public NotEnoughEventDungeonTicketsException()
```

### NotEnoughEventDungeonTicketsException(SerializationInfo, StreamingContext)

```
protected NotEnoughEventDungeonTicketsException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## NotEnoughEventDungeonTicketsException(string)

```
public NotEnoughEventDungeonTicketsException(string message)
```

### Parameters

`message` string

## NotEnoughEventDungeonTicketsException(string, Exception)

```
public NotEnoughEventDungeonTicketsException(string message,
Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

## NotEnoughEventDungeonTicketsException(string, string, int, int)

```
public NotEnoughEventDungeonTicketsException(string actionType, string addressesHex,
int requiredAmount, int currentAmount)
```

### Parameters

`actionType` string

`addressesHex` string

`requiredAmount` int

`currentAmount` int

# Class NotEnoughItemException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughItemException : Exception, ISerializable
```

### Inheritance

object ← Exception ← NotEnoughItemException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughItemException()

```
public NotEnoughItemException()
```

### NotEnoughItemException(Address, HashDigest<SHA256>, int, int, Exception?)

```
public NotEnoughItemException(Address inventoryAddr, HashDigest<SHA256> fungibleId,
int expectedCount, int actualCount, Exception? innerException = null)
```

### Parameters

`inventoryAddr` Address

`fungibleId` HashDigest<SHA256>

`expectedCount` int

`actualCount` int

`innerException` Exception

## NotEnoughItemException(SerializationInfo, StreamingContext)

```
protected NotEnoughItemException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## NotEnoughItemException(string?)

```
public NotEnoughItemException(string? message)
```

### Parameters

`message` string

## NotEnoughItemException(string?, Exception?)

```
public NotEnoughItemException(string? message, Exception? innerException)
```

### Parameters

`message` string

`innerException` Exception

# Class PetCostNotFoundException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class PetCostNotFoundException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← PetCostNotFoundException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### PetCostNotFoundException(string)

```
public PetCostNotFoundException(string message)
```

#### Parameters

`message` string

### PetCostNotFoundException(string, Exception)

```
public PetCostNotFoundException(string message, Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

## PetCostNotFoundException(string, string, string, Exception)

```
public PetCostNotFoundException(string actionType, string addressesHex, string
message, Exception innerException = null)
```

### Parameters

`actionType` string

`addressesHex` string

`message` string

`innerException` Exception

# Class StateNullException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class StateNullException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← StateNullException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### StateNullException()

```
public StateNullException()
```

### StateNullException(Address, Address, Exception?)

```
public StateNullException(Address accountAddress, Address address, Exception?
innerException = null)
```

#### Parameters

`accountAddress` Address

`address` Address

`innerException` Exception

## StateNullException(SerializationInfo, StreamingContext)

```
protected StateNullException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## StateNullException(string?)

```
public StateNullException(string? message)
```

### Parameters

`message` string

## StateNullException(string?, Exception?)

```
public StateNullException(string? message, Exception? innerException)
```

### Parameters

`message` string

`innerException` Exception

# Class UnsupportedStateException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class UnsupportedStateException : Exception, ISerializable
```

### Inheritance

object ← Exception ← UnsupportedStateException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### UnsupportedStateException(int, int, Exception)

```
public UnsupportedStateException(int expectedVersion, int actualVersion, Exception
innerException = null)
```

#### Parameters

`expectedVersion` int

`actualVersion` int

`innerException` Exception

# UnsupportedStateException(SerializationInfo, StreamingContext)

```
protected UnsupportedStateException(SerializationInfo info,
StreamingContext context)
```

## Parameters

`info` SerializationInfo

`context` StreamingContext

# UnsupportedStateException(string)

```
public UnsupportedStateException(string msg)
```

## Parameters

`msg` string

# UnsupportedStateException(string, int, string, int, Exception)

```
public UnsupportedStateException(string expectedName, int expectedVersion, string
actualName, int actualVersion, Exception innerException = null)
```

## Parameters

`expectedName` string

`expectedVersion` int

`actualName` string

`actualVersion` int

`innerException` Exception

# Namespace Nekoyume.Extensions

## Classes

ActionContextExtensions

AgentAddressExtensions

CombinationSlotStateExtensions

EquipmentExtensions

EventDungeonExtensions

EventRecipeExtensions

EventScheduleExtensions

IntegerExtensions

MaterialItemSheetExtensions

RandomExtensions

SheetsExtensions

WorldExtensions

WorldInformationExtensions

# Class ActionContextExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class ActionContextExtensions
```

**Inheritance**

object ← ActionContextExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetAgentAddress(IActionContext)

```
public static AgentAddress GetAgentAddress(this IActionContext context)
```

#### Parameters

`context` IActionContext

#### Returns

AgentAddress

# Class AgentAddressExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class AgentAddressExtensions
```

**Inheritance**

object ← AgentAddressExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetPledgeAddress(AgentAddress)

```
public static PledgeAddress GetPledgeAddress(this AgentAddress address)
```

### Parameters

`address` AgentAddress

### Returns

PledgeAddress

# Class CombinationSlotStateExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class CombinationSlotStateExtensions
```

**Inheritance**

object ← CombinationSlotStateExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### TryGetMail(CombinationSlotState, long, long, out CombinationMail, out ItemEnhanceMail)

```
public static bool TryGetMail(this CombinationSlotState state, long
blockIndex, long requiredBlockIndex, out CombinationMail combinationMail, out
ItemEnhanceMail itemEnhanceMail)
```

#### Parameters

`state` CombinationSlotState

`blockIndex` long

`requiredBlockIndex` long

`combinationMail` CombinationMail

`itemEnhanceMail` ItemEnhanceMail

#### Returns

bool

## TryGetResultId(CombinationSlotState, out Guid)

```
public static bool TryGetResultId(this CombinationSlotState state, out
Guid resultId)
```

### Parameters

`state` CombinationSlotState

`resultId` Guid

### Returns

bool

## ValidateFromAction(CombinationSlotState, long, AvatarState, int, string, string)

```
public static void ValidateFromAction(this CombinationSlotState slotState,
long blockIndex, AvatarState avatarState, int slotIndex, string actionTypeText,
string addressesHex)
```

### Parameters

`slotState` CombinationSlotState

`blockIndex` long

`avatarState` AvatarState

`slotIndex` int

`actionTypeText` string

`addressesHex` string

# Class EquipmentExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class EquipmentExtensions
```

**Inheritance**

object ← EquipmentExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

```
public static int GetRequirementLevel(this Equipment equipment,
ItemRequirementSheet requirementSheet, EquipmentItemRecipeSheet recipeSheet,
EquipmentItemSubRecipeSheetV2 subRecipeSheet, EquipmentItemOptionSheet
itemOptionSheet)
```

#### Parameters

`equipment` Equipment

`requirementSheet` ItemRequirementSheet

`recipeSheet` EquipmentItemRecipeSheet

`subRecipeSheet` EquipmentItemSubRecipeSheetV2

`itemOptionSheet` EquipmentItemOptionSheet

## Returns

int

# IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

```
public static bool IsMadeWithMimisbrunnrRecipe(this Equipment equipment,
EquipmentItemRecipeSheet recipeSheet, EquipmentItemSubRecipeSheetV2 subRecipeSheet,
EquipmentItemOptionSheet itemOptionSheet)
```

## Parameters

`equipment` Equipment

`recipeSheet` EquipmentItemRecipeSheet

`subRecipeSheet` EquipmentItemSubRecipeSheetV2

`itemOptionSheet` EquipmentItemOptionSheet

## Returns

bool

# Class EventDungeonExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class EventDungeonExtensions
```

**Inheritance**

object ← EventDungeonExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetRemainingTicketsConsiderReset(EventDungeonInfo, Row, long)

```
public static int GetRemainingTicketsConsiderReset(this EventDungeonInfo info,
EventScheduleSheet.Row eventScheduleRow, long currentBlockIndex)
```

#### Parameters

`info` EventDungeonInfo

`eventScheduleRow` EventScheduleSheet.Row

`currentBlockIndex` long

#### Returns

int

### GetStageNumber(Row)

```
public static int GetStageNumber(this EventDungeonStageSheet.Row row)
```

### Parameters

`row` EventDungeonStageSheet.Row

### Returns

int

## GetStageRows(EventDungeonStageSheet, int, int)

```
public static List<EventDungeonStageSheet.Row> GetStageRows(this
EventDungeonStageSheet sheet, int beginningStageId, int endStageId)
```

### Parameters

`sheet` EventDungeonStageSheet

`beginningStageId` int

`endStageId` int

### Returns

List<EventDungeonStageSheet.Row>

## GetStageWaveRows(EventDungeonStageWaveSheet, int, int)

```
public static List<EventDungeonStageWaveSheet.Row> GetStageWaveRows(this
EventDungeonStageWaveSheet sheet, int beginningStageId, int endStageId)
```

### Parameters

`sheet` EventDungeonStageWaveSheet

`beginningStageId` int

`endStageId` int

### Returns

List<EventDungeonStageWaveSheet.Row>

## ToEventDungeonStageNumber(int)

```
public static int ToEventDungeonStageNumber(this int eventDungeonStageId)
```

### Parameters

`eventDungeonStageId` int

### Returns

int

## TryGetRowByEventDungeonStageId(EventDungeonSheet, int, out Row)

```
public static bool TryGetRowByEventDungeonStageId(this EventDungeonSheet sheet, int
eventDungeonStageId, out EventDungeonSheet.Row row)
```

### Parameters

`sheet` EventDungeonSheet

`eventDungeonStageId` int

`row` EventDungeonSheet.Row

### Returns

bool

## TryGetRowByEventScheduleId(EventDungeonSheet, int, out Row)

```
public static bool TryGetRowByEventScheduleId(this EventDungeonSheet sheet, int
eventScheduleId, out EventDungeonSheet.Row row)
```

### Parameters

`sheet` EventDungeonSheet

`eventScheduleId` int

`row` EventDungeonSheet.Row

### Returns

bool

## ValidateFromAction(EventDungeonSheet, int, int, string, string)

```
public static EventDungeonSheet.Row ValidateFromAction(this EventDungeonSheet
sheet, int eventDungeonId, int eventDungeonStageId, string actionTypeText,
string addressesHex)
```

### Parameters

`sheet` EventDungeonSheet

`eventDungeonId` int

`eventDungeonStageId` int

`actionTypeText` string

`addressesHex` string

### Returns

EventDungeonSheet.Row

# ValidateFromAction(EventDungeonStageSheet, int, string, string)

```
public static EventDungeonStageSheet.Row ValidateFromAction(this
EventDungeonStageSheet sheet, int eventDungeonStageId, string actionTypeText,
string addressesHex)
```

## Parameters

`sheet` EventDungeonStageSheet

`eventDungeonStageId` int

`actionTypeText` string

`addressesHex` string

## Returns

EventDungeonStageSheet.Row

# Class EventRecipeExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class EventRecipeExtensions
```

**Inheritance**

object ← EventRecipeExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetRecipeRows(EventConsumableItemRecipeSheet, int)

```
public static List<EventConsumableItemRecipeSheet.Row> GetRecipeRows(this
EventConsumableItemRecipeSheet sheet, int eventScheduleId)
```

### Parameters

`sheet` EventConsumableItemRecipeSheet

`eventScheduleId` int

### Returns

List<EventConsumableItemRecipeSheet.Row>

## GetRecipeRows(EventMaterialItemRecipeSheet, int)

```
public static List<EventMaterialItemRecipeSheet.Row> GetRecipeRows(this
EventMaterialItemRecipeSheet sheet, int eventScheduleId)
```

### Parameters

`sheet` EventMaterialItemRecipeSheet

`eventScheduleId` int

### Returns

List<EventMaterialItemRecipeSheet.Row>

## ValidateFromAction(EventConsumableItemRecipeSheet, int, string, string)

```
public static EventConsumableItemRecipeSheet.Row ValidateFromAction(this
EventConsumableItemRecipeSheet sheet, int eventConsumableItemRecipeId, string
actionTypeText, string addressesHex)
```

### Parameters

`sheet` EventConsumableItemRecipeSheet

`eventConsumableItemRecipeId` int

`actionTypeText` string

`addressesHex` string

### Returns

EventConsumableItemRecipeSheet.Row

## ValidateFromAction(EventMaterialItemRecipeSheet, int, string, string)

```
public static EventMaterialItemRecipeSheet.Row ValidateFromAction(this
EventMaterialItemRecipeSheet sheet, int eventMaterialItemRecipeId, string
actionTypeText, string addressesHex)
```

### Parameters

`sheet` EventMaterialItemRecipeSheet

`eventMaterialItemRecipeId` int

`actionTypeText` string

`addressesHex` string

### Returns

EventMaterialItemRecipeSheet.Row

## ValidateFromAction(Row, MaterialItemSheet, Dictionary<int, int>, string, string)

```
public static void ValidateFromAction(this EventMaterialItemRecipeSheet.Row
recipeRow, MaterialItemSheet materialItemSheet, Dictionary<int, int> materialsToUse,
string actionTypeText, string addressesHex)
```

### Parameters

`recipeRow` EventMaterialItemRecipeSheet.Row

`materialItemSheet` MaterialItemSheet

`materialsToUse` Dictionary<int, int>

`actionTypeText` string

`addressesHex` string

# Class EventScheduleExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class EventScheduleExtensions
```

**Inheritance**

object ← EventScheduleExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetDungeonTicketCost(Row, int, Currency)

```
public static FungibleAssetValue GetDungeonTicketCost(this EventScheduleSheet.Row
row, int numberOfTicketPurchases, Currency currency)
```

### Parameters

`row` EventScheduleSheet.Row

`numberOfTicketPurchases` int

`currency` Currency

### Returns

FungibleAssetValue

## GetDungeonTicketCostV1(Row, int)

```
[Obsolete("Use `GetDungeonTicketCost()` instead.")]
public static long GetDungeonTicketCostV1(this EventScheduleSheet.Row row,
int numberOfTicketPurchases)
```

### Parameters

`row` EventScheduleSheet.Row

`numberOfTicketPurchases` int

### Returns

long

## GetStageExp(Row, int, int)

```
public static int GetStageExp(this EventScheduleSheet.Row row, int stageNumber, int
multiplier = 1)
```

### Parameters

`row` EventScheduleSheet.Row

`stageNumber` int

`multiplier` int

### Returns

int

## ToEventScheduleId(int)

```
public static int ToEventScheduleId(this int eventDungeonOrRecipeId)
```

### Parameters

`eventDungeonOrRecipeId` int

### Returns

int

## TryGetRowForDungeon(EventScheduleSheet, long, out Row)

```
public static bool TryGetRowForDungeon(this EventScheduleSheet sheet, long
blockIndex, out EventScheduleSheet.Row row)
```

### Parameters

`sheet` EventScheduleSheet

`blockIndex` long

`row` EventScheduleSheet.Row

### Returns

bool

## TryGetRowForRecipe(EventScheduleSheet, long, out Row)

```
public static bool TryGetRowForRecipe(this EventScheduleSheet sheet, long
blockIndex, out EventScheduleSheet.Row row)
```

### Parameters

`sheet` EventScheduleSheet

`blockIndex` long

`row` EventScheduleSheet.Row

### Returns

bool

## ValidateFromActionForDungeon(EventScheduleSheet, long, int, int, string, string)

```
public static EventScheduleSheet.Row ValidateFromActionForDungeon(this
EventScheduleSheet scheduleSheet, long blockIndex, int eventScheduleId, int
eventDungeonId, string actionTypeText, string addressesHex)
```

### Parameters

`scheduleSheet` EventScheduleSheet

`blockIndex` long

`eventScheduleId` int

`eventDungeonId` int

`actionTypeText` string

`addressesHex` string

### Returns

EventScheduleSheet.Row

## ValidateFromActionForRecipe(EventScheduleSheet, long, int, int, string, string)

```
public static EventScheduleSheet.Row ValidateFromActionForRecipe(this
EventScheduleSheet scheduleSheet, long blockIndex, int eventScheduleId, int
eventRecipeId, string actionTypeText, string addressesHex)
```

### Parameters

`scheduleSheet` EventScheduleSheet

`blockIndex` long

`eventScheduleId` int

`eventRecipeId` int

`actionTypeText` string

`addressesHex` string

## Returns

EventScheduleSheet.Row

# Class IntegerExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class IntegerExtensions
```

**Inheritance**

object ← IntegerExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### NormalizeFromTenThousandths(int)

```
public static decimal NormalizeFromTenThousandths(this int value)
```

#### Parameters

`value` int

#### Returns

decimal

# Class MaterialItemSheetExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class MaterialItemSheetExtensions
```

**Inheritance**

object ← MaterialItemSheetExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### ValidateFromAction(MaterialItemSheet, List<MaterialInfo>, Dictionary<int, int>, string)

```
public static void ValidateFromAction(this MaterialItemSheet materialItemSheet,
List<EquipmentItemSubRecipeSheet.MaterialInfo> materialInfos, Dictionary<int, int>
requiredFungibleItems, string addressesHex)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`materialInfos` List<EquipmentItemSubRecipeSheet.MaterialInfo>

`requiredFungibleItems` Dictionary<int, int>

`addressesHex` string

# Class RandomExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class RandomExtensions
```

**Inheritance**

object ← RandomExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GenerateAddress(IRandom)

```
public static Address GenerateAddress(this IRandom random)
```

#### Parameters

`random` IRandom

#### Returns

Address

# Class SheetsExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class SheetsExtensions
```

**Inheritance**

object ← SheetsExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

```
public static int FindLevelByStakedAmount(this IStakeRewardSheet sheet, Address
agentAddress, FungibleAssetValue balance)
```

#### Parameters

`sheet` IStakeRewardSheet

`agentAddress` Address

`balance` FungibleAssetValue

#### Returns

int

## FindPreviousRaidIdByBlockIndex(WorldBossListSheet, long)

```
public static int FindPreviousRaidIdByBlockIndex(this WorldBossListSheet sheet,
long blockIndex)
```

### Parameters

`sheet` WorldBossListSheet

`blockIndex` long

### Returns

int

## FindPreviousRowByBlockIndex(WorldBossListSheet, long)

```
public static WorldBossListSheet.Row FindPreviousRowByBlockIndex(this
WorldBossListSheet sheet, long blockIndex)
```

### Parameters

`sheet` WorldBossListSheet

`blockIndex` long

### Returns

WorldBossListSheet.Row

## FindRaidIdByBlockIndex(WorldBossListSheet, long)

```
public static int FindRaidIdByBlockIndex(this WorldBossListSheet sheet,
long blockIndex)
```

### Parameters

`sheet` WorldBossListSheet

`blockIndex` long

### Returns

int

## FindRowByBlockIndex(WorldBossListSheet, long)

```
public static WorldBossListSheet.Row FindRowByBlockIndex(this WorldBossListSheet
sheet, long blockIndex)
```

### Parameters

`sheet` WorldBossListSheet

`blockIndex` long

### Returns

WorldBossListSheet.Row

## GetActionPointByStaking(StakeActionPointCoefficientSheet, int, int, int)

```
public static int GetActionPointByStaking(this StakeActionPointCoefficientSheet
sheet, int originAp, int playCount, int level)
```

### Parameters

`sheet` StakeActionPointCoefficientSheet

`originAp` int

`playCount` int

`level` int

### Returns

int

## GetAddress(Dictionary<Type, (Address address, ISheet sheet)>, Type)

```
public static Address GetAddress(this Dictionary<Type, (Address address, ISheet
sheet)> sheets, Type type)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`type` Type

### Returns

Address

## GetAddress<T>(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static Address GetAddress<T>(this Dictionary<Type, (Address address, ISheet
sheet)> sheets) where T : ISheet
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

Address

### Type Parameters

T

## GetArenaSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static ArenaSimulatorSheets GetArenaSimulatorSheets(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

ArenaSimulatorSheets

## GetArenaSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static ArenaSimulatorSheetsV1 GetArenaSimulatorSheetsV1(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

ArenaSimulatorSheetsV1

## GetArenaSimulatorSheets_v100291(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static ArenaSimulatorSheetsV1 GetArenaSimulatorSheets_v100291(this
Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

ArenaSimulatorSheetsV1

## GetAvatarSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static AvatarSheets GetAvatarSheets(this Dictionary<Type, (Address address,
ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

AvatarSheets

## GetItemSheet(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static ItemSheet GetItemSheet(this Dictionary<Type, (Address address, ISheet
sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

ItemSheet

# GetQuestSheet(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static QuestSheet GetQuestSheet(this Dictionary<Type, (Address address,
ISheet sheet)> sheets)
```

## Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

## Returns

QuestSheet

# GetRaidSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RaidSimulatorSheets GetRaidSimulatorSheets(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

## Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

## Returns

RaidSimulatorSheets

# GetRaidSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RaidSimulatorSheetsV1 GetRaidSimulatorSheetsV1(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

## Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

RaidSimulatorSheetsV1

## GetRankingSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RankingSimulatorSheets GetRankingSimulatorSheets(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

RankingSimulatorSheets

## GetRankingSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RankingSimulatorSheetsV1 GetRankingSimulatorSheetsV1(this
Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

RankingSimulatorSheetsV1

# GetRankingSimulatorSheetsV100291(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RankingSimulatorSheetsV1 GetRankingSimulatorSheetsV100291(this
Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

## Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

## Returns

RankingSimulatorSheetsV1

# GetSheet(Dictionary<Type, (Address address, ISheet sheet)>, Type)

```
public static ISheet GetSheet(this Dictionary<Type, (Address address, ISheet sheet)>
sheets, Type sheetType)
```

## Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`sheetType` Type

## Returns

ISheet

# GetSheet<T>(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static T GetSheet<T>(this Dictionary<Type, (Address address, ISheet sheet)>
sheets) where T : ISheet
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

T

### Type Parameters

`T`

## GetSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static SimulatorSheets GetSimulatorSheets(this Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

SimulatorSheets

## GetSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static SimulatorSheetsV1 GetSimulatorSheetsV1(this Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

SimulatorSheetsV1

## GetSimulatorSheetsV100291(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static SimulatorSheetsV1 GetSimulatorSheetsV100291(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

SimulatorSheetsV1

## GetStageSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static StageSimulatorSheets GetStageSimulatorSheets(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

StageSimulatorSheets

## GetStageSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static StageSimulatorSheetsV1 GetStageSimulatorSheetsV1(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

StageSimulatorSheetsV1

## GetStageSimulatorSheetsV100291(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static StageSimulatorSheetsV1 GetStageSimulatorSheetsV100291(this
Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

### Returns

StageSimulatorSheetsV1

## TryGetAddress(Dictionary<Type, (Address address, ISheet sheet)>, Type, out Address)

```
public static bool TryGetAddress(this Dictionary<Type, (Address address, ISheet
sheet)> sheets, Type type, out Address address)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`type` Type

`address` Address

### Returns

bool

## TryGetAddress<T>(Dictionary<Type, (Address address, ISheet sheet)>, out Address)

```
public static bool TryGetAddress<T>(this Dictionary<Type, (Address address, ISheet sheet)> sheets, out Address address) where T : ISheet
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`address` Address

### Returns

bool

### Type Parameters

`T`

## TryGetSheet(Dictionary<Type, (Address address, ISheet sheet)>, Type, out ISheet)

```
public static bool TryGetSheet(this Dictionary<Type, (Address address, ISheet sheet)> sheets, Type type, out ISheet sheet)
```

### Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`type` Type

`sheet` ISheet

### Returns

bool

# TryGetSheet<T>(Dictionary<Type, (Address address, ISheet sheet)>, out T)

```
public static bool TryGetSheet<T>(this Dictionary<Type, (Address address, ISheet sheet)> sheets, out T sheet) where T : ISheet
```

## Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`sheet` T

## Returns

bool

## Type Parameters

`T`

# Class WorldExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class WorldExtensions
```

**Inheritance**

object ← WorldExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetFeeAddress(IWorld, long)

Retrieves the fee address based on the block index and migration height.

```
public static Address GetFeeAddress(this IWorld world, long blockIndex)
```

### Parameters

`world` IWorld

The current world instance.

`blockIndex` long

The block index for which the fee address is required.

### Returns

Address

The fee address, which may vary based on migration height and arena data.

# MutateAccount(IWorld, Address, Func<IAccount, IAccount>)

```
public static IWorld MutateAccount(this IWorld world, Address accountAddress,
Func<IAccount, IAccount> mutateFn)
```

## Parameters

`world` IWorld

`accountAddress` Address

`mutateFn` Func<IAccount, IAccount>

## Returns

IWorld

# Class WorldInformationExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class WorldInformationExtensions
```

**Inheritance**

object ← WorldInformationExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### ValidateFromAction(WorldInformation, int, string, string)

```
public static void ValidateFromAction(this WorldInformation worldInformation, int
stageId, string actionTypeText, string addressesHex)
```

#### Parameters

`worldInformation` WorldInformation

`stageId` int

`actionTypeText` string

`addressesHex` string

# Namespace Nekoyume.Helper

## Classes

AdventureBossHelper

AvatarStateExtensions

CrystalCalculator

CustomCraftHelper

InventoryExtensions

ItemOptionHelper

ItemOptionInfo

NumberConversionHelper
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PetHelper

RuneEnhancementCostBoost
A BoostScheduleSheet already checked to apply to the enhancement cost of one rune, RuneId: an enhancement cost row was active and the rune was summonable when it was resolved. Only ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>) creates one, and TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long) and ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long) ignore it for any other rune, so a boost only ever discounts the rune it was resolved for — neither handing over the raw patched sheet nor reusing a boost after switching runes can skip the check.

RuneHelper

SummonHelper

SynthesizeSimulator
A class that simulates the synthesis of items. SynthesizeSimulator.InputData

WorldBossHelper

## Structs

SynthesizeResult
Represents the result of the synthesis.

SynthesizeSimulator.InputData

Simulate the synthesis of items.

# Class AdventureBossHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class AdventureBossHelper
```

**Inheritance**

object ← AdventureBossHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Fields

### FixedRewardRatio

```
public const decimal FixedRewardRatio = 0.7
```

#### Field Value

decimal

### RaffleRewardPercent

```
public const int RaffleRewardPercent = 5
```

#### Field Value

int

### RandomRewardRatio

```
public const decimal RandomRewardRatio = 0.3
```

### Field Value

decimal

## TotalRewardMultiplier

```
public const decimal TotalRewardMultiplier = 1.2
```

### Field Value

decimal

# Methods

## AddExploreRewards(IActionContext, IWorld, Address, Inventory, IEnumerable<RewardAmountData>)

```
public static IWorld AddExploreRewards(IActionContext context, IWorld states,
Address avatarAddress, Inventory inventory,
IEnumerable<AdventureBossSheet.RewardAmountData> rewardList)
```

### Parameters

`context` IActionContext

`states` IWorld

`avatarAddress` Address

`inventory` Inventory

`rewardList` IEnumerable<AdventureBossSheet.RewardAmountData>

### Returns

# ApplyRewardBoost(RewardAmountData, Row)

Applies a BoostScheduleSheet row to one explore or sweep reward entry.

```
public static AdventureBossSheet.RewardAmountData
ApplyRewardBoost(AdventureBossSheet.RewardAmountData reward,
BoostScheduleSheet.Row rewardBoost)
```

## Parameters

`reward` AdventureBossSheet.RewardAmountData

The reward as the sheets and the random draws decided it.

`rewardBoost` BoostScheduleSheet.Row

The row adjusting this floor's reward at the current block — see AdventureBossFloorReward and AdventureBossFirstClearReward — or `null` for none. A caller replaying an action must pass what the action resolved.

## Returns

AdventureBossSheet.RewardAmountData

`reward` itself when there is no row or nothing to adjust, otherwise a copy whose Amount is adjusted.

## Remarks

Only the granted amount changes: the entry was already drawn, so a boost never adds or reorders random draws. A boost adjusts a reward but never creates one, so an entry with no amount stays as it is; and it never takes a reward below 1, because granting 0 of an item or a currency fails the action.

# CalculateExploreReward(ClaimableReward, BountyBoard, ExploreBoard, Explorer, Address,

# AdventureBossNcgRewardRatioSheet, decimal, decimal, bool, out FungibleAssetValue)

Calculate reward for adventure boss explorers. This method only calculates reward for given avatar, not actually give rewards.

```
public static AdventureBossGameData.ClaimableReward
CalculateExploreReward(AdventureBossGameData.ClaimableReward reward, BountyBoard
bountyBoard, ExploreBoard exploreBoard, Explorer explorer, Address avatarAddress,
AdventureBossNcgRewardRatioSheet sheet, decimal ncgApRatio, decimal ncgRuneRatio,
bool isReal, out FungibleAssetValue ncgReward)
```

## Parameters

`reward` AdventureBossGameData.ClaimableReward

Claimable reward for this avatar so far.

`bountyBoard` BountyBoard

Bounty board for this season. NCG reward is based on totalBounty on this board.

`exploreBoard` ExploreBoard

Explore board for this season. Total reward amount is base on usedApPotion on this board.

`explorer` Explorer

Target explorer to calculate reward

`avatarAddress` Address

Target avatar address to calculate reward.

`sheet` AdventureBossNcgRewardRatioSheet

NCG to reward exchange ratio sheet. Calculate total reward amount based on this sheet.

`ncgApRatio` decimal

Exchange ratio between used AP potion to NCG. Used to set total reward amount.

`ncgRuneRatio` decimal

If a reward is rune, use this fixed ratio, not in sheet.

`isReal` bool

`ncgReward` FungibleAssetValue

out value: calculated NCG reward in this function. We must handle NCG reward separately because NCG reward must be transferred from each season's bounty address.

### Returns

AdventureBossGameData.ClaimableReward

Updated Claimable reward after calculation.

## CalculateRaffleReward(BountyBoard)

```
public static FungibleAssetValue CalculateRaffleReward(BountyBoard bountyBoard)
```

### Parameters

`bountyBoard` BountyBoard

### Returns

FungibleAssetValue

## CalculateWantedReward(ClaimableReward, BountyBoard, Address, AdventureBossNcgRewardRatioSheet, decimal, out FungibleAssetValue)

Calculate reward for adventure boss operators. This method only calculates reward for given avatar, not actually give rewards.

```
public static AdventureBossGameData.ClaimableReward
CalculateWantedReward(AdventureBossGameData.ClaimableReward reward, BountyBoard
```

```
bountyBoard, Address avatarAddress, AdventureBossNcgRewardRatioSheet sheet, decimal
ncgRuneRatio, out FungibleAssetValue ncgReward)
```

## Parameters

`reward` AdventureBossGameData.ClaimableReward

Claimable reward for this avatar so far.

`bountyBoard` BountyBoard

Bounty board for this season. All the reward amount is based on totalBounty on this board.

`avatarAddress` Address

Target avatar address to calculate reward.

`sheet` AdventureBossNcgRewardRatioSheet

NCG to reward exchange ratio sheet. Calculate total reward amount based on this sheet.

`ncgRuneRatio` decimal

If a reward is rune, use this fixed ratio, not in sheet.

`ncgReward` FungibleAssetValue

out value: calculated NCG reward in this function. We must handle NCG reward separately because NCG reward must be transferred from each season's bounty address.

## Returns

AdventureBossGameData.ClaimableReward

Updated Claimable reward after calculation.

# CollectExploreReward(ClaimableReward, GameConfigState, AdventureBossNcgRewardRatioSheet, SeasonInfo, BountyBoard, ExploreBoard, Explorer, long, Address, ref ClaimableReward, out FungibleAssetValue)

```
public static bool CollectExploreReward(AdventureBossGameData.ClaimableReward
reward, GameConfigState gameConfig, AdventureBossNcgRewardRatioSheet
ncgRewardRatioSheet, SeasonInfo seasonInfo, BountyBoard bountyBoard,
ExploreBoard exploreBoard, Explorer explorer, long currentBlockIndex, Address
avatarAddress, ref AdventureBossGameData.ClaimableReward updatedReward, out
FungibleAssetValue ncgReward)
```

## Parameters

`reward` AdventureBossGameData.ClaimableReward

`gameConfig` GameConfigState

`ncgRewardRatioSheet` AdventureBossNcgRewardRatioSheet

`seasonInfo` SeasonInfo

`bountyBoard` BountyBoard

`exploreBoard` ExploreBoard

`explorer` Explorer

`currentBlockIndex` long

`avatarAddress` Address

`updatedReward` AdventureBossGameData.ClaimableReward

`ncgReward` FungibleAssetValue

## Returns

bool

# CollectWantedReward(ClaimableReward, GameConfigState, AdventureBossNcgRewardRatioSheet, SeasonInfo, BountyBoard, Investor, long, Address, ref ClaimableReward)

```
public static bool CollectWantedReward(AdventureBossGameData.ClaimableReward reward,
GameConfigState gameConfig, AdventureBossNcgRewardRatioSheet ncgRewardRatioSheet,
SeasonInfo seasonInfo, BountyBoard bountyBoard, Investor investor, long
currentBlockIndex, Address avatarAddress, ref AdventureBossGameData.ClaimableReward
updatedReward)
```

### Parameters

`reward` AdventureBossGameData.ClaimableReward

`gameConfig` GameConfigState

`ncgRewardRatioSheet` AdventureBossNcgRewardRatioSheet

`seasonInfo` SeasonInfo

`bountyBoard` BountyBoard

`investor` Investor

`currentBlockIndex` long

`avatarAddress` Address

`updatedReward` AdventureBossGameData.ClaimableReward

### Returns

bool

## GetSeasonAsAddressForm(long)

```
public static string GetSeasonAsAddressForm(long season)
```

### Parameters

`season` long

### Returns

string

## PickExploreRaffle(BountyBoard, ExploreBoard, ExplorerList, IRandom)

```
public static ExploreBoard PickExploreRaffle(BountyBoard bountyBoard, ExploreBoard
exploreBoard, ExplorerList explorerList, IRandom random)
```

### Parameters

`bountyBoard` BountyBoard

`exploreBoard` ExploreBoard

`explorerList` ExplorerList

`random` IRandom

### Returns

ExploreBoard

## PickReward(IRandom, IEnumerable<RewardRatioData>)

```
public static (int?, int?) PickReward(IRandom random,
IEnumerable<AdventureBossSheet.RewardRatioData> rewardData)
```

### Parameters

`random` IRandom

`rewardData` IEnumerable<AdventureBossSheet.RewardRatioData>

### Returns

(int?, int?)

# Class AvatarStateExtensions

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class AvatarStateExtensions
```

**Inheritance**

object ← AvatarStateExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetLevelAndExp(AvatarState, CharacterLevelSheet, int, int)

```
public static (int, long) GetLevelAndExp(this AvatarState avatarState,
CharacterLevelSheet characterLevelSheet, int stageId, int repeatCount)
```

#### Parameters

`avatarState` AvatarState

`characterLevelSheet` CharacterLevelSheet

`stageId` int

`repeatCount` int

#### Returns

(int, long)

# GetLevelAndExpForEventDungeon(AvatarState, CharacterLevelSheet, Row, int, int)

Calculates level and experience for event dungeon sweep with level-up consideration. This method handles experience accumulation correctly without subtracting experience during level up.

```
public static (int, long) GetLevelAndExpForEventDungeon(this AvatarState
avatarState, CharacterLevelSheet characterLevelSheet, EventScheduleSheet.Row
scheduleRow, int stageNumber, int repeatCount)
```

## Parameters

`avatarState` AvatarState

The avatar state to calculate experience for

`characterLevelSheet` CharacterLevelSheet

Character level sheet for level information

`scheduleRow` EventScheduleSheet.Row

Event schedule row containing dungeon experience configuration

`stageNumber` int

Event dungeon stage number

`repeatCount` int

Number of times to repeat the battle

## Returns

(int, long)

Tuple containing final level and experience

# GetLevelAndExpV1(AvatarState, CharacterLevelSheet, int, int)

```
[Obsolete("Use GetLevelAndExp")]
public static (int, long) GetLevelAndExpV1(this AvatarState avatarState,
CharacterLevelSheet characterLevelSheet, int stageId, int repeatCount)
```

### Parameters

`avatarState` AvatarState

`characterLevelSheet` CharacterLevelSheet

`stageId` int

`repeatCount` int

### Returns

(int, long)

## UpdateExp(AvatarState, int, long)

```
public static void UpdateExp(this AvatarState avatarState, int level, long exp)
```

### Parameters

`avatarState` AvatarState

`level` int

`exp` long

## UpdateInventory(AvatarState, List<ItemBase>)

```
public static void UpdateInventory(this AvatarState avatarState,
List<ItemBase> rewards)
```

### Parameters

`avatarState` AvatarState

`rewards` List<ItemBase>

## UpdateMonsterMap(AvatarState, StageWaveSheet, int)

```
public static void UpdateMonsterMap(this AvatarState avatarState, StageWaveSheet
stageWaveSheet, int stageId)
```

### Parameters

`avatarState` AvatarState

`stageWaveSheet` StageWaveSheet

`stageId` int

## ValidEquipmentAndCostume(AvatarState, IEnumerable<Guid>, List<Guid>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, long, string)

```
public static void ValidEquipmentAndCostume(this AvatarState avatarState,
IEnumerable<Guid> costumeIds, List<Guid> equipmentIds, ItemRequirementSheet
itemRequirementSheet, EquipmentItemRecipeSheet equipmentItemRecipeSheet,
EquipmentItemSubRecipeSheetV2 equipmentItemSubRecipeSheetV2,
EquipmentItemOptionSheet equipmentItemOptionSheet, long blockIndex,
string addressesHex)
```

### Parameters

`avatarState` AvatarState

`costumeIds` IEnumerable<Guid>

`equipmentIds` List<Guid>

`itemRequirementSheet` ItemRequirementSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`equipmentItemSubRecipeSheetV2` EquipmentItemSubRecipeSheetV2

`equipmentItemOptionSheet` EquipmentItemOptionSheet

`blockIndex` long

`addressesHex` string

## ValidEquipmentAndCostumeV2(AvatarState, IEnumerable<Guid>, List<Guid>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, long, string, GameConfigState)

```
public static (List<Equipment> equipments, List<Costume> costumes)
ValidEquipmentAndCostumeV2(this AvatarState avatarState, IEnumerable<Guid>
costumeIds, List<Guid> equipmentIds, ItemRequirementSheet itemRequirementSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2
equipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet equipmentItemOptionSheet,
long blockIndex, string addressesHex, GameConfigState gameConfigState)
```

### Parameters

`avatarState` AvatarState

`costumeIds` IEnumerable<Guid>

`equipmentIds` List<Guid>

`itemRequirementSheet` ItemRequirementSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`equipmentItemSubRecipeSheetV2` EquipmentItemSubRecipeSheetV2

`equipmentItemOptionSheet` EquipmentItemOptionSheet

`blockIndex` long

`addressesHex` string

`gameConfigState` GameConfigState

## Returns

(List<Equipment> equipments, List<Costume> costumes)

# Class CrystalCalculator

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class CrystalCalculator
```

**Inheritance**

object ← CrystalCalculator

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## CRYSTAL

```
public static readonly Currency CRYSTAL
```

### Field Value

Currency

## CrystalLimit

```
public const long CrystalLimit = 100000000
```

### Field Value

long

## MaxLevelExponent

```
public const int MaxLevelExponent = 5
```

### Field Value

int

## Methods

### CalculateBuffGachaCost(int, bool, CrystalStageBuffGachaSheet)

```
public static FungibleAssetValue CalculateBuffGachaCost(int stageId, bool
advancedGacha, CrystalStageBuffGachaSheet stageBuffGachaSheet)
```

### Parameters

`stageId` int

`advancedGacha` bool

`stageBuffGachaSheet` CrystalStageBuffGachaSheet

### Returns

FungibleAssetValue

### CalculateCombinationCost(FungibleAssetValue, Row, CrystalCostState, CrystalCostState)

```
public static FungibleAssetValue CalculateCombinationCost(FungibleAssetValue
crystal, CrystalFluctuationSheet.Row row, CrystalCostState prevWeeklyCostState =
null, CrystalCostState beforePrevWeeklyCostState = null)
```

### Parameters

`crystal` FungibleAssetValue

`row` CrystalFluctuationSheet.Row

`prevWeeklyCostState` CrystalCostState

`beforePrevWeeklyCostState` CrystalCostState

### Returns

FungibleAssetValue

## CalculateCrystal(Address, IEnumerable<Equipment>, FungibleAssetValue, bool, CrystalEquipmentGrindingSheet, CrystalMonsterCollectionMultiplierSheet, StakeRegularRewardSheet)

```
public static FungibleAssetValue CalculateCrystal(Address agentAddress,
IEnumerable<Equipment> equipmentList, FungibleAssetValue stakedAmount, bool
enhancementFailed, CrystalEquipmentGrindingSheet crystalEquipmentGrindingSheet,
CrystalMonsterCollectionMultiplierSheet crystalMonsterCollectionMultiplierSheet,
StakeRegularRewardSheet stakeRegularRewardSheet)
```

### Parameters

`agentAddress` Address

`equipmentList` IEnumerable<Equipment>

`stakedAmount` FungibleAssetValue

`enhancementFailed` bool

`crystalEquipmentGrindingSheet` CrystalEquipmentGrindingSheet

`crystalMonsterCollectionMultiplierSheet` CrystalMonsterCollectionMultiplierSheet

`stakeRegularRewardSheet` StakeRegularRewardSheet

### Returns

FungibleAssetValue

## CalculateCrystal(IEnumerable<Equipment>, bool, CrystalEquipmentGrindingSheet, CrystalMonsterCollectionMultiplierSheet, int)

```
public static FungibleAssetValue CalculateCrystal(IEnumerable<Equipment>
equipmentList, bool enhancementFailed, CrystalEquipmentGrindingSheet
crystalEquipmentGrindingSheet, CrystalMonsterCollectionMultiplierSheet
crystalMonsterCollectionMultiplierSheet, int stakingLevel)
```

### Parameters

`equipmentList` IEnumerable<Equipment>

`enhancementFailed` bool

`crystalEquipmentGrindingSheet` CrystalEquipmentGrindingSheet

`crystalMonsterCollectionMultiplierSheet` CrystalMonsterCollectionMultiplierSheet

`stakingLevel` int

### Returns

FungibleAssetValue

## CalculateEntranceFee(int, BigInteger)

```
public static FungibleAssetValue CalculateEntranceFee(int level,
BigInteger entranceFee)
```

### Parameters

`level` int

`entranceFee` BigInteger

### Returns

FungibleAssetValue

## CalculateMaterialCost(int, int, CrystalMaterialCostSheet)

```
public static FungibleAssetValue CalculateMaterialCost(int materialId, int
materialCount, CrystalMaterialCostSheet crystalMaterialCostSheet)
```

### Parameters

`materialId` int

`materialCount` int

`crystalMaterialCostSheet` CrystalMaterialCostSheet

### Returns

FungibleAssetValue

## CalculateRecipeUnlockCost(IEnumerable<int>, EquipmentItemRecipeSheet)

```
public static FungibleAssetValue CalculateRecipeUnlockCost(IEnumerable<int>
recipeIds, EquipmentItemRecipeSheet equipmentItemRecipeSheet)
```

### Parameters

`recipeIds` IEnumerable<int>

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

### Returns

FungibleAssetValue

# CalculateWorldUnlockCost(IEnumerable<int>, WorldUnlockSheet)

```
public static FungibleAssetValue CalculateWorldUnlockCost(IEnumerable<int> worldIds,
WorldUnlockSheet worldUnlockSheet)
```

## Parameters

`worldIds` IEnumerable<int>

`worldUnlockSheet` WorldUnlockSheet

## Returns

FungibleAssetValue

# Class CustomCraftHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class CustomCraftHelper
```

**Inheritance**

object ← CustomCraftHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### CalculateAdditionalCost(int, CustomEquipmentCraftRelationshipSheet)

```
public static (BigInteger, IDictionary<int, int>)? CalculateAdditionalCost(int
relationship, CustomEquipmentCraftRelationshipSheet relationshipSheet)
```

#### Parameters

`relationship` int

`relationshipSheet` CustomEquipmentCraftRelationshipSheet

#### Returns

(BigInteger, IDictionary<int, int>)?

### CalculateCraftCost(int, int, MaterialItemSheet, Row, Row, decimal)

```
public static (BigInteger, IDictionary<int, int>) CalculateCraftCost(int iconId, int
relationship, MaterialItemSheet materialItemSheet,
CustomEquipmentCraftRecipeSheet.Row recipeRow,
CustomEquipmentCraftRelationshipSheet.Row relationshipRow,
decimal iconCostMultiplier)
```

### Parameters

`iconId` int

`relationship` int

`materialItemSheet` MaterialItemSheet

`recipeRow` CustomEquipmentCraftRecipeSheet.Row

`relationshipRow` CustomEquipmentCraftRelationshipSheet.Row

`iconCostMultiplier` decimal

### Returns

(BigInteger, IDictionary<int, int>)

## SelectCp(Row, IRandom)

```
public static int SelectCp(CustomEquipmentCraftRelationshipSheet.Row
relationshipRow, IRandom random)
```

### Parameters

`relationshipRow` CustomEquipmentCraftRelationshipSheet.Row

`random` IRandom

### Returns

int

# Class InventoryExtensions

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class InventoryExtensions
```

**Inheritance**

object ← InventoryExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetEquippedFullCostumeOrArmorId(Inventory)

```
public static int GetEquippedFullCostumeOrArmorId(this Inventory inventory)
```

#### Parameters

`inventory` Inventory

#### Returns

int

### MintItem(Inventory, Row, int, bool, IRandom)

Create and Add item.

```
public static void MintItem(this Inventory inventory, ItemSheet.Row itemRow, int
quantity, bool tradable, IRandom random)
```

### Parameters

`inventory` Inventory

Inventory>

`itemRow` ItemSheet.Row

ItemSheet.Row

`quantity` int

create item count

`tradable` bool

item is tradable

`random` IRandom

Libplanet.Action.IRandom>

## UseActionPoint(Inventory, long, int, bool, MaterialItemSheet, long)

```
public static long UseActionPoint(this Inventory inventory, long originalAp, int
requiredAp, bool chargeAp, MaterialItemSheet materialItemSheet, long blockIndex)
```

### Parameters

`inventory` Inventory

`originalAp` long

`requiredAp` int

`chargeAp` bool

`materialItemSheet` MaterialItemSheet

`blockIndex` long

### Returns

long

# Class ItemOptionHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class ItemOptionHelper
```

**Inheritance**

object ← ItemOptionHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetStatOptionRows(Row, EquipmentItemOptionSheet, ItemUsable)

```
public static List<EquipmentItemOptionSheet.Row>
GetStatOptionRows(EquipmentItemSubRecipeSheetV2.Row subRecipeRow,
EquipmentItemOptionSheet itemOptionSheet, ItemUsable itemUsable)
```

#### Parameters

`subRecipeRow` EquipmentItemSubRecipeSheetV2.Row

`itemOptionSheet` EquipmentItemOptionSheet

`itemUsable` ItemUsable

#### Returns

List<EquipmentItemOptionSheet.Row>

## GetStatOptionRows(long, ItemUsable, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

```
public static List<EquipmentItemOptionSheet.Row> GetStatOptionRows(long subRecipeId,
ItemUsable itemUsable, EquipmentItemSubRecipeSheetV2 subRecipeSheet,
EquipmentItemOptionSheet itemOptionSheet)
```

### Parameters

`subRecipeId` long

`itemUsable` ItemUsable

`subRecipeSheet` EquipmentItemSubRecipeSheetV2

`itemOptionSheet` EquipmentItemOptionSheet

### Returns

List<EquipmentItemOptionSheet.Row>

## TryGet(ItemUsable, out ItemOptionInfo)

```
public static bool TryGet(ItemUsable itemUsable, out ItemOptionInfo itemOptionInfo)
```

### Parameters

`itemUsable` ItemUsable

`itemOptionInfo` ItemOptionInfo

### Returns

bool

# Class ItemOptionInfo

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public class ItemOptionInfo
```

**Inheritance**

object ← ItemOptionInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ItemOptionInfo(Equipment)

```
public ItemOptionInfo(Equipment equipment)
```

#### Parameters

`equipment` Equipment

### ItemOptionInfo(ItemUsable)

```
public ItemOptionInfo(ItemUsable itemUsable)
```

#### Parameters

`itemUsable` ItemUsable

## Fields

## CP

```
public readonly long CP
```

### Field Value

long

## MainStat

```
public readonly (StatType type, long baseValue, long totalValue) MainStat
```

### Field Value

(StatType type, long baseValue, long totalValue)

## OptionCountFromCombination

```
public readonly int OptionCountFromCombination
```

### Field Value

int

## SkillOptions

```
public readonly List<(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType refStatType)> SkillOptions
```

### Field Value

List<(SkillSheet.Row skillRow, long power, int chance, int statPowerRatio, StatType refStatType)>

# StatOptions

```
public readonly List<(StatType type, long value, int count)> StatOptions
```

## Field Value

List<(StatType type, long value, int count)>

# Class NumberConversionHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

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```
public static class NumberConversionHelper
```

**Inheritance**

object ← NumberConversionHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## SafeBigIntegerToInt32(BigInteger)

Safely converts BigInteger to int32, clamping to int32 bounds to prevent overflow.

```
public static int SafeBigIntegerToInt32(BigInteger value)
```

### Parameters

`value` BigInteger

The BigInteger value to convert.

### Returns

int

The converted int32 value, clamped to int32 bounds.

# SafeDecimalToInt32(decimal)

Safely converts decimal to int32, clamping to int32 bounds to prevent overflow.

```
public static int SafeDecimalToInt32(decimal value)
```

## Parameters

`value` decimal

The decimal value to convert.

## Returns

int

The converted int32 value, clamped to int32 bounds.

# SafeDecimalToInt64(decimal)

Safely converts decimal to long, clamping to long bounds to prevent overflow.

```
public static long SafeDecimalToInt64(decimal value)
```

## Parameters

`value` decimal

The decimal value to convert.

## Returns

long

The converted long value, clamped to long bounds.

# Class PetHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class PetHelper
```

**Inheritance**

object ← PetHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### CalculateDiscountedHourglass(long, int, PetState, PetOptionSheet)

```
public static int CalculateDiscountedHourglass(long diff, int hourglassPerBlock,
PetState petState, PetOptionSheet petOptionSheet)
```

#### Parameters

`diff` long

`hourglassPerBlock` int

`petState` PetState

`petOptionSheet` PetOptionSheet

#### Returns

int

# CalculateDiscountedMaterialCost(FungibleAssetValue, PetState, PetOptionSheet)

```
public static FungibleAssetValue CalculateDiscountedMaterialCost(FungibleAssetValue
originalCost, PetState petState, PetOptionSheet petOptionSheet)
```

## Parameters

`originalCost` FungibleAssetValue

`petState` PetState

`petOptionSheet` PetOptionSheet

## Returns

FungibleAssetValue

# CalculateEnhancementCost(PetCostSheet, int, int, int)

```
public static (int ncgQuantity, int soulStoneQuantity)
CalculateEnhancementCost(PetCostSheet costSheet, int petId, int currentLevel,
int targetLevel)
```

## Parameters

`costSheet` PetCostSheet

`petId` int

`currentLevel` int

`targetLevel` int

## Returns

(int challengerScoreDelta, int defenderScoreDelta)

# CalculateReducedBlockOnCraft(long, long, PetState, PetOptionSheet)

```
public static long CalculateReducedBlockOnCraft(long originalBlock, long
minimumBlock, PetState petState, PetOptionSheet petOptionSheet)
```

## Parameters

`originalBlock` long

`minimumBlock` long

`petState` PetState

`petOptionSheet` PetOptionSheet

## Returns

long

# GetBonusOptionProbability(int, PetState, PetOptionSheet)

```
public static int GetBonusOptionProbability(int originalRatio, PetState petState,
PetOptionSheet petOptionSheet)
```

## Parameters

`originalRatio` int

`petState` PetState

`petOptionSheet` PetOptionSheet

## Returns

int

# GetSoulstoneCurrency(string)

```
public static Currency GetSoulstoneCurrency(string ticker)
```

## Parameters

`ticker` string

## Returns

Currency

# Class RuneEnhancementCostBoost

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

A BoostScheduleSheet already checked to apply to the enhancement cost of one rune, RuneId: an enhancement cost row was active and the rune was summonable when it was resolved. Only ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>) creates one, and TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long) and ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long) ignore it for any other rune, so a boost only ever discounts the rune it was resolved for — neither handing over the raw patched sheet nor reusing a boost after switching runes can skip the check.

```
public sealed class RuneEnhancementCostBoost
```

**Inheritance**

object ← RuneEnhancementCostBoost

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## RuneId

The rune this boost was resolved for, and the only one it adjusts.

```
public int RuneId { get; }
```

### Property Value

int

# Class RuneHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class RuneHelper
```

**Inheritance**

object ← RuneHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## DailyRewardRune

```
public static readonly Currency DailyRewardRune
```

### Field Value

Currency

## StakeRune

```
public static readonly Currency StakeRune
```

### Field Value

Currency

# Methods

# ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long)

Applies the SummonRuneEnhancementCrystalCost and SummonRuneEnhancementRuneStoneCost rows active at `blockIndex` to the cost of one try that reaches `targetLevel`.

```
public static RuneCostSheet.RuneCostData
ApplyEnhancementCostBoost(RuneCostSheet.RuneCostData cost, int runeId, int
targetLevel, RuneEnhancementCostBoost? boost, long blockIndex)
```

## Parameters

`cost` RuneCostSheet.RuneCostData

The sheet's cost for `targetLevel`.

`runeId` int

The rune `cost` belongs to. `boost` applies only when it was resolved for this rune.

`targetLevel` int

The level the try attempts to reach, matched against `target_id`.

`boost` RuneEnhancementCostBoost

What ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>) returned for the rune, or `null` for none. Ignored when resolved for a rune other than `runeId`.

`blockIndex` long

The block being evaluated.

## Returns

RuneCostSheet.RuneCostData

`cost` itself when no row applies; otherwise a copy whose crystal and rune stone quantities are adjusted by ApplyEnhancementCostBoostQuantity(Row?, int). NCG quantity and success rate are always copied unchanged.

## Remarks

Callers showing a cost — or a maximum try count — should pass every per-level cost through this, with the boost from ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>), so that they agree with RuneEnhancement.

## ApplyEnhancementCostBoostQuantity(Row?, int)

Applies one BoostScheduleSheet row to one enhancement cost quantity.

```
public static int ApplyEnhancementCostBoostQuantity(BoostScheduleSheet.Row? boost,
int baseCost)
```

### Parameters

`boost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`baseCost` int

The sheet's quantity.

### Returns

int

The adjusted quantity, kept within `[1, baseCost]` when `baseCost` is positive. A non-positive `baseCost` is returned unchanged.

### Remarks

Discount only: a row can never raise a cost. Raising one is not an event anyone has asked for, and it would let a mistyped row push the per-try sum in TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long) past MaxValue, where it wraps negative and the cost is skipped instead of charged.

At least 1: rounding down a small cost (4 stones × 0.2 = 0.8) would otherwise make the try free, turning a discount into a giveaway that the row never declared. A free enhancement has to be configured in RuneCostSheet itself. A cost the sheet already sets to 0 stays 0, so a boost never creates a cost either.

## CalculateRuneLevelBonus(AllRuneState, RuneListSheet, RuneLevelBonusSheet)

```
public static int CalculateRuneLevelBonus(AllRuneState allRuneState, RuneListSheet
runeListSheet, RuneLevelBonusSheet runeLevelBonusSheet)
```

### Parameters

`allRuneState` AllRuneState

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

### Returns

int

## CalculateStakeReward(FungibleAssetValue, int)

```
public static FungibleAssetValue CalculateStakeReward(FungibleAssetValue
stakeAmount, int rate)
```

### Parameters

`stakeAmount` FungibleAssetValue

`rate` int

### Returns

FungibleAssetValue

## GetRuneOptions(IEnumerable<RuneSlotInfo>, AllRuneState, RuneOptionSheet)

```
public static List<RuneOptionSheet.Row.RuneOptionInfo>
GetRuneOptions(IEnumerable<RuneSlotInfo> runeInfos, AllRuneState runeStates,
```

```
RuneOptionSheet runeOptionSheet)
```

### Parameters

`runeInfos` IEnumerable<RuneSlotInfo>

`runeStates` AllRuneState

`runeOptionSheet` RuneOptionSheet

### Returns

List<RuneOptionSheet.Row.RuneOptionInfo>

## GetRuneOptions(IEnumerable<RuneInfo>, AllRuneState, RuneOptionSheet)

```
public static List<RuneOptionSheet.Row.RuneOptionInfo>
GetRuneOptions(IEnumerable<RuneWeightSheet.RuneInfo> runeInfos, AllRuneState
runeStates, RuneOptionSheet runeOptionSheet)
```

### Parameters

`runeInfos` IEnumerable<RuneWeightSheet.RuneInfo>

`runeStates` AllRuneState

`runeOptionSheet` RuneOptionSheet

### Returns

List<RuneOptionSheet.Row.RuneOptionInfo>

## ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>)

Decides whether a rune's enhancement cost may be adjusted at all, and returns what to pass to TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCost

Boost?, long) and ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancement CostBoost?, long).

```
public static RuneEnhancementCostBoost?
ResolveEnhancementCostBoost(BoostScheduleSheet? boostScheduleSheet, int runeId, long
blockIndex, Func<RuneSummonSheet?> getRuneSummonSheet)
```

## Parameters

`boostScheduleSheet` BoostScheduleSheet

The patched BoostScheduleSheet, or `null` when the chain has none.

`runeId` int

The rune being enhanced.

`blockIndex` long

The block being evaluated.

`getRuneSummonSheet` Func<RuneSummonSheet>

Loads the RuneSummonSheet, returning `null` when it is absent or unreadable. Called only when an enhancement cost row is active at `blockIndex`, so that evaluation without one reads no extra state.

## Returns

RuneEnhancementCostBoost

A RuneEnhancementCostBoost over `boostScheduleSheet` when a SummonRune EnhancementCrystalCost or SummonRuneEnhancementRuneStoneCost row is active at `blockIndex` and `runeId` is a recipe id of any RuneSummonSheet group; otherwise `null`, meaning no adjustment.

## Remarks

The summon pool is derived from the sheet rather than listed, so the adjustment follows later changes to it. A recipe counts whatever its ratio and whether or not its group is currently sold. A missing RuneSummonSheet means no rune is summonable, never an error.

## ToCurrency(Row)

```
public static Currency ToCurrency(RuneSheet.Row runeRow)
```

### Parameters

`runeRow` RuneSheet.Row

### Returns

Currency

## ToFungibleAssetValue(Row, int)

```
public static FungibleAssetValue ToFungibleAssetValue(RuneSheet.Row runeRow,
int quantity)
```

### Parameters

`runeRow` RuneSheet.Row

`quantity` int

### Returns

FungibleAssetValue

## TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long)

Simulates `tryCount` enhancement tries from `startRuneLevel`, summing the cost of every try and drawing one random number per try for its success.

```
public static bool TryEnhancement(int startRuneLevel, RuneCostSheet.Row costRow,
IRandom random, int tryCount, out RuneEnhancement.LevelUpResult levelUpResult,
RuneEnhancementCostBoost? boost = null, long blockIndex = 0)
```

## Parameters

`startRuneLevel` int

The rune level before the first try.

`costRow` RuneCostSheet.Row

The rune's cost row.

`random` IRandom

The random source; one draw per try.

`tryCount` int

The number of tries.

`levelUpResult` RuneEnhancement.LevelUpResult

The level-ups and the summed cost.

`boost` RuneEnhancementCostBoost

What ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>) returned for this rune, or `null` for none. Adjusts each try's crystal and rune stone cost; see ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long). Ignored, not rejected, when it was resolved for a rune other than `costRow`'s, so a preview holding on to a stale boost shows the undiscounted cost the chain would charge.

`blockIndex` long

The block being evaluated, against which `boost` rows are matched. Ignored when `boost` is `null`.

## Returns

bool

`false` when a try reaches a level `costRow` has no cost for.

# Class SummonHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class SummonHelper
```

**Inheritance**

object ← SummonHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AllowedSummonCount

Array of allowed summon counts. Only these values are permitted for summon actions.

```
public static readonly int[] AllowedSummonCount
```

### Field Value

int[]

# Methods

## ApplyGuaranteeBoost(Row, int, int)

Applies a BoostScheduleSheet row to a configured guarantee count.

```
public static int ApplyGuaranteeBoost(BoostScheduleSheet.Row guaranteeBoost, int
guaranteeCount, int summonCount)
```

### Parameters

`guaranteeBoost` BoostScheduleSheet.Row

The row to apply.

`guaranteeCount` int

The guarantee count the summon sheet configures.

`summonCount` int

The number of summons, bonus included.

### Returns

int

The adjusted count, kept below `summonCount` so that at least one draw stays a normal draw — unless the sheet itself already configures more.

### Remarks

A boost adjusts a guarantee but never creates one, since a guarantee that exists only during an event reads as one taken away when the event ends. That also covers a group configured with a count of 0: its guaranteed-grade path has never run, so starting it mid-event could surface a grade with no eligible recipe and fail every summon.

## CalculateSummonCount(int)

Calculates the actual summon count by applying the 10+1 bonus rule. For every 10 summons, 1 additional summon is granted as a bonus.

```
public static int CalculateSummonCount(int count)
```

### Parameters

`count` int

Base summon count

### Returns

int

Total summon count including bonus

## CheckSummonCountIsValid(int)

Validates if the given summon count is allowed. Only counts in AllowedSummonCount array (1, 10, 100) are permitted.

```
public static bool CheckSummonCountIsValid(int count)
```

### Parameters

`count` int

Summon count to validate

### Returns

bool

True if the count is allowed, false otherwise

## GetGuaranteeSettings(Row, int, Row)

Determines guarantee settings based on summon count and SummonSheet.Row configuration.

```
public static (bool useGuarantee, int minimumGrade, int guaranteeCount)
GetGuaranteeSettings(SummonSheet.Row summonRow, int summonCount,
BoostScheduleSheet.Row guaranteeBoost = null)
```

### Parameters

`summonRow` SummonSheet.Row

The summon row containing guarantee settings

`summonCount` int

The number of summons

`guaranteeBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this group's guarantee count at the current block, or `null` for none. See ApplyGuaranteeBoost(Row, int, int).

### Returns

(bool useGuarantee, int minimumGrade, int guaranteeCount)

A tuple containing (useGuarantee, minimumGrade, guaranteeCount)

## GetSummonRecipeIdByRandom(Row, IRandom)

Gets a random recipe ID from the summon row based on weighted ratios. Uses cumulative ratio calculation to select recipes according to their configured probabilities.

```
public static int GetSummonRecipeIdByRandom(SummonSheet.Row summonRow,
IRandom random)
```

### Parameters

`summonRow` SummonSheet.Row

Summon sheet row containing recipes and their ratios

`random` IRandom

Random number generator

### Returns

int

Selected recipe ID

## GetSummonRecipeIdsWithGradeGuarantee(Row, int, IRandom, CostumeItemSheet, EquipmentItemRecipeSheet, Row)

Gets summon recipe IDs with grade guarantee system for costumes. Applies different guarantee settings for 11 summons vs 110 summons. This method processes items one by one to maintain the same random call order as the original implementation. Uses grade guarantee settings from SummonSheet.Row.

```
public static List<int> GetSummonRecipeIdsWithGradeGuarantee(SummonSheet.Row
summonRow, int summonCount, IRandom random, CostumeItemSheet costumeItemSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet, BoostScheduleSheet.Row
guaranteeBoost = null)
```

## Parameters

`summonRow` SummonSheet.Row

Summon sheet row containing recipes, ratios, and grade guarantee settings

`summonCount` int

Number of items to summon

`random` IRandom

Random number generator

`costumeItemSheet` CostumeItemSheet

Costume item sheet to check grades

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

Equipment item recipe sheet (not used for costumes, can be null)

`guaranteeBoost` BoostScheduleSheet.Row

Guarantee count adjustment, or `null` for none.

## Returns

List<int>

List of recipe IDs with grade guarantee applied

# GetSummonRecipeIdsWithGradeGuarantee(Row, int, IRandom, EquipmentItemSheet, EquipmentItemRecipeSheet, Row)

Gets summon recipe IDs with grade guarantee system based on summon count. Applies different guarantee settings for 11 summons vs 110 summons. This method processes items one by one to maintain the same random call order as the original implementation. Uses grade guarantee settings from SummonSheet.Row.

```
public static List<int> GetSummonRecipeIdsWithGradeGuarantee(SummonSheet.Row
summonRow, int summonCount, IRandom random, EquipmentItemSheet equipmentItemSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet, BoostScheduleSheet.Row
guaranteeBoost = null)
```

## Parameters

`summonRow` SummonSheet.Row

Summon sheet row containing recipes, ratios, and grade guarantee settings

`summonCount` int

Number of items to summon

`random` IRandom

Random number generator

`equipmentItemSheet` EquipmentItemSheet

Equipment item sheet to check grades

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

Equipment item recipe sheet to map recipe IDs to equipment IDs

`guaranteeBoost` BoostScheduleSheet.Row

Guarantee count adjustment, or `null` for none.

## Returns

List<int>

List of recipe IDs with grade guarantee applied

# Struct SynthesizeResult

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

Represents the result of the synthesis.

```
public struct SynthesizeResult
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

# Fields

## IsSuccess

```
public bool IsSuccess
```

### Field Value

bool

## ItemBase

```
public ItemBase ItemBase
```

### Field Value

ItemBase

## RecipeId

```
public int RecipeId
```

### Field Value

int

## SubRecipeId

```
public int SubRecipeId
```

### Field Value

int

# Properties

## IsEquipment

```
public bool IsEquipment { get; }
```

### Property Value

bool

# Class SynthesizeSimulator

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

A class that simulates the synthesis of items. SynthesizeSimulator.InputData

```
public static class SynthesizeSimulator
```

**Inheritance**

object ← SynthesizeSimulator

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetCostumeMaterialList(List<Guid>, AvatarState, Grade, ItemSubType, string)

```
public static List<ItemBase> GetCostumeMaterialList(List<Guid> materialIds,
AvatarState avatarState, Grade grade, ItemSubType itemSubType, string addressesHex)
```

#### Parameters

`materialIds` List<Guid>

`avatarState` AvatarState

`grade` Grade

`itemSubType` ItemSubType

`addressesHex` string

#### Returns

List<ItemBase>

## GetEquipmentMaterialList(List<Guid>, AvatarState, long, Grade, ItemSubType, string)

```
public static List<ItemBase> GetEquipmentMaterialList(List<Guid> materialIds,
AvatarState avatarState, long blockIndex, Grade grade, ItemSubType itemSubType,
string addressesHex)
```

### Parameters

`materialIds` List<Guid>

`avatarState` AvatarState

`blockIndex` long

`grade` Grade

`itemSubType` ItemSubType

`addressesHex` string

### Returns

List<ItemBase>

## GetItemGuid(ItemBase)

Get the item id from the item.

```
public static List<Guid> GetItemGuid(ItemBase itemBase)
```

### Parameters

`itemBase` ItemBase

item to get id

### Returns

List<Guid>

item id

### Exceptions

ArgumentException

## GetItemGuids(IEnumerable<ItemBase>)

Get the item id from the item.

```
public static List<Guid> GetItemGuids(IEnumerable<ItemBase> itemBases)
```

### Parameters

`itemBases` IEnumerable<ItemBase>

items to get id

### Returns

List<Guid>

item id list

### Exceptions

ArgumentException

## GetMaterialList(List<Guid>, AvatarState, long, Grade, ItemSubType, string)

```
public static List<ItemBase> GetMaterialList(List<Guid> materialIds,
AvatarState avatarState, long blockIndex, Grade grade, ItemSubType itemSubType,
string addressesHex)
```

### Parameters

`materialIds` List<Guid>

`avatarState` AvatarState

`blockIndex` long

`grade` Grade

`itemSubType` ItemSubType

`addressesHex` string

### Returns

List<ItemBase>

## GetRequiredCount(int, Row?)

Returns how many materials one synthesis consumes, after a BoostScheduleSheet adjustment.

```
public static int GetRequiredCount(int requiredCount, BoostScheduleSheet.Row?
requiredCountBoost)
```

### Parameters

`requiredCount` int

The `required_count` that SynthesizeSheet configures.

`requiredCountBoost` BoostScheduleSheet.Row

The row found for SynthesizeRequiredCount and the material grade at the current block, or `null` for none.

### Returns

int

`requiredCount` itself when there is no row or it is not positive; otherwise the adjusted count, rounded down but never below 1.

### Remarks

The floor of 1 keeps a discount from making a synthesis free (and the division by the count from failing): `1 × 0.6` stays `1`. A count the sheet configures as 0 or less is left alone so that it fails exactly as it did without the boost.

## GetSynthesizeResultPool(Grade, ItemSubType, CostumeItemSheet)

get expected result item pool (Costume)

```
public static HashSet<int> GetSynthesizeResultPool(Grade sourceGrade, ItemSubType subType, CostumeItemSheet sheet)
```

### Parameters

`sourceGrade` Grade

grade of material items

`subType` ItemSubType

subtype of material items

`sheet` CostumeItemSheet

costume item sheet to use

### Returns

HashSet<int>

expected result item pool of costume

## GetSynthesizeResultPool(Grade, ItemSubType, EquipmentItemSheet)

get expected result item pool (Equipment)

```
public static HashSet<int> GetSynthesizeResultPool(Grade sourceGrade, ItemSubType
subType, EquipmentItemSheet sheet)
```

### Parameters

`sourceGrade` Grade

grade of material items

`subType` ItemSubType

subtype of material items

`sheet` EquipmentItemSheet

equipment item sheet to use

### Returns

HashSet<int>

expected result item pool of equipment

## GetSynthesizeResultPool(HashSet<Grade>, ItemSubType, CostumeItemSheet)

Returns a list of items that may come out as a result of that synthesis.

```
public static HashSet<(int, Grade)> GetSynthesizeResultPool(HashSet<Grade>
sourceGrades, ItemSubType subType, CostumeItemSheet sheet)
```

### Parameters

`sourceGrades` HashSet<Grade>

grades of material items

`subType` ItemSubType

excepted FullCostume,Title

`sheet` CostumeItemSheet

CostumeItemSheet to use

### Returns

HashSet<(int, Grade)>

list of items key(int), grade(Grade) tuple

## GetSynthesizeResultPool(HashSet<Grade>, ItemSubType, EquipmentItemSheet)

Returns a list of items that may come out as a result of that synthesis.

```
public static HashSet<(int, Grade)> GetSynthesizeResultPool(HashSet<Grade>
sourceGrades, ItemSubType subType, EquipmentItemSheet sheet)
```

### Parameters

`sourceGrades` HashSet<Grade>

grades of material items

`subType` ItemSubType

excepted Grimoire,Aura

`sheet` EquipmentItemSheet

EquipmentItemSheet to use

### Returns

HashSet<(int, Grade)>

list of items key(int), grade(Grade) tuple

# GetTargetGrade(Grade)

Get the target grade of the item: the grade one step above the given one.

```
public static Grade GetTargetGrade(Grade grade)
```

## Parameters

`grade` Grade

grade of the item

## Returns

Grade

target grade

## Remarks

Since there is no upper bound (see the int overload), the returned value is one step above whatever it is given and so need not be a defined Grade member. Callers that hand it to a sheet lookup are fine — an unlisted grade simply matches no row. Callers that switch on it should keep a default arm.

## Exceptions

ArgumentOutOfRangeException

Thrown when `grade` is below the first grade.

# GetTargetGrade(int)

Get the target grade of the item: the grade id one step above the given one.

```
public static int GetTargetGrade(int gradeId)
```

## Parameters

`gradeId` int

grade id of the item

### Returns

int

target grade id

### Remarks

There is deliberately no upper bound here. Whether the grade above exists is a question the item sheets answer, and both callers inside this class ask them: the result-pool lookups fall back to the source grade when the pool for the grade above comes back empty. Capping here as well would only mean editing this method every time a grade is added. Callers outside this class have to arrange that fallback themselves.

### Exceptions

ArgumentOutOfRangeException

Thrown when `gradeId` is below the first grade, or so high that the grade above it does not fit in an int.

## GetUpgradeGrade(Grade, ItemSubType, CostumeItemSheet)

Returns the grade of the item that can be obtained by synthesizing the item. (Costume)

```
public static Grade GetUpgradeGrade(Grade grade, ItemSubType subType,
CostumeItemSheet sheet)
```

### Parameters

`grade` Grade

source grade

`subType` ItemSubType

subtype of material items

`sheet` CostumeItemSheet

costume item sheet to use

### Returns

Grade

grade of the item that can be obtained by synthesizing the item

## GetUpgradeGrade(Grade, ItemSubType, EquipmentItemSheet)

Returns the grade of the item that can be obtained by synthesizing the item. (Equipment)

```
public static Grade GetUpgradeGrade(Grade grade, ItemSubType subType,
EquipmentItemSheet sheet)
```

### Parameters

`grade` Grade

source grade

`subType` ItemSubType

subtype of material items

`sheet` EquipmentItemSheet

equipment item sheet to use

### Returns

Grade

grade of the item that can be obtained by synthesizing the item

## GetWeight(int, SynthesizeWeightSheet)

Returns the weight of the item that can be obtained by synthesizing the item.

```
public static int GetWeight(int itemId, SynthesizeWeightSheet sheet)
```

### Parameters

`itemId` int

item id of material item

`sheet` SynthesizeWeightSheet

SynthesizeWeightSheet to use

### Returns

int

weight of the item that can be obtained by synthesizing the item

## Simulate(InputData)

Simulate the synthesis of items.

```
public static List<SynthesizeResult> Simulate(SynthesizeSimulator.InputData
inputData)
```

### Parameters

`inputData` SynthesizeSimulator.InputData

SynthesizeSimulator.InputData

### Returns

List<SynthesizeResult>

### Exceptions

SheetRowNotFoundException

NotEnoughMaterialException

# Struct SynthesizeSimulator.InputData

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

Simulate the synthesis of items.

```
public struct SynthesizeSimulator.InputData
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

# Fields

## BlockIndex

The block index when action is executed.

```
public long BlockIndex
```

### Field Value

long

## CostumeItemSheet

The sheet that contains the costume item information. CostumeItemSheet

```
public CostumeItemSheet CostumeItemSheet
```

### Field Value

CostumeItemSheet

## EquipmentItemOptionSheet

The sheet that contains the equipment item option information. EquipmentItemOptionSheet

```
public EquipmentItemOptionSheet EquipmentItemOptionSheet
```

### Field Value

EquipmentItemOptionSheet

## EquipmentItemRecipeSheet

The sheet that contains the equipment item recipe information. EquipmentItemRecipeSheet

```
public EquipmentItemRecipeSheet EquipmentItemRecipeSheet
```

### Field Value

EquipmentItemRecipeSheet

## EquipmentItemSheet

The sheet that contains the equipment item information. EquipmentItemSheet

```
public EquipmentItemSheet EquipmentItemSheet
```

### Field Value

EquipmentItemSheet

## EquipmentItemSubRecipeSheetV2

The sheet that contains the equipment item sub recipe information. EquipmentItemSubRecipeSheetV2

```
public EquipmentItemSubRecipeSheetV2 EquipmentItemSubRecipeSheetV2
```

### Field Value

EquipmentItemSubRecipeSheetV2

## Grade

The grade of the material item.

```
public Grade Grade
```

### Field Value

Grade

## ItemSubType

The subtype of the material item.

```
public ItemSubType ItemSubType
```

### Field Value

ItemSubType

## MaterialCount

The number of materials.

```
public int MaterialCount
```

### Field Value

int

## RandomObject

The random object to use. Caution: Must have the same seed as when the action is executed

```
public IRandom RandomObject
```

### Field Value

IRandom

## RequiredCountBoost

The BoostScheduleSheet row adjusting the required material count of Grade at BlockIndex, or `null` for none. See GetRequiredCount(int, Row?). Caution: Must be the row the action resolved, or the material count may not divide.

```
public BoostScheduleSheet.Row? RequiredCountBoost
```

### Field Value

BoostScheduleSheet.Row

## SkillSheet

The sheet that contains the skill information. SkillSheet

```
public SkillSheet SkillSheet
```

### Field Value

SkillSheet

## SynthesizeSheet

The sheet that contains the synthesis information. SynthesizeSheet

```
public SynthesizeSheet SynthesizeSheet
```

### Field Value

SynthesizeSheet

## SynthesizeWeightSheet

The sheet that contains the synthesis weight information. SynthesizeWeightSheet

```
public SynthesizeWeightSheet SynthesizeWeightSheet
```

### Field Value

SynthesizeWeightSheet

# Class WorldBossHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class WorldBossHelper
```

**Inheritance**

object ← WorldBossHelper

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## MaxChallengeCount

```
public const int MaxChallengeCount = 3
```

### Field Value

int

## RefillInterval

```
[Obsolete("Use GameConfigState.DailyWorldBossInterval")]
public const long RefillInterval = 7200
```

### Field Value

long

# Methods

## ApplyRewardBoost(Row, int)

Applies a BoostScheduleSheet row to one granted amount of a world boss reward: the count of one drawn rune, the crystal or the circles.

```
public static int ApplyRewardBoost(BoostScheduleSheet.Row rewardBoost, int amount)
```

### Parameters

`rewardBoost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`amount` int

The amount the reward row and the draws grant.

### Returns

int

`amount` unchanged when there is no row or nothing is granted; otherwise Apply(int) of it. A result of 0 means the amount is not granted at all.

### Remarks

A boost adjusts what a reward grants but never adds a kind the row does not grant — a row with no crystal stays without crystal under `ADD` — since a reward that exists only during an event reads as one taken away when the event ends. `ADD` applies to every amount separately, i.e. once per distinct rune drawn; use `MUL` to scale a reward as a whole.

## CalculateContribution(BigInteger, long)

Calculates the contribution percentage based on total damage and individual damage.

```
public static decimal CalculateContribution(BigInteger totalDamage, long myDamage)
```

### Parameters

`totalDamage` BigInteger

The total damage dealt.

`myDamage` long

The damage dealt by the individual.

### Returns

decimal

The contribution percentage rounded to four decimal places.

### Exceptions

ArgumentOutOfRangeException

Thrown when totalDamage is lower than 0.

## CalculateContributionReward(Row, decimal)

Calculates the contribution reward based on the given contribution percentage.

```
public static (List<(int id, int count)>, List<FungibleAssetValue>)
CalculateContributionReward(WorldBossContributionRewardSheet.Row row,
decimal contribution)
```

### Parameters

`row` WorldBossContributionRewardSheet.Row

The row from the WorldBossContributionRewardSheet containing reward details.

`contribution` decimal

The contribution percentage of the player.

### Returns

(List<(int id, int count)>, List<FungibleAssetValue>)

A tuple containing a list of item rewards and a list of fungible asset values.

## CalculateRank(Row, long)

```
public static int CalculateRank(WorldBossCharacterSheet.Row row, long score)
```

### Parameters

`row` WorldBossCharacterSheet.Row

`score` long

### Returns

int

## CalculateReward(int, int, RuneWeightSheet, IWorldBossRewardSheet, RuneSheet, MaterialItemSheet, IRandom, Row)

Calculates a world boss reward — runes drawn by RuneWeightSheet, crystal and circles — for `rank` of `bossId`.

```
public static (List<FungibleAssetValue> assets, Dictionary<TradableMaterial, int>
materials) CalculateReward(int rank, int bossId, RuneWeightSheet sheet,
IWorldBossRewardSheet rewardSheet, RuneSheet runeSheet, MaterialItemSheet
materialSheet, IRandom random, BoostScheduleSheet.Row rewardBoost = null)
```

### Parameters

`rank` int

The rank whose reward row is granted.

`bossId` int

The boss id, as in `WorldBossListSheet.boss_id`.

`sheet` RuneWeightSheet

Weights the drawn runes.

`rewardSheet` IWorldBossRewardSheet

The battle, kill or rank reward sheet the reward row is taken from.

`runeSheet` RuneSheet

Maps a drawn rune id to its currency.

`materialSheet` MaterialItemSheet

Supplies the circle material.

`random` IRandom

Draws the rune count, for a ranged row, and each rune.

`rewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this reward at the current block, or `null` for none. A caller replaying a reward must pass what the action resolved. See ApplyRewardBoost(Row, int).

### Returns

(List<FungibleAssetValue> assets, Dictionary<TradableMaterial, int> materials)

The granted assets, runes first and crystal last, and circles.

### Remarks

`rewardBoost` scales what was drawn rather than how many draws are made, so `random` is consumed exactly as without it and everything drawn after this reward (e.g. the next kill reward of the same action) stays the same.

## CalculateTicketPrice(Row, RaiderState, Currency)

```
public static FungibleAssetValue CalculateTicketPrice(WorldBossListSheet.Row row,
RaiderState raiderState, Currency currency)
```

### Parameters

`row` WorldBossListSheet.Row

`raiderState` RaiderState

`currency` Currency

### Returns

FungibleAssetValue

## CanRefillTicket(long, long, long, int)

```
public static bool CanRefillTicket(long blockIndex, long refilledIndex, long
startedIndex, int refillInterval)
```

### Parameters

`blockIndex` long

`refilledIndex` long

`startedIndex` long

`refillInterval` int

### Returns

bool

## CanRefillTicketV1(long, long, long)

```
public static bool CanRefillTicketV1(long blockIndex, long refilledIndex,
long startedIndex)
```

### Parameters

`blockIndex` long

`refilledIndex` long

`startedIndex` long

## Returns

bool

# Namespace Nekoyume.Model

## Classes

ArenaCharacter

ArenaSkills

CharacterBase

CollectionMap

Represents a collection map with key-value pairs. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. entries - List of key-value pairs ordered by key

Enemy

EnemyPlayer

FailedAddWorldException

FailedToUnlockWorldException

Player

Player.ExpData

RaidBoss

Skills

WorldInformation

## Structs

ActivationKey

ArenaPlayerDigest

Introduced at https://github.com/planetarium/lib9c/pull/1156

FungibleItemValue

# Interfaces

IArena

IStage

# Struct ActivationKey

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public struct ActivationKey
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Fields

### DeriveKey

```
public const string DeriveKey = "activated"
```

#### Field Value

string

## Properties

### PendingAddress

```
public readonly Address PendingAddress { get; }
```

#### Property Value

Address

### PrivateKey

```
public readonly PrivateKey PrivateKey { get; }
```

### Property Value

PrivateKey

## Methods

### Create(PrivateKey, byte[])

```
public static (ActivationKey, PendingActivationState) Create(PrivateKey privateKey,
byte[] nonce)
```

### Parameters

`privateKey` PrivateKey

`nonce` byte[]

### Returns

(ActivationKey, PendingActivationState)

### Decode(string)

```
public static ActivationKey Decode(string hexWithSlash)
```

### Parameters

`hexWithSlash` string

### Returns

ActivationKey

# Encode()

```
public string Encode()
```

## Returns

string

# Class ArenaCharacter

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public class ArenaCharacter : ICloneable
```

**Inheritance**

object ← ArenaCharacter

**Implements**

ICloneable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaCharacter(IArenaSimulator, ArenaPlayerDigest, ArenaSimulatorSheets, bool)

```
public ArenaCharacter(IArenaSimulator simulator, ArenaPlayerDigest digest,
ArenaSimulatorSheets sheets, bool isEnemy = false)
```

#### Parameters

`simulator` IArenaSimulator

`digest` ArenaPlayerDigest

`sheets` ArenaSimulatorSheets

`isEnemy` bool

# ArenaCharacter(IArenaSimulator, ArenaPlayerDigest, ArenaSimulatorSheets, long, List<StatModifier>, bool, bool)

```
public ArenaCharacter(IArenaSimulator simulator, ArenaPlayerDigest
digest, ArenaSimulatorSheets sheets, long hpModifier, List<StatModifier>
collectionModifiers, bool isEnemy = false, bool setExtraValueBuffBeforeGetBuffs
= false)
```

## Parameters

`simulator` IArenaSimulator

`digest` ArenaPlayerDigest

`sheets` ArenaSimulatorSheets

`hpModifier` long

`collectionModifiers` List<StatModifier>

`isEnemy` bool

`setExtraValueBuffBeforeGetBuffs` bool

# Fields

## CriticalMultiplier

```
public const decimal CriticalMultiplier = 1.5
```

## Field Value

decimal

## RuneSkillCooldownMap

```
public readonly Dictionary<int, int> RuneSkillCooldownMap
```

### Field Value

Dictionary<int, int>

## Simulator

```
public readonly IArenaSimulator Simulator
```

### Field Value

IArenaSimulator

## _runeSkills

```
public readonly ArenaSkills _runeSkills
```

### Field Value

ArenaSkills

## usedSkill

```
public ArenaSkill usedSkill
```

### Field Value

ArenaSkill

# Properties

## ATK

```
public long ATK { get; }
```

### Property Value

long

## ActionBuffs

```
public IEnumerable<ActionBuff> ActionBuffs { get; }
```

### Property Value

IEnumerable<ActionBuff>

## AdditionalHP

```
public long AdditionalHP { get; }
```

### Property Value

long

## ArmorPenetration

```
public long ArmorPenetration { get; }
```

### Property Value

long

## AttackCount

```
public int AttackCount { get; }
```

### Property Value

int

## AttackRange

```
public float AttackRange { get; }
```

### Property Value

float

## Buffs

```
public Dictionary<int, Buff> Buffs { get; }
```

### Property Value

Dictionary<int, Buff>

## CDMG

```
public long CDMG { get; }
```

### Property Value

long

## CRI

```
public long CRI { get; }
```

### Property Value

long

## CharacterId

```
public int CharacterId { get; }
```

### Property Value

int

## CurrentHP

```
public long CurrentHP { get; set; }
```

### Property Value

long

## DEF

```
public long DEF { get; }
```

### Property Value

long

## DRR

```
public long DRR { get; }
```

### Property Value

long

## DRV

```
public long DRV { get; }
```

### Property Value

long

## DefenseElementalType

```
public ElementalType DefenseElementalType { get; }
```

### Property Value

ElementalType

## HIT

```
public long HIT { get; }
```

### Property Value

long

## HP

```
public long HP { get; }
```

### Property Value

long

## Id

```
public Guid Id { get; }
```

### Property Value

Guid

## IsDead

```
public bool IsDead { get; }
```

### Property Value

bool

## IsEnemy

```
public bool IsEnemy { get; }
```

### Property Value

bool

## Level

```
public int Level { get; set; }
```

### Property Value

int

## OffensiveElementalType

```
public ElementalType OffensiveElementalType { get; }
```

### Property Value

ElementalType

## RunSpeed

```
public float RunSpeed { get; }
```

### Property Value

float

## SPD

```
public long SPD { get; }
```

### Property Value

long

## SizeType

```
public SizeType SizeType { get; }
```

### Property Value

SizeType

## SkillLog

```
public ArenaSkill SkillLog { get; }
```

### Property Value

ArenaSkill

## StatBuffs

```
public IEnumerable<StatBuff> StatBuffs { get; }
```

### Property Value

IEnumerable<StatBuff>

## Stats

```
public CharacterStats Stats { get; }
```

### Property Value

CharacterStats

## Thorn

```
public long Thorn { get; }
```

### Property Value

long

# Methods

## AddBuff(Buff, bool)

```
public IEnumerable<Buff> AddBuff(Buff buff, bool updateImmediate = true)
```

### Parameters

`buff` Buff

`updateImmediate` bool

### Returns

IEnumerable<Buff>

## Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## GetDamage(long, bool)

```
public long GetDamage(long damage, bool considerAttackCount = true)
```

### Parameters

`damage` long

`considerAttackCount` bool

### Returns

long

## Heal(long)

```
public void Heal(long heal)
```

### Parameters

`heal` long

## IsCritical(bool)

```
public bool IsCritical(bool considerAttackCount = true)
```

### Parameters

`considerAttackCount` bool

### Returns

bool

## IsHit(ArenaCharacter)

```
public virtual bool IsHit(ArenaCharacter caster)
```

### Parameters

`caster` ArenaCharacter

### Returns

bool

## OnPostSkill(ArenaSkill)

```
protected virtual void OnPostSkill(ArenaSkill usedSkill)
```

### Parameters

`usedSkill` ArenaSkill

## OnPreSkill()

```
protected virtual bool OnPreSkill()
```

### Returns

bool

## RemoveActionBuff(ActionBuff)

```
public void RemoveActionBuff(ActionBuff removedBuff)
```

### Parameters

`removedBuff` ActionBuff

## RemoveAllStatBuffs()

Removes all positive stat buffs from the character. Only buffs with a positive value are removed; debuffs (negative value) are preserved.

```
public void RemoveAllStatBuffs()
```

## RemoveRecentStatBuff()

```
public void RemoveRecentStatBuff()
```

## RemoveStatBuff(StatBuff)

```
public void RemoveStatBuff(StatBuff removedBuff)
```

### Parameters

`removedBuff` StatBuff

## ResetCurrentHP()

```
protected void ResetCurrentHP()
```

## SetRuneSkills(IEnumerable<RuneState>, RuneOptionSheet, SkillSheet)

```
public void SetRuneSkills(IEnumerable<RuneState> runes, RuneOptionSheet
runeOptionSheet, SkillSheet skillSheet)
```

### Parameters

`runes` IEnumerable<RuneState>

`runeOptionSheet` RuneOptionSheet

`skillSheet` SkillSheet

## SetRuneStats(IEnumerable<RuneState>, RuneOptionSheet, int)

```
public void SetRuneStats(IEnumerable<RuneState> runes, RuneOptionSheet
runeOptionSheet, int runeLevelBonus)
```

### Parameters

`runes` IEnumerable<RuneState>

`runeOptionSheet` RuneOptionSheet

runeLevelBonus int

## SetRuneV1(List<RuneState>, RuneOptionSheet, SkillSheet)

```
[Obsolete("Use SetRune instead.")]
public void SetRuneV1(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
SkillSheet skillSheet)
```

### Parameters

runes List<RuneState>

runeOptionSheet RuneOptionSheet

skillSheet SkillSheet

## SetRuneV2(List<RuneState>, RuneOptionSheet, SkillSheet)

```
[Obsolete("Use SetRune instead.")]
public void SetRuneV2(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
SkillSheet skillSheet)
```

### Parameters

runes List<RuneState>

runeOptionSheet RuneOptionSheet

skillSheet SkillSheet

## Spawn(ArenaCharacter)

```
public void Spawn(ArenaCharacter target)
```

### Parameters

`target` ArenaCharacter

## SpawnV2(ArenaCharacter)

```
[Obsolete("Use Spawn")]
public void SpawnV2(ArenaCharacter target)
```

### Parameters

`target` ArenaCharacter

## Tick()

```
public void Tick()
```

# Struct ArenaPlayerDigest

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1156

```
public readonly struct ArenaPlayerDigest : IState
```

**Implements**
IState

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### ArenaPlayerDigest(List)

```
public ArenaPlayerDigest(List serialized)
```

#### Parameters

`serialized` List

### ArenaPlayerDigest(AvatarState, AllRuneState, RuneSlotState)

```
public ArenaPlayerDigest(AvatarState avatarState, AllRuneState runes,
RuneSlotState runeSlotState)
```

#### Parameters

`avatarState` AvatarState

`runes` AllRuneState

`runeSlotState` RuneSlotState

## ArenaPlayerDigest(AvatarState, ArenaAvatarState)

```
[Obsolete("Do not use it")]
public ArenaPlayerDigest(AvatarState avatarState, ArenaAvatarState arenaAvatarState)
```

### Parameters

`avatarState` AvatarState

`arenaAvatarState` ArenaAvatarState

## ArenaPlayerDigest(AvatarState, List<Costume>, List<Equipment>, AllRuneState, RuneSlotState)

```
public ArenaPlayerDigest(AvatarState avatarState, List<Costume> costumes,
List<Equipment> equipments, AllRuneState runes, RuneSlotState runeSlotState)
```

### Parameters

`avatarState` AvatarState

`costumes` List<Costume>

`equipments` List<Equipment>

`runes` AllRuneState

`runeSlotState` RuneSlotState

## ArenaPlayerDigest(AvatarState, List<Guid>, List<Guid>, AllRuneState, RuneSlotState)

```
public ArenaPlayerDigest(AvatarState avatarState, List<Guid> equipments, List<Guid>
costumes, AllRuneState runes, RuneSlotState runeSlotState)
```

### Parameters

`avatarState` AvatarState

`equipments` List<Guid>

`costumes` List<Guid>

`runes` AllRuneState

`runeSlotState` RuneSlotState

## Fields

### CharacterId

```
public readonly int CharacterId
```

#### Field Value

int

### Costumes

```
public readonly List<Costume> Costumes
```

#### Field Value

List<Costume>

### EarIndex

```
public readonly int EarIndex
```

### Field Value

int

## Equipments

```
public readonly List<Equipment> Equipments
```

### Field Value

List<Equipment>

## HairIndex

```
public readonly int HairIndex
```

### Field Value

int

## LensIndex

```
public readonly int LensIndex
```

### Field Value

int

## Level

```
public readonly int Level
```

### Field Value

int

## NameWithHash

```
public readonly string NameWithHash
```

### Field Value

string

## RuneSlotState

```
public readonly RuneSlotState RuneSlotState
```

### Field Value

RuneSlotState

## Runes

```
public readonly AllRuneState Runes
```

### Field Value

AllRuneState

## TailIndex

```
public readonly int TailIndex
```

### Field Value

int

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class ArenaSkills

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSkills : IEnumerable<ArenaSkill>, IEnumerable
```

**Inheritance**

object ← ArenaSkills

**Implements**

IEnumerable<ArenaSkill>, IEnumerable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## Add(ArenaSkill)

```
public void Add(ArenaSkill skill)
```

### Parameters

`skill` ArenaSkill

## Clear()

```
public void Clear()
```

## GetCooldown(int)

```
public int GetCooldown(int skillId)
```

### Parameters

`skillId` int

### Returns

int

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<ArenaSkill> GetEnumerator()
```

### Returns

IEnumerator<ArenaSkill>

An enumerator that can be used to iterate through the collection.

## ReduceCooldown()

```
public void ReduceCooldown()
```

## Select(IRandom)

```
public ArenaSkill Select(IRandom random)
```

### Parameters

`random` IRandom

### Returns

ArenaSkill

## SelectWithoutDefaultAttack(IRandom)

```
public ArenaSkill SelectWithoutDefaultAttack(IRandom random)
```

### Parameters

`random` IRandom

### Returns

ArenaSkill

## SetCooldown(int, int)

```
public void SetCooldown(int skillId, int cooldown)
```

### Parameters

`skillId` int

`cooldown` int

# Class CharacterBase

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public abstract class CharacterBase : ICloneable
```

**Inheritance**

object ← CharacterBase

**Implements**

ICloneable

**Derived**

Enemy, Player

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CharacterBase(Simulator, CharacterStats, int, ElementalType, Row, SizeType, float, float)

```
protected CharacterBase(Simulator simulator, CharacterStats stat, int characterId,
ElementalType elementalType, CharacterSheet.Row rowData = null, SizeType sizeType =
SizeType.XL, float attackRange = 4, float runSpeed = 0.3)
```

#### Parameters

`simulator` Simulator

`stat` CharacterStats

`characterId` int

`elementalType` ElementalType

`rowData` CharacterSheet.Row

`sizeType` SizeType

`attackRange` float

`runSpeed` float

## CharacterBase(Simulator, CharacterSheet, int, int, IEnumerable<StatModifier>)

```
protected CharacterBase(Simulator simulator, CharacterSheet characterSheet, int
characterId, int level, IEnumerable<StatModifier> optionalStatModifiers = null)
```

### Parameters

`simulator` Simulator

`characterSheet` CharacterSheet

`characterId` int

`level` int

`optionalStatModifiers` IEnumerable<StatModifier>

## CharacterBase(CharacterBase)

```
protected CharacterBase(CharacterBase value)
```

### Parameters

`value` CharacterBase

## CharacterBase(Row)

```
protected CharacterBase(CharacterSheet.Row row)
```

### Parameters

`row` CharacterSheet.Row

## Fields

### BuffSkills

```
public readonly Skills BuffSkills
```

#### Field Value

Skills

### Buffs

```
public readonly Dictionary<int, Buff> Buffs
```

#### Field Value

Dictionary<int, Buff>

### CriticalMultiplier

```
public const decimal CriticalMultiplier = 1.5
```

#### Field Value

decimal

### Id

```
public readonly Guid Id
```

### Field Value

Guid

## Simulator

```
[NonSerialized]
public Simulator Simulator
```

### Field Value

Simulator

## Skills

```
public readonly Skills Skills
```

### Field Value

Skills

## Targets

```
public readonly List<CharacterBase> Targets
```

### Field Value

List<CharacterBase>

## atkElementType

```
public ElementalType atkElementType
```

### Field Value

ElementalType

## attackRange

```
public float attackRange
```

### Field Value

float

## defElementType

```
public ElementalType defElementType
```

### Field Value

ElementalType

# Properties

## ATK

```
public long ATK { get; }
```

### Property Value

long

## ActionBuffs

```
public IEnumerable<ActionBuff> ActionBuffs { get; }
```

### Property Value

IEnumerable<ActionBuff>

## AdditionalHP

```
public long AdditionalHP { get; }
```

### Property Value

long

## ArmorPenetration

```
public long ArmorPenetration { get; }
```

### Property Value

long

## AttackCount

```
public int AttackCount { get; set; }
```

### Property Value

int

## AttackCountMax

```
public int AttackCountMax { get; protected set; }
```

### Property Value

int

## CDMG

```
public long CDMG { get; }
```

### Property Value

long

## CRI

```
public long CRI { get; }
```

### Property Value

long

## CharacterId

```
public int CharacterId { get; }
```

### Property Value

int

## CurrentHP

```
public long CurrentHP { get; set; }
```

### Property Value

long

## DEF

```
public long DEF { get; }
```

### Property Value

long

## DRR

```
public long DRR { get; }
```

### Property Value

long

## DRV

```
public long DRV { get; }
```

### Property Value

long

## HIT

```
public long HIT { get; }
```

### Property Value

long

## HP

```
public long HP { get; }
```

### Property Value

long

## IsDead

```
public bool IsDead { get; }
```

### Property Value

bool

## Level

```
public int Level { get; set; }
```

### Property Value

int

## RowData

```
public CharacterSheet.Row RowData { get; }
```

### Property Value

CharacterSheet.Row

## RunSpeed

```
public float RunSpeed { get; }
```

### Property Value

float

## SPD

```
public long SPD { get; }
```

### Property Value

long

## SizeType

```
public SizeType SizeType { get; }
```

### Property Value

SizeType

## StatBuffs

```
public IEnumerable<StatBuff> StatBuffs { get; }
```

### Property Value

IEnumerable<StatBuff>

## Stats

```
public CharacterStats Stats { get; }
```

### Property Value

CharacterStats

## Thorn

```
public long Thorn { get; }
```

### Property Value

long

## usedSkill

```
public Skill usedSkill { get; set; }
```

### Property Value

Skill

# Methods

## AddBuff(Buff, bool)

Add buff/debuff to target; it means buff/debuff is used by caster. When `Dispel` is used, it can remove prev. debuffs on target. All the removed debuffs will be returned and saved in battle log.

```
public IEnumerable<Buff> AddBuff(Buff buff, bool updateImmediate = true)
```

### Parameters

`buff` Buff

`updateImmediate` bool

### Returns

IEnumerable<Buff>

An enumerable of removed debuffs from target. `null` will be returned if nothing eliminated.

## Clone()

Creates a new object that is a copy of the current instance.

```
public abstract object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## Die()

```
public void Die()
```

## EndTurn()

```
protected virtual void EndTurn()
```

## GetChance(int)

```
public bool GetChance(int chance)
```

### Parameters

`chance` int

### Returns

bool

## GetDamage(long, bool)

```
public long GetDamage(long damage, bool considerAttackCount = true)
```

### Parameters

`damage` long

`considerAttackCount` bool

### Returns

long

## Heal(long)

```
public void Heal(long heal)
```

### Parameters

`heal` long

## InitAI()

```
public virtual void InitAI()
```

## InitAIV1()

```
[Obsolete("Use InitAI")]
public void InitAIV1()
```

## InitAIV2()

```
[Obsolete("Use InitAI")]
public void InitAIV2()
```

## IsCritical(bool)

```
public bool IsCritical(bool considerAttackCount = true)
```

### Parameters

`considerAttackCount` bool

### Returns

bool

## IsHit(CharacterBase)

```
public virtual bool IsHit(CharacterBase caster)
```

### Parameters

`caster` CharacterBase

### Returns

bool

## IsHit(ElementalResult)

```
public bool IsHit(ElementalResult result)
```

### Parameters

`result` ElementalResult

### Returns

bool

## OnDead()

```
protected virtual void OnDead()
```

## OnPostSkill(Skill)

```
protected virtual void OnPostSkill(Skill usedSkill)
```

### Parameters

`usedSkill` Skill

## OnPreSkill()

```
protected virtual bool OnPreSkill()
```

### Returns

bool

## ReduceSkillCooldown()

```
protected virtual void ReduceSkillCooldown()
```

## RemoveActionBuff(ActionBuff)

```
public void RemoveActionBuff(ActionBuff removedBuff)
```

### Parameters

`removedBuff` ActionBuff

## RemoveAllStatBuffs()

Removes all positive stat buffs from the character. Only buffs with a positive value are removed; debuffs (negative value) are preserved.

```
public void RemoveAllStatBuffs()
```

## RemoveRecentStatBuff()

```
public void RemoveRecentStatBuff()
```

## RemoveStatBuff(StatBuff)

```
public void RemoveStatBuff(StatBuff removedBuff)
```

### Parameters

`removedBuff` StatBuff

## ResetCurrentHP()

```
public void ResetCurrentHP()
```

## SetSkill()

```
protected virtual void SetSkill()
```

## Tick()

```
public void Tick()
```

## UseSkill()

```
protected virtual Skill UseSkill()
```

### Returns

Skill

# Class CollectionMap

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

Represents a collection map with key-value pairs. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. entries - List of key-value pairs ordered by key

```
[Serializable]
public class CollectionMap : IState, IDictionary<int, int>,
ICollection<KeyValuePair<int, int>>, IEnumerable<KeyValuePair<int, int>>,
IEnumerable, ISerializable
```

**Inheritance**

object ← CollectionMap

**Implements**

IState, IDictionary<int, int>, ICollection<KeyValuePair<int, int>>, IEnumerable<KeyValuePair<int, int>>, IEnumerable, ISerializable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

This class implements dual serialization support to ensure smooth migration from the legacy Dictionary format to the new List format. The List format provides better performance and smaller serialized data size.

Example usage:

```
// Create collection map
var map = new CollectionMap();
map.Add(1, 100);
```

```
map.Add(2, 200);
// Serialize to List format (new)
var serialized = map.Serialize(); // Returns List
// Deserialize from any format
var deserialized = new CollectionMap(serialized); // Supports both Dictionary
and List
```

# Constructors

## CollectionMap()

```
public CollectionMap()
```

## CollectionMap(Dictionary)

Constructor for backward compatibility with Dictionary format.

```
public CollectionMap(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

Dictionary containing serialized data

## CollectionMap(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public CollectionMap(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

### Exceptions

ArgumentNullException

Thrown when serialized is null

ArgumentException

Thrown when serialized format is not supported

## Properties

## Count

Gets the number of elements contained in the ICollection<T>.

```
public int Count { get; }
```

### Property Value

int

The number of elements contained in the ICollection<T>.

## IsReadOnly

Gets a value indicating whether the ICollection<T> is read-only.

```
public bool IsReadOnly { get; }
```

### Property Value

bool

true if the ICollection<T> is read-only; otherwise, false.

## this[int]

Gets or sets the element with the specified key.

```
public int this[int key] { get; set; }
```

### Parameters

`key` int

The key of the element to get or set.

### Property Value

int

The element with the specified key.

### Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

NotSupportedException

The property is set and the IDictionary<TKey, TValue> is read-only.

## Keys

Gets an ICollection<T> containing the keys of the IDictionary<TKey, TValue>.

```
public ICollection<int> Keys { get; }
```

### Property Value

ICollection<int>

An ICollection<T> containing the keys of the object that implements IDictionary<TKey, TValue>.

## Values

Gets an ICollection<T> containing the values in the IDictionary<TKey, TValue>.

```
public ICollection<int> Values { get; }
```

### Property Value

ICollection<int>

An ICollection<T> containing the values in the object that implements IDictionary<TKey, TValue>.

# Methods

## Add(KeyValuePair<int, int>)

Adds an item to the ICollection<T>.

```
public void Add(KeyValuePair<int, int> item)
```

### Parameters

`item` KeyValuePair<int, int>

The object to add to the ICollection<T>.

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Add(int, int)

Adds an element with the provided key and value to the IDictionary<TKey, TValue>.

```
public void Add(int key, int value)
```

### Parameters

`key` int

The object to use as the key of the element to add.

`value` int

The object to use as the value of the element to add.

### Exceptions

ArgumentNullException

`key` is null.

ArgumentException

An element with the same key already exists in the IDictionary<TKey, TValue>.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

## Clear()

Removes all items from the ICollection<T>.

```
public void Clear()
```

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Contains(KeyValuePair<int, int>)

Determines whether the ICollection<T> contains a specific value.

```
public bool Contains(KeyValuePair<int, int> item)
```

### Parameters

`item` KeyValuePair<int, int>

The object to locate in the ICollection<T>.

### Returns

bool

true if `item` is found in the ICollection<T>; otherwise, false.

## ContainsKey(int)

Determines whether the IDictionary<TKey, TValue> contains an element with the specified key.

```
public bool ContainsKey(int key)
```

### Parameters

`key` int

The key to locate in the IDictionary<TKey, TValue>.

### Returns

bool

true if the IDictionary<TKey, TValue> contains an element with the key; otherwise, false.

### Exceptions

ArgumentNullException

`key` is null.

## CopyTo(KeyValuePair<int, int>[], int)

Copies the elements of the ICollection<T> to an Array, starting at a particular Array index.

```
public void CopyTo(KeyValuePair<int, int>[] array, int arrayIndex)
```

## Parameters

`array` KeyValuePair<int, int>[]

The one-dimensional Array that is the destination of the elements copied from ICollection<T>. The Array must have zero-based indexing.

`arrayIndex` int

The zero-based index in `array` at which copying begins.

## Exceptions

ArgumentNullException

`array` is null.

ArgumentOutOfRangeException

`arrayIndex` is less than 0.

ArgumentException

The number of elements in the source ICollection<T> is greater than the available space from `arrayIndex` to the end of the destination `array`.

# GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<KeyValuePair<int, int>> GetEnumerator()
```

## Returns

IEnumerator<KeyValuePair<int, int>>

An enumerator that can be used to iterate through the collection.

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## Remove(KeyValuePair<int, int>)

Removes the first occurrence of a specific object from the ICollection<T>.

```
public bool Remove(KeyValuePair<int, int> item)
```

### Parameters

`item` KeyValuePair<int, int>

The object to remove from the ICollection<T>.

### Returns

bool

true if `item` was successfully removed from the ICollection<T>; otherwise, false. This method also returns false if `item` is not found in the original ICollection<T>.

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Remove(int)

Removes the element with the specified key from the IDictionary<TKey, TValue>.

```
public bool Remove(int key)
```

### Parameters

`key` int

The key of the element to remove.

### Returns

bool

true if the element is successfully removed; otherwise, false. This method also returns false if `key` was not found in the original IDictionary<TKey, TValue>.

### Exceptions

ArgumentNullException

`key` is null.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

## Serialize()

Serializes the CollectionMap to List format (new format). Order: [version, entries]

```
public IValue Serialize()
```

### Returns

IValue

List containing serialized data

## TryGetValue(int, out int)

Gets the value associated with the specified key.

```
public bool TryGetValue(int key, out int value)
```

### Parameters

`key` int

The key whose value to get.

`value` int

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

### Returns

bool

true if the object that implements IDictionary<TKey, TValue> contains an element with the specified key; otherwise, false.

### Exceptions

ArgumentNullException

`key` is null.

# Class Enemy

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class Enemy : CharacterBase, ICloneable
```

### Inheritance

object ← CharacterBase ← Enemy

### Implements

ICloneable

### Derived

RaidBoss

### Inherited Members

CharacterBase.CriticalMultiplier , CharacterBase.Id , CharacterBase.Simulator , CharacterBase.atkElementType , CharacterBase.attackRange , CharacterBase.defElementType , CharacterBase.Skills , CharacterBase.BuffSkills , CharacterBase.Buffs , CharacterBase.StatBuffs , CharacterBase.ActionBuffs , CharacterBase.Targets , CharacterBase.RowData , CharacterBase.CharacterId , CharacterBase.SizeType , CharacterBase.RunSpeed , CharacterBase.Stats , CharacterBase.Level , CharacterBase.HP , CharacterBase.AdditionalHP , CharacterBase.ATK , CharacterBase.DEF , CharacterBase.CRI , CharacterBase.HIT , CharacterBase.SPD , CharacterBase.DRV , CharacterBase.DRR , CharacterBase.CDMG , CharacterBase.ArmorPenetration , CharacterBase.Thorn , CharacterBase.CurrentHP , CharacterBase.IsDead , CharacterBase.AttackCount , CharacterBase.AttackCountMax , CharacterBase.usedSkill , CharacterBase.InitAI() , CharacterBase.InitAIV1() , CharacterBase.InitAIV2() , CharacterBase.Tick() , CharacterBase.ReduceSkillCooldown() , CharacterBase.UseSkill() , CharacterBase.EndTurn() , CharacterBase.AddBuff(Buff, bool) , CharacterBase.RemoveActionBuff(ActionBuff) , CharacterBase.RemoveStatBuff(StatBuff) , CharacterBase.RemoveRecentStatBuff() , CharacterBase.RemoveAllStatBuffs() , CharacterBase.ResetCurrentHP() , CharacterBase.IsCritical(bool) , CharacterBase.IsHit(ElementalResult) , CharacterBase.IsHit(CharacterBase) , CharacterBase.GetDamage(long, bool) , CharacterBase.Die() , CharacterBase.Heal(long) , CharacterBase.GetChance(int) , CharacterBase.OnPreSkill() , CharacterBase.OnPostSkill(Skill) , object.Equals(object) , object.Equals(object, object) ,

object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Enemy(CharacterBase, CharacterStats, Row, ElementalType)

```
public Enemy(CharacterBase player, CharacterStats stat, CharacterSheet.Row rowData,
ElementalType elementalType)
```

### Parameters

`player` CharacterBase

`stat` CharacterStats

`rowData` CharacterSheet.Row

`elementalType` ElementalType

## Enemy(CharacterBase, CharacterStats, int, ElementalType)

```
public Enemy(CharacterBase player, CharacterStats stat, int characterId,
ElementalType elementalType)
```

### Parameters

`player` CharacterBase

`stat` CharacterStats

`characterId` int

`elementalType` ElementalType

## Enemy(CharacterBase, Row, int, IEnumerable<StatModifier>)

```
public Enemy(CharacterBase player, CharacterSheet.Row rowData, int monsterLevel,
IEnumerable<StatModifier> optionalStatModifiers = null)
```

### Parameters

`player` CharacterBase

`rowData` CharacterSheet.Row

`monsterLevel` int

`optionalStatModifiers` IEnumerable<StatModifier>

## Enemy(Enemy)

```
public Enemy(Enemy value)
```

### Parameters

`value` Enemy

## Enemy(Row)

```
public Enemy(CharacterSheet.Row rowData)
```

### Parameters

`rowData` CharacterSheet.Row

# Fields

## spawnIndex

```
public int spawnIndex
```

### Field Value

int

## Methods

### Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

### OnDead()

```
protected override void OnDead()
```

### SetSkill()

```
protected override void SetSkill()
```

# Class EnemyPlayer

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class EnemyPlayer : Player, ICloneable
```

### Inheritance

object ← CharacterBase ← Player ← EnemyPlayer

### Implements

ICloneable

### Inherited Members

Player.Exp , Player.Inventory , Player.worldInformation , Player.weapon , Player.armor , Player.belt , Player.necklace , Player.ring , Player.aura , Player.Grimoire , Player.monsterMap , Player.eventMap , Player.monsterMapForBeforeV100310 , Player.eventMapForBeforeV100310 , Player.hairIndex , Player.lensIndex , Player.earIndex , Player.tailIndex , Player.characterLevelSheet , Player.costumes , Player.equipments , Player.RuneSkills , Player.RuneSkillCooldownMap , Player.Costumes , Player.Equipments , Player.IsHit(CharacterBase) , Player.RemoveTarget(Enemy) , Player.RemoveTarget(EnemyPlayer) , Player.SetEquipmentStat(EquipmentItemSetEffectSheet) , Player.GetExp(long, bool) , Player.GetExp2(long, bool) , Player.GetExp3(long, bool) , Player.GetRewards(List<ItemBase>) , Player.Use(List<Guid>) , Player.OverrideSkill(Skill) , Player.AddSkill(Skill) , Player.UseSkill() , Player.ReduceSkillCooldown() , Player.SetCostumeStat(CostumeStatSheet) , Player.SetRuneStats(List<RuneState>, RuneOptionSheet, int) , Player.SetRuneSkills(List<RuneState>, RuneOptionSheet, SkillSheet) , Player.SetCollections(IEnumerable<StatModifier>) , Player.ConfigureStats(CostumeStatSheet, List<RuneState>, RuneOptionSheet, int, SkillSheet, List<StatModifier>) , Player.SetRuneV1(List<RuneState>, RuneOptionSheet, SkillSheet) , Player.EndTurn() , CharacterBase.CriticalMultiplier , CharacterBase.Id , CharacterBase.Simulator , CharacterBase.atkElementType , CharacterBase.attackRange , CharacterBase.defElementType , CharacterBase.Skills , CharacterBase.BuffSkills , CharacterBase.Buffs , CharacterBase.StatBuffs , CharacterBase.ActionBuffs , CharacterBase.Targets , CharacterBase.RowData , CharacterBase.CharacterId ,

CharacterBase.SizeType , CharacterBase.RunSpeed , CharacterBase.Stats , CharacterBase.Level , CharacterBase.HP , CharacterBase.AdditionalHP , CharacterBase.ATK , CharacterBase.DEF , CharacterBase.CRI , CharacterBase.HIT , CharacterBase.SPD , CharacterBase.DRV , CharacterBase.DRR , CharacterBase.CDMG , CharacterBase.ArmorPenetration , CharacterBase.Thorn , CharacterBase.CurrentHP , CharacterBase.IsDead , CharacterBase.AttackCount , CharacterBase.AttackCountMax , CharacterBase.usedSkill , CharacterBase.InitAI() , CharacterBase.InitAIV1() , CharacterBase.InitAIV2() , CharacterBase.Tick() , CharacterBase.AddBuff(Buff, bool) , CharacterBase.RemoveActionBuff(ActionBuff) , CharacterBase.RemoveStatBuff(StatBuff) , CharacterBase.RemoveRecentStatBuff() , CharacterBase.RemoveAllStatBuffs() , CharacterBase.ResetCurrentHP() , CharacterBase.IsCritical(bool) , CharacterBase.IsHit(ElementalResult) , CharacterBase.GetDamage(long, bool) , CharacterBase.Die() , CharacterBase.Heal(long) , CharacterBase.SetSkill() , CharacterBase.GetChance(int) , CharacterBase.OnPreSkill() , CharacterBase.OnPostSkill(Skill) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EnemyPlayer(EnemyPlayerDigest, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public EnemyPlayer(EnemyPlayerDigest enemyPlayerDigest, CharacterSheet
characterSheet, CharacterLevelSheet levelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet)
```

### Parameters

`enemyPlayerDigest` EnemyPlayerDigest

`characterSheet` CharacterSheet

`levelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

## EnemyPlayer(ArenaPlayerDigest, ArenaSimulatorSheetsV1)

```
public EnemyPlayer(ArenaPlayerDigest arenaPlayerDigest, ArenaSimulatorSheetsV1
simulatorSheets)
```

### Parameters

`arenaPlayerDigest` ArenaPlayerDigest

`simulatorSheets` ArenaSimulatorSheetsV1

## EnemyPlayer(AvatarState, Simulator)

```
public EnemyPlayer(AvatarState avatarState, Simulator simulator)
```

### Parameters

`avatarState` AvatarState

`simulator` Simulator

## EnemyPlayer(AvatarState, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public EnemyPlayer(AvatarState avatarState, CharacterSheet characterSheet,
CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet)
```

### Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

## EnemyPlayer(AvatarState, SimulatorSheetsV1)

```
public EnemyPlayer(AvatarState avatarState, SimulatorSheetsV1 simulatorSheets)
```

### Parameters

`avatarState` AvatarState

`simulatorSheets` SimulatorSheetsV1

## EnemyPlayer(int, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public EnemyPlayer(int level, CharacterSheet characterSheet, CharacterLevelSheet
characterLevelSheet, EquipmentItemSetEffectSheet equipmentItemSetEffectSheet)
```

### Parameters

`level` int

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

# Fields

## NameWithHash

```
public readonly string NameWithHash
```

### Field Value

string

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## OnDead()

```
protected override void OnDead()
```

## Spawn()

```
public override void Spawn()
```

## SpawnV1()

```
[Obsolete("Use Spawn")]
public override void SpawnV1()
```

## SpawnV2()

```
[Obsolete("Use Spawn")]
```

```
public override void SpawnV2()
```

# Class FailedAddWorldException

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class FailedAddWorldException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← FailedAddWorldException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### FailedAddWorldException(SerializationInfo, StreamingContext)

```
protected FailedAddWorldException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# FailedAddWorldException(string)

```
public FailedAddWorldException(string message)
```

## Parameters

`message` string

# Class FailedToUnlockWorldException

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class FailedToUnlockWorldException : Exception, ISerializable
```

### Inheritance

object ← Exception ← FailedToUnlockWorldException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### FailedToUnlockWorldException(SerializationInfo, StreamingContext)

```
protected FailedToUnlockWorldException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# FailedToUnlockWorldException(string)

```
public FailedToUnlockWorldException(string message)
```

## Parameters

`message` string

# Struct FungibleItemValue

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public readonly struct FungibleItemValue
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### FungibleItemValue(List)

```
public FungibleItemValue(List bencoded)
```

#### Parameters

`bencoded` List

### FungibleItemValue(HashDigest<SHA256>, int)

```
public FungibleItemValue(HashDigest<SHA256> id, int count)
```

#### Parameters

`id` HashDigest<SHA256>

`count` int

## Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## Id

```
public HashDigest<SHA256> Id { get; }
```

### Property Value

HashDigest<SHA256>

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Interface IArena

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public interface IArena
```

## Methods

### CoAreaAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoAreaAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

#### Returns

IEnumerator

### CoBlowAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoBlowAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
```

```
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

### Returns

IEnumerator

## CoBuff(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoBuff(ArenaCharacter caster, IEnumerable<ArenaSkill.ArenaSkillInfo>
skillInfos, IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

### Returns

IEnumerator

## CoBuffRemovalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoBuffRemovalAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
```

```
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

### Returns

IEnumerator

## CoCustomEvent(ArenaCharacter, ArenaEventBase)

```
IEnumerator CoCustomEvent(ArenaCharacter caster, ArenaEventBase eventBase)
```

### Parameters

`caster` ArenaCharacter

`eventBase` ArenaEventBase

### Returns

IEnumerator

## CoDead(ArenaCharacter)

```
IEnumerator CoDead(ArenaCharacter caster)
```

### Parameters

`caster` ArenaCharacter

### Returns

IEnumerator

## CoDoubleAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoDoubleAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

### Returns

IEnumerator

## CoDoubleAttackWithCombo(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoDoubleAttackWithCombo(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

## Returns

IEnumerator

# CoFullBuffRemovalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

Coroutine for full buff removal attack that removes all positive stat buffs.

```
IEnumerator CoFullBuffRemovalAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

## Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

## Returns

IEnumerator

# CoHeal(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoHeal(ArenaCharacter caster, IEnumerable<ArenaSkill.ArenaSkillInfo>
skillInfos, IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

## Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

### Returns

IEnumerator

## CoNormalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoNormalAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

### Returns

IEnumerator

## CoRemoveBuffs(ArenaCharacter)

```
IEnumerator CoRemoveBuffs(ArenaCharacter caster)
```

### Parameters

`caster` ArenaCharacter

### Returns

IEnumerator

## CoShatterStrike(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoShatterStrike(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

### Returns

IEnumerator

## CoSpawnCharacter(ArenaCharacter)

```
IEnumerator CoSpawnCharacter(ArenaCharacter character)
```

### Parameters

`character` ArenaCharacter

### Returns

IEnumerator

## CoTickDamage(ArenaCharacter, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoTickDamage(ArenaCharacter affectedCharacter,
```

```
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos)
```

### Parameters

`affectedCharacter` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

### Returns

IEnumerator

## CoTurnEnd(int)

```
IEnumerator CoTurnEnd(int turnNumber)
```

### Parameters

`turnNumber` int

### Returns

IEnumerator

# Interface IStage

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public interface IStage
```

## Methods

### CoAreaAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoAreaAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

#### Returns

IEnumerator

### CoBlowAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoBlowAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoBreakthrough(CharacterBase, int, List<MonsterData>)

```
IEnumerator CoBreakthrough(CharacterBase character, int floorId,
List<AdventureBossFloorWaveSheet.MonsterData> monsters)
```

### Parameters

`character` CharacterBase

`floorId` int

`monsters` List<AdventureBossFloorWaveSheet.MonsterData>

### Returns

IEnumerator

## CoBuff(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoBuff(CharacterBase caster, int skillId, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoBuffRemovalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoBuffRemovalAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoCustomEvent(CharacterBase, EventBase)

```
IEnumerator CoCustomEvent(CharacterBase character, EventBase eventBase)
```

### Parameters

`character` CharacterBase

`eventBase` EventBase

### Returns

IEnumerator

## CoDead(CharacterBase)

```
IEnumerator CoDead(CharacterBase character)
```

### Parameters

`character` CharacterBase

### Returns

IEnumerator

## CoDoubleAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoDoubleAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoDoubleAttackWithCombo(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoDoubleAttackWithCombo(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoDropBox(List<ItemBase>)

```
IEnumerator CoDropBox(List<ItemBase> items)
```

### Parameters

`items` List<ItemBase>

### Returns

IEnumerator

## CoFullBuffRemovalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

Coroutine for full buff removal attack that removes all positive stat buffs.

```
IEnumerator CoFullBuffRemovalAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoGetExp(long)

```
IEnumerator CoGetExp(long exp)
```

### Parameters

`exp` long

### Returns

IEnumerator

## CoGetReward(List<ItemBase>, Dictionary<string, int>)

```
IEnumerator CoGetReward(List<ItemBase> rewards, Dictionary<string, int> favRewards)
```

### Parameters

`rewards` List<ItemBase>

`favRewards` Dictionary<string, int>

### Returns

IEnumerator

## CoHeal(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoHeal(CharacterBase caster, int skillId, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoNormalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoNormalAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoRemoveBuffs(CharacterBase)

```
IEnumerator CoRemoveBuffs(CharacterBase caster)
```

### Parameters

`caster` CharacterBase

### Returns

IEnumerator

## CoShatterStrike(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoShatterStrike(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoSpawnEnemyPlayer(EnemyPlayer)

```
IEnumerator CoSpawnEnemyPlayer(EnemyPlayer character)
```

### Parameters

`character` EnemyPlayer

### Returns

IEnumerator

## CoSpawnPlayer(Player)

```
IEnumerator CoSpawnPlayer(Player character)
```

### Parameters

`character` Player

### Returns

IEnumerator

## CoSpawnWave(int, int, List<Enemy>, bool)

```
IEnumerator CoSpawnWave(int waveNumber, int waveTurn, List<Enemy> enemies,
bool hasBoss)
```

### Parameters

`waveNumber` int

`waveTurn` int

`enemies` List<Enemy>

`hasBoss` bool

### Returns

IEnumerator

## CoStageBuff(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoStageBuff(CharacterBase affected, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`affected` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoTickDamage(CharacterBase, int, IEnumerable<SkillInfo>)

```
IEnumerator CoTickDamage(CharacterBase affectedCharacter, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos)
```

### Parameters

`affectedCharacter` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

### Returns

IEnumerator

## CoWaveTurnEnd(int, int)

```
IEnumerator CoWaveTurnEnd(int turnNumber, int waveTurn)
```

### Parameters

`turnNumber` int

`waveTurn` int

### Returns

IEnumerator

# Class Player

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class Player : CharacterBase, ICloneable
```

### Inheritance

object ← CharacterBase ← Player

### Implements

ICloneable

### Derived

EnemyPlayer

### Inherited Members

CharacterBase.CriticalMultiplier , CharacterBase.Id , CharacterBase.Simulator , CharacterBase.atkElementType , CharacterBase.attackRange , CharacterBase.defElementType , CharacterBase.Skills , CharacterBase.BuffSkills , CharacterBase.Buffs , CharacterBase.StatBuffs , CharacterBase.ActionBuffs , CharacterBase.Targets , CharacterBase.RowData , CharacterBase.CharacterId , CharacterBase.SizeType , CharacterBase.RunSpeed , CharacterBase.Stats , CharacterBase.Level , CharacterBase.HP , CharacterBase.AdditionalHP , CharacterBase.ATK , CharacterBase.DEF , CharacterBase.CRI , CharacterBase.HIT , CharacterBase.SPD , CharacterBase.DRV , CharacterBase.DRR , CharacterBase.CDMG , CharacterBase.ArmorPenetration , CharacterBase.Thorn , CharacterBase.CurrentHP , CharacterBase.IsDead , CharacterBase.AttackCount , CharacterBase.AttackCountMax , CharacterBase.usedSkill , CharacterBase.InitAI() , CharacterBase.InitAIV1() , CharacterBase.InitAIV2() , CharacterBase.Tick() , CharacterBase.AddBuff(Buff, bool) , CharacterBase.RemoveActionBuff(ActionBuff) , CharacterBase.RemoveStatBuff(StatBuff) , CharacterBase.RemoveRecentStatBuff() , CharacterBase.RemoveAllStatBuffs() , CharacterBase.ResetCurrentHP() , CharacterBase.IsCritical(bool) , CharacterBase.IsHit(ElementalResult) , CharacterBase.GetDamage(long, bool) , CharacterBase.Die() , CharacterBase.Heal(long) , CharacterBase.SetSkill() , CharacterBase.GetChance(int) , CharacterBase.OnPreSkill() , CharacterBase.OnPostSkill(Skill) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Player(ArenaPlayerDigest, ArenaSimulatorSheetsV1)

```
public Player(ArenaPlayerDigest enemyArenaPlayerDigest, ArenaSimulatorSheetsV1
simulatorSheets)
```

### Parameters

`enemyArenaPlayerDigest` ArenaPlayerDigest

`simulatorSheets` ArenaSimulatorSheetsV1

## Player(Player)

```
protected Player(Player value)
```

### Parameters

`value` Player

## Player(AvatarState, Simulator)

```
public Player(AvatarState avatarState, Simulator simulator)
```

### Parameters

`avatarState` AvatarState

`simulator` Simulator

## Player(AvatarState, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public Player(AvatarState avatarState, CharacterSheet characterSheet,
CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet)
```

### Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

## Player(AvatarState, SimulatorSheetsV1)

```
public Player(AvatarState avatarState, SimulatorSheetsV1 simulatorSheets)
```

### Parameters

`avatarState` AvatarState

`simulatorSheets` SimulatorSheetsV1

## Player(int, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public Player(int level, CharacterSheet characterSheet, CharacterLevelSheet
characterLevelSheet, EquipmentItemSetEffectSheet equipmentItemSetEffectSheet)
```

### Parameters

`level` int

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

# Fields

## Exp

```
public readonly Player.ExpData Exp
```

### Field Value

Player.ExpData

## Grimoire

```
public Grimoire Grimoire
```

### Field Value

Grimoire

## Inventory

```
public readonly Inventory Inventory
```

### Field Value

Inventory

## RuneSkillCooldownMap

```
public readonly Dictionary<int, int> RuneSkillCooldownMap
```

### Field Value

Dictionary<int, int>

## RuneSkills

```
public readonly Skills RuneSkills
```

### Field Value

Skills

## armor

```
public Armor armor
```

### Field Value

Armor

## aura

```
public Aura aura
```

### Field Value

Aura

## belt

```
public Belt belt
```

### Field Value

Belt

## characterLevelSheet

```
public CharacterLevelSheet characterLevelSheet
```

### Field Value

CharacterLevelSheet

## costumes

```
protected List<Costume> costumes
```

### Field Value

List<Costume>

## earIndex

```
public int earIndex
```

### Field Value

int

## equipments

```
protected List<Equipment> equipments
```

### Field Value

List<Equipment>

## eventMap

```
public CollectionMap eventMap
```

### Field Value

CollectionMap

## eventMapForBeforeV100310

```
public CollectionMap eventMapForBeforeV100310
```

### Field Value

CollectionMap

## hairIndex

```
public int hairIndex
```

### Field Value

int

## lensIndex

```
public int lensIndex
```

### Field Value

int

## monsterMap

```
public CollectionMap monsterMap
```

### Field Value

CollectionMap

## monsterMapForBeforeV100310

```
public CollectionMap monsterMapForBeforeV100310
```

### Field Value

CollectionMap

## necklace

```
public Necklace necklace
```

### Field Value

Necklace

## ring

```
public Ring ring
```

### Field Value

Ring

## tailIndex

```
public int tailIndex
```

### Field Value

int

## weapon

```
public Weapon weapon
```

### Field Value

Weapon

## worldInformation

```
public WorldInformation worldInformation
```

### Field Value

WorldInformation

# Properties

## Costumes

```
public IReadOnlyList<Costume> Costumes { get; }
```

### Property Value

IReadOnlyList<Costume>

## Equipments

```
public IReadOnlyList<Equipment> Equipments { get; }
```

### Property Value

IReadOnlyList<Equipment>

## Methods

## AddSkill(Skill)

```
public void AddSkill(Skill skill)
```

### Parameters

`skill` Skill

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## ConfigureStats(CostumeStatSheet, List<RuneState>, RuneOptionSheet, int, SkillSheet, List<StatModifier>)

```
public void ConfigureStats(CostumeStatSheet costumeStatSheet, List<RuneState>
runeStates, RuneOptionSheet runeOptionSheet, int runeLevelBonus, SkillSheet
skillSheet, List<StatModifier> collectionModifiers)
```

### Parameters

`costumeStatSheet` CostumeStatSheet

`runeStates` List<RuneState>

`runeOptionSheet` RuneOptionSheet

`runeLevelBonus` int

`skillSheet` SkillSheet

`collectionModifiers` List<StatModifier>

## EndTurn()

```
protected override void EndTurn()
```

## GetExp(long, bool)

```
public void GetExp(long waveExp, bool log = false)
```

### Parameters

`waveExp` long

`log` bool

## GetExp2(long, bool)

```
[Obsolete("Use GetExp")]
public void GetExp2(long waveExp, bool log = false)
```

### Parameters

`waveExp` long

`log` bool

## GetExp3(long, bool)

```
[Obsolete("Use GetExp")]
public void GetExp3(long waveExp, bool log = false)
```

### Parameters

`waveExp` long

`log` bool

## GetRewards(List<ItemBase>)

```
public CollectionMap GetRewards(List<ItemBase> items)
```

### Parameters

`items` List<ItemBase>

### Returns

CollectionMap

## IsHit(CharacterBase)

```
public override bool IsHit(CharacterBase caster)
```

### Parameters

`caster` CharacterBase

### Returns

bool

## OnDead()

```
protected override void OnDead()
```

## OverrideSkill(Skill)

```
public void OverrideSkill(Skill skill)
```

### Parameters

`skill` Skill

## ReduceSkillCooldown()

```
protected override void ReduceSkillCooldown()
```

## RemoveTarget(Enemy)

```
public void RemoveTarget(Enemy enemy)
```

### Parameters

`enemy` Enemy

## RemoveTarget(EnemyPlayer)

```
public void RemoveTarget(EnemyPlayer enemy)
```

### Parameters

`enemy` EnemyPlayer

## SetCollections(IEnumerable<StatModifier>)

```
public void SetCollections(IEnumerable<StatModifier> statModifiers)
```

### Parameters

`statModifiers` IEnumerable<StatModifier>

## SetCostumeStat(CostumeStatSheet)

```
public void SetCostumeStat(CostumeStatSheet costumeStatSheet)
```

### Parameters

`costumeStatSheet` CostumeStatSheet

## SetEquipmentStat(EquipmentItemSetEffectSheet)

```
protected void SetEquipmentStat(EquipmentItemSetEffectSheet sheet)
```

### Parameters

`sheet` EquipmentItemSetEffectSheet

## SetRuneSkills(List<RuneState>, RuneOptionSheet, SkillSheet)

Sets the rune skills for the player.

```
public void SetRuneSkills(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
SkillSheet skillSheet)
```

### Parameters

`runes` List<RuneState>

The list of rune states.

`runeOptionSheet` RuneOptionSheet

The rune option sheet.

`skillSheet` SkillSheet

The skill sheet.

## SetRuneStats(List<RuneState>, RuneOptionSheet, int)

Sets the rune stats for a player character.

```
public void SetRuneStats(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
int runeLevelBonus)
```

### Parameters

`runes` List<RuneState>

The AllRuneState for the player character.

`runeOptionSheet` RuneOptionSheet

The rune option sheet that contains information about rune options.

`runeLevelBonus` int

The rune level bonus value from RuneLevelBonusSheet. This enhances equipped rune stats.

## SetRuneV1(List<RuneState>, RuneOptionSheet, SkillSheet)

```
[Obsolete("Use SetRune")]
public void SetRuneV1(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
```

```
  SkillSheet skillSheet)
```

### Parameters

`runes` List<RuneState>

`runeOptionSheet` RuneOptionSheet

`skillSheet` SkillSheet

## Spawn()

```
  public virtual void Spawn()
```

## SpawnV1()

```
  [Obsolete("Use Spawn")]
  public virtual void SpawnV1()
```

## SpawnV2()

```
  [Obsolete("Use Spawn")]
  public virtual void SpawnV2()
```

## Use(List<Guid>)

```
  public void Use(List<Guid> consumableIds)
```

### Parameters

`consumableIds` List<Guid>

# UseSkill()

```
protected override Skill UseSkill()
```

## Returns

Skill

# Class Player.ExpData

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class Player.ExpData : ICloneable
```

### Inheritance

object ← Player.ExpData

### Implements

ICloneable

### Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

## ExpData()

```
public ExpData()
```

## ExpData(ExpData)

```
protected ExpData(Player.ExpData value)
```

### Parameters

`value` Player.ExpData

## Properties

## Current

```
public long Current { get; set; }
```

### Property Value

long

## Max

```
public long Max { get; }
```

### Property Value

long

## Need

```
public long Need { get; }
```

### Property Value

long

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## Set(Row)

```
public void Set(CharacterLevelSheet.Row row)
```

### Parameters

`row` CharacterLevelSheet.Row

# Class RaidBoss

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public class RaidBoss : Enemy, ICloneable
```

### Inheritance

object ← CharacterBase ← Enemy ← RaidBoss

### Implements

ICloneable

### Inherited Members

Enemy.spawnIndex , Enemy.OnDead() , CharacterBase.CriticalMultiplier , CharacterBase.Id , CharacterBase.Simulator , CharacterBase.atkElementType , CharacterBase.attackRange , CharacterBase.defElementType , CharacterBase.Skills , CharacterBase.BuffSkills , CharacterBase.Buffs , CharacterBase.StatBuffs , CharacterBase.ActionBuffs , CharacterBase.Targets , CharacterBase.CharacterId , CharacterBase.SizeType , CharacterBase.RunSpeed , CharacterBase.Stats , CharacterBase.Level , CharacterBase.HP , CharacterBase.AdditionalHP , CharacterBase.ATK , CharacterBase.DEF , CharacterBase.CRI , CharacterBase.HIT , CharacterBase.SPD , CharacterBase.DRV , CharacterBase.DRR , CharacterBase.CDMG , CharacterBase.ArmorPenetration , CharacterBase.Thorn , CharacterBase.CurrentHP , CharacterBase.IsDead , CharacterBase.AttackCount , CharacterBase.AttackCountMax , CharacterBase.usedSkill , CharacterBase.InitAI() , CharacterBase.InitAIV1() , CharacterBase.InitAIV2() , CharacterBase.Tick() , CharacterBase.ReduceSkillCooldown() , CharacterBase.AddBuff(Buff, bool) , CharacterBase.RemoveActionBuff(ActionBuff) , CharacterBase.RemoveStatBuff(StatBuff) , CharacterBase.RemoveRecentStatBuff() , CharacterBase.RemoveAllStatBuffs() , CharacterBase.ResetCurrentHP() , CharacterBase.IsCritical(bool) , CharacterBase.IsHit(ElementalResult) , CharacterBase.GetDamage(long, bool) , CharacterBase.Die() , CharacterBase.Heal(long) , CharacterBase.GetChance(int) , CharacterBase.OnPreSkill() , CharacterBase.OnPostSkill(Skill) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RaidBoss(CharacterBase, Row, Row, WaveStatData, bool)

```
public RaidBoss(CharacterBase player, WorldBossCharacterSheet.Row characterRow,
WorldBossActionPatternSheet.Row patternRow, WorldBossCharacterSheet.WaveStatData
statData, bool ignoreLevelCorrectionOnHit)
```

### Parameters

`player` CharacterBase

`characterRow` WorldBossCharacterSheet.Row

`patternRow` WorldBossActionPatternSheet.Row

`statData` WorldBossCharacterSheet.WaveStatData

`ignoreLevelCorrectionOnHit` bool

## RaidBoss(RaidBoss)

```
public RaidBoss(RaidBoss value)
```

### Parameters

`value` RaidBoss

# Properties

## Enraged

```
public bool Enraged { get; protected set; }
```

### Property Value

bool

## IgnoreLevelCorrectionOnHit

```
public bool IgnoreLevelCorrectionOnHit { get; protected set; }
```

### Property Value

bool

## PatternRowData

```
public WorldBossActionPatternSheet.Row PatternRowData { get; }
```

### Property Value

WorldBossActionPatternSheet.Row

## RowData

```
public WorldBossCharacterSheet.Row RowData { get; }
```

### Property Value

WorldBossCharacterSheet.Row

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## EndTurn()

```
protected override void EndTurn()
```

## Enrage()

```
public void Enrage()
```

## IsHit(CharacterBase)

```
public override bool IsHit(CharacterBase caster)
```

### Parameters

`caster` CharacterBase

### Returns

bool

## SetSkill()

```
protected override void SetSkill()
```

## UseSkill()

```
protected override Skill UseSkill()
```

### Returns

Skill

# Class Skills

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class Skills : IEnumerable<Skill>, IEnumerable
```

### Inheritance

object ← Skills

### Implements

IEnumerable<Skill>, IEnumerable

### Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### Add(Skill)

```
public void Add(Skill skill)
```

#### Parameters

`skill` Skill

### Clear()

```
public void Clear()
```

### GetCooldown(int)

```
public int GetCooldown(int skillId)
```

### Parameters

`skillId` int

### Returns

int

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Skill> GetEnumerator()
```

### Returns

IEnumerator<Skill>

An enumerator that can be used to iterate through the collection.

## ReduceCooldown()

```
public void ReduceCooldown()
```

## ReduceCooldownV1()

```
[Obsolete("ReduceCooldown")]
public void ReduceCooldownV1()
```

## Select(IRandom)

```
public Skill Select(IRandom random)
```

### Parameters

`random` IRandom

### Returns

Skill

## SelectV1(IRandom)

```
[Obsolete("Use Select")]
public Skill SelectV1(IRandom random)
```

### Parameters

`random` IRandom

### Returns

Skill

## SelectV2(IRandom)

```
[Obsolete("Use Select")]
public Skill SelectV2(IRandom random)
```

### Parameters

`random` IRandom

### Returns

Skill

## SelectWithoutDefaultAttack(IRandom)

```
public Skill SelectWithoutDefaultAttack(IRandom random)
```

### Parameters

`random` IRandom

### Returns

Skill

## SetCooldown(int, int)

```
public void SetCooldown(int skillId, int cooldown)
```

### Parameters

`skillId` int

`cooldown` int

# Class WorldInformation

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class WorldInformation : IState, ISerializable, ICloneable
```

**Inheritance**

object ← WorldInformation

**Implements**

IState, ISerializable, ICloneable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

WorldInformationExtensions.ValidateFromAction(WorldInformation, int, string, string)

## Constructors

### WorldInformation(Dictionary)

```
public WorldInformation(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### WorldInformation(long, WorldSheet, bool, string)

```
public WorldInformation(long blockIndex, WorldSheet worldSheet, bool
openAllOfWorldsAndStages = false, string avatarName = "")
```

### Parameters

`blockIndex` long

`worldSheet` WorldSheet

`openAllOfWorldsAndStages` bool

`avatarName` string

## WorldInformation(long, WorldSheet, int)

```
public WorldInformation(long blockIndex, WorldSheet worldSheet, int clearStageId
= 0)
```

### Parameters

`blockIndex` long

`worldSheet` WorldSheet

`clearStageId` int

# Methods

## AddAndUnlockMimisbrunnrWorld(Row, long, WorldSheet, WorldUnlockSheet)

```
public void AddAndUnlockMimisbrunnrWorld(WorldSheet.Row worldRow, long unlockedAt,
WorldSheet worldSheet, WorldUnlockSheet worldUnlockSheet)
```

### Parameters

`worldRow` WorldSheet.Row

`unlockedAt` long

`worldSheet` WorldSheet

`worldUnlockSheet` WorldUnlockSheet

## AddAndUnlockNewWorld(Row, long, WorldSheet)

```
public void AddAndUnlockNewWorld(WorldSheet.Row worldRow, long unlockedAt,
WorldSheet worldSheet)
```

### Parameters

`worldRow` WorldSheet.Row

`unlockedAt` long

`worldSheet` WorldSheet

## ClearStage(int, int, long, WorldSheet, WorldUnlockSheet)

Clear a specific stage. And consider world unlock.

```
public void ClearStage(int worldId, int stageId, long clearedAt, WorldSheet
worldSheet, WorldUnlockSheet worldUnlockSheet)
```

### Parameters

`worldId` int

`stageId` int

`clearedAt` long

`worldSheet` WorldSheet

`worldUnlockSheet` WorldUnlockSheet

### Exceptions

FailedToUnlockWorldException

## Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## IsStageCleared(int)

```
public bool IsStageCleared(int stageId)
```

### Parameters

`stageId` int

### Returns

bool

## IsWorldUnlocked(int)

```
public bool IsWorldUnlocked(int worldId)
```

### Parameters

`worldId` int

### Returns

bool

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## TryAddWorld(Row, out World)

```
public bool TryAddWorld(WorldSheet.Row worldRow, out WorldInformation.World world)
```

### Parameters

`worldRow` WorldSheet.Row

`world` WorldInformation.World

### Returns

bool

## TryGetFirstWorld(out World)

```
public bool TryGetFirstWorld(out WorldInformation.World world)
```

### Parameters

`world` WorldInformation.World

### Returns

bool

## TryGetLastClearedMimisbrunnrStageId(out int)

```
public bool TryGetLastClearedMimisbrunnrStageId(out int stageId)
```

### Parameters

`stageId` int

### Returns

bool

## TryGetLastClearedStageId(out int)

Get stage id of the most recent cleared.(ignore mimisbrunnr)

```
public bool TryGetLastClearedStageId(out int stageId)
```

### Parameters

`stageId` int

### Returns

bool

## TryGetUnlockedWorldByStageClearedBlockIndex(out World)

Get `World` object that contains the most recent stage clear.

```
public bool TryGetUnlockedWorldByStageClearedBlockIndex(out
WorldInformation.World world)
```

### Parameters

`world` WorldInformation.World

### Returns

bool

## TryGetWorld(int, out World)

Get `World` object that equals to `worldId` argument.

```
public bool TryGetWorld(int worldId, out WorldInformation.World world)
```

### Parameters

`worldId` int

`world` WorldInformation.World

### Returns

bool

### Exceptions

KeyNotFoundException

## TryGetWorldByStageId(int, out World)

Get `World` object that contains `stageId` argument.

```
public bool TryGetWorldByStageId(int stageId, out WorldInformation.World world)
```

### Parameters

`stageId` int

`world` WorldInformation.World

### Returns

bool

## UnlockWorld(int, long, WorldSheet)

Unlock a specific world.

```
public void UnlockWorld(int worldId, long unlockedAt, WorldSheet worldSheet)
```

### Parameters

`worldId` int

`unlockedAt` long

`worldSheet` WorldSheet

### Exceptions

FailedToUnlockWorldException

# UpdateWorld(Row)

```
public void UpdateWorld(WorldSheet.Row worldRow)
```

## Parameters

`worldRow` WorldSheet.Row

# Struct WorldInformation.World

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public struct WorldInformation.World : IState
```

**Implements**
IState

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### World(Dictionary)

```
public World(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### World(World, long)

```
public World(WorldInformation.World world, long unlockedBlockIndex = -1)
```

#### Parameters

`world` WorldInformation.World

`unlockedBlockIndex` long

## World(World, long, int)

```
public World(WorldInformation.World world, long stageClearedBlockIndex,
int stageClearedId)
```

### Parameters

`world` WorldInformation.World

`stageClearedBlockIndex` long

`stageClearedId` int

## World(Row, long, long, int)

```
public World(WorldSheet.Row worldRow, long unlockedBlockIndex = -1, long
stageClearedBlockIndex = -1, int stageClearedId = -1)
```

### Parameters

`worldRow` WorldSheet.Row

`unlockedBlockIndex` long

`stageClearedBlockIndex` long

`stageClearedId` int

# Fields

## Id

```
public readonly int Id
```

### Field Value

int

## Name

```
public readonly string Name
```

### Field Value

string

## StageBegin

```
public readonly int StageBegin
```

### Field Value

int

## StageClearedBlockIndex

```
public readonly long StageClearedBlockIndex
```

### Field Value

long

## StageClearedId

```
public readonly int StageClearedId
```

### Field Value

int

## StageEnd

```
public readonly int StageEnd
```

### Field Value

int

## UnlockedBlockIndex

```
public readonly long UnlockedBlockIndex
```

### Field Value

long

# Properties

## IsStageCleared

```
public bool IsStageCleared { get; }
```

### Property Value

bool

## IsUnlocked

```
public bool IsUnlocked { get; }
```

### Property Value

bool

# Methods

## ContainsStageId(int)

```
public bool ContainsStageId(int stageId)
```

### Parameters

`stageId` int

### Returns

bool

## GetNextStageId()

```
public int GetNextStageId()
```

### Returns

int

## GetNextStageIdForPlay()

```
public int GetNextStageIdForPlay()
```

### Returns

int

## IsPlayable(int)

```
public bool IsPlayable(int stageId)
```

### Parameters

`stageId` int

## Returns

bool

# Serialize()

```
public IValue Serialize()
```

## Returns

IValue

# Namespace Nekoyume.Model.Adventure Boss

## Classes

BountyBoard

ExploreBoard

Explorer

ExplorerList

Investor

SeasonInfo

# Class BountyBoard

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class BountyBoard
```

**Inheritance**

object ← BountyBoard

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BountyBoard(List)

```
public BountyBoard(List bencoded)
```

#### Parameters

`bencoded` List

### BountyBoard(long)

```
public BountyBoard(long season)
```

#### Parameters

`season` long

## Fields

## FixedRewardFavId

```
public int? FixedRewardFavId
```

### Field Value

int?

## FixedRewardItemId

```
public int? FixedRewardItemId
```

### Field Value

int?

## Investors

```
public List<Investor> Investors
```

### Field Value

List<Investor>

## RandomRewardFavId

```
public int? RandomRewardFavId
```

### Field Value

int?

## RandomRewardItemId

```
public int? RandomRewardItemId
```

### Field Value

int?

## Season

```
public long Season
```

### Field Value

long

# Properties

## Bencoded

```
public IValue Bencoded { get; }
```

### Property Value

IValue

# Methods

## AddOrUpdate(Address, string, FungibleAssetValue)

```
public void AddOrUpdate(Address avatarAddress, string name,
FungibleAssetValue price)
```

### Parameters

avatarAddress Address

`name` string

`price` FungibleAssetValue

## SetReward(Row, IRandom)

```
public void SetReward(AdventureBossWantedRewardSheet.Row rewardInfo, IRandom random)
```

### Parameters

`rewardInfo` AdventureBossWantedRewardSheet.Row

`random` IRandom

## totalBounty()

```
public FungibleAssetValue totalBounty()
```

### Returns

FungibleAssetValue

# Class ExploreBoard

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class ExploreBoard
```

**Inheritance**

object ← ExploreBoard

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ExploreBoard(List)

```
public ExploreBoard(List bencoded)
```

#### Parameters

`bencoded` List

### ExploreBoard(long)

```
public ExploreBoard(long season)
```

#### Parameters

`season` long

## Fields

## ExplorerCount

```
public int ExplorerCount
```

### Field Value

int

## FixedRewardFavId

```
public int? FixedRewardFavId
```

### Field Value

int?

## FixedRewardItemId

```
public int? FixedRewardItemId
```

### Field Value

int?

## RaffleReward

```
public FungibleAssetValue? RaffleReward
```

### Field Value

FungibleAssetValue?

## RaffleWinner

```
public Address? RaffleWinner
```

### Field Value

Address?

## RaffleWinnerName

```
public string RaffleWinnerName
```

### Field Value

string

## Season

```
public long Season
```

### Field Value

long

## TotalPoint

```
public long TotalPoint
```

### Field Value

long

## UsedApPotion

```
public long UsedApPotion
```

### Field Value

long

## UsedGoldenDust

```
public long UsedGoldenDust
```

### Field Value

long

## UsedNcg

```
public BigInteger UsedNcg
```

### Field Value

BigInteger

# Methods

## Bencoded()

```
public IValue Bencoded()
```

### Returns

IValue

## SetReward(Row, IRandom)

```
public void SetReward(AdventureBossContributionRewardSheet.Row rewardInfo,
IRandom random)
```

## Parameters

`rewardInfo` AdventureBossContributionRewardSheet.Row

`random` IRandom

# Class Explorer

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class Explorer : IBencodable
```

**Inheritance**

object ← Explorer

**Implements**

IBencodable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Explorer(IValue)

```
public Explorer(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### Explorer(Address, string)

```
public Explorer(Address avatarAddress, string name)
```

#### Parameters

`avatarAddress` Address

name string

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Claimed

```
public bool Claimed
```

### Field Value

bool

## Floor

```
public int Floor
```

### Field Value

int

## MaxFloor

```
public int MaxFloor
```

### Field Value

int

## Name

```
public string Name
```

### Field Value

string

## Score

```
public int Score
```

### Field Value

int

## UsedApPotion

```
public int UsedApPotion
```

### Field Value

int

## UsedGoldenDust

```
public int UsedGoldenDust
```

### Field Value

int

## UsedNcg

```
public BigInteger UsedNcg
```

### Field Value

BigInteger

# Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

# Class ExplorerList

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class ExplorerList
```

**Inheritance**

object ← ExplorerList

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ExplorerList(List)

```
public ExplorerList(List bencoded)
```

#### Parameters

`bencoded` List

### ExplorerList(long)

```
public ExplorerList(long season)
```

#### Parameters

`season` long

## Fields

## Explorers

```
public HashSet<(Address, string)> Explorers
```

### Field Value

HashSet<(Address, string)>

## Season

```
public long Season
```

### Field Value

long

# Properties

## Bencoded

```
public IValue Bencoded { get; }
```

### Property Value

IValue

# Methods

## AddExplorer(Address, string)

```
public void AddExplorer(Address avatarAddress, string name)
```

### Parameters

`avatarAddress` Address

`name` string

# Class Investor

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class Investor : IBencodable
```

**Inheritance**

object ← Investor

**Implements**

IBencodable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Investor(IValue)

```
public Investor(IValue value)
```

#### Parameters

`value` IValue

### Investor(Address, string, FungibleAssetValue)

```
public Investor(Address avatarAddress, string name, FungibleAssetValue price)
```

#### Parameters

`avatarAddress` Address

`name` string

`price` FungibleAssetValue

# Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## Claimed

```
public bool Claimed
```

### Field Value

bool

## Count

```
public int Count
```

### Field Value

int

## MaxInvestmentCount

```
public const int MaxInvestmentCount = 3
```

### Field Value

int

## Name

```
public string Name
```

### Field Value

string

## Price

```
public FungibleAssetValue Price
```

### Field Value

FungibleAssetValue

# Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

# Class SeasonInfo

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class SeasonInfo
```

**Inheritance**

object ← SeasonInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SeasonInfo(List)

```
public SeasonInfo(List serialized)
```

#### Parameters

`serialized` List

### SeasonInfo(long, long, long, long, long?, long?)

```
public SeasonInfo(long season, long startBlockIndex, long activeInterval, long
inactiveInterval, long? endBlockIndex = null, long? nextStartBlockIndex = null)
```

#### Parameters

`season` long

`startBlockIndex` long

`activeInterval` long

`inactiveInterval` long

`endBlockIndex` long?

`nextStartBlockIndex` long?

# Fields

## BossId

```
public int BossId
```

### Field Value

int

## EndBlockIndex

```
public readonly long EndBlockIndex
```

### Field Value

long

## NextStartBlockIndex

```
public readonly long NextStartBlockIndex
```

### Field Value

long

## Season

```
public readonly long Season
```

### Field Value

long

## StartBlockIndex

```
public readonly long StartBlockIndex
```

### Field Value

long

# Properties

## Bencoded

```
public IValue Bencoded { get; }
```

### Property Value

IValue

# Namespace Nekoyume.Model.Arena

## Classes

AddressNotFoundInArenaParticipantsException

ArenaAvatarStateNotFoundException

ArenaInformation
Introduced at https://github.com/planetarium/lib9c/pull/1029

ArenaInformationAlreadyContainsException

ArenaInformationNotFoundException

ArenaParticipant
This class combines the information from ArenaScore and ArenaInformation, and brings together information about the arena participant, including additional information.

ArenaParticipants
Introduced at https://github.com/planetarium/lib9c/pull/1027

ArenaParticipantsNotFoundException

ArenaResult
temp

ArenaScore
Introduced at https://github.com/planetarium/lib9c/pull/1027

ArenaScoreAlreadyContainsException

ArenaScoreNotFoundException

CoolDownBlockException

ExceedPlayCountException

ExceedTicketPurchaseLimitDuringIntervalException

ExceedTicketPurchaseLimitException

InvalidSeasonException

NotEnoughTicketException

RoundNotFoundException

ThisArenaIsClosedException

TicketPurchaseLimitExceedException

ValidateScoreDifferenceException

# Class AddressNotFoundInArenaParticipantsException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class AddressNotFoundInArenaParticipantsException : Exception, ISerializable
```

### Inheritance

object ← Exception ← AddressNotFoundInArenaParticipantsException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### AddressNotFoundInArenaParticipantsException(SerializationInfo, StreamingContext)

```
protected AddressNotFoundInArenaParticipantsException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## AddressNotFoundInArenaParticipantsException(string)

```
public AddressNotFoundInArenaParticipantsException(string message)
```

### Parameters

`message` string

# Class ArenaAvatarStateNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaAvatarStateNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ArenaAvatarStateNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ArenaAvatarStateNotFoundException(SerializationInfo, StreamingContext)

```
protected ArenaAvatarStateNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ArenaAvatarStateNotFoundException(string)

```
public ArenaAvatarStateNotFoundException(string message)
```

## Parameters

`message` string

# Class ArenaInformation

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1029

```
public class ArenaInformation : IState
```

**Inheritance**

object ← ArenaInformation

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaInformation(List)

```
public ArenaInformation(List serialized)
```

#### Parameters

`serialized` List

### ArenaInformation(Address, int, int)

```
public ArenaInformation(Address avatarAddress, int championshipId, int round)
```

#### Parameters

`avatarAddress` Address

`championshipId` int

`round` int

# Fields

## Address

```
public Address Address
```

### Field Value

Address

## MaxTicketCount

```
public const int MaxTicketCount = 8
```

### Field Value

int

# Properties

## Lose

```
public int Lose { get; }
```

### Property Value

int

## PurchasedTicketCount

```
public int PurchasedTicketCount { get; }
```

### Property Value

int

## Ticket

```
public int Ticket { get; }
```

### Property Value

int

## TicketResetCount

```
public int TicketResetCount { get; }
```

### Property Value

int

## Win

```
public int Win { get; }
```

### Property Value

int

# Methods

## BuyTicket(long)

```
public void BuyTicket(long maxCount)
```

### Parameters

`maxCount` long

## DeriveAddress(Address, int, int)

```
public static Address DeriveAddress(Address avatarAddress, int championshipId,
int round)
```

### Parameters

`avatarAddress` Address

`championshipId` int

`round` int

### Returns

Address

## ResetTicket(int)

```
public void ResetTicket(int resetCount)
```

### Parameters

`resetCount` int

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## UpdateRecord(int, int)

```
public void UpdateRecord(int win, int lose)
```

### Parameters

`win` int

`lose` int

## UseTicket(int)

```
public void UseTicket(int ticketCount)
```

### Parameters

`ticketCount` int

# Class ArenaInformationAlreadyContainsException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaInformationAlreadyContainsException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ArenaInformationAlreadyContainsException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ArenaInformationAlreadyContainsException(SerializationInfo, StreamingContext)

```
protected ArenaInformationAlreadyContainsException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## ArenaInformationAlreadyContainsException(string)

```
public ArenaInformationAlreadyContainsException(string message)
```

### Parameters

`message` string

# Class ArenaInformationNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaInformationNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ArenaInformationNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ArenaInformationNotFoundException(SerializationInfo, StreamingContext)

```
protected ArenaInformationNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ArenaInformationNotFoundException(string)

```
public ArenaInformationNotFoundException(string message)
```

## Parameters

`message` string

# Class ArenaParticipant

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

This class combines the information from ArenaScore and ArenaInformation, and brings together information about the arena participant, including additional information.

```
public class ArenaParticipant : IBencodable, IState
```

**Inheritance**

object ← ArenaParticipant

**Implements**

IBencodable, IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaParticipant()

```
public ArenaParticipant()
```

### ArenaParticipant(IValue)

```
public ArenaParticipant(IValue bencoded)
```

#### Parameters

`bencoded` IValue

## ArenaParticipant(Address)

```
public ArenaParticipant(Address avatarAddr)
```

### Parameters

`avatarAddr` Address

# Fields

## AvatarAddr

```
public readonly Address AvatarAddr
```

### Field Value

Address

## Cp

```
public long Cp
```

### Field Value

long

## DefaultScore

```
public const int DefaultScore = 1000
```

### Field Value

int

## LastBattleBlockIndex

```
public long LastBattleBlockIndex
```

### Field Value

long

## Level

```
public int Level
```

### Field Value

int

## Lose

```
public int Lose
```

### Field Value

int

## MaxTicketCount

```
public const int MaxTicketCount = 8
```

### Field Value

int

## Name

If you need to know NameWithHash, check PostConstructor() method of the AvatarState class and you can find the relevant information there. It provides a formatted string that includes the avatar's name and a shortened version of their address.

```
public string Name
```

### Field Value

string

### Examples

```
$"{name} <size=80%><color=#A68F7E>#{address.ToHex().Substring(0, 4)}
</color></size>";
```

## PortraitId

```
public int PortraitId
```

### Field Value

int

## PurchasedTicketCount

```
public int PurchasedTicketCount
```

### Field Value

int

## Score

```
public int Score
```

### Field Value

int

## Ticket

```
public int Ticket
```

### Field Value

int

## TicketResetCount

```
public int TicketResetCount
```

### Field Value

int

## Win

```
public int Win
```

### Field Value

int

# Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

## Methods

## AddScore(int)

```
public void AddScore(int score)
```

### Parameters

`score` int

## BuyTicket(long)

```
public void BuyTicket(long maxCount)
```

### Parameters

`maxCount` long

## ResetTicket(int)

```
public void ResetTicket(int resetCount)
```

### Parameters

`resetCount` int

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## UpdateRecord(int, int)

```
public void UpdateRecord(int win, int lose)
```

### Parameters

`win` int

`lose` int

## UseTicket(int)

```
public void UseTicket(int ticketCount)
```

### Parameters

`ticketCount` int

# Class ArenaParticipants

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1027

```
public class ArenaParticipants : IState
```

**Inheritance**

object ← ArenaParticipants

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaParticipants(List)

```
public ArenaParticipants(List serialized)
```

#### Parameters

`serialized` List

### ArenaParticipants(int, int)

```
public ArenaParticipants(int championshipId, int round)
```

#### Parameters

`championshipId` int

`round` int

# Fields

## Address

```
public Address Address
```

### Field Value

Address

## AvatarAddresses

```
public readonly List<Address> AvatarAddresses
```

### Field Value

List<Address>

# Methods

## Add(Address)

```
public void Add(Address address)
```

### Parameters

`address` Address

## DeriveAddress(int, int)

```
public static Address DeriveAddress(int championshipId, int round)
```

### Parameters

`championshipId` int

`round` int

### Returns

Address

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class ArenaParticipantsNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaParticipantsNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ArenaParticipantsNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ArenaParticipantsNotFoundException(SerializationInfo, StreamingContext)

```
protected ArenaParticipantsNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ArenaParticipantsNotFoundException(string)

```
public ArenaParticipantsNotFoundException(string message)
```

## Parameters

`message` string

# Class ArenaResult

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

temp

```
public class ArenaResult : IBencodable, IState
```

**Inheritance**

object ← ArenaResult

**Implements**

IBencodable, IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaResult(IValue)

```
public ArenaResult(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### ArenaResult(bool, int, int, long)

```
public ArenaResult(bool isVictory, int portraitId, int level, long cp)
```

#### Parameters

`isVictory` bool

`portraitId` int

`level` int

`cp` long

# Fields

## Cp

```
public long Cp
```

### Field Value

long

## IsVictory

```
public bool IsVictory
```

### Field Value

bool

## Level

```
public int Level
```

### Field Value

int

## PortraitId

```
public int PortraitId
```

### Field Value

int

## Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

## Methods

## DeriveAddress(Address, TxId)

```
public static Address DeriveAddress(Address avatarAddress, TxId txId)
```

### Parameters

`avatarAddress` Address

`txId` TxId

### Returns

Address

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class ArenaScore

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1027

```
public class ArenaScore : IState
```

**Inheritance**

object ← ArenaScore

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaScore(List)

```
public ArenaScore(List serialized)
```

#### Parameters

`serialized` List

### ArenaScore(Address, int, int, int)

```
public ArenaScore(Address avatarAddress, int championshipId, int round, int score
= 1000)
```

#### Parameters

`avatarAddress` Address

`championshipId` int

`round` int

`score` int

# Fields

## Address

```
public Address Address
```

### Field Value

Address

## ArenaScoreDefault

```
public const int ArenaScoreDefault = 1000
```

### Field Value

int

# Properties

## Score

```
public int Score { get; }
```

### Property Value

int

# Methods

## AddScore(int)

```
public void AddScore(int score)
```

### Parameters

`score` int

## DeriveAddress(Address, int, int)

```
public static Address DeriveAddress(Address avatarAddress, int championshipId,
int round)
```

### Parameters

`avatarAddress` Address

`championshipId` int

`round` int

### Returns

Address

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class ArenaScoreAlreadyContainsException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaScoreAlreadyContainsException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← ArenaScoreAlreadyContainsException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ArenaScoreAlreadyContainsException(SerializationInfo, StreamingContext)

```
protected ArenaScoreAlreadyContainsException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ArenaScoreAlreadyContainsException(string)

```
public ArenaScoreAlreadyContainsException(string message)
```

## Parameters

`message` string

# Class ArenaScoreNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaScoreNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ArenaScoreNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ArenaScoreNotFoundException(SerializationInfo, StreamingContext)

```
protected ArenaScoreNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ArenaScoreNotFoundException(string)

```
public ArenaScoreNotFoundException(string message)
```

## Parameters

`message` string

# Class CoolDownBlockException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class CoolDownBlockException : Exception, ISerializable
```

### Inheritance

object ← Exception ← CoolDownBlockException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### CoolDownBlockException(SerializationInfo, StreamingContext)

```
protected CoolDownBlockException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# CoolDownBlockException(string)

```
public CoolDownBlockException(string message)
```

## Parameters

`message` string

# Class ExceedPlayCountException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ExceedPlayCountException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ExceedPlayCountException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ExceedPlayCountException(SerializationInfo, StreamingContext)

```
protected ExceedPlayCountException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ExceedPlayCountException(string)

```
public ExceedPlayCountException(string message)
```

## Parameters

`message` string

# Class ExceedTicketPurchaseLimitDuringIntervalException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ExceedTicketPurchaseLimitDuringIntervalException :
Exception, ISerializable
```

### Inheritance

object ← Exception ← ExceedTicketPurchaseLimitDuringIntervalException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ExceedTicketPurchaseLimitDuringIntervalException(SerializationInfo, StreamingContext)

```
protected ExceedTicketPurchaseLimitDuringIntervalException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ExceedTicketPurchaseLimitDuringIntervalException(string)

```
public ExceedTicketPurchaseLimitDuringIntervalException(string message)
```

## Parameters

`message` string

# Class ExceedTicketPurchaseLimitException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ExceedTicketPurchaseLimitException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← ExceedTicketPurchaseLimitException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ExceedTicketPurchaseLimitException(SerializationInfo, StreamingContext)

```
protected ExceedTicketPurchaseLimitException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ExceedTicketPurchaseLimitException(string)

```
public ExceedTicketPurchaseLimitException(string message)
```

## Parameters

`message` string

# Class InvalidSeasonException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidSeasonException : Exception, ISerializable
```

### Inheritance

object ← Exception ← InvalidSeasonException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidSeasonException(SerializationInfo, StreamingContext)

```
protected InvalidSeasonException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# InvalidSeasonException(string)

```
public InvalidSeasonException(string message)
```

## Parameters

`message` string

# Class NotEnoughTicketException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughTicketException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← NotEnoughTicketException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### NotEnoughTicketException(SerializationInfo, StreamingContext)

```
protected NotEnoughTicketException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# NotEnoughTicketException(string)

```
public NotEnoughTicketException(string message)
```

## Parameters

`message` string

# Class RoundNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class RoundNotFoundException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← RoundNotFoundException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RoundNotFoundException(SerializationInfo, StreamingContext)

```
protected RoundNotFoundException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# RoundNotFoundException(string)

```
public RoundNotFoundException(string message)
```

## Parameters

`message` string

# Class ThisArenaIsClosedException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ThisArenaIsClosedException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ThisArenaIsClosedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ThisArenaIsClosedException(SerializationInfo, StreamingContext)

```
protected ThisArenaIsClosedException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ThisArenaIsClosedException(string)

```
public ThisArenaIsClosedException(string message)
```

## Parameters

`message` string

# Class TicketPurchaseLimitExceedException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
public class TicketPurchaseLimitExceedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← TicketPurchaseLimitExceedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### TicketPurchaseLimitExceedException(SerializationInfo, StreamingContext)

```
protected TicketPurchaseLimitExceedException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# TicketPurchaseLimitExceedException(string)

```
public TicketPurchaseLimitExceedException(string message)
```

## Parameters

`message` string

# Class ValidateScoreDifferenceException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ValidateScoreDifferenceException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← ValidateScoreDifferenceException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ValidateScoreDifferenceException(SerializationInfo, StreamingContext)

```
protected ValidateScoreDifferenceException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ValidateScoreDifferenceException(string)

```
public ValidateScoreDifferenceException(string message)
```

## Parameters

`message` string

# Namespace Nekoyume.Model.BattleStatus

## Classes

AreaAttack

BattleLog

BlowAttack

Buff

BuffRemovalAttack

Dead

DoubleAttack

DoubleAttackWithCombo

DropBox

EventBase

FullBuffRemovalAttack
: Battle status event for the full buff removal attack skill. Triggers the CoFullBuffRemovalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>) coroutine for stage rendering.

GetExp

GetReward
: Event that represents a character receiving rewards after battle. Supports both item rewards and fungible asset rewards.

HealSkill

NormalAttack

RemoveBuffs

ShatterStrike

Skill

Skill.SkillInfo

SpawnEnemyPlayer

SpawnPlayer

SpawnWave

Tick

TickDamage

WaveTurnEnd

## Enums

BattleLog.Result

# Class AreaAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class AreaAttack : Skill
```

**Inheritance**

object ← EventBase ← Skill ← AreaAttack

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AreaAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public AreaAttack(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class BattleLog

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class BattleLog : IEnumerable<EventBase>, IEnumerable
```

**Inheritance**

object ← BattleLog

**Implements**

IEnumerable<EventBase>, IEnumerable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## clearedWaveNumber

```
public int clearedWaveNumber
```

### Field Value

int

## diffScore

```
public int diffScore
```

### Field Value

int

## events

```
public List<EventBase> events
```

### Field Value

List<EventBase>

## id

```
public Guid id
```

### Field Value

Guid

## newlyCleared

```
public bool newlyCleared
```

### Field Value

bool

## result

```
public BattleLog.Result result
```

### Field Value

BattleLog.Result

## score

```
public int score
```

### Field Value

int

## stageId

```
public int stageId
```

### Field Value

int

## waveCount

```
public int waveCount
```

### Field Value

int

## worldId

```
public int worldId
```

### Field Value

int

# Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## IsClear

```
public bool IsClear { get; }
```

### Property Value

bool

# Methods

## Add(EventBase)

```
public void Add(EventBase e)
```

### Parameters

e EventBase

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<EventBase> GetEnumerator()
```

### Returns

IEnumerator<EventBase>

An enumerator that can be used to iterate through the collection.

# Enum BattleLog.Result

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
public enum BattleLog.Result
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Lose = 1`

`TimeOver = 2`

`Win = 0`

# Class BlowAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class BlowAttack : Skill
```

**Inheritance**

object ← EventBase ← Skill ← BlowAttack

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BlowAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public BlowAttack(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class Buff

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class Buff : Skill
```

**Inheritance**

object ← EventBase ← Skill ← Buff

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Buff(int, CharacterBase, IEnumerable<SkillInfo>)

```
public Buff(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

## Methods

### CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class BuffRemovalAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class BuffRemovalAttack : Skill
```

**Inheritance**

object ← EventBase ← Skill ← BuffRemovalAttack

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BuffRemovalAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public BuffRemovalAttack(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class Dead

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class Dead : EventBase
```

**Inheritance**

object ← EventBase ← Dead

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Dead(CharacterBase)

```
public Dead(CharacterBase character)
```

#### Parameters

`character` CharacterBase

## Methods

### CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

#### Parameters

`stage` IStage

## Returns

IEnumerator

# Class DoubleAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class DoubleAttack : Skill
```

**Inheritance**

object ← EventBase ← Skill ← DoubleAttack

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### DoubleAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public DoubleAttack(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class DoubleAttackWithCombo

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class DoubleAttackWithCombo : Skill
```

**Inheritance**

object ← EventBase ← Skill ← DoubleAttackWithCombo

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### DoubleAttackWithCombo(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public DoubleAttackWithCombo(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class DropBox

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class DropBox : EventBase
```

**Inheritance**

object ← EventBase ← DropBox

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### DropBox(CharacterBase, List<ItemBase>)

```
public DropBox(CharacterBase character, List<ItemBase> items)
```

#### Parameters

`character` CharacterBase

`items` List<ItemBase>

## Fields

### Items

```
public readonly List<ItemBase> Items
```

#### Field Value

List<ItemBase>

# Methods

## CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

### Parameters

`stage` IStage

### Returns

IEnumerator

# Class EventBase

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public abstract class EventBase
```

**Inheritance**

object ← EventBase

**Derived**

Breakthrough, Dead, DropBox, GetExp, GetReward, RemoveBuffs, Skill, SpawnEnemyPlayer, SpawnPlayer, SpawnWave, WaveTurnEnd

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EventBase(CharacterBase)

```
protected EventBase(CharacterBase character)
```

### Parameters

`character` CharacterBase

# Fields

## Character

```
public readonly CharacterBase Character
```

### Field Value

CharacterBase

## Methods

### CoExecute(IStage)

```
public abstract IEnumerator CoExecute(IStage stage)
```

#### Parameters

`stage` IStage

#### Returns

IEnumerator

# Class FullBuffRemovalAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

Battle status event for the full buff removal attack skill. Triggers the CoFullBuffRemovalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>) coroutine for stage rendering.

```
[Serializable]
public class FullBuffRemovalAttack : Skill
```

**Inheritance**

object ← EventBase ← Skill ← FullBuffRemovalAttack

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### FullBuffRemovalAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public FullBuffRemovalAttack(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

# Methods

## CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

### Parameters

`stage` IStage

### Returns

IEnumerator

# Class GetExp

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class GetExp : EventBase
```

**Inheritance**

object ← EventBase ← GetExp

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GetExp(CharacterBase, long)

```
public GetExp(CharacterBase character, long exp)
```

#### Parameters

`character` CharacterBase

`exp` long

## Properties

### Exp

```
public long Exp { get; }
```

#### Property Value

long

# Methods

## CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

### Parameters

`stage` IStage

### Returns

IEnumerator

# Class GetReward

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

Event that represents a character receiving rewards after battle. Supports both item rewards and fungible asset rewards.

```
[Serializable]
public class GetReward : EventBase
```

**Inheritance**

object ← EventBase ← GetReward

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GetReward(CharacterBase, List<ItemBase>)

Initializes a new instance of the GetReward class with item rewards only.

```
public GetReward(CharacterBase character, List<ItemBase> rewards)
```

#### Parameters

`character` CharacterBase

The character receiving the rewards.

`rewards` List<ItemBase>

List of item rewards.

## GetReward(CharacterBase, List<ItemBase>, Dictionary<string, int>)

Initializes a new instance of the GetReward class with both item and fungible asset rewards.

```
public GetReward(CharacterBase character, List<ItemBase> rewards, Dictionary<string,
int> fungibleAssetRewards)
```

### Parameters

`character` CharacterBase

The character receiving the rewards.

`rewards` List<ItemBase>

List of item rewards.

`fungibleAssetRewards` Dictionary<string, int>

Dictionary of fungible asset rewards.

# Fields

## FungibleAssetRewards

Dictionary of fungible asset rewards (e.g., NCG, Crystal) received by the character.

```
public readonly Dictionary<string, int> FungibleAssetRewards
```

### Field Value

Dictionary<string, int>

## Rewards

List of item rewards received by the character.

```
public readonly List<ItemBase> Rewards
```

### Field Value

List<ItemBase>

## Methods

### CoExecute(IStage)

Executes the reward event by giving rewards to the character.

```
public override IEnumerator CoExecute(IStage stage)
```

### Parameters

`stage` IStage

The stage context for reward processing.

### Returns

IEnumerator

Coroutine enumerator for reward execution.

# Class HealSkill

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class HealSkill : Skill
```

**Inheritance**

object ← EventBase ← Skill ← HealSkill

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### HealSkill(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public HealSkill(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class NormalAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class NormalAttack : Skill
```

**Inheritance**

object ← EventBase ← Skill ← NormalAttack

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### NormalAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public NormalAttack(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class RemoveBuffs

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class RemoveBuffs : EventBase
```

**Inheritance**

object ← EventBase ← RemoveBuffs

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RemoveBuffs(CharacterBase)

```
public RemoveBuffs(CharacterBase character)
```

#### Parameters

`character` CharacterBase

## Methods

### CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

#### Parameters

`stage` IStage

## Returns

IEnumerator

# Class ShatterStrike

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class ShatterStrike : Skill
```

**Inheritance**

object ← EventBase ← Skill ← ShatterStrike

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ShatterStrike(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public ShatterStrike(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class Skill

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Skill : EventBase
```

### Inheritance

object ← EventBase ← Skill

### Derived

StageBuff, AreaAttack, BlowAttack, Buff, BuffRemovalAttack, DoubleAttack, DoubleAttackWithCombo, FullBuffRemovalAttack, HealSkill, NormalAttack, ShatterStrike, Tick, TickDamage

### Inherited Members

EventBase.Character , EventBase.CoExecute(IStage) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

## Skill(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
protected Skill(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

# Fields

## BuffInfos

```
public readonly IEnumerable<Skill.SkillInfo>? BuffInfos
```

### Field Value

IEnumerable<Skill.SkillInfo>

## SkillId

```
public readonly int SkillId
```

### Field Value

int

## SkillInfos

```
public readonly IEnumerable<Skill.SkillInfo> SkillInfos
```

### Field Value

IEnumerable<Skill.SkillInfo>

# Class Skill.SkillInfo

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class Skill.SkillInfo
```

**Inheritance**

object ← Skill.SkillInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SkillInfo(Guid, bool, long, long, bool, SkillCategory, int, ElementalType, SkillTargetType, Buff?, CharacterBase?, bool, IEnumerable<Buff>?, IceShield?)

```
public SkillInfo(Guid characterId, bool isDead, long thorn, long effect, bool
critical, SkillCategory skillCategory, int waveTurn, ElementalType elementalType =
ElementalType.Normal, SkillTargetType targetType = SkillTargetType.Enemy, Buff? buff
= null, CharacterBase? target = null, bool affected = true, IEnumerable<Buff>?
dispelList = null, IceShield? iceShield = null)
```

#### Parameters

`characterId` Guid

`isDead` bool

`thorn` long

`effect` long

`critical` bool

`skillCategory` SkillCategory

`waveTurn` int

`elementalType` ElementalType

`targetType` SkillTargetType

`buff` Buff

`target` CharacterBase

`affected` bool

`dispelList` IEnumerable<Buff>

`iceShield` IceShield

# Fields

## Affected

```
public readonly bool Affected
```

### Field Value

bool

## Buff

```
public readonly Buff? Buff
```

### Field Value

Buff

## CharacterId

```
public readonly Guid CharacterId
```

### Field Value

Guid

## Critical

```
public readonly bool Critical
```

### Field Value

bool

## DispelList

```
public readonly IEnumerable<Buff>? DispelList
```

### Field Value

IEnumerable<Buff>

## Effect

```
public readonly long Effect
```

### Field Value

long

## ElementalType

```
public readonly ElementalType ElementalType
```

### Field Value

ElementalType

## IceShield

```
public readonly IceShield? IceShield
```

### Field Value

IceShield

## IsDead

```
public readonly bool IsDead
```

### Field Value

bool

## SkillCategory

```
public readonly SkillCategory SkillCategory
```

### Field Value

SkillCategory

## SkillTargetType

```
public readonly SkillTargetType SkillTargetType
```

### Field Value

SkillTargetType

## Target

```
public readonly CharacterBase? Target
```

### Field Value

CharacterBase

## Thorn

```
public readonly long Thorn
```

### Field Value

long

## WaveTurn

```
public readonly int WaveTurn
```

### Field Value

int

# Class SpawnEnemyPlayer

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class SpawnEnemyPlayer : EventBase
```

**Inheritance**

object ← EventBase ← SpawnEnemyPlayer

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SpawnEnemyPlayer(CharacterBase)

```
public SpawnEnemyPlayer(CharacterBase character)
```

#### Parameters

`character` CharacterBase

## Methods

### CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

#### Parameters

`stage` IStage

## Returns

IEnumerator

# Class SpawnPlayer

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class SpawnPlayer : EventBase
```

**Inheritance**

object ← EventBase ← SpawnPlayer

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SpawnPlayer(CharacterBase)

```
public SpawnPlayer(CharacterBase character)
```

#### Parameters

`character` CharacterBase

## Methods

### CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

#### Parameters

`stage` IStage

## Returns

IEnumerator

# Class SpawnWave

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class SpawnWave : EventBase
```

**Inheritance**

object ← EventBase ← SpawnWave

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SpawnWave(CharacterBase, int, int, List<Enemy>, bool)

```
public SpawnWave(CharacterBase character, int waveNumber, int waveTurn, List<Enemy>
enemies, bool hasBoss)
```

#### Parameters

`character` CharacterBase

`waveNumber` int

`waveTurn` int

`enemies` List<Enemy>

`hasBoss` bool

## Fields

## Enemies

```
public readonly List<Enemy> Enemies
```

### Field Value

List<Enemy>

## HasBoss

```
public readonly bool HasBoss
```

### Field Value

bool

## WaveNumber

```
public readonly int WaveNumber
```

### Field Value

int

## WaveTurn

```
public readonly int WaveTurn
```

### Field Value

int

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class Tick

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class Tick : Skill
```

**Inheritance**

object ← EventBase ← Skill ← Tick

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Tick(CharacterBase)

```
public Tick(CharacterBase character)
```

#### Parameters

`character` CharacterBase

### Tick(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public Tick(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

# Methods

## CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

### Parameters

`stage` IStage

### Returns

IEnumerator

# Class TickDamage

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class TickDamage : Skill
```

**Inheritance**

object ← EventBase ← Skill ← TickDamage

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### TickDamage(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public TickDamage(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Class WaveTurnEnd

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class WaveTurnEnd : EventBase
```

**Inheritance**

object ← EventBase ← WaveTurnEnd

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### WaveTurnEnd(CharacterBase, int, int)

```
public WaveTurnEnd(CharacterBase character, int turnNumber, int waveTurn)
```

#### Parameters

`character` CharacterBase

`turnNumber` int

`waveTurn` int

## Fields

### TurnNumber

```
public readonly int TurnNumber
```

### Field Value

int

## WaveTurn

```
public readonly int WaveTurn
```

### Field Value

int

# Methods

## CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

### Parameters

`stage` IStage

### Returns

IEnumerator

# Namespace Nekoyume.Model.Battle Status.AdventureBoss

## Classes

Breakthrough

StageBuff

# Class Breakthrough

Namespace: Nekoyume.Model.BattleStatus.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class Breakthrough : EventBase
```

**Inheritance**

object ← EventBase ← Breakthrough

**Inherited Members**

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Breakthrough(CharacterBase, int, List<MonsterData>)

```
public Breakthrough(CharacterBase character, int floorId,
List<AdventureBossFloorWaveSheet.MonsterData> monsters)
```

#### Parameters

`character` CharacterBase

`floorId` int

`monsters` List<AdventureBossFloorWaveSheet.MonsterData>

## Fields

### FloorId

```
public readonly int FloorId
```

### Field Value

int

## Monsters

```
public readonly List<AdventureBossFloorWaveSheet.MonsterData> Monsters
```

### Field Value

List<AdventureBossFloorWaveSheet.MonsterData>

# Methods

## CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

### Parameters

`stage` IStage

### Returns

IEnumerator

# Class StageBuff

Namespace: Nekoyume.Model.BattleStatus.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class StageBuff : Skill
```

**Inheritance**

object ← EventBase ← Skill ← StageBuff

**Inherited Members**

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StageBuff(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public StageBuff(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

## Methods

# CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

## Parameters

`stage` IStage

## Returns

IEnumerator

# Namespace Nekoyume.Model.Battle Status.Arena

## Classes

ArenaAreaAttack

ArenaBlowAttack

ArenaBuff

ArenaBuffRemovalAttack

ArenaDead

ArenaDoubleAttack

ArenaDoubleAttackWithCombo

ArenaEventBase

ArenaFullBuffRemovalAttack

Arena battle status event for the full buff removal attack skill. Triggers the CoFullBuffRemovalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>) coroutine for arena rendering.

ArenaHeal

ArenaLog

ArenaNormalAttack

ArenaRemoveBuffs

ArenaShatterStrike

ArenaSkill

ArenaSkill.ArenaSkillInfo

ArenaSpawnCharacter

ArenaTick

ArenaTickDamage

ArenaTurnEnd

## Enums

ArenaLog.ArenaResult

# Class ArenaAreaAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaAreaAttack : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaAreaAttack

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaAreaAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaAreaAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaBlowAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBlowAttack : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaBlowAttack

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaBlowAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaBlowAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaBuff

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBuff : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaBuff

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaBuff(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>)

```
public ArenaBuff(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

## Methods

### CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

## Parameters

`arena` IArena

## Returns

IEnumerator

# Class ArenaBuffRemovalAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBuffRemovalAttack : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaBuffRemovalAttack

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaBuffRemovalAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaBuffRemovalAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaDead

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaDead : ArenaEventBase
```

**Inheritance**

object ← ArenaEventBase ← ArenaDead

**Inherited Members**

ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaDead(ArenaCharacter)

```
public ArenaDead(ArenaCharacter character)
```

#### Parameters

`character` ArenaCharacter

## Methods

### CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

#### Parameters

`arena` IArena

## Returns

IEnumerator

# Class ArenaDoubleAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaDoubleAttack : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaDoubleAttack

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaDoubleAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaDoubleAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaDoubleAttackWithCombo

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaDoubleAttackWithCombo : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaDoubleAttackWithCombo

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaDoubleAttackWithCombo(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaDoubleAttackWithCombo(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaEventBase

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ArenaEventBase
```

**Inheritance**

object ← ArenaEventBase

**Derived**

ArenaDead, ArenaRemoveBuffs, ArenaSkill, ArenaSpawnCharacter, ArenaTurnEnd

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaEventBase(ArenaCharacter)

```
protected ArenaEventBase(ArenaCharacter character)
```

#### Parameters

`character` ArenaCharacter

## Fields

### Character

```
public readonly ArenaCharacter Character
```

### Field Value

ArenaCharacter

## Methods

### CoExecute(IArena)

```
public abstract IEnumerator CoExecute(IArena arena)
```

#### Parameters

`arena` IArena

#### Returns

IEnumerator

# Class ArenaFullBuffRemovalAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

Arena battle status event for the full buff removal attack skill. Triggers the CoFullBuffRemovalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>) coroutine for arena rendering.

```
[Serializable]
public class ArenaFullBuffRemovalAttack : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaFullBuffRemovalAttack

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaFullBuffRemovalAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaFullBuffRemovalAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaHeal

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaHeal : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaHeal

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaHeal(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaHeal(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaLog

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaLog : IEnumerable<ArenaEventBase>, IEnumerable
```

### Inheritance

object ← ArenaLog

### Implements

IEnumerable<ArenaEventBase>, IEnumerable

### Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Fields

## Events

```
public List<ArenaEventBase> Events
```

### Field Value

List<ArenaEventBase>

## Id

```
public Guid Id
```

### Field Value

Guid

## Result

```
public ArenaLog.ArenaResult Result
```

### Field Value

ArenaLog.ArenaResult

## Score

```
public int Score
```

### Field Value

int

# Methods

## Add(ArenaEventBase)

```
public void Add(ArenaEventBase e)
```

### Parameters

e ArenaEventBase

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<ArenaEventBase> GetEnumerator()
```

## Returns

IEnumerator<ArenaEventBase>

An enumerator that can be used to iterate through the collection.

# Enum ArenaLog.ArenaResult

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
public enum ArenaLog.ArenaResult
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Lose = 1`

`Win = 0`

# Class ArenaNormalAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaNormalAttack : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaNormalAttack

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaNormalAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaNormalAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaRemoveBuffs

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaRemoveBuffs : ArenaEventBase
```

**Inheritance**

object ← ArenaEventBase ← ArenaRemoveBuffs

**Inherited Members**

ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaRemoveBuffs(ArenaCharacter)

```
public ArenaRemoveBuffs(ArenaCharacter character)
```

#### Parameters

`character` ArenaCharacter

## Methods

### CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

#### Parameters

`arena` IArena

## Returns

IEnumerator

# Class ArenaShatterStrike

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaShatterStrike : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaShatterStrike

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaShatterStrike(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaShatterStrike(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaSkill

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ArenaSkill : ArenaEventBase
```

### Inheritance

object ← ArenaEventBase ← ArenaSkill

### Derived

ArenaAreaAttack, ArenaBlowAttack, ArenaBuff, ArenaBuffRemovalAttack, ArenaDoubleAttack, ArenaDoubleAttackWithCombo, ArenaFullBuffRemovalAttack, ArenaHeal, ArenaNormalAttack, ArenaShatterStrike, ArenaTick, ArenaTickDamage

### Inherited Members

ArenaEventBase.Character , ArenaEventBase.CoExecute(IArena) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

## ArenaSkill(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
protected ArenaSkill(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Fields

## BuffInfos

```
public readonly IEnumerable<ArenaSkill.ArenaSkillInfo>? BuffInfos
```

### Field Value

IEnumerable<ArenaSkill.ArenaSkillInfo>

## SkillId

```
public readonly int SkillId
```

### Field Value

int

## SkillInfos

```
public readonly IEnumerable<ArenaSkill.ArenaSkillInfo> SkillInfos
```

### Field Value

IEnumerable<ArenaSkill.ArenaSkillInfo>

# Class ArenaSkill.ArenaSkillInfo

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSkill.ArenaSkillInfo
```

**Inheritance**

object ← ArenaSkill.ArenaSkillInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaSkillInfo(ArenaCharacter, long, bool, SkillCategory, int, ElementalType, SkillTargetType, Buff?, bool, IEnumerable<Buff>?, IceShield?)

```
public ArenaSkillInfo(ArenaCharacter character, long effect, bool critical,
SkillCategory skillCategory, int turn, ElementalType elementalType =
ElementalType.Normal, SkillTargetType targetType = SkillTargetType.Enemy, Buff? buff
= null, bool affected = true, IEnumerable<Buff>? dispelList = null, IceShield?
iceShield = null)
```

#### Parameters

`character` ArenaCharacter

`effect` long

`critical` bool

`skillCategory` SkillCategory

`turn` int

`elementalType` ElementalType

`targetType` SkillTargetType

`buff` Buff

`affected` bool

`dispelList` IEnumerable<Buff>

`iceShield` IceShield

# Fields

## Affected

```
public readonly bool Affected
```

### Field Value

bool

## Buff

```
public readonly Buff? Buff
```

### Field Value

Buff

## Critical

```
public readonly bool Critical
```

### Field Value

bool

## DispelList

```
public readonly IEnumerable<Buff>? DispelList
```

### Field Value

IEnumerable<Buff>

## Effect

```
public readonly long Effect
```

### Field Value

long

## ElementalType

```
public readonly ElementalType ElementalType
```

### Field Value

ElementalType

## IceShield

```
public readonly IceShield? IceShield
```

### Field Value

IceShield

## SkillCategory

```
public readonly SkillCategory SkillCategory
```

### Field Value

SkillCategory

## SkillTargetType

```
public readonly SkillTargetType SkillTargetType
```

### Field Value

SkillTargetType

## Target

```
public readonly ArenaCharacter Target
```

### Field Value

ArenaCharacter

## Turn

```
public readonly int Turn
```

### Field Value

int

# Class ArenaSpawnCharacter

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
public class ArenaSpawnCharacter : ArenaEventBase
```

**Inheritance**

object ← ArenaEventBase ← ArenaSpawnCharacter

**Inherited Members**

ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaSpawnCharacter(ArenaCharacter)

```
public ArenaSpawnCharacter(ArenaCharacter character)
```

#### Parameters

`character` ArenaCharacter

## Methods

### CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

#### Parameters

`arena` IArena

## Returns

IEnumerator

# Class ArenaTick

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaTick : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaTick

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaTick(ArenaCharacter)

```
public ArenaTick(ArenaCharacter character)
```

#### Parameters

`character` ArenaCharacter

### ArenaTick(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaTick(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

## Methods

### CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaTickDamage

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaTickDamage : ArenaSkill
```

**Inheritance**

object ← ArenaEventBase ← ArenaSkill ← ArenaTickDamage

**Inherited Members**

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaTickDamage(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaTickDamage(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

#### Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Class ArenaTurnEnd

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaTurnEnd : ArenaEventBase
```

**Inheritance**

object ← ArenaEventBase ← ArenaTurnEnd

**Inherited Members**

ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaTurnEnd(ArenaCharacter, int)

```
public ArenaTurnEnd(ArenaCharacter character, int turnNumber)
```

#### Parameters

`character` ArenaCharacter

`turnNumber` int

## Fields

### TurnNumber

```
public readonly int TurnNumber
```

#### Field Value

int

# Methods

## CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

### Parameters

`arena` IArena

### Returns

IEnumerator

# Namespace Nekoyume.Model.Buff

## Classes

ActionBuff

Bleed

Buff

BuffFactory

Dispel

Focus

IceShield

StatBuff

Stun

Vampiric

## Structs

BuffInfo

# Class ActionBuff

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ActionBuff : Buff, ICloneable
```

**Inheritance**

object ← Buff ← ActionBuff

**Implements**

ICloneable

**Derived**

Bleed, Dispel, Focus, IceShield, Stun, Vampiric

**Inherited Members**

Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , Buff.Clone() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ActionBuff(ActionBuff)

```
protected ActionBuff(ActionBuff value)
```

### Parameters

`value` ActionBuff

## ActionBuff(SkillCustomField, Row)

```
public ActionBuff(SkillCustomField customField, ActionBuffSheet.Row row)
```

### Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

## ActionBuff(Row)

```
public ActionBuff(ActionBuffSheet.Row row)
```

### Parameters

`row` ActionBuffSheet.Row

# Properties

## CustomField

```
public SkillCustomField? CustomField { get; }
```

### Property Value

SkillCustomField?

## RowData

```
public ActionBuffSheet.Row RowData { get; }
```

### Property Value

ActionBuffSheet.Row

# Methods

## IsBuff()

```
public override bool IsBuff()
```

### Returns

bool

## IsDebuff()

```
public override bool IsDebuff()
```

### Returns

bool

# Class Bleed

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Bleed : ActionBuff, ICloneable
```

**Inheritance**

object ← Buff ← ActionBuff ← Bleed

**Implements**

ICloneable

**Inherited Members**

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Bleed(Bleed)

```
protected Bleed(Bleed value)
```

#### Parameters

`value` Bleed

### Bleed(SkillCustomField, Row)

```
public Bleed(SkillCustomField customField, ActionBuffSheet.Row row)
```

### Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

## Bleed(Row, long)

```
public Bleed(ActionBuffSheet.Row row, long power)
```

### Parameters

`row` ActionBuffSheet.Row

`power` long

# Properties

## Power

```
public long Power { get; }
```

### Property Value

long

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## GiveEffect(CharacterBase, int, bool)

```
public Skill GiveEffect(CharacterBase affectedCharacter, int simulatorWaveTurn, bool
copyCharacter = true)
```

### Parameters

`affectedCharacter` CharacterBase

`simulatorWaveTurn` int

`copyCharacter` bool

### Returns

Skill

## GiveEffectForArena(ArenaCharacter, int)

```
public ArenaSkill GiveEffectForArena(ArenaCharacter affectedCharacter,
int simulatorWaveTurn)
```

### Parameters

`affectedCharacter` ArenaCharacter

`simulatorWaveTurn` int

### Returns

ArenaSkill

# Class Buff

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Buff : ICloneable
```

### Inheritance

object ← Buff

### Implements

ICloneable

### Derived

ActionBuff, StatBuff

### Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Buff(Buff)

```
protected Buff(Buff value)
```

#### Parameters

`value` Buff

### Buff(BuffInfo)

```
protected Buff(BuffInfo buffInfo)
```

### Parameters

`buffInfo` BuffInfo

## Properties

### BuffInfo

```
public BuffInfo BuffInfo { get; }
```

#### Property Value

BuffInfo

### OriginalDuration

```
public int OriginalDuration { get; }
```

#### Property Value

int

### RemainedDuration

```
public int RemainedDuration { get; set; }
```

#### Property Value

int

## Methods

### Clone()

Creates a new object that is a copy of the current instance.

```
public abstract object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## GetTarget(CharacterBase)

```
public virtual IEnumerable<CharacterBase> GetTarget(CharacterBase caster)
```

### Parameters

`caster` CharacterBase

### Returns

IEnumerable<CharacterBase>

## IsBuff()

```
public abstract bool IsBuff()
```

### Returns

bool

## IsDebuff()

```
public abstract bool IsDebuff()
```

### Returns

bool

# Class BuffFactory

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
public static class BuffFactory
```

**Inheritance**

object ← BuffFactory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetActionBuff(Stats, Row)

```
public static ActionBuff GetActionBuff(Stats stat, ActionBuffSheet.Row row)
```

#### Parameters

`stat` Stats

`row` ActionBuffSheet.Row

#### Returns

ActionBuff

### GetBuffs(CharacterStats, ISkill, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, bool)

```
public static IList<Buff> GetBuffs(CharacterStats stats, ISkill skill,
SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet, SkillActionBuffSheet
skillActionBuffSheet, ActionBuffSheet actionBuffSheet, bool hasExtraValueBuff
= false)
```

### Parameters

`stats` CharacterStats

`skill` ISkill

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`hasExtraValueBuff` bool

### Returns

IList<Buff>

## GetCustomActionBuff(SkillCustomField, Row)

```
public static ActionBuff GetCustomActionBuff(SkillCustomField customField,
ActionBuffSheet.Row row)
```

### Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

### Returns

ActionBuff

## GetCustomStatBuff(Row, SkillCustomField)

```
public static StatBuff GetCustomStatBuff(StatBuffSheet.Row row,
SkillCustomField customField)
```

### Parameters

`row` StatBuffSheet.Row

`customField` SkillCustomField

### Returns

StatBuff

## GetStatBuff(Row)

```
public static StatBuff GetStatBuff(StatBuffSheet.Row row)
```

### Parameters

`row` StatBuffSheet.Row

### Returns

StatBuff

# Struct BuffInfo

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public struct BuffInfo
```

**Inherited Members**

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### BuffInfo(int, int, int, int, SkillTargetType)

```
public BuffInfo(int id, int groupId, int chance, int duration,
SkillTargetType skillTargetType)
```

#### Parameters

`id` int

`groupId` int

`chance` int

`duration` int

`skillTargetType` SkillTargetType

## Fields

### Chance

```
public int Chance
```

### Field Value

int

## Duration

```
public int Duration
```

### Field Value

int

## GroupId

```
public int GroupId
```

### Field Value

int

## Id

```
public int Id
```

### Field Value

int

## SkillTargetType

```
public SkillTargetType SkillTargetType
```

## Field Value

SkillTargetType

# Class Dispel

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Dispel : ActionBuff, ICloneable
```

**Inheritance**

object ← Buff ← ActionBuff ← Dispel

**Implements**

ICloneable

**Inherited Members**

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Dispel(ActionBuff)

```
protected Dispel(ActionBuff value)
```

#### Parameters

`value` ActionBuff

### Dispel(SkillCustomField, Row)

```
public Dispel(SkillCustomField customField, ActionBuffSheet.Row row)
```

### Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

## Dispel(Row)

```
public Dispel(ActionBuffSheet.Row row)
```

### Parameters

`row` ActionBuffSheet.Row

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

# Class Focus

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Focus : ActionBuff, ICloneable
```

**Inheritance**

object ← Buff ← ActionBuff ← Focus

**Implements**

ICloneable

**Inherited Members**

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Focus(Focus)

```
protected Focus(Focus value)
```

#### Parameters

`value` Focus

### Focus(SkillCustomField, Row)

```
public Focus(SkillCustomField customField, ActionBuffSheet.Row row)
```

### Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

## Focus(Row)

```
public Focus(ActionBuffSheet.Row row)
```

### Parameters

`row` ActionBuffSheet.Row

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

# Class IceShield

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class IceShield : ActionBuff, ICloneable
```

**Inheritance**

object ← Buff ← ActionBuff ← IceShield

**Implements**

ICloneable

**Inherited Members**

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### IceShield(IceShield)

```
protected IceShield(IceShield value)
```

#### Parameters

`value` IceShield

### IceShield(SkillCustomField, Row)

```
public IceShield(SkillCustomField customField, ActionBuffSheet.Row row)
```

### Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

## IceShield(Row)

```
public IceShield(ActionBuffSheet.Row row)
```

### Parameters

`row` ActionBuffSheet.Row

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## FrostBite(StatBuffSheet, BuffLinkSheet)

```
public StatBuff FrostBite(StatBuffSheet statBuffSheet, BuffLinkSheet buffLinkSheet)
```

### Parameters

`statBuffSheet` StatBuffSheet

`buffLinkSheet` BuffLinkSheet

## Returns

StatBuff

# Class StatBuff

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class StatBuff : Buff, ICloneable
```

**Inheritance**

object ← Buff ← StatBuff

**Implements**

ICloneable

**Inherited Members**

Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StatBuff(StatBuff)

```
protected StatBuff(StatBuff value)
```

#### Parameters

`value` StatBuff

### StatBuff(SkillCustomField, Row)

```
public StatBuff(SkillCustomField customField, StatBuffSheet.Row row)
```

#### Parameters

`customField` SkillCustomField

`row` StatBuffSheet.Row

## StatBuff(Row)

```
public StatBuff(StatBuffSheet.Row row)
```

### Parameters

`row` StatBuffSheet.Row

# Properties

## CustomField

```
public SkillCustomField? CustomField { get; }
```

### Property Value

SkillCustomField?

## RowData

```
public StatBuffSheet.Row RowData { get; }
```

### Property Value

StatBuffSheet.Row

## Stack

```
public int Stack { get; }
```

### Property Value

int

## Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## GetModifier()

```
public StatModifier GetModifier()
```

### Returns

StatModifier

## IsBuff()

```
public override bool IsBuff()
```

### Returns

bool

## IsDebuff()

```
public override bool IsDebuff()
```

### Returns

bool

## SetStack(int)

```
public void SetStack(int stack)
```

### Parameters

`stack` int

# Class Stun

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Stun : ActionBuff, ICloneable
```

**Inheritance**

object ← Buff ← ActionBuff ← Stun

**Implements**

ICloneable

**Inherited Members**

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Stun(ActionBuff)

```
protected Stun(ActionBuff value)
```

#### Parameters

`value` ActionBuff

### Stun(SkillCustomField, Row)

```
public Stun(SkillCustomField customField, ActionBuffSheet.Row row)
```

### Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

## Stun(Row)

```
public Stun(ActionBuffSheet.Row row)
```

### Parameters

`row` ActionBuffSheet.Row

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

# Class Vampiric

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Vampiric : ActionBuff, ICloneable
```

**Inheritance**

object ← Buff ← ActionBuff ← Vampiric

**Implements**

ICloneable

**Inherited Members**

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Vampiric(Vampiric)

```
protected Vampiric(Vampiric value)
```

#### Parameters

`value` Vampiric

### Vampiric(SkillCustomField, Row)

```
public Vampiric(SkillCustomField customField, ActionBuffSheet.Row row)
```

### Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

## Vampiric(Row, long)

```
public Vampiric(ActionBuffSheet.Row row, long basisPoint)
```

### Parameters

`row` ActionBuffSheet.Row

`basisPoint` long

# Properties

## BasisPoint

```
public long BasisPoint { get; }
```

### Property Value

long

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## GiveEffect(CharacterBase, SkillInfo, int, bool)

```
public Skill GiveEffect(CharacterBase affectedCharacter, Skill.SkillInfo skillInfo,
int simulatorWaveTurn, bool copyCharacter = true)
```

### Parameters

`affectedCharacter` CharacterBase

`skillInfo` Skill.SkillInfo

`simulatorWaveTurn` int

`copyCharacter` bool

### Returns

Skill

## GiveEffectForArena(ArenaCharacter, ArenaSkillInfo, int)

```
public ArenaSkill GiveEffectForArena(ArenaCharacter affectedCharacter,
ArenaSkill.ArenaSkillInfo skillInfo, int simulatorWaveTurn)
```

### Parameters

`affectedCharacter` ArenaCharacter

`skillInfo` ArenaSkill.ArenaSkillInfo

`simulatorWaveTurn` int

### Returns

ArenaSkill

# Namespace Nekoyume.Model.Character

## Enums

SizeType

# Enum SizeType

Namespace: Nekoyume.Model.Character

Assembly: Lib9c.dll

```
public enum SizeType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`L = 3`

`M = 2`

`S = 1`

`XL = 4`

`XS = 0`

# Namespace Nekoyume.Model.Collection

## Classes

CollectionFactory

FungibleCollectionMaterial

NonFungibleCollectionMaterial

## Interfaces

ICollectionMaterial

## Enums

MaterialType

# Class CollectionFactory

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public static class CollectionFactory
```

**Inheritance**

object ← CollectionFactory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### DeserializeMaterial(List)

```
public static ICollectionMaterial DeserializeMaterial(List serialized)
```

#### Parameters

`serialized` List

#### Returns

ICollectionMaterial

# Class FungibleCollectionMaterial

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public class FungibleCollectionMaterial : ICollectionMaterial, IBencodable
```

**Inheritance**

object ← FungibleCollectionMaterial

**Implements**

ICollectionMaterial, IBencodable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### FungibleCollectionMaterial()

```
public FungibleCollectionMaterial()
```

### FungibleCollectionMaterial(IValue)

```
public FungibleCollectionMaterial(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### FungibleCollectionMaterial(List)

```
public FungibleCollectionMaterial(List serialized)
```

### Parameters

`serialized` List

## Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

## ItemCount

```
public int ItemCount { get; set; }
```

### Property Value

int

## ItemId

```
public int ItemId { get; set; }
```

### Property Value

int

## Type

```
public MaterialType Type { get; }
```

### Property Value

MaterialType

# Methods

## BurnMaterial(Row, Inventory, long)

Burns the material from the inventory based on the given item row, inventory, and block index.

```
public void BurnMaterial(ItemSheet.Row itemRow, Inventory inventory,
long blockIndex)
```

### Parameters

`itemRow` ItemSheet.Row

The item row from the item sheet.

`inventory` Inventory

The inventory object.

`blockIndex` long

The block index.

# Interface ICollectionMaterial

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public interface ICollectionMaterial : IBencodable
```

**Inherited Members**
IBencodable.Bencoded

## Properties

### ItemCount

```
int ItemCount { get; set; }
```

#### Property Value

int

### ItemId

```
int ItemId { get; set; }
```

#### Property Value

int

### Type

```
MaterialType Type { get; }
```

#### Property Value

MaterialType

# Enum MaterialType

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public enum MaterialType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Fungible = 0`

`NonFungible = 1`

# Class NonFungibleCollectionMaterial

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public class NonFungibleCollectionMaterial : ICollectionMaterial, IBencodable
```

**Inheritance**

object ← NonFungibleCollectionMaterial

**Implements**

ICollectionMaterial, IBencodable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### NonFungibleCollectionMaterial()

```
public NonFungibleCollectionMaterial()
```

### NonFungibleCollectionMaterial(List)

```
public NonFungibleCollectionMaterial(List serialized)
```

#### Parameters

`serialized` List

## Properties

# Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

## Property Value

IValue

## Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

# ItemCount

```
public int ItemCount { get; set; }
```

## Property Value

int

# ItemId

```
public int ItemId { get; set; }
```

## Property Value

int

## Level

```
public int Level { get; set; }
```

### Property Value

int

## NonFungibleId

```
public Guid NonFungibleId { get; set; }
```

### Property Value

Guid

## SkillContains

```
public bool SkillContains { get; set; }
```

### Property Value

bool

## Type

```
public MaterialType Type { get; }
```

### Property Value

MaterialType

## Methods

# BurnMaterial(Row, Inventory, RequiredMaterial, long)

Burns the specified material from the inventory based on the item type.

```
public void BurnMaterial(ItemSheet.Row itemRow, Inventory inventory,
CollectionSheet.RequiredMaterial materialInfo, long blockIndex)
```

## Parameters

`itemRow` ItemSheet.Row

The ItemSheet.Row object representing the item.

`inventory` Inventory

The Inventory object representing the player's inventory.

`materialInfo` CollectionSheet.RequiredMaterial

The CollectionSheet.RequiredMaterial object representing the material info.

`blockIndex` long

The block index where the burn operation is taking place.

## Exceptions

ItemDoesNotExistException

Thrown when the material item does not exist in the inventory.

InvalidItemTypeException

Thrown when the item type is not supported by NonFungibleCollectionMaterial.

# Namespace Nekoyume.Model.Coupons

## Structs

Coupon

RewardSet

RewardSet.Comparer

# Struct Coupon

Namespace: Nekoyume.Model.Coupons

Assembly: Lib9c.dll

```
public readonly struct Coupon : IOrderedEnumerable<(int ItemId, uint Quantity)>,
IEnumerable<(int ItemId, uint Quantity)>, IEnumerable, IEquatable<Coupon>
```

**Implements**
IOrderedEnumerable<(int ItemId, uint Quantity)>, IEnumerable<(int ItemId, uint Quantity)>, IEnumerable, IEquatable<Coupon>

**Inherited Members**
ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### Coupon(IValue)

```
public Coupon(IValue serialized)
```

#### Parameters

`serialized` IValue

### Coupon(Guid, in RewardSet)

```
public Coupon(Guid id, in RewardSet rewards)
```

#### Parameters

`id` Guid

`rewards` RewardSet

## Coupon(Guid, IEnumerable<(int ItemId, uint Quantity)>)

```
public Coupon(Guid id, IEnumerable<(int ItemId, uint Quantity)> rewards)
```

### Parameters

`id` Guid

`rewards` IEnumerable<(int ItemId, uint Quantity)>

## Coupon(Guid, params (int ItemId, uint Quantity)[])

```
public Coupon(Guid id, params (int ItemId, uint Quantity)[] rewards)
```

### Parameters

`id` Guid

`rewards` (int ItemId, uint Quantity)[]

# Fields

## Id

```
public readonly Guid Id
```

### Field Value

Guid

## Rewards

```
public readonly RewardSet Rewards
```

### Field Value

RewardSet

## Methods

## Equals(Coupon)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(Coupon other)
```

### Parameters

`other` Coupon

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current instance.

### Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

## GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

### Returns

int

A 32-bit signed integer that is the hash code for this instance.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Struct RewardSet

Namespace: Nekoyume.Model.Coupons

Assembly: Lib9c.dll

```
public readonly struct RewardSet : IImmutableDictionary<int, uint>,
IReadOnlyDictionary<int, uint>, IReadOnlyCollection<KeyValuePair<int, uint>>,
IEnumerable<KeyValuePair<int, uint>>, IEnumerable, IEquatable<RewardSet>
```

**Implements**

IImmutableDictionary<int, uint>, IReadOnlyDictionary<int, uint>, IReadOnlyCollection<KeyValuePair<int, uint>>, IEnumerable<KeyValuePair<int, uint>>, IEnumerable, IEquatable<RewardSet>

**Inherited Members**

ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### RewardSet(Dictionary)

```
public RewardSet(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### RewardSet(IEnumerable<(int ItemId, uint Quantity)>)

```
public RewardSet(IEnumerable<(int ItemId, uint Quantity)> rewards)
```

#### Parameters

`rewards` IEnumerable<(int ItemId, uint Quantity)>

## RewardSet(in ImmutableDictionary<int, uint>)

```
public RewardSet(in ImmutableDictionary<int, uint> rewards)
```

### Parameters

`rewards` ImmutableDictionary<int, uint>

## RewardSet(params (int ItemId, uint Quantity)[])

```
public RewardSet(params (int ItemId, uint Quantity)[] rewards)
```

### Parameters

`rewards` (int ItemId, uint Quantity)[]

# Properties

## Count

Gets the number of elements in the collection.

```
public int Count { get; }
```

### Property Value

int

The number of elements in the collection.

## this[int]

Gets the element that has the specified key in the read-only dictionary.

```
public uint this[int key] { get; }
```

### Parameters

`key` int

The key to locate.

### Property Value

uint

The element that has the specified key in the read-only dictionary.

### Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

## Keys

Gets an enumerable collection that contains the keys in the read-only dictionary.

```
public IEnumerable<int> Keys { get; }
```

### Property Value

IEnumerable<int>

An enumerable collection that contains the keys in the read-only dictionary.

## Values

Gets an enumerable collection that contains the values in the read-only dictionary.

```
public IEnumerable<uint> Values { get; }
```

### Property Value

IEnumerable<uint>

An enumerable collection that contains the values in the read-only dictionary.

## Methods

## Add(int, uint)

Adds an element with the specified key and value to the dictionary.

```
public IImmutableDictionary<int, uint> Add(int key, uint value)
```

### Parameters

`key` int

The key of the element to add.

`value` uint

The value of the element to add.

### Returns

IImmutableDictionary<int, uint>

A new immutable dictionary that contains the additional key/value pair.

### Exceptions

ArgumentException

The given key already exists in the dictionary but has a different value.

## AddRange(IEnumerable<KeyValuePair<int, uint>>)

Adds the specified key/value pairs to the dictionary.

```
public IImmutableDictionary<int, uint> AddRange(IEnumerable<KeyValuePair<int,
uint>> pairs)
```

### Parameters

`pairs` IEnumerable<KeyValuePair<int, uint>>

The key/value pairs to add.

### Returns

IImmutableDictionary<int, uint>

A new immutable dictionary that contains the additional key/value pairs.

### Exceptions

ArgumentException

One of the given keys already exists in the dictionary but has a different value.

## Clear()

Retrieves an empty dictionary that has the same ordering and key/value comparison rules as this dictionary instance.

```
public IImmutableDictionary<int, uint> Clear()
```

### Returns

IImmutableDictionary<int, uint>

An empty dictionary with equivalent ordering and key/value comparison rules.

## Contains(KeyValuePair<int, uint>)

Determines whether the immutable dictionary contains the specified key/value pair.

```
public bool Contains(KeyValuePair<int, uint> pair)
```

### Parameters

`pair` KeyValuePair<int, uint>

The key/value pair to locate.

### Returns

bool

true if the specified key/value pair is found in the dictionary; otherwise, false.

## ContainsKey(int)

Determines whether the read-only dictionary contains an element that has the specified key.

```
public bool ContainsKey(int key)
```

### Parameters

`key` int

The key to locate.

### Returns

bool

true if the read-only dictionary contains an element that has the specified key; otherwise, false.

### Exceptions

ArgumentNullException

`key` is null.

## Equals(RewardSet)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(RewardSet other)
```

### Parameters

`other` RewardSet

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current instance.

### Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<KeyValuePair<int, uint>> GetEnumerator()
```

### Returns

IEnumerator<KeyValuePair<int, uint>>

An enumerator that can be used to iterate through the collection.

## GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

### Returns

int

A 32-bit signed integer that is the hash code for this instance.

## Remove(int)

Removes the element with the specified key from the immutable dictionary.

```
public IImmutableDictionary<int, uint> Remove(int key)
```

### Parameters

`key` int

The key of the element to remove.

### Returns

IImmutableDictionary<int, uint>

A new immutable dictionary with the specified element removed; or this instance if the specified key cannot be found in the dictionary.

## RemoveRange(IEnumerable<int>)

Removes the elements with the specified keys from the immutable dictionary.

```
public IImmutableDictionary<int, uint> RemoveRange(IEnumerable<int> keys)
```

### Parameters

`keys` IEnumerable<int>

The keys of the elements to remove.

### Returns

IImmutableDictionary<int, uint>

A new immutable dictionary with the specified keys removed; or this instance if the specified keys cannot be found in the dictionary.

## Serialize()

```
public Dictionary Serialize()
```

### Returns

Dictionary

## SetItem(int, uint)

Sets the specified key and value in the immutable dictionary, possibly overwriting an existing value for the key.

```
public IImmutableDictionary<int, uint> SetItem(int key, uint value)
```

### Parameters

`key` int

The key of the entry to add.

`value` uint

The key value to set.

### Returns

IImmutableDictionary<int, uint>

A new immutable dictionary that contains the specified key/value pair.

## SetItems(IEnumerable<KeyValuePair<int, uint>>)

Sets the specified key/value pairs in the immutable dictionary, possibly overwriting existing values for the keys.

```
public IImmutableDictionary<int, uint> SetItems(IEnumerable<KeyValuePair<int,
uint>> items)
```

### Parameters

`items` IEnumerable<KeyValuePair<int, uint>>

The key/value pairs to set in the dictionary. If any of the keys already exist in the dictionary, this method will overwrite their previous values.

### Returns

IImmutableDictionary<int, uint>

A new immutable dictionary that contains the specified key/value pairs.

## TryGetKey(int, out int)

Determines whether this dictionary contains a specified key.

```
public bool TryGetKey(int equalKey, out int actualKey)
```

### Parameters

`equalKey` int

The key to search for.

`actualKey` int

The matching key located in the dictionary if found, or `equalkey` if no match is found.

### Returns

bool

true if a match for `equalKey` is found; otherwise, false.

## TryGetValue(int, out uint)

Gets the value that is associated with the specified key.

```
public bool TryGetValue(int key, out uint value)
```

### Parameters

`key` int

The key to locate.

`value` uint

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

### Returns

bool

true if the object that implements the IReadOnlyDictionary<TKey, TValue> interface contains an element that has the specified key; otherwise, false.

## Exceptions

### ArgumentNullException

`key` is null.

# Struct RewardSet.Comparer

Namespace: Nekoyume.Model.Coupons

Assembly: Lib9c.dll

```
public struct RewardSet.Comparer : IComparer<RewardSet>
```

**Implements**
IComparer<RewardSet>

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

# Methods

## Compare(RewardSet, RewardSet)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public int Compare(RewardSet x, RewardSet y)
```

### Parameters

x RewardSet

The first object to compare.

y RewardSet

The second object to compare.

### Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

| Value | Meaning |
| --- | --- |
| **Less than zero** | x is less than y. |
| **Zero** | x equals y. |
| **Greater than zero** | x is greater than y. |

# Namespace Nekoyume.Model.Elemental

## Classes

ElementalTypeComparer

ElementalTypeExtension

## Enums

ElementalResult

ElementalType

# Enum ElementalResult

Namespace: Nekoyume.Model.Elemental

Assembly: Lib9c.dll

```
public enum ElementalResult
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Draw = 1`

`Lose = 2`

`Win = 0`

# Enum ElementalType

Namespace: Nekoyume.Model.Elemental

Assembly: Lib9c.dll

```
public enum ElementalType
```

**Extension Methods**
ElementalTypeExtension.GetBattleResult(ElementalType, ElementalType) ,
ElementalTypeExtension.GetDamage(ElementalType, ElementalType, long) ,
ElementalTypeExtension.GetMultiplier(ElementalType, ElementalType) ,
ElementalTypeExtension.TryGetLoseCase(ElementalType, out ElementalType) ,
ElementalTypeExtension.TryGetWinCase(ElementalType, out ElementalType) ,
StateExtensions.Serialize(Enum)

## Fields

`Fire = 1`

`Land = 3`

`Normal = 0`

`Water = 2`

`Wind = 4`

# Class ElementalTypeComparer

Namespace: Nekoyume.Model.Elemental

Assembly: Lib9c.dll

```
public class ElementalTypeComparer : IEqualityComparer<ElementalType>
```

**Inheritance**

object ← ElementalTypeComparer

**Implements**

IEqualityComparer<ElementalType>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Instance

```
public static readonly ElementalTypeComparer Instance
```

### Field Value

ElementalTypeComparer

# Methods

## Equals(ElementalType, ElementalType)

Determines whether the specified objects are equal.

```
public bool Equals(ElementalType x, ElementalType y)
```

### Parameters

`x` ElementalType

The first object of type `T` to compare.

`y` ElementalType

The second object of type `T` to compare.

### Returns

bool

[true] if the specified objects are equal; otherwise, [false].

## GetHashCode(ElementalType)

Returns a hash code for the specified object.

```
public int GetHashCode(ElementalType obj)
```

### Parameters

`obj` ElementalType

The [object] for which a hash code is to be returned.

### Returns

int

A hash code for the specified object.

### Exceptions

ArgumentNullException

The type of `obj` is a reference type and `obj` is [null].

# Class ElementalTypeExtension

Namespace: Nekoyume.Model.Elemental

Assembly: Lib9c.dll

```
public static class ElementalTypeExtension
```

**Inheritance**

object ← ElementalTypeExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## WinMultiplier

```
public const decimal WinMultiplier = 1.2
```

### Field Value

decimal

# Methods

## GetAllTypes()

```
public static List<ElementalType> GetAllTypes()
```

### Returns

List<ElementalType>

## GetBattleResult(ElementalType, ElementalType)

```
public static ElementalResult GetBattleResult(this ElementalType from,
ElementalType to)
```

### Parameters

`from` ElementalType

`to` ElementalType

### Returns

ElementalResult

1: Win 0: Draw -1: Lose

## GetDamage(ElementalType, ElementalType, long)

```
public static long GetDamage(this ElementalType from, ElementalType to, long damage)
```

### Parameters

`from` ElementalType

`to` ElementalType

`damage` long

### Returns

long

## GetMultiplier(ElementalType, ElementalType)

```
public static decimal GetMultiplier(this ElementalType from, ElementalType to)
```

### Parameters

`from` ElementalType

`to` ElementalType

### Returns

decimal

## TryGetLoseCase(ElementalType, out ElementalType)

```
public static bool TryGetLoseCase(this ElementalType lose, out ElementalType win)
```

### Parameters

`lose` ElementalType

`win` ElementalType

### Returns

bool

## TryGetWinCase(ElementalType, out ElementalType)

```
public static bool TryGetWinCase(this ElementalType win, out ElementalType lose)
```

### Parameters

`win` ElementalType

`lose` ElementalType

### Returns

bool

# Namespace Nekoyume.Model.EnumType

## Classes

BattleTypeExtensions

## Enums

ArenaType

BattleType

CraftType

Grade
Enumeration of item grades.

RuneSlotType

RuneType

RuneUsePlace

StatReferenceType

TradeType

# Enum ArenaType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum ArenaType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Championship = 2`

`OffSeason = 0`

`Season = 1`

# Enum BattleType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum BattleType
```

**Extension Methods**
BattleTypeExtensions.IsEquippableRune(BattleType, RuneUsePlace) ,
StateExtensions.Serialize(Enum)

## Fields

`Adventure = 1`

`Arena = 2`

`End = 5`

`InfiniteTower = 4`

`Raid = 3`

# Class BattleTypeExtensions

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public static class BattleTypeExtensions
```

**Inheritance**

object ← BattleTypeExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## IsEquippableRune(BattleType, RuneUsePlace)

```
public static bool IsEquippableRune(this BattleType battleType,
RuneUsePlace runePlace)
```

### Parameters

`battleType` BattleType

`runePlace` RuneUsePlace

### Returns

bool

# Enum CraftType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum CraftType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Mimisbrunnr = 2`

`Normal = 0`

`Premium = 1`

# Enum Grade

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

Enumeration of item grades.

```
public enum Grade
```

**Extension Methods**
StateExtensions.Serialize(Enum)

# Fields

`Divinity = 6`

Divinity grade.

`Epic = 3`

Epic grade.

`Legendary = 5`

Legendary grade.

`Mythic = 7`

Mythic grade.

`Normal = 1`

Normal grade.

`Rare = 2`

Rare grade.

`Transcendent = 8`

Transcendent grade.

`Ultimate = 9`

Ultimate grade.

```
Unique = 4
```

Unique grade.

# Enum RuneSlotType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum RuneSlotType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Crystal = 4`

`Default = 1`

`Ncg = 2`

`Stake = 3`

# Enum RuneType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum RuneType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Skill = 2`

`Stat = 1`

# Enum RuneUsePlace

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum RuneUsePlace
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Adventure = 1`

`AdventureAndArena = 3`

`All = 7`

`Arena = 2`

`Raid = 4`

`RaidAndAdventure = 5`

`RaidAndArena = 6`

# Enum StatReferenceType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum StatReferenceType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Caster = 0`

`Target = 1`

# Enum TradeType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum TradeType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Buy = 0`

`Sell = 1`

# Namespace Nekoyume.Model.Event

## Classes

EventDungeonInfo

# Class EventDungeonInfo

Namespace: Nekoyume.Model.Event

Assembly: Lib9c.dll

```
public class EventDungeonInfo : IState
```

**Inheritance**

object ← EventDungeonInfo

**Implements**

IState

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

EventDungeonExtensions.GetRemainingTicketsConsiderReset(EventDungeonInfo, EventScheduleSheet.Row, long)

## Constructors

### EventDungeonInfo(IValue)

```
public EventDungeonInfo(IValue serialized)
```

#### Parameters

`serialized` IValue

### EventDungeonInfo(List)

```
public EventDungeonInfo(List serialized)
```

#### Parameters

serialized List

## EventDungeonInfo(int, int, int, int)

```
public EventDungeonInfo(int resetTicketsInterval = 0, int remainingTickets = 0, int
numberOfTicketPurchases = 0, int clearedStageId = 0)
```

### Parameters

resetTicketsInterval int

remainingTickets int

numberOfTicketPurchases int

clearedStageId int

# Properties

## ClearedStageId

```
public int ClearedStageId { get; }
```

### Property Value

int

## NumberOfTicketPurchases

```
public int NumberOfTicketPurchases { get; }
```

### Property Value

int

## RemainingTickets

```
public int RemainingTickets { get; }
```

### Property Value

int

## ResetTicketsInterval

```
public int ResetTicketsInterval { get; }
```

### Property Value

int

# Methods

## ClearStage(int)

```
public void ClearStage(int stageId)
```

### Parameters

`stageId` int

## DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address address, int dungeonId)
```

### Parameters

`address` Address

`dungeonId` int

### Returns

Address

## Equals(EventDungeonInfo)

```
protected bool Equals(EventDungeonInfo other)
```

### Parameters

`other` EventDungeonInfo

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## HasTickets(int)

```
public bool HasTickets(int tickets)
```

### Parameters

`tickets` int

### Returns

bool

## IncreaseNumberOfTicketPurchases()

```
public void IncreaseNumberOfTicketPurchases()
```

## IsCleared(int)

```
public bool IsCleared(int stageId)
```

### Parameters

`stageId` int

### Returns

bool

## ResetTickets(int, int)

```
public void ResetTickets(int interval, int tickets)
```

### Parameters

`interval` int

`tickets` int

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## TryUseTickets(int)

```
public bool TryUseTickets(int tickets)
```

### Parameters

`tickets` int

### Returns

bool

# Namespace Nekoyume.Model.Faucet

## Classes

FaucetRuneInfo

# Class FaucetRuneInfo

Namespace: Nekoyume.Model.Faucet

Assembly: Lib9c.dll

```
[Serializable]
public class FaucetRuneInfo
```

**Inheritance**

object ← FaucetRuneInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### FaucetRuneInfo(List)

```
public FaucetRuneInfo(List serialized)
```

#### Parameters

`serialized` List

### FaucetRuneInfo(int, int)

```
public FaucetRuneInfo(int runeId, int amount)
```

#### Parameters

`runeId` int

`amount` int

# Properties

## Amount

```
public int Amount { get; }
```

### Property Value

int

## RuneId

```
public int RuneId { get; }
```

### Property Value

int

# Methods

## Equals(FaucetRuneInfo)

```
protected bool Equals(FaucetRuneInfo other)
```

### Parameters

`other` FaucetRuneInfo

### Returns

bool

## Serialize()

```
public IValue Serialize()
```

## Returns

IValue

# Namespace Nekoyume.Model.Garages

## Classes

FungibleItemGarage

## Interfaces

IFungibleItemGarage

IGarage<T1, T2>

# Class FungibleItemGarage

Namespace: Nekoyume.Model.Garages

Assembly: Lib9c.dll

```
public class FungibleItemGarage : IFungibleItemGarage, IGarage<IFungibleItemGarage,
int>
```

**Inheritance**

object ← FungibleItemGarage

**Implements**

IFungibleItemGarage, IGarage<IFungibleItemGarage, int>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### FungibleItemGarage(IValue?)

```
public FungibleItemGarage(IValue? serialized)
```

#### Parameters

`serialized` IValue

### FungibleItemGarage(IFungibleItem, int)

```
public FungibleItemGarage(IFungibleItem item, int count)
```

#### Parameters

`item` IFungibleItem

`count` int

# Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## Item

```
public IFungibleItem Item { get; }
```

### Property Value

IFungibleItem

# Methods

## Deliver(IFungibleItemGarage, int)

```
public void Deliver(IFungibleItemGarage to, int count)
```

### Parameters

`to` IFungibleItemGarage

`count` int

## Load(int)

```
public void Load(int count)
```

### Parameters

`count` int

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Unload(int)

```
public void Unload(int count)
```

### Parameters

`count` int

# Interface IFungibleItemGarage

Namespace: Nekoyume.Model.Garages

Assembly: Lib9c.dll

```
public interface IFungibleItemGarage : IGarage<IFungibleItemGarage, int>
```

**Inherited Members**
IGarage<IFungibleItemGarage, int>.Load(int) ,
IGarage<IFungibleItemGarage, int>.Deliver(IFungibleItemGarage, int) ,
IGarage<IFungibleItemGarage, int>.Unload(int)

## Properties

### Count

```
int Count { get; }
```

#### Property Value

int

### Item

```
IFungibleItem Item { get; }
```

#### Property Value

IFungibleItem

# Interface IGarage<T1, T2>

Namespace: Nekoyume.Model.Garages

Assembly: Lib9c.dll

```
public interface IGarage<in T1, in T2> where T1 : IGarage<in T1, in T2>
```

### Type Parameters

T1

T2

## Methods

## Deliver(T1, T2)

```
void Deliver(T1 to, T2 amount)
```

### Parameters

to T1

amount T2

## Load(T2)

```
void Load(T2 amount)
```

### Parameters

amount T2

## Unload(T2)

```
void Unload(T2 amount)
```

## Parameters

`amount` T2

# Namespace Nekoyume.Model.GrandFinale

## Classes

GrandFinaleInformation

# Class GrandFinaleInformation

Namespace: Nekoyume.Model.GrandFinale

Assembly: Lib9c.dll

```
public class GrandFinaleInformation : IState
```

**Inheritance**

object ← GrandFinaleInformation

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GrandFinaleInformation(List)

```
public GrandFinaleInformation(List serialized)
```

#### Parameters

`serialized` List

### GrandFinaleInformation(Address, int)

```
public GrandFinaleInformation(Address avatarAddress, int grandFinaleId)
```

#### Parameters

`avatarAddress` Address

`grandFinaleId` int

# Fields

## Address

```
public Address Address
```

### Field Value

Address

# Methods

## DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address avatarAddress, int grandFinaleId)
```

### Parameters

`avatarAddress` Address

`grandFinaleId` int

### Returns

Address

## GetBattleRecordList()

```
public List<KeyValuePair<Address, bool>> GetBattleRecordList()
```

### Returns

List<KeyValuePair<Address, bool>>

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## TryGetBattleRecord(Address, out bool)

```
public bool TryGetBattleRecord(Address enemyAddress, out bool win)
```

### Parameters

`enemyAddress` Address

`win` bool

### Returns

bool

## UpdateRecord(Address, bool)

```
public void UpdateRecord(Address enemyAddress, bool win)
```

### Parameters

`enemyAddress` Address

`win` bool

# Namespace Nekoyume.Model.Guild

## Classes

Guild

GuildDelegatee

GuildDelegator

GuildParticipant

GuildRepository

# Class Guild

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class Guild : IBencodable, IEquatable<Guild>
```

**Inheritance**

object ← Guild

**Implements**

IBencodable, IEquatable<Guild>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Guild(GuildAddress, IValue, GuildRepository)

```
public Guild(GuildAddress address, IValue bencoded, GuildRepository repository)
```

#### Parameters

`address` GuildAddress

`bencoded` IValue

`repository` GuildRepository

### Guild(GuildAddress, AgentAddress, Address, GuildRepository)

```
public Guild(GuildAddress address, AgentAddress guildMasterAddress, Address
validatorAddress, GuildRepository repository)
```

### Parameters

`address` GuildAddress

`guildMasterAddress` AgentAddress

`validatorAddress` Address

`repository` GuildRepository

## Fields

### GuildMasterAddress

```
public readonly AgentAddress GuildMasterAddress
```

#### Field Value

AgentAddress

### ValidatorAddress

```
public readonly Address ValidatorAddress
```

#### Field Value

Address

## Properties

### Address

```
public GuildAddress Address { get; }
```

#### Property Value

GuildAddress

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## Repository

```
public GuildRepository Repository { get; }
```

### Property Value

GuildRepository

## Methods

## ClaimReward(Address, long)

```
public void ClaimReward(Address validatorAddress, long height)
```

### Parameters

`validatorAddress` Address

`height` long

## Equals(Guild?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(Guild? other)
```

### Parameters

`other` Guild

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

## Returns

int

A hash code for the current object.

# Class GuildDelegatee

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class GuildDelegatee : Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>, IDelegatee, IEquatable<GuildDelegatee>
```

### Inheritance

object ← Delegatee<GuildRepository, GuildDelegatee, GuildDelegator> ← GuildDelegatee

### Implements

IDelegatee, IEquatable<GuildDelegatee>

### Inherited Members

Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Metadata ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Repository ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Address ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.AccountAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.MetadataAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.DelegationCurrency ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.RewardCurrencies ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.DelegationPoolAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.RewardPoolAddress ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.RewardRemainderPoolAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.SlashedPoolAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.UnbondingPeriod ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.MaxUnbondLockInEntries ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.MaxRebondGraceEntries ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.TotalDelegated ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.TotalShares ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Jailed ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.JailedUntil ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Tombstoned ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.MetadataBencoded ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.ShareFromFAV(FungibleAssetValue) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.FAVFromShare(BigInteger) ,

Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Jail(long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Unjail(long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Tombstone() ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.BondAddress(Address) ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.UnbondLockInAddress(Address) ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.RebondGraceAddress(Address) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.DistributionPoolAddress() ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.CurrentRewardBaseAddress() ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.RewardBaseAddress(long) ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.CurrentLumpSumRewardsRecordAddress() ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.LumpSumRewardsRecordAddress(long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Bond(GuildDelegator,
FungibleAssetValue, long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Unbond(GuildDelegator,
BigInteger, long) ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.DistributeReward(GuildDelegator, long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.CollectRewards(long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Slash(BigInteger, long, long)
,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.StartNewRewardPeriod(long)
,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.CalculateRewards(BigInteger, RewardBase, RewardBase) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.OnDelegationChanged(long)
,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.OnEnjailed() ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.OnUnjailed() ,
object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GuildDelegatee(Address, GuildRepository)

```
public GuildDelegatee(Address address, GuildRepository repository)
```

### Parameters

`address` Address

`repository` GuildRepository

## GuildDelegatee(Address, IEnumerable<Currency>, GuildRepository)

```
public GuildDelegatee(Address address, IEnumerable<Currency> rewardCurrencies,
GuildRepository repository)
```

### Parameters

`address` Address

`rewardCurrencies` IEnumerable<Currency>

`repository` GuildRepository

# Properties

## GuildMaxRebondGraceEntries

```
public static int GuildMaxRebondGraceEntries { get; }
```

### Property Value

int

## GuildMaxUnbondLockInEntries

```
public static int GuildMaxUnbondLockInEntries { get; }
```

### Property Value

int

## Methods

## Activate()

```
public void Activate()
```

## Deactivate()

```
public void Deactivate()
```

## Equals(GuildDelegatee?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GuildDelegatee? other)
```

### Parameters

`other` GuildDelegatee

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Slash(BigInteger, long, long)

```
public override void Slash(BigInteger slashFactor, long infractionHeight,
long height)
```

### Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

# Class GuildDelegator

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class GuildDelegator : Delegator<GuildRepository, GuildDelegatee,
GuildDelegator>, IDelegator, IEquatable<GuildDelegator>
```

### Inheritance

object ← Delegator<GuildRepository, GuildDelegatee, GuildDelegator> ← GuildDelegator

### Implements

IDelegator, IEquatable<GuildDelegator>

### Inherited Members

Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Metadata ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Repository ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Address ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.AccountAddress ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.MetadataAddress ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.DelegationPoolAddress ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.RewardAddress ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Delegatees ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.UnbondingRefs ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.MetadataBencoded ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Delegate(GuildDelegatee, FungibleAssetValue, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Undelegate(GuildDelegatee, BigInteger, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Redelegate(GuildDelegatee, GuildDelegatee, BigInteger, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.CancelUndelegate(GuildDelegatee, FungibleAssetValue, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.ClaimReward(GuildDelegatee, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.ReleaseUnbondings(long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.OnUnbondingReleased(long, IUnbonding, FungibleAssetValue?) ,

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GuildDelegator(Address, Address, GuildRepository)

```
public GuildDelegator(Address address, Address delegationPoolAddress,
GuildRepository repository)
```

### Parameters

`address` Address

`delegationPoolAddress` Address

`repository` GuildRepository

## GuildDelegator(Address, GuildRepository)

```
public GuildDelegator(Address address, GuildRepository repository)
```

### Parameters

`address` Address

`repository` GuildRepository

# Methods

## Equals(GuildDelegator?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GuildDelegator? other)
```

### Parameters

`other` GuildDelegator

An object to compare with this object.

### Returns

bool

[true] if the current object is equal to the `other` parameter; otherwise, [false].

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

[true] if the specified object is equal to the current object; otherwise, [false].

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## OnUnbondingReleased(long, IUnbonding, FungibleAssetValue?)

```
protected override void OnUnbondingReleased(long height, IUnbonding
releasedUnbonding, FungibleAssetValue? releasedFAV)
```

### Parameters

`height` long

`releasedUnbonding` IUnbonding

`releasedFAV` FungibleAssetValue?

# Class GuildParticipant

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class GuildParticipant : IBencodable, IEquatable<GuildParticipant>
```

**Inheritance**

object ← GuildParticipant

**Implements**

IBencodable, IEquatable<GuildParticipant>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GuildParticipant(AgentAddress, IValue, GuildRepository)

```
public GuildParticipant(AgentAddress address, IValue bencoded,
GuildRepository repository)
```

#### Parameters

`address` AgentAddress

`bencoded` IValue

`repository` GuildRepository

### GuildParticipant(AgentAddress, GuildAddress, GuildRepository)

```
public GuildParticipant(AgentAddress address, GuildAddress guildAddress,
```

```
  GuildRepository repository)
```

### Parameters

`address` AgentAddress

`guildAddress` GuildAddress

`repository` GuildRepository

## Fields

## GuildAddress

```
  public readonly GuildAddress GuildAddress
```

### Field Value

GuildAddress

## Properties

## Address

```
  public AgentAddress Address { get; }
```

### Property Value

AgentAddress

## Bencoded

```
  public List Bencoded { get; }
```

### Property Value

List

## DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

### Property Value

Address

## Repository

```
public GuildRepository Repository { get; }
```

### Property Value

GuildRepository

## RewardAddress

```
public Address RewardAddress { get; }
```

### Property Value

Address

# Methods

## ClaimReward(Guild, long)

```
public void ClaimReward(Guild guild, long height)
```

### Parameters

`guild` Guild

`height` long

## Delegate(Guild, FungibleAssetValue, long)

```
public void Delegate(Guild guild, FungibleAssetValue fav, long height)
```

### Parameters

`guild` Guild

`fav` FungibleAssetValue

`height` long

## Equals(GuildParticipant?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GuildParticipant? other)
```

### Parameters

`other` GuildParticipant

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Redelegate(Guild, Guild, BigInteger, long)

```
public void Redelegate(Guild srcGuild, Guild dstGuild, BigInteger share,
long height)
```

### Parameters

`srcGuild` Guild

`dstGuild` Guild

`share` BigInteger

`height` long

## Undelegate(Guild, BigInteger, long)

```
public void Undelegate(Guild guild, BigInteger share, long height)
```

### Parameters

`guild` Guild

`share` BigInteger

`height` long

# Class GuildRepository

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class GuildRepository : DelegationRepository<GuildRepository, GuildDelegatee,
GuildDelegator>, IDelegationRepository
```

### Inheritance

object ← DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator> ← GuildRepository

### Implements

IDelegationRepository

### Inherited Members

DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.previousWorld ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.delegateeAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.delegatorAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.delegateeMetadataAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.delegatorMetadataAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.bondAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.unbondLockInAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.rebondGraceAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.unbondingSetAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.rewardBaseAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.lumpSumRewardsRecordAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.World ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.ActionContext ,

DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.DelegateeAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.DelegatorAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.DelegateeMetadataAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.DelegatorMetadataAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.BondAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.UnbondLockInAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.RebondGraceAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.UnbondingSetAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.RewardBaseAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.LumpSumRewardsRecordAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetDelegatee(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetDelegator(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetDelegatee(GuildDelegatee) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetDelegator(GuildDelegator) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetDelegateeMetadata(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetDelegatorMetadata(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetBond(GuildDelegatee, Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetUnbondLockIn(GuildDelegatee, Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetUnlimitedUnbondLockIn(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetRebondGrace(GuildDelegatee, Address) ,

DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetUnlimitedRebondGrace(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetCurrentRewardBase(GuildDelegatee) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetRewardBase(GuildDelegatee, long) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetLumpSumRewardsRecord(GuildDelegatee, long) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetCurrentLumpSumRewardsRecord(GuildDelegatee) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetBalance(Address, Currency) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetDelegateeMetadata(DelegateeMetadata) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetDelegatorMetadata(DelegatorMetadata) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetBond(Bond) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetUnbondLockIn(UnbondLockIn) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetRebondGrace(RebondGrace) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetRewardBase(RewardBase) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetLumpSumRewardsRecord(LumpSumRewardsRecord) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.RemoveLumpSumRewardsRecord(LumpSumRewardsRecord) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.TransferAsset(Address, Address, FungibleAssetValue) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.UpdateWorld(IWorld) ,
object.Equals(object) , object.Equals(object, object) , object.GetHashCode() ,
object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) ,
object.ToString()

**Extension Methods**

GuildBanModule.Ban(GuildRepository, AgentAddress, AgentAddress) ,
GuildBanModule.IsBanned(GuildRepository, GuildAddress, Address) ,
GuildBanModule.RemoveBanList(GuildRepository, GuildAddress) ,
GuildBanModule.Unban(GuildRepository, AgentAddress, Address) ,
GuildDelegateeModule.CreateDelegatee(GuildRepository, Address) ,

GuildDelegateeModule.TryGetDelegatee(GuildRepository, Address, out GuildDelegatee?) ,
GuildMemberCounterModule.DecreaseGuildMemberCount(GuildRepository, GuildAddress) ,
GuildMemberCounterModule.GetGuildMemberCount(GuildRepository, GuildAddress) ,
GuildMemberCounterModule.IncreaseGuildMemberCount(GuildRepository, GuildAddress) ,
GuildModule.GetDelegationInfo(GuildRepository, Address) ,
GuildModule.MakeGuild(GuildRepository, GuildAddress, Address) ,
GuildModule.RemoveGuild(GuildRepository) ,
GuildModule.TryGetGuild(GuildRepository, GuildAddress, out Guild?) ,
GuildParticipantModule.GetJoinedGuild(GuildRepository, AgentAddress) ,
GuildParticipantModule.JoinGuild(GuildRepository, GuildAddress, AgentAddress) ,
GuildParticipantModule.LeaveGuild(GuildRepository, AgentAddress) ,
GuildParticipantModule.MoveGuild(GuildRepository, AgentAddress, GuildAddress) ,
GuildParticipantModule.TryGetGuildParticipant(GuildRepository, AgentAddress, out GuildParticipant?)

# Constructors

## GuildRepository(IWorld, IActionContext)

```
public GuildRepository(IWorld world, IActionContext actionContext)
```

### Parameters

`world` IWorld

`actionContext` IActionContext

## GuildRepository(IDelegationRepository)

```
public GuildRepository(IDelegationRepository repository)
```

### Parameters

`repository` IDelegationRepository

# Properties

## World

```
public override IWorld World { get; }
```

### Property Value

IWorld

# Methods

## GetDelegatee(Address)

```
public override GuildDelegatee GetDelegatee(Address address)
```

### Parameters

`address` Address

### Returns

GuildDelegatee

## GetDelegator(Address)

```
public override GuildDelegator GetDelegator(Address address)
```

### Parameters

`address` Address

### Returns

GuildDelegator

## GetGuild(Address)

```
public Guild GetGuild(Address address)
```

### Parameters

`address` Address

### Returns

Guild

## GetGuildParticipant(Address)

```
public GuildParticipant GetGuildParticipant(Address address)
```

### Parameters

`address` Address

### Returns

GuildParticipant

## RemoveGuildParticipant(Address)

```
public void RemoveGuildParticipant(Address guildParticipantAddress)
```

### Parameters

`guildParticipantAddress` Address

## SetDelegatee(GuildDelegatee)

```
public override void SetDelegatee(GuildDelegatee delegatee)
```

### Parameters

`delegatee` GuildDelegatee

## SetDelegator(GuildDelegator)

```
public override void SetDelegator(GuildDelegator delegator)
```

### Parameters

`delegator` GuildDelegator

## SetGuild(Guild)

```
public void SetGuild(Guild guild)
```

### Parameters

`guild` Guild

## SetGuildParticipant(GuildParticipant)

```
public void SetGuildParticipant(GuildParticipant guildParticipant)
```

### Parameters

`guildParticipant` GuildParticipant

## UpdateWorld(IWorld)

```
public override void UpdateWorld(IWorld world)
```

### Parameters

```
world IWorld
```

# Namespace Nekoyume.Model.InfiniteTower

## Classes

InfiniteTowerBattleCondition

Infinite tower battle condition for managing floor-specific restrictions and requirements. Represents various types of battle conditions that can be applied to infinite tower floors.

InfiniteTowerCondition

Infinite tower condition for managing battle conditions and restrictions. Uses StatType, SkillTargetType, and StatModifier for flexible stat modifications.

InfiniteTowerInfo

Infinite tower info for managing player's tower progress and tickets.

## Enums

BattleConditionType

Represents different types of battle conditions that can be applied to infinite tower floors.

# Enum BattleConditionType

Namespace: Nekoyume.Model.InfiniteTower

Assembly: Lib9c.dll

Represents different types of battle conditions that can be applied to infinite tower floors.

```
public enum BattleConditionType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`CP = 0`

Combat Power restrictions (RequiredCp, MaxCp)

`ForbiddenItemSubTypes = 5`

Forbidden Item Sub Types restrictions

`ForbiddenRuneTypes = 3`

Forbidden Rune Types restrictions

`ItemGrade = 1`

Item Grade restrictions (MinItemGrade, MaxItemGrade)

`ItemLevel = 2`

Item Level restrictions (MinItemLevel, MaxItemLevel)

`RequiredElementalType = 4`

Required Elemental Type restrictions

# Class InfiniteTowerBattleCondition

Namespace: Nekoyume.Model.InfiniteTower

Assembly: Lib9c.dll

Infinite tower battle condition for managing floor-specific restrictions and requirements. Represents various types of battle conditions that can be applied to infinite tower floors.

```
[Serializable]
public class InfiniteTowerBattleCondition
```

**Inheritance**

object ← InfiniteTowerBattleCondition

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InfiniteTowerBattleCondition(List<ElementalType>, bool)

Initializes a new instance of the InfiniteTowerBattleCondition class for Required Elemental Type.

```
public InfiniteTowerBattleCondition(List<ElementalType> requiredElementalTypes,
bool isElementalType)
```

#### Parameters

`requiredElementalTypes` List<ElementalType>

List of required elemental types

`isElementalType` bool

Must be true to indicate this is for elemental type

# InfiniteTowerBattleCondition(List<RuneType>)

Initializes a new instance of the InfiniteTowerBattleCondition class for Forbidden Rune Types.

```
public InfiniteTowerBattleCondition(List<RuneType> forbiddenRuneTypes)
```

## Parameters

`forbiddenRuneTypes` List<RuneType>

List of forbidden rune types

# InfiniteTowerBattleCondition(List<ItemSubType>, bool)

Initializes a new instance of the InfiniteTowerBattleCondition class for Forbidden Item Sub Types.

```
public InfiniteTowerBattleCondition(List<ItemSubType> forbiddenItemSubTypes,
bool isItemSubTypes)
```

## Parameters

`forbiddenItemSubTypes` List<ItemSubType>

List of forbidden item sub types

`isItemSubTypes` bool

Must be true to indicate this is for item sub types

# InfiniteTowerBattleCondition(int?, int?)

Initializes a new instance of the InfiniteTowerBattleCondition class for Item Level restrictions.

```
public InfiniteTowerBattleCondition(int? minItemLevel, int? maxItemLevel)
```

### Parameters

`minItemLevel` int?

Minimum item level

`maxItemLevel` int?

Maximum item level

## InfiniteTowerBattleCondition(int?, int?, bool)

Initializes a new instance of the InfiniteTowerBattleCondition class for Item Grade restrictions.

```
public InfiniteTowerBattleCondition(int? minItemGrade, int? maxItemGrade,
bool isItemGrade)
```

### Parameters

`minItemGrade` int?

Minimum item grade

`maxItemGrade` int?

Maximum item grade

`isItemGrade` bool

Must be true to indicate this is for item grade

## InfiniteTowerBattleCondition(long?, long?)

Initializes a new instance of the InfiniteTowerBattleCondition class for CP restrictions.

```
public InfiniteTowerBattleCondition(long? requiredCp, long? maxCp)
```

### Parameters

`requiredCp` long?

Minimum required combat power

`maxCp` long?

Maximum allowed combat power

# Properties

## ForbiddenItemSubTypes

Gets the list of forbidden item sub types (for ForbiddenItemSubTypes type).

```
public List<ItemSubType> ForbiddenItemSubTypes { get; }
```

### Property Value

List<ItemSubType>

## ForbiddenRuneTypes

Gets the list of forbidden rune types (for ForbiddenRuneTypes type).

```
public List<RuneType> ForbiddenRuneTypes { get; }
```

### Property Value

List<RuneType>

## MaxCp

Gets the maximum allowed combat power (for CP type).

```
public long? MaxCp { get; }
```

### Property Value

long?

## MaxItemGrade

Gets the maximum item grade (for ItemGrade type).

```
public int? MaxItemGrade { get; }
```

### Property Value

int?

## MaxItemLevel

Gets the maximum item level (for ItemLevel type).

```
public int? MaxItemLevel { get; }
```

### Property Value

int?

## MinItemGrade

Gets the minimum item grade (for ItemGrade type).

```
public int? MinItemGrade { get; }
```

### Property Value

int?

## MinItemLevel

Gets the minimum item level (for ItemLevel type).

```
public int? MinItemLevel { get; }
```

### Property Value

int?

## RequiredCp

Gets the minimum required combat power (for CP type).

```
public long? RequiredCp { get; }
```

### Property Value

long?

## RequiredElementalTypes

Gets the list of required elemental types (for RequiredElementalType type).

```
public List<ElementalType> RequiredElementalTypes { get; }
```

### Property Value

List<ElementalType>

## Type

Gets the type of this battle condition.

```
public BattleConditionType Type { get; }
```

### Property Value

BattleConditionType

# Methods

## HasRestrictions()

Checks if this battle condition has any restrictions set.

```
public bool HasRestrictions()
```

### Returns

bool

True if the condition has restrictions, false otherwise

## ToString()

Gets a description of this battle condition.

```
public override string ToString()
```

### Returns

string

String description of the condition

# Class InfiniteTowerCondition

Namespace: Nekoyume.Model.InfiniteTower

Assembly: Lib9c.dll

Infinite tower condition for managing battle conditions and restrictions. Uses StatType, SkillTargetType, and StatModifier for flexible stat modifications.

```
[Serializable]
public class InfiniteTowerCondition
```

**Inheritance**

object ← InfiniteTowerCondition

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## InfiniteTowerCondition(Row)

Initializes a new instance of the InfiniteTowerCondition class.

```
public InfiniteTowerCondition(InfiniteTowerConditionSheet.Row row)
```

### Parameters

`row` InfiniteTowerConditionSheet.Row

The condition sheet row containing the condition data.

# Properties

## Id

Gets the unique identifier for this condition.

```
public int Id { get; }
```

### Property Value

int

## OperationType

Gets the operation type for the stat modification.

```
public StatModifier.OperationType OperationType { get; }
```

### Property Value

StatModifier.OperationType

## StatType

Gets the stat type that this condition affects.

```
public StatType StatType { get; }
```

### Property Value

StatType

## TargetType

Gets the list of target types for this condition (Self, Enemy, etc.).

```
public List<SkillTargetType> TargetType { get; }
```

### Property Value

List<SkillTargetType>

## Value

Gets the value for the stat modification.

```
public int Value { get; }
```

### Property Value

int

# Methods

## GetStatModifier()

Creates a StatModifier from this condition.

```
public StatModifier GetStatModifier()
```

### Returns

StatModifier

## GetValue()

Gets the value of this condition.

```
public int GetValue()
```

### Returns

int

The condition value.

# IsApplicableToEquipment(Equipment)

Checks if this condition applies to the given equipment for stat modification. This is only used for stat modifiers, not equipment restrictions.

```
public bool IsApplicableToEquipment(Equipment equipment)
```

## Parameters

`equipment` Equipment

## Returns

bool

# IsApplicableToRune(RuneState)

Checks if this condition applies to the given rune for stat modification. This is only used for stat modifiers, not rune restrictions.

```
public bool IsApplicableToRune(RuneState runeState)
```

## Parameters

`runeState` RuneState

## Returns

bool

# Class InfiniteTowerInfo

Namespace: Nekoyume.Model.InfiniteTower

Assembly: Lib9c.dll

Infinite tower info for managing player's tower progress and tickets.

```
[Serializable]
public class InfiniteTowerInfo : IState
```

**Inheritance**

object ← InfiniteTowerInfo

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### InfiniteTowerInfo(List)

Initializes a new instance of the InfiniteTowerInfo class from serialized data.

```
public InfiniteTowerInfo(List serialized)
```

#### Parameters

`serialized` List

The serialized data.

### InfiniteTowerInfo(Address, int, int)

Initializes a new instance of the InfiniteTowerInfo class.

```
public InfiniteTowerInfo(Address address, int infiniteTowerId, int initialTickets)
```

### Parameters

`address` Address

The avatar address.

`infiniteTowerId` int

The infinite tower ID.

`initialTickets` int

The initial number of tickets to grant on creation.

## Properties

### Address

Gets the avatar address for this infinite tower info.

```
public Address Address { get; }
```

### Property Value

Address

### ClearedFloor

Gets the highest floor that has been cleared by the player.

```
public int ClearedFloor { get; }
```

### Property Value

int

## InfiniteTowerId

Gets the infinite tower ID this info belongs to.

```
public int InfiniteTowerId { get; }
```

### Property Value

int

## LastResetBlockIndex

Gets the block index when the tower was last reset.

```
public long LastResetBlockIndex { get; }
```

### Property Value

long

## LastTicketRefillBlockIndex

Gets the block index when tickets were last refilled.

```
public long LastTicketRefillBlockIndex { get; }
```

### Property Value

long

## NumberOfTicketPurchases

Gets the number of times the player has purchased tickets.

```
public int NumberOfTicketPurchases { get; }
```

### Property Value

int

## RemainingTickets

Gets the number of remaining tickets for the player.

```
public int RemainingTickets { get; }
```

### Property Value

int

## TotalTicketsUsed

Gets the total number of tickets used by the player.

```
public int TotalTicketsUsed { get; }
```

### Property Value

int

# Methods

## AddTickets(int)

```
public void AddTickets(int count)
```

### Parameters

`count` int

## ClearFloor(int)

```
public void ClearFloor(int floor)
```

### Parameters

`floor` int

## GetScheduleStatus(long, long, long)

Gets the schedule status for the infinite tower.

```
public string GetScheduleStatus(long currentBlockIndex, long startBlockIndex,
long endBlockIndex)
```

### Parameters

`currentBlockIndex` long

Current block index

`startBlockIndex` long

Schedule start block index

`endBlockIndex` long

Schedule end block index

### Returns

string

Schedule status string

## IncreaseNumberOfTicketPurchases()

```
public void IncreaseNumberOfTicketPurchases()
```

# IsAccessible(long, long, long)

Checks if the infinite tower is currently accessible based on schedule.

```
public bool IsAccessible(long currentBlockIndex, long startBlockIndex,
long endBlockIndex)
```

## Parameters

`currentBlockIndex` long

Current block index

`startBlockIndex` long

Schedule start block index

`endBlockIndex` long

Schedule end block index

## Returns

bool

True if accessible, false otherwise

# IsCleared(int)

```
public bool IsCleared(int floor)
```

## Parameters

`floor` int

## Returns

bool

## PerformSeasonReset(long, int, int)

Performs a season reset of the infinite tower progress. This is called when a new season starts. Grants exactly 1 ticket on season start.

```
public void PerformSeasonReset(long currentBlockIndex, int dailyFreeTickets,
int maxTickets)
```

### Parameters

`currentBlockIndex` long

Current block index

`dailyFreeTickets` int

Ignored for initial grant; kept for backward-compatibility

`maxTickets` int

Ignored; capacity is fixed to 10

## ResetTickets(int)

```
public void ResetTickets(int maxTickets)
```

### Parameters

`maxTickets` int

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# TryRefillDailyTickets(int, int, long, int)

Refills daily free tickets according to elapsed block periods since the last refill. Caps the total number of tickets to a fixed capacity (10).

```
public bool TryRefillDailyTickets(int dailyFreeTickets, int maxTickets, long
currentBlockIndex, int blocksPerDay = 10800)
```

## Parameters

`dailyFreeTickets` int

Number of free tickets to give per period (from schedule's DailyFreeTickets)

`maxTickets` int

Maximum number of tickets player can hold (from schedule's MaxTickets)

`currentBlockIndex` long

Current block index

`blocksPerDay` int

Number of blocks per period (from schedule's ResetIntervalBlocks, default: 10800)

## Returns

bool

True if tickets were refilled, false otherwise

# TryUseTickets(int)

```
public bool TryUseTickets(int count)
```

## Parameters

`count` int

## Returns

bool

# Namespace Nekoyume.Model.Item

## Classes

Armor

Represents armor equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Aura

Represents aura equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Belt

Represents belt equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Consumable

Represents consumable items that can be used by characters. Supports both Dictionary and List serialization formats for backward compatibility.

Costume

Represents costume items that can be equipped by characters. Supports both Dictionary and List serialization formats for backward compatibility.

Equipment

Represents equipment items that can be equipped by characters. Supports both Dictionary and List serialization formats for backward compatibility.

Field Order (List Format): Base fields (0~10): 11 fields from ItemUsable Equipment fields (11~22): equipped, level, stat, setId, spineResourcePath, iconId, byCustomCraft, craftWithRandom, hasRandomOnlyIcon, optionCountFromCombination, madeWithMimisbrunnrRecipe, exp

Equipment Properties: - Equipped: Whether the equipment is currently equipped - Level: Equipment enhancement level - Stat: Primary stat of the equipment - SetId: ID for equipment set bonuses - SpineResourcePath: Path to spine animation resource - IconId: Icon identifier for UI display - ByCustomCraft: Whether crafted through custom crafting - CraftWithRandom: Whether random options were applied during crafting - HasRandomOnlyIcon: Whether has random-only icon - OptionCountFromCombination: Number of options from combination - MadeWithMimisbrunnrRecipe: Whether made with Mimisbrunnr recipe - Exp: Experience points for leveling

Grimoire

Represents grimoire equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Inventory

Inventory.Item

ItemBase

Base class for all items in the game. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. id - Item ID (Binary format) 3. itemType - Item type as string 4. itemSubType - Item sub-type as string 5. grade - Item grade 6. elementalType - Elemental type as string

ItemFactory

Factory class for creating and deserializing items. Supports both Dictionary and List serialization formats for backward compatibility.

ItemSubTypeComparer

ItemTypeComparer

ItemUsable

Material

Represents a material item that can be used for crafting and other purposes. Supports both Dictionary and List serialization formats for backward compatibility.

Field Order (List Format): Base fields (0~5): version, id, itemType, itemSubType, grade, elementalType Material fields (6): itemId

Material Properties: - ItemId: Unique identifier for the material item

Necklace

Represents necklace equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Ring

Represents ring equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

ShopItem

TradableMaterial

Represents a tradable material item that can be traded between players. Supports both Dictionary and List serialization formats for backward compatibility.

Weapon

Represents weapon equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

## Structs

OrderLock

## Interfaces

IEquippableItem

IFungibleItem

IItem

ILock

INonFungibleItem

ITradableFungibleItem

ITradableItem

## Enums

ItemSubType

ItemType

LockType

# Class Armor

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents armor equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Armor : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

### Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Armor

### Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

### Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

**Extension Methods**
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

# Constructors

## Armor(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Armor(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Armor(Row, Guid, long, bool)

```
public Armor(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

### Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

## Armor(SerializationInfo, StreamingContext)

```
protected Armor(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

## Properties

### TradableId

```
public Guid TradableId { get; }
```

### Property Value

Guid

# Class Aura

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents aura equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Aura : Equipment, INonFungibleItem, IEquippableItem, IItem, IState
```

### Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Aura

### Implements

INonFungibleItem, IEquippableItem, IItem, IState

### Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

**Extension Methods**
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

# Constructors

## Aura(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Aura(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Aura(Row, Guid, long, bool)

```
public Aura(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

### Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

## Aura(SerializationInfo, StreamingContext)

```
protected Aura(SerializationInfo info, StreamingContext _)
```

## Parameters

`info` SerializationInfo

`_` StreamingContext

# Class Belt

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents belt equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Belt : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

### Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Belt

### Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

### Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

**Extension Methods**
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

# Constructors

## Belt(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Belt(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Belt(Row, Guid, long, bool)

```
public Belt(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

### Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

## Belt(SerializationInfo, StreamingContext)

```
protected Belt(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

## Properties

### TradableId

```
public Guid TradableId { get; }
```

### Property Value

Guid

# Class Consumable

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents consumable items that can be used by characters. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Consumable : ItemUsable, INonFungibleItem, ITradableItem, IItem, IState
```

**Inheritance**

object ← ItemBase ← ItemUsable ← Consumable

**Implements**

INonFungibleItem, ITradableItem, IItem, IState

**Inherited Members**

ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.Equals(object) , ItemUsable.GetHashCode() , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemUsable.GetFieldNames() , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Constructors

## Consumable(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Consumable(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Consumable(Row, Guid, long)

```
public Consumable(ConsumableItemSheet.Row data, Guid id, long requiredBlockIndex)
```

### Parameters

`data` ConsumableItemSheet.Row

`id` Guid

`requiredBlockIndex` long

## Consumable(SerializationInfo, StreamingContext)

```
protected Consumable(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

# Properties

## MainStat

```
public StatType MainStat { get; }
```

### Property Value

StatType

## Stats

```
public List<DecimalStat> Stats { get; set; }
```

### Property Value

List<DecimalStat>

## TradableId

```
public Guid TradableId { get; }
```

### Property Value

Guid

# Methods

## Serialize()

Serializes the consumable to List format (new format). Order: [baseData..., stats]

```
public override IValue Serialize()
```

### Returns

IValue

List containing serialized data

# Class Costume

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents costume items that can be equipped by characters. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Costume : ItemBase, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

**Inheritance**

object ← ItemBase ← Costume

**Implements**

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

**Inherited Members**

ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , ItemBase.GetFieldNames() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Constructors

## Costume(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Costume(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Costume(Row, Guid)

```
public Costume(CostumeItemSheet.Row data, Guid itemId)
```

### Parameters

`data` CostumeItemSheet.Row

`itemId` Guid

## Costume(SerializationInfo, StreamingContext)

```
protected Costume(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

# Fields

## equipped

```
public bool equipped
```

### Field Value

bool

# Properties

## Equipped

```
public bool Equipped { get; }
```

### Property Value

bool

## ItemId

```
public Guid ItemId { get; }
```

### Property Value

Guid

## NonFungibleId

```
public Guid NonFungibleId { get; }
```

### Property Value

Guid

## RequiredBlockIndex

```
public long RequiredBlockIndex { get; set; }
```

### Property Value

long

## SpineResourcePath

```
public string SpineResourcePath { get; set; }
```

### Property Value

string

## TradableId

```
public Guid TradableId { get; }
```

### Property Value

Guid

# Methods

## Equals(Costume)

```
protected bool Equals(Costume other)
```

### Parameters

`other` Costume

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## Equip()

```
public void Equip()
```

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

Serializes the costume to List format (new format). Order: [baseData..., equipped, spineResourcePath, itemId, requiredBlockIndex]

```
public override IValue Serialize()
```

### Returns

IValue

List containing serialized data

## Unequip()

```
public void Unequip()
```

## Update(long)

```
public void Update(long blockIndex)
```

### Parameters

`blockIndex` long

# Class Equipment

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents equipment items that can be equipped by characters. Supports both Dictionary and List serialization formats for backward compatibility.

Field Order (List Format): Base fields (0~10): 11 fields from ItemUsable Equipment fields (11~22): equipped, level, stat, setId, spineResourcePath, iconId, byCustomCraft, craftWithRandom, hasRandomOnlyIcon, optionCountFromCombination, madeWithMimisbrunnrRecipe, exp

Equipment Properties: - Equipped: Whether the equipment is currently equipped - Level: Equipment enhancement level - Stat: Primary stat of the equipment - SetId: ID for equipment set bonuses - SpineResourcePath: Path to spine animation resource - IconId: Icon identifier for UI display - ByCustomCraft: Whether crafted through custom crafting - CraftWithRandom: Whether random options were applied during crafting - HasRandomOnlyIcon: Whether has random-only icon - OptionCountFromCombination: Number of options from combination - MadeWithMimisbrunnrRecipe: Whether made with Mimisbrunnr recipe - Exp: Experience points for leveling

```
[Serializable]
public class Equipment : ItemUsable, INonFungibleItem, IEquippableItem,
IItem, IState
```

### Inheritance

object ← ItemBase ← ItemUsable ← Equipment

### Implements

INonFungibleItem, IEquippableItem, IItem, IState

### Derived

Armor, Aura, Belt, Grimoire, Necklace, Ring, Weapon

### Inherited Members

ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade ,

ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

**Extension Methods**
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) , EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

# Remarks

Equipment items can be enhanced, equipped, and provide various stats and bonuses. The equipment system supports both regular crafting and custom crafting with random options.

Example usage:

```
// Create equipment
var equipment = new Equipment(equipmentRow, Guid.NewGuid(), 1000L);
// Enhance equipment
equipment.LevelUp();
// Equip/unequip
equipment.Equipped = true;
// Check stats
var statValue = equipment.Stat.StatType;
var setBonus = equipment.SetId;
```

# Constructors

## Equipment(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Equipment(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Equipment(Row, Guid, long, bool, int)

```
public Equipment(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false, int iconId = 0)
```

### Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

`iconId` int

## Equipment(SerializationInfo, StreamingContext)

```
protected Equipment(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

# Fields

## ByCustomCraft

```
public bool ByCustomCraft
```

### Field Value

bool

## CraftWithRandom

```
public bool CraftWithRandom
```

### Field Value

bool

## Exp

```
public long Exp
```

### Field Value

long

## HasRandomOnlyIcon

```
public bool HasRandomOnlyIcon
```

### Field Value

bool

## IconId

```
public int IconId
```

### Field Value

int

## equipped

```
public bool equipped
```

### Field Value

bool

## level

```
public int level
```

### Field Value

int

## optionCountFromCombination

```
public int optionCountFromCombination
```

### Field Value

int

# Properties

## Equipped

```
public bool Equipped { get; }
```

### Property Value

bool

## MadeWithMimisbrunnrRecipe

```
public bool MadeWithMimisbrunnrRecipe { get; set; }
```

### Property Value

bool

## SetId

```
public int SetId { get; }
```

### Property Value

int

## SpineResourcePath

```
public string SpineResourcePath { get; }
```

### Property Value

string

## Stat

```
public DecimalStat Stat { get; }
```

### Property Value

DecimalStat

## UniqueStatType

```
public StatType UniqueStatType { get; }
```

### Property Value

StatType

## Methods

### Equals(Equipment)

```
protected bool Equals(Equipment other)
```

#### Parameters

`other` Equipment

#### Returns

bool

### Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

#### Parameters

`obj` object

The object to compare with the current object.

#### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## Equip()

```
public void Equip()
```

## GetFieldNames()

Gets the field names for serialization in order.

```
protected override string[] GetFieldNames()
```

### Returns

string[]

Array of field names

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## GetIncrementAmountOfEnhancement()

```
public decimal GetIncrementAmountOfEnhancement()
```

### Returns

decimal

## GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3)

```
public long GetRealExp(EquipmentItemSheet itemSheet, EnhancementCostSheetV3
costSheet)
```

### Parameters

`itemSheet` EquipmentItemSheet

`costSheet` EnhancementCostSheetV3

### Returns

long

## LevelUp(IRandom, Row, bool)

```
[Obsolete("Since ItemEnhancement12, Use `SetLevel` instead.")]
public void LevelUp(IRandom random, EnhancementCostSheetV2.Row row,
bool isGreatSuccess)
```

### Parameters

`random` IRandom

`row` EnhancementCostSheetV2.Row

`isGreatSuccess` bool

## LevelUpV1()

```
[Obsolete("Use LevelUp")]
public void LevelUpV1()
```

## Serialize()

Serializes the equipment to List format (new format). Order: [baseData..., equipped, level, stat, setId, spineResourcePath, iconId, byCustomCraft, craftWithRandom, hasRandomOnlyIcon, optionCountFromCombination, madeWithMimisbrunnrRecipe, exp]

```
public override IValue Serialize()
```

### Returns

IValue

List containing serialized data

## SetLevel(IRandom, int, EnhancementCostSheetV3)

```
public void SetLevel(IRandom random, int targetLevel, EnhancementCostSheetV3 sheet)
```

### Parameters

`random` IRandom

`targetLevel` int

`sheet` EnhancementCostSheetV3

## Unequip()

```
public void Unequip()
```

# Class Grimoire

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents grimoire equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Grimoire : Equipment, INonFungibleItem, IEquippableItem, IItem, IState
```

### Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Grimoire

### Implements

INonFungibleItem, IEquippableItem, IItem, IState

### Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

**Extension Methods**
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

# Constructors

## Grimoire(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Grimoire(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Grimoire(Row, Guid, long, bool)

```
public Grimoire(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

### Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

## Grimoire(SerializationInfo, StreamingContext)

```
protected Grimoire(SerializationInfo info, StreamingContext _)
```

## Parameters

`info` SerializationInfo

`_` StreamingContext

# Interface IEquippableItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface IEquippableItem : IItem, IState
```

**Inherited Members**
IItem.ItemType , IItem.ItemSubType , IState.Serialize()

## Properties

### Equipped

```
bool Equipped { get; }
```

#### Property Value

bool

## Methods

### Equip()

```
void Equip()
```

### Unequip()

```
void Unequip()
```

# Interface IFungibleItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface IFungibleItem : IItem, IState
```

**Inherited Members**
IItem.ItemType , IItem.ItemSubType , IState.Serialize()

## Properties

### FungibleId

```
HashDigest<SHA256> FungibleId { get; }
```

#### Property Value

HashDigest<SHA256>

# Interface IItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface IItem : IState
```

**Inherited Members**
IState.Serialize()

## Properties

### ItemSubType

```
ItemSubType ItemSubType { get; }
```

#### Property Value

ItemSubType

### ItemType

```
ItemType ItemType { get; }
```

#### Property Value

ItemType

# Interface ILock

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface ILock
```

## Properties

### Type

```
LockType Type { get; }
```

#### Property Value

LockType

## Methods

### Serialize()

```
IValue Serialize()
```

#### Returns

IValue

# Interface INonFungibleItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface INonFungibleItem : IItem, IState
```

**Inherited Members**
IItem.ItemType , IItem.ItemSubType , IState.Serialize()

## Properties

### NonFungibleId

```
Guid NonFungibleId { get; }
```

#### Property Value

Guid

### RequiredBlockIndex

```
long RequiredBlockIndex { get; set; }
```

#### Property Value

long

# Interface ITradableFungibleItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface ITradableFungibleItem : ITradableItem, IFungibleItem, IItem,
IState, ICloneable
```

**Inherited Members**

ITradableItem.TradableId , ITradableItem.RequiredBlockIndex , IFungibleItem.FungibleId , IItem.ItemType , IItem.ItemSubType , IState.Serialize() , ICloneable.Clone()

# Interface ITradableItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface ITradableItem : IItem, IState
```

**Inherited Members**
IItem.ItemType , IItem.ItemSubType , IState.Serialize()

## Properties

### RequiredBlockIndex

```
long RequiredBlockIndex { get; set; }
```

#### Property Value

long

### TradableId

```
Guid TradableId { get; }
```

#### Property Value

Guid

# Class Inventory

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public class Inventory : IState, ICloneable
```

**Inheritance**

object ← Inventory

**Implements**

IState, ICloneable

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

InventoryExtensions.GetEquippedFullCostumeOrArmorId(Inventory) , InventoryExtensions.MintItem(Inventory, ItemSheet.Row, int, bool, IRandom) , InventoryExtensions.UseActionPoint(Inventory, long, int, bool, MaterialItemSheet, long)

# Constructors

## Inventory()

```
public Inventory()
```

## Inventory(List)

```
public Inventory(List serialized)
```

### Parameters

`serialized` List

# Properties

## Consumables

```
public IEnumerable<Consumable> Consumables { get; }
```

### Property Value

IEnumerable<Consumable>

## Costumes

```
public IEnumerable<Costume> Costumes { get; }
```

### Property Value

IEnumerable<Costume>

## Equipments

```
public IEnumerable<Equipment> Equipments { get; }
```

### Property Value

IEnumerable<Equipment>

## Items

```
public IReadOnlyList<Inventory.Item> Items { get; }
```

### Property Value

IReadOnlyList<Inventory.Item>

## Materials

```
public IEnumerable<Material> Materials { get; }
```

### Property Value

IEnumerable<Material>

# Methods

## AddFungibleItem(ItemBase, int, ILock)

```
public Inventory.Item AddFungibleItem(ItemBase itemBase, int count = 1, ILock iLock
= null)
```

### Parameters

`itemBase` ItemBase

`count` int

`iLock` ILock

### Returns

Inventory.Item

## AddItem(ItemBase, int, ILock)

```
public KeyValuePair<int, int> AddItem(ItemBase itemBase, int count = 1, ILock iLock
= null)
```

### Parameters

`itemBase` ItemBase

`count` int

`iLock` ILock

### Returns

KeyValuePair<int, int>

## AddNonFungibleItem(ItemBase, ILock)

```
public Inventory.Item AddNonFungibleItem(ItemBase itemBase, ILock iLock = null)
```

### Parameters

`itemBase` ItemBase

`iLock` ILock

### Returns

Inventory.Item

## AddTradableFungibleItem(ITradableFungibleItem, int)

Add tradable fungible item. If there is same item which is not locked, add count to it. The same item is item which has same item sheet id, fungible id, tradable id and same required block index. If there is no same item, add new item.

```
public Inventory.Item AddTradableFungibleItem(ITradableFungibleItem
tradableFungibleItem, int count)
```

### Parameters

`tradableFungibleItem` ITradableFungibleItem

`count` int

### Returns

Inventory.Item

### Exceptions

ArgumentNullException

ArgumentOutOfRangeException

## Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## Equals(Inventory)

```
protected bool Equals(Inventory other)
```

### Parameters

`other` Inventory

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## FilterConsumables(int, long)

```
public List<Consumable> FilterConsumables(int id, long blockIndex)
```

### Parameters

`id` int

`blockIndex` long

### Returns

List<Consumable>

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## HasFungibleItem(HashDigest<SHA256>, long, int)

```
public bool HasFungibleItem(HashDigest<SHA256> fungibleId, long blockIndex, int
count = 1)
```

### Parameters

`fungibleId` HashDigest<SHA256>

`blockIndex` long

`count` int

### Returns

bool

## HasItem(int, int)

```
public bool HasItem(int rowId, int count = 1)
```

### Parameters

`rowId` int

`count` int

### Returns

bool

## HasNonFungibleItem(Guid)

```
public bool HasNonFungibleItem(Guid nonFungibleId)
```

### Parameters

`nonFungibleId` Guid

### Returns

bool

## HasTradableFungibleItem(HashDigest<SHA256>, long?, long?, int)

```
public bool HasTradableFungibleItem(HashDigest<SHA256> fungibleId, long?
requiredBlockIndex, long? blockIndex, int count)
```

### Parameters

`fungibleId` HashDigest<SHA256>

`requiredBlockIndex` long?

`blockIndex` long?

`count` int

### Returns

bool

## HasTradableItem(Guid, long, int)

```
public bool HasTradableItem(Guid tradableId, long blockIndex, int count)
```

### Parameters

`tradableId` Guid

`blockIndex` long

`count` int

### Returns

bool

## IsMaterialRemovable(Item, int)

Checks if the given item is non-tradable material and can be removed from the inventory.

```
public static bool IsMaterialRemovable(Inventory.Item item, int id)
```

### Parameters

`item` Inventory.Item

The item to check for removal.

`id` int

The ID of the material to remove.

### Returns

bool

Returns a boolean indicating whether the item is removable.

## IsMaterialRemovable(Item, int, long)

Checks if the given item is material and can be removed from the inventory.

```
public static bool IsMaterialRemovable(Inventory.Item item, int id, long blockIndex)
```

### Parameters

`item` Inventory.Item

The item to check for removal.

`id` int

The ID of the material to remove.

`blockIndex` long

The block index.

### Returns

bool

Returns a boolean indicating whether the item is removable.

## LockByReferringToDigestList(OrderDigestListState, Guid, long)

```
public void LockByReferringToDigestList(OrderDigestListState digestList, Guid
tradableId, long blockIndex)
```

### Parameters

`digestList` OrderDigestListState

`tradableId` Guid

`blockIndex` long

## ReconfigureFungibleItem(OrderDigestListState, Guid)

```
public void ReconfigureFungibleItem(OrderDigestListState digestList,
Guid tradableId)
```

### Parameters

`digestList` OrderDigestListState

`tradableId` Guid

# RemoveConsumable(int, long, int)

```
public bool RemoveConsumable(int id, long blockIndex, int count = 1)
```

## Parameters

`id` int

`blockIndex` long

`count` int

## Returns

bool

# RemoveFungibleItem(HashDigest<SHA256>, long, int, bool)

```
public bool RemoveFungibleItem(HashDigest<SHA256> fungibleId, long blockIndex, int
count = 1, bool onlyTradableItem = false)
```

## Parameters

`fungibleId` HashDigest<SHA256>

`blockIndex` long

`count` int

`onlyTradableItem` bool

## Returns

bool

# RemoveFungibleItem(IFungibleItem, long, int, bool)

```
public bool RemoveFungibleItem(IFungibleItem fungibleItem, long blockIndex, int
count = 1, bool onlyTradableItem = false)
```

### Parameters

`fungibleItem` IFungibleItem

`blockIndex` long

`count` int

`onlyTradableItem` bool

### Returns

bool

## RemoveFungibleItem2(HashDigest<SHA256>, int, bool)

```
[Obsolete("Use RemoveFungibleItem")]
public bool RemoveFungibleItem2(HashDigest<SHA256> fungibleId, int count = 1, bool
onlyTradableItem = false)
```

### Parameters

`fungibleId` HashDigest<SHA256>

`count` int

`onlyTradableItem` bool

### Returns

bool

## RemoveFungibleItem2(IFungibleItem, int, bool)

```
[Obsolete("Use RemoveFungibleItem")]
public bool RemoveFungibleItem2(IFungibleItem fungibleItem, int count = 1, bool
```

```
  onlyTradableItem = false)
```

### Parameters

`fungibleItem` IFungibleItem

`count` int

`onlyTradableItem` bool

### Returns

bool

## RemoveItem(Item)

```
  public void RemoveItem(Inventory.Item item)
```

### Parameters

`item` Inventory.Item

## RemoveMaterial(int, long, int)

Remove a material from the inventory. contains tradable materials.

```
  public bool RemoveMaterial(int id, long blockIndex, int count = 1)
```

### Parameters

`id` int

The ID of the material item to remove.

`blockIndex` long

The block index.

`count` int

The number of materials to remove. Default value is 1.

### Returns

bool

True if the material is successfully removed, false otherwise.

## RemoveNonFungibleItem(INonFungibleItem)

```
public bool RemoveNonFungibleItem(INonFungibleItem nonFungibleItem)
```

### Parameters

`nonFungibleItem` INonFungibleItem

### Returns

bool

## RemoveNonFungibleItem(Guid)

```
public bool RemoveNonFungibleItem(Guid nonFungibleId)
```

### Parameters

`nonFungibleId` Guid

### Returns

bool

## RemoveNonFungibleItem2(Guid)

```
[Obsolete("Use RemoveNonFungibleItem(Guid nonFungibleId)")]
public bool RemoveNonFungibleItem2(Guid nonFungibleId)
```

### Parameters

`nonFungibleId` Guid

### Returns

bool

## RemoveNonTradableMaterial(int, int)

Remove a non-tradable material from the inventory.

```
public bool RemoveNonTradableMaterial(int id, int count = 1)
```

### Parameters

`id` int

The ID of the material item to remove.

`count` int

The number of materials to remove. Default value is 1.

### Returns

bool

True if the material is successfully removed, false otherwise.

## RemoveTradableFungibleItem(HashDigest<SHA256>, long?, long?, int)

Remove tradable item from inventory. Use the requiredBlockIndex when you want to remove the item that has same requiredBlockIndex. Use the blockIndex when you want to remove the item that has requiredBlockIndex less than or equal to blockIndex. Do not use both requiredBlockIndex and blockIndex at the same time.

```
public bool RemoveTradableFungibleItem(HashDigest<SHA256> fungibleId, long?
```

```
requiredBlockIndex, long? blockIndex, int count)
```

### Parameters

`fungibleId` HashDigest<SHA256>

`requiredBlockIndex` long?

`blockIndex` long?

`count` int

### Returns

bool

### Exceptions

ArgumentNullException

## RemoveTradableItem(Guid, long, int)

```
public bool RemoveTradableItem(Guid tradableId, long blockIndex, int count = 1)
```

### Parameters

`tradableId` Guid

`blockIndex` long

`count` int

### Returns

bool

## RemoveTradableItemV1(ITradableItem, int)

```
[Obsolete("Use RemoveTradableItem()")]
public bool RemoveTradableItemV1(ITradableItem tradableItem, int count = 1)
```

### Parameters

`tradableItem` ITradableItem

`count` int

### Returns

bool

## RemoveTradableItemV1(Guid, long, int)

```
[Obsolete("Use RemoveTradableItem()")]
public bool RemoveTradableItemV1(Guid tradableId, long blockIndex, int count = 1)
```

### Parameters

`tradableId` Guid

`blockIndex` long

`count` int

### Returns

bool

## RemoveTradableMaterial(int, long, int)

Remove a tradable material from the inventory.

```
public bool RemoveTradableMaterial(int materialId, long blockIndex, int count = 1)
```

### Parameters

`materialId` int

The ID of the material item to remove.

`blockIndex` long

The block index.

`count` int

The number of materials to remove. Default value is 1.

### Returns

bool

True if the material is successfully removed, false otherwise.

## SellItem(Guid, long, int)

```
public ITradableItem SellItem(Guid tradableId, long blockIndex, int count)
```

### Parameters

`tradableId` Guid

`blockIndex` long

`count` int

### Returns

ITradableItem

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## TryGetCostume(int, out Costume)

```
[Obsolete("Use public bool TryGetNonFungibleItem<T>(Guid itemId, out
T outNonFungibleItem)")]
public bool TryGetCostume(int rowId, out Costume outCostume)
```

### Parameters

`rowId` int

`outCostume` Costume

### Returns

bool

## TryGetFungibleItems(HashDigest<SHA256>, out List<Item>)

```
public bool TryGetFungibleItems(HashDigest<SHA256> fungibleId, out
List<Inventory.Item> outItems)
```

### Parameters

`fungibleId` HashDigest<SHA256>

`outItems` List<Inventory.Item>

### Returns

bool

## TryGetItem(int, out Item)

```
public bool TryGetItem(int rowId, out Inventory.Item outItem)
```

### Parameters

`rowId` int

`outItem` Inventory.Item

### Returns

bool

## TryGetLockedItem(ILock, out Item)

```
public bool TryGetLockedItem(ILock iLock, out Inventory.Item outItem)
```

### Parameters

`iLock` ILock

`outItem` Inventory.Item

### Returns

bool

## TryGetNonFungibleItem(Guid, out Item)

```
public bool TryGetNonFungibleItem(Guid nonFungibleId, out Inventory.Item outItem)
```

### Parameters

`nonFungibleId` Guid

`outItem` Inventory.Item

### Returns

bool

## TryGetNonFungibleItem<T>(Guid, out T, long)

Tries to get a non-fungible item from the inventory with the specified item ID.

```
public bool TryGetNonFungibleItem<T>(Guid itemId, out T outNonFungibleItem, long
blockIndex = 9223372036854775807) where T : INonFungibleItem
```

### Parameters

`itemId` Guid

The ID of the non-fungible item to retrieve.

`outNonFungibleItem` T

When this method returns, contains the non-fungible item with the specified ID, if found; otherwise, the default value for the type.

`blockIndex` long

The block index for the non-fungible item to be considered valid. Defaults to MaxValue.

### Returns

bool

`true` if a non-fungible item with the specified ID is found in the inventory; otherwise, `false`.

### Type Parameters

`T`

The type of non-fungible item to retrieve.

## TryGetNonFungibleItem<T>(T, out T)

```
public bool TryGetNonFungibleItem<T>(T nonFungibleItem, out T outNonFungibleItem)
where T : INonFungibleItem
```

### Parameters

`nonFungibleItem` T

`outNonFungibleItem` T

### Returns

bool

### Type Parameters

`T`

## TryGetTradableFungibleItems(HashDigest<SHA256>, long?, long?, out IEnumerable<Item>)

Get ITradableFungibleItem from the inventory. This method check the item is not locked, and the item's required block index is less than or equal to the given block index. Do not use both requiredBlockIndex and blockIndex at the same time.

```
public bool TryGetTradableFungibleItems(HashDigest<SHA256> fungibleId, long?
requiredBlockIndex, long? blockIndex, out IEnumerable<Inventory.Item> outItems)
```

### Parameters

`fungibleId` HashDigest<SHA256>

`requiredBlockIndex` long?

`blockIndex` long?

`outItems` IEnumerable<Inventory.Item>

### Returns

bool

### Exceptions

ArgumentNullException

## TryGetTradableItem(Guid, long, int, out Item)

```
public bool TryGetTradableItem(Guid tradeId, long blockIndex, int count, out
Inventory.Item outItem)
```

### Parameters

`tradeId` Guid

`blockIndex` long

`count` int

`outItem` Inventory.Item

### Returns

bool

## TryGetTradableItems(Guid, long, int, out List<Item>)

```
public bool TryGetTradableItems(Guid tradeId, long blockIndex, int count, out
List<Inventory.Item> outItem)
```

### Parameters

`tradeId` Guid

`blockIndex` long

`count` int

`outItem` List<Inventory.Item>

### Returns

bool

## UnlockInvalidSlot(OrderDigestListState, Address, Address)

```
public void UnlockInvalidSlot(OrderDigestListState digestListState, Address
agentAddress, Address avatarAddress)
```

### Parameters

`digestListState` OrderDigestListState

`agentAddress` Address

`avatarAddress` Address

## UpdateTradableItem(Guid, long, int, long)

```
public ITradableItem UpdateTradableItem(Guid tradableId, long blockIndex, int count,
long requiredBlockIndex)
```

### Parameters

`tradableId` Guid

`blockIndex` long

`count` int

`requiredBlockIndex` long

### Returns

ITradableItem

# Class Inventory.Item

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public class Inventory.Item : IState, IComparer<Inventory.Item>,
IComparable<Inventory.Item>
```

**Inheritance**

object ← Inventory.Item

**Implements**

IState, IComparer<Inventory.Item>, IComparable<Inventory.Item>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Item(Dictionary)

```
public Item(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### Item(ItemBase, int)

```
public Item(ItemBase itemBase, int count = 1)
```

#### Parameters

`itemBase` ItemBase

count int

# Fields

## Lock

```
public ILock Lock
```

### Field Value

ILock

## count

```
public int count
```

### Field Value

int

## item

```
public ItemBase item
```

### Field Value

ItemBase

# Properties

## Locked

```
public bool Locked { get; }
```

### Property Value

bool

# Methods

## Compare(Item, Item)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public int Compare(Inventory.Item x, Inventory.Item y)
```

### Parameters

x Inventory.Item

The first object to compare.

y Inventory.Item

The second object to compare.

### Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

| Value | Meaning |
| --- | --- |
| **Less than zero** | x is less than y. |
| **Zero** | x equals y. |
| **Greater than zero** | x is greater than y. |

## CompareTo(Item)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(Inventory.Item other)
```

### Parameters

`other` Inventory.Item

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `other` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `other`. |
| **Greater than zero** | This instance follows `other` in the sort order. |

## Equals(Item)

```
protected bool Equals(Inventory.Item other)
```

### Parameters

`other` Inventory.Item

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## LockUp(ILock)

```
public void LockUp(ILock iLock)
```

### Parameters

`iLock` ILock

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Unlock()

```
public void Unlock()
```

# Class ItemBase

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Base class for all items in the game. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. id - Item ID (Binary format) 3. itemType - Item type as string 4. itemSubType - Item sub-type as string 5. grade - Item grade 6. elementalType - Elemental type as string

```
[Serializable]
public abstract class ItemBase : IItem, IState
```

**Inheritance**

object ← ItemBase

**Implements**

IItem, IState

**Derived**

Costume, ItemUsable, Material

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Remarks

This class implements dual serialization support to ensure smooth migration from the legacy Dictionary format to the new List format. The List format provides better performance and smaller serialized data size.

Example usage:

```
// Create item
var material = new Material(materialRow);
// Serialize to List format (new)
```

```
var serialized = material.Serialize(); // Returns List
// Deserialize from any format
var deserialized = new Material(serialized); // Supports both Dictionary and List
```

# Constructors

## ItemBase(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
protected ItemBase(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

### Exceptions

ArgumentNullException

Thrown when serialized is null

ArgumentException

Thrown when serialized format is not supported or has insufficient fields

## ItemBase(ItemBase)

```
protected ItemBase(ItemBase other)
```

### Parameters

`other` ItemBase

## ItemBase(Row)

```
protected ItemBase(ItemSheet.Row data)
```

### Parameters

`data` ItemSheet.Row

## ItemBase(SerializationInfo, StreamingContext)

```
protected ItemBase(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

# Fields

## BaseFieldCount

```
protected const int BaseFieldCount = 6
```

### Field Value

int

## Codec

```
protected static readonly Codec Codec
```

### Field Value

Codec

## SerializationVersion

```
public const int SerializationVersion = 2
```

### Field Value

int

# Properties

## ElementalType

```
public ElementalType ElementalType { get; set; }
```

### Property Value

ElementalType

## Grade

```
public int Grade { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## ItemSubType

```
public ItemSubType ItemSubType { get; }
```

### Property Value

ItemSubType

## ItemType

```
public ItemType ItemType { get; }
```

### Property Value

ItemType

# Methods

## Equals(ItemBase)

```
protected bool Equals(ItemBase other)
```

### Parameters

`other` ItemBase

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetFieldNames()

Gets the field names for serialization in order.

```
protected virtual string[] GetFieldNames()
```

### Returns

string[]

Array of field names

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## GetObjectData(SerializationInfo, StreamingContext)

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## Serialize()

Serializes the item to List format (new format). Order: [version, id, itemType, itemSubType, grade, elementalType]

```
public virtual IValue Serialize()
```

### Returns

IValue

List containing serialized data

## ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

### Returns

string

A string that represents the current object.

# Class ItemFactory

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Factory class for creating and deserializing items. Supports both Dictionary and List serialization formats for backward compatibility.

```
public static class ItemFactory
```

**Inheritance**

object ← ItemFactory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## CreateCostume(Row, Guid)

Creates a costume item.

```
public static Costume CreateCostume(CostumeItemSheet.Row row, Guid itemId)
```

### Parameters

`row` CostumeItemSheet.Row

The costume item sheet row data

`itemId` Guid

The unique ID for the costume

### Returns

Costume

The created costume

## CreateItem(Row, IRandom)

Creates an item from the given row data.

```
public static ItemBase CreateItem(ItemSheet.Row row, IRandom random)
```

### Parameters

`row` ItemSheet.Row

The item sheet row data

`random` IRandom

Random number generator

### Returns

ItemBase

The created item

## CreateItemUsable(Row, Guid, long, int, bool)

Creates a usable item (consumable or equipment) from the given row data.

```
public static ItemUsable CreateItemUsable(ItemSheet.Row itemRow, Guid id, long
requiredBlockIndex, int level = 0, bool madeWithMimisbrunnrRecipe = false)
```

### Parameters

`itemRow` ItemSheet.Row

The item sheet row data

`id` Guid

The unique ID for the item

`requiredBlockIndex` long

The required block index for using the item

`level` int

The level of the equipment (default: 0)

`madeWithMimisbrunnrRecipe` bool

Whether the item was made with Mimisbrunnr recipe (default: false)

### Returns

ItemUsable

The created usable item

## CreateItemUsableV2(Row, Guid, long, int, IRandom, Row, bool, bool)

```
public static ItemUsable CreateItemUsableV2(ItemSheet.Row itemRow, Guid id, long
requiredBlockIndex, int level, IRandom random, EnhancementCostSheetV2.Row row, bool
isGreatSuccess, bool madeWithMimisbrunnrRecipe = false)
```

### Parameters

`itemRow` ItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`level` int

`random` IRandom

`row` EnhancementCostSheetV2.Row

`isGreatSuccess` bool

`madeWithMimisbrunnrRecipe` bool

### Returns

ItemUsable

## CreateMaterial(MaterialItemSheet, int)

Creates a material item from the sheet by ID.

```
public static Material CreateMaterial(MaterialItemSheet sheet, int itemId)
```

### Parameters

`sheet` MaterialItemSheet

The material item sheet

`itemId` int

The material item ID

### Returns

Material

The created material, or null if not found

## CreateMaterial(Row)

Creates a material item from the given row data.

```
public static Material CreateMaterial(MaterialItemSheet.Row row)
```

### Parameters

`row` MaterialItemSheet.Row

The material item sheet row data

### Returns

Material

The created material

## CreateTradableMaterial(Row)

Creates a tradable material item from the given row data.

```
public static TradableMaterial CreateTradableMaterial(MaterialItemSheet.Row row)
```

### Parameters

`row` MaterialItemSheet.Row

The material item sheet row data

### Returns

TradableMaterial

The created tradable material

## Deserialize(IValue)

Deserializes an item from serialized data. Supports both Dictionary and List formats for backward compatibility.

```
public static ItemBase Deserialize(IValue serialized)
```

### Parameters

`serialized` IValue

The serialized item data

### Returns

ItemBase

The deserialized item

### Exceptions

ArgumentException

Thrown when the serialization format is not supported

## SelectIconId(int, bool, Row, int, CustomEquipmentCraftIconSheet, IRandom)

```
public static (int, bool) SelectIconId(int iconId, bool isRandom,
EquipmentItemSheet.Row equipmentRow, int relationship, CustomEquipmentCraftIconSheet
iconSheet, IRandom random)
```

### Parameters

`iconId` int

`isRandom` bool

`equipmentRow` EquipmentItemSheet.Row

`relationship` int

`iconSheet` CustomEquipmentCraftIconSheet

`random` IRandom

### Returns

(int, bool)

## SelectOption(ItemSubType, CustomEquipmentCraftOptionSheet, IRandom)

```
public static CustomEquipmentCraftOptionSheet.Row SelectOption(ItemSubType
itemSubType, CustomEquipmentCraftOptionSheet optionSheet, IRandom random)
```

## Parameters

`itemSubType` ItemSubType

`optionSheet` CustomEquipmentCraftOptionSheet

`random` IRandom

## Returns

CustomEquipmentCraftOptionSheet.Row

# Enum ItemSubType

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public enum ItemSubType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

ApStone = 16

Armor = 7

Aura = 19

Belt = 8

[Obsolete("ItemSubType.Chest has never been used outside the MaterialItemSheet. And we won't use it in the future until we have a specific reason.")] Chest = 17

Circle = 22

EarCostume = 3

EquipmentMaterial = 11

EyeCostume = 4

Food = 0

FoodMaterial = 12

FullCostume = 1

Grimoire = 20

HairCostume = 2

Hourglass = 15

MonsterPart = 13

Necklace = 9

NormalMaterial = 14

Ring = 10

Scroll = 21

TailCostume = 5

Title = 18

Weapon = 6

# Class ItemSubTypeComparer

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public class ItemSubTypeComparer : IEqualityComparer<ItemSubType>
```

**Inheritance**

object ← ItemSubTypeComparer

**Implements**

IEqualityComparer<ItemSubType>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Instance

```
public static readonly ItemSubTypeComparer Instance
```

### Field Value

ItemSubTypeComparer

# Methods

## Equals(ItemSubType, ItemSubType)

Determines whether the specified objects are equal.

```
public bool Equals(ItemSubType x, ItemSubType y)
```

### Parameters

`x` ItemSubType

The first object of type `T` to compare.

`y` ItemSubType

The second object of type `T` to compare.

### Returns

bool

[true] if the specified objects are equal; otherwise, [false].

## GetHashCode(ItemSubType)

Returns a hash code for the specified object.

```
public int GetHashCode(ItemSubType obj)
```

### Parameters

`obj` ItemSubType

The [object] for which a hash code is to be returned.

### Returns

int

A hash code for the specified object.

### Exceptions

ArgumentNullException

The type of `obj` is a reference type and `obj` is [null].

# Enum ItemType

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public enum ItemType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Consumable = 0`

`Costume = 1`

`Equipment = 2`

`Material = 3`

# Class ItemTypeComparer

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public class ItemTypeComparer : IEqualityComparer<ItemType>
```

**Inheritance**

object ← ItemTypeComparer

**Implements**

IEqualityComparer<ItemType>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Instance

```
public static readonly ItemTypeComparer Instance
```

### Field Value

ItemTypeComparer

# Methods

## Equals(ItemType, ItemType)

Determines whether the specified objects are equal.

```
public bool Equals(ItemType x, ItemType y)
```

### Parameters

`x` ItemType

The first object of type `T` to compare.

`y` ItemType

The second object of type `T` to compare.

### Returns

bool

[true] if the specified objects are equal; otherwise, [false].

## GetHashCode(ItemType)

Returns a hash code for the specified object.

```
public int GetHashCode(ItemType obj)
```

### Parameters

`obj` ItemType

The [object] for which a hash code is to be returned.

### Returns

int

A hash code for the specified object.

### Exceptions

ArgumentNullException

The type of `obj` is a reference type and `obj` is [null].

# Class ItemUsable

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ItemUsable : ItemBase, INonFungibleItem, IItem, IState
```

**Inheritance**

object ← ItemBase ← ItemUsable

**Implements**

INonFungibleItem, IItem, IState

**Derived**

Consumable, Equipment

**Inherited Members**

ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Constructors

## ItemUsable(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
protected ItemUsable(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## ItemUsable(Row, Guid, long)

```
protected ItemUsable(ItemSheet.Row data, Guid id, long requiredBlockIndex)
```

### Parameters

`data` ItemSheet.Row

`id` Guid

`requiredBlockIndex` long

## ItemUsable(SerializationInfo, StreamingContext)

```
protected ItemUsable(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

# Fields

## ITEM_USABLE_FIELD_COUNT

```
protected const int ITEM_USABLE_FIELD_COUNT = 11
```

### Field Value

int

# Properties

## BuffSkills

```
public List<BuffSkill> BuffSkills { get; }
```

### Property Value

List<BuffSkill>

## ItemId

```
public Guid ItemId { get; }
```

### Property Value

Guid

## NonFungibleId

```
public Guid NonFungibleId { get; }
```

### Property Value

Guid

## RequiredBlockIndex

```
public long RequiredBlockIndex { get; set; }
```

### Property Value

long

## Skills

```
public List<Skill> Skills { get; }
```

### Property Value

List<Skill>

## StatsMap

```
public StatsMap StatsMap { get; }
```

### Property Value

StatsMap

# Methods

## Equals(ItemUsable)

```
protected bool Equals(ItemUsable other)
```

### Parameters

`other` ItemUsable

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetFieldNames()

Gets the field names for serialization in order.

```
protected override string[] GetFieldNames()
```

### Returns

string[]

Array of field names

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## GetOptionCount()

```
public int GetOptionCount()
```

### Returns

int

## Serialize()

Serializes the item to List format (new format). Order: [baseData..., itemId, statsMap, skills, buffSkills, requiredBlockIndex]

```
public override IValue Serialize()
```

### Returns

IValue

List containing serialized data

## Update(long)

```
public void Update(long blockIndex)
```

### Parameters

`blockIndex` long

# Enum LockType

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public enum LockType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Order = 0`

# Class Material

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents a material item that can be used for crafting and other purposes. Supports both Dictionary and List serialization formats for backward compatibility.

Field Order (List Format): Base fields (0~5): version, id, itemType, itemSubType, grade, elementalType Material fields (6): itemId

Material Properties: - ItemId: Unique identifier for the material item

```
[Serializable]
public class Material : ItemBase, ISerializable, IFungibleItem, IItem, IState
```

**Inheritance**

object ← ItemBase ← Material

**Implements**

ISerializable, IFungibleItem, IItem, IState

**Derived**

TradableMaterial

**Inherited Members**

ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Remarks

Material items are fungible items used primarily for crafting other items. They support both Binary and Text formats for ItemId during deserialization to maintain compatibility with different serialization formats.

Example usage:

```
// Create material
var material = new Material(materialRow);
// Use in crafting
var requiredMaterials = new Dictionary<HashDigest<SHA256>, int>
{
{ material.ItemId, 5 } // 5 of this material
};
// Check material properties
var grade = material.Grade;
var itemType = material.ItemType;
```

# Constructors

## Material(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Material(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Material(Material)

```
public Material(Material other)
```

### Parameters

`other` Material

## Material(Row)

```
public Material(MaterialItemSheet.Row data)
```

### Parameters

`data` MaterialItemSheet.Row

## Material(SerializationInfo, StreamingContext)

```
protected Material(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

# Properties

## FungibleId

```
public HashDigest<SHA256> FungibleId { get; }
```

### Property Value

HashDigest<SHA256>

## ItemId

```
public HashDigest<SHA256> ItemId { get; }
```

### Property Value

HashDigest<SHA256>

# Methods

## Equals(Material)

```
protected bool Equals(Material other)
```

### Parameters

`other` Material

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetFieldNames()

Gets the field names for serialization in order.

```
protected override string[] GetFieldNames()
```

### Returns

string[]

Array of field names

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

Serializes the material to List format (new format). Order: [baseData..., itemId]

```
public override IValue Serialize()
```

### Returns

IValue

List containing serialized data

## ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

### Returns

string

A string that represents the current object.

# Class Necklace

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents necklace equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Necklace : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

### Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Necklace

### Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

### Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

**Extension Methods**
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) , EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

# Constructors

## Necklace(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Necklace(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Necklace(Row, Guid, long, bool)

```
public Necklace(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

### Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

## Necklace(SerializationInfo, StreamingContext)

```
protected Necklace(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

## Properties

### TradableId

```
public Guid TradableId { get; }
```

### Property Value

Guid

# Struct OrderLock

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public struct OrderLock : ILock
```

**Implements**
ILock

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### OrderLock(List)

```
public OrderLock(List serialized)
```

#### Parameters

`serialized` List

### OrderLock(Guid)

```
public OrderLock(Guid orderId)
```

#### Parameters

`orderId` Guid

# Fields

## OrderId

```
public readonly Guid OrderId
```

### Field Value

Guid

# Properties

## Type

```
public LockType Type { get; }
```

### Property Value

LockType

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class Ring

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents ring equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Ring : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

### Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Ring

### Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

### Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

**Extension Methods**
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

# Constructors

## Ring(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Ring(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Ring(Row, Guid, long, bool)

```
public Ring(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

### Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

## Ring(SerializationInfo, StreamingContext)

```
protected Ring(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

## Properties

### TradableId

```
public Guid TradableId { get; }
```

### Property Value

Guid

# Class ShopItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public class ShopItem
```

**Inheritance**

object ← ShopItem

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ShopItem(Dictionary)

```
public ShopItem(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ShopItem(Address, Address, Guid, FungibleAssetValue, ITradableItem)

```
public ShopItem(Address sellerAgentAddress, Address sellerAvatarAddress, Guid
productId, FungibleAssetValue price, ITradableItem tradableItem)
```

#### Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`productId` Guid

`price` FungibleAssetValue

`tradableItem` ITradableItem

## ShopItem(Address, Address, Guid, FungibleAssetValue, long, ITradableItem, int)

```
public ShopItem(Address sellerAgentAddress, Address sellerAvatarAddress, Guid
productId, FungibleAssetValue price, long expiredBlockIndex, ITradableItem
tradableItem, int tradableItemCount = 1)
```

### Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`productId` Guid

`price` FungibleAssetValue

`expiredBlockIndex` long

`tradableItem` ITradableItem

`tradableItemCount` int

## ShopItem(SerializationInfo, StreamingContext)

```
protected ShopItem(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

# Fields

## Codec

```
protected static readonly Codec Codec
```

### Field Value

Codec

## Costume

```
public readonly Costume Costume
```

### Field Value

Costume

## ExpiredBlockIndexKey

```
public const string ExpiredBlockIndexKey = "ebi"
```

### Field Value

string

## ItemUsable

```
public readonly ItemUsable ItemUsable
```

### Field Value

ItemUsable

## Price

```
public readonly FungibleAssetValue Price
```

### Field Value

FungibleAssetValue

## ProductId

```
public readonly Guid ProductId
```

### Field Value

Guid

## SellerAgentAddress

```
public readonly Address SellerAgentAddress
```

### Field Value

Address

## SellerAvatarAddress

```
public readonly Address SellerAvatarAddress
```

### Field Value

Address

## TradableFungibleItem

```
public readonly ITradableFungibleItem TradableFungibleItem
```

### Field Value

ITradableFungibleItem

## TradableFungibleItemCount

```
public readonly int TradableFungibleItemCount
```

### Field Value

int

# Properties

## ExpiredBlockIndex

```
public long ExpiredBlockIndex { get; }
```

### Property Value

long

# Methods

## Equals(ShopItem)

```
protected bool Equals(ShopItem other)
```

### Parameters

`other` ShopItem

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## GetObjectData(SerializationInfo, StreamingContext)

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## SerializeBackup1()

```
public IValue SerializeBackup1()
```

### Returns

IValue

# Class TradableMaterial

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents a tradable material item that can be traded between players. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class TradableMaterial : Material, ISerializable, ITradableFungibleItem,
ITradableItem, IFungibleItem, IItem, IState, ICloneable
```

**Inheritance**

object ← ItemBase ← Material ← TradableMaterial

**Implements**

ISerializable, ITradableFungibleItem, ITradableItem, IFungibleItem, IItem, IState, ICloneable

**Inherited Members**

Material.ItemId , Material.FungibleId , Material.Equals(Material) , Material.GetFieldNames() , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

# Constructors

## TradableMaterial(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public TradableMaterial(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## TradableMaterial(Row)

Initializes a new instance of the TradableMaterial class.

```
public TradableMaterial(MaterialItemSheet.Row data)
```

### Parameters

`data` MaterialItemSheet.Row

The material item sheet row data

## TradableMaterial(SerializationInfo, StreamingContext)

```
protected TradableMaterial(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

# Properties

## RequiredBlockIndex

Gets or sets the required block index for trading this material.

```
public long RequiredBlockIndex { get; set; }
```

### Property Value

long

## TradableId

Gets the tradable ID of this material.

```
public Guid TradableId { get; }
```

### Property Value

Guid

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## DeriveTradableId(HashDigest<SHA256>)

Derives a tradable ID from a fungible ID.

```
public static Guid DeriveTradableId(HashDigest<SHA256> fungibleId)
```

### Parameters

`fungibleId` HashDigest<SHA256>

The fungible ID to derive from

### Returns

Guid

The derived tradable ID

## Equals(TradableMaterial)

```
protected bool Equals(TradableMaterial other)
```

### Parameters

`other` TradableMaterial

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

Serializes the tradable material to List format (new format). Order: [baseData..., requiredBlockIndex]

```
public override IValue Serialize()
```

### Returns

IValue

List containing serialized data

## ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

### Returns

string

A string that represents the current object.

# Class Weapon

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents weapon equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Weapon : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

### Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Weapon

### Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

### Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

**Extension Methods**
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

# Constructors

## Weapon(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Weapon(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

## Weapon(Row, Guid, long, bool)

```
public Weapon(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

### Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

## Weapon(SerializationInfo, StreamingContext)

```
protected Weapon(SerializationInfo info, StreamingContext _)
```

### Parameters

`info` SerializationInfo

`_` StreamingContext

## Properties

### TradableId

```
public Guid TradableId { get; }
```

### Property Value

Guid

# Namespace Nekoyume.Model.Mail

## Classes

AdventureBossRaffleWinnerMail

AttachmentMail

BuyerMail

CancelOrderMail

ClaimItemsMail

CombinationMail

CustomCraftMail

DailyRewardMail

GrindingMail

ItemEnhanceMail

Mail

MailBox

MaterialCraftMail

MonsterCollectionMail

OrderBuyerMail

OrderExpirationMail

OrderSellerMail

PatrolRewardMail

Represents a patrol reward mail that contains rewards such as fungible assets and items.

ProductBuyerMail

ProductCancelMail

ProductSellerMail

SellCancelMail

SellerMail

UnloadFromMyGaragesRecipientMail

WorldBossRewardMail

Represents a worldboss reward mail that contains rewards such as fungible assets and items.

## Interfaces

IMail

## Enums

MailType

# Class AdventureBossRaffleWinnerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class AdventureBossRaffleWinnerMail : Mail, IState
```

**Inheritance**

object ← Mail ← AdventureBossRaffleWinnerMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.MailType , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AdventureBossRaffleWinnerMail(Dictionary)

```
public AdventureBossRaffleWinnerMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### AdventureBossRaffleWinnerMail(long, Guid, long, long, FungibleAssetValue)

```
public AdventureBossRaffleWinnerMail(long blockIndex, Guid id, long
requiredBlockIndex, long season, FungibleAssetValue reward)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`season` long

`reward` FungibleAssetValue

## Fields

## Reward

```
public FungibleAssetValue Reward
```

### Field Value

FungibleAssetValue

## Season

```
public readonly long Season
```

### Field Value

long

## Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

## Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class AttachmentMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public abstract class AttachmentMail : Mail, IState
```

**Inheritance**

object ← Mail ← AttachmentMail

**Implements**

IState

**Derived**

BuyerMail, CombinationMail, DailyRewardMail, ItemEnhanceMail, MonsterCollectionMail, SellCancelMail, SellerMail

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.MailType , Mail.requiredBlockIndex , Mail.Read(IMail) , Mail.TypeId , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AttachmentMail(Dictionary)

```
public AttachmentMail(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## AttachmentMail(AttachmentActionResult, long, Guid, long)

```
protected AttachmentMail(AttachmentActionResult attachmentActionResult, long
blockIndex, Guid id, long requiredBlockIndex)
```

### Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Fields

### attachment

```
public AttachmentActionResult attachment
```

### Field Value

AttachmentActionResult

## Methods

### Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class BuyerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class BuyerMail : AttachmentMail, IState
```

**Inheritance**

object ← Mail ← AttachmentMail ← BuyerMail

**Implements**

IState

**Inherited Members**

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BuyerMail(Dictionary)

```
public BuyerMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### BuyerMail(AttachmentActionResult, long, Guid, long)

```
public BuyerMail(AttachmentActionResult attachmentActionResult, long blockIndex,
Guid id, long requiredBlockIndex)
```

### Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Properties

### MailType

```
public override MailType MailType { get; }
```

#### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

#### Property Value

string

## Methods

### Read(IMail)

```
public override void Read(IMail mail)
```

#### Parameters

`mail` IMail

# Class CancelOrderMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class CancelOrderMail : Mail, IState
```

**Inheritance**

object ← Mail ← CancelOrderMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CancelOrderMail(Dictionary)

```
public CancelOrderMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### CancelOrderMail(long, Guid, long, Guid)

```
public CancelOrderMail(long blockIndex, Guid id, long requiredBlockIndex,
Guid orderId)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`orderId` Guid

## Fields

### OrderId

```
public readonly Guid OrderId
```

#### Field Value

Guid

## Properties

### MailType

```
public override MailType MailType { get; }
```

#### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

#### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ClaimItemsMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class ClaimItemsMail : Mail, IState
```

**Inheritance**

object ← Mail ← ClaimItemsMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ClaimItemsMail(Dictionary)

```
public ClaimItemsMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ClaimItemsMail(long, Guid, long, List<FungibleAssetValue>, List<(int id, int count)>, string)

```
public ClaimItemsMail(long blockIndex, Guid id, long requiredBlockIndex,
List<FungibleAssetValue> fungibleAssetValues, List<(int id, int count)> items,
```

```
  string memo)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`fungibleAssetValues` List<FungibleAssetValue>

`items` List<(int id, int count)>

`memo` string

## Fields

## FungibleAssetValues

```
  public List<FungibleAssetValue> FungibleAssetValues
```

### Field Value

List<FungibleAssetValue>

## Items

```
  public List<(int id, int count)> Items
```

### Field Value

List<(int id, int count)>

## Memo

```
public string Memo
```

### Field Value

string

## Properties

### MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

# Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Class CombinationMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationMail : AttachmentMail, IState
```

**Inheritance**

object ← Mail ← AttachmentMail ← CombinationMail

**Implements**

IState

**Inherited Members**

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CombinationMail(Dictionary)

```
public CombinationMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### CombinationMail(ResultModel, long, Guid, long)

```
public CombinationMail(CombinationConsumable5.ResultModel attachmentActionResult,
long blockIndex, Guid id, long requiredBlockIndex)
```

### Parameters

`attachmentActionResult` CombinationConsumable5.ResultModel

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Properties

### MailType

```
public override MailType MailType { get; }
```

#### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

#### Property Value

string

## Methods

### Read(IMail)

```
public override void Read(IMail mail)
```

#### Parameters

`mail` IMail

# Class CustomCraftMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class CustomCraftMail : Mail, IState
```

**Inheritance**

object ← Mail ← CustomCraftMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CustomCraftMail(Dictionary)

```
public CustomCraftMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### CustomCraftMail(long, Guid, long, Equipment)

```
public CustomCraftMail(long blockIndex, Guid id, long requiredBlockIndex,
Equipment equipment)
```

#### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`equipment` Equipment

# Fields

## Equipment

```
public Equipment Equipment
```

### Field Value

Equipment

# Properties

## MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class DailyRewardMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class DailyRewardMail : AttachmentMail, IState
```

**Inheritance**

object ← Mail ← AttachmentMail ← DailyRewardMail

**Implements**

IState

**Inherited Members**

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### DailyRewardMail(Dictionary)

```
public DailyRewardMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### DailyRewardMail(AttachmentActionResult, long, Guid, long)

```
public DailyRewardMail(AttachmentActionResult attachmentActionResult, long
```

```
  blockIndex, Guid id, long requiredBlockIndex)
```

### Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Properties

### MailType

```
  public override MailType MailType { get; }
```

### Property Value

MailType

### TypeId

```
  protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Read(IMail)

```
  public override void Read(IMail mail)
```

## Parameters

`mail` IMail

# Class GrindingMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class GrindingMail : Mail, IState
```

**Inheritance**

object ← Mail ← GrindingMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GrindingMail(Dictionary)

```
public GrindingMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### GrindingMail(long, Guid, long, int, FungibleAssetValue, int)

```
public GrindingMail(long blockIndex, Guid id, long requiredBlockIndex, int
itemCount, FungibleAssetValue asset, int rewardMaterialCount)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`itemCount` int

`asset` FungibleAssetValue

`rewardMaterialCount` int

## Fields

### Asset

```
public FungibleAssetValue Asset
```

#### Field Value

FungibleAssetValue

### ItemCount

```
public readonly int ItemCount
```

#### Field Value

int

### RewardMaterialCount

```
public readonly int RewardMaterialCount
```

### Field Value

int

## Properties

### MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

### Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Interface IMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public interface IMail
```

## Methods

### Read(AdventureBossRaffleWinnerMail)

```
void Read(AdventureBossRaffleWinnerMail adventureBossRaffleWinnerMail)
```

#### Parameters

`adventureBossRaffleWinnerMail` AdventureBossRaffleWinnerMail

### Read(BuyerMail)

```
void Read(BuyerMail buyerMail)
```

#### Parameters

`buyerMail` BuyerMail

### Read(CancelOrderMail)

```
void Read(CancelOrderMail cancelOrderMail)
```

#### Parameters

`cancelOrderMail` CancelOrderMail

## Read(ClaimItemsMail)

```
void Read(ClaimItemsMail claimItemsMail)
```

### Parameters

`claimItemsMail` ClaimItemsMail

## Read(CombinationMail)

```
void Read(CombinationMail mail)
```

### Parameters

`mail` CombinationMail

## Read(CustomCraftMail)

```
void Read(CustomCraftMail customCraftMail)
```

### Parameters

`customCraftMail` CustomCraftMail

## Read(DailyRewardMail)

```
void Read(DailyRewardMail dailyRewardMail)
```

### Parameters

`dailyRewardMail` DailyRewardMail

## Read(GrindingMail)

```
void Read(GrindingMail grindingMail)
```

### Parameters

`grindingMail` GrindingMail

## Read(ItemEnhanceMail)

```
void Read(ItemEnhanceMail itemEnhanceMail)
```

### Parameters

`itemEnhanceMail` ItemEnhanceMail

## Read(MaterialCraftMail)

```
void Read(MaterialCraftMail materialCraftMail)
```

### Parameters

`materialCraftMail` MaterialCraftMail

## Read(MonsterCollectionMail)

```
void Read(MonsterCollectionMail monsterCollectionMail)
```

### Parameters

`monsterCollectionMail` MonsterCollectionMail

## Read(OrderBuyerMail)

```
void Read(OrderBuyerMail orderBuyerMail)
```

### Parameters

`orderBuyerMail` OrderBuyerMail

## Read(OrderExpirationMail)

```
void Read(OrderExpirationMail orderExpirationMail)
```

### Parameters

`orderExpirationMail` OrderExpirationMail

## Read(OrderSellerMail)

```
void Read(OrderSellerMail orderSellerMail)
```

### Parameters

`orderSellerMail` OrderSellerMail

## Read(PatrolRewardMail)

```
void Read(PatrolRewardMail patrolRewardMail)
```

### Parameters

`patrolRewardMail` PatrolRewardMail

## Read(ProductBuyerMail)

```
void Read(ProductBuyerMail productBuyerMail)
```

### Parameters

`productBuyerMail` ProductBuyerMail

## Read(ProductCancelMail)

```
void Read(ProductCancelMail productCancelMail)
```

### Parameters

`productCancelMail` ProductCancelMail

## Read(ProductSellerMail)

```
void Read(ProductSellerMail productSellerMail)
```

### Parameters

`productSellerMail` ProductSellerMail

## Read(SellCancelMail)

```
void Read(SellCancelMail mail)
```

### Parameters

`mail` SellCancelMail

## Read(SellerMail)

```
void Read(SellerMail sellerMail)
```

### Parameters

`sellerMail` SellerMail

## Read(UnloadFromMyGaragesRecipientMail)

```
void Read(UnloadFromMyGaragesRecipientMail unloadFromMyGaragesRecipientMail)
```

### Parameters

`unloadFromMyGaragesRecipientMail` UnloadFromMyGaragesRecipientMail

## Read(WorldBossRewardMail)

```
void Read(WorldBossRewardMail worldBossRewardMail)
```

### Parameters

`worldBossRewardMail` WorldBossRewardMail

# Class ItemEnhanceMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhanceMail : AttachmentMail, IState
```

**Inheritance**

object ← Mail ← AttachmentMail ← ItemEnhanceMail

**Implements**

IState

**Inherited Members**

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ItemEnhanceMail(Dictionary)

```
public ItemEnhanceMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ItemEnhanceMail(AttachmentActionResult, long, Guid, long)

```
public ItemEnhanceMail(AttachmentActionResult attachmentActionResult, long
```

```
  blockIndex, Guid id, long requiredBlockIndex)
```

### Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Properties

### MailType

```
  public override MailType MailType { get; }
```

### Property Value

MailType

### TypeId

```
  protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Read(IMail)

```
  public override void Read(IMail mail)
```

## Parameters

`mail` IMail

# Class Mail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Mail : IState
```

**Inheritance**

object ← Mail

**Implements**

IState

**Derived**

AdventureBossRaffleWinnerMail, AttachmentMail, CancelOrderMail, ClaimItemsMail, CustomCraftMail, GrindingMail, MaterialCraftMail, OrderBuyerMail, OrderExpirationMail, OrderSellerMail, PatrolRewardMail, ProductBuyerMail, ProductCancelMail, ProductSellerMail, UnloadFromMyGaragesRecipientMail, WorldBossRewardMail

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Mail(Dictionary)

```
protected Mail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### Mail(long, Guid, long)

```
protected Mail(long blockIndex, Guid id, long requiredBlockIndex)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Fields

### New

```
public bool New
```

#### Field Value

bool

### blockIndex

```
public long blockIndex
```

#### Field Value

long

### id

```
public Guid id
```

#### Field Value

Guid

## requiredBlockIndex

```
public long requiredBlockIndex
```

### Field Value

long

# Properties

## MailType

```
public virtual MailType MailType { get; }
```

### Property Value

MailType

## TypeId

```
protected abstract string TypeId { get; }
```

### Property Value

string

# Methods

## Deserialize(Dictionary)

```
public static Mail Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

Mail

## Read(IMail)

```
public abstract void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public virtual IValue Serialize()
```

### Returns

IValue

# Class MailBox

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class MailBox : IEnumerable<Mail>, IEnumerable, IState
```

**Inheritance**

object ← MailBox

**Implements**

IEnumerable<Mail>, IEnumerable, IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MailBox()

```
public MailBox()
```

### MailBox(List)

```
public MailBox(List serialized)
```

#### Parameters

`serialized` List

## Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## this[int]

```
public Mail this[int idx] { get; }
```

### Parameters

`idx` int

### Property Value

Mail

# Methods

## Add(Mail)

```
public void Add(Mail mail)
```

### Parameters

`mail` Mail

## CleanUp()

```
public void CleanUp()
```

## CleanUpTemp(long)

```
[Obsolete("No longer in use.")]
public void CleanUpTemp(long blockIndex)
```

### Parameters

`blockIndex` long

## CleanUpV1()

```
[Obsolete("Use CleanUp")]
public void CleanUpV1()
```

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Mail> GetEnumerator()
```

### Returns

IEnumerator<Mail>

An enumerator that can be used to iterate through the collection.

## Remove(Mail)

```
public void Remove(Mail mail)
```

### Parameters

`mail` Mail

# Serialize()

```
public IValue Serialize()
```

## Returns

IValue

# Enum MailType

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public enum MailType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Auction = 2`

`CustomCraft = 6`

`Grinding = 4`

`Summon = 5`

`System = 3`

`Workshop = 1`

# Class MaterialCraftMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class MaterialCraftMail : Mail, IState
```

**Inheritance**

object ← Mail ← MaterialCraftMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MaterialCraftMail(Dictionary)

```
public MaterialCraftMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### MaterialCraftMail(long, Guid, long, int, int)

```
public MaterialCraftMail(long blockIndex, Guid id, long requiredBlockIndex, int
itemCount, int itemId)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`itemCount` int

`itemId` int

## Fields

### ItemCount

```
public int ItemCount
```

#### Field Value

int

### ItemId

```
public int ItemId
```

#### Field Value

int

## Properties

### MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class MonsterCollectionMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionMail : AttachmentMail, IState
```

**Inheritance**

object ← Mail ← AttachmentMail ← MonsterCollectionMail

**Implements**

IState

**Inherited Members**

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MonsterCollectionMail(Dictionary)

```
public MonsterCollectionMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### MonsterCollectionMail(AttachmentActionResult, long, Guid, long)

```
public MonsterCollectionMail(AttachmentActionResult attachmentActionResult, long
```

```
  blockIndex, Guid id, long requiredBlockIndex)
```

### Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Properties

### MailType

```
  public override MailType MailType { get; }
```

### Property Value

MailType

### TypeId

```
  protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Read(IMail)

```
  public override void Read(IMail mail)
```

## Parameters

`mail` IMail

# Class OrderBuyerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class OrderBuyerMail : Mail, IState
```

**Inheritance**

object ← Mail ← OrderBuyerMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### OrderBuyerMail(Dictionary)

```
public OrderBuyerMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### OrderBuyerMail(long, Guid, long, Guid)

```
public OrderBuyerMail(long blockIndex, Guid id, long requiredBlockIndex,
Guid orderId)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`orderId` Guid

## Fields

### OrderId

```
public readonly Guid OrderId
```

### Field Value

Guid

## Properties

### MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class OrderExpirationMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class OrderExpirationMail : Mail, IState
```

**Inheritance**

object ← Mail ← OrderExpirationMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### OrderExpirationMail(Dictionary)

```
public OrderExpirationMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### OrderExpirationMail(long, Guid, long, Guid)

```
public OrderExpirationMail(long blockIndex, Guid id, long requiredBlockIndex,
Guid orderId)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`orderId` Guid

## Fields

### OrderId

```
public readonly Guid OrderId
```

#### Field Value

Guid

## Properties

### MailType

```
public override MailType MailType { get; }
```

#### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

#### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class OrderSellerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class OrderSellerMail : Mail, IState
```

**Inheritance**

object ← Mail ← OrderSellerMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### OrderSellerMail(Dictionary)

```
public OrderSellerMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### OrderSellerMail(long, Guid, long, Guid)

```
public OrderSellerMail(long blockIndex, Guid id, long requiredBlockIndex,
Guid orderId)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`orderId` Guid

## Fields

### OrderId

```
public readonly Guid OrderId
```

### Field Value

Guid

## Properties

### MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class PatrolRewardMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

Represents a patrol reward mail that contains rewards such as fungible assets and items.

```
public class PatrolRewardMail : Mail, IState
```

**Inheritance**

object ← Mail ← PatrolRewardMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PatrolRewardMail(Dictionary)

Constructor for deserializing a PatrolRewardMail from a Bencodex dictionary.

```
public PatrolRewardMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

The serialized dictionary representation of the mail.

### PatrolRewardMail(long, Guid, long, List<FungibleAssetValue>, List<(int id, int count)>)

Constructor for creating a PatrolRewardMail with specified parameters.

```
public PatrolRewardMail(long blockIndex, Guid id, long requiredBlockIndex,
List<FungibleAssetValue> fungibleAssetValues, List<(int id, int count)> items)
```

### Parameters

`blockIndex` long

The block index at which the mail is created.

`id` Guid

The unique identifier for the mail.

`requiredBlockIndex` long

The block index required to claim the mail.

`fungibleAssetValues` List<FungibleAssetValue>

The list of fungible asset rewards.

`items` List<(int id, int count)>

The list of item rewards.

## Fields

## FungibleAssetValues

A list of fungible asset values included in the reward.

```
public List<FungibleAssetValue> FungibleAssetValues
```

### Field Value

List<FungibleAssetValue>

## Items

A list of item rewards where each entry contains the item ID and count.

```
public List<(int id, int count)> Items
```

### Field Value

List<(int id, int count)>

## Properties

### MailType

The type of mail. Defaults to Auction.

```
public override MailType MailType { get; }
```

### Property Value

MailType

### TypeId

The type identifier for the PatrolRewardMail class.

```
protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Read(IMail)

Reads the mail contents into the specified mail object.

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

The mail object to populate with this mail's data.

## Serialize()

Serializes the PatrolRewardMail into a Bencodex-compatible format.

```
public override IValue Serialize()
```

### Returns

IValue

A serialized representation of the mail.

# Class ProductBuyerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class ProductBuyerMail : Mail, IState
```

**Inheritance**

object ← Mail ← ProductBuyerMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ProductBuyerMail(Dictionary)

```
public ProductBuyerMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ProductBuyerMail(long, Guid, long, Guid, Product)

```
public ProductBuyerMail(long blockIndex, Guid id, long requiredBlockIndex, Guid
productId, Product product)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`productId` Guid

`product` Product

## Fields

### Product

```
public readonly Product Product
```

#### Field Value

Product

### ProductId

```
public readonly Guid ProductId
```

#### Field Value

Guid

### ProductKey

```
public const string ProductKey = "p"
```

#### Field Value

string

# Properties

## MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

# Returns

IValue

# Class ProductCancelMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class ProductCancelMail : Mail, IState
```

**Inheritance**

object ← Mail ← ProductCancelMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ProductCancelMail(Dictionary)

```
public ProductCancelMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ProductCancelMail(long, Guid, long, Guid, Product)

```
public ProductCancelMail(long blockIndex, Guid id, long requiredBlockIndex, Guid
productId, Product product)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`productId` Guid

`product` Product

## Fields

### Product

```
public readonly Product Product
```

#### Field Value

Product

### ProductId

```
public readonly Guid ProductId
```

#### Field Value

Guid

### ProductKey

```
public const string ProductKey = "p"
```

#### Field Value

string

# Properties

## MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

# Returns

IValue

# Class ProductSellerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class ProductSellerMail : Mail, IState
```

**Inheritance**

object ← Mail ← ProductSellerMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ProductSellerMail(Dictionary)

```
public ProductSellerMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ProductSellerMail(long, Guid, long, Guid, Product)

```
public ProductSellerMail(long blockIndex, Guid id, long requiredBlockIndex, Guid
productId, Product product)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`productId` Guid

`product` Product

## Fields

### Product

```
public readonly Product Product
```

#### Field Value

Product

### ProductId

```
public readonly Guid ProductId
```

#### Field Value

Guid

### ProductKey

```
public const string ProductKey = "p"
```

#### Field Value

string

# Properties

## MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

# Returns

IValue

# Class SellCancelMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class SellCancelMail : AttachmentMail, IState
```

**Inheritance**

object ← Mail ← AttachmentMail ← SellCancelMail

**Implements**

IState

**Inherited Members**

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SellCancelMail(Dictionary)

```
public SellCancelMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### SellCancelMail(Result, long, Guid, long)

```
public SellCancelMail(SellCancellation.Result attachmentActionResult, long
blockIndex, Guid id, long requiredBlockIndex)
```

### Parameters

`attachmentActionResult` SellCancellation.Result

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Properties

### MailType

```
public override MailType MailType { get; }
```

#### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

#### Property Value

string

## Methods

### Read(IMail)

```
public override void Read(IMail mail)
```

#### Parameters

`mail` IMail

# Class SellerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class SellerMail : AttachmentMail, IState
```

**Inheritance**

object ← Mail ← AttachmentMail ← SellerMail

**Implements**

IState

**Inherited Members**

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SellerMail(Dictionary)

```
public SellerMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### SellerMail(AttachmentActionResult, long, Guid, long)

```
public SellerMail(AttachmentActionResult attachmentActionResult, long blockIndex,
Guid id, long requiredBlockIndex)
```

### Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

## Properties

### MailType

```
public override MailType MailType { get; }
```

#### Property Value

MailType

### TypeId

```
protected override string TypeId { get; }
```

#### Property Value

string

## Methods

### Read(IMail)

```
public override void Read(IMail mail)
```

#### Parameters

`mail` IMail

# Class UnloadFromMyGaragesRecipientMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class UnloadFromMyGaragesRecipientMail : Mail, IState
```

**Inheritance**

object ← Mail ← UnloadFromMyGaragesRecipientMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### UnloadFromMyGaragesRecipientMail(Dictionary)

```
public UnloadFromMyGaragesRecipientMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### UnloadFromMyGaragesRecipientMail(long, Guid, long, IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?, string?)

```
public UnloadFromMyGaragesRecipientMail(long blockIndex, Guid id, long
requiredBlockIndex, IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?
fungibleAssetValue, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
fungibleIdAndCount, string? memo)
```

### Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`fungibleAssetValue` IEnumerable<(Address balanceAddr, FungibleAssetValue value)>

`fungibleIdAndCount` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

`memo` string

## Fields

## FungibleAssetValues

```
public readonly IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue
value)>? FungibleAssetValues
```

### Field Value

IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>

## FungibleIdAndCounts

```
public readonly IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int
count)>? FungibleIdAndCounts
```

### Field Value

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

## Memo

```
public readonly string? Memo
```

### Field Value

string

# Properties

## MailType

```
public override MailType MailType { get; }
```

### Property Value

MailType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class WorldBossRewardMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

Represents a worldboss reward mail that contains rewards such as fungible assets and items.

```
public class WorldBossRewardMail : Mail, IState
```

**Inheritance**

object ← Mail ← WorldBossRewardMail

**Implements**

IState

**Inherited Members**

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### WorldBossRewardMail(Dictionary)

Constructor for deserializing a WorldBossRewardMail from a Bencodex dictionary.

```
public WorldBossRewardMail(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

The serialized dictionary representation of the mail.

# WorldBossRewardMail(long, Guid, long, List<FungibleAssetValue>, List<(int id, int count)>)

Constructor for creating a WorldBossRewardMail with specified parameters.

```
public WorldBossRewardMail(long blockIndex, Guid id, long requiredBlockIndex,
List<FungibleAssetValue> fungibleAssetValues, List<(int id, int count)> items)
```

## Parameters

`blockIndex` long

The block index at which the mail is created.

`id` Guid

The unique identifier for the mail.

`requiredBlockIndex` long

The block index required to claim the mail.

`fungibleAssetValues` List<FungibleAssetValue>

The list of fungible asset rewards.

`items` List<(int id, int count)>

The list of item rewards.

# Fields

# FungibleAssetValues

A list of fungible asset values included in the reward.

```
public List<FungibleAssetValue> FungibleAssetValues
```

## Field Value

List<FungibleAssetValue>

## Items

A list of item rewards where each entry contains the item ID and count.

```
public List<(int id, int count)> Items
```

### Field Value

List<(int id, int count)>

# Properties

## MailType

The type of mail.

```
public override MailType MailType { get; }
```

### Property Value

MailType

## TypeId

The type identifier for the WorldBossRewardMail class.

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Read(IMail)

Reads the mail contents into the specified mail object.

```
public override void Read(IMail mail)
```

### Parameters

`mail` IMail

The mail object to populate with this mail's data.

## Serialize()

Serializes the WorldBossRewardMail into a Bencodex-compatible format.

```
public override IValue Serialize()
```

### Returns

IValue

A serialized representation of the mail.

# Namespace Nekoyume.Model.Market

## Classes

FavProduct

ItemProduct

Product

ProductFactory

ProductReceipt

ProductsState

## Enums

ProductType

# Class FavProduct

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class FavProduct : Product
```

**Inheritance**

object ← Product ← FavProduct

**Inherited Members**

Product.DeriveAddress(Guid) , Product.ProductId , Product.Type , Product.Price , Product.RegisteredBlockIndex , Product.SellerAvatarAddress , Product.SellerAgentAddress , Product.Validate(IProductInfo) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## FavProduct()

```
public FavProduct()
```

## FavProduct(List)

```
public FavProduct(List serialized)
```

### Parameters

`serialized` List

# Fields

## Asset

```
public FungibleAssetValue Asset
```

### Field Value

FungibleAssetValue

## Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ItemProduct

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class ItemProduct : Product
```

**Inheritance**

object ← Product ← ItemProduct

**Inherited Members**

Product.DeriveAddress(Guid) , Product.ProductId , Product.Type , Product.Price , Product.RegisteredBlockIndex , Product.SellerAvatarAddress , Product.SellerAgentAddress , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemProduct()

```
public ItemProduct()
```

## ItemProduct(List)

```
public ItemProduct(List serialized)
```

### Parameters

`serialized` List

# Fields

## ItemCount

```
public int ItemCount
```

### Field Value

int

## TradableItem

```
public ITradableItem TradableItem
```

### Field Value

ITradableItem

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Validate(IProductInfo)

```
public override void Validate(IProductInfo productInfo)
```

### Parameters

`productInfo` IProductInfo

# Class Product

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class Product
```

**Inheritance**

object ← Product

**Derived**

FavProduct, ItemProduct

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Product()

```
protected Product()
```

### Product(List)

```
protected Product(List serialized)
```

#### Parameters

`serialized` List

## Fields

## Price

```
public FungibleAssetValue Price
```

### Field Value

FungibleAssetValue

## ProductId

```
public Guid ProductId
```

### Field Value

Guid

## RegisteredBlockIndex

```
public long RegisteredBlockIndex
```

### Field Value

long

## SellerAgentAddress

```
public Address SellerAgentAddress
```

### Field Value

Address

## SellerAvatarAddress

```
public Address SellerAvatarAddress
```

### Field Value

Address

## Type

```
public ProductType Type
```

### Field Value

ProductType

## Methods

## DeriveAddress(Guid)

```
public static Address DeriveAddress(Guid productId)
```

### Parameters

`productId` Guid

### Returns

Address

## Serialize()

```
public virtual IValue Serialize()
```

### Returns

# Validate(IProductInfo)

```
public virtual void Validate(IProductInfo productInfo)
```

## Parameters

`productInfo` IProductInfo

# Class ProductFactory

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public static class ProductFactory
```

**Inheritance**

object ← ProductFactory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## DeserializeProduct(List)

```
public static Product DeserializeProduct(List serialized)
```

### Parameters

`serialized` List

### Returns

Product

## DeserializeProductInfo(List)

```
public static IProductInfo DeserializeProductInfo(List serialized)
```

### Parameters

`serialized` List

### Returns

IProductInfo

## DeserializeRegisterInfo(List)

```
public static IRegisterInfo DeserializeRegisterInfo(List serialized)
```

### Parameters

`serialized` List

### Returns

IRegisterInfo

# Class ProductReceipt

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class ProductReceipt
```

**Inheritance**

object ← ProductReceipt

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ProductReceipt(List)

```
public ProductReceipt(List serialized)
```

#### Parameters

`serialized` List

### ProductReceipt(Guid, Address, Address, FungibleAssetValue, long)

```
public ProductReceipt(Guid productId, Address sellerAvatarAddress, Address
buyerAvatarAddress, FungibleAssetValue price, long purchasedBlockIndex)
```

#### Parameters

`productId` Guid

`sellerAvatarAddress` Address

`buyerAvatarAddress` Address

`price` FungibleAssetValue

`purchasedBlockIndex` long

# Fields

## BuyerAvatarAddress

```
public readonly Address BuyerAvatarAddress
```

### Field Value

Address

## Price

```
public FungibleAssetValue Price
```

### Field Value

FungibleAssetValue

## ProductId

```
public readonly Guid ProductId
```

### Field Value

Guid

## PurchasedBlockIndex

```
public long PurchasedBlockIndex
```

### Field Value

long

## SellerAvatarAddress

```
public readonly Address SellerAvatarAddress
```

### Field Value

Address

# Methods

## DeriveAddress(Guid)

```
public static Address DeriveAddress(Guid productId)
```

### Parameters

`productId` Guid

### Returns

Address

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Enum ProductType

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public enum ProductType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Fungible = 0`

`FungibleAssetValue = 1`

`NonFungible = 2`

# Class ProductsState

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class ProductsState
```

**Inheritance**

object ← ProductsState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ProductsState()

```
public ProductsState()
```

### ProductsState(List)

```
public ProductsState(List serialized)
```

#### Parameters

`serialized` List

## Fields

### ProductIds

```
public List<Guid> ProductIds
```

### Field Value

List<Guid>

## Methods

### DeriveAddress(Address)

```
public static Address DeriveAddress(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

### Returns

Address

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Namespace Nekoyume.Model.Pet

## Enums

PetOptionType

# Enum PetOptionType

Namespace: Nekoyume.Model.Pet

Assembly: Lib9c.dll

```
public enum PetOptionType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`AdditionalOptionRate = 2`

`AdditionalOptionRateByFixedValue = 3`

`DiscountMaterialCostCrystal = 5`

`IncreaseBlockPerHourglass = 4`

`ReduceRequiredBlock = 0`

`ReduceRequiredBlockByFixedValue = 1`

# Namespace Nekoyume.Model.Quest

## Classes

CollectQuest

CombinationEquipmentQuest

CombinationQuest

GeneralQuest

GoldQuest

ItemEnhancementQuest

ItemGradeQuest

ItemTypeCollectQuest

MonsterQuest

Quest

QuestList

QuestReward

TradeQuest

UpdateListQuestsCountException

UpdateListVersionException

WorldQuest

## Enums

QuestEventType

QuestType

# Class CollectQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class CollectQuest : Quest, IState
```

### Inheritance

object ← Quest ← CollectQuest

### Implements

IState

### Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CollectQuest(Dictionary)

```
public CollectQuest(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### CollectQuest(List)

```
public CollectQuest(List serialized)
```

### Parameters

`serialized` List

## CollectQuest(Row, QuestReward)

```
public CollectQuest(CollectQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` CollectQuestSheet.Row

`reward` QuestReward

# Fields

## ItemId

```
public readonly int ItemId
```

### Field Value

int

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public override IValue SerializeList()
```

### Returns

IValue

## Update(CollectionMap)

```
public void Update(CollectionMap itemMap)
```

### Parameters

`itemMap` CollectionMap

# Class CombinationEquipmentQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
public class CombinationEquipmentQuest : Quest, IState
```

**Inheritance**

object ← Quest ← CombinationEquipmentQuest

**Implements**

IState

**Inherited Members**

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CombinationEquipmentQuest(Dictionary)

```
public CombinationEquipmentQuest(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### CombinationEquipmentQuest(List)

```
public CombinationEquipmentQuest(List serialized)
```

### Parameters

`serialized` List

## CombinationEquipmentQuest(Row, QuestReward, int)

```
public CombinationEquipmentQuest(QuestSheet.Row data, QuestReward reward,
int stageId)
```

### Parameters

`data` QuestSheet.Row

`reward` QuestReward

`stageId` int

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## RecipeId

```
public int RecipeId { get; }
```

### Property Value

int

## StageId

```
public int StageId { get; }
```

### Property Value

int

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public override IValue SerializeList()
```

### Returns

IValue

## Update(int)

```
public void Update(int recipeId)
```

### Parameters

`recipeId` int

# Class CombinationQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationQuest : Quest, IState
```

**Inheritance**

object ← Quest ← CombinationQuest

**Implements**

IState

**Inherited Members**

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CombinationQuest(Dictionary)

```
public CombinationQuest(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## CombinationQuest(List)

```
public CombinationQuest(List serialized)
```

### Parameters

`serialized` List

## CombinationQuest(Row, QuestReward)

```
public CombinationQuest(CombinationQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` CombinationQuestSheet.Row

`reward` QuestReward

# Properties

## ItemSubType

```
public ItemSubType ItemSubType { get; }
```

### Property Value

ItemSubType

## ItemType

```
public ItemType ItemType { get; }
```

### Property Value

ItemType

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public override IValue SerializeList()
```

### Returns

IValue

## Update(List<ItemBase>)

```
public void Update(List<ItemBase> items)
```

### Parameters

`items` List<ItemBase>

# Class GeneralQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class GeneralQuest : Quest, IState
```

**Inheritance**

object ← Quest ← GeneralQuest

**Implements**

IState

**Inherited Members**

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GeneralQuest(Dictionary)

```
public GeneralQuest(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## GeneralQuest(List)

```
public GeneralQuest(List serialized)
```

### Parameters

`serialized` List

## GeneralQuest(Row, QuestReward)

```
public GeneralQuest(GeneralQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` GeneralQuestSheet.Row

`reward` QuestReward

# Fields

## Event

```
public readonly QuestEventType Event
```

### Field Value

QuestEventType

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public override IValue SerializeList()
```

### Returns

IValue

## Update(CollectionMap)

```
public void Update(CollectionMap eventMap)
```

### Parameters

`eventMap` CollectionMap

# Class GoldQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class GoldQuest : Quest, IState
```

### Inheritance

object ← Quest ← GoldQuest

### Implements

IState

### Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GoldQuest(Dictionary)

```
public GoldQuest(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### GoldQuest(List)

```
public GoldQuest(List serialized)
```

### Parameters

`serialized` List

## GoldQuest(Row, QuestReward)

```
public GoldQuest(GoldQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` GoldQuestSheet.Row

`reward` QuestReward

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## Type

```
public TradeType Type { get; }
```

### Property Value

TradeType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Check()

```
public override void Check()
```

### GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

### Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

### SerializeList()

```
public override IValue SerializeList()
```

## Returns

IValue

# Update(FungibleAssetValue)

```
public void Update(FungibleAssetValue gold)
```

## Parameters

`gold` FungibleAssetValue

# Class ItemEnhancementQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancementQuest : Quest, IState
```

**Inheritance**

object ← Quest ← ItemEnhancementQuest

**Implements**

IState

**Inherited Members**

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ItemEnhancementQuest(Dictionary)

```
public ItemEnhancementQuest(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ItemEnhancementQuest(List)

```
public ItemEnhancementQuest(List serialized)
```

### Parameters

`serialized` List

## ItemEnhancementQuest(Row, QuestReward)

```
public ItemEnhancementQuest(ItemEnhancementQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` ItemEnhancementQuestSheet.Row

`reward` QuestReward

# Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## Grade

```
public int Grade { get; }
```

### Property Value

int

## Progress

```
public override float Progress { get; }
```

### Property Value

float

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public override IValue SerializeList()
```

### Returns

IValue

## Update(Equipment)

```
public void Update(Equipment equipment)
```

### Parameters

`equipment` Equipment

# Class ItemGradeQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class ItemGradeQuest : Quest, IState
```

**Inheritance**

object ← Quest ← ItemGradeQuest

**Implements**

IState

**Inherited Members**

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemGradeQuest(Dictionary)

```
public ItemGradeQuest(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## ItemGradeQuest(List)

```
public ItemGradeQuest(List serialized)
```

### Parameters

`serialized` List

## ItemGradeQuest(Row, QuestReward)

```
public ItemGradeQuest(ItemGradeQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` ItemGradeQuestSheet.Row

`reward` QuestReward

# Fields

## Grade

```
public readonly int Grade
```

### Field Value

int

## ItemIds

```
public readonly List<int> ItemIds
```

### Field Value

List<int>

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public override IValue SerializeList()
```

### Returns

IValue

## Update(ItemUsable)

```
public void Update(ItemUsable itemUsable)
```

### Parameters

`itemUsable` ItemUsable

# Class ItemTypeCollectQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class ItemTypeCollectQuest : Quest, IState
```

**Inheritance**

object ← Quest ← ItemTypeCollectQuest

**Implements**

IState

**Inherited Members**

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemTypeCollectQuest(Dictionary)

```
public ItemTypeCollectQuest(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## ItemTypeCollectQuest(List)

```
public ItemTypeCollectQuest(List serialized)
```

### Parameters

`serialized` List

## ItemTypeCollectQuest(Row, QuestReward)

```
public ItemTypeCollectQuest(ItemTypeCollectQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` ItemTypeCollectQuestSheet.Row

`reward` QuestReward

# Fields

## ItemIds

```
public readonly List<int> ItemIds
```

### Field Value

List<int>

## ItemType

```
public readonly ItemType ItemType
```

### Field Value

ItemType

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public override IValue SerializeList()
```

### Returns

IValue

## Update(ItemBase)

```
public void Update(ItemBase item)
```

### Parameters

`item` ItemBase

# Class MonsterQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterQuest : Quest, IState
```

### Inheritance

object ← Quest ← MonsterQuest

### Implements

IState

### Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MonsterQuest(Dictionary)

```
public MonsterQuest(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### MonsterQuest(List)

```
public MonsterQuest(List serialized)
```

### Parameters

`serialized` List

## MonsterQuest(Row, QuestReward)

```
public MonsterQuest(MonsterQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` MonsterQuestSheet.Row

`reward` QuestReward

# Fields

## MonsterId

```
public readonly int MonsterId
```

### Field Value

int

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public override IValue SerializeList()
```

### Returns

IValue

## Update(CollectionMap)

```
public void Update(CollectionMap monsterMap)
```

### Parameters

`monsterMap` CollectionMap

# Class Quest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Quest : IState
```

**Inheritance**

object ← Quest

**Implements**

IState

**Derived**

CollectQuest, CombinationEquipmentQuest, CombinationQuest, GeneralQuest, GoldQuest, ItemEnhancementQuest, ItemGradeQuest, ItemTypeCollectQuest, MonsterQuest, TradeQuest, WorldQuest

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Quest(Dictionary)

```
protected Quest(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## Quest(List)

```
protected Quest(List serialized)
```

### Parameters

`serialized` List

## Quest(Row, QuestReward)

```
protected Quest(QuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` QuestSheet.Row

`reward` QuestReward

# Fields

## GoalFormat

```
public const string GoalFormat = "({0}/{1})"
```

### Field Value

string

## _current

```
protected int _current
```

### Field Value

int

## isReceivable

```
[NonSerialized]
public bool isReceivable
```

### Field Value

bool

# Properties

## Complete

```
public bool Complete { get; protected set; }
```

### Property Value

bool

## Goal

```
public int Goal { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

# IsPaidInAction

이미 퀘스트 보상이 액션에서 지급되었는가?

```
public bool IsPaidInAction { get; set; }
```

## Property Value

bool

# Progress

```
public virtual float Progress { get; }
```

## Property Value

float

# QuestType

```
public abstract QuestType QuestType { get; }
```

## Property Value

QuestType

# Reward

```
public QuestReward Reward { get; }
```

## Property Value

QuestReward

## TypeId

```
protected abstract string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public abstract void Check()
```

## Deserialize(Dictionary)

```
public static Quest Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

Quest

## Deserialize(IValue)

```
public static Quest Deserialize(IValue arg)
```

### Parameters

`arg` IValue

### Returns

Quest

## DeserializeList(List)

```
public static Quest DeserializeList(List serialized)
```

### Parameters

`serialized` List

### Returns

Quest

## GetProgressText()

```
public abstract string GetProgressText()
```

### Returns

string

## Serialize()

```
public virtual IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public virtual IValue SerializeList()
```

## Returns

IValue

# Enum QuestEventType

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
public enum QuestEventType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Complete = 4`

`Consumable = 6`

`Create = 0`

`Die = 3`

`Enhancement = 1`

`Equipment = 5`

`Level = 2`

# Class QuestList

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class QuestList : IEnumerable<Quest>, IEnumerable, IState, ICloneable
```

**Inheritance**

object ← QuestList

**Implements**

IEnumerable<Quest>, IEnumerable, IState, ICloneable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### QuestList(Dictionary)

```
public QuestList(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### QuestList(List)

```
public QuestList(List serialized)
```

#### Parameters

`serialized` List

## QuestList(QuestSheet, QuestRewardSheet, QuestItemRewardSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheet)

```
public QuestList(QuestSheet questSheet, QuestRewardSheet questRewardSheet,
QuestItemRewardSheet questItemRewardSheet, EquipmentItemRecipeSheet
equipmentItemRecipeSheet, EquipmentItemSubRecipeSheet equipmentItemSubRecipeSheet)
```

### Parameters

`questSheet` QuestSheet

`questRewardSheet` QuestRewardSheet

`questItemRewardSheet` QuestItemRewardSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`equipmentItemSubRecipeSheet` EquipmentItemSubRecipeSheet

# Fields

## CompletedQuestIdsKey

```
public const string CompletedQuestIdsKey = "c"
```

### Field Value

string

## ListVersionKey

```
public const string ListVersionKey = "v"
```

### Field Value

string

## QuestsKey

```
public const string QuestsKey = "q"
```

### Field Value

string

## completedQuestIds

```
public List<int> completedQuestIds
```

### Field Value

List<int>

# Properties

## ListVersion

```
public int ListVersion { get; }
```

### Property Value

int

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Quest> GetEnumerator()
```

### Returns

IEnumerator<Quest>

An enumerator that can be used to iterate through the collection.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## SerializeDictionary()

```
[Obsolete("Dictionary type is obsolete.")]
public IValue SerializeDictionary()
```

### Returns

IValue

## SerializeList()

```
public IValue SerializeList()
```

### Returns

IValue

## UpdateCollectQuest(CollectionMap)

```
public void UpdateCollectQuest(CollectionMap itemMap)
```

### Parameters

`itemMap` CollectionMap

## UpdateCombinationEquipmentQuest(int)

```
public void UpdateCombinationEquipmentQuest(int recipeId)
```

### Parameters

`recipeId` int

## UpdateCombinationQuest(ItemUsable)

```
public void UpdateCombinationQuest(ItemUsable itemUsable)
```

### Parameters

`itemUsable` ItemUsable

## UpdateCompletedQuest(CollectionMap)

```
public CollectionMap UpdateCompletedQuest(CollectionMap eventMap)
```

### Parameters

`eventMap` CollectionMap

### Returns

CollectionMap

## UpdateGeneralQuest(IEnumerable<QuestEventType>, CollectionMap)

```
public CollectionMap UpdateGeneralQuest(IEnumerable<QuestEventType> types,
CollectionMap eventMap)
```

### Parameters

`types` IEnumerable<QuestEventType>

`eventMap` CollectionMap

### Returns

CollectionMap

## UpdateItemEnhancementQuest(Equipment)

```
public void UpdateItemEnhancementQuest(Equipment equipment)
```

### Parameters

`equipment` Equipment

## UpdateItemGradeQuest(ItemUsable)

```
public void UpdateItemGradeQuest(ItemUsable itemUsable)
```

### Parameters

`itemUsable` ItemUsable

## UpdateItemTypeCollectQuest(IEnumerable<ItemBase>)

```
public void UpdateItemTypeCollectQuest(IEnumerable<ItemBase> items)
```

### Parameters

`items` IEnumerable<ItemBase>

## UpdateList(QuestSheet, QuestRewardSheet, QuestItemRewardSheet, EquipmentItemRecipeSheet, ICollection<int>)

```
public void UpdateList(QuestSheet questSheet, QuestRewardSheet questRewardSheet,
QuestItemRewardSheet questItemRewardSheet, EquipmentItemRecipeSheet
equipmentItemRecipeSheet, ICollection<int> newIds)
```

### Parameters

`questSheet` QuestSheet

`questRewardSheet` QuestRewardSheet

`questItemRewardSheet` QuestItemRewardSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`newIds` ICollection<int>

# UpdateListV1(int, QuestSheet, QuestRewardSheet, QuestItemRewardSheet, EquipmentItemRecipeSheet)

```
[Obsolete("Use UpdateList()")]
public void UpdateListV1(int listVersion, QuestSheet questSheet, QuestRewardSheet
questRewardSheet, QuestItemRewardSheet questItemRewardSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet)
```

## Parameters

`listVersion` int

`questSheet` QuestSheet

`questRewardSheet` QuestRewardSheet

`questItemRewardSheet` QuestItemRewardSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

## Exceptions

UpdateListVersionException

UpdateListQuestsCountException

Exception

# UpdateMonsterQuest(CollectionMap)

```
public void UpdateMonsterQuest(CollectionMap monsterMap)
```

## Parameters

`monsterMap` CollectionMap

# UpdateStageQuest(CollectionMap)

```
public void UpdateStageQuest(CollectionMap stageMap)
```

### Parameters

`stageMap` CollectionMap

## UpdateTradeQuest(TradeType, FungibleAssetValue)

```
public void UpdateTradeQuest(TradeType type, FungibleAssetValue price)
```

### Parameters

`type` TradeType

`price` FungibleAssetValue

# Class QuestReward

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class QuestReward : IState
```

### Inheritance

object ← QuestReward

### Implements

IState

### Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### QuestReward(Dictionary)

```
public QuestReward(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### QuestReward(Dictionary<int, int>)

```
public QuestReward(Dictionary<int, int> map)
```

#### Parameters

`map` Dictionary<int, int>

# Fields

## ItemMap

```
public readonly IEnumerable<Tuple<int, int>> ItemMap
```

### Field Value

IEnumerable<Tuple<int, int>>

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Enum QuestType

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
public enum QuestType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Adventure = 0`

`Craft = 2`

`Exchange = 3`

`Obtain = 1`

# Class TradeQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class TradeQuest : Quest, IState
```

**Inheritance**

object ← Quest ← TradeQuest

**Implements**

IState

**Inherited Members**

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## TradeQuest(Dictionary)

```
public TradeQuest(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## TradeQuest(List)

```
public TradeQuest(List serialized)
```

### Parameters

`serialized` List

## TradeQuest(Row, QuestReward)

```
public TradeQuest(TradeQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` TradeQuestSheet.Row

`reward` QuestReward

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## Type

```
public TradeType Type { get; }
```

### Property Value

TradeType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

## Methods

### Check()

```
public override void Check()
```

### GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

### Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

### SerializeList()

```
public override IValue SerializeList()
```

# Returns

IValue

# Class UpdateListQuestsCountException

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class UpdateListQuestsCountException : ArgumentException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← ArgumentException ← UpdateListQuestsCountException

### Implements

ISerializable

### Inherited Members

ArgumentException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### UpdateListQuestsCountException()

```
public UpdateListQuestsCountException()
```

### UpdateListQuestsCountException(int, int)

```
public UpdateListQuestsCountException(int expected, int actual)
```

### Parameters

`expected` int

`actual` int

## UpdateListQuestsCountException(SerializationInfo, StreamingContext)

```
protected UpdateListQuestsCountException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## UpdateListQuestsCountException(string)

```
public UpdateListQuestsCountException(string s)
```

### Parameters

`s` string

# Class UpdateListVersionException

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class UpdateListVersionException : ArgumentOutOfRangeException, ISerializable
```

**Inheritance**

object ← Exception ← SystemException ← ArgumentException ← ArgumentOutOfRangeException ← UpdateListVersionException

**Implements**

ISerializable

**Inherited Members**

ArgumentOutOfRangeException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentOutOfRangeException.ActualValue , ArgumentOutOfRangeException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### UpdateListVersionException()

```
public UpdateListVersionException()
```

### UpdateListVersionException(int, int)

```
public UpdateListVersionException(int expected, int actual)
```

### Parameters

`expected` int

`actual` int

## UpdateListVersionException(SerializationInfo, StreamingContext)

```
protected UpdateListVersionException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## UpdateListVersionException(string)

```
public UpdateListVersionException(string s)
```

### Parameters

`s` string

# Class WorldQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class WorldQuest : Quest, IState
```

### Inheritance

object ← Quest ← WorldQuest

### Implements

IState

### Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Serialize() , Quest.SerializeList() , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### WorldQuest(Dictionary)

```
public WorldQuest(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### WorldQuest(List)

```
public WorldQuest(List serialized)
```

### Parameters

`serialized` List

## WorldQuest(Row, QuestReward)

```
public WorldQuest(WorldQuestSheet.Row data, QuestReward reward)
```

### Parameters

`data` WorldQuestSheet.Row

`reward` QuestReward

# Properties

## QuestType

```
public override QuestType QuestType { get; }
```

### Property Value

QuestType

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Check()

```
public override void Check()
```

## GetProgressText()

```
public override string GetProgressText()
```

### Returns

string

## Update(CollectionMap)

```
public void Update(CollectionMap stageMap)
```

### Parameters

`stageMap` CollectionMap

# Namespace Nekoyume.Model.Rune

## Classes

DuplicatedRuneIdException

DuplicatedRuneSlotIndexException

ForbiddenRuneEquippedException

ForbiddenRuneTypeEquippedException

IsEquippableRuneException

IsUsableSlotException

MismatchRuneSlotTypeException

RuneCostDataNotFoundException

RuneCostNotFoundException

RuneInfosIsEmptyException

RuneListNotFoundException

RuneNotFoundException

RuneSlot

RuneStateNotFoundException

SlotIsAlreadyUnlockedException

SlotIsLockedException

SlotNotFoundException

SlotRuneTypeException

TryCountIsZeroException

# Class DuplicatedRuneIdException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicatedRuneIdException : Exception, ISerializable
```

### Inheritance

object ← Exception ← DuplicatedRuneIdException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### DuplicatedRuneIdException(SerializationInfo, StreamingContext)

```
protected DuplicatedRuneIdException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicatedRuneIdException(string)

```
public DuplicatedRuneIdException(string message)
```

## Parameters

`message` string

# Class DuplicatedRuneSlotIndexException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicatedRuneSlotIndexException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← DuplicatedRuneSlotIndexException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### DuplicatedRuneSlotIndexException(SerializationInfo, StreamingContext)

```
protected DuplicatedRuneSlotIndexException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicatedRuneSlotIndexException(string)

```
public DuplicatedRuneSlotIndexException(string message)
```

## Parameters

`message` string

# Class ForbiddenRuneEquippedException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class ForbiddenRuneEquippedException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ForbiddenRuneEquippedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ForbiddenRuneEquippedException(SerializationInfo, StreamingContext)

```
protected ForbiddenRuneEquippedException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## ForbiddenRuneEquippedException(string)

```
public ForbiddenRuneEquippedException(string message)
```

### Parameters

`message` string

## ForbiddenRuneEquippedException(string, Exception)

```
public ForbiddenRuneEquippedException(string message, Exception innerException)
```

### Parameters

`message` string

`innerException` Exception

# Class ForbiddenRuneTypeEquippedException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class ForbiddenRuneTypeEquippedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← ForbiddenRuneTypeEquippedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ForbiddenRuneTypeEquippedException()

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException()
```

### ForbiddenRuneTypeEquippedException(List<RuneType>, List<RuneType>)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException(List<RuneType> forbiddenRuneTypes,
List<RuneType> equippedRuneTypes)
```

### Parameters

`forbiddenRuneTypes` List<RuneType>

The list of forbidden rune types.

`equippedRuneTypes` List<RuneType>

The list of equipped rune types that are forbidden.

## ForbiddenRuneTypeEquippedException(SerializationInfo, StreamingContext)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class for serialization.

```
protected ForbiddenRuneTypeEquippedException(SerializationInfo info,
StreamingContext context)
```

### Parameters

`info` SerializationInfo

The serialization info.

`context` StreamingContext

The streaming context.

## ForbiddenRuneTypeEquippedException(string)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException(string message)
```

### Parameters

`message` string

The error message.

## ForbiddenRuneTypeEquippedException(string, List<RuneType>, List<RuneType>)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException(string message, List<RuneType>
forbiddenRuneTypes, List<RuneType> equippedRuneTypes)
```

### Parameters

`message` string

The error message.

`forbiddenRuneTypes` List<RuneType>

The list of forbidden rune types.

`equippedRuneTypes` List<RuneType>

The list of equipped rune types that are forbidden.

## ForbiddenRuneTypeEquippedException(string, Exception)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException(string message, Exception innerException)
```

### Parameters

`message` string

The error message.

`innerException` Exception

The inner exception.

# Properties

## EquippedRuneTypes

Gets the list of equipped rune types that are forbidden.

```
public List<RuneType> EquippedRuneTypes { get; }
```

### Property Value

List<RuneType>

## ForbiddenRuneTypes

Gets the list of forbidden rune types that were equipped.

```
public List<RuneType> ForbiddenRuneTypes { get; }
```

### Property Value

List<RuneType>

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

Gets object data for serialization.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The serialization info.

`context` StreamingContext

The streaming context.

# Class IsEquippableRuneException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class IsEquippableRuneException : Exception, ISerializable
```

### Inheritance

object ← Exception ← IsEquippableRuneException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### IsEquippableRuneException(SerializationInfo, StreamingContext)

```
protected IsEquippableRuneException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# IsEquippableRuneException(string)

```
public IsEquippableRuneException(string message)
```

## Parameters

`message` string

# Class IsUsableSlotException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class IsUsableSlotException : Exception, ISerializable
```

### Inheritance

object ← Exception ← IsUsableSlotException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### IsUsableSlotException(SerializationInfo, StreamingContext)

```
protected IsUsableSlotException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# IsUsableSlotException(string)

```
public IsUsableSlotException(string message)
```

## Parameters

`message` string

# Class MismatchRuneSlotTypeException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class MismatchRuneSlotTypeException : Exception, ISerializable
```

### Inheritance

object ← Exception ← MismatchRuneSlotTypeException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### MismatchRuneSlotTypeException(SerializationInfo, StreamingContext)

```
protected MismatchRuneSlotTypeException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# MismatchRuneSlotTypeException(string)

```
public MismatchRuneSlotTypeException(string message)
```

## Parameters

`message` string

# Class RuneCostDataNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneCostDataNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← RuneCostDataNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RuneCostDataNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneCostDataNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# RuneCostDataNotFoundException(string)

```
public RuneCostDataNotFoundException(string message)
```

## Parameters

`message` string

# Class RuneCostNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneCostNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← RuneCostNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RuneCostNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneCostNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# RuneCostNotFoundException(string)

```
public RuneCostNotFoundException(string message)
```

## Parameters

`message` string

# Class RuneInfosIsEmptyException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneInfosIsEmptyException : Exception, ISerializable
```

### Inheritance

object ← Exception ← RuneInfosIsEmptyException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RuneInfosIsEmptyException(SerializationInfo, StreamingContext)

```
protected RuneInfosIsEmptyException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# RuneInfosIsEmptyException(string)

```
public RuneInfosIsEmptyException(string message)
```

## Parameters

`message` string

# Class RuneListNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneListNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← RuneListNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RuneListNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneListNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# RuneListNotFoundException(string)

```
public RuneListNotFoundException(string message)
```

## Parameters

`message` string

# Class RuneNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← RuneNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RuneNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneNotFoundException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# RuneNotFoundException(string)

```
public RuneNotFoundException(string message)
```

## Parameters

`message` string

# Class RuneSlot

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
public class RuneSlot : IState
```

**Inheritance**

object ← RuneSlot

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RuneSlot(List)

```
public RuneSlot(List serialized)
```

#### Parameters

`serialized` List

### RuneSlot(int, RuneSlotType, RuneType, bool)

```
public RuneSlot(int index, RuneSlotType runeSlotType, RuneType runeType,
bool isLock)
```

#### Parameters

`index` int

`runeSlotType` RuneSlotType

`runeType` RuneType

`isLock` bool

# Properties

## Index

```
public int Index { get; }
```

### Property Value

int

## IsLock

```
public bool IsLock { get; }
```

### Property Value

bool

## RuneId

```
public int? RuneId { get; }
```

### Property Value

int?

## RuneSlotType

```
public RuneSlotType RuneSlotType { get; }
```

### Property Value

RuneSlotType

## RuneType

```
public RuneType RuneType { get; }
```

### Property Value

RuneType

# Methods

## Equip(int)

```
public void Equip(int runeId)
```

### Parameters

`runeId` int

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Unequip()

```
public void Unequip()
```

## Unlock()

```
public void Unlock()
```

# Class RuneStateNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneStateNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← RuneStateNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### RuneStateNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneStateNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# RuneStateNotFoundException(string)

```
public RuneStateNotFoundException(string message)
```

## Parameters

`message` string

# Class SlotIsAlreadyUnlockedException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class SlotIsAlreadyUnlockedException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← SlotIsAlreadyUnlockedException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SlotIsAlreadyUnlockedException(SerializationInfo, StreamingContext)

```
protected SlotIsAlreadyUnlockedException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# SlotIsAlreadyUnlockedException(string)

```
public SlotIsAlreadyUnlockedException(string message)
```

## Parameters

`message` string

# Class SlotIsLockedException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class SlotIsLockedException : Exception, ISerializable
```

### Inheritance

object ← Exception ← SlotIsLockedException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SlotIsLockedException(SerializationInfo, StreamingContext)

```
protected SlotIsLockedException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# SlotIsLockedException(string)

```
public SlotIsLockedException(string message)
```

## Parameters

`message` string

# Class SlotNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class SlotNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← SlotNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SlotNotFoundException(SerializationInfo, StreamingContext)

```
protected SlotNotFoundException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# SlotNotFoundException(string)

```
public SlotNotFoundException(string message)
```

## Parameters

`message` string

# Class SlotRuneTypeException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class SlotRuneTypeException : Exception, ISerializable
```

### Inheritance

object ← Exception ← SlotRuneTypeException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SlotRuneTypeException(SerializationInfo, StreamingContext)

```
protected SlotRuneTypeException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# SlotRuneTypeException(string)

```
public SlotRuneTypeException(string message)
```

## Parameters

`message` string

# Class TryCountIsZeroException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class TryCountIsZeroException : Exception, ISerializable
```

### Inheritance

object ← Exception ← TryCountIsZeroException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### TryCountIsZeroException(SerializationInfo, StreamingContext)

```
public TryCountIsZeroException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# TryCountIsZeroException(string)

```
public TryCountIsZeroException(string msg)
```

## Parameters

`msg` string

# Namespace Nekoyume.Model.Skill

## Classes

AreaAttack

AttackSkill

BlowAttack

BuffRemovalAttack

BuffSkill

DoubleAttack

FullBuffRemovalAttack
An attack skill that deals damage and removes all positive stat buffs from the target. Unlike BuffRemovalAttack which removes only the most recent buff, this skill removes every positive stat buff at once.

HealSkill

NormalAttack

ShatterStrike

Skill
Abstract base class for all skills in the game. Provides common functionality for skill execution and serialization.

SkillFactory

SkillTargetTypeExtension

UnexpectedOperationException

## Structs

SkillCustomField

## Interfaces

ISkill

## Enums

ActionBuffType

SkillCategory

SkillTargetType

SkillType

# Enum ActionBuffType

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public enum ActionBuffType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Bleed = 0`

`Dispel = 4`

`Focus = 3`

`IceShield = 5`

`Stun = 1`

`Vampiric = 2`

# Class AreaAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class AreaAttack : AttackSkill, IState, ISkill
```

### Inheritance

object ← Skill ← AttackSkill ← AreaAttack

### Implements

IState, ISkill

### Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AreaAttack(Row, long, int, int, StatType)

```
public AreaAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Class AttackSkill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public abstract class AttackSkill : Skill, IState, ISkill
```

### Inheritance

object ← Skill ← AttackSkill

### Implements

IState, ISkill

### Derived

AreaAttack, BlowAttack, BuffRemovalAttack, DoubleAttack, FullBuffRemovalAttack, NormalAttack, ShatterStrike

### Inherited Members

Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Use(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AttackSkill(Row, long, int, int, StatType)

```
protected AttackSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## GetMultiplier(int, decimal)

Calculates damage multipliers for multi-hit skills. For single hit skills, returns the total damage as is. For multi-hit skills, distributes the total damage across hits with the last hit dealing 30% more damage. This creates a pattern where the final hit is more impactful than the preceding hits.

```
protected static decimal[] GetMultiplier(int hitCount, decimal totalDamage)
```

### Parameters

`hitCount` int

Number of hits for the skill

`totalDamage` decimal

Total damage to be distributed across hits

### Returns

decimal[]

Array of damage multipliers for each hit

## ProcessDamage(CharacterBase, int, bool, bool)

Processes damage calculation for attack skills with improved large value handling. Uses NumberConversionHelper.SafeDecimalToInt64 to prevent overflow when dealing with large

stat values. Implements DamageHelper.GetFinalDefense and DamageHelper.GetReducedDamage for consistent damage calculation.

todo: 캐릭터 스탯에 반영된 버프 효과가 스킬의 순수 데미지에는 영향을 주지 않는 로직. todo: 타겟의 회피 여부와 상관없이 버프의 확률로 발동되고 있음. 고민이 필요함.

```
protected IEnumerable<Skill.SkillInfo> ProcessDamage(CharacterBase caster, int
simulatorWaveTurn, bool isNormalAttack = false, bool copyCharacter = true)
```

## Parameters

`caster` CharacterBase

The character casting the skill

`simulatorWaveTurn` int

Current wave turn in the battle simulator

`isNormalAttack` bool

Whether this is a normal attack

`copyCharacter` bool

Whether to copy the target character for skill info

## Returns

IEnumerable<Skill.SkillInfo>

Collection of skill information including damage and effects

# Class BlowAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class BlowAttack : AttackSkill, IState, ISkill
```

### Inheritance

object ← Skill ← AttackSkill ← BlowAttack

### Implements

IState, ISkill

### Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BlowAttack(Row, long, int, int, StatType)

```
public BlowAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Class BuffRemovalAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class BuffRemovalAttack : AttackSkill, IState, ISkill
```

**Inheritance**

object ← Skill ← AttackSkill ← BuffRemovalAttack

**Implements**

IState, ISkill

**Inherited Members**

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## BuffRemovalAttack(Row, long, int, int, StatType)

```
public BuffRemovalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Class BuffSkill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class BuffSkill : Skill, IState, ISkill
```

### Inheritance

object ← Skill ← BuffSkill

### Implements

IState, ISkill

### Inherited Members

Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### BuffSkill(Row, long, int, int, StatType)

```
public BuffSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Class DoubleAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class DoubleAttack : AttackSkill, IState, ISkill
```

### Inheritance

object ← Skill ← AttackSkill ← DoubleAttack

### Implements

IState, ISkill

### Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### DoubleAttack(Row, long, int, int, StatType)

```
public DoubleAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Class FullBuffRemovalAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

An attack skill that deals damage and removes all positive stat buffs from the target. Unlike BuffRemovalAttack which removes only the most recent buff, this skill removes every positive stat buff at once.

```
[Serializable]
public class FullBuffRemovalAttack : AttackSkill, IState, ISkill
```

**Inheritance**

object ← Skill ← AttackSkill ← FullBuffRemovalAttack

**Implements**

IState, ISkill

**Inherited Members**

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## FullBuffRemovalAttack(Row, long, int, int, StatType)

```
public FullBuffRemovalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Class HealSkill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class HealSkill : Skill, IState, ISkill
```

### Inheritance

object ← Skill ← HealSkill

### Implements

IState, ISkill

### Inherited Members

Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### HealSkill(Row, long, int, int, StatType)

```
public HealSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## ProcessHeal(CharacterBase, int, bool)

```
protected IEnumerable<Skill.SkillInfo> ProcessHeal(CharacterBase caster, int
simulatorWaveTurn, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`copyCharacter` bool

### Returns

IEnumerable<Skill.SkillInfo>

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Interface ISkill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public interface ISkill
```

## Properties

### Chance

```
int Chance { get; }
```

#### Property Value

int

### CustomField

```
SkillCustomField? CustomField { get; }
```

#### Property Value

SkillCustomField?

### Power

Determines damage of `AttackSkill`. Determines effect of `BuffSkill`.

```
long Power { get; }
```

#### Property Value

long

## ReferencedStatType

```
StatType ReferencedStatType { get; }
```

### Property Value

StatType

## SkillRow

```
SkillSheet.Row SkillRow { get; }
```

### Property Value

SkillSheet.Row

## StatPowerRatio

```
int StatPowerRatio { get; }
```

### Property Value

int

# Methods

## IsBuff()

```
bool IsBuff()
```

### Returns

bool

## IsDebuff()

```
bool IsDebuff()
```

### Returns

bool

## Update(int, long, int)

```
void Update(int chance, long power, int statPowerRatio)
```

### Parameters

`chance` int

`power` long

`statPowerRatio` int

# Class NormalAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class NormalAttack : AttackSkill, IState, ISkill
```

### Inheritance

object ← Skill ← AttackSkill ← NormalAttack

### Implements

IState, ISkill

### Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### NormalAttack(Row, long, int, int, StatType)

```
public NormalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Class ShatterStrike

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class ShatterStrike : AttackSkill, IState, ISkill
```

### Inheritance

object ← Skill ← AttackSkill ← ShatterStrike

### Implements

IState, ISkill

### Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ShatterStrike(Row, long, int, int, StatType)

```
public ShatterStrike(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Class Skill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

Abstract base class for all skills in the game. Provides common functionality for skill execution and serialization.

```
[Serializable]
public abstract class Skill : IState, ISkill
```

**Inheritance**

object ← Skill

**Implements**

IState, ISkill

**Derived**

AttackSkill, BuffSkill, HealSkill

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

Skills are categorized by their type (Attack, Heal, Buff, Debuff) and category (NormalAttack, DoubleAttack, BlowAttack, etc.). Each skill has power, chance, and optional stat-based damage ratio.

Serialization format has been migrated from Dictionary to List for better performance and consistency. The new format stores data in the following order: [0] SkillRow (serialized SkillSheet.Row) [1] Power (long) [2] Chance (int) [3] StatPowerRatio (int, optional - only included if > 0) [4] ReferencedStatType (StatType, optional - only included if StatPowerRatio > 0)

Backward compatibility is maintained through SkillFactory.Deserialize() which automatically detects and handles both Dictionary and List formats.

# Constructors

## Skill(Row, long, int, int, StatType)

```
protected Skill(SkillSheet.Row skillRow, long power, int chance, int statPowerRatio,
StatType referencedStatType)
```

### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Properties

## Chance

```
public int Chance { get; }
```

### Property Value

int

## CustomField

```
public SkillCustomField? CustomField { get; set; }
```

### Property Value

SkillCustomField?

## Power

Determines damage of `AttackSkill`. Determines effect of `BuffSkill`.

```
public long Power { get; }
```

### Property Value

long

## ReferencedStatType

```
public StatType ReferencedStatType { get; }
```

### Property Value

StatType

## SkillRow

```
public SkillSheet.Row SkillRow { get; }
```

### Property Value

SkillSheet.Row

## StatPowerRatio

```
public int StatPowerRatio { get; }
```

### Property Value

int

# Methods

## Equals(Skill)

```
protected bool Equals(Skill other)
```

### Parameters

`other` Skill

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## IsBuff()

Determines if this skill is a buff skill.

```
public bool IsBuff()
```

### Returns

bool

True if the skill type is Buff, false otherwise.

## IsDebuff()

Determines if this skill is a debuff skill.

```
public bool IsDebuff()
```

### Returns

bool

True if the skill type is Debuff, false otherwise.

## ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool)

```
protected IEnumerable<Skill.SkillInfo> ProcessBuff(CharacterBase caster, int
simulatorWaveTurn, IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

IEnumerable<Skill.SkillInfo>

## Serialize()

Serializes the skill to a List format for storage and transmission.

```
public IValue Serialize()
```

### Returns

IValue

A List containing the serialized skill data in the following order: [0] SkillRow (serialized SkillSheet.Row) [1] Power (long) [2] Chance (int) [3] StatPowerRatio (int, optional - only included if > 0) [4] ReferencedStatType (StatType, optional - only included if StatPowerRatio > 0)

### Remarks

This method implements the new List-based serialization format for better performance and consistency. The format is more compact than the previous Dictionary format and maintains all necessary data.

Optional fields (StatPowerRatio and ReferencedStatType) are only included when they have non-default values to minimize serialization overhead.

For backward compatibility, use SkillFactory.Deserialize() which can handle both the new List format and the legacy Dictionary format.

## Update(int, long, int)

```
public void Update(int chance, long power, int statPowerRatio)
```

### Parameters

`chance` int

`power` long

`statPowerRatio` int

## Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public abstract Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

### Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

### Returns

Skill

# Enum SkillCategory

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public enum SkillCategory
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`AreaAttack = 3`

`AttackBuff = 8`

`BlowAttack = 1`

`Buff = 15`

`BuffRemovalAttack = 4`

`CriticalBuff = 10`

`CriticalDamageBuff = 14`

`DamageReductionBuff = 13`

`Debuff = 16`

`DefenseBuff = 9`

`Dispel = 19`

`DoubleAttack = 2`

`Focus = 18`

`FullBuffRemovalAttack = 20`

Attack enemy and remove ALL positive stat buffs.

`HPBuff = 7`

```
Heal = 6

HitBuff = 11

NormalAttack = 0

ShatterStrike = 5

SpeedBuff = 12

TickDamage = 17
```

# Struct SkillCustomField

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public struct SkillCustomField
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Properties

### BuffDuration

```
public int BuffDuration { readonly get; set; }
```

#### Property Value

int

### BuffValue

```
public long BuffValue { readonly get; set; }
```

#### Property Value

long

# Class SkillFactory

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public static class SkillFactory
```

**Inheritance**

object ← SkillFactory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## Deserialize(IValue)

Deserializes a skill from serialized data. Supports both Dictionary and List formats for backward compatibility.

```
public static Skill Deserialize(IValue serialized)
```

### Parameters

`serialized` IValue

The serialized skill data

### Returns

Skill

The deserialized skill

### Remarks

This method automatically detects the serialization format and delegates to the appropriate deserialization method. It supports both the new List-based format and the legacy Dictionary format to ensure backward compatibility with existing data.

List format (new): [SkillRow, Power, Chance, StatPowerRatio?, ReferencedStatType?]
Dictionary format (legacy): {"skillRow": ..., "power": ..., "chance": ..., "stat_power_ratio": ..., "referenced_stat_type": ...}

### Exceptions

ArgumentException

Thrown when the serialization format is not supported

## DeserializeFromDictionary(Dictionary)

Deserializes a skill from the legacy Dictionary-based serialization format. This method is maintained for backward compatibility with existing data.

```
[Obsolete("Use Deserialize(IValue) instead.")]
public static Skill DeserializeFromDictionary(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

The serialized skill data in Dictionary format

### Returns

Skill

The deserialized skill

### Remarks

Expected Dictionary format: { "skillRow": SkillSheet.Row, "power": long, "chance": int, "stat_power_ratio": int (optional), "referenced_stat_type": StatType (optional) }

This method is marked as obsolete and should only be used for backward compatibility. New code should use the List-based format through Deserialize(IValue) or

DeserializeFromList(List).

## DeserializeFromList(List)

Deserializes a skill from the new List-based serialization format.

```
public static Skill DeserializeFromList(List serialized)
```

### Parameters

`serialized` List

The serialized skill data in List format

### Returns

Skill

The deserialized skill

### Remarks

Expected List format: [SkillRow, Power, Chance, StatPowerRatio?, ReferencedStatType?] Where StatPowerRatio and ReferencedStatType are optional and only included if StatPowerRatio > 0.

This method handles the new List-based serialization format which is more compact and performant than the previous Dictionary format.

## Get(Row, long, int, int, StatType)

```
public static Skill Get(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

### Returns

Skill

## GetForArena(Row, long, int, int, StatType)

```
public static ArenaSkill GetForArena(SkillSheet.Row skillRow, long power, int
chance, int statPowerRatio, StatType referencedStatType)
```

### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

### Returns

ArenaSkill

## GetV1(Row, long, int)

```
[Obsolete("Use Get() instead.")]
public static Skill GetV1(SkillSheet.Row skillRow, long power, int chance)
```

### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

### Returns

Skill

## SelectSkill(ItemSubType, CustomEquipmentCraftRecipeSkillSheet, EquipmentItemOptionSheet, SkillSheet, IRandom)

```
public static Skill SelectSkill(ItemSubType itemSubType,
CustomEquipmentCraftRecipeSkillSheet recipeSkillSheet, EquipmentItemOptionSheet
itemOptionSheet, SkillSheet skillSheet, IRandom random)
```

### Parameters

`itemSubType` ItemSubType

`recipeSkillSheet` CustomEquipmentCraftRecipeSkillSheet

`itemOptionSheet` EquipmentItemOptionSheet

`skillSheet` SkillSheet

`random` IRandom

### Returns

Skill

# Enum SkillTargetType

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public enum SkillTargetType
```

**Extension Methods**
SkillTargetTypeExtension.GetTarget(SkillTargetType, CharacterBase) ,
StateExtensions.Serialize(Enum)

## Fields

`Ally = 3`

`Enemies = 1`

`Enemy = 0`

`Self = 2`

# Class SkillTargetTypeExtension

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public static class SkillTargetTypeExtension
```

**Inheritance**

object ← SkillTargetTypeExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetTarget(SkillTargetType, CharacterBase)

```
public static IEnumerable<CharacterBase> GetTarget(this SkillTargetType value,
CharacterBase caster)
```

### Parameters

`value` SkillTargetType

`caster` CharacterBase

### Returns

IEnumerable<CharacterBase>

# Enum SkillType

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public enum SkillType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Attack = 0`

`Buff = 2`

`Debuff = 3`

`Heal = 1`

# Class UnexpectedOperationException

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public class UnexpectedOperationException : Exception, ISerializable
```

**Inheritance**

object ← Exception ← UnexpectedOperationException

**Implements**

ISerializable

**Inherited Members**

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### UnexpectedOperationException(string)

```
public UnexpectedOperationException(string message)
```

#### Parameters

`message` string

# Namespace Nekoyume.Model.Skill.Arena

## Classes

ArenaAreaAttack

ArenaAttackSkill

ArenaBlowAttack

ArenaBuffRemovalAttack

ArenaBuffSkill

ArenaDoubleAttack

ArenaFullBuffRemovalAttack

Arena variant of FullBuffRemovalAttack. Deals damage and removes all positive stat buffs from the target.

ArenaHealSkill

ArenaNormalAttack

ArenaShatterStrike

ArenaSkill

# Class ArenaAreaAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaAreaAttack : ArenaAttackSkill, ISkill
```

### Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaAreaAttack

### Implements

ISkill

### Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaAreaAttack(Row, long, int, int, StatType)

```
public ArenaAreaAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaAttackSkill

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ArenaAttackSkill : ArenaSkill, ISkill
```

### Inheritance

object ← ArenaSkill ← ArenaAttackSkill

### Implements

ISkill

### Derived

ArenaAreaAttack, ArenaBlowAttack, ArenaBuffRemovalAttack, ArenaDoubleAttack, ArenaFullBuffRemovalAttack, ArenaNormalAttack, ArenaShatterStrike

### Inherited Members

ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaAttackSkill(Row, long, int, int, StatType)

```
protected ArenaAttackSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## GetMultiplier(int, decimal)

Calculates damage multipliers for multi-hit arena skills. For single hit skills, returns the total damage as is. For multi-hit skills, distributes the total damage across hits with the last hit dealing 30% more damage. This creates a pattern where the final hit is more impactful than the preceding hits.

```
protected static decimal[] GetMultiplier(int hitCount, decimal totalDamage)
```

### Parameters

`hitCount` int

Number of hits for the arena skill

`totalDamage` decimal

Total damage to be distributed across hits

### Returns

decimal[]

Array of damage multipliers for each hit

## ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool)

Processes damage calculation for arena attack skills with improved large value handling. Uses NumberConversionHelper.SafeDecimalToInt64 to prevent overflow when dealing with

large stat values. Implements DamageHelper.GetFinalDefense and DamageHelper.GetReducedDamage for consistent damage calculation.

```
protected IEnumerable<ArenaSkill.ArenaSkillInfo> ProcessDamage(ArenaCharacter
caster, ArenaCharacter target, int simulatorWaveTurn, bool isNormalAttack = false)
```

## Parameters

`caster` ArenaCharacter

The arena character casting the skill

`target` ArenaCharacter

The target arena character

`simulatorWaveTurn` int

Current wave turn in the arena simulator

`isNormalAttack` bool

Whether this is a normal attack

## Returns

IEnumerable<ArenaSkill.ArenaSkillInfo>

Collection of arena skill information including damage and effects

# Class ArenaBlowAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBlowAttack : ArenaAttackSkill, ISkill
```

### Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaBlowAttack

### Implements

ISkill

### Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaBlowAttack(Row, long, int, int, StatType)

```
public ArenaBlowAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaBuffRemovalAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBuffRemovalAttack : ArenaAttackSkill, ISkill
```

### Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaBuffRemovalAttack

### Implements

ISkill

### Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaBuffRemovalAttack(Row, long, int, int, StatType)

```
public ArenaBuffRemovalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaBuffSkill

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBuffSkill : ArenaSkill, ISkill
```

### Inheritance

object ← ArenaSkill ← ArenaBuffSkill

### Implements

ISkill

### Inherited Members

ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaBuffSkill(Row, long, int, int, StatType)

```
public ArenaBuffSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaDoubleAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaDoubleAttack : ArenaAttackSkill, ISkill
```

### Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaDoubleAttack

### Implements

ISkill

### Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaDoubleAttack(Row, long, int, int, StatType)

```
public ArenaDoubleAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaFullBuffRemovalAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

Arena variant of FullBuffRemovalAttack. Deals damage and removes all positive stat buffs from the target.

```
[Serializable]
public class ArenaFullBuffRemovalAttack : ArenaAttackSkill, ISkill
```

**Inheritance**

object ← ArenaSkill ← ArenaAttackSkill ← ArenaFullBuffRemovalAttack

**Implements**

ISkill

**Inherited Members**

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaFullBuffRemovalAttack(Row, long, int, int, StatType)

```
public ArenaFullBuffRemovalAttack(SkillSheet.Row skillRow, long power, int chance,
int statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaHealSkill

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaHealSkill : ArenaSkill, ISkill
```

### Inheritance

object ← ArenaSkill ← ArenaHealSkill

### Implements

ISkill

### Inherited Members

ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaHealSkill(Row, long, int, int, StatType)

```
public ArenaHealSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaNormalAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaNormalAttack : ArenaAttackSkill, ISkill
```

### Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaNormalAttack

### Implements

ISkill

### Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaNormalAttack(Row, long, int, int, StatType)

```
public ArenaNormalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaShatterStrike

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaShatterStrike : ArenaAttackSkill, ISkill
```

**Inheritance**

object ← ArenaSkill ← ArenaAttackSkill ← ArenaShatterStrike

**Implements**

ISkill

**Inherited Members**

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ArenaShatterStrike(Row, long, int, int, StatType)

```
public ArenaShatterStrike(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Methods

## Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

ArenaSkill

# Class ArenaSkill

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ArenaSkill : ISkill
```

**Inheritance**

object ← ArenaSkill

**Implements**

ISkill

**Derived**

ArenaAttackSkill, ArenaBuffSkill, ArenaHealSkill

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaSkill(Row, long, int, int, StatType)

```
protected ArenaSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

#### Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

# Properties

## Chance

```
public int Chance { get; }
```

### Property Value

int

## CustomField

```
public SkillCustomField? CustomField { get; set; }
```

### Property Value

SkillCustomField?

## Power

Determines damage of `AttackSkill`. Determines effect of `BuffSkill`.

```
public long Power { get; }
```

### Property Value

long

## ReferencedStatType

```
public StatType ReferencedStatType { get; }
```

### Property Value

StatType

## SkillRow

```
public SkillSheet.Row SkillRow { get; }
```

### Property Value

SkillSheet.Row

## StatPowerRatio

```
public int StatPowerRatio { get; }
```

### Property Value

int

# Methods

## Equals(Skill)

```
protected bool Equals(Skill other)
```

### Parameters

`other` Skill

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## IsBuff()

```
public bool IsBuff()
```

### Returns

bool

## IsDebuff()

```
public bool IsDebuff()
```

### Returns

bool

## ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
protected IEnumerable<ArenaSkill.ArenaSkillInfo> ProcessBuff(ArenaCharacter caster,
ArenaCharacter target, int turn, IEnumerable<Buff> buffs)
```

### Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

### Returns

IEnumerable<ArenaSkill.ArenaSkillInfo>

## Update(int, long, int)

```
public void Update(int chance, long power, int statPowerRatio)
```

### Parameters

`chance` int

`power` long

`statPowerRatio` int

# Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public abstract ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

## Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

## Returns

ArenaSkill

# Namespace Nekoyume.Model.Stake

## Classes

Contract

StakeStateUtils

## Structs

StakeState

# Class Contract

Namespace: Nekoyume.Model.Stake

Assembly: Lib9c.dll

```
public class Contract
```

**Inheritance**

object ← Contract

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Contract(IValue)

```
public Contract(IValue serialized)
```

#### Parameters

`serialized` IValue

### Contract(StakePolicySheet)

```
public Contract(StakePolicySheet stakePolicySheet)
```

#### Parameters

`stakePolicySheet` StakePolicySheet

### Contract(string, string, long, long)

```
public Contract(string stakeRegularFixedRewardSheetTableName, string
stakeRegularRewardSheetTableName, long rewardInterval, long lockupInterval)
```

### Parameters

`stakeRegularFixedRewardSheetTableName` string

`stakeRegularRewardSheetTableName` string

`rewardInterval` long

`lockupInterval` long

## Fields

### StakeRegularFixedRewardSheetPrefix

```
public const string StakeRegularFixedRewardSheetPrefix =
"StakeRegularFixedRewardSheet_"
```

#### Field Value

string

### StakeRegularRewardSheetPrefix

```
public const string StakeRegularRewardSheetPrefix = "StakeRegularRewardSheet_"
```

#### Field Value

string

### StateTypeName

```
public const string StateTypeName = "stake_contract"
```

### Field Value

string

## StateTypeVersion

```
public const long StateTypeVersion = 1
```

### Field Value

long

# Properties

## LockupInterval

```
[Obsolete("Not used because of guild system")]
public long LockupInterval { get; }
```

### Property Value

long

## RewardInterval

```
public long RewardInterval { get; }
```

### Property Value

long

## StakeRegularFixedRewardSheetTableName

```
public string StakeRegularFixedRewardSheetTableName { get; }
```

### Property Value

string

## StakeRegularRewardSheetTableName

```
public string StakeRegularRewardSheetTableName { get; }
```

### Property Value

string

# Methods

## Equals(Contract)

```
protected bool Equals(Contract other)
```

### Parameters

`other` Contract

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

```
public List Serialize()
```

### Returns

List

# Struct StakeState

Namespace: Nekoyume.Model.Stake

Assembly: Lib9c.dll

```
public readonly struct StakeState : IState
```

**Implements**
IState

**Inherited Members**
ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### StakeState(IValue)

```
public StakeState(IValue serialized)
```

#### Parameters

`serialized` IValue

### StakeState(Contract, long, long, int)

```
public StakeState(Contract contract, long startedBlockIndex, long receivedBlockIndex
= 0, int stateVersion = 3)
```

#### Parameters

`contract` Contract

`startedBlockIndex` long

`receivedBlockIndex` long

`stateVersion` int

## StakeState(LegacyStakeState, Contract)

```
public StakeState(LegacyStakeState legacyStakeState, Contract contract)
```

### Parameters

`legacyStakeState` LegacyStakeState

`contract` Contract

# Fields

## Contract

```
public readonly Contract Contract
```

### Field Value

Contract

## LatestStateTypeVersion

```
public const int LatestStateTypeVersion = 3
```

### Field Value

int

## ReceivedBlockIndex

```
public readonly long ReceivedBlockIndex
```

### Field Value

long

## StartedBlockIndex

```
public readonly long StartedBlockIndex
```

### Field Value

long

## StateTypeName

```
public const string StateTypeName = "stake_state"
```

### Field Value

string

## StateVersion

```
public readonly int StateVersion
```

### Field Value

int

# Properties

## CancellableBlockIndex

```
public long CancellableBlockIndex { get; }
```

### Property Value

long

## ClaimableBlockIndex

```
public long ClaimableBlockIndex { get; }
```

### Property Value

long

## ClaimedBlockIndex

```
public long ClaimedBlockIndex { get; }
```

### Property Value

long

# Methods

## DeriveAddress(Address)

```
public static Address DeriveAddress(Address address)
```

### Parameters

`address` Address

### Returns

Address

## Equals(StakeState)

```
public bool Equals(StakeState other)
```

### Parameters

`other` StakeState

### Returns

bool

## Equals(object)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current instance.

### Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

## GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

### Returns

int

A 32-bit signed integer that is the hash code for this instance.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Operators

## operator ==(StakeState, StakeState)

```
public static bool operator ==(StakeState left, StakeState right)
```

### Parameters

`left` StakeState

`right` StakeState

### Returns

bool

## operator !=(StakeState, StakeState)

```
public static bool operator !=(StakeState left, StakeState right)
```

### Parameters

`left` StakeState

`right` StakeState

## Returns

bool

# Class StakeStateUtils

Namespace: Nekoyume.Model.Stake

Assembly: Lib9c.dll

```
public static class StakeStateUtils
```

**Inheritance**

object ← StakeStateUtils

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### MigrateV1ToV2(IValue, GameConfigState)

```
public static StakeState? MigrateV1ToV2(IValue serialized,
GameConfigState gameConfigState)
```

#### Parameters

`serialized` IValue

`gameConfigState` GameConfigState

#### Returns

StakeState?

### TryMigrateV1ToV2(IValue, GameConfigState, out StakeState)

```
public static bool TryMigrateV1ToV2(IValue serialized, GameConfigState
gameConfigState, out StakeState stakeState)
```

### Parameters

`serialized` IValue

`gameConfigState` GameConfigState

`stakeState` StakeState

### Returns

bool

## TryMigrateV1ToV2(IWorldState, Address, out StakeState)

```
public static bool TryMigrateV1ToV2(IWorldState state, Address stakeStateAddr, out
StakeState stakeState)
```

### Parameters

`state` IWorldState

`stakeStateAddr` Address

`stakeState` StakeState

### Returns

bool

## TryMigrateV2ToV3(IActionContext, IWorld, Address, StakeState, out (IWorld world, StakeState newStakeState)?)

```
public static bool TryMigrateV2ToV3(IActionContext context, IWorld world,
Address stakeStateAddr, StakeState stakeState, out (IWorld world, StakeState
```

```
newStakeState)? result)
```

## Parameters

`context` IActionContext

`world` IWorld

`stakeStateAddr` Address

`stakeState` StakeState

`result` (IWorld world, StakeState newStakeState)?

## Returns

bool

# Namespace Nekoyume.Model.Stat

## Classes

CharacterStats

Stat is built with _baseStats based on level, _equipmentStats based on equipments, _consumableStats based on consumables, _buffStats based on buffs and _optionalStats for runes, etc... Stat of character is built with total of these stats.

DecimalStat

Represents a single character statistic with base and additional values. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. statType - The type of statistic 3. baseValue - Base value of the statistic 4. additionalValue - Additional value of the statistic

StatMap

Represents a collection of character statistics. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. stats - List of DecimalStat objects ordered by StatType

StatModifier

StatTypeComparer

StatTypeExtension

Stats

StatsMap

## Interfaces

IBaseAndAdditionalStats

IStats

# Enums

StatModifier.OperationType

StatType

# Class CharacterStats

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

Stat is built with _baseStats based on level, _equipmentStats based on equipments, _consumableStats based on consumables, _buffStats based on buffs and _optionalStats for runes, etc... Stat of character is built with total of these stats.

```
[Serializable]
public class CharacterStats : Stats, IStats, IBaseAndAdditionalStats, ICloneable
```

**Inheritance**

object ← Stats ← CharacterStats

**Implements**

IStats, IBaseAndAdditionalStats, ICloneable

**Inherited Members**

Stats._statMap , Stats.HP , Stats.ATK , Stats.DEF , Stats.CRI , Stats.HIT , Stats.SPD , Stats.DRV , Stats.DRR , Stats.CDMG , Stats.ArmorPenetration , Stats.Thorn , Stats.LegacyDecimalStatTypes , Stats.Reset() , Stats.Set(StatMap, params Stats[]) , Stats.Set(StatsMap) , Stats.Modify(IEnumerable<StatModifier>) , Stats.Set(IEnumerable<StatModifier>, params Stats[]) , Stats.GetStatAsLong(StatType) , Stats.GetStat(StatType) , Stats.SetStatForTest(StatType, decimal) , Stats.GetStats(bool) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CharacterStats(CharacterStats)

Initializes a new instance of the CharacterStats class by copying from another instance.

```
public CharacterStats(CharacterStats value)
```

### Parameters

`value` CharacterStats

The character stats to copy from.

## CharacterStats(Row, int, IReadOnlyList<StatModifier>)

Initializes a new instance of the CharacterStats class.

```
public CharacterStats(CharacterSheet.Row row, int level, IReadOnlyList<StatModifier>
initialStatModifiers = null)
```

### Parameters

`row` CharacterSheet.Row

The character sheet row data.

`level` int

The character level.

`initialStatModifiers` IReadOnlyList<StatModifier>

Optional initial stat modifiers.

## CharacterStats(WaveStatData)

Initializes a new instance of the CharacterStats class from world boss wave stat data.

```
public CharacterStats(WorldBossCharacterSheet.WaveStatData stat)
```

### Parameters

`stat` WorldBossCharacterSheet.WaveStatData

The world boss wave stat data.

## Fields

## StatWithoutBuffs

Stat map without buff effects for arena calculations.

```
public readonly StatMap StatWithoutBuffs
```

### Field Value

StatMap

# Properties

## AdditionalATK

Gets the additional ATK stat from equipment, buffs, etc.

```
public long AdditionalATK { get; }
```

### Property Value

long

## AdditionalArmorPenetration

Gets the additional ArmorPenetration stat from equipment, buffs, etc.

```
public long AdditionalArmorPenetration { get; }
```

### Property Value

long

## AdditionalCDMG

Gets the additional CDMG stat from equipment, buffs, etc.

```
public long AdditionalCDMG { get; }
```

### Property Value

long

## AdditionalCRI

Gets the additional CRI stat from equipment, buffs, etc.

```
public long AdditionalCRI { get; }
```

### Property Value

long

## AdditionalDEF

Gets the additional DEF stat from equipment, buffs, etc.

```
public long AdditionalDEF { get; }
```

### Property Value

long

## AdditionalDRR

Gets the additional DRR stat from equipment, buffs, etc.

```
public long AdditionalDRR { get; }
```

### Property Value

long

## AdditionalDRV

Gets the additional DRV stat from equipment, buffs, etc.

```
public long AdditionalDRV { get; }
```

### Property Value

long

## AdditionalHIT

Gets the additional HIT stat from equipment, buffs, etc.

```
public long AdditionalHIT { get; }
```

### Property Value

long

## AdditionalHP

Gets the additional HP stat from equipment, buffs, etc.

```
public long AdditionalHP { get; }
```

### Property Value

long

## AdditionalSPD

Gets the additional SPD stat from equipment, buffs, etc.

```
public long AdditionalSPD { get; }
```

### Property Value

long

## AdditionalThorn

Gets the additional Thorn stat from equipment, buffs, etc.

```
public long AdditionalThorn { get; }
```

### Property Value

long

## BaseATK

Gets the base ATK stat.

```
public long BaseATK { get; }
```

### Property Value

long

## BaseArmorPenetration

Gets the base ArmorPenetration stat.

```
public long BaseArmorPenetration { get; }
```

### Property Value

long

## BaseCDMG

Gets the base CDMG stat.

```
public long BaseCDMG { get; }
```

### Property Value

long

## BaseCRI

Gets the base CRI stat.

```
public long BaseCRI { get; }
```

### Property Value

long

## BaseDEF

Gets the base DEF stat.

```
public long BaseDEF { get; }
```

### Property Value

long

## BaseDRR

Gets the base DRR stat.

```
public long BaseDRR { get; }
```

### Property Value

long

## BaseDRV

Gets the base DRV stat.

```
public long BaseDRV { get; }
```

### Property Value

long

## BaseHIT

Gets the base HIT stat.

```
public long BaseHIT { get; }
```

### Property Value

long

## BaseHP

Gets the base HP stat.

```
public long BaseHP { get; }
```

### Property Value

long

## BaseSPD

Gets the base SPD stat.

```
public long BaseSPD { get; }
```

### Property Value

long

## BaseStats

Gets the base stats of the character.

```
public IStats BaseStats { get; }
```

### Property Value

IStats

## BaseThorn

Gets the base Thorn stat.

```
public long BaseThorn { get; }
```

### Property Value

long

## BuffStats

Gets the buff stats of the character.

```
public IStats BuffStats { get; }
```

### Property Value

IStats

## CollectionStats

Gets the collection stats of the character.

```
public IStats CollectionStats { get; }
```

### Property Value

IStats

## ConditionStats

Gets the condition stats of the character.

```
public IStats ConditionStats { get; }
```

### Property Value

IStats

## ConsumableStats

Gets the consumable stats of the character.

```
public IStats ConsumableStats { get; }
```

### Property Value

IStats

## CostumeStats

Gets the costume stats of the character.

```
public IStats CostumeStats { get; }
```

### Property Value

IStats

## EquipmentStats

Gets the equipment stats of the character.

```
public IStats EquipmentStats { get; }
```

### Property Value

IStats

## HpIncreasingModifier

Gets or sets the HP increasing modifier for arena battles.

```
public long HpIncreasingModifier { set; }
```

### Property Value

long

## IsArenaCharacter

Gets or sets whether this character is used in arena battles.

```
public bool IsArenaCharacter { set; }
```

### Property Value

bool

## Level

Gets the character level.

```
public int Level { get; }
```

### Property Value

int

## RuneStats

Gets the rune stats of the character.

```
public IStats RuneStats { get; }
```

### Property Value

IStats

# Methods

## AddBuff(StatBuff, BuffLimitSheet, bool)

```
public void AddBuff(StatBuff buff, BuffLimitSheet buffLimitSheet, bool
updateImmediate = true)
```

### Parameters

`buff` StatBuff

`buffLimitSheet` BuffLimitSheet

`updateImmediate` bool

## AddCostume(IEnumerable<StatModifier>)

```
public void AddCostume(IEnumerable<StatModifier> statModifiers)
```

### Parameters

`statModifiers` IEnumerable<StatModifier>

## AddRune(IEnumerable<StatModifier>)

```
public void AddRune(IEnumerable<StatModifier> statModifiers)
```

### Parameters

`statModifiers` IEnumerable<StatModifier>

## AddRuneStat(RuneOptionInfo, int)

```
public void AddRuneStat(RuneOptionSheet.Row.RuneOptionInfo optionInfo,
int runeLevelBonus)
```

### Parameters

`optionInfo` RuneOptionSheet.Row.RuneOptionInfo

`runeLevelBonus` int

## ClearConditions(bool)

Clear all condition stat modifiers. This should be called after simulation ends to remove temporary conditions.

```
public CharacterStats ClearConditions(bool updateImmediate = true)
```

### Parameters

`updateImmediate` bool

Whether to update stats immediately

### Returns

CharacterStats

This instance for method chaining

## Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## ConfigureStats(IReadOnlyCollection<Equipment>, IReadOnlyCollection<Costume>, IReadOnlyCollection<RuneOptionInfo>, CostumeStatSheet, List<StatModifier>, int)

```
public void ConfigureStats(IReadOnlyCollection<Equipment> equipments,
IReadOnlyCollection<Costume> costumes,
IReadOnlyCollection<RuneOptionSheet.Row.RuneOptionInfo> runeOptions,
CostumeStatSheet costumeStatSheet, List<StatModifier> collectionStatModifiers,
int runeLevelBonus)
```

### Parameters

`equipments` IReadOnlyCollection<Equipment>

`costumes` IReadOnlyCollection<Costume>

`runeOptions` IReadOnlyCollection<RuneOptionSheet.Row.RuneOptionInfo>

`costumeStatSheet` CostumeStatSheet

`collectionStatModifiers` List<StatModifier>

`runeLevelBonus` int

## GetAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long additionalValue)>
GetAdditionalStats(bool ignoreZero = false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## GetBaseAndAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long baseValue, long additionalValue)>
GetBaseAndAdditionalStats(bool ignoreZero = false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue, long additionalValue)>

## GetBaseStats(bool)

```
public IEnumerable<(StatType statType, long baseValue)> GetBaseStats(bool ignoreZero
= false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## IncreaseHpForArena()

Increases the HP of the character for the arena.

```
public void IncreaseHpForArena()
```

## RemoveBuff(StatBuff, bool)

```
public void RemoveBuff(StatBuff buff, bool updateImmediate = true)
```

### Parameters

`buff` StatBuff

`updateImmediate` bool

## SetBuffs(IEnumerable<StatBuff>, BuffLimitSheet, bool)

Set stats based on buffs.

```
public CharacterStats SetBuffs(IEnumerable<StatBuff> value, BuffLimitSheet
buffLimitSheet, bool updateImmediate = true)
```

### Parameters

`value` IEnumerable<StatBuff>

`buffLimitSheet` BuffLimitSheet

`updateImmediate` bool

### Returns

CharacterStats

## SetCollections(IEnumerable<StatModifier>, bool)

```
public CharacterStats SetCollections(IEnumerable<StatModifier> statModifiers, bool
updateImmediate = true)
```

### Parameters

`statModifiers` IEnumerable<StatModifier>

`updateImmediate` bool

### Returns

CharacterStats

## SetConditions(IEnumerable<StatModifier>, bool)

Set stats based on infinite tower conditions. This is applied after buffs and is temporary for simulation only.

```
public CharacterStats SetConditions(IEnumerable<StatModifier> statModifiers, bool
updateImmediate = true)
```

### Parameters

`statModifiers` IEnumerable<StatModifier>

Condition stat modifiers

`updateImmediate` bool

Whether to update stats immediately

### Returns

CharacterStats

This instance for method chaining

## SetConsumables(IEnumerable<Consumable>, bool)

Set stats based on consumables. Also recalculates stats from buffs.

```
public CharacterStats SetConsumables(IEnumerable<Consumable> consumables, bool
updateImmediate = true)
```

### Parameters

`consumables` IEnumerable<Consumable>

`updateImmediate` bool

### Returns

CharacterStats

## SetCostume(IEnumerable<StatModifier>)

```
public void SetCostume(IEnumerable<StatModifier> statModifiers)
```

### Parameters

`statModifiers` IEnumerable<StatModifier>

## SetCostumeStat(IReadOnlyCollection<Costume>, CostumeStatSheet)

```
public void SetCostumeStat(IReadOnlyCollection<Costume> costumes,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`costumes` IReadOnlyCollection<Costume>

`costumeStatSheet` CostumeStatSheet

## SetEquipments(IEnumerable<Equipment>, EquipmentItemSetEffectSheet, bool)

Set stats based on equipments. Also recalculates stats from consumables and buffs.

```
public CharacterStats SetEquipments(IEnumerable<Equipment> equipments,
EquipmentItemSetEffectSheet sheet, bool updateImmediate = true)
```

### Parameters

`equipments` IEnumerable<Equipment>

`sheet` EquipmentItemSetEffectSheet

`updateImmediate` bool

### Returns

CharacterStats

## SetRunes(IEnumerable<StatModifier>, bool)

Set stats based on runes.

```
public CharacterStats SetRunes(IEnumerable<StatModifier> value, bool updateImmediate
= true)
```

### Parameters

`value` IEnumerable<StatModifier>

`updateImmediate` bool

### Returns

CharacterStats

## SetStats(int, bool)

Set base stats based on character level.

```
public CharacterStats SetStats(int level, bool updateImmediate = true)
```

### Parameters

`level` int

`updateImmediate` bool

### Returns

CharacterStats

# Class DecimalStat

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

Represents a single character statistic with base and additional values. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. statType - The type of statistic 3. baseValue - Base value of the statistic 4. additionalValue - Additional value of the statistic

```
[Serializable]
public class DecimalStat : ICloneable, IState
```

**Inheritance**

object ← DecimalStat

**Implements**

ICloneable, IState

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

StateExtensions.SerializeForLegacyEquipmentStat(DecimalStat)

# Remarks

This class implements dual serialization support to ensure smooth migration from the legacy Dictionary format to the new List format. The List format provides better performance and smaller serialized data size.

Example usage:

```
// Create decimal stat
var stat = new DecimalStat(StatType.HP, 100, 50);
// Serialize to List format (new)
```

```
var serialized = stat.Serialize(); // Returns List
// Deserialize from any format
var deserialized = new DecimalStat(serialized); // Supports both Dictionary and List
```

# Constructors

## DecimalStat(Dictionary)

Constructor for backward compatibility with Dictionary format.

```
public DecimalStat(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

Dictionary containing serialized data

## DecimalStat(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public DecimalStat(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

### Exceptions

ArgumentNullException

Thrown when serialized is null

ArgumentException

Thrown when serialized format is not supported

## DecimalStat(DecimalStat)

```
protected DecimalStat(DecimalStat value)
```

### Parameters

`value` DecimalStat

## DecimalStat(StatType, decimal, decimal)

```
public DecimalStat(StatType type, decimal value = 0, decimal additionalValue = 0)
```

### Parameters

`type` StatType

`value` decimal

`additionalValue` decimal

# Fields

## StatType

```
public StatType StatType
```

### Field Value

StatType

# Properties

## AdditionalValue

```
public decimal AdditionalValue { get; }
```

### Property Value

decimal

## AdditionalValueAsLong

```
public long AdditionalValueAsLong { get; }
```

### Property Value

long

## BaseValue

```
public decimal BaseValue { get; }
```

### Property Value

decimal

## BaseValueAsLong

```
public long BaseValueAsLong { get; }
```

### Property Value

long

## HasAdditionalValue

```
public bool HasAdditionalValue { get; }
```

### Property Value

bool

## HasAdditionalValueAsLong

```
public bool HasAdditionalValueAsLong { get; }
```

### Property Value

bool

## HasBaseValue

```
public bool HasBaseValue { get; }
```

### Property Value

bool

## HasBaseValueAsLong

```
public bool HasBaseValueAsLong { get; }
```

### Property Value

bool

## HasTotalValueAsLong

```
public bool HasTotalValueAsLong { get; }
```

### Property Value

bool

## TotalValue

```
public decimal TotalValue { get; }
```

### Property Value

decimal

## TotalValueAsLong

```
[Obsolete("For legacy equipments. (Before world 7 patch)")]
public long TotalValueAsLong { get; }
```

### Property Value

long

# Methods

## AddAdditionalValue(decimal)

```
public void AddAdditionalValue(decimal value)
```

### Parameters

`value` decimal

## AddBaseValue(decimal)

```
public void AddBaseValue(decimal value)
```

### Parameters

`value` decimal

## Clone()

Creates a new object that is a copy of the current instance.

```
public virtual object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## Deserialize(Dictionary)

Deserializes data from Dictionary format (legacy support).

```
public virtual void Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

Dictionary containing serialized data

## Equals(DecimalStat)

```
protected bool Equals(DecimalStat other)
```

### Parameters

`other` DecimalStat

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Reset()

```
public virtual void Reset()
```

## Serialize()

Serializes the DecimalStat to List format (new format). Order: [version, statType, baseValue, additionalValue]

```
public virtual IValue Serialize()
```

### Returns

IValue

List containing serialized data

## SerializeWithoutAdditional()

```
public IValue SerializeWithoutAdditional()
```

### Returns

IValue

## SetAdditionalValue(decimal)

```
public void SetAdditionalValue(decimal value)
```

### Parameters

`value` decimal

## SetBaseValue(decimal)

```
public void SetBaseValue(decimal value)
```

### Parameters

`value` decimal

# Interface IBaseAndAdditionalStats

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public interface IBaseAndAdditionalStats
```

## Properties

### AdditionalATK

```
long AdditionalATK { get; }
```

#### Property Value

long

### AdditionalArmorPenetration

```
long AdditionalArmorPenetration { get; }
```

#### Property Value

long

### AdditionalCDMG

```
long AdditionalCDMG { get; }
```

#### Property Value

long

## AdditionalCRI

```
long AdditionalCRI { get; }
```

### Property Value

long

## AdditionalDEF

```
long AdditionalDEF { get; }
```

### Property Value

long

## AdditionalDRR

```
long AdditionalDRR { get; }
```

### Property Value

long

## AdditionalDRV

```
long AdditionalDRV { get; }
```

### Property Value

long

## AdditionalHIT

```
long AdditionalHIT { get; }
```

### Property Value

long

## AdditionalHP

```
long AdditionalHP { get; }
```

### Property Value

long

## AdditionalSPD

```
long AdditionalSPD { get; }
```

### Property Value

long

## AdditionalThorn

```
long AdditionalThorn { get; }
```

### Property Value

long

## BaseATK

```
long BaseATK { get; }
```

### Property Value

long

## BaseArmorPenetration

```
long BaseArmorPenetration { get; }
```

### Property Value

long

## BaseCDMG

```
long BaseCDMG { get; }
```

### Property Value

long

## BaseCRI

```
long BaseCRI { get; }
```

### Property Value

long

## BaseDEF

```
long BaseDEF { get; }
```

### Property Value

long

## BaseDRR

```
long BaseDRR { get; }
```

### Property Value

long

## BaseDRV

```
long BaseDRV { get; }
```

### Property Value

long

## BaseHIT

```
long BaseHIT { get; }
```

### Property Value

long

## BaseHP

```
long BaseHP { get; }
```

### Property Value

long

## BaseSPD

```
long BaseSPD { get; }
```

### Property Value

long

## BaseThorn

```
long BaseThorn { get; }
```

### Property Value

long

# Methods

## GetAdditionalStats(bool)

```
IEnumerable<(StatType statType, long additionalValue)> GetAdditionalStats(bool
ignoreZero = false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## GetBaseAndAdditionalStats(bool)

```
IEnumerable<(StatType statType, long baseValue, long additionalValue)>
GetBaseAndAdditionalStats(bool ignoreZero = false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue, long additionalValue)>

## GetBaseStats(bool)

```
IEnumerable<(StatType statType, long baseValue)> GetBaseStats(bool ignoreZero
= false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

# Interface IStats

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public interface IStats
```

## Properties

### ATK

```
long ATK { get; }
```

#### Property Value

long

### ArmorPenetration

```
long ArmorPenetration { get; }
```

#### Property Value

long

### CDMG

```
long CDMG { get; }
```

#### Property Value

long

## CRI

```
long CRI { get; }
```

### Property Value

long

## DEF

```
long DEF { get; }
```

### Property Value

long

## DRR

```
long DRR { get; }
```

### Property Value

long

## DRV

```
long DRV { get; }
```

### Property Value

long

## HIT

```
long HIT { get; }
```

### Property Value

long

## HP

```
long HP { get; }
```

### Property Value

long

## SPD

```
long SPD { get; }
```

### Property Value

long

## Thorn

```
long Thorn { get; }
```

### Property Value

long

# Methods

## GetStats(bool)

```
IEnumerable<(StatType statType, long value)> GetStats(bool ignoreZero = false)
```

## Parameters

`ignoreZero` bool

## Returns

IEnumerable<(StatType statType, long baseValue)>

# Class StatMap

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

Represents a collection of character statistics. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. stats - List of DecimalStat objects ordered by StatType

```
[Serializable]
public class StatMap : IStats, IBaseAndAdditionalStats, IState
```

**Inheritance**

object ← StatMap

**Implements**

IStats, IBaseAndAdditionalStats, IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

This class implements dual serialization support to ensure smooth migration from the legacy Dictionary format to the new List format. The List format provides better performance and smaller serialized data size.

Example usage:

```
// Create stat map
var statMap = new StatMap();
statMap[StatType.HP].SetBaseValue(100);
statMap[StatType.ATK].SetBaseValue(50);
// Serialize to List format (new)
var serialized = statMap.Serialize(); // Returns List
```

```
// Deserialize from any format
var deserialized = new StatMap(serialized); // Supports both Dictionary and List
```

# Constructors

## StatMap()

```
public StatMap()
```

## StatMap(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public StatMap(IValue serialized)
```

### Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

### Exceptions

ArgumentNullException

Thrown when serialized is null

ArgumentException

Thrown when serialized format is not supported

## StatMap(StatMap)

```
public StatMap(StatMap statMap)
```

### Parameters

`statMap` StatMap

# Properties

## ATK

```
public long ATK { get; }
```

### Property Value

long

## AdditionalATK

```
public long AdditionalATK { get; }
```

### Property Value

long

## AdditionalArmorPenetration

```
public long AdditionalArmorPenetration { get; }
```

### Property Value

long

## AdditionalCDMG

```
public long AdditionalCDMG { get; }
```

### Property Value

long

## AdditionalCRI

```
public long AdditionalCRI { get; }
```

### Property Value

long

## AdditionalDEF

```
public long AdditionalDEF { get; }
```

### Property Value

long

## AdditionalDRR

```
public long AdditionalDRR { get; }
```

### Property Value

long

## AdditionalDRV

```
public long AdditionalDRV { get; }
```

### Property Value

long

## AdditionalHIT

```
public long AdditionalHIT { get; }
```

### Property Value

long

## AdditionalHP

```
public long AdditionalHP { get; }
```

### Property Value

long

## AdditionalSPD

```
public long AdditionalSPD { get; }
```

### Property Value

long

## AdditionalThorn

```
public long AdditionalThorn { get; }
```

### Property Value

long

## ArmorPenetration

```
public long ArmorPenetration { get; }
```

### Property Value

long

## BaseATK

```
public long BaseATK { get; }
```

### Property Value

long

## BaseArmorPenetration

```
public long BaseArmorPenetration { get; }
```

### Property Value

long

## BaseCDMG

```
public long BaseCDMG { get; }
```

### Property Value

long

## BaseCRI

```
public long BaseCRI { get; }
```

### Property Value

long

## BaseDEF

```
public long BaseDEF { get; }
```

### Property Value

long

## BaseDRR

```
public long BaseDRR { get; }
```

### Property Value

long

## BaseDRV

```
public long BaseDRV { get; }
```

### Property Value

long

## BaseHIT

```
public long BaseHIT { get; }
```

### Property Value

long

## BaseHP

```
public long BaseHP { get; }
```

### Property Value

long

## BaseSPD

```
public long BaseSPD { get; }
```

### Property Value

long

## BaseThorn

```
public long BaseThorn { get; }
```

### Property Value

long

## CDMG

```
public long CDMG { get; }
```

### Property Value

long

## CRI

```
public long CRI { get; }
```

### Property Value

long

## DEF

```
public long DEF { get; }
```

### Property Value

long

## DRR

```
public long DRR { get; }
```

### Property Value

long

## DRV

```
public long DRV { get; }
```

### Property Value

long

## HIT

```
public long HIT { get; }
```

### Property Value

long

## HP

```
public long HP { get; }
```

### Property Value

long

## this[StatType]

```
public DecimalStat this[StatType type] { get; }
```

### Parameters

`type` StatType

### Property Value

DecimalStat

## SPD

```
public long SPD { get; }
```

### Property Value

long

## Thorn

```
public long Thorn { get; }
```

### Property Value

long

# Methods

## GetAdditionalStat(StatType)

```
public long GetAdditionalStat(StatType statType)
```

### Parameters

`statType` StatType

### Returns

long

## GetAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long additionalValue)>
GetAdditionalStats(bool ignoreZero = false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## GetBaseAndAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long baseValue, long additionalValue)>
GetBaseAndAdditionalStats(bool ignoreZero = false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue, long additionalValue)>

## GetBaseStat(StatType)

```
public long GetBaseStat(StatType statType)
```

### Parameters

`statType` StatType

### Returns

long

## GetBaseStats(bool)

```
public IEnumerable<(StatType statType, long baseValue)> GetBaseStats(bool ignoreZero
= false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## GetDecimalStats(bool)

```
public IEnumerable<DecimalStat> GetDecimalStats(bool ignoreZero)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<DecimalStat>

## GetStat(StatType)

```
public decimal GetStat(StatType statType)
```

### Parameters

`statType` StatType

### Returns

decimal

## GetStatAsLong(StatType)

```
public long GetStatAsLong(StatType statType)
```

### Parameters

`statType` StatType

### Returns

long

## GetStats(bool)

```
public IEnumerable<(StatType statType, long value)> GetStats(bool ignoreZero
= false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## Reset()

```
public void Reset()
```

## Serialize()

Serializes the StatMap to List format (new format). Order: [version, stats]

```
public IValue Serialize()
```

## Returns

IValue

List containing serialized data

# Class StatModifier

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
[Serializable]
public class StatModifier
```

**Inheritance**

object ← StatModifier

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### StatModifier(DecimalStat)

```
public StatModifier(DecimalStat decimalStat)
```

#### Parameters

`decimalStat` DecimalStat

### StatModifier(StatType, OperationType, long)

```
public StatModifier(StatType statType, StatModifier.OperationType operation,
long value)
```

#### Parameters

`statType` StatType

`operation` StatModifier.OperationType

`value` long

# Properties

## Operation

```
public StatModifier.OperationType Operation { get; }
```

### Property Value

StatModifier.OperationType

## StatType

```
public StatType StatType { get; }
```

### Property Value

StatType

## Value

```
public long Value { get; }
```

### Property Value

long

# Methods

## GetModifiedAll(decimal)

value와 함께 value를 바탕으로 변경시킨 값의 합을 리턴한다.

```
public decimal GetModifiedAll(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

## GetModifiedAll(long)

value와 함께 value를 바탕으로 변경시킨 값의 합을 리턴한다.

```
public long GetModifiedAll(long value)
```

### Parameters

`value` long

### Returns

long

## GetModifiedValue(decimal)

value를 바탕으로 변경시킨 값만을 리턴한다.

```
public decimal GetModifiedValue(decimal value)
```

### Parameters

`value` decimal

### Returns

decimal

### Exceptions

ArgumentOutOfRangeException

## GetModifiedValue(int)

value를 바탕으로 변경시킨 값만을 리턴한다.

```
public long GetModifiedValue(int value)
```

### Parameters

`value` int

### Returns

long

### Exceptions

ArgumentOutOfRangeException

## Modify(DecimalStat)

value를 변경시킨다.

```
public void Modify(DecimalStat value)
```

### Parameters

`value` DecimalStat

### Exceptions

ArgumentOutOfRangeException

## SetForTest(int)

```
public void SetForTest(int value)
```

### Parameters

`value` int

## ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

### Returns

string

A string that represents the current object.

# Enum StatModifier.OperationType

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public enum StatModifier.OperationType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Add = 0`

`Percentage = 1`

# Enum StatType

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public enum StatType
```

**Extension Methods**
StatTypeExtension.Serialize(StatType) , StateExtensions.Serialize(Enum)

# Fields

`ATK = 2`

Attack Power

`ArmorPenetration = 10`

Armor penetration

`CDMG = 9`

Crit Damage (Permyriad)

`CRI = 4`

Critical Chance

`DEF = 3`

Defence

`DRR = 8`

Damage Reduction Rate. Multiplied on original damage. (Permyriad)

`DRV = 7`

Damage Reduction Value. Subtracted from original damage.

`HIT = 5`

Hit Chance

`HP = 1`

Health Point

`NONE = 0`

Default, It's same with null.

`SPD = 6`

Speed

`Thorn = 11`

Apply {Thorn} damage to the attacker when an attack is attempted.

# Class StatTypeComparer

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
[Serializable]
public class StatTypeComparer : IEqualityComparer<StatType>
```

**Inheritance**

object ← StatTypeComparer

**Implements**

IEqualityComparer<StatType>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Instance

```
public static readonly StatTypeComparer Instance
```

### Field Value

StatTypeComparer

# Methods

## Equals(StatType, StatType)

Determines whether the specified objects are equal.

```
public bool Equals(StatType x, StatType y)
```

### Parameters

`x` StatType

The first object of type `T` to compare.

`y` StatType

The second object of type `T` to compare.

### Returns

bool

[true] if the specified objects are equal; otherwise, [false].

## GetHashCode(StatType)

Returns a hash code for the specified object.

```
public int GetHashCode(StatType obj)
```

### Parameters

`obj` StatType

The [object] for which a hash code is to be returned.

### Returns

int

A hash code for the specified object.

### Exceptions

ArgumentNullException

The type of `obj` is a reference type and `obj` is [null].

# Class StatTypeExtension

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public static class StatTypeExtension
```

**Inheritance**

object ← StatTypeExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### Deserialize(Binary)

```
public static StatType Deserialize(Binary serialized)
```

#### Parameters

`serialized` Binary

#### Returns

StatType

### Serialize(StatType)

```
public static IKey Serialize(this StatType statType)
```

#### Parameters

`statType` StatType

## Returns

IKey

# Class Stats

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
[Serializable]
public class Stats : IStats, ICloneable
```

**Inheritance**

object ← Stats

**Implements**

IStats, ICloneable

**Derived**

CharacterStats

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Stats()

```
public Stats()
```

### Stats(Stats)

```
public Stats(Stats value)
```

#### Parameters

`value` Stats

# Fields

## LegacyDecimalStatTypes

```
protected readonly HashSet<StatType> LegacyDecimalStatTypes
```

### Field Value

HashSet<StatType>

## _statMap

```
protected readonly StatMap _statMap
```

### Field Value

StatMap

# Properties

## ATK

```
public long ATK { get; }
```

### Property Value

long

## ArmorPenetration

```
public long ArmorPenetration { get; }
```

### Property Value

long

## CDMG

```
public long CDMG { get; }
```

### Property Value

long

## CRI

```
public long CRI { get; }
```

### Property Value

long

## DEF

```
public long DEF { get; }
```

### Property Value

long

## DRR

```
public long DRR { get; }
```

### Property Value

long

## DRV

```
public long DRV { get; }
```

### Property Value

long

## HIT

```
public long HIT { get; }
```

### Property Value

long

## HP

```
public long HP { get; }
```

### Property Value

long

## SPD

```
public long SPD { get; }
```

### Property Value

long

## Thorn

```
public long Thorn { get; }
```

### Property Value

long

## Methods

### Clone()

Creates a new object that is a copy of the current instance.

```
public virtual object Clone()
```

### Returns

object

A new object that is a copy of this instance.

### GetStat(StatType)

```
public decimal GetStat(StatType statType)
```

### Parameters

`statType` StatType

### Returns

decimal

### GetStatAsLong(StatType)

```
public long GetStatAsLong(StatType statType)
```

### Parameters

`statType` StatType

### Returns

long

## GetStats(bool)

```
public IEnumerable<(StatType statType, long value)> GetStats(bool ignoreZero
= false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## Modify(IEnumerable<StatModifier>)

```
public void Modify(IEnumerable<StatModifier> statModifiers)
```

### Parameters

`statModifiers` IEnumerable<StatModifier>

## Reset()

```
public void Reset()
```

## Set(StatMap, params Stats[])

```
public void Set(StatMap statMap, params Stats[] statsArray)
```

### Parameters

`statMap` StatMap

`statsArray` Stats[]

## Set(StatsMap)

```
public void Set(StatsMap value)
```

### Parameters

`value` StatsMap

## Set(IEnumerable<StatModifier>, params Stats[])

```
public void Set(IEnumerable<StatModifier> statModifiers, params Stats[] baseStats)
```

### Parameters

`statModifiers` IEnumerable<StatModifier>

`baseStats` Stats[]

## SetStatForTest(StatType, decimal)

Use this only for testing.

```
public void SetStatForTest(StatType statType, decimal value)
```

### Parameters

`statType` StatType

`value` [decimal]

# Class StatsMap

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
[Serializable]
public class StatsMap : IStats, IBaseAndAdditionalStats, IState
```

**Inheritance**

object ← StatsMap

**Implements**

IStats, IBaseAndAdditionalStats, IState

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### ATK

```
public long ATK { get; }
```

#### Property Value

long

### AdditionalATK

```
public long AdditionalATK { get; }
```

#### Property Value

long

## AdditionalArmorPenetration

```
public long AdditionalArmorPenetration { get; }
```

### Property Value

long

## AdditionalCDMG

```
public long AdditionalCDMG { get; }
```

### Property Value

long

## AdditionalCRI

```
public long AdditionalCRI { get; }
```

### Property Value

long

## AdditionalDEF

```
public long AdditionalDEF { get; }
```

### Property Value

long

## AdditionalDRR

```
public long AdditionalDRR { get; }
```

### Property Value

long

## AdditionalDRV

```
public long AdditionalDRV { get; }
```

### Property Value

long

## AdditionalHIT

```
public long AdditionalHIT { get; }
```

### Property Value

long

## AdditionalHP

```
public long AdditionalHP { get; }
```

### Property Value

long

## AdditionalSPD

```
public long AdditionalSPD { get; }
```

### Property Value

long

## AdditionalThorn

```
public long AdditionalThorn { get; }
```

### Property Value

long

## ArmorPenetration

```
public long ArmorPenetration { get; }
```

### Property Value

long

## BaseATK

```
public long BaseATK { get; }
```

### Property Value

long

## BaseArmorPenetration

```
public long BaseArmorPenetration { get; }
```

### Property Value

long

## BaseCDMG

```
public long BaseCDMG { get; }
```

### Property Value

long

## BaseCRI

```
public long BaseCRI { get; }
```

### Property Value

long

## BaseDEF

```
public long BaseDEF { get; }
```

### Property Value

long

## BaseDRR

```
public long BaseDRR { get; }
```

### Property Value

long

## BaseDRV

```
public long BaseDRV { get; }
```

### Property Value

long

## BaseHIT

```
public long BaseHIT { get; }
```

### Property Value

long

## BaseHP

```
public long BaseHP { get; }
```

### Property Value

long

## BaseSPD

```
public long BaseSPD { get; }
```

### Property Value

long

## BaseThorn

```
public long BaseThorn { get; }
```

### Property Value

long

## CDMG

```
public long CDMG { get; }
```

### Property Value

long

## CRI

```
public long CRI { get; }
```

### Property Value

long

## DEF

```
public long DEF { get; }
```

### Property Value

long

## DRR

```
public long DRR { get; }
```

### Property Value

long

## DRV

```
public long DRV { get; }
```

### Property Value

long

## HIT

```
public long HIT { get; }
```

### Property Value

long

## HP

```
public long HP { get; }
```

### Property Value

long

## SPD

```
public long SPD { get; }
```

### Property Value

long

## Thorn

```
public long Thorn { get; }
```

### Property Value

long

# Methods

## AddStatAdditionalValue(StatModifier)

```
public void AddStatAdditionalValue(StatModifier statModifier)
```

### Parameters

`statModifier` StatModifier

## AddStatAdditionalValue(StatType, decimal)

```
public void AddStatAdditionalValue(StatType key, decimal additionalValue)
```

### Parameters

`key` StatType

`additionalValue` decimal

## AddStatValue(StatType, decimal)

```
public void AddStatValue(StatType key, decimal value)
```

### Parameters

`key` StatType

`value` decimal

## Deserialize(IValue)

```
public void Deserialize(IValue serialized)
```

### Parameters

`serialized` IValue

## Equals(StatsMap)

```
protected bool Equals(StatsMap other)
```

### Parameters

`other` StatsMap

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetAdditionalStat(StatType)

```
public long GetAdditionalStat(StatType statType)
```

### Parameters

`statType` StatType

### Returns

long

## GetAdditionalStats()

```
public IEnumerable<DecimalStat> GetAdditionalStats()
```

### Returns

IEnumerable<DecimalStat>

## GetAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long additionalValue)>
GetAdditionalStats(bool ignoreZero = false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## GetBaseAndAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long baseValue, long additionalValue)>
GetBaseAndAdditionalStats(bool ignoreZero = false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue, long additionalValue)>

## GetBaseStat(StatType)

```
public long GetBaseStat(StatType statType)
```

### Parameters

`statType` StatType

### Returns

long

## GetBaseStats(bool)

```
public IEnumerable<(StatType statType, long baseValue)> GetBaseStats(bool ignoreZero
= false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## GetDecimalStats(bool)

```
public IEnumerable<DecimalStat> GetDecimalStats(bool ignoreZero)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<DecimalStat>

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## GetStat(StatType)

```
public long GetStat(StatType statType)
```

### Parameters

`statType` StatType

### Returns

long

## GetStats(bool)

```
public IEnumerable<(StatType statType, long value)> GetStats(bool ignoreZero
= false)
```

### Parameters

`ignoreZero` bool

### Returns

IEnumerable<(StatType statType, long baseValue)>

## Serialize()

```
public IValue Serialize()
```

### Returns

# SetStatAdditionalValue(StatType, decimal)

```
public void SetStatAdditionalValue(StatType key, decimal additionalValue)
```

## Parameters

`key` StatType

`additionalValue` decimal

# Namespace Nekoyume.Model.State

## Classes

ActivatedAccountsState

AdminState

AgentState
Agent□ □□ □□□□.

AllCombinationSlotState
This is new version of combination slot State. This state stores all combinationSlot states of an avatar. AllCombinationSlotState has all combinationSlotStates as dictionary and has methods to get/set/update each combinationSlotState. Use this with CombinationSlotStateModule.

AllRuneState
This is new version of rune state. This state stores all rune states of an avatar. AllRuneState has all RuneStates as dictionary and has methods to get/set/update each RuneState. Use this with RuneStateModule.

ArenaAvatarState
Introduced at https://github.com/planetarium/lib9c/pull/1156

ArenaInfo

ArenaInfo.Record

AuthorizedMinersState

AvatarState
Agent□ □□□□ □ Avatar□ □□ □□□□.

CollectionState
Represents the state of a collection.

CombinationSlotState

CpState
CpState is a state that stores the cp of the player.

CreditsState

CrystalCostState

CrystalRandomSkillState

DuplicateRedeemException

FailedToUnregisterInShopStateException

GameConfigState

GoldBalanceState

GoldCurrencyState

HammerPointState

InfiniteTowerBoardState

Represents the board state for infinite tower, tracking clear counts per floor.

InvalidRedeemCodeException

ItemSlotState

LazyState<TState, TEncoding>

LegacyStakeState

LegacyStakeState.StakeAchievements

MarketState

MonsterCollectionResult

MonsterCollectionState

MonsterCollectionState0

PendingActivationState

PetState

RaiderState

RankingInfo

RankingMapState

RankingState

RankingState0

RankingState1

RedeemCodeState

RedeemCodeState.Reward

RuneSlotState

RuneState

ShardedShopState

ShardedShopStateV2

ShopState
This is a model class of shop state.

ShopStateAlreadyContainsException

StakeRewardCalculator
Provides methods for calculating staking rewards from reward sheets.

State

StateExtensions

WeeklyArenaState

WorldBossKillRewardRecord

WorldBossState
Represents the state of a World Boss in the game.

## Interfaces

IState

## Delegates

StateExtensions.IValueTryParseDelegate<T>

# Class ActivatedAccountsState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class ActivatedAccountsState : State, IState, ISerializable
```

**Inheritance**

object ← State ← ActivatedAccountsState

**Implements**

IState, ISerializable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ActivatedAccountsState()

```
public ActivatedAccountsState()
```

### ActivatedAccountsState(Dictionary)

```
public ActivatedAccountsState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## ActivatedAccountsState(IImmutableSet<Address>)

```
public ActivatedAccountsState(IImmutableSet<Address> accounts)
```

### Parameters

`accounts` IImmutableSet<Address>

## ActivatedAccountsState(SerializationInfo, StreamingContext)

```
protected ActivatedAccountsState(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

# Properties

## Accounts

```
public IImmutableSet<Address> Accounts { get; }
```

### Property Value

IImmutableSet<Address>

## Methods

## AddAccount(Address)

```
public ActivatedAccountsState AddAccount(Address account)
```

### Parameters

`account` Address

### Returns

ActivatedAccountsState

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

# Remove(Address)

```
public void Remove(Address account)
```

## Parameters

`account` Address

# Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Class AdminState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class AdminState : State, IState
```

**Inheritance**

object ← State ← AdminState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AdminState(Dictionary)

```
public AdminState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### AdminState(Address, long)

```
public AdminState(Address adminAddress, long validUntil)
```

#### Parameters

`adminAddress` Address

`validUntil` long

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

# Properties

## AdminAddress

```
public Address AdminAddress { get; }
```

### Property Value

Address

## ValidUntil

```
public long ValidUntil { get; }
```

### Property Value

long

# Methods

# Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Class AgentState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Agent□ □□ □□□□.

```
[Serializable]
public class AgentState : State, IState, ICloneable
```

**Inheritance**

object ← State ← AgentState

**Implements**

IState, ICloneable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AgentState(Dictionary)

```
public AgentState(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## AgentState(List)

```
public AgentState(List serialized)
```

### Parameters

`serialized` List

## AgentState(Address)

```
public AgentState(Address address)
```

### Parameters

`address` Address

# Fields

## CurrentVersion

```
public const int CurrentVersion = 1
```

### Field Value

int

## avatarAddresses

```
public readonly Dictionary<int, Address> avatarAddresses
```

### Field Value

Dictionary<int, Address>

# Properties

## MonsterCollectionRound

```
public int MonsterCollectionRound { get; }
```

### Property Value

int

## Version

```
public int Version { get; }
```

### Property Value

int

# Methods

## Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## IncreaseCollectionRound()

```
public void IncreaseCollectionRound()
```

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public IValue SerializeList()
```

### Returns

IValue

# Class AllCombinationSlotState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

This is new version of combination slot State. This state stores all combinationSlot states of an avatar. AllCombinationSlotState has all combinationSlotStates as dictionary and has methods to get/set/update each combinationSlotState. Use this with CombinationSlotState Module.

```
public class AllCombinationSlotState : IState, IEnumerable<CombinationSlotState>,
IEnumerable
```

**Inheritance**

object ← AllCombinationSlotState

**Implements**

IState, IEnumerable<CombinationSlotState>, IEnumerable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AllCombinationSlotState()

```
public AllCombinationSlotState()
```

### AllCombinationSlotState(List)

```
public AllCombinationSlotState(List serialized)
```

#### Parameters

`serialized` List

# Properties

## CombinationSlots

```
public Dictionary<int, CombinationSlotState> CombinationSlots { get; }
```

### Property Value

Dictionary<int, CombinationSlotState>

# Methods

## AddSlot(Address, int)

```
public void AddSlot(Address address, int index = 0)
```

### Parameters

`address` Address

`index` int

## AddSlot(CombinationSlotState)

```
public void AddSlot(CombinationSlotState combinationSlotState)
```

### Parameters

`combinationSlotState` CombinationSlotState

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<CombinationSlotState> GetEnumerator()
```

### Returns

IEnumerator<CombinationSlotState>

An enumerator that can be used to iterate through the collection.

## GetSlot(int)

```
public CombinationSlotState GetSlot(int slotStateIndex)
```

### Parameters

`slotStateIndex` int

### Returns

CombinationSlotState

## MigrationLegacySlotState(IWorldState, Address)

만약 AllCombinationSlotState가 없다면, 슬롯 확장 업데이트 전 4개의 슬롯을 가져와서 채워넣는다.

```
public static AllCombinationSlotState MigrationLegacySlotState(IWorldState
worldState, Address avatarAddress)
```

### Parameters

`worldState` IWorldState

Slot 상태를 가져올 world state

`avatarAddress` Address

Migration을 진행할 아바타

## Returns

AllCombinationSlotState

Migration AllCombinationSlotState

# MigrationLegacySlotState(Func<int, CombinationSlotState?>, Address)

AllCombinationSlotState , , 4 .

```
public static AllCombinationSlotState MigrationLegacySlotState(Func<int,
CombinationSlotState?> stateFactory, Address avatarAddress)
```

## Parameters

`stateFactory` Func<int, CombinationSlotState>

CombinationSlotState

`avatarAddress` Address

Migration

## Returns

AllCombinationSlotState

Migration AllCombinationSlotState

# Serialize()

```
public IValue Serialize()
```

## Returns

IValue

## SetSlot(CombinationSlotState)

```
public void SetSlot(CombinationSlotState combinationSlotState)
```

### Parameters

`combinationSlotState` CombinationSlotState

## TryGetSlot(int, out CombinationSlotState?)

```
public bool TryGetSlot(int slotStateIndex, out CombinationSlotState?
combinationSlotState)
```

### Parameters

`slotStateIndex` int

`combinationSlotState` CombinationSlotState

### Returns

bool

## UnlockSlot(Address, int)

```
public void UnlockSlot(Address avatarAddress, int index)
```

### Parameters

`avatarAddress` Address

`index` int

# Class AllRuneState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

This is new version of rune state. This state stores all rune states of an avatar. AllRuneState has all RuneStates as dictionary and has methods to get/set/update each RuneState. Use this with RuneStateModule.

```
public class AllRuneState : IState
```

**Inheritance**

object ← AllRuneState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AllRuneState()

```
public AllRuneState()
```

### AllRuneState(List)

```
public AllRuneState(List serialized)
```

#### Parameters

`serialized` List

## AllRuneState(int, int)

```
public AllRuneState(int runeId, int level = 0)
```

### Parameters

`runeId` int

`level` int

# Properties

## Runes

```
public Dictionary<int, RuneState> Runes { get; }
```

### Property Value

Dictionary<int, RuneState>

# Methods

## AddRuneState(RuneState)

```
public void AddRuneState(RuneState runeState)
```

### Parameters

`runeState` RuneState

## AddRuneState(int, int)

```
public void AddRuneState(int runeId, int level = 0)
```

### Parameters

`runeId` int

`level` int

## GetRuneState(int)

```
public RuneState GetRuneState(int runeId)
```

### Parameters

`runeId` int

### Returns

RuneState

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## SetRuneState(RuneState)

```
public void SetRuneState(RuneState runeState)
```

### Parameters

`runeState` RuneState

## SetRuneState(int, int)

```
public void SetRuneState(int runeId, int level)
```

### Parameters

`runeId` int

`level` int

## TryGetRuneState(int, out RuneState)

```
public bool TryGetRuneState(int runeId, out RuneState runeState)
```

### Parameters

`runeId` int

`runeState` RuneState

### Returns

bool

# Class ArenaAvatarState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1156

```
public class ArenaAvatarState : IState
```

**Inheritance**

object ← ArenaAvatarState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaAvatarState(List)

```
public ArenaAvatarState(List serialized)
```

#### Parameters

`serialized` List

### ArenaAvatarState(AvatarState)

```
public ArenaAvatarState(AvatarState avatarState)
```

#### Parameters

`avatarState` AvatarState

# Fields

## Address

```
public Address Address
```

### Field Value

Address

## LastBattleBlockIndex

```
public long LastBattleBlockIndex
```

### Field Value

long

# Properties

## Costumes

```
public List<Guid> Costumes { get; }
```

### Property Value

List<Guid>

## Equipments

```
public List<Guid> Equipments { get; }
```

### Property Value

List<Guid>

## Methods

## DeriveAddress(Address)

```
public static Address DeriveAddress(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

### Returns

Address

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## UpdateCostumes(List<Guid>)

```
public void UpdateCostumes(List<Guid> costumes)
```

### Parameters

`costumes` List<Guid>

# UpdateEquipment(List<Guid>)

```
public void UpdateEquipment(List<Guid> equipments)
```

## Parameters

`equipments` List<Guid>

# Class ArenaInfo

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class ArenaInfo : IState
```

**Inheritance**

object ← ArenaInfo

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaInfo(Dictionary)

```
public ArenaInfo(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ArenaInfo(ArenaInfo)

```
public ArenaInfo(ArenaInfo prevInfo)
```

#### Parameters

`prevInfo` ArenaInfo

## ArenaInfo(AvatarState, CharacterSheet, CostumeStatSheet, bool)

```
public ArenaInfo(AvatarState avatarState, CharacterSheet characterSheet,
CostumeStatSheet costumeStatSheet, bool active)
```

### Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

`active` bool

## ArenaInfo(AvatarState, CharacterSheet, bool)

```
public ArenaInfo(AvatarState avatarState, CharacterSheet characterSheet,
bool active)
```

### Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`active` bool

# Fields

## AgentAddress

```
public readonly Address AgentAddress
```

### Field Value

Address

## ArenaRecord

```
public readonly ArenaInfo.Record ArenaRecord
```

### Field Value

ArenaInfo.Record

## ArmorId

```
[Obsolete("Not used anymore since v100070")]
public int ArmorId
```

### Field Value

int

## AvatarAddress

```
public readonly Address AvatarAddress
```

### Field Value

Address

## AvatarName

```
public readonly string AvatarName
```

### Field Value

string

## CombatPoint

```
[Obsolete("Not used anymore since v100070")]
public long CombatPoint
```

### Field Value

long

## Level

```
[Obsolete("Not used anymore since v100070")]
public int Level
```

### Field Value

int

## Receive

```
[Obsolete("Not used anymore since v100070")]
public bool Receive
```

### Field Value

bool

# Properties

## Active

```
public bool Active { get; }
```

### Property Value

bool

## DailyChallengeCount

```
public int DailyChallengeCount { get; }
```

### Property Value

int

## Score

```
public int Score { get; }
```

### Property Value

int

# Methods

## Activate()

```
public void Activate()
```

## Clone()

```
public ArenaInfo Clone()
```

### Returns

ArenaInfo

## GetRewardCount()

```
public int GetRewardCount()
```

### Returns

int

## IsActive(Dictionary)

```
public static bool IsActive(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

bool

## ResetCount()

```
public void ResetCount()
```

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Update(ArenaInfo, Result, Func<int, int, Result, (int challengerScore, int defenderScore)>)

```
public int Update(ArenaInfo enemyInfo, BattleLog.Result result, Func<int, int,
BattleLog.Result, (int challengerScore, int defenderScore)> scoreGetter)
```

### Parameters

`enemyInfo` ArenaInfo

`result` BattleLog.Result

`scoreGetter` Func<int, int, BattleLog.Result, (int challengerScore, int defenderScore)>

### Returns

int

## UpdateV1(AvatarState, CharacterSheet)

```
[Obsolete("Use Update()")]
public void UpdateV1(AvatarState state, CharacterSheet characterSheet)
```

### Parameters

`state` AvatarState

`characterSheet` CharacterSheet

## UpdateV2(AvatarState, CharacterSheet, CostumeStatSheet)

```
[Obsolete("Use Update()")]
public void UpdateV2(AvatarState state, CharacterSheet characterSheet,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`state` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

## UpdateV3(AvatarState, ArenaInfo, Result)

```
[Obsolete("Use Update()")]
public int UpdateV3(AvatarState avatarState, ArenaInfo enemyInfo,
BattleLog.Result result)
```

### Parameters

`avatarState` AvatarState

`enemyInfo` ArenaInfo

`result` BattleLog.Result

### Returns

int

## UpdateV4(ArenaInfo, Result)

```
[Obsolete("Use Update()")]
public int UpdateV4(ArenaInfo enemyInfo, BattleLog.Result result)
```

### Parameters

`enemyInfo` ArenaInfo

`result` BattleLog.Result

### Returns

int

## UpdateV5(ArenaInfo, Result)

```
[Obsolete("Use Update()")]
public int UpdateV5(ArenaInfo enemyInfo, BattleLog.Result result)
```

### Parameters

`enemyInfo` ArenaInfo

`result` BattleLog.Result

### Returns

int

# Class ArenaInfo.Record

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class ArenaInfo.Record : IState
```

**Inheritance**

object ← ArenaInfo.Record

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Record()

```
public Record()
```

### Record(Dictionary)

```
public Record(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## Fields

## Draw

```
public int Draw
```

### Field Value

int

## Lose

```
public int Lose
```

### Field Value

int

## Win

```
public int Win
```

### Field Value

int

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class AuthorizedMinersState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class AuthorizedMinersState : State, IState
```

**Inheritance**

object ← State ← AuthorizedMinersState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AuthorizedMinersState(Dictionary)

```
public AuthorizedMinersState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### AuthorizedMinersState(IEnumerable<Address>, long, long)

```
public AuthorizedMinersState(IEnumerable<Address> miners, long interval,
long validUntil)
```

### Parameters

`miners` IEnumerable<Address>

`interval` long

`validUntil` long

## Fields

### Address

```
public static readonly Address Address
```

#### Field Value

Address

## Properties

### Interval

```
public long Interval { get; }
```

#### Property Value

long

### Miners

```
public ImmutableHashSet<Address> Miners { get; }
```

#### Property Value

ImmutableHashSet<Address>

## ValidUntil

```
public long ValidUntil { get; }
```

### Property Value

long

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class AvatarState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Agent□ □□□□ □ Avatar□ □□ □□□□.

```
[Serializable]
public class AvatarState : State, IState, ICloneable
```

### Inheritance

object ← State ← AvatarState

### Implements

IState, ICloneable

### Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

### Extension Methods

AvatarStateExtensions.GetLevelAndExp(AvatarState, CharacterLevelSheet, int, int) , AvatarStateExtensions.GetLevelAndExpForEventDungeon(AvatarState, CharacterLevelSheet, EventScheduleSheet.Row, int, int) , AvatarStateExtensions.GetLevelAndExpV1(AvatarState, CharacterLevelSheet, int, int) , AvatarStateExtensions.UpdateExp(AvatarState, int, long) , AvatarStateExtensions.UpdateInventory(AvatarState, List<ItemBase>) , AvatarStateExtensions.UpdateMonsterMap(AvatarState, StageWaveSheet, int) , AvatarStateExtensions.ValidEquipmentAndCostume(AvatarState, IEnumerable<Guid>, List<Guid>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, long, string) , AvatarStateExtensions.ValidEquipmentAndCostumeV2(AvatarState, IEnumerable<Guid>, List<Guid>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, long, string, GameConfigState)

# Constructors

## AvatarState(Dictionary)

```
public AvatarState(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## AvatarState(List)

```
public AvatarState(List serialized)
```

### Parameters

`serialized` List

## AvatarState(Address, Address, long, QuestList, WorldInformation, Address, string)

```
public AvatarState(Address address, Address agentAddress, long blockIndex, QuestList
questList, WorldInformation worldInformation, Address rankingMapAddress, string name
= null)
```

### Parameters

`address` Address

`agentAddress` Address

`blockIndex` long

`questList` QuestList

`worldInformation` WorldInformation

`rankingMapAddress` Address

`name` string

## AvatarState(AvatarState)

```
public AvatarState(AvatarState avatarState)
```

### Parameters

`avatarState` AvatarState

# Fields

## CombinationSlotCapacity

```
public const int CombinationSlotCapacity = 8
```

### Field Value

int

## CurrentVersion

```
public const int CurrentVersion = 2
```

### Field Value

int

## DefaultCombinationSlotCount

```
public const int DefaultCombinationSlotCount = 4
```

### Field Value

int

## RankingMapAddress

```
[Obsolete("don't use this field.")]
public readonly Address RankingMapAddress
```

### Field Value

Address

## actionPoint

```
public int actionPoint
```

### Field Value

int

## agentAddress

```
public Address agentAddress
```

### Field Value

Address

## blockIndex

```
public long blockIndex
```

### Field Value

long

## characterId

```
public int characterId
```

### Field Value

int

## combinationSlotAddresses

```
[Obsolete("don't use this field, use AllCombinationSlotState instead.")]
public List<Address> combinationSlotAddresses
```

### Field Value

List<Address>

## dailyRewardReceivedIndex

```
public long dailyRewardReceivedIndex
```

### Field Value

long

## ear

```
public int ear
```

### Field Value

int

## eventMap

```
public CollectionMap eventMap
```

### Field Value

CollectionMap

## exp

```
public long exp
```

### Field Value

long

## hair

```
public int hair
```

### Field Value

int

## inventory

```
public Inventory inventory
```

### Field Value

Inventory

## itemMap

```
public CollectionMap itemMap
```

### Field Value

CollectionMap

## lens

```
public int lens
```

### Field Value

int

## level

```
public int level
```

### Field Value

int

## mailBox

```
public MailBox mailBox
```

### Field Value

MailBox

## monsterMap

```
public CollectionMap monsterMap
```

### Field Value

CollectionMap

## name

```
public string name
```

### Field Value

string

## questList

```
public QuestList questList
```

### Field Value

QuestList

## stageMap

```
public CollectionMap stageMap
```

### Field Value

CollectionMap

## tail

```
public int tail
```

### Field Value

int

## updatedAt

```
public long updatedAt
```

### Field Value

long

## worldInformation

```
public WorldInformation worldInformation
```

### Field Value

WorldInformation

# Properties

## NameWithHash

```
public string NameWithHash { get; }
```

### Property Value

string

## Nonce

```
public int Nonce { get; }
```

### Property Value

int

## Version

```
public int Version { get; }
```

### Property Value

int

# Methods

## Apply(Player, long)

```
public void Apply(Player player, long index)
```

### Parameters

`player` Player

`index` long

## Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

### Returns

object

A new object that is a copy of this instance.

## Create(Address, Address, long, AvatarSheets, Address, string)

```
public static AvatarState Create(Address address, Address agentAddress, long
blockIndex, AvatarSheets avatarSheets, Address rankingMapAddress, string name
= null)
```

### Parameters

`address` Address

`agentAddress` Address

`blockIndex` long

`avatarSheets` AvatarSheets

`rankingMapAddress` Address

`name` string

### Returns

AvatarState

## CreateAvatarAddress()

```
public static Address CreateAvatarAddress()
```

### Returns

Address

## Customize(int, int, int, int)

```
public void Customize(int hair, int lens, int ear, int tail)
```

### Parameters

`hair` int

`lens` int

`ear` int

`tail` int

## EquipItems(IEnumerable<Guid>)

```
public void EquipItems(IEnumerable<Guid> itemIds)
```

### Parameters

`itemIds` IEnumerable<Guid>

## GetArmorId()

```
public int GetArmorId()
```

### Returns

int

## GetNonFungibleItems<T>(List<Guid>)

```
public List<T> GetNonFungibleItems<T>(List<Guid> itemIds)
```

### Parameters

`itemIds` List<Guid>

### Returns

List<T>

### Type Parameters

T

## GetPortraitId()

```
public int GetPortraitId()
```

### Returns

int

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeList()

```
public IValue SerializeList()
```

### Returns

IValue

## Update(IStageSimulator)

```
public void Update(IStageSimulator stageSimulator)
```

### Parameters

`stageSimulator` IStageSimulator

## Update(Mail)

```
public void Update(Mail mail)
```

### Parameters

`mail` Mail

## Update2(Mail)

```
[Obsolete("Use Update")]
public void Update2(Mail mail)
```

### Parameters

`mail` Mail

## Update3(Mail)

```
[Obsolete("Use Update")]
public void Update3(Mail mail)
```

### Parameters

`mail` Mail

## UpdateFromAddCostume(Costume)

```
public void UpdateFromAddCostume(Costume costume)
```

### Parameters

`costume` Costume

## UpdateFromAddItem(ItemBase, int, bool)

```
public void UpdateFromAddItem(ItemBase itemUsable, int count, bool canceled)
```

### Parameters

`itemUsable` ItemBase

`count` int

`canceled` bool

## UpdateFromAddItem(ItemUsable, bool)

```
public void UpdateFromAddItem(ItemUsable itemUsable, bool canceled)
```

### Parameters

`itemUsable` ItemUsable

`canceled` bool

## UpdateFromCombination(ItemUsable)

```
public void UpdateFromCombination(ItemUsable itemUsable)
```

### Parameters

`itemUsable` ItemUsable

## UpdateFromItemEnhancement(Equipment)

```
public void UpdateFromItemEnhancement(Equipment equipment)
```

### Parameters

`equipment` Equipment

## UpdateFromQuestReward(Quest, MaterialItemSheet)

```
public void UpdateFromQuestReward(Quest quest, MaterialItemSheet materialItemSheet)
```

### Parameters

`quest` Quest

`materialItemSheet` MaterialItemSheet

## UpdateFromRapidCombination(ResultModel, long)

```
public void UpdateFromRapidCombination(CombinationConsumable5.ResultModel result,
long requiredIndex)
```

### Parameters

`result` CombinationConsumable5.ResultModel

`requiredIndex` long

## UpdateFromRapidCombinationV2(ResultModel, long)

```
public void UpdateFromRapidCombinationV2(RapidCombination5.ResultModel result,
long requiredIndex)
```

### Parameters

`result` RapidCombination5.ResultModel

`requiredIndex` long

## UpdateGeneralQuest(IEnumerable<QuestEventType>)

```
public void UpdateGeneralQuest(IEnumerable<QuestEventType> types)
```

### Parameters

`types` IEnumerable<QuestEventType>

## UpdateQuestRewards(MaterialItemSheet)

```
public void UpdateQuestRewards(MaterialItemSheet materialItemSheet)
```

### Parameters

`materialItemSheet` MaterialItemSheet

## UpdateTemp(Mail, long)

```
[Obsolete("No longer in use.")]
public void UpdateTemp(Mail mail, long currentBlockIndex)
```

### Parameters

`mail` Mail

`currentBlockIndex` long

## ValidateConsumable(List<Guid>, long)

```
public List<int> ValidateConsumable(List<Guid> consumableIds,
long currentBlockIndex)
```

### Parameters

`consumableIds` List<Guid>

`currentBlockIndex` long

### Returns

List<int>

## ValidateConsumableV2(List<Guid>, long, GameConfigState)

```
public List<int> ValidateConsumableV2(List<Guid> consumableIds, long
currentBlockIndex, GameConfigState gameConfigState)
```

### Parameters

`consumableIds` List<Guid>

`currentBlockIndex` long

`gameConfigState` GameConfigState

### Returns

List<int>

## ValidateCostume(HashSet<int>)

```
public void ValidateCostume(HashSet<int> costumeIds)
```

### Parameters

`costumeIds` HashSet<int>

## ValidateCostume(IEnumerable<Guid>)

```
public List<int> ValidateCostume(IEnumerable<Guid> costumeIds)
```

### Parameters

`costumeIds` IEnumerable<Guid>

### Returns

List<int>

## ValidateCostumeV2(IEnumerable<Guid>, GameConfigState)

```
public List<Costume> ValidateCostumeV2(IEnumerable<Guid> costumeIds,
GameConfigState gameConfigState)
```

### Parameters

`costumeIds` IEnumerable<Guid>

`gameConfigState` GameConfigState

### Returns

List<Costume>

## ValidateEquipments(List<Guid>, long)

```
public void ValidateEquipments(List<Guid> equipmentIds, long blockIndex)
```

### Parameters

`equipmentIds` List<Guid>

`blockIndex` long

## ValidateEquipmentsV2(List<Guid>, long)

```
public List<Equipment> ValidateEquipmentsV2(List<Guid> equipmentIds,
long blockIndex)
```

### Parameters

`equipmentIds` List<Guid>

`blockIndex` long

### Returns

List<Equipment>

## ValidateEquipmentsV3(List<Guid>, long, GameConfigState)

```
public List<Equipment> ValidateEquipmentsV3(List<Guid> equipmentIds, long
blockIndex, GameConfigState gameConfigState)
```

### Parameters

`equipmentIds` List<Guid>

`blockIndex` long

`gameConfigState` GameConfigState

### Returns

List<Equipment>

# ValidateItemRequirement(List<int>, List<Equipment>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, string)

```
public void ValidateItemRequirement(List<int> itemIds, List<Equipment> equipments,
ItemRequirementSheet requirementSheet, EquipmentItemRecipeSheet recipeSheet,
EquipmentItemSubRecipeSheetV2 subRecipeSheet, EquipmentItemOptionSheet
itemOptionSheet, string addressesHex)
```

## Parameters

`itemIds` List<int>

`equipments` List<Equipment>

`requirementSheet` ItemRequirementSheet

`recipeSheet` EquipmentItemRecipeSheet

`subRecipeSheet` EquipmentItemSubRecipeSheetV2

`itemOptionSheet` EquipmentItemOptionSheet

`addressesHex` string

# Class CollectionState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Represents the state of a collection.

```
public class CollectionState : IBencodable
```

**Inheritance**

object ← CollectionState

**Implements**

IBencodable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CollectionState()

```
public CollectionState()
```

### CollectionState(IValue)

```
public CollectionState(IValue bencoded)
```

#### Parameters

`bencoded` IValue

## CollectionState(List)

```
public CollectionState(List serialized)
```

### Parameters

`serialized` List

# Fields

## Ids

```
public SortedSet<int> Ids
```

### Field Value

SortedSet<int>

# Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that

additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

# Methods

## GetModifiers(CollectionSheet)

```
public List<StatModifier> GetModifiers(CollectionSheet collectionSheet)
```

### Parameters

`collectionSheet` CollectionSheet

### Returns

List<StatModifier>

# Class CombinationSlotState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class CombinationSlotState : State, IState
```

**Inheritance**

object ← State ← CombinationSlotState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

CombinationSlotStateExtensions.TryGetMail(CombinationSlotState, long, long, out CombinationMail, out ItemEnhanceMail) , CombinationSlotStateExtensions.TryGetResultId(CombinationSlotState, out Guid) , CombinationSlotStateExtensions.ValidateFromAction(CombinationSlotState, long, AvatarState, int, string, string)

# Constructors

## CombinationSlotState(Dictionary)

```
public CombinationSlotState(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## CombinationSlotState(Address, int)

```
public CombinationSlotState(Address address, int index = 0)
```

### Parameters

`address` Address

`index` int

# Fields

## DeriveFormat

```
public const string DeriveFormat = "combination-slot-{0}"
```

### Field Value

string

# Properties

## Index

It is a CombinationSlot index. start from 0.

```
public int Index { get; set; }
```

### Property Value

int

## IsUnlocked

```
public bool IsUnlocked { get; }
```

### Property Value

bool

## PetId

```
public int? PetId { get; }
```

### Property Value

int?

## RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

### Property Value

long

## Result

```
public AttachmentActionResult Result { get; }
```

### Property Value

AttachmentActionResult

## WorkCompleteBlockIndex

Serialize key is "unlockBlockIndex".

```
public long WorkCompleteBlockIndex { get; }
```

### Property Value

long

## WorkStartBlockIndex

Serialize key is "startBlockIndex".

```
public long WorkStartBlockIndex { get; }
```

### Property Value

long

# Methods

## DeriveAddress(Address, int)

Is only used for migration legacy state.

```
public static Address DeriveAddress(Address address, int slotIndex)
```

### Parameters

`address` Address

`slotIndex` int

### Returns

Address

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Unlock()

```
public bool Unlock()
```

### Returns

bool

## Update(AttachmentActionResult, long, long, int?)

```
public void Update(AttachmentActionResult result, long blockIndex, long
workCompleteBlockIndex, int? petId = null)
```

### Parameters

`result` AttachmentActionResult

`blockIndex` long

`workCompleteBlockIndex` long

`petId` int?

## Update(long)

```
public void Update(long blockIndex)
```

### Parameters

`blockIndex` long

## Update(long, Material, int)

```
public void Update(long blockIndex, Material material, int count)
```

### Parameters

`blockIndex` long

`material` Material

`count` int

## UpdateV2(long, Material, int)

```
public void UpdateV2(long blockIndex, Material material, int count)
```

### Parameters

`blockIndex` long

`material` Material

`count` int

## Validate(AvatarState, long)

```
[Obsolete("Use ValidateV2")]
public bool Validate(AvatarState avatarState, long blockIndex)
```

### Parameters

`avatarState` AvatarState

`blockIndex` long

### Returns

bool

## ValidateSlotIndex(int)

```
public static bool ValidateSlotIndex(int index)
```

### Parameters

`index` int

### Returns

bool

## ValidateV2(long)

```
public bool ValidateV2(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

bool

# Class CpState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

CpState is a state that stores the cp of the player.

```
public class CpState : IBencodable, IState
```

**Inheritance**

object ← CpState

**Implements**

IBencodable, IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CpState(IValue)

```
public CpState(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### CpState(long)

```
public CpState(long cp)
```

#### Parameters

cp long

# Fields

## Cp

```
public long Cp
```

### Field Value

long

# Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class CreditsState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class CreditsState : State, IState
```

**Inheritance**

object ← State ← CreditsState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CreditsState(Dictionary)

```
public CreditsState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### CreditsState(IEnumerable<string>)

```
public CreditsState(IEnumerable<string> names)
```

#### Parameters

names IEnumerable<string>

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

# Properties

## Names

```
public IImmutableList<string> Names { get; }
```

### Property Value

IImmutableList<string>

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class CrystalCostState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalCostState : IState
```

**Inheritance**

object ← CrystalCostState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CrystalCostState(Address, List)

```
public CrystalCostState(Address address, List serialized)
```

#### Parameters

`address` Address

`serialized` List

### CrystalCostState(Address, FungibleAssetValue)

```
public CrystalCostState(Address address, FungibleAssetValue crystal)
```

#### Parameters

`address` Address

`crystal` FungibleAssetValue

# Fields

## Address

```
public Address Address
```

### Field Value

Address

## CRYSTAL

```
public FungibleAssetValue CRYSTAL
```

### Field Value

FungibleAssetValue

## Count

```
public int Count
```

### Field Value

int

## DailyIntervalIndex

```
public const long DailyIntervalIndex = 7200
```

### Field Value

long

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class CrystalRandomSkillState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class CrystalRandomSkillState : IState
```

**Inheritance**

object ← CrystalRandomSkillState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CrystalRandomSkillState(Address, List)

```
public CrystalRandomSkillState(Address address, List serialized)
```

#### Parameters

`address` Address

`serialized` List

### CrystalRandomSkillState(Address, int)

```
public CrystalRandomSkillState(Address address, int stageId)
```

#### Parameters

`address` Address

`stageId` int

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## SkillIds

```
public List<int> SkillIds { get; }
```

### Property Value

List<int>

## StageId

```
public int StageId { get; }
```

### Property Value

int

## StarCount

```
public int StarCount { get; }
```

### Property Value

int

## Methods

### GetHighestRankSkill(CrystalRandomBuffSheet)

```
public int GetHighestRankSkill(CrystalRandomBuffSheet crystalRandomBuffSheet)
```

#### Parameters

`crystalRandomBuffSheet` CrystalRandomBuffSheet

#### Returns

int

### GetSkill(int, CrystalRandomBuffSheet, SkillSheet)

```
public static Skill GetSkill(int skillId, CrystalRandomBuffSheet
crystalRandomBuffSheet, SkillSheet skillSheet)
```

#### Parameters

`skillId` int

`crystalRandomBuffSheet` CrystalRandomBuffSheet

`skillSheet` SkillSheet

#### Returns

Skill

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Update(List<int>)

```
public void Update(List<int> skillIds)
```

### Parameters

`skillIds` List<int>

## Update(int, CrystalStageBuffGachaSheet)

```
public void Update(int gotStarCount, CrystalStageBuffGachaSheet sheet)
```

### Parameters

`gotStarCount` int

`sheet` CrystalStageBuffGachaSheet

# Class DuplicateRedeemException

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateRedeemException : InvalidOperationException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateRedeemException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

## DuplicateRedeemException(SerializationInfo, StreamingContext)

```
public DuplicateRedeemException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicateRedeemException(string)

```
public DuplicateRedeemException(string s)
```

## Parameters

s [string]

# Class FailedToUnregisterInShopStateException

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class FailedToUnregisterInShopStateException : Exception, ISerializable
```

### Inheritance

object ← Exception ← FailedToUnregisterInShopStateException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### FailedToUnregisterInShopStateException(SerializationInfo, StreamingContext)

```
protected FailedToUnregisterInShopStateException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# FailedToUnregisterInShopStateException(string)

```
public FailedToUnregisterInShopStateException(string message)
```

## Parameters

`message` string

# Class GameConfigState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class GameConfigState : State, IState
```

**Inheritance**

object ← State ← GameConfigState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GameConfigState()

```
public GameConfigState()
```

### GameConfigState(Dictionary)

```
public GameConfigState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## GameConfigState(string)

```
public GameConfigState(string csv)
```

### Parameters

`csv` string

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

# Properties

## ActionPointMax

```
public int ActionPointMax { get; }
```

### Property Value

int

## AdventureBossActiveInterval

```
public long AdventureBossActiveInterval { get; }
```

### Property Value

long

## AdventureBossClaimInterval

```
public long AdventureBossClaimInterval { get; }
```

### Property Value

long

## AdventureBossInactiveInterval

```
public long AdventureBossInactiveInterval { get; }
```

### Property Value

long

## AdventureBossMinBounty

```
public BigInteger AdventureBossMinBounty { get; }
```

### Property Value

BigInteger

## AdventureBossNcgApRatio

```
public decimal AdventureBossNcgApRatio { get; }
```

### Property Value

decimal

## AdventureBossNcgRuneRatio

```
public decimal AdventureBossNcgRuneRatio { get; }
```

### Property Value

decimal

## AdventureBossWantedRequiredStakingLevel

```
public int AdventureBossWantedRequiredStakingLevel { get; }
```

### Property Value

int

## BattleArenaInterval

```
public int BattleArenaInterval { get; }
```

### Property Value

int

## CustomEquipmentCraftIconCostMultiplier

```
public long CustomEquipmentCraftIconCostMultiplier { get; }
```

### Property Value

long

## DailyArenaInterval

```
public int DailyArenaInterval { get; }
```

### Property Value

int

## DailyRewardInterval

```
public int DailyRewardInterval { get; }
```

### Property Value

int

## DailyRuneRewardAmount

```
public int DailyRuneRewardAmount { get; }
```

### Property Value

int

## DailyWorldBossInterval

```
public int DailyWorldBossInterval { get; }
```

### Property Value

int

## HourglassPerBlock

```
public int HourglassPerBlock { get; }
```

### Property Value

int

## RequireCharacterLevel_ConsumableSlot1

```
public int RequireCharacterLevel_ConsumableSlot1 { get; }
```

### Property Value

int

## RequireCharacterLevel_ConsumableSlot2

```
public int RequireCharacterLevel_ConsumableSlot2 { get; }
```

### Property Value

int

## RequireCharacterLevel_ConsumableSlot3

```
public int RequireCharacterLevel_ConsumableSlot3 { get; }
```

### Property Value

int

## RequireCharacterLevel_ConsumableSlot4

```
public int RequireCharacterLevel_ConsumableSlot4 { get; }
```

### Property Value

int

## RequireCharacterLevel_ConsumableSlot5

```
public int RequireCharacterLevel_ConsumableSlot5 { get; }
```

### Property Value

int

## RequireCharacterLevel_EarCostumeSlot

```
public int RequireCharacterLevel_EarCostumeSlot { get; }
```

### Property Value

int

## RequireCharacterLevel_EquipmentSlotArmor

```
public int RequireCharacterLevel_EquipmentSlotArmor { get; }
```

### Property Value

int

## RequireCharacterLevel_EquipmentSlotAura

```
public int RequireCharacterLevel_EquipmentSlotAura { get; }
```

### Property Value

int

## RequireCharacterLevel_EquipmentSlotBelt

```
public int RequireCharacterLevel_EquipmentSlotBelt { get; }
```

### Property Value

int

## RequireCharacterLevel_EquipmentSlotGrimoire

```
public int RequireCharacterLevel_EquipmentSlotGrimoire { get; }
```

### Property Value

int

## RequireCharacterLevel_EquipmentSlotNecklace

```
public int RequireCharacterLevel_EquipmentSlotNecklace { get; }
```

### Property Value

int

## RequireCharacterLevel_EquipmentSlotRing1

```
public int RequireCharacterLevel_EquipmentSlotRing1 { get; }
```

### Property Value

int

## RequireCharacterLevel_EquipmentSlotRing2

```
public int RequireCharacterLevel_EquipmentSlotRing2 { get; }
```

### Property Value

int

## RequireCharacterLevel_EquipmentSlotWeapon

```
public int RequireCharacterLevel_EquipmentSlotWeapon { get; }
```

### Property Value

int

## RequireCharacterLevel_EyeCostumeSlot

```
public int RequireCharacterLevel_EyeCostumeSlot { get; }
```

### Property Value

int

## RequireCharacterLevel_FullCostumeSlot

```
public int RequireCharacterLevel_FullCostumeSlot { get; }
```

### Property Value

int

## RequireCharacterLevel_HairCostumeSlot

```
public int RequireCharacterLevel_HairCostumeSlot { get; }
```

### Property Value

int

## RequireCharacterLevel_TailCostumeSlot

```
public int RequireCharacterLevel_TailCostumeSlot { get; }
```

### Property Value

int

## RequireCharacterLevel_TitleSlot

```
public int RequireCharacterLevel_TitleSlot { get; }
```

### Property Value

int

## RuneSkillSlotCrystalUnlockCost

```
public int RuneSkillSlotCrystalUnlockCost { get; }
```

### Property Value

int

## RuneSkillSlotUnlockCost

```
public int RuneSkillSlotUnlockCost { get; }
```

### Property Value

int

## RuneStatSlotCrystalUnlockCost

```
public int RuneStatSlotCrystalUnlockCost { get; }
```

### Property Value

int

## RuneStatSlotUnlockCost

```
public int RuneStatSlotUnlockCost { get; }
```

### Property Value

int

## ShatterStrikeMaxDamage

```
public long ShatterStrikeMaxDamage { get; }
```

### Property Value

long

## StakeRegularFixedRewardSheet_V2_StartBlockIndex

```
public long StakeRegularFixedRewardSheet_V2_StartBlockIndex { get; }
```

### Property Value

long

## StakeRegularRewardSheet_V2_StartBlockIndex

```
public long StakeRegularRewardSheet_V2_StartBlockIndex { get; }
```

### Property Value

long

## StakeRegularRewardSheet_V3_StartBlockIndex

```
public long StakeRegularRewardSheet_V3_StartBlockIndex { get; }
```

### Property Value

long

## StakeRegularRewardSheet_V4_StartBlockIndex

```
public long StakeRegularRewardSheet_V4_StartBlockIndex { get; }
```

### Property Value

long

## StakeRegularRewardSheet_V5_StartBlockIndex

```
public long StakeRegularRewardSheet_V5_StartBlockIndex { get; }
```

### Property Value

long

## WeeklyArenaInterval

```
public int WeeklyArenaInterval { get; }
```

### Property Value

int

## WorldBossRequiredInterval

```
public int WorldBossRequiredInterval { get; }
```

### Property Value

int

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Set(GameConfigSheet)

```
public void Set(GameConfigSheet sheet)
```

### Parameters

`sheet` GameConfigSheet

## Update(Row)

```
public void Update(GameConfigSheet.Row row)
```

### Parameters

`row` GameConfigSheet.Row

# Class GoldBalanceState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class GoldBalanceState : State, IState, ICloneable
```

**Inheritance**

object ← State ← GoldBalanceState

**Implements**

IState, ICloneable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GoldBalanceState(Dictionary)

```
public GoldBalanceState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### GoldBalanceState(IValue)

```
public GoldBalanceState(IValue serialized)
```

#### Parameters

`serialized` IValue

## GoldBalanceState(Address, FungibleAssetValue)

```
public GoldBalanceState(Address address, FungibleAssetValue gold)
```

### Parameters

`address` Address

`gold` FungibleAssetValue

# Fields

## Gold

```
public readonly FungibleAssetValue Gold
```

### Field Value

FungibleAssetValue

# Methods

## Add(FungibleAssetValue)

```
public GoldBalanceState Add(FungibleAssetValue adder)
```

### Parameters

`adder` FungibleAssetValue

### Returns

GoldBalanceState

# Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

## Returns

object

A new object that is a copy of this instance.

# Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Class GoldCurrencyState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class GoldCurrencyState : State, IState, ISerializable
```

**Inheritance**

object ← State ← GoldCurrencyState

**Implements**

IState, ISerializable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### GoldCurrencyState(Dictionary)

```
public GoldCurrencyState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### GoldCurrencyState(Currency)

```
public GoldCurrencyState(Currency currency)
```

#### Parameters

`currency` Currency

## GoldCurrencyState(Currency, long)

```
public GoldCurrencyState(Currency currency, long initialSupply)
```

### Parameters

`currency` Currency

`initialSupply` long

## GoldCurrencyState(SerializationInfo, StreamingContext)

```
protected GoldCurrencyState(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

## DEFAULT_INITIAL_SUPPLY

```
public const long DEFAULT_INITIAL_SUPPLY = 1000000000
```

### Field Value

long

## Properties

### Currency

```
public Currency Currency { get; }
```

### Property Value

Currency

### InitialSupply

```
public long InitialSupply { get; }
```

### Property Value

long

## Methods

### GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class HammerPointState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class HammerPointState : IState
```

**Inheritance**

object ← HammerPointState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### HammerPointState(Address, List)

```
public HammerPointState(Address address, List serialized)
```

#### Parameters

`address` Address

`serialized` List

### HammerPointState(Address, int)

```
public HammerPointState(Address address, int recipeId)
```

#### Parameters

`address` Address

`recipeId` int

# Properties

## Address

```
public Address Address { get; }
```

### Property Value

Address

## HammerPoint

```
public int HammerPoint { get; }
```

### Property Value

int

## RecipeId

```
public int RecipeId { get; }
```

### Property Value

int

# Methods

## AddHammerPoint(int, CrystalHammerPointSheet)

```
public void AddHammerPoint(int point, CrystalHammerPointSheet sheet = null)
```

### Parameters

`point` int

`sheet` CrystalHammerPointSheet

## ResetHammerPoint()

```
public void ResetHammerPoint()
```

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Interface IState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public interface IState
```

## Methods

### Serialize()

```
IValue Serialize()
```

#### Returns

IValue

# Class InfiniteTowerBoardState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Represents the board state for infinite tower, tracking clear counts per floor.

```
[Serializable]
public class InfiniteTowerBoardState : IState
```

**Inheritance**

object ← InfiniteTowerBoardState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### InfiniteTowerBoardState(List)

Creates an infinite tower board state from serialized data.

```
public InfiniteTowerBoardState(List serialized)
```

#### Parameters

`serialized` List

The serialized data.

### InfiniteTowerBoardState(int)

Creates a new infinite tower board state.

```
public InfiniteTowerBoardState(int infiniteTowerId)
```

### Parameters

`infiniteTowerId` int

The infinite tower ID.

## Properties

## FloorClearCounts

Dictionary mapping floor ID to the number of players who cleared it.

```
public Dictionary<int, int> FloorClearCounts { get; }
```

### Property Value

Dictionary<int, int>

## InfiniteTowerId

The infinite tower ID this board belongs to.

```
public int InfiniteTowerId { get; }
```

### Property Value

int

## LastUpdatedBlockIndex

The block index when this board was last updated.

```
public long LastUpdatedBlockIndex { get; }
```

### Property Value

long

## Methods

## GetFloorClearCount(int)

Gets the clear count for a specific floor.

```
public int GetFloorClearCount(int floorId)
```

### Parameters

`floorId` int

The floor ID.

### Returns

int

The number of players who cleared this floor.

## GetStatistics()

Gets statistics about the board state.

```
public Dictionary<string, object> GetStatistics()
```

### Returns

Dictionary<string, object>

A dictionary containing various statistics.

## GetTopFloorsByClearCount(int)

Gets the top floors by clear count.

```
public List<int> GetTopFloorsByClearCount(int limit = 10)
```

### Parameters

`limit` int

The maximum number of floors to return.

### Returns

List<int>

A list of floor IDs sorted by clear count (descending).

## GetTotalClearedFloors()

Gets the total number of floors that have been cleared by at least one player.

```
public int GetTotalClearedFloors()
```

### Returns

int

The number of floors with at least one clear.

## RecordFloorClear(int, long)

Records a floor clear for a player.

```
public void RecordFloorClear(int floorId, long blockIndex)
```

### Parameters

`floorId` int

The floor ID that was cleared.

`blockIndex` long

The current block index.

## Serialize()

Serializes the state to Bencodex format.

```
public IValue Serialize()
```

### Returns

IValue

The serialized state.

# Class InvalidRedeemCodeException

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidRedeemCodeException : KeyNotFoundException, ISerializable
```

### Inheritance

object ← Exception ← SystemException ← KeyNotFoundException ← InvalidRedeemCodeException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### InvalidRedeemCodeException()

```
public InvalidRedeemCodeException()
```

### InvalidRedeemCodeException(SerializationInfo, StreamingContext)

```
public InvalidRedeemCodeException(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

## InvalidRedeemCodeException(string)

```
public InvalidRedeemCodeException(string s)
```

### Parameters

`s` string

# Class ItemSlotState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class ItemSlotState : IState
```

**Inheritance**

object ← ItemSlotState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ItemSlotState(List)

```
public ItemSlotState(List serialized)
```

#### Parameters

`serialized` List

### ItemSlotState(BattleType)

```
public ItemSlotState(BattleType battleType)
```

#### Parameters

`battleType` BattleType

# Properties

## BattleType

```
public BattleType BattleType { get; }
```

### Property Value

BattleType

## Costumes

```
public List<Guid> Costumes { get; }
```

### Property Value

List<Guid>

## Equipments

```
public List<Guid> Equipments { get; }
```

### Property Value

List<Guid>

# Methods

## DeriveAddress(Address, BattleType)

```
public static Address DeriveAddress(Address avatarAddress, BattleType battleType)
```

### Parameters

`avatarAddress` Address

`battleType` BattleType

### Returns

Address

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## UpdateCostumes(List<Guid>)

```
public void UpdateCostumes(List<Guid> costumes)
```

### Parameters

`costumes` List<Guid>

## UpdateEquipment(List<Guid>)

```
public void UpdateEquipment(List<Guid> equipments)
```

### Parameters

`equipments` List<Guid>

# Class LazyState<TState, TEncoding>

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public sealed class LazyState<TState, TEncoding> : IState, ISerializable where
TState : IState where TEncoding : IValue
```

## Type Parameters

`TState`

`TEncoding`

**Inheritance**

object ← LazyState<TState, TEncoding>

**Implements**

IState, ISerializable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## LazyState(TState)

```
public LazyState(TState loadedValue)
```

### Parameters

`loadedValue` TState

## LazyState(TEncoding, Func<TEncoding, TState>)

```
public LazyState(TEncoding serialized, Func<TEncoding, TState> loader)
```

### Parameters

`serialized` TEncoding

`loader` Func<TEncoding, TState>

## Properties

### State

```
public TState State { get; }
```

### Property Value

TState

## Methods

### GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## GetStateOrSerializedEncoding(out TState, out TEncoding)

```
public bool GetStateOrSerializedEncoding(out TState loadedState, out
TEncoding serialized)
```

### Parameters

`loadedState` TState

`serialized` TEncoding

### Returns

bool

## LoadState(LazyState<TState, TEncoding>)

```
public static TState LoadState(LazyState<TState, TEncoding> lazyState)
```

### Parameters

`lazyState` LazyState<TState, TEncoding>

### Returns

TState

## LoadStatePair<T>(KeyValuePair<T, LazyState<TState, TEncoding>>)

```
public static KeyValuePair<T, TState> LoadStatePair<T>(KeyValuePair<T,
LazyState<TState, TEncoding>> lazyPair)
```

### Parameters

`lazyPair` KeyValuePair<T, LazyState<TState, TEncoding>>

### Returns

KeyValuePair<T, TState>

### Type Parameters

`T`

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class LegacyStakeState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class LegacyStakeState : State, IState
```

**Inheritance**

object ← State ← LegacyStakeState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### LegacyStakeState(Dictionary)

```
public LegacyStakeState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### LegacyStakeState(Address, long)

```
public LegacyStakeState(Address address, long startedBlockIndex)
```

#### Parameters

`address` Address

`startedBlockIndex` long

## LegacyStakeState(Address, long, long, long, StakeAchievements)

```
public LegacyStakeState(Address address, long startedBlockIndex, long
receivedBlockIndex, long cancellableBlockIndex, LegacyStakeState.StakeAchievements
achievements)
```

### Parameters

`address` Address

`startedBlockIndex` long

`receivedBlockIndex` long

`cancellableBlockIndex` long

`achievements` LegacyStakeState.StakeAchievements

# Fields

## CurrencyAsRewardStartIndex

```
public const long CurrencyAsRewardStartIndex = 6910000
```

### Field Value

long

## LockupInterval

```
public const long LockupInterval = 201600
```

### Field Value

long

## RewardInterval

```
public const long RewardInterval = 50400
```

### Field Value

long

## StakeRewardSheetV2Index

```
public const long StakeRewardSheetV2Index = 6700000
```

### Field Value

long

## StakeRewardSheetV3Index

```
public const long StakeRewardSheetV3Index = 7650000
```

### Field Value

long

# Properties

## Achievements

```
public LegacyStakeState.StakeAchievements Achievements { get; }
```

### Property Value

LegacyStakeState.StakeAchievements

## CancellableBlockIndex

```
public long CancellableBlockIndex { get; }
```

### Property Value

long

## ReceivedBlockIndex

```
public long ReceivedBlockIndex { get; }
```

### Property Value

long

## StartedBlockIndex

```
public long StartedBlockIndex { get; }
```

### Property Value

long

# Methods

## CalculateAccumulatedCurrencyCrystalRewards(long, out int, out int, out int)

```
public int CalculateAccumulatedCurrencyCrystalRewards(long blockIndex, out int
v1Step, out int v2Step, out int v3Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

`v3Step` int

### Returns

int

## CalculateAccumulatedCurrencyRewards(long, out int, out int, out int)

```
public int CalculateAccumulatedCurrencyRewards(long blockIndex, out int v1Step, out int v2Step, out int v3Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

`v3Step` int

### Returns

int

## CalculateAccumulatedCurrencyRewardsV1(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedCurrencyRewards() instead.")]
public int CalculateAccumulatedCurrencyRewardsV1(long blockIndex, out int v1Step,
```

```
out int v2Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

### Returns

int

## CalculateAccumulatedCurrencyRewardsV2(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedCurrencyRewards() instead.")]
public int CalculateAccumulatedCurrencyRewardsV2(long blockIndex, out int v1Step,
out int v2Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

### Returns

int

## CalculateAccumulatedItemRewards(long, out int, out int, out int)

```
public int CalculateAccumulatedItemRewards(long blockIndex, out int v1Step, out int
v2Step, out int v3Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

`v3Step` int

### Returns

int

## CalculateAccumulatedItemRewardsV1(long)

```
[Obsolete("Use CalculateAccumulatedItemRewards() instead.")]
public int CalculateAccumulatedItemRewardsV1(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

int

## CalculateAccumulatedItemRewardsV1(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedItemRewards() instead.")]
public int CalculateAccumulatedItemRewardsV1(long blockIndex, out int v1Step, out
int v2Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

### Returns

int

## CalculateAccumulatedItemRewardsV2(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedItemRewards() instead.")]
public int CalculateAccumulatedItemRewardsV2(long blockIndex, out int v1Step, out int v2Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

### Returns

int

## CalculateAccumulatedItemRewardsV3(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedItemRewards() instead.")]
public int CalculateAccumulatedItemRewardsV3(long blockIndex, out int v1Step, out int v2Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

### Returns

int

## CalculateAccumulatedRuneRewards(long)

```
public int CalculateAccumulatedRuneRewards(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

int

## CalculateAccumulatedRuneRewards(long, out int, out int, out int)

```
public int CalculateAccumulatedRuneRewards(long blockIndex, out int v1Step, out int
v2Step, out int v3Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

`v3Step` int

### Returns

int

# CalculateAccumulatedRuneRewardsV1(long)

```
[Obsolete("Use CalculateAccumulatedRuneRewards() instead.")]
public int CalculateAccumulatedRuneRewardsV1(long blockIndex)
```

## Parameters

`blockIndex` long

## Returns

int

# CalculateAccumulatedRuneRewardsV1(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedRuneRewards() instead.")]
public int CalculateAccumulatedRuneRewardsV1(long blockIndex, out int v1Step, out
int v2Step)
```

## Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

## Returns

int

# CalculateAccumulatedRuneRewardsV2(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedRuneRewards() instead.")]
public int CalculateAccumulatedRuneRewardsV2(long blockIndex, out int v1Step, out
```

```
int v2Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

### Returns

int

## CalculateAccumulatedRuneRewardsV3(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedRuneRewards() instead.")]
public int CalculateAccumulatedRuneRewardsV3(long blockIndex, out int v1Step, out
int v2Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

### Returns

int

## Claim(long)

```
public void Claim(long blockIndex)
```

### Parameters

`blockIndex` long

## DeriveAddress(Address)

```
public static Address DeriveAddress(Address agentAddress)
```

### Parameters

`agentAddress` Address

### Returns

Address

## GetClaimableBlockIndex(long)

```
public long GetClaimableBlockIndex(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

long

## GetRewardStep(long, long?)

Calculate accumulated rewards step.

```
public int GetRewardStep(long currentBlockIndex, long? rewardStartBlockIndex)
```

### Parameters

`currentBlockIndex` long

The block index of the current block.

`rewardStartBlockIndex` long?

The block index of the reward start block. If not null, the return value is calculated differently. StakeStateTest.GetRewardStep()

### Returns

int

The accumulated rewards step.

## IsCancellable(long)

```
public bool IsCancellable(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

bool

## IsClaimable(long)

```
public bool IsClaimable(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

bool

## IsClaimable(long, out int, out int)

```
public bool IsClaimable(long blockIndex, out int v1Step, out int v2Step)
```

### Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

### Returns

bool

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeV2()

```
public IValue SerializeV2()
```

### Returns

IValue

# Class LegacyStakeState.StakeAchievements

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class LegacyStakeState.StakeAchievements
```

**Inheritance**

object ← LegacyStakeState.StakeAchievements

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StakeAchievements(Dictionary)

```
public StakeAchievements(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### StakeAchievements(Dictionary<int, int>)

```
public StakeAchievements(Dictionary<int, int> achievements = null)
```

#### Parameters

`achievements` Dictionary<int, int>

# Methods

## Achieve(int, int)

```
public void Achieve(int level, int step)
```

### Parameters

`level` int

`step` int

## Check(int, int)

```
public bool Check(int level, int step)
```

### Parameters

`level` int

`step` int

### Returns

bool

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class MarketState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class MarketState
```

**Inheritance**

object ← MarketState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MarketState()

```
public MarketState()
```

### MarketState(IValue)

```
public MarketState(IValue rawList)
```

#### Parameters

`rawList` IValue

## Fields

### AvatarAddresses

```
public List<Address> AvatarAddresses
```

### Field Value

List<Address>

## Methods

### Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class MonsterCollectionResult

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionResult : AttachmentActionResult, IState
```

**Inheritance**

object ← AttachmentActionResult ← MonsterCollectionResult

**Implements**

IState

**Inherited Members**

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MonsterCollectionResult(Dictionary)

```
public MonsterCollectionResult(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### MonsterCollectionResult(Guid, Address, List<RewardInfo>)

```
public MonsterCollectionResult(Guid guid, Address address,
List<MonsterCollectionRewardSheet.RewardInfo> rewardInfos)
```

### Parameters

`guid` Guid

`address` Address

`rewardInfos` List<MonsterCollectionRewardSheet.RewardInfo>

## Fields

## avatarAddress

```
public Address avatarAddress
```

### Field Value

Address

## id

```
public Guid id
```

### Field Value

Guid

## rewards

```
public List<MonsterCollectionRewardSheet.RewardInfo> rewards
```

### Field Value

List<MonsterCollectionRewardSheet.RewardInfo>

# Properties

## TypeId

```
protected override string TypeId { get; }
```

### Property Value

string

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class MonsterCollectionState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionState : State, IState
```

**Inheritance**

object ← State ← MonsterCollectionState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MonsterCollectionState(Dictionary)

```
public MonsterCollectionState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### MonsterCollectionState(Address, int, long)

```
public MonsterCollectionState(Address address, int level, long blockIndex)
```

#### Parameters

`address` Address

`level` int

`blockIndex` long

## MonsterCollectionState(Address, int, long, MonsterCollectionRewardSheet)

```
public MonsterCollectionState(Address address, int level, long blockIndex,
MonsterCollectionRewardSheet monsterCollectionRewardSheet)
```

### Parameters

`address` Address

`level` int

`blockIndex` long

`monsterCollectionRewardSheet` MonsterCollectionRewardSheet

# Fields

## DeriveFormat

```
public const string DeriveFormat = "monster-collection-{0}"
```

### Field Value

string

## LockUpInterval

```
public const long LockUpInterval = 201600
```

### Field Value

long

## RewardInterval

```
public const long RewardInterval = 50400
```

### Field Value

long

# Properties

## ExpiredBlockIndex

```
public long ExpiredBlockIndex { get; }
```

### Property Value

long

## Level

```
public int Level { get; }
```

### Property Value

int

## ReceivedBlockIndex

```
public long ReceivedBlockIndex { get; }
```

### Property Value

long

## RewardLevel

```
public long RewardLevel { get; }
```

### Property Value

long

## StartedBlockIndex

```
public long StartedBlockIndex { get; }
```

### Property Value

long

# Methods

## CalculateRewards(MonsterCollectionRewardSheet, long)

```
public List<MonsterCollectionRewardSheet.RewardInfo>
CalculateRewards(MonsterCollectionRewardSheet sheet, long blockIndex)
```

### Parameters

`sheet` MonsterCollectionRewardSheet

`blockIndex` long

### Returns

List<MonsterCollectionRewardSheet.RewardInfo>

## CalculateStep(long)

```
public int CalculateStep(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

int

## Claim(long)

```
public void Claim(long blockIndex)
```

### Parameters

`blockIndex` long

## DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address baseAddress, int round)
```

### Parameters

`baseAddress` Address

`round` int

### Returns

Address

## Equals(MonsterCollectionState)

```
protected bool Equals(MonsterCollectionState other)
```

### Parameters

`other` MonsterCollectionState

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## IsLocked(long)

```
public bool IsLocked(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

bool

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class MonsterCollectionState0

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Obsolete("This class is reserved for backwards compatibility with
MonsterCollection0 only.", false)]
[Serializable]
public class MonsterCollectionState0 : State, IState
```

### Inheritance

object ← State ← MonsterCollectionState0

### Implements

IState

### Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MonsterCollectionState0(Dictionary)

```
public MonsterCollectionState0(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### MonsterCollectionState0(Address, int, long)

```
public MonsterCollectionState0(Address address, int level, long blockIndex)
```

#### Parameters

`address` Address

`level` int

`blockIndex` long

## MonsterCollectionState0(Address, int, long, MonsterCollectionRewardSheet)

```
public MonsterCollectionState0(Address address, int level, long blockIndex,
MonsterCollectionRewardSheet monsterCollectionRewardSheet)
```

### Parameters

`address` Address

`level` int

`blockIndex` long

`monsterCollectionRewardSheet` MonsterCollectionRewardSheet

# Fields

## DeriveFormat

```
public const string DeriveFormat = "monster-collection-{0}"
```

### Field Value

string

## ExpirationIndex

```
public const long ExpirationIndex = 201600
```

### Field Value

long

## LockUpInterval

```
public const long LockUpInterval = 201600
```

### Field Value

long

## RewardCapacity

```
public const int RewardCapacity = 4
```

### Field Value

int

## RewardInterval

```
public const long RewardInterval = 50400
```

### Field Value

long

# Properties

## End

```
public bool End { get; }
```

### Property Value

bool

## ExpiredBlockIndex

```
public long ExpiredBlockIndex { get; }
```

### Property Value

long

## Level

```
public int Level { get; }
```

### Property Value

int

## ReceivedBlockIndex

```
public long ReceivedBlockIndex { get; }
```

### Property Value

long

## RewardLevel

```
public long RewardLevel { get; }
```

### Property Value

long

## RewardLevelMap

```
public Dictionary<long, List<MonsterCollectionRewardSheet.RewardInfo>>
RewardLevelMap { get; }
```

### Property Value

Dictionary<long, List<MonsterCollectionRewardSheet.RewardInfo>>

## RewardMap

```
public Dictionary<long, MonsterCollectionResult> RewardMap { get; }
```

### Property Value

Dictionary<long, MonsterCollectionResult>

## StartedBlockIndex

```
public long StartedBlockIndex { get; }
```

### Property Value

long

# Methods

## CanReceive(long)

```
public bool CanReceive(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

bool

## DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address baseAddress, int collectRound)
```

### Parameters

`baseAddress` Address

`collectRound` int

### Returns

Address

## Equals(MonsterCollectionState0)

```
protected bool Equals(MonsterCollectionState0 other)
```

### Parameters

`other` MonsterCollectionState0

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## GetRewardLevel(long)

```
public long GetRewardLevel(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

long

## IsLock(long)

```
public bool IsLock(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

bool

## Receive(long)

```
public void Receive(long blockIndex)
```

### Parameters

`blockIndex` long

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## SerializeV2()

```
public IValue SerializeV2()
```

### Returns

IValue

## Update(int, long, MonsterCollectionRewardSheet)

```
public void Update(int level, long rewardLevel, MonsterCollectionRewardSheet
monsterCollectionRewardSheet)
```

### Parameters

`level` int

`rewardLevel` long

`monsterCollectionRewardSheet` MonsterCollectionRewardSheet

## UpdateRewardMap(long, MonsterCollectionResult, long)

```
public void UpdateRewardMap(long rewardLevel, MonsterCollectionResult avatarAddress,
long blockIndex)
```

### Parameters

`rewardLevel` long

`avatarAddress` MonsterCollectionResult

`blockIndex` long

# Class PendingActivationState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class PendingActivationState : State, IState, ISerializable
```

**Inheritance**

object ← State ← PendingActivationState

**Implements**

IState, ISerializable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PendingActivationState(Dictionary)

```
public PendingActivationState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### PendingActivationState(byte[], PublicKey)

```
public PendingActivationState(byte[] nonce, PublicKey publicKey)
```

#### Parameters

`nonce` byte[]

`publicKey` PublicKey

## PendingActivationState(SerializationInfo, StreamingContext)

```
protected PendingActivationState(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Properties

## Nonce

```
public byte[] Nonce { get; }
```

### Property Value

byte[]

## PublicKey

```
public PublicKey PublicKey { get; }
```

### Property Value

PublicKey

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Verify(byte[])

```
public bool Verify(byte[] signature)
```

### Parameters

`signature` byte[]

## Returns

bool

# Class PetState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class PetState : IState
```

**Inheritance**

object ← PetState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PetState(List)

```
public PetState(List serialized)
```

#### Parameters

`serialized` List

### PetState(int)

```
public PetState(int petId)
```

#### Parameters

`petId` int

# Properties

## Level

```
public int Level { get; }
```

### Property Value

int

## PetId

```
public int PetId { get; }
```

### Property Value

int

## UnlockedBlockIndex

```
public long UnlockedBlockIndex { get; }
```

### Property Value

long

# Methods

## DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address avatarAddress, int petId)
```

### Parameters

`avatarAddress` Address

`petId` int

### Returns

Address

## LevelUp()

```
public void LevelUp()
```

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Update(long)

```
public void Update(long unlockedIndex)
```

### Parameters

`unlockedIndex` long

## Validate(long)

```
public bool Validate(long blockIndex)
```

### Parameters

`blockIndex` long

## Returns

bool

# Class RaiderState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RaiderState : IState
```

**Inheritance**

object ← RaiderState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RaiderState()

```
public RaiderState()
```

### RaiderState(List)

```
public RaiderState(List rawState)
```

#### Parameters

`rawState` List

## Fields

## AvatarAddress

```
public Address AvatarAddress
```

### Field Value

Address

## AvatarName

```
public string AvatarName
```

### Field Value

string

## ClaimedBlockIndex

```
public long ClaimedBlockIndex
```

### Field Value

long

## Cp

```
public long Cp
```

### Field Value

long

## HasClaimedReward

```
public bool HasClaimedReward
```

### Field Value

bool

## HighScore

```
public long HighScore
```

### Field Value

long

## IconId

```
public int IconId
```

### Field Value

int

## LatestBossLevel

```
public int LatestBossLevel
```

### Field Value

int

## LatestRewardRank

```
public int LatestRewardRank
```

### Field Value

int

## Level

```
public int Level
```

### Field Value

int

## PurchaseCount

```
public int PurchaseCount
```

### Field Value

int

## RefillBlockIndex

```
public long RefillBlockIndex
```

### Field Value

long

## RemainChallengeCount

```
public int RemainChallengeCount
```

### Field Value

int

## TotalChallengeCount

```
public int TotalChallengeCount
```

### Field Value

int

## TotalScore

```
public long TotalScore
```

### Field Value

long

## UpdatedBlockIndex

```
public long UpdatedBlockIndex
```

### Field Value

long

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Update(AvatarState, long, long, bool, long)

```
public void Update(AvatarState avatarState, long cp, long score, bool payNcg,
long blockIndex)
```

### Parameters

`avatarState` AvatarState

`cp` long

`score` long

`payNcg` bool

`blockIndex` long

# Class RankingInfo

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RankingInfo : IState
```

**Inheritance**

object ← RankingInfo

**Implements**

IState

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingInfo(Dictionary)

```
public RankingInfo(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### RankingInfo(AvatarState)

```
public RankingInfo(AvatarState avatarState)
```

#### Parameters

`avatarState` AvatarState

# Fields

## AgentAddress

```
public readonly Address AgentAddress
```

### Field Value

Address

## ArmorId

```
public readonly int ArmorId
```

### Field Value

int

## AvatarAddress

```
public readonly Address AvatarAddress
```

### Field Value

Address

## AvatarName

```
public readonly string AvatarName
```

### Field Value

string

## Exp

```
public readonly long Exp
```

### Field Value

long

## Level

```
public readonly int Level
```

### Field Value

int

## StageClearedBlockIndex

```
public readonly long StageClearedBlockIndex
```

### Field Value

long

## UpdatedAt

```
public readonly long UpdatedAt
```

### Field Value

long

## Methods

## Equals(RankingInfo)

```
protected bool Equals(RankingInfo other)
```

### Parameters

`other` RankingInfo

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class RankingMapState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RankingMapState : State, IState
```

**Inheritance**

object ← State ← RankingMapState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingMapState(Dictionary)

```
public RankingMapState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### RankingMapState(Address)

```
public RankingMapState(Address address)
```

#### Parameters

`address` Address

# Fields

## Capacity

```
public const int Capacity = 500
```

### Field Value

int

# Methods

## GetRankingInfos(long?)

```
public List<RankingInfo> GetRankingInfos(long? blockOffset)
```

### Parameters

`blockOffset` long?

### Returns

List<RankingInfo>

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Update(AvatarState)

```
public void Update(AvatarState state)
```

## Parameters

`state` AvatarState

# Class RankingState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RankingState : State, IState, ISerializable
```

**Inheritance**

object ← State ← RankingState

**Implements**

IState, ISerializable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingState()

```
public RankingState()
```

### RankingState(Dictionary)

```
public RankingState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## RankingState(SerializationInfo, StreamingContext)

```
public RankingState(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

## RankingMapCapacity

```
public const int RankingMapCapacity = 150
```

### Field Value

int

# Properties

## RankingMap

```
public Dictionary<Address, ImmutableHashSet<Address>> RankingMap { get; }
```

### Property Value

Dictionary<Address, ImmutableHashSet<Address>>

## Methods

## Derive(int)

```
public static Address Derive(int index)
```

### Parameters

`index` int

### Returns

Address

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## UpdateRankingMap(Address)

```
public Address UpdateRankingMap(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

### Returns

Address

# Class RankingState0

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RankingState0 : State, IState, ISerializable
```

**Inheritance**

object ← State ← RankingState0

**Implements**

IState, ISerializable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingState0()

```
public RankingState0()
```

### RankingState0(Dictionary)

```
public RankingState0(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## RankingState0(SerializationInfo, StreamingContext)

```
public RankingState0(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

## RankingMapCapacity

```
public const int RankingMapCapacity = 100
```

### Field Value

int

# Properties

## RankingMap

```
public Dictionary<Address, ImmutableHashSet<Address>> RankingMap { get; }
```

### Property Value

Dictionary<Address, ImmutableHashSet<Address>>

## Methods

## Derive(int)

```
public static Address Derive(int index)
```

### Parameters

`index` int

### Returns

Address

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## UpdateRankingMap(Address)

```
public Address UpdateRankingMap(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

### Returns

Address

# Class RankingState1

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RankingState1 : State, IState, ISerializable
```

**Inheritance**

object ← State ← RankingState1

**Implements**

IState, ISerializable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingState1()

```
public RankingState1()
```

### RankingState1(Dictionary)

```
public RankingState1(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## RankingState1(SerializationInfo, StreamingContext)

```
public RankingState1(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

## RankingMapCapacity

```
public const int RankingMapCapacity = 150
```

### Field Value

int

# Properties

## RankingMap

```
public Dictionary<Address, ImmutableHashSet<Address>> RankingMap { get; }
```

### Property Value

Dictionary<Address, ImmutableHashSet<Address>>

## Methods

## Derive(int)

```
public static Address Derive(int index)
```

### Parameters

`index` int

### Returns

Address

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## UpdateRankingMap(Address)

```
public Address UpdateRankingMap(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

### Returns

Address

# Class RedeemCodeState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemCodeState : State, IState
```

**Inheritance**

object ← State ← RedeemCodeState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RedeemCodeState(Dictionary)

```
public RedeemCodeState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### RedeemCodeState(RedeemCodeListSheet)

```
public RedeemCodeState(RedeemCodeListSheet sheet)
```

#### Parameters

`sheet` RedeemCodeListSheet

## RedeemCodeState(Dictionary<PublicKey, Reward>)

```
public RedeemCodeState(Dictionary<PublicKey, RedeemCodeState.Reward> rewardMap)
```

### Parameters

`rewardMap` Dictionary<PublicKey, RedeemCodeState.Reward>

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

# Properties

## Map

```
public IReadOnlyDictionary<PublicKey, RedeemCodeState.Reward> Map { get; }
```

### Property Value

IReadOnlyDictionary<PublicKey, RedeemCodeState.Reward>

# Methods

## Redeem(string, Address)

```
public int Redeem(string code, Address userAddress)
```

### Parameters

`code` string

`userAddress` Address

### Returns

int

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Update(RedeemCodeListSheet)

```
public void Update(RedeemCodeListSheet sheet)
```

### Parameters

`sheet` RedeemCodeListSheet

# Class RedeemCodeState.Reward

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RedeemCodeState.Reward
```

**Inheritance**

object ← RedeemCodeState.Reward

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Reward(Dictionary)

```
public Reward(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### Reward(int)

```
public Reward(int rewardId)
```

#### Parameters

`rewardId` int

## Fields

## RewardId

```
public readonly int RewardId
```

### Field Value

int

## UserAddress

```
public Address? UserAddress
```

### Field Value

Address?

# Methods

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class RuneSlotState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RuneSlotState : IState
```

**Inheritance**

object ← RuneSlotState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RuneSlotState(List)

```
public RuneSlotState(List serialized)
```

#### Parameters

`serialized` List

### RuneSlotState(BattleType)

```
public RuneSlotState(BattleType battleType)
```

#### Parameters

`battleType` BattleType

# Properties

## BattleType

```
public BattleType BattleType { get; }
```

### Property Value

BattleType

# Methods

## DeriveAddress(Address, BattleType)

```
public static Address DeriveAddress(Address avatarAddress, BattleType battleType)
```

### Parameters

`avatarAddress` Address

`battleType` BattleType

### Returns

Address

## GetEquippedRuneSlotInfos()

```
public List<RuneSlotInfo> GetEquippedRuneSlotInfos()
```

### Returns

List<RuneSlotInfo>

## GetRuneSlot()

```
public List<RuneSlot> GetRuneSlot()
```

### Returns

List<RuneSlot>

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Unlock(int)

```
public void Unlock(int index)
```

### Parameters

`index` int

## UpdateSlot(List<RuneSlotInfo>, RuneListSheet)

```
public void UpdateSlot(List<RuneSlotInfo> runeInfos, RuneListSheet runeListSheet)
```

### Parameters

`runeInfos` List<RuneSlotInfo>

`runeListSheet` RuneListSheet

## UpdateSlotV2(List<RuneSlotInfo>, RuneListSheet)

```
[Obsolete("UpdateSlotV2")]
public void UpdateSlotV2(List<RuneSlotInfo> runeInfos, RuneListSheet runeListSheet)
```

## Parameters

`runeInfos` List<RuneSlotInfo>

`runeListSheet` RuneListSheet

# Class RuneState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RuneState : IState
```

**Inheritance**

object ← RuneState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RuneState(List)

```
public RuneState(List serialized)
```

#### Parameters

`serialized` List

### RuneState(int)

```
public RuneState(int runeId)
```

#### Parameters

`runeId` int

## RuneState(int, int)

```
public RuneState(int runeId, int level)
```

### Parameters

`runeId` int

`level` int

# Properties

## Level

```
public int Level { get; }
```

### Property Value

int

## RuneId

```
public int RuneId { get; }
```

### Property Value

int

# Methods

## DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address avatarAddress, int runeId)
```

### Parameters

`avatarAddress` Address

`runeId` int

### Returns

Address

## LevelUp(int)

```
public void LevelUp(int level = 1)
```

### Parameters

`level` int

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

# Class ShardedShopState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class ShardedShopState : State, IState
```

**Inheritance**

object ← State ← ShardedShopState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ShardedShopState(Dictionary)

```
public ShardedShopState(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ShardedShopState(Address)

```
public ShardedShopState(Address address)
```

#### Parameters

`address` Address

# Fields

## AddressKeys

```
public static readonly IReadOnlyList<string> AddressKeys
```

### Field Value

IReadOnlyList<string>

## Products

```
public readonly Dictionary<Guid, ShopItem> Products
```

### Field Value

Dictionary<Guid, ShopItem>

# Methods

## DeriveAddress(ItemSubType, Guid)

```
public static Address DeriveAddress(ItemSubType itemSubType, Guid productId)
```

### Parameters

`itemSubType` ItemSubType

`productId` Guid

### Returns

Address

## DeriveAddress(ItemSubType, string)

```
public static Address DeriveAddress(ItemSubType itemSubType, string nonce)
```

### Parameters

`itemSubType` ItemSubType

`nonce` string

### Returns

Address

## Register(ShopItem)

```
public void Register(ShopItem shopItem)
```

### Parameters

`shopItem` ShopItem

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ShardedShopStateV2

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class ShardedShopStateV2 : State, IState
```

**Inheritance**

object ← State ← ShardedShopStateV2

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ShardedShopStateV2(Dictionary)

```
public ShardedShopStateV2(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### ShardedShopStateV2(Address)

```
public ShardedShopStateV2(Address address)
```

#### Parameters

`address` Address

# Properties

## OrderDigestList

```
public IReadOnlyList<OrderDigest> OrderDigestList { get; }
```

### Property Value

IReadOnlyList<OrderDigest>

# Methods

## Add(OrderDigest, long)

```
public void Add(OrderDigest orderDigest, long blockIndex)
```

### Parameters

`orderDigest` OrderDigest

`blockIndex` long

## DeriveAddress(ItemSubType, Guid)

```
public static Address DeriveAddress(ItemSubType itemSubType, Guid orderId)
```

### Parameters

`itemSubType` ItemSubType

`orderId` Guid

### Returns

Address

## DeriveAddress(ItemSubType, string)

```
public static Address DeriveAddress(ItemSubType itemSubType, string nonce)
```

### Parameters

`itemSubType` ItemSubType

`nonce` string

### Returns

Address

## Remove(Order, long)

```
public void Remove(Order order, long blockIndex)
```

### Parameters

`order` Order

`blockIndex` long

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

# Class ShopState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

This is a model class of shop state.

```
[Serializable]
public class ShopState : State, IState
```

**Inheritance**

object ← State ← ShopState

**Implements**

IState

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ShopState()

```
public ShopState()
```

## ShopState(Dictionary)

```
public ShopState(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

# Fields

## Address

```
public static readonly Address Address
```

### Field Value

Address

# Properties

## Products

```
public IReadOnlyDictionary<Guid, ShopItem> Products { get; }
```

### Property Value

IReadOnlyDictionary<Guid, ShopItem>

# Methods

## Register(ShopItem)

```
public void Register(ShopItem shopItem)
```

### Parameters

`shopItem` ShopItem

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## TryGet(Address, Guid, out ShopItem)

```
public bool TryGet(Address sellerAgentAddress, Guid productId, out
ShopItem shopItem)
```

### Parameters

`sellerAgentAddress` Address

`productId` Guid

`shopItem` ShopItem

### Returns

bool

## TryUnregister(Guid, out ShopItem)

```
public bool TryUnregister(Guid productId, out ShopItem unregisteredItem)
```

### Parameters

`productId` Guid

`unregisteredItem` ShopItem

### Returns

bool

## Unregister(ShopItem)

```
public void Unregister(ShopItem shopItem)
```

### Parameters

`shopItem` ShopItem

## Unregister(Guid)

```
public void Unregister(Guid productId)
```

### Parameters

`productId` Guid

# Class ShopStateAlreadyContainsException

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class ShopStateAlreadyContainsException : Exception, ISerializable
```

### Inheritance

object ← Exception ← ShopStateAlreadyContainsException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### ShopStateAlreadyContainsException(SerializationInfo, StreamingContext)

```
protected ShopStateAlreadyContainsException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# ShopStateAlreadyContainsException(string)

```
public ShopStateAlreadyContainsException(string message)
```

## Parameters

`message` string

# Class StakeRewardCalculator

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Provides methods for calculating staking rewards from reward sheets.

```
public static class StakeRewardCalculator
```

**Inheritance**

object ← StakeRewardCalculator

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## CalculateFixedRewards(int, IRandom, StakeRegularFixedRewardSheet, ItemSheet, int)

Calculates fixed item rewards for a given stake level. Fixed rewards have a predetermined item count (unlike rate-based rewards in CalculateRewards(Currency, FungibleAssetValue, int, int, StakeRegularRewardSheet, ItemSheet, IRandom)), and the `tradable` flag on each StakeRegularFixedRewardSheet.RewardInfo controls whether material items are created as tradable or non-tradable.

```
public static Dictionary<ItemBase, int> CalculateFixedRewards(int stakeLevel,
IRandom random, StakeRegularFixedRewardSheet stakeRegularFixedRewardSheet, ItemSheet
itemSheet, int rewardSteps)
```

### Parameters

`stakeLevel` int

The current stake level used to look up the reward row.

`random` IRandom

Random source used for non-material item creation.

`stakeRegularFixedRewardSheet` StakeRegularFixedRewardSheet

Sheet containing fixed reward definitions.

`itemSheet` ItemSheet

Sheet used to look up item metadata by ID.

`rewardSteps` int

The number of reward steps accumulated since the last claim. Each reward's count is multiplied by this value.

### Returns

Dictionary<ItemBase, int>

A dictionary mapping each rewarded ItemBase to its total count.

## CalculateRewards(Currency, FungibleAssetValue, int, int, StakeRegularRewardSheet, ItemSheet, IRandom)

Calculates rate-based item and currency rewards for a given staking level. Unlike CalculateFixedRewards(int, IRandom, StakeRegularFixedRewardSheet, ItemSheet, int), reward quantities are derived from the ratio of the staked NCG amount to each reward's Decimal Rate. Rewards whose computed quantity is zero or negative are skipped.

```
public static (Dictionary<ItemBase, int> itemResult, List<FungibleAssetValue>
favResult) CalculateRewards(Currency ncg, FungibleAssetValue stakedNcg, int
stakingLevel, int rewardSteps, StakeRegularRewardSheet stakeRegularRewardSheet,
ItemSheet itemSheet, IRandom random)
```

### Parameters

`ncg` Currency

The NCG currency definition.

`stakedNcg` FungibleAssetValue

The total amount of staked NCG.

`stakingLevel` int

The current staking level used to look up the reward row.

`rewardSteps` int

The number of reward steps accumulated since the last claim. Each reward quantity is multiplied by this value.

`stakeRegularRewardSheet` StakeRegularRewardSheet

Sheet containing rate-based reward definitions.

`itemSheet` ItemSheet

Sheet used to look up item metadata by ID.

`random` IRandom

Random source used for non-material item creation.

## Returns

(Dictionary<ItemBase, int> itemResult, List<FungibleAssetValue> favResult)

A tuple of:

- `itemResult`: a dictionary mapping each rewarded ItemBase to its total count.
- `favResult`: a list of Libplanet.Types.Assets.FungibleAssetValue for rune and currency rewards.

# Class State

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public abstract class State : IState
```

### Inheritance

object ← State

### Implements

IState

### Derived

ActivatedAccountsState, AdminState, AgentState, AuthorizedMinersState, AvatarState, CombinationSlotState, CreditsState, GameConfigState, GoldBalanceState, GoldCurrencyState, LegacyStakeState, MonsterCollectionState, MonsterCollectionState0, PendingActivationState, RankingMapState, RankingState, RankingState0, RankingState1, RedeemCodeState, ShardedShopState, ShardedShopStateV2, ShopState, WeeklyArenaState

### Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### State(IValue)

```
protected State(IValue iValue)
```

#### Parameters

`iValue` IValue

## State(Address)

```
protected State(Address address)
```

### Parameters

`address` Address

# Fields

## address

```
public Address address
```

### Field Value

Address

# Methods

## Serialize()

```
public abstract IValue Serialize()
```

### Returns

IValue

## SerializeBase()

```
protected IValue SerializeBase()
```

### Returns

IValue

## SerializeListBase()

```
protected IValue SerializeListBase()
```

### Returns

IValue

## SerializeV2Base()

```
protected IValue SerializeV2Base()
```

### Returns

IValue

# Class StateExtensions

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public static class StateExtensions
```

**Inheritance**

object ← StateExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## Deserialize<T>(Func<IValue, T>, IValue)

```
public static T? Deserialize<T>(Func<IValue, T> deserializer, IValue serialized)
where T : struct
```

### Parameters

`deserializer` Func<IValue, T>

`serialized` IValue

### Returns

T?

### Type Parameters

`T`

# GetAddress(Dictionary, string, Address)

```
public static Address GetAddress(this Dictionary serialized, string key, Address
defaultValue = default)
```

## Parameters

`serialized` Dictionary

`key` string

`defaultValue` Address

## Returns

Address

# GetBoolean(Dictionary, string, bool)

```
public static bool GetBoolean(this Dictionary serialized, string key, bool
defaultValue = false)
```

## Parameters

`serialized` Dictionary

`key` string

`defaultValue` bool

## Returns

bool

# GetDateTimeOffset(Dictionary, string, DateTimeOffset)

```
public static DateTimeOffset GetDateTimeOffset(this Dictionary serialized, string
key, DateTimeOffset defaultValue = default)
```

### Parameters

`serialized` Dictionary

`key` string

`defaultValue` DateTimeOffset

### Returns

DateTimeOffset

## GetDecimal(Dictionary, string, decimal)

```
public static decimal GetDecimal(this Dictionary serialized, string key, decimal
defaultValue = 0)
```

### Parameters

`serialized` Dictionary

`key` string

`defaultValue` decimal

### Returns

decimal

## GetDecimalStat(Dictionary, string, DecimalStat)

```
public static DecimalStat GetDecimalStat(this Dictionary serialized, string key,
DecimalStat defaultValue = null)
```

### Parameters

`serialized` Dictionary

`key` string

`defaultValue` DecimalStat

### Returns

DecimalStat

## GetGuid(Dictionary, string, Guid)

```
public static Guid GetGuid(this Dictionary serialized, string key, Guid defaultValue
= default)
```

### Parameters

`serialized` Dictionary

`key` string

`defaultValue` Guid

### Returns

Guid

## GetInteger(Dictionary, string, int)

```
public static int GetInteger(this Dictionary serialized, string key, int
defaultValue = 0)
```

### Parameters

`serialized` Dictionary

`key` string

`defaultValue` int

### Returns

int

## GetLong(Dictionary, string, long)

```
public static long GetLong(this Dictionary serialized, string key, long defaultValue
= 0)
```

### Parameters

`serialized` Dictionary

`key` string

`defaultValue` long

### Returns

long

## GetStat(Dictionary)

```
public static (StatType statType, decimal baseValue, decimal additionalValue)
GetStat(this Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

(StatType statType, decimal baseValue, decimal additionalValue)

## GetString(Dictionary, string, string)

```
public static string GetString(this Dictionary serialized, string key, string
defaultValue = "")
```

### Parameters

`serialized` Dictionary

`key` string

`defaultValue` string

### Returns

string

## GetValue<T>(Dictionary, string, T, IValueTryParseDelegate<T>)

```
public static T GetValue<T>(this Dictionary serialized, string key, T defaultValue,
StateExtensions.IValueTryParseDelegate<T> tryParser)
```

### Parameters

`serialized` Dictionary

`key` string

`defaultValue` T

`tryParser` StateExtensions.IValueTryParseDelegate<T>

### Returns

T

### Type Parameters

`T`

## Serialize(HashDigest<SHA256>)

```
public static IValue Serialize(this HashDigest<SHA256> hashDigest)
```

### Parameters

`hashDigest` HashDigest<SHA256>

### Returns

IValue

## Serialize(Address)

```
public static IValue Serialize(this Address address)
```

### Parameters

`address` Address

### Returns

IValue

## Serialize(PublicKey)

```
public static IValue Serialize(this PublicKey key)
```

### Parameters

`key` PublicKey

### Returns

IValue

## Serialize(FungibleAssetValue)

```
public static IValue Serialize(this FungibleAssetValue value)
```

### Parameters

`value` FungibleAssetValue

### Returns

IValue

## Serialize(bool)

```
public static IValue Serialize(this bool boolean)
```

### Parameters

`boolean` bool

### Returns

IValue

## Serialize(Dictionary<Material, int>)

```
public static IValue Serialize(this Dictionary<Material, int> value)
```

### Parameters

`value` Dictionary<Material, int>

### Returns

IValue

## Serialize(IEnumerable<(Address, Address, IValue)>)

```
public static IValue Serialize(this IEnumerable<(Address, Address, IValue)> value)
```

### Parameters

`value` IEnumerable<(Address, Address, IValue)>

### Returns

IValue

## Serialize(DateTimeOffset)

```
public static IValue Serialize(this DateTimeOffset dateTime)
```

### Parameters

`dateTime` DateTimeOffset

### Returns

IValue

## Serialize(decimal)

```
public static IValue Serialize(this decimal number)
```

### Parameters

`number` decimal

### Returns

IValue

## Serialize(Enum)

```
public static IValue Serialize(this Enum type)
```

### Parameters

`type` Enum

### Returns

IValue

## Serialize(Guid)

```
public static IValue Serialize(this Guid number)
```

### Parameters

`number` Guid

### Returns

IValue

## Serialize(int)

```
public static IValue Serialize(this int number)
```

### Parameters

`number` int

### Returns

IValue

## Serialize(long)

```
public static IValue Serialize(this long number)
```

### Parameters

`number` long

### Returns

IValue

## Serialize(Address?)

```
public static IValue Serialize(this Address? address)
```

### Parameters

`address` Address?

### Returns

IValue

## Serialize(FungibleAssetValue?)

```
public static IValue Serialize(this FungibleAssetValue? value)
```

### Parameters

`value` FungibleAssetValue?

### Returns

IValue

## Serialize(bool?)

```
public static IValue Serialize(this bool? boolean)
```

### Parameters

`boolean` bool?

### Returns

IValue

## Serialize(DateTimeOffset?)

```
public static IValue Serialize(this DateTimeOffset? dateTime)
```

### Parameters

`dateTime` DateTimeOffset?

### Returns

IValue

## Serialize(decimal?)

```
public static IValue Serialize(this decimal? number)
```

### Parameters

`number` decimal?

### Returns

IValue

## Serialize(Guid?)

```
public static IValue Serialize(this Guid? number)
```

### Parameters

`number` Guid?

### Returns

IValue

## Serialize(int?)

```
public static IValue Serialize(this int? number)
```

### Parameters

`number` int?

### Returns

IValue

## Serialize(long?)

```
public static IValue Serialize(this long? number)
```

### Parameters

`number` long?

### Returns

IValue

## Serialize(BigInteger?)

```
public static IValue Serialize(this BigInteger? number)
```

### Parameters

`number` BigInteger?

### Returns

IValue

## Serialize(BigInteger)

```
public static IValue Serialize(this BigInteger number)
```

### Parameters

`number` BigInteger

### Returns

IValue

## Serialize(string)

```
public static IValue Serialize(this string text)
```

### Parameters

`text` string

### Returns

IValue

## SerializeForLegacyEquipmentStat(DecimalStat)

```
public static IValue SerializeForLegacyEquipmentStat(this DecimalStat decimalStat)
```

### Parameters

`decimalStat` DecimalStat

### Returns

IValue

## Serialize<T>(IEnumerable<T>)

```
public static IValue Serialize<T>(this IEnumerable<T> values) where T : IValue
```

### Parameters

`values` IEnumerable<T>

### Returns

IValue

### Type Parameters

`T`

## Serialize<T>(Func<T, IValue>, T?)

```
public static IValue Serialize<T>(Func<T, IValue> serializer, T? value) where T
: struct
```

### Parameters

`serializer` Func<T, IValue>

`value` T?

### Returns

IValue

### Type Parameters

`T`

## ToAddress(IValue)

```
public static Address ToAddress(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

Address

## ToArray<T>(IValue, Func<IValue, T>)

```
public static T[] ToArray<T>(this IValue serialized, Func<IValue, T> deserializer)
```

### Parameters

`serialized` IValue

`deserializer` Func<IValue, T>

### Returns

T[]

### Type Parameters

`T`

## ToBigInteger(IValue)

```
public static BigInteger ToBigInteger(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

BigInteger

## ToBoolean(IValue)

```
public static bool ToBoolean(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

bool

## ToDateTimeOffset(IValue)

```
public static DateTimeOffset ToDateTimeOffset(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

DateTimeOffset

## ToDecimal(IValue)

```
public static decimal ToDecimal(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

decimal

## ToDecimalStat(Dictionary)

```
public static DecimalStat ToDecimalStat(this Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

DecimalStat

## ToDecimalStat(IValue)

```
public static DecimalStat ToDecimalStat(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

DecimalStat

## ToDictionary_Material_int(IValue)

```
public static Dictionary<Material, int> ToDictionary_Material_int(this
IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

Dictionary<Material, int>

## ToDotnetString(IValue)

```
public static string ToDotnetString(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

string

## ToEnum<T>(IValue)

```
public static T ToEnum<T>(this IValue serialized) where T : struct
```

### Parameters

`serialized` IValue

### Returns

T

### Type Parameters

T

## ToEnumerable<T>(IValue, Func<IValue, T>)

```
public static IEnumerable<T> ToEnumerable<T>(this IValue serialized, Func<IValue,
T> deserializer)
```

### Parameters

serialized IValue

deserializer Func<IValue, T>

### Returns

IEnumerable<T>

### Type Parameters

T

## ToFungibleAssetValue(IValue)

```
public static FungibleAssetValue ToFungibleAssetValue(this IValue serialized)
```

### Parameters

serialized IValue

### Returns

FungibleAssetValue

## ToGuid(IValue)

```
public static Guid ToGuid(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

Guid

## ToHashSet<T>(IValue, Func<IValue, T>)

```
public static HashSet<T> ToHashSet<T>(this IValue serialized, Func<IValue,
T> deserializer)
```

### Parameters

`serialized` IValue

`deserializer` Func<IValue, T>

### Returns

HashSet<T>

### Type Parameters

`T`

## ToImmutableHashSet<T>(IValue, Func<IValue, T>)

```
public static ImmutableHashSet<T> ToImmutableHashSet<T>(this IValue serialized,
Func<IValue, T> deserializer)
```

### Parameters

`serialized` IValue

`deserializer` Func<IValue, T>

### Returns

ImmutableHashSet<T>

### Type Parameters

T

## ToInteger(IValue)

```
public static int ToInteger(this IValue serialized)
```

### Parameters

serialized IValue

### Returns

int

## ToItemId(IValue)

```
public static HashDigest<SHA256> ToItemId(this IValue serialized)
```

### Parameters

serialized IValue

### Returns

HashDigest<SHA256>

## ToList<T>(IValue, Func<IValue, T>)

```
public static List<T> ToList<T>(this IValue serialized, Func<IValue,
T> deserializer)
```

### Parameters

`serialized` IValue

`deserializer` Func<IValue, T>

### Returns

List<T>

### Type Parameters

`T`

## ToLock(IValue)

```
public static ILock ToLock(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

ILock

## ToLong(IValue)

```
public static long ToLong(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

long

## ToNullableAddress(IValue)

```
public static Address? ToNullableAddress(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

Address?

## ToNullableBigInteger(IValue)

```
public static BigInteger? ToNullableBigInteger(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

BigInteger?

## ToNullableBoolean(IValue)

```
public static bool? ToNullableBoolean(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

bool?

## ToNullableDateTimeOffset(IValue)

```
public static DateTimeOffset? ToNullableDateTimeOffset(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

DateTimeOffset?

## ToNullableDecimal(IValue)

```
public static decimal? ToNullableDecimal(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

decimal?

## ToNullableFungibleAssetValue(IValue)

```
public static FungibleAssetValue? ToNullableFungibleAssetValue(this
IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

FungibleAssetValue?

## ToNullableGuid(IValue)

```
public static Guid? ToNullableGuid(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

Guid?

## ToNullableInteger(IValue)

```
public static int? ToNullableInteger(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

int?

## ToNullableLong(IValue)

```
public static long? ToNullableLong(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

long?

## ToPublicKey(IValue)

```
public static PublicKey ToPublicKey(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

PublicKey

## ToPurchaseInfo(IValue)

```
public static PurchaseInfo0 ToPurchaseInfo(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

PurchaseInfo0

## ToPurchaseInfoLegacy(IValue)

```
public static BuyMultiple.PurchaseInfo ToPurchaseInfoLegacy(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

BuyMultiple.PurchaseInfo

## ToPurchaseResult(IValue)

```
public static Buy7.PurchaseResult ToPurchaseResult(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

Buy7.PurchaseResult

## ToPurchaseResultLegacy(IValue)

```
public static BuyMultiple.PurchaseResult ToPurchaseResultLegacy(this
IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

BuyMultiple.PurchaseResult

## ToSellerResult(IValue)

```
public static Buy7.SellerResult ToSellerResult(this IValue serialized)
```

### Parameters

`serialized` IValue

### Returns

Buy7.SellerResult

# ToStateList(IValue)

```
public static List<(Address, Address, IValue)> ToStateList(this IValue serialized)
```

## Parameters

`serialized` IValue

## Returns

List<(Address, Address, IValue)>

# Delegate StateExtensions.IValueTryParseDelegate<T>

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public delegate bool StateExtensions.IValueTryParseDelegate<T>(IValue input, out T output)
```

## Parameters

`input` IValue

`output` T

## Returns

bool

## Type Parameters

`T`

# Class WeeklyArenaState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaState : State, IState, IDictionary<Address, ArenaInfo>,
ICollection<KeyValuePair<Address, ArenaInfo>>, IEnumerable<KeyValuePair<Address,
ArenaInfo>>, IEnumerable, ISerializable
```

**Inheritance**

object ← State ← WeeklyArenaState

**Implements**

IState, IDictionary<Address, ArenaInfo>, ICollection<KeyValuePair<Address, ArenaInfo>>, IEnumerable<KeyValuePair<Address, ArenaInfo>>, IEnumerable, ISerializable

**Inherited Members**

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WeeklyArenaState(Dictionary)

```
public WeeklyArenaState(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## WeeklyArenaState(IValue)

```
public WeeklyArenaState(IValue iValue)
```

### Parameters

`iValue` IValue

## WeeklyArenaState(Address)

```
public WeeklyArenaState(Address address)
```

### Parameters

`address` Address

## WeeklyArenaState(int)

```
public WeeklyArenaState(int index)
```

### Parameters

`index` int

## WeeklyArenaState(SerializationInfo, StreamingContext)

```
protected WeeklyArenaState(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Fields

## Ended

```
public bool Ended
```

### Field Value

bool

## ResetIndex

```
public long ResetIndex
```

### Field Value

long

# Properties

## Count

Gets the number of elements contained in the ICollection<T>.

```
public int Count { get; }
```

### Property Value

int

The number of elements contained in the ICollection<T>.

## IsReadOnly

Gets a value indicating whether the ICollection<T> is read-only.

```
public bool IsReadOnly { get; }
```

### Property Value

bool

true if the ICollection<T> is read-only; otherwise, false.

## this[Address]

Gets or sets the element with the specified key.

```
public ArenaInfo this[Address key] { get; set; }
```

### Parameters

`key` Address

The key of the element to get or set.

### Property Value

ArenaInfo

The element with the specified key.

### Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

NotSupportedException

The property is set and the IDictionary<TKey, TValue> is read-only.

## Keys

Gets an ICollection<T> containing the keys of the IDictionary<TKey, TValue>.

```
public ICollection<Address> Keys { get; }
```

### Property Value

ICollection<Address>

An ICollection<T> containing the keys of the object that implements IDictionary<TKey, TValue>.

## Map

```
public IReadOnlyDictionary<Address, LazyState<ArenaInfo, Dictionary>> Map { get; }
```

### Property Value

IReadOnlyDictionary<Address, LazyState<ArenaInfo, Dictionary>>

## OrderedArenaInfos

```
public List<ArenaInfo> OrderedArenaInfos { get; }
```

### Property Value

List<ArenaInfo>

## Values

Gets an ICollection<T> containing the values in the IDictionary<TKey, TValue>.

```
public ICollection<ArenaInfo> Values { get; }
```

### Property Value

ICollection<ArenaInfo>

An ICollection<T> containing the values in the object that implements IDictionary<TKey, TValue>.

# Methods

## Add(Address, ArenaInfo)

Adds an element with the provided key and value to the IDictionary<TKey, TValue>.

```
public void Add(Address key, ArenaInfo value)
```

### Parameters

`key` Address

The object to use as the key of the element to add.

`value` ArenaInfo

The object to use as the value of the element to add.

### Exceptions

ArgumentNullException

`key` is null.

ArgumentException

An element with the same key already exists in the IDictionary<TKey, TValue>.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

## Add(KeyValuePair<Address, ArenaInfo>)

Adds an item to the ICollection<T>.

```
public void Add(KeyValuePair<Address, ArenaInfo> item)
```

### Parameters

`item` KeyValuePair<Address, ArenaInfo>

The object to add to the ICollection<T>.

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Clear()

Removes all items from the ICollection<T>.

```
public void Clear()
```

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Contains(KeyValuePair<Address, ArenaInfo>)

Determines whether the ICollection<T> contains a specific value.

```
public bool Contains(KeyValuePair<Address, ArenaInfo> item)
```

### Parameters

`item` KeyValuePair<Address, ArenaInfo>

The object to locate in the ICollection<T>.

### Returns

bool

true if `item` is found in the ICollection<T>; otherwise, false.

## ContainsKey(Address)

Determines whether the IDictionary<TKey, TValue> contains an element with the specified key.

```
public bool ContainsKey(Address key)
```

### Parameters

`key` Address

The key to locate in the IDictionary<TKey, TValue>.

### Returns

bool

true if the IDictionary<TKey, TValue> contains an element with the key; otherwise, false.

### Exceptions

ArgumentNullException

`key` is null.

## CopyTo(KeyValuePair<Address, ArenaInfo>[], int)

Copies the elements of the ICollection<T> to an Array, starting at a particular Array index.

```
public void CopyTo(KeyValuePair<Address, ArenaInfo>[] array, int arrayIndex)
```

### Parameters

`array` KeyValuePair<Address, ArenaInfo>[]

The one-dimensional Array that is the destination of the elements copied from ICollection<T>. The Array must have zero-based indexing.

`arrayIndex` int

The zero-based index in `array` at which copying begins.

### Exceptions

ArgumentNullException

`array` is null.

ArgumentOutOfRangeException

`arrayIndex` is less than 0.

ArgumentException

The number of elements in the source ICollection<T> is greater than the available space from `arrayIndex` to the end of the destination `array`.

## DeriveAddress(int)

```
public static Address DeriveAddress(int index)
```

### Parameters

`index` int

### Returns

Address

## End()

```
public void End()
```

# GetArenaInfo(Address)

```
public ArenaInfo GetArenaInfo(Address avatarAddress)
```

## Parameters

`avatarAddress` Address

## Returns

ArenaInfo

# GetArenaInfos(Address, int, int)

Get arena rank information.

```
public List<(int rank, ArenaInfo arenaInfo)> GetArenaInfos(Address avatarAddress,
int upperRange = 10, int lowerRange = 10)
```

## Parameters

`avatarAddress` Address

The base value of the range that want to get.

`upperRange` int

The upper range than base value in the ranges that want to get.

`lowerRange` int

The lower range than base value in the ranges that want to get.

## Returns

List<(int rank, ArenaInfo arenaInfo)>

A list of tuples that contains `int` and `ArenaInfo`.

# GetArenaInfos(int, int?)

Get arena rank information.

```
public List<(int rank, ArenaInfo arenaInfo)> GetArenaInfos(int firstRank = 1, int?
count = null)
```

## Parameters

`firstRank` int

The first rank in the range that want to get.

`count` int?

The count of the range that want to get.

## Returns

List<(int rank, ArenaInfo arenaInfo)>

A list of tuples that contains `int` and `ArenaInfo`.

# GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<KeyValuePair<Address, ArenaInfo>> GetEnumerator()
```

## Returns

IEnumerator<KeyValuePair<Address, ArenaInfo>>

An enumerator that can be used to iterate through the collection.

# GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## Remove(Address)

Removes the element with the specified key from the IDictionary<TKey, TValue>.

```
public bool Remove(Address key)
```

### Parameters

`key` Address

The key of the element to remove.

### Returns

bool

true if the element is successfully removed; otherwise, false. This method also returns false if `key` was not found in the original IDictionary<TKey, TValue>.

### Exceptions

ArgumentNullException

`key` is null.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

## Remove(KeyValuePair<Address, ArenaInfo>)

Removes the first occurrence of a specific object from the ICollection<T>.

```
public bool Remove(KeyValuePair<Address, ArenaInfo> item)
```

### Parameters

`item` KeyValuePair<Address, ArenaInfo>

The object to remove from the ICollection<T>.

### Returns

bool

true if `item` was successfully removed from the ICollection<T>; otherwise, false. This method also returns false if `item` is not found in the original ICollection<T>.

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## ResetCount(long)

```
public void ResetCount(long ctxBlockIndex)
```

### Parameters

`ctxBlockIndex` long

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Set(AvatarState, CharacterSheet)

```
public void Set(AvatarState avatarState, CharacterSheet characterSheet)
```

### Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

## SetReceive(Address)

```
public void SetReceive(Address avatarAddress)
```

### Parameters

`avatarAddress` Address

## SetV2(AvatarState, CharacterSheet, CostumeStatSheet)

```
public void SetV2(AvatarState avatarState, CharacterSheet characterSheet,
CostumeStatSheet costumeStatSheet)
```

### Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

## TryGetValue(Address, out ArenaInfo)

Gets the value associated with the specified key.

```
public bool TryGetValue(Address key, out ArenaInfo value)
```

### Parameters

`key` Address

The key whose value to get.

`value` ArenaInfo

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

### Returns

bool

true if the object that implements IDictionary<TKey, TValue> contains an element with the specified key; otherwise, false.

### Exceptions

ArgumentNullException

`key` is null.

## Update(ArenaInfo)

```
public void Update(ArenaInfo info)
```

### Parameters

`info` ArenaInfo

## Update(WeeklyArenaState, long)

```
public void Update(WeeklyArenaState prevState, long index)
```

### Parameters

`prevState` WeeklyArenaState

`index` long

# Class WorldBossKillRewardRecord

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class WorldBossKillRewardRecord : IDictionary<int, bool>,
ICollection<KeyValuePair<int, bool>>, IEnumerable<KeyValuePair<int, bool>>,
IEnumerable, IState
```

**Inheritance**

object ← WorldBossKillRewardRecord

**Implements**

IDictionary<int, bool>, ICollection<KeyValuePair<int, bool>>, IEnumerable<KeyValuePair<int, bool>>, IEnumerable, IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### WorldBossKillRewardRecord()

```
public WorldBossKillRewardRecord()
```

### WorldBossKillRewardRecord(List)

```
public WorldBossKillRewardRecord(List serialized)
```

#### Parameters

`serialized` List

# Properties

## Count

Gets the number of elements contained in the ICollection<T>.

```
public int Count { get; }
```

### Property Value

int

The number of elements contained in the ICollection<T>.

## IsReadOnly

Gets a value indicating whether the ICollection<T> is read-only.

```
public bool IsReadOnly { get; }
```

### Property Value

bool

true if the ICollection<T> is read-only; otherwise, false.

## this[int]

Gets or sets the element with the specified key.

```
public bool this[int key] { get; set; }
```

### Parameters

`key` int

The key of the element to get or set.

### Property Value

bool

The element with the specified key.

### Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

NotSupportedException

The property is set and the IDictionary<TKey, TValue> is read-only.

## Keys

Gets an ICollection<T> containing the keys of the IDictionary<TKey, TValue>.

```
public ICollection<int> Keys { get; }
```

### Property Value

ICollection<int>

An ICollection<T> containing the keys of the object that implements IDictionary<TKey, TValue>.

## Values

Gets an ICollection<T> containing the values in the IDictionary<TKey, TValue>.

```
public ICollection<bool> Values { get; }
```

### Property Value

ICollection<bool>

An ICollection<T> containing the values in the object that implements IDictionary<TKey, TValue>.

# Methods

## Add(KeyValuePair<int, bool>)

Adds an item to the ICollection<T>.

```
public void Add(KeyValuePair<int, bool> item)
```

### Parameters

`item` KeyValuePair<int, bool>

The object to add to the ICollection<T>.

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Add(int, bool)

Adds an element with the provided key and value to the IDictionary<TKey, TValue>.

```
public void Add(int key, bool value)
```

### Parameters

`key` int

The object to use as the key of the element to add.

`value` bool

The object to use as the value of the element to add.

### Exceptions

ArgumentNullException

`key` is null.

ArgumentException

An element with the same key already exists in the IDictionary<TKey, TValue>.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

## Clear()

Removes all items from the ICollection<T>.

```
public void Clear()
```

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Contains(KeyValuePair<int, bool>)

Determines whether the ICollection<T> contains a specific value.

```
public bool Contains(KeyValuePair<int, bool> item)
```

### Parameters

`item` KeyValuePair<int, bool>

The object to locate in the ICollection<T>.

### Returns

bool

true if `item` is found in the ICollection<T>; otherwise, false.

## ContainsKey(int)

Determines whether the IDictionary<TKey, TValue> contains an element with the specified key.

```
public bool ContainsKey(int key)
```

### Parameters

`key` int

The key to locate in the IDictionary<TKey, TValue>.

### Returns

bool

true if the IDictionary<TKey, TValue> contains an element with the key; otherwise, false.

### Exceptions

ArgumentNullException

`key` is null.

## CopyTo(KeyValuePair<int, bool>[], int)

Copies the elements of the ICollection<T> to an Array, starting at a particular Array index.

```
public void CopyTo(KeyValuePair<int, bool>[] array, int arrayIndex)
```

### Parameters

`array` KeyValuePair<int, bool>[]

The one-dimensional Array that is the destination of the elements copied from ICollection<T>. The Array must have zero-based indexing.

`arrayIndex` int

The zero-based index in `array` at which copying begins.

### Exceptions

ArgumentNullException

`array` is null.

ArgumentOutOfRangeException

`arrayIndex` is less than 0.

ArgumentException

The number of elements in the source ICollection<T> is greater than the available space from `arrayIndex` to the end of the destination `array`.

## GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<KeyValuePair<int, bool>> GetEnumerator()
```

### Returns

IEnumerator<KeyValuePair<int, bool>>

An enumerator that can be used to iterate through the collection.

## IsClaimable(int)

```
public bool IsClaimable(int level)
```

### Parameters

`level` int

### Returns

bool

## Remove(KeyValuePair<int, bool>)

Removes the first occurrence of a specific object from the ICollection<T>.

```
public bool Remove(KeyValuePair<int, bool> item)
```

### Parameters

`item` KeyValuePair<int, bool>

The object to remove from the ICollection<T>.

### Returns

bool

true if `item` was successfully removed from the ICollection<T>; otherwise, false. This method also returns false if `item` is not found in the original ICollection<T>.

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Remove(int)

Removes the element with the specified key from the IDictionary<TKey, TValue>.

```
public bool Remove(int key)
```

### Parameters

`key` int

The key of the element to remove.

### Returns

bool

true if the element is successfully removed; otherwise, false. This method also returns false if `key` was not found in the original IDictionary<TKey, TValue>.

### Exceptions

ArgumentNullException

`key` is null.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## TryGetValue(int, out bool)

Gets the value associated with the specified key.

```
public bool TryGetValue(int key, out bool value)
```

### Parameters

`key` int

The key whose value to get.

`value` bool

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

## Returns

bool

true if the object that implements IDictionary<TKey, TValue> contains an element with the specified key; otherwise, false.

## Exceptions

ArgumentNullException

`key` is null.

# Class WorldBossState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Represents the state of a World Boss in the game.

```
[Serializable]
public class WorldBossState : IState
```

**Inheritance**

object ← WorldBossState

**Implements**

IState

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### WorldBossState(List)

Initializes a new instance of the WorldBossState class from serialized data.

```
public WorldBossState(List serialized)
```

#### Parameters

`serialized` List

The serialized data for the World Boss state.

### WorldBossState(Row, Row)

Initializes a new instance of the WorldBossState class using the specified row data.

```
public WorldBossState(WorldBossListSheet.Row row, WorldBossGlobalHpSheet.Row hpRow)
```

### Parameters

`row` WorldBossListSheet.Row

The row containing the World Boss data.

`hpRow` WorldBossGlobalHpSheet.Row

The row containing the World Boss HP data.

## Fields

## CurrentHp

Current hit points of the World Boss.

```
public BigInteger CurrentHp
```

### Field Value

BigInteger

## EndedBlockIndex

Block index when the World Boss season ended.

```
public long EndedBlockIndex
```

### Field Value

long

## Id

The boss id of the World Boss from WorldBossListSheet.

```
public int Id
```

### Field Value

int

## Level

Current boss level.

```
public int Level
```

### Field Value

int

## StartedBlockIndex

Block index when the World Boss season started.

```
public long StartedBlockIndex
```

### Field Value

long

## TotalDamage

Total damage dealt to the World Boss in season.

```
public BigInteger TotalDamage
```

### Field Value

BigInteger

# Methods

## Serialize()

Serializes the current state of the World Boss into a serializable format.

```
public IValue Serialize()
```

### Returns

IValue

A serialized representation of the World Boss state.

# Namespace Nekoyume.Model.Swap

## Classes

SwapPool

Swap pool for swap between currencies.

# Class SwapPool

Namespace: Nekoyume.Model.Swap

Assembly: Lib9c.dll

Swap pool for swap between currencies.

```
public class SwapPool
```

**Inheritance**

object ← SwapPool

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SwapPool(SwapRateSheet)

Constructor for SwapPool.

```
public SwapPool(SwapRateSheet swapRateSheet)
```

### Parameters

`swapRateSheet` SwapRateSheet

Swap rate sheet to refer.

# Properties

## Address

Address of the swap pool.

```
public static Address Address { get; }
```

### Property Value

Address

## SwapRateSheet

Swap rate sheet to refer.

```
public SwapRateSheet SwapRateSheet { get; }
```

### Property Value

SwapRateSheet

# Methods

## ConvertToSwapFAV(FungibleAssetValue, Currency, out FungibleAssetValue)

Convert `from` to `to` with swap rate.

```
public FungibleAssetValue ConvertToSwapFAV(FungibleAssetValue from, Currency to, out
FungibleAssetValue remainder)
```

### Parameters

`from` FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue to swap from.

`to` Currency

Libplanet.Types.Assets.Currency to swap to.

`remainder` FungibleAssetValue

Remainder of the swap. This will be returned to the Address who requested swap.

### Returns

FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue after swapping.

### Exceptions

SheetRowNotFoundException

Thrown when the swap rate sheet does not contain the rate for `from` and `to`.

## ConvertToSwapFAV(FungibleAssetValue, Currency, BigInteger, BigInteger, out FungibleAssetValue)

Convert `from` to `to` with swap rate.

```
public static FungibleAssetValue ConvertToSwapFAV(FungibleAssetValue from,
Currency to, BigInteger rateNumerator, BigInteger rateDenominator, out
FungibleAssetValue remainder)
```

### Parameters

`from` FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue to swap from.

`to` Currency

Libplanet.Types.Assets.Currency to swap to.

`rateNumerator` BigInteger

Numerator of the swap rate.

`rateDenominator` BigInteger

Denominator of the swap rate.

`remainder` FungibleAssetValue

Remainder of the swap. This will be returned to the Address who requested swap.

### Returns

FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue after swapping.

### Exceptions

ArgumentException

Thrown when `from` and `to` are the same currency.

## Swap(IWorld, IActionContext, FungibleAssetValue, Currency)

Swap the currency from `from` to `to`.

```
public IWorld Swap(IWorld world, IActionContext context, FungibleAssetValue from,
Currency to)
```

### Parameters

`world` IWorld

Libplanet.Action.State.IWorld to swap the currency.

`context` IActionContext

Libplanet.Action.IActionContext to swap the currency.

`from` FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue to swap from.

`to` Currency

Libplanet.Types.Assets.Currency to swap to.

### Returns

IWorld

New Libplanet.Action.State.IWorld after swapping the currency.

## Exceptions

ArgumentException

Thrown when `from` and `to` are the same currency.

# Namespace Nekoyume.Module

## Classes

ActionPointModule

AdventureBossModule

AgentModule

ArenaModule

AvatarModule

CollectionModule
Provides utility methods for working with collection states in the world state.

CombinationSlotStateModule
CombinationSlotStateModule is the module to use
CombinationSlotState/AllCombinationSlotState with account.

DailyRewardModule

GiftModule

InfiniteTowerModule

LegacyModule

PatrolRewardModule
Provides utility methods for handling patrol reward states.

RelationshipModule

RuneStateModule
RuneStateModule is the module to use RuneState/AllRuneState with account.

StateResolver

# Class ActionPointModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class ActionPointModule
```

**Inheritance**

object ← ActionPointModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetActionPoint(IWorldState, Address)

```
public static long GetActionPoint(this IWorldState worldState,
Address avatarAddress)
```

### Parameters

`worldState` IWorldState

`avatarAddress` Address

### Returns

long

## SetActionPoint(IWorld, Address, long)

```
public static IWorld SetActionPoint(this IWorld world, Address avatarAddress,
long actionPoint)
```

### Parameters

`world` IWorld

`avatarAddress` Address

`actionPoint` long

### Returns

IWorld

## TryGetActionPoint(IWorldState, Address, out long)

```
public static bool TryGetActionPoint(this IWorldState worldState, Address
avatarAddress, out long actionPoint)
```

### Parameters

`worldState` IWorldState

`avatarAddress` Address

`actionPoint` long

### Returns

bool

# Class AdventureBossModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class AdventureBossModule
```

**Inheritance**

object ← AdventureBossModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## LatestSeasonAddress

```
public static readonly Address LatestSeasonAddress
```

### Field Value

Address

# Methods

## GetBountyBoard(IWorldState, long)

```
public static BountyBoard GetBountyBoard(this IWorldState worldState, long season)
```

### Parameters

`worldState` IWorldState

`season` long

### Returns

BountyBoard

## GetExploreBoard(IWorldState, long)

```
public static ExploreBoard GetExploreBoard(this IWorldState worldState, long season)
```

### Parameters

`worldState` IWorldState

`season` long

### Returns

ExploreBoard

## GetExplorer(IWorldState, long, Address)

```
public static Explorer GetExplorer(this IWorldState worldState, long season,
Address avatarAddress)
```

### Parameters

`worldState` IWorldState

`season` long

`avatarAddress` Address

### Returns

Explorer

# GetExplorerList(IWorldState, long)

```
public static ExplorerList GetExplorerList(this IWorldState worldState, long season)
```

## Parameters

`worldState` IWorldState

`season` long

## Returns

ExplorerList

# GetLatestAdventureBossSeason(IWorldState)

Get brief adventure boss season info of latest season. This only has brief, readonly data, so we must use SeasonInfo to handle season itself.

```
public static SeasonInfo GetLatestAdventureBossSeason(this IWorldState worldState)
```

## Parameters

`worldState` IWorldState

## Returns

SeasonInfo

LatestSeason state. If no season is started, return a state with default value(0)

# GetSeasonInfo(IWorldState, long)

```
public static SeasonInfo GetSeasonInfo(this IWorldState worldState, long season)
```

## Parameters

`worldState` IWorldState

`season` long

### Returns

SeasonInfo

## SetBountyBoard(IWorld, long, BountyBoard)

```
public static IWorld SetBountyBoard(this IWorld world, long season,
BountyBoard bountyBoard)
```

### Parameters

`world` IWorld

`season` long

`bountyBoard` BountyBoard

### Returns

IWorld

## SetExploreBoard(IWorld, long, ExploreBoard)

```
public static IWorld SetExploreBoard(this IWorld world, long season,
ExploreBoard exploreBoard)
```

### Parameters

`world` IWorld

`season` long

`exploreBoard` ExploreBoard

### Returns

IWorld

## SetExplorer(IWorld, long, Explorer)

```
public static IWorld SetExplorer(this IWorld world, long season, Explorer explorer)
```

### Parameters

`world` IWorld

`season` long

`explorer` Explorer

### Returns

IWorld

## SetExplorerList(IWorld, long, ExplorerList)

```
public static IWorld SetExplorerList(this IWorld world, long season,
ExplorerList explorerList)
```

### Parameters

`world` IWorld

`season` long

`explorerList` ExplorerList

### Returns

IWorld

## SetLatestAdventureBossSeason(IWorld, SeasonInfo)

```
public static IWorld SetLatestAdventureBossSeason(this IWorld world,
SeasonInfo latestSeasonInfo)
```

### Parameters

`world` IWorld

`latestSeasonInfo` SeasonInfo

### Returns

IWorld

## SetSeasonInfo(IWorld, SeasonInfo)

```
public static IWorld SetSeasonInfo(this IWorld world, SeasonInfo seasonInfo)
```

### Parameters

`world` IWorld

`seasonInfo` SeasonInfo

### Returns

IWorld

## TryGetExploreBoard(IWorldState, long, out ExploreBoard)

```
public static bool TryGetExploreBoard(this IWorldState worldState, long season, out
ExploreBoard exploreBoard)
```

### Parameters

`worldState` IWorldState

`season` long

`exploreBoard` ExploreBoard

### Returns

bool

## TryGetExplorer(IWorldState, long, Address, out Explorer)

```
public static bool TryGetExplorer(this IWorldState worldState, long season, Address
avatarAddress, out Explorer explorer)
```

### Parameters

`worldState` IWorldState

`season` long

`avatarAddress` Address

`explorer` Explorer

### Returns

bool

# Class AgentModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class AgentModule
```

**Inheritance**

object ← AgentModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetAgentState(IWorldState, Address)

```
public static AgentState? GetAgentState(this IWorldState worldState,
Address address)
```

### Parameters

`worldState` IWorldState

`address` Address

### Returns

AgentState

## SetAgentState(IWorld, Address, AgentState)

```
public static IWorld SetAgentState(this IWorld world, Address agent,
AgentState state)
```

## Parameters

`world` IWorld

`agent` Address

`state` AgentState

## Returns

IWorld

# Class ArenaModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class ArenaModule
```

**Inheritance**

object ← ArenaModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetArenaParticipant(IWorldState, int, int, Address)

```
public static ArenaParticipant GetArenaParticipant(this IWorldState worldState, int
championshipId, int round, Address avatarAddress)
```

### Parameters

`worldState` IWorldState

`championshipId` int

`round` int

`avatarAddress` Address

### Returns

ArenaParticipant

# GetArenaParticipantState(IWorldState, Address, Address)

```
public static IValue GetArenaParticipantState(this IWorldState worldState, Address
accountAddress, Address stateAddress)
```

## Parameters

`worldState` IWorldState

`accountAddress` Address

`stateAddress` Address

## Returns

IValue

# GetArenaParticipantState(IWorldState, int, int, Address)

```
public static IValue GetArenaParticipantState(this IWorldState worldState, int
championshipId, int round, Address avatarAddress)
```

## Parameters

`worldState` IWorldState

`championshipId` int

`round` int

`avatarAddress` Address

## Returns

IValue

# SetArenaParticipant(IWorld, Address, Address, IValue)

```
public static IWorld SetArenaParticipant(this IWorld world, Address accountAddress,
Address stateAddress, IValue arenaParticipantState)
```

### Parameters

`world` IWorld

`accountAddress` Address

`stateAddress` Address

`arenaParticipantState` IValue

### Returns

IWorld

## SetArenaParticipant(IWorld, int, int, Address, IValue)

```
public static IWorld SetArenaParticipant(this IWorld world, int championshipId, int
round, Address avatarAddress, IValue arenaParticipantState)
```

### Parameters

`world` IWorld

`championshipId` int

`round` int

`avatarAddress` Address

`arenaParticipantState` IValue

### Returns

IWorld

# SetArenaParticipant(IWorld, int, int, Address, ArenaParticipant)

```
public static IWorld SetArenaParticipant(this IWorld world, int championshipId, int
round, Address avatarAddress, ArenaParticipant arenaParticipant)
```

## Parameters

`world` IWorld

`championshipId` int

`round` int

`avatarAddress` Address

`arenaParticipant` ArenaParticipant

## Returns

IWorld

# Class AvatarModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class AvatarModule
```

**Inheritance**

object ← AvatarModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetAvatarState(IWorldState, Address, bool, bool, bool)

```
public static AvatarState GetAvatarState(this IWorldState worldState, Address
address, bool getInventory = true, bool getWorldInformation = true, bool
getQuestList = true)
```

### Parameters

`worldState` IWorldState

`address` Address

`getInventory` bool

`getWorldInformation` bool

`getQuestList` bool

### Returns

AvatarState

# GetEnemyAvatarState(IWorldState, Address)

```
public static AvatarState GetEnemyAvatarState(this IWorldState worldState,
Address avatarAddress)
```

## Parameters

`worldState` IWorldState

`avatarAddress` Address

## Returns

AvatarState

# SetAvatarState(IWorld, Address, AvatarState, bool, bool, bool, bool)

```
public static IWorld SetAvatarState(this IWorld world, Address avatarAddress,
AvatarState state, bool setAvatar = true, bool setInventory = true, bool
setWorldInformation = true, bool setQuestList = true)
```

## Parameters

`world` IWorld

`avatarAddress` Address

`state` AvatarState

`setAvatar` bool

`setInventory` bool

`setWorldInformation` bool

`setQuestList` bool

## Returns

# TryGetAvatarState(IWorldState, Address, Address, out AvatarState)

```
public static bool TryGetAvatarState(this IWorldState worldState, Address
agentAddress, Address avatarAddress, out AvatarState avatarState)
```

## Parameters

`worldState` IWorldState

`agentAddress` Address

`avatarAddress` Address

`avatarState` AvatarState

## Returns

bool

# Class CollectionModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

Provides utility methods for working with collection states in the world state.

```
public static class CollectionModule
```

**Inheritance**

object ← CollectionModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetCollectionState(IWorldState, Address)

Get the CollectionState for a given address from the world state.

```
public static CollectionState GetCollectionState(this IWorldState worldState,
Address address)
```

### Parameters

`worldState` IWorldState

The world state object.

`address` Address

The address to retrieve the CollectionState.

### Returns

CollectionState

The CollectionState corresponding to the address.

### Exceptions

FailedLoadStateException

Thrown when the Collection state for the given address is not found in the world state.

InvalidCastException

Thrown when the serialized Collection state is not in the correct format.

## GetCollectionStates(IWorldState, IReadOnlyList<Address>)

Retrieves the collection states for the given addresses from the world state.

```
public static Dictionary<Address, CollectionState> GetCollectionStates(this
IWorldState worldState, IReadOnlyList<Address> addresses)
```

### Parameters

`worldState` IWorldState

The world state used to retrieve the collection states.

`addresses` IReadOnlyList<Address>

The list of addresses to retrieve the collection states for.

### Returns

Dictionary<Address, CollectionState>

A dictionary of Address and CollectionState pairs representing the collection states for the given addresses, or an empty dictionary for addresses that do not have a collection state.

## SetCollectionState(IWorld, Address, CollectionState)

Sets the state of a collection in the world.

```
public static IWorld SetCollectionState(this IWorld world, Address collection,
CollectionState state)
```

### Parameters

`world` IWorld

`collection` Address

The address of the collection.

`state` CollectionState

The state of the collection.

### Returns

IWorld

The updated world with the updated collection state.

## TryGetCollectionState(IWorldState, Address, out CollectionState)

Tries to get the collection state for a specific address from the given world state.

```
public static bool TryGetCollectionState(this IWorldState worldState, Address
address, out CollectionState collectionState)
```

### Parameters

`worldState` IWorldState

The world state from which to get the collection state.

`address` Address

The address for which to retrieve the collection state.

`collectionState` CollectionState

The resulting collection state, if found.

## Returns

bool

True if the collection state is found, otherwise false.

# Class CombinationSlotStateModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

CombinationSlotStateModule is the module to use CombinationSlotState/AllCombinationSlotState with account.

```
public static class CombinationSlotStateModule
```

**Inheritance**

object ← CombinationSlotStateModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetAllCombinationSlotState(IWorldState, Address)

```
public static AllCombinationSlotState GetAllCombinationSlotState(this IWorldState worldState, Address avatarAddress)
```

### Parameters

`worldState` IWorldState

`avatarAddress` Address

### Returns

AllCombinationSlotState

# SetCombinationSlotState(IWorld, Address, AllCombinationSlotState)

```
public static IWorld SetCombinationSlotState(this IWorld world, Address
avatarAddress, AllCombinationSlotState combinationSlotState)
```

## Parameters

`world` IWorld

`avatarAddress` Address

`combinationSlotState` AllCombinationSlotState

## Returns

IWorld

# Class DailyRewardModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class DailyRewardModule
```

**Inheritance**

object ← DailyRewardModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetDailyRewardReceivedBlockIndex(IWorldState, Address)

```
public static long GetDailyRewardReceivedBlockIndex(this IWorldState worldState,
Address avatarAddress)
```

#### Parameters

`worldState` IWorldState

`avatarAddress` Address

#### Returns

long

### SetDailyRewardReceivedBlockIndex(IWorld, Address, long)

```
public static IWorld SetDailyRewardReceivedBlockIndex(this IWorld world, Address
avatarAddress, long blockIndex)
```

### Parameters

`world` IWorld

`avatarAddress` Address

`blockIndex` long

### Returns

IWorld

## TryGetDailyRewardReceivedBlockIndex(IWorldState, Address, out long)

```
public static bool TryGetDailyRewardReceivedBlockIndex(this IWorldState worldState,
Address avatarAddress, out long receivedBlockIndex)
```

### Parameters

`worldState` IWorldState

`avatarAddress` Address

`receivedBlockIndex` long

### Returns

bool

# Class GiftModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class GiftModule
```

**Inheritance**

object ← GiftModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetClaimedGifts(IWorldState, Address)

```
public static List<int> GetClaimedGifts(this IWorldState state,
Address avatarAddress)
```

#### Parameters

`state` IWorldState

`avatarAddress` Address

#### Returns

List<int>

### SetClaimedGifts(IWorld, Address, List<int>)

```
public static IWorld SetClaimedGifts(this IWorld world, Address avatarAddress,
List<int> claimedGiftIds)
```

## Parameters

`world` IWorld

`avatarAddress` Address

`claimedGiftIds` List<int>

## Returns

IWorld

# Class InfiniteTowerModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class InfiniteTowerModule
```

**Inheritance**

object ← InfiniteTowerModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetFloorClearCount(IWorld, int, int)

Get the clear count for a specific floor in a specific tower.

```
public static int GetFloorClearCount(this IWorld worldState, int infiniteTowerId,
int floorId)
```

### Parameters

`worldState` IWorld

The world state.

`infiniteTowerId` int

The infinite tower ID.

`floorId` int

The floor ID.

### Returns

int

The number of times the floor has been cleared.

## GetInfiniteTowerBoardState(IWorld, int)

Get infinite tower board state for a specific tower.

```
public static InfiniteTowerBoardState GetInfiniteTowerBoardState(this IWorld
worldState, int infiniteTowerId)
```

### Parameters

`worldState` IWorld

The world state.

`infiniteTowerId` int

The infinite tower ID.

### Returns

InfiniteTowerBoardState

InfiniteTowerBoardState for the tower.

## GetInfiniteTowerInfo(IWorld, Address, int)

Get infinite tower info for a specific avatar and tower.

```
public static InfiniteTowerInfo GetInfiniteTowerInfo(this IWorld worldState, Address
avatarAddress, int infiniteTowerId)
```

### Parameters

`worldState` IWorld

The world state.

`avatarAddress` Address

The avatar address.

`infiniteTowerId` int

The infinite tower ID.

### Returns

InfiniteTowerInfo

InfiniteTowerInfo for the avatar and tower.

## RecordFloorClear(IWorld, int, int, long)

Update infinite tower board state by recording a floor clear.

```
public static IWorld RecordFloorClear(this IWorld world, int infiniteTowerId, int
floorId, long blockIndex)
```

### Parameters

`world` IWorld

The world.

`infiniteTowerId` int

The infinite tower ID.

`floorId` int

The floor ID that was cleared.

`blockIndex` long

The block index when the floor was cleared.

### Returns

IWorld

Updated world with the recorded floor clear.

## SetInfiniteTowerBoardState(IWorld, InfiniteTowerBoardState)

Set infinite tower board state for a specific tower.

```
public static IWorld SetInfiniteTowerBoardState(this IWorld world,
InfiniteTowerBoardState infiniteTowerBoardState)
```

### Parameters

`world` IWorld

The world.

`infiniteTowerBoardState` InfiniteTowerBoardState

The infinite tower board state to set.

### Returns

IWorld

Updated world with the infinite tower board state.

## SetInfiniteTowerInfo(IWorld, Address, InfiniteTowerInfo)

Set infinite tower info for a specific avatar and tower.

```
public static IWorld SetInfiniteTowerInfo(this IWorld world, Address avatarAddress,
InfiniteTowerInfo infiniteTowerInfo)
```

### Parameters

`world` IWorld

The world.

`avatarAddress` Address

The avatar address.

`infiniteTowerInfo` InfiniteTowerInfo

The infinite tower info to set.

## Returns

IWorld

Updated world with the infinite tower info.

# TryGetInfiniteTowerBoardState(IWorld, int, out InfiniteTowerBoardState)

Try to get infinite tower board state for a specific tower.

```
public static bool TryGetInfiniteTowerBoardState(this IWorld worldState, int
infiniteTowerId, out InfiniteTowerBoardState infiniteTowerBoardState)
```

## Parameters

`worldState` IWorld

The world state.

`infiniteTowerId` int

The infinite tower ID.

`infiniteTowerBoardState` InfiniteTowerBoardState

The infinite tower board state if found.

## Returns

bool

True if the infinite tower board state was found, false otherwise.

# TryGetInfiniteTowerInfo(IWorld, Address, int, out InfiniteTowerInfo)

Try to get infinite tower info for a specific avatar and tower.

```
public static bool TryGetInfiniteTowerInfo(this IWorld worldState, Address
avatarAddress, int infiniteTowerId, out InfiniteTowerInfo infiniteTowerInfo)
```

## Parameters

`worldState` IWorld

The world state.

`avatarAddress` Address

The avatar address.

`infiniteTowerId` int

The infinite tower ID.

`infiniteTowerInfo` InfiniteTowerInfo

The infinite tower info if found.

## Returns

bool

True if the infinite tower info was found, false otherwise.

# Class LegacyModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class LegacyModule
```

**Inheritance**

object ← LegacyModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetArenaAvatarState(IWorldState, Address, AvatarState)

```
public static ArenaAvatarState GetArenaAvatarState(this IWorldState worldState,
Address arenaAvatarStateAddress, AvatarState avatarState)
```

#### Parameters

`worldState` IWorldState

`arenaAvatarStateAddress` Address

`avatarState` AvatarState

#### Returns

ArenaAvatarState

# GetArenaInfo(IWorldState, Address, AvatarState, CharacterSheet, CostumeStatSheet)

```
public static (Address arenaInfoAddress, ArenaInfo arenaInfo, bool isNewArenaInfo)
GetArenaInfo(this IWorldState worldState, Address weeklyArenaAddress, AvatarState
avatarState, CharacterSheet characterSheet, CostumeStatSheet costumeStatSheet)
```

## Parameters

`worldState` IWorldState

`weeklyArenaAddress` Address

`avatarState` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

## Returns

(Address arenaInfoAddress, ArenaInfo arenaInfo, bool isNewArenaInfo)

# GetArenaParticipants(IWorldState, Address, int, int)

```
public static ArenaParticipants GetArenaParticipants(this IWorldState worldState,
Address arenaParticipantsAddress, int id, int round)
```

## Parameters

`worldState` IWorldState

`arenaParticipantsAddress` Address

`id` int

`round` int

## Returns

ArenaParticipants

## GetAvatarSheets(IWorldState)

```
public static AvatarSheets GetAvatarSheets(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

AvatarSheets

## GetCombinationSlotStateLegacy(IWorldState, Address, int)

```
[Obsolete("Use AllCombinationSlotState.GetRuneState() instead.")]
public static CombinationSlotState GetCombinationSlotStateLegacy(this IWorldState
worldState, Address avatarAddress, int index)
```

### Parameters

`worldState` IWorldState

`avatarAddress` Address

`index` int

### Returns

CombinationSlotState

## GetCouponWallet(IWorldState, Address)

```
public static IImmutableDictionary<Guid, Coupon> GetCouponWallet(this IWorldState
```

```
worldState, Address agentAddress)
```

### Parameters

`worldState` IWorldState

`agentAddress` Address

### Returns

IImmutableDictionary<Guid, Coupon>

## GetCrystalCostState(IWorldState, Address)

```
public static CrystalCostState GetCrystalCostState(this IWorldState worldState,
Address address)
```

### Parameters

`worldState` IWorldState

`address` Address

### Returns

CrystalCostState

## GetCrystalCostStates(IWorldState, long, long)

```
public static (CrystalCostState DailyCostState, CrystalCostState WeeklyCostState,
CrystalCostState PrevWeeklyCostState, CrystalCostState BeforePrevWeeklyCostState)
GetCrystalCostStates(this IWorldState worldState, long blockIndex, long interval)
```

### Parameters

`worldState` IWorldState

`blockIndex` long

`interval` long

### Returns

(CrystalCostState DailyCostState, CrystalCostState WeeklyCostState, CrystalCostState PrevWeeklyCostState, CrystalCostState BeforePrevWeeklyCostState)

## GetGameConfigState(IWorldState)

```
public static GameConfigState GetGameConfigState(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

GameConfigState

## GetGoldBalanceState(IWorldState, Address, Currency)

```
public static GoldBalanceState GetGoldBalanceState(this IWorldState worldState,
Address address, Currency currency)
```

### Parameters

`worldState` IWorldState

`address` Address

`currency` Currency

### Returns

GoldBalanceState

## GetGoldCurrency(IWorldState)

```
public static Currency GetGoldCurrency(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

Currency

## GetGoldDistribution(IWorldState)

```
public static IEnumerable<GoldDistribution> GetGoldDistribution(this
IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

IEnumerable<GoldDistribution>

## GetItemSheet(IWorldState)

```
public static ItemSheet GetItemSheet(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

ItemSheet

## GetLegacyState(IWorldState, Address)

```
public static IValue GetLegacyState(this IWorldState worldState, Address address)
```

### Parameters

`worldState` IWorldState

`address` Address

### Returns

IValue

## GetLegacyStates(IWorldState, IReadOnlyList<Address>)

```
public static IReadOnlyList<IValue?> GetLegacyStates(this IWorldState worldState,
IReadOnlyList<Address> addresses)
```

### Parameters

`worldState` IWorldState

`addresses` IReadOnlyList<Address>

### Returns

IReadOnlyList<IValue>

## GetLegacyStatesAsDict(IWorldState, params Address[])

```
public static Dictionary<Address, IValue> GetLegacyStatesAsDict(this IWorldState
worldState, params Address[] addresses)
```

### Parameters

`worldState` IWorldState

`addresses` Address[]

### Returns

Dictionary<Address, IValue>

## GetQuestSheet(IWorldState)

```
public static QuestSheet GetQuestSheet(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

QuestSheet

## GetRaiderState(IWorldState, Address)

```
public static RaiderState GetRaiderState(this IWorldState worldState,
Address raiderAddress)
```

### Parameters

`worldState` IWorldState

`raiderAddress` Address

### Returns

RaiderState

## GetRaiderState(IWorldState, Address, int)

```
public static RaiderState GetRaiderState(this IWorldState worldState, Address
avatarAddress, int raidId)
```

### Parameters

`worldState` IWorldState

`avatarAddress` Address

`raidId` int

### Returns

RaiderState

## GetRankingSimulatorSheets(IWorldState)

```
public static RankingSimulatorSheets GetRankingSimulatorSheets(this
IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

RankingSimulatorSheets

## GetRankingSimulatorSheetsV1(IWorldState)

```
public static RankingSimulatorSheetsV1 GetRankingSimulatorSheetsV1(this
IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

RankingSimulatorSheetsV1

## GetRankingState(IWorldState)

```
public static RankingState GetRankingState(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

RankingState

## GetRankingState0(IWorldState)

```
public static RankingState0 GetRankingState0(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

RankingState0

## GetRankingState1(IWorldState)

```
public static RankingState1 GetRankingState1(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

RankingState1

## GetRedeemCodeState(IWorldState)

```
public static RedeemCodeState GetRedeemCodeState(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

RedeemCodeState

## GetSheetCsv(IWorldState, Address)

```
public static string GetSheetCsv(this IWorldState worldState, Address address)
```

### Parameters

`worldState` IWorldState

`address` Address

### Returns

string

## GetSheetCsv<T>(IWorldState)

```
public static string GetSheetCsv<T>(this IWorldState worldState) where T :
ISheet, new()
```

### Parameters

`worldState` IWorldState

### Returns

string

### Type Parameters

`T`

## GetSheet<T>(IWorldState)

```
public static T GetSheet<T>(this IWorldState worldState) where T : ISheet, new()
```

### Parameters

`worldState` IWorldState

### Returns

T

### Type Parameters

`T`

## GetSheet<T>(IWorldState, Address)

```
public static T GetSheet<T>(this IWorldState worldState, Address sheetAddr) where T
: ISheet, new()
```

### Parameters

`worldState` IWorldState

`sheetAddr` Address

## Returns

T

## Type Parameters

T

# GetSheets(IWorldState, bool, bool, bool, bool, bool, bool, bool, bool, bool, IEnumerable<Type>)

```
public static Dictionary<Type, (Address address, ISheet sheet)> GetSheets(this
IWorldState worldState, bool containAvatarSheets = false, bool containItemSheet =
false, bool containQuestSheet = false, bool containSimulatorSheets = false, bool
containStageSimulatorSheets = false, bool containRankingSimulatorSheets = false,
bool containArenaSimulatorSheets = false, bool containValidateItemRequirementSheets
= false, bool containRaidSimulatorSheets = false, IEnumerable<Type> sheetTypes
= null)
```

## Parameters

worldState IWorldState

containAvatarSheets bool

containItemSheet bool

containQuestSheet bool

containSimulatorSheets bool

containStageSimulatorSheets bool

containRankingSimulatorSheets bool

containArenaSimulatorSheets bool

containValidateItemRequirementSheets bool

containRaidSimulatorSheets bool

sheetTypes IEnumerable<Type>

### Returns

Dictionary<Type, (Address address, ISheet sheet)>

## GetSheets(IWorldState, params Type[])

```
public static Dictionary<Type, (Address address, ISheet sheet)> GetSheets(this
IWorldState worldState, params Type[] sheetTypes)
```

### Parameters

`worldState` IWorldState

`sheetTypes` Type[]

### Returns

Dictionary<Type, (Address address, ISheet sheet)>

## GetSheets(IWorldState, params (Type sheetType, string sheetName)[])

```
public static Dictionary<Type, (Address address, ISheet sheet)> GetSheets(this
IWorldState worldState, params (Type sheetType, string sheetName)[] sheetTuples)
```

### Parameters

`worldState` IWorldState

`sheetTuples` (Type sheetType, string sheetName)[]

### Returns

Dictionary<Type, (Address address, ISheet sheet)>

# GetSheetsV1(IWorldState, bool, bool, bool, bool, bool, bool, bool, bool, bool, IEnumerable<Type>)

```
public static Dictionary<Type, (Address address, ISheet sheet)> GetSheetsV1(this
IWorldState worldState, bool containAvatarSheets = false, bool containItemSheet =
false, bool containQuestSheet = false, bool containSimulatorSheets = false, bool
containStageSimulatorSheets = false, bool containRankingSimulatorSheets = false,
bool containArenaSimulatorSheets = false, bool containValidateItemRequirementSheets
= false, bool containRaidSimulatorSheets = false, IEnumerable<Type> sheetTypes
= null)
```

## Parameters

`worldState` IWorldState

`containAvatarSheets` bool

`containItemSheet` bool

`containQuestSheet` bool

`containSimulatorSheets` bool

`containStageSimulatorSheets` bool

`containRankingSimulatorSheets` bool

`containArenaSimulatorSheets` bool

`containValidateItemRequirementSheets` bool

`containRaidSimulatorSheets` bool

`sheetTypes` IEnumerable<Type>

## Returns

Dictionary<Type, (Address address, ISheet sheet)>

# GetSheetsV100291(IWorldState, bool, bool, bool, bool, bool, bool, bool, bool, bool, IEnumerable<Type>)

```
public static Dictionary<Type, (Address address, ISheet sheet)>
GetSheetsV100291(this IWorldState worldState, bool containAvatarSheets = false, bool
containItemSheet = false, bool containQuestSheet = false, bool
containSimulatorSheets = false, bool containStageSimulatorSheets = false, bool
containRankingSimulatorSheets = false, bool containArenaSimulatorSheets = false,
bool containValidateItemRequirementSheets = false, IEnumerable<Type> sheetTypes
= null)
```

### Parameters

`worldState` IWorldState

`containAvatarSheets` bool

`containItemSheet` bool

`containQuestSheet` bool

`containSimulatorSheets` bool

`containStageSimulatorSheets` bool

`containRankingSimulatorSheets` bool

`containArenaSimulatorSheets` bool

`containValidateItemRequirementSheets` bool

`sheetTypes` IEnumerable<Type>

### Returns

Dictionary<Type, (Address address, ISheet sheet)>

## GetShopState(IWorldState)

```
public static ShopState GetShopState(this IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

ShopState

## GetStageSimulatorSheets(IWorldState)

```
public static StageSimulatorSheets GetStageSimulatorSheets(this
IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

StageSimulatorSheets

## GetStageSimulatorSheetsV1(IWorldState)

```
public static StageSimulatorSheetsV1 GetStageSimulatorSheetsV1(this
IWorldState worldState)
```

### Parameters

`worldState` IWorldState

### Returns

StageSimulatorSheetsV1

## GetWeeklyArenaState(IWorldState, Address)

```
public static WeeklyArenaState GetWeeklyArenaState(this IWorldState worldState,
Address address)
```

### Parameters

`worldState` IWorldState

`address` Address

### Returns

WeeklyArenaState

## GetWeeklyArenaState(IWorldState, int)

```
public static WeeklyArenaState GetWeeklyArenaState(this IWorldState worldState,
int index)
```

### Parameters

`worldState` IWorldState

`index` int

### Returns

WeeklyArenaState

## MarkBalanceChanged(IWorld, IActionContext, Currency, params Address[])

```
public static IWorld MarkBalanceChanged(this IWorld world, IActionContext context,
Currency currency, params Address[] accounts)
```

### Parameters

`world` IWorld

`context` IActionContext

`currency` Currency

`accounts` Address[]

### Returns

IWorld

## RemoveLegacyState(IWorld, Address)

```
public static IWorld RemoveLegacyState(this IWorld world, Address address)
```

### Parameters

`world` IWorld

`address` Address

### Returns

IWorld

## SetCouponWallet(IWorld, Address, IImmutableDictionary<Guid, Coupon>)

```
public static IWorld SetCouponWallet(this IWorld world, Address agentAddress,
IImmutableDictionary<Guid, Coupon> couponWallet)
```

### Parameters

`world` IWorld

`agentAddress` Address

`couponWallet` IImmutableDictionary<Guid, Coupon>

### Returns

IWorld

## SetLegacyState(IWorld, Address, IValue)

```
public static IWorld SetLegacyState(this IWorld world, Address address,
IValue state)
```

### Parameters

`world` IWorld

`address` Address

`state` IValue

### Returns

IWorld

## SetWorldBossKillReward(IWorld, IActionContext, Address, WorldBossKillRewardRecord, int, WorldBossState, RuneWeightSheet, WorldBossKillRewardSheet, RuneSheet, MaterialItemSheet, IRandom, Inventory, Address, Address, Row)

```
public static IWorld SetWorldBossKillReward(this IWorld world, IActionContext
context, Address rewardInfoAddress, WorldBossKillRewardRecord rewardRecord, int
rank, WorldBossState bossState, RuneWeightSheet runeWeightSheet,
WorldBossKillRewardSheet worldBossKillRewardSheet, RuneSheet runeSheet,
MaterialItemSheet materialItemSheet, IRandom random, Inventory inventory, Address
avatarAddress, Address agentAddress, BoostScheduleSheet.Row killRewardBoost = null)
```

### Parameters

`world` IWorld

`context` IActionContext

`rewardInfoAddress` Address

`rewardRecord` WorldBossKillRewardRecord

`rank` int

`bossState` WorldBossState

`runeWeightSheet` RuneWeightSheet

`worldBossKillRewardSheet` WorldBossKillRewardSheet

`runeSheet` RuneSheet

`materialItemSheet` MaterialItemSheet

`random` IRandom

`inventory` Inventory

`avatarAddress` Address

`agentAddress` Address

`killRewardBoost` BoostScheduleSheet.Row

### Returns

IWorld

## TryGetArenaAvatarState(IWorldState, Address, out ArenaAvatarState)

```
public static bool TryGetArenaAvatarState(this IWorldState worldState, Address
arenaAvatarStateAddress, out ArenaAvatarState arenaAvatarState)
```

### Parameters

`worldState` IWorldState

`arenaAvatarStateAddress` Address

`arenaAvatarState` ArenaAvatarState

### Returns

bool

## TryGetArenaInformation(IWorldState, Address, out ArenaInformation)

```
public static bool TryGetArenaInformation(this IWorldState worldState, Address arenaInformationAddress, out ArenaInformation arenaInformation)
```

### Parameters

`worldState` IWorldState

`arenaInformationAddress` Address

`arenaInformation` ArenaInformation

### Returns

bool

## TryGetArenaParticipants(IWorldState, Address, out ArenaParticipants)

```
public static bool TryGetArenaParticipants(this IWorldState worldState, Address arenaParticipantsAddress, out ArenaParticipants arenaParticipants)
```

### Parameters

`worldState` IWorldState

`arenaParticipantsAddress` Address

`arenaParticipants` ArenaParticipants

### Returns

bool

## TryGetArenaScore(IWorldState, Address, out ArenaScore)

```
public static bool TryGetArenaScore(this IWorldState worldState, Address
arenaScoreAddress, out ArenaScore arenaScore)
```

### Parameters

`worldState` IWorldState

`arenaScoreAddress` Address

`arenaScore` ArenaScore

### Returns

bool

## TryGetGoldBalance(IWorldState, Address, Currency, out FungibleAssetValue)

```
public static bool TryGetGoldBalance(this IWorldState worldState, Address address,
Currency currency, out FungibleAssetValue balance)
```

### Parameters

`worldState` IWorldState

`address` Address

`currency` Currency

`balance` FungibleAssetValue

### Returns

bool

## TryGetLegacyStakeState(IWorldState, Address, out LegacyStakeState)

```
public static bool TryGetLegacyStakeState(this IWorldState worldState, Address agentAddress, out LegacyStakeState legacyStakeState)
```

### Parameters

`worldState` IWorldState

`agentAddress` Address

`legacyStakeState` LegacyStakeState

### Returns

bool

## TryGetLegacyState<T>(IWorldState, Address, out T)

```
public static bool TryGetLegacyState<T>(this IWorldState worldState, Address address, out T result) where T : IValue
```

### Parameters

`worldState` IWorldState

`address` Address

`result` T

### Returns

bool

### Type Parameters

`T`

## TryGetPatchedSheet<T>(IWorldState, out T)

Loads `T` only when the chain has already been patched with it, and lets a malformed sheet throw instead of reporting it as missing.

```
public static bool TryGetPatchedSheet<T>(this IWorldState worldState, out T sheet)
where T : ISheet, new()
```

### Parameters

`worldState` IWorldState

The world to read the sheet from.

`sheet` T

The parsed sheet, or null when absent.

### Returns

bool

false when the sheet has never been patched onto this chain.

### Type Parameters

`T`

The sheet to load.

### Remarks

Unlike TryGetSheet<T>(IWorldState, out T) this does not swallow a parse failure. A sheet that carries restrictions must not be able to switch itself off by becoming unreadable; only genuine absence may fall back, and that is also what keeps re-evaluation of blocks older than the patch identical.

## TryGetSheet<T>(IWorldState, Address, out T)

```
public static bool TryGetSheet<T>(this IWorldState worldState, Address address, out
T sheet) where T : ISheet, new()
```

### Parameters

`worldState` IWorldState

`address` Address

`sheet` T

### Returns

bool

### Type Parameters

`T`

## TryGetSheet<T>(IWorldState, out T)

```
public static bool TryGetSheet<T>(this IWorldState worldState, out T sheet) where T
: ISheet, new()
```

### Parameters

`worldState` IWorldState

`sheet` T

### Returns

bool

### Type Parameters

`T`

# TryGetStakeState(IWorldState, Address, out StakeState)

```
public static bool TryGetStakeState(this IWorldState worldState, Address agentAddr,
out StakeState stakeState)
```

## Parameters

`worldState` IWorldState

`agentAddr` Address

`stakeState` StakeState

## Returns

bool

# ValidateWorldId(IWorldState, Address, int)

```
public static void ValidateWorldId(this IWorldState worldState, Address
avatarAddress, int worldId)
```

## Parameters

`worldState` IWorldState

`avatarAddress` Address

`worldId` int

# Class PatrolRewardModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

Provides utility methods for handling patrol reward states.

```
public static class PatrolRewardModule
```

**Inheritance**

object ← PatrolRewardModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetPatrolRewardClaimedBlockIndex(IWorldState, Address)

Retrieves the block index at which the patrol reward was last claimed for a given avatar.

```
public static long GetPatrolRewardClaimedBlockIndex(this IWorldState worldState,
Address avatarAddress)
```

#### Parameters

`worldState` IWorldState

The current world state.

`avatarAddress` Address

The address of the avatar.

#### Returns

long

The block index when the patrol reward was last claimed.

### Exceptions

FailedLoadStateException

Thrown if the state could not be loaded.

## SetPatrolRewardClaimedBlockIndex(IWorld, Address, long)

Sets the block index at which the patrol reward was last claimed for a given avatar.

```
public static IWorld SetPatrolRewardClaimedBlockIndex(this IWorld world, Address
avatarAddress, long blockIndex)
```

### Parameters

`world` IWorld

The current world instance.

`avatarAddress` Address

The address of the avatar.

`blockIndex` long

The block index to set.

### Returns

IWorld

The updated world instance.

## TryGetPatrolRewardClaimedBlockIndex(IWorldState, Address, out long)

Tries to retrieve the block index at which the patrol reward was last claimed for a given avatar.

```
public static bool TryGetPatrolRewardClaimedBlockIndex(this IWorldState worldState,
Address avatarAddress, out long blockIndex)
```

## Parameters

`worldState` IWorldState

The current world state.

`avatarAddress` Address

The address of the avatar.

`blockIndex` long

Outputs the block index when the patrol reward was last claimed.

## Returns

bool

`true` if the block index was successfully retrieved; otherwise, `false`.

# Class RelationshipModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class RelationshipModule
```

**Inheritance**

object ← RelationshipModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetRelationship(IWorldState, Address)

```
public static int GetRelationship(this IWorldState state, Address avatarAddress)
```

### Parameters

`state` IWorldState

`avatarAddress` Address

### Returns

int

## SetRelationship(IWorld, Address, int)

```
public static IWorld SetRelationship(this IWorld world, Address avatarAddress,
int relationship)
```

## Parameters

`world` IWorld

`avatarAddress` Address

`relationship` int

## Returns

IWorld

# Class RuneStateModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

RuneStateModule is the module to use RuneState/AllRuneState with account.

```
public static class RuneStateModule
```

**Inheritance**

object ← RuneStateModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetRuneState(IWorldState, Address, out bool)

```
public static AllRuneState GetRuneState(this IWorldState worldState, Address
avatarAddress, out bool migrateRequired)
```

### Parameters

`worldState` IWorldState

`avatarAddress` Address

`migrateRequired` bool

### Returns

AllRuneState

## SetRuneState(IWorld, Address, AllRuneState)

```
public static IWorld SetRuneState(this IWorld world, Address avatarAddress,
AllRuneState allRuneState)
```

## Parameters

`world` IWorld

`avatarAddress` Address

`allRuneState` AllRuneState

## Returns

IWorld

# Class StateResolver

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class StateResolver
```

**Inheritance**

object ← StateResolver

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetResolvedState(IWorldState, Address, Address)

Gets the account of address `address` if it exists. If not, returns from the legacy account.

```
public static IValue? GetResolvedState(this IWorldState world, Address address,
Address accountAddress)
```

### Parameters

`world` IWorldState

An Libplanet.Action.State.IWorld instance of to get state from.

`address` Address

The Libplanet.Crypto.Address of the state to get.

`accountAddress` Address

The Libplanet.Crypto.Address of the account to get.

### Returns

IValue

An Libplanet.Action.State.IAccountState instance of address `address` if exits. If not, legacy account.

## GetResolvedState(IWorldState, Address, Address, Address)

Gets the account of address `address` if it exists. If not, returns from the legacy account with `legacyAddress`.

```
public static IValue? GetResolvedState(this IWorldState world, Address address,
Address accountAddress, Address legacyAddress)
```

### Parameters

`world` IWorldState

An Libplanet.Action.State.IWorld instance of to get state from.

`address` Address

The Libplanet.Crypto.Address of the state to get.

`accountAddress` Address

The Libplanet.Crypto.Address of the account to get.

`legacyAddress` Address

The Libplanet.Crypto.Address of the state to get from the legacy account.

### Returns

IValue

An Libplanet.Action.State.IAccountState instance of address `address` if exits. If not, legacy account.

# Namespace Nekoyume.Module.CombinationSlot

## Classes

CombinationSlotNotFoundException

DuplicatedCombinationSlotIndexException

# Class CombinationSlotNotFoundException

Namespace: Nekoyume.Module.CombinationSlot

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationSlotNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← CombinationSlotNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### CombinationSlotNotFoundException(SerializationInfo, StreamingContext)

```
protected CombinationSlotNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# CombinationSlotNotFoundException(string)

```
public CombinationSlotNotFoundException(string message)
```

## Parameters

`message` string

# Class DuplicatedCombinationSlotIndexException

Namespace: Nekoyume.Module.CombinationSlot

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicatedCombinationSlotIndexException : Exception, ISerializable
```

### Inheritance

object ← Exception ← DuplicatedCombinationSlotIndexException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### DuplicatedCombinationSlotIndexException(SerializationInfo, StreamingContext)

```
protected DuplicatedCombinationSlotIndexException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# DuplicatedCombinationSlotIndexException(string)

```
public DuplicatedCombinationSlotIndexException(string message)
```

## Parameters

`message` string

# Namespace Nekoyume.Module.Guild

## Classes

GuildBanModule

GuildDelegateeModule

GuildMemberCounterModule

GuildModule

GuildParticipantModule

# Class GuildBanModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildBanModule
```

**Inheritance**

object ← GuildBanModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### Ban(GuildRepository, AgentAddress, AgentAddress)

```
public static void Ban(this GuildRepository repository, AgentAddress signer,
AgentAddress target)
```

#### Parameters

`repository` GuildRepository

`signer` AgentAddress

`target` AgentAddress

### IsBanned(GuildRepository, GuildAddress, Address)

```
public static bool IsBanned(this GuildRepository repository, GuildAddress
guildAddress, Address agentAddress)
```

#### Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

`agentAddress` Address

### Returns

bool

## RemoveBanList(GuildRepository, GuildAddress)

```
public static void RemoveBanList(this GuildRepository repository,
GuildAddress guildAddress)
```

### Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

## Unban(GuildRepository, AgentAddress, Address)

```
public static void Unban(this GuildRepository repository, AgentAddress signer,
Address target)
```

### Parameters

`repository` GuildRepository

`signer` AgentAddress

`target` Address

# Class GuildDelegateeModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildDelegateeModule
```

**Inheritance**

object ← GuildDelegateeModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## CreateDelegatee(GuildRepository, Address)

```
public static GuildDelegatee CreateDelegatee(this GuildRepository repository,
Address address)
```

### Parameters

`repository` GuildRepository

`address` Address

### Returns

GuildDelegatee

## TryGetDelegatee(GuildRepository, Address, out GuildDelegatee?)

```
public static bool TryGetDelegatee(this GuildRepository repository, Address address,
out GuildDelegatee? guildDelegatee)
```

## Parameters

`repository` GuildRepository

`address` Address

`guildDelegatee` GuildDelegatee

## Returns

bool

# Class GuildMemberCounterModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildMemberCounterModule
```

**Inheritance**

object ← GuildMemberCounterModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### DecreaseGuildMemberCount(GuildRepository, GuildAddress)

```
public static GuildRepository DecreaseGuildMemberCount(this GuildRepository
repository, GuildAddress guildAddress)
```

#### Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

#### Returns

GuildRepository

### GetGuildMemberCount(GuildRepository, GuildAddress)

```
public static BigInteger GetGuildMemberCount(this GuildRepository repository,
GuildAddress guildAddress)
```

### Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

### Returns

BigInteger

## IncreaseGuildMemberCount(GuildRepository, GuildAddress)

```
public static GuildRepository IncreaseGuildMemberCount(this GuildRepository
repository, GuildAddress guildAddress)
```

### Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

### Returns

GuildRepository

# Class GuildModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildModule
```

**Inheritance**

object ← GuildModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### ConvertCurrency(FungibleAssetValue, Currency)

```
public static (FungibleAssetValue TargetFAV, FungibleAssetValue Remainder)
ConvertCurrency(FungibleAssetValue sourceFAV, Currency targetCurrency)
```

#### Parameters

`sourceFAV` FungibleAssetValue

`targetCurrency` Currency

#### Returns

(FungibleAssetValue TargetFAV, FungibleAssetValue Remainder)

### FAVFromShare(BigInteger, BigInteger, FungibleAssetValue)

```
public static FungibleAssetValue FAVFromShare(BigInteger share, BigInteger
totalShares, FungibleAssetValue totalDelegated)
```

### Parameters

`share` BigInteger

`totalShares` BigInteger

`totalDelegated` FungibleAssetValue

### Returns

FungibleAssetValue

## GetDelegated(IWorld, Address)

```
public static FungibleAssetValue GetDelegated(this IWorld world,
Address agentAddress)
```

### Parameters

`world` IWorld

`agentAddress` Address

### Returns

FungibleAssetValue

## GetDelegationInfo(IWorld, Address)

```
public static (BigInteger Share, BigInteger TotalShare, FungibleAssetValue
TotalDelegated) GetDelegationInfo(this IWorld world, Address agentAddress)
```

### Parameters

`world` IWorld

`agentAddress` Address

### Returns

(BigInteger Share, BigInteger TotalShare, FungibleAssetValue TotalDelegated)

## GetDelegationInfo(IWorldState, Address)

```
public static (BigInteger Share, BigInteger TotalShare, FungibleAssetValue
TotalDelegated) GetDelegationInfo(this IWorldState worldState, Address agentAddress)
```

### Parameters

`worldState` IWorldState

`agentAddress` Address

### Returns

(BigInteger Share, BigInteger TotalShare, FungibleAssetValue TotalDelegated)

## GetDelegationInfo(GuildRepository, Address)

```
public static (BigInteger Share, BigInteger TotalShare, FungibleAssetValue
TotalDelegated) GetDelegationInfo(this GuildRepository repository,
Address agentAddress)
```

### Parameters

`repository` GuildRepository

`agentAddress` Address

### Returns

(BigInteger Share, BigInteger TotalShare, FungibleAssetValue TotalDelegated)

## GetGuildRepository(IWorld, IActionContext)

```
public static GuildRepository GetGuildRepository(this IWorld world,
IActionContext context)
```

### Parameters

`world` IWorld

`context` IActionContext

### Returns

GuildRepository

## GetStaked(IWorld, Address)

```
public static FungibleAssetValue GetStaked(this IWorld world, Address agentAddress)
```

### Parameters

`world` IWorld

`agentAddress` Address

### Returns

FungibleAssetValue

## GetStaked(IWorldState, Address)

```
public static FungibleAssetValue GetStaked(this IWorldState worldState,
Address agentAddress)
```

### Parameters

`worldState` IWorldState

`agentAddress` Address

### Returns

FungibleAssetValue

## MakeGuild(GuildRepository, GuildAddress, Address)

```
public static GuildRepository MakeGuild(this GuildRepository repository,
GuildAddress guildAddress, Address validatorAddress)
```

### Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

`validatorAddress` Address

### Returns

GuildRepository

## RemoveGuild(GuildRepository)

```
public static GuildRepository RemoveGuild(this GuildRepository repository)
```

### Parameters

`repository` GuildRepository

### Returns

GuildRepository

## TryGetGuild(GuildRepository, GuildAddress, out Guild?)

```
public static bool TryGetGuild(this GuildRepository repository, GuildAddress
guildAddress, out Guild? guild)
```

## Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

`guild` Guild

## Returns

bool

# Class GuildParticipantModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildParticipantModule
```

**Inheritance**

object ← GuildParticipantModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetJoinedGuild(GuildRepository, AgentAddress)

```
public static GuildAddress? GetJoinedGuild(this GuildRepository repository,
AgentAddress agentAddress)
```

#### Parameters

`repository` GuildRepository

`agentAddress` AgentAddress

#### Returns

GuildAddress?

### JoinGuild(GuildRepository, GuildAddress, AgentAddress)

```
public static GuildRepository JoinGuild(this GuildRepository repository,
GuildAddress guildAddress, AgentAddress target)
```

### Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

`target` AgentAddress

### Returns

GuildRepository

## LeaveGuild(GuildRepository, AgentAddress)

```
public static GuildRepository LeaveGuild(this GuildRepository repository,
AgentAddress agentAddress)
```

### Parameters

`repository` GuildRepository

`agentAddress` AgentAddress

### Returns

GuildRepository

## MoveGuild(GuildRepository, AgentAddress, GuildAddress)

```
public static GuildRepository MoveGuild(this GuildRepository repository,
AgentAddress guildParticipantAddress, GuildAddress dstGuildAddress)
```

### Parameters

`repository` GuildRepository

`guildParticipantAddress` AgentAddress

`dstGuildAddress` GuildAddress

### Returns

GuildRepository

## TryGetGuildParticipant(GuildRepository, AgentAddress, out GuildParticipant?)

```
public static bool TryGetGuildParticipant(this GuildRepository repository,
AgentAddress agentAddress, out GuildParticipant? guildParticipant)
```

### Parameters

`repository` GuildRepository

`agentAddress` AgentAddress

`guildParticipant` GuildParticipant

### Returns

bool

# Namespace Nekoyume.Module.Validator Delegation

## Classes

AbstainHistoryModule

ValidatorDelegateeModule

ValidatorDelegatorModule

ValidatorListModule

# Class AbstainHistoryModule

Namespace: Nekoyume.Module.ValidatorDelegation

Assembly: Lib9c.dll

```
public static class AbstainHistoryModule
```

**Inheritance**

object ← AbstainHistoryModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetAbstainHistory(ValidatorRepository)

```
public static AbstainHistory GetAbstainHistory(this ValidatorRepository repository)
```

#### Parameters

`repository` ValidatorRepository

#### Returns

AbstainHistory

### SetAbstainHistory(ValidatorRepository, AbstainHistory)

```
public static ValidatorRepository SetAbstainHistory(this ValidatorRepository
repository, AbstainHistory abstainHistory)
```

#### Parameters

`repository` ValidatorRepository

`abstainHistory` AbstainHistory

## Returns

ValidatorRepository

# Class ValidatorDelegateeModule

Namespace: Nekoyume.Module.ValidatorDelegation

Assembly: Lib9c.dll

```
public static class ValidatorDelegateeModule
```

**Inheritance**

object ← ValidatorDelegateeModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### CreateDelegatee(ValidatorRepository, PublicKey, BigInteger)

```
public static ValidatorDelegatee CreateDelegatee(this ValidatorRepository
repository, PublicKey publicKey, BigInteger commissionPercentage)
```

#### Parameters

`repository` ValidatorRepository

`publicKey` PublicKey

`commissionPercentage` BigInteger

#### Returns

ValidatorDelegatee

# TryGetDelegatee(ValidatorRepository, Address, out ValidatorDelegatee?)

```
public static bool TryGetDelegatee(this ValidatorRepository repository, Address address, out ValidatorDelegatee? validatorDelegatee)
```

## Parameters

`repository` ValidatorRepository

`address` Address

`validatorDelegatee` ValidatorDelegatee

## Returns

bool

# Class ValidatorDelegatorModule

Namespace: Nekoyume.Module.ValidatorDelegation

Assembly: Lib9c.dll

```
public static class ValidatorDelegatorModule
```

**Inheritance**

object ← ValidatorDelegatorModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### TryGetDelegator(ValidatorRepository, Address, out ValidatorDelegator?)

```
public static bool TryGetDelegator(this ValidatorRepository repository, Address
address, out ValidatorDelegator? validatorDelegator)
```

#### Parameters

`repository` ValidatorRepository

`address` Address

`validatorDelegator` ValidatorDelegator

#### Returns

bool

# Class ValidatorListModule

Namespace: Nekoyume.Module.ValidatorDelegation

Assembly: Lib9c.dll

```
public static class ValidatorListModule
```

**Inheritance**

object ← ValidatorListModule

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## GetProposerInfo(ValidatorRepository)

```
public static ProposerInfo GetProposerInfo(this ValidatorRepository repository)
```

### Parameters

`repository` ValidatorRepository

### Returns

ProposerInfo

## GetValidatorList(ValidatorRepository)

```
public static ValidatorList GetValidatorList(this ValidatorRepository repository)
```

### Parameters

`repository` ValidatorRepository

### Returns

ValidatorList

## SetProposerInfo(ValidatorRepository, ProposerInfo)

```
public static ValidatorRepository SetProposerInfo(this ValidatorRepository
repository, ProposerInfo proposerInfo)
```

### Parameters

`repository` ValidatorRepository

`proposerInfo` ProposerInfo

### Returns

ValidatorRepository

## TryGetProposerInfo(ValidatorRepository, Address, out ProposerInfo?)

```
public static bool TryGetProposerInfo(this ValidatorRepository repository, Address
address, out ProposerInfo? proposerInfo)
```

### Parameters

`repository` ValidatorRepository

`address` Address

`proposerInfo` ProposerInfo

### Returns

bool

# TryGetValidatorList(ValidatorRepository, out ValidatorList?)

```
public static bool TryGetValidatorList(this ValidatorRepository repository, out
ValidatorList? validatorList)
```

## Parameters

`repository` ValidatorRepository

`validatorList` ValidatorList

## Returns

bool

# Namespace Nekoyume.TableData

## Classes

ActionBuffSheet

ActionBuffSheet.Row

ArenaSheet

ArenaSheet.RoundData

ArenaSheet.Row

ArenaSimulatorSheets

ArenaSimulatorSheetsV1

ArenaSimulatorSheetsV100291

AvatarSheets

BoostScheduleSheet

Schedules a temporary adjustment of a game value — a reward amount, a cost, a guarantee count — for a block range, so that an event can be run with `patch_table_sheet` instead of rewriting the value's own sheet and patching it back afterwards.

Columns: `id`, `target`, `target_id`, `op`, `value`, `start_block`, `end_block`. A `_memo` column, like any `_` prefixed column, is dropped on load and may be used for notes.

`target_id` is a single id (`50`), an inclusive range (`1~50`) or `*` for every id. `~` rather than `-` keeps a spreadsheet from reading a range as a date on its way to the CSV.

`target` names what is adjusted and is matched as a plain string by the call site that owns it, so the sheet itself never needs to know which targets exist. A node that predates a target simply finds no match for it instead of failing to read the sheet. BoostScheduleSheet.Targets lists the targets this version applies.

A row applies in `[start_block, end_block)`. When several rows match, the one with the lowest `id` wins; rows are never combined, so overlapping rows cannot stack into a larger adjustment than any single row declares.

Reading is total: a row that does not parse is kept but marked invalid and never matches, so a typo fails toward "no adjustment" instead of aborting every action that reads the

sheet. ValidateCsv(string) rejects such a row at patch time so it should not reach the chain in the first place.

Callers must tolerate the sheet being absent — it reaches an existing chain only by `patch_table_sheet`. Absence means no adjustment, which is also what keeps re-evaluation of pre-patch blocks identical.

BoostScheduleSheet.Row
: One scheduled adjustment.

BoostScheduleSheet.TargetInfo
: How to write a row for one target: reference text for people and tools, never read by an action. See Infos.

BoostScheduleSheet.Targets
: The targets this version applies. A row naming any other target is valid data but has no effect until a version that applies it is running.

BuffLimitSheet

BuffLimitSheet.Row

BuffLinkSheet

BuffLinkSheet.Row

BuffSheet

BuffSheet.Row

CharacterLevelSheet

CharacterLevelSheet.Row

CharacterSheet

CharacterSheet.Row

CharacterSheetExtension

ClaimableGiftsSheet
: Sheet for claimable gifts that can be claimed by players during specific block index ranges. Each row contains gift ID, block index range, and a list of items to be given.

ClaimableGiftsSheet.Row

Represents a single row in the ClaimableGiftsSheet. Contains gift ID, block index range, and items to be given.

CollectQuestSheet

CollectQuestSheet.Row

CollectionSheet

CollectionSheet.RequiredMaterial

CollectionSheet.Row

CombinationEquipmentQuestSheet

CombinationEquipmentQuestSheet.Row

CombinationQuestSheet

CombinationQuestSheet.Row

ConsumableItemRecipeSheet

ConsumableItemRecipeSheet.Row

ConsumableItemSheet

ConsumableItemSheet.Row

CostumeItemSheet

CostumeItemSheet.Row

CostumeStatSheet

CostumeStatSheet.Row

CreateAvatarFavSheet

CreateAvatarFavSheet.Row

CreateAvatarItemSheet

CreateAvatarItemSheet.Row

EnemySkillSheet

EnemySkillSheet.Row

EnhancementCostSheet

EnhancementCostSheet.Row

EnhancementCostSheetV2

EnhancementCostSheetV2.Row

EnhancementCostSheetV3

EnhancementCostSheetV3.Row

EquipmentItemOptionSheet

EquipmentItemOptionSheet.Row

EquipmentItemRecipeSheet

EquipmentItemRecipeSheet.Row

EquipmentItemSetEffectSheet

EquipmentItemSetEffectSheet.Row

EquipmentItemSheet

EquipmentItemSheet.Row

EquipmentItemSubRecipeSheet

EquipmentItemSubRecipeSheet.Row

EquipmentItemSubRecipeSheetV2

EquipmentItemSubRecipeSheetV2.Row

GameConfigSheet

GameConfigSheet.Row

GeneralQuestSheet

GeneralQuestSheet.Row

GoldQuestSheet

GoldQuestSheet.Row

InfiniteTowerConditionSheet

Infinite tower condition sheet for managing battle conditions and restrictions. Uses StatType, SkillTargetType, and StatModifier.OperationType for flexible stat modifications.

InfiniteTowerConditionSheet.Row

Represents a row in the InfiniteTowerConditionSheet containing condition data.

InfiniteTowerFloorSheet

Infinite tower floor sheet for managing tower floors and their properties.

InfiniteTowerFloorSheet.Row

Represents a row in the InfiniteTowerFloorSheet containing floor data.

InfiniteTowerFloorWaveSheet

Infinite tower floor wave sheet for managing wave configurations. Supports multiple monster types per wave, similar to StageWaveSheet.

InfiniteTowerFloorWaveSheet.MonsterData

Represents a single monster type within a wave.

InfiniteTowerFloorWaveSheet.Row

Represents a row in the InfiniteTowerFloorWaveSheet containing wave data for a specific floor.

InfiniteTowerFloorWaveSheet.WaveData

Represents a single wave within a floor, containing multiple monster types.

InfiniteTowerScheduleSheet

Infinite tower schedule sheet for managing season schedules and ticket policies.

InfiniteTowerScheduleSheet.Row

Represents a row in the InfiniteTowerScheduleSheet containing schedule data.

ItemConfigForGradeSheet

ItemConfigForGradeSheet.Row

ItemEnhancementQuestSheet

ItemEnhancementQuestSheet.Row

ItemGradeQuestSheet

ItemGradeQuestSheet.Row

ItemRequirementSheet

ItemRequirementSheet.Row

ItemSheet

ItemSheet.Row

ItemSheet.RowBase

ItemTypeCollectQuestSheet

ItemTypeCollectQuestSheet.Row

MaterialItemSheet

MaterialItemSheet.Row

MimisbrunnrSheet

MimisbrunnrSheet.Row

MonsterCollectionRewardSheet

MonsterCollectionRewardSheet.RewardInfo

MonsterCollectionRewardSheet.Row

MonsterCollectionSheet

MonsterCollectionSheet.Row

MonsterQuestSheet

MonsterQuestSheet.Row

PetCostSheet

PetCostSheet.PetCostData

PetCostSheet.Row

PetSheet

PetSheet.Row

QuestItemRewardSheet

QuestItemRewardSheet.Row

QuestRewardSheet

QuestRewardSheet.Row

QuestSheet

QuestSheet.Row

RaidSimulatorSheets

RaidSimulatorSheetsV1

RankingSimulatorSheets

RankingSimulatorSheetsV1

RankingSimulatorSheetsV100291

RedeemCodeListSheet

RedeemCodeListSheet.Row

RedeemRewardSheet

RedeemRewardSheet.RewardInfo

RedeemRewardSheet.Row

RestrictionSheet

Declares what a given item or currency may not be used for, so that a restriction can be adjusted with `patch_table_sheet` instead of a headless and client release.

A row targets either an item id (numeric key) or a currency ticker (non-numeric key). Which one is looked up is decided by the call site — a market action already holds an item id or a ticker — so the sheet needs no discriminator column.

Columns: `key`, `market_registrable`, `synthesize_material`. The first governs the market registration path only — a fungible item can still reach another account through a garage or by being wrapped into a currency, so a row here does not make its target account bound by itself. The second closes the conversion path that would otherwise turn a restricted item into an unrestricted one.

An empty cell means "unspecified" and leaves the behavior that predates this sheet untouched. An unparsable cell is read as `false`(= restricted) on purpose: a typo then fails toward blocking, which is recoverable, rather than toward leaking. ValidateCsv(string) rejects such a cell at patch time so it should not reach the chain in the first place.

A key is an item id, not an item instance, so a row applies to every copy that exists — give a restricted item a new id rather than reusing one players already hold. A false `market_registrable` also stops `ReRegisterProduct`, leaving an existing seller able to cancel but not to reprice, and it does not retract listings that are already up: `BuyProduct` does not consult this sheet, so a restriction added after the fact still needs those products cancelled. A false `synthesize_material` only stops the item from being consumed as a material — an item that must not be handed out by synthesis is kept out of the result pool by leaving it out of `SynthesizeWeightSheet`, which is a separate sheet from this one.

Callers must tolerate the sheet being absent — it reaches an existing chain only by `patch_table_sheet`, so loading it as a required sheet would fail every affected action until each chain is patched. Absence has to fall back to the restrictions that are still hardcoded, which is also what keeps re-evaluation of pre-patch blocks identical.

RestrictionSheet.Row
: One target and what it may not be used for.

RuneCostSheet
: Represents a sheet containing rune cost data organized by level ranges. This sheet stores cost information for rune enhancement across different level intervals, allowing for efficient data compression and retrieval.

RuneCostSheet.Row
: Represents a row in the RuneCostSheet containing cost data for a specific rune. Each row can contain multiple cost data entries for different level ranges.

RuneCostSheet.RuneCostData
: Represents cost data for a specific level range of rune enhancement. Contains information about required materials and success rates for a given level interval.

RuneOptionSheet

RuneOptionSheet.Row

RuneOptionSheet.Row.RuneOptionInfo

RuneSheet

RuneSheet.Row

RuneWeightSheet

RuneWeightSheet.Row

RuneWeightSheet.RuneInfo

SetEffectExtension

SheetRowColumnException

SheetRowNotFoundException

SheetRowValidateException

SheetRow<T>

Sheet<TKey, TValue>

SimulatorSheets

SimulatorSheetsV1

SimulatorSheetsV100291

SkillActionBuffSheet

SkillActionBuffSheet.Row

SkillBuffSheet

SkillBuffSheet.Row

SkillSheet
Sheet containing skill data loaded from CSV files. Provides access to skill definitions and their properties.

SkillSheet.Row
Represents a single skill row from the skill sheet. Contains skill metadata and properties used for skill creation and execution.

StageDialogSheet

StageDialogSheet.Row

StageSheet

StageSheet.FavRewardData

StageSheet.RewardData

StageSheet.Row

StageSimulatorSheets

StageSimulatorSheetsV1

StageSimulatorSheetsV100291

StageWaveSheet

StageWaveSheet.MonsterData

StageWaveSheet.Row

StageWaveSheet.WaveData

StakeAchievementRewardSheet

StakeAchievementRewardSheet.RewardInfo

StakeAchievementRewardSheet.Row

StakeAchievementRewardSheet.Step

StakeActionPointCoefficientSheet

StakeActionPointCoefficientSheet.Row

StakeRegularFixedRewardSheet

This sheet is used for setting the regular rewards for staking. The difference between this sheet and StakeRegularRewardSheet is that the StakeRegularFixedRewardSheet.RewardInfo of this sheet has a fixed count of reward.

CSV column order: level, required_gold, item_id, count[, tradable]

The optional `tradable` column controls whether material items are created as tradable (TradableMaterial) or non-tradable (Material). Omitting the column defaults to `true`, preserving backward compatibility with existing CSV data.

StakeRegularFixedRewardSheet.RewardInfo

StakeRegularFixedRewardSheet.Row

StakeRegularRewardSheet

This sheet is used for setting the regular rewards for staking. The difference between this sheet and StakeRegularFixedRewardSheet is that the StakeRegularRewardSheet.RewardInfo of this sheet has a rate and a reward type. The count of reward is calculated by the rate and the amount of staked currency.

StakeRegularRewardSheet.RewardInfo

StakeRegularRewardSheet.Row

StatBuffSheet

StatBuffSheet.Row

SweepRequiredCPSheet

SweepRequiredCPSheet.Row

SynthesizeSheet
Represents a SynthesizeSheet.

SynthesizeSheet.Row

SynthesizeWeightSheet
Represents a SynthesizeWeightSheet.

SynthesizeWeightSheet.Row

TableExtensions

TradeQuestSheet

TradeQuestSheet.Row

UnlockCombinationSlotCostSheet

UnlockCombinationSlotCostSheet.Row

WeeklyArenaRewardSheet

WeeklyArenaRewardSheet.RewardData

WeeklyArenaRewardSheet.Row

WorldBossActionPatternSheet

WorldBossActionPatternSheet.ActionPatternData

WorldBossActionPatternSheet.Row

WorldBossBattleRewardSheet

WorldBossBattleRewardSheet.Row

WorldBossCharacterSheet

WorldBossCharacterSheet.Row

WorldBossCharacterSheet.WaveStatData

WorldBossContributionRewardSheet

Represents the reward sheet for world boss contributions.

WorldBossContributionRewardSheet.RewardModel

WorldBossContributionRewardSheet.Row

Represents each row in the sheet.

WorldBossGlobalHpSheet

WorldBossGlobalHpSheet.Row

WorldBossKillRewardSheet

WorldBossKillRewardSheet.Row

WorldBossListSheet

WorldBossListSheet.Row

WorldBossRankRewardSheet

WorldBossRankRewardSheet.Row

WorldBossRankingRewardSheet

WorldBossRankingRewardSheet.Row

WorldBossStatSheetExtension

WorldQuestSheet

WorldQuestSheet.Row

WorldSheet

WorldSheet.Row

WorldUnlockSheet

WorldUnlockSheet.Row

## Structs

EquipmentItemSubRecipeSheet.MaterialInfo

EquipmentItemSubRecipeSheet.OptionInfo

EquipmentItemSubRecipeSheetV2.OptionInfo

SynthesizeSheet.SynthesizeData
synthesize data for each subtype

WorldBossRankingRewardSheet.Row.RuneInfo

## Interfaces

ISheet

IStakeRewardRow

IStakeRewardSheet

IWorldBossRewardRow

IWorldBossRewardSheet

## Enums

BoostScheduleSheet.Operation
How a row's Value is combined with the value it adjusts.

CreateAvatarFavSheet.Target

RewardType

StakeRegularRewardSheet.StakeRewardType
The reward type of stake. Do not use the whole element of this enum at once in actions. (e.g., count, loop) Do versioning when you change or remove the elements of this enum.

# Class ActionBuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ActionBuffSheet : Sheet<int, ActionBuffSheet.Row>, IDictionary<int,
ActionBuffSheet.Row>, ICollection<KeyValuePair<int, ActionBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, ActionBuffSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ActionBuffSheet.Row> ← ActionBuffSheet

### Implements

IDictionary<int, ActionBuffSheet.Row>,
ICollection<KeyValuePair<int, ActionBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, ActionBuffSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, ActionBuffSheet.Row>.Name , Sheet<int, ActionBuffSheet.Row>.OrderedList ,
Sheet<int, ActionBuffSheet.Row>.First , Sheet<int, ActionBuffSheet.Row>.Last ,
Sheet<int, ActionBuffSheet.Row>.Keys , Sheet<int, ActionBuffSheet.Row>.Values ,
Sheet<int, ActionBuffSheet.Row>.Count , Sheet<int, ActionBuffSheet.Row>.IsReadOnly ,
Sheet<int, ActionBuffSheet.Row>.this[int] ,
Sheet<int, ActionBuffSheet.Row>.Set(string, bool) ,
Sheet<int, ActionBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ActionBuffSheet.Row>.GetEnumerator() ,
Sheet<int, ActionBuffSheet.Row>.TryGetValue(int, out ActionBuffSheet.Row, bool) ,
Sheet<int, ActionBuffSheet.Row>.AddRow(int, ActionBuffSheet.Row) ,
Sheet<int, ActionBuffSheet.Row>.Add(int, ActionBuffSheet.Row) ,
Sheet<int, ActionBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, ActionBuffSheet.Row>.Remove(int) ,
Sheet<int, ActionBuffSheet.Row>.TryGetValue(int, out ActionBuffSheet.Row) ,
Sheet<int, ActionBuffSheet.Row>.Add(KeyValuePair<int, ActionBuffSheet.Row>) ,
Sheet<int, ActionBuffSheet.Row>.Clear() ,
Sheet<int, ActionBuffSheet.Row>.Contains(KeyValuePair<int, ActionBuffSheet.Row>) ,
Sheet<int, ActionBuffSheet.Row>.CopyTo(KeyValuePair<int, ActionBuffSheet.Row>[], int) ,
Sheet<int, ActionBuffSheet.Row>.Remove(KeyValuePair<int, ActionBuffSheet.Row>) ,
Sheet<int, ActionBuffSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ActionBuffSheet()

```
public ActionBuffSheet()
```

# Class ActionBuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ActionBuffSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← ActionBuffSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ATKPowerRatio

```
public decimal ATKPowerRatio { get; }
```

### Property Value

decimal

## ActionBuffType

```
public ActionBuffType ActionBuffType { get; }
```

### Property Value

ActionBuffType

## Chance

```
public int Chance { get; }
```

### Property Value

int

## Duration

```
public int Duration { get; }
```

### Property Value

int

## ElementalType

```
public ElementalType ElementalType { get; }
```

### Property Value

ElementalType

## GroupId

```
public int GroupId { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## TargetType

```
public SkillTargetType TargetType { get; }
```

### Property Value

SkillTargetType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class ArenaSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSheet : Sheet<int, ArenaSheet.Row>, IDictionary<int,
ArenaSheet.Row>, ICollection<KeyValuePair<int, ArenaSheet.Row>>,
IEnumerable<KeyValuePair<int, ArenaSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ArenaSheet.Row> ← ArenaSheet

### Implements

IDictionary<int, ArenaSheet.Row>,
ICollection<KeyValuePair<int, ArenaSheet.Row>>,
IEnumerable<KeyValuePair<int, ArenaSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, ArenaSheet.Row>.Name , Sheet<int, ArenaSheet.Row>.OrderedList ,
Sheet<int, ArenaSheet.Row>.First , Sheet<int, ArenaSheet.Row>.Last ,
Sheet<int, ArenaSheet.Row>.Keys , Sheet<int, ArenaSheet.Row>.Values ,
Sheet<int, ArenaSheet.Row>.Count , Sheet<int, ArenaSheet.Row>.IsReadOnly ,
Sheet<int, ArenaSheet.Row>.this[int] , Sheet<int, ArenaSheet.Row>.Set(string, bool) ,
Sheet<int, ArenaSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ArenaSheet.Row>.GetEnumerator() ,
Sheet<int, ArenaSheet.Row>.TryGetValue(int, out ArenaSheet.Row, bool) ,
Sheet<int, ArenaSheet.Row>.AddRow(int, ArenaSheet.Row) ,
Sheet<int, ArenaSheet.Row>.Add(int, ArenaSheet.Row) ,
Sheet<int, ArenaSheet.Row>.ContainsKey(int) , Sheet<int, ArenaSheet.Row>.Remove(int) ,
Sheet<int, ArenaSheet.Row>.TryGetValue(int, out ArenaSheet.Row) ,
Sheet<int, ArenaSheet.Row>.Add(KeyValuePair<int, ArenaSheet.Row>) ,
Sheet<int, ArenaSheet.Row>.Clear() ,
Sheet<int, ArenaSheet.Row>.Contains(KeyValuePair<int, ArenaSheet.Row>) ,
Sheet<int, ArenaSheet.Row>.CopyTo(KeyValuePair<int, ArenaSheet.Row>[], int) ,
Sheet<int, ArenaSheet.Row>.Remove(KeyValuePair<int, ArenaSheet.Row>) ,
Sheet<int, ArenaSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ArenaSheet()

```
public ArenaSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, ArenaSheet.Row value)
```

### Parameters

`key` int

`value` ArenaSheet.Row

## GetRoundByBlockIndex(long)

```
public ArenaSheet.RoundData GetRoundByBlockIndex(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

ArenaSheet.RoundData

## GetRowByBlockIndex(long)

```
public ArenaSheet.Row GetRowByBlockIndex(long blockIndex)
```

## Parameters

`blockIndex` long

## Returns

ArenaSheet.Row

# Class ArenaSheet.RoundData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSheet.RoundData
```

**Inheritance**

object ← ArenaSheet.RoundData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RoundData(int, int, ArenaType, long, long, int, long, long, long, int, int, int)

```
public RoundData(int championshipId, int round, ArenaType arenaType, long
startBlockIndex, long endBlockIndex, int requiredMedalCount, long entranceFee, long
ticketPrice, long additionalTicketPrice, int maxPurchaseCount, int
maxPurchaseCountWithInterval, int medalId = 0)
```

#### Parameters

`championshipId` int

`round` int

`arenaType` ArenaType

`startBlockIndex` long

`endBlockIndex` long

`requiredMedalCount` int

`entranceFee` long

`ticketPrice` long

`additionalTicketPrice` long

`maxPurchaseCount` int

`maxPurchaseCountWithInterval` int

`medalId` int

# Properties

## AdditionalTicketPrice

```
public long AdditionalTicketPrice { get; }
```

### Property Value

long

## ArenaType

```
public ArenaType ArenaType { get; }
```

### Property Value

ArenaType

## ChampionshipId

```
public int ChampionshipId { get; }
```

### Property Value

int

## EndBlockIndex

```
public long EndBlockIndex { get; }
```

### Property Value

long

## EntranceFee

```
public long EntranceFee { get; }
```

### Property Value

long

## MaxPurchaseCount

```
public int MaxPurchaseCount { get; }
```

### Property Value

int

## MaxPurchaseCountWithInterval

```
public int MaxPurchaseCountWithInterval { get; }
```

### Property Value

int

## MedalId

```
public int MedalId { get; }
```

### Property Value

int

## RequiredMedalCount

```
public int RequiredMedalCount { get; }
```

### Property Value

int

## Round

```
public int Round { get; }
```

### Property Value

int

## StartBlockIndex

```
public long StartBlockIndex { get; }
```

### Property Value

long

## TicketPrice

```
public long TicketPrice { get; }
```

### Property Value

long

## Methods

### IsTheRoundOpened(long)

```
public bool IsTheRoundOpened(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

bool

# Class ArenaSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← ArenaSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ChampionshipId

```
public int ChampionshipId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Round

```
public List<ArenaSheet.RoundData> Round { get; }
```

### Property Value

List<ArenaSheet.RoundData>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

## TryGetChampionshipRound(out RoundData)

```
public bool TryGetChampionshipRound(out ArenaSheet.RoundData roundData)
```

### Parameters

`roundData` ArenaSheet.RoundData

### Returns

bool

## TryGetRound(int, out RoundData)

```
public bool TryGetRound(int round, out ArenaSheet.RoundData roundData)
```

## Parameters

`round` int

`roundData` ArenaSheet.RoundData

## Returns

bool

# Class ArenaSimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ArenaSimulatorSheets : SimulatorSheets
```

**Inheritance**

object ← SimulatorSheetsV1 ← SimulatorSheets ← ArenaSimulatorSheets

**Inherited Members**

SimulatorSheets.RuneOptionSheet , SimulatorSheets.RuneListSheet , SimulatorSheets.RuneLevelBonusSheet , SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaSimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, CostumeStatSheet, WeeklyArenaRewardSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public ArenaSimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, CostumeStatSheet
costumeStatSheet, WeeklyArenaRewardSheet weeklyArenaRewardSheet,
```

```
RuneOptionSheet runeOptionSheet, RuneListSheet runeListSheet,
RuneLevelBonusSheet runeLevelBonusSheet)
```

### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`costumeStatSheet` CostumeStatSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

## Properties

## CostumeStatSheet

```
public CostumeStatSheet CostumeStatSheet { get; }
```

### Property Value

CostumeStatSheet

# WeeklyArenaRewardSheet

```
public WeeklyArenaRewardSheet WeeklyArenaRewardSheet { get; }
```

## Property Value

WeeklyArenaRewardSheet

# Class ArenaSimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ArenaSimulatorSheetsV1 : SimulatorSheetsV1
```

**Inheritance**

object ← SimulatorSheetsV1 ← ArenaSimulatorSheetsV1

**Inherited Members**

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaSimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, CostumeStatSheet, WeeklyArenaRewardSheet)

```
public ArenaSimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, CostumeStatSheet
costumeStatSheet, WeeklyArenaRewardSheet weeklyArenaRewardSheet)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`costumeStatSheet` CostumeStatSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

# Properties

## CostumeStatSheet

```
public CostumeStatSheet CostumeStatSheet { get; }
```

### Property Value

CostumeStatSheet

## WeeklyArenaRewardSheet

```
public WeeklyArenaRewardSheet WeeklyArenaRewardSheet { get; }
```

### Property Value

WeeklyArenaRewardSheet

# Class ArenaSimulatorSheetsV100291

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ArenaSimulatorSheetsV100291 : SimulatorSheetsV100291
```

**Inheritance**

object ← SimulatorSheetsV100291 ← ArenaSimulatorSheetsV100291

**Inherited Members**

SimulatorSheetsV100291.MaterialItemSheet , SimulatorSheetsV100291.SkillSheet , SimulatorSheetsV100291.SkillBuffSheet , SimulatorSheetsV100291.BuffSheet , SimulatorSheetsV100291.CharacterSheet , SimulatorSheetsV100291.CharacterLevelSheet , SimulatorSheetsV100291.EquipmentItemSetEffectSheet , SimulatorSheetsV100291.ToSimulatorSheetsV1() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ArenaSimulatorSheetsV100291(MaterialItemSheet, SkillSheet, SkillBuffSheet, BuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, CostumeStatSheet, WeeklyArenaRewardSheet)

```
public ArenaSimulatorSheetsV100291(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, BuffSheet buffSheet, CharacterSheet
characterSheet, CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet, CostumeStatSheet costumeStatSheet,
WeeklyArenaRewardSheet weeklyArenaRewardSheet)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`buffSheet` BuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`costumeStatSheet` CostumeStatSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

# Properties

## CostumeStatSheet

```
public CostumeStatSheet CostumeStatSheet { get; }
```

### Property Value

CostumeStatSheet

## WeeklyArenaRewardSheet

```
public WeeklyArenaRewardSheet WeeklyArenaRewardSheet { get; }
```

### Property Value

WeeklyArenaRewardSheet

# Methods

## ToArenaSimulatorSheetsV1()

```
public ArenaSimulatorSheetsV1 ToArenaSimulatorSheetsV1()
```

## Returns

ArenaSimulatorSheetsV1

# Class AvatarSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class AvatarSheets
```

**Inheritance**

object ← AvatarSheets

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AvatarSheets(WorldSheet, QuestSheet, QuestRewardSheet, QuestItemRewardSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheet)

```
public AvatarSheets(WorldSheet worldSheet, QuestSheet questSheet,
QuestRewardSheet questRewardSheet, QuestItemRewardSheet questItemRewardSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet, EquipmentItemSubRecipeSheet
equipmentItemSubRecipeSheet)
```

#### Parameters

`worldSheet` WorldSheet

`questSheet` QuestSheet

`questRewardSheet` QuestRewardSheet

`questItemRewardSheet` QuestItemRewardSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`equipmentItemSubRecipeSheet` EquipmentItemSubRecipeSheet

# Fields

## EquipmentItemRecipeSheet

```
public readonly EquipmentItemRecipeSheet EquipmentItemRecipeSheet
```

### Field Value

EquipmentItemRecipeSheet

## EquipmentItemSubRecipeSheet

```
public readonly EquipmentItemSubRecipeSheet EquipmentItemSubRecipeSheet
```

### Field Value

EquipmentItemSubRecipeSheet

## QuestItemRewardSheet

```
public readonly QuestItemRewardSheet QuestItemRewardSheet
```

### Field Value

QuestItemRewardSheet

## QuestRewardSheet

```
public readonly QuestRewardSheet QuestRewardSheet
```

### Field Value

QuestRewardSheet

## QuestSheet

```
public readonly QuestSheet QuestSheet
```

### Field Value

QuestSheet

## WorldSheet

```
public readonly WorldSheet WorldSheet
```

### Field Value

WorldSheet

# Class BoostScheduleSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Schedules a temporary adjustment of a game value — a reward amount, a cost, a guarantee count — for a block range, so that an event can be run with `patch_table_sheet` instead of rewriting the value's own sheet and patching it back afterwards.

Columns: `id`, `target`, `target_id`, `op`, `value`, `start_block`, `end_block`. A `_memo` column, like any `_` prefixed column, is dropped on load and may be used for notes.

`target_id` is a single id (`50`), an inclusive range (`1~50`) or `*` for every id. `~` rather than `-` keeps a spreadsheet from reading a range as a date on its way to the CSV.

`target` names what is adjusted and is matched as a plain string by the call site that owns it, so the sheet itself never needs to know which targets exist. A node that predates a target simply finds no match for it instead of failing to read the sheet. BoostScheduleSheet.Targets lists the targets this version applies.

A row applies in `[start_block, end_block)`. When several rows match, the one with the lowest `id` wins; rows are never combined, so overlapping rows cannot stack into a larger adjustment than any single row declares.

Reading is total: a row that does not parse is kept but marked invalid and never matches, so a typo fails toward "no adjustment" instead of aborting every action that reads the sheet. ValidateCsv(string) rejects such a row at patch time so it should not reach the chain in the first place.

Callers must tolerate the sheet being absent — it reaches an existing chain only by `patch_table_sheet`. Absence means no adjustment, which is also what keeps re-evaluation of pre-patch blocks identical.

```
[Serializable]
public class BoostScheduleSheet : Sheet<int, BoostScheduleSheet.Row>,
IDictionary<int, BoostScheduleSheet.Row>, ICollection<KeyValuePair<int,
BoostScheduleSheet.Row>>, IEnumerable<KeyValuePair<int, BoostScheduleSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, BoostScheduleSheet.Row> ← BoostScheduleSheet

**Implements**
IDictionary<int, BoostScheduleSheet.Row>,
ICollection<KeyValuePair<int, BoostScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, BoostScheduleSheet.Row>>, IEnumerable, ISheet

**Inherited Members**
Sheet<int, BoostScheduleSheet.Row>.Name ,
Sheet<int, BoostScheduleSheet.Row>.OrderedList ,
Sheet<int, BoostScheduleSheet.Row>.First , Sheet<int, BoostScheduleSheet.Row>.Last ,
Sheet<int, BoostScheduleSheet.Row>.Keys ,
Sheet<int, BoostScheduleSheet.Row>.Values ,
Sheet<int, BoostScheduleSheet.Row>.Count ,
Sheet<int, BoostScheduleSheet.Row>.IsReadOnly ,
Sheet<int, BoostScheduleSheet.Row>.this[int] ,
Sheet<int, BoostScheduleSheet.Row>.Set(string, bool) ,
Sheet<int, BoostScheduleSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, BoostScheduleSheet.Row>.GetEnumerator() ,
Sheet<int, BoostScheduleSheet.Row>.TryGetValue(int, out BoostScheduleSheet.Row, bool)
,
Sheet<int, BoostScheduleSheet.Row>.AddRow(int, BoostScheduleSheet.Row) ,
Sheet<int, BoostScheduleSheet.Row>.Add(int, BoostScheduleSheet.Row) ,
Sheet<int, BoostScheduleSheet.Row>.ContainsKey(int) ,
Sheet<int, BoostScheduleSheet.Row>.Remove(int) ,
Sheet<int, BoostScheduleSheet.Row>.TryGetValue(int, out BoostScheduleSheet.Row) ,
Sheet<int, BoostScheduleSheet.Row>.Add(KeyValuePair<int, BoostScheduleSheet.Row>) ,
Sheet<int, BoostScheduleSheet.Row>.Clear() ,
Sheet<int, BoostScheduleSheet.Row>.Contains(KeyValuePair<int,
BoostScheduleSheet.Row>) ,
Sheet<int, BoostScheduleSheet.Row>.CopyTo(KeyValuePair<int, BoostScheduleSheet.Row>
[], int) ,
Sheet<int, BoostScheduleSheet.Row>.Remove(KeyValuePair<int,
BoostScheduleSheet.Row>) ,
Sheet<int, BoostScheduleSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## BoostScheduleSheet()

Creates an empty sheet.

```
public BoostScheduleSheet()
```

# Fields

## AnyTargetId

The `target_id` cell that matches every id of a target.

```
public const string AnyTargetId = "*"
```

### Field Value

string

## ColumnNames

Column names in their expected order, after `_` prefixed columns are dropped.

```
public static readonly string[] ColumnNames
```

### Field Value

string[]

## MaxAbsoluteValue

Upper bound of `|value|`. Keeps every Apply(int) inside decimal range so that applying a row can never throw.

```
public const decimal MaxAbsoluteValue = 2147483647
```

### Field Value

decimal

## TargetIdRangeSeparator

Separates the two ends of a `target_id` range, as in `1~50`.

```
public const char TargetIdRangeSeparator = '~'
```

### Field Value

char

# Methods

## AddRow(int, Row)

Adds a parsed row to the sheet.

```
protected override void AddRow(int key, BoostScheduleSheet.Row value)
```

### Parameters

`key` int

The row's key.

`value` BoostScheduleSheet.Row

The row to add.

### Remarks

A row without a usable id is dropped and a duplicated id keeps the first row instead of throwing, because a throw here would fail every action that reads the sheet.

## FindActive(string, int, long)

Finds the row that applies to `targetId` of `target` at `blockIndex`.

```
public BoostScheduleSheet.Row FindActive(string target, int targetId,
long blockIndex)
```

### Parameters

`target` string

The target being adjusted. See BoostScheduleSheet.Targets.

`targetId` int

The id within the target.

`blockIndex` long

The block being evaluated.

### Returns

BoostScheduleSheet.Row

The matching row with the lowest id, or `null` when none applies.

## ValidateCsv(string)

Parses `csv` strictly so that a malformed schedule fails the `patch_table_sheet` transaction instead of landing on the chain as a row that silently never applies.

```
public static void ValidateCsv(string csv)
```

### Parameters

`csv` string

The sheet's CSV, as the patch would write it.

### Remarks

Whether a target is applied by any running code is deliberately not checked here: the set of targets grows with releases, and a patch must not be accepted by one node version and rejected by another. Compare against Applied off-chain.

## Exceptions

SheetRowValidateException

The header does not match ColumnNames, or a row has an unusable or duplicated id, an empty target, an unparsable cell, a value out of range for its operation, or a block range that is empty.

# Enum BoostScheduleSheet.Operation

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

How a row's Value is combined with the value it adjusts.

```
public enum BoostScheduleSheet.Operation
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Add = 1`

`base + value`. `value` must be an integer.

`Mul = 0`

`floor(base × value)`.

# Class BoostScheduleSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One scheduled adjustment.

```
[Serializable]
public class BoostScheduleSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← BoostScheduleSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## EndBlockIndex

First block index this row no longer applies to.

```
public long EndBlockIndex { get; }
```

### Property Value

long

## Id

The row id. Also the precedence among overlapping rows: lower wins.

```
public int Id { get; }
```

### Property Value

int

## IsValid

Whether every cell parsed. An invalid row never matches.

```
public bool IsValid { get; }
```

### Property Value

bool

## Key

```
public override int Key { get; }
```

### Property Value

int

## Op

How Value is combined with the adjusted value.

```
public BoostScheduleSheet.Operation Op { get; }
```

### Property Value

BoostScheduleSheet.Operation

## StartBlockIndex

First block index this row applies to.

```
public long StartBlockIndex { get; }
```

### Property Value

long

## Target

What this row adjusts. See BoostScheduleSheet.Targets.

```
public string Target { get; }
```

### Property Value

string

## TargetIdBegin

First id within Target this row applies to, or `null` for all.

```
public int? TargetIdBegin { get; }
```

### Property Value

int?

## TargetIdEnd

Last id within Target this row applies to, inclusive, or `null` for all. Equals TargetIdBegin for a single id.

```
public int? TargetIdEnd { get; }
```

### Property Value

int?

## Value

The operand of Op.

```
public decimal Value { get; }
```

### Property Value

decimal

# Methods

## Apply(int)

Applies this row to `baseValue`, rounding down and clamping the result to `[0, int.MaxValue]`.

```
public int Apply(int baseValue)
```

### Parameters

`baseValue` int

The value before adjustment.

### Returns

int

The adjusted value.

### Remarks

Rounding down means a small base can be left unchanged by a multiplier — e.g. `1 × 1.5` stays `1`. Use Add when every target must actually move.

# Apply(long)

Apply(int) for 64-bit values: same rounding (down) and lower bound (`0`), clamped to Max Value instead of MaxValue so that large bases such as hammer experience are not cut.

```
public long Apply(long baseValue)
```

## Parameters

`baseValue` long

The value before adjustment.

## Returns

long

The adjusted value.

# Matches(string, int, long)

Whether this row applies to `targetId` of `target` at `blockIndex`.

```
public bool Matches(string target, int targetId, long blockIndex)
```

## Parameters

`target` string

The target being adjusted.

`targetId` int

The id within the target.

`blockIndex` long

The block being evaluated.

## Returns

bool

`true` when the row is valid and covers all three.

## Set(IReadOnlyList<string>)

Reads one row.

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

The row's cells, with "_" prefixed columns already dropped.

### Remarks

Total by design: Sheet<TKey, TValue> does not guard this call, so a throw here would fail every action that reads the sheet.

# Class BoostScheduleSheet.TargetInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

How to write a row for one target: reference text for people and tools, never read by an action. See Infos.

```
public sealed class BoostScheduleSheet.TargetInfo
```

**Inheritance**

object ← BoostScheduleSheet.TargetInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### TargetInfo(string, string, string, string, string, string, string, string)

Creates a description.

```
public TargetInfo(string target, string adjusts, string targetIdMeaning, string
targetIdSheet, string usualOp, string valueRule, string notes, string example)
```

#### Parameters

`target` string

`adjusts` string

`targetIdMeaning` string

`targetIdSheet` string

`usualOp` string

`valueRule` string

`notes` string

`example` string

# Properties

## Adjusts

What a row changes.

```
public string Adjusts { get; }
```

### Property Value

string

## Example

An example row, with the block range left as placeholders.

```
public string Example { get; }
```

### Property Value

string

## Notes

What the row does not cover, and what to release first.

```
public string Notes { get; }
```

### Property Value

string

## Target

The `target` cell.

```
public string Target { get; }
```

### Property Value

string

## TargetIdMeaning

What the `target_id` cell names (a description, not a value).

```
public string TargetIdMeaning { get; }
```

### Property Value

string

## TargetIdSheet

The sheet whose ids `target_id` takes.

```
public string TargetIdSheet { get; }
```

### Property Value

string

## UsualOp

The `op` and value direction an event normally uses.

```
public string UsualOp { get; }
```

### Property Value

string

## ValueRule

How the value is applied: rounding and bounds.

```
public string ValueRule { get; }
```

### Property Value

string

# Class BoostScheduleSheet.Targets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

The targets this version applies. A row naming any other target is valid data but has no effect until a version that applies it is running.

```
public static class BoostScheduleSheet.Targets
```

**Inheritance**

object ← BoostScheduleSheet.Targets

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AdventureBossFirstClearReward

Amount of each `AdventureBossFloorFirstRewardSheet` reward, granted by `ExploreAdventureBoss` when a floor is cleared for the first time in a season. `target_id` is the floor number, as for AdventureBossFloorReward.

```
public const string AdventureBossFirstClearReward =
"ADVENTURE_BOSS_FIRST_CLEAR_REWARD"
```

### Field Value

string

## AdventureBossFloorReward

Amount of the reward an adventure boss floor drops on every clear — the one entry drawn from `AdventureBossFloorSheet` rewards per floor, by both `ExploreAdventureBoss` and

`SweepAdventureBoss`. `target_id` is the floor number (`AdventureBossFloorSheetfloor`, 1 for the first floor of every boss), not the floor row id.

```
public const string AdventureBossFloorReward = "ADVENTURE_BOSS_FLOOR_REWARD"
```

### Field Value

string

## Applied

Every target above, for tooling that warns about rows no running code applies.

```
public static readonly IReadOnlyCollection<string> Applied
```

### Field Value

IReadOnlyCollection<string>

## CostumeSummonGuarantee

Guarantee count of a `CostumeSummonSheet` group. `target_id` is the group id.

```
public const string CostumeSummonGuarantee = "COSTUME_SUMMON_GUARANTEE"
```

### Field Value

string

## EnhancementHammerExp

Experience one enhancement hammer gives an equipment in `ItemEnhancement` (`EnhancementCostSheetV3` hammer rows). `target_id` is the hammer item id (`600301~600308`), so grades can get different multipliers.

The adjusted experience is stored on the equipment like any other experience: the equipment ends up exactly as if it had gathered that much normally, so it neither owes the

boost back on its next enhancement nor is worth less as a material. The result is kept between the configured experience and `ItemEnhancement.MaxHammerExpBoostFactor` times it — a row can only raise it. See `ItemEnhancement.GetHammerExp`.

Patch a row only after a client that previews with the adjusted experience is released: an old client shows a lower target level and a lower NCG cost than the action charges (cost follows the level reached), so an avatar holding only the shown NCG fails, and it lets players add more hammers than the max level needs.

```
public const string EnhancementHammerExp = "ENHANCEMENT_HAMMER_EXP"
```

### Field Value

string

## EquipmentSummonGuarantee

Guarantee count of an `EquipmentSummonSheet` group. `target_id` is the group id.

```
public const string EquipmentSummonGuarantee = "EQUIPMENT_SUMMON_GUARANTEE"
```

### Field Value

string

## Infos

How to write a row for each target above, for tooling that helps people fill the sheet (the backoffice serves this as is). It is reference text only: no action reads it, so changing it never changes what a row does. Rules shared by every row are in RowRules.

```
public static readonly IReadOnlyList<BoostScheduleSheet.TargetInfo> Infos
```

### Field Value

IReadOnlyList<BoostScheduleSheet.TargetInfo>

## RowRules

Rules every row follows whatever its target, for the same tooling as Infos. Reference text only.

```
public static readonly IReadOnlyList<string> RowRules
```

### Field Value

IReadOnlyList<string>

## RuneSummonGuarantee

Guarantee count of a `RuneSummonSheet` group. `target_id` is the group id.

```
public const string RuneSummonGuarantee = "RUNE_SUMMON_GUARANTEE"
```

### Field Value

string

## StageFavReward

Fungible asset rewards (e.g. crystal, rune stones) of an adventure stage clear (`HackAndSlash`, `HackAndSlashSweep`). `target_id` is the `StageSheet` stage id. Event dungeon stages are not covered.

Applied per clear to the amount of each drawn ticker after it is drawn, so the random draws are the same as without the row. See `StageSimulator.ApplyFavRewardBoost`.

```
public const string StageFavReward = "STAGE_FAV_REWARD"
```

### Field Value

string

# StageItemReward

Material item rewards of an adventure stage clear (`HackAndSlash`, `HackAndSlashSweep`). `target_id` is the `StageSheet` stage id, so `1~300` covers stages 1 through 300. Event dungeon stages are not covered.

Applied per clear to the count of each distinct item after the items are drawn, so the random draws are the same as without the row: `MUL 2` grants exactly twice each drawn item and `ADD 1` one more of each. See `StageSimulator.GetBoostedItemRewardCount`.

Boosted items are ordinary rewards everywhere else too: they count toward collect quest progress, and a circle stays tradable.

Rounding is down, per item id per clear, with one exception: a `MUL` above 1 turns an item drawn once in a clear (how most stages drop) into at least 2, so `MUL 1.5` grants 2 of a single drop. A count of 2 or more keeps the plain floor (`MUL 1.2` on 2 stays 2, `MUL 1.5` on 2 becomes 3); this is not a general "at least one more". `ADD` and a `MUL` of 1 or below are unaffected. Beware that any `MUL` just above 1 (e.g. `MUL 1.01`) also turns a single drop into 2 — effectively a 100% boost for those items, however small the value. StageFavReward amounts are large and always plainly floored.

```
public const string StageItemReward = "STAGE_ITEM_REWARD"
```

## Field Value

string

# SummonRuneEnhancementCrystalCost

Crystal cost of one `RuneEnhancement` try, per `RuneCostSheet` level, for **summonable runes only**: a rune whose id is a recipe id of any `RuneSummonSheet` row, whatever its ratio and whether or not that group is currently sold. Other runes are never adjusted. The summon pool is read from the chain when the try is evaluated, so a rune added to or removed from `RuneSummonSheet` gains or loses the adjustment with it.

`target_id` is the rune level the try attempts to reach — the current level plus one — so `1~50` adjusts the tries from level 0 up to level 50. It cannot narrow the row to particular rune ids. The adjusted cost never exceeds the sheet's cost and never drops below 1 while the sheet's cost is positive. NCG cost and success rate are not adjustable.

```
public const string SummonRuneEnhancementCrystalCost =
"SUMMON_RUNE_ENHANCEMENT_CRYSTAL_COST"
```

### Field Value

string

## SummonRuneEnhancementRuneStoneCost

Rune stone cost of one `RuneEnhancement` try, per `RuneCostSheet` level, for summonable runes only. Scope, `target_id` and bounds as in SummonRuneEnhancementCrystalCost.

```
public const string SummonRuneEnhancementRuneStoneCost =
"SUMMON_RUNE_ENHANCEMENT_RUNE_STONE_COST"
```

### Field Value

string

## SynthesizeRequiredCount

`required_count` of a `SynthesizeSheet` grade — how many materials one synthesis consumes. `target_id` is the material grade id (`grade_id`, e.g. `1~4`), and a row applies to every `item_sub_type` of that grade alike. The adjusted count never drops below 1 — which keeps synthesis from being free, not from being cheap: a typo such as `MUL 0.06` brings every grade down to 1, so keep values within the range planning intended.

Patch a row only after a client that picks materials by the adjusted count is released: the action still requires the material count to be a multiple of it, so a client choosing by the configured count fails every synthesis while the row is active.

```
public const string SynthesizeRequiredCount = "SYNTHESIZE_REQUIRED_COUNT"
```

### Field Value

string

# WorldBossBattleReward

Every amount — each drawn rune, crystal, circle — of the `WorldBossBattleRewardSheet` reward a `Raid` grants for its battle. `target_id` is the boss id (`WorldBossListSheet.boss_id`, e.g. `900001`), and the block is the one the raid is played in.

```
public const string WorldBossBattleReward = "WORLD_BOSS_BATTLE_REWARD"
```

## Field Value

string

# WorldBossKillReward

Every amount — each drawn rune, crystal, circle — of each `WorldBossKillRewardSheet` reward, granted either by the `Raid` that raises the boss level or by `ClaimWordBossKillReward`. `target_id` is the boss id (`WorldBossListSheet.boss_id`), and the block is the one the reward is granted in, not the one the boss was killed in: a kill left unclaimed until the event is boosted when it is claimed during the event.

```
public const string WorldBossKillReward = "WORLD_BOSS_KILL_REWARD"
```

## Field Value

string

# WorldBossRankReward

Every amount — each drawn rune, crystal, circle — of each `WorldBossRankRewardSheet` reward `ClaimRaidReward` grants. `target_id` is the boss id (`WorldBossListSheet.boss_id`), and the block is the one the reward is claimed in, not the one the rank was reached in: a rank left unclaimed until the event is boosted when it is claimed during the event.

```
public const string WorldBossRankReward = "WORLD_BOSS_RANK_REWARD"
```

## Field Value

string

# Class BuffLimitSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class BuffLimitSheet : Sheet<int, BuffLimitSheet.Row>, IDictionary<int,
BuffLimitSheet.Row>, ICollection<KeyValuePair<int, BuffLimitSheet.Row>>,
IEnumerable<KeyValuePair<int, BuffLimitSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, BuffLimitSheet.Row> ← BuffLimitSheet

### Implements

IDictionary<int, BuffLimitSheet.Row>,
ICollection<KeyValuePair<int, BuffLimitSheet.Row>>,
IEnumerable<KeyValuePair<int, BuffLimitSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, BuffLimitSheet.Row>.Name , Sheet<int, BuffLimitSheet.Row>.OrderedList ,
Sheet<int, BuffLimitSheet.Row>.First , Sheet<int, BuffLimitSheet.Row>.Last ,
Sheet<int, BuffLimitSheet.Row>.Keys , Sheet<int, BuffLimitSheet.Row>.Values ,
Sheet<int, BuffLimitSheet.Row>.Count , Sheet<int, BuffLimitSheet.Row>.IsReadOnly ,
Sheet<int, BuffLimitSheet.Row>.this[int] ,
Sheet<int, BuffLimitSheet.Row>.Set(string, bool) ,
Sheet<int, BuffLimitSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, BuffLimitSheet.Row>.GetEnumerator() ,
Sheet<int, BuffLimitSheet.Row>.TryGetValue(int, out BuffLimitSheet.Row, bool) ,
Sheet<int, BuffLimitSheet.Row>.AddRow(int, BuffLimitSheet.Row) ,
Sheet<int, BuffLimitSheet.Row>.Add(int, BuffLimitSheet.Row) ,
Sheet<int, BuffLimitSheet.Row>.ContainsKey(int) ,
Sheet<int, BuffLimitSheet.Row>.Remove(int) ,
Sheet<int, BuffLimitSheet.Row>.TryGetValue(int, out BuffLimitSheet.Row) ,
Sheet<int, BuffLimitSheet.Row>.Add(KeyValuePair<int, BuffLimitSheet.Row>) ,
Sheet<int, BuffLimitSheet.Row>.Clear() ,
Sheet<int, BuffLimitSheet.Row>.Contains(KeyValuePair<int, BuffLimitSheet.Row>) ,
Sheet<int, BuffLimitSheet.Row>.CopyTo(KeyValuePair<int, BuffLimitSheet.Row>[], int) ,
Sheet<int, BuffLimitSheet.Row>.Remove(KeyValuePair<int, BuffLimitSheet.Row>) ,
Sheet<int, BuffLimitSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## BuffLimitSheet()

```
public BuffLimitSheet()
```

# Class BuffLimitSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class BuffLimitSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← BuffLimitSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## GroupId

```
public int GroupId { get; set; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## StatModifier

```
public StatModifier.OperationType StatModifier { get; }
```

### Property Value

StatModifier.OperationType

## Value

```
public int Value { get; set; }
```

### Property Value

int

# Methods

## GetModifier(StatType)

```
public StatModifier GetModifier(StatType statType)
```

### Parameters

`statType` StatType

### Returns

StatModifier

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class BuffLinkSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class BuffLinkSheet : Sheet<int, BuffLinkSheet.Row>, IDictionary<int,
BuffLinkSheet.Row>, ICollection<KeyValuePair<int, BuffLinkSheet.Row>>,
IEnumerable<KeyValuePair<int, BuffLinkSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, BuffLinkSheet.Row> ← BuffLinkSheet

### Implements

IDictionary<int, BuffLinkSheet.Row>,
ICollection<KeyValuePair<int, BuffLinkSheet.Row>>,
IEnumerable<KeyValuePair<int, BuffLinkSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, BuffLinkSheet.Row>.Name , Sheet<int, BuffLinkSheet.Row>.OrderedList ,
Sheet<int, BuffLinkSheet.Row>.First , Sheet<int, BuffLinkSheet.Row>.Last ,
Sheet<int, BuffLinkSheet.Row>.Keys , Sheet<int, BuffLinkSheet.Row>.Values ,
Sheet<int, BuffLinkSheet.Row>.Count , Sheet<int, BuffLinkSheet.Row>.IsReadOnly ,
Sheet<int, BuffLinkSheet.Row>.this[int] , Sheet<int, BuffLinkSheet.Row>.Set(string, bool) ,
Sheet<int, BuffLinkSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, BuffLinkSheet.Row>.GetEnumerator() ,
Sheet<int, BuffLinkSheet.Row>.TryGetValue(int, out BuffLinkSheet.Row, bool) ,
Sheet<int, BuffLinkSheet.Row>.AddRow(int, BuffLinkSheet.Row) ,
Sheet<int, BuffLinkSheet.Row>.Add(int, BuffLinkSheet.Row) ,
Sheet<int, BuffLinkSheet.Row>.ContainsKey(int) ,
Sheet<int, BuffLinkSheet.Row>.Remove(int) ,
Sheet<int, BuffLinkSheet.Row>.TryGetValue(int, out BuffLinkSheet.Row) ,
Sheet<int, BuffLinkSheet.Row>.Add(KeyValuePair<int, BuffLinkSheet.Row>) ,
Sheet<int, BuffLinkSheet.Row>.Clear() ,
Sheet<int, BuffLinkSheet.Row>.Contains(KeyValuePair<int, BuffLinkSheet.Row>) ,
Sheet<int, BuffLinkSheet.Row>.CopyTo(KeyValuePair<int, BuffLinkSheet.Row>[], int) ,
Sheet<int, BuffLinkSheet.Row>.Remove(KeyValuePair<int, BuffLinkSheet.Row>) ,
Sheet<int, BuffLinkSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## BuffLinkSheet()

```
public BuffLinkSheet()
```

# Class BuffLinkSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class BuffLinkSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← BuffLinkSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## BuffId

```
public int BuffId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## LinkedBuffId

```
public int LinkedBuffId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class BuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class BuffSheet : Sheet<int, BuffSheet.Row>, IDictionary<int, BuffSheet.Row>,
ICollection<KeyValuePair<int, BuffSheet.Row>>, IEnumerable<KeyValuePair<int,
BuffSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, BuffSheet.Row> ← BuffSheet

### Implements

IDictionary<int, BuffSheet.Row>, ICollection<KeyValuePair<int, BuffSheet.Row>>, IEnumerable<KeyValuePair<int, BuffSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, BuffSheet.Row>.Name , Sheet<int, BuffSheet.Row>.OrderedList , Sheet<int, BuffSheet.Row>.First , Sheet<int, BuffSheet.Row>.Last , Sheet<int, BuffSheet.Row>.Keys , Sheet<int, BuffSheet.Row>.Values , Sheet<int, BuffSheet.Row>.Count , Sheet<int, BuffSheet.Row>.IsReadOnly , Sheet<int, BuffSheet.Row>.this[int] , Sheet<int, BuffSheet.Row>.Set(string, bool) , Sheet<int, BuffSheet.Row>.Set<T>(Sheet<int, T>, bool) , Sheet<int, BuffSheet.Row>.GetEnumerator() , Sheet<int, BuffSheet.Row>.TryGetValue(int, out BuffSheet.Row, bool) , Sheet<int, BuffSheet.Row>.AddRow(int, BuffSheet.Row) , Sheet<int, BuffSheet.Row>.Add(int, BuffSheet.Row) , Sheet<int, BuffSheet.Row>.ContainsKey(int) , Sheet<int, BuffSheet.Row>.Remove(int) , Sheet<int, BuffSheet.Row>.TryGetValue(int, out BuffSheet.Row) , Sheet<int, BuffSheet.Row>.Add(KeyValuePair<int, BuffSheet.Row>) , Sheet<int, BuffSheet.Row>.Clear() , Sheet<int, BuffSheet.Row>.Contains(KeyValuePair<int, BuffSheet.Row>) , Sheet<int, BuffSheet.Row>.CopyTo(KeyValuePair<int, BuffSheet.Row>[], int) , Sheet<int, BuffSheet.Row>.Remove(KeyValuePair<int, BuffSheet.Row>) , Sheet<int, BuffSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## BuffSheet()

```
public BuffSheet()
```

# Class BuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class BuffSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← BuffSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Chance

100: 100%

```
public int Chance { get; }
```

### Property Value

int

## Duration

Turn count.

```
public int Duration { get; }
```

### Property Value

int

## GroupId

GroupId는 같은 GroupId를 공유하는 버프 중 가장 작은 값으로 함. Id가 다르더라도 GroupId가 같으면, 서로 덮어씀. 이때의 우선순위는 Id가 큰 것이 남는 것으로 구현함.

```
public int GroupId { get; }
```

### Property Value

int

## IconResource

```
public string IconResource { get; }
```

### Property Value

string

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## StatModifier

```
public StatModifier StatModifier { get; }
```

### Property Value

StatModifier

## TargetType

```
public SkillTargetType TargetType { get; }
```

### Property Value

SkillTargetType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CharacterLevelSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CharacterLevelSheet : Sheet<int, CharacterLevelSheet.Row>,
IDictionary<int, CharacterLevelSheet.Row>, ICollection<KeyValuePair<int,
CharacterLevelSheet.Row>>, IEnumerable<KeyValuePair<int, CharacterLevelSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CharacterLevelSheet.Row> ← CharacterLevelSheet

### Implements

IDictionary<int, CharacterLevelSheet.Row>,
ICollection<KeyValuePair<int, CharacterLevelSheet.Row>>,
IEnumerable<KeyValuePair<int, CharacterLevelSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CharacterLevelSheet.Row>.Name ,
Sheet<int, CharacterLevelSheet.Row>.OrderedList ,
Sheet<int, CharacterLevelSheet.Row>.First , Sheet<int, CharacterLevelSheet.Row>.Last ,
Sheet<int, CharacterLevelSheet.Row>.Keys ,
Sheet<int, CharacterLevelSheet.Row>.Values ,
Sheet<int, CharacterLevelSheet.Row>.Count ,
Sheet<int, CharacterLevelSheet.Row>.IsReadOnly ,
Sheet<int, CharacterLevelSheet.Row>.this[int] ,
Sheet<int, CharacterLevelSheet.Row>.Set(string, bool) ,
Sheet<int, CharacterLevelSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CharacterLevelSheet.Row>.GetEnumerator() ,
Sheet<int, CharacterLevelSheet.Row>.TryGetValue(int, out CharacterLevelSheet.Row, bool)
,
Sheet<int, CharacterLevelSheet.Row>.AddRow(int, CharacterLevelSheet.Row) ,
Sheet<int, CharacterLevelSheet.Row>.Add(int, CharacterLevelSheet.Row) ,
Sheet<int, CharacterLevelSheet.Row>.ContainsKey(int) ,
Sheet<int, CharacterLevelSheet.Row>.Remove(int) ,
Sheet<int, CharacterLevelSheet.Row>.TryGetValue(int, out CharacterLevelSheet.Row) ,
Sheet<int, CharacterLevelSheet.Row>.Add(KeyValuePair<int, CharacterLevelSheet.Row>) ,
Sheet<int, CharacterLevelSheet.Row>.Clear() ,

Sheet<int, CharacterLevelSheet.Row>.Contains(KeyValuePair<int, CharacterLevelSheet.Row>) , Sheet<int, CharacterLevelSheet.Row>.CopyTo(KeyValuePair<int, CharacterLevelSheet.Row>[], int) , Sheet<int, CharacterLevelSheet.Row>.Remove(KeyValuePair<int, CharacterLevelSheet.Row>) , Sheet<int, CharacterLevelSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CharacterLevelSheet()

```
public CharacterLevelSheet()
```

# Methods

## GetLevel(long)

```
public int GetLevel(long exp)
```

### Parameters

exp long

### Returns

int

## TryGetLevel(long, out int)

```
public bool TryGetLevel(long exp, out int level)
```

### Parameters

`exp` long

`level` int

## Returns

bool

# Class CharacterLevelSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CharacterLevelSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CharacterLevelSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Exp

```
public long Exp { get; }
```

### Property Value

long

## ExpNeed

```
public long ExpNeed { get; }
```

### Property Value

long

## Key

```
public override int Key { get; }
```

### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CharacterSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CharacterSheet : Sheet<int, CharacterSheet.Row>, IDictionary<int,
CharacterSheet.Row>, ICollection<KeyValuePair<int, CharacterSheet.Row>>,
IEnumerable<KeyValuePair<int, CharacterSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CharacterSheet.Row> ← CharacterSheet

### Implements

IDictionary<int, CharacterSheet.Row>,
ICollection<KeyValuePair<int, CharacterSheet.Row>>,
IEnumerable<KeyValuePair<int, CharacterSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CharacterSheet.Row>.Name , Sheet<int, CharacterSheet.Row>.OrderedList ,
Sheet<int, CharacterSheet.Row>.First , Sheet<int, CharacterSheet.Row>.Last ,
Sheet<int, CharacterSheet.Row>.Keys , Sheet<int, CharacterSheet.Row>.Values ,
Sheet<int, CharacterSheet.Row>.Count , Sheet<int, CharacterSheet.Row>.IsReadOnly ,
Sheet<int, CharacterSheet.Row>.this[int] ,
Sheet<int, CharacterSheet.Row>.Set(string, bool) ,
Sheet<int, CharacterSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CharacterSheet.Row>.GetEnumerator() ,
Sheet<int, CharacterSheet.Row>.TryGetValue(int, out CharacterSheet.Row, bool) ,
Sheet<int, CharacterSheet.Row>.AddRow(int, CharacterSheet.Row) ,
Sheet<int, CharacterSheet.Row>.Add(int, CharacterSheet.Row) ,
Sheet<int, CharacterSheet.Row>.ContainsKey(int) ,
Sheet<int, CharacterSheet.Row>.Remove(int) ,
Sheet<int, CharacterSheet.Row>.TryGetValue(int, out CharacterSheet.Row) ,
Sheet<int, CharacterSheet.Row>.Add(KeyValuePair<int, CharacterSheet.Row>) ,
Sheet<int, CharacterSheet.Row>.Clear() ,
Sheet<int, CharacterSheet.Row>.Contains(KeyValuePair<int, CharacterSheet.Row>) ,
Sheet<int, CharacterSheet.Row>.CopyTo(KeyValuePair<int, CharacterSheet.Row>[], int) ,
Sheet<int, CharacterSheet.Row>.Remove(KeyValuePair<int, CharacterSheet.Row>) ,
Sheet<int, CharacterSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CharacterSheet()

```
public CharacterSheet()
```

# Class CharacterSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CharacterSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CharacterSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

CharacterSheetExtension.ToStats(CharacterSheet.Row, int)

# Properties

## ATK

```
public decimal ATK { get; }
```

### Property Value

decimal

## AttackRange

```
public float AttackRange { get; }
```

### Property Value

float

## CRI

```
public decimal CRI { get; }
```

### Property Value

decimal

## DEF

```
public decimal DEF { get; }
```

### Property Value

decimal

## ElementalType

```
public ElementalType ElementalType { get; }
```

### Property Value

ElementalType

## HIT

```
public decimal HIT { get; }
```

### Property Value

decimal

## HP

```
public decimal HP { get; }
```

### Property Value

decimal

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## LvATK

```
public decimal LvATK { get; }
```

### Property Value

decimal

## LvCRI

```
public decimal LvCRI { get; }
```

### Property Value

decimal

## LvDEF

```
public decimal LvDEF { get; }
```

### Property Value

decimal

## LvHIT

```
public decimal LvHIT { get; }
```

### Property Value

decimal

## LvHP

```
public decimal LvHP { get; }
```

### Property Value

decimal

## LvSPD

```
public decimal LvSPD { get; }
```

### Property Value

decimal

## RunSpeed

```
public float RunSpeed { get; }
```

### Property Value

float

## SPD

```
public decimal SPD { get; }
```

### Property Value

decimal

## SizeType

```
public SizeType SizeType { get; }
```

### Property Value

SizeType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class CharacterSheetExtension

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public static class CharacterSheetExtension
```

**Inheritance**

object ← CharacterSheetExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### ToStats(Row, int)

```
public static StatsMap ToStats(this CharacterSheet.Row row, int level)
```

#### Parameters

`row` CharacterSheet.Row

`level` int

#### Returns

StatsMap

# Class ClaimableGiftsSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Sheet for claimable gifts that can be claimed by players during specific block index ranges. Each row contains gift ID, block index range, and a list of items to be given.

```
public class ClaimableGiftsSheet : Sheet<int, ClaimableGiftsSheet.Row>,
IDictionary<int, ClaimableGiftsSheet.Row>, ICollection<KeyValuePair<int,
ClaimableGiftsSheet.Row>>, IEnumerable<KeyValuePair<int, ClaimableGiftsSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ClaimableGiftsSheet.Row> ← ClaimableGiftsSheet

### Implements

IDictionary<int, ClaimableGiftsSheet.Row>,
ICollection<KeyValuePair<int, ClaimableGiftsSheet.Row>>,
IEnumerable<KeyValuePair<int, ClaimableGiftsSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, ClaimableGiftsSheet.Row>.Name ,
Sheet<int, ClaimableGiftsSheet.Row>.OrderedList ,
Sheet<int, ClaimableGiftsSheet.Row>.First , Sheet<int, ClaimableGiftsSheet.Row>.Last ,
Sheet<int, ClaimableGiftsSheet.Row>.Keys , Sheet<int, ClaimableGiftsSheet.Row>.Values ,
Sheet<int, ClaimableGiftsSheet.Row>.Count ,
Sheet<int, ClaimableGiftsSheet.Row>.IsReadOnly ,
Sheet<int, ClaimableGiftsSheet.Row>.this[int] ,
Sheet<int, ClaimableGiftsSheet.Row>.Set(string, bool) ,
Sheet<int, ClaimableGiftsSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ClaimableGiftsSheet.Row>.GetEnumerator() ,
Sheet<int, ClaimableGiftsSheet.Row>.TryGetValue(int, out ClaimableGiftsSheet.Row, bool) ,
Sheet<int, ClaimableGiftsSheet.Row>.AddRow(int, ClaimableGiftsSheet.Row) ,
Sheet<int, ClaimableGiftsSheet.Row>.Add(int, ClaimableGiftsSheet.Row) ,
Sheet<int, ClaimableGiftsSheet.Row>.ContainsKey(int) ,
Sheet<int, ClaimableGiftsSheet.Row>.Remove(int) ,
Sheet<int, ClaimableGiftsSheet.Row>.TryGetValue(int, out ClaimableGiftsSheet.Row) ,
Sheet<int, ClaimableGiftsSheet.Row>.Add(KeyValuePair<int, ClaimableGiftsSheet.Row>) ,
Sheet<int, ClaimableGiftsSheet.Row>.Clear() ,

Sheet<int, ClaimableGiftsSheet.Row>.Contains(KeyValuePair<int, ClaimableGiftsSheet.Row>) , Sheet<int, ClaimableGiftsSheet.Row>.CopyTo(KeyValuePair<int, ClaimableGiftsSheet.Row>[], int) , Sheet<int, ClaimableGiftsSheet.Row>.Remove(KeyValuePair<int, ClaimableGiftsSheet.Row>) , Sheet<int, ClaimableGiftsSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ClaimableGiftsSheet()

Initializes a new instance of the ClaimableGiftsSheet class.

```
public ClaimableGiftsSheet()
```

# Methods

## TryFindRowByBlockIndex(long, out Row)

Tries to find a row that is claimable at the given block index.

```
public bool TryFindRowByBlockIndex(long blockIndex, out ClaimableGiftsSheet.Row row)
```

### Parameters

`blockIndex` long

Block index to search for

`row` ClaimableGiftsSheet.Row

When this method returns, contains the row if found; otherwise, null

### Returns

bool

True if a claimable row is found; otherwise, false

# Class ClaimableGiftsSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a single row in the ClaimableGiftsSheet. Contains gift ID, block index range, and items to be given.

```
public class ClaimableGiftsSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← ClaimableGiftsSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## EndedBlockIndex

Block index when this gift becomes unclaimable (inclusive).

```
public long EndedBlockIndex
```

### Field Value

long

## Id

Gift ID that uniquely identifies this gift.

```
public int Id
```

### Field Value

int

## Items

List of items to be given when this gift is claimed. Each tuple contains (itemId, quantity, tradable).

```
public List<(int itemId, int quantity, bool tradable)> Items
```

### Field Value

List<(int id, int count, bool tradable)>

## StartedBlockIndex

Block index when this gift becomes claimable (inclusive).

```
public long StartedBlockIndex
```

### Field Value

long

## Properties

### Key

Gets the key for this row, which is the gift ID.

```
public override int Key { get; }
```

### Property Value

int

## Methods

### Set(IReadOnlyList<string>)

Sets the row data from CSV fields. Expected format:
id,started_block_index,ended_block_index,item_1_id,item_1_quantity,item_1_tradable,...,item_5_id,item_5_quantity,item_5_tradable
Supports up to 5 items. Empty or zero values for item_id or quantity will be skipped.

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

CSV field values

### Validate(long)

Validates whether the given block index is within the claimable range.

```
public bool Validate(long blockIndex)
```

### Parameters

`blockIndex` long

Block index to validate

### Returns

bool

True if the block index is within [StartedBlockIndex, EndedBlockIndex], false otherwise

# Class CollectQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CollectQuestSheet : Sheet<int, CollectQuestSheet.Row>, IDictionary<int,
CollectQuestSheet.Row>, ICollection<KeyValuePair<int, CollectQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, CollectQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CollectQuestSheet.Row> ← CollectQuestSheet

### Implements

IDictionary<int, CollectQuestSheet.Row>,
ICollection<KeyValuePair<int, CollectQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, CollectQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CollectQuestSheet.Row>.Name ,
Sheet<int, CollectQuestSheet.Row>.OrderedList ,
Sheet<int, CollectQuestSheet.Row>.First , Sheet<int, CollectQuestSheet.Row>.Last ,
Sheet<int, CollectQuestSheet.Row>.Keys , Sheet<int, CollectQuestSheet.Row>.Values ,
Sheet<int, CollectQuestSheet.Row>.Count ,
Sheet<int, CollectQuestSheet.Row>.IsReadOnly ,
Sheet<int, CollectQuestSheet.Row>.this[int] ,
Sheet<int, CollectQuestSheet.Row>.Set(string, bool) ,
Sheet<int, CollectQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CollectQuestSheet.Row>.GetEnumerator() ,
Sheet<int, CollectQuestSheet.Row>.TryGetValue(int, out CollectQuestSheet.Row, bool) ,
Sheet<int, CollectQuestSheet.Row>.AddRow(int, CollectQuestSheet.Row) ,
Sheet<int, CollectQuestSheet.Row>.Add(int, CollectQuestSheet.Row) ,
Sheet<int, CollectQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, CollectQuestSheet.Row>.Remove(int) ,
Sheet<int, CollectQuestSheet.Row>.TryGetValue(int, out CollectQuestSheet.Row) ,
Sheet<int, CollectQuestSheet.Row>.Add(KeyValuePair<int, CollectQuestSheet.Row>) ,
Sheet<int, CollectQuestSheet.Row>.Clear() ,
Sheet<int, CollectQuestSheet.Row>.Contains(KeyValuePair<int, CollectQuestSheet.Row>) ,

Sheet<int, CollectQuestSheet.Row>.CopyTo(KeyValuePair<int, CollectQuestSheet.Row>[], int) ,
Sheet<int, CollectQuestSheet.Row>.Remove(KeyValuePair<int, CollectQuestSheet.Row>) ,
Sheet<int, CollectQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CollectQuestSheet()

```
public CollectQuestSheet()
```

# Class CollectQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CollectQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← CollectQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ItemId

```
public int ItemId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class CollectionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CollectionSheet : Sheet<int, CollectionSheet.Row>, IDictionary<int,
CollectionSheet.Row>, ICollection<KeyValuePair<int, CollectionSheet.Row>>,
IEnumerable<KeyValuePair<int, CollectionSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CollectionSheet.Row> ← CollectionSheet

### Implements

IDictionary<int, CollectionSheet.Row>,
ICollection<KeyValuePair<int, CollectionSheet.Row>>,
IEnumerable<KeyValuePair<int, CollectionSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CollectionSheet.Row>.Name , Sheet<int, CollectionSheet.Row>.OrderedList ,
Sheet<int, CollectionSheet.Row>.First , Sheet<int, CollectionSheet.Row>.Last ,
Sheet<int, CollectionSheet.Row>.Keys , Sheet<int, CollectionSheet.Row>.Values ,
Sheet<int, CollectionSheet.Row>.Count , Sheet<int, CollectionSheet.Row>.IsReadOnly ,
Sheet<int, CollectionSheet.Row>.this[int] ,
Sheet<int, CollectionSheet.Row>.Set(string, bool) ,
Sheet<int, CollectionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CollectionSheet.Row>.GetEnumerator() ,
Sheet<int, CollectionSheet.Row>.TryGetValue(int, out CollectionSheet.Row, bool) ,
Sheet<int, CollectionSheet.Row>.AddRow(int, CollectionSheet.Row) ,
Sheet<int, CollectionSheet.Row>.Add(int, CollectionSheet.Row) ,
Sheet<int, CollectionSheet.Row>.ContainsKey(int) ,
Sheet<int, CollectionSheet.Row>.Remove(int) ,
Sheet<int, CollectionSheet.Row>.TryGetValue(int, out CollectionSheet.Row) ,
Sheet<int, CollectionSheet.Row>.Add(KeyValuePair<int, CollectionSheet.Row>) ,
Sheet<int, CollectionSheet.Row>.Clear() ,
Sheet<int, CollectionSheet.Row>.Contains(KeyValuePair<int, CollectionSheet.Row>) ,
Sheet<int, CollectionSheet.Row>.CopyTo(KeyValuePair<int, CollectionSheet.Row>[], int) ,
Sheet<int, CollectionSheet.Row>.Remove(KeyValuePair<int, CollectionSheet.Row>) ,
Sheet<int, CollectionSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CollectionSheet()

```
public CollectionSheet()
```

# Class CollectionSheet.RequiredMaterial

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CollectionSheet.RequiredMaterial
```

**Inheritance**

object ← CollectionSheet.RequiredMaterial

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Count

```
public int Count
```

### Field Value

int

## ItemId

```
public int ItemId
```

### Field Value

int

## Level

```
public int Level
```

### Field Value

int

## SkillContains

```
public bool SkillContains
```

### Field Value

bool

# Methods

## GetMaterial(IEnumerable<ICollectionMaterial>)

Retrieves the ICollectionMaterial object from the given collection of materials based on the item ID and count.

```
public ICollectionMaterial GetMaterial(IEnumerable<ICollectionMaterial> materials)
```

### Parameters

`materials` IEnumerable<ICollectionMaterial>

The collection of materials to search.

### Returns

ICollectionMaterial

The ICollectionMaterial object if found; otherwise, an exception is thrown.

## Validate(INonFungibleItem)

```
public bool Validate(INonFungibleItem nonFungibleItem)
```

## Parameters

`nonFungibleItem` INonFungibleItem

## Returns

bool

# Class CollectionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CollectionSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CollectionSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Materials

```
public List<CollectionSheet.RequiredMaterial> Materials
```

### Field Value

List<CollectionSheet.RequiredMaterial>

## StatModifiers

```
public List<StatModifier> StatModifiers
```

### Field Value

List<StatModifier>

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CombinationEquipmentQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationEquipmentQuestSheet : Sheet<int,
CombinationEquipmentQuestSheet.Row>, IDictionary<int,
CombinationEquipmentQuestSheet.Row>, ICollection<KeyValuePair<int,
CombinationEquipmentQuestSheet.Row>>, IEnumerable<KeyValuePair<int,
CombinationEquipmentQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CombinationEquipmentQuestSheet.Row> ← CombinationEquipmentQuestSheet

### Implements

IDictionary<int, CombinationEquipmentQuestSheet.Row>, ICollection<KeyValuePair<int, CombinationEquipmentQuestSheet.Row>>, IEnumerable<KeyValuePair<int, CombinationEquipmentQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CombinationEquipmentQuestSheet.Row>.Name ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.OrderedList ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.First ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Last ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Keys ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Values ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Count ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.IsReadOnly ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.this[int] ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Set(string, bool) ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.GetEnumerator() ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.TryGetValue(int, out CombinationEquipmentQuestSheet.Row, bool) ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.AddRow(int, CombinationEquipmentQuestSheet.Row) ,

Sheet<int, CombinationEquipmentQuestSheet.Row>.Add(int, CombinationEquipmentQuestSheet.Row) , Sheet<int, CombinationEquipmentQuestSheet.Row>.ContainsKey(int) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Remove(int) , Sheet<int, CombinationEquipmentQuestSheet.Row>.TryGetValue(int, out CombinationEquipmentQuestSheet.Row) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Add(KeyValuePair<int, CombinationEquipmentQuestSheet.Row>) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Clear() , Sheet<int, CombinationEquipmentQuestSheet.Row>.Contains(KeyValuePair<int, CombinationEquipmentQuestSheet.Row>) , Sheet<int, CombinationEquipmentQuestSheet.Row>.CopyTo(KeyValuePair<int, CombinationEquipmentQuestSheet.Row>[], int) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Remove(KeyValuePair<int, CombinationEquipmentQuestSheet.Row>) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CombinationEquipmentQuestSheet()

```
public CombinationEquipmentQuestSheet()
```

# Class CombinationEquipmentQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationEquipmentQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← CombinationEquipmentQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## RecipeId

```
public int RecipeId { get; }
```

### Property Value

int

# Methods

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class CombinationQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationQuestSheet : Sheet<int, CombinationQuestSheet.Row>,
IDictionary<int, CombinationQuestSheet.Row>, ICollection<KeyValuePair<int,
CombinationQuestSheet.Row>>, IEnumerable<KeyValuePair<int,
CombinationQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CombinationQuestSheet.Row> ← CombinationQuestSheet

### Implements

IDictionary<int, CombinationQuestSheet.Row>,
ICollection<KeyValuePair<int, CombinationQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, CombinationQuestSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, CombinationQuestSheet.Row>.Name ,
Sheet<int, CombinationQuestSheet.Row>.OrderedList ,
Sheet<int, CombinationQuestSheet.Row>.First ,
Sheet<int, CombinationQuestSheet.Row>.Last ,
Sheet<int, CombinationQuestSheet.Row>.Keys ,
Sheet<int, CombinationQuestSheet.Row>.Values ,
Sheet<int, CombinationQuestSheet.Row>.Count ,
Sheet<int, CombinationQuestSheet.Row>.IsReadOnly ,
Sheet<int, CombinationQuestSheet.Row>.this[int] ,
Sheet<int, CombinationQuestSheet.Row>.Set(string, bool) ,
Sheet<int, CombinationQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CombinationQuestSheet.Row>.GetEnumerator() ,
Sheet<int, CombinationQuestSheet.Row>.TryGetValue(int, out
CombinationQuestSheet.Row, bool) ,
Sheet<int, CombinationQuestSheet.Row>.AddRow(int, CombinationQuestSheet.Row) ,
Sheet<int, CombinationQuestSheet.Row>.Add(int, CombinationQuestSheet.Row) ,
Sheet<int, CombinationQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, CombinationQuestSheet.Row>.Remove(int) ,

Sheet<int, CombinationQuestSheet.Row>.TryGetValue(int, out CombinationQuestSheet.Row) , Sheet<int, CombinationQuestSheet.Row>.Add(KeyValuePair<int, CombinationQuestSheet.Row>) , Sheet<int, CombinationQuestSheet.Row>.Clear() , Sheet<int, CombinationQuestSheet.Row>.Contains(KeyValuePair<int, CombinationQuestSheet.Row>) , Sheet<int, CombinationQuestSheet.Row>.CopyTo(KeyValuePair<int, CombinationQuestSheet.Row>[], int) , Sheet<int, CombinationQuestSheet.Row>.Remove(KeyValuePair<int, CombinationQuestSheet.Row>) , Sheet<int, CombinationQuestSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CombinationQuestSheet()

```
public CombinationQuestSheet()
```

# Class CombinationQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← CombinationQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ItemSubType

```
public ItemSubType ItemSubType { get; }
```

### Property Value

ItemSubType

## ItemType

```
public ItemType ItemType { get; }
```

### Property Value

ItemType

## Methods

### Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class ConsumableItemRecipeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableItemRecipeSheet : Sheet<int, ConsumableItemRecipeSheet.Row>,
IDictionary<int, ConsumableItemRecipeSheet.Row>, ICollection<KeyValuePair<int,
ConsumableItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
ConsumableItemRecipeSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ConsumableItemRecipeSheet.Row> ← ConsumableItemRecipeSheet

### Implements

IDictionary<int, ConsumableItemRecipeSheet.Row>,
ICollection<KeyValuePair<int, ConsumableItemRecipeSheet.Row>>,
IEnumerable<KeyValuePair<int, ConsumableItemRecipeSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, ConsumableItemRecipeSheet.Row>.Name ,
Sheet<int, ConsumableItemRecipeSheet.Row>.OrderedList ,
Sheet<int, ConsumableItemRecipeSheet.Row>.First ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Last ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Keys ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Values ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Count ,
Sheet<int, ConsumableItemRecipeSheet.Row>.IsReadOnly ,
Sheet<int, ConsumableItemRecipeSheet.Row>.this[int] ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, ConsumableItemRecipeSheet.Row>.TryGetValue(int, out
ConsumableItemRecipeSheet.Row, bool) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.AddRow(int,
ConsumableItemRecipeSheet.Row) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Add(int, ConsumableItemRecipeSheet.Row)
,
Sheet<int, ConsumableItemRecipeSheet.Row>.ContainsKey(int) ,

Sheet<int, ConsumableItemRecipeSheet.Row>.Remove(int) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.TryGetValue(int, out
ConsumableItemRecipeSheet.Row) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Add(KeyValuePair<int,
ConsumableItemRecipeSheet.Row>) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Clear() ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Contains(KeyValuePair<int,
ConsumableItemRecipeSheet.Row>) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.CopyTo(KeyValuePair<int,
ConsumableItemRecipeSheet.Row>[], int) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Remove(KeyValuePair<int,
ConsumableItemRecipeSheet.Row>) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ConsumableItemRecipeSheet()

```
public ConsumableItemRecipeSheet()
```

# Class ConsumableItemRecipeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableItemRecipeSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← ConsumableItemRecipeSheet.Row

**Derived**

EventConsumableItemRecipeSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Id

```
public int Id { get; }
```

#### Property Value

int

### Key

```
public override int Key { get; }
```

#### Property Value

int

## Materials

```
public List<EquipmentItemSubRecipeSheet.MaterialInfo> Materials { get; }
```

### Property Value

List<EquipmentItemSubRecipeSheet.MaterialInfo>

## RequiredActionPoint

```
public int RequiredActionPoint { get; }
```

### Property Value

int

## RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

### Property Value

long

## RequiredGold

```
public decimal RequiredGold { get; }
```

### Property Value

decimal

## ResultConsumableItemId

```
public int ResultConsumableItemId { get; }
```

### Property Value

int

# Methods

## GetAllMaterials()

```
public IEnumerable<EquipmentItemSubRecipeSheet.MaterialInfo> GetAllMaterials()
```

### Returns

IEnumerable<EquipmentItemSubRecipeSheet.MaterialInfo>

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class ConsumableItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableItemSheet : Sheet<int, ConsumableItemSheet.Row>,
IDictionary<int, ConsumableItemSheet.Row>, ICollection<KeyValuePair<int,
ConsumableItemSheet.Row>>, IEnumerable<KeyValuePair<int, ConsumableItemSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ConsumableItemSheet.Row> ← ConsumableItemSheet

### Implements

IDictionary<int, ConsumableItemSheet.Row>,
ICollection<KeyValuePair<int, ConsumableItemSheet.Row>>,
IEnumerable<KeyValuePair<int, ConsumableItemSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, ConsumableItemSheet.Row>.Name ,
Sheet<int, ConsumableItemSheet.Row>.OrderedList ,
Sheet<int, ConsumableItemSheet.Row>.First ,
Sheet<int, ConsumableItemSheet.Row>.Last ,
Sheet<int, ConsumableItemSheet.Row>.Keys ,
Sheet<int, ConsumableItemSheet.Row>.Values ,
Sheet<int, ConsumableItemSheet.Row>.Count ,
Sheet<int, ConsumableItemSheet.Row>.IsReadOnly ,
Sheet<int, ConsumableItemSheet.Row>.this[int] ,
Sheet<int, ConsumableItemSheet.Row>.Set(string, bool) ,
Sheet<int, ConsumableItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ConsumableItemSheet.Row>.GetEnumerator() ,
Sheet<int, ConsumableItemSheet.Row>.TryGetValue(int, out ConsumableItemSheet.Row, bool) ,
Sheet<int, ConsumableItemSheet.Row>.AddRow(int, ConsumableItemSheet.Row) ,
Sheet<int, ConsumableItemSheet.Row>.Add(int, ConsumableItemSheet.Row) ,
Sheet<int, ConsumableItemSheet.Row>.ContainsKey(int) ,
Sheet<int, ConsumableItemSheet.Row>.Remove(int) ,
Sheet<int, ConsumableItemSheet.Row>.TryGetValue(int, out ConsumableItemSheet.Row) ,

Sheet<int, ConsumableItemSheet.Row>.Add(KeyValuePair<int, ConsumableItemSheet.Row>) ,
Sheet<int, ConsumableItemSheet.Row>.Clear() ,
Sheet<int, ConsumableItemSheet.Row>.Contains(KeyValuePair<int, ConsumableItemSheet.Row>) ,
Sheet<int, ConsumableItemSheet.Row>.CopyTo(KeyValuePair<int, ConsumableItemSheet.Row>[], int) ,
Sheet<int, ConsumableItemSheet.Row>.Remove(KeyValuePair<int, ConsumableItemSheet.Row>) ,
Sheet<int, ConsumableItemSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ConsumableItemSheet()

```
public ConsumableItemSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, ConsumableItemSheet.Row value)
```

### Parameters

`key` int

`value` ConsumableItemSheet.Row

# Class ConsumableItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableItemSheet.Row : ItemSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row ← ConsumableItemSheet.Row

**Implements**

IState

**Inherited Members**

ItemSheet.Row.Key , ItemSheet.Row.Id , ItemSheet.Row.ItemSubType , ItemSheet.Row.Grade , ItemSheet.Row.ElementalType , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Row()

```
public Row()
```

## Row(Dictionary)

```
public Row(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

# Properties

## ItemType

```
public override ItemType ItemType { get; }
```

### Property Value

ItemType

## Stats

```
public List<DecimalStat> Stats { get; }
```

### Property Value

List<DecimalStat>

# Methods

## Deserialize(Dictionary)

```
public static ConsumableItemSheet.Row Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

ConsumableItemSheet.Row

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CostumeItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeItemSheet : Sheet<int, CostumeItemSheet.Row>, IDictionary<int,
CostumeItemSheet.Row>, ICollection<KeyValuePair<int, CostumeItemSheet.Row>>,
IEnumerable<KeyValuePair<int, CostumeItemSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CostumeItemSheet.Row> ← CostumeItemSheet

### Implements

IDictionary<int, CostumeItemSheet.Row>,
ICollection<KeyValuePair<int, CostumeItemSheet.Row>>,
IEnumerable<KeyValuePair<int, CostumeItemSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CostumeItemSheet.Row>.Name ,
Sheet<int, CostumeItemSheet.Row>.OrderedList ,
Sheet<int, CostumeItemSheet.Row>.First , Sheet<int, CostumeItemSheet.Row>.Last ,
Sheet<int, CostumeItemSheet.Row>.Keys , Sheet<int, CostumeItemSheet.Row>.Values ,
Sheet<int, CostumeItemSheet.Row>.Count ,
Sheet<int, CostumeItemSheet.Row>.IsReadOnly ,
Sheet<int, CostumeItemSheet.Row>.this[int] ,
Sheet<int, CostumeItemSheet.Row>.Set(string, bool) ,
Sheet<int, CostumeItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CostumeItemSheet.Row>.GetEnumerator() ,
Sheet<int, CostumeItemSheet.Row>.TryGetValue(int, out CostumeItemSheet.Row, bool) ,
Sheet<int, CostumeItemSheet.Row>.AddRow(int, CostumeItemSheet.Row) ,
Sheet<int, CostumeItemSheet.Row>.Add(int, CostumeItemSheet.Row) ,
Sheet<int, CostumeItemSheet.Row>.ContainsKey(int) ,
Sheet<int, CostumeItemSheet.Row>.Remove(int) ,
Sheet<int, CostumeItemSheet.Row>.TryGetValue(int, out CostumeItemSheet.Row) ,
Sheet<int, CostumeItemSheet.Row>.Add(KeyValuePair<int, CostumeItemSheet.Row>) ,
Sheet<int, CostumeItemSheet.Row>.Clear() ,
Sheet<int, CostumeItemSheet.Row>.Contains(KeyValuePair<int, CostumeItemSheet.Row>)
,

Sheet<int, CostumeItemSheet.Row>.CopyTo(KeyValuePair<int, CostumeItemSheet.Row>[], int) ,
Sheet<int, CostumeItemSheet.Row>.Remove(KeyValuePair<int, CostumeItemSheet.Row>) ,
Sheet<int, CostumeItemSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CostumeItemSheet()

```
public CostumeItemSheet()
```

# Class CostumeItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeItemSheet.Row : ItemSheet.Row
```

**Inheritance**

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row ← CostumeItemSheet.Row

**Inherited Members**

ItemSheet.Row.Key , ItemSheet.Row.Id , ItemSheet.Row.ItemSubType , ItemSheet.Row.Grade , ItemSheet.Row.ElementalType , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Row()

```
public Row()
```

### Row(Dictionary)

```
public Row(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## Properties

## ItemType

```
public override ItemType ItemType { get; }
```

### Property Value

ItemType

## SpineResourcePath

```
public string SpineResourcePath { get; }
```

### Property Value

string

# Methods

## Deserialize(Dictionary)

```
public static CostumeItemSheet.Row Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

CostumeItemSheet.Row

## Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class CostumeStatSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeStatSheet : Sheet<int, CostumeStatSheet.Row>, IDictionary<int,
CostumeStatSheet.Row>, ICollection<KeyValuePair<int, CostumeStatSheet.Row>>,
IEnumerable<KeyValuePair<int, CostumeStatSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CostumeStatSheet.Row> ← CostumeStatSheet

### Implements

IDictionary<int, CostumeStatSheet.Row>,
ICollection<KeyValuePair<int, CostumeStatSheet.Row>>,
IEnumerable<KeyValuePair<int, CostumeStatSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CostumeStatSheet.Row>.Name ,
Sheet<int, CostumeStatSheet.Row>.OrderedList ,
Sheet<int, CostumeStatSheet.Row>.First , Sheet<int, CostumeStatSheet.Row>.Last ,
Sheet<int, CostumeStatSheet.Row>.Keys , Sheet<int, CostumeStatSheet.Row>.Values ,
Sheet<int, CostumeStatSheet.Row>.Count ,
Sheet<int, CostumeStatSheet.Row>.IsReadOnly ,
Sheet<int, CostumeStatSheet.Row>.this[int] ,
Sheet<int, CostumeStatSheet.Row>.Set(string, bool) ,
Sheet<int, CostumeStatSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CostumeStatSheet.Row>.GetEnumerator() ,
Sheet<int, CostumeStatSheet.Row>.TryGetValue(int, out CostumeStatSheet.Row, bool) ,
Sheet<int, CostumeStatSheet.Row>.AddRow(int, CostumeStatSheet.Row) ,
Sheet<int, CostumeStatSheet.Row>.Add(int, CostumeStatSheet.Row) ,
Sheet<int, CostumeStatSheet.Row>.ContainsKey(int) ,
Sheet<int, CostumeStatSheet.Row>.Remove(int) ,
Sheet<int, CostumeStatSheet.Row>.TryGetValue(int, out CostumeStatSheet.Row) ,
Sheet<int, CostumeStatSheet.Row>.Add(KeyValuePair<int, CostumeStatSheet.Row>) ,
Sheet<int, CostumeStatSheet.Row>.Clear() ,
Sheet<int, CostumeStatSheet.Row>.Contains(KeyValuePair<int, CostumeStatSheet.Row>) ,

Sheet<int, CostumeStatSheet.Row>.CopyTo(KeyValuePair<int, CostumeStatSheet.Row>[], int) ,
Sheet<int, CostumeStatSheet.Row>.Remove(KeyValuePair<int, CostumeStatSheet.Row>) ,
Sheet<int, CostumeStatSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CostumeStatSheet()

```
public CostumeStatSheet()
```

# Class CostumeStatSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeStatSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CostumeStatSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## CostumeId

```
public int CostumeId { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Stat

```
public decimal Stat { get; }
```

### Property Value

decimal

## StatType

```
public StatType StatType { get; }
```

### Property Value

StatType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CreateAvatarFavSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CreateAvatarFavSheet : Sheet<string, CreateAvatarFavSheet.Row>,
IDictionary<string, CreateAvatarFavSheet.Row>, ICollection<KeyValuePair<string,
CreateAvatarFavSheet.Row>>, IEnumerable<KeyValuePair<string,
CreateAvatarFavSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<string, CreateAvatarFavSheet.Row> ← CreateAvatarFavSheet

### Implements

IDictionary<string, CreateAvatarFavSheet.Row>,
ICollection<KeyValuePair<string, CreateAvatarFavSheet.Row>>,
IEnumerable<KeyValuePair<string, CreateAvatarFavSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<string, CreateAvatarFavSheet.Row>.Name ,
Sheet<string, CreateAvatarFavSheet.Row>.OrderedList ,
Sheet<string, CreateAvatarFavSheet.Row>.First ,
Sheet<string, CreateAvatarFavSheet.Row>.Last ,
Sheet<string, CreateAvatarFavSheet.Row>.Keys ,
Sheet<string, CreateAvatarFavSheet.Row>.Values ,
Sheet<string, CreateAvatarFavSheet.Row>.Count ,
Sheet<string, CreateAvatarFavSheet.Row>.IsReadOnly ,
Sheet<string, CreateAvatarFavSheet.Row>.this[string] ,
Sheet<string, CreateAvatarFavSheet.Row>.Set(string, bool) ,
Sheet<string, CreateAvatarFavSheet.Row>.Set<T>(Sheet<string, T>, bool) ,
Sheet<string, CreateAvatarFavSheet.Row>.GetEnumerator() ,
Sheet<string, CreateAvatarFavSheet.Row>.TryGetValue(string, out
CreateAvatarFavSheet.Row, bool) ,
Sheet<string, CreateAvatarFavSheet.Row>.AddRow(string, CreateAvatarFavSheet.Row) ,
Sheet<string, CreateAvatarFavSheet.Row>.Add(string, CreateAvatarFavSheet.Row) ,
Sheet<string, CreateAvatarFavSheet.Row>.ContainsKey(string) ,
Sheet<string, CreateAvatarFavSheet.Row>.Remove(string) ,

Sheet<string, CreateAvatarFavSheet.Row>.TryGetValue(string, out CreateAvatarFavSheet.Row) , Sheet<string, CreateAvatarFavSheet.Row>.Add(KeyValuePair<string, CreateAvatarFavSheet.Row>) , Sheet<string, CreateAvatarFavSheet.Row>.Clear() , Sheet<string, CreateAvatarFavSheet.Row>.Contains(KeyValuePair<string, CreateAvatarFavSheet.Row>) , Sheet<string, CreateAvatarFavSheet.Row>.CopyTo(KeyValuePair<string, CreateAvatarFavSheet.Row>[], int) , Sheet<string, CreateAvatarFavSheet.Row>.Remove(KeyValuePair<string, CreateAvatarFavSheet.Row>) , Sheet<string, CreateAvatarFavSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CreateAvatarFavSheet()

```
public CreateAvatarFavSheet()
```

# Class CreateAvatarFavSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CreateAvatarFavSheet.Row : SheetRow<string>
```

**Inheritance**

object ← SheetRow<string> ← CreateAvatarFavSheet.Row

**Inherited Members**

SheetRow<string>.Validate() , SheetRow<string>.EndOfSheetInitialize() , SheetRow<string>.Equals(object) , SheetRow<string>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Currency

```
public Currency Currency { get; }
```

### Property Value

Currency

## Key

```
public override string Key { get; }
```

### Property Value

string

## Quantity

```
public int Quantity { get; }
```

### Property Value

int

## Target

```
public CreateAvatarFavSheet.Target Target { get; }
```

### Property Value

CreateAvatarFavSheet.Target

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Enum CreateAvatarFavSheet.Target

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public enum CreateAvatarFavSheet.Target
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Agent = 0`

`Avatar = 1`

# Class CreateAvatarItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CreateAvatarItemSheet : Sheet<int, CreateAvatarItemSheet.Row>,
IDictionary<int, CreateAvatarItemSheet.Row>, ICollection<KeyValuePair<int,
CreateAvatarItemSheet.Row>>, IEnumerable<KeyValuePair<int,
CreateAvatarItemSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CreateAvatarItemSheet.Row> ← CreateAvatarItemSheet

### Implements

IDictionary<int, CreateAvatarItemSheet.Row>,
ICollection<KeyValuePair<int, CreateAvatarItemSheet.Row>>,
IEnumerable<KeyValuePair<int, CreateAvatarItemSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, CreateAvatarItemSheet.Row>.Name ,
Sheet<int, CreateAvatarItemSheet.Row>.OrderedList ,
Sheet<int, CreateAvatarItemSheet.Row>.First ,
Sheet<int, CreateAvatarItemSheet.Row>.Last ,
Sheet<int, CreateAvatarItemSheet.Row>.Keys ,
Sheet<int, CreateAvatarItemSheet.Row>.Values ,
Sheet<int, CreateAvatarItemSheet.Row>.Count ,
Sheet<int, CreateAvatarItemSheet.Row>.IsReadOnly ,
Sheet<int, CreateAvatarItemSheet.Row>.this[int] ,
Sheet<int, CreateAvatarItemSheet.Row>.Set(string, bool) ,
Sheet<int, CreateAvatarItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CreateAvatarItemSheet.Row>.GetEnumerator() ,
Sheet<int, CreateAvatarItemSheet.Row>.TryGetValue(int, out CreateAvatarItemSheet.Row,
bool) ,
Sheet<int, CreateAvatarItemSheet.Row>.AddRow(int, CreateAvatarItemSheet.Row) ,
Sheet<int, CreateAvatarItemSheet.Row>.Add(int, CreateAvatarItemSheet.Row) ,
Sheet<int, CreateAvatarItemSheet.Row>.ContainsKey(int) ,
Sheet<int, CreateAvatarItemSheet.Row>.Remove(int) ,

Sheet<int, CreateAvatarItemSheet.Row>.TryGetValue(int, out CreateAvatarItemSheet.Row) , Sheet<int, CreateAvatarItemSheet.Row>.Add(KeyValuePair<int, CreateAvatarItemSheet.Row>) , Sheet<int, CreateAvatarItemSheet.Row>.Clear() , Sheet<int, CreateAvatarItemSheet.Row>.Contains(KeyValuePair<int, CreateAvatarItemSheet.Row>) , Sheet<int, CreateAvatarItemSheet.Row>.CopyTo(KeyValuePair<int, CreateAvatarItemSheet.Row>[], int) , Sheet<int, CreateAvatarItemSheet.Row>.Remove(KeyValuePair<int, CreateAvatarItemSheet.Row>) , Sheet<int, CreateAvatarItemSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CreateAvatarItemSheet()

```
public CreateAvatarItemSheet()
```

# Class CreateAvatarItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CreateAvatarItemSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CreateAvatarItemSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## ItemId

```
public int ItemId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EnemySkillSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EnemySkillSheet : Sheet<int, EnemySkillSheet.Row>, IDictionary<int,
EnemySkillSheet.Row>, ICollection<KeyValuePair<int, EnemySkillSheet.Row>>,
IEnumerable<KeyValuePair<int, EnemySkillSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EnemySkillSheet.Row> ← EnemySkillSheet

### Implements

IDictionary<int, EnemySkillSheet.Row>,
ICollection<KeyValuePair<int, EnemySkillSheet.Row>>,
IEnumerable<KeyValuePair<int, EnemySkillSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EnemySkillSheet.Row>.Name , Sheet<int, EnemySkillSheet.Row>.OrderedList ,
Sheet<int, EnemySkillSheet.Row>.First , Sheet<int, EnemySkillSheet.Row>.Last ,
Sheet<int, EnemySkillSheet.Row>.Keys , Sheet<int, EnemySkillSheet.Row>.Values ,
Sheet<int, EnemySkillSheet.Row>.Count , Sheet<int, EnemySkillSheet.Row>.IsReadOnly ,
Sheet<int, EnemySkillSheet.Row>.this[int] ,
Sheet<int, EnemySkillSheet.Row>.Set(string, bool) ,
Sheet<int, EnemySkillSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EnemySkillSheet.Row>.GetEnumerator() ,
Sheet<int, EnemySkillSheet.Row>.TryGetValue(int, out EnemySkillSheet.Row, bool) ,
Sheet<int, EnemySkillSheet.Row>.AddRow(int, EnemySkillSheet.Row) ,
Sheet<int, EnemySkillSheet.Row>.Add(int, EnemySkillSheet.Row) ,
Sheet<int, EnemySkillSheet.Row>.ContainsKey(int) ,
Sheet<int, EnemySkillSheet.Row>.Remove(int) ,
Sheet<int, EnemySkillSheet.Row>.TryGetValue(int, out EnemySkillSheet.Row) ,
Sheet<int, EnemySkillSheet.Row>.Add(KeyValuePair<int, EnemySkillSheet.Row>) ,
Sheet<int, EnemySkillSheet.Row>.Clear() ,
Sheet<int, EnemySkillSheet.Row>.Contains(KeyValuePair<int, EnemySkillSheet.Row>) ,
Sheet<int, EnemySkillSheet.Row>.CopyTo(KeyValuePair<int, EnemySkillSheet.Row>[], int) ,
Sheet<int, EnemySkillSheet.Row>.Remove(KeyValuePair<int, EnemySkillSheet.Row>) ,
Sheet<int, EnemySkillSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EnemySkillSheet()

```
public EnemySkillSheet()
```

# Class EnemySkillSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EnemySkillSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EnemySkillSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## characterId

```
public int characterId
```

### Field Value

int

## id

```
public int id
```

### Field Value

int

## skillId

```
public int skillId
```

### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EnhancementCostSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheet : Sheet<int, EnhancementCostSheet.Row>,
IDictionary<int, EnhancementCostSheet.Row>, ICollection<KeyValuePair<int,
EnhancementCostSheet.Row>>, IEnumerable<KeyValuePair<int,
EnhancementCostSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EnhancementCostSheet.Row> ← EnhancementCostSheet

### Implements

IDictionary<int, EnhancementCostSheet.Row>,
ICollection<KeyValuePair<int, EnhancementCostSheet.Row>>,
IEnumerable<KeyValuePair<int, EnhancementCostSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, EnhancementCostSheet.Row>.Name ,
Sheet<int, EnhancementCostSheet.Row>.OrderedList ,
Sheet<int, EnhancementCostSheet.Row>.First ,
Sheet<int, EnhancementCostSheet.Row>.Last ,
Sheet<int, EnhancementCostSheet.Row>.Keys ,
Sheet<int, EnhancementCostSheet.Row>.Values ,
Sheet<int, EnhancementCostSheet.Row>.Count ,
Sheet<int, EnhancementCostSheet.Row>.IsReadOnly ,
Sheet<int, EnhancementCostSheet.Row>.this[int] ,
Sheet<int, EnhancementCostSheet.Row>.Set(string, bool) ,
Sheet<int, EnhancementCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EnhancementCostSheet.Row>.GetEnumerator() ,
Sheet<int, EnhancementCostSheet.Row>.TryGetValue(int, out
EnhancementCostSheet.Row, bool) ,
Sheet<int, EnhancementCostSheet.Row>.AddRow(int, EnhancementCostSheet.Row) ,
Sheet<int, EnhancementCostSheet.Row>.Add(int, EnhancementCostSheet.Row) ,
Sheet<int, EnhancementCostSheet.Row>.ContainsKey(int) ,
Sheet<int, EnhancementCostSheet.Row>.Remove(int) ,

Sheet<int, EnhancementCostSheet.Row>.TryGetValue(int, out EnhancementCostSheet.Row) , Sheet<int, EnhancementCostSheet.Row>.Add(KeyValuePair<int, EnhancementCostSheet.Row>) , Sheet<int, EnhancementCostSheet.Row>.Clear() , Sheet<int, EnhancementCostSheet.Row>.Contains(KeyValuePair<int, EnhancementCostSheet.Row>) , Sheet<int, EnhancementCostSheet.Row>.CopyTo(KeyValuePair<int, EnhancementCostSheet.Row>[], int) , Sheet<int, EnhancementCostSheet.Row>.Remove(KeyValuePair<int, EnhancementCostSheet.Row>) , Sheet<int, EnhancementCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EnhancementCostSheet()

```
public EnhancementCostSheet()
```

# Class EnhancementCostSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EnhancementCostSheet.Row

**Derived**

EnhancementCostSheetV2.Row, EnhancementCostSheetV3.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Cost

```
public BigInteger Cost { get; }
```

### Property Value

BigInteger

## Grade

```
public int Grade { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## ItemSubType

```
public ItemSubType ItemSubType { get; }
```

### Property Value

ItemSubType

## Key

```
public override int Key { get; }
```

### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EnhancementCostSheetV2

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheetV2 : Sheet<int, EnhancementCostSheetV2.Row>,
IDictionary<int, EnhancementCostSheetV2.Row>, ICollection<KeyValuePair<int,
EnhancementCostSheetV2.Row>>, IEnumerable<KeyValuePair<int,
EnhancementCostSheetV2.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EnhancementCostSheetV2.Row> ← EnhancementCostSheetV2

### Implements

IDictionary<int, EnhancementCostSheetV2.Row>,
ICollection<KeyValuePair<int, EnhancementCostSheetV2.Row>>,
IEnumerable<KeyValuePair<int, EnhancementCostSheetV2.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, EnhancementCostSheetV2.Row>.Name ,
Sheet<int, EnhancementCostSheetV2.Row>.OrderedList ,
Sheet<int, EnhancementCostSheetV2.Row>.First ,
Sheet<int, EnhancementCostSheetV2.Row>.Last ,
Sheet<int, EnhancementCostSheetV2.Row>.Keys ,
Sheet<int, EnhancementCostSheetV2.Row>.Values ,
Sheet<int, EnhancementCostSheetV2.Row>.Count ,
Sheet<int, EnhancementCostSheetV2.Row>.IsReadOnly ,
Sheet<int, EnhancementCostSheetV2.Row>.this[int] ,
Sheet<int, EnhancementCostSheetV2.Row>.Set(string, bool) ,
Sheet<int, EnhancementCostSheetV2.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EnhancementCostSheetV2.Row>.GetEnumerator() ,
Sheet<int, EnhancementCostSheetV2.Row>.TryGetValue(int, out
EnhancementCostSheetV2.Row, bool) ,
Sheet<int, EnhancementCostSheetV2.Row>.AddRow(int, EnhancementCostSheetV2.Row) ,
Sheet<int, EnhancementCostSheetV2.Row>.Add(int, EnhancementCostSheetV2.Row) ,
Sheet<int, EnhancementCostSheetV2.Row>.ContainsKey(int) ,
Sheet<int, EnhancementCostSheetV2.Row>.Remove(int) ,

Sheet<int, EnhancementCostSheetV2.Row>.TryGetValue(int, out EnhancementCostSheetV2.Row) ,
Sheet<int, EnhancementCostSheetV2.Row>.Add(KeyValuePair<int, EnhancementCostSheetV2.Row>) ,
Sheet<int, EnhancementCostSheetV2.Row>.Clear() ,
Sheet<int, EnhancementCostSheetV2.Row>.Contains(KeyValuePair<int, EnhancementCostSheetV2.Row>) ,
Sheet<int, EnhancementCostSheetV2.Row>.CopyTo(KeyValuePair<int, EnhancementCostSheetV2.Row>[], int) ,
Sheet<int, EnhancementCostSheetV2.Row>.Remove(KeyValuePair<int, EnhancementCostSheetV2.Row>) ,
Sheet<int, EnhancementCostSheetV2.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EnhancementCostSheetV2()

```
public EnhancementCostSheetV2()
```

# Class EnhancementCostSheetV2.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheetV2.Row : EnhancementCostSheet.Row
```

**Inheritance**

object ← SheetRow<int> ← EnhancementCostSheet.Row ← EnhancementCostSheetV2.Row

**Inherited Members**

EnhancementCostSheet.Row.Key , EnhancementCostSheet.Row.Id , EnhancementCostSheet.Row.ItemSubType , EnhancementCostSheet.Row.Grade , EnhancementCostSheet.Row.Level , EnhancementCostSheet.Row.Cost , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## BaseStatGrowthMax

```
public int BaseStatGrowthMax { get; }
```

### Property Value

int

## BaseStatGrowthMin

```
public int BaseStatGrowthMin { get; }
```

### Property Value

int

## ExtraSkillChanceGrowthMax

```
public int ExtraSkillChanceGrowthMax { get; }
```

### Property Value

int

## ExtraSkillChanceGrowthMin

```
public int ExtraSkillChanceGrowthMin { get; }
```

### Property Value

int

## ExtraSkillDamageGrowthMax

```
public int ExtraSkillDamageGrowthMax { get; }
```

### Property Value

int

## ExtraSkillDamageGrowthMin

```
public int ExtraSkillDamageGrowthMin { get; }
```

### Property Value

int

## ExtraStatGrowthMax

```
public int ExtraStatGrowthMax { get; }
```

### Property Value

int

## ExtraStatGrowthMin

```
public int ExtraStatGrowthMin { get; }
```

### Property Value

int

## FailRatio

```
public int FailRatio { get; }
```

### Property Value

int

## FailRequiredBlockIndex

```
public int FailRequiredBlockIndex { get; }
```

### Property Value

int

## GreatSuccessRatio

```
public int GreatSuccessRatio { get; }
```

### Property Value

int

## GreatSuccessRequiredBlockIndex

```
public int GreatSuccessRequiredBlockIndex { get; }
```

### Property Value

int

## SuccessRatio

```
public int SuccessRatio { get; }
```

### Property Value

int

## SuccessRequiredBlockIndex

```
public int SuccessRequiredBlockIndex { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class EnhancementCostSheetV3

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheetV3 : Sheet<int, EnhancementCostSheetV3.Row>,
IDictionary<int, EnhancementCostSheetV3.Row>, ICollection<KeyValuePair<int,
EnhancementCostSheetV3.Row>>, IEnumerable<KeyValuePair<int,
EnhancementCostSheetV3.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EnhancementCostSheetV3.Row> ← EnhancementCostSheetV3

### Implements

IDictionary<int, EnhancementCostSheetV3.Row>,
ICollection<KeyValuePair<int, EnhancementCostSheetV3.Row>>,
IEnumerable<KeyValuePair<int, EnhancementCostSheetV3.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, EnhancementCostSheetV3.Row>.Name ,
Sheet<int, EnhancementCostSheetV3.Row>.OrderedList ,
Sheet<int, EnhancementCostSheetV3.Row>.First ,
Sheet<int, EnhancementCostSheetV3.Row>.Last ,
Sheet<int, EnhancementCostSheetV3.Row>.Keys ,
Sheet<int, EnhancementCostSheetV3.Row>.Values ,
Sheet<int, EnhancementCostSheetV3.Row>.Count ,
Sheet<int, EnhancementCostSheetV3.Row>.IsReadOnly ,
Sheet<int, EnhancementCostSheetV3.Row>.this[int] ,
Sheet<int, EnhancementCostSheetV3.Row>.Set(string, bool) ,
Sheet<int, EnhancementCostSheetV3.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EnhancementCostSheetV3.Row>.GetEnumerator() ,
Sheet<int, EnhancementCostSheetV3.Row>.TryGetValue(int, out
EnhancementCostSheetV3.Row, bool) ,
Sheet<int, EnhancementCostSheetV3.Row>.AddRow(int, EnhancementCostSheetV3.Row) ,
Sheet<int, EnhancementCostSheetV3.Row>.Add(int, EnhancementCostSheetV3.Row) ,
Sheet<int, EnhancementCostSheetV3.Row>.ContainsKey(int) ,
Sheet<int, EnhancementCostSheetV3.Row>.Remove(int) ,

Sheet<int, EnhancementCostSheetV3.Row>.TryGetValue(int, out EnhancementCostSheetV3.Row) , Sheet<int, EnhancementCostSheetV3.Row>.Add(KeyValuePair<int, EnhancementCostSheetV3.Row>) , Sheet<int, EnhancementCostSheetV3.Row>.Clear() , Sheet<int, EnhancementCostSheetV3.Row>.Contains(KeyValuePair<int, EnhancementCostSheetV3.Row>) , Sheet<int, EnhancementCostSheetV3.Row>.CopyTo(KeyValuePair<int, EnhancementCostSheetV3.Row>[], int) , Sheet<int, EnhancementCostSheetV3.Row>.Remove(KeyValuePair<int, EnhancementCostSheetV3.Row>) , Sheet<int, EnhancementCostSheetV3.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EnhancementCostSheetV3()

```
public EnhancementCostSheetV3()
```

# Methods

## GetHammerExp(int)

```
public long GetHammerExp(int hammerId)
```

### Parameters

`hammerId` int

### Returns

long

# Class EnhancementCostSheetV3.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheetV3.Row : EnhancementCostSheet.Row
```

**Inheritance**

object ← SheetRow<int> ← EnhancementCostSheet.Row ← EnhancementCostSheetV3.Row

**Inherited Members**

EnhancementCostSheet.Row.Key , EnhancementCostSheet.Row.Id , EnhancementCostSheet.Row.ItemSubType , EnhancementCostSheet.Row.Grade , EnhancementCostSheet.Row.Level , EnhancementCostSheet.Row.Cost , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## BaseStatGrowthMax

```
public int BaseStatGrowthMax { get; }
```

### Property Value

int

## BaseStatGrowthMin

```
public int BaseStatGrowthMin { get; }
```

### Property Value

int

## Exp

```
public long Exp { get; }
```

### Property Value

long

## ExtraSkillChanceGrowthMax

```
public int ExtraSkillChanceGrowthMax { get; }
```

### Property Value

int

## ExtraSkillChanceGrowthMin

```
public int ExtraSkillChanceGrowthMin { get; }
```

### Property Value

int

## ExtraSkillDamageGrowthMax

```
public int ExtraSkillDamageGrowthMax { get; }
```

### Property Value

int

## ExtraSkillDamageGrowthMin

```
public int ExtraSkillDamageGrowthMin { get; }
```

### Property Value

int

## ExtraStatGrowthMax

```
public int ExtraStatGrowthMax { get; }
```

### Property Value

int

## ExtraStatGrowthMin

```
public int ExtraStatGrowthMin { get; }
```

### Property Value

int

## RequiredBlockIndex

```
public int RequiredBlockIndex { get; }
```

### Property Value

int

# Methods

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class EquipmentItemOptionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemOptionSheet : Sheet<int, EquipmentItemOptionSheet.Row>,
IDictionary<int, EquipmentItemOptionSheet.Row>, ICollection<KeyValuePair<int,
EquipmentItemOptionSheet.Row>>, IEnumerable<KeyValuePair<int,
EquipmentItemOptionSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EquipmentItemOptionSheet.Row> ← EquipmentItemOptionSheet

### Implements

IDictionary<int, EquipmentItemOptionSheet.Row>, ICollection<KeyValuePair<int, EquipmentItemOptionSheet.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemOptionSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EquipmentItemOptionSheet.Row>.Name ,
Sheet<int, EquipmentItemOptionSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemOptionSheet.Row>.First ,
Sheet<int, EquipmentItemOptionSheet.Row>.Last ,
Sheet<int, EquipmentItemOptionSheet.Row>.Keys ,
Sheet<int, EquipmentItemOptionSheet.Row>.Values ,
Sheet<int, EquipmentItemOptionSheet.Row>.Count ,
Sheet<int, EquipmentItemOptionSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemOptionSheet.Row>.this[int] ,
Sheet<int, EquipmentItemOptionSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemOptionSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemOptionSheet.Row>.TryGetValue(int, out EquipmentItemOptionSheet.Row, bool) ,
Sheet<int, EquipmentItemOptionSheet.Row>.AddRow(int, EquipmentItemOptionSheet.Row) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Add(int, EquipmentItemOptionSheet.Row) ,
Sheet<int, EquipmentItemOptionSheet.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Remove(int) ,

Sheet<int, EquipmentItemOptionSheet.Row>.TryGetValue(int, out EquipmentItemOptionSheet.Row) , Sheet<int, EquipmentItemOptionSheet.Row>.Add(KeyValuePair<int, EquipmentItemOptionSheet.Row>) , Sheet<int, EquipmentItemOptionSheet.Row>.Clear() , Sheet<int, EquipmentItemOptionSheet.Row>.Contains(KeyValuePair<int, EquipmentItemOptionSheet.Row>) , Sheet<int, EquipmentItemOptionSheet.Row>.CopyTo(KeyValuePair<int, EquipmentItemOptionSheet.Row>[], int) , Sheet<int, EquipmentItemOptionSheet.Row>.Remove(KeyValuePair<int, EquipmentItemOptionSheet.Row>) , Sheet<int, EquipmentItemOptionSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EquipmentItemOptionSheet()

```
public EquipmentItemOptionSheet()
```

# Class EquipmentItemOptionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemOptionSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EquipmentItemOptionSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## ReferencedStatType

```
public StatType ReferencedStatType { get; }
```

### Property Value

StatType

## SkillChanceMax

```
public int SkillChanceMax { get; }
```

### Property Value

int

## SkillChanceMin

```
public int SkillChanceMin { get; }
```

### Property Value

int

## SkillDamageMax

```
public int SkillDamageMax { get; }
```

### Property Value

int

## SkillDamageMin

```
public int SkillDamageMin { get; }
```

### Property Value

int

## SkillId

```
public int SkillId { get; }
```

### Property Value

int

## StatDamageRatioMax

```
public int StatDamageRatioMax { get; }
```

### Property Value

int

## StatDamageRatioMin

```
public int StatDamageRatioMin { get; }
```

### Property Value

int

## StatMax

```
public int StatMax { get; }
```

### Property Value

int

## StatMin

```
public int StatMin { get; }
```

### Property Value

int

## StatType

```
public StatType StatType { get; }
```

### Property Value

StatType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EquipmentItemRecipeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemRecipeSheet : Sheet<int, EquipmentItemRecipeSheet.Row>,
IDictionary<int, EquipmentItemRecipeSheet.Row>, ICollection<KeyValuePair<int,
EquipmentItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
EquipmentItemRecipeSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EquipmentItemRecipeSheet.Row> ← EquipmentItemRecipeSheet

### Implements

IDictionary<int, EquipmentItemRecipeSheet.Row>,
ICollection<KeyValuePair<int, EquipmentItemRecipeSheet.Row>>,
IEnumerable<KeyValuePair<int, EquipmentItemRecipeSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, EquipmentItemRecipeSheet.Row>.Name ,
Sheet<int, EquipmentItemRecipeSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemRecipeSheet.Row>.First ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Last ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Keys ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Values ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Count ,
Sheet<int, EquipmentItemRecipeSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemRecipeSheet.Row>.this[int] ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemRecipeSheet.Row>.TryGetValue(int, out
EquipmentItemRecipeSheet.Row, bool) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.AddRow(int, EquipmentItemRecipeSheet.Row)
,
Sheet<int, EquipmentItemRecipeSheet.Row>.Add(int, EquipmentItemRecipeSheet.Row) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Remove(int) ,

Sheet<int, EquipmentItemRecipeSheet.Row>.TryGetValue(int, out EquipmentItemRecipeSheet.Row) , Sheet<int, EquipmentItemRecipeSheet.Row>.Add(KeyValuePair<int, EquipmentItemRecipeSheet.Row>) , Sheet<int, EquipmentItemRecipeSheet.Row>.Clear() , Sheet<int, EquipmentItemRecipeSheet.Row>.Contains(KeyValuePair<int, EquipmentItemRecipeSheet.Row>) , Sheet<int, EquipmentItemRecipeSheet.Row>.CopyTo(KeyValuePair<int, EquipmentItemRecipeSheet.Row>[], int) , Sheet<int, EquipmentItemRecipeSheet.Row>.Remove(KeyValuePair<int, EquipmentItemRecipeSheet.Row>) , Sheet<int, EquipmentItemRecipeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EquipmentItemRecipeSheet()

```
public EquipmentItemRecipeSheet()
```

# Class EquipmentItemRecipeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemRecipeSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EquipmentItemRecipeSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### CRYSTAL

```
public int CRYSTAL { get; }
```

#### Property Value

int

### Id

```
public int Id { get; }
```

#### Property Value

int

## ItemSubType

```
public ItemSubType? ItemSubType { get; }
```

### Property Value

ItemSubType?

## Key

```
public override int Key { get; }
```

### Property Value

int

## MaterialCount

```
public int MaterialCount { get; }
```

### Property Value

int

## MaterialId

```
public int MaterialId { get; }
```

### Property Value

int

## RequiredActionPoint

```
public int RequiredActionPoint { get; }
```

### Property Value

int

## RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

### Property Value

long

## RequiredGold

```
public long RequiredGold { get; }
```

### Property Value

long

## ResultEquipmentId

```
public int ResultEquipmentId { get; }
```

### Property Value

int

## SubRecipeIds

```
public List<int> SubRecipeIds { get; }
```

### Property Value

List<int>

## UnlockStage

```
public int UnlockStage { get; }
```

### Property Value

int

## Methods

## GetAllMaterials(EquipmentItemSubRecipeSheetV2, CraftType)

```
public IEnumerable<EquipmentItemSubRecipeSheet.MaterialInfo>
GetAllMaterials(EquipmentItemSubRecipeSheetV2 sheet, CraftType subRecipeType
= CraftType.Normal)
```

### Parameters

`sheet` EquipmentItemSubRecipeSheetV2

`subRecipeType` CraftType

### Returns

IEnumerable<EquipmentItemSubRecipeSheet.MaterialInfo>

## IsMimisBrunnrSubRecipe(int?)

```
[Obsolete("It is obsoleted. Check
EquipmentItemSubRecipeSheetV2.IsMimisbrunnrSubRecipe.")]
public bool IsMimisBrunnrSubRecipe(int? subRecipeId)
```

### Parameters

`subRecipeId` int?

### Returns

bool

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EquipmentItemSetEffectSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentItemSetEffectSheet : Sheet<int,
EquipmentItemSetEffectSheet.Row>, IDictionary<int, EquipmentItemSetEffectSheet.Row>,
ICollection<KeyValuePair<int, EquipmentItemSetEffectSheet.Row>>,
IEnumerable<KeyValuePair<int, EquipmentItemSetEffectSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EquipmentItemSetEffectSheet.Row> ← EquipmentItemSetEffectSheet

### Implements

IDictionary<int, EquipmentItemSetEffectSheet.Row>, ICollection<KeyValuePair<int, EquipmentItemSetEffectSheet.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemSetEffectSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EquipmentItemSetEffectSheet.Row>.Name ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.First ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Last ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Keys ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Values ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Count ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.this[int] ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.TryGetValue(int, out EquipmentItemSetEffectSheet.Row, bool) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.AddRow(int, EquipmentItemSetEffectSheet.Row) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Add(int, EquipmentItemSetEffectSheet.Row) ,

Sheet<int, EquipmentItemSetEffectSheet.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Remove(int) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.TryGetValue(int, out EquipmentItemSetEffectSheet.Row) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Add(KeyValuePair<int, EquipmentItemSetEffectSheet.Row>) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Clear() ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Contains(KeyValuePair<int, EquipmentItemSetEffectSheet.Row>) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.CopyTo(KeyValuePair<int, EquipmentItemSetEffectSheet.Row>[], int) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Remove(KeyValuePair<int, EquipmentItemSetEffectSheet.Row>) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
SetEffectExtension.GetSetEffectRows(EquipmentItemSetEffectSheet, IEnumerable<Equipment>)

# Constructors

## EquipmentItemSetEffectSheet()

```
public EquipmentItemSetEffectSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, EquipmentItemSetEffectSheet.Row value)
```

### Parameters

`key` int

`value` EquipmentItemSetEffectSheet.Row

# Class EquipmentItemSetEffectSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentItemSetEffectSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EquipmentItemSetEffectSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## StatModifiers

```
public Dictionary<int, StatModifier> StatModifiers { get; }
```

### Property Value

Dictionary<int, StatModifier>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EquipmentItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentItemSheet : Sheet<int, EquipmentItemSheet.Row>,
IDictionary<int, EquipmentItemSheet.Row>, ICollection<KeyValuePair<int,
EquipmentItemSheet.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EquipmentItemSheet.Row> ← EquipmentItemSheet

### Implements

IDictionary<int, EquipmentItemSheet.Row>,
ICollection<KeyValuePair<int, EquipmentItemSheet.Row>>,
IEnumerable<KeyValuePair<int, EquipmentItemSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EquipmentItemSheet.Row>.Name ,
Sheet<int, EquipmentItemSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemSheet.Row>.First , Sheet<int, EquipmentItemSheet.Row>.Last ,
Sheet<int, EquipmentItemSheet.Row>.Keys ,
Sheet<int, EquipmentItemSheet.Row>.Values ,
Sheet<int, EquipmentItemSheet.Row>.Count ,
Sheet<int, EquipmentItemSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemSheet.Row>.this[int] ,
Sheet<int, EquipmentItemSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemSheet.Row>.TryGetValue(int, out EquipmentItemSheet.Row, bool) ,
Sheet<int, EquipmentItemSheet.Row>.AddRow(int, EquipmentItemSheet.Row) ,
Sheet<int, EquipmentItemSheet.Row>.Add(int, EquipmentItemSheet.Row) ,
Sheet<int, EquipmentItemSheet.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemSheet.Row>.Remove(int) ,
Sheet<int, EquipmentItemSheet.Row>.TryGetValue(int, out EquipmentItemSheet.Row) ,
Sheet<int, EquipmentItemSheet.Row>.Add(KeyValuePair<int, EquipmentItemSheet.Row>) ,
Sheet<int, EquipmentItemSheet.Row>.Clear() ,

Sheet<int, EquipmentItemSheet.Row>.Contains(KeyValuePair<int, EquipmentItemSheet.Row>) , Sheet<int, EquipmentItemSheet.Row>.CopyTo(KeyValuePair<int, EquipmentItemSheet.Row>[], int) , Sheet<int, EquipmentItemSheet.Row>.Remove(KeyValuePair<int, EquipmentItemSheet.Row>) , Sheet<int, EquipmentItemSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EquipmentItemSheet()

```
public EquipmentItemSheet()
```

# Class EquipmentItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentItemSheet.Row : ItemSheet.Row
```

**Inheritance**

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row ← EquipmentItemSheet.Row

**Inherited Members**

ItemSheet.Row.Key , ItemSheet.Row.Id , ItemSheet.Row.ItemSubType , ItemSheet.Row.Grade , ItemSheet.Row.ElementalType , ItemSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Row()

```
public Row()
```

### Row(Dictionary)

```
public Row(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

# Properties

## AttackRange

```
public decimal AttackRange { get; }
```

### Property Value

decimal

## Exp

```
public long? Exp { get; }
```

### Property Value

long?

## ItemType

```
public override ItemType ItemType { get; }
```

### Property Value

ItemType

## SetId

```
public int SetId { get; }
```

### Property Value

int

## SpineResourcePath

```
public string SpineResourcePath { get; }
```

### Property Value

string

## Stat

```
public DecimalStat Stat { get; }
```

### Property Value

DecimalStat

# Methods

## Serialize()

```
public override IValue Serialize()
```

### Returns

IValue

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EquipmentItemSubRecipeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemSubRecipeSheet : Sheet<int,
EquipmentItemSubRecipeSheet.Row>, IDictionary<int, EquipmentItemSubRecipeSheet.Row>,
ICollection<KeyValuePair<int, EquipmentItemSubRecipeSheet.Row>>,
IEnumerable<KeyValuePair<int, EquipmentItemSubRecipeSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EquipmentItemSubRecipeSheet.Row> ← EquipmentItemSubRecipeSheet

### Implements

IDictionary<int, EquipmentItemSubRecipeSheet.Row>, ICollection<KeyValuePair<int, EquipmentItemSubRecipeSheet.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemSubRecipeSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EquipmentItemSubRecipeSheet.Row>.Name ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.First ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Last ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Keys ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Values ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Count ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.this[int] ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.TryGetValue(int, out EquipmentItemSubRecipeSheet.Row, bool) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.AddRow(int, EquipmentItemSubRecipeSheet.Row) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Add(int, EquipmentItemSubRecipeSheet.Row) ,

Sheet<int, EquipmentItemSubRecipeSheet.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Remove(int) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.TryGetValue(int, out
EquipmentItemSubRecipeSheet.Row) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Add(KeyValuePair<int,
EquipmentItemSubRecipeSheet.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Clear() ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Contains(KeyValuePair<int,
EquipmentItemSubRecipeSheet.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.CopyTo(KeyValuePair<int,
EquipmentItemSubRecipeSheet.Row>[], int) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Remove(KeyValuePair<int,
EquipmentItemSubRecipeSheet.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EquipmentItemSubRecipeSheet()

```
public EquipmentItemSubRecipeSheet()
```

# Struct EquipmentItemSubRecipeSheet.MaterialInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public readonly struct EquipmentItemSubRecipeSheet.MaterialInfo
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### MaterialInfo(int, int)

```
public MaterialInfo(int id, int count)
```

#### Parameters

`id` int

`count` int

## Fields

### Count

```
public readonly int Count
```

#### Field Value

int

# Id

```
public readonly int Id
```

## Field Value

int

# Struct EquipmentItemSubRecipeSheet.OptionInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public readonly struct EquipmentItemSubRecipeSheet.OptionInfo
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### OptionInfo(int, decimal)

```
public OptionInfo(int id, decimal ratio)
```

#### Parameters

`id` int

`ratio` decimal

## Fields

### Id

```
public readonly int Id
```

#### Field Value

int

# Ratio

```
public readonly decimal Ratio
```

## Field Value

decimal

# Class EquipmentItemSubRecipeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemSubRecipeSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EquipmentItemSubRecipeSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Id

```
public int Id { get; }
```

#### Property Value

int

### Key

```
public override int Key { get; }
```

#### Property Value

int

## Materials

```
public List<EquipmentItemSubRecipeSheet.MaterialInfo> Materials { get; }
```

### Property Value

List<EquipmentItemSubRecipeSheet.MaterialInfo>

## MaxOptionLimit

```
public int MaxOptionLimit { get; }
```

### Property Value

int

## Options

```
public List<EquipmentItemSubRecipeSheet.OptionInfo> Options { get; }
```

### Property Value

List<EquipmentItemSubRecipeSheet.OptionInfo>

## RequiredActionPoint

```
public int RequiredActionPoint { get; }
```

### Property Value

int

## RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

### Property Value

long

## RequiredGold

```
public long RequiredGold { get; }
```

### Property Value

long

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EquipmentItemSubRecipeSheetV2

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemSubRecipeSheetV2 : Sheet<int,
EquipmentItemSubRecipeSheetV2.Row>, IDictionary<int,
EquipmentItemSubRecipeSheetV2.Row>, ICollection<KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>>, IEnumerable<KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EquipmentItemSubRecipeSheetV2.Row> ← EquipmentItemSubRecipeSheetV2

### Implements

IDictionary<int, EquipmentItemSubRecipeSheetV2.Row>, ICollection<KeyValuePair<int, EquipmentItemSubRecipeSheetV2.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemSubRecipeSheetV2.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Name ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.OrderedList ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.First ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Last ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Keys ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Values ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Count ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.IsReadOnly ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.this[int] ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.TryGetValue(int, out EquipmentItemSubRecipeSheetV2.Row, bool) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.AddRow(int, EquipmentItemSubRecipeSheetV2.Row) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Add(int, EquipmentItemSubRecipeSheetV2.Row) ,

Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Remove(int) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.TryGetValue(int, out
EquipmentItemSubRecipeSheetV2.Row) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Add(KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Clear() ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Contains(KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.CopyTo(KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>[], int) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Remove(KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## EquipmentItemSubRecipeSheetV2()

```
public EquipmentItemSubRecipeSheetV2()
```

# Struct EquipmentItemSubRecipeSheetV2.OptionInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public readonly struct EquipmentItemSubRecipeSheetV2.OptionInfo
```

**Inherited Members**

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### OptionInfo(int, int, int)

```
public OptionInfo(int id, int ratio, int requiredBlockIndex = 0)
```

#### Parameters

`id` int

`ratio` int

`requiredBlockIndex` int

## Fields

### Id

```
public readonly int Id
```

### Field Value

int

## Ratio

```
public readonly int Ratio
```

### Field Value

int

## RequiredBlockIndex

```
public readonly int RequiredBlockIndex
```

### Field Value

int

# Class EquipmentItemSubRecipeSheetV2.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemSubRecipeSheetV2.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EquipmentItemSubRecipeSheetV2.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Id

```
public int Id { get; }
```

#### Property Value

int

### IsMimisbrunnrSubRecipe

```
public bool? IsMimisbrunnrSubRecipe { get; }
```

#### Property Value

bool?

## Key

```
public override int Key { get; }
```

### Property Value

int

## Materials

```
public List<EquipmentItemSubRecipeSheet.MaterialInfo> Materials { get; }
```

### Property Value

List<EquipmentItemSubRecipeSheet.MaterialInfo>

## Options

```
public List<EquipmentItemSubRecipeSheetV2.OptionInfo> Options { get; }
```

### Property Value

List<EquipmentItemSubRecipeSheetV2.OptionInfo>

## RequiredActionPoint

```
public int RequiredActionPoint { get; }
```

### Property Value

int

## RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

### Property Value

long

## RequiredGold

```
public long RequiredGold { get; }
```

### Property Value

long

## RewardHammerPoint

```
public int? RewardHammerPoint { get; }
```

### Property Value

int?

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class GameConfigSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GameConfigSheet : Sheet<string, GameConfigSheet.Row>,
IDictionary<string, GameConfigSheet.Row>, ICollection<KeyValuePair<string,
GameConfigSheet.Row>>, IEnumerable<KeyValuePair<string, GameConfigSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<string, GameConfigSheet.Row> ← GameConfigSheet

### Implements

IDictionary<string, GameConfigSheet.Row>,
ICollection<KeyValuePair<string, GameConfigSheet.Row>>,
IEnumerable<KeyValuePair<string, GameConfigSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<string, GameConfigSheet.Row>.Name ,
Sheet<string, GameConfigSheet.Row>.OrderedList ,
Sheet<string, GameConfigSheet.Row>.First , Sheet<string, GameConfigSheet.Row>.Last ,
Sheet<string, GameConfigSheet.Row>.Keys ,
Sheet<string, GameConfigSheet.Row>.Values ,
Sheet<string, GameConfigSheet.Row>.Count ,
Sheet<string, GameConfigSheet.Row>.IsReadOnly ,
Sheet<string, GameConfigSheet.Row>.this[string] ,
Sheet<string, GameConfigSheet.Row>.Set(string, bool) ,
Sheet<string, GameConfigSheet.Row>.Set<T>(Sheet<string, T>, bool) ,
Sheet<string, GameConfigSheet.Row>.GetEnumerator() ,
Sheet<string, GameConfigSheet.Row>.TryGetValue(string, out GameConfigSheet.Row, bool) ,
Sheet<string, GameConfigSheet.Row>.AddRow(string, GameConfigSheet.Row) ,
Sheet<string, GameConfigSheet.Row>.Add(string, GameConfigSheet.Row) ,
Sheet<string, GameConfigSheet.Row>.ContainsKey(string) ,
Sheet<string, GameConfigSheet.Row>.Remove(string) ,
Sheet<string, GameConfigSheet.Row>.TryGetValue(string, out GameConfigSheet.Row) ,
Sheet<string, GameConfigSheet.Row>.Add(KeyValuePair<string, GameConfigSheet.Row>)
,

Sheet<string, GameConfigSheet.Row>.Clear() ,
Sheet<string, GameConfigSheet.Row>.Contains(KeyValuePair<string, GameConfigSheet.Row>) ,
Sheet<string, GameConfigSheet.Row>.CopyTo(KeyValuePair<string, GameConfigSheet.Row>[], int) ,
Sheet<string, GameConfigSheet.Row>.Remove(KeyValuePair<string, GameConfigSheet.Row>) ,
Sheet<string, GameConfigSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GameConfigSheet()

```
public GameConfigSheet()
```

# Class GameConfigSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GameConfigSheet.Row : SheetRow<string>
```

**Inheritance**

object ← SheetRow<string> ← GameConfigSheet.Row

**Inherited Members**

SheetRow<string>.Validate() , SheetRow<string>.EndOfSheetInitialize() , SheetRow<string>.Equals(object) , SheetRow<string>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

```
public override string Key { get; }
```

### Property Value

string

## Value

```
public string Value { get; }
```

### Property Value

string

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class GeneralQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GeneralQuestSheet : Sheet<int, GeneralQuestSheet.Row>, IDictionary<int,
GeneralQuestSheet.Row>, ICollection<KeyValuePair<int, GeneralQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, GeneralQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, GeneralQuestSheet.Row> ← GeneralQuestSheet

### Implements

IDictionary<int, GeneralQuestSheet.Row>,
ICollection<KeyValuePair<int, GeneralQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, GeneralQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, GeneralQuestSheet.Row>.Name ,
Sheet<int, GeneralQuestSheet.Row>.OrderedList ,
Sheet<int, GeneralQuestSheet.Row>.First , Sheet<int, GeneralQuestSheet.Row>.Last ,
Sheet<int, GeneralQuestSheet.Row>.Keys , Sheet<int, GeneralQuestSheet.Row>.Values ,
Sheet<int, GeneralQuestSheet.Row>.Count ,
Sheet<int, GeneralQuestSheet.Row>.IsReadOnly ,
Sheet<int, GeneralQuestSheet.Row>.this[int] ,
Sheet<int, GeneralQuestSheet.Row>.Set(string, bool) ,
Sheet<int, GeneralQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, GeneralQuestSheet.Row>.GetEnumerator() ,
Sheet<int, GeneralQuestSheet.Row>.TryGetValue(int, out GeneralQuestSheet.Row, bool) ,
Sheet<int, GeneralQuestSheet.Row>.AddRow(int, GeneralQuestSheet.Row) ,
Sheet<int, GeneralQuestSheet.Row>.Add(int, GeneralQuestSheet.Row) ,
Sheet<int, GeneralQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, GeneralQuestSheet.Row>.Remove(int) ,
Sheet<int, GeneralQuestSheet.Row>.TryGetValue(int, out GeneralQuestSheet.Row) ,
Sheet<int, GeneralQuestSheet.Row>.Add(KeyValuePair<int, GeneralQuestSheet.Row>) ,
Sheet<int, GeneralQuestSheet.Row>.Clear() ,
Sheet<int, GeneralQuestSheet.Row>.Contains(KeyValuePair<int,
GeneralQuestSheet.Row>) ,

Sheet<int, GeneralQuestSheet.Row>.CopyTo(KeyValuePair<int, GeneralQuestSheet.Row>[], int) ,
Sheet<int, GeneralQuestSheet.Row>.Remove(KeyValuePair<int, GeneralQuestSheet.Row>) ,
Sheet<int, GeneralQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GeneralQuestSheet()

```
public GeneralQuestSheet()
```

# Class GeneralQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GeneralQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← GeneralQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Event

```
public QuestEventType Event { get; }
```

### Property Value

QuestEventType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class GoldQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GoldQuestSheet : Sheet<int, GoldQuestSheet.Row>, IDictionary<int,
GoldQuestSheet.Row>, ICollection<KeyValuePair<int, GoldQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, GoldQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, GoldQuestSheet.Row> ← GoldQuestSheet

### Implements

IDictionary<int, GoldQuestSheet.Row>,
ICollection<KeyValuePair<int, GoldQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, GoldQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, GoldQuestSheet.Row>.Name , Sheet<int, GoldQuestSheet.Row>.OrderedList ,
Sheet<int, GoldQuestSheet.Row>.First , Sheet<int, GoldQuestSheet.Row>.Last ,
Sheet<int, GoldQuestSheet.Row>.Keys , Sheet<int, GoldQuestSheet.Row>.Values ,
Sheet<int, GoldQuestSheet.Row>.Count , Sheet<int, GoldQuestSheet.Row>.IsReadOnly ,
Sheet<int, GoldQuestSheet.Row>.this[int] ,
Sheet<int, GoldQuestSheet.Row>.Set(string, bool) ,
Sheet<int, GoldQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, GoldQuestSheet.Row>.GetEnumerator() ,
Sheet<int, GoldQuestSheet.Row>.TryGetValue(int, out GoldQuestSheet.Row, bool) ,
Sheet<int, GoldQuestSheet.Row>.AddRow(int, GoldQuestSheet.Row) ,
Sheet<int, GoldQuestSheet.Row>.Add(int, GoldQuestSheet.Row) ,
Sheet<int, GoldQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, GoldQuestSheet.Row>.Remove(int) ,
Sheet<int, GoldQuestSheet.Row>.TryGetValue(int, out GoldQuestSheet.Row) ,
Sheet<int, GoldQuestSheet.Row>.Add(KeyValuePair<int, GoldQuestSheet.Row>) ,
Sheet<int, GoldQuestSheet.Row>.Clear() ,
Sheet<int, GoldQuestSheet.Row>.Contains(KeyValuePair<int, GoldQuestSheet.Row>) ,
Sheet<int, GoldQuestSheet.Row>.CopyTo(KeyValuePair<int, GoldQuestSheet.Row>[], int) ,
Sheet<int, GoldQuestSheet.Row>.Remove(KeyValuePair<int, GoldQuestSheet.Row>) ,
Sheet<int, GoldQuestSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GoldQuestSheet()

```
public GoldQuestSheet()
```

# Class GoldQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GoldQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← GoldQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Type

```
public TradeType Type { get; }
```

### Property Value

TradeType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Interface ISheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface ISheet
```

## Methods

### Set(string, bool)

```
void Set(string csv, bool isReversed = false)
```

#### Parameters

`csv` string

`isReversed` bool

# Interface IStakeRewardRow

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface IStakeRewardRow
```

## Properties

### Level

```
int Level { get; }
```

#### Property Value

int

### RequiredGold

```
long RequiredGold { get; }
```

#### Property Value

long

# Interface IStakeRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface IStakeRewardSheet : ISheet
```

**Inherited Members**
ISheet.Set(string, bool)

**Extension Methods**
SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

## Properties

### OrderedRows

```
IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

#### Property Value

IReadOnlyList<IStakeRewardRow>

# Interface IWorldBossRewardRow

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface IWorldBossRewardRow
```

## Properties

### BossId

```
int BossId { get; }
```

#### Property Value

int

### Circle

```
int Circle { get; }
```

#### Property Value

int

### Crystal

```
int Crystal { get; }
```

#### Property Value

int

# Rank

```
int Rank { get; }
```

## Property Value

int

# Rune

```
int Rune { get; }
```

## Property Value

int

# Interface IWorldBossRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface IWorldBossRewardSheet : ISheet
```

**Inherited Members**
ISheet.Set(string, bool)

## Properties

### OrderedRows

```
IReadOnlyList<IWorldBossRewardRow> OrderedRows { get; }
```

#### Property Value

IReadOnlyList<IWorldBossRewardRow>

# Class InfiniteTowerConditionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Infinite tower condition sheet for managing battle conditions and restrictions. Uses StatType, SkillTargetType, and StatModifier.OperationType for flexible stat modifications.

```
[Serializable]
public class InfiniteTowerConditionSheet : Sheet<int,
InfiniteTowerConditionSheet.Row>, IDictionary<int, InfiniteTowerConditionSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerConditionSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerConditionSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, InfiniteTowerConditionSheet.Row> ← InfiniteTowerConditionSheet

### Implements

IDictionary<int, InfiniteTowerConditionSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerConditionSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerConditionSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, InfiniteTowerConditionSheet.Row>.Name ,
Sheet<int, InfiniteTowerConditionSheet.Row>.OrderedList ,
Sheet<int, InfiniteTowerConditionSheet.Row>.First ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Last ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Keys ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Values ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Count ,
Sheet<int, InfiniteTowerConditionSheet.Row>.IsReadOnly ,
Sheet<int, InfiniteTowerConditionSheet.Row>.this[int] ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Set(string, bool) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.GetEnumerator() ,
Sheet<int, InfiniteTowerConditionSheet.Row>.TryGetValue(int, out
InfiniteTowerConditionSheet.Row, bool) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.AddRow(int,
InfiniteTowerConditionSheet.Row) ,

Sheet<int, InfiniteTowerConditionSheet.Row>.Add(int, InfiniteTowerConditionSheet.Row) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.ContainsKey(int) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Remove(int) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.TryGetValue(int, out
InfiniteTowerConditionSheet.Row) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Add(KeyValuePair<int,
InfiniteTowerConditionSheet.Row>) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Clear() ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Contains(KeyValuePair<int,
InfiniteTowerConditionSheet.Row>) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.CopyTo(KeyValuePair<int,
InfiniteTowerConditionSheet.Row>[], int) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Remove(KeyValuePair<int,
InfiniteTowerConditionSheet.Row>) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## InfiniteTowerConditionSheet()

```
public InfiniteTowerConditionSheet()
```

# Class InfiniteTowerConditionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the InfiniteTowerConditionSheet containing condition data.

```
[Serializable]
public class InfiniteTowerConditionSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← InfiniteTowerConditionSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Id

Gets the unique identifier for this condition.

```
public int Id { get; }
```

### Property Value

int

## Key

Gets the condition ID as the key for this row.

```
public override int Key { get; }
```

### Property Value

int

## OperationType

Gets the operation type for the stat modification.

```
public StatModifier.OperationType OperationType { get; }
```

### Property Value

StatModifier.OperationType

## StatType

Gets the stat type that this condition affects.

```
public StatType StatType { get; }
```

### Property Value

StatType

## TargetType

Gets the list of target types for this condition (Self, Enemy, etc.).

```
public List<SkillTargetType> TargetType { get; }
```

### Property Value

List<SkillTargetType>

## Value

Gets the value for the stat modification.

```
public int Value { get; }
```

### Property Value

int

## Methods

### GetStatModifier()

Creates a StatModifier from this condition row.

```
public StatModifier GetStatModifier()
```

### Returns

StatModifier

### Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class InfiniteTowerFloorSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Infinite tower floor sheet for managing tower floors and their properties.

```
[Serializable]
public class InfiniteTowerFloorSheet : Sheet<int, InfiniteTowerFloorSheet.Row>,
IDictionary<int, InfiniteTowerFloorSheet.Row>, ICollection<KeyValuePair<int,
InfiniteTowerFloorSheet.Row>>, IEnumerable<KeyValuePair<int,
InfiniteTowerFloorSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, InfiniteTowerFloorSheet.Row> ← InfiniteTowerFloorSheet

### Implements

IDictionary<int, InfiniteTowerFloorSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerFloorSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerFloorSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, InfiniteTowerFloorSheet.Row>.Name ,
Sheet<int, InfiniteTowerFloorSheet.Row>.OrderedList ,
Sheet<int, InfiniteTowerFloorSheet.Row>.First ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Last ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Keys ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Values ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Count ,
Sheet<int, InfiniteTowerFloorSheet.Row>.IsReadOnly ,
Sheet<int, InfiniteTowerFloorSheet.Row>.this[int] ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Set(string, bool) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.GetEnumerator() ,
Sheet<int, InfiniteTowerFloorSheet.Row>.TryGetValue(int, out InfiniteTowerFloorSheet.Row,
bool) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.AddRow(int, InfiniteTowerFloorSheet.Row) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Add(int, InfiniteTowerFloorSheet.Row) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.ContainsKey(int) ,

Sheet<int, InfiniteTowerFloorSheet.Row>.Remove(int) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.TryGetValue(int, out InfiniteTowerFloorSheet.Row)
,
Sheet<int, InfiniteTowerFloorSheet.Row>.Add(KeyValuePair<int,
InfiniteTowerFloorSheet.Row>) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Clear() ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Contains(KeyValuePair<int,
InfiniteTowerFloorSheet.Row>) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.CopyTo(KeyValuePair<int,
InfiniteTowerFloorSheet.Row>[], int) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Remove(KeyValuePair<int,
InfiniteTowerFloorSheet.Row>) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## InfiniteTowerFloorSheet()

```
public InfiniteTowerFloorSheet()
```

# Class InfiniteTowerFloorSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the InfiniteTowerFloorSheet containing floor data.

```
[Serializable]
public class InfiniteTowerFloorSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← InfiniteTowerFloorSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## EnemyInitialStatModifiers

Gets the list of initial stat modifiers applied to enemies on this floor.

```
public List<StatModifier> EnemyInitialStatModifiers { get; }
```

### Property Value

List<StatModifier>

## Floor

Gets the floor number.

```
public int Floor { get; }
```

### Property Value

int

## ForbiddenItemSubTypes

Gets the list of forbidden item subtypes for this floor.

```
public List<ItemSubType> ForbiddenItemSubTypes { get; }
```

### Property Value

List<ItemSubType>

## ForbiddenRuneTypes

Gets the list of forbidden rune types for this floor.

```
public List<RuneType> ForbiddenRuneTypes { get; }
```

### Property Value

List<RuneType>

## FungibleAssetRewardAmount1

Gets the amount of the first fungible asset reward.

```
public int? FungibleAssetRewardAmount1 { get; }
```

### Property Value

int?

## FungibleAssetRewardAmount2

Gets the amount of the second fungible asset reward.

```
public int? FungibleAssetRewardAmount2 { get; }
```

### Property Value

int?

## FungibleAssetRewardAmount3

Gets the amount of the third fungible asset reward.

```
public int? FungibleAssetRewardAmount3 { get; }
```

### Property Value

int?

## FungibleAssetRewardAmount4

Gets the amount of the fourth fungible asset reward.

```
public int? FungibleAssetRewardAmount4 { get; }
```

### Property Value

int?

## FungibleAssetRewardAmount5

Gets the amount of the fifth fungible asset reward.

```
public int? FungibleAssetRewardAmount5 { get; }
```

### Property Value

int?

## FungibleAssetRewardTicker1

Gets the first fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker1 { get; }
```

### Property Value

string

## FungibleAssetRewardTicker2

Gets the second fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker2 { get; }
```

### Property Value

string

## FungibleAssetRewardTicker3

Gets the third fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker3 { get; }
```

### Property Value

string

## FungibleAssetRewardTicker4

Gets the fourth fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker4 { get; }
```

### Property Value

string

## FungibleAssetRewardTicker5

Gets the fifth fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker5 { get; }
```

### Property Value

string

## GuaranteedConditionId

Gets the guaranteed condition ID that will always be applied to this floor.

```
public int GuaranteedConditionId { get; }
```

### Property Value

int

## Id

Gets the unique identifier for this floor.

```
public int Id { get; }
```

### Property Value

int

## ItemRewardCount1

Gets the count of the first item reward.

```
public int? ItemRewardCount1 { get; }
```

### Property Value

int?

## ItemRewardCount2

Gets the count of the second item reward.

```
public int? ItemRewardCount2 { get; }
```

### Property Value

int?

## ItemRewardCount3

Gets the count of the third item reward.

```
public int? ItemRewardCount3 { get; }
```

### Property Value

int?

## ItemRewardCount4

Gets the count of the fourth item reward.

```
public int? ItemRewardCount4 { get; }
```

### Property Value

int?

## ItemRewardCount5

Gets the count of the fifth item reward.

```
public int? ItemRewardCount5 { get; }
```

### Property Value

int?

## ItemRewardId1

Gets the first item reward ID for this floor.

```
public int? ItemRewardId1 { get; }
```

### Property Value

int?

## ItemRewardId2

Gets the second item reward ID for this floor.

```
public int? ItemRewardId2 { get; }
```

### Property Value

int?

## ItemRewardId3

Gets the third item reward ID for this floor.

```
public int? ItemRewardId3 { get; }
```

### Property Value

int?

## ItemRewardId4

Gets the fourth item reward ID for this floor.

```
public int? ItemRewardId4 { get; }
```

### Property Value

int?

## ItemRewardId5

Gets the fifth item reward ID for this floor.

```
public int? ItemRewardId5 { get; }
```

### Property Value

int?

## Key

Gets the floor ID as the key for this row.

```
public override int Key { get; }
```

### Property Value

int

## MaterialCostCount

Gets the amount of material required for purchasing tickets.

```
public int? MaterialCostCount { get; }
```

### Property Value

int?

## MaterialCostId

Gets the material item ID required for purchasing tickets.

```
public int? MaterialCostId { get; }
```

### Property Value

int?

## MaxCp

Gets the maximum allowed combat power for this floor.

```
public long? MaxCp { get; }
```

### Property Value

long?

## MaxItemGrade

Gets the maximum item grade allowed for this floor.

```
public int? MaxItemGrade { get; }
```

### Property Value

int?

## MaxItemLevel

Gets the maximum item level allowed for this floor.

```
public int? MaxItemLevel { get; }
```

### Property Value

int?

## MaxRandomConditions

Gets the maximum number of random conditions to apply.

```
public int MaxRandomConditions { get; }
```

### Property Value

int

## MinItemGrade

Gets the minimum item grade allowed for this floor.

```
public int? MinItemGrade { get; }
```

### Property Value

int?

## MinItemLevel

Gets the minimum item level allowed for this floor.

```
public int? MinItemLevel { get; }
```

### Property Value

int?

## MinRandomConditions

Gets the minimum number of random conditions to apply.

```
public int MinRandomConditions { get; }
```

### Property Value

int

## NcgCost

Gets the NCG cost for purchasing tickets for this floor.

```
public int? NcgCost { get; }
```

### Property Value

int?

## RandomConditionId1

Gets the first random condition ID for weighted selection.

```
public int? RandomConditionId1 { get; }
```

### Property Value

int?

## RandomConditionId2

Gets the second random condition ID for weighted selection.

```
public int? RandomConditionId2 { get; }
```

### Property Value

int?

## RandomConditionId3

Gets the third random condition ID for weighted selection.

```
public int? RandomConditionId3 { get; }
```

### Property Value

int?

## RandomConditionId4

Gets the fourth random condition ID for weighted selection.

```
public int? RandomConditionId4 { get; }
```

### Property Value

int?

## RandomConditionId5

Gets the fifth random condition ID for weighted selection.

```
public int? RandomConditionId5 { get; }
```

### Property Value

int?

## RandomConditionWeight1

Gets the weight for the first random condition.

```
public int? RandomConditionWeight1 { get; }
```

### Property Value

int?

## RandomConditionWeight2

Gets the weight for the second random condition.

```
public int? RandomConditionWeight2 { get; }
```

### Property Value

int?

## RandomConditionWeight3

Gets the weight for the third random condition.

```
public int? RandomConditionWeight3 { get; }
```

### Property Value

int?

## RandomConditionWeight4

Gets the weight for the fourth random condition.

```
public int? RandomConditionWeight4 { get; }
```

### Property Value

int?

## RandomConditionWeight5

Gets the weight for the fifth random condition.

```
public int? RandomConditionWeight5 { get; }
```

### Property Value

int?

## RequiredCp

Gets the minimum required combat power for this floor.

```
public long? RequiredCp { get; }
```

### Property Value

long?

## RequiredElementalTypes

Gets the list of required elemental types for equipment on this floor.

```
public List<ElementalType> RequiredElementalTypes { get; }
```

### Property Value

List<ElementalType>

## Methods

### GetBattleCondition(BattleConditionType)

Gets battle conditions of a specific type from this floor's restrictions. This method can be used by clients for simulation and replay purposes.

```
public InfiniteTowerBattleCondition? GetBattleCondition(BattleConditionType
conditionType)
```

### Parameters

`conditionType` BattleConditionType

The type of battle condition to retrieve

### Returns

InfiniteTowerBattleCondition

Battle condition of the specified type, or null if not found

### GetBattleConditions()

Creates a list of battle conditions from this floor's restrictions. This method can be used by clients for simulation and replay purposes.

```
public List<InfiniteTowerBattleCondition> GetBattleConditions()
```

### Returns

List<InfiniteTowerBattleCondition>

List of battle conditions representing this floor's restrictions

## GetFungibleAssetRewards()

Gets all fungible asset rewards as a list of tuples (ticker, amount).

```
public List<(string ticker, int amount)> GetFungibleAssetRewards()
```

### Returns

List<(string ticker, int amount)>

## GetItemRewards()

Gets all item rewards as a list of tuples (itemId, count).

```
public List<(int itemId, int count)> GetItemRewards()
```

### Returns

List<(int id, int count)>

## GetRandomConditions(InfiniteTowerConditionSheet, IRandom, int?)

Selects random conditions from all available conditions using uniform random selection. Uses Fisher-Yates shuffle algorithm to avoid index issues when removing items. This method can be used by clients for simulation and replay purposes.

```
public List<InfiniteTowerCondition> GetRandomConditions(InfiniteTowerConditionSheet
conditionSheet, IRandom random, int? guaranteedConditionId = null)
```

### Parameters

`conditionSheet` InfiniteTowerConditionSheet

The condition sheet containing all available conditions

`random` IRandom

Random number generator

`guaranteedConditionId` int?

Optional guaranteed condition ID to exclude from random selection

### Returns

List<InfiniteTowerCondition>

List of selected random conditions

### Exceptions

InvalidOperationException

Thrown when available conditions are insufficient for minimum count requirement

## GetRandomConditionsWithWeights()

Gets random conditions with weights for weighted selection.

```
public List<(int conditionId, int weight)> GetRandomConditionsWithWeights()
```

### Returns

List<(int id, int count)>

## GetRandomConditionsWithWeights(InfiniteTowerConditionSheet, IRandom, int?)

Selects random conditions using weighted selection for this floor. This method can be used by clients for simulation and replay purposes.

```
public List<InfiniteTowerCondition>
GetRandomConditionsWithWeights(InfiniteTowerConditionSheet conditionSheet, IRandom
```

```
  random, int? guaranteedConditionId = null)
```

## Parameters

`conditionSheet` InfiniteTowerConditionSheet

The condition sheet containing all available conditions

`random` IRandom

Random number generator

`guaranteedConditionId` int?

Optional guaranteed condition ID to exclude from random selection

## Returns

List<InfiniteTowerCondition>

List of selected random conditions

# ProcessRewards(IActionContext, IWorld, AvatarState, Address, EquipmentItemSheet, MaterialItemSheet, ConsumableItemSheet, CostumeItemSheet, IRandom)

Processes rewards for successful floor completion.

```
  public IWorld ProcessRewards(IActionContext context, IWorld states, AvatarState
  avatarState, Address avatarAddress, EquipmentItemSheet equipmentSheet,
  MaterialItemSheet materialSheet, ConsumableItemSheet consumableSheet,
  CostumeItemSheet costumeSheet, IRandom random)
```

## Parameters

`context` IActionContext

The action context

`states` IWorld

The world state

`avatarState` AvatarState

The avatar state to update

`avatarAddress` Address

The avatar address

`equipmentSheet` EquipmentItemSheet

The equipment item sheet

`materialSheet` MaterialItemSheet

The material item sheet

`consumableSheet` ConsumableItemSheet

The consumable item sheet

`costumeSheet` CostumeItemSheet

The costume item sheet

`random` IRandom

Random number generator for creating items

## Returns

IWorld

Updated world state

## Exceptions

InvalidItemIdException

Thrown when item ID is not found in any item sheet

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

## ValidateCpRequirements(long)

Validates CP requirements (minimum and maximum) for the floor.

```
public void ValidateCpRequirements(long currentCp)
```

### Parameters

`currentCp` long

## ValidateCurrencyForTicketPurchase(IActionContext, IWorld, Address, bool, string)

Validates that the player has enough currency to purchase a ticket.

```
public void ValidateCurrencyForTicketPurchase(IActionContext context, IWorld states,
Address avatarAddress, bool useNcgForTicket, string addressesHex)
```

### Parameters

`context` IActionContext

The action context

`states` IWorld

The world state

`avatarAddress` Address

The avatar address

`useNcgForTicket` bool

Whether to use NCG for ticket purchase

`addressesHex` string

Addresses hex for logging

### Exceptions

InvalidOperationException

Thrown when cost configuration is missing

InsufficientBalanceException

Thrown when NCG balance is insufficient

NotEnoughMaterialException

Thrown when material count is insufficient

## ValidateEquipmentElementalType(List<Equipment>?)

Validates equipment elemental types against the floor's requirements. This method can be used by clients for simulation and replay purposes.

```
public void ValidateEquipmentElementalType(List<Equipment>? equipments)
```

### Parameters

`equipments` List<Equipment>

List of equipment to validate

### Exceptions

InvalidElementalException

Thrown when equipment has invalid elemental type

# ValidateEquipmentElementalType<T>(List<T>)

Validates equipment elemental type restrictions for the floor.

```
public void ValidateEquipmentElementalType<T>(List<T> equipmentList) where T
: ItemBase
```

## Parameters

`equipmentList` List<T>

## Type Parameters

`T`

# ValidateFloorRestrictions(List<Equipment>, List<Costume>)

Validates floor-specific restrictions for equipment and costumes. This method can be used by clients for simulation and replay purposes.

```
public void ValidateFloorRestrictions(List<Equipment> equipmentList,
List<Costume> costumeList)
```

## Parameters

`equipmentList` List<Equipment>

List of equipment to validate

`costumeList` List<Costume>

List of costumes to validate

## Exceptions

Exception

Thrown when equipment or costume restrictions are violated

# ValidateItemGradeRestrictions<T>(List<T>)

Validates item grade restrictions for the floor (applies to both equipment and costumes).

```
public void ValidateItemGradeRestrictions<T>(List<T> itemList) where T : ItemBase
```

## Parameters

`itemList` List<T>

## Type Parameters

`T`

# ValidateItemLevelRestrictions<T>(List<T>)

Validates item level restrictions for the floor (applies to both equipment and costumes).

```
public void ValidateItemLevelRestrictions<T>(List<T> itemList) where T : ItemBase
```

## Parameters

`itemList` List<T>

## Type Parameters

`T`

# ValidateItemTypeRestrictions<T>(List<T>)

Validates item subtype restrictions for the floor (applies to both equipment and costumes).

```
public void ValidateItemTypeRestrictions<T>(List<T> itemList) where T : ItemBase
```

## Parameters

`itemList` List<T>

### Type Parameters

`T`

## ValidateRuneTypes(List<RuneSlotInfo>?, RuneListSheet)

Validates equipped rune types against the floor's forbidden list. This method can be used by clients for simulation and replay purposes.

```
public void ValidateRuneTypes(List<RuneSlotInfo>? runeInfos,
RuneListSheet runeListSheet)
```

### Parameters

`runeInfos` List<RuneSlotInfo>

List of rune slot information

`runeListSheet` RuneListSheet

Rune list sheet for type lookup

### Exceptions

ForbiddenRuneTypeEquippedException

Thrown when forbidden rune types are equipped

# Class InfiniteTowerFloorWaveSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Infinite tower floor wave sheet for managing wave configurations. Supports multiple monster types per wave, similar to StageWaveSheet.

```
[Serializable]
public class InfiniteTowerFloorWaveSheet : Sheet<int,
InfiniteTowerFloorWaveSheet.Row>, IDictionary<int, InfiniteTowerFloorWaveSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, InfiniteTowerFloorWaveSheet.Row> ← InfiniteTowerFloorWaveSheet

### Implements

IDictionary<int, InfiniteTowerFloorWaveSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Name ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.OrderedList ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.First ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Last ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Keys ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Values ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Count ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.IsReadOnly ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.this[int] ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Set(string, bool) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.GetEnumerator() ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.TryGetValue(int, out
InfiniteTowerFloorWaveSheet.Row, bool) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.AddRow(int,
InfiniteTowerFloorWaveSheet.Row) ,

Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Add(int, InfiniteTowerFloorWaveSheet.Row) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.ContainsKey(int) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Remove(int) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.TryGetValue(int, out InfiniteTowerFloorWaveSheet.Row) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Add(KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Clear() ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Contains(KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.CopyTo(KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>[], int) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Remove(KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## InfiniteTowerFloorWaveSheet()

Initializes a new instance of the InfiniteTowerFloorWaveSheet class.

```
public InfiniteTowerFloorWaveSheet()
```

# Methods

## AddRow(int, Row)

Adds a row to the sheet, merging waves if the floor already exists.

```
protected override void AddRow(int key, InfiniteTowerFloorWaveSheet.Row value)
```

### Parameters

`key` int

The floor ID key.

`value` InfiniteTowerFloorWaveSheet.Row

The row data to add.

# Class InfiniteTowerFloorWaveSheet.MonsterData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a single monster type within a wave.

```
[Serializable]
public class InfiniteTowerFloorWaveSheet.MonsterData
```

**Inheritance**

object ← InfiniteTowerFloorWaveSheet.MonsterData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MonsterData(int, int, int)

Initializes a new instance of the MonsterData class.

```
public MonsterData(int characterId, int level, int count)
```

#### Parameters

`characterId` int

The character ID.

`level` int

The monster level.

`count` int

The number of monsters.

# Properties

## CharacterId

Gets the character ID of the monster.

```
public int CharacterId { get; }
```

### Property Value

int

## Count

Gets the number of monsters of this type in the wave.

```
public int Count { get; }
```

### Property Value

int

## Level

Gets the level of the monster.

```
public int Level { get; }
```

### Property Value

int

# Class InfiniteTowerFloorWaveSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the InfiniteTowerFloorWaveSheet containing wave data for a specific floor.

```
[Serializable]
public class InfiniteTowerFloorWaveSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← InfiniteTowerFloorWaveSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## FloorId

Gets the floor ID this row represents.

```
public int FloorId { get; }
```

### Property Value

int

## HasBoss

Gets whether any wave in this floor contains a boss monster.

```
public bool HasBoss { get; }
```

### Property Value

bool

## Key

Gets the floor ID as the key for this row.

```
public override int Key { get; }
```

### Property Value

int

## TotalMonsterIds

Gets the list of all unique monster character IDs across all waves in this floor.

```
public List<int> TotalMonsterIds { get; }
```

### Property Value

List<int>

## Waves

Gets the list of waves for this floor.

```
public List<InfiniteTowerFloorWaveSheet.WaveData> Waves { get; }
```

### Property Value

List<InfiniteTowerFloorWaveSheet.WaveData>

# Methods

# EndOfSheetInitialize()

Performs post-initialization tasks after all rows have been loaded. Sorts waves by number and populates derived properties.

```
public override void EndOfSheetInitialize()
```

# Set(IReadOnlyList<string>)

Sets the row data from CSV fields. Parses multiple monster types per wave from the CSV data.

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

CSV field data.

# Class InfiniteTowerFloorWaveSheet.WaveData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a single wave within a floor, containing multiple monster types.

```
[Serializable]
public class InfiniteTowerFloorWaveSheet.WaveData
```

**Inheritance**

object ← InfiniteTowerFloorWaveSheet.WaveData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### WaveData(int, List<MonsterData>, bool)

Initializes a new instance of the WaveData class.

```
public WaveData(int number, List<InfiniteTowerFloorWaveSheet.MonsterData> monsters,
bool hasBoss)
```

#### Parameters

`number` int

The wave number.

`monsters` List<InfiniteTowerFloorWaveSheet.MonsterData>

List of monsters in this wave.

`hasBoss` bool

Whether this wave has a boss.

# Properties

## HasBoss

Gets whether this wave contains a boss monster.

```
public bool HasBoss { get; }
```

### Property Value

bool

## Monsters

Gets the list of monster types in this wave.

```
public List<InfiniteTowerFloorWaveSheet.MonsterData> Monsters { get; }
```

### Property Value

List<InfiniteTowerFloorWaveSheet.MonsterData>

## Number

Gets the wave number within the floor.

```
public int Number { get; }
```

### Property Value

int

# Class InfiniteTowerScheduleSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Infinite tower schedule sheet for managing season schedules and ticket policies.

```
[Serializable]
public class InfiniteTowerScheduleSheet : Sheet<int,
InfiniteTowerScheduleSheet.Row>, IDictionary<int, InfiniteTowerScheduleSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerScheduleSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, InfiniteTowerScheduleSheet.Row> ← InfiniteTowerScheduleSheet

### Implements

IDictionary<int, InfiniteTowerScheduleSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerScheduleSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, InfiniteTowerScheduleSheet.Row>.Name ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.OrderedList ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.First ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Last ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Keys ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Values ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Count ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.IsReadOnly ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.this[int] ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Set(string, bool) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.GetEnumerator() ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.TryGetValue(int, out
InfiniteTowerScheduleSheet.Row, bool) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.AddRow(int, InfiniteTowerScheduleSheet.Row)
,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Add(int, InfiniteTowerScheduleSheet.Row) ,

Sheet<int, InfiniteTowerScheduleSheet.Row>.ContainsKey(int) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Remove(int) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.TryGetValue(int, out
InfiniteTowerScheduleSheet.Row) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Add(KeyValuePair<int,
InfiniteTowerScheduleSheet.Row>) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Clear() ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Contains(KeyValuePair<int,
InfiniteTowerScheduleSheet.Row>) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.CopyTo(KeyValuePair<int,
InfiniteTowerScheduleSheet.Row>[], int) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Remove(KeyValuePair<int,
InfiniteTowerScheduleSheet.Row>) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## InfiniteTowerScheduleSheet()

```
public InfiniteTowerScheduleSheet()
```

# Class InfiniteTowerScheduleSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the InfiniteTowerScheduleSheet containing schedule data.

```
[Serializable]
public class InfiniteTowerScheduleSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← InfiniteTowerScheduleSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## DailyFreeTickets

Gets the number of free tickets given daily.

```
public int DailyFreeTickets { get; }
```

### Property Value

int

## EndBlockIndex

Gets the block index when the schedule ends.

```
public long EndBlockIndex { get; }
```

### Property Value

long

## FloorBegin

Gets the starting floor number for this schedule.

```
public int FloorBegin { get; }
```

### Property Value

int

## FloorEnd

Gets the ending floor number for this schedule.

```
public int FloorEnd { get; }
```

### Property Value

int

## FloorsCount

Gets the total number of floors in this schedule.

```
public int FloorsCount { get; }
```

### Property Value

int

## Id

Gets the unique identifier for this schedule.

```
public int Id { get; }
```

### Property Value

int

## InfiniteTowerId

Gets the infinite tower ID this schedule belongs to.

```
public int InfiniteTowerId { get; }
```

### Property Value

int

## Key

Gets the schedule ID as the key for this row.

```
public override int Key { get; }
```

### Property Value

int

## MaxTickets

Gets the maximum number of tickets a player can hold.

```
public int MaxTickets { get; }
```

### Property Value

int

## ResetIntervalBlocks

Gets the number of blocks between ticket resets.

```
public int ResetIntervalBlocks { get; }
```

### Property Value

int

## StartBlockIndex

Gets the block index when the schedule starts.

```
public long StartBlockIndex { get; }
```

### Property Value

long

# Methods

## GetRemainingBlocks(long)

Gets the remaining blocks until the schedule ends.

```
public long GetRemainingBlocks(long currentBlockIndex)
```

### Parameters

`currentBlockIndex` long

Current block index

### Returns

long

Number of blocks remaining, or 0 if already ended

## HasEnded(long)

Checks if the infinite tower schedule has ended.

```
public bool HasEnded(long currentBlockIndex)
```

### Parameters

`currentBlockIndex` long

Current block index

### Returns

bool

True if the schedule has ended, false otherwise

## HasStarted(long)

Checks if the infinite tower schedule has started.

```
public bool HasStarted(long currentBlockIndex)
```

### Parameters

`currentBlockIndex` long

Current block index

### Returns

bool

True if the schedule has started, false otherwise

## IsActive(long)

Checks if the infinite tower schedule is currently active.

```
public bool IsActive(long currentBlockIndex)
```

### Parameters

`currentBlockIndex` long

Current block index

### Returns

bool

True if the schedule is active, false otherwise

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

## ValidateFloorRange(int, string)

Validates that the floor ID is within this schedule's floor range.

```
public void ValidateFloorRange(int floorId, string addressesHex)
```

### Parameters

`floorId` int

Floor ID to validate

`addressesHex` string

Addresses hex for logging

### Exceptions

InvalidOperationException

Thrown when floor ID is out of range

## ValidateInfiniteTowerId(int, string)

Validates that the infinite tower ID matches this schedule's configuration.

```
public void ValidateInfiniteTowerId(int expectedInfiniteTowerId,
string addressesHex)
```

### Parameters

`expectedInfiniteTowerId` int

The expected infinite tower ID

`addressesHex` string

Addresses hex for logging

### Exceptions

SheetRowNotFoundException

Thrown when infinite tower ID mismatch

## ValidateScheduleTiming(long, int, string)

Validates schedule timing for the infinite tower.

```
public void ValidateScheduleTiming(long currentBlockIndex, int infiniteTowerId,
```

```
  string addressesHex)
```

## Parameters

`currentBlockIndex` long

Current block index

`infiniteTowerId` int

The infinite tower ID for error messages

`addressesHex` string

Addresses hex for logging

## Exceptions

InfiniteTowerScheduleNotStartedException

Thrown when schedule has not started

InfiniteTowerScheduleEndedException

Thrown when schedule has ended

InfiniteTowerScheduleNotActiveException

Thrown when schedule is not active

# Class ItemConfigForGradeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemConfigForGradeSheet : Sheet<int, ItemConfigForGradeSheet.Row>,
IDictionary<int, ItemConfigForGradeSheet.Row>, ICollection<KeyValuePair<int,
ItemConfigForGradeSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemConfigForGradeSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ItemConfigForGradeSheet.Row> ← ItemConfigForGradeSheet

### Implements

IDictionary<int, ItemConfigForGradeSheet.Row>,
ICollection<KeyValuePair<int, ItemConfigForGradeSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemConfigForGradeSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, ItemConfigForGradeSheet.Row>.Name ,
Sheet<int, ItemConfigForGradeSheet.Row>.OrderedList ,
Sheet<int, ItemConfigForGradeSheet.Row>.First ,
Sheet<int, ItemConfigForGradeSheet.Row>.Last ,
Sheet<int, ItemConfigForGradeSheet.Row>.Keys ,
Sheet<int, ItemConfigForGradeSheet.Row>.Values ,
Sheet<int, ItemConfigForGradeSheet.Row>.Count ,
Sheet<int, ItemConfigForGradeSheet.Row>.IsReadOnly ,
Sheet<int, ItemConfigForGradeSheet.Row>.this[int] ,
Sheet<int, ItemConfigForGradeSheet.Row>.Set(string, bool) ,
Sheet<int, ItemConfigForGradeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemConfigForGradeSheet.Row>.GetEnumerator() ,
Sheet<int, ItemConfigForGradeSheet.Row>.TryGetValue(int, out
ItemConfigForGradeSheet.Row, bool) ,
Sheet<int, ItemConfigForGradeSheet.Row>.AddRow(int, ItemConfigForGradeSheet.Row) ,
Sheet<int, ItemConfigForGradeSheet.Row>.Add(int, ItemConfigForGradeSheet.Row) ,
Sheet<int, ItemConfigForGradeSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemConfigForGradeSheet.Row>.Remove(int) ,

Sheet<int, ItemConfigForGradeSheet.Row>.TryGetValue(int, out ItemConfigForGradeSheet.Row) , Sheet<int, ItemConfigForGradeSheet.Row>.Add(KeyValuePair<int, ItemConfigForGradeSheet.Row>) , Sheet<int, ItemConfigForGradeSheet.Row>.Clear() , Sheet<int, ItemConfigForGradeSheet.Row>.Contains(KeyValuePair<int, ItemConfigForGradeSheet.Row>) , Sheet<int, ItemConfigForGradeSheet.Row>.CopyTo(KeyValuePair<int, ItemConfigForGradeSheet.Row>[], int) , Sheet<int, ItemConfigForGradeSheet.Row>.Remove(KeyValuePair<int, ItemConfigForGradeSheet.Row>) , Sheet<int, ItemConfigForGradeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemConfigForGradeSheet()

```
public ItemConfigForGradeSheet()
```

# Class ItemConfigForGradeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemConfigForGradeSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← ItemConfigForGradeSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## EnhancementLimit

```
public int EnhancementLimit { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## MonsterPartsCountForCombination

```
public int MonsterPartsCountForCombination { get; }
```

### Property Value

int

## MonsterPartsCountForCombinationWithNCG

```
public int MonsterPartsCountForCombinationWithNCG { get; }
```

### Property Value

int

## RandomBuffSkillMaxCountForCombination

```
public int RandomBuffSkillMaxCountForCombination { get; }
```

### Property Value

int

## RandomBuffSkillMinCountForCombination

```
public int RandomBuffSkillMinCountForCombination { get; }
```

### Property Value

int

## Methods

### Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class ItemEnhancementQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancementQuestSheet : Sheet<int, ItemEnhancementQuestSheet.Row>,
IDictionary<int, ItemEnhancementQuestSheet.Row>, ICollection<KeyValuePair<int,
ItemEnhancementQuestSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemEnhancementQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ItemEnhancementQuestSheet.Row> ← ItemEnhancementQuestSheet

### Implements

IDictionary<int, ItemEnhancementQuestSheet.Row>, ICollection<KeyValuePair<int, ItemEnhancementQuestSheet.Row>>, IEnumerable<KeyValuePair<int, ItemEnhancementQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, ItemEnhancementQuestSheet.Row>.Name ,
Sheet<int, ItemEnhancementQuestSheet.Row>.OrderedList ,
Sheet<int, ItemEnhancementQuestSheet.Row>.First ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Last ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Keys ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Values ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Count ,
Sheet<int, ItemEnhancementQuestSheet.Row>.IsReadOnly ,
Sheet<int, ItemEnhancementQuestSheet.Row>.this[int] ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Set(string, bool) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.GetEnumerator() ,
Sheet<int, ItemEnhancementQuestSheet.Row>.TryGetValue(int, out ItemEnhancementQuestSheet.Row, bool) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.AddRow(int, ItemEnhancementQuestSheet.Row) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Add(int, ItemEnhancementQuestSheet.Row) ,

Sheet<int, ItemEnhancementQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Remove(int) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.TryGetValue(int, out
ItemEnhancementQuestSheet.Row) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Add(KeyValuePair<int,
ItemEnhancementQuestSheet.Row>) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Clear() ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Contains(KeyValuePair<int,
ItemEnhancementQuestSheet.Row>) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.CopyTo(KeyValuePair<int,
ItemEnhancementQuestSheet.Row>[], int) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Remove(KeyValuePair<int,
ItemEnhancementQuestSheet.Row>) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemEnhancementQuestSheet()

```
public ItemEnhancementQuestSheet()
```

# Class ItemEnhancementQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancementQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← ItemEnhancementQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## Grade

```
public int Grade { get; }
```

### Property Value

int

## Methods

### Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

#### Parameters

`fields` IReadOnlyList<string>

# Class ItemGradeQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemGradeQuestSheet : Sheet<int, ItemGradeQuestSheet.Row>,
IDictionary<int, ItemGradeQuestSheet.Row>, ICollection<KeyValuePair<int,
ItemGradeQuestSheet.Row>>, IEnumerable<KeyValuePair<int, ItemGradeQuestSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ItemGradeQuestSheet.Row> ← ItemGradeQuestSheet

### Implements

IDictionary<int, ItemGradeQuestSheet.Row>,
ICollection<KeyValuePair<int, ItemGradeQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemGradeQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, ItemGradeQuestSheet.Row>.Name ,
Sheet<int, ItemGradeQuestSheet.Row>.OrderedList ,
Sheet<int, ItemGradeQuestSheet.Row>.First ,
Sheet<int, ItemGradeQuestSheet.Row>.Last ,
Sheet<int, ItemGradeQuestSheet.Row>.Keys ,
Sheet<int, ItemGradeQuestSheet.Row>.Values ,
Sheet<int, ItemGradeQuestSheet.Row>.Count ,
Sheet<int, ItemGradeQuestSheet.Row>.IsReadOnly ,
Sheet<int, ItemGradeQuestSheet.Row>.this[int] ,
Sheet<int, ItemGradeQuestSheet.Row>.Set(string, bool) ,
Sheet<int, ItemGradeQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemGradeQuestSheet.Row>.GetEnumerator() ,
Sheet<int, ItemGradeQuestSheet.Row>.TryGetValue(int, out ItemGradeQuestSheet.Row, bool) ,
Sheet<int, ItemGradeQuestSheet.Row>.AddRow(int, ItemGradeQuestSheet.Row) ,
Sheet<int, ItemGradeQuestSheet.Row>.Add(int, ItemGradeQuestSheet.Row) ,
Sheet<int, ItemGradeQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemGradeQuestSheet.Row>.Remove(int) ,
Sheet<int, ItemGradeQuestSheet.Row>.TryGetValue(int, out ItemGradeQuestSheet.Row) ,

Sheet<int, ItemGradeQuestSheet.Row>.Add(KeyValuePair<int, ItemGradeQuestSheet.Row>) ,
Sheet<int, ItemGradeQuestSheet.Row>.Clear() ,
Sheet<int, ItemGradeQuestSheet.Row>.Contains(KeyValuePair<int, ItemGradeQuestSheet.Row>) ,
Sheet<int, ItemGradeQuestSheet.Row>.CopyTo(KeyValuePair<int, ItemGradeQuestSheet.Row>[], int) ,
Sheet<int, ItemGradeQuestSheet.Row>.Remove(KeyValuePair<int, ItemGradeQuestSheet.Row>) ,
Sheet<int, ItemGradeQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemGradeQuestSheet()

```
public ItemGradeQuestSheet()
```

# Class ItemGradeQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemGradeQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← ItemGradeQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Grade

```
public int Grade { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class ItemRequirementSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ItemRequirementSheet : Sheet<int, ItemRequirementSheet.Row>,
IDictionary<int, ItemRequirementSheet.Row>, ICollection<KeyValuePair<int,
ItemRequirementSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemRequirementSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ItemRequirementSheet.Row> ← ItemRequirementSheet

### Implements

IDictionary<int, ItemRequirementSheet.Row>,
ICollection<KeyValuePair<int, ItemRequirementSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemRequirementSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, ItemRequirementSheet.Row>.Name ,
Sheet<int, ItemRequirementSheet.Row>.OrderedList ,
Sheet<int, ItemRequirementSheet.Row>.First ,
Sheet<int, ItemRequirementSheet.Row>.Last ,
Sheet<int, ItemRequirementSheet.Row>.Keys ,
Sheet<int, ItemRequirementSheet.Row>.Values ,
Sheet<int, ItemRequirementSheet.Row>.Count ,
Sheet<int, ItemRequirementSheet.Row>.IsReadOnly ,
Sheet<int, ItemRequirementSheet.Row>.this[int] ,
Sheet<int, ItemRequirementSheet.Row>.Set(string, bool) ,
Sheet<int, ItemRequirementSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemRequirementSheet.Row>.GetEnumerator() ,
Sheet<int, ItemRequirementSheet.Row>.TryGetValue(int, out ItemRequirementSheet.Row,
bool) ,
Sheet<int, ItemRequirementSheet.Row>.AddRow(int, ItemRequirementSheet.Row) ,
Sheet<int, ItemRequirementSheet.Row>.Add(int, ItemRequirementSheet.Row) ,
Sheet<int, ItemRequirementSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemRequirementSheet.Row>.Remove(int) ,
Sheet<int, ItemRequirementSheet.Row>.TryGetValue(int, out ItemRequirementSheet.Row) ,

Sheet<int, ItemRequirementSheet.Row>.Add(KeyValuePair<int, ItemRequirementSheet.Row>) ,
Sheet<int, ItemRequirementSheet.Row>.Clear() ,
Sheet<int, ItemRequirementSheet.Row>.Contains(KeyValuePair<int, ItemRequirementSheet.Row>) ,
Sheet<int, ItemRequirementSheet.Row>.CopyTo(KeyValuePair<int, ItemRequirementSheet.Row>[], int) ,
Sheet<int, ItemRequirementSheet.Row>.Remove(KeyValuePair<int, ItemRequirementSheet.Row>) ,
Sheet<int, ItemRequirementSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemRequirementSheet()

```
public ItemRequirementSheet()
```

# Class ItemRequirementSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ItemRequirementSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← ItemRequirementSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### ItemId

```
public int ItemId { get; }
```

#### Property Value

int

### Key

```
public override int Key { get; }
```

#### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

## MimisLevel

```
public int MimisLevel { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class ItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemSheet : Sheet<int, ItemSheet.Row>, IDictionary<int, ItemSheet.Row>,
ICollection<KeyValuePair<int, ItemSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ItemSheet.Row> ← ItemSheet

### Implements

IDictionary<int, ItemSheet.Row>,
ICollection<KeyValuePair<int, ItemSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, ItemSheet.Row>.Name , Sheet<int, ItemSheet.Row>.OrderedList ,
Sheet<int, ItemSheet.Row>.First , Sheet<int, ItemSheet.Row>.Last ,
Sheet<int, ItemSheet.Row>.Keys , Sheet<int, ItemSheet.Row>.Values ,
Sheet<int, ItemSheet.Row>.Count , Sheet<int, ItemSheet.Row>.IsReadOnly ,
Sheet<int, ItemSheet.Row>.this[int] , Sheet<int, ItemSheet.Row>.Set(string, bool) ,
Sheet<int, ItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemSheet.Row>.GetEnumerator() ,
Sheet<int, ItemSheet.Row>.TryGetValue(int, out ItemSheet.Row, bool) ,
Sheet<int, ItemSheet.Row>.AddRow(int, ItemSheet.Row) ,
Sheet<int, ItemSheet.Row>.Add(int, ItemSheet.Row) ,
Sheet<int, ItemSheet.Row>.ContainsKey(int) , Sheet<int, ItemSheet.Row>.Remove(int) ,
Sheet<int, ItemSheet.Row>.TryGetValue(int, out ItemSheet.Row) ,
Sheet<int, ItemSheet.Row>.Add(KeyValuePair<int, ItemSheet.Row>) ,
Sheet<int, ItemSheet.Row>.Clear() ,
Sheet<int, ItemSheet.Row>.Contains(KeyValuePair<int, ItemSheet.Row>) ,
Sheet<int, ItemSheet.Row>.CopyTo(KeyValuePair<int, ItemSheet.Row>[], int) ,
Sheet<int, ItemSheet.Row>.Remove(KeyValuePair<int, ItemSheet.Row>) ,
Sheet<int, ItemSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemSheet()

```
public ItemSheet()
```

# Class ItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemSheet.Row : ItemSheet.RowBase
```

**Inheritance**

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row

**Derived**

ConsumableItemSheet.Row, CostumeItemSheet.Row, EquipmentItemSheet.Row, MaterialItemSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Row()

```
public Row()
```

### Row(Dictionary)

```
public Row(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

# Properties

## ElementalType

```
public ElementalType ElementalType { get; }
```

### Property Value

ElementalType

## Grade

```
public int Grade { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## ItemSubType

```
public ItemSubType ItemSubType { get; protected set; }
```

### Property Value

ItemSubType

## ItemType

```
public virtual ItemType ItemType { get; }
```

### Property Value

ItemType

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Deserialize(Dictionary)

```
public static ItemSheet.Row Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

ItemSheet.Row

## Serialize()

```
public override IValue Serialize()
```

## Returns

IValue

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class ItemSheet.RowBase

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ItemSheet.RowBase : SheetRow<int>
```

### Inheritance

object ← SheetRow<int> ← ItemSheet.RowBase

### Derived

ItemSheet.Row

### Inherited Members

SheetRow<int>.Key , SheetRow<int>.Set(IReadOnlyList<string>) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

## Serialize()

```
public abstract IValue Serialize()
```

### Returns

IValue

# Class ItemTypeCollectQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemTypeCollectQuestSheet : Sheet<int, ItemTypeCollectQuestSheet.Row>,
IDictionary<int, ItemTypeCollectQuestSheet.Row>, ICollection<KeyValuePair<int,
ItemTypeCollectQuestSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemTypeCollectQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, ItemTypeCollectQuestSheet.Row> ← ItemTypeCollectQuestSheet

### Implements

IDictionary<int, ItemTypeCollectQuestSheet.Row>,
ICollection<KeyValuePair<int, ItemTypeCollectQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemTypeCollectQuestSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, ItemTypeCollectQuestSheet.Row>.Name ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.OrderedList ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.First ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Last ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Keys ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Values ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Count ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.IsReadOnly ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.this[int] ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Set(string, bool) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.GetEnumerator() ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.TryGetValue(int, out
ItemTypeCollectQuestSheet.Row, bool) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.AddRow(int, ItemTypeCollectQuestSheet.Row)
,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Add(int, ItemTypeCollectQuestSheet.Row) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Remove(int) ,

Sheet<int, ItemTypeCollectQuestSheet.Row>.TryGetValue(int, out ItemTypeCollectQuestSheet.Row) , Sheet<int, ItemTypeCollectQuestSheet.Row>.Add(KeyValuePair<int, ItemTypeCollectQuestSheet.Row>) , Sheet<int, ItemTypeCollectQuestSheet.Row>.Clear() , Sheet<int, ItemTypeCollectQuestSheet.Row>.Contains(KeyValuePair<int, ItemTypeCollectQuestSheet.Row>) , Sheet<int, ItemTypeCollectQuestSheet.Row>.CopyTo(KeyValuePair<int, ItemTypeCollectQuestSheet.Row>[], int) , Sheet<int, ItemTypeCollectQuestSheet.Row>.Remove(KeyValuePair<int, ItemTypeCollectQuestSheet.Row>) , Sheet<int, ItemTypeCollectQuestSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ItemTypeCollectQuestSheet()

```
public ItemTypeCollectQuestSheet()
```

# Class ItemTypeCollectQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemTypeCollectQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← ItemTypeCollectQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ItemType

```
public ItemType ItemType { get; }
```

### Property Value

ItemType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class MaterialItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MaterialItemSheet : Sheet<int, MaterialItemSheet.Row>, IDictionary<int,
MaterialItemSheet.Row>, ICollection<KeyValuePair<int, MaterialItemSheet.Row>>,
IEnumerable<KeyValuePair<int, MaterialItemSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, MaterialItemSheet.Row> ← MaterialItemSheet

### Implements

IDictionary<int, MaterialItemSheet.Row>,
ICollection<KeyValuePair<int, MaterialItemSheet.Row>>,
IEnumerable<KeyValuePair<int, MaterialItemSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, MaterialItemSheet.Row>.Name ,
Sheet<int, MaterialItemSheet.Row>.OrderedList ,
Sheet<int, MaterialItemSheet.Row>.First , Sheet<int, MaterialItemSheet.Row>.Last ,
Sheet<int, MaterialItemSheet.Row>.Keys , Sheet<int, MaterialItemSheet.Row>.Values ,
Sheet<int, MaterialItemSheet.Row>.Count ,
Sheet<int, MaterialItemSheet.Row>.IsReadOnly ,
Sheet<int, MaterialItemSheet.Row>.this[int] ,
Sheet<int, MaterialItemSheet.Row>.Set(string, bool) ,
Sheet<int, MaterialItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MaterialItemSheet.Row>.GetEnumerator() ,
Sheet<int, MaterialItemSheet.Row>.TryGetValue(int, out MaterialItemSheet.Row, bool) ,
Sheet<int, MaterialItemSheet.Row>.AddRow(int, MaterialItemSheet.Row) ,
Sheet<int, MaterialItemSheet.Row>.Add(int, MaterialItemSheet.Row) ,
Sheet<int, MaterialItemSheet.Row>.ContainsKey(int) ,
Sheet<int, MaterialItemSheet.Row>.Remove(int) ,
Sheet<int, MaterialItemSheet.Row>.TryGetValue(int, out MaterialItemSheet.Row) ,
Sheet<int, MaterialItemSheet.Row>.Add(KeyValuePair<int, MaterialItemSheet.Row>) ,
Sheet<int, MaterialItemSheet.Row>.Clear() ,
Sheet<int, MaterialItemSheet.Row>.Contains(KeyValuePair<int, MaterialItemSheet.Row>) ,

Sheet<int, MaterialItemSheet.Row>.CopyTo(KeyValuePair<int, MaterialItemSheet.Row>[], int) ,
Sheet<int, MaterialItemSheet.Row>.Remove(KeyValuePair<int, MaterialItemSheet.Row>) ,
Sheet<int, MaterialItemSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
MaterialItemSheetExtensions.ValidateFromAction(MaterialItemSheet, List<EquipmentItemSubRecipeSheet.MaterialInfo>, Dictionary<int, int>, string)

# Constructors

## MaterialItemSheet()

```
public MaterialItemSheet()
```

# Class MaterialItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MaterialItemSheet.Row : ItemSheet.Row, ISerializable
```

### Inheritance

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row ← MaterialItemSheet.Row

### Implements

ISerializable

### Inherited Members

ItemSheet.Row.Key , ItemSheet.Row.Id , ItemSheet.Row.ItemSubType , ItemSheet.Row.Grade , ItemSheet.Row.ElementalType , ItemSheet.Row.Serialize() , ItemSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Row()

```
public Row()
```

### Row(Dictionary)

```
public Row(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

## Row(SerializationInfo, StreamingContext)

```
protected Row(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

`context` StreamingContext

# Properties

## ItemId

```
public HashDigest<SHA256> ItemId { get; }
```

### Property Value

HashDigest<SHA256>

## ItemType

```
public override ItemType ItemType { get; }
```

### Property Value

ItemType

# Methods

## GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

### Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

### Exceptions

SecurityException

The caller does not have the required permission.

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class MimisbrunnrSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MimisbrunnrSheet : Sheet<int, MimisbrunnrSheet.Row>, IDictionary<int,
MimisbrunnrSheet.Row>, ICollection<KeyValuePair<int, MimisbrunnrSheet.Row>>,
IEnumerable<KeyValuePair<int, MimisbrunnrSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, MimisbrunnrSheet.Row> ← MimisbrunnrSheet

### Implements

IDictionary<int, MimisbrunnrSheet.Row>,
ICollection<KeyValuePair<int, MimisbrunnrSheet.Row>>,
IEnumerable<KeyValuePair<int, MimisbrunnrSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, MimisbrunnrSheet.Row>.Name ,
Sheet<int, MimisbrunnrSheet.Row>.OrderedList ,
Sheet<int, MimisbrunnrSheet.Row>.First , Sheet<int, MimisbrunnrSheet.Row>.Last ,
Sheet<int, MimisbrunnrSheet.Row>.Keys , Sheet<int, MimisbrunnrSheet.Row>.Values ,
Sheet<int, MimisbrunnrSheet.Row>.Count ,
Sheet<int, MimisbrunnrSheet.Row>.IsReadOnly ,
Sheet<int, MimisbrunnrSheet.Row>.this[int] ,
Sheet<int, MimisbrunnrSheet.Row>.Set(string, bool) ,
Sheet<int, MimisbrunnrSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MimisbrunnrSheet.Row>.GetEnumerator() ,
Sheet<int, MimisbrunnrSheet.Row>.TryGetValue(int, out MimisbrunnrSheet.Row, bool) ,
Sheet<int, MimisbrunnrSheet.Row>.AddRow(int, MimisbrunnrSheet.Row) ,
Sheet<int, MimisbrunnrSheet.Row>.Add(int, MimisbrunnrSheet.Row) ,
Sheet<int, MimisbrunnrSheet.Row>.ContainsKey(int) ,
Sheet<int, MimisbrunnrSheet.Row>.Remove(int) ,
Sheet<int, MimisbrunnrSheet.Row>.TryGetValue(int, out MimisbrunnrSheet.Row) ,
Sheet<int, MimisbrunnrSheet.Row>.Add(KeyValuePair<int, MimisbrunnrSheet.Row>) ,
Sheet<int, MimisbrunnrSheet.Row>.Clear() ,
Sheet<int, MimisbrunnrSheet.Row>.Contains(KeyValuePair<int, MimisbrunnrSheet.Row>) ,

Sheet<int, MimisbrunnrSheet.Row>.CopyTo(KeyValuePair<int, MimisbrunnrSheet.Row>[], int) ,
Sheet<int, MimisbrunnrSheet.Row>.Remove(KeyValuePair<int, MimisbrunnrSheet.Row>) ,
Sheet<int, MimisbrunnrSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## MimisbrunnrSheet()

```
public MimisbrunnrSheet()
```

# Class MimisbrunnrSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MimisbrunnrSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← MimisbrunnrSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ElementalTypes

```
public List<ElementalType> ElementalTypes { get; }
```

### Property Value

List<ElementalType>

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class MonsterCollectionRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionRewardSheet : Sheet<int,
MonsterCollectionRewardSheet.Row>, IDictionary<int,
MonsterCollectionRewardSheet.Row>, ICollection<KeyValuePair<int,
MonsterCollectionRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
MonsterCollectionRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, MonsterCollectionRewardSheet.Row> ← MonsterCollectionRewardSheet

### Implements

IDictionary<int, MonsterCollectionRewardSheet.Row>, ICollection<KeyValuePair<int, MonsterCollectionRewardSheet.Row>>, IEnumerable<KeyValuePair<int, MonsterCollectionRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, MonsterCollectionRewardSheet.Row>.Name ,
Sheet<int, MonsterCollectionRewardSheet.Row>.OrderedList ,
Sheet<int, MonsterCollectionRewardSheet.Row>.First ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Last ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Keys ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Values ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Count ,
Sheet<int, MonsterCollectionRewardSheet.Row>.IsReadOnly ,
Sheet<int, MonsterCollectionRewardSheet.Row>.this[int] ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Set(string, bool) ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MonsterCollectionRewardSheet.Row>.GetEnumerator() ,
Sheet<int, MonsterCollectionRewardSheet.Row>.TryGetValue(int, out MonsterCollectionRewardSheet.Row, bool) ,
Sheet<int, MonsterCollectionRewardSheet.Row>.AddRow(int, MonsterCollectionRewardSheet.Row) ,

Sheet<int, MonsterCollectionRewardSheet.Row>.Add(int, MonsterCollectionRewardSheet.Row) , Sheet<int, MonsterCollectionRewardSheet.Row>.ContainsKey(int) , Sheet<int, MonsterCollectionRewardSheet.Row>.Remove(int) , Sheet<int, MonsterCollectionRewardSheet.Row>.TryGetValue(int, out MonsterCollectionRewardSheet.Row) , Sheet<int, MonsterCollectionRewardSheet.Row>.Add(KeyValuePair<int, MonsterCollectionRewardSheet.Row>) , Sheet<int, MonsterCollectionRewardSheet.Row>.Clear() , Sheet<int, MonsterCollectionRewardSheet.Row>.Contains(KeyValuePair<int, MonsterCollectionRewardSheet.Row>) , Sheet<int, MonsterCollectionRewardSheet.Row>.CopyTo(KeyValuePair<int, MonsterCollectionRewardSheet.Row>[], int) , Sheet<int, MonsterCollectionRewardSheet.Row>.Remove(KeyValuePair<int, MonsterCollectionRewardSheet.Row>) , Sheet<int, MonsterCollectionRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## MonsterCollectionRewardSheet()

```
public MonsterCollectionRewardSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, MonsterCollectionRewardSheet.Row value)
```

### Parameters

`key` int

`value` MonsterCollectionRewardSheet.Row

# Class MonsterCollectionRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionRewardSheet.RewardInfo
```

**Inheritance**

object ← MonsterCollectionRewardSheet.RewardInfo

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardInfo(Dictionary)

```
public RewardInfo(Dictionary dictionary)
```

#### Parameters

`dictionary` Dictionary

### RewardInfo(int, int)

```
public RewardInfo(int itemId, int quantity)
```

#### Parameters

`itemId` int

`quantity` int

## RewardInfo(params string[])

```
public RewardInfo(params string[] fields)
```

### Parameters

`fields` string[]

# Fields

## ItemId

```
public readonly int ItemId
```

### Field Value

int

## Quantity

```
public readonly int Quantity
```

### Field Value

int

# Methods

## Equals(RewardInfo)

```
protected bool Equals(MonsterCollectionRewardSheet.RewardInfo other)
```

### Parameters

`other` MonsterCollectionRewardSheet.RewardInfo

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

# Serialize()

```
public IValue Serialize()
```

## Returns

IValue

# Class MonsterCollectionRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← MonsterCollectionRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Key

```
public override int Key { get; }
```

#### Property Value

int

### MonsterCollectionLevel

```
public int MonsterCollectionLevel { get; }
```

#### Property Value

int

## Rewards

```
public List<MonsterCollectionRewardSheet.RewardInfo> Rewards { get; }
```

### Property Value

List<MonsterCollectionRewardSheet.RewardInfo>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class MonsterCollectionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionSheet : Sheet<int, MonsterCollectionSheet.Row>,
IDictionary<int, MonsterCollectionSheet.Row>, ICollection<KeyValuePair<int,
MonsterCollectionSheet.Row>>, IEnumerable<KeyValuePair<int,
MonsterCollectionSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, MonsterCollectionSheet.Row> ← MonsterCollectionSheet

### Implements

IDictionary<int, MonsterCollectionSheet.Row>,
ICollection<KeyValuePair<int, MonsterCollectionSheet.Row>>,
IEnumerable<KeyValuePair<int, MonsterCollectionSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, MonsterCollectionSheet.Row>.Name ,
Sheet<int, MonsterCollectionSheet.Row>.OrderedList ,
Sheet<int, MonsterCollectionSheet.Row>.First ,
Sheet<int, MonsterCollectionSheet.Row>.Last ,
Sheet<int, MonsterCollectionSheet.Row>.Keys ,
Sheet<int, MonsterCollectionSheet.Row>.Values ,
Sheet<int, MonsterCollectionSheet.Row>.Count ,
Sheet<int, MonsterCollectionSheet.Row>.IsReadOnly ,
Sheet<int, MonsterCollectionSheet.Row>.this[int] ,
Sheet<int, MonsterCollectionSheet.Row>.Set(string, bool) ,
Sheet<int, MonsterCollectionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MonsterCollectionSheet.Row>.GetEnumerator() ,
Sheet<int, MonsterCollectionSheet.Row>.TryGetValue(int, out MonsterCollectionSheet.Row,
bool) ,
Sheet<int, MonsterCollectionSheet.Row>.AddRow(int, MonsterCollectionSheet.Row) ,
Sheet<int, MonsterCollectionSheet.Row>.Add(int, MonsterCollectionSheet.Row) ,
Sheet<int, MonsterCollectionSheet.Row>.ContainsKey(int) ,
Sheet<int, MonsterCollectionSheet.Row>.Remove(int) ,

Sheet<int, MonsterCollectionSheet.Row>.TryGetValue(int, out MonsterCollectionSheet.Row) , Sheet<int, MonsterCollectionSheet.Row>.Add(KeyValuePair<int, MonsterCollectionSheet.Row>) , Sheet<int, MonsterCollectionSheet.Row>.Clear() , Sheet<int, MonsterCollectionSheet.Row>.Contains(KeyValuePair<int, MonsterCollectionSheet.Row>) , Sheet<int, MonsterCollectionSheet.Row>.CopyTo(KeyValuePair<int, MonsterCollectionSheet.Row>[], int) , Sheet<int, MonsterCollectionSheet.Row>.Remove(KeyValuePair<int, MonsterCollectionSheet.Row>) , Sheet<int, MonsterCollectionSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## MonsterCollectionSheet()

```
public MonsterCollectionSheet()
```

# Class MonsterCollectionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← MonsterCollectionSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

## RequiredGold

```
public int RequiredGold { get; }
```

### Property Value

int

## RewardId

```
public int RewardId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class MonsterQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterQuestSheet : Sheet<int, MonsterQuestSheet.Row>, IDictionary<int,
MonsterQuestSheet.Row>, ICollection<KeyValuePair<int, MonsterQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, MonsterQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, MonsterQuestSheet.Row> ← MonsterQuestSheet

### Implements

IDictionary<int, MonsterQuestSheet.Row>,
ICollection<KeyValuePair<int, MonsterQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, MonsterQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, MonsterQuestSheet.Row>.Name ,
Sheet<int, MonsterQuestSheet.Row>.OrderedList ,
Sheet<int, MonsterQuestSheet.Row>.First , Sheet<int, MonsterQuestSheet.Row>.Last ,
Sheet<int, MonsterQuestSheet.Row>.Keys , Sheet<int, MonsterQuestSheet.Row>.Values ,
Sheet<int, MonsterQuestSheet.Row>.Count ,
Sheet<int, MonsterQuestSheet.Row>.IsReadOnly ,
Sheet<int, MonsterQuestSheet.Row>.this[int] ,
Sheet<int, MonsterQuestSheet.Row>.Set(string, bool) ,
Sheet<int, MonsterQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MonsterQuestSheet.Row>.GetEnumerator() ,
Sheet<int, MonsterQuestSheet.Row>.TryGetValue(int, out MonsterQuestSheet.Row, bool) ,
Sheet<int, MonsterQuestSheet.Row>.AddRow(int, MonsterQuestSheet.Row) ,
Sheet<int, MonsterQuestSheet.Row>.Add(int, MonsterQuestSheet.Row) ,
Sheet<int, MonsterQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, MonsterQuestSheet.Row>.Remove(int) ,
Sheet<int, MonsterQuestSheet.Row>.TryGetValue(int, out MonsterQuestSheet.Row) ,
Sheet<int, MonsterQuestSheet.Row>.Add(KeyValuePair<int, MonsterQuestSheet.Row>) ,
Sheet<int, MonsterQuestSheet.Row>.Clear() ,
Sheet<int, MonsterQuestSheet.Row>.Contains(KeyValuePair<int,
MonsterQuestSheet.Row>) ,

Sheet<int, MonsterQuestSheet.Row>.CopyTo(KeyValuePair<int, MonsterQuestSheet.Row>[], int) ,
Sheet<int, MonsterQuestSheet.Row>.Remove(KeyValuePair<int, MonsterQuestSheet.Row>) ,
Sheet<int, MonsterQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## MonsterQuestSheet()

```
public MonsterQuestSheet()
```

# Class MonsterQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← MonsterQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## MonsterId

```
public int MonsterId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class PetCostSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class PetCostSheet : Sheet<int, PetCostSheet.Row>, IDictionary<int,
PetCostSheet.Row>, ICollection<KeyValuePair<int, PetCostSheet.Row>>,
IEnumerable<KeyValuePair<int, PetCostSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, PetCostSheet.Row> ← PetCostSheet

### Implements

IDictionary<int, PetCostSheet.Row>,
ICollection<KeyValuePair<int, PetCostSheet.Row>>,
IEnumerable<KeyValuePair<int, PetCostSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, PetCostSheet.Row>.Name , Sheet<int, PetCostSheet.Row>.OrderedList ,
Sheet<int, PetCostSheet.Row>.First , Sheet<int, PetCostSheet.Row>.Last ,
Sheet<int, PetCostSheet.Row>.Keys , Sheet<int, PetCostSheet.Row>.Values ,
Sheet<int, PetCostSheet.Row>.Count , Sheet<int, PetCostSheet.Row>.IsReadOnly ,
Sheet<int, PetCostSheet.Row>.this[int] , Sheet<int, PetCostSheet.Row>.Set(string, bool) ,
Sheet<int, PetCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, PetCostSheet.Row>.GetEnumerator() ,
Sheet<int, PetCostSheet.Row>.TryGetValue(int, out PetCostSheet.Row, bool) ,
Sheet<int, PetCostSheet.Row>.AddRow(int, PetCostSheet.Row) ,
Sheet<int, PetCostSheet.Row>.Add(int, PetCostSheet.Row) ,
Sheet<int, PetCostSheet.Row>.ContainsKey(int) ,
Sheet<int, PetCostSheet.Row>.Remove(int) ,
Sheet<int, PetCostSheet.Row>.TryGetValue(int, out PetCostSheet.Row) ,
Sheet<int, PetCostSheet.Row>.Add(KeyValuePair<int, PetCostSheet.Row>) ,
Sheet<int, PetCostSheet.Row>.Clear() ,
Sheet<int, PetCostSheet.Row>.Contains(KeyValuePair<int, PetCostSheet.Row>) ,
Sheet<int, PetCostSheet.Row>.CopyTo(KeyValuePair<int, PetCostSheet.Row>[], int) ,
Sheet<int, PetCostSheet.Row>.Remove(KeyValuePair<int, PetCostSheet.Row>) ,
Sheet<int, PetCostSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## PetCostSheet()

```
public PetCostSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, PetCostSheet.Row value)
```

### Parameters

`key` int

`value` PetCostSheet.Row

# Class PetCostSheet.PetCostData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class PetCostSheet.PetCostData
```

**Inheritance**

object ← PetCostSheet.PetCostData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PetCostData(int, int, int)

```
public PetCostData(int level, int soulStoneQuantity, int ncgQuantity)
```

#### Parameters

`level` int

`soulStoneQuantity` int

`ncgQuantity` int

## Properties

### Level

```
public int Level { get; }
```

### Property Value

int

## NcgQuantity

```
public int NcgQuantity { get; }
```

### Property Value

int

## SoulStoneQuantity

```
public int SoulStoneQuantity { get; }
```

### Property Value

int

# Class PetCostSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class PetCostSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← PetCostSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Cost

```
public List<PetCostSheet.PetCostData> Cost { get; }
```

### Property Value

List<PetCostSheet.PetCostData>

## Key

```
public override int Key { get; }
```

### Property Value

int

## PetId

```
public int PetId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

## TryGetCost(int, out PetCostData)

```
public bool TryGetCost(int level, out PetCostSheet.PetCostData costData)
```

### Parameters

`level` int

`costData` PetCostSheet.PetCostData

### Returns

bool

# Class PetSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class PetSheet : Sheet<int, PetSheet.Row>, IDictionary<int, PetSheet.Row>,
ICollection<KeyValuePair<int, PetSheet.Row>>, IEnumerable<KeyValuePair<int,
PetSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, PetSheet.Row> ← PetSheet

### Implements

IDictionary<int, PetSheet.Row>, ICollection<KeyValuePair<int, PetSheet.Row>>, IEnumerable<KeyValuePair<int, PetSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, PetSheet.Row>.Name , Sheet<int, PetSheet.Row>.OrderedList , Sheet<int, PetSheet.Row>.First , Sheet<int, PetSheet.Row>.Last , Sheet<int, PetSheet.Row>.Keys , Sheet<int, PetSheet.Row>.Values , Sheet<int, PetSheet.Row>.Count , Sheet<int, PetSheet.Row>.IsReadOnly , Sheet<int, PetSheet.Row>.this[int] , Sheet<int, PetSheet.Row>.Set(string, bool) , Sheet<int, PetSheet.Row>.Set<T>(Sheet<int, T>, bool) , Sheet<int, PetSheet.Row>.GetEnumerator() , Sheet<int, PetSheet.Row>.TryGetValue(int, out PetSheet.Row, bool) , Sheet<int, PetSheet.Row>.AddRow(int, PetSheet.Row) , Sheet<int, PetSheet.Row>.Add(int, PetSheet.Row) , Sheet<int, PetSheet.Row>.ContainsKey(int) , Sheet<int, PetSheet.Row>.Remove(int) , Sheet<int, PetSheet.Row>.TryGetValue(int, out PetSheet.Row) , Sheet<int, PetSheet.Row>.Add(KeyValuePair<int, PetSheet.Row>) , Sheet<int, PetSheet.Row>.Clear() , Sheet<int, PetSheet.Row>.Contains(KeyValuePair<int, PetSheet.Row>) , Sheet<int, PetSheet.Row>.CopyTo(KeyValuePair<int, PetSheet.Row>[], int) , Sheet<int, PetSheet.Row>.Remove(KeyValuePair<int, PetSheet.Row>) , Sheet<int, PetSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## PetSheet()

```
public PetSheet()
```

# Class PetSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class PetSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← PetSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Grade

```
public int Grade { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## SoulStoneTicker

```
public string SoulStoneTicker { get; }
```

### Property Value

string

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class QuestItemRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestItemRewardSheet : Sheet<int, QuestItemRewardSheet.Row>,
IDictionary<int, QuestItemRewardSheet.Row>, ICollection<KeyValuePair<int,
QuestItemRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
QuestItemRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, QuestItemRewardSheet.Row> ← QuestItemRewardSheet

### Implements

IDictionary<int, QuestItemRewardSheet.Row>,
ICollection<KeyValuePair<int, QuestItemRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestItemRewardSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, QuestItemRewardSheet.Row>.Name ,
Sheet<int, QuestItemRewardSheet.Row>.OrderedList ,
Sheet<int, QuestItemRewardSheet.Row>.First ,
Sheet<int, QuestItemRewardSheet.Row>.Last ,
Sheet<int, QuestItemRewardSheet.Row>.Keys ,
Sheet<int, QuestItemRewardSheet.Row>.Values ,
Sheet<int, QuestItemRewardSheet.Row>.Count ,
Sheet<int, QuestItemRewardSheet.Row>.IsReadOnly ,
Sheet<int, QuestItemRewardSheet.Row>.this[int] ,
Sheet<int, QuestItemRewardSheet.Row>.Set(string, bool) ,
Sheet<int, QuestItemRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, QuestItemRewardSheet.Row>.GetEnumerator() ,
Sheet<int, QuestItemRewardSheet.Row>.TryGetValue(int, out QuestItemRewardSheet.Row,
bool) ,
Sheet<int, QuestItemRewardSheet.Row>.AddRow(int, QuestItemRewardSheet.Row) ,
Sheet<int, QuestItemRewardSheet.Row>.Add(int, QuestItemRewardSheet.Row) ,
Sheet<int, QuestItemRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, QuestItemRewardSheet.Row>.Remove(int) ,

Sheet<int, QuestItemRewardSheet.Row>.TryGetValue(int, out QuestItemRewardSheet.Row) , Sheet<int, QuestItemRewardSheet.Row>.Add(KeyValuePair<int, QuestItemRewardSheet.Row>) , Sheet<int, QuestItemRewardSheet.Row>.Clear() , Sheet<int, QuestItemRewardSheet.Row>.Contains(KeyValuePair<int, QuestItemRewardSheet.Row>) , Sheet<int, QuestItemRewardSheet.Row>.CopyTo(KeyValuePair<int, QuestItemRewardSheet.Row>[], int) , Sheet<int, QuestItemRewardSheet.Row>.Remove(KeyValuePair<int, QuestItemRewardSheet.Row>) , Sheet<int, QuestItemRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## QuestItemRewardSheet()

```
public QuestItemRewardSheet()
```

# Class QuestItemRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestItemRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← QuestItemRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Count

```
public int Count { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## ItemId

```
public int ItemId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class QuestRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestRewardSheet : Sheet<int, QuestRewardSheet.Row>, IDictionary<int,
QuestRewardSheet.Row>, ICollection<KeyValuePair<int, QuestRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, QuestRewardSheet.Row> ← QuestRewardSheet

### Implements

IDictionary<int, QuestRewardSheet.Row>,
ICollection<KeyValuePair<int, QuestRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, QuestRewardSheet.Row>.Name ,
Sheet<int, QuestRewardSheet.Row>.OrderedList ,
Sheet<int, QuestRewardSheet.Row>.First , Sheet<int, QuestRewardSheet.Row>.Last ,
Sheet<int, QuestRewardSheet.Row>.Keys , Sheet<int, QuestRewardSheet.Row>.Values ,
Sheet<int, QuestRewardSheet.Row>.Count ,
Sheet<int, QuestRewardSheet.Row>.IsReadOnly ,
Sheet<int, QuestRewardSheet.Row>.this[int] ,
Sheet<int, QuestRewardSheet.Row>.Set(string, bool) ,
Sheet<int, QuestRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, QuestRewardSheet.Row>.GetEnumerator() ,
Sheet<int, QuestRewardSheet.Row>.TryGetValue(int, out QuestRewardSheet.Row, bool) ,
Sheet<int, QuestRewardSheet.Row>.AddRow(int, QuestRewardSheet.Row) ,
Sheet<int, QuestRewardSheet.Row>.Add(int, QuestRewardSheet.Row) ,
Sheet<int, QuestRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, QuestRewardSheet.Row>.Remove(int) ,
Sheet<int, QuestRewardSheet.Row>.TryGetValue(int, out QuestRewardSheet.Row) ,
Sheet<int, QuestRewardSheet.Row>.Add(KeyValuePair<int, QuestRewardSheet.Row>) ,
Sheet<int, QuestRewardSheet.Row>.Clear() ,
Sheet<int, QuestRewardSheet.Row>.Contains(KeyValuePair<int, QuestRewardSheet.Row>)
,

Sheet<int, QuestRewardSheet.Row>.CopyTo(KeyValuePair<int, QuestRewardSheet.Row>[], int) ,
Sheet<int, QuestRewardSheet.Row>.Remove(KeyValuePair<int, QuestRewardSheet.Row>) ,
Sheet<int, QuestRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## QuestRewardSheet()

```
public QuestRewardSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, QuestRewardSheet.Row value)
```

### Parameters

`key` int

`value` QuestRewardSheet.Row

# Class QuestRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← QuestRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## RewardIds

```
public List<int> RewardIds { get; }
```

### Property Value

List<int>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class QuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestSheet : Sheet<int, QuestSheet.Row>, IDictionary<int,
QuestSheet.Row>, ICollection<KeyValuePair<int, QuestSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, QuestSheet.Row> ← QuestSheet

### Implements

IDictionary<int, QuestSheet.Row>,
ICollection<KeyValuePair<int, QuestSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, QuestSheet.Row>.Name , Sheet<int, QuestSheet.Row>.OrderedList ,
Sheet<int, QuestSheet.Row>.First , Sheet<int, QuestSheet.Row>.Last ,
Sheet<int, QuestSheet.Row>.Keys , Sheet<int, QuestSheet.Row>.Values ,
Sheet<int, QuestSheet.Row>.Count , Sheet<int, QuestSheet.Row>.IsReadOnly ,
Sheet<int, QuestSheet.Row>.this[int] , Sheet<int, QuestSheet.Row>.Set(string, bool) ,
Sheet<int, QuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, QuestSheet.Row>.GetEnumerator() ,
Sheet<int, QuestSheet.Row>.TryGetValue(int, out QuestSheet.Row, bool) ,
Sheet<int, QuestSheet.Row>.AddRow(int, QuestSheet.Row) ,
Sheet<int, QuestSheet.Row>.Add(int, QuestSheet.Row) ,
Sheet<int, QuestSheet.Row>.ContainsKey(int) , Sheet<int, QuestSheet.Row>.Remove(int) ,
Sheet<int, QuestSheet.Row>.TryGetValue(int, out QuestSheet.Row) ,
Sheet<int, QuestSheet.Row>.Add(KeyValuePair<int, QuestSheet.Row>) ,
Sheet<int, QuestSheet.Row>.Clear() ,
Sheet<int, QuestSheet.Row>.Contains(KeyValuePair<int, QuestSheet.Row>) ,
Sheet<int, QuestSheet.Row>.CopyTo(KeyValuePair<int, QuestSheet.Row>[], int) ,
Sheet<int, QuestSheet.Row>.Remove(KeyValuePair<int, QuestSheet.Row>) ,
Sheet<int, QuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## QuestSheet()

```
public QuestSheet()
```

# Class QuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestSheet.Row : SheetRow<int>, IState
```

### Inheritance

object ← SheetRow<int> ← QuestSheet.Row

### Implements

IState

### Derived

CollectQuestSheet.Row, CombinationEquipmentQuestSheet.Row, CombinationQuestSheet.Row, GeneralQuestSheet.Row, GoldQuestSheet.Row, ItemEnhancementQuestSheet.Row, ItemGradeQuestSheet.Row, ItemTypeCollectQuestSheet.Row, MonsterQuestSheet.Row, TradeQuestSheet.Row, WorldQuestSheet.Row

### Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Goal

```
public int Goal { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## QuestRewardId

```
public int QuestRewardId { get; }
```

### Property Value

int

# Methods

## Deserialize(Dictionary)

```
public static QuestSheet.Row Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

QuestSheet.Row

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class RaidSimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RaidSimulatorSheets : SimulatorSheets
```

**Inheritance**

object ← SimulatorSheetsV1 ← SimulatorSheets ← RaidSimulatorSheets

**Inherited Members**

SimulatorSheets.RuneOptionSheet , SimulatorSheets.RuneListSheet , SimulatorSheets.RuneLevelBonusSheet , SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RaidSimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WorldBossCharacterSheet, WorldBossActionPatternSheet, WorldBossBattleRewardSheet, RuneWeightSheet, RuneSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public RaidSimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
```

```
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, WorldBossCharacterSheet
worldBossCharacterSheet, WorldBossActionPatternSheet worldBossActionPatternSheet,
WorldBossBattleRewardSheet worldBossBattleRewardSheet, RuneWeightSheet
runeWeightSheet, RuneSheet runeSheet, RuneOptionSheet runeOptionSheet, RuneListSheet
runeListSheet, RuneLevelBonusSheet runeLevelBonusSheet)
```

## Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`worldBossCharacterSheet` WorldBossCharacterSheet

`worldBossActionPatternSheet` WorldBossActionPatternSheet

`worldBossBattleRewardSheet` WorldBossBattleRewardSheet

`runeWeightSheet` RuneWeightSheet

`runeSheet` RuneSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

# Properties

## RuneSheet

```
public RuneSheet RuneSheet { get; }
```

### Property Value

RuneSheet

## RuneWeightSheet

```
public RuneWeightSheet RuneWeightSheet { get; }
```

### Property Value

RuneWeightSheet

## WorldBossActionPatternSheet

```
public WorldBossActionPatternSheet WorldBossActionPatternSheet { get; }
```

### Property Value

WorldBossActionPatternSheet

## WorldBossBattleRewardSheet

```
public WorldBossBattleRewardSheet WorldBossBattleRewardSheet { get; }
```

### Property Value

WorldBossBattleRewardSheet

## WorldBossCharacterSheet

```
public WorldBossCharacterSheet WorldBossCharacterSheet { get; }
```

## Property Value

WorldBossCharacterSheet

# Class RaidSimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RaidSimulatorSheetsV1 : SimulatorSheetsV1
```

**Inheritance**

object ← SimulatorSheetsV1 ← RaidSimulatorSheetsV1

**Inherited Members**

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RaidSimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WorldBossCharacterSheet, WorldBossActionPatternSheet, WorldBossBattleRewardSheet, RuneWeightSheet, RuneSheet)

```
public RaidSimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, WorldBossCharacterSheet
worldBossCharacterSheet, WorldBossActionPatternSheet worldBossActionPatternSheet,
```

```
WorldBossBattleRewardSheet worldBossBattleRewardSheet, RuneWeightSheet
runeWeightSheet, RuneSheet runeSheet)
```

### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`worldBossCharacterSheet` WorldBossCharacterSheet

`worldBossActionPatternSheet` WorldBossActionPatternSheet

`worldBossBattleRewardSheet` WorldBossBattleRewardSheet

`runeWeightSheet` RuneWeightSheet

`runeSheet` RuneSheet

## Properties

## RuneSheet

```
public RuneSheet RuneSheet { get; }
```

### Property Value

RuneSheet

## RuneWeightSheet

```
public RuneWeightSheet RuneWeightSheet { get; }
```

### Property Value

RuneWeightSheet

## WorldBossActionPatternSheet

```
public WorldBossActionPatternSheet WorldBossActionPatternSheet { get; }
```

### Property Value

WorldBossActionPatternSheet

## WorldBossBattleRewardSheet

```
public WorldBossBattleRewardSheet WorldBossBattleRewardSheet { get; }
```

### Property Value

WorldBossBattleRewardSheet

## WorldBossCharacterSheet

```
public WorldBossCharacterSheet WorldBossCharacterSheet { get; }
```

### Property Value

WorldBossCharacterSheet

# Class RankingSimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RankingSimulatorSheets : SimulatorSheets
```

**Inheritance**

object ← SimulatorSheetsV1 ← SimulatorSheets ← RankingSimulatorSheets

**Inherited Members**

SimulatorSheets.RuneOptionSheet , SimulatorSheets.RuneListSheet , SimulatorSheets.RuneLevelBonusSheet , SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingSimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WeeklyArenaRewardSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public RankingSimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, WeeklyArenaRewardSheet
```

```
  weeklyArenaRewardSheet, RuneOptionSheet runeOptionSheet, RuneListSheet
  runeListSheet, RuneLevelBonusSheet runeLevelBonusSheet)
```

### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

## Fields

## WeeklyArenaRewardSheet

```
  public readonly WeeklyArenaRewardSheet WeeklyArenaRewardSheet
```

### Field Value

WeeklyArenaRewardSheet

# Class RankingSimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RankingSimulatorSheetsV1 : SimulatorSheetsV1
```

**Inheritance**

object ← SimulatorSheetsV1 ← RankingSimulatorSheetsV1

**Inherited Members**

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingSimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WeeklyArenaRewardSheet)

```
public RankingSimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, WeeklyArenaRewardSheet
weeklyArenaRewardSheet)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

# Fields

## WeeklyArenaRewardSheet

```
public readonly WeeklyArenaRewardSheet WeeklyArenaRewardSheet
```

### Field Value

WeeklyArenaRewardSheet

# Class RankingSimulatorSheetsV100291

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RankingSimulatorSheetsV100291 : SimulatorSheetsV100291
```

**Inheritance**

object ← SimulatorSheetsV100291 ← RankingSimulatorSheetsV100291

**Inherited Members**

SimulatorSheetsV100291.MaterialItemSheet , SimulatorSheetsV100291.SkillSheet , SimulatorSheetsV100291.SkillBuffSheet , SimulatorSheetsV100291.BuffSheet , SimulatorSheetsV100291.CharacterSheet , SimulatorSheetsV100291.CharacterLevelSheet , SimulatorSheetsV100291.EquipmentItemSetEffectSheet , SimulatorSheetsV100291.ToSimulatorSheetsV1() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RankingSimulatorSheetsV100291(MaterialItemSheet, SkillSheet, SkillBuffSheet, BuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WeeklyArenaRewardSheet)

```
public RankingSimulatorSheetsV100291(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, BuffSheet buffSheet, CharacterSheet
characterSheet, CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet, WeeklyArenaRewardSheet weeklyArenaRewardSheet)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`buffSheet` BuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

# Fields

## WeeklyArenaRewardSheet

```
public readonly WeeklyArenaRewardSheet WeeklyArenaRewardSheet
```

### Field Value

WeeklyArenaRewardSheet

# Methods

## ToRankingSimulatorSheetsV1()

```
public RankingSimulatorSheetsV1 ToRankingSimulatorSheetsV1()
```

### Returns

RankingSimulatorSheetsV1

# Class RedeemCodeListSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemCodeListSheet : Sheet<int, RedeemCodeListSheet.Row>,
IDictionary<int, RedeemCodeListSheet.Row>, ICollection<KeyValuePair<int,
RedeemCodeListSheet.Row>>, IEnumerable<KeyValuePair<int, RedeemCodeListSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, RedeemCodeListSheet.Row> ← RedeemCodeListSheet

### Implements

IDictionary<int, RedeemCodeListSheet.Row>,
ICollection<KeyValuePair<int, RedeemCodeListSheet.Row>>,
IEnumerable<KeyValuePair<int, RedeemCodeListSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, RedeemCodeListSheet.Row>.Name ,
Sheet<int, RedeemCodeListSheet.Row>.OrderedList ,
Sheet<int, RedeemCodeListSheet.Row>.First ,
Sheet<int, RedeemCodeListSheet.Row>.Last ,
Sheet<int, RedeemCodeListSheet.Row>.Keys ,
Sheet<int, RedeemCodeListSheet.Row>.Values ,
Sheet<int, RedeemCodeListSheet.Row>.Count ,
Sheet<int, RedeemCodeListSheet.Row>.IsReadOnly ,
Sheet<int, RedeemCodeListSheet.Row>.this[int] ,
Sheet<int, RedeemCodeListSheet.Row>.Set(string, bool) ,
Sheet<int, RedeemCodeListSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RedeemCodeListSheet.Row>.GetEnumerator() ,
Sheet<int, RedeemCodeListSheet.Row>.TryGetValue(int, out RedeemCodeListSheet.Row, bool) ,
Sheet<int, RedeemCodeListSheet.Row>.AddRow(int, RedeemCodeListSheet.Row) ,
Sheet<int, RedeemCodeListSheet.Row>.Add(int, RedeemCodeListSheet.Row) ,
Sheet<int, RedeemCodeListSheet.Row>.ContainsKey(int) ,
Sheet<int, RedeemCodeListSheet.Row>.Remove(int) ,
Sheet<int, RedeemCodeListSheet.Row>.TryGetValue(int, out RedeemCodeListSheet.Row) ,

Sheet<int, RedeemCodeListSheet.Row>.Add(KeyValuePair<int, RedeemCodeListSheet.Row>) , Sheet<int, RedeemCodeListSheet.Row>.Clear() , Sheet<int, RedeemCodeListSheet.Row>.Contains(KeyValuePair<int, RedeemCodeListSheet.Row>) , Sheet<int, RedeemCodeListSheet.Row>.CopyTo(KeyValuePair<int, RedeemCodeListSheet.Row>[], int) , Sheet<int, RedeemCodeListSheet.Row>.Remove(KeyValuePair<int, RedeemCodeListSheet.Row>) , Sheet<int, RedeemCodeListSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RedeemCodeListSheet()

```
public RedeemCodeListSheet()
```

# Class RedeemCodeListSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemCodeListSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← RedeemCodeListSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## PublicKey

```
public PublicKey PublicKey { get; }
```

### Property Value

PublicKey

## PublicKeyBinary

```
public Binary PublicKeyBinary { get; }
```

### Property Value

Binary

## RewardId

```
public int RewardId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class RedeemRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemRewardSheet : Sheet<int, RedeemRewardSheet.Row>, IDictionary<int,
RedeemRewardSheet.Row>, ICollection<KeyValuePair<int, RedeemRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, RedeemRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, RedeemRewardSheet.Row> ← RedeemRewardSheet

### Implements

IDictionary<int, RedeemRewardSheet.Row>,
ICollection<KeyValuePair<int, RedeemRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, RedeemRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, RedeemRewardSheet.Row>.Name ,
Sheet<int, RedeemRewardSheet.Row>.OrderedList ,
Sheet<int, RedeemRewardSheet.Row>.First , Sheet<int, RedeemRewardSheet.Row>.Last ,
Sheet<int, RedeemRewardSheet.Row>.Keys ,
Sheet<int, RedeemRewardSheet.Row>.Values ,
Sheet<int, RedeemRewardSheet.Row>.Count ,
Sheet<int, RedeemRewardSheet.Row>.IsReadOnly ,
Sheet<int, RedeemRewardSheet.Row>.this[int] ,
Sheet<int, RedeemRewardSheet.Row>.Set(string, bool) ,
Sheet<int, RedeemRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RedeemRewardSheet.Row>.GetEnumerator() ,
Sheet<int, RedeemRewardSheet.Row>.TryGetValue(int, out RedeemRewardSheet.Row,
bool) ,
Sheet<int, RedeemRewardSheet.Row>.AddRow(int, RedeemRewardSheet.Row) ,
Sheet<int, RedeemRewardSheet.Row>.Add(int, RedeemRewardSheet.Row) ,
Sheet<int, RedeemRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, RedeemRewardSheet.Row>.Remove(int) ,
Sheet<int, RedeemRewardSheet.Row>.TryGetValue(int, out RedeemRewardSheet.Row) ,
Sheet<int, RedeemRewardSheet.Row>.Add(KeyValuePair<int, RedeemRewardSheet.Row>)
,

Sheet<int, RedeemRewardSheet.Row>.Clear() ,
Sheet<int, RedeemRewardSheet.Row>.Contains(KeyValuePair<int,
RedeemRewardSheet.Row>) ,
Sheet<int, RedeemRewardSheet.Row>.CopyTo(KeyValuePair<int,
RedeemRewardSheet.Row>[], int) ,
Sheet<int, RedeemRewardSheet.Row>.Remove(KeyValuePair<int,
RedeemRewardSheet.Row>) ,
Sheet<int, RedeemRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RedeemRewardSheet()

```
public RedeemRewardSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, RedeemRewardSheet.Row value)
```

### Parameters

`key` int

`value` RedeemRewardSheet.Row

# Class RedeemRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemRewardSheet.RewardInfo : IState
```

**Inheritance**

object ← RedeemRewardSheet.RewardInfo

**Implements**

IState

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardInfo(Dictionary)

```
public RewardInfo(Dictionary serialized)
```

#### Parameters

`serialized` Dictionary

### RewardInfo(params string[])

```
public RewardInfo(params string[] fields)
```

#### Parameters

`fields` string[]

# Fields

## ItemId

```
public readonly int? ItemId
```

### Field Value

int?

## Quantity

```
public readonly int Quantity
```

### Field Value

int

## Type

```
public readonly RewardType Type
```

### Field Value

RewardType

# Methods

## Equals(RewardInfo)

```
protected bool Equals(RedeemRewardSheet.RewardInfo other)
```

### Parameters

`other` RedeemRewardSheet.RewardInfo

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Serialize()

```
public IValue Serialize()
```

## Returns

IValue

# Class RedeemRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← RedeemRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

# Rewards

```
public List<RedeemRewardSheet.RewardInfo> Rewards { get; }
```

## Property Value

List<RedeemRewardSheet.RewardInfo>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class RestrictionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Declares what a given item or currency may not be used for, so that a restriction can be adjusted with `patch_table_sheet` instead of a headless and client release.

A row targets either an item id (numeric key) or a currency ticker (non-numeric key). Which one is looked up is decided by the call site — a market action already holds an item id or a ticker — so the sheet needs no discriminator column.

Columns: `key`, `market_registrable`, `synthesize_material`. The first governs the market registration path only — a fungible item can still reach another account through a garage or by being wrapped into a currency, so a row here does not make its target account bound by itself. The second closes the conversion path that would otherwise turn a restricted item into an unrestricted one.

An empty cell means "unspecified" and leaves the behavior that predates this sheet untouched. An unparsable cell is read as `false`(= restricted) on purpose: a typo then fails toward blocking, which is recoverable, rather than toward leaking. ValidateCsv(string) rejects such a cell at patch time so it should not reach the chain in the first place.

A key is an item id, not an item instance, so a row applies to every copy that exists — give a restricted item a new id rather than reusing one players already hold. A false `market_registrable` also stops `ReRegisterProduct`, leaving an existing seller able to cancel but not to reprice, and it does not retract listings that are already up: `BuyProduct` does not consult this sheet, so a restriction added after the fact still needs those products cancelled. A false `synthesize_material` only stops the item from being consumed as a material — an item that must not be handed out by synthesis is kept out of the result pool by leaving it out of `SynthesizeWeightSheet`, which is a separate sheet from this one.

Callers must tolerate the sheet being absent — it reaches an existing chain only by `patch_table_sheet`, so loading it as a required sheet would fail every affected action until each chain is patched. Absence has to fall back to the restrictions that are still hardcoded, which is also what keeps re-evaluation of pre-patch blocks identical.

```
[Serializable]
public class RestrictionSheet : Sheet<string, RestrictionSheet.Row>,
IDictionary<string, RestrictionSheet.Row>, ICollection<KeyValuePair<string,
```

```
RestrictionSheet.Row>>, IEnumerable<KeyValuePair<string, RestrictionSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<string, RestrictionSheet.Row> ← RestrictionSheet

### Implements

IDictionary<string, RestrictionSheet.Row>,
ICollection<KeyValuePair<string, RestrictionSheet.Row>>,
IEnumerable<KeyValuePair<string, RestrictionSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<string, RestrictionSheet.Row>.Name ,
Sheet<string, RestrictionSheet.Row>.OrderedList ,
Sheet<string, RestrictionSheet.Row>.First , Sheet<string, RestrictionSheet.Row>.Last ,
Sheet<string, RestrictionSheet.Row>.Keys , Sheet<string, RestrictionSheet.Row>.Values ,
Sheet<string, RestrictionSheet.Row>.Count ,
Sheet<string, RestrictionSheet.Row>.IsReadOnly ,
Sheet<string, RestrictionSheet.Row>.this[string] ,
Sheet<string, RestrictionSheet.Row>.Set(string, bool) ,
Sheet<string, RestrictionSheet.Row>.Set<T>(Sheet<string, T>, bool) ,
Sheet<string, RestrictionSheet.Row>.GetEnumerator() ,
Sheet<string, RestrictionSheet.Row>.TryGetValue(string, out RestrictionSheet.Row, bool) ,
Sheet<string, RestrictionSheet.Row>.AddRow(string, RestrictionSheet.Row) ,
Sheet<string, RestrictionSheet.Row>.Add(string, RestrictionSheet.Row) ,
Sheet<string, RestrictionSheet.Row>.ContainsKey(string) ,
Sheet<string, RestrictionSheet.Row>.Remove(string) ,
Sheet<string, RestrictionSheet.Row>.TryGetValue(string, out RestrictionSheet.Row) ,
Sheet<string, RestrictionSheet.Row>.Add(KeyValuePair<string, RestrictionSheet.Row>) ,
Sheet<string, RestrictionSheet.Row>.Clear() ,
Sheet<string, RestrictionSheet.Row>.Contains(KeyValuePair<string,
RestrictionSheet.Row>) ,
Sheet<string, RestrictionSheet.Row>.CopyTo(KeyValuePair<string, RestrictionSheet.Row>
[], int) ,
Sheet<string, RestrictionSheet.Row>.Remove(KeyValuePair<string, RestrictionSheet.Row>)
,
Sheet<string, RestrictionSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RestrictionSheet()

Creates an empty sheet.

```
public RestrictionSheet()
```

# Fields

## ColumnNames

Column names in their expected order, after _ prefixed columns are dropped.

```
public static readonly string[] ColumnNames
```

### Field Value

string[]

# Methods

## AddRow(string, Row)

Adds a parsed row to the sheet.

```
protected override void AddRow(string key, RestrictionSheet.Row value)
```

### Parameters

`key` string

The row's key.

`value` RestrictionSheet.Row

The row to add.

### Remarks

Blank keys are dropped and duplicate keys are merged toward the restrictive value instead of throwing, because a throw here would take the entire policy offline and with it every restriction the sheet carries.

## IsCurrencyMarketRegistrable(string)

Whether the currency may be registered on the market as a fungible asset product. True when unlisted or unspecified.

```
public bool IsCurrencyMarketRegistrable(string ticker)
```

### Parameters

`ticker` string

### Returns

bool

## IsItemMarketRegistrable(int)

Whether the item may be registered on the market. True when unlisted or unspecified.

```
public bool IsItemMarketRegistrable(int itemId)
```

### Parameters

`itemId` int

### Returns

bool

## IsItemSynthesizeMaterial(int)

Whether the item may be consumed as a synthesis material. True when unlisted or unspecified.

```
public bool IsItemSynthesizeMaterial(int itemId)
```

### Parameters

`itemId` int

### Returns

bool

## ToKey(int)

Renders an item id the way a row key spells it.

```
public static string ToKey(int itemId)
```

### Parameters

`itemId` int

### Returns

string

## ValidateCsv(string)

Parses `csv` strictly so that a malformed policy fails the `patch_table_sheet` transaction instead of silently landing on the chain, where a row that no lookup can match reads as "unrestricted".

```
public static void ValidateCsv(string csv)
```

### Parameters

csv string

The sheet's CSV, as the patch would write it.

## Exceptions

SheetRowValidateException

The header does not match ColumnNames, or a row has no key, a key no lookup can produce, an unparsable policy cell, no policy at all, a synthesis policy on a currency, or a key an earlier row already claimed.

# Class RestrictionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One target and what it may not be used for.

```
[Serializable]
public class RestrictionSheet.Row : SheetRow<string>
```

**Inheritance**

object ← SheetRow<string> ← RestrictionSheet.Row

**Inherited Members**

SheetRow<string>.Validate() , SheetRow<string>.EndOfSheetInitialize() , SheetRow<string>.Equals(object) , SheetRow<string>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

The item id or ticker this row restricts, which is also how it is looked up.

```
public override string Key { get; }
```

### Property Value

string

## MarketRegistrable

Whether the target may be registered on the market. `null` when the cell is empty(= unspecified).

```
public bool? MarketRegistrable { get; }
```

### Property Value

bool?

## SynthesizeMaterial

Whether the target may be consumed as a synthesis material. `null` when the cell is empty(= unspecified).

```
public bool? SynthesizeMaterial { get; }
```

### Property Value

bool?

## Target

An item id in its string form, or a currency ticker.

```
public string Target { get; }
```

### Property Value

string

# Methods

## Set(IReadOnlyList<string>)

Reads one row.

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

The row's cells, with "_" prefixed columns already dropped.

## Remarks

Total by design: Sheet<TKey, TValue> does not guard this call, so a throw here would abort the whole sheet and take every restriction with it.

# Enum RewardType

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public enum RewardType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Gold = 1`

`Item = 0`

# Class RuneCostSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a sheet containing rune cost data organized by level ranges. This sheet stores cost information for rune enhancement across different level intervals, allowing for efficient data compression and retrieval.

```
[Serializable]
public class RuneCostSheet : Sheet<int, RuneCostSheet.Row>, IDictionary<int,
RuneCostSheet.Row>, ICollection<KeyValuePair<int, RuneCostSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneCostSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, RuneCostSheet.Row> ← RuneCostSheet

### Implements

IDictionary<int, RuneCostSheet.Row>,
ICollection<KeyValuePair<int, RuneCostSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneCostSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, RuneCostSheet.Row>.Name , Sheet<int, RuneCostSheet.Row>.OrderedList ,
Sheet<int, RuneCostSheet.Row>.First , Sheet<int, RuneCostSheet.Row>.Last ,
Sheet<int, RuneCostSheet.Row>.Keys , Sheet<int, RuneCostSheet.Row>.Values ,
Sheet<int, RuneCostSheet.Row>.Count , Sheet<int, RuneCostSheet.Row>.IsReadOnly ,
Sheet<int, RuneCostSheet.Row>.this[int] ,
Sheet<int, RuneCostSheet.Row>.Set(string, bool) ,
Sheet<int, RuneCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneCostSheet.Row>.GetEnumerator() ,
Sheet<int, RuneCostSheet.Row>.TryGetValue(int, out RuneCostSheet.Row, bool) ,
Sheet<int, RuneCostSheet.Row>.AddRow(int, RuneCostSheet.Row) ,
Sheet<int, RuneCostSheet.Row>.Add(int, RuneCostSheet.Row) ,
Sheet<int, RuneCostSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneCostSheet.Row>.Remove(int) ,
Sheet<int, RuneCostSheet.Row>.TryGetValue(int, out RuneCostSheet.Row) ,
Sheet<int, RuneCostSheet.Row>.Add(KeyValuePair<int, RuneCostSheet.Row>) ,
Sheet<int, RuneCostSheet.Row>.Clear() ,
Sheet<int, RuneCostSheet.Row>.Contains(KeyValuePair<int, RuneCostSheet.Row>) ,

Sheet<int, RuneCostSheet.Row>.CopyTo(KeyValuePair<int, RuneCostSheet.Row>[], int) , Sheet<int, RuneCostSheet.Row>.Remove(KeyValuePair<int, RuneCostSheet.Row>) , Sheet<int, RuneCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RuneCostSheet()

Initializes a new instance of the RuneCostSheet class.

```
public RuneCostSheet()
```

# Methods

## AddRow(int, Row)

Adds a row to the sheet, handling cases where the key already exists. If a row with the same key exists, the cost data is merged.

```
protected override void AddRow(int key, RuneCostSheet.Row value)
```

### Parameters

`key` int

The unique identifier for the row

`value` RuneCostSheet.Row

The row data to add

# Class RuneCostSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the RuneCostSheet containing cost data for a specific rune. Each row can contain multiple cost data entries for different level ranges.

```
[Serializable]
public class RuneCostSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← RuneCostSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Cost

Gets the collection of cost data entries for different level ranges.

```
public List<RuneCostSheet.RuneCostData> Cost { get; }
```

### Property Value

List<RuneCostSheet.RuneCostData>

## Key

Gets the unique identifier for this row, which is the rune ID.

```
public override int Key { get; }
```

### Property Value

int

## RuneId

Gets the rune identifier associated with this cost data.

```
public int RuneId { get; }
```

### Property Value

int

# Methods

## GetCostsInRange(int, int)

Retrieves all cost data entries that overlap with the specified level range.

```
public IEnumerable<RuneCostSheet.RuneCostData> GetCostsInRange(int startLevel,
int endLevel)
```

### Parameters

`startLevel` int

The starting level of the range (inclusive)

`endLevel` int

The ending level of the range (inclusive)

### Returns

IEnumerable<RuneCostSheet.RuneCostData>

An enumerable collection of cost data entries that overlap with the specified range

## Set(IReadOnlyList<string>)

Sets the row data from the provided CSV fields.

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

The CSV fields containing rune cost information

## TryGetCost(int, out RuneCostData)

Attempts to find cost data for a specific level.

```
public bool TryGetCost(int level, out RuneCostSheet.RuneCostData costData)
```

### Parameters

`level` int

The level to search for

`costData` RuneCostSheet.RuneCostData

When this method returns, contains the cost data for the specified level, if found; otherwise, null

### Returns

bool

true if cost data was found for the specified level; otherwise, false

# Class RuneCostSheet.RuneCostData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents cost data for a specific level range of rune enhancement. Contains information about required materials and success rates for a given level interval.

```
[Serializable]
public class RuneCostSheet.RuneCostData
```

**Inheritance**

object ← RuneCostSheet.RuneCostData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RuneCostData(int, int, int, int, int, int)

Initializes a new instance of the RuneCostData class with specified parameters.

```
public RuneCostData(int levelStart, int levelEnd, int runeStoneQuantity, int
crystalQuantity, int ncgQuantity, int levelUpSuccessRate)
```

#### Parameters

`levelStart` int

The starting level of the cost range

`levelEnd` int

The ending level of the cost range

`runeStoneQuantity` int

Required rune stone quantity

`crystalQuantity` int

Required crystal quantity

`ncgQuantity` int

Required NCG quantity

`levelUpSuccessRate` int

Success rate for enhancement

# Properties

## CrystalQuantity

The quantity of crystals required for enhancement

```
public int CrystalQuantity { get; }
```

### Property Value

int

## LevelEnd

The ending level of this cost range (inclusive)

```
public int LevelEnd { get; }
```

### Property Value

int

## LevelStart

The starting level of this cost range (inclusive)

```
public int LevelStart { get; }
```

### Property Value

int

## LevelUpSuccessRate

The success rate percentage for rune enhancement (in basis points, e.g., 10000 = 100%)

```
public int LevelUpSuccessRate { get; }
```

### Property Value

int

## NcgQuantity

The quantity of NCG (Nine Chronicles Gold) required for enhancement

```
public int NcgQuantity { get; }
```

### Property Value

int

## RuneStoneQuantity

The quantity of rune stones required for enhancement

```
public int RuneStoneQuantity { get; }
```

### Property Value

int

# Methods

## ContainsLevel(int)

Determines whether the specified level falls within this cost range.

```
public bool ContainsLevel(int level)
```

### Parameters

`level` int

The level to check

### Returns

bool

true if the level is within the range (inclusive); otherwise, false

# Class RuneOptionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RuneOptionSheet : Sheet<int, RuneOptionSheet.Row>, IDictionary<int,
RuneOptionSheet.Row>, ICollection<KeyValuePair<int, RuneOptionSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneOptionSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, RuneOptionSheet.Row> ← RuneOptionSheet

### Implements

IDictionary<int, RuneOptionSheet.Row>,
ICollection<KeyValuePair<int, RuneOptionSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneOptionSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, RuneOptionSheet.Row>.Name ,
Sheet<int, RuneOptionSheet.Row>.OrderedList , Sheet<int, RuneOptionSheet.Row>.First ,
Sheet<int, RuneOptionSheet.Row>.Last , Sheet<int, RuneOptionSheet.Row>.Keys ,
Sheet<int, RuneOptionSheet.Row>.Values , Sheet<int, RuneOptionSheet.Row>.Count ,
Sheet<int, RuneOptionSheet.Row>.IsReadOnly ,
Sheet<int, RuneOptionSheet.Row>.this[int] ,
Sheet<int, RuneOptionSheet.Row>.Set(string, bool) ,
Sheet<int, RuneOptionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneOptionSheet.Row>.GetEnumerator() ,
Sheet<int, RuneOptionSheet.Row>.TryGetValue(int, out RuneOptionSheet.Row, bool) ,
Sheet<int, RuneOptionSheet.Row>.AddRow(int, RuneOptionSheet.Row) ,
Sheet<int, RuneOptionSheet.Row>.Add(int, RuneOptionSheet.Row) ,
Sheet<int, RuneOptionSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneOptionSheet.Row>.Remove(int) ,
Sheet<int, RuneOptionSheet.Row>.TryGetValue(int, out RuneOptionSheet.Row) ,
Sheet<int, RuneOptionSheet.Row>.Add(KeyValuePair<int, RuneOptionSheet.Row>) ,
Sheet<int, RuneOptionSheet.Row>.Clear() ,
Sheet<int, RuneOptionSheet.Row>.Contains(KeyValuePair<int, RuneOptionSheet.Row>) ,
Sheet<int, RuneOptionSheet.Row>.CopyTo(KeyValuePair<int, RuneOptionSheet.Row>[],
int) ,

Sheet<int, RuneOptionSheet.Row>.Remove(KeyValuePair<int, RuneOptionSheet.Row>) , Sheet<int, RuneOptionSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RuneOptionSheet()

```
public RuneOptionSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, RuneOptionSheet.Row value)
```

### Parameters

`key` int

`value` RuneOptionSheet.Row

## TryGetOptionInfo(int, int, out RuneOptionInfo)

```
public bool TryGetOptionInfo(int runeId, int level, out
RuneOptionSheet.Row.RuneOptionInfo optionInfo)
```

### Parameters

`runeId` int

`level` int

`optionInfo` RuneOptionSheet.Row.RuneOptionInfo

## Returns

bool

# Class RuneOptionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RuneOptionSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← RuneOptionSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

## LevelOptionMap

```
public Dictionary<int, RuneOptionSheet.Row.RuneOptionInfo> LevelOptionMap { get; }
```

### Property Value

Dictionary<int, RuneOptionSheet.Row.RuneOptionInfo>

## RuneId

```
public int RuneId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class RuneOptionSheet.Row.RuneOptionInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneOptionSheet.Row.RuneOptionInfo
```

**Inheritance**

object ← RuneOptionSheet.Row.RuneOptionInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RuneOptionInfo(long, List<(DecimalStat, OperationType)>)

```
public RuneOptionInfo(long cp, List<(DecimalStat, StatModifier.OperationType)>
stats)
```

#### Parameters

`cp` long

`stats` List<(DecimalStat stat, StatModifier.OperationType operationType)>

### RuneOptionInfo(long, List<(DecimalStat, OperationType)>, int, int, int, decimal, OperationType, StatType, StatReferenceType, int)

```
public RuneOptionInfo(long cp, List<(DecimalStat, StatModifier.OperationType)>
stats, int skillId, int skillCooldown, int skillChance, decimal skillValue,
StatModifier.OperationType skillValueType, StatType skillStatType, StatReferenceType
statReferenceType, int buffDuration)
```

### Parameters

`cp` long

`stats` List<(DecimalStat stat, StatModifier.OperationType operationType)>

`skillId` int

`skillCooldown` int

`skillChance` int

`skillValue` decimal

`skillValueType` StatModifier.OperationType

`skillStatType` StatType

`statReferenceType` StatReferenceType

`buffDuration` int

## Properties

## BuffDuration

```
public int BuffDuration { get; set; }
```

### Property Value

int

## Cp

```
public long Cp { get; }
```

### Property Value

long

## SkillChance

```
public int SkillChance { get; set; }
```

### Property Value

int

## SkillCooldown

```
public int SkillCooldown { get; set; }
```

### Property Value

int

## SkillId

```
public int SkillId { get; set; }
```

### Property Value

int

## SkillStatType

```
public StatType SkillStatType { get; set; }
```

### Property Value

StatType

## SkillValue

```
public decimal SkillValue { get; set; }
```

### Property Value

decimal

## SkillValueType

```
public StatModifier.OperationType SkillValueType { get; set; }
```

### Property Value

StatModifier.OperationType

## StatReferenceType

```
public StatReferenceType StatReferenceType { get; set; }
```

### Property Value

StatReferenceType

## Stats

```
public List<(DecimalStat stat, StatModifier.OperationType operationType)> Stats {
get; set; }
```

## Property Value

List<(DecimalStat stat, StatModifier.OperationType operationType)>

# Class RuneSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneSheet : Sheet<int, RuneSheet.Row>, IDictionary<int, RuneSheet.Row>,
ICollection<KeyValuePair<int, RuneSheet.Row>>, IEnumerable<KeyValuePair<int,
RuneSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, RuneSheet.Row> ← RuneSheet

### Implements

IDictionary<int, RuneSheet.Row>,
ICollection<KeyValuePair<int, RuneSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, RuneSheet.Row>.Name , Sheet<int, RuneSheet.Row>.OrderedList ,
Sheet<int, RuneSheet.Row>.First , Sheet<int, RuneSheet.Row>.Last ,
Sheet<int, RuneSheet.Row>.Keys , Sheet<int, RuneSheet.Row>.Values ,
Sheet<int, RuneSheet.Row>.Count , Sheet<int, RuneSheet.Row>.IsReadOnly ,
Sheet<int, RuneSheet.Row>.this[int] , Sheet<int, RuneSheet.Row>.Set(string, bool) ,
Sheet<int, RuneSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneSheet.Row>.GetEnumerator() ,
Sheet<int, RuneSheet.Row>.TryGetValue(int, out RuneSheet.Row, bool) ,
Sheet<int, RuneSheet.Row>.AddRow(int, RuneSheet.Row) ,
Sheet<int, RuneSheet.Row>.Add(int, RuneSheet.Row) ,
Sheet<int, RuneSheet.Row>.ContainsKey(int) , Sheet<int, RuneSheet.Row>.Remove(int) ,
Sheet<int, RuneSheet.Row>.TryGetValue(int, out RuneSheet.Row) ,
Sheet<int, RuneSheet.Row>.Add(KeyValuePair<int, RuneSheet.Row>) ,
Sheet<int, RuneSheet.Row>.Clear() ,
Sheet<int, RuneSheet.Row>.Contains(KeyValuePair<int, RuneSheet.Row>) ,
Sheet<int, RuneSheet.Row>.CopyTo(KeyValuePair<int, RuneSheet.Row>[], int) ,
Sheet<int, RuneSheet.Row>.Remove(KeyValuePair<int, RuneSheet.Row>) ,
Sheet<int, RuneSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RuneSheet()

```
public RuneSheet()
```

# Class RuneSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← RuneSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Id

```
public int Id
```

### Field Value

int

## Ticker

```
public string Ticker
```

### Field Value

string

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class RuneWeightSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneWeightSheet : Sheet<int, RuneWeightSheet.Row>, IDictionary<int,
RuneWeightSheet.Row>, ICollection<KeyValuePair<int, RuneWeightSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneWeightSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, RuneWeightSheet.Row> ← RuneWeightSheet

### Implements

IDictionary<int, RuneWeightSheet.Row>,
ICollection<KeyValuePair<int, RuneWeightSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneWeightSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, RuneWeightSheet.Row>.Name ,
Sheet<int, RuneWeightSheet.Row>.OrderedList , Sheet<int, RuneWeightSheet.Row>.First ,
Sheet<int, RuneWeightSheet.Row>.Last , Sheet<int, RuneWeightSheet.Row>.Keys ,
Sheet<int, RuneWeightSheet.Row>.Values , Sheet<int, RuneWeightSheet.Row>.Count ,
Sheet<int, RuneWeightSheet.Row>.IsReadOnly ,
Sheet<int, RuneWeightSheet.Row>.this[int] ,
Sheet<int, RuneWeightSheet.Row>.Set(string, bool) ,
Sheet<int, RuneWeightSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneWeightSheet.Row>.GetEnumerator() ,
Sheet<int, RuneWeightSheet.Row>.TryGetValue(int, out RuneWeightSheet.Row, bool) ,
Sheet<int, RuneWeightSheet.Row>.AddRow(int, RuneWeightSheet.Row) ,
Sheet<int, RuneWeightSheet.Row>.Add(int, RuneWeightSheet.Row) ,
Sheet<int, RuneWeightSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneWeightSheet.Row>.Remove(int) ,
Sheet<int, RuneWeightSheet.Row>.TryGetValue(int, out RuneWeightSheet.Row) ,
Sheet<int, RuneWeightSheet.Row>.Add(KeyValuePair<int, RuneWeightSheet.Row>) ,
Sheet<int, RuneWeightSheet.Row>.Clear() ,
Sheet<int, RuneWeightSheet.Row>.Contains(KeyValuePair<int, RuneWeightSheet.Row>) ,
Sheet<int, RuneWeightSheet.Row>.CopyTo(KeyValuePair<int, RuneWeightSheet.Row>[],
int) ,
Sheet<int, RuneWeightSheet.Row>.Remove(KeyValuePair<int, RuneWeightSheet.Row>) ,

Sheet<int, RuneWeightSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RuneWeightSheet()

```
public RuneWeightSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, RuneWeightSheet.Row value)
```

### Parameters

`key` int

`value` RuneWeightSheet.Row

# Class RuneWeightSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneWeightSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← RuneWeightSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BossId

```
public int BossId
```

### Field Value

int

## Id

```
public int Id
```

### Field Value

int

## Rank

```
public int Rank
```

### Field Value

int

## RuneInfos

```
public List<RuneWeightSheet.RuneInfo> RuneInfos
```

### Field Value

List<RuneWeightSheet.RuneInfo>

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class RuneWeightSheet.RuneInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneWeightSheet.RuneInfo
```

**Inheritance**

object ← RuneWeightSheet.RuneInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RuneInfo(int, decimal)

```
public RuneInfo(int runeId, decimal weight)
```

#### Parameters

`runeId` int

`weight` decimal

## Fields

### RuneId

```
public int RuneId
```

#### Field Value

int

## Weight

```
public decimal Weight
```

### Field Value

decimal

# Class SetEffectExtension

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public static class SetEffectExtension
```

**Inheritance**

object ← SetEffectExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### GetSetEffectRows(EquipmentItemSetEffectSheet, IEnumerable<Equipment>)

```
public static List<EquipmentItemSetEffectSheet.Row> GetSetEffectRows(this
EquipmentItemSetEffectSheet sheet, IEnumerable<Equipment> equipments)
```

#### Parameters

`sheet` EquipmentItemSetEffectSheet

`equipments` IEnumerable<Equipment>

#### Returns

List<EquipmentItemSetEffectSheet.Row>

# Class SheetRowColumnException

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SheetRowColumnException : Exception, ISerializable
```

### Inheritance

object ← Exception ← SheetRowColumnException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SheetRowColumnException(SerializationInfo, StreamingContext)

```
protected SheetRowColumnException(SerializationInfo info, StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# SheetRowColumnException(string)

```
public SheetRowColumnException(string message)
```

## Parameters

`message` string

# Class SheetRowNotFoundException

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SheetRowNotFoundException : Exception, ISerializable
```

### Inheritance

object ← Exception ← SheetRowNotFoundException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SheetRowNotFoundException(SerializationInfo, StreamingContext)

```
protected SheetRowNotFoundException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

## SheetRowNotFoundException(string)

```
public SheetRowNotFoundException(string message)
```

### Parameters

`message` string

## SheetRowNotFoundException(string, int)

```
public SheetRowNotFoundException(string sheetName, int intKey)
```

### Parameters

`sheetName` string

`intKey` int

## SheetRowNotFoundException(string, long)

```
public SheetRowNotFoundException(string sheetName, long longKey)
```

### Parameters

`sheetName` string

`longKey` long

## SheetRowNotFoundException(string, string)

```
public SheetRowNotFoundException(string sheetName, string key)
```

### Parameters

`sheetName` string

key string

## SheetRowNotFoundException(string, string, int)

```
public SheetRowNotFoundException(string addressesHex, string sheetName, int intKey)
```

### Parameters

addressesHex string

sheetName string

intKey int

## SheetRowNotFoundException(string, string, string)

```
public SheetRowNotFoundException(string sheetName, string condition, string value)
```

### Parameters

sheetName string

condition string

value string

## SheetRowNotFoundException(string, string, string, int)

```
public SheetRowNotFoundException(string actionType, string addressesHex, string
sheetName, int intKey)
```

### Parameters

actionType string

addressesHex string

`sheetName` string

`intKey` int

# Class SheetRowValidateException

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SheetRowValidateException : Exception, ISerializable
```

### Inheritance

object ← Exception ← SheetRowValidateException

### Implements

ISerializable

### Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

## Constructors

### SheetRowValidateException(SerializationInfo, StreamingContext)

```
protected SheetRowValidateException(SerializationInfo info,
StreamingContext context)
```

#### Parameters

`info` SerializationInfo

`context` StreamingContext

# SheetRowValidateException(string)

```
public SheetRowValidateException(string message)
```

## Parameters

`message` string

# Class SheetRow<T>

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public abstract class SheetRow<T>
```

## Type Parameters

T

### Inheritance

object ← SheetRow<T>

### Derived

ActionBuffSheet.Row, AdventureBossContributionRewardSheet.Row, AdventureBossFloorFirstRewardSheet.Row, AdventureBossFloorPointSheet.Row, AdventureBossFloorSheet.Row, AdventureBossFloorWaveSheet.Row, AdventureBossNcgRewardRatioSheet.Row, AdventureBossSheet.Row, AdventureBossUnlockFloorCostSheet.Row, AdventureBossWantedRewardSheet.Row, ArenaSheet.Row, BoostScheduleSheet.Row, BuffLimitSheet.Row, BuffLinkSheet.Row, BuffSheet.Row, CharacterLevelSheet.Row, CharacterSheet.Row, ClaimableGiftsSheet.Row, CollectionSheet.Row, ConsumableItemRecipeSheet.Row, CostumeStatSheet.Row, CreateAvatarFavSheet.Row, CreateAvatarItemSheet.Row, CrystalEquipmentGrindingSheet.Row, CrystalFluctuationSheet.Row, CrystalHammerPointSheet.Row, CrystalMaterialCostSheet.Row, CrystalMonsterCollectionMultiplierSheet.Row, CrystalRandomBuffSheet.Row, CrystalStageBuffGachaSheet.Row, CustomEquipmentCraftIconSheet.Row, CustomEquipmentCraftOptionSheet.Row, CustomEquipmentCraftRecipeSheet.Row, CustomEquipmentCraftRecipeSkillSheet.Row, CustomEquipmentCraftRelationshipSheet.Row, EnemySkillSheet.Row, EnhancementCostSheet.Row, EquipmentItemOptionSheet.Row, EquipmentItemRecipeSheet.Row, EquipmentItemSetEffectSheet.Row, EquipmentItemSubRecipeSheet.Row, EquipmentItemSubRecipeSheetV2.Row, EventMaterialItemRecipeSheet.Row, EventScheduleSheet.Row, PatrolRewardSheet.Row, GameConfigSheet.Row, LoadIntoMyGaragesCostSheet.Row, GrandFinaleParticipantsSheet.Row, GrandFinaleScheduleSheet.Row, InfiniteTowerConditionSheet.Row, InfiniteTowerFloorSheet.Row,

InfiniteTowerFloorWaveSheet.Row, InfiniteTowerScheduleSheet.Row, ItemConfigForGradeSheet.Row, ItemRequirementSheet.Row, ItemSheet.RowBase, MimisbrunnrSheet.Row, MonsterCollectionRewardSheet.Row, MonsterCollectionSheet.Row, PetOptionSheet.Row, PetCostSheet.Row, PetSheet.Row, QuestItemRewardSheet.Row, QuestRewardSheet.Row, QuestSheet.Row, RedeemCodeListSheet.Row, RedeemRewardSheet.Row, RestrictionSheet.Row, RuneLevelBonusSheet.Row, RuneListSheet.Row, RuneCostSheet.Row, RuneOptionSheet.Row, RuneSheet.Row, RuneWeightSheet.Row, SkillActionBuffSheet.Row, SkillBuffSheet.Row, SkillSheet.Row, StageDialogSheet.Row, StageSheet.Row, StageWaveSheet.Row, StakePolicySheet.Row, StakeAchievementRewardSheet.Row, StakeActionPointCoefficientSheet.Row, StakeRegularFixedRewardSheet.Row, StakeRegularRewardSheet.Row, StatBuffSheet.Row, SummonSheet.Row, SwapRateSheet.Row, SweepRequiredCPSheet.Row, SynthesizeSheet.Row, SynthesizeWeightSheet.Row, UnlockCombinationSlotCostSheet.Row, WeeklyArenaRewardSheet.Row, WorldBossActionPatternSheet.Row, WorldBossBattleRewardSheet.Row, WorldBossCharacterSheet.Row, WorldBossContributionRewardSheet.Row, WorldBossGlobalHpSheet.Row, WorldBossKillRewardSheet.Row, WorldBossListSheet.Row, WorldBossRankRewardSheet.Row, WorldBossRankingRewardSheet.Row, WorldSheet.Row, WorldUnlockSheet.Row

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

```
public abstract T Key { get; }
```

### Property Value

T

# Methods

## EndOfSheetInitialize()

```
public virtual void EndOfSheetInitialize()
```

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## Set(IReadOnlyList<string>)

```
public abstract void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

## Validate()

```
public virtual void Validate()
```

# Class Sheet<TKey, TValue>

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Sheet<TKey, TValue> : IDictionary<TKey, TValue>,
ICollection<KeyValuePair<TKey, TValue>>, IEnumerable<KeyValuePair<TKey, TValue>>,
IEnumerable, ISheet where TKey : notnull where TValue : SheetRow<TKey>, new()
```

## Type Parameters

TKey

TValue

### Inheritance

object ← Sheet<TKey, TValue>

### Implements

IDictionary<TKey, TValue>, ICollection<KeyValuePair<TKey, TValue>>, IEnumerable<KeyValuePair<TKey, TValue>>, IEnumerable, ISheet

### Derived

ActionBuffSheet, AdventureBossContributionRewardSheet, AdventureBossFloorFirstRewardSheet, AdventureBossFloorPointSheet, AdventureBossFloorSheet, AdventureBossFloorWaveSheet, AdventureBossNcgRewardRatioSheet, AdventureBossSheet, AdventureBossUnlockFloorCostSheet, AdventureBossWantedRewardSheet, ArenaSheet, BoostScheduleSheet, BuffLimitSheet, BuffLinkSheet, BuffSheet, CharacterLevelSheet, CharacterSheet, ClaimableGiftsSheet, CollectQuestSheet, CollectionSheet, CombinationEquipmentQuestSheet, CombinationQuestSheet, ConsumableItemRecipeSheet, ConsumableItemSheet, CostumeItemSheet, CostumeStatSheet, CreateAvatarFavSheet, CreateAvatarItemSheet, CrystalEquipmentGrindingSheet, CrystalFluctuationSheet, CrystalHammerPointSheet, CrystalMaterialCostSheet, CrystalMonsterCollectionMultiplierSheet, CrystalRandomBuffSheet, CrystalStageBuffGachaSheet, CustomEquipmentCraftIconSheet, CustomEquipmentCraftOptionSheet, CustomEquipmentCraftRecipeSheet, CustomEquipmentCraftRecipeSkillSheet, CustomEquipmentCraftRelationshipSheet, EnemySkillSheet, EnhancementCostSheet, EnhancementCostSheetV2,

EnhancementCostSheetV3, EquipmentItemOptionSheet, EquipmentItemRecipeSheet, EquipmentItemSetEffectSheet, EquipmentItemSheet, EquipmentItemSubRecipeSheet, EquipmentItemSubRecipeSheetV2, EventConsumableItemRecipeSheet, EventDungeonSheet, EventDungeonStageSheet, EventDungeonStageWaveSheet, EventMaterialItemRecipeSheet, EventScheduleSheet, PatrolRewardSheet, GameConfigSheet, LoadIntoMyGaragesCostSheet, GeneralQuestSheet, GoldQuestSheet, GrandFinaleParticipantsSheet, GrandFinaleScheduleSheet, InfiniteTowerConditionSheet, InfiniteTowerFloorSheet, InfiniteTowerFloorWaveSheet, InfiniteTowerScheduleSheet, ItemConfigForGradeSheet, ItemEnhancementQuestSheet, ItemGradeQuestSheet, ItemRequirementSheet, ItemSheet, ItemTypeCollectQuestSheet, MaterialItemSheet, MimisbrunnrSheet, MonsterCollectionRewardSheet, MonsterCollectionSheet, MonsterQuestSheet, PetOptionSheet, PetCostSheet, PetSheet, QuestItemRewardSheet, QuestRewardSheet, QuestSheet, RedeemCodeListSheet, RedeemRewardSheet, RestrictionSheet, RuneLevelBonusSheet, RuneListSheet, RuneCostSheet, RuneOptionSheet, RuneSheet, RuneWeightSheet, SkillActionBuffSheet, SkillBuffSheet, SkillSheet, StageDialogSheet, StageSheet, StageWaveSheet, StakePolicySheet, StakeAchievementRewardSheet, StakeActionPointCoefficientSheet, StakeRegularFixedRewardSheet, StakeRegularRewardSheet, StatBuffSheet, SummonSheet, SwapRateSheet, SweepRequiredCPSheet, SynthesizeSheet, SynthesizeWeightSheet, TradeQuestSheet, UnlockCombinationSlotCostSheet, WeeklyArenaRewardSheet, WorldBossActionPatternSheet, WorldBossBattleRewardSheet, WorldBossCharacterSheet, WorldBossContributionRewardSheet, WorldBossGlobalHpSheet, WorldBossKillRewardSheet, WorldBossListSheet, WorldBossRankRewardSheet, WorldBossRankingRewardSheet, WorldQuestSheet, WorldSheet, WorldUnlockSheet

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## Sheet(string)

```
protected Sheet(string name)
```

### Parameters

`name` string

# Properties

## Count

Gets the number of elements contained in the ICollection<T>.

```
public int Count { get; }
```

### Property Value

int

The number of elements contained in the ICollection<T>.

## First

```
public TValue? First { get; }
```

### Property Value

TValue

## IsReadOnly

Gets a value indicating whether the ICollection<T> is read-only.

```
public bool IsReadOnly { get; }
```

### Property Value

bool

true if the ICollection<T> is read-only; otherwise, false.

# this[TKey]

Gets or sets the element with the specified key.

```
public TValue this[TKey key] { get; set; }
```

## Parameters

`key` TKey

The key of the element to get or set.

## Property Value

TValue

The element with the specified key.

## Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

NotSupportedException

The property is set and the IDictionary<TKey, TValue> is read-only.

# Keys

Gets an ICollection<T> containing the keys of the IDictionary<TKey, TValue>.

```
public ICollection<TKey> Keys { get; }
```

## Property Value

ICollection<TKey>

An ICollection<T> containing the keys of the object that implements IDictionary<TKey, TValue>.

## Last

```
public TValue? Last { get; }
```

### Property Value

TValue

## Name

```
public string Name { get; }
```

### Property Value

string

## OrderedList

```
public IReadOnlyList<TValue>? OrderedList { get; }
```

### Property Value

IReadOnlyList<TValue>

## Values

Gets an ICollection<T> containing the values in the IDictionary<TKey, TValue>.

```
public ICollection<TValue> Values { get; }
```

### Property Value

ICollection<TValue>

An ICollection<T> containing the values in the object that implements IDictionary<TKey, TValue>.

## Methods

## Add(KeyValuePair<TKey, TValue>)

Adds an item to the ICollection<T>.

```
public void Add(KeyValuePair<TKey, TValue> item)
```

### Parameters

`item` KeyValuePair<TKey, TValue>

The object to add to the ICollection<T>.

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Add(TKey, TValue)

Adds an element with the provided key and value to the IDictionary<TKey, TValue>.

```
public void Add(TKey key, TValue value)
```

### Parameters

`key` TKey

The object to use as the key of the element to add.

`value` TValue

The object to use as the value of the element to add.

### Exceptions

ArgumentNullException

`key` is null.

ArgumentException

An element with the same key already exists in the IDictionary<TKey, TValue>.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

## AddRow(TKey, TValue)

```
protected virtual void AddRow(TKey key, TValue value)
```

### Parameters

`key` TKey

`value` TValue

## Clear()

Removes all items from the ICollection<T>.

```
public void Clear()
```

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

# Contains(KeyValuePair<TKey, TValue>)

Determines whether the ICollection<T> contains a specific value.

```
public bool Contains(KeyValuePair<TKey, TValue> item)
```

## Parameters

`item` KeyValuePair<TKey, TValue>

The object to locate in the ICollection<T>.

## Returns

bool

true if `item` is found in the ICollection<T>; otherwise, false.

# ContainsKey(TKey)

Determines whether the IDictionary<TKey, TValue> contains an element with the specified key.

```
public bool ContainsKey(TKey key)
```

## Parameters

`key` TKey

The key to locate in the IDictionary<TKey, TValue>.

## Returns

bool

true if the IDictionary<TKey, TValue> contains an element with the key; otherwise, false.

## Exceptions

ArgumentNullException

`key` is null.

## CopyTo(KeyValuePair<TKey, TValue>[], int)

Copies the elements of the ICollection<T> to an Array, starting at a particular Array index.

```
public void CopyTo(KeyValuePair<TKey, TValue>[] array, int arrayIndex)
```

### Parameters

`array` KeyValuePair<TKey, TValue>[]

The one-dimensional Array that is the destination of the elements copied from ICollection<T>. The Array must have zero-based indexing.

`arrayIndex` int

The zero-based index in `array` at which copying begins.

### Exceptions

ArgumentNullException

`array` is null.

ArgumentOutOfRangeException

`arrayIndex` is less than 0.

ArgumentException

The number of elements in the source ICollection<T> is greater than the available space from `arrayIndex` to the end of the destination `array`.

## GetEnumerator()

```
public IEnumerator<TValue> GetEnumerator()
```

### Returns

IEnumerator<TValue>

## Remove(KeyValuePair<TKey, TValue>)

Removes the first occurrence of a specific object from the ICollection<T>.

```
public bool Remove(KeyValuePair<TKey, TValue> item)
```

### Parameters

`item` KeyValuePair<TKey, TValue>

The object to remove from the ICollection<T>.

### Returns

bool

true if `item` was successfully removed from the ICollection<T>; otherwise, false. This method also returns false if `item` is not found in the original ICollection<T>.

### Exceptions

NotSupportedException

The ICollection<T> is read-only.

## Remove(TKey)

Removes the element with the specified key from the IDictionary<TKey, TValue>.

```
public bool Remove(TKey key)
```

### Parameters

`key` TKey

The key of the element to remove.

### Returns

bool

true if the element is successfully removed; otherwise, false. This method also returns false if `key` was not found in the original IDictionary<TKey, TValue>.

### Exceptions

ArgumentNullException

`key` is null.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Set(string, bool)

```
public virtual void Set(string csv, bool isReversed = false)
```

### Parameters

`csv` string

`isReversed` bool

true: csv값의 column과 row가 뒤집혀서 작성되어 있다고 판단합니다.

### Exceptions

ArgumentNullException

InvalidDataException

## Set<T>(Sheet<TKey, T>, bool)

```
public void Set<T>(Sheet<TKey, T> sheet, bool executePostSet = true) where T :
TValue, new()
```

### Parameters

`sheet` Sheet<TKey, T>

`executePostSet` bool

### Type Parameters

`T`

## TryGetValue(TKey, out TValue)

Gets the value associated with the specified key.

```
public bool TryGetValue(TKey key, out TValue value)
```

### Parameters

`key` TKey

The key whose value to get.

`value` TValue

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

### Returns

bool

true if the object that implements IDictionary<TKey, TValue> contains an element with the specified key; otherwise, false.

### Exceptions

ArgumentNullException

`key` is null.

## TryGetValue(TKey, out TValue, bool)

```
public bool TryGetValue(TKey key, out TValue value, bool throwException)
```

### Parameters

`key` TKey

`value` TValue

`throwException` bool

### Returns

bool

# Class SimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SimulatorSheets : SimulatorSheetsV1
```

**Inheritance**

object ← SimulatorSheetsV1 ← SimulatorSheets

**Derived**

ArenaSimulatorSheets, RaidSimulatorSheets, RankingSimulatorSheets, StageSimulatorSheets

**Inherited Members**

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public SimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet skillSheet,
SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet, SkillActionBuffSheet
skillActionBuffSheet, ActionBuffSheet actionBuffSheet, CharacterSheet
characterSheet, CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet, RuneOptionSheet runeOptionSheet, RuneListSheet
runeListSheet, RuneLevelBonusSheet runeLevelBonusSheet)
```

### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

## Fields

## RuneLevelBonusSheet

```
public readonly RuneLevelBonusSheet RuneLevelBonusSheet
```

### Field Value

RuneLevelBonusSheet

## RuneListSheet

```
public readonly RuneListSheet RuneListSheet
```

### Field Value

RuneListSheet

## RuneOptionSheet

```
public readonly RuneOptionSheet RuneOptionSheet
```

### Field Value

RuneOptionSheet

# Class SimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SimulatorSheetsV1
```

**Inheritance**

object ← SimulatorSheetsV1

**Derived**

ArenaSimulatorSheetsV1, RaidSimulatorSheetsV1, RankingSimulatorSheetsV1, SimulatorSheets, StageSimulatorSheetsV1

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public SimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet skillSheet,
SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

# Fields

## ActionBuffSheet

```
public readonly ActionBuffSheet ActionBuffSheet
```

### Field Value

ActionBuffSheet

## CharacterLevelSheet

```
public readonly CharacterLevelSheet CharacterLevelSheet
```

### Field Value

CharacterLevelSheet

## CharacterSheet

```
public readonly CharacterSheet CharacterSheet
```

### Field Value

CharacterSheet

## EquipmentItemSetEffectSheet

```
public readonly EquipmentItemSetEffectSheet EquipmentItemSetEffectSheet
```

### Field Value

EquipmentItemSetEffectSheet

## MaterialItemSheet

```
public readonly MaterialItemSheet MaterialItemSheet
```

### Field Value

MaterialItemSheet

## SkillActionBuffSheet

```
public readonly SkillActionBuffSheet SkillActionBuffSheet
```

### Field Value

SkillActionBuffSheet

## SkillBuffSheet

```
public readonly SkillBuffSheet SkillBuffSheet
```

### Field Value

SkillBuffSheet

## SkillSheet

```
public readonly SkillSheet SkillSheet
```

### Field Value

SkillSheet

## StatBuffSheet

```
public readonly StatBuffSheet StatBuffSheet
```

### Field Value

StatBuffSheet

# Class SimulatorSheetsV100291

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SimulatorSheetsV100291
```

**Inheritance**

object ← SimulatorSheetsV100291

**Derived**

ArenaSimulatorSheetsV100291, RankingSimulatorSheetsV100291, StageSimulatorSheetsV100291

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### SimulatorSheetsV100291(MaterialItemSheet, SkillSheet, SkillBuffSheet, BuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public SimulatorSheetsV100291(MaterialItemSheet materialItemSheet,
SkillSheet skillSheet, SkillBuffSheet skillBuffSheet, BuffSheet buffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`buffSheet` BuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

# Fields

## BuffSheet

```
public readonly BuffSheet BuffSheet
```

### Field Value

BuffSheet

## CharacterLevelSheet

```
public readonly CharacterLevelSheet CharacterLevelSheet
```

### Field Value

CharacterLevelSheet

## CharacterSheet

```
public readonly CharacterSheet CharacterSheet
```

### Field Value

CharacterSheet

## EquipmentItemSetEffectSheet

```
public readonly EquipmentItemSetEffectSheet EquipmentItemSetEffectSheet
```

### Field Value

EquipmentItemSetEffectSheet

## MaterialItemSheet

```
public readonly MaterialItemSheet MaterialItemSheet
```

### Field Value

MaterialItemSheet

## SkillBuffSheet

```
public readonly SkillBuffSheet SkillBuffSheet
```

### Field Value

SkillBuffSheet

## SkillSheet

```
public readonly SkillSheet SkillSheet
```

### Field Value

SkillSheet

# Methods

## ToSimulatorSheetsV1()

```
public SimulatorSheetsV1 ToSimulatorSheetsV1()
```

## Returns

SimulatorSheetsV1

# Class SkillActionBuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SkillActionBuffSheet : Sheet<int, SkillActionBuffSheet.Row>,
IDictionary<int, SkillActionBuffSheet.Row>, ICollection<KeyValuePair<int,
SkillActionBuffSheet.Row>>, IEnumerable<KeyValuePair<int,
SkillActionBuffSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SkillActionBuffSheet.Row> ← SkillActionBuffSheet

### Implements

IDictionary<int, SkillActionBuffSheet.Row>,
ICollection<KeyValuePair<int, SkillActionBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, SkillActionBuffSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, SkillActionBuffSheet.Row>.Name ,
Sheet<int, SkillActionBuffSheet.Row>.OrderedList ,
Sheet<int, SkillActionBuffSheet.Row>.First , Sheet<int, SkillActionBuffSheet.Row>.Last ,
Sheet<int, SkillActionBuffSheet.Row>.Keys , Sheet<int, SkillActionBuffSheet.Row>.Values ,
Sheet<int, SkillActionBuffSheet.Row>.Count ,
Sheet<int, SkillActionBuffSheet.Row>.IsReadOnly ,
Sheet<int, SkillActionBuffSheet.Row>.this[int] ,
Sheet<int, SkillActionBuffSheet.Row>.Set(string, bool) ,
Sheet<int, SkillActionBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SkillActionBuffSheet.Row>.GetEnumerator() ,
Sheet<int, SkillActionBuffSheet.Row>.TryGetValue(int, out SkillActionBuffSheet.Row, bool) ,
Sheet<int, SkillActionBuffSheet.Row>.AddRow(int, SkillActionBuffSheet.Row) ,
Sheet<int, SkillActionBuffSheet.Row>.Add(int, SkillActionBuffSheet.Row) ,
Sheet<int, SkillActionBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, SkillActionBuffSheet.Row>.Remove(int) ,
Sheet<int, SkillActionBuffSheet.Row>.TryGetValue(int, out SkillActionBuffSheet.Row) ,
Sheet<int, SkillActionBuffSheet.Row>.Add(KeyValuePair<int, SkillActionBuffSheet.Row>) ,
Sheet<int, SkillActionBuffSheet.Row>.Clear() ,
Sheet<int, SkillActionBuffSheet.Row>.Contains(KeyValuePair<int,
SkillActionBuffSheet.Row>) ,

Sheet<int, SkillActionBuffSheet.Row>.CopyTo(KeyValuePair<int, SkillActionBuffSheet.Row>[], int) ,
Sheet<int, SkillActionBuffSheet.Row>.Remove(KeyValuePair<int, SkillActionBuffSheet.Row>) ,
Sheet<int, SkillActionBuffSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SkillActionBuffSheet()

```
public SkillActionBuffSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, SkillActionBuffSheet.Row value)
```

### Parameters

`key` int

`value` SkillActionBuffSheet.Row

# Class SkillActionBuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SkillActionBuffSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← SkillActionBuffSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## BuffIds

```
public List<int> BuffIds { get; }
```

### Property Value

List<int>

## Key

```
public override int Key { get; }
```

### Property Value

int

## SkillId

```
public int SkillId { get; }
```

### Property Value

int

# Methods

## EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class SkillBuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SkillBuffSheet : Sheet<int, SkillBuffSheet.Row>, IDictionary<int,
SkillBuffSheet.Row>, ICollection<KeyValuePair<int, SkillBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, SkillBuffSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SkillBuffSheet.Row> ← SkillBuffSheet

### Implements

IDictionary<int, SkillBuffSheet.Row>,
ICollection<KeyValuePair<int, SkillBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, SkillBuffSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, SkillBuffSheet.Row>.Name , Sheet<int, SkillBuffSheet.Row>.OrderedList ,
Sheet<int, SkillBuffSheet.Row>.First , Sheet<int, SkillBuffSheet.Row>.Last ,
Sheet<int, SkillBuffSheet.Row>.Keys , Sheet<int, SkillBuffSheet.Row>.Values ,
Sheet<int, SkillBuffSheet.Row>.Count , Sheet<int, SkillBuffSheet.Row>.IsReadOnly ,
Sheet<int, SkillBuffSheet.Row>.this[int] , Sheet<int, SkillBuffSheet.Row>.Set(string, bool) ,
Sheet<int, SkillBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SkillBuffSheet.Row>.GetEnumerator() ,
Sheet<int, SkillBuffSheet.Row>.TryGetValue(int, out SkillBuffSheet.Row, bool) ,
Sheet<int, SkillBuffSheet.Row>.AddRow(int, SkillBuffSheet.Row) ,
Sheet<int, SkillBuffSheet.Row>.Add(int, SkillBuffSheet.Row) ,
Sheet<int, SkillBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, SkillBuffSheet.Row>.Remove(int) ,
Sheet<int, SkillBuffSheet.Row>.TryGetValue(int, out SkillBuffSheet.Row) ,
Sheet<int, SkillBuffSheet.Row>.Add(KeyValuePair<int, SkillBuffSheet.Row>) ,
Sheet<int, SkillBuffSheet.Row>.Clear() ,
Sheet<int, SkillBuffSheet.Row>.Contains(KeyValuePair<int, SkillBuffSheet.Row>) ,
Sheet<int, SkillBuffSheet.Row>.CopyTo(KeyValuePair<int, SkillBuffSheet.Row>[], int) ,
Sheet<int, SkillBuffSheet.Row>.Remove(KeyValuePair<int, SkillBuffSheet.Row>) ,
Sheet<int, SkillBuffSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SkillBuffSheet()

```
public SkillBuffSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, SkillBuffSheet.Row value)
```

### Parameters

`key` int

`value` SkillBuffSheet.Row

# Class SkillBuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SkillBuffSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← SkillBuffSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## BuffIds

```
public List<int> BuffIds { get; }
```

### Property Value

List<int>

## Key

```
public override int Key { get; }
```

### Property Value

int

## SkillId

```
public int SkillId { get; }
```

### Property Value

int

# Methods

## EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class SkillSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Sheet containing skill data loaded from CSV files. Provides access to skill definitions and their properties.

```
[Serializable]
public class SkillSheet : Sheet<int, SkillSheet.Row>, IDictionary<int,
SkillSheet.Row>, ICollection<KeyValuePair<int, SkillSheet.Row>>,
IEnumerable<KeyValuePair<int, SkillSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SkillSheet.Row> ← SkillSheet

### Implements

IDictionary<int, SkillSheet.Row>, ICollection<KeyValuePair<int, SkillSheet.Row>>, IEnumerable<KeyValuePair<int, SkillSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, SkillSheet.Row>.Name , Sheet<int, SkillSheet.Row>.OrderedList , Sheet<int, SkillSheet.Row>.First , Sheet<int, SkillSheet.Row>.Last , Sheet<int, SkillSheet.Row>.Keys , Sheet<int, SkillSheet.Row>.Values , Sheet<int, SkillSheet.Row>.Count , Sheet<int, SkillSheet.Row>.IsReadOnly , Sheet<int, SkillSheet.Row>.this[int] , Sheet<int, SkillSheet.Row>.Set(string, bool) , Sheet<int, SkillSheet.Row>.Set<T>(Sheet<int, T>, bool) , Sheet<int, SkillSheet.Row>.GetEnumerator() , Sheet<int, SkillSheet.Row>.TryGetValue(int, out SkillSheet.Row, bool) , Sheet<int, SkillSheet.Row>.AddRow(int, SkillSheet.Row) , Sheet<int, SkillSheet.Row>.Add(int, SkillSheet.Row) , Sheet<int, SkillSheet.Row>.ContainsKey(int) , Sheet<int, SkillSheet.Row>.Remove(int) , Sheet<int, SkillSheet.Row>.TryGetValue(int, out SkillSheet.Row) , Sheet<int, SkillSheet.Row>.Add(KeyValuePair<int, SkillSheet.Row>) , Sheet<int, SkillSheet.Row>.Clear() , Sheet<int, SkillSheet.Row>.Contains(KeyValuePair<int, SkillSheet.Row>) , Sheet<int, SkillSheet.Row>.CopyTo(KeyValuePair<int, SkillSheet.Row>[], int) , Sheet<int, SkillSheet.Row>.Remove(KeyValuePair<int, SkillSheet.Row>) , Sheet<int, SkillSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SkillSheet()

```
public SkillSheet()
```

# Class SkillSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a single skill row from the skill sheet. Contains skill metadata and properties used for skill creation and execution.

```
[Serializable]
public class SkillSheet.Row : SheetRow<int>, IState
```

**Inheritance**

object ← SheetRow<int> ← SkillSheet.Row

**Implements**

IState

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Remarks

SkillSheet.Row maps 1:1 to the following 9 fields in SkillSheet.csv: [0] id [1] _name [2] elemental_type [3] skill_type [4] skill_category [5] skill_target_type [6] hit_count [7] cooldown [8] combo

Serialization and deserialization are performed as a List in this order. (Dictionary is also supported for backward compatibility.)

# Constructors

## Row()

```
public Row()
```

## Row(Dictionary)

```
public Row(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

## Row(List)

```
public Row(List serialized)
```

### Parameters

`serialized` List

# Properties

## Combo

```
public bool Combo { get; }
```

### Property Value

bool

## Cooldown

```
public int Cooldown { get; }
```

### Property Value

int

## ElementalType

```
public ElementalType ElementalType { get; }
```

### Property Value

ElementalType

## HitCount

```
public int HitCount { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## SkillCategory

```
public SkillCategory SkillCategory { get; }
```

### Property Value

SkillCategory

## SkillTargetType

```
public SkillTargetType SkillTargetType { get; }
```

### Property Value

SkillTargetType

## SkillType

```
public SkillType SkillType { get; }
```

### Property Value

SkillType

# Methods

## Deserialize(Dictionary)

```
[Obsolete("Use Deserialize(IValue) instead.")]
public static SkillSheet.Row Deserialize(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

SkillSheet.Row

## Deserialize(IValue)

Deserializes a skill row from serialized data. Supports both Dictionary and List formats for backward compatibility.

```
public static SkillSheet.Row Deserialize(IValue serialized)
```

### Parameters

`serialized` IValue

The serialized skill row data

### Returns

SkillSheet.Row

The deserialized skill row

### Exceptions

ArgumentException

Thrown when the serialization format is not supported

## DeserializeFromDictionary(Dictionary)

```
public static SkillSheet.Row DeserializeFromDictionary(Dictionary serialized)
```

### Parameters

`serialized` Dictionary

### Returns

SkillSheet.Row

## DeserializeFromList(List)

```
public static SkillSheet.Row DeserializeFromList(List serialized)
```

### Parameters

`serialized` List

### Returns

SkillSheet.Row

## Serialize()

```
public IValue Serialize()
```

### Returns

IValue

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class StageDialogSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageDialogSheet : Sheet<int, StageDialogSheet.Row>, IDictionary<int,
StageDialogSheet.Row>, ICollection<KeyValuePair<int, StageDialogSheet.Row>>,
IEnumerable<KeyValuePair<int, StageDialogSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, StageDialogSheet.Row> ← StageDialogSheet

### Implements

IDictionary<int, StageDialogSheet.Row>,
ICollection<KeyValuePair<int, StageDialogSheet.Row>>,
IEnumerable<KeyValuePair<int, StageDialogSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, StageDialogSheet.Row>.Name ,
Sheet<int, StageDialogSheet.Row>.OrderedList , Sheet<int, StageDialogSheet.Row>.First ,
Sheet<int, StageDialogSheet.Row>.Last , Sheet<int, StageDialogSheet.Row>.Keys ,
Sheet<int, StageDialogSheet.Row>.Values , Sheet<int, StageDialogSheet.Row>.Count ,
Sheet<int, StageDialogSheet.Row>.IsReadOnly ,
Sheet<int, StageDialogSheet.Row>.this[int] ,
Sheet<int, StageDialogSheet.Row>.Set(string, bool) ,
Sheet<int, StageDialogSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StageDialogSheet.Row>.GetEnumerator() ,
Sheet<int, StageDialogSheet.Row>.TryGetValue(int, out StageDialogSheet.Row, bool) ,
Sheet<int, StageDialogSheet.Row>.AddRow(int, StageDialogSheet.Row) ,
Sheet<int, StageDialogSheet.Row>.Add(int, StageDialogSheet.Row) ,
Sheet<int, StageDialogSheet.Row>.ContainsKey(int) ,
Sheet<int, StageDialogSheet.Row>.Remove(int) ,
Sheet<int, StageDialogSheet.Row>.TryGetValue(int, out StageDialogSheet.Row) ,
Sheet<int, StageDialogSheet.Row>.Add(KeyValuePair<int, StageDialogSheet.Row>) ,
Sheet<int, StageDialogSheet.Row>.Clear() ,
Sheet<int, StageDialogSheet.Row>.Contains(KeyValuePair<int, StageDialogSheet.Row>) ,
Sheet<int, StageDialogSheet.Row>.CopyTo(KeyValuePair<int, StageDialogSheet.Row>[],
int) ,

Sheet<int, StageDialogSheet.Row>.Remove(KeyValuePair<int, StageDialogSheet.Row>) , Sheet<int, StageDialogSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## StageDialogSheet()

```
public StageDialogSheet()
```

# Class StageDialogSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageDialogSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← StageDialogSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## DialogId

```
public int DialogId { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## StageId

```
public int StageId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class StageSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageSheet : Sheet<int, StageSheet.Row>, IDictionary<int,
StageSheet.Row>, ICollection<KeyValuePair<int, StageSheet.Row>>,
IEnumerable<KeyValuePair<int, StageSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, StageSheet.Row> ← StageSheet

### Implements

IDictionary<int, StageSheet.Row>,
ICollection<KeyValuePair<int, StageSheet.Row>>,
IEnumerable<KeyValuePair<int, StageSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, StageSheet.Row>.Name , Sheet<int, StageSheet.Row>.OrderedList ,
Sheet<int, StageSheet.Row>.First , Sheet<int, StageSheet.Row>.Last ,
Sheet<int, StageSheet.Row>.Keys , Sheet<int, StageSheet.Row>.Values ,
Sheet<int, StageSheet.Row>.Count , Sheet<int, StageSheet.Row>.IsReadOnly ,
Sheet<int, StageSheet.Row>.this[int] , Sheet<int, StageSheet.Row>.Set(string, bool) ,
Sheet<int, StageSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StageSheet.Row>.GetEnumerator() ,
Sheet<int, StageSheet.Row>.TryGetValue(int, out StageSheet.Row, bool) ,
Sheet<int, StageSheet.Row>.AddRow(int, StageSheet.Row) ,
Sheet<int, StageSheet.Row>.Add(int, StageSheet.Row) ,
Sheet<int, StageSheet.Row>.ContainsKey(int) , Sheet<int, StageSheet.Row>.Remove(int) ,
Sheet<int, StageSheet.Row>.TryGetValue(int, out StageSheet.Row) ,
Sheet<int, StageSheet.Row>.Add(KeyValuePair<int, StageSheet.Row>) ,
Sheet<int, StageSheet.Row>.Clear() ,
Sheet<int, StageSheet.Row>.Contains(KeyValuePair<int, StageSheet.Row>) ,
Sheet<int, StageSheet.Row>.CopyTo(KeyValuePair<int, StageSheet.Row>[], int) ,
Sheet<int, StageSheet.Row>.Remove(KeyValuePair<int, StageSheet.Row>) ,
Sheet<int, StageSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## StageSheet()

```
public StageSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, StageSheet.Row value)
```

### Parameters

`key` int

`value` StageSheet.Row

# Class StageSheet.FavRewardData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageSheet.FavRewardData
```

**Inheritance**

object ← StageSheet.FavRewardData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### FavRewardData(string, decimal, int, int)

Initializes a new instance of StageSheet.FavRewardData.

```
public FavRewardData(string ticker, decimal ratio, int min, int max)
```

#### Parameters

`ticker` string

The fungible asset ticker.

`ratio` decimal

The selection weight.

`min` int

Minimum granted amount (inclusive).

`max` int

Maximum granted amount (inclusive).

# Properties

## Max

Gets the maximum amount granted when selected.

```
public int Max { get; }
```

### Property Value

int

## Min

Gets the minimum amount granted when selected.

```
public int Min { get; }
```

### Property Value

int

## Ratio

Gets the weight used for random selection.

```
public decimal Ratio { get; }
```

### Property Value

decimal

## Ticker

Gets the currency ticker of the fungible asset reward (e.g., `CRYSTAL`).

```
public string Ticker { get; }
```

## Property Value

string

# Class StageSheet.RewardData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageSheet.RewardData
```

**Inheritance**

object ← StageSheet.RewardData

**Derived**

WeeklyArenaRewardSheet.RewardData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardData(int, decimal, int, int)

```
public RewardData(int itemId, decimal ratio, int min, int max)
```

#### Parameters

`itemId` int

`ratio` decimal

`min` int

`max` int

## Properties

## ItemId

```
public int ItemId { get; }
```

### Property Value

int

## Max

```
public int Max { get; }
```

### Property Value

int

## Min

```
public int Min { get; }
```

### Property Value

int

## Ratio

```
public decimal Ratio { get; }
```

### Property Value

decimal

# Class StageSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← StageSheet.Row

**Derived**

EventDungeonStageSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## BGM

```
public string BGM { get; }
```

### Property Value

string

## Background

```
public string Background { get; }
```

### Property Value

string

## CostAP

```
public int CostAP { get; }
```

### Property Value

int

## DropItemMax

```
public int DropItemMax { get; }
```

### Property Value

int

## DropItemMin

```
public int DropItemMin { get; }
```

### Property Value

int

## EnemyInitialStatModifiers

```
public List<StatModifier> EnemyInitialStatModifiers { get; }
```

### Property Value

List<StatModifier>

## EntryCostItemCount

Gets the additional entry cost material item count required per play for this stage. A value of 0 means no additional entry material cost is required.

```
public int EntryCostItemCount { get; }
```

### Property Value

int

## EntryCostItemId

Gets the additional entry cost material item ID required to play this stage. A value of 0 means no additional entry material cost is required.

```
public int EntryCostItemId { get; }
```

### Property Value

int

## FavDropMax

Gets the maximum number of draws from FavRewards (inclusive).

```
public int FavDropMax { get; }
```

### Property Value

int

## FavDropMin

Gets the minimum number of draws from FavRewards (inclusive).

```
public int FavDropMin { get; }
```

### Property Value

int

## FavRewards

Gets the pool of fungible asset reward candidates for this stage. Rewards are selected by weight (Ratio) and granted as (ticker, amount) pairs.

```
public List<StageSheet.FavRewardData> FavRewards { get; }
```

### Property Value

List<StageSheet.FavRewardData>

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Rewards

```
public List<StageSheet.RewardData> Rewards { get; }
```

### Property Value

List<StageSheet.RewardData>

## TurnLimit

```
public int TurnLimit { get; }
```

### Property Value

int

# Methods

## CloneWithId(int)

Creates a shallow clone of this row with a new stage ID.

```
public StageSheet.Row CloneWithId(int newId)
```

### Parameters

`newId` int

The stage ID to assign to the cloned row.

### Returns

StageSheet.Row

A new StageSheet.Row with copied values and the specified ID.

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class StageSimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StageSimulatorSheets : SimulatorSheets
```

**Inheritance**

object ← SimulatorSheetsV1 ← SimulatorSheets ← StageSimulatorSheets

**Inherited Members**

SimulatorSheets.RuneOptionSheet , SimulatorSheets.RuneListSheet , SimulatorSheets.RuneLevelBonusSheet , SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StageSimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, StageSheet, StageWaveSheet, EnemySkillSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public StageSimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, StageSheet
stageSheet, StageWaveSheet stageWaveSheet, EnemySkillSheet enemySkillSheet,
```

```
RuneOptionSheet runeOptionSheet, RuneListSheet runeListSheet,
RuneLevelBonusSheet runeLevelBonusSheet)
```

### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`stageSheet` StageSheet

`stageWaveSheet` StageWaveSheet

`enemySkillSheet` EnemySkillSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

## Fields

## EnemySkillSheet

```
public readonly EnemySkillSheet EnemySkillSheet
```

### Field Value

EnemySkillSheet

## StageSheet

```
public readonly StageSheet StageSheet
```

### Field Value

StageSheet

## StageWaveSheet

```
public readonly StageWaveSheet StageWaveSheet
```

### Field Value

StageWaveSheet

# Class StageSimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StageSimulatorSheetsV1 : SimulatorSheetsV1
```

**Inheritance**

object ← SimulatorSheetsV1 ← StageSimulatorSheetsV1

**Inherited Members**

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StageSimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, StageSheet, StageWaveSheet, EnemySkillSheet)

```
public StageSimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, StageSheet stageSheet,
StageWaveSheet stageWaveSheet, EnemySkillSheet enemySkillSheet)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`stageSheet` StageSheet

`stageWaveSheet` StageWaveSheet

`enemySkillSheet` EnemySkillSheet

# Fields

## EnemySkillSheet

```
public readonly EnemySkillSheet EnemySkillSheet
```

### Field Value

EnemySkillSheet

## StageSheet

```
public readonly StageSheet StageSheet
```

### Field Value

StageSheet

# StageWaveSheet

```
public readonly StageWaveSheet StageWaveSheet
```

## Field Value

StageWaveSheet

# Class StageSimulatorSheetsV100291

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StageSimulatorSheetsV100291 : SimulatorSheetsV100291
```

**Inheritance**

object ← SimulatorSheetsV100291 ← StageSimulatorSheetsV100291

**Inherited Members**

SimulatorSheetsV100291.MaterialItemSheet , SimulatorSheetsV100291.SkillSheet , SimulatorSheetsV100291.SkillBuffSheet , SimulatorSheetsV100291.BuffSheet , SimulatorSheetsV100291.CharacterSheet , SimulatorSheetsV100291.CharacterLevelSheet , SimulatorSheetsV100291.EquipmentItemSetEffectSheet , SimulatorSheetsV100291.ToSimulatorSheetsV1() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### StageSimulatorSheetsV100291(MaterialItemSheet, SkillSheet, SkillBuffSheet, BuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, StageSheet, StageWaveSheet, EnemySkillSheet)

```
public StageSimulatorSheetsV100291(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, BuffSheet buffSheet, CharacterSheet
characterSheet, CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet, StageSheet stageSheet, StageWaveSheet stageWaveSheet,
EnemySkillSheet enemySkillSheet)
```

#### Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`buffSheet` BuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`stageSheet` StageSheet

`stageWaveSheet` StageWaveSheet

`enemySkillSheet` EnemySkillSheet

# Fields

## EnemySkillSheet

```
public readonly EnemySkillSheet EnemySkillSheet
```

### Field Value

EnemySkillSheet

## StageSheet

```
public readonly StageSheet StageSheet
```

### Field Value

StageSheet

## StageWaveSheet

```
public readonly StageWaveSheet StageWaveSheet
```

### Field Value

StageWaveSheet

## Methods

### ToStageSimulatorSheetsV1()

```
public StageSimulatorSheetsV1 ToStageSimulatorSheetsV1()
```

### Returns

StageSimulatorSheetsV1

# Class StageWaveSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageWaveSheet : Sheet<int, StageWaveSheet.Row>, IDictionary<int,
StageWaveSheet.Row>, ICollection<KeyValuePair<int, StageWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, StageWaveSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, StageWaveSheet.Row> ← StageWaveSheet

### Implements

IDictionary<int, StageWaveSheet.Row>,
ICollection<KeyValuePair<int, StageWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, StageWaveSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, StageWaveSheet.Row>.Name , Sheet<int, StageWaveSheet.Row>.OrderedList ,
Sheet<int, StageWaveSheet.Row>.First , Sheet<int, StageWaveSheet.Row>.Last ,
Sheet<int, StageWaveSheet.Row>.Keys , Sheet<int, StageWaveSheet.Row>.Values ,
Sheet<int, StageWaveSheet.Row>.Count , Sheet<int, StageWaveSheet.Row>.IsReadOnly ,
Sheet<int, StageWaveSheet.Row>.this[int] ,
Sheet<int, StageWaveSheet.Row>.Set(string, bool) ,
Sheet<int, StageWaveSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StageWaveSheet.Row>.GetEnumerator() ,
Sheet<int, StageWaveSheet.Row>.TryGetValue(int, out StageWaveSheet.Row, bool) ,
Sheet<int, StageWaveSheet.Row>.AddRow(int, StageWaveSheet.Row) ,
Sheet<int, StageWaveSheet.Row>.Add(int, StageWaveSheet.Row) ,
Sheet<int, StageWaveSheet.Row>.ContainsKey(int) ,
Sheet<int, StageWaveSheet.Row>.Remove(int) ,
Sheet<int, StageWaveSheet.Row>.TryGetValue(int, out StageWaveSheet.Row) ,
Sheet<int, StageWaveSheet.Row>.Add(KeyValuePair<int, StageWaveSheet.Row>) ,
Sheet<int, StageWaveSheet.Row>.Clear() ,
Sheet<int, StageWaveSheet.Row>.Contains(KeyValuePair<int, StageWaveSheet.Row>) ,
Sheet<int, StageWaveSheet.Row>.CopyTo(KeyValuePair<int, StageWaveSheet.Row>[], int)
,
Sheet<int, StageWaveSheet.Row>.Remove(KeyValuePair<int, StageWaveSheet.Row>) ,

Sheet<int, StageWaveSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## StageWaveSheet()

```
public StageWaveSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, StageWaveSheet.Row value)
```

### Parameters

`key` int

`value` StageWaveSheet.Row

# Class StageWaveSheet.MonsterData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageWaveSheet.MonsterData
```

**Inheritance**

object ← StageWaveSheet.MonsterData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MonsterData(int, int, int)

```
public MonsterData(int characterId, int level, int count)
```

#### Parameters

`characterId` int

`level` int

`count` int

## Properties

### CharacterId

```
public int CharacterId { get; }
```

### Property Value

int

## Count

```
public int Count { get; }
```

### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

# Class StageWaveSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageWaveSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← StageWaveSheet.Row

**Derived**

EventDungeonStageWaveSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## HasBoss

```
public bool HasBoss { get; }
```

### Property Value

bool

## Key

```
public override int Key { get; }
```

### Property Value

int

## StageId

```
public int StageId { get; }
```

### Property Value

int

## TotalMonsterIds

```
public List<int> TotalMonsterIds { get; }
```

### Property Value

List<int>

## Waves

```
public List<StageWaveSheet.WaveData> Waves { get; }
```

### Property Value

List<StageWaveSheet.WaveData>

# Methods

## CloneWithStageId(int)

Creates a shallow clone of this row with a new stage ID.

```
public StageWaveSheet.Row CloneWithStageId(int newStageId)
```

### Parameters

`newStageId` int

The stage ID to assign to the cloned row.

### Returns

StageWaveSheet.Row

A new StageWaveSheet.Row with copied wave data and the specified stage ID.

## EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class StageWaveSheet.WaveData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageWaveSheet.WaveData
```

**Inheritance**

object ← StageWaveSheet.WaveData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### WaveData(int, List<MonsterData>, bool)

```
public WaveData(int number, List<StageWaveSheet.MonsterData> monsters, bool hasBoss)
```

#### Parameters

`number` int

`monsters` List<StageWaveSheet.MonsterData>

`hasBoss` bool

## Properties

### HasBoss

```
public bool HasBoss { get; }
```

### Property Value

bool

## Monsters

```
public List<StageWaveSheet.MonsterData> Monsters { get; }
```

### Property Value

List<StageWaveSheet.MonsterData>

## Number

```
public int Number { get; }
```

### Property Value

int

# Class StakeAchievementRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeAchievementRewardSheet : Sheet<int,
StakeAchievementRewardSheet.Row>, IDictionary<int, StakeAchievementRewardSheet.Row>,
ICollection<KeyValuePair<int, StakeAchievementRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, StakeAchievementRewardSheet.Row>>, IEnumerable,
IStakeRewardSheet, ISheet
```

### Inheritance

object ← Sheet<int, StakeAchievementRewardSheet.Row> ← StakeAchievementRewardSheet

### Implements

IDictionary<int, StakeAchievementRewardSheet.Row>,
ICollection<KeyValuePair<int, StakeAchievementRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, StakeAchievementRewardSheet.Row>>,
IEnumerable, IStakeRewardSheet, ISheet

### Inherited Members

Sheet<int, StakeAchievementRewardSheet.Row>.Name ,
Sheet<int, StakeAchievementRewardSheet.Row>.OrderedList ,
Sheet<int, StakeAchievementRewardSheet.Row>.First ,
Sheet<int, StakeAchievementRewardSheet.Row>.Last ,
Sheet<int, StakeAchievementRewardSheet.Row>.Keys ,
Sheet<int, StakeAchievementRewardSheet.Row>.Values ,
Sheet<int, StakeAchievementRewardSheet.Row>.Count ,
Sheet<int, StakeAchievementRewardSheet.Row>.IsReadOnly ,
Sheet<int, StakeAchievementRewardSheet.Row>.this[int] ,
Sheet<int, StakeAchievementRewardSheet.Row>.Set(string, bool) ,
Sheet<int, StakeAchievementRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StakeAchievementRewardSheet.Row>.GetEnumerator() ,
Sheet<int, StakeAchievementRewardSheet.Row>.TryGetValue(int, out StakeAchievementRewardSheet.Row, bool) ,
Sheet<int, StakeAchievementRewardSheet.Row>.AddRow(int, StakeAchievementRewardSheet.Row) ,

Sheet<int, StakeAchievementRewardSheet.Row>.Add(int, StakeAchievementRewardSheet.Row) , Sheet<int, StakeAchievementRewardSheet.Row>.ContainsKey(int) , Sheet<int, StakeAchievementRewardSheet.Row>.Remove(int) , Sheet<int, StakeAchievementRewardSheet.Row>.TryGetValue(int, out StakeAchievementRewardSheet.Row) , Sheet<int, StakeAchievementRewardSheet.Row>.Add(KeyValuePair<int, StakeAchievementRewardSheet.Row>) , Sheet<int, StakeAchievementRewardSheet.Row>.Clear() , Sheet<int, StakeAchievementRewardSheet.Row>.Contains(KeyValuePair<int, StakeAchievementRewardSheet.Row>) , Sheet<int, StakeAchievementRewardSheet.Row>.CopyTo(KeyValuePair<int, StakeAchievementRewardSheet.Row>[], int) , Sheet<int, StakeAchievementRewardSheet.Row>.Remove(KeyValuePair<int, StakeAchievementRewardSheet.Row>) , Sheet<int, StakeAchievementRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

# Constructors

## StakeAchievementRewardSheet()

```
public StakeAchievementRewardSheet()
```

# Properties

## OrderedRows

```
public IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

### Property Value

IReadOnlyList<IStakeRewardRow>

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, StakeAchievementRewardSheet.Row value)
```

### Parameters

`key` int

`value` StakeAchievementRewardSheet.Row

## FindStep(int, long)

```
public int FindStep(int level, long stakedBlockPeriod)
```

### Parameters

`level` int

`stakedBlockPeriod` long

### Returns

int

# Class StakeAchievementRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeAchievementRewardSheet.RewardInfo
```

**Inheritance**

object ← StakeAchievementRewardSheet.RewardInfo

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardInfo(Dictionary)

```
public RewardInfo(Dictionary dictionary)
```

#### Parameters

`dictionary` Dictionary

### RewardInfo(int, int)

```
public RewardInfo(int itemId, int quantity)
```

#### Parameters

`itemId` int

quantity int

## RewardInfo(params string[])

```
public RewardInfo(params string[] fields)
```

### Parameters

fields string[]

# Fields

## ItemId

```
public readonly int ItemId
```

### Field Value

int

## Quantity

```
public readonly int Quantity
```

### Field Value

int

# Methods

## Equals(RewardInfo)

```
protected bool Equals(StakeAchievementRewardSheet.RewardInfo other)
```

### Parameters

`other` StakeAchievementRewardSheet.RewardInfo

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

# Serialize()

```
public IValue Serialize()
```

## Returns

IValue

# Class StakeAchievementRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeAchievementRewardSheet.Row : SheetRow<int>, IStakeRewardRow
```

**Inheritance**

object ← SheetRow<int> ← StakeAchievementRewardSheet.Row

**Implements**

IStakeRewardRow

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Key

```
public override int Key { get; }
```

#### Property Value

int

### Level

```
public int Level { get; }
```

#### Property Value

int

## RequiredGold

```
public long RequiredGold { get; }
```

### Property Value

long

## Steps

```
public List<StakeAchievementRewardSheet.Step> Steps { get; }
```

### Property Value

List<StakeAchievementRewardSheet.Step>

## Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class StakeAchievementRewardSheet.Step

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StakeAchievementRewardSheet.Step
```

**Inheritance**

object ← StakeAchievementRewardSheet.Step

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Step(long, long, List<RewardInfo>)

```
public Step(long requiredGold, long requiredBlockIndex,
List<StakeAchievementRewardSheet.RewardInfo> rewards)
```

#### Parameters

`requiredGold` long

`requiredBlockIndex` long

`rewards` List<StakeAchievementRewardSheet.RewardInfo>

## Properties

### RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

### Property Value

long

## RequiredGold

```
public long RequiredGold { get; }
```

### Property Value

long

## Rewards

```
public List<StakeAchievementRewardSheet.RewardInfo> Rewards { get; }
```

### Property Value

List<StakeAchievementRewardSheet.RewardInfo>

# Class StakeActionPointCoefficientSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeActionPointCoefficientSheet : Sheet<int,
StakeActionPointCoefficientSheet.Row>, IDictionary<int,
StakeActionPointCoefficientSheet.Row>, ICollection<KeyValuePair<int,
StakeActionPointCoefficientSheet.Row>>, IEnumerable<KeyValuePair<int,
StakeActionPointCoefficientSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet
```

### Inheritance

object ← Sheet<int, StakeActionPointCoefficientSheet.Row> ← StakeActionPointCoefficientSheet

### Implements

IDictionary<int, StakeActionPointCoefficientSheet.Row>, ICollection<KeyValuePair<int, StakeActionPointCoefficientSheet.Row>>, IEnumerable<KeyValuePair<int, StakeActionPointCoefficientSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet

### Inherited Members

Sheet<int, StakeActionPointCoefficientSheet.Row>.Name ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.OrderedList ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.First ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Last ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Keys ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Values ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Count ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.IsReadOnly ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.this[int] ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Set(string, bool) ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.GetEnumerator() ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.TryGetValue(int, out StakeActionPointCoefficientSheet.Row, bool) ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.AddRow(int, StakeActionPointCoefficientSheet.Row) ,

Sheet<int, StakeActionPointCoefficientSheet.Row>.Add(int, StakeActionPointCoefficientSheet.Row) , Sheet<int, StakeActionPointCoefficientSheet.Row>.ContainsKey(int) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Remove(int) , Sheet<int, StakeActionPointCoefficientSheet.Row>.TryGetValue(int, out StakeActionPointCoefficientSheet.Row) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Add(KeyValuePair<int, StakeActionPointCoefficientSheet.Row>) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Clear() , Sheet<int, StakeActionPointCoefficientSheet.Row>.Contains(KeyValuePair<int, StakeActionPointCoefficientSheet.Row>) , Sheet<int, StakeActionPointCoefficientSheet.Row>.CopyTo(KeyValuePair<int, StakeActionPointCoefficientSheet.Row>[], int) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Remove(KeyValuePair<int, StakeActionPointCoefficientSheet.Row>) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue) , SheetsExtensions.GetActionPointByStaking(StakeActionPointCoefficientSheet, int, int, int)

# Constructors

## StakeActionPointCoefficientSheet()

```
public StakeActionPointCoefficientSheet()
```

# Properties

## OrderedRows

```
public IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

### Property Value

IReadOnlyList<IStakeRewardRow>

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, StakeActionPointCoefficientSheet.Row value)
```

### Parameters

`key` int

`value` StakeActionPointCoefficientSheet.Row

# Class StakeActionPointCoefficientSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StakeActionPointCoefficientSheet.Row : SheetRow<int>, IStakeRewardRow
```

**Inheritance**

object ← SheetRow<int> ← StakeActionPointCoefficientSheet.Row

**Implements**

IStakeRewardRow

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Coefficient

```
public int Coefficient { get; }
```

#### Property Value

int

### Key

```
public override int Key { get; }
```

#### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

## RequiredGold

```
public long RequiredGold { get; }
```

### Property Value

long

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class StakeRegularFixedRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

This sheet is used for setting the regular rewards for staking. The difference between this sheet and StakeRegularRewardSheet is that the StakeRegularFixedRewardSheet.RewardInfo of this sheet has a fixed count of reward.

CSV column order: level, required_gold, item_id, count[, tradable]

The optional `tradable` column controls whether material items are created as tradable (TradableMaterial) or non-tradable (Material). Omitting the column defaults to `true`, preserving backward compatibility with existing CSV data.

```
[Serializable]
public class StakeRegularFixedRewardSheet : Sheet<int,
StakeRegularFixedRewardSheet.Row>, IDictionary<int,
StakeRegularFixedRewardSheet.Row>, ICollection<KeyValuePair<int,
StakeRegularFixedRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
StakeRegularFixedRewardSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet
```

### Inheritance

object ← Sheet<int, StakeRegularFixedRewardSheet.Row> ← StakeRegularFixedRewardSheet

### Implements

IDictionary<int, StakeRegularFixedRewardSheet.Row>, ICollection<KeyValuePair<int, StakeRegularFixedRewardSheet.Row>>, IEnumerable<KeyValuePair<int, StakeRegularFixedRewardSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet

### Inherited Members

Sheet<int, StakeRegularFixedRewardSheet.Row>.Name ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.OrderedList ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.First ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Last ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Keys ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Values ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Count ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.IsReadOnly ,

Sheet<int, StakeRegularFixedRewardSheet.Row>.this[int] ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Set(string, bool) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.GetEnumerator() ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.TryGetValue(int, out StakeRegularFixedRewardSheet.Row, bool) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.AddRow(int, StakeRegularFixedRewardSheet.Row) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Add(int, StakeRegularFixedRewardSheet.Row) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Remove(int) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.TryGetValue(int, out StakeRegularFixedRewardSheet.Row) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Add(KeyValuePair<int, StakeRegularFixedRewardSheet.Row>) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Clear() ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Contains(KeyValuePair<int, StakeRegularFixedRewardSheet.Row>) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.CopyTo(KeyValuePair<int, StakeRegularFixedRewardSheet.Row>[], int) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Remove(KeyValuePair<int, StakeRegularFixedRewardSheet.Row>) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

# Constructors

## StakeRegularFixedRewardSheet()

```
public StakeRegularFixedRewardSheet()
```

# Properties

## OrderedRows

```
public IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

### Property Value

IReadOnlyList<IStakeRewardRow>

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, StakeRegularFixedRewardSheet.Row value)
```

### Parameters

`key` int

`value` StakeRegularFixedRewardSheet.Row

# Class StakeRegularFixedRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeRegularFixedRewardSheet.RewardInfo
```

**Inheritance**

object ← StakeRegularFixedRewardSheet.RewardInfo

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardInfo(int, int, bool)

```
public RewardInfo(int itemId, int count, bool tradable = true)
```

#### Parameters

`itemId` int

The item ID to reward.

`count` int

The fixed count of items to reward.

`tradable` bool

Whether the rewarded material item should be tradable. Defaults to `true`.

## RewardInfo(params string[])

Initializes a StakeRegularFixedRewardSheet.RewardInfo from CSV field tokens. Expected order: item_id, count[, tradable].

```
public RewardInfo(params string[] fields)
```

### Parameters

`fields` string[]

# Fields

## Count

```
public readonly int Count
```

### Field Value

int

## ItemId

```
public readonly int ItemId
```

### Field Value

int

## Tradable

Whether the rewarded item is tradable. If `true`, materials are created as TradableMaterial; otherwise they are created as non-tradable Material. Defaults to `true` when the column is omitted in the CSV.

```
public readonly bool Tradable
```

### Field Value

bool

## Methods

### Equals(RewardInfo)

```
protected bool Equals(StakeRegularFixedRewardSheet.RewardInfo other)
```

### Parameters

`other` StakeRegularFixedRewardSheet.RewardInfo

### Returns

bool

### Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

# GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

## Returns

int

A hash code for the current object.

# Class StakeRegularFixedRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeRegularFixedRewardSheet.Row : SheetRow<int>, IStakeRewardRow
```

**Inheritance**

object ← SheetRow<int> ← StakeRegularFixedRewardSheet.Row

**Implements**

IStakeRewardRow

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

## RequiredGold

```
public long RequiredGold { get; }
```

### Property Value

long

## Rewards

```
public List<StakeRegularFixedRewardSheet.RewardInfo> Rewards { get; }
```

### Property Value

List<StakeRegularFixedRewardSheet.RewardInfo>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class StakeRegularRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

This sheet is used for setting the regular rewards for staking. The difference between this sheet and StakeRegularFixedRewardSheet is that the StakeRegularRewardSheet.RewardInfo of this sheet has a rate and a reward type. The count of reward is calculated by the rate and the amount of staked currency.

```
[Serializable]
public class StakeRegularRewardSheet : Sheet<int, StakeRegularRewardSheet.Row>,
IDictionary<int, StakeRegularRewardSheet.Row>, ICollection<KeyValuePair<int,
StakeRegularRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
StakeRegularRewardSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet
```

### Inheritance

object ← Sheet<int, StakeRegularRewardSheet.Row> ← StakeRegularRewardSheet

### Implements

IDictionary<int, StakeRegularRewardSheet.Row>,
ICollection<KeyValuePair<int, StakeRegularRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, StakeRegularRewardSheet.Row>>, IEnumerable,
IStakeRewardSheet, ISheet

### Inherited Members

Sheet<int, StakeRegularRewardSheet.Row>.Name ,
Sheet<int, StakeRegularRewardSheet.Row>.OrderedList ,
Sheet<int, StakeRegularRewardSheet.Row>.First ,
Sheet<int, StakeRegularRewardSheet.Row>.Last ,
Sheet<int, StakeRegularRewardSheet.Row>.Keys ,
Sheet<int, StakeRegularRewardSheet.Row>.Values ,
Sheet<int, StakeRegularRewardSheet.Row>.Count ,
Sheet<int, StakeRegularRewardSheet.Row>.IsReadOnly ,
Sheet<int, StakeRegularRewardSheet.Row>.this[int] ,
Sheet<int, StakeRegularRewardSheet.Row>.Set(string, bool) ,
Sheet<int, StakeRegularRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StakeRegularRewardSheet.Row>.GetEnumerator() ,
Sheet<int, StakeRegularRewardSheet.Row>.TryGetValue(int, out
StakeRegularRewardSheet.Row, bool) ,

Sheet<int, StakeRegularRewardSheet.Row>.AddRow(int, StakeRegularRewardSheet.Row) ,
Sheet<int, StakeRegularRewardSheet.Row>.Add(int, StakeRegularRewardSheet.Row) ,
Sheet<int, StakeRegularRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, StakeRegularRewardSheet.Row>.Remove(int) ,
Sheet<int, StakeRegularRewardSheet.Row>.TryGetValue(int, out StakeRegularRewardSheet.Row) ,
Sheet<int, StakeRegularRewardSheet.Row>.Add(KeyValuePair<int, StakeRegularRewardSheet.Row>) ,
Sheet<int, StakeRegularRewardSheet.Row>.Clear() ,
Sheet<int, StakeRegularRewardSheet.Row>.Contains(KeyValuePair<int, StakeRegularRewardSheet.Row>) ,
Sheet<int, StakeRegularRewardSheet.Row>.CopyTo(KeyValuePair<int, StakeRegularRewardSheet.Row>[], int) ,
Sheet<int, StakeRegularRewardSheet.Row>.Remove(KeyValuePair<int, StakeRegularRewardSheet.Row>) ,
Sheet<int, StakeRegularRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

# Constructors

## StakeRegularRewardSheet()

```
public StakeRegularRewardSheet()
```

# Properties

## OrderedRows

```
public IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

### Property Value

IReadOnlyList<IStakeRewardRow>

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, StakeRegularRewardSheet.Row value)
```

### Parameters

`key` int

`value` StakeRegularRewardSheet.Row

# Class StakeRegularRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeRegularRewardSheet.RewardInfo
```

**Inheritance**

object ← StakeRegularRewardSheet.RewardInfo

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardInfo(int, int, StakeRewardType, string, int?, decimal)

```
public RewardInfo(int itemId, int rate = 0, StakeRegularRewardSheet.StakeRewardType
type = StakeRewardType.Item, string currencyTicker = null, int?
currencyDecimalPlaces = null, decimal decimalRate = 0)
```

#### Parameters

`itemId` int

`rate` int

`type` StakeRegularRewardSheet.StakeRewardType

`currencyTicker` string

`currencyDecimalPlaces` int?

`decimalRate` decimal

## RewardInfo(params string[])

```
public RewardInfo(params string[] fields)
```

### Parameters

`fields` string[]

# Fields

## CurrencyDecimalPlaces

The decimal places of currency. This field is only used when Type is Currency.

```
public readonly int? CurrencyDecimalPlaces
```

### Field Value

int?

## CurrencyTicker

The ticker of currency. This field is only used when Type is Currency.

```
public readonly string CurrencyTicker
```

### Field Value

string

## DecimalRate

The decimal rate of reward. This field is used from `V3`(in blockchain state) or later. This field is set to `Rate` when `ClaimStakeReward.V2` or earlier.

```
public readonly decimal DecimalRate
```

### Field Value

decimal

## ItemId

```
public readonly int ItemId
```

### Field Value

int

## Rate

The rate of reward.

```
[Obsolete("This field is used from `ClaimStakeReward.V2` or earlier. Use
`DecimalRate` instead. Deprecated this field because we need to support
decimal rate.")]
public readonly int Rate
```

### Field Value

int

## Tradable

```
public readonly bool Tradable
```

### Field Value

bool

## Type

```
public readonly StakeRegularRewardSheet.StakeRewardType Type
```

### Field Value

StakeRegularRewardSheet.StakeRewardType

# Methods

## Equals(RewardInfo)

```
protected bool Equals(StakeRegularRewardSheet.RewardInfo other)
```

### Parameters

`other` StakeRegularRewardSheet.RewardInfo

### Returns

bool

## Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

# Class StakeRegularRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeRegularRewardSheet.Row : SheetRow<int>, IStakeRewardRow
```

**Inheritance**

object ← SheetRow<int> ← StakeRegularRewardSheet.Row

**Implements**

IStakeRewardRow

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

## RequiredGold

```
public long RequiredGold { get; }
```

### Property Value

long

## Rewards

```
public List<StakeRegularRewardSheet.RewardInfo> Rewards { get; }
```

### Property Value

List<StakeRegularRewardSheet.RewardInfo>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Enum StakeRegularRewardSheet.StakeRewardType

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

The reward type of stake. Do not use the whole element of this enum at once in actions. (e.g., count, loop) Do versioning when you change or remove the elements of this enum.

```
public enum StakeRegularRewardSheet.StakeRewardType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Currency = 2`

`Item = 0`

`Rune = 1`

# Class StatBuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StatBuffSheet : Sheet<int, StatBuffSheet.Row>, IDictionary<int,
StatBuffSheet.Row>, ICollection<KeyValuePair<int, StatBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, StatBuffSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, StatBuffSheet.Row> ← StatBuffSheet

### Implements

IDictionary<int, StatBuffSheet.Row>,
ICollection<KeyValuePair<int, StatBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, StatBuffSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, StatBuffSheet.Row>.Name , Sheet<int, StatBuffSheet.Row>.OrderedList ,
Sheet<int, StatBuffSheet.Row>.First , Sheet<int, StatBuffSheet.Row>.Last ,
Sheet<int, StatBuffSheet.Row>.Keys , Sheet<int, StatBuffSheet.Row>.Values ,
Sheet<int, StatBuffSheet.Row>.Count , Sheet<int, StatBuffSheet.Row>.IsReadOnly ,
Sheet<int, StatBuffSheet.Row>.this[int] , Sheet<int, StatBuffSheet.Row>.Set(string, bool) ,
Sheet<int, StatBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StatBuffSheet.Row>.GetEnumerator() ,
Sheet<int, StatBuffSheet.Row>.TryGetValue(int, out StatBuffSheet.Row, bool) ,
Sheet<int, StatBuffSheet.Row>.AddRow(int, StatBuffSheet.Row) ,
Sheet<int, StatBuffSheet.Row>.Add(int, StatBuffSheet.Row) ,
Sheet<int, StatBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, StatBuffSheet.Row>.Remove(int) ,
Sheet<int, StatBuffSheet.Row>.TryGetValue(int, out StatBuffSheet.Row) ,
Sheet<int, StatBuffSheet.Row>.Add(KeyValuePair<int, StatBuffSheet.Row>) ,
Sheet<int, StatBuffSheet.Row>.Clear() ,
Sheet<int, StatBuffSheet.Row>.Contains(KeyValuePair<int, StatBuffSheet.Row>) ,
Sheet<int, StatBuffSheet.Row>.CopyTo(KeyValuePair<int, StatBuffSheet.Row>[], int) ,
Sheet<int, StatBuffSheet.Row>.Remove(KeyValuePair<int, StatBuffSheet.Row>) ,
Sheet<int, StatBuffSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## StatBuffSheet()

```
public StatBuffSheet()
```

# Class StatBuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StatBuffSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← StatBuffSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Chance

100: 100%

```
public int Chance { get; }
```

### Property Value

int

## Duration

Turn count.

```
public int Duration { get; }
```

### Property Value

int

## GroupId

GroupId는 같은 GroupId를 공유하는 버프 중 가장 작은 값으로 함. Id가 다르더라도 GroupId가 같으면, 서로 덮어씀. 이때의 우선순위는 Id가 큰 것이 남는 것으로 구현함.

```
public int GroupId { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## IsEnhanceable

```
public bool IsEnhanceable { get; }
```

### Property Value

bool

## Key

```
public override int Key { get; }
```

### Property Value

int

## MaxStack

```
public int MaxStack { get; }
```

### Property Value

int

## OperationType

```
public StatModifier.OperationType OperationType { get; }
```

### Property Value

StatModifier.OperationType

## StatType

```
public StatType StatType { get; }
```

### Property Value

StatType

## TargetType

```
public SkillTargetType TargetType { get; }
```

### Property Value

SkillTargetType

## Value

```
public long Value { get; }
```

### Property Value

long

## Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class SweepRequiredCPSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SweepRequiredCPSheet : Sheet<int, SweepRequiredCPSheet.Row>,
IDictionary<int, SweepRequiredCPSheet.Row>, ICollection<KeyValuePair<int,
SweepRequiredCPSheet.Row>>, IEnumerable<KeyValuePair<int,
SweepRequiredCPSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SweepRequiredCPSheet.Row> ← SweepRequiredCPSheet

### Implements

IDictionary<int, SweepRequiredCPSheet.Row>,
ICollection<KeyValuePair<int, SweepRequiredCPSheet.Row>>,
IEnumerable<KeyValuePair<int, SweepRequiredCPSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, SweepRequiredCPSheet.Row>.Name ,
Sheet<int, SweepRequiredCPSheet.Row>.OrderedList ,
Sheet<int, SweepRequiredCPSheet.Row>.First ,
Sheet<int, SweepRequiredCPSheet.Row>.Last ,
Sheet<int, SweepRequiredCPSheet.Row>.Keys ,
Sheet<int, SweepRequiredCPSheet.Row>.Values ,
Sheet<int, SweepRequiredCPSheet.Row>.Count ,
Sheet<int, SweepRequiredCPSheet.Row>.IsReadOnly ,
Sheet<int, SweepRequiredCPSheet.Row>.this[int] ,
Sheet<int, SweepRequiredCPSheet.Row>.Set(string, bool) ,
Sheet<int, SweepRequiredCPSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SweepRequiredCPSheet.Row>.GetEnumerator() ,
Sheet<int, SweepRequiredCPSheet.Row>.TryGetValue(int, out SweepRequiredCPSheet.Row,
bool) ,
Sheet<int, SweepRequiredCPSheet.Row>.AddRow(int, SweepRequiredCPSheet.Row) ,
Sheet<int, SweepRequiredCPSheet.Row>.Add(int, SweepRequiredCPSheet.Row) ,
Sheet<int, SweepRequiredCPSheet.Row>.ContainsKey(int) ,
Sheet<int, SweepRequiredCPSheet.Row>.Remove(int) ,

Sheet<int, SweepRequiredCPSheet.Row>.TryGetValue(int, out SweepRequiredCPSheet.Row) , Sheet<int, SweepRequiredCPSheet.Row>.Add(KeyValuePair<int, SweepRequiredCPSheet.Row>) , Sheet<int, SweepRequiredCPSheet.Row>.Clear() , Sheet<int, SweepRequiredCPSheet.Row>.Contains(KeyValuePair<int, SweepRequiredCPSheet.Row>) , Sheet<int, SweepRequiredCPSheet.Row>.CopyTo(KeyValuePair<int, SweepRequiredCPSheet.Row>[], int) , Sheet<int, SweepRequiredCPSheet.Row>.Remove(KeyValuePair<int, SweepRequiredCPSheet.Row>) , Sheet<int, SweepRequiredCPSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SweepRequiredCPSheet()

```
public SweepRequiredCPSheet()
```

# Class SweepRequiredCPSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SweepRequiredCPSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← SweepRequiredCPSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

## RequiredCP

```
public long RequiredCP { get; }
```

### Property Value

long

## StageId

```
public int StageId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class SynthesizeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a SynthesizeSheet.

```
[Serializable]
public class SynthesizeSheet : Sheet<int, SynthesizeSheet.Row>, IDictionary<int,
SynthesizeSheet.Row>, ICollection<KeyValuePair<int, SynthesizeSheet.Row>>,
IEnumerable<KeyValuePair<int, SynthesizeSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SynthesizeSheet.Row> ← SynthesizeSheet

### Implements

IDictionary<int, SynthesizeSheet.Row>,
ICollection<KeyValuePair<int, SynthesizeSheet.Row>>,
IEnumerable<KeyValuePair<int, SynthesizeSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, SynthesizeSheet.Row>.Name , Sheet<int, SynthesizeSheet.Row>.OrderedList ,
Sheet<int, SynthesizeSheet.Row>.First , Sheet<int, SynthesizeSheet.Row>.Last ,
Sheet<int, SynthesizeSheet.Row>.Keys , Sheet<int, SynthesizeSheet.Row>.Values ,
Sheet<int, SynthesizeSheet.Row>.Count , Sheet<int, SynthesizeSheet.Row>.IsReadOnly ,
Sheet<int, SynthesizeSheet.Row>.this[int] ,
Sheet<int, SynthesizeSheet.Row>.Set(string, bool) ,
Sheet<int, SynthesizeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SynthesizeSheet.Row>.GetEnumerator() ,
Sheet<int, SynthesizeSheet.Row>.TryGetValue(int, out SynthesizeSheet.Row, bool) ,
Sheet<int, SynthesizeSheet.Row>.AddRow(int, SynthesizeSheet.Row) ,
Sheet<int, SynthesizeSheet.Row>.Add(int, SynthesizeSheet.Row) ,
Sheet<int, SynthesizeSheet.Row>.ContainsKey(int) ,
Sheet<int, SynthesizeSheet.Row>.Remove(int) ,
Sheet<int, SynthesizeSheet.Row>.TryGetValue(int, out SynthesizeSheet.Row) ,
Sheet<int, SynthesizeSheet.Row>.Add(KeyValuePair<int, SynthesizeSheet.Row>) ,
Sheet<int, SynthesizeSheet.Row>.Clear() ,
Sheet<int, SynthesizeSheet.Row>.Contains(KeyValuePair<int, SynthesizeSheet.Row>) ,
Sheet<int, SynthesizeSheet.Row>.CopyTo(KeyValuePair<int, SynthesizeSheet.Row>[], int) ,
Sheet<int, SynthesizeSheet.Row>.Remove(KeyValuePair<int, SynthesizeSheet.Row>) ,

Sheet<int, SynthesizeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SynthesizeSheet()

```
public SynthesizeSheet()
```

# Fields

## SucceedRateMax

```
public const int SucceedRateMax = 10000
```

### Field Value

int

## SucceedRateMin

```
public const int SucceedRateMin = 0
```

### Field Value

int

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, SynthesizeSheet.Row value)
```

## Parameters

`key` int

`value` SynthesizeSheet.Row

# Class SynthesizeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SynthesizeSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← SynthesizeSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## GradeId

```
public int GradeId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## RequiredCountDict

```
public Dictionary<ItemSubType, SynthesizeSheet.SynthesizeData> RequiredCountDict {
get; }
```

### Property Value

Dictionary<ItemSubType, SynthesizeSheet.SynthesizeData>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Struct SynthesizeSheet.SynthesizeData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

synthesize data for each subtype

```
public struct SynthesizeSheet.SynthesizeData
```

**Inherited Members**
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Fields

### RequiredCount

```
public int RequiredCount
```

#### Field Value

int

### SucceedRate

```
public int SucceedRate
```

#### Field Value

int

# Class SynthesizeWeightSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a SynthesizeWeightSheet.

```
[Serializable]
public class SynthesizeWeightSheet : Sheet<int, SynthesizeWeightSheet.Row>,
IDictionary<int, SynthesizeWeightSheet.Row>, ICollection<KeyValuePair<int,
SynthesizeWeightSheet.Row>>, IEnumerable<KeyValuePair<int,
SynthesizeWeightSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SynthesizeWeightSheet.Row> ← SynthesizeWeightSheet

### Implements

IDictionary<int, SynthesizeWeightSheet.Row>,
ICollection<KeyValuePair<int, SynthesizeWeightSheet.Row>>,
IEnumerable<KeyValuePair<int, SynthesizeWeightSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, SynthesizeWeightSheet.Row>.Name ,
Sheet<int, SynthesizeWeightSheet.Row>.OrderedList ,
Sheet<int, SynthesizeWeightSheet.Row>.First ,
Sheet<int, SynthesizeWeightSheet.Row>.Last ,
Sheet<int, SynthesizeWeightSheet.Row>.Keys ,
Sheet<int, SynthesizeWeightSheet.Row>.Values ,
Sheet<int, SynthesizeWeightSheet.Row>.Count ,
Sheet<int, SynthesizeWeightSheet.Row>.IsReadOnly ,
Sheet<int, SynthesizeWeightSheet.Row>.this[int] ,
Sheet<int, SynthesizeWeightSheet.Row>.Set(string, bool) ,
Sheet<int, SynthesizeWeightSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SynthesizeWeightSheet.Row>.GetEnumerator() ,
Sheet<int, SynthesizeWeightSheet.Row>.TryGetValue(int, out SynthesizeWeightSheet.Row,
bool) ,
Sheet<int, SynthesizeWeightSheet.Row>.AddRow(int, SynthesizeWeightSheet.Row) ,
Sheet<int, SynthesizeWeightSheet.Row>.Add(int, SynthesizeWeightSheet.Row) ,
Sheet<int, SynthesizeWeightSheet.Row>.ContainsKey(int) ,

Sheet<int, SynthesizeWeightSheet.Row>.Remove(int) ,
Sheet<int, SynthesizeWeightSheet.Row>.TryGetValue(int, out SynthesizeWeightSheet.Row)
,
Sheet<int, SynthesizeWeightSheet.Row>.Add(KeyValuePair<int,
SynthesizeWeightSheet.Row>) ,
Sheet<int, SynthesizeWeightSheet.Row>.Clear() ,
Sheet<int, SynthesizeWeightSheet.Row>.Contains(KeyValuePair<int,
SynthesizeWeightSheet.Row>) ,
Sheet<int, SynthesizeWeightSheet.Row>.CopyTo(KeyValuePair<int,
SynthesizeWeightSheet.Row>[], int) ,
Sheet<int, SynthesizeWeightSheet.Row>.Remove(KeyValuePair<int,
SynthesizeWeightSheet.Row>) ,
Sheet<int, SynthesizeWeightSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SynthesizeWeightSheet()

Creates an empty sheet.

```
public SynthesizeWeightSheet()
```

# Fields

## DefaultWeight

Weight of an item this sheet does not list, or lists with an unreadable weight.

```
public const int DefaultWeight = 0
```

### Field Value

int

### Remarks

This used to be 10000, which meant a result pool held every item of its grade and sub type until someone remembered to write a 0 here. An item sheet row was therefore enough to put a costume into circulation, which is how four grade 8 costumes were handed out in 2026-08. A pool is spelled out now instead: what this sheet does not list is not drawn.

# Class SynthesizeWeightSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SynthesizeWeightSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← SynthesizeWeightSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ItemId

```
public int ItemId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Weight

```
public int Weight { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class TableExtensions

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public static class TableExtensions
```

**Inheritance**

object ← TableExtensions

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### ParseBigInteger(string)

```
public static BigInteger ParseBigInteger(string value)
```

#### Parameters

`value` string

#### Returns

BigInteger

### ParseBool(string, bool)

```
public static bool ParseBool(string value, bool defaultValue)
```

#### Parameters

`value` string

`defaultValue` bool

### Returns

bool

## ParseCurrency(string)

Parse a currency from a string.

```
public static Currency ParseCurrency(string value)
```

### Parameters

`value` string

The string to parse. Currency should be in the form of "ticker;decimalPlaces;minters;totalSupplyTractable;maximumSupply".

### Returns

Currency

The parsed currency.

### Exceptions

ArgumentException

The string is not a valid currency.

## ParseDecimal(string)

```
public static decimal ParseDecimal(string value)
```

### Parameters

`value` string

### Returns

decimal

## ParseDecimal(string, decimal)

```
public static decimal ParseDecimal(string value, decimal defaultValue)
```

### Parameters

`value` string

`defaultValue` decimal

### Returns

decimal

## ParseElementalTypes(string)

Parses a colon-separated list of elemental types. Accepts numeric values matching ElementalType enum. Returns an empty list if input is null/empty or no valid items.

```
public static List<ElementalType> ParseElementalTypes(string value)
```

### Parameters

`value` string

### Returns

List<ElementalType>

## ParseFraction(string)

Parse a fraction from a string.

```
public static SwapRateSheet.Fraction ParseFraction(string value)
```

### Parameters

`value` string

The string to parse. Fraction should be in the form of "numerator/denominator".

### Returns

SwapRateSheet.Fraction

The parsed fraction.

### Exceptions

ArgumentException

The string is not a valid fraction.

## ParseInt(string)

```
public static int ParseInt(string value)
```

### Parameters

`value` string

### Returns

int

## ParseInt(string, int)

```
public static int ParseInt(string value, int defaultValue)
```

### Parameters

`value` string

`defaultValue` int

### Returns

int

## ParseIntList(string)

Parses a colon-separated list of integers. Returns an empty list for null/empty input. Invalid tokens are ignored.

```
public static List<int> ParseIntList(string value)
```

### Parameters

`value` string

### Returns

List<int>

## ParseIntOrNull(string)

Parses an integer or returns null for empty input.

```
public static int? ParseIntOrNull(string value)
```

### Parameters

`value` string

### Returns

int?

## ParseItemSubTypes(string)

Parses a colon-separated list of item subtypes. Accepts numeric values matching ItemSubType enum. Returns an empty list if input is null/empty or no valid items.

```
public static List<ItemSubType> ParseItemSubTypes(string value)
```

### Parameters

`value` string

### Returns

List<ItemSubType>

## ParseLong(string)

```
public static long ParseLong(string value)
```

### Parameters

`value` string

### Returns

long

## ParseLong(string, long)

```
public static long ParseLong(string value, long defaultValue)
```

### Parameters

`value` string

`defaultValue` long

### Returns

long

## ParseRuneTypes(string)

Parses a comma-separated list of rune types from a string. Returns an empty list if input is null/empty or no valid items.

```
public static List<RuneType> ParseRuneTypes(string value)
```

### Parameters

`value` string

### Returns

List<RuneType>

## ParseSkillTargetTypes(string)

Parses a colon-separated list of skill target types. Accepts numeric values matching SkillTargetType enum. Returns an empty list if input is null/empty or no valid items.

```
public static List<SkillTargetType> ParseSkillTargetTypes(string value)
```

### Parameters

`value` string

### Returns

List<SkillTargetType>

## ParseStringOrNull(string)

Parses a trimmed string or returns null for empty input.

```
public static string? ParseStringOrNull(string value)
```

### Parameters

`value` string

### Returns

string

## TryParseBigInteger(string, out BigInteger)

```
public static bool TryParseBigInteger(string value, out BigInteger result)
```

### Parameters

`value` string

`result` BigInteger

### Returns

bool

## TryParseCurrency(string, out Currency)

Try to parse a currency from a string.

```
public static bool TryParseCurrency(string value, out Currency currency)
```

### Parameters

`value` string

> The string to parse. Currency should be in the form of "ticker;decimalPlaces;minters;totalSupplyTractable;maximumSupply".

`currency` Currency

The parsed currency.

### Returns

bool

Whether the parsing is successful.

## TryParseDecimal(string, out decimal)

```
public static bool TryParseDecimal(string value, out decimal result)
```

### Parameters

`value` string

`result` decimal

### Returns

bool

## TryParseFloat(string, out float)

```
public static bool TryParseFloat(string value, out float result)
```

### Parameters

`value` string

`result` float

### Returns

bool

## TryParseFraction(string, out Fraction)

Try to parse a fraction from a string.

```
public static bool TryParseFraction(string value, out SwapRateSheet.Fraction result)
```

### Parameters

`value` string

The string to parse. Fraction should be in the form of "numerator/denominator".

`result` SwapRateSheet.Fraction

The parsed fraction.

### Returns

bool

## TryParseInt(string, out int)

```
public static bool TryParseInt(string value, out int result)
```

### Parameters

`value` string

`result` int

### Returns

bool

## TryParseLong(string, out long)

```
public static bool TryParseLong(string value, out long result)
```

## Parameters

`value` string

`result` long

## Returns

bool

# Class TradeQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class TradeQuestSheet : Sheet<int, TradeQuestSheet.Row>, IDictionary<int,
TradeQuestSheet.Row>, ICollection<KeyValuePair<int, TradeQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, TradeQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, TradeQuestSheet.Row> ← TradeQuestSheet

### Implements

IDictionary<int, TradeQuestSheet.Row>,
ICollection<KeyValuePair<int, TradeQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, TradeQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, TradeQuestSheet.Row>.Name , Sheet<int, TradeQuestSheet.Row>.OrderedList ,
Sheet<int, TradeQuestSheet.Row>.First , Sheet<int, TradeQuestSheet.Row>.Last ,
Sheet<int, TradeQuestSheet.Row>.Keys , Sheet<int, TradeQuestSheet.Row>.Values ,
Sheet<int, TradeQuestSheet.Row>.Count , Sheet<int, TradeQuestSheet.Row>.IsReadOnly ,
Sheet<int, TradeQuestSheet.Row>.this[int] ,
Sheet<int, TradeQuestSheet.Row>.Set(string, bool) ,
Sheet<int, TradeQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, TradeQuestSheet.Row>.GetEnumerator() ,
Sheet<int, TradeQuestSheet.Row>.TryGetValue(int, out TradeQuestSheet.Row, bool) ,
Sheet<int, TradeQuestSheet.Row>.AddRow(int, TradeQuestSheet.Row) ,
Sheet<int, TradeQuestSheet.Row>.Add(int, TradeQuestSheet.Row) ,
Sheet<int, TradeQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, TradeQuestSheet.Row>.Remove(int) ,
Sheet<int, TradeQuestSheet.Row>.TryGetValue(int, out TradeQuestSheet.Row) ,
Sheet<int, TradeQuestSheet.Row>.Add(KeyValuePair<int, TradeQuestSheet.Row>) ,
Sheet<int, TradeQuestSheet.Row>.Clear() ,
Sheet<int, TradeQuestSheet.Row>.Contains(KeyValuePair<int, TradeQuestSheet.Row>) ,
Sheet<int, TradeQuestSheet.Row>.CopyTo(KeyValuePair<int, TradeQuestSheet.Row>[], int)
,
Sheet<int, TradeQuestSheet.Row>.Remove(KeyValuePair<int, TradeQuestSheet.Row>) ,

Sheet<int, TradeQuestSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## TradeQuestSheet()

```
public TradeQuestSheet()
```

# Class TradeQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class TradeQuestSheet.Row : QuestSheet.Row, IState
```

**Inheritance**

object ← SheetRow<int> ← QuestSheet.Row ← TradeQuestSheet.Row

**Implements**

IState

**Inherited Members**

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Type

```
public TradeType Type { get; }
```

### Property Value

TradeType

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class UnlockCombinationSlotCostSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class UnlockCombinationSlotCostSheet : Sheet<int,
UnlockCombinationSlotCostSheet.Row>, IDictionary<int,
UnlockCombinationSlotCostSheet.Row>, ICollection<KeyValuePair<int,
UnlockCombinationSlotCostSheet.Row>>, IEnumerable<KeyValuePair<int,
UnlockCombinationSlotCostSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, UnlockCombinationSlotCostSheet.Row> ← UnlockCombinationSlotCostSheet

### Implements

IDictionary<int, UnlockCombinationSlotCostSheet.Row>, ICollection<KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>>, IEnumerable<KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, UnlockCombinationSlotCostSheet.Row>.Name ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.OrderedList ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.First ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Last ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Keys ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Values ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Count ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.IsReadOnly ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.this[int] ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Set(string, bool) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.GetEnumerator() ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.TryGetValue(int, out UnlockCombinationSlotCostSheet.Row, bool) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.AddRow(int, UnlockCombinationSlotCostSheet.Row) ,

Sheet<int, UnlockCombinationSlotCostSheet.Row>.Add(int, UnlockCombinationSlotCostSheet.Row) , Sheet<int, UnlockCombinationSlotCostSheet.Row>.ContainsKey(int) , Sheet<int, UnlockCombinationSlotCostSheet.Row>.Remove(int) , Sheet<int, UnlockCombinationSlotCostSheet.Row>.TryGetValue(int, out UnlockCombinationSlotCostSheet.Row) , Sheet<int, UnlockCombinationSlotCostSheet.Row>.Add(KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>) , Sheet<int, UnlockCombinationSlotCostSheet.Row>.Clear() , Sheet<int, UnlockCombinationSlotCostSheet.Row>.Contains(KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>) , Sheet<int, UnlockCombinationSlotCostSheet.Row>.CopyTo(KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>[], int) , Sheet<int, UnlockCombinationSlotCostSheet.Row>.Remove(KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>) , Sheet<int, UnlockCombinationSlotCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## UnlockCombinationSlotCostSheet()

```
public UnlockCombinationSlotCostSheet()
```

# Class UnlockCombinationSlotCostSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class UnlockCombinationSlotCostSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← UnlockCombinationSlotCostSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## CrystalPrice

```
public int CrystalPrice
```

### Field Value

int

## GoldenDustPrice

```
public int GoldenDustPrice
```

### Field Value

int

## NcgPrice

```
public BigInteger NcgPrice
```

### Field Value

BigInteger

## RubyDustPrice

```
public int RubyDustPrice
```

### Field Value

int

## SlotId

```
public int SlotId
```

### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class WeeklyArenaRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaRewardSheet : Sheet<int, WeeklyArenaRewardSheet.Row>,
IDictionary<int, WeeklyArenaRewardSheet.Row>, ICollection<KeyValuePair<int,
WeeklyArenaRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
WeeklyArenaRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WeeklyArenaRewardSheet.Row> ← WeeklyArenaRewardSheet

### Implements

IDictionary<int, WeeklyArenaRewardSheet.Row>,
ICollection<KeyValuePair<int, WeeklyArenaRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WeeklyArenaRewardSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, WeeklyArenaRewardSheet.Row>.Name ,
Sheet<int, WeeklyArenaRewardSheet.Row>.OrderedList ,
Sheet<int, WeeklyArenaRewardSheet.Row>.First ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Last ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Keys ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Values ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Count ,
Sheet<int, WeeklyArenaRewardSheet.Row>.IsReadOnly ,
Sheet<int, WeeklyArenaRewardSheet.Row>.this[int] ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WeeklyArenaRewardSheet.Row>.TryGetValue(int, out
WeeklyArenaRewardSheet.Row, bool) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.AddRow(int, WeeklyArenaRewardSheet.Row) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Add(int, WeeklyArenaRewardSheet.Row) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Remove(int) ,

Sheet<int, WeeklyArenaRewardSheet.Row>.TryGetValue(int, out WeeklyArenaRewardSheet.Row) , Sheet<int, WeeklyArenaRewardSheet.Row>.Add(KeyValuePair<int, WeeklyArenaRewardSheet.Row>) , Sheet<int, WeeklyArenaRewardSheet.Row>.Clear() , Sheet<int, WeeklyArenaRewardSheet.Row>.Contains(KeyValuePair<int, WeeklyArenaRewardSheet.Row>) , Sheet<int, WeeklyArenaRewardSheet.Row>.CopyTo(KeyValuePair<int, WeeklyArenaRewardSheet.Row>[], int) , Sheet<int, WeeklyArenaRewardSheet.Row>.Remove(KeyValuePair<int, WeeklyArenaRewardSheet.Row>) , Sheet<int, WeeklyArenaRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WeeklyArenaRewardSheet()

```
public WeeklyArenaRewardSheet()
```

# Class WeeklyArenaRewardSheet.RewardData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaRewardSheet.RewardData : StageSheet.RewardData
```

**Inheritance**

object ← StageSheet.RewardData ← WeeklyArenaRewardSheet.RewardData

**Inherited Members**

StageSheet.RewardData.ItemId , StageSheet.RewardData.Ratio , StageSheet.RewardData.Min , StageSheet.RewardData.Max , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardData(int, decimal, int, int, int)

```
public RewardData(int itemId, decimal ratio, int min, int max, int requiredLevel)
```

#### Parameters

`itemId` int

`ratio` decimal

`min` int

`max` int

`requiredLevel` int

# Properties

## RequiredLevel

```
public int RequiredLevel { get; }
```

### Property Value

int

# Class WeeklyArenaRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WeeklyArenaRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

# Reward

```
public WeeklyArenaRewardSheet.RewardData Reward { get; }
```

## Property Value

WeeklyArenaRewardSheet.RewardData

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class WorldBossActionPatternSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossActionPatternSheet : Sheet<int,
WorldBossActionPatternSheet.Row>, IDictionary<int, WorldBossActionPatternSheet.Row>,
ICollection<KeyValuePair<int, WorldBossActionPatternSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossActionPatternSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossActionPatternSheet.Row> ← WorldBossActionPatternSheet

### Implements

IDictionary<int, WorldBossActionPatternSheet.Row>,
ICollection<KeyValuePair<int, WorldBossActionPatternSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossActionPatternSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, WorldBossActionPatternSheet.Row>.Name ,
Sheet<int, WorldBossActionPatternSheet.Row>.OrderedList ,
Sheet<int, WorldBossActionPatternSheet.Row>.First ,
Sheet<int, WorldBossActionPatternSheet.Row>.Last ,
Sheet<int, WorldBossActionPatternSheet.Row>.Keys ,
Sheet<int, WorldBossActionPatternSheet.Row>.Values ,
Sheet<int, WorldBossActionPatternSheet.Row>.Count ,
Sheet<int, WorldBossActionPatternSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossActionPatternSheet.Row>.this[int] ,
Sheet<int, WorldBossActionPatternSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossActionPatternSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossActionPatternSheet.Row>.TryGetValue(int, out
WorldBossActionPatternSheet.Row, bool) ,
Sheet<int, WorldBossActionPatternSheet.Row>.AddRow(int,
WorldBossActionPatternSheet.Row) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Add(int, WorldBossActionPatternSheet.Row)
,
Sheet<int, WorldBossActionPatternSheet.Row>.ContainsKey(int) ,

Sheet<int, WorldBossActionPatternSheet.Row>.Remove(int) ,
Sheet<int, WorldBossActionPatternSheet.Row>.TryGetValue(int, out
WorldBossActionPatternSheet.Row) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Add(KeyValuePair<int,
WorldBossActionPatternSheet.Row>) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Clear() ,
Sheet<int, WorldBossActionPatternSheet.Row>.Contains(KeyValuePair<int,
WorldBossActionPatternSheet.Row>) ,
Sheet<int, WorldBossActionPatternSheet.Row>.CopyTo(KeyValuePair<int,
WorldBossActionPatternSheet.Row>[], int) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Remove(KeyValuePair<int,
WorldBossActionPatternSheet.Row>) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldBossActionPatternSheet()

```
public WorldBossActionPatternSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, WorldBossActionPatternSheet.Row value)
```

### Parameters

`key` int

`value` WorldBossActionPatternSheet.Row

# Class WorldBossActionPatternSheet.ActionPatternData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossActionPatternSheet.ActionPatternData
```

**Inheritance**

object ← WorldBossActionPatternSheet.ActionPatternData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## SkillIds

```
public readonly List<int> SkillIds
```

### Field Value

List<int>

## Wave

```
public int Wave
```

### Field Value

int

# Class WorldBossActionPatternSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossActionPatternSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WorldBossActionPatternSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BossId

```
public int BossId
```

### Field Value

int

## Patterns

```
public List<WorldBossActionPatternSheet.ActionPatternData> Patterns
```

### Field Value

List<WorldBossActionPatternSheet.ActionPatternData>

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class WorldBossBattleRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossBattleRewardSheet : Sheet<int,
WorldBossBattleRewardSheet.Row>, IDictionary<int, WorldBossBattleRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossBattleRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossBattleRewardSheet.Row>>, IEnumerable,
IWorldBossRewardSheet, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossBattleRewardSheet.Row> ← WorldBossBattleRewardSheet

### Implements

IDictionary<int, WorldBossBattleRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossBattleRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossBattleRewardSheet.Row>>, IEnumerable,
IWorldBossRewardSheet, ISheet

### Inherited Members

Sheet<int, WorldBossBattleRewardSheet.Row>.Name ,
Sheet<int, WorldBossBattleRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossBattleRewardSheet.Row>.First ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Last ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Keys ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Values ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Count ,
Sheet<int, WorldBossBattleRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossBattleRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossBattleRewardSheet.Row>.TryGetValue(int, out
WorldBossBattleRewardSheet.Row, bool) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.AddRow(int,
WorldBossBattleRewardSheet.Row) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Add(int, WorldBossBattleRewardSheet.Row)
,
Sheet<int, WorldBossBattleRewardSheet.Row>.ContainsKey(int) ,

Sheet<int, WorldBossBattleRewardSheet.Row>.Remove(int) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.TryGetValue(int, out
WorldBossBattleRewardSheet.Row) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Add(KeyValuePair<int,
WorldBossBattleRewardSheet.Row>) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Clear() ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Contains(KeyValuePair<int,
WorldBossBattleRewardSheet.Row>) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.CopyTo(KeyValuePair<int,
WorldBossBattleRewardSheet.Row>[], int) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Remove(KeyValuePair<int,
WorldBossBattleRewardSheet.Row>) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldBossBattleRewardSheet()

```
public WorldBossBattleRewardSheet()
```

# Properties

## OrderedRows

```
public IReadOnlyList<IWorldBossRewardRow> OrderedRows { get; }
```

### Property Value

IReadOnlyList<IWorldBossRewardRow>

# Class WorldBossBattleRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossBattleRewardSheet.Row : SheetRow<int>, IWorldBossRewardRow
```

**Inheritance**

object ← SheetRow<int> ← WorldBossBattleRewardSheet.Row

**Implements**

IWorldBossRewardRow

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Id

```
public int Id
```

### Field Value

int

## RuneMax

```
public int RuneMax
```

### Field Value

int

## RuneMin

```
public int RuneMin
```

### Field Value

int

# Properties

## BossId

```
public int BossId { get; }
```

### Property Value

int

## Circle

```
public int Circle { get; }
```

### Property Value

int

## Crystal

```
public int Crystal { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Rank

```
public int Rank { get; }
```

### Property Value

int

## Rune

```
public int Rune { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

## SetRune(IRandom)

```
public void SetRune(IRandom random)
```

### Parameters

`random` IRandom

# Class WorldBossCharacterSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossCharacterSheet : Sheet<int, WorldBossCharacterSheet.Row>,
IDictionary<int, WorldBossCharacterSheet.Row>, ICollection<KeyValuePair<int,
WorldBossCharacterSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossCharacterSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossCharacterSheet.Row> ← WorldBossCharacterSheet

### Implements

IDictionary<int, WorldBossCharacterSheet.Row>,
ICollection<KeyValuePair<int, WorldBossCharacterSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossCharacterSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, WorldBossCharacterSheet.Row>.Name ,
Sheet<int, WorldBossCharacterSheet.Row>.OrderedList ,
Sheet<int, WorldBossCharacterSheet.Row>.First ,
Sheet<int, WorldBossCharacterSheet.Row>.Last ,
Sheet<int, WorldBossCharacterSheet.Row>.Keys ,
Sheet<int, WorldBossCharacterSheet.Row>.Values ,
Sheet<int, WorldBossCharacterSheet.Row>.Count ,
Sheet<int, WorldBossCharacterSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossCharacterSheet.Row>.this[int] ,
Sheet<int, WorldBossCharacterSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossCharacterSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossCharacterSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossCharacterSheet.Row>.TryGetValue(int, out
WorldBossCharacterSheet.Row, bool) ,
Sheet<int, WorldBossCharacterSheet.Row>.AddRow(int, WorldBossCharacterSheet.Row) ,
Sheet<int, WorldBossCharacterSheet.Row>.Add(int, WorldBossCharacterSheet.Row) ,
Sheet<int, WorldBossCharacterSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossCharacterSheet.Row>.Remove(int) ,

Sheet<int, WorldBossCharacterSheet.Row>.TryGetValue(int, out WorldBossCharacterSheet.Row) , Sheet<int, WorldBossCharacterSheet.Row>.Add(KeyValuePair<int, WorldBossCharacterSheet.Row>) , Sheet<int, WorldBossCharacterSheet.Row>.Clear() , Sheet<int, WorldBossCharacterSheet.Row>.Contains(KeyValuePair<int, WorldBossCharacterSheet.Row>) , Sheet<int, WorldBossCharacterSheet.Row>.CopyTo(KeyValuePair<int, WorldBossCharacterSheet.Row>[], int) , Sheet<int, WorldBossCharacterSheet.Row>.Remove(KeyValuePair<int, WorldBossCharacterSheet.Row>) , Sheet<int, WorldBossCharacterSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldBossCharacterSheet()

```
public WorldBossCharacterSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, WorldBossCharacterSheet.Row value)
```

### Parameters

`key` int

`value` WorldBossCharacterSheet.Row

# Class WorldBossCharacterSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossCharacterSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WorldBossCharacterSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## BossId

```
public int BossId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## WaveStats

```
public List<WorldBossCharacterSheet.WaveStatData> WaveStats { get; }
```

### Property Value

List<WorldBossCharacterSheet.WaveStatData>

# Methods

## EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class WorldBossCharacterSheet.WaveStatData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossCharacterSheet.WaveStatData
```

**Inheritance**

object ← WorldBossCharacterSheet.WaveStatData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

WorldBossStatSheetExtension.ToStats(WorldBossCharacterSheet.WaveStatData)

## Properties

### ATK

```
public decimal ATK { get; set; }
```

#### Property Value

decimal

### CRI

```
public decimal CRI { get; set; }
```

#### Property Value

decimal

## DEF

```
public decimal DEF { get; set; }
```

### Property Value

decimal

## ElementalType

```
public ElementalType ElementalType { get; set; }
```

### Property Value

ElementalType

## EnrageSkillId

```
public int EnrageSkillId { get; set; }
```

### Property Value

int

## EnrageTurn

```
public int EnrageTurn { get; set; }
```

### Property Value

int

## HIT

```
public decimal HIT { get; set; }
```

### Property Value

decimal

## HP

```
public decimal HP { get; set; }
```

### Property Value

decimal

## Level

```
public int Level { get; set; }
```

### Property Value

int

## SPD

```
public decimal SPD { get; set; }
```

### Property Value

decimal

## TurnLimit

```
public int TurnLimit { get; set; }
```

### Property Value

int

## Wave

```
public int Wave { get; set; }
```

### Property Value

int

# Class WorldBossContributionRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents the reward sheet for world boss contributions.

```
public class WorldBossContributionRewardSheet : Sheet<int,
WorldBossContributionRewardSheet.Row>, IDictionary<int,
WorldBossContributionRewardSheet.Row>, ICollection<KeyValuePair<int,
WorldBossContributionRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossContributionRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossContributionRewardSheet.Row> ← WorldBossContributionRewardSheet

### Implements

IDictionary<int, WorldBossContributionRewardSheet.Row>, ICollection<KeyValuePair<int, WorldBossContributionRewardSheet.Row>>, IEnumerable<KeyValuePair<int, WorldBossContributionRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, WorldBossContributionRewardSheet.Row>.Name ,
Sheet<int, WorldBossContributionRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossContributionRewardSheet.Row>.First ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Last ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Keys ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Values ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Count ,
Sheet<int, WorldBossContributionRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossContributionRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossContributionRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossContributionRewardSheet.Row>.TryGetValue(int, out WorldBossContributionRewardSheet.Row, bool) ,
Sheet<int, WorldBossContributionRewardSheet.Row>.AddRow(int, WorldBossContributionRewardSheet.Row) ,

Sheet<int, WorldBossContributionRewardSheet.Row>.Add(int, WorldBossContributionRewardSheet.Row) , Sheet<int, WorldBossContributionRewardSheet.Row>.ContainsKey(int) , Sheet<int, WorldBossContributionRewardSheet.Row>.Remove(int) , Sheet<int, WorldBossContributionRewardSheet.Row>.TryGetValue(int, out WorldBossContributionRewardSheet.Row) , Sheet<int, WorldBossContributionRewardSheet.Row>.Add(KeyValuePair<int, WorldBossContributionRewardSheet.Row>) , Sheet<int, WorldBossContributionRewardSheet.Row>.Clear() , Sheet<int, WorldBossContributionRewardSheet.Row>.Contains(KeyValuePair<int, WorldBossContributionRewardSheet.Row>) , Sheet<int, WorldBossContributionRewardSheet.Row>.CopyTo(KeyValuePair<int, WorldBossContributionRewardSheet.Row>[], int) , Sheet<int, WorldBossContributionRewardSheet.Row>.Remove(KeyValuePair<int, WorldBossContributionRewardSheet.Row>) , Sheet<int, WorldBossContributionRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldBossContributionRewardSheet()

Initializes a new instance of the WorldBossContributionRewardSheet class.

```
public WorldBossContributionRewardSheet()
```

# Class WorldBossContributionRewardSheet.RewardModel

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossContributionRewardSheet.RewardModel
```

**Inheritance**

object ← WorldBossContributionRewardSheet.RewardModel

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardModel(BigInteger, int, string)

Initializes a new instance of the WorldBossContributionRewardSheet.RewardModel class.

```
public RewardModel(BigInteger count, int itemId, string ticker)
```

#### Parameters

`count` BigInteger

The quantity of the reward.

`itemId` int

The ID of the item if reward is ItemBase.

`ticker` string

The ticker of the item if reward is FungibleAssetValue.

# Fields

## Count

Gets or sets the quantity of the reward.

```
public BigInteger Count
```

### Field Value

BigInteger

## ItemId

Gets or sets the ID of the item.

```
public int ItemId
```

### Field Value

int

## Ticker

Gets or sets the ticker of the item.

```
public string Ticker
```

### Field Value

string

# Class WorldBossContributionRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents each row in the sheet.

```
public class WorldBossContributionRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WorldBossContributionRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BossId

Gets or sets the ID of the boss.

```
public int BossId
```

### Field Value

int

## Rewards

Gets the list of rewards.

```
public readonly List<WorldBossContributionRewardSheet.RewardModel> Rewards
```

### Field Value

List<WorldBossContributionRewardSheet.RewardModel>

## Properties

### Key

Gets the key value of the row.

```
public override int Key { get; }
```

### Property Value

int

## Methods

### Set(IReadOnlyList<string>)

Sets the row based on given fields.

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

The fields to set the row.

# Class WorldBossGlobalHpSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossGlobalHpSheet : Sheet<int, WorldBossGlobalHpSheet.Row>,
IDictionary<int, WorldBossGlobalHpSheet.Row>, ICollection<KeyValuePair<int,
WorldBossGlobalHpSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossGlobalHpSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossGlobalHpSheet.Row> ← WorldBossGlobalHpSheet

### Implements

IDictionary<int, WorldBossGlobalHpSheet.Row>,
ICollection<KeyValuePair<int, WorldBossGlobalHpSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossGlobalHpSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, WorldBossGlobalHpSheet.Row>.Name ,
Sheet<int, WorldBossGlobalHpSheet.Row>.OrderedList ,
Sheet<int, WorldBossGlobalHpSheet.Row>.First ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Last ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Keys ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Values ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Count ,
Sheet<int, WorldBossGlobalHpSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossGlobalHpSheet.Row>.this[int] ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossGlobalHpSheet.Row>.TryGetValue(int, out
WorldBossGlobalHpSheet.Row, bool) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.AddRow(int, WorldBossGlobalHpSheet.Row) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Add(int, WorldBossGlobalHpSheet.Row) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Remove(int) ,

Sheet<int, WorldBossGlobalHpSheet.Row>.TryGetValue(int, out WorldBossGlobalHpSheet.Row) , Sheet<int, WorldBossGlobalHpSheet.Row>.Add(KeyValuePair<int, WorldBossGlobalHpSheet.Row>) , Sheet<int, WorldBossGlobalHpSheet.Row>.Clear() , Sheet<int, WorldBossGlobalHpSheet.Row>.Contains(KeyValuePair<int, WorldBossGlobalHpSheet.Row>) , Sheet<int, WorldBossGlobalHpSheet.Row>.CopyTo(KeyValuePair<int, WorldBossGlobalHpSheet.Row>[], int) , Sheet<int, WorldBossGlobalHpSheet.Row>.Remove(KeyValuePair<int, WorldBossGlobalHpSheet.Row>) , Sheet<int, WorldBossGlobalHpSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldBossGlobalHpSheet()

```
public WorldBossGlobalHpSheet()
```

# Class WorldBossGlobalHpSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossGlobalHpSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WorldBossGlobalHpSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Hp

```
public BigInteger Hp
```

### Field Value

BigInteger

## Level

```
public int Level
```

### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class WorldBossKillRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossKillRewardSheet : Sheet<int, WorldBossKillRewardSheet.Row>,
IDictionary<int, WorldBossKillRewardSheet.Row>, ICollection<KeyValuePair<int,
WorldBossKillRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossKillRewardSheet.Row>>, IEnumerable, IWorldBossRewardSheet, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossKillRewardSheet.Row> ← WorldBossKillRewardSheet

### Implements

IDictionary<int, WorldBossKillRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossKillRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossKillRewardSheet.Row>>, IEnumerable,
IWorldBossRewardSheet, ISheet

### Inherited Members

Sheet<int, WorldBossKillRewardSheet.Row>.Name ,
Sheet<int, WorldBossKillRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossKillRewardSheet.Row>.First ,
Sheet<int, WorldBossKillRewardSheet.Row>.Last ,
Sheet<int, WorldBossKillRewardSheet.Row>.Keys ,
Sheet<int, WorldBossKillRewardSheet.Row>.Values ,
Sheet<int, WorldBossKillRewardSheet.Row>.Count ,
Sheet<int, WorldBossKillRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossKillRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossKillRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossKillRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossKillRewardSheet.Row>.TryGetValue(int, out
WorldBossKillRewardSheet.Row, bool) ,
Sheet<int, WorldBossKillRewardSheet.Row>.AddRow(int, WorldBossKillRewardSheet.Row) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Add(int, WorldBossKillRewardSheet.Row) ,
Sheet<int, WorldBossKillRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Remove(int) ,

Sheet<int, WorldBossKillRewardSheet.Row>.TryGetValue(int, out WorldBossKillRewardSheet.Row) , Sheet<int, WorldBossKillRewardSheet.Row>.Add(KeyValuePair<int, WorldBossKillRewardSheet.Row>) , Sheet<int, WorldBossKillRewardSheet.Row>.Clear() , Sheet<int, WorldBossKillRewardSheet.Row>.Contains(KeyValuePair<int, WorldBossKillRewardSheet.Row>) , Sheet<int, WorldBossKillRewardSheet.Row>.CopyTo(KeyValuePair<int, WorldBossKillRewardSheet.Row>[], int) , Sheet<int, WorldBossKillRewardSheet.Row>.Remove(KeyValuePair<int, WorldBossKillRewardSheet.Row>) , Sheet<int, WorldBossKillRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldBossKillRewardSheet()

```
public WorldBossKillRewardSheet()
```

# Properties

## OrderedRows

```
public IReadOnlyList<IWorldBossRewardRow> OrderedRows { get; }
```

### Property Value

IReadOnlyList<IWorldBossRewardRow>

# Class WorldBossKillRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossKillRewardSheet.Row : SheetRow<int>, IWorldBossRewardRow
```

**Inheritance**

object ← SheetRow<int> ← WorldBossKillRewardSheet.Row

**Implements**

IWorldBossRewardRow

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Id

```
public int Id
```

### Field Value

int

## RuneMax

```
public int RuneMax
```

### Field Value

int

## RuneMin

```
public int RuneMin
```

### Field Value

int

# Properties

## BossId

```
public int BossId { get; }
```

### Property Value

int

## Circle

```
public int Circle { get; }
```

### Property Value

int

## Crystal

```
public int Crystal { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Rank

```
public int Rank { get; }
```

### Property Value

int

## Rune

```
public int Rune { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

## SetRune(IRandom)

```
public void SetRune(IRandom random)
```

### Parameters

`random` IRandom

# Class WorldBossListSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossListSheet : Sheet<int, WorldBossListSheet.Row>,
IDictionary<int, WorldBossListSheet.Row>, ICollection<KeyValuePair<int,
WorldBossListSheet.Row>>, IEnumerable<KeyValuePair<int, WorldBossListSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossListSheet.Row> ← WorldBossListSheet

### Implements

IDictionary<int, WorldBossListSheet.Row>,
ICollection<KeyValuePair<int, WorldBossListSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossListSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, WorldBossListSheet.Row>.Name ,
Sheet<int, WorldBossListSheet.Row>.OrderedList ,
Sheet<int, WorldBossListSheet.Row>.First , Sheet<int, WorldBossListSheet.Row>.Last ,
Sheet<int, WorldBossListSheet.Row>.Keys , Sheet<int, WorldBossListSheet.Row>.Values ,
Sheet<int, WorldBossListSheet.Row>.Count ,
Sheet<int, WorldBossListSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossListSheet.Row>.this[int] ,
Sheet<int, WorldBossListSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossListSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossListSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossListSheet.Row>.TryGetValue(int, out WorldBossListSheet.Row, bool) ,
Sheet<int, WorldBossListSheet.Row>.AddRow(int, WorldBossListSheet.Row) ,
Sheet<int, WorldBossListSheet.Row>.Add(int, WorldBossListSheet.Row) ,
Sheet<int, WorldBossListSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossListSheet.Row>.Remove(int) ,
Sheet<int, WorldBossListSheet.Row>.TryGetValue(int, out WorldBossListSheet.Row) ,
Sheet<int, WorldBossListSheet.Row>.Add(KeyValuePair<int, WorldBossListSheet.Row>) ,
Sheet<int, WorldBossListSheet.Row>.Clear() ,
Sheet<int, WorldBossListSheet.Row>.Contains(KeyValuePair<int,
WorldBossListSheet.Row>) ,

Sheet<int, WorldBossListSheet.Row>.CopyTo(KeyValuePair<int, WorldBossListSheet.Row>[], int) ,
Sheet<int, WorldBossListSheet.Row>.Remove(KeyValuePair<int, WorldBossListSheet.Row>) ,
Sheet<int, WorldBossListSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
SheetsExtensions.FindPreviousRaidIdByBlockIndex(WorldBossListSheet, long) ,
SheetsExtensions.FindPreviousRowByBlockIndex(WorldBossListSheet, long) ,
SheetsExtensions.FindRaidIdByBlockIndex(WorldBossListSheet, long) ,
SheetsExtensions.FindRowByBlockIndex(WorldBossListSheet, long)

# Constructors

## WorldBossListSheet()

```
public WorldBossListSheet()
```

# Class WorldBossListSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossListSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WorldBossListSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AdditionalTicketPrice

```
public int AdditionalTicketPrice
```

### Field Value

int

## BossId

```
public int BossId
```

### Field Value

int

## EndedBlockIndex

```
public long EndedBlockIndex
```

### Field Value

long

## EntranceFee

```
public int EntranceFee
```

### Field Value

int

## Id

```
public int Id
```

### Field Value

int

## MaxPurchaseCount

```
public int MaxPurchaseCount
```

### Field Value

int

## StartedBlockIndex

```
public long StartedBlockIndex
```

### Field Value

long

## TicketPrice

```
public int TicketPrice
```

### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class WorldBossRankRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossRankRewardSheet : Sheet<int, WorldBossRankRewardSheet.Row>,
IDictionary<int, WorldBossRankRewardSheet.Row>, ICollection<KeyValuePair<int,
WorldBossRankRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossRankRewardSheet.Row>>, IEnumerable, IWorldBossRewardSheet, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossRankRewardSheet.Row> ← WorldBossRankRewardSheet

### Implements

IDictionary<int, WorldBossRankRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossRankRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossRankRewardSheet.Row>>, IEnumerable,
IWorldBossRewardSheet, ISheet

### Inherited Members

Sheet<int, WorldBossRankRewardSheet.Row>.Name ,
Sheet<int, WorldBossRankRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossRankRewardSheet.Row>.First ,
Sheet<int, WorldBossRankRewardSheet.Row>.Last ,
Sheet<int, WorldBossRankRewardSheet.Row>.Keys ,
Sheet<int, WorldBossRankRewardSheet.Row>.Values ,
Sheet<int, WorldBossRankRewardSheet.Row>.Count ,
Sheet<int, WorldBossRankRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossRankRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossRankRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossRankRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossRankRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossRankRewardSheet.Row>.TryGetValue(int, out
WorldBossRankRewardSheet.Row, bool) ,
Sheet<int, WorldBossRankRewardSheet.Row>.AddRow(int,
WorldBossRankRewardSheet.Row) ,
Sheet<int, WorldBossRankRewardSheet.Row>.Add(int, WorldBossRankRewardSheet.Row) ,
Sheet<int, WorldBossRankRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossRankRewardSheet.Row>.Remove(int) ,

Sheet<int, WorldBossRankRewardSheet.Row>.TryGetValue(int, out WorldBossRankRewardSheet.Row) , Sheet<int, WorldBossRankRewardSheet.Row>.Add(KeyValuePair<int, WorldBossRankRewardSheet.Row>) , Sheet<int, WorldBossRankRewardSheet.Row>.Clear() , Sheet<int, WorldBossRankRewardSheet.Row>.Contains(KeyValuePair<int, WorldBossRankRewardSheet.Row>) , Sheet<int, WorldBossRankRewardSheet.Row>.CopyTo(KeyValuePair<int, WorldBossRankRewardSheet.Row>[], int) , Sheet<int, WorldBossRankRewardSheet.Row>.Remove(KeyValuePair<int, WorldBossRankRewardSheet.Row>) , Sheet<int, WorldBossRankRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldBossRankRewardSheet()

```
public WorldBossRankRewardSheet()
```

# Properties

## OrderedRows

```
public IReadOnlyList<IWorldBossRewardRow> OrderedRows { get; }
```

### Property Value

IReadOnlyList<IWorldBossRewardRow>

# Class WorldBossRankRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossRankRewardSheet.Row : SheetRow<int>, IWorldBossRewardRow
```

**Inheritance**

object ← SheetRow<int> ← WorldBossRankRewardSheet.Row

**Implements**

IWorldBossRewardRow

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Id

```
public int Id
```

### Field Value

int

# Properties

## BossId

```
public int BossId { get; }
```

### Property Value

int

## Circle

```
public int Circle { get; }
```

### Property Value

int

## Crystal

```
public int Crystal { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Rank

```
public int Rank { get; }
```

### Property Value

int

## Rune

```
public int Rune { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class WorldBossRankingRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossRankingRewardSheet : Sheet<int,
WorldBossRankingRewardSheet.Row>, IDictionary<int, WorldBossRankingRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossRankingRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossRankingRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldBossRankingRewardSheet.Row> ← WorldBossRankingRewardSheet

### Implements

IDictionary<int, WorldBossRankingRewardSheet.Row>, ICollection<KeyValuePair<int, WorldBossRankingRewardSheet.Row>>, IEnumerable<KeyValuePair<int, WorldBossRankingRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, WorldBossRankingRewardSheet.Row>.Name ,
Sheet<int, WorldBossRankingRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossRankingRewardSheet.Row>.First ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Last ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Keys ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Values ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Count ,
Sheet<int, WorldBossRankingRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossRankingRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossRankingRewardSheet.Row>.TryGetValue(int, out WorldBossRankingRewardSheet.Row, bool) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.AddRow(int, WorldBossRankingRewardSheet.Row) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Add(int, WorldBossRankingRewardSheet.Row) ,

Sheet<int, WorldBossRankingRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Remove(int) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.TryGetValue(int, out
WorldBossRankingRewardSheet.Row) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Add(KeyValuePair<int,
WorldBossRankingRewardSheet.Row>) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Clear() ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Contains(KeyValuePair<int,
WorldBossRankingRewardSheet.Row>) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.CopyTo(KeyValuePair<int,
WorldBossRankingRewardSheet.Row>[], int) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Remove(KeyValuePair<int,
WorldBossRankingRewardSheet.Row>) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldBossRankingRewardSheet()

```
public WorldBossRankingRewardSheet()
```

# Methods

## FindRow(int, int, int)

```
public WorldBossRankingRewardSheet.Row FindRow(int bossId, int ranking, int rate)
```

### Parameters

`bossId` int

`ranking` int

`rate` int

## Returns

WorldBossRankingRewardSheet.Row

# Class WorldBossRankingRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossRankingRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WorldBossRankingRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BossId

```
public int BossId
```

### Field Value

int

## Crystal

```
public int Crystal
```

### Field Value

int

# Id

```
public int Id
```

## Field Value

int

# Materials

```
public List<(int itemId, int quantity)> Materials
```

## Field Value

List<(int id, int count)>

# RankingMax

```
public int RankingMax
```

## Field Value

int

# RankingMin

```
public int RankingMin
```

## Field Value

int

# RateMax

```
public int RateMax
```

### Field Value

int

## RateMin

```
public int RateMin
```

### Field Value

int

## Runes

```
public List<WorldBossRankingRewardSheet.Row.RuneInfo> Runes
```

### Field Value

List<WorldBossRankingRewardSheet.Row.RuneInfo>

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

# GetRewards(RuneSheet, MaterialItemSheet)

```
public List<FungibleAssetValue> GetRewards(RuneSheet runeSheet,
MaterialItemSheet materialSheet)
```

## Parameters

`runeSheet` RuneSheet

`materialSheet` MaterialItemSheet

## Returns

List<FungibleAssetValue>

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Struct WorldBossRankingRewardSheet.Row.RuneInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public struct WorldBossRankingRewardSheet.Row.RuneInfo
```

**Inherited Members**

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### RuneInfo(int, int)

```
public RuneInfo(int id, int qty)
```

#### Parameters

`id` int

`qty` int

## Fields

### RuneId

```
public int RuneId
```

#### Field Value

int

## RuneQty

```
public int RuneQty
```

### Field Value

int

# Class WorldBossStatSheetExtension

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public static class WorldBossStatSheetExtension
```

**Inheritance**

object ← WorldBossStatSheetExtension

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Methods

### ToStats(WaveStatData)

```
public static StatsMap ToStats(this WorldBossCharacterSheet.WaveStatData statData)
```

#### Parameters

`statData` WorldBossCharacterSheet.WaveStatData

#### Returns

StatsMap

# Class WorldQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldQuestSheet : Sheet<int, WorldQuestSheet.Row>, IDictionary<int,
WorldQuestSheet.Row>, ICollection<KeyValuePair<int, WorldQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldQuestSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldQuestSheet.Row> ← WorldQuestSheet

### Implements

IDictionary<int, WorldQuestSheet.Row>,
ICollection<KeyValuePair<int, WorldQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldQuestSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, WorldQuestSheet.Row>.Name , Sheet<int, WorldQuestSheet.Row>.OrderedList ,
Sheet<int, WorldQuestSheet.Row>.First , Sheet<int, WorldQuestSheet.Row>.Last ,
Sheet<int, WorldQuestSheet.Row>.Keys , Sheet<int, WorldQuestSheet.Row>.Values ,
Sheet<int, WorldQuestSheet.Row>.Count , Sheet<int, WorldQuestSheet.Row>.IsReadOnly ,
Sheet<int, WorldQuestSheet.Row>.this[int] ,
Sheet<int, WorldQuestSheet.Row>.Set(string, bool) ,
Sheet<int, WorldQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldQuestSheet.Row>.GetEnumerator() ,
Sheet<int, WorldQuestSheet.Row>.TryGetValue(int, out WorldQuestSheet.Row, bool) ,
Sheet<int, WorldQuestSheet.Row>.AddRow(int, WorldQuestSheet.Row) ,
Sheet<int, WorldQuestSheet.Row>.Add(int, WorldQuestSheet.Row) ,
Sheet<int, WorldQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldQuestSheet.Row>.Remove(int) ,
Sheet<int, WorldQuestSheet.Row>.TryGetValue(int, out WorldQuestSheet.Row) ,
Sheet<int, WorldQuestSheet.Row>.Add(KeyValuePair<int, WorldQuestSheet.Row>) ,
Sheet<int, WorldQuestSheet.Row>.Clear() ,
Sheet<int, WorldQuestSheet.Row>.Contains(KeyValuePair<int, WorldQuestSheet.Row>) ,
Sheet<int, WorldQuestSheet.Row>.CopyTo(KeyValuePair<int, WorldQuestSheet.Row>[], int)
,
Sheet<int, WorldQuestSheet.Row>.Remove(KeyValuePair<int, WorldQuestSheet.Row>) ,

Sheet<int, WorldQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldQuestSheet()

```
public WorldQuestSheet()
```

# Class WorldQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldQuestSheet.Row : QuestSheet.Row, IState
```

### Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← WorldQuestSheet.Row

### Implements

IState

### Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Set(IReadOnlyList<string>) , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Class WorldSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldSheet : Sheet<int, WorldSheet.Row>, IDictionary<int,
WorldSheet.Row>, ICollection<KeyValuePair<int, WorldSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldSheet.Row> ← WorldSheet

### Implements

IDictionary<int, WorldSheet.Row>,
ICollection<KeyValuePair<int, WorldSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, WorldSheet.Row>.Name , Sheet<int, WorldSheet.Row>.OrderedList ,
Sheet<int, WorldSheet.Row>.First , Sheet<int, WorldSheet.Row>.Last ,
Sheet<int, WorldSheet.Row>.Keys , Sheet<int, WorldSheet.Row>.Values ,
Sheet<int, WorldSheet.Row>.Count , Sheet<int, WorldSheet.Row>.IsReadOnly ,
Sheet<int, WorldSheet.Row>.this[int] , Sheet<int, WorldSheet.Row>.Set(string, bool) ,
Sheet<int, WorldSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldSheet.Row>.GetEnumerator() ,
Sheet<int, WorldSheet.Row>.TryGetValue(int, out WorldSheet.Row, bool) ,
Sheet<int, WorldSheet.Row>.AddRow(int, WorldSheet.Row) ,
Sheet<int, WorldSheet.Row>.Add(int, WorldSheet.Row) ,
Sheet<int, WorldSheet.Row>.ContainsKey(int) , Sheet<int, WorldSheet.Row>.Remove(int) ,
Sheet<int, WorldSheet.Row>.TryGetValue(int, out WorldSheet.Row) ,
Sheet<int, WorldSheet.Row>.Add(KeyValuePair<int, WorldSheet.Row>) ,
Sheet<int, WorldSheet.Row>.Clear() ,
Sheet<int, WorldSheet.Row>.Contains(KeyValuePair<int, WorldSheet.Row>) ,
Sheet<int, WorldSheet.Row>.CopyTo(KeyValuePair<int, WorldSheet.Row>[], int) ,
Sheet<int, WorldSheet.Row>.Remove(KeyValuePair<int, WorldSheet.Row>) ,
Sheet<int, WorldSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldSheet()

```
public WorldSheet()
```

# Class WorldSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WorldSheet.Row

**Derived**

EventDungeonSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Name

```
public string Name { get; }
```

### Property Value

string

## StageBegin

```
public int StageBegin { get; }
```

### Property Value

int

## StageEnd

```
public int StageEnd { get; }
```

### Property Value

int

## StagesCount

```
public int StagesCount { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class WorldUnlockSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldUnlockSheet : Sheet<int, WorldUnlockSheet.Row>, IDictionary<int,
WorldUnlockSheet.Row>, ICollection<KeyValuePair<int, WorldUnlockSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldUnlockSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, WorldUnlockSheet.Row> ← WorldUnlockSheet

### Implements

IDictionary<int, WorldUnlockSheet.Row>,
ICollection<KeyValuePair<int, WorldUnlockSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldUnlockSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, WorldUnlockSheet.Row>.Name ,
Sheet<int, WorldUnlockSheet.Row>.OrderedList , Sheet<int, WorldUnlockSheet.Row>.First ,
Sheet<int, WorldUnlockSheet.Row>.Last , Sheet<int, WorldUnlockSheet.Row>.Keys ,
Sheet<int, WorldUnlockSheet.Row>.Values , Sheet<int, WorldUnlockSheet.Row>.Count ,
Sheet<int, WorldUnlockSheet.Row>.IsReadOnly ,
Sheet<int, WorldUnlockSheet.Row>.this[int] ,
Sheet<int, WorldUnlockSheet.Row>.Set(string, bool) ,
Sheet<int, WorldUnlockSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldUnlockSheet.Row>.GetEnumerator() ,
Sheet<int, WorldUnlockSheet.Row>.TryGetValue(int, out WorldUnlockSheet.Row, bool) ,
Sheet<int, WorldUnlockSheet.Row>.AddRow(int, WorldUnlockSheet.Row) ,
Sheet<int, WorldUnlockSheet.Row>.Add(int, WorldUnlockSheet.Row) ,
Sheet<int, WorldUnlockSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldUnlockSheet.Row>.Remove(int) ,
Sheet<int, WorldUnlockSheet.Row>.TryGetValue(int, out WorldUnlockSheet.Row) ,
Sheet<int, WorldUnlockSheet.Row>.Add(KeyValuePair<int, WorldUnlockSheet.Row>) ,
Sheet<int, WorldUnlockSheet.Row>.Clear() ,
Sheet<int, WorldUnlockSheet.Row>.Contains(KeyValuePair<int, WorldUnlockSheet.Row>) ,
Sheet<int, WorldUnlockSheet.Row>.CopyTo(KeyValuePair<int, WorldUnlockSheet.Row>[],
int) ,

Sheet<int, WorldUnlockSheet.Row>.Remove(KeyValuePair<int, WorldUnlockSheet.Row>) , Sheet<int, WorldUnlockSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## WorldUnlockSheet()

```
public WorldUnlockSheet()
```

# Methods

## TryGetUnlockedInformation(int, int, out List<int>)

```
public bool TryGetUnlockedInformation(int clearedWorldId, int clearedStageId, out
List<int> worldIdsToUnlock)
```

### Parameters

`clearedWorldId` int

`clearedStageId` int

`worldIdsToUnlock` List<int>

### Returns

bool

# Class WorldUnlockSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldUnlockSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← WorldUnlockSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## CRYSTAL

```
public int CRYSTAL { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## StageId

```
public int StageId { get; }
```

### Property Value

int

## WorldId

```
public int WorldId { get; }
```

### Property Value

int

## WorldIdToUnlock

```
public int WorldIdToUnlock { get; }
```

### Property Value

int

# Methods

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Namespace Nekoyume.TableData.AdventureBoss

## Classes

AdventureBossContributionRewardSheet

AdventureBossContributionRewardSheet.Row

AdventureBossFloorFirstRewardSheet

AdventureBossFloorFirstRewardSheet.Row

AdventureBossFloorPointSheet

AdventureBossFloorPointSheet.Row

AdventureBossFloorSheet

AdventureBossFloorSheet.RewardData

AdventureBossFloorSheet.Row

AdventureBossFloorWaveSheet

AdventureBossFloorWaveSheet.MonsterData

AdventureBossFloorWaveSheet.Row

AdventureBossFloorWaveSheet.WaveData

AdventureBossNcgRewardRatioSheet

AdventureBossNcgRewardRatioSheet.Row

AdventureBossSheet

AdventureBossSheet.RewardAmountData

AdventureBossSheet.RewardRatioData

AdventureBossSheet.Row

AdventureBossUnlockFloorCostSheet

AdventureBossUnlockFloorCostSheet.Row

AdventureBossWantedRewardSheet

AdventureBossWantedRewardSheet.Row

# Class AdventureBossContributionRewardSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossContributionRewardSheet : Sheet<int,
AdventureBossContributionRewardSheet.Row>, IDictionary<int,
AdventureBossContributionRewardSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossContributionRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossContributionRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossContributionRewardSheet.Row> ← AdventureBossContributionRewardSheet

### Implements

IDictionary<int, AdventureBossContributionRewardSheet.Row>, ICollection<KeyValuePair<int, AdventureBossContributionRewardSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossContributionRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, AdventureBossContributionRewardSheet.Row>.Name ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.OrderedList ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.First ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Last ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Keys ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Values ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Count ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.this[int] ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.TryGetValue(int, out AdventureBossContributionRewardSheet.Row, bool) ,

Sheet<int, AdventureBossContributionRewardSheet.Row>.AddRow(int, AdventureBossContributionRewardSheet.Row) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Add(int, AdventureBossContributionRewardSheet.Row) , Sheet<int, AdventureBossContributionRewardSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Remove(int) , Sheet<int, AdventureBossContributionRewardSheet.Row>.TryGetValue(int, out AdventureBossContributionRewardSheet.Row) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Add(KeyValuePair<int, AdventureBossContributionRewardSheet.Row>) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Clear() , Sheet<int, AdventureBossContributionRewardSheet.Row>.Contains(KeyValuePair<int, AdventureBossContributionRewardSheet.Row>) , Sheet<int, AdventureBossContributionRewardSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossContributionRewardSheet.Row>[], int) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Remove(KeyValuePair<int, AdventureBossContributionRewardSheet.Row>) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossContributionRewardSheet()

```
public AdventureBossContributionRewardSheet()
```

# Class AdventureBossContributionRewardSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossContributionRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossContributionRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AdventureBossId

```
public int AdventureBossId
```

### Field Value

int

## Id

```
public int Id
```

### Field Value

int

## Rewards

```
public List<AdventureBossSheet.RewardRatioData> Rewards
```

### Field Value

List<AdventureBossSheet.RewardRatioData>

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class AdventureBossFloorFirstRewardSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorFirstRewardSheet : Sheet<int,
AdventureBossFloorFirstRewardSheet.Row>, IDictionary<int,
AdventureBossFloorFirstRewardSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossFloorFirstRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossFloorFirstRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossFloorFirstRewardSheet.Row> ← AdventureBossFloorFirstRewardSheet

### Implements

IDictionary<int, AdventureBossFloorFirstRewardSheet.Row>, ICollection<KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Name ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.OrderedList ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.First ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Last ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Keys ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Values ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Count ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.this[int] ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.TryGetValue(int, out AdventureBossFloorFirstRewardSheet.Row, bool) ,

Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.AddRow(int, AdventureBossFloorFirstRewardSheet.Row) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Add(int, AdventureBossFloorFirstRewardSheet.Row) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Remove(int) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.TryGetValue(int, out AdventureBossFloorFirstRewardSheet.Row) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Add(KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Clear() , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Contains(KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>[], int) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Remove(KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossFloorFirstRewardSheet()

```
public AdventureBossFloorFirstRewardSheet()
```

# Class AdventureBossFloorFirstRewardSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorFirstRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossFloorFirstRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## FloorId

```
public int FloorId
```

### Field Value

int

## Rewards

```
public List<AdventureBossSheet.RewardAmountData> Rewards
```

### Field Value

List<AdventureBossSheet.RewardAmountData>

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class AdventureBossFloorPointSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorPointSheet : Sheet<int,
AdventureBossFloorPointSheet.Row>, IDictionary<int,
AdventureBossFloorPointSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossFloorPointSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossFloorPointSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossFloorPointSheet.Row> ← AdventureBossFloorPointSheet

### Implements

IDictionary<int, AdventureBossFloorPointSheet.Row>, ICollection<KeyValuePair<int, AdventureBossFloorPointSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossFloorPointSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, AdventureBossFloorPointSheet.Row>.Name ,
Sheet<int, AdventureBossFloorPointSheet.Row>.OrderedList ,
Sheet<int, AdventureBossFloorPointSheet.Row>.First ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Last ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Keys ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Values ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Count ,
Sheet<int, AdventureBossFloorPointSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossFloorPointSheet.Row>.this[int] ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossFloorPointSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossFloorPointSheet.Row>.TryGetValue(int, out AdventureBossFloorPointSheet.Row, bool) ,
Sheet<int, AdventureBossFloorPointSheet.Row>.AddRow(int, AdventureBossFloorPointSheet.Row) ,

Sheet<int, AdventureBossFloorPointSheet.Row>.Add(int, AdventureBossFloorPointSheet.Row) , Sheet<int, AdventureBossFloorPointSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossFloorPointSheet.Row>.Remove(int) , Sheet<int, AdventureBossFloorPointSheet.Row>.TryGetValue(int, out AdventureBossFloorPointSheet.Row) , Sheet<int, AdventureBossFloorPointSheet.Row>.Add(KeyValuePair<int, AdventureBossFloorPointSheet.Row>) , Sheet<int, AdventureBossFloorPointSheet.Row>.Clear() , Sheet<int, AdventureBossFloorPointSheet.Row>.Contains(KeyValuePair<int, AdventureBossFloorPointSheet.Row>) , Sheet<int, AdventureBossFloorPointSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossFloorPointSheet.Row>[], int) , Sheet<int, AdventureBossFloorPointSheet.Row>.Remove(KeyValuePair<int, AdventureBossFloorPointSheet.Row>) , Sheet<int, AdventureBossFloorPointSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossFloorPointSheet()

```
public AdventureBossFloorPointSheet()
```

# Class AdventureBossFloorPointSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorPointSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossFloorPointSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## FloorId

```
public int FloorId
```

### Field Value

int

## MaxPoint

```
public int MaxPoint
```

### Field Value

int

## MinPoint

```
public int MinPoint
```

### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class AdventureBossFloorSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorSheet : Sheet<int, AdventureBossFloorSheet.Row>,
IDictionary<int, AdventureBossFloorSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossFloorSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossFloorSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossFloorSheet.Row> ← AdventureBossFloorSheet

### Implements

IDictionary<int, AdventureBossFloorSheet.Row>,
ICollection<KeyValuePair<int, AdventureBossFloorSheet.Row>>,
IEnumerable<KeyValuePair<int, AdventureBossFloorSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, AdventureBossFloorSheet.Row>.Name ,
Sheet<int, AdventureBossFloorSheet.Row>.OrderedList ,
Sheet<int, AdventureBossFloorSheet.Row>.First ,
Sheet<int, AdventureBossFloorSheet.Row>.Last ,
Sheet<int, AdventureBossFloorSheet.Row>.Keys ,
Sheet<int, AdventureBossFloorSheet.Row>.Values ,
Sheet<int, AdventureBossFloorSheet.Row>.Count ,
Sheet<int, AdventureBossFloorSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossFloorSheet.Row>.this[int] ,
Sheet<int, AdventureBossFloorSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossFloorSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossFloorSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossFloorSheet.Row>.TryGetValue(int, out
AdventureBossFloorSheet.Row, bool) ,
Sheet<int, AdventureBossFloorSheet.Row>.AddRow(int, AdventureBossFloorSheet.Row) ,
Sheet<int, AdventureBossFloorSheet.Row>.Add(int, AdventureBossFloorSheet.Row) ,
Sheet<int, AdventureBossFloorSheet.Row>.ContainsKey(int) ,
Sheet<int, AdventureBossFloorSheet.Row>.Remove(int) ,

Sheet<int, AdventureBossFloorSheet.Row>.TryGetValue(int, out AdventureBossFloorSheet.Row) , Sheet<int, AdventureBossFloorSheet.Row>.Add(KeyValuePair<int, AdventureBossFloorSheet.Row>) , Sheet<int, AdventureBossFloorSheet.Row>.Clear() , Sheet<int, AdventureBossFloorSheet.Row>.Contains(KeyValuePair<int, AdventureBossFloorSheet.Row>) , Sheet<int, AdventureBossFloorSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossFloorSheet.Row>[], int) , Sheet<int, AdventureBossFloorSheet.Row>.Remove(KeyValuePair<int, AdventureBossFloorSheet.Row>) , Sheet<int, AdventureBossFloorSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossFloorSheet()

```
public AdventureBossFloorSheet()
```

# Class AdventureBossFloorSheet.RewardData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorSheet.RewardData
```

**Inheritance**

object ← AdventureBossFloorSheet.RewardData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardData(string, int, int, int, int)

```
public RewardData(string itemType, int itemId, int min, int max, int ratio)
```

#### Parameters

`itemType` string

`itemId` int

`min` int

`max` int

`ratio` int

## Properties

## ItemId

```
public int ItemId { get; }
```

### Property Value

int

## ItemType

```
public string ItemType { get; }
```

### Property Value

string

## Max

```
public int Max { get; }
```

### Property Value

int

## Min

```
public int Min { get; }
```

### Property Value

int

## Ratio

```
public int Ratio { get; }
```

## Property Value

int

# Class AdventureBossFloorSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossFloorSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## AdventureBossId

```
public int AdventureBossId { get; }
```

### Property Value

int

## BGM

```
public string BGM { get; }
```

### Property Value

string

## Background

```
public string Background { get; }
```

### Property Value

string

## EnemyInitialStatModifiers

```
public List<StatModifier> EnemyInitialStatModifiers { get; }
```

### Property Value

List<StatModifier>

## Floor

```
public int Floor { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## MaxDropItem

```
public int MaxDropItem { get; }
```

### Property Value

int

## MinDropItem

```
public int MinDropItem { get; }
```

### Property Value

int

## Rewards

```
public List<AdventureBossFloorSheet.RewardData> Rewards { get; }
```

### Property Value

List<AdventureBossFloorSheet.RewardData>

## StageBuffSkillId

```
public int StageBuffSkillId { get; }
```

### Property Value

int

## TurnLimit

```
public int TurnLimit { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class AdventureBossFloorWaveSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorWaveSheet : Sheet<int,
AdventureBossFloorWaveSheet.Row>, IDictionary<int, AdventureBossFloorWaveSheet.Row>,
ICollection<KeyValuePair<int, AdventureBossFloorWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, AdventureBossFloorWaveSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossFloorWaveSheet.Row> ← AdventureBossFloorWaveSheet

### Implements

IDictionary<int, AdventureBossFloorWaveSheet.Row>, ICollection<KeyValuePair<int, AdventureBossFloorWaveSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossFloorWaveSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, AdventureBossFloorWaveSheet.Row>.Name ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.OrderedList ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.First ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Last ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Keys ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Values ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Count ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.this[int] ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.TryGetValue(int, out AdventureBossFloorWaveSheet.Row, bool) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.AddRow(int, AdventureBossFloorWaveSheet.Row) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Add(int, AdventureBossFloorWaveSheet.Row) ,

Sheet<int, AdventureBossFloorWaveSheet.Row>.ContainsKey(int) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Remove(int) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.TryGetValue(int, out
AdventureBossFloorWaveSheet.Row) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Add(KeyValuePair<int,
AdventureBossFloorWaveSheet.Row>) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Clear() ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Contains(KeyValuePair<int,
AdventureBossFloorWaveSheet.Row>) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.CopyTo(KeyValuePair<int,
AdventureBossFloorWaveSheet.Row>[], int) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Remove(KeyValuePair<int,
AdventureBossFloorWaveSheet.Row>) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossFloorWaveSheet()

```
public AdventureBossFloorWaveSheet()
```

# Class AdventureBossFloorWaveSheet.MonsterData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorWaveSheet.MonsterData
```

**Inheritance**

object ← AdventureBossFloorWaveSheet.MonsterData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### MonsterData(int, int, int)

```
public MonsterData(int characterId, int level, int count)
```

#### Parameters

`characterId` int

`level` int

`count` int

## Properties

### CharacterId

```
public int CharacterId { get; }
```

### Property Value

int

## Count

```
public int Count { get; }
```

### Property Value

int

## Level

```
public int Level { get; }
```

### Property Value

int

# Class AdventureBossFloorWaveSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorWaveSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossFloorWaveSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## FloorId

```
public int FloorId { get; }
```

### Property Value

int

## HasBoss

```
public bool HasBoss { get; }
```

### Property Value

bool

## Key

```
public override int Key { get; }
```

### Property Value

int

## TotalMonsterIds

```
public List<int> TotalMonsterIds { get; }
```

### Property Value

List<int>

## Waves

```
public List<AdventureBossFloorWaveSheet.WaveData> Waves { get; }
```

### Property Value

List<AdventureBossFloorWaveSheet.WaveData>

# Methods

## EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class AdventureBossFloorWaveSheet.WaveData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorWaveSheet.WaveData
```

**Inheritance**

object ← AdventureBossFloorWaveSheet.WaveData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### WaveData(int, List<MonsterData>, bool)

```
public WaveData(int number, List<AdventureBossFloorWaveSheet.MonsterData> monsters,
bool hasBoss)
```

#### Parameters

`number` int

`monsters` List<AdventureBossFloorWaveSheet.MonsterData>

`hasBoss` bool

## Properties

### HasBoss

```
public bool HasBoss { get; }
```

### Property Value

bool

## Monsters

```
public List<AdventureBossFloorWaveSheet.MonsterData> Monsters { get; }
```

### Property Value

List<AdventureBossFloorWaveSheet.MonsterData>

## Number

```
public int Number { get; }
```

### Property Value

int

# Class AdventureBossNcgRewardRatioSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossNcgRewardRatioSheet : Sheet<int,
AdventureBossNcgRewardRatioSheet.Row>, IDictionary<int,
AdventureBossNcgRewardRatioSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossNcgRewardRatioSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossNcgRewardRatioSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossNcgRewardRatioSheet.Row> ← AdventureBossNcgRewardRatioSheet

### Implements

IDictionary<int, AdventureBossNcgRewardRatioSheet.Row>, ICollection<KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Name ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.OrderedList ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.First ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Last ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Keys ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Values ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Count ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.this[int] ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.TryGetValue(int, out AdventureBossNcgRewardRatioSheet.Row, bool) ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.AddRow(int, AdventureBossNcgRewardRatioSheet.Row) ,

Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Add(int, AdventureBossNcgRewardRatioSheet.Row) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Remove(int) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.TryGetValue(int, out AdventureBossNcgRewardRatioSheet.Row) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Add(KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Clear() , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Contains(KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>[], int) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Remove(KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossNcgRewardRatioSheet()

```
public AdventureBossNcgRewardRatioSheet()
```

# Class AdventureBossNcgRewardRatioSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossNcgRewardRatioSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossNcgRewardRatioSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## ItemId

```
public int ItemId
```

### Field Value

int

## Ratio

```
public decimal Ratio
```

### Field Value

decimal

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class AdventureBossSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossSheet : Sheet<int, AdventureBossSheet.Row>,
IDictionary<int, AdventureBossSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossSheet.Row> ← AdventureBossSheet

### Implements

IDictionary<int, AdventureBossSheet.Row>,
ICollection<KeyValuePair<int, AdventureBossSheet.Row>>,
IEnumerable<KeyValuePair<int, AdventureBossSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, AdventureBossSheet.Row>.Name ,
Sheet<int, AdventureBossSheet.Row>.OrderedList ,
Sheet<int, AdventureBossSheet.Row>.First , Sheet<int, AdventureBossSheet.Row>.Last ,
Sheet<int, AdventureBossSheet.Row>.Keys ,
Sheet<int, AdventureBossSheet.Row>.Values ,
Sheet<int, AdventureBossSheet.Row>.Count ,
Sheet<int, AdventureBossSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossSheet.Row>.this[int] ,
Sheet<int, AdventureBossSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossSheet.Row>.TryGetValue(int, out AdventureBossSheet.Row, bool)
,
Sheet<int, AdventureBossSheet.Row>.AddRow(int, AdventureBossSheet.Row) ,
Sheet<int, AdventureBossSheet.Row>.Add(int, AdventureBossSheet.Row) ,
Sheet<int, AdventureBossSheet.Row>.ContainsKey(int) ,
Sheet<int, AdventureBossSheet.Row>.Remove(int) ,
Sheet<int, AdventureBossSheet.Row>.TryGetValue(int, out AdventureBossSheet.Row) ,
Sheet<int, AdventureBossSheet.Row>.Add(KeyValuePair<int, AdventureBossSheet.Row>) ,
Sheet<int, AdventureBossSheet.Row>.Clear() ,

Sheet<int, AdventureBossSheet.Row>.Contains(KeyValuePair<int, AdventureBossSheet.Row>) , Sheet<int, AdventureBossSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossSheet.Row>[], int) , Sheet<int, AdventureBossSheet.Row>.Remove(KeyValuePair<int, AdventureBossSheet.Row>) , Sheet<int, AdventureBossSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossSheet()

```
public AdventureBossSheet()
```

# Class AdventureBossSheet.RewardAmountData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossSheet.RewardAmountData
```

**Inheritance**

object ← AdventureBossSheet.RewardAmountData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardAmountData(string, int, int)

```
public RewardAmountData(string itemType, int itemId, int amount)
```

#### Parameters

`itemType` string

`itemId` int

`amount` int

## Properties

### Amount

```
public int Amount { get; }
```

### Property Value

int

## ItemId

```
public int ItemId { get; }
```

### Property Value

int

## ItemType

```
public string ItemType { get; }
```

### Property Value

string

# Class AdventureBossSheet.RewardRatioData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossSheet.RewardRatioData
```

**Inheritance**

object ← AdventureBossSheet.RewardRatioData

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardRatioData(string, int, int)

```
public RewardRatioData(string itemType, int itemId, int ratio)
```

#### Parameters

`itemType` string

`itemId` int

`ratio` int

## Properties

### ItemId

```
public int ItemId { get; }
```

### Property Value

int

## ItemType

```
public string ItemType { get; }
```

### Property Value

string

## Ratio

```
public int Ratio { get; }
```

### Property Value

int

# Class AdventureBossSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## BossId

```
public int BossId { get; }
```

### Property Value

int

## ExploreAp

```
public int ExploreAp { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## SweepAp

```
public int SweepAp { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class AdventureBossUnlockFloorCostSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossUnlockFloorCostSheet : Sheet<int,
AdventureBossUnlockFloorCostSheet.Row>, IDictionary<int,
AdventureBossUnlockFloorCostSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossUnlockFloorCostSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossUnlockFloorCostSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossUnlockFloorCostSheet.Row> ← AdventureBossUnlockFloorCostSheet

### Implements

IDictionary<int, AdventureBossUnlockFloorCostSheet.Row>, ICollection<KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Name ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.OrderedList ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.First ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Last ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Keys ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Values ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Count ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.this[int] ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.TryGetValue(int, out AdventureBossUnlockFloorCostSheet.Row, bool) ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.AddRow(int, AdventureBossUnlockFloorCostSheet.Row) ,

Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Add(int, AdventureBossUnlockFloorCostSheet.Row) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Remove(int) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.TryGetValue(int, out AdventureBossUnlockFloorCostSheet.Row) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Add(KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Clear() , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Contains(KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>[], int) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Remove(KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossUnlockFloorCostSheet()

```
public AdventureBossUnlockFloorCostSheet()
```

# Class AdventureBossUnlockFloorCostSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossUnlockFloorCostSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossUnlockFloorCostSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## FloorId

```
public int FloorId
```

### Field Value

int

## GoldenDustPrice

```
public int GoldenDustPrice
```

### Field Value

int

## NcgPrice

```
public BigInteger NcgPrice
```

### Field Value

BigInteger

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class AdventureBossWantedRewardSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossWantedRewardSheet : Sheet<int,
AdventureBossWantedRewardSheet.Row>, IDictionary<int,
AdventureBossWantedRewardSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossWantedRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossWantedRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, AdventureBossWantedRewardSheet.Row> ← AdventureBossWantedRewardSheet

### Implements

IDictionary<int, AdventureBossWantedRewardSheet.Row>, ICollection<KeyValuePair<int, AdventureBossWantedRewardSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossWantedRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, AdventureBossWantedRewardSheet.Row>.Name ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.OrderedList ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.First ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Last ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Keys ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Values ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Count ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.this[int] ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.TryGetValue(int, out AdventureBossWantedRewardSheet.Row, bool) ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.AddRow(int, AdventureBossWantedRewardSheet.Row) ,

Sheet<int, AdventureBossWantedRewardSheet.Row>.Add(int, AdventureBossWantedRewardSheet.Row) , Sheet<int, AdventureBossWantedRewardSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Remove(int) , Sheet<int, AdventureBossWantedRewardSheet.Row>.TryGetValue(int, out AdventureBossWantedRewardSheet.Row) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Add(KeyValuePair<int, AdventureBossWantedRewardSheet.Row>) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Clear() , Sheet<int, AdventureBossWantedRewardSheet.Row>.Contains(KeyValuePair<int, AdventureBossWantedRewardSheet.Row>) , Sheet<int, AdventureBossWantedRewardSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossWantedRewardSheet.Row>[], int) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Remove(KeyValuePair<int, AdventureBossWantedRewardSheet.Row>) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## AdventureBossWantedRewardSheet()

```
public AdventureBossWantedRewardSheet()
```

# Class AdventureBossWantedRewardSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossWantedRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← AdventureBossWantedRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AdventureBossId

```
public int AdventureBossId
```

### Field Value

int

## FixedReward

```
public AdventureBossSheet.RewardRatioData FixedReward
```

### Field Value

AdventureBossSheet.RewardRatioData

## Id

```
public int Id
```

### Field Value

int

## RandomRewards

```
public List<AdventureBossSheet.RewardRatioData> RandomRewards
```

### Field Value

List<AdventureBossSheet.RewardRatioData>

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Namespace Nekoyume.TableData.Crystal

## Classes

CrystalEquipmentGrindingSheet

CrystalEquipmentGrindingSheet.Row

CrystalFluctuationSheet

CrystalFluctuationSheet.Row

CrystalHammerPointSheet

CrystalHammerPointSheet.Row

CrystalMaterialCostSheet

CrystalMaterialCostSheet.Row

CrystalMonsterCollectionMultiplierSheet

CrystalMonsterCollectionMultiplierSheet.Row

CrystalRandomBuffSheet

CrystalRandomBuffSheet.Row

CrystalStageBuffGachaSheet

CrystalStageBuffGachaSheet.Row

## Enums

CrystalFluctuationSheet.ServiceType

CrystalRandomBuffSheet.Row.BuffRank

# Class CrystalEquipmentGrindingSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalEquipmentGrindingSheet : Sheet<int,
CrystalEquipmentGrindingSheet.Row>, IDictionary<int,
CrystalEquipmentGrindingSheet.Row>, ICollection<KeyValuePair<int,
CrystalEquipmentGrindingSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalEquipmentGrindingSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CrystalEquipmentGrindingSheet.Row> ← CrystalEquipmentGrindingSheet

### Implements

IDictionary<int, CrystalEquipmentGrindingSheet.Row>, ICollection<KeyValuePair<int, CrystalEquipmentGrindingSheet.Row>>, IEnumerable<KeyValuePair<int, CrystalEquipmentGrindingSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CrystalEquipmentGrindingSheet.Row>.Name ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.OrderedList ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.First ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Last ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Keys ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Values ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Count ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.IsReadOnly ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.this[int] ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.TryGetValue(int, out CrystalEquipmentGrindingSheet.Row, bool) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.AddRow(int, CrystalEquipmentGrindingSheet.Row) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Add(int, CrystalEquipmentGrindingSheet.Row) ,

Sheet<int, CrystalEquipmentGrindingSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Remove(int) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.TryGetValue(int, out
CrystalEquipmentGrindingSheet.Row) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Add(KeyValuePair<int,
CrystalEquipmentGrindingSheet.Row>) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Clear() ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Contains(KeyValuePair<int,
CrystalEquipmentGrindingSheet.Row>) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.CopyTo(KeyValuePair<int,
CrystalEquipmentGrindingSheet.Row>[], int) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Remove(KeyValuePair<int,
CrystalEquipmentGrindingSheet.Row>) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CrystalEquipmentGrindingSheet()

```
public CrystalEquipmentGrindingSheet()
```

# Class CrystalEquipmentGrindingSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalEquipmentGrindingSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CrystalEquipmentGrindingSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## CRYSTAL

```
public int CRYSTAL
```

### Field Value

int

## EnchantBaseId

```
public int EnchantBaseId
```

### Field Value

int

## ItemId

```
public int ItemId
```

### Field Value

int

## RewardMaterials

```
public List<(int materialId, int count)> RewardMaterials
```

### Field Value

List<(int id, int count)>

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CrystalFluctuationSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalFluctuationSheet : Sheet<int, CrystalFluctuationSheet.Row>,
IDictionary<int, CrystalFluctuationSheet.Row>, ICollection<KeyValuePair<int,
CrystalFluctuationSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalFluctuationSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CrystalFluctuationSheet.Row> ← CrystalFluctuationSheet

### Implements

IDictionary<int, CrystalFluctuationSheet.Row>,
ICollection<KeyValuePair<int, CrystalFluctuationSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalFluctuationSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, CrystalFluctuationSheet.Row>.Name ,
Sheet<int, CrystalFluctuationSheet.Row>.OrderedList ,
Sheet<int, CrystalFluctuationSheet.Row>.First ,
Sheet<int, CrystalFluctuationSheet.Row>.Last ,
Sheet<int, CrystalFluctuationSheet.Row>.Keys ,
Sheet<int, CrystalFluctuationSheet.Row>.Values ,
Sheet<int, CrystalFluctuationSheet.Row>.Count ,
Sheet<int, CrystalFluctuationSheet.Row>.IsReadOnly ,
Sheet<int, CrystalFluctuationSheet.Row>.this[int] ,
Sheet<int, CrystalFluctuationSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalFluctuationSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalFluctuationSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalFluctuationSheet.Row>.TryGetValue(int, out CrystalFluctuationSheet.Row,
bool) ,
Sheet<int, CrystalFluctuationSheet.Row>.AddRow(int, CrystalFluctuationSheet.Row) ,
Sheet<int, CrystalFluctuationSheet.Row>.Add(int, CrystalFluctuationSheet.Row) ,
Sheet<int, CrystalFluctuationSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalFluctuationSheet.Row>.Remove(int) ,

Sheet<int, CrystalFluctuationSheet.Row>.TryGetValue(int, out CrystalFluctuationSheet.Row) , Sheet<int, CrystalFluctuationSheet.Row>.Add(KeyValuePair<int, CrystalFluctuationSheet.Row>) , Sheet<int, CrystalFluctuationSheet.Row>.Clear() , Sheet<int, CrystalFluctuationSheet.Row>.Contains(KeyValuePair<int, CrystalFluctuationSheet.Row>) , Sheet<int, CrystalFluctuationSheet.Row>.CopyTo(KeyValuePair<int, CrystalFluctuationSheet.Row>[], int) , Sheet<int, CrystalFluctuationSheet.Row>.Remove(KeyValuePair<int, CrystalFluctuationSheet.Row>) , Sheet<int, CrystalFluctuationSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CrystalFluctuationSheet()

```
public CrystalFluctuationSheet()
```

# Class CrystalFluctuationSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalFluctuationSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CrystalFluctuationSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BlockInterval

```
public long BlockInterval
```

### Field Value

long

## Id

```
public int Id
```

### Field Value

int

## MaximumRate

```
public int MaximumRate
```

### Field Value

int

## MinimumRate

```
public int MinimumRate
```

### Field Value

int

## Type

```
public CrystalFluctuationSheet.ServiceType Type
```

### Field Value

CrystalFluctuationSheet.ServiceType

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Enum CrystalFluctuationSheet.ServiceType

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public enum CrystalFluctuationSheet.ServiceType
```

**Extension Methods**
StateExtensions.Serialize(Enum)

## Fields

`Combination = 0`

# Class CrystalHammerPointSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalHammerPointSheet : Sheet<int, CrystalHammerPointSheet.Row>,
IDictionary<int, CrystalHammerPointSheet.Row>, ICollection<KeyValuePair<int,
CrystalHammerPointSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalHammerPointSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CrystalHammerPointSheet.Row> ← CrystalHammerPointSheet

### Implements

IDictionary<int, CrystalHammerPointSheet.Row>,
ICollection<KeyValuePair<int, CrystalHammerPointSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalHammerPointSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, CrystalHammerPointSheet.Row>.Name ,
Sheet<int, CrystalHammerPointSheet.Row>.OrderedList ,
Sheet<int, CrystalHammerPointSheet.Row>.First ,
Sheet<int, CrystalHammerPointSheet.Row>.Last ,
Sheet<int, CrystalHammerPointSheet.Row>.Keys ,
Sheet<int, CrystalHammerPointSheet.Row>.Values ,
Sheet<int, CrystalHammerPointSheet.Row>.Count ,
Sheet<int, CrystalHammerPointSheet.Row>.IsReadOnly ,
Sheet<int, CrystalHammerPointSheet.Row>.this[int] ,
Sheet<int, CrystalHammerPointSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalHammerPointSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalHammerPointSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalHammerPointSheet.Row>.TryGetValue(int, out
CrystalHammerPointSheet.Row, bool) ,
Sheet<int, CrystalHammerPointSheet.Row>.AddRow(int, CrystalHammerPointSheet.Row) ,
Sheet<int, CrystalHammerPointSheet.Row>.Add(int, CrystalHammerPointSheet.Row) ,
Sheet<int, CrystalHammerPointSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalHammerPointSheet.Row>.Remove(int) ,

Sheet<int, CrystalHammerPointSheet.Row>.TryGetValue(int, out CrystalHammerPointSheet.Row) , Sheet<int, CrystalHammerPointSheet.Row>.Add(KeyValuePair<int, CrystalHammerPointSheet.Row>) , Sheet<int, CrystalHammerPointSheet.Row>.Clear() , Sheet<int, CrystalHammerPointSheet.Row>.Contains(KeyValuePair<int, CrystalHammerPointSheet.Row>) , Sheet<int, CrystalHammerPointSheet.Row>.CopyTo(KeyValuePair<int, CrystalHammerPointSheet.Row>[], int) , Sheet<int, CrystalHammerPointSheet.Row>.Remove(KeyValuePair<int, CrystalHammerPointSheet.Row>) , Sheet<int, CrystalHammerPointSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CrystalHammerPointSheet()

```
public CrystalHammerPointSheet()
```

# Class CrystalHammerPointSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalHammerPointSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CrystalHammerPointSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### CRYSTAL

```
public int CRYSTAL { get; }
```

#### Property Value

int

### Key

```
public override int Key { get; }
```

#### Property Value

int

## MaxPoint

```
public int MaxPoint { get; }
```

### Property Value

int

## RecipeId

```
public int RecipeId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CrystalMaterialCostSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalMaterialCostSheet : Sheet<int, CrystalMaterialCostSheet.Row>,
IDictionary<int, CrystalMaterialCostSheet.Row>, ICollection<KeyValuePair<int,
CrystalMaterialCostSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalMaterialCostSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CrystalMaterialCostSheet.Row> ← CrystalMaterialCostSheet

### Implements

IDictionary<int, CrystalMaterialCostSheet.Row>,
ICollection<KeyValuePair<int, CrystalMaterialCostSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalMaterialCostSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, CrystalMaterialCostSheet.Row>.Name ,
Sheet<int, CrystalMaterialCostSheet.Row>.OrderedList ,
Sheet<int, CrystalMaterialCostSheet.Row>.First ,
Sheet<int, CrystalMaterialCostSheet.Row>.Last ,
Sheet<int, CrystalMaterialCostSheet.Row>.Keys ,
Sheet<int, CrystalMaterialCostSheet.Row>.Values ,
Sheet<int, CrystalMaterialCostSheet.Row>.Count ,
Sheet<int, CrystalMaterialCostSheet.Row>.IsReadOnly ,
Sheet<int, CrystalMaterialCostSheet.Row>.this[int] ,
Sheet<int, CrystalMaterialCostSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalMaterialCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalMaterialCostSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalMaterialCostSheet.Row>.TryGetValue(int, out
CrystalMaterialCostSheet.Row, bool) ,
Sheet<int, CrystalMaterialCostSheet.Row>.AddRow(int, CrystalMaterialCostSheet.Row) ,
Sheet<int, CrystalMaterialCostSheet.Row>.Add(int, CrystalMaterialCostSheet.Row) ,
Sheet<int, CrystalMaterialCostSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalMaterialCostSheet.Row>.Remove(int) ,

Sheet<int, CrystalMaterialCostSheet.Row>.TryGetValue(int, out CrystalMaterialCostSheet.Row) , Sheet<int, CrystalMaterialCostSheet.Row>.Add(KeyValuePair<int, CrystalMaterialCostSheet.Row>) , Sheet<int, CrystalMaterialCostSheet.Row>.Clear() , Sheet<int, CrystalMaterialCostSheet.Row>.Contains(KeyValuePair<int, CrystalMaterialCostSheet.Row>) , Sheet<int, CrystalMaterialCostSheet.Row>.CopyTo(KeyValuePair<int, CrystalMaterialCostSheet.Row>[], int) , Sheet<int, CrystalMaterialCostSheet.Row>.Remove(KeyValuePair<int, CrystalMaterialCostSheet.Row>) , Sheet<int, CrystalMaterialCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CrystalMaterialCostSheet()

```
public CrystalMaterialCostSheet()
```

# Class CrystalMaterialCostSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalMaterialCostSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CrystalMaterialCostSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### CRYSTAL

```
public int CRYSTAL { get; }
```

#### Property Value

int

### ItemId

```
public int ItemId { get; }
```

#### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CrystalMonsterCollectionMultiplierSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalMonsterCollectionMultiplierSheet : Sheet<int,
CrystalMonsterCollectionMultiplierSheet.Row>, IDictionary<int,
CrystalMonsterCollectionMultiplierSheet.Row>, ICollection<KeyValuePair<int,
CrystalMonsterCollectionMultiplierSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalMonsterCollectionMultiplierSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row> ← CrystalMonsterCollectionMultiplierSheet

### Implements

IDictionary<int, CrystalMonsterCollectionMultiplierSheet.Row>, ICollection<KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>>, IEnumerable<KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Name ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.OrderedList ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.First ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Last ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Keys ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Values ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Count ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.IsReadOnly ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.this[int] ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.TryGetValue(int, out CrystalMonsterCollectionMultiplierSheet.Row, bool) ,

Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.AddRow(int, CrystalMonsterCollectionMultiplierSheet.Row) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Add(int, CrystalMonsterCollectionMultiplierSheet.Row) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.ContainsKey(int) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Remove(int) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.TryGetValue(int, out CrystalMonsterCollectionMultiplierSheet.Row) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Add(KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Clear() , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Contains(KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.CopyTo(KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>[], int) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Remove(KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CrystalMonsterCollectionMultiplierSheet()

```
public CrystalMonsterCollectionMultiplierSheet()
```

# Class CrystalMonsterCollectionMultiplierSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalMonsterCollectionMultiplierSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CrystalMonsterCollectionMultiplierSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Fields

### Level

```
public int Level
```

#### Field Value

int

### Multiplier

```
public int Multiplier
```

#### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CrystalRandomBuffSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalRandomBuffSheet : Sheet<int, CrystalRandomBuffSheet.Row>,
IDictionary<int, CrystalRandomBuffSheet.Row>, ICollection<KeyValuePair<int,
CrystalRandomBuffSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalRandomBuffSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CrystalRandomBuffSheet.Row> ← CrystalRandomBuffSheet

### Implements

IDictionary<int, CrystalRandomBuffSheet.Row>,
ICollection<KeyValuePair<int, CrystalRandomBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalRandomBuffSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, CrystalRandomBuffSheet.Row>.Name ,
Sheet<int, CrystalRandomBuffSheet.Row>.OrderedList ,
Sheet<int, CrystalRandomBuffSheet.Row>.First ,
Sheet<int, CrystalRandomBuffSheet.Row>.Last ,
Sheet<int, CrystalRandomBuffSheet.Row>.Keys ,
Sheet<int, CrystalRandomBuffSheet.Row>.Values ,
Sheet<int, CrystalRandomBuffSheet.Row>.Count ,
Sheet<int, CrystalRandomBuffSheet.Row>.IsReadOnly ,
Sheet<int, CrystalRandomBuffSheet.Row>.this[int] ,
Sheet<int, CrystalRandomBuffSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalRandomBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalRandomBuffSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalRandomBuffSheet.Row>.TryGetValue(int, out
CrystalRandomBuffSheet.Row, bool) ,
Sheet<int, CrystalRandomBuffSheet.Row>.AddRow(int, CrystalRandomBuffSheet.Row) ,
Sheet<int, CrystalRandomBuffSheet.Row>.Add(int, CrystalRandomBuffSheet.Row) ,
Sheet<int, CrystalRandomBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalRandomBuffSheet.Row>.Remove(int) ,

Sheet<int, CrystalRandomBuffSheet.Row>.TryGetValue(int, out CrystalRandomBuffSheet.Row) , Sheet<int, CrystalRandomBuffSheet.Row>.Add(KeyValuePair<int, CrystalRandomBuffSheet.Row>) , Sheet<int, CrystalRandomBuffSheet.Row>.Clear() , Sheet<int, CrystalRandomBuffSheet.Row>.Contains(KeyValuePair<int, CrystalRandomBuffSheet.Row>) , Sheet<int, CrystalRandomBuffSheet.Row>.CopyTo(KeyValuePair<int, CrystalRandomBuffSheet.Row>[], int) , Sheet<int, CrystalRandomBuffSheet.Row>.Remove(KeyValuePair<int, CrystalRandomBuffSheet.Row>) , Sheet<int, CrystalRandomBuffSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CrystalRandomBuffSheet()

```
public CrystalRandomBuffSheet()
```

# Class CrystalRandomBuffSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalRandomBuffSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CrystalRandomBuffSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Id

```
public int Id
```

### Field Value

int

## Rank

```
public CrystalRandomBuffSheet.Row.BuffRank Rank
```

### Field Value

CrystalRandomBuffSheet.Row.BuffRank

## Ratio

```
public decimal Ratio
```

### Field Value

decimal

## SkillId

```
public int SkillId
```

### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Enum CrystalRandomBuffSheet.Row.BuffRank

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public enum CrystalRandomBuffSheet.Row.BuffRank
```

**Extension Methods**

StateExtensions.Serialize(Enum)

## Fields

A = 3

B = 4

S = 2

SS = 1

# Class CrystalStageBuffGachaSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalStageBuffGachaSheet : Sheet<int,
CrystalStageBuffGachaSheet.Row>, IDictionary<int, CrystalStageBuffGachaSheet.Row>,
ICollection<KeyValuePair<int, CrystalStageBuffGachaSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalStageBuffGachaSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CrystalStageBuffGachaSheet.Row> ← CrystalStageBuffGachaSheet

### Implements

IDictionary<int, CrystalStageBuffGachaSheet.Row>,
ICollection<KeyValuePair<int, CrystalStageBuffGachaSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalStageBuffGachaSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, CrystalStageBuffGachaSheet.Row>.Name ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.OrderedList ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.First ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Last ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Keys ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Values ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Count ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.IsReadOnly ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.this[int] ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.TryGetValue(int, out
CrystalStageBuffGachaSheet.Row, bool) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.AddRow(int,
CrystalStageBuffGachaSheet.Row) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Add(int, CrystalStageBuffGachaSheet.Row) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Remove(int) ,

Sheet<int, CrystalStageBuffGachaSheet.Row>.TryGetValue(int, out CrystalStageBuffGachaSheet.Row) , Sheet<int, CrystalStageBuffGachaSheet.Row>.Add(KeyValuePair<int, CrystalStageBuffGachaSheet.Row>) , Sheet<int, CrystalStageBuffGachaSheet.Row>.Clear() , Sheet<int, CrystalStageBuffGachaSheet.Row>.Contains(KeyValuePair<int, CrystalStageBuffGachaSheet.Row>) , Sheet<int, CrystalStageBuffGachaSheet.Row>.CopyTo(KeyValuePair<int, CrystalStageBuffGachaSheet.Row>[], int) , Sheet<int, CrystalStageBuffGachaSheet.Row>.Remove(KeyValuePair<int, CrystalStageBuffGachaSheet.Row>) , Sheet<int, CrystalStageBuffGachaSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CrystalStageBuffGachaSheet()

```
public CrystalStageBuffGachaSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, CrystalStageBuffGachaSheet.Row value)
```

### Parameters

`key` int

`value` CrystalStageBuffGachaSheet.Row

# Class CrystalStageBuffGachaSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalStageBuffGachaSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CrystalStageBuffGachaSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## AdvancedCost

```
public int AdvancedCost
```

### Field Value

int

## MaxStar

```
public int MaxStar
```

### Field Value

int

## NormalCost

```
public int NormalCost
```

### Field Value

int

## StageId

```
public int StageId
```

### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Namespace Nekoyume.TableData.Custom EquipmentCraft

## Classes

CustomEquipmentCraftIconSheet

CustomEquipmentCraftIconSheet.Row

CustomEquipmentCraftOptionSheet

CustomEquipmentCraftOptionSheet.Row

CustomEquipmentCraftRecipeSheet

CustomEquipmentCraftRecipeSheet.Row

CustomEquipmentCraftRecipeSkillSheet

CustomEquipmentCraftRecipeSkillSheet.Row

CustomEquipmentCraftRelationshipSheet

CustomEquipmentCraftRelationshipSheet.Row

## Structs

CustomEquipmentCraftOptionSheet.SubStat

CustomEquipmentCraftRelationshipSheet.CpGroup

CustomEquipmentCraftRelationshipSheet.MaterialCost

# Class CustomEquipmentCraftIconSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftIconSheet : Sheet<int,
CustomEquipmentCraftIconSheet.Row>, IDictionary<int,
CustomEquipmentCraftIconSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftIconSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftIconSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CustomEquipmentCraftIconSheet.Row> ← CustomEquipmentCraftIconSheet

### Implements

IDictionary<int, CustomEquipmentCraftIconSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CustomEquipmentCraftIconSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.TryGetValue(int, out CustomEquipmentCraftIconSheet.Row, bool) ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.AddRow(int, CustomEquipmentCraftIconSheet.Row) ,

Sheet<int, CustomEquipmentCraftIconSheet.Row>.Add(int, CustomEquipmentCraftIconSheet.Row) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.TryGetValue(int, out CustomEquipmentCraftIconSheet.Row) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CustomEquipmentCraftIconSheet()

```
public CustomEquipmentCraftIconSheet()
```

# Class CustomEquipmentCraftIconSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftIconSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CustomEquipmentCraftIconSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### IconId

```
public int IconId { get; }
```

#### Property Value

int

### Id

```
public int Id { get; }
```

#### Property Value

int

## ItemSubType

```
public ItemSubType ItemSubType { get; }
```

### Property Value

ItemSubType

## Key

```
public override int Key { get; }
```

### Property Value

int

## RandomOnly

```
public bool RandomOnly { get; }
```

### Property Value

bool

## Ratio

```
public int Ratio { get; }
```

### Property Value

int

# RequiredRelationship

```
public int RequiredRelationship { get; }
```

## Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class CustomEquipmentCraftOptionSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftOptionSheet : Sheet<int,
CustomEquipmentCraftOptionSheet.Row>, IDictionary<int,
CustomEquipmentCraftOptionSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftOptionSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftOptionSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CustomEquipmentCraftOptionSheet.Row> ← CustomEquipmentCraftOptionSheet

### Implements

IDictionary<int, CustomEquipmentCraftOptionSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.TryGetValue(int, out CustomEquipmentCraftOptionSheet.Row, bool) ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.AddRow(int, CustomEquipmentCraftOptionSheet.Row) ,

Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Add(int, CustomEquipmentCraftOptionSheet.Row) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.TryGetValue(int, out CustomEquipmentCraftOptionSheet.Row) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CustomEquipmentCraftOptionSheet()

```
public CustomEquipmentCraftOptionSheet()
```

# Class CustomEquipmentCraftOptionSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftOptionSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CustomEquipmentCraftOptionSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Id

```
public int Id { get; }
```

#### Property Value

int

### ItemSubType

```
public ItemSubType ItemSubType { get; }
```

#### Property Value

ItemSubType

## Key

```
public override int Key { get; }
```

### Property Value

int

## Ratio

```
public int Ratio { get; }
```

### Property Value

int

## SubStatData

```
public List<CustomEquipmentCraftOptionSheet.SubStat> SubStatData { get; }
```

### Property Value

List<CustomEquipmentCraftOptionSheet.SubStat>

## TotalOptionRatio

```
public int TotalOptionRatio { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Struct CustomEquipmentCraftOptionSheet.SubStat

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
public struct CustomEquipmentCraftOptionSheet.SubStat
```

**Inherited Members**

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Fields

### Ratio

```
public int Ratio
```

#### Field Value

int

### StatType

```
public StatType StatType
```

#### Field Value

StatType

# Class CustomEquipmentCraftRecipeSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRecipeSheet : Sheet<int,
CustomEquipmentCraftRecipeSheet.Row>, IDictionary<int,
CustomEquipmentCraftRecipeSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftRecipeSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CustomEquipmentCraftRecipeSheet.Row> ← CustomEquipmentCraftRecipeSheet

### Implements

IDictionary<int, CustomEquipmentCraftRecipeSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRecipeSheet.Row, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.AddRow(int, CustomEquipmentCraftRecipeSheet.Row) ,

Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Add(int, CustomEquipmentCraftRecipeSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRecipeSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CustomEquipmentCraftRecipeSheet()

```
public CustomEquipmentCraftRecipeSheet()
```

# Class CustomEquipmentCraftRecipeSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRecipeSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CustomEquipmentCraftRecipeSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## CircleAmount

```
public int CircleAmount { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## ItemSubType

```
public ItemSubType ItemSubType { get; }
```

### Property Value

ItemSubType

## Key

```
public override int Key { get; }
```

### Property Value

int

## RequiredBlock

```
public long RequiredBlock { get; }
```

### Property Value

long

## ScrollAmount

```
public int ScrollAmount { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CustomEquipmentCraftRecipeSkillSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRecipeSkillSheet : Sheet<int,
CustomEquipmentCraftRecipeSkillSheet.Row>, IDictionary<int,
CustomEquipmentCraftRecipeSkillSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftRecipeSkillSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftRecipeSkillSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row> ← CustomEquipmentCraftRecipeSkillSheet

### Implements

IDictionary<int, CustomEquipmentCraftRecipeSkillSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRecipeSkillSheet.Row, bool) ,

Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.AddRow(int, CustomEquipmentCraftRecipeSkillSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Add(int, CustomEquipmentCraftRecipeSkillSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRecipeSkillSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CustomEquipmentCraftRecipeSkillSheet()

```
public CustomEquipmentCraftRecipeSkillSheet()
```

# Class CustomEquipmentCraftRecipeSkillSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRecipeSkillSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CustomEquipmentCraftRecipeSkillSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### Id

```
public int Id { get; }
```

#### Property Value

int

### ItemOptionId

```
public int ItemOptionId { get; }
```

#### Property Value

int

## ItemSubType

```
public ItemSubType ItemSubType { get; }
```

### Property Value

ItemSubType

## Key

```
public override int Key { get; }
```

### Property Value

int

## Ratio

```
public int Ratio { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class CustomEquipmentCraftRelationshipSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRelationshipSheet : Sheet<int,
CustomEquipmentCraftRelationshipSheet.Row>, IDictionary<int,
CustomEquipmentCraftRelationshipSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftRelationshipSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftRelationshipSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, CustomEquipmentCraftRelationshipSheet.Row> ← CustomEquipmentCraftRelationshipSheet

### Implements

IDictionary<int, CustomEquipmentCraftRelationshipSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRelationshipSheet.Row, bool) ,

Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.AddRow(int, CustomEquipmentCraftRelationshipSheet.Row) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Add(int, CustomEquipmentCraftRelationshipSheet.Row) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRelationshipSheet.Row) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## CustomEquipmentCraftRelationshipSheet()

```
public CustomEquipmentCraftRelationshipSheet()
```

# Struct CustomEquipmentCraftRelationshipSheet.CpGroup

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
public struct CustomEquipmentCraftRelationshipSheet.CpGroup
```

**Inherited Members**

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Fields

### MaxCp

```
public int MaxCp
```

#### Field Value

int

### MinCp

```
public int MinCp
```

#### Field Value

int

## Ratio

```
public int Ratio
```

### Field Value

int

# Methods

## SelectCp(IRandom)

```
public int SelectCp(IRandom random)
```

### Parameters

`random` IRandom

### Returns

int

# Struct CustomEquipmentCraftRelationshipSheet.MaterialCost

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
public struct CustomEquipmentCraftRelationshipSheet.MaterialCost
```

**Inherited Members**

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

# Fields

## Amount

```
public int Amount
```

### Field Value

int

## ItemId

```
public int ItemId
```

### Field Value

int

# Class CustomEquipmentCraftRelationshipSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRelationshipSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← CustomEquipmentCraftRelationshipSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## ArmorItemId

```
public int ArmorItemId { get; }
```

### Property Value

int

## BeltItemId

```
public int BeltItemId { get; }
```

### Property Value

int

## CostMultiplier

```
public long CostMultiplier { get; }
```

### Property Value

long

## CpGroups

```
public List<CustomEquipmentCraftRelationshipSheet.CpGroup> CpGroups { get; }
```

### Property Value

List<CustomEquipmentCraftRelationshipSheet.CpGroup>

## GoldAmount

```
public BigInteger GoldAmount { get; }
```

### Property Value

BigInteger

## Key

```
public override int Key { get; }
```

### Property Value

int

## MaterialCosts

```
public List<CustomEquipmentCraftRelationshipSheet.MaterialCost> MaterialCosts {
get; }
```

### Property Value

List<CustomEquipmentCraftRelationshipSheet.MaterialCost>

## NecklaceItemId

```
public int NecklaceItemId { get; }
```

### Property Value

int

## Relationship

```
public int Relationship { get; }
```

### Property Value

int

## RequiredBlockMultiplier

```
public long RequiredBlockMultiplier { get; }
```

### Property Value

long

## RingItemId

```
public int RingItemId { get; }
```

### Property Value

int

## WeaponItemId

```
public int WeaponItemId { get; }
```

### Property Value

int

# Methods

## GetItemId(ItemSubType)

```
public int GetItemId(ItemSubType itemSubType)
```

### Parameters

`itemSubType` ItemSubType

### Returns

int

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Namespace Nekoyume.TableData.Event

## Classes

EventConsumableItemRecipeSheet

EventConsumableItemRecipeSheet.Row

EventDungeonSheet

EventDungeonSheet.Row

EventDungeonStageSheet

EventDungeonStageSheet.Row

EventDungeonStageWaveSheet

EventDungeonStageWaveSheet.Row

EventMaterialItemRecipeSheet

EventMaterialItemRecipeSheet.Row

EventScheduleSheet

EventScheduleSheet.Row

PatrolRewardSheet
: Represents the PatrolRewardSheet, which defines patrol reward policies and rules.

PatrolRewardSheet.RewardModel
: Represents a reward model consisting of item counts, IDs, and tickers.

PatrolRewardSheet.Row
: Represents a row in the PatrolRewardSheet, defining patrol reward policies for specific levels and intervals.

# Class EventConsumableItemRecipeSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventConsumableItemRecipeSheet : Sheet<int,
EventConsumableItemRecipeSheet.Row>, IDictionary<int,
EventConsumableItemRecipeSheet.Row>, ICollection<KeyValuePair<int,
EventConsumableItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
EventConsumableItemRecipeSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EventConsumableItemRecipeSheet.Row> ← EventConsumableItemRecipeSheet

### Implements

IDictionary<int, EventConsumableItemRecipeSheet.Row>, ICollection<KeyValuePair<int, EventConsumableItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int, EventConsumableItemRecipeSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EventConsumableItemRecipeSheet.Row>.Name ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.OrderedList ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.First ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Last ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Keys ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Values ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Count ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.IsReadOnly ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.this[int] ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.TryGetValue(int, out EventConsumableItemRecipeSheet.Row, bool) ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.AddRow(int, EventConsumableItemRecipeSheet.Row) ,

Sheet<int, EventConsumableItemRecipeSheet.Row>.Add(int, EventConsumableItemRecipeSheet.Row) , Sheet<int, EventConsumableItemRecipeSheet.Row>.ContainsKey(int) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Remove(int) , Sheet<int, EventConsumableItemRecipeSheet.Row>.TryGetValue(int, out EventConsumableItemRecipeSheet.Row) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Add(KeyValuePair<int, EventConsumableItemRecipeSheet.Row>) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Clear() , Sheet<int, EventConsumableItemRecipeSheet.Row>.Contains(KeyValuePair<int, EventConsumableItemRecipeSheet.Row>) , Sheet<int, EventConsumableItemRecipeSheet.Row>.CopyTo(KeyValuePair<int, EventConsumableItemRecipeSheet.Row>[], int) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Remove(KeyValuePair<int, EventConsumableItemRecipeSheet.Row>) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

EventRecipeExtensions.GetRecipeRows(EventConsumableItemRecipeSheet, int) , EventRecipeExtensions.ValidateFromAction(EventConsumableItemRecipeSheet, int, string, string)

# Constructors

## EventConsumableItemRecipeSheet()

```
public EventConsumableItemRecipeSheet()
```

# Class EventConsumableItemRecipeSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventConsumableItemRecipeSheet.Row : ConsumableItemRecipeSheet.Row
```

### Inheritance

object ← SheetRow<int> ← ConsumableItemRecipeSheet.Row ← EventConsumableItemRecipeSheet.Row

### Inherited Members

ConsumableItemRecipeSheet.Row.Key , ConsumableItemRecipeSheet.Row.Id , ConsumableItemRecipeSheet.Row.RequiredBlockIndex , ConsumableItemRecipeSheet.Row.RequiredActionPoint , ConsumableItemRecipeSheet.Row.RequiredGold , ConsumableItemRecipeSheet.Row.Materials , ConsumableItemRecipeSheet.Row.ResultConsumableItemId , ConsumableItemRecipeSheet.Row.Set(IReadOnlyList<string>) , ConsumableItemRecipeSheet.Row.GetAllMaterials() , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Class EventDungeonSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonSheet : Sheet<int, EventDungeonSheet.Row>, IDictionary<int,
EventDungeonSheet.Row>, ICollection<KeyValuePair<int, EventDungeonSheet.Row>>,
IEnumerable<KeyValuePair<int, EventDungeonSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EventDungeonSheet.Row> ← EventDungeonSheet

### Implements

IDictionary<int, EventDungeonSheet.Row>,
ICollection<KeyValuePair<int, EventDungeonSheet.Row>>,
IEnumerable<KeyValuePair<int, EventDungeonSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EventDungeonSheet.Row>.Name ,
Sheet<int, EventDungeonSheet.Row>.OrderedList ,
Sheet<int, EventDungeonSheet.Row>.First , Sheet<int, EventDungeonSheet.Row>.Last ,
Sheet<int, EventDungeonSheet.Row>.Keys , Sheet<int, EventDungeonSheet.Row>.Values ,
Sheet<int, EventDungeonSheet.Row>.Count ,
Sheet<int, EventDungeonSheet.Row>.IsReadOnly ,
Sheet<int, EventDungeonSheet.Row>.this[int] ,
Sheet<int, EventDungeonSheet.Row>.Set(string, bool) ,
Sheet<int, EventDungeonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventDungeonSheet.Row>.GetEnumerator() ,
Sheet<int, EventDungeonSheet.Row>.TryGetValue(int, out EventDungeonSheet.Row, bool)
,
Sheet<int, EventDungeonSheet.Row>.AddRow(int, EventDungeonSheet.Row) ,
Sheet<int, EventDungeonSheet.Row>.Add(int, EventDungeonSheet.Row) ,
Sheet<int, EventDungeonSheet.Row>.ContainsKey(int) ,
Sheet<int, EventDungeonSheet.Row>.Remove(int) ,
Sheet<int, EventDungeonSheet.Row>.TryGetValue(int, out EventDungeonSheet.Row) ,
Sheet<int, EventDungeonSheet.Row>.Add(KeyValuePair<int, EventDungeonSheet.Row>) ,
Sheet<int, EventDungeonSheet.Row>.Clear() ,

Sheet<int, EventDungeonSheet.Row>.Contains(KeyValuePair<int, EventDungeonSheet.Row>) ,
Sheet<int, EventDungeonSheet.Row>.CopyTo(KeyValuePair<int, EventDungeonSheet.Row>[], int) ,
Sheet<int, EventDungeonSheet.Row>.Remove(KeyValuePair<int, EventDungeonSheet.Row>) ,
Sheet<int, EventDungeonSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
EventDungeonExtensions.TryGetRowByEventDungeonStageId(EventDungeonSheet, int, out EventDungeonSheet.Row) ,
EventDungeonExtensions.TryGetRowByEventScheduleId(EventDungeonSheet, int, out EventDungeonSheet.Row) ,
EventDungeonExtensions.ValidateFromAction(EventDungeonSheet, int, int, string, string)

# Constructors

## EventDungeonSheet()

```
public EventDungeonSheet()
```

# Class EventDungeonSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonSheet.Row : WorldSheet.Row
```

### Inheritance

object ← SheetRow<int> ← WorldSheet.Row ← EventDungeonSheet.Row

### Inherited Members

WorldSheet.Row.Key , WorldSheet.Row.Id , WorldSheet.Row.Name , WorldSheet.Row.StageBegin , WorldSheet.Row.StageEnd , WorldSheet.Row.StagesCount , WorldSheet.Row.Set(IReadOnlyList<string>) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Class EventDungeonStageSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonStageSheet : Sheet<int, EventDungeonStageSheet.Row>,
IDictionary<int, EventDungeonStageSheet.Row>, ICollection<KeyValuePair<int,
EventDungeonStageSheet.Row>>, IEnumerable<KeyValuePair<int,
EventDungeonStageSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EventDungeonStageSheet.Row> ← EventDungeonStageSheet

### Implements

IDictionary<int, EventDungeonStageSheet.Row>,
ICollection<KeyValuePair<int, EventDungeonStageSheet.Row>>,
IEnumerable<KeyValuePair<int, EventDungeonStageSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, EventDungeonStageSheet.Row>.Name ,
Sheet<int, EventDungeonStageSheet.Row>.OrderedList ,
Sheet<int, EventDungeonStageSheet.Row>.First ,
Sheet<int, EventDungeonStageSheet.Row>.Last ,
Sheet<int, EventDungeonStageSheet.Row>.Keys ,
Sheet<int, EventDungeonStageSheet.Row>.Values ,
Sheet<int, EventDungeonStageSheet.Row>.Count ,
Sheet<int, EventDungeonStageSheet.Row>.IsReadOnly ,
Sheet<int, EventDungeonStageSheet.Row>.this[int] ,
Sheet<int, EventDungeonStageSheet.Row>.Set(string, bool) ,
Sheet<int, EventDungeonStageSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventDungeonStageSheet.Row>.GetEnumerator() ,
Sheet<int, EventDungeonStageSheet.Row>.TryGetValue(int, out
EventDungeonStageSheet.Row, bool) ,
Sheet<int, EventDungeonStageSheet.Row>.AddRow(int, EventDungeonStageSheet.Row) ,
Sheet<int, EventDungeonStageSheet.Row>.Add(int, EventDungeonStageSheet.Row) ,
Sheet<int, EventDungeonStageSheet.Row>.ContainsKey(int) ,
Sheet<int, EventDungeonStageSheet.Row>.Remove(int) ,

Sheet<int, EventDungeonStageSheet.Row>.TryGetValue(int, out EventDungeonStageSheet.Row) , Sheet<int, EventDungeonStageSheet.Row>.Add(KeyValuePair<int, EventDungeonStageSheet.Row>) , Sheet<int, EventDungeonStageSheet.Row>.Clear() , Sheet<int, EventDungeonStageSheet.Row>.Contains(KeyValuePair<int, EventDungeonStageSheet.Row>) , Sheet<int, EventDungeonStageSheet.Row>.CopyTo(KeyValuePair<int, EventDungeonStageSheet.Row>[], int) , Sheet<int, EventDungeonStageSheet.Row>.Remove(KeyValuePair<int, EventDungeonStageSheet.Row>) , Sheet<int, EventDungeonStageSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

EventDungeonExtensions.GetStageRows(EventDungeonStageSheet, int, int) , EventDungeonExtensions.ValidateFromAction(EventDungeonStageSheet, int, string, string)

# Constructors

## EventDungeonStageSheet()

```
public EventDungeonStageSheet()
```

# Class EventDungeonStageSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonStageSheet.Row : StageSheet.Row
```

### Inheritance

object ← SheetRow<int> ← StageSheet.Row ← EventDungeonStageSheet.Row

### Inherited Members

StageSheet.Row.Key , StageSheet.Row.Id , StageSheet.Row.CostAP , StageSheet.Row.EntryCostItemId , StageSheet.Row.EntryCostItemCount , StageSheet.Row.TurnLimit , StageSheet.Row.EnemyInitialStatModifiers , StageSheet.Row.Background , StageSheet.Row.BGM , StageSheet.Row.Rewards , StageSheet.Row.DropItemMin , StageSheet.Row.DropItemMax , StageSheet.Row.FavRewards , StageSheet.Row.FavDropMin , StageSheet.Row.FavDropMax , StageSheet.Row.Set(IReadOnlyList<string>) , StageSheet.Row.CloneWithId(int) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

### Extension Methods

EventDungeonExtensions.GetStageNumber(EventDungeonStageSheet.Row)

# Class EventDungeonStageWaveSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonStageWaveSheet : Sheet<int,
EventDungeonStageWaveSheet.Row>, IDictionary<int, EventDungeonStageWaveSheet.Row>,
ICollection<KeyValuePair<int, EventDungeonStageWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, EventDungeonStageWaveSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EventDungeonStageWaveSheet.Row> ← EventDungeonStageWaveSheet

### Implements

IDictionary<int, EventDungeonStageWaveSheet.Row>, ICollection<KeyValuePair<int, EventDungeonStageWaveSheet.Row>>, IEnumerable<KeyValuePair<int, EventDungeonStageWaveSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EventDungeonStageWaveSheet.Row>.Name ,
Sheet<int, EventDungeonStageWaveSheet.Row>.OrderedList ,
Sheet<int, EventDungeonStageWaveSheet.Row>.First ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Last ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Keys ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Values ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Count ,
Sheet<int, EventDungeonStageWaveSheet.Row>.IsReadOnly ,
Sheet<int, EventDungeonStageWaveSheet.Row>.this[int] ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Set(string, bool) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.GetEnumerator() ,
Sheet<int, EventDungeonStageWaveSheet.Row>.TryGetValue(int, out EventDungeonStageWaveSheet.Row, bool) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.AddRow(int, EventDungeonStageWaveSheet.Row) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Add(int, EventDungeonStageWaveSheet.Row) ,

Sheet<int, EventDungeonStageWaveSheet.Row>.ContainsKey(int) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Remove(int) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.TryGetValue(int, out
EventDungeonStageWaveSheet.Row) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Add(KeyValuePair<int,
EventDungeonStageWaveSheet.Row>) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Clear() ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Contains(KeyValuePair<int,
EventDungeonStageWaveSheet.Row>) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.CopyTo(KeyValuePair<int,
EventDungeonStageWaveSheet.Row>[], int) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Remove(KeyValuePair<int,
EventDungeonStageWaveSheet.Row>) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
EventDungeonExtensions.GetStageWaveRows(EventDungeonStageWaveSheet, int, int)

# Constructors

## EventDungeonStageWaveSheet()

```
public EventDungeonStageWaveSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, EventDungeonStageWaveSheet.Row value)
```

### Parameters

`key` int

`value` EventDungeonStageWaveSheet.Row

# Class EventDungeonStageWaveSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonStageWaveSheet.Row : StageWaveSheet.Row
```

### Inheritance

object ← SheetRow<int> ← StageWaveSheet.Row ← EventDungeonStageWaveSheet.Row

### Inherited Members

StageWaveSheet.Row.Key , StageWaveSheet.Row.StageId , StageWaveSheet.Row.Waves , StageWaveSheet.Row.HasBoss , StageWaveSheet.Row.TotalMonsterIds , StageWaveSheet.Row.Set(IReadOnlyList<string>) , StageWaveSheet.Row.CloneWithStageId(int) , StageWaveSheet.Row.EndOfSheetInitialize() , SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Class EventMaterialItemRecipeSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventMaterialItemRecipeSheet : Sheet<int,
EventMaterialItemRecipeSheet.Row>, IDictionary<int,
EventMaterialItemRecipeSheet.Row>, ICollection<KeyValuePair<int,
EventMaterialItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
EventMaterialItemRecipeSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EventMaterialItemRecipeSheet.Row> ← EventMaterialItemRecipeSheet

### Implements

IDictionary<int, EventMaterialItemRecipeSheet.Row>, ICollection<KeyValuePair<int, EventMaterialItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int, EventMaterialItemRecipeSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EventMaterialItemRecipeSheet.Row>.Name ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.OrderedList ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.First ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Last ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Keys ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Values ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Count ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.IsReadOnly ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.this[int] ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.TryGetValue(int, out EventMaterialItemRecipeSheet.Row, bool) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.AddRow(int, EventMaterialItemRecipeSheet.Row) ,

Sheet<int, EventMaterialItemRecipeSheet.Row>.Add(int, EventMaterialItemRecipeSheet.Row) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.ContainsKey(int) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Remove(int) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.TryGetValue(int, out EventMaterialItemRecipeSheet.Row) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Add(KeyValuePair<int, EventMaterialItemRecipeSheet.Row>) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Clear() ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Contains(KeyValuePair<int, EventMaterialItemRecipeSheet.Row>) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.CopyTo(KeyValuePair<int, EventMaterialItemRecipeSheet.Row>[], int) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Remove(KeyValuePair<int, EventMaterialItemRecipeSheet.Row>) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
EventRecipeExtensions.GetRecipeRows(EventMaterialItemRecipeSheet, int) ,
EventRecipeExtensions.ValidateFromAction(EventMaterialItemRecipeSheet, int, string, string)

# Constructors

## EventMaterialItemRecipeSheet()

```
public EventMaterialItemRecipeSheet()
```

# Class EventMaterialItemRecipeSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventMaterialItemRecipeSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EventMaterialItemRecipeSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

EventRecipeExtensions.ValidateFromAction(EventMaterialItemRecipeSheet.Row, MaterialItemSheet, Dictionary<int, int>, string, string)

# Properties

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## RequiredMaterialsCount

```
public int RequiredMaterialsCount { get; }
```

### Property Value

int

## RequiredMaterialsId

```
public List<int> RequiredMaterialsId { get; }
```

### Property Value

List<int>

## ResultMaterialItemCount

```
public int ResultMaterialItemCount { get; }
```

### Property Value

int

## ResultMaterialItemId

```
public int ResultMaterialItemId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class EventScheduleSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
public class EventScheduleSheet : Sheet<int, EventScheduleSheet.Row>,
IDictionary<int, EventScheduleSheet.Row>, ICollection<KeyValuePair<int,
EventScheduleSheet.Row>>, IEnumerable<KeyValuePair<int, EventScheduleSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, EventScheduleSheet.Row> ← EventScheduleSheet

### Implements

IDictionary<int, EventScheduleSheet.Row>,
ICollection<KeyValuePair<int, EventScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, EventScheduleSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, EventScheduleSheet.Row>.Name ,
Sheet<int, EventScheduleSheet.Row>.OrderedList ,
Sheet<int, EventScheduleSheet.Row>.First , Sheet<int, EventScheduleSheet.Row>.Last ,
Sheet<int, EventScheduleSheet.Row>.Keys ,
Sheet<int, EventScheduleSheet.Row>.Values ,
Sheet<int, EventScheduleSheet.Row>.Count ,
Sheet<int, EventScheduleSheet.Row>.IsReadOnly ,
Sheet<int, EventScheduleSheet.Row>.this[int] ,
Sheet<int, EventScheduleSheet.Row>.Set(string, bool) ,
Sheet<int, EventScheduleSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventScheduleSheet.Row>.GetEnumerator() ,
Sheet<int, EventScheduleSheet.Row>.TryGetValue(int, out EventScheduleSheet.Row, bool)
,
Sheet<int, EventScheduleSheet.Row>.AddRow(int, EventScheduleSheet.Row) ,
Sheet<int, EventScheduleSheet.Row>.Add(int, EventScheduleSheet.Row) ,
Sheet<int, EventScheduleSheet.Row>.ContainsKey(int) ,
Sheet<int, EventScheduleSheet.Row>.Remove(int) ,
Sheet<int, EventScheduleSheet.Row>.TryGetValue(int, out EventScheduleSheet.Row) ,
Sheet<int, EventScheduleSheet.Row>.Add(KeyValuePair<int, EventScheduleSheet.Row>) ,
Sheet<int, EventScheduleSheet.Row>.Clear() ,

Sheet<int, EventScheduleSheet.Row>.Contains(KeyValuePair<int, EventScheduleSheet.Row>) ,
Sheet<int, EventScheduleSheet.Row>.CopyTo(KeyValuePair<int, EventScheduleSheet.Row>[], int) ,
Sheet<int, EventScheduleSheet.Row>.Remove(KeyValuePair<int, EventScheduleSheet.Row>) ,
Sheet<int, EventScheduleSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**
EventScheduleExtensions.TryGetRowForDungeon(EventScheduleSheet, long, out EventScheduleSheet.Row) ,
EventScheduleExtensions.TryGetRowForRecipe(EventScheduleSheet, long, out EventScheduleSheet.Row) ,
EventScheduleExtensions.ValidateFromActionForDungeon(EventScheduleSheet, long, int, int, string, string) ,
EventScheduleExtensions.ValidateFromActionForRecipe(EventScheduleSheet, long, int, int, string, string)

# Constructors

## EventScheduleSheet()

```
public EventScheduleSheet()
```

# Class EventScheduleSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventScheduleSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← EventScheduleSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

EventScheduleExtensions.GetDungeonTicketCost(EventScheduleSheet.Row, int, Currency) , EventScheduleExtensions.GetDungeonTicketCostV1(EventScheduleSheet.Row, int) , EventScheduleExtensions.GetStageExp(EventScheduleSheet.Row, int, int)

# Properties

## DungeonEndBlockIndex

```
public long DungeonEndBlockIndex { get; }
```

### Property Value

long

## DungeonExpSeedValue

```
public int DungeonExpSeedValue { get; }
```

### Property Value

int

## DungeonTicketAdditionalPrice

This value is divided by 10 and used. e.g., DecimalMath.DecimalEx.Floor(row.DungeonTicketAdditionalPrice * .1m, 2);

```
public int DungeonTicketAdditionalPrice { get; }
```

### Property Value

int

## DungeonTicketPrice

This value is divided by 10 and used. e.g., DecimalMath.DecimalEx.Floor(row.DungeonTicketPrice * .1m, 2);

```
public int DungeonTicketPrice { get; }
```

### Property Value

int

## DungeonTicketsMax

```
public int DungeonTicketsMax { get; }
```

### Property Value

int

## DungeonTicketsResetIntervalBlockRange

```
public int DungeonTicketsResetIntervalBlockRange { get; }
```

### Property Value

int

## Id

```
public int Id { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## RecipeEndBlockIndex

```
public long RecipeEndBlockIndex { get; }
```

### Property Value

long

## StartBlockIndex

```
public long StartBlockIndex { get; }
```

### Property Value

long

## Methods

### Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class PatrolRewardSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

Represents the PatrolRewardSheet, which defines patrol reward policies and rules.

```
[Serializable]
public class PatrolRewardSheet : Sheet<int, PatrolRewardSheet.Row>, IDictionary<int,
PatrolRewardSheet.Row>, ICollection<KeyValuePair<int, PatrolRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, PatrolRewardSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, PatrolRewardSheet.Row> ← PatrolRewardSheet

### Implements

IDictionary<int, PatrolRewardSheet.Row>,
ICollection<KeyValuePair<int, PatrolRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, PatrolRewardSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, PatrolRewardSheet.Row>.Name ,
Sheet<int, PatrolRewardSheet.Row>.OrderedList ,
Sheet<int, PatrolRewardSheet.Row>.First , Sheet<int, PatrolRewardSheet.Row>.Last ,
Sheet<int, PatrolRewardSheet.Row>.Keys , Sheet<int, PatrolRewardSheet.Row>.Values ,
Sheet<int, PatrolRewardSheet.Row>.Count ,
Sheet<int, PatrolRewardSheet.Row>.IsReadOnly ,
Sheet<int, PatrolRewardSheet.Row>.this[int] ,
Sheet<int, PatrolRewardSheet.Row>.Set(string, bool) ,
Sheet<int, PatrolRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, PatrolRewardSheet.Row>.GetEnumerator() ,
Sheet<int, PatrolRewardSheet.Row>.TryGetValue(int, out PatrolRewardSheet.Row, bool) ,
Sheet<int, PatrolRewardSheet.Row>.AddRow(int, PatrolRewardSheet.Row) ,
Sheet<int, PatrolRewardSheet.Row>.Add(int, PatrolRewardSheet.Row) ,
Sheet<int, PatrolRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, PatrolRewardSheet.Row>.Remove(int) ,
Sheet<int, PatrolRewardSheet.Row>.TryGetValue(int, out PatrolRewardSheet.Row) ,
Sheet<int, PatrolRewardSheet.Row>.Add(KeyValuePair<int, PatrolRewardSheet.Row>) ,
Sheet<int, PatrolRewardSheet.Row>.Clear() ,

Sheet<int, PatrolRewardSheet.Row>.Contains(KeyValuePair<int, PatrolRewardSheet.Row>) , Sheet<int, PatrolRewardSheet.Row>.CopyTo(KeyValuePair<int, PatrolRewardSheet.Row>[], int) , Sheet<int, PatrolRewardSheet.Row>.Remove(KeyValuePair<int, PatrolRewardSheet.Row>) , Sheet<int, PatrolRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## PatrolRewardSheet()

Initializes a new instance of the PatrolRewardSheet class.

```
public PatrolRewardSheet()
```

# Methods

## FindByLevel(int, long)

Finds the reward policy row based on avatar level and block index.

```
public PatrolRewardSheet.Row FindByLevel(int level, long blockIndex)
```

### Parameters

`level` int

The avatar's level.

`blockIndex` long

The current block index.

### Returns

PatrolRewardSheet.Row

The corresponding reward policy row.

## Exceptions

InvalidOperationException

Thrown if no activated policy matches the criteria.

# Class PatrolRewardSheet.RewardModel

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

Represents a reward model consisting of item counts, IDs, and tickers.

```
[Serializable]
public class PatrolRewardSheet.RewardModel
```

**Inheritance**

object ← PatrolRewardSheet.RewardModel

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### RewardModel(int, int, string)

```
public RewardModel(int count, int itemId, string ticker)
```

#### Parameters

`count` int

`itemId` int

`ticker` string

## Fields

### Count

```
public int Count
```

### Field Value

int

## ItemId

```
public int ItemId
```

### Field Value

int

## Ticker

```
public string Ticker
```

### Field Value

string

# Class PatrolRewardSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

Represents a row in the PatrolRewardSheet, defining patrol reward policies for specific levels and intervals.

```
[Serializable]
public class PatrolRewardSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← PatrolRewardSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## MaxRewardCount

```
public const int MaxRewardCount = 100
```

### Field Value

int

# Properties

## EndedBlockIndex

```
public long EndedBlockIndex { get; set; }
```

### Property Value

long

## Id

```
public int Id { get; set; }
```

### Property Value

int

## Interval

```
public long Interval { get; set; }
```

### Property Value

long

## Key

```
public override int Key { get; }
```

### Property Value

int

## MaxLevel

```
public int? MaxLevel { get; set; }
```

### Property Value

int?

## MinimumLevel

```
public int MinimumLevel { get; set; }
```

### Property Value

int

## Rewards

```
public List<PatrolRewardSheet.RewardModel> Rewards { get; set; }
```

### Property Value

List<PatrolRewardSheet.RewardModel>

## StartedBlockIndex

```
public long StartedBlockIndex { get; set; }
```

### Property Value

long

# Methods

## Set(IReadOnlyList<string>)

Sets the row data using a list of string fields.

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

The fields to parse into the row.

# Namespace Nekoyume.TableData.Garages

## Classes

LoadIntoMyGaragesCostSheet

LoadIntoMyGaragesCostSheet.Row

# Class LoadIntoMyGaragesCostSheet

Namespace: Nekoyume.TableData.Garages

Assembly: Lib9c.dll

```
[Serializable]
public class LoadIntoMyGaragesCostSheet : Sheet<int,
LoadIntoMyGaragesCostSheet.Row>, IDictionary<int, LoadIntoMyGaragesCostSheet.Row>,
ICollection<KeyValuePair<int, LoadIntoMyGaragesCostSheet.Row>>,
IEnumerable<KeyValuePair<int, LoadIntoMyGaragesCostSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, LoadIntoMyGaragesCostSheet.Row> ← LoadIntoMyGaragesCostSheet

### Implements

IDictionary<int, LoadIntoMyGaragesCostSheet.Row>, ICollection<KeyValuePair<int, LoadIntoMyGaragesCostSheet.Row>>, IEnumerable<KeyValuePair<int, LoadIntoMyGaragesCostSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Name ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.OrderedList ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.First ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Last ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Keys ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Values ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Count ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.IsReadOnly ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.this[int] ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Set(string, bool) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.GetEnumerator() ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.TryGetValue(int, out LoadIntoMyGaragesCostSheet.Row, bool) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.AddRow(int, LoadIntoMyGaragesCostSheet.Row) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Add(int, LoadIntoMyGaragesCostSheet.Row) ,

Sheet<int, LoadIntoMyGaragesCostSheet.Row>.ContainsKey(int) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Remove(int) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.TryGetValue(int, out
LoadIntoMyGaragesCostSheet.Row) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Add(KeyValuePair<int,
LoadIntoMyGaragesCostSheet.Row>) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Clear() ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Contains(KeyValuePair<int,
LoadIntoMyGaragesCostSheet.Row>) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.CopyTo(KeyValuePair<int,
LoadIntoMyGaragesCostSheet.Row>[], int) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Remove(KeyValuePair<int,
LoadIntoMyGaragesCostSheet.Row>) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## LoadIntoMyGaragesCostSheet()

```
public LoadIntoMyGaragesCostSheet()
```

# Methods

## GetGarageCost(HashDigest<SHA256>, int)

```
public FungibleAssetValue GetGarageCost(HashDigest<SHA256> fungibleId, int count)
```

### Parameters

`fungibleId` HashDigest<SHA256>

`count` int

### Returns

## GetGarageCost(FungibleAssetValue)

```
public FungibleAssetValue GetGarageCost(FungibleAssetValue fav)
```

### Parameters

`fav` FungibleAssetValue

### Returns

FungibleAssetValue

## GetGarageCost(IEnumerable<FungibleAssetValue>, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>)

```
public FungibleAssetValue GetGarageCost(IEnumerable<FungibleAssetValue>
fungibleAssetValues, IEnumerable<(HashDigest<SHA256> fungibleId, int
count)> fungibleIdAndCounts)
```

### Parameters

`fungibleAssetValues` IEnumerable<FungibleAssetValue>

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

### Returns

FungibleAssetValue

## GetGarageCost(IEnumerable<FungibleAssetValue>, IEnumerable<(int itemId, int count, bool tradable)>)

```
public FungibleAssetValue GetGarageCost(IEnumerable<FungibleAssetValue>
fungibleAssetValues, IEnumerable<(int itemId, int count, bool
tradable)> itemIdAndCounts)
```

### Parameters

`fungibleAssetValues` IEnumerable<FungibleAssetValue>

`itemIdAndCounts` IEnumerable<(int itemId, int count, bool tradable)>

### Returns

FungibleAssetValue

## GetGarageCost(int, int)

```
public FungibleAssetValue GetGarageCost(int itemId, int count)
```

### Parameters

`itemId` int

`count` int

### Returns

FungibleAssetValue

## GetGarageCostPerUnit(HashDigest<SHA256>)

```
public decimal GetGarageCostPerUnit(HashDigest<SHA256> fungibleId)
```

### Parameters

`fungibleId` HashDigest<SHA256>

### Returns

decimal

## GetGarageCostPerUnit(int)

```
public decimal GetGarageCostPerUnit(int itemId)
```

### Parameters

`itemId` int

### Returns

decimal

## GetGarageCostPerUnit(string)

```
public decimal GetGarageCostPerUnit(string currencyTicker)
```

### Parameters

`currencyTicker` string

### Returns

decimal

## HasCost(HashDigest<SHA256>)

```
public bool HasCost(HashDigest<SHA256> fungibleId)
```

### Parameters

`fungibleId` HashDigest<SHA256>

### Returns

bool

## HasCost(string)

```
public bool HasCost(string currencyTicker)
```

### Parameters

`currencyTicker` string

### Returns

bool

# Class LoadIntoMyGaragesCostSheet.Row

Namespace: Nekoyume.TableData.Garages

Assembly: Lib9c.dll

```
[Serializable]
public class LoadIntoMyGaragesCostSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← LoadIntoMyGaragesCostSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## CurrencyTicker

```
public string CurrencyTicker { get; }
```

### Property Value

string

## FungibleId

```
public HashDigest<SHA256>? FungibleId { get; }
```

### Property Value

HashDigest<SHA256>?

## GarageCostPerUnit

```
public decimal GarageCostPerUnit { get; }
```

### Property Value

decimal

## Id

```
public int Id { get; }
```

### Property Value

int

## ItemId

```
public int ItemId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## Methods

# Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Namespace Nekoyume.TableData.GrandFinale

## Classes

GrandFinaleParticipantsSheet

GrandFinaleParticipantsSheet.Row

GrandFinaleScheduleSheet

GrandFinaleScheduleSheet.Row

# Class GrandFinaleParticipantsSheet

Namespace: Nekoyume.TableData.GrandFinale

Assembly: Lib9c.dll

```
[Serializable]
public class GrandFinaleParticipantsSheet : Sheet<int,
GrandFinaleParticipantsSheet.Row>, IDictionary<int,
GrandFinaleParticipantsSheet.Row>, ICollection<KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>>, IEnumerable<KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, GrandFinaleParticipantsSheet.Row> ← GrandFinaleParticipantsSheet

### Implements

IDictionary<int, GrandFinaleParticipantsSheet.Row>,
ICollection<KeyValuePair<int, GrandFinaleParticipantsSheet.Row>>,
IEnumerable<KeyValuePair<int, GrandFinaleParticipantsSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, GrandFinaleParticipantsSheet.Row>.Name ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.OrderedList ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.First ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Last ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Keys ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Values ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Count ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.IsReadOnly ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.this[int] ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Set(string, bool) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.GetEnumerator() ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.TryGetValue(int, out
GrandFinaleParticipantsSheet.Row, bool) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.AddRow(int,
GrandFinaleParticipantsSheet.Row) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Add(int, GrandFinaleParticipantsSheet.Row)
,

Sheet<int, GrandFinaleParticipantsSheet.Row>.ContainsKey(int) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Remove(int) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.TryGetValue(int, out
GrandFinaleParticipantsSheet.Row) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Add(KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Clear() ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Contains(KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.CopyTo(KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>[], int) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Remove(KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GrandFinaleParticipantsSheet()

```
public GrandFinaleParticipantsSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, GrandFinaleParticipantsSheet.Row value)
```

### Parameters

`key` int

`value` GrandFinaleParticipantsSheet.Row

# Class GrandFinaleParticipantsSheet.Row

Namespace: Nekoyume.TableData.GrandFinale

Assembly: Lib9c.dll

```
[Serializable]
public class GrandFinaleParticipantsSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← GrandFinaleParticipantsSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## GrandFinaleId

```
public int GrandFinaleId { get; }
```

### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

# Participants

```
public List<Address> Participants { get; }
```

## Property Value

List<Address>

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Class GrandFinaleScheduleSheet

Namespace: Nekoyume.TableData.GrandFinale

Assembly: Lib9c.dll

```
[Serializable]
public class GrandFinaleScheduleSheet : Sheet<int, GrandFinaleScheduleSheet.Row>,
IDictionary<int, GrandFinaleScheduleSheet.Row>, ICollection<KeyValuePair<int,
GrandFinaleScheduleSheet.Row>>, IEnumerable<KeyValuePair<int,
GrandFinaleScheduleSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, GrandFinaleScheduleSheet.Row> ← GrandFinaleScheduleSheet

### Implements

IDictionary<int, GrandFinaleScheduleSheet.Row>,
ICollection<KeyValuePair<int, GrandFinaleScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, GrandFinaleScheduleSheet.Row>>, IEnumerable,
ISheet

### Inherited Members

Sheet<int, GrandFinaleScheduleSheet.Row>.Name ,
Sheet<int, GrandFinaleScheduleSheet.Row>.OrderedList ,
Sheet<int, GrandFinaleScheduleSheet.Row>.First ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Last ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Keys ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Values ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Count ,
Sheet<int, GrandFinaleScheduleSheet.Row>.IsReadOnly ,
Sheet<int, GrandFinaleScheduleSheet.Row>.this[int] ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Set(string, bool) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.GetEnumerator() ,
Sheet<int, GrandFinaleScheduleSheet.Row>.TryGetValue(int, out
GrandFinaleScheduleSheet.Row, bool) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.AddRow(int, GrandFinaleScheduleSheet.Row) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Add(int, GrandFinaleScheduleSheet.Row) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.ContainsKey(int) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Remove(int) ,

Sheet<int, GrandFinaleScheduleSheet.Row>.TryGetValue(int, out GrandFinaleScheduleSheet.Row) , Sheet<int, GrandFinaleScheduleSheet.Row>.Add(KeyValuePair<int, GrandFinaleScheduleSheet.Row>) , Sheet<int, GrandFinaleScheduleSheet.Row>.Clear() , Sheet<int, GrandFinaleScheduleSheet.Row>.Contains(KeyValuePair<int, GrandFinaleScheduleSheet.Row>) , Sheet<int, GrandFinaleScheduleSheet.Row>.CopyTo(KeyValuePair<int, GrandFinaleScheduleSheet.Row>[], int) , Sheet<int, GrandFinaleScheduleSheet.Row>.Remove(KeyValuePair<int, GrandFinaleScheduleSheet.Row>) , Sheet<int, GrandFinaleScheduleSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## GrandFinaleScheduleSheet()

```
public GrandFinaleScheduleSheet()
```

# Methods

## GetRowByBlockIndex(long)

```
public GrandFinaleScheduleSheet.Row GetRowByBlockIndex(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

GrandFinaleScheduleSheet.Row

# Class GrandFinaleScheduleSheet.Row

Namespace: Nekoyume.TableData.GrandFinale

Assembly: Lib9c.dll

```
[Serializable]
public class GrandFinaleScheduleSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← GrandFinaleScheduleSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Properties

### EndBlockIndex

```
public long EndBlockIndex { get; }
```

#### Property Value

long

### Id

```
public int Id { get; }
```

#### Property Value

int

## Key

```
public override int Key { get; }
```

### Property Value

int

## StartBlockIndex

```
public long StartBlockIndex { get; }
```

### Property Value

long

# Methods

## IsOpened(long)

```
public bool IsOpened(long blockIndex)
```

### Parameters

`blockIndex` long

### Returns

bool

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

## Parameters

`fields` IReadOnlyList<string>

# Namespace Nekoyume.TableData.Pet

## Classes

PetOptionSheet

PetOptionSheet.Row

PetOptionSheet.Row.PetOptionInfo

# Class PetOptionSheet

Namespace: Nekoyume.TableData.Pet

Assembly: Lib9c.dll

```
[Serializable]
public class PetOptionSheet : Sheet<int, PetOptionSheet.Row>, IDictionary<int,
PetOptionSheet.Row>, ICollection<KeyValuePair<int, PetOptionSheet.Row>>,
IEnumerable<KeyValuePair<int, PetOptionSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, PetOptionSheet.Row> ← PetOptionSheet

### Implements

IDictionary<int, PetOptionSheet.Row>,
ICollection<KeyValuePair<int, PetOptionSheet.Row>>,
IEnumerable<KeyValuePair<int, PetOptionSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, PetOptionSheet.Row>.Name , Sheet<int, PetOptionSheet.Row>.OrderedList ,
Sheet<int, PetOptionSheet.Row>.First , Sheet<int, PetOptionSheet.Row>.Last ,
Sheet<int, PetOptionSheet.Row>.Keys , Sheet<int, PetOptionSheet.Row>.Values ,
Sheet<int, PetOptionSheet.Row>.Count , Sheet<int, PetOptionSheet.Row>.IsReadOnly ,
Sheet<int, PetOptionSheet.Row>.this[int] ,
Sheet<int, PetOptionSheet.Row>.Set(string, bool) ,
Sheet<int, PetOptionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, PetOptionSheet.Row>.GetEnumerator() ,
Sheet<int, PetOptionSheet.Row>.TryGetValue(int, out PetOptionSheet.Row, bool) ,
Sheet<int, PetOptionSheet.Row>.AddRow(int, PetOptionSheet.Row) ,
Sheet<int, PetOptionSheet.Row>.Add(int, PetOptionSheet.Row) ,
Sheet<int, PetOptionSheet.Row>.ContainsKey(int) ,
Sheet<int, PetOptionSheet.Row>.Remove(int) ,
Sheet<int, PetOptionSheet.Row>.TryGetValue(int, out PetOptionSheet.Row) ,
Sheet<int, PetOptionSheet.Row>.Add(KeyValuePair<int, PetOptionSheet.Row>) ,
Sheet<int, PetOptionSheet.Row>.Clear() ,
Sheet<int, PetOptionSheet.Row>.Contains(KeyValuePair<int, PetOptionSheet.Row>) ,
Sheet<int, PetOptionSheet.Row>.CopyTo(KeyValuePair<int, PetOptionSheet.Row>[], int) ,
Sheet<int, PetOptionSheet.Row>.Remove(KeyValuePair<int, PetOptionSheet.Row>) ,
Sheet<int, PetOptionSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## PetOptionSheet()

```
public PetOptionSheet()
```

# Methods

## AddRow(int, Row)

```
protected override void AddRow(int key, PetOptionSheet.Row value)
```

### Parameters

`key` int

`value` PetOptionSheet.Row

# Class PetOptionSheet.Row

Namespace: Nekoyume.TableData.Pet

Assembly: Lib9c.dll

```
[Serializable]
public class PetOptionSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← PetOptionSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

## LevelOptionMap

```
public Dictionary<int, PetOptionSheet.Row.PetOptionInfo> LevelOptionMap { get; }
```

### Property Value

Dictionary<int, PetOptionSheet.Row.PetOptionInfo>

## PetId

```
public int PetId { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class PetOptionSheet.Row.PetOptionInfo

Namespace: Nekoyume.TableData.Pet

Assembly: Lib9c.dll

```
public class PetOptionSheet.Row.PetOptionInfo
```

**Inheritance**

object ← PetOptionSheet.Row.PetOptionInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### PetOptionInfo(PetOptionType, decimal)

```
public PetOptionInfo(PetOptionType optionType, decimal optionValue)
```

#### Parameters

`optionType` PetOptionType

`optionValue` decimal

## Properties

### OptionType

```
public PetOptionType OptionType { get; }
```

#### Property Value

PetOptionType

# OptionValue

```
public decimal OptionValue { get; }
```

## Property Value

decimal

# Namespace Nekoyume.TableData.Rune

## Classes

RuneLevelBonusSheet

RuneLevelBonusSheet.Row

RuneListSheet

RuneListSheet.Row

# Class RuneLevelBonusSheet

Namespace: Nekoyume.TableData.Rune

Assembly: Lib9c.dll

```
public class RuneLevelBonusSheet : Sheet<int, RuneLevelBonusSheet.Row>,
IDictionary<int, RuneLevelBonusSheet.Row>, ICollection<KeyValuePair<int,
RuneLevelBonusSheet.Row>>, IEnumerable<KeyValuePair<int, RuneLevelBonusSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, RuneLevelBonusSheet.Row> ← RuneLevelBonusSheet

### Implements

IDictionary<int, RuneLevelBonusSheet.Row>,
ICollection<KeyValuePair<int, RuneLevelBonusSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneLevelBonusSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, RuneLevelBonusSheet.Row>.Name ,
Sheet<int, RuneLevelBonusSheet.Row>.OrderedList ,
Sheet<int, RuneLevelBonusSheet.Row>.First ,
Sheet<int, RuneLevelBonusSheet.Row>.Last ,
Sheet<int, RuneLevelBonusSheet.Row>.Keys ,
Sheet<int, RuneLevelBonusSheet.Row>.Values ,
Sheet<int, RuneLevelBonusSheet.Row>.Count ,
Sheet<int, RuneLevelBonusSheet.Row>.IsReadOnly ,
Sheet<int, RuneLevelBonusSheet.Row>.this[int] ,
Sheet<int, RuneLevelBonusSheet.Row>.Set(string, bool) ,
Sheet<int, RuneLevelBonusSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneLevelBonusSheet.Row>.GetEnumerator() ,
Sheet<int, RuneLevelBonusSheet.Row>.TryGetValue(int, out RuneLevelBonusSheet.Row, bool) ,
Sheet<int, RuneLevelBonusSheet.Row>.AddRow(int, RuneLevelBonusSheet.Row) ,
Sheet<int, RuneLevelBonusSheet.Row>.Add(int, RuneLevelBonusSheet.Row) ,
Sheet<int, RuneLevelBonusSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneLevelBonusSheet.Row>.Remove(int) ,
Sheet<int, RuneLevelBonusSheet.Row>.TryGetValue(int, out RuneLevelBonusSheet.Row) ,

Sheet<int, RuneLevelBonusSheet.Row>.Add(KeyValuePair<int, RuneLevelBonusSheet.Row>) , Sheet<int, RuneLevelBonusSheet.Row>.Clear() , Sheet<int, RuneLevelBonusSheet.Row>.Contains(KeyValuePair<int, RuneLevelBonusSheet.Row>) , Sheet<int, RuneLevelBonusSheet.Row>.CopyTo(KeyValuePair<int, RuneLevelBonusSheet.Row>[], int) , Sheet<int, RuneLevelBonusSheet.Row>.Remove(KeyValuePair<int, RuneLevelBonusSheet.Row>) , Sheet<int, RuneLevelBonusSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RuneLevelBonusSheet()

```
public RuneLevelBonusSheet()
```

# Class RuneLevelBonusSheet.Row

Namespace: Nekoyume.TableData.Rune

Assembly: Lib9c.dll

```
public class RuneLevelBonusSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← RuneLevelBonusSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## Bonus

```
public int Bonus
```

### Field Value

int

## Id

```
public int Id
```

### Field Value

int

## RuneLevel

```
public int RuneLevel
```

### Field Value

int

# Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Class RuneListSheet

Namespace: Nekoyume.TableData.Rune

Assembly: Lib9c.dll

```
public class RuneListSheet : Sheet<int, RuneListSheet.Row>, IDictionary<int,
RuneListSheet.Row>, ICollection<KeyValuePair<int, RuneListSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneListSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, RuneListSheet.Row> ← RuneListSheet

### Implements

IDictionary<int, RuneListSheet.Row>,
ICollection<KeyValuePair<int, RuneListSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneListSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, RuneListSheet.Row>.Name , Sheet<int, RuneListSheet.Row>.OrderedList ,
Sheet<int, RuneListSheet.Row>.First , Sheet<int, RuneListSheet.Row>.Last ,
Sheet<int, RuneListSheet.Row>.Keys , Sheet<int, RuneListSheet.Row>.Values ,
Sheet<int, RuneListSheet.Row>.Count , Sheet<int, RuneListSheet.Row>.IsReadOnly ,
Sheet<int, RuneListSheet.Row>.this[int] ,
Sheet<int, RuneListSheet.Row>.Set(string, bool) ,
Sheet<int, RuneListSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneListSheet.Row>.GetEnumerator() ,
Sheet<int, RuneListSheet.Row>.TryGetValue(int, out RuneListSheet.Row, bool) ,
Sheet<int, RuneListSheet.Row>.AddRow(int, RuneListSheet.Row) ,
Sheet<int, RuneListSheet.Row>.Add(int, RuneListSheet.Row) ,
Sheet<int, RuneListSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneListSheet.Row>.Remove(int) ,
Sheet<int, RuneListSheet.Row>.TryGetValue(int, out RuneListSheet.Row) ,
Sheet<int, RuneListSheet.Row>.Add(KeyValuePair<int, RuneListSheet.Row>) ,
Sheet<int, RuneListSheet.Row>.Clear() ,
Sheet<int, RuneListSheet.Row>.Contains(KeyValuePair<int, RuneListSheet.Row>) ,
Sheet<int, RuneListSheet.Row>.CopyTo(KeyValuePair<int, RuneListSheet.Row>[], int) ,
Sheet<int, RuneListSheet.Row>.Remove(KeyValuePair<int, RuneListSheet.Row>) ,
Sheet<int, RuneListSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## RuneListSheet()

```
public RuneListSheet()
```

# Class RuneListSheet.Row

Namespace: Nekoyume.TableData.Rune

Assembly: Lib9c.dll

```
public class RuneListSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← RuneListSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## BonusCoef

```
public int BonusCoef
```

### Field Value

int

## Grade

```
public int Grade
```

### Field Value

int

## Id

```
public int Id
```

### Field Value

int

## RequiredLevel

```
public int RequiredLevel
```

### Field Value

int

## RuneType

```
public int RuneType
```

### Field Value

int

## UsePlace

```
public int UsePlace
```

### Field Value

int

## Properties

## Key

```
public override int Key { get; }
```

### Property Value

int

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Namespace Nekoyume.TableData.Stake

## Classes

StakePolicySheet

StakePolicySheet.Row

# Class StakePolicySheet

Namespace: Nekoyume.TableData.Stake

Assembly: Lib9c.dll

```
public class StakePolicySheet : Sheet<string, StakePolicySheet.Row>,
IDictionary<string, StakePolicySheet.Row>, ICollection<KeyValuePair<string,
StakePolicySheet.Row>>, IEnumerable<KeyValuePair<string, StakePolicySheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<string, StakePolicySheet.Row> ← StakePolicySheet

### Implements

IDictionary<string, StakePolicySheet.Row>,
ICollection<KeyValuePair<string, StakePolicySheet.Row>>,
IEnumerable<KeyValuePair<string, StakePolicySheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<string, StakePolicySheet.Row>.Name ,
Sheet<string, StakePolicySheet.Row>.OrderedList ,
Sheet<string, StakePolicySheet.Row>.First , Sheet<string, StakePolicySheet.Row>.Last ,
Sheet<string, StakePolicySheet.Row>.Keys , Sheet<string, StakePolicySheet.Row>.Values ,
Sheet<string, StakePolicySheet.Row>.Count ,
Sheet<string, StakePolicySheet.Row>.IsReadOnly ,
Sheet<string, StakePolicySheet.Row>.this[string] ,
Sheet<string, StakePolicySheet.Row>.Set(string, bool) ,
Sheet<string, StakePolicySheet.Row>.Set<T>(Sheet<string, T>, bool) ,
Sheet<string, StakePolicySheet.Row>.GetEnumerator() ,
Sheet<string, StakePolicySheet.Row>.TryGetValue(string, out StakePolicySheet.Row, bool) ,
Sheet<string, StakePolicySheet.Row>.AddRow(string, StakePolicySheet.Row) ,
Sheet<string, StakePolicySheet.Row>.Add(string, StakePolicySheet.Row) ,
Sheet<string, StakePolicySheet.Row>.ContainsKey(string) ,
Sheet<string, StakePolicySheet.Row>.Remove(string) ,
Sheet<string, StakePolicySheet.Row>.TryGetValue(string, out StakePolicySheet.Row) ,
Sheet<string, StakePolicySheet.Row>.Add(KeyValuePair<string, StakePolicySheet.Row>) ,
Sheet<string, StakePolicySheet.Row>.Clear() ,
Sheet<string, StakePolicySheet.Row>.Contains(KeyValuePair<string,
StakePolicySheet.Row>) ,

Sheet<string, StakePolicySheet.Row>.CopyTo(KeyValuePair<string, StakePolicySheet.Row>[], int) ,
Sheet<string, StakePolicySheet.Row>.Remove(KeyValuePair<string, StakePolicySheet.Row>) ,
Sheet<string, StakePolicySheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## StakePolicySheet()

```
public StakePolicySheet()
```

# Fields

## RequiredAttrNames

```
public static readonly string[] RequiredAttrNames
```

### Field Value

string[]

## SheetPrefixRules

```
public static readonly (string attrName, string value)[] SheetPrefixRules
```

### Field Value

(string attrName, string value)[]

# Properties

## LockupIntervalValue

```
public long LockupIntervalValue { get; }
```

### Property Value

long

## RewardIntervalValue

```
public long RewardIntervalValue { get; }
```

### Property Value

long

## StakeRegularFixedRewardSheetValue

```
public string StakeRegularFixedRewardSheetValue { get; }
```

### Property Value

string

## StakeRegularRewardSheetValue

```
public string StakeRegularRewardSheetValue { get; }
```

### Property Value

string

# Methods

# Set(string, bool)

```
public override void Set(string csv, bool isReversed = false)
```

## Parameters

`csv` string

`isReversed` bool

true: csv값의 column과 row가 뒤집혀서 작성되어 있다고 판단합니다.

## Exceptions

ArgumentNullException

InvalidDataException

# Class StakePolicySheet.Row

Namespace: Nekoyume.TableData.Stake

Assembly: Lib9c.dll

```
[Serializable]
public class StakePolicySheet.Row : SheetRow<string>
```

**Inheritance**

object ← SheetRow<string> ← StakePolicySheet.Row

**Inherited Members**

SheetRow<string>.EndOfSheetInitialize() , SheetRow<string>.Equals(object) , SheetRow<string>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Properties

## AttrName

```
public string AttrName { get; }
```

### Property Value

string

## Key

```
public override string Key { get; }
```

### Property Value

string

## Value

```
public string Value { get; }
```

### Property Value

string

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

## Validate()

```
public override void Validate()
```

# Namespace Nekoyume.TableData.Summon

## Classes

CostumeSummonSheet

EquipmentSummonSheet

RuneSummonSheet

SummonSheet

SummonSheet.Row

# Class CostumeSummonSheet

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class CostumeSummonSheet : SummonSheet, IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>, IEnumerable<KeyValuePair<int,
SummonSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SummonSheet.Row> ← SummonSheet ← CostumeSummonSheet

### Implements

IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, SummonSheet.Row>.Name , Sheet<int, SummonSheet.Row>.OrderedList ,
Sheet<int, SummonSheet.Row>.First , Sheet<int, SummonSheet.Row>.Last ,
Sheet<int, SummonSheet.Row>.Keys , Sheet<int, SummonSheet.Row>.Values ,
Sheet<int, SummonSheet.Row>.Count , Sheet<int, SummonSheet.Row>.IsReadOnly ,
Sheet<int, SummonSheet.Row>.this[int] ,
Sheet<int, SummonSheet.Row>.Set(string, bool) ,
Sheet<int, SummonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SummonSheet.Row>.GetEnumerator() ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row, bool) ,
Sheet<int, SummonSheet.Row>.AddRow(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.ContainsKey(int) ,
Sheet<int, SummonSheet.Row>.Remove(int) ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Clear() ,
Sheet<int, SummonSheet.Row>.Contains(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.CopyTo(KeyValuePair<int, SummonSheet.Row>[], int) ,
Sheet<int, SummonSheet.Row>.Remove(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Class EquipmentSummonSheet

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class EquipmentSummonSheet : SummonSheet, IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>, IEnumerable<KeyValuePair<int,
SummonSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SummonSheet.Row> ← SummonSheet ← EquipmentSummonSheet

### Implements

IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, SummonSheet.Row>.Name , Sheet<int, SummonSheet.Row>.OrderedList ,
Sheet<int, SummonSheet.Row>.First , Sheet<int, SummonSheet.Row>.Last ,
Sheet<int, SummonSheet.Row>.Keys , Sheet<int, SummonSheet.Row>.Values ,
Sheet<int, SummonSheet.Row>.Count , Sheet<int, SummonSheet.Row>.IsReadOnly ,
Sheet<int, SummonSheet.Row>.this[int] ,
Sheet<int, SummonSheet.Row>.Set(string, bool) ,
Sheet<int, SummonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SummonSheet.Row>.GetEnumerator() ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row, bool) ,
Sheet<int, SummonSheet.Row>.AddRow(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.ContainsKey(int) ,
Sheet<int, SummonSheet.Row>.Remove(int) ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Clear() ,
Sheet<int, SummonSheet.Row>.Contains(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.CopyTo(KeyValuePair<int, SummonSheet.Row>[], int) ,
Sheet<int, SummonSheet.Row>.Remove(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Class RuneSummonSheet

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class RuneSummonSheet : SummonSheet, IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>, IEnumerable<KeyValuePair<int,
SummonSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SummonSheet.Row> ← SummonSheet ← RuneSummonSheet

### Implements

IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet

### Inherited Members

Sheet<int, SummonSheet.Row>.Name , Sheet<int, SummonSheet.Row>.OrderedList ,
Sheet<int, SummonSheet.Row>.First , Sheet<int, SummonSheet.Row>.Last ,
Sheet<int, SummonSheet.Row>.Keys , Sheet<int, SummonSheet.Row>.Values ,
Sheet<int, SummonSheet.Row>.Count , Sheet<int, SummonSheet.Row>.IsReadOnly ,
Sheet<int, SummonSheet.Row>.this[int] ,
Sheet<int, SummonSheet.Row>.Set(string, bool) ,
Sheet<int, SummonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SummonSheet.Row>.GetEnumerator() ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row, bool) ,
Sheet<int, SummonSheet.Row>.AddRow(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.ContainsKey(int) ,
Sheet<int, SummonSheet.Row>.Remove(int) ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Clear() ,
Sheet<int, SummonSheet.Row>.Contains(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.CopyTo(KeyValuePair<int, SummonSheet.Row>[], int) ,
Sheet<int, SummonSheet.Row>.Remove(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Class SummonSheet

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class SummonSheet : Sheet<int, SummonSheet.Row>, IDictionary<int,
SummonSheet.Row>, ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet
```

### Inheritance

object ← Sheet<int, SummonSheet.Row> ← SummonSheet

### Implements

IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet

### Derived

CostumeSummonSheet, EquipmentSummonSheet, RuneSummonSheet

### Inherited Members

Sheet<int, SummonSheet.Row>.Name , Sheet<int, SummonSheet.Row>.OrderedList ,
Sheet<int, SummonSheet.Row>.First , Sheet<int, SummonSheet.Row>.Last ,
Sheet<int, SummonSheet.Row>.Keys , Sheet<int, SummonSheet.Row>.Values ,
Sheet<int, SummonSheet.Row>.Count , Sheet<int, SummonSheet.Row>.IsReadOnly ,
Sheet<int, SummonSheet.Row>.this[int] ,
Sheet<int, SummonSheet.Row>.Set(string, bool) ,
Sheet<int, SummonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SummonSheet.Row>.GetEnumerator() ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row, bool) ,
Sheet<int, SummonSheet.Row>.AddRow(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.ContainsKey(int) ,
Sheet<int, SummonSheet.Row>.Remove(int) ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Clear() ,
Sheet<int, SummonSheet.Row>.Contains(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.CopyTo(KeyValuePair<int, SummonSheet.Row>[], int) ,
Sheet<int, SummonSheet.Row>.Remove(KeyValuePair<int, SummonSheet.Row>) ,

Sheet<int, SummonSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SummonSheet()

```
public SummonSheet()
```

# Class SummonSheet.Row

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class SummonSheet.Row : SheetRow<int>
```

**Inheritance**

object ← SheetRow<int> ← SummonSheet.Row

**Inherited Members**

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Fields

## MaxRecipeCount

Maximum number of recipes allowed per summon group.

```
public const int MaxRecipeCount = 100
```

### Field Value

int

## Recipes

List of recipes available for this summon group. Each tuple contains (recipeId, ratio) where ratio determines the probability of selection.

```
public readonly List<(int, int)> Recipes
```

### Field Value

List<(int id, int count)>

# Properties

## CostMaterial

Material ID required for summoning.

```
public int CostMaterial { get; }
```

### Property Value

int

## CostMaterialCount

Quantity of material required for summoning.

```
public int CostMaterialCount { get; }
```

### Property Value

int

## CostNcg

NCG (Nine Chronicles Gold) cost for summoning.

```
public int CostNcg { get; }
```

### Property Value

int

## GroupId

Unique identifier for this summon group.

```
public int GroupId { get; }
```

### Property Value

int

## GuaranteeCount11

Number of items to guarantee for 11 summons. Null if not configured.

```
public int? GuaranteeCount11 { get; }
```

### Property Value

int?

## GuaranteeCount110

Number of items to guarantee for 110 summons. Null if not configured.

```
public int? GuaranteeCount110 { get; }
```

### Property Value

int?

## Key

```
public override int Key { get; }
```

### Property Value

int

## MinimumGrade11

Minimum grade to guarantee for 11 summons. Null if not configured.

```
public int? MinimumGrade11 { get; }
```

### Property Value

int?

## MinimumGrade110

Minimum grade to guarantee for 110 summons. Null if not configured.

```
public int? MinimumGrade110 { get; }
```

### Property Value

int?

# Methods

## CumulativeRatio(int)

Calculates the cumulative ratio up to the specified recipe index. Used for weighted random selection of recipes.

```
public int CumulativeRatio(int index)
```

### Parameters

`index` int

Recipe index (1-based)

### Returns

int

Cumulative ratio up to the specified index

### Exceptions

IndexOutOfRangeException

Thrown when index is out of valid range

## Set(IReadOnlyList<string>)

Sets the row data from CSV fields, supporting both legacy and new formats with grade guarantee settings. Legacy format: groupID,cost_material,cost_material_count,cost_ncg,recipe1ID,recipe1ratio,... New format: groupID,cost_material,cost_material_count,cost_ncg,GUARANTEE,min_grade_11,count_11,min_grade_110,count_110,recipe1ID,recipe1rati

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

CSV field values

## TotalRatio()

Calculates the total ratio of all recipes in this summon group.

```
public int TotalRatio()
```

### Returns

int

Sum of all recipe ratios

## UseGradeGuarantee(int)

Determines if grade guarantee is enabled for this summon group based on summon count. Returns true if guarantee settings are configured for the specified summon count. For 11 summons: checks if MinimumGrade11 and GuaranteeCount11 are configured. For 110 summons: checks if MinimumGrade110 and GuaranteeCount110 are configured. For other counts: returns false.

```
public bool UseGradeGuarantee(int summonCount)
```

### Parameters

`summonCount` int

Number of summons to check guarantee for

### Returns

bool

True if grade guarantee is configured for the specified summon count, false otherwise

# Namespace Nekoyume.TableData.Swap

## Classes

SwapRateSheet

Represents a sheet of swap rates.

SwapRateSheet.CurrencyPair

Represents a currency pair for swap.

SwapRateSheet.Fraction

Represents a fraction.

SwapRateSheet.Row

Represents a row of SwapRateSheet.

# Class SwapRateSheet

Namespace: Nekoyume.TableData.Swap

Assembly: Lib9c.dll

Represents a sheet of swap rates.

```
public class SwapRateSheet : Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>,
IDictionary<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>,
ICollection<KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>>,
IEnumerable<KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>>,
IEnumerable, ISheet
```

### Inheritance

object ← Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row> ← SwapRateSheet

### Implements

IDictionary<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>,
ICollection<KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>>,
IEnumerable<KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>>,
IEnumerable, ISheet

### Inherited Members

Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Name ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.OrderedList ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.First ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Last ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Keys ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Values ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Count ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.IsReadOnly ,
Sheet<SwapRateSheet.CurrencyPair,
SwapRateSheet.Row>.this[SwapRateSheet.CurrencyPair] ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Set(string, bool) ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Set<T>
(Sheet<SwapRateSheet.CurrencyPair, T>, bool) ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.GetEnumerator() ,
Sheet<SwapRateSheet.CurrencyPair,
SwapRateSheet.Row>.TryGetValue(SwapRateSheet.CurrencyPair, out SwapRateSheet.Row,
bool) ,

Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.AddRow(SwapRateSheet.CurrencyPair, SwapRateSheet.Row) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Add(SwapRateSheet.CurrencyPair, SwapRateSheet.Row) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.ContainsKey(SwapRateSheet.CurrencyPair) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Remove(SwapRateSheet.CurrencyPair) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.TryGetValue(SwapRateSheet.CurrencyPair, out SwapRateSheet.Row) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Add(KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Clear() , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Contains(KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.CopyTo(KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>[], int) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Remove(KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## SwapRateSheet()

Initializes a new instance of the SwapRateSheet class.

```
public SwapRateSheet()
```

# Class SwapRateSheet.CurrencyPair

Namespace: Nekoyume.TableData.Swap

Assembly: Lib9c.dll

Represents a currency pair for swap.

```
public class SwapRateSheet.CurrencyPair : IEquatable<SwapRateSheet.CurrencyPair>,
IComparable<SwapRateSheet.CurrencyPair>
```

**Inheritance**

object ← SwapRateSheet.CurrencyPair

**Implements**

IEquatable<SwapRateSheet.CurrencyPair>,
IComparable<SwapRateSheet.CurrencyPair>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### CurrencyPair(Currency, Currency)

Initializes a new instance of the SwapRateSheet.CurrencyPair class.

```
public CurrencyPair(Currency from, Currency to)
```

#### Parameters

`from` Currency

The currency to swap from.

`to` Currency

The currency to swap to.

# Properties

## From

The currency to swap from.

```
public Currency From { get; }
```

### Property Value

Currency

## To

The currency to swap to.

```
public Currency To { get; }
```

### Property Value

Currency

# Methods

## CompareTo(CurrencyPair?)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(SwapRateSheet.CurrencyPair? other)
```

### Parameters

`other` SwapRateSheet.CurrencyPair

An object to compare with this instance.

### Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

| Value | Meaning |
| --- | --- |
| **Less than zero** | This instance precedes `other` in the sort order. |
| **Zero** | This instance occurs in the same position in the sort order as `other`. |
| **Greater than zero** | This instance follows `other` in the sort order. |

## Equals(CurrencyPair?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(SwapRateSheet.CurrencyPair? other)
```

### Parameters

`other` SwapRateSheet.CurrencyPair

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

# Operators

## operator ==(CurrencyPair, CurrencyPair)

Determines whether two specified instances of SwapRateSheet.CurrencyPair are equal.

```
public static bool operator ==(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

### Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

The second SwapRateSheet.CurrencyPair to compare.

### Returns

bool

true if `left` and `right` represent the same value;

## operator >(CurrencyPair, CurrencyPair)

Determines whether one specified SwapRateSheet.CurrencyPair is greater than another specified SwapRateSheet.CurrencyPair.

```
public static bool operator >(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

### Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

The second SwapRateSheet.CurrencyPair to compare.

### Returns

bool

true if `left` is greater than `right`;

## operator >=(CurrencyPair, CurrencyPair)

Determines whether one specified SwapRateSheet.CurrencyPair is greater than or equal to another specified SwapRateSheet.CurrencyPair.

```
public static bool operator >=(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

### Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

The second SwapRateSheet.CurrencyPair to compare.

### Returns

bool

true if `left` is greater than or equal to `right`;

## operator !=(CurrencyPair, CurrencyPair)

Determines whether two specified instances of SwapRateSheet.CurrencyPair are not equal.

```
public static bool operator !=(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

### Parameters

`left` SwapRateSheet.CurrencyPair

`right` SwapRateSheet.CurrencyPair

### Returns

bool

## operator <(CurrencyPair, CurrencyPair)

Determines whether one specified SwapRateSheet.CurrencyPair is less than another specified SwapRateSheet.CurrencyPair.

```
public static bool operator <(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

### Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

### Returns

bool

## operator <=(CurrencyPair, CurrencyPair)

Determines whether one specified SwapRateSheet.CurrencyPair is less than or equal to another specified SwapRateSheet.CurrencyPair.

```
public static bool operator <=(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

### Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

The second SwapRateSheet.CurrencyPair to compare.

### Returns

bool

true if `left` is less than or equal to `right`;

# Class SwapRateSheet.Fraction

Namespace: Nekoyume.TableData.Swap

Assembly: Lib9c.dll

Represents a fraction.

```
public class SwapRateSheet.Fraction : IEquatable<SwapRateSheet.Fraction>
```

**Inheritance**

object ← SwapRateSheet.Fraction

**Implements**

IEquatable<SwapRateSheet.Fraction>

**Inherited Members**

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### Fraction(BigInteger, BigInteger)

Initializes a new instance of the SwapRateSheet.Fraction class.

```
public Fraction(BigInteger numerator, BigInteger denominator)
```

#### Parameters

`numerator` BigInteger

The numerator of the fraction.

`denominator` BigInteger

The denominator of the fraction.

## Properties

## Denominator

The denominator of the fraction.

```
public BigInteger Denominator { get; }
```

### Property Value

BigInteger

## Numerator

The numerator of the fraction.

```
public BigInteger Numerator { get; }
```

### Property Value

BigInteger

# Methods

## Equals(Fraction?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(SwapRateSheet.Fraction? other)
```

### Parameters

`other` SwapRateSheet.Fraction

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

# Class SwapRateSheet.Row

Namespace: Nekoyume.TableData.Swap

Assembly: Lib9c.dll

Represents a row of SwapRateSheet.

```
public class SwapRateSheet.Row : SheetRow<SwapRateSheet.CurrencyPair>
```

**Inheritance**

object ← SheetRow<SwapRateSheet.CurrencyPair> ← SwapRateSheet.Row

**Inherited Members**

SheetRow<SwapRateSheet.CurrencyPair>.Validate() ,
SheetRow<SwapRateSheet.CurrencyPair>.EndOfSheetInitialize() ,
SheetRow<SwapRateSheet.CurrencyPair>.Equals(object) ,
SheetRow<SwapRateSheet.CurrencyPair>.GetHashCode() , object.Equals(object, object) ,
object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) ,
object.ToString()

# Properties

## Key

```
public override SwapRateSheet.CurrencyPair Key { get; }
```

### Property Value

SwapRateSheet.CurrencyPair

## Pair

The currency pair for swap.

```
public SwapRateSheet.CurrencyPair? Pair { get; }
```

### Property Value

SwapRateSheet.CurrencyPair

## Rate

The swap rate represented by SwapRateSheet.Fraction.

```
public SwapRateSheet.Fraction? Rate { get; }
```

### Property Value

SwapRateSheet.Fraction

# Methods

## Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

### Parameters

`fields` IReadOnlyList<string>

# Namespace Nekoyume.TypedAddress

## Structs

AgentAddress

GuildAddress

PledgeAddress

# Struct AgentAddress

Namespace: Nekoyume.TypedAddress

Assembly: Lib9c.dll

```
public readonly struct AgentAddress : IBencodable
```

**Implements**
IBencodable

**Inherited Members**
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

**Extension Methods**
AgentAddressExtensions.GetPledgeAddress(AgentAddress)

## Constructors

### AgentAddress(IValue)

```
public AgentAddress(IValue value)
```

#### Parameters

`value` IValue

### AgentAddress(Address)

```
public AgentAddress(Address address)
```

#### Parameters

`address` Address

## AgentAddress(byte[])

```
public AgentAddress(byte[] bytes)
```

### Parameters

`bytes` byte[]

## AgentAddress(string)

```
public AgentAddress(string hex)
```

### Parameters

`hex` string

# Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

# Methods

## Equals(AgentAddress)

```
public bool Equals(AgentAddress other)
```

### Parameters

`other` AgentAddress

### Returns

bool

## Equals(object)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current instance.

### Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

## GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

### Returns

int

A 32-bit signed integer that is the hash code for this instance.

## ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

### Returns

string

The fully qualified type name.

# Operators

## operator ==(AgentAddress, AgentAddress)

```
public static bool operator ==(AgentAddress left, AgentAddress right)
```

### Parameters

`left` AgentAddress

`right` AgentAddress

### Returns

bool

## implicit operator Address(AgentAddress)

```
public static implicit operator Address(AgentAddress agentAddress)
```

### Parameters

`agentAddress` AgentAddress

### Returns

Address

## operator !=(AgentAddress, AgentAddress)

```
public static bool operator !=(AgentAddress left, AgentAddress right)
```

### Parameters

`left` AgentAddress

`right` AgentAddress

### Returns

bool

# Struct GuildAddress

Namespace: Nekoyume.TypedAddress

Assembly: Lib9c.dll

```
public readonly struct GuildAddress : IBencodable, IEquatable<GuildAddress>
```

**Implements**
IBencodable, IEquatable<GuildAddress>

**Inherited Members**
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### GuildAddress(IValue)

```
public GuildAddress(IValue value)
```

#### Parameters

`value` IValue

### GuildAddress(Address)

```
public GuildAddress(Address address)
```

#### Parameters

`address` Address

### GuildAddress(byte[])

```
public GuildAddress(byte[] bytes)
```

### Parameters

`bytes` byte[]

## GuildAddress(string)

```
public GuildAddress(string hex)
```

### Parameters

`hex` string

# Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

# Methods

## Equals(GuildAddress)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GuildAddress other)
```

### Parameters

`other` GuildAddress

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current instance.

### Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

## GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

### Returns

int

A 32-bit signed integer that is the hash code for this instance.

## ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

### Returns

string

The fully qualified type name.

# Operators

## operator ==(GuildAddress, GuildAddress)

```
public static bool operator ==(GuildAddress left, GuildAddress right)
```

### Parameters

`left` GuildAddress

`right` GuildAddress

### Returns

bool

## implicit operator Address(GuildAddress)

```
public static implicit operator Address(GuildAddress guildAddress)
```

### Parameters

`guildAddress` GuildAddress

### Returns

Address

## operator !=(GuildAddress, GuildAddress)

```
public static bool operator !=(GuildAddress left, GuildAddress right)
```

### Parameters

`left` GuildAddress

`right` GuildAddress

### Returns

bool

# Struct PledgeAddress

Namespace: Nekoyume.TypedAddress

Assembly: Lib9c.dll

```
public readonly struct PledgeAddress : IBencodable
```

**Implements**
IBencodable

**Inherited Members**
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

## Constructors

### PledgeAddress(IValue)

```
public PledgeAddress(IValue value)
```

#### Parameters

`value` IValue

### PledgeAddress(Address)

```
public PledgeAddress(Address address)
```

#### Parameters

`address` Address

### PledgeAddress(byte[])

```
public PledgeAddress(byte[] bytes)
```

### Parameters

`bytes` byte[]

## PledgeAddress(string)

```
public PledgeAddress(string hex)
```

### Parameters

`hex` string

# Properties

## Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

### Property Value

IValue

### Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

# Methods

## Equals(PledgeAddress)

```
public bool Equals(PledgeAddress other)
```

### Parameters

`other` PledgeAddress

### Returns

bool

## Equals(object)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object obj)
```

### Parameters

`obj` object

The object to compare with the current instance.

### Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

## GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

### Returns

int

A 32-bit signed integer that is the hash code for this instance.

## ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

### Returns

string

The fully qualified type name.

# Operators

## operator ==(PledgeAddress, PledgeAddress)

```
public static bool operator ==(PledgeAddress left, PledgeAddress right)
```

### Parameters

`left` PledgeAddress

`right` PledgeAddress

### Returns

bool

## implicit operator Address(PledgeAddress)

```
public static implicit operator Address(PledgeAddress agentAddress)
```

### Parameters

`agentAddress` PledgeAddress

### Returns

Address

## operator !=(PledgeAddress, PledgeAddress)

```
public static bool operator !=(PledgeAddress left, PledgeAddress right)
```

### Parameters

`left` PledgeAddress

`right` PledgeAddress

### Returns

bool

# Namespace Nekoyume.Validator Delegation

## Classes

AbstainHistory

PayMaster

PayMaster is a contract that manages the proxy of mortgage and refund of gas.

ProposerInfo

ValidatorComparer

ValidatorDelegatee

ValidatorDelegator

ValidatorList

ValidatorRepository

# Class AbstainHistory

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class AbstainHistory
```

**Inheritance**

object ← AbstainHistory

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### AbstainHistory()

```
public AbstainHistory()
```

### AbstainHistory(IValue)

```
public AbstainHistory(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### AbstainHistory(List)

```
public AbstainHistory(List bencoded)
```

#### Parameters

`bencoded` List

# Properties

## Address

```
public static Address Address { get; }
```

### Property Value

Address

## Bencoded

```
public IValue Bencoded { get; }
```

### Property Value

IValue

## History

```
public SortedDictionary<PublicKey, List<long>> History { get; }
```

### Property Value

SortedDictionary<PublicKey, List<long>>

## MaxAbstainAllowance

Maximum abstain allowance to slash and jail.

```
public static int MaxAbstainAllowance { get; }
```

### Property Value

int

## WindowSize

```
public static int WindowSize { get; }
```

### Property Value

int

# Methods

## FindToSlashAndAdd(IEnumerable<PublicKey>, long)

```
public List<PublicKey> FindToSlashAndAdd(IEnumerable<PublicKey> abstainList,
long height)
```

### Parameters

`abstainList` IEnumerable<PublicKey>

`height` long

### Returns

List<PublicKey>

# Class PayMaster

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

PayMaster is a contract that manages the proxy of mortgage and refund of gas.

```
public static class PayMaster
```

**Inheritance**

object ← PayMaster

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## Mortgage(IWorld, IActionContext, Address, Address, FungibleAssetValue)

Mortgage gas with the PayMaster.

```
public static IWorld Mortgage(this IWorld world, IActionContext context, Address
address, Address payMaster, FungibleAssetValue gasToMortgage)
```

### Parameters

`world` IWorld

The Libplanet.Action.State.IWorld where the mortgage happen.

`context` IActionContext

The Libplanet.Action.IActionContext where the mortgage happen.

`address` Address

The Libplanet.Crypto.Address of the user to mortgage gas.

`payMaster` Address

The Libplanet.Crypto.Address of the paymaster.

`gasToMortgage` FungibleAssetValue

The Libplanet.Types.Assets.FungibleAssetValue of gas to mortgage.

### Returns

IWorld

The updated Libplanet.Action.State.IWorld.

## Refund(IWorld, IActionContext, Address, FungibleAssetValue)

Refund gas with the PayMaster.

```
public static IWorld Refund(this IWorld world, IActionContext context, Address
address, FungibleAssetValue gasToRefund)
```

### Parameters

`world` IWorld

The Libplanet.Action.State.IWorld where the refund happen.

`context` IActionContext

The Libplanet.Action.IActionContext where the refund happen.

`address` Address

The Libplanet.Crypto.Address of the user to refund gas.

`gasToRefund` FungibleAssetValue

The Libplanet.Types.Assets.FungibleAssetValue of gas to refund.

### Returns

## IWorld

The updated Libplanet.Action.State.IWorld.

# Class ProposerInfo

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public class ProposerInfo
```

**Inheritance**

object ← ProposerInfo

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ProposerInfo(IValue)

```
public ProposerInfo(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### ProposerInfo(List)

```
public ProposerInfo(List bencoded)
```

#### Parameters

`bencoded` List

### ProposerInfo(long, Address)

```
public ProposerInfo(long blockIndex, Address proposer)
```

### Parameters

`blockIndex` long

`proposer` Address

## Properties

### Address

```
public static Address Address { get; }
```

#### Property Value

Address

### Bencoded

```
public IValue Bencoded { get; }
```

#### Property Value

IValue

### BlockIndex

```
public long BlockIndex { get; }
```

#### Property Value

long

# Proposer

```
public Address Proposer { get; }
```

## Property Value

Address

# Class ValidatorComparer

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public class ValidatorComparer : IComparer<Validator>
```

**Inheritance**

object ← ValidatorComparer

**Implements**

IComparer<Validator>

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

# Methods

## Compare(Validator?, Validator?)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public int Compare(Validator? x, Validator? y)
```

### Parameters

x Validator

The first object to compare.

y Validator

The second object to compare.

### Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

| Value | Meaning |
| --- | --- |
| **Less than zero** | x is less than y. |
| **Zero** | x equals y. |
| **Greater than zero** | x is greater than y. |

# Class ValidatorDelegatee

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class ValidatorDelegatee : Delegatee<ValidatorRepository,
ValidatorDelegatee, ValidatorDelegator>, IDelegatee, IEquatable<ValidatorDelegatee>,
IBencodable
```

### Inheritance

object ← Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator> ← ValidatorDelegatee

### Implements

IDelegatee, IEquatable<ValidatorDelegatee>, IBencodable

### Inherited Members

Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Metadata ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Repository ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Address ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.AccountAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.MetadataAddress
,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.DelegationCurrency ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RewardCurrencies
,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.DelegationPoolAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.RewardPoolAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.RewardRemainderPoolAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.SlashedPoolAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondingPeriod
,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.MaxUnbondLockInEntries ,

Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.MaxRebondGraceEntries ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.TotalDelegated ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.TotalShares ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Jailed ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.JailedUntil ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Tombstoned ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.MetadataBencoded ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.ShareFromFAV(FungibleAssetValue) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.FAVFromShare(BigInteger) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Jail(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Unjail(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Tombstone() ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.BondAddress(Address) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondLockInAddress(Address) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RebondGraceAddress(Address) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DistributionPoolAddress() ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.CurrentRewardBaseAddress() ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RewardBaseAddress(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.CurrentLumpSumRewardsRecordAddress() ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.LumpSumRewardsRecordAddress(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Bond(ValidatorDelegator, FungibleAssetValue, long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Unbond(ValidatorDelegator, BigInteger, long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DistributeReward(ValidatorDelegator, long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.CollectRewards(long) ,

Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Slash(BigInteger, long, long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.StartNewRewardPeriod(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.CalculateRewards(BigInteger, RewardBase, RewardBase) ,
object.Equals(object, object) , object.GetType() ,
object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ValidatorDelegatee(Address, IValue, ValidatorRepository)

```
public ValidatorDelegatee(Address address, IValue bencoded,
ValidatorRepository repository)
```

### Parameters

`address` Address

`bencoded` IValue

`repository` ValidatorRepository

## ValidatorDelegatee(Address, List, ValidatorRepository)

```
public ValidatorDelegatee(Address address, List bencoded, ValidatorRepository
repository)
```

### Parameters

`address` Address

`bencoded` List

`repository` ValidatorRepository

# ValidatorDelegatee(Address, PublicKey, BigInteger, long, IEnumerable<Currency>, ValidatorRepository)

```
public ValidatorDelegatee(Address address, PublicKey publicKey, BigInteger
commissionPercentage, long creationHeight, IEnumerable<Currency> rewardCurrencies,
ValidatorRepository repository)
```

## Parameters

`address` Address

`publicKey` PublicKey

`commissionPercentage` BigInteger

`creationHeight` long

`rewardCurrencies` IEnumerable<Currency>

`repository` ValidatorRepository

# Properties

## ActiveDelegationPoolAddress

```
public static Address ActiveDelegationPoolAddress { get; }
```

### Property Value

Address

## BaseProposerRewardPercentage

```
public static BigInteger BaseProposerRewardPercentage { get; }
```

### Property Value

BigInteger

## Bencoded

```
public List Bencoded { get; }
```

### Property Value

List

## BonusProposerRewardPercentage

```
public static BigInteger BonusProposerRewardPercentage { get; }
```

### Property Value

BigInteger

## CommissionPercentage

```
public BigInteger CommissionPercentage { get; }
```

### Property Value

BigInteger

## CommissionPercentageLastUpdateHeight

```
public long CommissionPercentageLastUpdateHeight { get; }
```

### Property Value

long

## CommissionPercentageMaxChange

```
public static BigInteger CommissionPercentageMaxChange { get; }
```

### Property Value

BigInteger

## CommissionPercentageUpdateCooldown

```
public static long CommissionPercentageUpdateCooldown { get; }
```

### Property Value

long

## DefaultCommissionPercentage

```
public static BigInteger DefaultCommissionPercentage { get; }
```

### Property Value

BigInteger

## DefaultUnbondingPeriod

```
public static long DefaultUnbondingPeriod { get; }
```

### Property Value

long

## InactiveDelegationPoolAddress

```
public static Address InactiveDelegationPoolAddress { get; }
```

### Property Value

Address

## IsActive

```
public bool IsActive { get; }
```

### Property Value

bool

## MaxCommissionPercentage

```
public static BigInteger MaxCommissionPercentage { get; }
```

### Property Value

BigInteger

## MinCommissionPercentage

```
public static BigInteger MinCommissionPercentage { get; }
```

### Property Value

BigInteger

## MinSelfDelegation

```
public FungibleAssetValue MinSelfDelegation { get; }
```

### Property Value

FungibleAssetValue

## Power

```
public BigInteger Power { get; }
```

### Property Value

BigInteger

## PublicKey

```
public PublicKey PublicKey { get; }
```

### Property Value

PublicKey

## Validator

```
public Validator Validator { get; }
```

### Property Value

Validator

## ValidatorDelegationCurrency

```
public static Currency ValidatorDelegationCurrency { get; }
```

### Property Value

Currency

## ValidatorMaxRebondGraceEntries

```
public static int ValidatorMaxRebondGraceEntries { get; }
```

### Property Value

int

## ValidatorMaxUnbondLockInEntries

```
public static int ValidatorMaxUnbondLockInEntries { get; }
```

### Property Value

int

# Methods

## Activate()

```
public void Activate()
```

## AllocateReward(FungibleAssetValue, BigInteger, BigInteger, Address, long)

```
public void AllocateReward(FungibleAssetValue rewardToAllocate, BigInteger
```

```
validatorPower, BigInteger validatorSetPower, Address RewardSource, long height)
```

### Parameters

`rewardToAllocate` FungibleAssetValue

`validatorPower` BigInteger

`validatorSetPower` BigInteger

`RewardSource` Address

`height` long

## Deactivate()

```
public void Deactivate()
```

## Equals(IDelegatee?)

```
public bool Equals(IDelegatee? other)
```

### Parameters

`other` IDelegatee

### Returns

bool

## Equals(ValidatorDelegatee?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(ValidatorDelegatee? other)
```

### Parameters

`other` ValidatorDelegatee

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

### Returns

int

A hash code for the current object.

## OnDelegationChanged(long)

```
protected override void OnDelegationChanged(long height)
```

### Parameters

`height` long

## OnEnjailed()

```
protected override void OnEnjailed()
```

## OnUnjailed()

```
protected override void OnUnjailed()
```

## SetCommissionPercentage(BigInteger, long)

```
public void SetCommissionPercentage(BigInteger percentage, long height)
```

### Parameters

`percentage` BigInteger

`height` long

## Unjail(long)

```
public void Unjail(long height)
```

## Parameters

`height` long

# Class ValidatorDelegator

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class ValidatorDelegator : Delegator<ValidatorRepository,
ValidatorDelegatee, ValidatorDelegator>, IDelegator, IEquatable<ValidatorDelegator>
```

### Inheritance

object ← Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator> ← ValidatorDelegator

### Implements

IDelegator, IEquatable<ValidatorDelegator>

### Inherited Members

Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Metadata ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Repository ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Address ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.AccountAddress ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.MetadataAddress
,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.DelegationPoolAddress ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RewardAddress ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Delegatees ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondingRefs ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.MetadataBencoded ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.Delegate(ValidatorDelegatee, FungibleAssetValue, long) ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.Undelegate(ValidatorDelegatee, BigInteger, long) ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.Redelegate(ValidatorDelegatee, ValidatorDelegatee, BigInteger, long) ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.CancelUndelegate(ValidatorDelegatee, FungibleAssetValue, long) ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.ClaimReward(ValidatorDelegatee, long) ,

Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.ReleaseUnbondings(long) , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

# Constructors

## ValidatorDelegator(Address, Address, ValidatorRepository)

```
public ValidatorDelegator(Address address, Address delegationPoolAddress,
ValidatorRepository repository)
```

### Parameters

`address` Address

`delegationPoolAddress` Address

`repository` ValidatorRepository

## ValidatorDelegator(Address, ValidatorRepository)

```
public ValidatorDelegator(Address address, ValidatorRepository repository)
```

### Parameters

`address` Address

`repository` ValidatorRepository

# Methods

## Equals(ValidatorDelegator?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(ValidatorDelegator? other)
```

### Parameters

`other` ValidatorDelegator

An object to compare with this object.

### Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

## Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

### Parameters

`obj` object

The object to compare with the current object.

### Returns

bool

true if the specified object is equal to the current object; otherwise, false.

## GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

## Returns

int

A hash code for the current object.

# Class ValidatorList

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class ValidatorList : IBencodable
```

**Inheritance**

object ← ValidatorList

**Implements**

IBencodable

**Inherited Members**

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

## Constructors

### ValidatorList()

```
public ValidatorList()
```

### ValidatorList(IValue)

```
public ValidatorList(IValue bencoded)
```

#### Parameters

`bencoded` IValue

### ValidatorList(List)

```
public ValidatorList(List bencoded)
```

### Parameters

`bencoded` List

## Properties

### Address

```
public static Address Address { get; }
```

#### Property Value

Address

### Bencoded

```
public List Bencoded { get; }
```

#### Property Value

List

### MaxActiveSetSize

```
public static int MaxActiveSetSize { get; }
```

#### Property Value

int

### Validators

```
public ImmutableList<Validator> Validators { get; }
```

### Property Value

ImmutableList<Validator>

## Methods

### ActiveSet()

```
public List<Validator> ActiveSet()
```

#### Returns

List<Validator>

### InActiveSet()

```
public List<Validator> InActiveSet()
```

#### Returns

List<Validator>

### RemoveValidator(PublicKey)

```
public ValidatorList RemoveValidator(PublicKey publicKey)
```

#### Parameters

`publicKey` PublicKey

#### Returns

ValidatorList

# SetValidator(Validator)

```
public ValidatorList SetValidator(Validator validator)
```

## Parameters

`validator` Validator

## Returns

ValidatorList

# Class ValidatorRepository

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class ValidatorRepository : DelegationRepository<ValidatorRepository,
ValidatorDelegatee, ValidatorDelegator>, IDelegationRepository
```

### Inheritance

object ←

DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator> ← ValidatorRepository

### Implements

IDelegationRepository

### Inherited Members

DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.World ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.ActionContext ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DelegateeAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DelegatorAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DelegateeMetadataAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DelegatorMetadataAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.BondAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondLockInAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RebondGraceAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondingSetAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RewardBaseAccountAddress ,

DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.LumpSumRewardsRecordAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetDelegatee(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetDelegator(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetDelegatee(ValidatorDelegatee) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetDelegator(ValidatorDelegator) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetDelegateeMetadata(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetDelegatorMetadata(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetBond(ValidatorDelegatee, Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetUnbondLockIn(ValidatorDelegatee, Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetUnlimitedUnbondLockIn(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetRebondGrace(ValidatorDelegatee, Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetUnlimitedRebondGrace(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetCurrentRewardBase(ValidatorDelegatee) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetRewardBase(ValidatorDelegatee, long) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetLumpSumRewardsRecord(ValidatorDelegatee, long) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetCurrentLumpSumRewardsRecord(ValidatorDelegatee) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetBalance(Address, Currency) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetDelegateeMetadata(DelegateeMetadata) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetDelegatorMetadata(DelegatorMetadata) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetBond(Bond) ,

DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetUnbondLockIn(UnbondLockIn) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetRebondGrace(RebondGrace) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetRewardBase(RewardBase) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetLumpSumRewardsRecord(LumpSumRewardsRecord) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RemoveLumpSumRewardsRecord(LumpSumRewardsRecord) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.TransferAsset(Address, Address, FungibleAssetValue) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UpdateWorld(IWorld) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

**Extension Methods**

AbstainHistoryModule.GetAbstainHistory(ValidatorRepository) , AbstainHistoryModule.SetAbstainHistory(ValidatorRepository, AbstainHistory) , ValidatorDelegateeModule.CreateDelegatee(ValidatorRepository, PublicKey, BigInteger) , ValidatorDelegateeModule.TryGetDelegatee(ValidatorRepository, Address, out ValidatorDelegatee?) , ValidatorDelegatorModule.TryGetDelegator(ValidatorRepository, Address, out ValidatorDelegator?) , ValidatorListModule.GetProposerInfo(ValidatorRepository) , ValidatorListModule.GetValidatorList(ValidatorRepository) , ValidatorListModule.SetProposerInfo(ValidatorRepository, ProposerInfo) , ValidatorListModule.TryGetProposerInfo(ValidatorRepository, Address, out ProposerInfo?) , ValidatorListModule.TryGetValidatorList(ValidatorRepository, out ValidatorList?)

# Constructors

## ValidatorRepository(IWorld, IActionContext)

```
public ValidatorRepository(IWorld world, IActionContext actionContext)
```

### Parameters

`world` IWorld

`actionContext` IActionContext

## ValidatorRepository(IDelegationRepository)

```
public ValidatorRepository(IDelegationRepository repository)
```

### Parameters

`repository` IDelegationRepository

# Properties

## World

```
public override IWorld World { get; }
```

### Property Value

IWorld

# Methods

## GetDelegatee(Address)

```
public override ValidatorDelegatee GetDelegatee(Address address)
```

### Parameters

`address` Address

### Returns

ValidatorDelegatee

## GetDelegator(Address)

```
public override ValidatorDelegator GetDelegator(Address address)
```

### Parameters

`address` Address

### Returns

ValidatorDelegator

## GetValidatorList()

```
public ValidatorList GetValidatorList()
```

### Returns

ValidatorList

## SetCommissionPercentage(Address, BigInteger, long)

```
public void SetCommissionPercentage(Address address, BigInteger
commissionPercentage, long height)
```

### Parameters

`address` Address

`commissionPercentage` BigInteger

`height` long

## SetDelegatee(ValidatorDelegatee)

```
public override void SetDelegatee(ValidatorDelegatee delegatee)
```

### Parameters

`delegatee` ValidatorDelegatee

## SetDelegator(ValidatorDelegator)

```
public override void SetDelegator(ValidatorDelegator delegator)
```

### Parameters

`delegator` ValidatorDelegator

## SetValidatorList(ValidatorList)

```
public void SetValidatorList(ValidatorList validatorList)
```

### Parameters

`validatorList` ValidatorList

## UpdateWorld(IWorld)

```
public override void UpdateWorld(IWorld world)
```

### Parameters

`world` IWorld